

CHAPTER 9

ELECTRICAL SYSTEMS

CHAPTER OVERVIEW

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SECTION 9-1

EQUIPMENT DESCRIPTION AND DATA

SECTION OVERVIEW

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9-1-1 AC ELECTRICAL SYSTEM DESCRIPTION AND DATA.

9-1-1.1 AC Electrical System Description.

The ac electrical system consists of primary ac power, auxiliary ac power, and external ac power (Figures 9-1-1-1 and 9-1-1-2). Primary ac power is provided by two independent generating systems, each consisting of a brushless generator, generator control unit (GCU), current transformer, and generator control switch. Auxiliary ac power is provided by a single generating system consisting of a brushless generator, generator control unit, current transformer and generator control switch. External ac power is supplied to the helicopter through an external power receptacle. It is controlled by an external power monitor panel and external power switch. AC power is distributed to the helicopter buses from the generating systems and external ac source through a series of contactors and relays. The system interfaces with the caution/advisory warning system where caution and advisory capsules monitor electrical system operation and status.

9-1-1.2 Generator Control Units.

The No. 1 and APU generator control units, on the upper left cabin, and the No. 2 generator control unit, on the upper right cabin, continuously monitor and regulate generator ac output to provide voltage regulation, current regulation, overvoltage protection, undervoltage protection, feeder fault protection, and underfrequency protection. Voltage and current regulation, overvoltage, undervoltage, and feeder fault protection are provided during all flight and ground operations. Underfrequency protection by the No. 1 and No. 2 generator control units is provided only during ground operation.

Power for the generator control unit and for generator field excitation is provided by the permanent magnet generator portion of the generator whenever the generator is turning. By controlling generator excitation, the voltage regulator maintains a preset voltage and current level of generator ac output under varying load conditions. The regulator senses the generator variations, amplifies the changes, and then varies the average current flow to the exciter control field. If the generator output level is above the preset level, a

decreased current flow is applied to the exciter control field. If the generator output is below the preset level, an increased current flow is applied to the exciter control field.

When the generator three-phase output rises above 124 to 126 vac, the overvoltage protection circuit senses this and, after a marginal time delay period to prevent nuisance trips, disables and disconnects the generator from ac loads. The length of time the overvoltage condition is permitted to exist is inversely proportional to the magnitude of the overvoltage condition.

When any single-phase voltage drops below 95 to 105 vac, the undervoltage protection circuit senses this and initially provides a marginal time delay period to prevent nuisance trips. If the voltage remains low for five to seven seconds, the protection circuit disables and disconnects the generator from the ac loads.

If the amount of current flow in the generator feeder lines differs from that in the generator main windings as monitored by two sets of current transformers, the difference is, most likely, the result of a leakage path to ground. The feeder fault protection circuit senses this differential current and immediately disables and disconnects the affected generator from the ac loads.

During ground operations, if the frequency of phase C of the permanent magnet generator drops below 1110 to 1130 Hz (equivalent to 370 to 380 Hz from main generator), the underfrequency protection circuit senses this and initially provides a marginal time delay to prevent nuisance trips. If the underfrequency condition is sustained for a period of one to three seconds, the protection circuit disables and disconnects the generator from the ac loads. Should the frequency rise to the acceptable level and 12 Hz (4 Hz from main generator) above the dropout frequency, the generator is automatically re-energized and reconnected to the ac loads. During flight operations, the underfrequency protection circuits in the No. 1 and No. 2 generator control units are intentionally disabled through action of the landing gear drag beam

switch. The right drag beam switch provides the underfrequency disable signal.

9-1-1.3 Generator Control Switches.

Generator outputs are connected to their associated generator control units and the ac primary bus contactors by the three-position control switches on the upper console (GENERATORS NO. 1, NO. 2, and APU). The ON position energizes the associated generator control unit (GCU) relay circuitry and the associated ac contactor to connect the generator main output to the primary buses. The TEST position disconnects the main output from the buses, but allows the GCU to control the generator output so that output can be monitored at the test connector. OFF/RESET either de-energizes the generator or permits generator recycling if the generator is disabled and disconnected from its loads. The control switch is manually positioned to RESET and then back to ON.

9-1-1.4 Junction Boxes.

The No. 1 junction box, on the upper left cabin, and the No. 2 junction box, on the upper right cabin, provide mounting space for ac power system components. The No. 1 junction box contains the No. 1 generator contactor, current transformer, current limiters, and a test receptacle. The No. 2 junction box contains the No. 2 generator contactor, APU/external power contactor, ac bus tie contactor, current transformers, current limiters, and test receptacles.

9-1-1.5 Generator Contactors.

The No. 1 and No. 2 generator contactors, on the No. 1 and No. 2 junction boxes, respectively, provide connections from the No. 1 and No. 2 generator three-phase feeder lines to the No. 1 and No. 2 ac primary buses for distribution of ac power. With the generator control switches placed ON, dc voltage from the generator control units is applied through the control switches to energize the generator contactor solenoids. Energized auxiliary contacts of No. 1 or No. 2 generator contactors complete the voltage path to the No. 2 hydraulic logic module when backup pump operation is required. Two sets of normally-open contacts

provide dual generators on voltage signals to the blade de-icing system (if installed).

When de-energized, auxiliary contacts of the contactors complete the dc voltage path to energize the ac bus tie contactor. This applies APU generator ac power or external ac power, through the de-energized main contacts of the generator contactors to the ac primary buses. The de-energized main contacts also provide a path to transfer power from a working generator to the ac primary bus of a failed generator. De-energized auxiliary contacts complete the dc voltage paths to the caution/advisory #1 GEN and #2 GEN capsules. Another set of normally-closed contacts provides a dual generators off voltage signal to the blade de-icing system (if installed).

9-1-1.6 APU/External Power Contactors.

The APU/external power contactor, on the No. 2 junction box, provides connection from the APU generator three-phase feeder lines or the ac external power receptacle to the ac primary buses for distribution. With the APU generator control switch placed ON, dc voltage from the APU generator control unit is applied through the control switch to energize the APU/external power contactor solenoid. The energized auxiliary contacts of the contactor are used to complete the dc voltage path from the generator control unit through the ac bus tie contactor to the caution/advisory panel APU GEN ON capsule. Voltage is also supplied to the environmental control system to energize the APU on relay K95. Additional auxiliary energized contacts of the APU/external power contactor provide paths which, through other contacts, supply voltage to the blade de-icing system (if installed). This voltage is also supplied to the secondary bus contactor when auxiliary power is required.

With external ac power connected to the helicopter, de-energized main contacts of the APU/external power contactor apply the external ac power to the energized main contacts of the ac bus tie contactor for distribution to the ac primary buses. Another set of normally-closed contacts, when de-energized, provides a path to energize the ac secondary bus contactor during single main generator operation.

9-1-1-2

9-1-1.7 AC Bus Tie Contactor.

The ac bus tie contactor, on the No. 2 junction box, provides three-phase ac power from the APU generator or external ac power receptacle to the ac primary buses. With the APU generator control switch placed ON, the ac bus tie contactor solenoid is energized by dc voltage from the APU generator control unit. This is accomplished through the de-energized auxiliary contacts of the No. 1 and No. 2 generator contactors. The energized main contacts of the ac bus tie contactor provide a connection from the APU/external power contactor to the ac primary buses. The energized auxiliary contacts of the ac bus tie contactor complete the dc voltage path to the caution/advisory panel APU GEN ON capsule.

With external ac power connected to the helicopter and the EXT PWR switch placed ON, the ac bus tie contactor solenoid is energized. It receives dc voltage from the external power monitor panel through the de-energized auxiliary contacts of the No. 1 and No. 2 generator contactors. The energized main contacts of the ac bus tie contactor provide a connection from the external ac power receptacle to the ac primary buses.

Auxiliary energized contacts of the ac bus tie contactor complete the voltage path to the No. 2 hydraulic logic module when backup pump operation is required. Additional sets of normally-open and normally-closed contacts provide additional logic to the hydraulic system and the blade de-icing system (if installed).

9-1-1.8 Current Transformers.

Three current transformers, one on the No. 1 junction box and two on the No. 2 junction box, are installed in the output legs of each generator to monitor generator output current and to detect feeder faults. Each current transformer consists of three individual windings, one winding on each of the generator feeder lines. The current transformers are connected in series with current limit sense circuits within each generator. As the output current of the generator increases during transient load conditions, the increased current through the transformer is sensed by the generator control unit. This decreases the generator exciter control field

and the generator output current. When a feeder fault (short circuit to ground) occurs between the generator and the generator contactor, a differential current develops between the current transformer windings and the current limit sense circuit in the generator. This current difference is sensed by the generator control unit and disables and disconnects the generator from its loads.

9-1-1.9 Current Limiters.

Current limiters CL1 through CL3, on the No. 2 junction box, and CL4 through CL6, on the No. 1 junction box, are 60 ampere fuses protecting the operating generator from excessive overloading during transfer operations. The current limiters are affected when one generator is used to carry both ac primary buses, permitting the loss of one or more phases of ac while the remaining phase or phases remain on-line. Current limiters CL1 through CL3 carry three-phase ac power from the No. 2 generator to the No. 1 ac primary bus in case of a No. 1 generator failure. Current limiters CL4 through CL6 carry three-phase ac power from the No. 1 generator to the No. 2 ac primary bus in case of a No. 2 generator failure.

9-1-1.10 Test Receptacles.

The No. 1 generator test receptacle, on the No. 1 junction box, and the No. 2 and APU generator test receptacles, on the No. 2 junction box, are provided for monitoring generator three-phase ac output. The generator three-phase ac output is applied to the test receptacle when the generator control switch is placed to ON or TEST.

9-1-1.11 AC Essential Bus Transfer Relay.

The AC ESENTL BUS XFR relay, on the pilot's circuit breaker panel, provides connection of 115 vac, B phase power to the ac essential bus. The relay solenoid is energized by 115 vac, B phase power from the No. 1 ac primary bus. When energized, the relay applies 115 vac, B phase power from the No. 1 ac primary bus to the ac essential bus. If there is a loss of B phase power on the No. 1 ac primary bus, the ac essential bus relay is de-energized, and 115 vac, B phase power from the No. 2 ac primary bus is applied through the de-energized contacts to the ac essential bus.

9-1-1.12 AC Essential Bus Fail Relay.

The ac essential bus fail relay, on the pilot's circuit breaker panel, is energized whenever there is power on the ac essential bus. If this power is lost, the relay is de-energized and its de-energized contacts complete the dc voltage path from the battery bus to light the caution/advisory panel AC ESS BUS OFF capsule.

9-1-1.13 External Power Receptacle.

The external power receptacle, on the front right-hand cabin, permits three-phase ac power from an external power cart to be applied to the helicopter electrical system during ground operations. The receptacle contains four pins for the application of power and two additional pins, which are jumpered in the external power cart plug. These jumpered pins form an interlock when the external power cart is connected and complete the path for applying battery bus voltage to light the caution/advisory panel EXT PWR CONNECTED capsule.

9-1-1.14 External Power Monitor Panel.

The external power monitor panel, on the upper right-hand cabin, monitors the external ac input for undervoltage, overvoltage, underfrequency, overfrequency, and correct phase rotation conditions. During external power application, if the monitor panel requirements are not met, external power is disconnected from the helicopter distribution system. It remains disconnected until the EXT PWR switch is positioned to RESET.

When external power drops below 100 to 105 vac, the power monitor panel undervoltage protection circuit senses this and initially provides a marginal time delay to prevent nuisance trips. If the voltage remains low for 0.85 to 2.55 seconds, the protection circuit opens the monitor panel output circuit. With the output circuit open, dc voltage to the ac bus tie contactor is removed and external power cannot be used.

When external power rises above 125 to 130 vac, the overvoltage protection circuit senses this. After a marginal time delay of 0.75 and 1.25 seconds to prevent nuisance trips, it disconnects external power as in the undervoltage condition. The length

of time the overvoltage condition is permitted to exist is inversely proportional to the magnitude of the overvoltage condition.

When external power frequency drops below 370 to 375 Hz, the underfrequency protection circuit senses this and provides a time delay to prevent nuisance trips. If the frequency remains low for 0.75 to 1.25 seconds, the protection circuit disconnects external power as in the undervoltage condition.

When external power frequency rises above 425 to 430 Hz, the overfrequency sensing circuit senses this and provides a time delay to prevent nuisance trips. If the frequency remains high for 0.75 to 1.25 seconds, the protection circuit disconnects external power as in the undervoltage condition.

When external power phase rotation (normal A-B-C) is incorrect, the phase rotation sensing circuit senses this and does not close the monitor panel output circuit. With the output circuit open, dc voltage is not applied to the ac bus tie contactor, and external power usage is not possible.

9-1-1.15 External Power Control Switch.

The external power control switch, on the upper console, is a three-position switch used to apply external power to the helicopter ac buses. The ON position completes the dc voltage path to energize the ac bus tie contactor to apply ac external power to the ac buses. The RESET position closes the external power monitor panel output circuit to reconnect external power to the ac buses if it has been disconnected as a result of an undervoltage, overvoltage, underfrequency, or overfrequency condition. The OFF position opens the dc voltage path to de-energize the ac bus tie contactor and remove external power from the ac buses.

9-1-1.16 No. 1 and No. 2 Generators.

The No. 1 and No. 2 oil-cooled, brushless generators, mounted on the left and right accessory modules of the main gear box, are driven whenever the main rotor head is turning (Figure 9-1-1-2). Each generator, rated at 30/45 KVA at 115/200 volts, is connected for a 4-wire output, with grounded neutral, furnishing three-phase alternat-

ing current. The generator drive shaft is normally driven at about 12,000 RPM to maintain an output frequency of 400 Hz. The generator output voltage is regulated by a voltage regulator within the GCU, which varies the exciter control field. Each generator output is controlled by its GCU. The GCU contains sensing circuits to disconnect the generator from its respective loads when an overvoltage, undervoltage, underfrequency, or feeder-fault condition occurs.

9-1-1.17 APU Generator.

The APU air-cooled, brushless generator, mounted on the APU, is driven whenever the APU is operat-

ing. The generator, rated at 20/30 KVA at 115/200 volts, is connected for a 4-wire Wye output, with grounded neutral, furnishing three-phase alternating current. The generator drive shaft is normally driven at about 12,000 RPM to maintain an output frequency of 400 Hz. The generator output voltage is regulated by a voltage regulator within the GCU which varies the exciter control field. The GCU contains sensing circuits to disconnect the generator from its respective loads when an overvoltage, undervoltage, underfrequency, or feeder-fault condition occurs. The generator is cooled by drawing air in at the front end and exhausting it at the drive end.

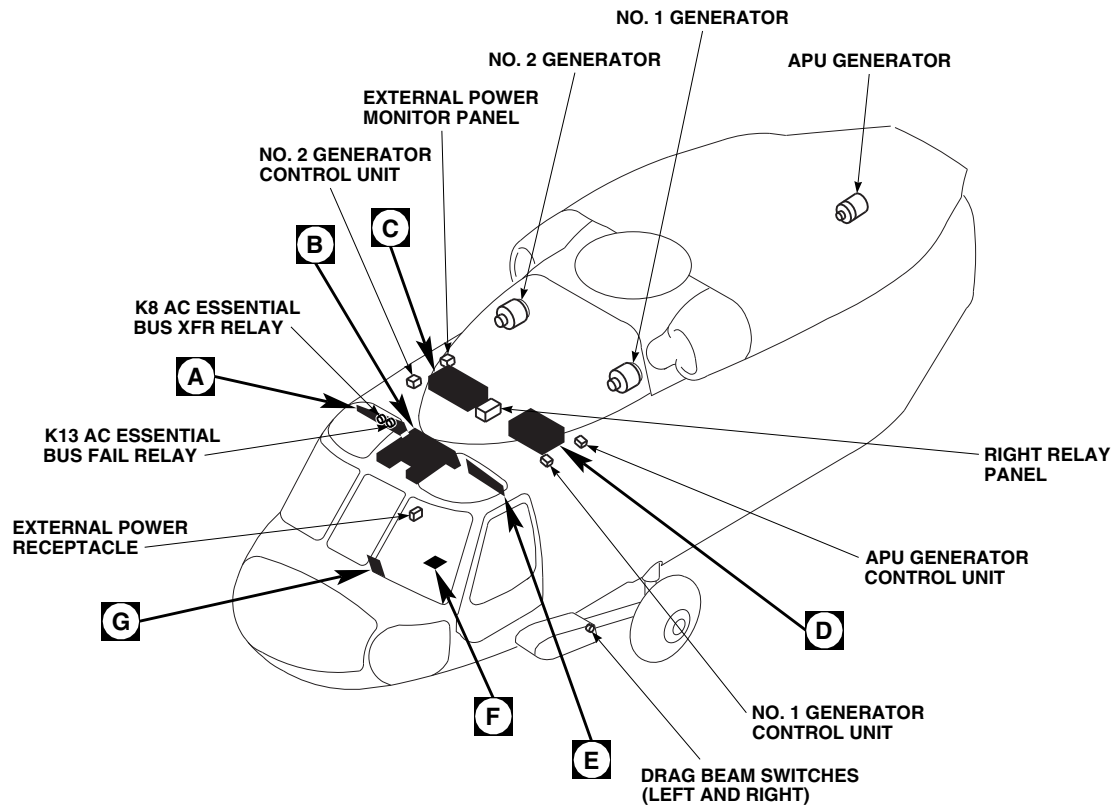
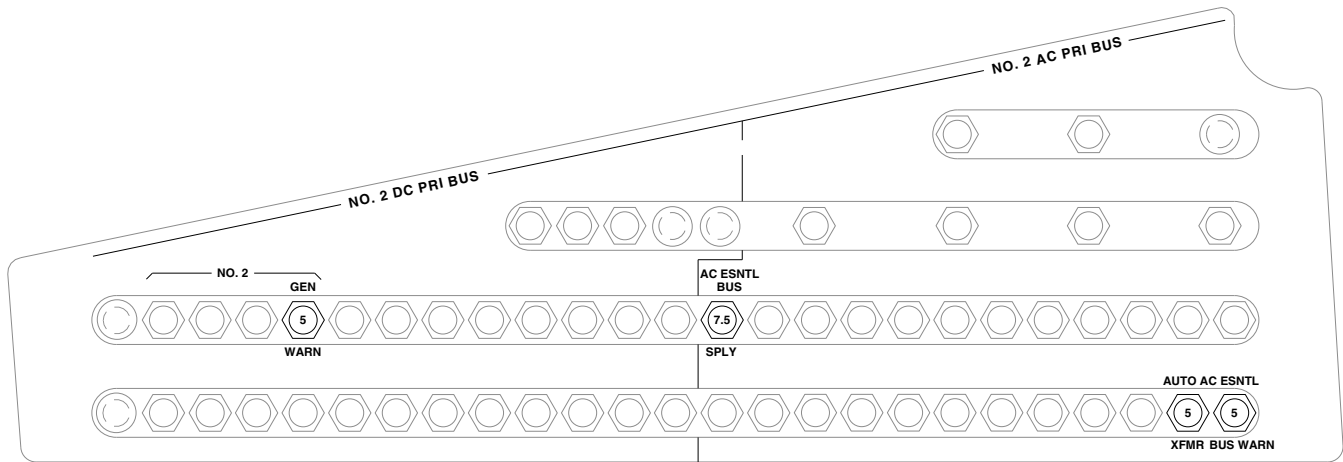
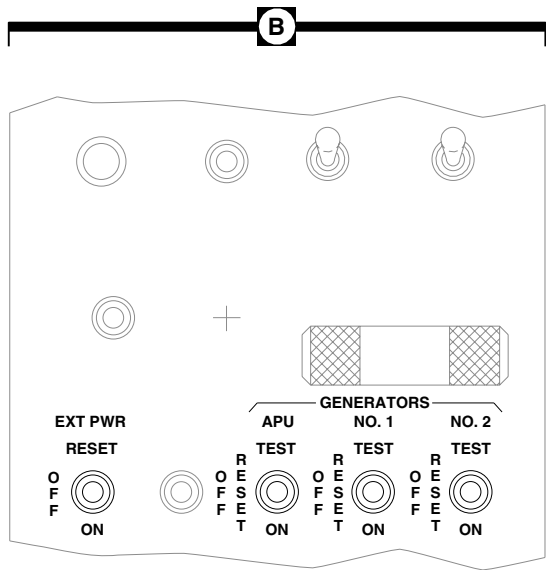
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FIGURE 9-1-1-1. AC ELECTRICAL SYSTEM LOCATION DIAGRAM (SHEET 1 OF 5)



PILOT'S CIRCUIT BREAKER PANEL

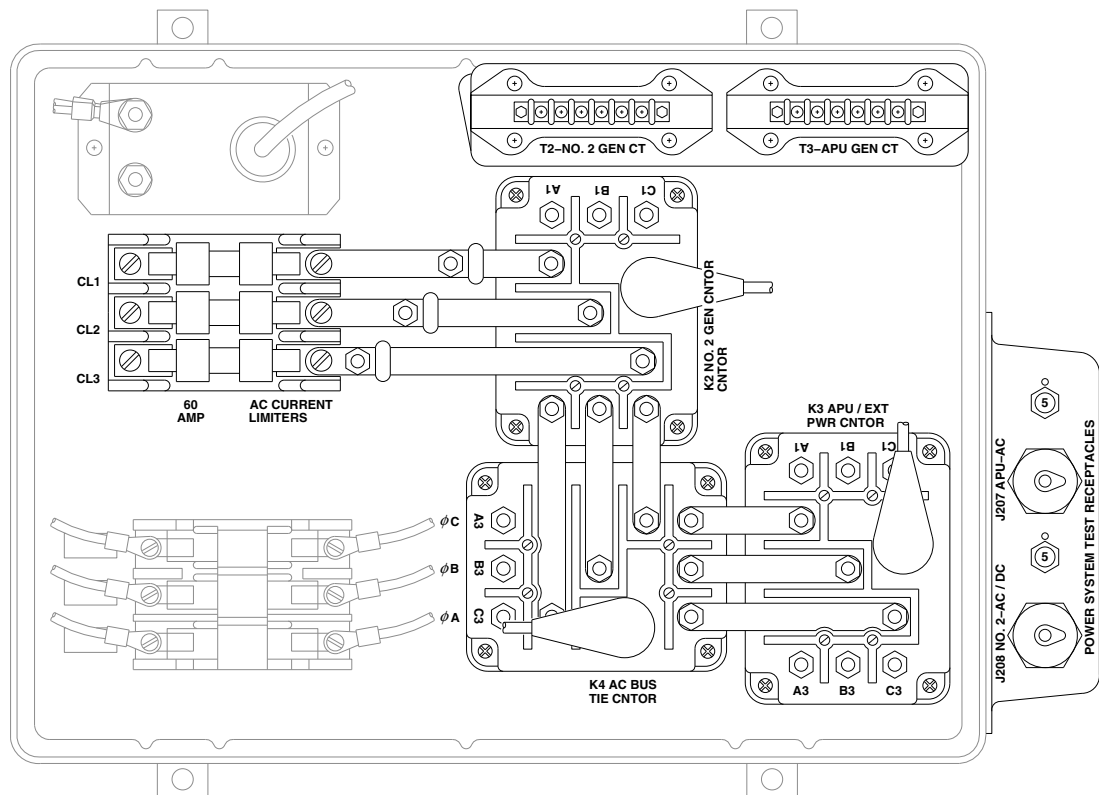
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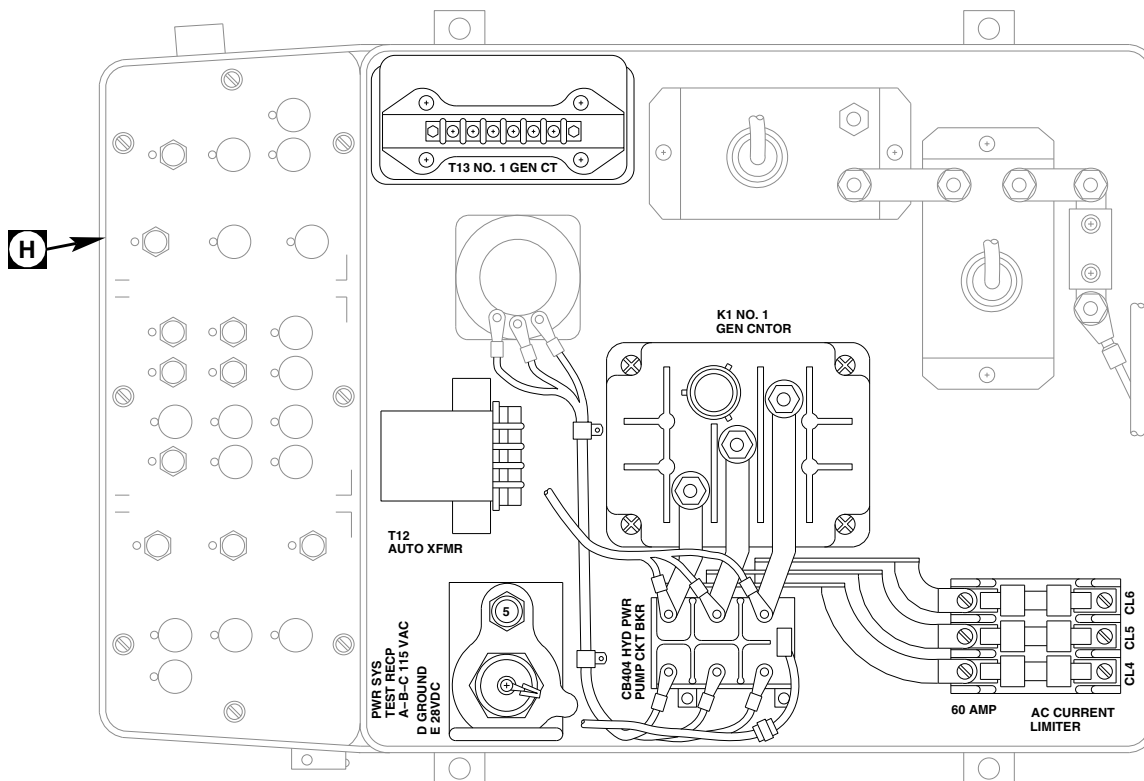
UPPER CONSOLE

FB2192_2B
SA

FIGURE 9-1-1-1. AC ELECTRICAL SYSTEM LOCATION DIAGRAM (SHEET 2 OF 5)



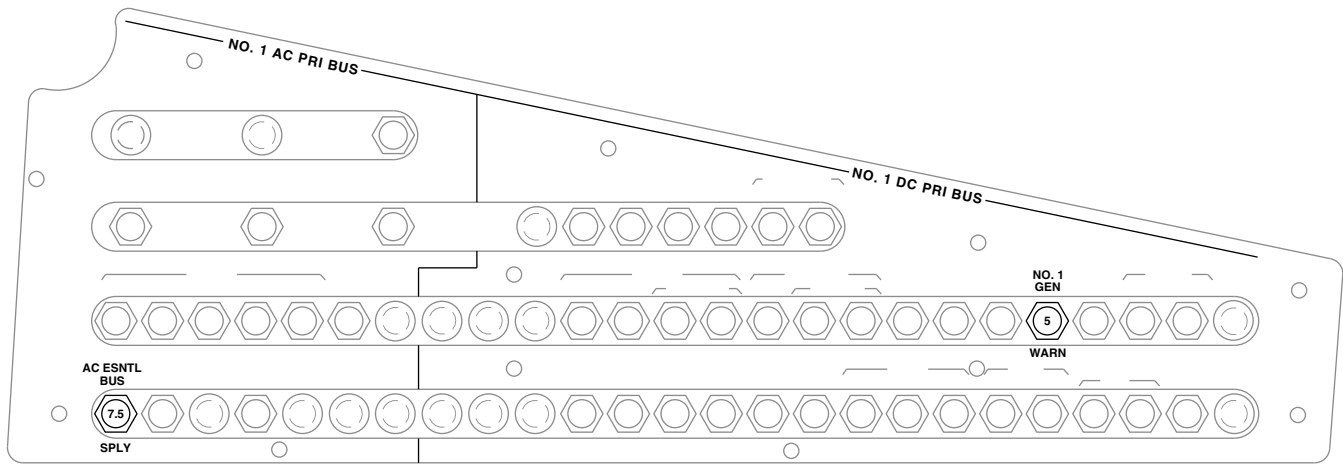
NO. 2 JUNCTION BOX



NO. 1 JUNCTION BOX

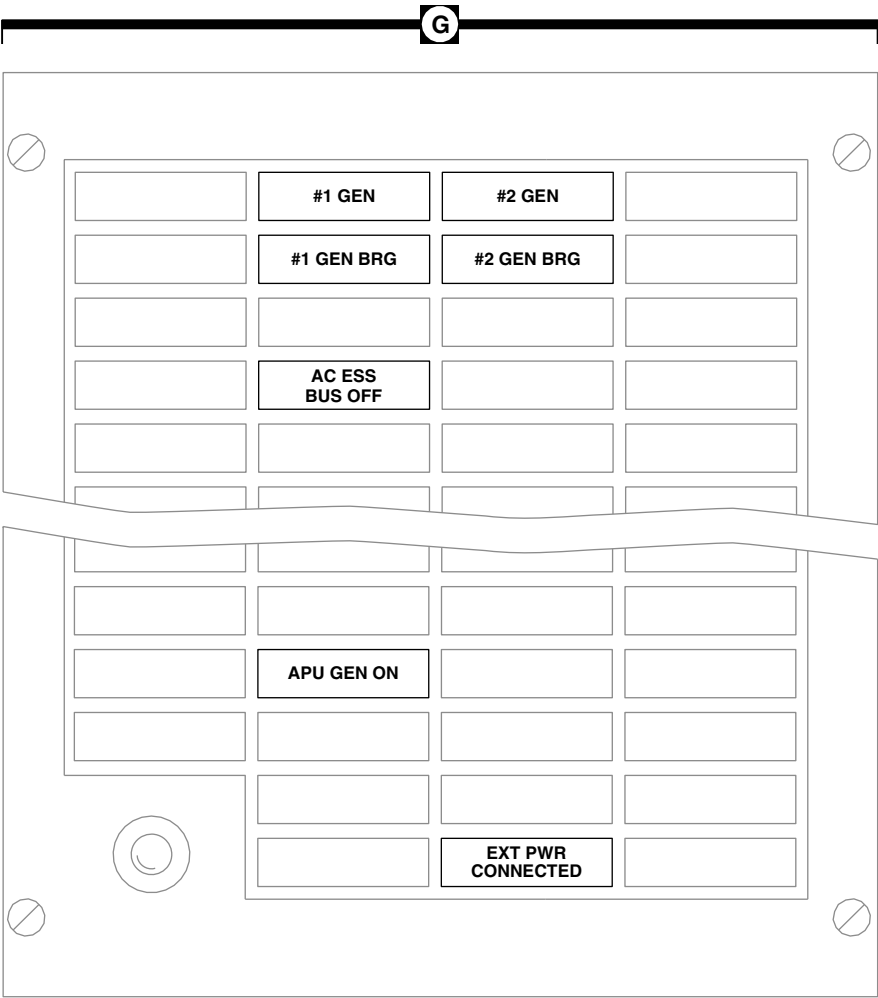
FIGURE 9-1-1-1. AC ELECTRICAL SYSTEM LOCATION DIAGRAM (SHEET 3 OF 5)

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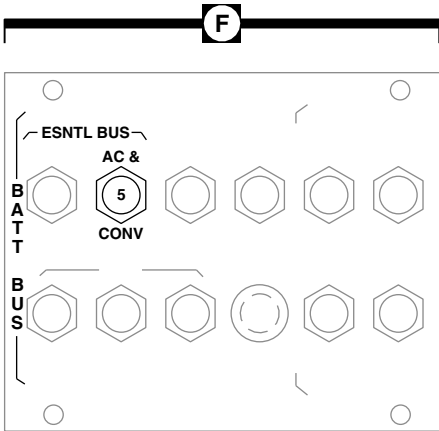


COPILOT'S CIRCUIT BREAKER PANEL

E



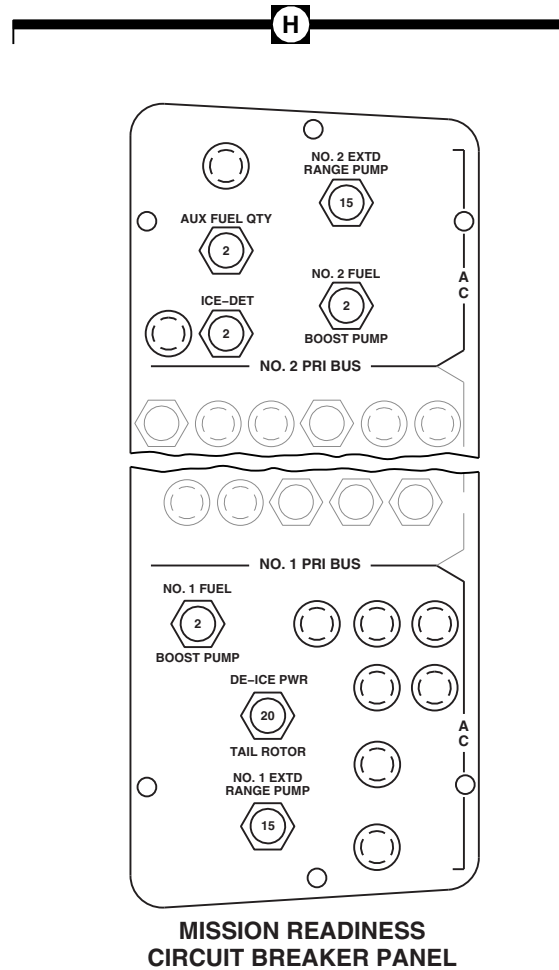
CAUTION/ADVISORY PANEL



LOWER CONSOLE
CIRCUIT BREAKER PANEL

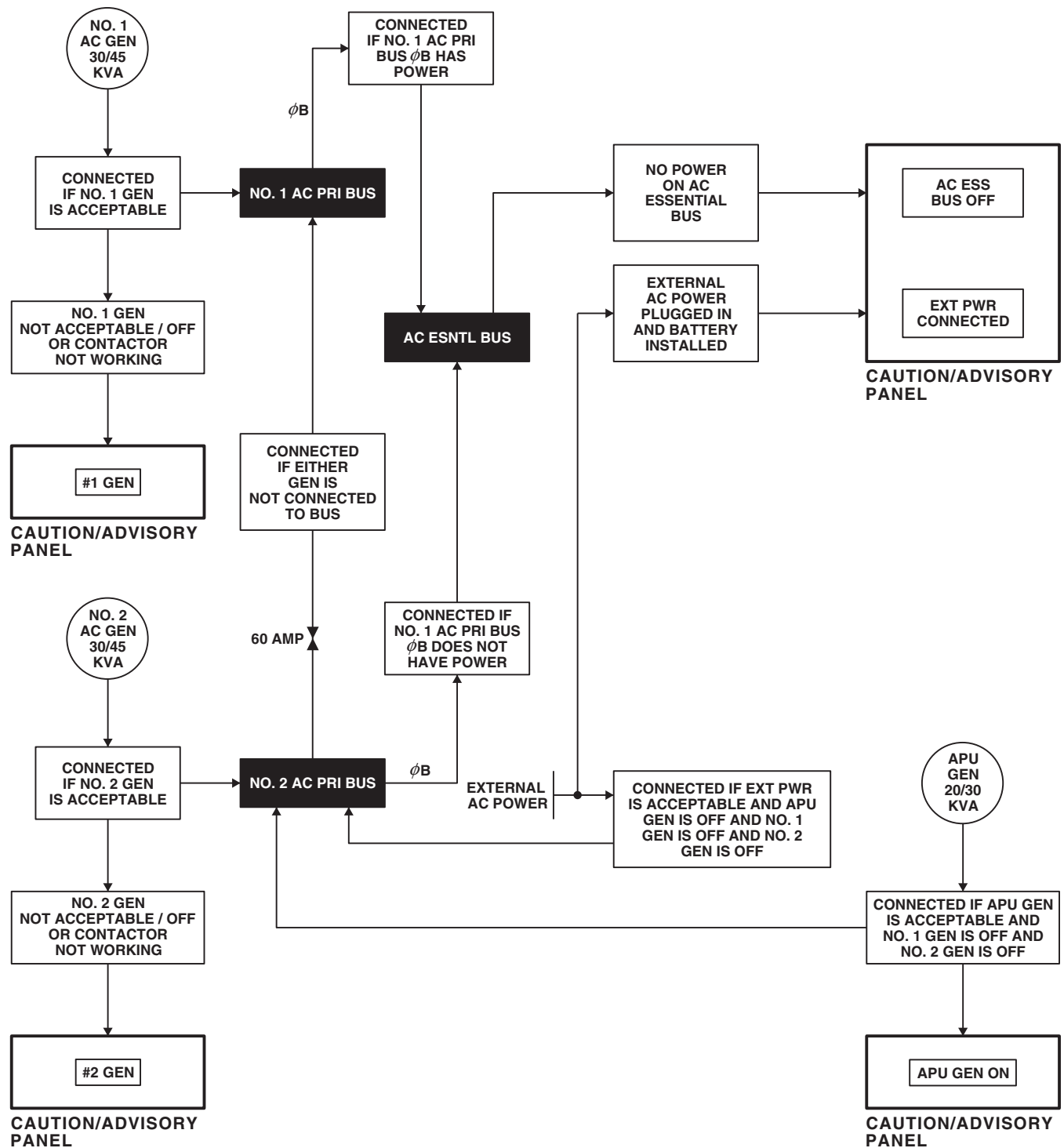
FIGURE 9-1-1-1. AC ELECTRICAL SYSTEM LOCATION DIAGRAM (SHEET 4 OF 5)

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FIGURE 9-1-1-1. AC ELECTRICAL SYSTEM LOCATION DIAGRAM (SHEET 5 OF 5)



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FIGURE 9-1-1-2. AC ELECTRICAL SYSTEM GENERAL BLOCK DIAGRAM

9-1-1-11/(9-1-1-12 Blank)

9-1-2 AC ELECTRICAL PRIMARY SYSTEM DESCRIPTION AND DATA.

9-1-2.1 Primary AC Power System.

Primary ac power is supplied by the No. 1 and No. 2 ac generators (Figure 9-1-2-3). Under normal operating conditions, with both generators operating and their outputs acceptable, the No. 1 generator supplies the No. 1 ac primary bus and the No. 2 generator supplies the No. 2 ac primary bus. Both generators are protected from overload by 60 amp current limiters. If the No. 2 generator output is not acceptable, if the generator is off, or if the associated contactor is not working, the #2 GEN capsule on the caution/advisory panel will light and both ac primary buses (No. 1 and No. 2) will be supplied from the No. 1 generator. Similarly, if the No. 1 generator output is not acceptable, if the generator is off, or if the associated contactor is not working, the #1 GEN capsule on the caution/advisory panel will light and both ac primary buses (No. 1 and No. 2) will be supplied from the No. 2 generator. The ac essential bus is normally supplied with B phase power from the No. 1 ac primary bus. If No. 1 ac primary bus B phase power is not available, the ac essential bus is supplied from B phase of the No. 2 ac primary bus. If no power is present on the ac essential bus, the AC ESS BUS OFF capsule on the caution/advisory panel will light. The No. 1 ac primary bus also supplies power to a 60 Hz converter (provisional) for two power receptacles. The No. 2 ac primary bus also supplies power to a cabin utility receptacle. For a complete illustration of all circuit breaker panels, see Figure 9-1-2-1.

9-1-2.2 Primary AC Power Distribution.

With the No. 1 and No. 2 generators operating and their outputs at rated value, the No. 1 and No. 2 generator contactors K1 and K2 are energized (Figures 9-1-2-2 and 9-1-2-4). The No. 1 and No. 2 generator outputs are applied through the main contacts of K1 and K2 to supply the No. 1 and No. 2 ac primary buses, respectively. The No. 1 ac primary bus supplies 115 vac, B phase power through the AC ESNTL BUS SPLY circuit breaker to energize AC ESNTL BUS XFR relay K8. With K8 energized, 115 vac, B phase power from the No. 1 ac primary bus is applied through the energized contacts of K8 to supply the ac essential bus.

If one generator fails due to overvoltage, undervoltage, or feeder fault, the generator is disabled and the associated ac generator contactor de-energizes to disconnect the generator from its loads. When this occurs, the operating generator will supply both ac primary buses. If the No. 1 generator fails, the No. 2 generator supplies the No. 1 ac primary bus through current limiters and the normally-closed main contacts of No. 1 generator contactor K1. If the No. 2 generator fails, the No. 1 generator supplies the No. 2 ac primary bus through current limiters and the normally-closed main contacts of ac bus tie contactor K4 and No. 2 generator contactor K2. The 60 amp current limiters protect the operating generator from too much overloading as a result of a feeder fault that may have disabled the failed generator. If there is a loss of B phase power on the No. 1 ac primary bus, AC ESNTL BUS XFR relay K8 will de-energize. When this occurs, 115 vac, B phase power from the No. 2 ac primary bus will be applied through the AC ESNTL BUS SPLY circuit breaker and the normally-closed contacts of relay K8, to supply the ac essential bus. If no power is available on the ac essential bus, AC ESNTL BUS FAIL relay K13 will de-energize. This closes the path between the battery bus and the caution/advisory panel AC ESS BUS OFF capsule, and the capsule will go on.

9-1-2.3 Power Generation.

Operation of the No. 1, No. 2, and auxiliary power unit (APU) generating systems is identical. With the generators driven at rated speed, the permanent magnet generator (PMG) section of each generator supplies ac voltage to the PMG rectifier of the associated generator control unit (GCU). The ac voltage is rectified and used as supply voltage for the GCU circuitry. When the GENERATORS NO. 1, NO. 2, or APU switch is placed ON, the voltage regulator section of the GCU controls the current flow to the exciter field of the generator, and the generator main output builds up. This output is monitored by the GCU and, when it reaches at least 95 vac at a frequency of between 370 and 380 Hz, power-ready sensing circuitry connects it to its loads. For each generator system, the main ac output and the primary dc bus output can be

monitored at a test connector located on the associated junction box. With the generator switch at TEST, the generator main output is disconnected from the ac primary bus. With all generators running, the #1 and #2 GEN capsules go off, and the APU GEN ON capsule goes on.

9-1-2.4 No. 1 and No. 2 Generator Description.

The No. 1 and No. 2 oil-cooled, brushless generators, mounted on the left and right accessory modules of the main gear box, are driven whenever the main rotor head is turning (Figures 9-1-2-3 and 9-1-2-4). Each generator, rated at 30/45 KVA at 115/200 volts, is connected for a four-wire output, with grounded neutral, furnishing three-phase alternating current. The generator drive shaft is normally driven at about 12,000 RPM to maintain an output frequency of 400 Hz. The generator output voltage is regulated by a voltage regulator within the generator control unit, which varies the exciter control field.

Each generator output is routed to the GCU. The GCU contains sensing circuits to disconnect the generator from its respective loads when an over-voltage, undervoltage, underfrequency, or feeder-fault condition occurs.

9-1-2.5 No. 1 and No. 2 Generator Operation.

9-1-2.5.1 Dual Generator Operation.

With the left and right accessory gear box modules driving the generators at rated speed, each generator's 12-pole PMG applies three-phase ac voltage to the PMG rectifier in the GCU (Figure 9-1-2-4). The PMG rectifier rectifies and filters the ac voltage and applies a dc voltage to the No. 1 and No. 2 generators switches and to a regulated dc power supply that supplies the GCU sensing circuits. When the generator control switch is placed ON, the dc voltage from the PMG rectifier is routed through AND gates 13 and 3. With no inhibit signal from OR gate 1, gate 3 is enabled, energizing generator control relay (GCR) K1. Energizing the GCR routes PMG power from the PMG rectifier to the voltage regulator, and also energizes the generator exciter control field. The

exciter is rotated in the magnetic field produced by the exciter control field, thereby producing three-phase ac voltages that are applied to diode rectifiers. A flywheel diode across the exciter control field input in the GCU allows the field to collapse rapidly, increasing response from the voltage regulator. The positive half cycle of each phase of the exciter voltage activates one of the diodes and causes a continuous flow of current in the rotating field. This moving field causes a voltage to be induced in the main generator winding, which supplies a three-phase ac output voltage.

The same dc voltage that is applied to GCU AND gates 13 and 3 through the ON position of the generator control switch is also applied to AND gates 7 and 8. AND gate 7 receives a second inhibiting input from OR gate 6, which senses an underfrequency or overvoltage condition from the respective sensing circuits. The voltage sensing circuit monitors the main generator's three-phase output for deviations in voltage from specified limits. The frequency sensing circuit monitors one of the three PMG voltages for frequency deviations. With no underfrequency, undervoltage, or overvoltage condition present, no inhibit signal is passed through OR gate 6. As a result, gate 7 is enabled and senses a power ready condition. With gate 7 satisfied, a signal through OR gate 9 satisfies one half of AND gate 11. With no signal from OR gate 2 or TD3 through OR gate 12, gate 11 is satisfied and the contactor control relay (CCR) K2 energizes. With the CCR energized, the logic 0 is removed from AND gates 8 and 4 and dc voltage from the PMG rectifier is applied through the contacts of No. 2 and No. 1 generators switches to energize No. 2 and No. 1 generator contactors K2 and K1, respectively. Energizing the CCR also opens the path between the No. 2 dc primary bus and the caution/advisory panel #2 GEN capsule, and between the No. 1 dc primary bus and the caution/advisory panel #1 GEN capsule. The capsules then go off.

The main generator is protected from transient current overloads and feeder faults. The current monitoring circuit consists of a current transformer, on the generator's feeder lines to the generator contactor, and an internal generator current limiter circuit. Current limiting is performed by the current transformer, connected in series with the current

limit circuit within the generator, to regulate the output voltage. As the load causes the current from the main generator to increase, the induced voltage in the current transformer windings increases proportionately to the load current. The GCU monitors this induced voltage and regulates the field current of the main generator. When the current from the current transformer winding is at the preset value, the GCU reduces the current in the exciter control field, thereby causing less voltage to be generated by the exciter. This action reduces the dc current through the rotating field, thereby lowering the induced voltage in the main generator. Less output results in a reduction of current in the field of the main generator, thereby returning the output current to the preset value. Under normal operation, the current flowing through the two current transformer windings around the main generator output are equal and out-of-phase, resulting in zero differential current. If a feeder fault (short circuit to ground) should occur, an unbalance condition is developed, producing a differential current in the current transformer loop. This differential current is sensed by the GCU feeder fault sensing circuit, which applies a signal directly to OR gate 2. The signal is routed through OR gate 2, through OR gate 1, to AND gate 3, inhibiting AND gate 3. With AND gate 3 inhibited, CGR relay K1 trips, and the generator is de-energized. The same signal from OR gate 2 inhibits gate 11 through gate 12, to immediately open CCR relay K2. The generator's load is disconnected and the caution/advisory panel #1 GEN or #2 GEN capsule goes on. The output of OR gate 2 is also applied to AND gate 13, through TD5, back into OR gate 2, causing OR gate 2 output to be continuous. The GCR and CCR relays cannot release until the signal to AND gate 13 is removed by placing the generator control switch to OFF/RESET.

If an overvoltage condition is sensed, a trip signal is developed through inverse time delay TD1; the greater the overvoltage, the shorter the time delay. This trip signal is applied to OR gate 2 and the GCR and CCR relays open as described above.

If an undervoltage condition is sensed, a GCR trip signal is developed through fixed time delay TD2 to OR gate 2, causing the GCR and CCR relays to open. At the same time an undervoltage is sensed, gate 7 is inhibited through gate 6. Thus, if the

generator control switch is cycled from OFF/RESET to ON, the generator is energized, but the CCR relay is not allowed to close and apply the underfrequency, undervoltage or overvoltage condition onto the load bus. The CCR relay closes only when the frequency and voltage become acceptable again.

If an underfrequency condition is sensed (during ground operation), it produces a trip signal through fixed time delay TD3 to OR gate 1. The signal from OR gate 1 inhibits gate 3, opening the GCR relay, thereby de-energizing the generator. The signal from TD3 also causes the CCR relay to open by inhibiting AND gate 11 via OR gate 12. When the CCR relay opens, a logic 0 inhibit signal is applied to AND gate 4. When the frequency is again within limits, the inhibit at gate 3 is removed and the GCR relay closes again. The input to AND gate 11 is not satisfied until the underfrequency inhibit signal at gate 7 through gate 6 disappears. This occurs when the power ready signal has been restored. When the frequency is within limits, the CCR relay again closes, energizing the bus load. During flight, with the helicopter's weight-off-wheels, the landing gear drag beam switch is closed. This applies a ground to the underfrequency sensing circuit, disabling it. The right drag beam switch provides the underfrequency disable signal. Thus, the generator will not be de-energized and disconnected from its load during an autorotative flare, when generator frequency decreases.

If too much wear of the generator main bearing is detected during generator operation, a caution/advisory panel indication is given. When the main bearing is worn a predetermined amount, a detector ring with a sensing lead contacts the outer race of the generator auxiliary bearing, grounding the sensing lead. The ground is applied from the generator to the caution/advisory panel, and the #1 GEN BRG or #2 GEN BRG capsule goes on.

With the No. 1 and No. 2 generators operating at rated value, the three-phase ac output of the No. 2 generator is applied to the No. 2 test receptacle for monitoring, and through the energized main contacts of No. 2 generator contactor K2 to supply the No. 2 ac primary bus. Similarly, the three-phase ac output of the No. 1 generator is applied to the No. 1 test receptacle and through the energized

main contacts of No. 1 generator contactor K1 to supply the No. 1 ac primary bus. The No. 1 ac primary bus supplies 115 vac, B phase power, through the AC ESNTL BUS SPLY circuit breaker, to the coil and contacts of AC ESSENTIAL BUS XFR relay K8. With relay K8 energized, the 115 vac, B phase power is supplied to the ac essential bus. The ac essential bus applies 115 vac, B phase power to an autotransformer that supplies 26 vac to navigation instrument loads. When the generator control switch is placed to TEST, system operation is the same as for the ON position, except that generator contactors K1 and K2 remain de-energized and GCU test relay K3 is energized. With K3 energized, its contacts open and the caution/advisory panel GEN capsules go off. The generator outputs are not applied to the ac buses but are applied to the test receptacles for monitoring.

9-1-2.5.2 Single No. 1 Generator Operation.

With only the GENERATORS NO. 1 switch ON, the No. 1 generator output builds up and is controlled by the No. 1 GCU (Figure 9-1-2-4). When the No. 1 generator output reaches rated value, No. 1 generator contactor K1 is energized and the three-phase ac generator output is applied through the energized main contacts of K1 to supply the No. 1 ac primary bus. The path between the No. 1 dc primary bus and the caution/advisory panel #1 GEN capsule is open through open auxiliary contacts of K1 and the capsule is off. The No. 1 ac primary bus supplies 115 vac, B phase power, through the AC ESNTL BUS SPLY circuit breaker, to the coil and contacts of AC ESNTL BUS XFR relay K8. With relay K8 energized, 115 vac, B phase power is supplied to the ac essential bus. The ac essential bus applies 115 vac, B phase power to an autotransformer, which supplies 26 vac to navigation instrument loads.

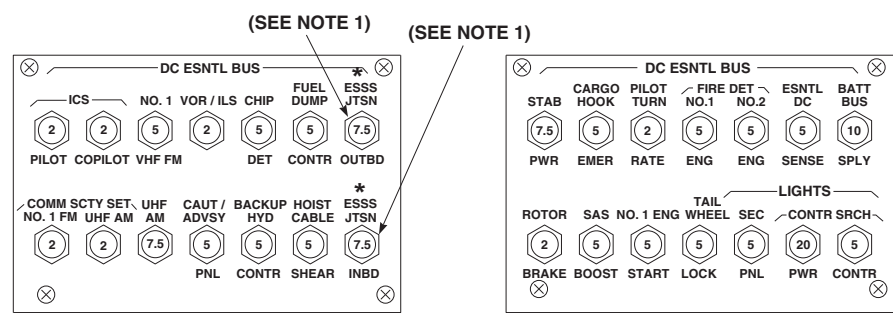
With GENERATORS NO. 2 switch OFF, the No. 2 generator output does not build up. No. 2 generator

contactor K2 is de-energized, the path between the No. 2 dc primary bus and the caution/advisory panel #2 GEN capsule is closed through normally-closed auxiliary contacts of K2, and the capsule is on. The three-phase output of the No. 1 generator is applied from the No. 1 ac primary bus, through current limiters CL4, 5, and 6, and through the normally-closed main contacts of ac bus tie contactor K4 and No. 2 generator contactor K2, to supply the No. 2 ac primary bus.

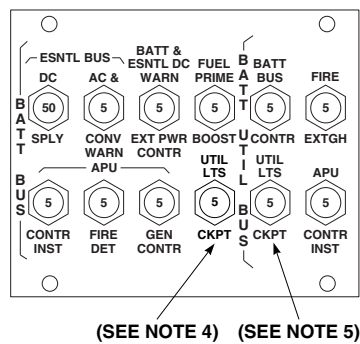
9-1-2.5.3 Single No. 2 Generator Operation.

With only the GENERATORS NO. 2 switch ON, the No. 2 generator output builds up and is controlled by the No. 2 GCU (Figure 9-1-2-4). When the No. 2 generator output reaches rated value, No. 2 generator contactor K2 is energized and the three-phase ac generator output is applied through the energized main contacts of K2 to supply the No. 2 ac primary bus. The path between the No. 2 dc primary bus and the caution/advisory panel #2 GEN capsule is open through the open auxiliary contacts of K2 and the capsule is off.

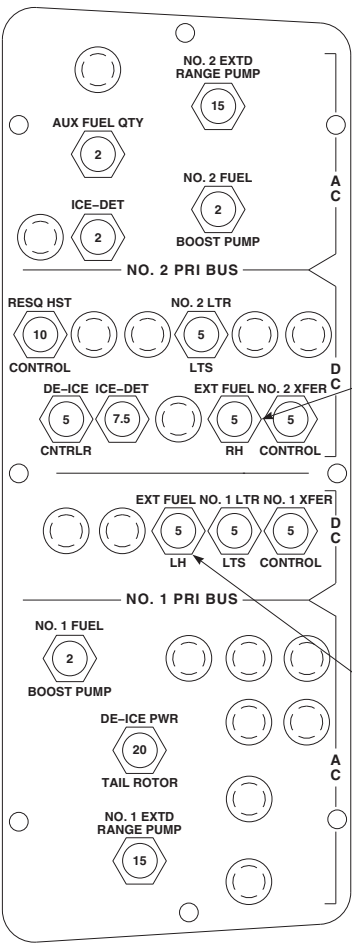
With GENERATORS NO. 1 switch OFF, the No. 1 generator output does not build up. No. 1 generator contactor K1 is de-energized, the path between the No. 1 dc primary bus and the caution/advisory panel #1 GEN capsule is closed through normally-closed auxiliary contacts of K1, and the capsule is on. The three-phase output of the No. 2 generator is applied from the No. 2 ac primary bus, through current limiters CL1, CL2, and CL3, and the normally-closed contacts of No. 1 generator contactor K1 to supply the No. 1 ac primary bus. The No. 1 ac primary bus supplies 115 vac, B phase power, through the AC ESNTL BUS SPLY circuit breaker, to the coil and contacts of AC ESNTL BUS XFR relay K8. With relay K8 energized, the 115 vac, B phase power is supplied to the ac essential bus. The ac essential bus applies 115 vac, B phase power to an autotransformer that supplies 26 vac to navigation instrument loads.



UPPER CONSOLE CIRCUIT BREAKER PANEL



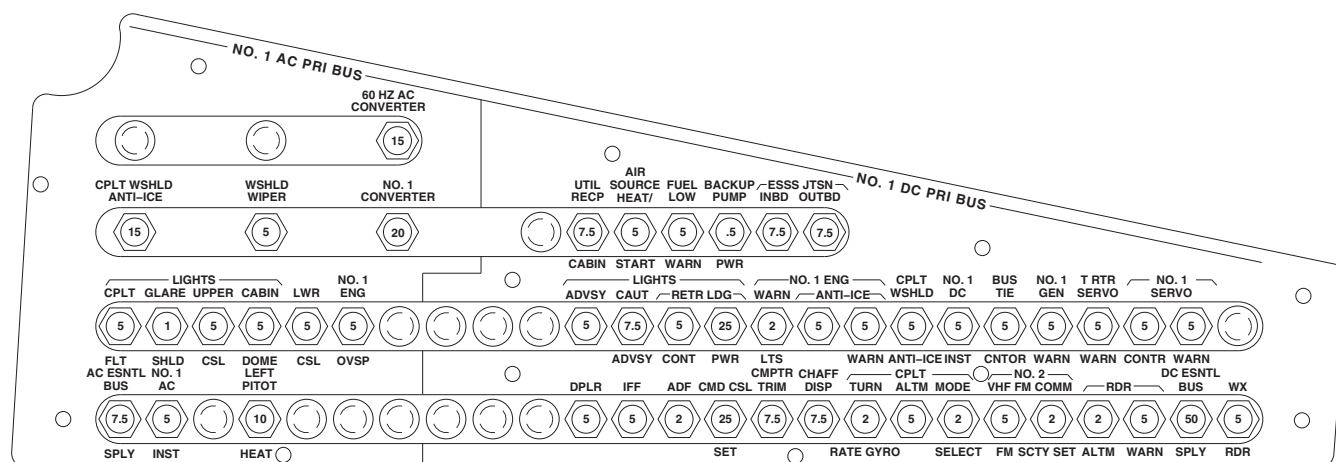
LOWER CONSOLE CIRCUIT BREAKER PANEL



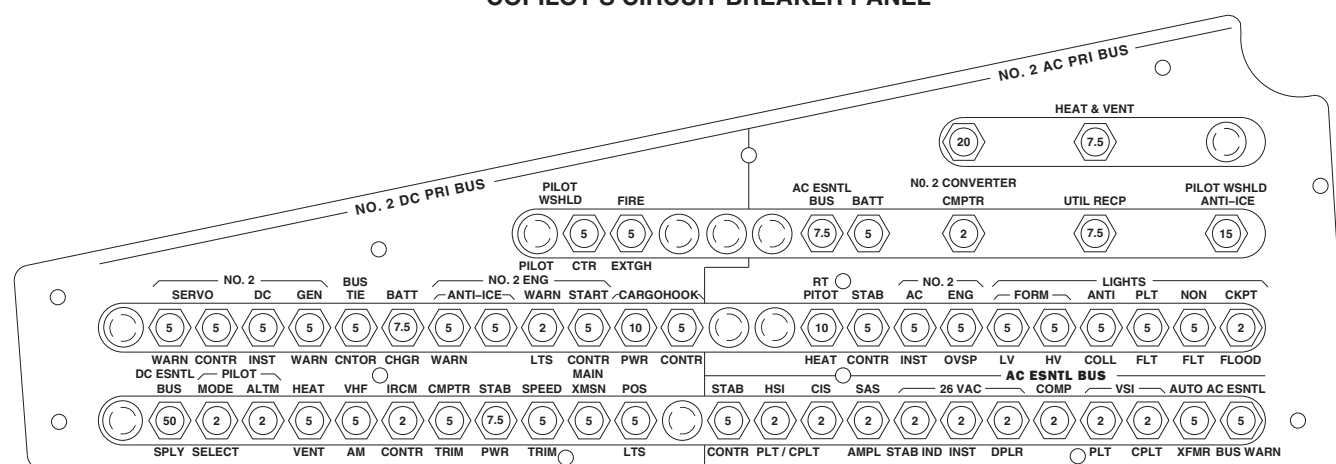
MISSION READINESS CIRCUIT BREAKER PANELS (CABIN)

NOTES	
1.	ESSS
2.	W/O SLS
3.	SLS
4.	702375-SUBQ
5.	70583-702374

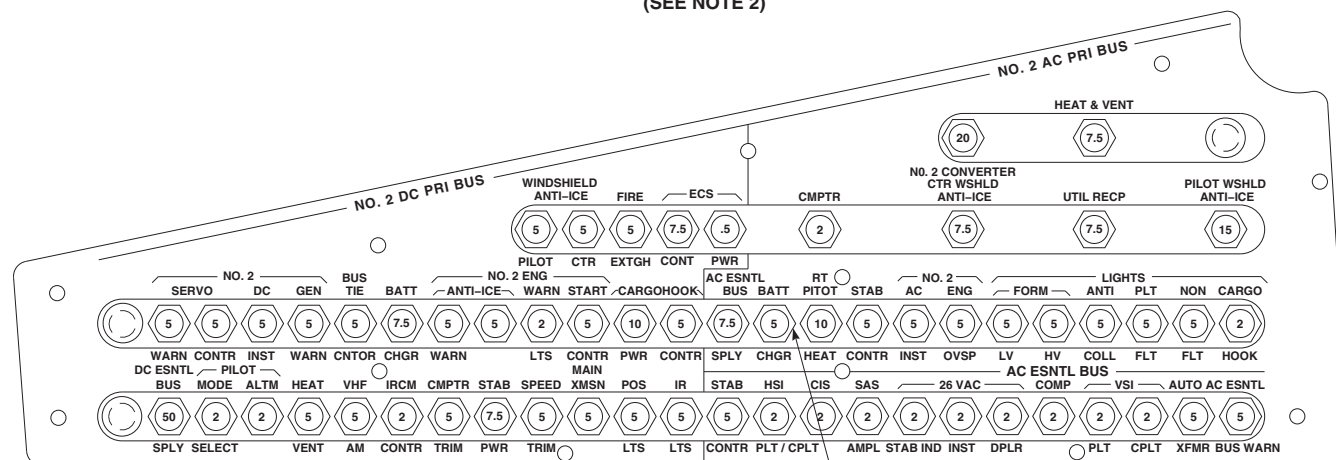
FIGURE 9-1-2-1. CIRCUIT BREAKER PANELS (SHEET 1 OF 2)



COPILOT'S CIRCUIT BREAKER PANEL



(SEE NOTE 2)



(SEE NOTE 3) (SEE NOTE 4)

PILOT'S CIRCUIT BREAKER PANEL

FB2252 2B
SA

FIGURE 9-1-2-1. CIRCUIT BREAKER PANELS (SHEET 2 OF 2)

LEGEND		AC POWER SOURCE				CONDITION	AC SYSTEM CONTACTORS/RELAY					
POWER SOURCE		A	B	C	D		K1	K2	K3	K4	K8	K13
A – NO. 1 MAIN GENERATOR		0	0	0	0	1	O	O	O	O	O	O
B – NO. 2 MAIN GENERATOR		1	0	0	0	2	X	O	O	O	X	X
C – APU GENERATOR		0	1	0	0	3	O	X	O	O	X	X
D –EXTERNAL AC POWER		1	1	0	0	4	X	X	O	O	X	X
1 – TURNED ON AND OPERATING		0	0	1	0	5	O	O	X	X	X	X
0 – TURNED OFF OR FAILED		1	0	1	0	6	X	O	X	O	X	X
CONTACTORS/RELAYS		0	1	1	0	7	O	X	X	O	X	X
K1 – NO. 1 GENERATOR CONTACTOR		1	1	1	0	8	X	X	X	O	X	X
K2 – NO. 2 GENERATOR CONTACTOR		0	0	0	1	9	O	O	O	X	X	X
K3 – APU / EXT PWR CONTACTOR		1	0	0	1	10	X	O	O	O	X	X
K4 – AC BUS TIE CONTACTOR		0	1	0	1	11	O	X	O	O	X	X
K8 – AC ESNTL BUS XFR RELAY		1	1	0	1	12	X	X	O	O	X	X
K13 – AC ESNTL BUS FAIL RELAY		0	0	1	1	13	O	O	X	X	X	X
X – ENERGIZED		1	0	1	1	14	X	O	X	O	X	X
O – DE-ENERGIZED		0	1	1	1	15	O	X	X	O	X	X
		1	1	1	1	16	X	X	X	O	X	X

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FIGURE 9-1-2-2. AC ELECTRICAL SYSTEM MATRIX

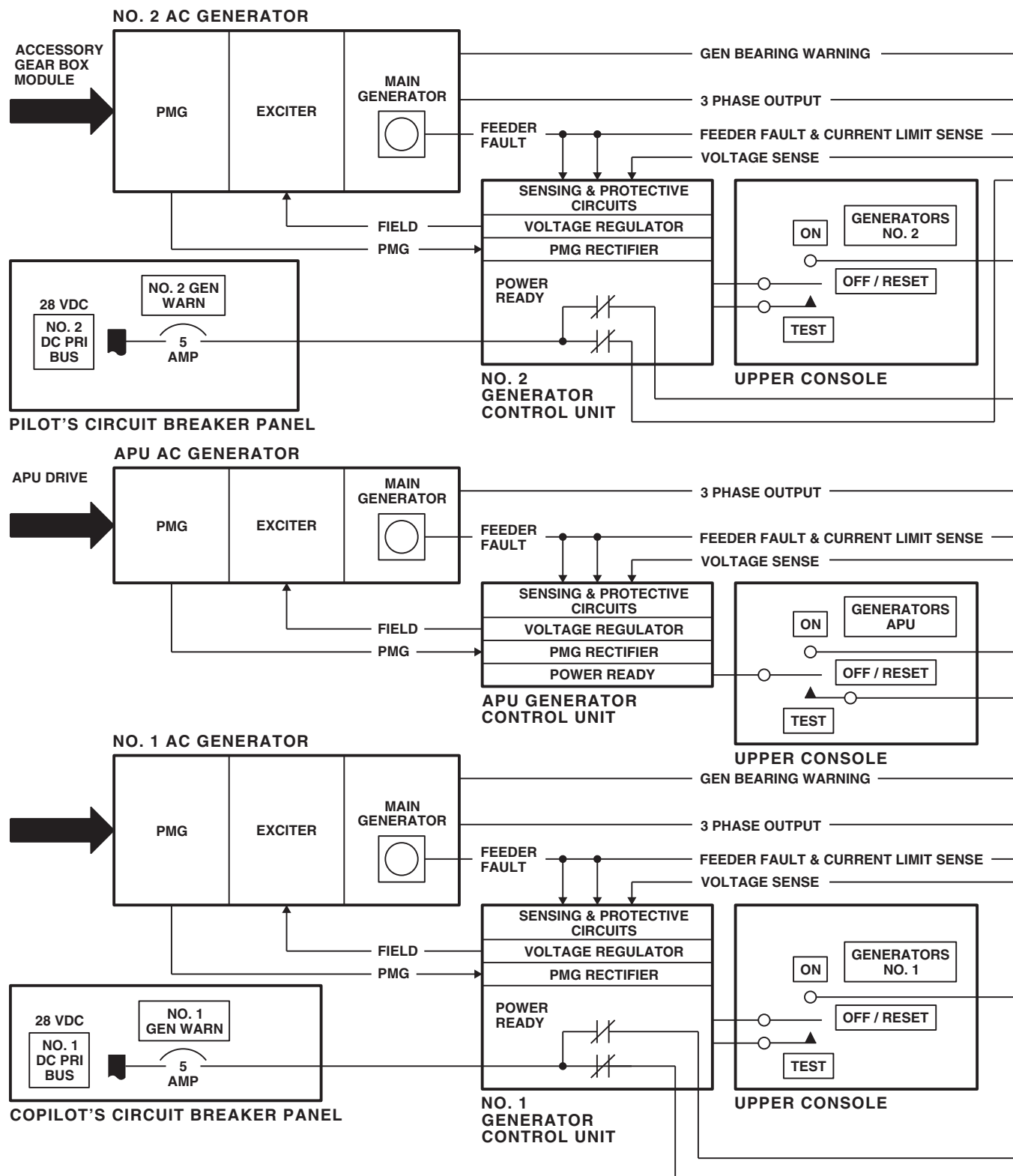
FB2253_1
SA

FIGURE 9-1-2-3. AC ELECTRICAL SYSTEM SIMPLIFIED BLOCK DIAGRAM (SHEET 1 OF 3)

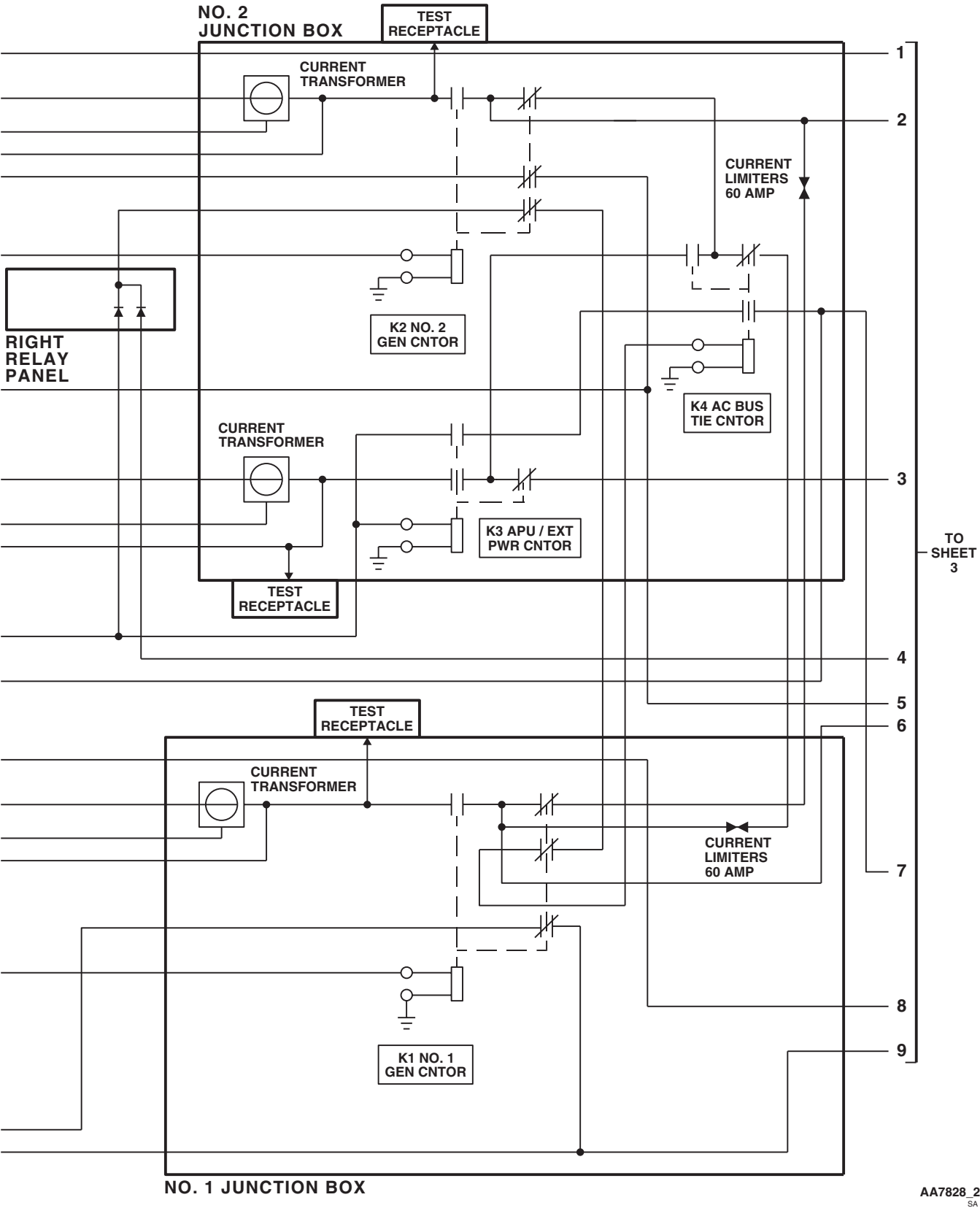


FIGURE 9-1-2-3. AC ELECTRICAL SYSTEM SIMPLIFIED BLOCK DIAGRAM (SHEET 2 OF 3)

AA7828 2
SA

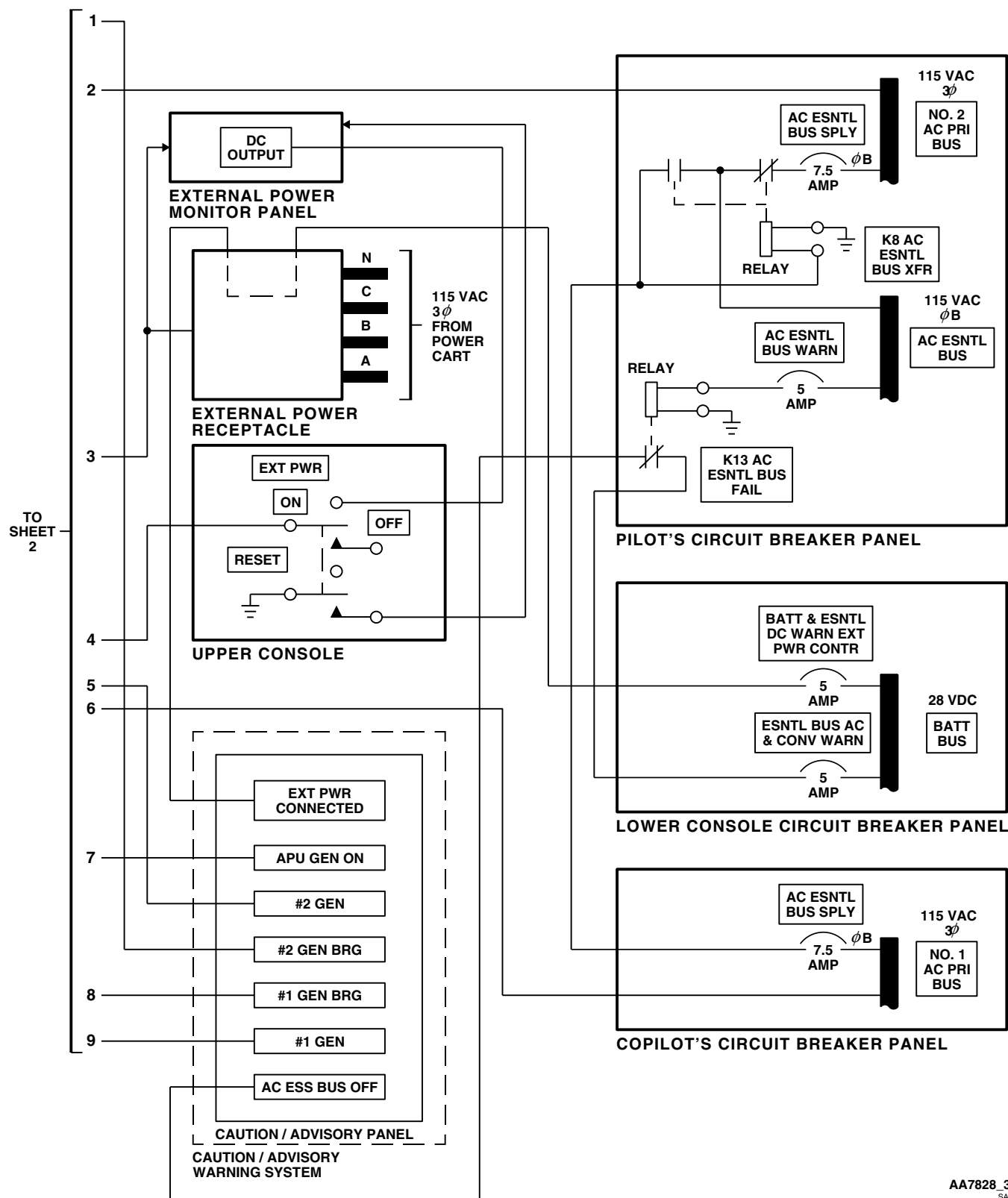
AA7828_3
SA

FIGURE 9-1-2-3. AC ELECTRICAL SYSTEM SIMPLIFIED BLOCK DIAGRAM (SHEET 3 OF 3)

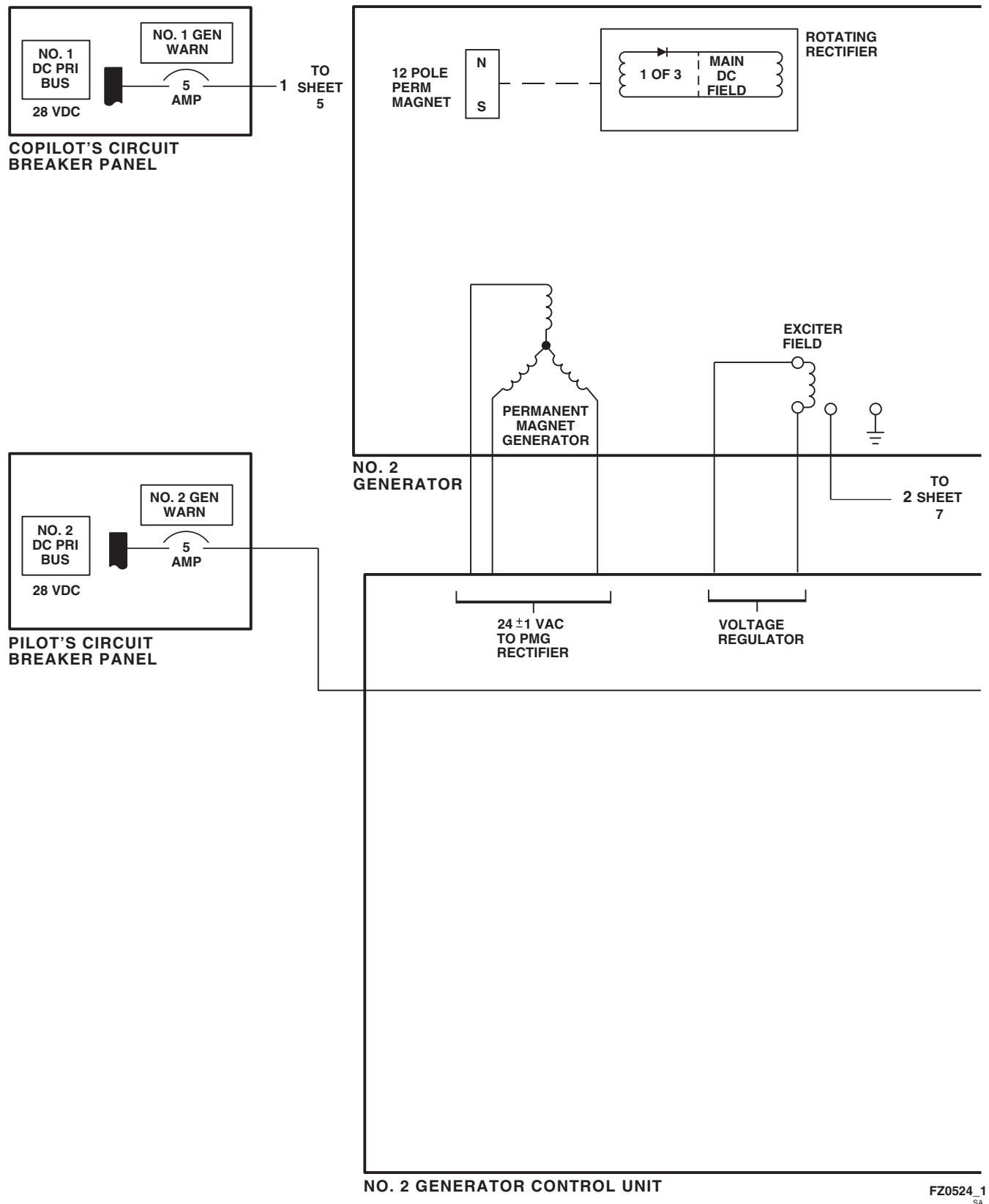


FIGURE 9-1-2-4. AC ELECTRICAL SYSTEM DETAILED BLOCK DIAGRAM (SHEET 1 OF 7)

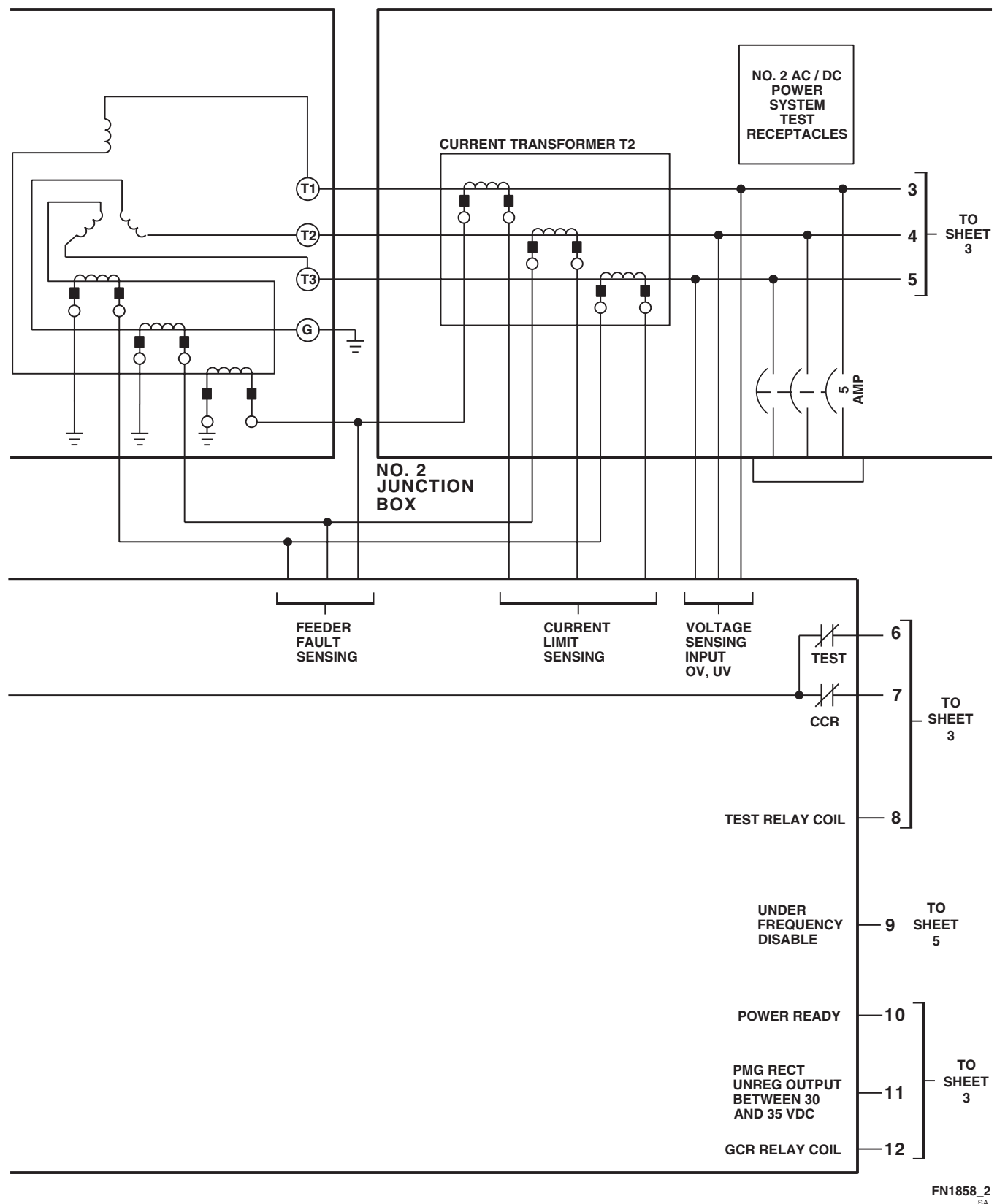


FIGURE 9-1-2-4. AC ELECTRICAL SYSTEM DETAILED BLOCK DIAGRAM (SHEET 2 OF 7)

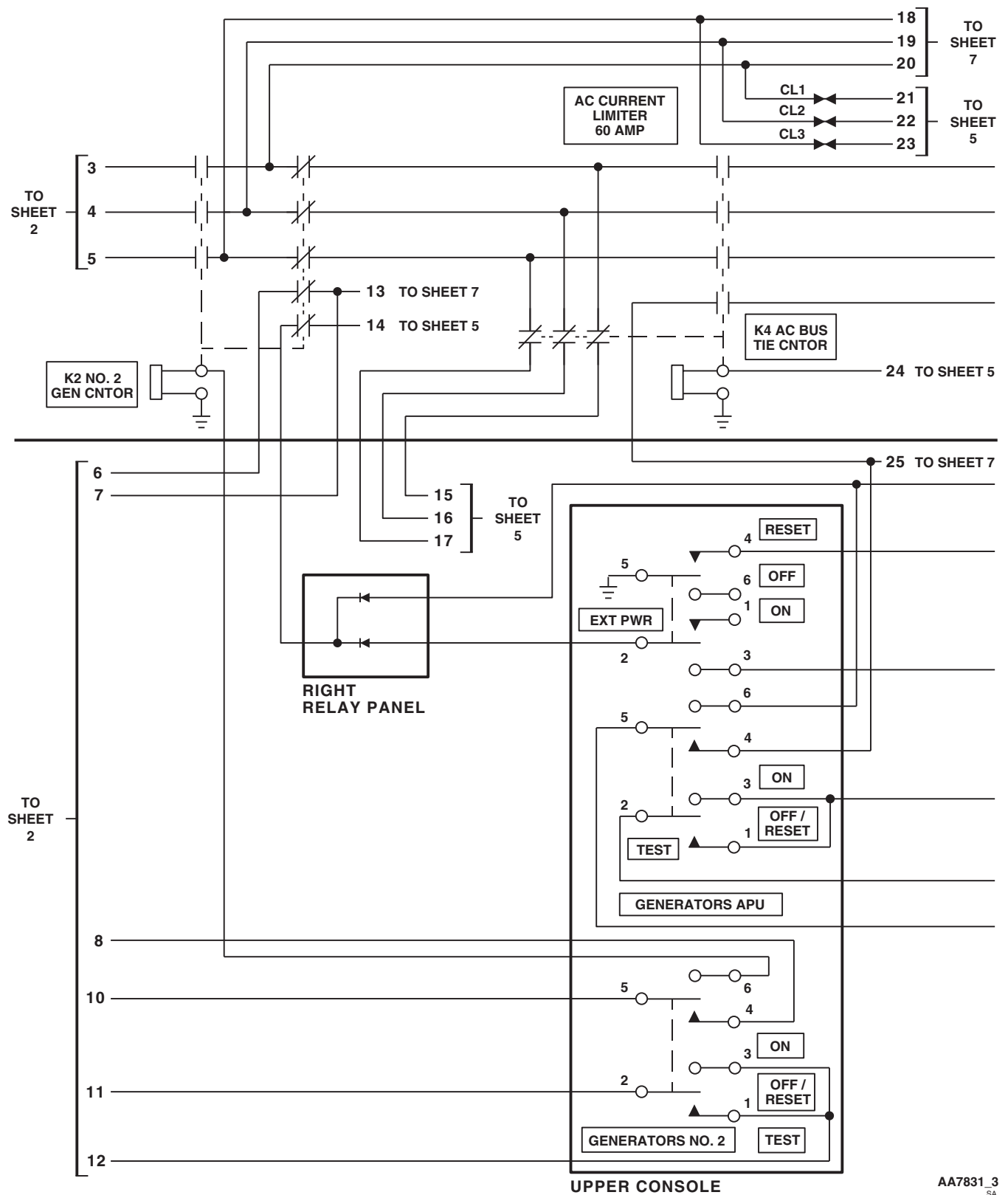


FIGURE 9-1-2-4. AC ELECTRICAL SYSTEM DETAILED BLOCK DIAGRAM (SHEET 3 OF 7)

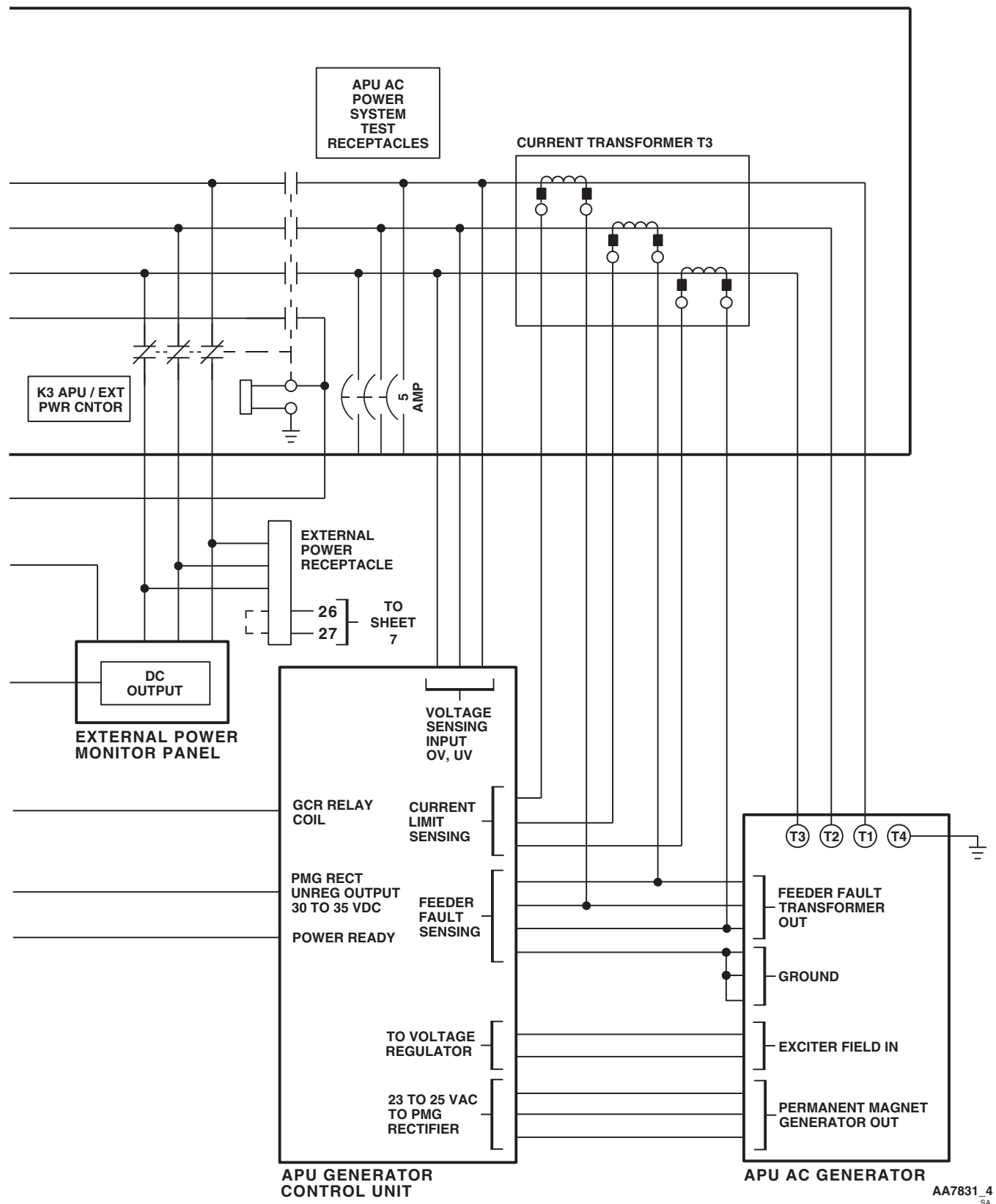


FIGURE 9-1-2-4. AC ELECTRICAL SYSTEM DETAILED BLOCK DIAGRAM (SHEET 4 OF 7)

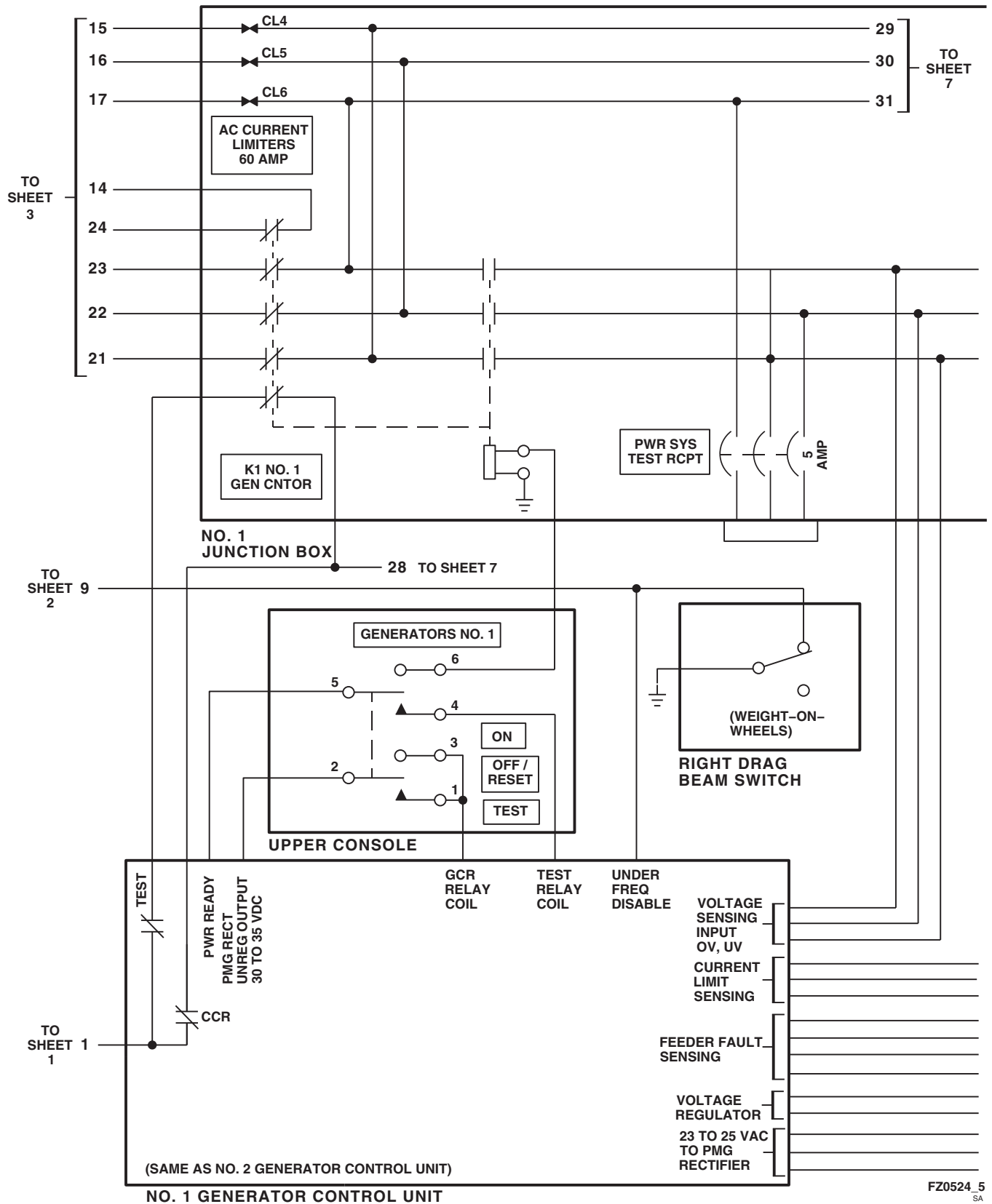
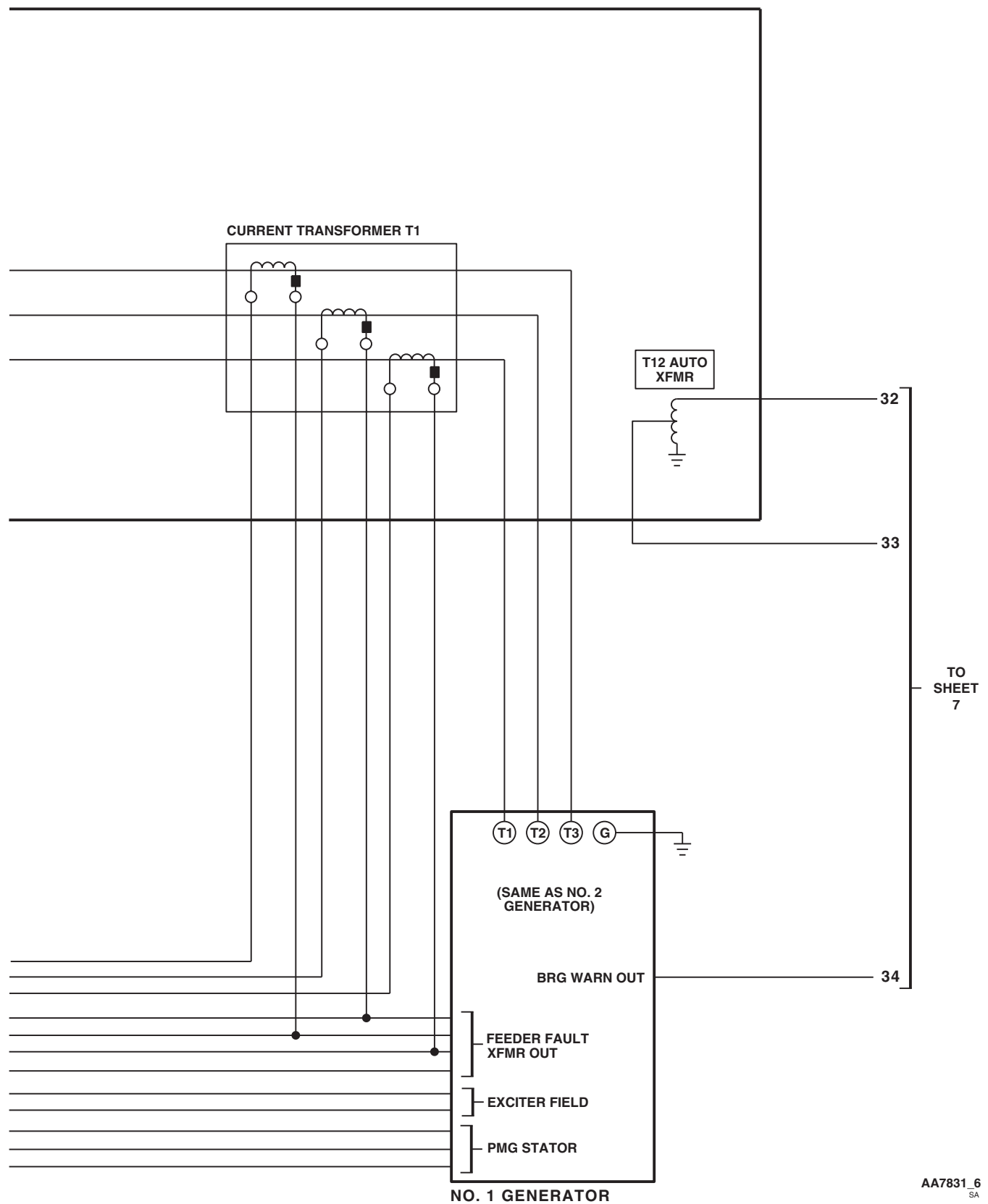
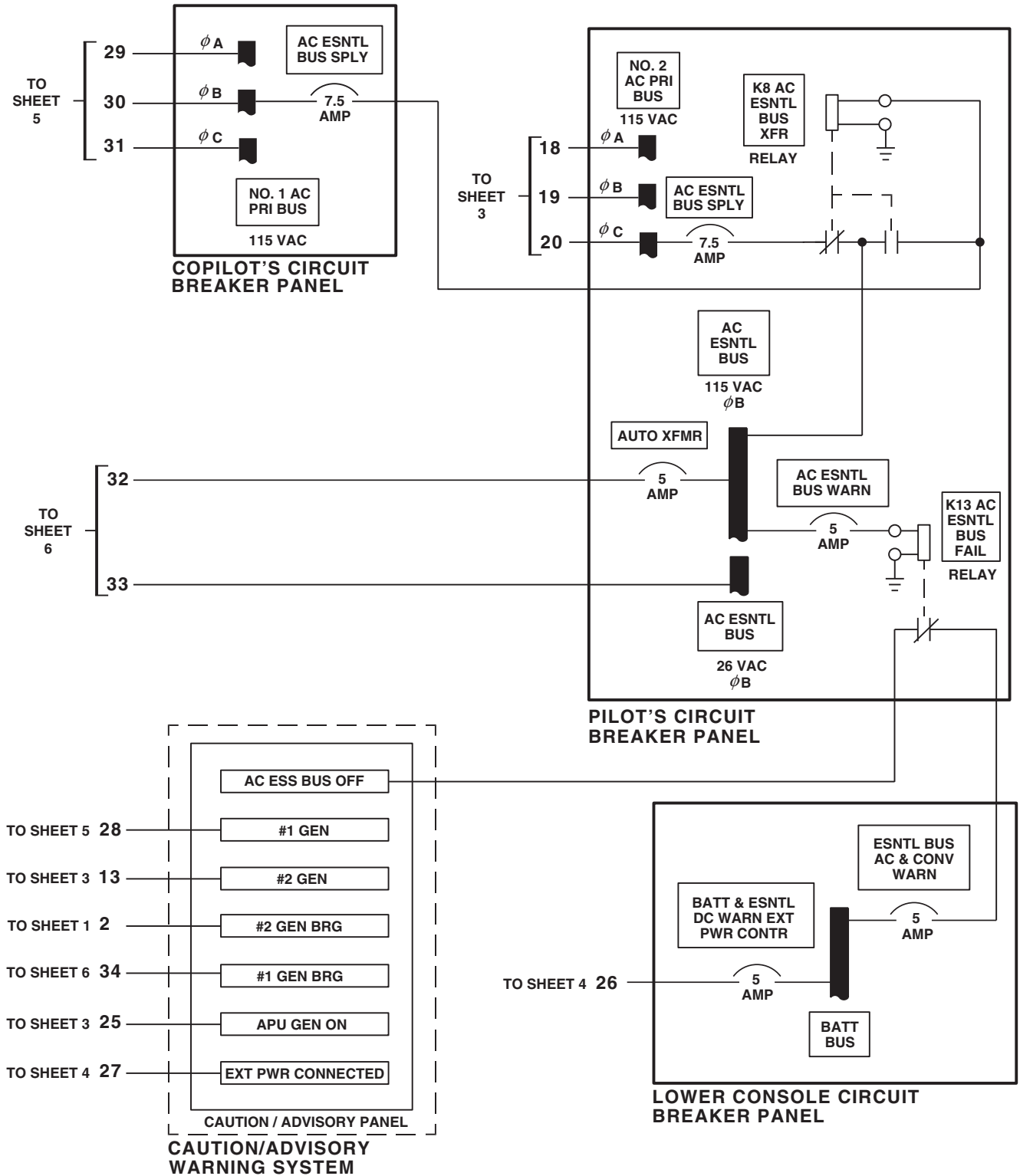


FIGURE 9-1-2-4. AC ELECTRICAL SYSTEM DETAILED BLOCK DIAGRAM (SHEET 5 OF 7)



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FIGURE 9-1-2-4. AC ELECTRICAL SYSTEM DETAILED BLOCK DIAGRAM (SHEET 6 OF 7)



AA7831_7
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FIGURE 9-1-2-4. AC ELECTRICAL SYSTEM DETAILED BLOCK DIAGRAM (SHEET 7 OF 7)

9-1-2-17/(9-1-2-18 Blank)

9-1-3 AC ELECTRICAL AUXILIARY POWER SYSTEM DESCRIPTION AND DATA.

9-1-3.1 Auxiliary Power Unit (APU) Generator Description.

The auxiliary power unit (APU) air-cooled, brushless generator, mounted on the APU, is driven whenever the APU is operating. The generator, rated at 20/30 KVA at 115/200 volts, is connected for a four-wire Wye output, with grounded neutral, furnishing three-phase alternating current. The generator drive shaft is normally driven at about 12,000 RPM to maintain an output frequency of 400 Hz. The generator output voltage is regulated by a voltage regulator within the generator control unit (GCU) which varies the exciter control field. The generator is cooled by drawing air in at the front end and exhausting it at the drive end. The GCU contains sensing circuits to disconnect the generator from its respective loads when an over-voltage, undervoltage, underfrequency, or feeder-fault condition occurs.

9-1-3.2 Auxiliary AC Power System.

Auxiliary ac power is supplied by the APU generator. When the No. 1 and No. 2 ac generators are off and the APU generator output is acceptable, the APU generator supplies power to the No. 1 and No.

2 ac primary buses. The APU GEN ON capsule on the caution/advisory panel will then light. When the No. 1, No. 2 and APU generators are off, external ac power can supply the No. 1 and No. 2 ac primary buses.

9-1-3.3 Auxiliary Power Distribution.

With the No. 1 and No. 2 generators OFF and the APU generator operating at its rated value, APU/external power contactor K3 is energized (Figures 9-1-2-2 and 9-1-2-4). DC voltage is applied through a blocking diode and the normally-closed auxiliary contacts of No. 2 and No. 1 generator contactors K2 and K1, to energize ac bus tie contactor K4. The three-phase ac output of the APU generator is applied through the energized main contacts of contactors K3 and K4 and the normally-closed main contacts of contactor K2 to supply the No. 2 ac primary bus. The No. 1 ac primary bus is supplied through current limiters and the normally-closed main contacts of contactor K1.

With either No. 1 or No. 2 generators on, the APU generator on, and the backup pump on, the ac secondary bus receives auxiliary power if the blade de-icing system (if installed) is off.

9-1-4 AC ELECTRICAL EXTERNAL POWER SYSTEM DESCRIPTION AND DATA.

9-1-4.1 External Power Distribution.

With the No. 1, No. 2, and APU generators off, a source of external ac power can be connected to supply the helicopter buses. When a source of 115 vac, three-phase power is connected to the ac external power receptacle and the battery is installed in the helicopter, dc voltage from the battery bus is applied through the power receptacle and jumpered pins in the external power cart plug to light the caution/advisory panel EXT PWR CONNECTED capsule. The external power is monitored for proper phase rotation, voltage, and frequency by the external power monitor panel. When external power is acceptable, the monitor panel provides a dc output voltage. When the EXT PWR switch is placed to ON, the monitor panel dc voltage is applied through the switch. It goes through a blocking diode and the normally-closed auxiliary contacts of No. 2 and No. 1 generator contactors K2 and K1 to energize ac bus tie contactor K4 (Figure 9-1-2-2). The three-phase external ac power is applied through the normally-closed main contacts of contactor K3, the energized main contacts of contactor K4 and the normally-closed main contacts of contactor K2 to supply the No. 2 ac primary bus. The No. 1 ac primary bus is supplied through current limiters and the normally-closed main contacts of contactor K1. If an overvoltage, undervoltage, overfrequency, or underfrequency fault occurs, the external power monitor panel dc output circuit opens; the ac bus tie contactor K4 will be de-energized; and the external ac power will be disconnected from the helicopter buses. When the fault is corrected, external power can be reconnected to the buses if the EXT PWR switch is placed to RESET and then back to ON.

9-1-4.2 External Power Operation.

With the No. 1, No. 2, and APU generators OFF, an external power source can be connected to supply all ac buses (Figures 9-1-2-3 and 9-1-2-4). A source of 115 vac, three-phase power is connected to the external power receptacle. With a battery installed in the helicopter and the BATT switch ON,

the battery supplies power to the battery bus. DC voltage from the battery bus is applied through the power receptacle and jumpered pins in the external power cart plug to light the caution/advisory panel EXT PWR CONNECTED capsule. The external power monitor panel monitors the external power for proper phase rotation, voltage, and frequency. One phase of external power is also supplied to a power supply in the monitor panel that provides dc voltage to the sensing circuits and to an output circuit. When the external power is acceptable, the monitor panel output circuit closes and applies dc voltage to the EXT PWR switch. With the No. 1, No. 2, and APU generators OFF, No. 1 and No. 2 generator contactors K1 and K2 and APU/external power contactor K3 are de-energized. When the EXT PWR switch is placed to ON, the dc voltage is applied through a blocking diode and the normally-closed auxiliary contacts of contactors K2 and K1 to energize ac bus tie contactor K4. The blocking diode prevents 28 vdc from being applied to the EXT PWR switch when the APU generator is on.

The three-phase ac power is applied from the external power receptacle through the normally-closed main contacts of contactor K3, the energized main contact of contactor K4, and the normally-closed main contacts of contactor K2 to supply the No. 2 ac primary bus. The No. 1 ac primary bus is supplied through current limiters CL1, CL2, and CL3 and the normally-closed main contacts of contactor K1. The No. 1 ac primary bus supplies 115 vac, B phase power, through the AC ESNTL BUS SPLY circuit breaker, to the coil and contacts of AC ESSENTIAL BUS XFR relay K8. With relay K8 energized, 115 vac, B phase power is supplied to the ac essential bus. The ac essential bus applies 115 vac, B phase power to an autotransformer that supplies 26 vac to navigation instrument loads.

An overvoltage, undervoltage, overfrequency, or underfrequency fault in the external power is sensed by the external power monitor panel. When this occurs, the monitor panel output circuit opens, de-energizing ac bus tie contactor K4 and removing the external power from the ac buses. When the

fault is corrected, contactor K4 can be re-energized and external power reconnected to the buses by set-

ting the EXT PWR switch to RESET and then back to ON.

9-1-5 DC ELECTRICAL SYSTEM DESCRIPTION AND DATA.

9-1-5.1 DC Electrical System Description.

Helicopter dc electrical power is provided by a primary dc system (Figures 9-1-5-1, 9-1-5-2, and 9-1-5-3). The primary dc system consists of two converters, supplied from the ac power system, and associated caution/advisory capsules.

70583-702374 The battery system consists of a nickel-cadmium battery, battery analyzer/conditioner, battery relay K7, battery switch S10, and caution/advisory panel capsules.■

702375 - SUBQ The battery system consists of a sealed lead acid battery, battery charging relay K200, battery low sensing relay K201, battery relay K7, battery switch S10, and caution/advisory panel capsules.■ DC power is distributed to the helicopter buses through a series of contactors and relays. The primary dc system supplies electrical power to the No. 1 and No. 2 dc primary buses, the dc essential bus, and the battery bus. The battery always supplies the battery utility bus. It can also supply power to the battery bus and dc essential bus when no ac electrical power is supplied to the helicopter buses. The battery bus controls APU starting on the ground, or in the air if both ac generators fail.

The No. 1 and No. 2 converters receive three-phase ac power from the No. 1 and No. 2 ac primary buses. With both converters operating normally, the No. 1 converter supplies the No. 1 dc primary bus and the No. 2 converter supplies the No. 2 dc primary bus. If the No. 2 (or No. 1, as applicable) converter has no output or if the associated contactor is not working, the #2 (or #1) CONV capsule on the caution/advisory panel will light and the No. 2 (or No. 1) dc primary bus will be supplied from the No. 1 (or No. 2) dc primary bus through a 100-amp current limiter. The dc essential bus is normally supplied by the No. 1 dc primary bus. If there is no power on the No. 1 dc primary bus, the dc essential bus is supplied by the No. 2 dc primary bus. With the dc essential bus supplied from either the No. 1 or No. 2 dc primary bus, the battery bus is supplied from the dc essential bus. The No. 1 and No. 2 primary bus also supplies power to a cabin utility receptacle and a command console power receptacle.

70583-702374 The battery is connected directly to and always supplies the battery utility bus. When both converters are off and the battery switch is ON, the battery also supplies the battery bus and the battery bus supplies the dc essential bus. Battery temperature and percent of charge are continuously monitored by the battery analyzer/conditioner. If a battery overtemperature or overcharge condition occurs, the BATTERY FAULT capsule on the caution/advisory panel will light. If the battery is less than 40% charged, the BATT LOW CHARGE capsule will light. If the battery charge drops below 35%, the dc essential bus will be dropped and the DC ESS BUS OFF capsule will light.

702375 - SUBQ The battery is connected directly to and always supplies the battery utility bus. When both converters are off and the battery switch is ON, the battery supplies the battery bus and the battery bus supplies the dc essential bus. If battery voltage drops to less than 23 vdc, relay K201 de-energizes and the BATT LOW CHARGE capsule on the caution/advisory panel will light. Battery charging is done via relay K200 when either converter is on line and the battery switch is on.

9-1-5.2 Converters.

The No. 1 converter, on the upper left-hand cabin, and the No. 2 converter, on the upper right-hand cabin, provide dc power for helicopter equipment. The converters operate from 115/200 vac, three-phase, 400 Hz power and provide outputs of 31 vdc at no load to 25 vdc at 200 amperes. The converters are cooled by internal fans.

9-1-5.3 Junction Boxes.

The No. 1 junction box, on the upper left-hand cabin, and the No. 2 junction box, on the upper right-hand cabin, provide mounting space for dc power system components. The No. 1 junction box contains the No. 1 dc primary bus contactor, the dc bus tie contactor, and a current limiter. The No. 2 junction box contains the No. 2 dc primary bus contactor.

9-1-5.4 DC Primary Bus Contactors.

The No. 1 and No. 2 dc primary bus contactors, on the No. 1 and No. 2 junction boxes, respectively, provide connections from the No. 1 and No. 2 converters to the No. 1 and No. 2 dc primary buses. With the converters operating, their dc outputs energize the contactor solenoids and supply the dc primary buses. When energized, one set of contactor auxiliary contacts open the path to the battery relay to prevent the battery from supplying power to the dc buses. If either converter fails, de-energizing its primary bus contactor, a set of auxiliary contacts closes, energizing the dc bus tie contactor, and another set of contacts closes, lighting the caution/advisory panel #1 or #2 CONV capsule. When both converters are off, normally closed contacts of both contactors provide paths for energizing the battery relay and the No. 2 dc essential bus supply relay to allow the battery to supply dc buses.

9-1-5.5 DC Bus Tie Contactor.

The dc bus tie contactor K15, on the No. 1 junction box, provides a connection between the No. 1 and No. 2 dc primary buses. If one converter fails, the path is closed from the primary bus of the operating converter to energize the solenoid of the dc bus tie contactor. The energized contactor connects the primary bus of the operating converter to the primary bus of the failed converter.

9-1-5.6 DC Bus Tie Current Limiter.

A current limiter on the No. 1 junction box is a 100-ampere fuse protecting the operation. This current limiter operates when one dc primary bus is supplying the other dc primary bus as a result of a converter failure caused by a load short circuit.

9-1-5.7 No. 1 DC Essential Bus Supply Relay.

The No. 1 dc essential bus supply relay K10, behind the pilot's circuit breaker panel, connects the dc essential bus to the No. 1 dc primary bus when energized, or the dc essential bus to the No. 2 dc primary bus or the battery bus when de-energized. When the converters are operating, power from the No. 1 dc primary bus energizes the

solenoid of the No. 1 dc essential bus supply relay. This power is applied through the energized relay contacts to supply the dc essential bus. If power is lost on the No. 1 dc primary bus, the relay is de-energized and power from the No. 2 dc primary bus is applied through the normally-closed relay contacts to supply the dc essential bus. When the converters are off and the BATT switch is ON, power from the battery bus is applied through the normally-closed contacts of the No. 1 dc essential bus supply relay to supply the dc essential bus.

9-1-5.8 No. 2 DC Essential Bus Supply Relay.

70583-702374 The No. 2 dc essential bus supply relay K9, behind the pilot's circuit breaker panel, connects the dc essential bus to the battery bus when energized, or the dc essential bus to the No. 2 dc primary bus when de-energized. When the converters are off, the BATT switch is ON, and the battery is more than 35% charged, the path is closed from the battery utility bus, through the battery analyzer/conditioner, to energize the solenoid of the No. 2 essential bus supply relay. Power from the battery bus is applied through the energized relay contacts to supply the dc essential bus. If the battery charge falls to 35%, the path through the analyzer/conditioner to the relay solenoid is opened and the relay de-energizes. The path from the battery bus to the dc essential bus is opened and the dc essential bus is dropped. If only the No. 2 converter is operating and there is no power on the No. 1 dc primary bus, power from the No. 2 dc primary bus is applied through the normally-closed contacts of the No. 2 dc essential bus supply relay and the No. 1 dc essential bus supply relay to supply the dc essential bus. **702375 - SUBQ** The No. 2 dc essential bus supply relay K9, behind the pilot's circuit breaker panel, connects the dc essential bus to the battery bus when energized, or the dc essential bus to the No. 2 dc primary bus when de-energized. When the converters are off, the BATT switch is ON, and battery voltage is greater than 22.7 vdc, the path is closed from the battery utility bus, to energize the solenoid of the No. 2 essential bus supply relay. Power from the battery bus is applied through the energized relay contacts to supply the dc essential bus. If battery voltage falls below 22.7 vdc, the path to the relay solenoid is opened

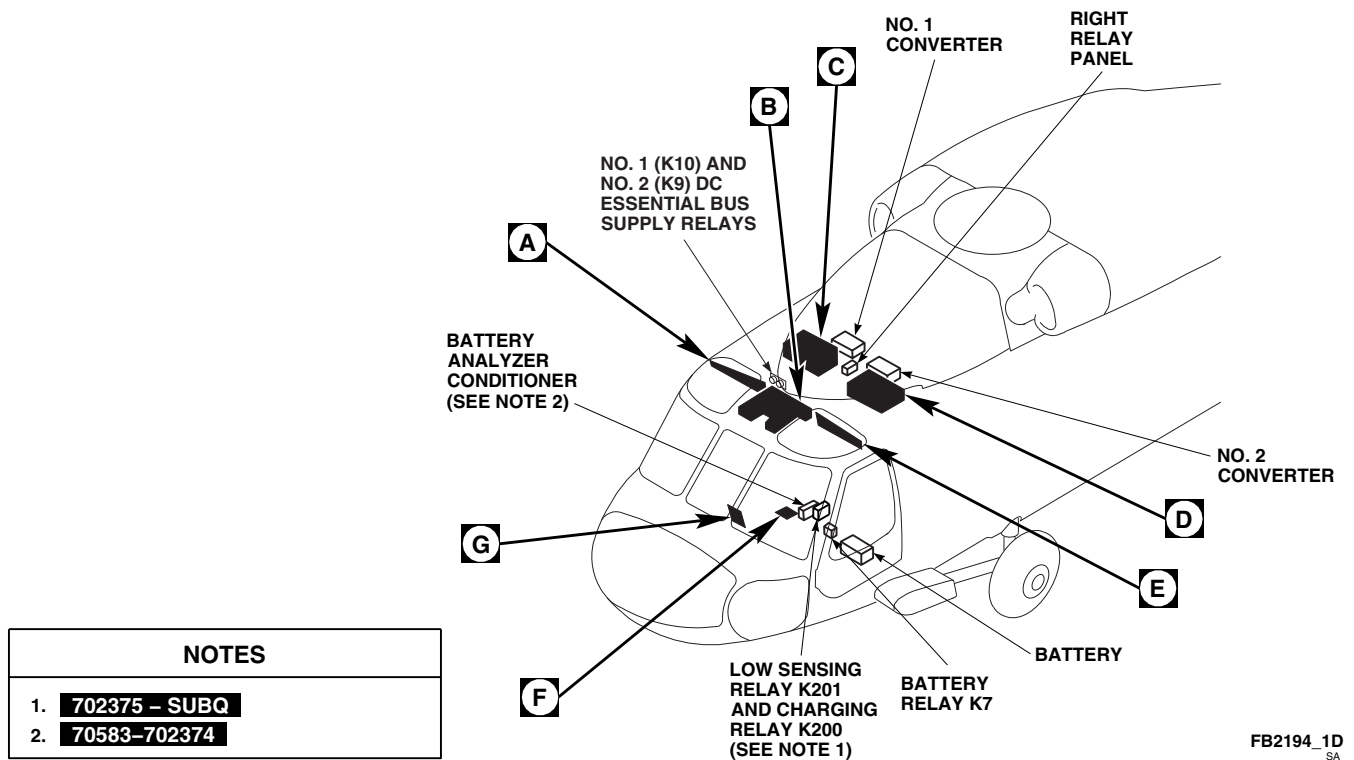


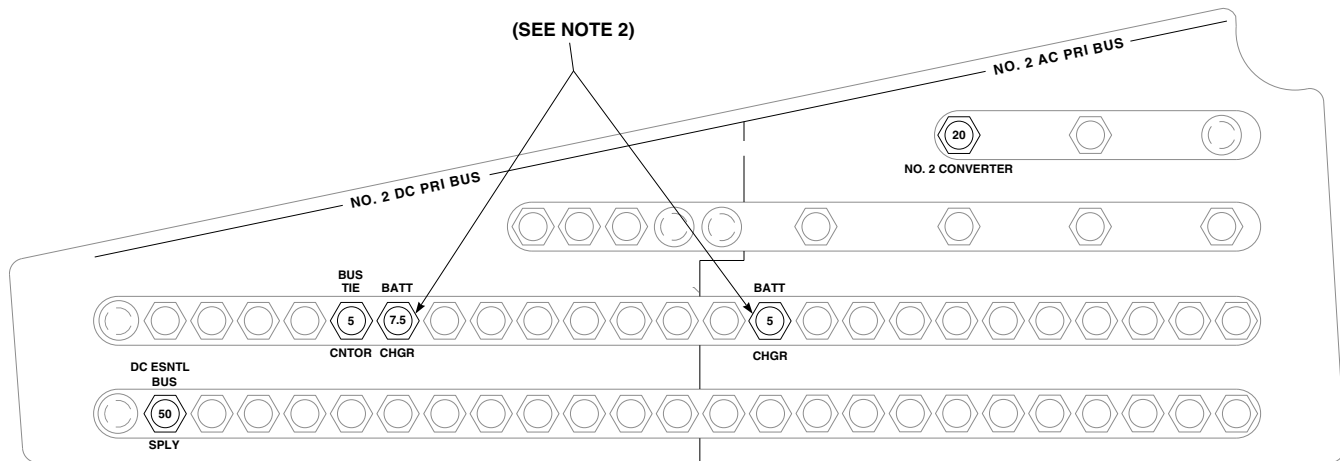
FIGURE 9-1-5-1. DC ELECTRICAL SYSTEM LOCATION DIAGRAM (SHEET 1 OF 5)

and the relay de-energizes. The path from the battery bus to the dc essential bus is opened and the dc essential bus is dropped. If only the No. 2 converter is operating and there is no power on the No. 1 dc primary bus, power from the No. 2 dc primary bus is applied through the normally-closed contacts of the No. 2 dc essential bus supply relay and the No. 1 dc essential bus supply relay to supply the dc essential bus. ■

9-1-5.9 DC Essential Bus Fail Relay.

The dc essential bus fail relay K20, on the right relay panel in the upper cabin, provides a path to

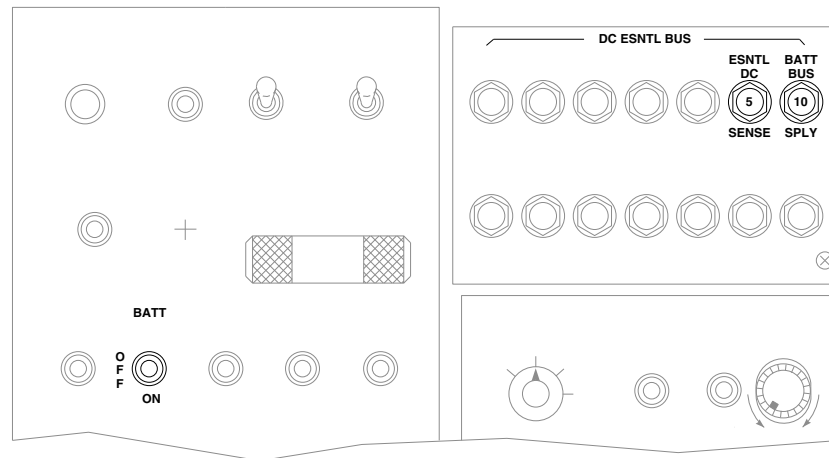
light the caution/advisory panel DC ESS BUS OFF capsule. Power from the dc essential bus energizes the relay solenoid opening the path between the battery bus and the caution/advisory panel capsule. When the battery is supplying the dc essential bus and the battery charge falls to 35% or battery voltage falls to less than 22.7 vdc, the bus is dropped. The relay is now de-energized and the normally-closed relay contacts close the path from the battery bus to light the DC ESS BUS OFF capsule.



PILOT'S CIRCUIT BREAKER PANEL

A

B



UPPER CONSOLE

FB2194_2B
SA

FIGURE 9-1-5-1. DC ELECTRICAL SYSTEM LOCATION DIAGRAM (SHEET 2 OF 5)

9-1-5-4

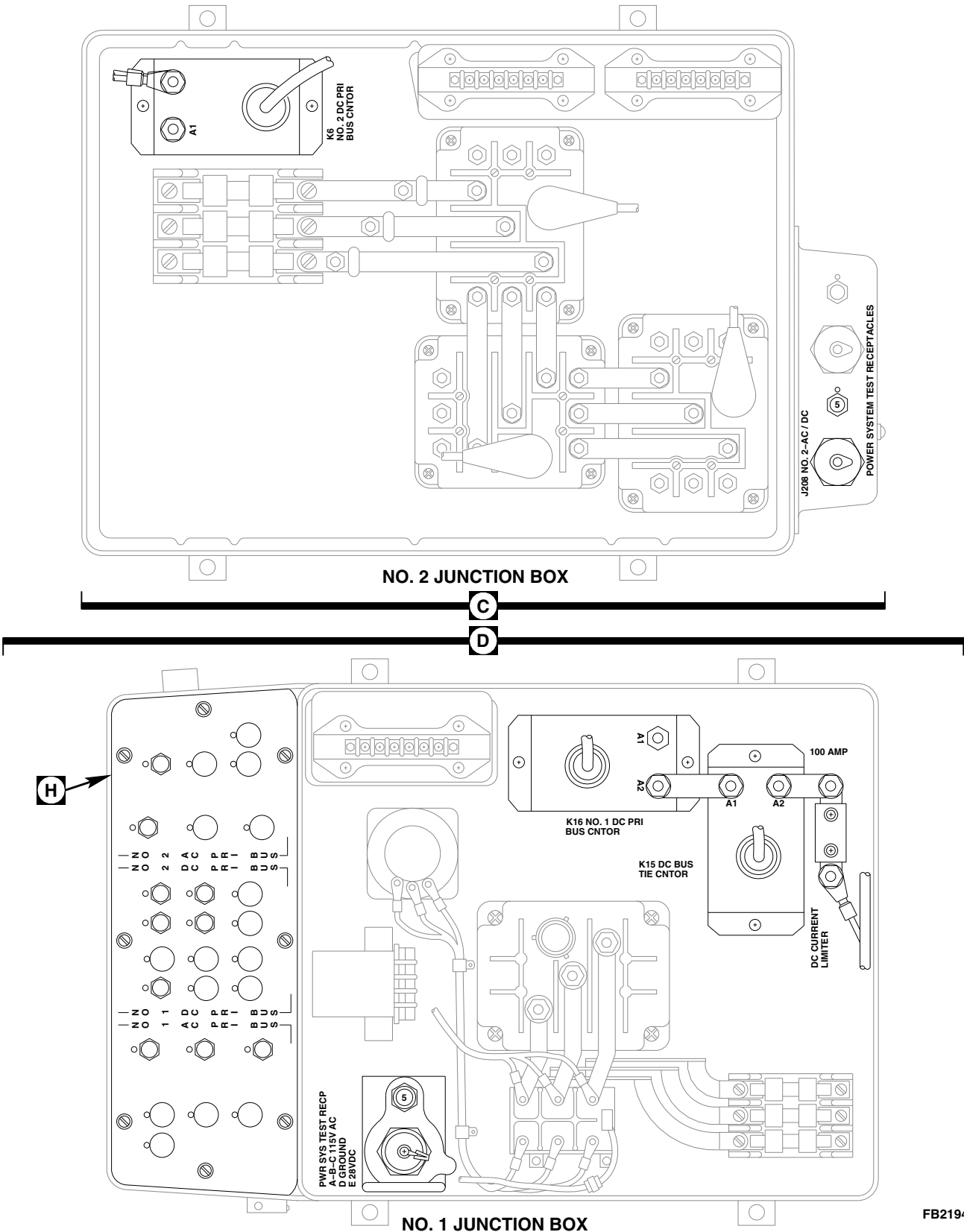
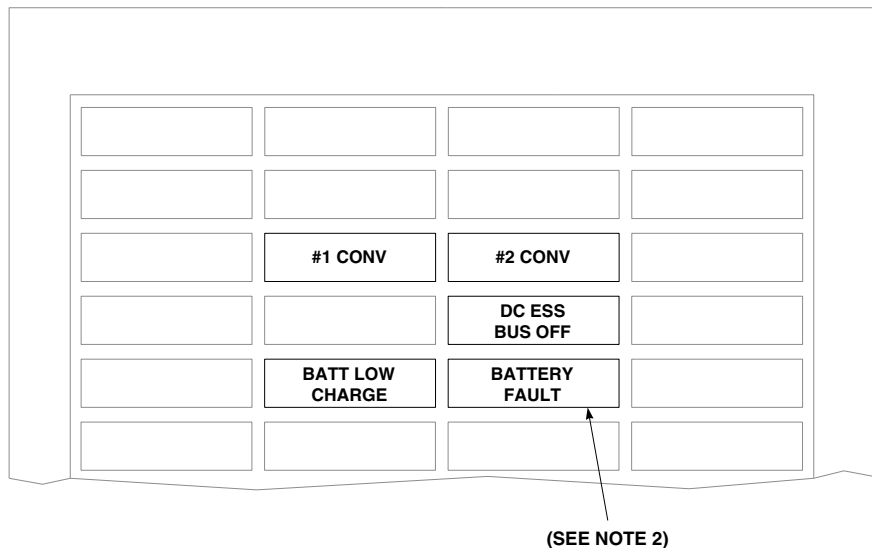
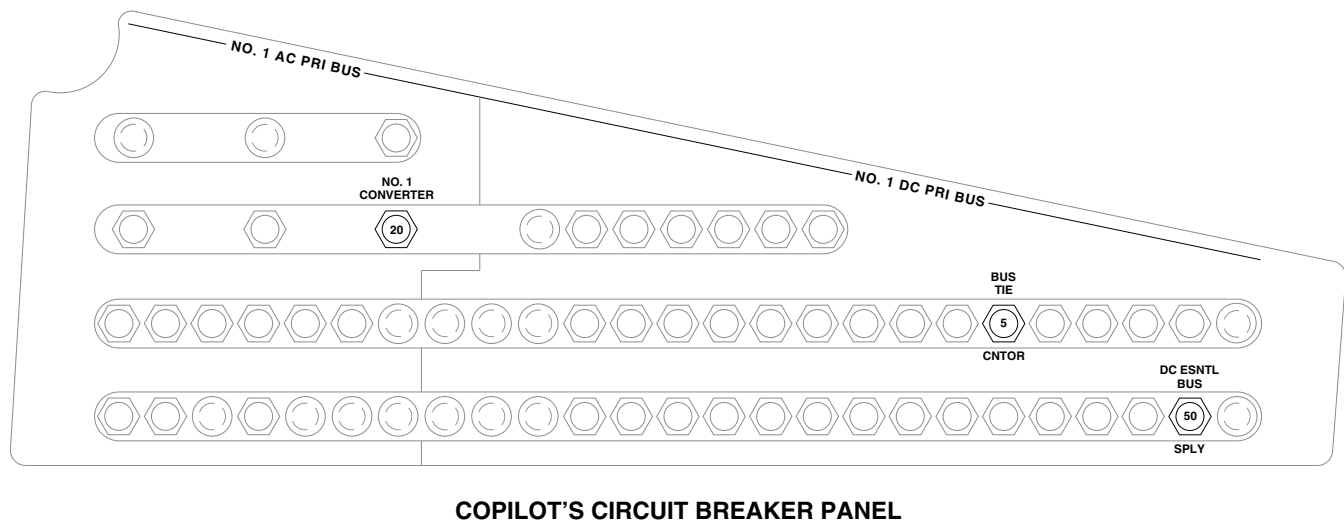
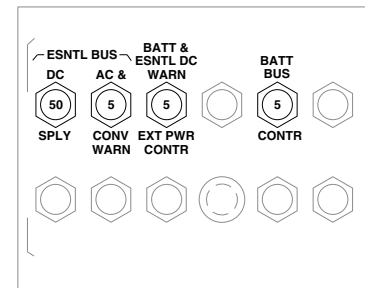


FIGURE 9-1-5-1. DC ELECTRICAL SYSTEM LOCATION DIAGRAM (SHEET 3 OF 5)

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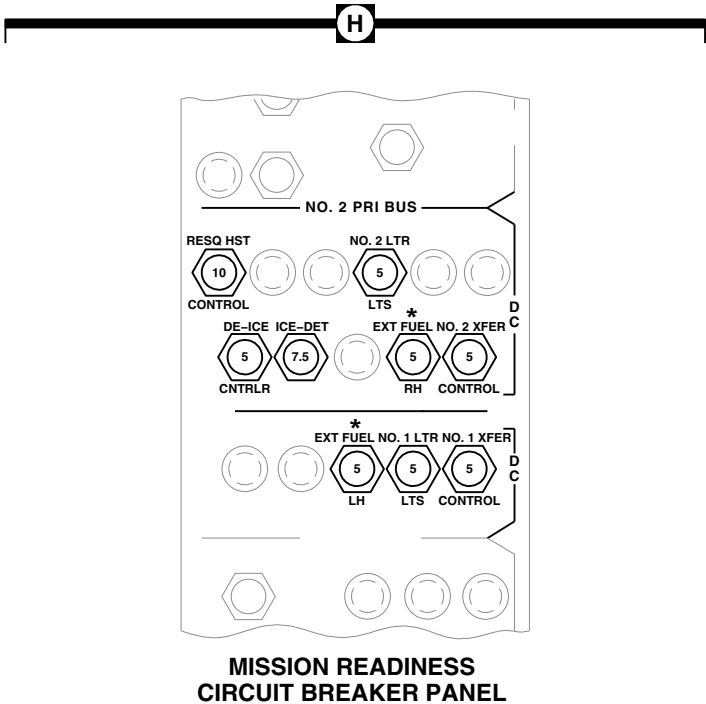
CAUTION/ADVISORY PANEL



LOWER CONSOLE
CIRCUIT BREAKER PANEL

FB2194_4A
SA

FIGURE 9-1-5-1. DC ELECTRICAL SYSTEM LOCATION DIAGRAM (SHEET 4 OF 5)



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FIGURE 9-1-5-1. DC ELECTRICAL SYSTEM LOCATION DIAGRAM (SHEET 5 OF 5)

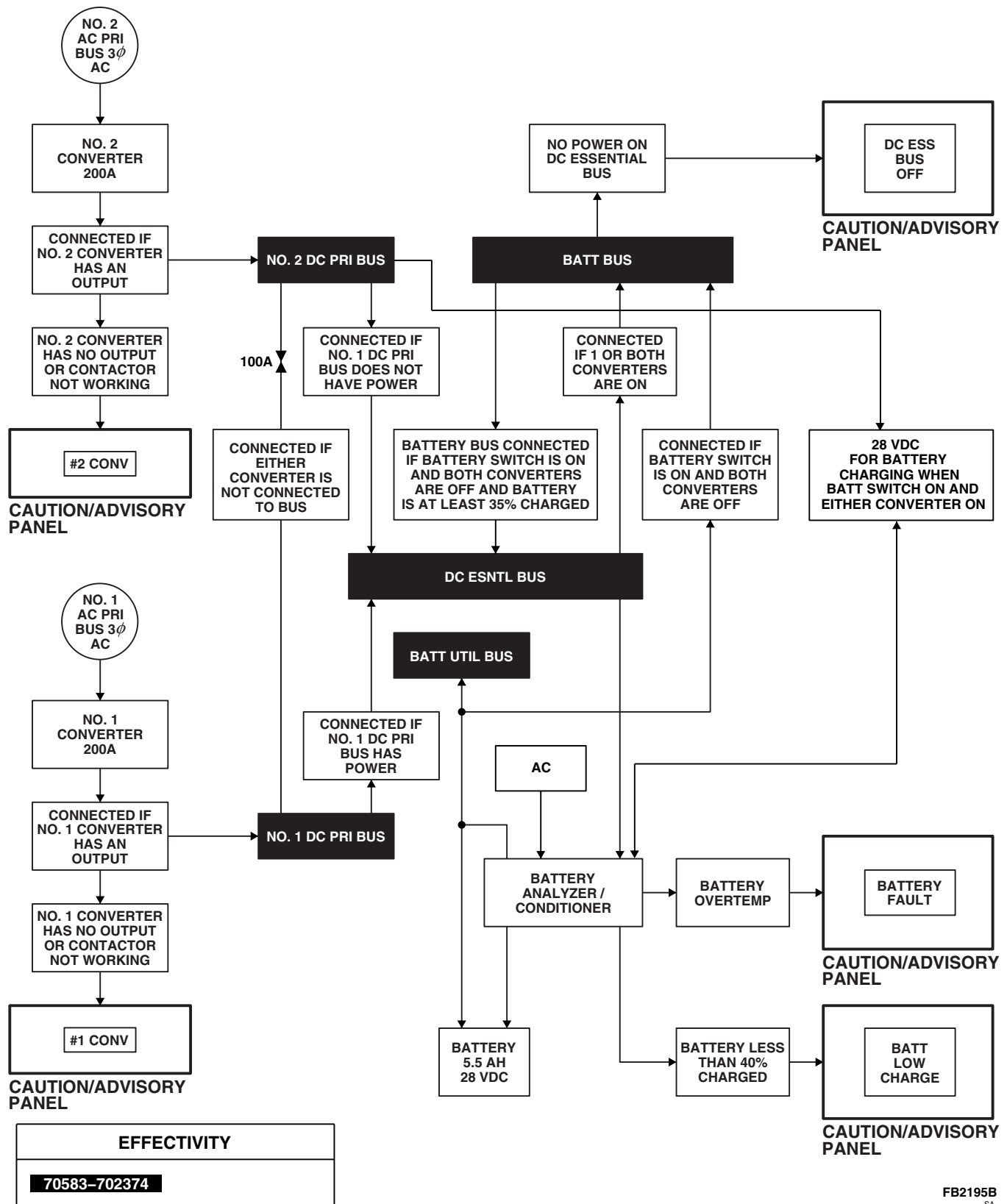


FIGURE 9-1-5-2. DC ELECTRICAL SYSTEM GENERAL BLOCK DIAGRAM

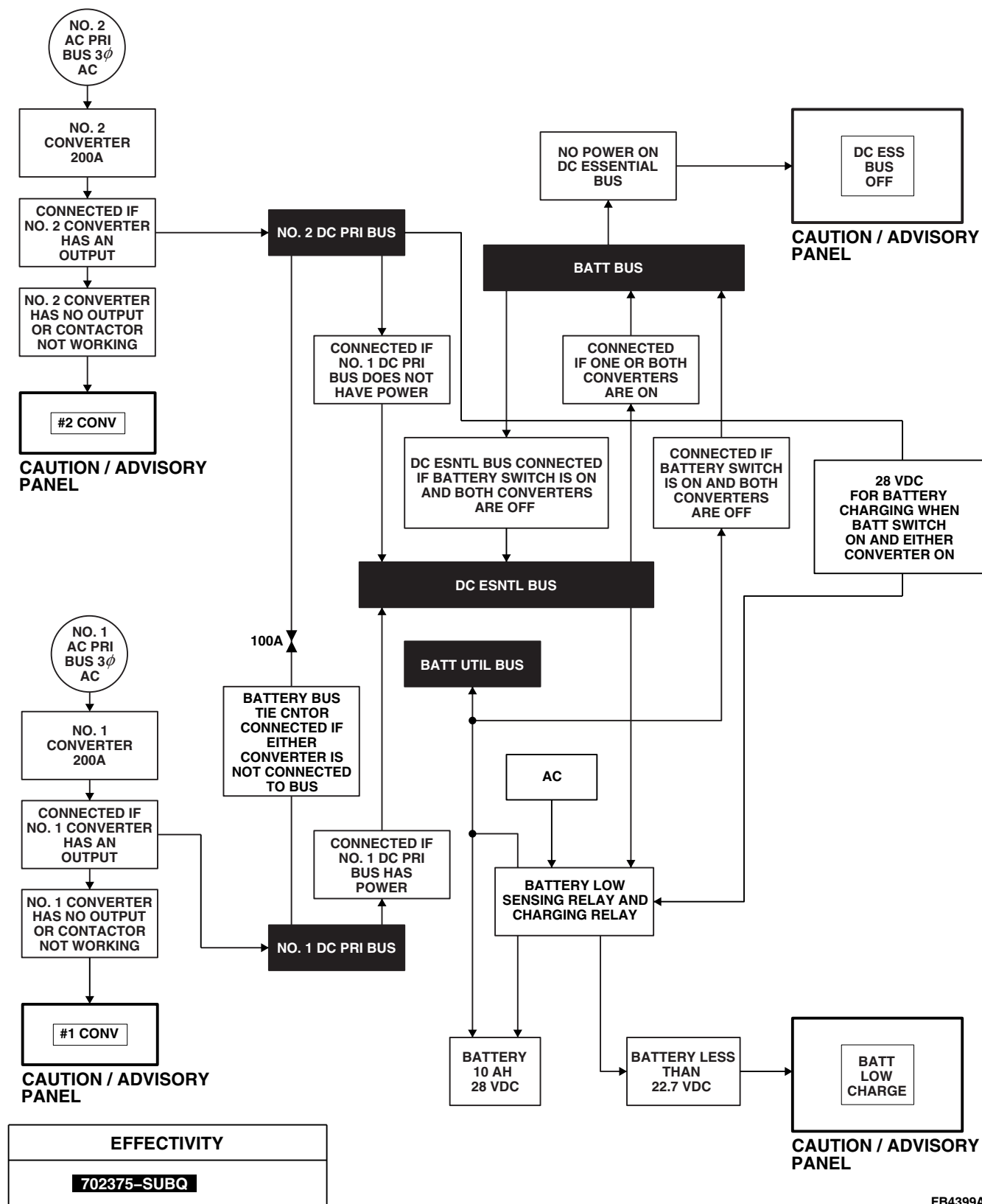


FIGURE 9-1-5-3. DC ELECTRICAL SYSTEM GENERAL BLOCK DIAGRAM

9-1-6 DC ELECTRICAL PRIMARY SYSTEM DESCRIPTION AND DATA.

9-1-6.1 DC Primary Power System.

Primary DC power is developed by the No. 1 and No. 2 converters using three-phase AC inputs from the NO. 1 and NO. 2 AC PRI BUSES (Figures 9-1-6-1 and 9-1-6-2). The DC outputs of the converters are applied through the energized contacts of NO. 1 and NO. 2 DC PRI BUS contactors K16 and K6 to supply the NO. 1 and NO. 2 PRI BUSES, respectively. If one converter fails, DC bus tie contactor K15 is energized and DC power from the primary bus of the operating converter is applied, through a current limiter, to the primary bus of the failed converter. The current limiter protects the operating converter from an excessive overload that may have disabled the failed converter.

DC power from the NO. 1 DC PRI BUS is applied through the energized contacts of the NO. 1 DC ESNTL BUS supply relay K10 to supply the DC ESNTL BUS. If there is no power on the NO. 1 DC PRI BUS, power from the NO. 2 DC PRI BUS is applied through the normally-closed contacts of the NO. 2 ESNTL BUS supply relay K9 and the NO. 1 DC ESNTL BUS supply relay K10, to supply the DC ESNTL BUS. The BATT BUS is supplied by the DC ESNTL BUS through the normally-closed contacts of battery relay K7.

When the converters are on and the battery switch is ON, the battery is charged by the primary buses through battery charging relay K200 (Figure 9-1-6-1).

9-1-6.2 Converter Operation.

The No. 1 converter is supplied with 115 VAC, three-phase power from the NO. 1 AC PRI BUS, through the NO. 1 CONVERTER circuit breaker. The No. 2 converter is supplied with 115 VAC, three-phase power from the NO. 2 AC PRI BUS, through the NO. 2 CONVERTER circuit breaker (Figure 9-1-6-2). The AC input is rectified and filtered by each converter to produce a DC output. The DC output of the No. 1 (or No. 2, as applicable) converter energizes NO. 1 (or NO. 2) DC PRI BUS contactor K16 (or K6) and is applied through the energized main contacts of contactor

K16 (or K6) (Figure 9-1-6-3) to supply the NO. 1 (or NO. 2) DC PRI BUS.

DC power from the NO. 1 DC PRI BUS, through the DC ESNTL BUS SPLY circuit breaker, energizes NO. 1 DC ESNTL BUS supply relay K10 and is applied through the energized contacts of relay K10 to supply the DC ESNTL BUS. The DC ESNTL BUS supplies DC power, through the BATT BUS SPLY circuit breaker and the normally-closed contacts of battery relay K7, to the battery bus. The DC ESNTL BUS also supplies DC power through the ESNTL DC SENSE circuit breaker to energize DC ESNTL BUS fail relay K20. With relay K20 energized, the path between the battery bus and the caution/advisory panel DC ESS BUS OFF capsule is open and the capsule is off.

If the No. 2 converter fails and its output drops to zero, NO. 2 DC PRI BUS contactor K6 de-energizes. DC power from the battery bus is applied through the ESNTL BUS AC & CONV WARN circuit breaker and the normally-closed auxiliary contacts of contactor K6 to light the caution/advisory panel #2 CONV capsule. DC power from the NO. 1 DC PRI BUS is applied through the BUS TIE CNTOR circuit breaker and the normally-closed auxiliary contacts of contactor K6 to energize DC bus tie contactor K15. With contactor K15 energized, DC power from the NO. 1 DC PRI BUS is applied through the 100 amp current limiter to supply the NO. 2 DC PRI BUS. If the No. 1 converter fails and its output drops to zero, NO. 1 DC PRI BUS contactor K16 de-energizes. DC power from the battery bus is applied through the ESNTL BUS AC & CONV WARN circuit breaker and the normally-closed auxiliary contacts of contactor K16 to light the caution/advisory panel #1 CONV capsule. DC power from the NO. 2 DC PRI BUS is applied through the BUS TIE CNTOR circuit breaker and the normally-closed auxiliary contacts of contactor K16 to energize DC bus tie contactor K15. With contactor K15 energized, DC power from the NO. 2 DC PRI BUS is applied through the 100-amp current limiter to supply the NO. 1 DC PRI BUS.

If a short circuit causes the No. 1 converter to be disabled and the 100-amp current limiter to open, all power is lost on the NO. 1 DC PRI BUS. If this occurs, NO. 1 DC ESNTL BUS supply relay K10 de-energizes. DC power from the NO. 2 DC PRI BUS is applied through the DC ESNTL BUS SPLY circuit breaker and the normally-closed contacts of relay K10 to supply the DC ESNTL BUS.

With the converters operating and the BATT switch ON, power is provided from the battery charging relay to charge the battery (Figure 9-1-6-1). DC power from the BATT UTIL BUS is applied through the BATT switch to energize relay K200 to charge the battery. When battery voltage drops below 22.7 VDC, the battery low sensing relay K201 lights the caution/advisory panel BATT LOW CHARGE capsule.

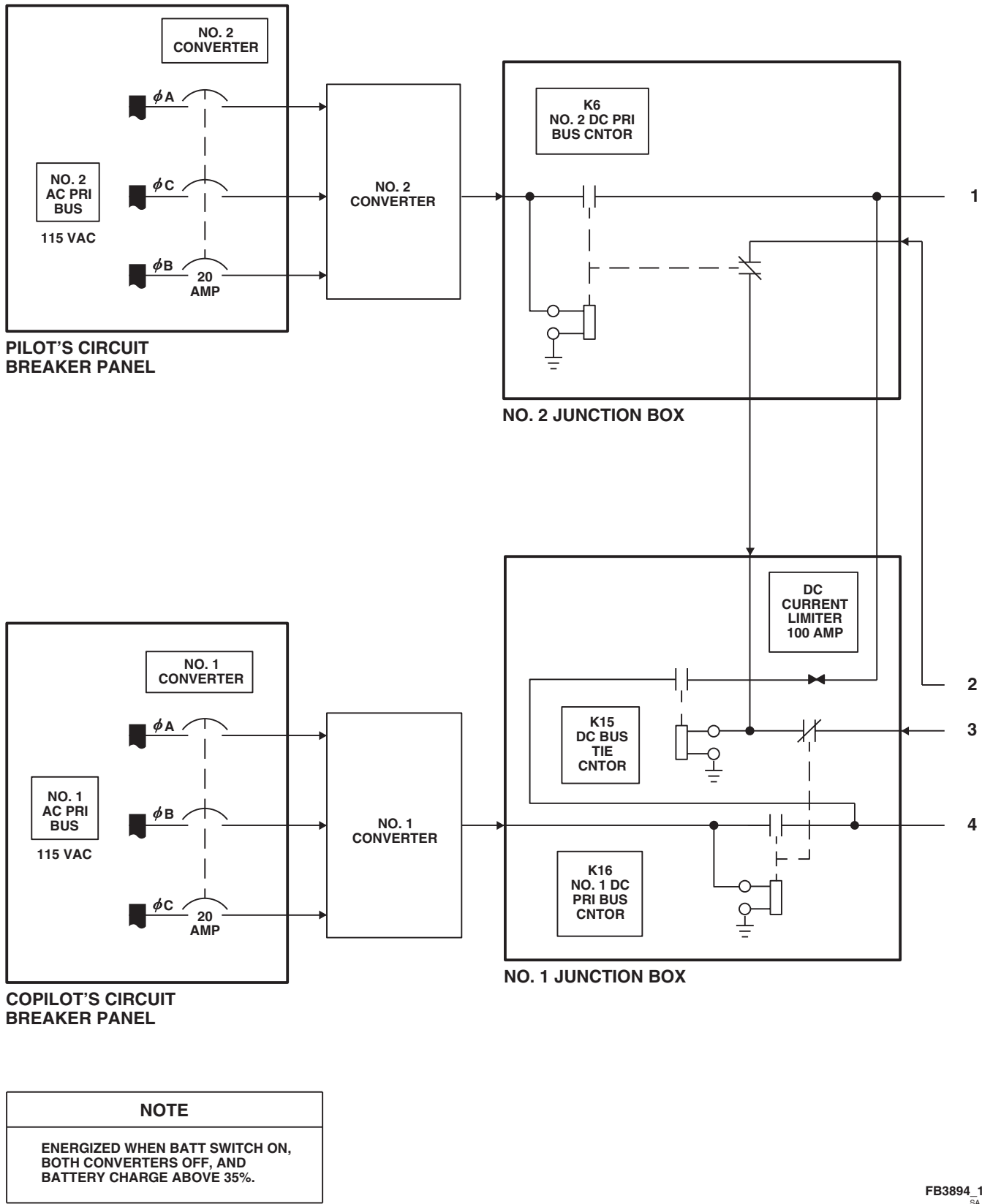


FIGURE 9-1-6-1. DC ELECTRICAL SYSTEM SIMPLIFIED BLOCK DIAGRAM (SHEET 1 OF 2)

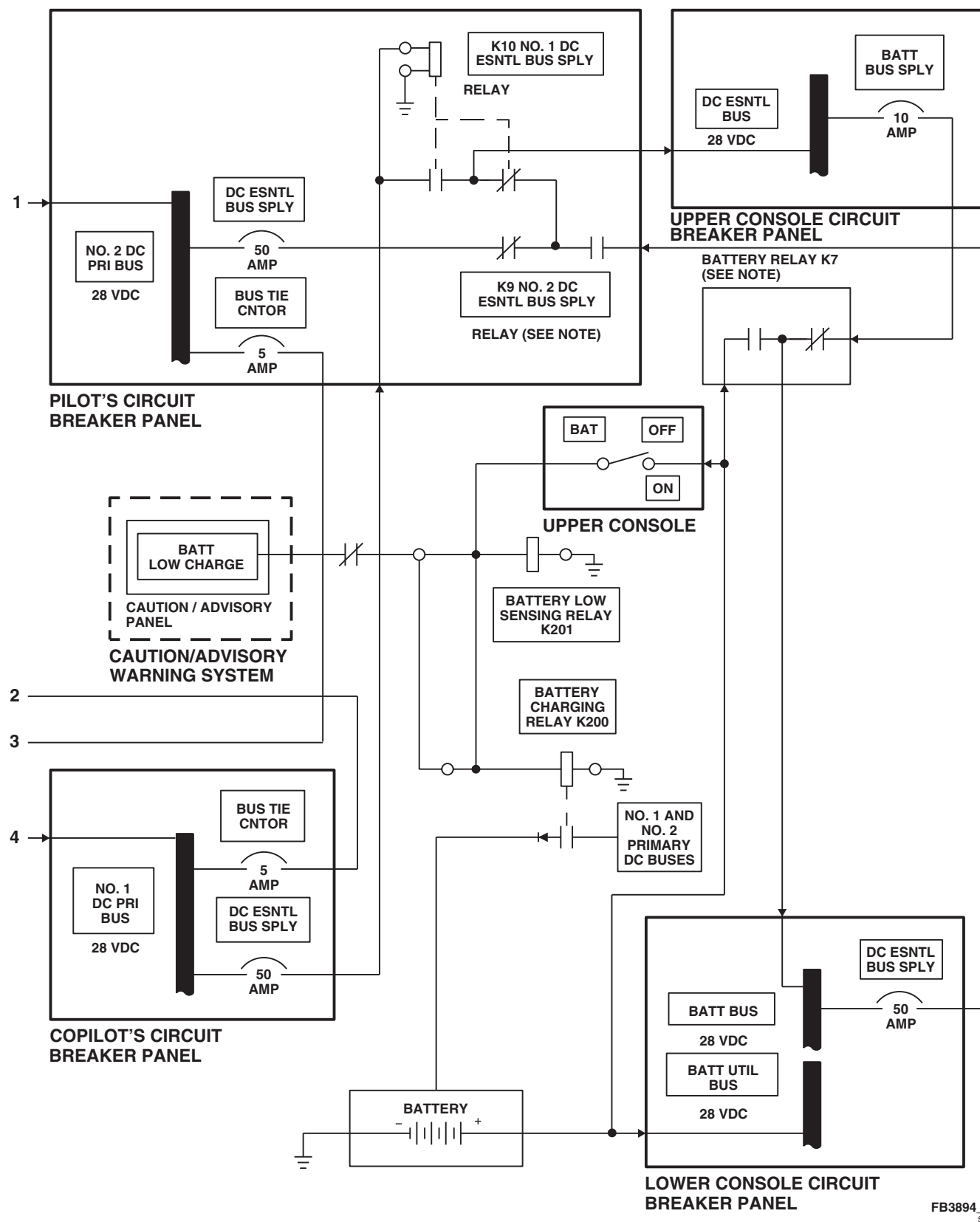


FIGURE 9-1-6-1. DC ELECTRICAL SYSTEM SIMPLIFIED BLOCK DIAGRAM (SHEET 2 OF 2)

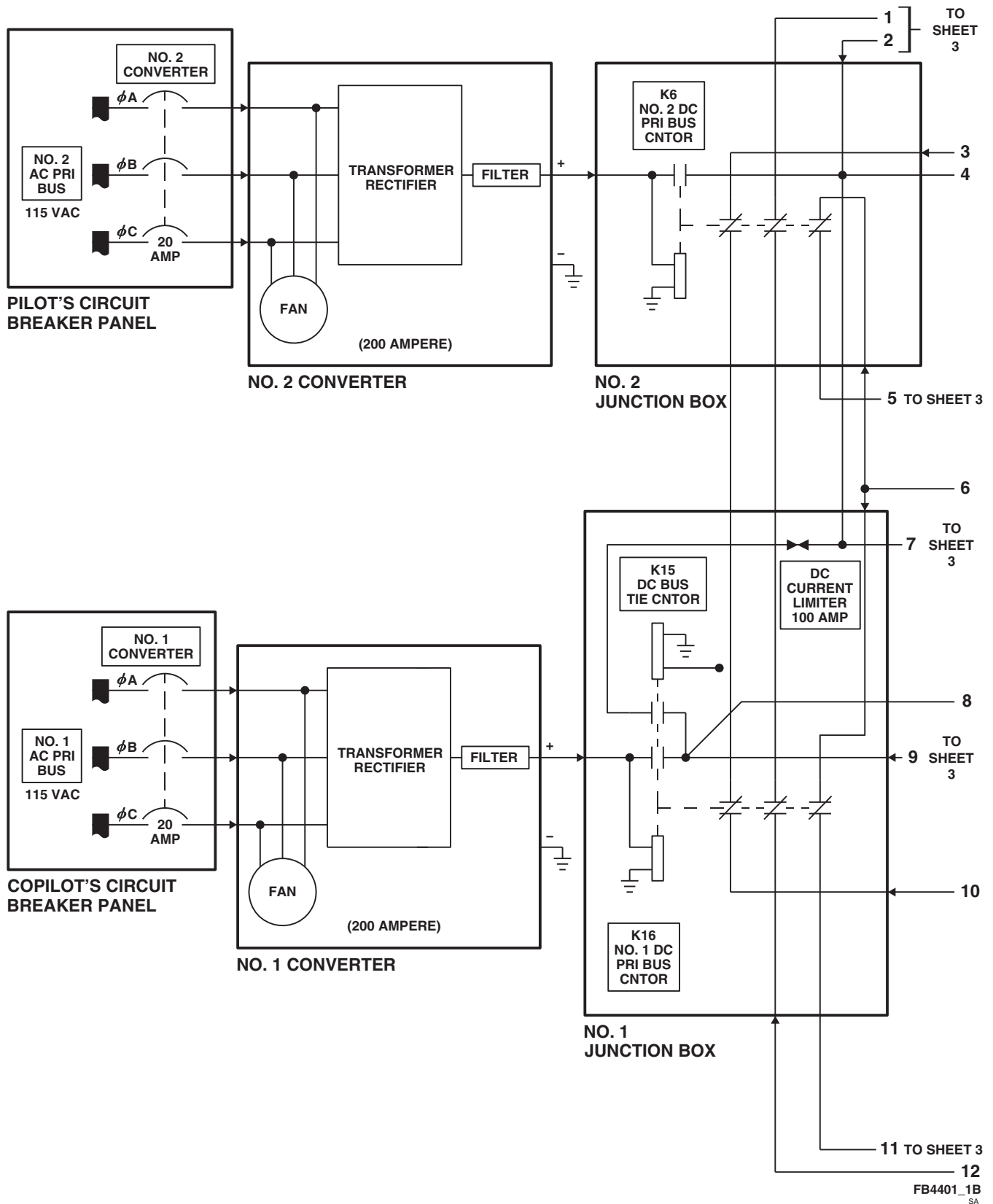
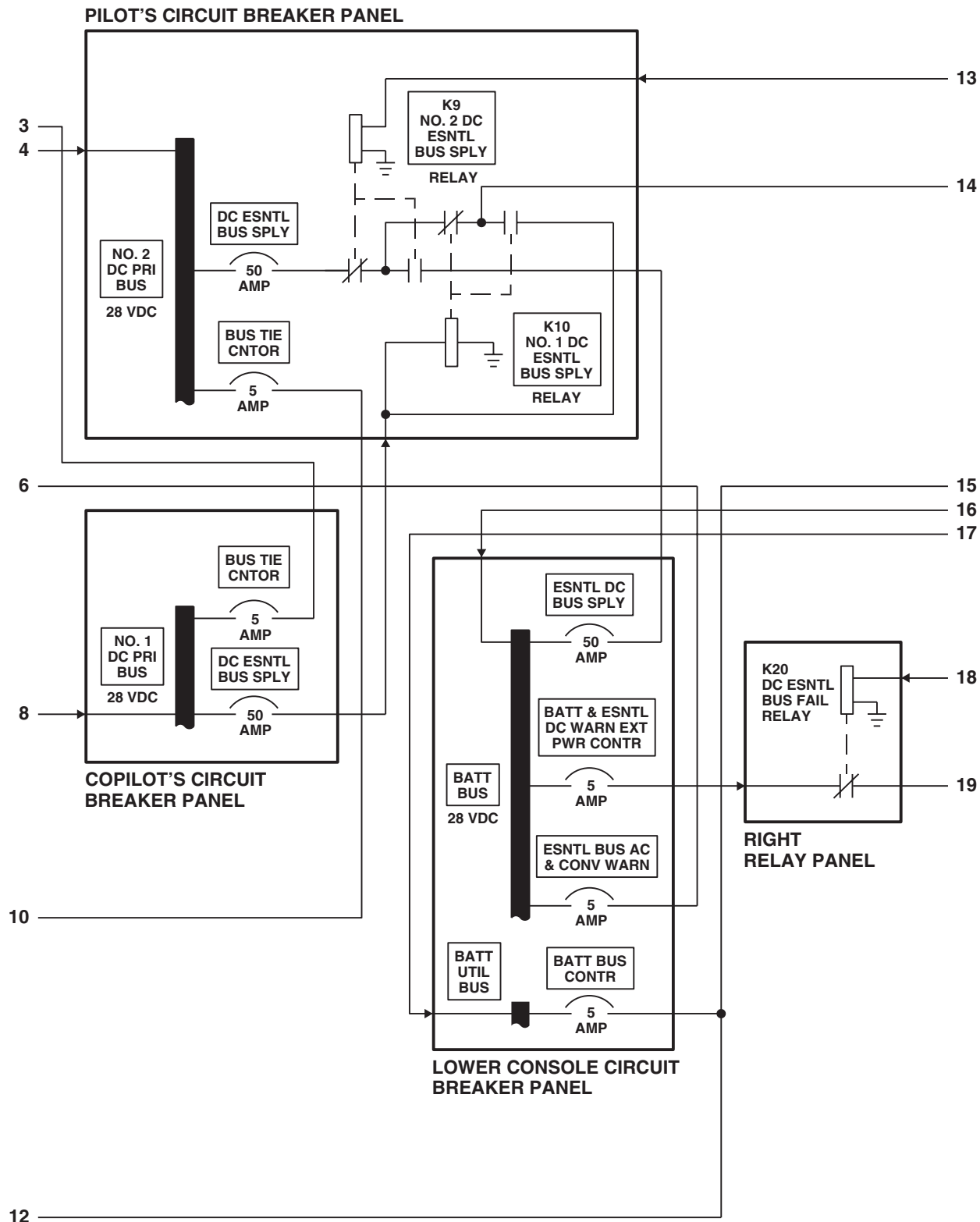
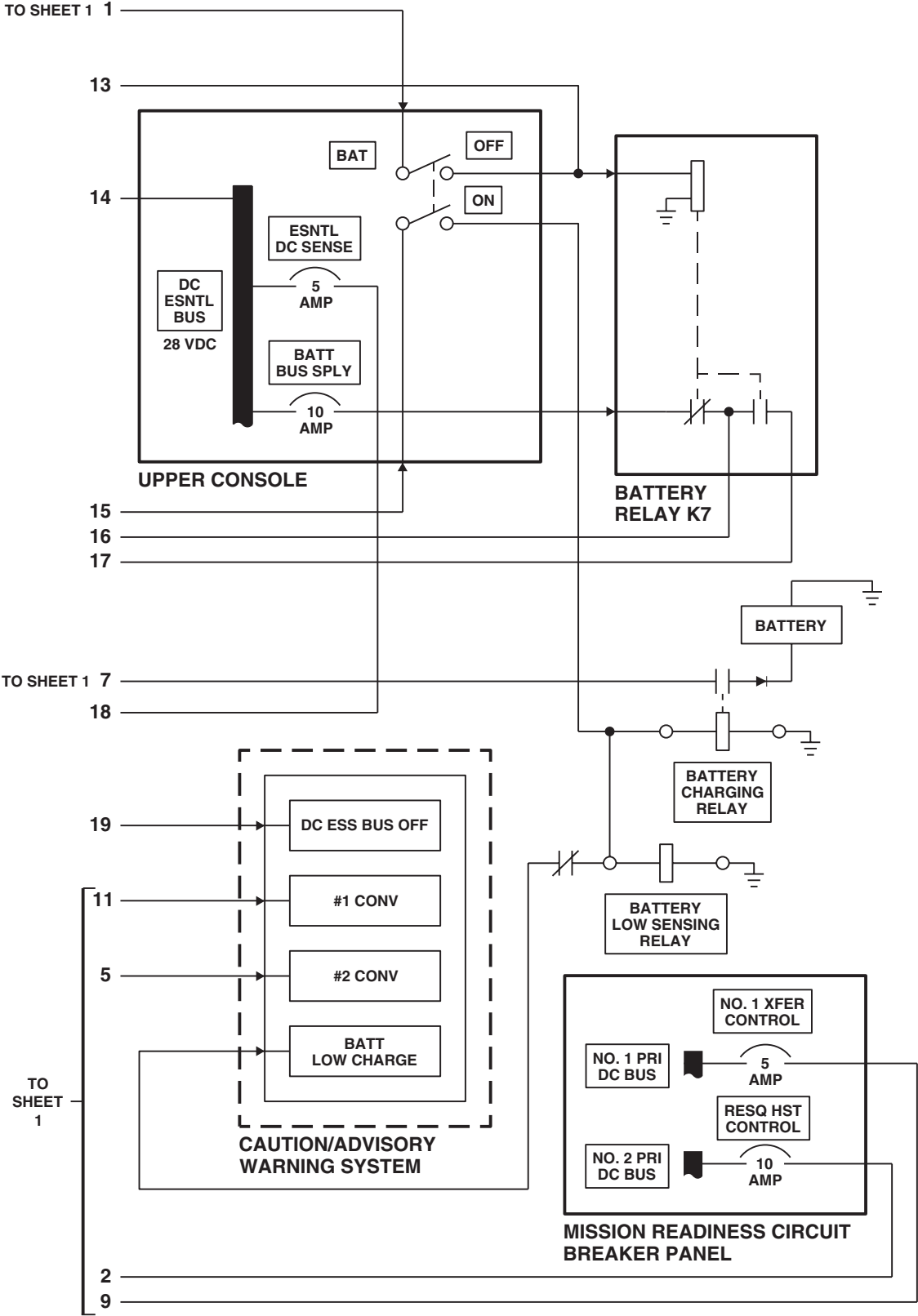


FIGURE 9-1-6-2. DC ELECTRICAL SYSTEM DETAILED BLOCK DIAGRAM (SHEET 1 OF 3)



FB4401_2B
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FIGURE 9-1-6-2. DC ELECTRICAL SYSTEM DETAILED BLOCK DIAGRAM (SHEET 2 OF 3)



FB4401_3B
SA

FIGURE 9-1-6-2. DC ELECTRICAL SYSTEM DETAILED BLOCK DIAGRAM (SHEET 3 OF 3)

DC POWER SOURCE			CONDITION	DC SYSTEM CONTACTORS/RELAYS								DC BUSES					
												NO.1 DC PRI	NO.2 DC PRI	DC ESNTL	BATT UTIL	BATT	
A	B	C		K7	K16	K6	K15	K10	K9	K20	K200	K201*					
0	0	0	1	O	O	O	O	O	O	O	O	O	O	O	O	O	O
1	0	0	2	O	X	O	X	X	O	X	X	X	X	X	X	X	X
0	1	0	3	O	O	X	X	X	O	X	X	X	X	X	X	X	X
1	1	0	4	O	X	X	O	X	O	X	X	X	X	X	X	X	X
0	0	1	5	X	O	O	O	O	X	X	O	O	O	O	X	X	X
1	0	1	6	O	X	O	X	X	O	X	X	O	X	X	X	X	X
0	1	1	7	O	O	X	X	X	O	X	X	O	X	X	X	X	X
1	1	1	8	O	X	X	O	X	O	X	X	O	X	X	X	X	X

LEGEND	
POWER SOURCE	CONTACTORS/RELAYS/BUSES
A = NO.1 CONVERTER	K7 = BATTERY RELAY
B = NO.2 CONVERTER	K16 = NO.1 DC PRI BUS CONTACTOR
C = BATTERY (GREATER THAN 35% CHARGED)	K6 = NO.2 DC PRI BUS CONTACTOR
1 = TURNED ON AND OPERATING	K15 = DC BUS TIE CONTACTOR
0 = TURNED OFF OR FAILED	K10 = NO.1 DC ESNTL BUS SPLY RELAY
* = BATTERY LESS THAN 35% CHARGED	K9 = NO.2 DC ESNTL BUS SPLY RELAY
	K20 = AC ESNTL BUS FAIL RELAY
	X = ENERGIZED
	O = DE-ENERGIZED
	K200 = BATTERY CHARGING RELAY
	K201 = BATTERY LOW SENSING RELAY

FO0516
SA

FIGURE 9-1-6-3. DC ELECTRICAL SYSTEM MATRIX

9-1-7 DC ELECTRICAL BATTERY SYSTEM DESCRIPTION AND DATA.

9-1-7.1 Battery.

The sealed lead acid battery is on the front cabin floor behind the copilot. It provides power for the DC ESNTL BUS on the ground when neither the engines nor the APU are operating. It also provides power for controlling APU starting on the ground, or in the air if both AC generators fail. The sealed lead acid battery has a capacity of ten ampere-hours, based on a one-hour discharge rate at 75°F (24°C) ambient temperature to a cutoff voltage of 18 volts.

9-1-7.2 Battery Switch.

The two-position BATT switch, on the upper console, controls Battery System operation. The ON position permits the battery to supply the BATT and DC ESNTL BUSES when the converters are off, or the battery to be charged by the battery charging relay K200 when the converters are on. The OFF position disables all Battery System operation, except supply of the BATT UTIL BUS.

With both helicopter converters off and the BATT switch placed to ON, DC power from the BATT UTIL BUS is applied through the BATT BUS CONTR circuit breaker, the normally-closed auxiliary contacts of NO. 1 and NO. 2 DC PRI BUS contactors K16 and K6, and the BATT switch, to energize battery relay K7. DC power from the battery is applied through the energized contacts of relay K7 to supply the BATT BUS.

9-1-7.3 Battery Relay.

The battery relay, on the lower front cabin bulkhead left-hand side, connects the battery to the BATT BUS when energized or the DC ESNTL BUS to the BATT BUS when de-energized. When the converters are off and the BATT switch is ON, the path is closed and the BATT UTIL BUS energizes the solenoid of the battery relay. The energized contacts of the battery relay apply power from the battery to the BATT BUS. When the converters are operating, the normally-closed contacts of the battery relay apply power from the DC ESNTL BUS to the BATT BUS. When the

converters are operating and the BATT switch is ON, the battery relay is de-energized.

9-1-7.4 Battery System Operation.

The helicopter battery is connected directly to and is always supplying the BATT UTIL BUS (PARA 9-1-6). With both helicopter converters off and the BATT switch placed to ON, DC power from the BATT UTIL BUS is applied through the BATT BUS CONTR circuit breaker, the normally-closed auxiliary contacts of NO. 1 and NO. 2 DC PRI BUS contactors K16 and K6, and the BATT switch, to energize battery relay K7. DC power from the battery is applied through the energized contacts of relay K7 to supply the BATT BUS. With the battery sufficiently charged, the NO. 2 DC ESNTL BUS supply relay K9 is energized and the BATT BUS supplies the DC ESNTL BUS through the energized contacts of relay K9 and the normally-closed contacts of the NO. 1 DC ESNTL BUS supply relay K10. The NO. 1 and NO. 2 DC PRI BUSES are not supplied by the battery.

The battery condition is continuously monitored by the battery low sensing relay K201. With a battery voltage of 22.7 VDC or higher, DC power from the BATT UTIL BUS is applied through the BATT BUS CONTR circuit breaker, normally-closed auxiliary contacts of NO. 1 and NO. 2 DC PRI BUS contactors K16 and K6, and the energized contacts of relay K1 to energize No. 2 DC ESNTL BUS supply relay K9. DC power from the BATT BUS is applied through the ESNTL DC SPLY circuit breaker, the energized contacts of relay K9, and the normally-closed contacts of No. 1 DC ESNTL BUS supply relay K10, to supply the DC ESNTL BUS. The DC ESNTL BUS applies power through the ESNTL DC SENSE circuit breaker to energize DC ESNTL BUS fail relay K20 to keep the caution/advisory panel DC ESS BUS OFF capsule off.

If battery voltage falls to less than 22.7 VDC, the battery low sensing relay K201 applies a voltage to light the caution/advisory panel BATT LOW CHARGE capsule. The removal of power from the DC ESNTL BUS de-energizes DC ESNTL BUS fail relay K20. This closes the path from the BATT

- I** BUS through the BATT & ESNTL DC WARN
EXT PWR CONTR circuit breaker to light the
caution/advisory panel DC ESS BUS OFF capsule.

9-1-8 INTERIOR LIGHTING DESCRIPTION AND DATA.

9-1-8.1 Interior Lighting.

The interior lighting systems are used for general interior illumination in the cockpit and/or in the cabin area. The interior lighting systems consists of instrument panel lights, console lights, instrument panel and consoles indicator lights dimming circuit, cabin dome lights, cockpit flood and secondary lights, and utility and maintenance lights.

9-1-8.2 Instrument Panel Lights.

The instrument panel lights consist of the pilot's flight instrument lights, the copilot's flight instrument lights, and the non-flight lights (Figures 9-1-8-1, 9-1-8-2, and 9-1-8-3). The lights are used to light instrument panel instruments and controls. Electrical power for the pilot's flight instruments is routed from the NO. 2 AC PRI BUS through the LIGHTS PLT FLT circuit breaker to the INST LT PILOT FLT control on the upper console. Electrical power for the copilot's flight instruments is routed from the NO. 1 AC PRI BUS through the LIGHTS CPLT FLT circuit breaker to the CPLT FLT INST LTS control on the upper console. Electrical power for the non-flight instruments is routed from the NO. 2 AC PRI BUS through the LIGHTS NON FLT circuit breaker to the INST LT NON FLT control, also on the upper console. Turning the dimming controls from OFF to BRT allows the pilot or copilot to vary the intensity of the flight and non-flight instrument lights. Lighted bezel assemblies provide lighting for the pilot's, copilot's, and non-flight indicators.

Electrical power (0 to 115 VAC) from the INST LT NON FLT dimming control is routed to the No. 1 signal data converter, the non-flight 5V/115V transformer, and the pilot's and copilot's collective stick grips. The output of the transformer (0 to 115 VAC) is routed to the icing rate meter light. Electrical power (0 to 5 VAC) from the INST LT PILOT FLT dimming control is routed to the pilot's flight instrument lights 5V/115V transformer and the pilot's airspeed indicator, clock, barometric altimeter, vertical speed indicator, vertical situation indicator, horizontal situation indicator, radar altimeter, and stabilator position

indicator. The output of the transformer (0 to 115 VAC) is routed to the pilot's HSI/VSI mode select panel, pilot's stabilator position placard, and the No. 2 signal data converter. The No. 2 signal data converter routes 0 to 115 VAC power to the pilot's display unit. Electrical power (0 to 5 VAC) from the CPLT FLT INST LTS dimming control is routed to the copilot's flight instrument lights 5V/115V transformer and the copilot's airspeed indicator, clock, barometric altimeter, vertical speed indicator, vertical situation indicator, horizontal situation indicator, radar altimeter, and stabilator position indicator. The output of the transformer (0 to 115 VAC) is routed to the copilot's HSI/VSI mode select panel, copilot's stabilator position placard, and the No. 1 signal data converter. The No. 1 signal data converter routes 0 to 115 VAC power to the central display unit.

9-1-8.3 Console Lighting **W/O SLS** .

Console lighting consists of the upper and lower console lights. Upper console lighting provides control for panel lighting of all control panels on the upper console (Figures 9-1-8-4 and 9-1-8-6). Lower console lighting provides control for panel lighting of all control panels on the lower console, left and right cabin ICS control panels, and mid-cabin ICS control panel. Console lighting uses separate circuits for the upper and lower consoles. The upper console lighting circuit is powered from the NO. 1 AC PRI BUS through the LIGHTS UPPER CSL circuit breaker on the copilot's circuit breaker panel. The lower console lighting circuit is powered from the NO. 1 DC PRI BUS through the LIGHTS LWR CSL circuit breaker on the copilot's circuit breaker panel. The upper console light intensity is controlled by the CONSOLE LT UPPER control on the upper console, which provides 0 to 5 VAC through a step-down auto transformer. Turning the control from OFF to BRT provides a variable 5 VAC to the upper console panels and cockpit flood and secondary lights panel. The lower console light intensity is controlled by the CONSOLE LT LOWER rheostat control on the upper console and relay K41 in the left relay panel. Turning the control from OFF to BRT provides a variable 28 VDC to the rescue hoist control panel,

the fuel boost pump control panel, and stabilator control/auto flight control panel, ESSS range extension system connector, and the dimming control power supply. The dimming control power supply provides a variable 5 VDC output to the Doppler computer display unit and the IFF receiver/transmitter control panel on the lower console. The variable 28 VDC is also fed through the junction box assembly to the lower console panels, the left cabin ICS control panel lights, the right cabin ICS control panel lights, and the mid-cabin ICS control panel lights. The PNL LTS switches on the pilot's and copilot's cyclic stick grips disable lower console lighting and select night dimming operation of the Caution/Advisory Warning System. When either switch is pressed to complete the circuit, 28 VDC is applied to relay K41 solenoid and to the caution/advisory panel.

9-1-8.4 Console Lighting **SLS** .

The console lighting consists of the upper and lower console lights (Figures 9-1-8-4 and 9-1-8-5). Upper console lighting provides control for panel lighting of all control panels on the upper console, cockpit flood and secondary lights panel, and engine controls quadrant. Lower console lighting provides control for panel lighting of some control panels on the lower console. Console lighting uses separate circuits for the upper and lower consoles. The upper console lighting circuit is powered from the NO. 1 AC PRI BUS through the LIGHTS UPPER CSL circuit breaker on the copilot's circuit breaker panel. The lower console lighting circuit is powered from the NO. 1 AC PRI BUS through the LIGHTS LWR CSL circuit breaker on the copilot's circuit breaker panel. The upper console light intensity is controlled by the CONSOLE LT UPPER control, on the upper console. Turning the control from OFF to BRT provides a variable voltage (0 to 115 VAC) to the upper console panels, cockpit flood and secondary lights panel, and engine controls quadrant panels.

The lower console light intensity is controlled by the CONSOLE LT LOWER control, also on the upper console. Turning the control from OFF to BRT provides a variable voltage (0 to 115 VAC) directly to the ESSS range extension system connector, the auxiliary searchlight control panel, the fuel boost pump control panel, the stabilator

control/auto flight control panel, the radio retransmission control panel, and a terminal junction board. From the terminal junction, voltage is supplied to the cabin dome lights dimmer panel, miscellaneous switch panel, compass set control panel, and the electrical balance control panel assembly. The No. 1 VHF/FM and No. 2 VHF/FM radio set control panels receive voltage for lighting through the EL balance control panel.

Turning the control from OFF to BRT also provides a variable voltage (0 to 115 VAC) directly to the low-voltage power supply. The low-voltage power supply feeds variable voltage (0 to 5 VAC) to the following lower console panels:

- Doppler computer display unit
- IFF receiver transmitter
- Civil navigation receiver control panel
- Rescue hoist control box
- Auxiliary rescue hoist control panel
- VHF/AM radio set control panel,
- UHF/AM radio set control panel
- LF/ADF radio set control panel

The low voltage power supply also feeds a variable voltage (0 to 28 VDC) through the audio junction box assembly to the following lower console panels:

- Pilot's ICS control panel
- Copilot's ICS control panel
- Mid-cabin ICS control panel
- Right cabin ICS control panel
- Left cabin ICS control panel
- VHF/AM radio set control panel

9-1-8.5 Instrument Panel and Consoles Indicator Lights Dimming **W/O SLS** .

The instrument panel and consoles indicator lights dimming circuit provides control for decreasing the

intensity of or turning off indicator lights on the instrument panel and upper and lower consoles, and engine and APU fire T-handles (Figures 9-1-8-7 and 9-1-8-9). The dimming circuit consists of relays, dimming resistors, and fuse-protected, solid-state dimming circuits in the left relay panel. Electrical power of 28 VDC is applied from the NO. 1 DC PRI BUS through the LIGHTS ADVSY circuit breaker on the copilot's circuit breaker panel to the left relay panel. With indicator lighting set for full brightness, relays K40, K41, and K43 are deenergized and the dimming circuit output is 28 VDC. The 28 VDC is applied through normally closed contacts of K41 and K40 to switch legend lights on the pilot's and copilot's HSI/VSI mode select panels, CIS mode select panel, and flight control panel. The 28 VDC is also applied to advisory lights on the pilot's and copilot's vertical situation indicators (VSIs), Doppler computer display, blade deice test and control panels, fuel boost control panel, rescue hoist control panel, the CARGO HOOK EMER REL TEST light, and the Tailwheel Lock Indicating System. Normally closed contacts of relay K41 also provide a ground path to light nondimmable switch lights on the flight control panel. In bright mode, the dimming circuit applies 28 VDC to warning and display lights on the pilot's and copilot's radar altimeters. Contacts of relay K43 apply the 28 VDC output of the No. 1 engine, No. 2 engine, and APU fire warning amplifiers directly to the respective fire warning T-handles.

When the caution/advisory panel BRT/DIM-TEST switch is actuated to dim mode, a ground is applied to the left relay panel to energize relays K40 and K43. With K40 energized, a ground is applied to the dimming circuit, causing the dimming circuit voltage to decrease. The decreased dimming circuit output is applied through energized contacts of K40 to light indicator lights with decreased intensity. The dimming circuit also applies decreased voltage directly to the pilot's and copilot's radar altimeters for warning and display light dimming. With K43 energized, the 28 VDC output of each fire warning amplifier is applied through dimming resistors R8, R9, and R10 to dim the T-handles should a fire warning occur. Relay K43 also supplies a dim command ground to the range extension control panel and ESSS range

extension system connector. When the pilot's or copilot's cyclic stick PNL LTS switch is pressed to turn off panel lighting, 28 VDC from the LIGHTS LWR CSL circuit breaker is applied through the switches to energize relay K41. With K41 energized, all indicator light power is removed and all indicator lights are turned off. The nondimmable switch lights on the flight control panel are also turned off by the opening of the ground path through contacts of K41. DC power is also applied to the caution/advisory panel and ESSS Range Extension System connector. This signal places the panel in the night vision goggle (SLS) mode of operation. This same signal disables the ground provided by the external dim control enable output of the caution/advisory panel. The loss of ground deenergizes relays K40 and K43. Deenergizing relay K43 places the No. 1 engine, No. 2 engine, and APU T-handle fire warning lamps and range extension control panel in the bright operating mode. Depending on instrument panel and lower console lighting switch positions, the T-handle lamps and range extension control panel may go on in the dim operating mode.

9-1-8.6 Instrument Panel and Console Indicator Lights Dimming **SLS** .

The instrument panel and consoles indicator lights dimming system provides control for decreasing the intensity of or turning off indicator lights on the instrument panel, upper and lower consoles, and engine and APU fire T-handles (Figures 9-1-8-7 and 9-1-8-8). The System consists of relays K40 and K43 as well as dimming resistors R8, R9, and R10 in the left relay panel, a LIGHTED SWITCHES dimmer control on the upper console, indicator lights dimmer forward of the lower console, pilot's and copilot's RAD ALT DIMMING controls on the instrument panel, and the BRT/DIM-TEST switch on the caution/advisory panel. Relay K43 and dimming resistors, R8, R9, and R10, are used in the engine and APU fire T-handle dimming circuit. Refer to the Fire Detection System (PARA 12-1-1).

The system is electrically powered by 28 VDC in four places. The primary part of the system is powered from the NO. 1 DC PRI BUS through the LIGHTS ADVSY circuit breaker on the copilot's circuit breaker panel. Relay K43 receives power

from the DC ESNTL BUS through the FIRE DET NO. 1 ENG circuit breaker in the upper console. The T-handles for the No. 1 engine, No. 2 engine, and APU receive power from their respective fire detector control amplifiers located in the cabin overhead. The caution/advisory panel is supplied by the DC ESNTL BUS through the CAUT/ADVSY PNL circuit breaker on the upper console circuit breaker panel.

In the bright mode of operation, relays K40 and K43 are de-energized and full voltage of 28 VDC is applied through normally-closed contacts of K40 to light indicator lights on the pilot's and copilot's HSI/VSI mode select panels, CIS mode select panel, ESSS control panel, the auxiliary searchlight control panel, stabilator control/auto flight control panel, the pilot's and copilot's vertical situation indicators (VSIs), the Doppler computer display, the blade de-ice test and control panels, the fuel boost pump control panel, the rescue hoist control panel, the CARGO HOOK EMERG REL TEST light, and the Tailwheel Lock Indicating System. The System also applies the 28 VDC to warning and display lights on the pilot's and copilot's radar altimeters. Normally-closed contacts of relay K43 apply full output voltage of the No. 1 engine, No. 2 engine, and APU fire detector control amplifiers directly to the respective fire warning T-handles.

When the BRT/DIM-TEST switch is placed to DIM, a ground is applied to the left relay panel to energize relays K40 and K43. Then, dimming voltages are routed through the energized contacts of relay K40 in the following way: from the indicator lights dimmer, 9 VDC is applied to the blade de-ice control and test panels, pilot's and copilot's VSI's, Doppler computer display, rescue hoist control panel, and the CARGO HOOK EMERG REL TEST light. From the LIGHTED SWITCHES dimmer control, a variable voltage (0 to 26 VDC) is routed to the pilot's and copilot's HSI/VSI mode select panels, CIS mode select panel, stabilator control/auto flight control panel, tailwheel lock system, ESSS control panel, auxiliary searchlight control panel, and fuel boost pump control panel. Also, through two RAD ALT DIMMING controls, a variable voltage (0 to 28 VDC) is directed to the pilot's and copilot's radar altimeters. While dimming is occurring through relay K40, the 28 VDC output power of each fire detector control amplifier

is applied through the energized contacts of relay K43 via dimming resistors R8, R9, and R10 to dim the T-handles should a fire warning occur. Relay K43 also supplies a dim command ground to the ESSS Range Extension System connector and auxiliary fuel management control panel.

When the BRT/DIM-TEST switch is actuated again to the bright mode, the ground is removed from the left relay panel. The loss of ground de-energizes relays K40 and K43 and all panel indicator lights mentioned above will brighten again.

9-1-8.7 Cabin Dome Lights **W/O SLS** .

Three dome light assemblies are installed in the cabin area to provide red or white lighting (Figures 9-1-8-10 and 9-1-8-12). Each dome light assembly contains a red light and a white light. System electrical power is supplied by the NO. 1 AC PRI BUS through the LIGHTS CABIN DOME circuit breaker on the copilot's circuit breaker panel. The lights are controlled by CABIN DOME LTS OFF-BRT control and the WHITE-OFF-RED switch on the upper console. The light intensity is controlled by the CABIN DOME LTS OFF-BRT control. The three-position switch, WHITE-OFF-RED, has on-off control and permits selection of color.

9-1-8.8 Cabin Dome Lights **SLS** .

Three dome light assemblies are installed in the cabin area to provide blue or white lighting (Figures 9-1-8-10 and 9-1-8-11). Each dome light assembly contains a blue light and a white light. System electrical power is supplied by the NO. 1 AC PRI BUS through the LIGHTS CABIN DOME circuit breaker on the copilot's circuit breaker panel. The lights are controlled by the CABIN DOME LT OFF-BRT dimmer unit behind the pilot's seat and the CABIN DOME LT WHITE-OFF-BLUE switch on the upper console. The three-position switch, WHITE-OFF-BLUE, has on-off control and permits selection of color.

9-1-8.9 Cockpit Flood and Secondary Lights **W/O SLS** .

The cockpit flood and secondary lights consist of two white floodlights and two red secondary lights on the cockpit overhead, six red glare shield lights

on the instrument panel, and standby compass light (Figures 9-1-8-13 and 9-1-8-15). The lighting is controlled by a RED/REMOTE/WHITE switch on the cockpit flood and secondary lights panel, and by SECONDARY LIGHTS-RED and CKPT FLOOD LIGHTS-WHITE controls on the upper console. Electrical power of 28 VDC is supplied to the RED/REMOTE/WHITE switch and to the standby compass switch by the DC ESNTL BUS through the LIGHTS SEC PNL circuit breaker on the upper console circuit breaker panel. Electrical power of 115 VAC is supplied to the CKPT FLOOD LIGHTS-WHITE control by the NO. 2 AC PRI BUS through the LIGHTS CKPT FLOOD circuit breaker on the pilot's circuit breaker panel. Electrical power of 115 VAC is supplied to the SECONDARY LIGHTS-RED control by the NO. 1 AC PRI BUS through the LIGHTS SEC circuit breaker on the copilot's circuit breaker panel. With the RED/REMOTE/WHITE switch placed to REMOTE, the two white floodlights are controlled from OFF to BRT by the CKPT FLOOD LIGHTS-WHITE control; one red secondary light and the glareshield lights are controlled from OFF to BRT by the SECONDARY LIGHTS-RED control. With the switch placed to RED, both secondary red lights are on bright, the glareshield lights are controlled by the SECONDARY LIGHTS-RED control, and the floodlights are controlled by the CKPT FLOOD LIGHTS-WHITE control. With the switch placed to WHITE, both floodlights are on bright, and one secondary light and the glareshield lights are controlled by the SECONDARY LIGHTS-RED control.

9-1-8.10 Cockpit Flood and Secondary Lights **SLS** .

The cockpit flood and secondary lights consist of two white flood lights and two blue secondary lights on the cockpit overhead, six blue glareshield lights on the instrument panel, and a standby compass light (Figures 9-1-8-13 and 9-1-8-14). The lighting is controlled by the BLUE/OFF/WHITE switch on the cockpit flood and secondary lights panel, and by the GLARESHIELD LIGHTS control on the upper console. Electrical power of 28 VDC is supplied to the BLUE/OFF/WHITE switch and to the standby compass switch by the DC ESNTL BUS through the LIGHTS SEC PNL

circuit breaker on the upper console. Electrical power of 115 VAC is supplied to the GLARESHIELD LIGHTS control by the NO. 1 AC PRI BUS through the GLARE SHLD circuit breaker on the copilot's circuit breaker panel. Both blue and white secondary overhead lights are controlled by the BLUE/OFF/WHITE switch located on the cockpit flood and secondary lights panel. The six glareshield lights are controlled by the GLARESHIELD LIGHTS control.

9-1-8.11 Utility and Maintenance Lights **W/O SLS** .

Two portable, hand-held utility lights with coil cords are installed on the upper console, one each for the pilot and copilot, to provide red or white lighting (Figures 9-1-8-16 and 9-1-8-17). System electrical power is supplied by a BATT UTIL BUS through the UTIL LTS CKPT circuit breaker on the lower console. The utility lights are attached to swivel-type detachable mountings and may be held to provide red or white flood or spotlighting. Depressing the lock button on the casing assembly and simultaneously turning the lens casing selects a light filter for red or white lighting. Color-coded dots on the lens casing, large for flood and small for spotlight, change the beam of light from a flood to a spotlight pattern. Each light is controlled by a rheostat and a momentary "on" switch on the end of each utility light assembly. The rheostats, with OFF, DIM and BRT settings, provide variable intensity control for lighting; the momentary "on" switch overrides the rheostat to provide maximum lighting intensity.

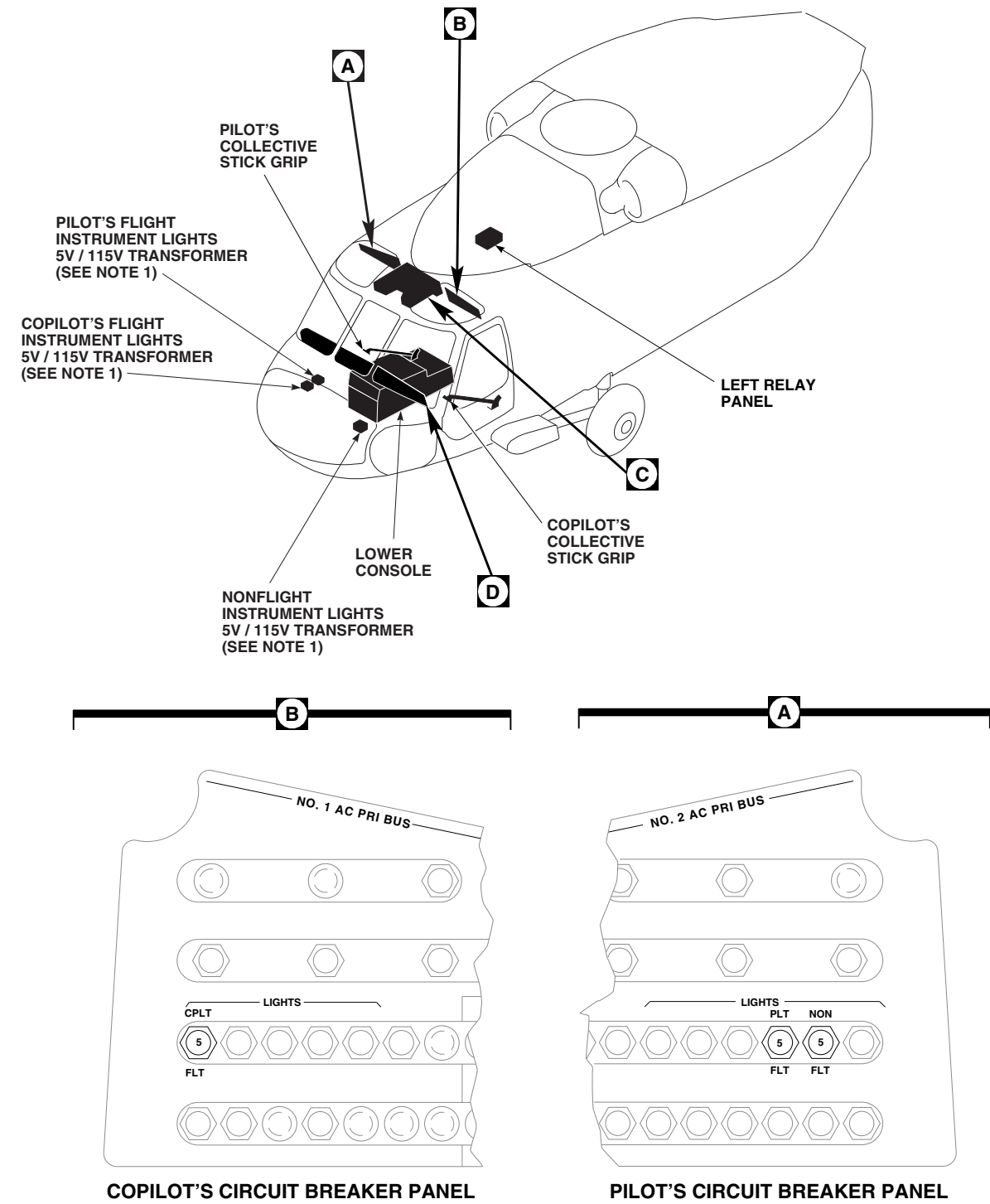
The maintenance light, stored in a stowage bag behind the pilot's seat, is a portable floodlight on a 20-foot cord. A DIM, OFF, and BRIGHT switch controls the brightness of the light. The cord of the light assembly can be connected to either of two maintenance light receptacles. One maintenance light receptacle is on the right side overhead in the forward cabin, while the other receptacle is on the bottom, outside of the tail cone rear of the tailwheel strut. Power to operate the maintenance light is provided from the BATT UTIL BUS through the UTIL LTS CKPT circuit breaker on the lower console circuit breaker panel.

9-1-8.12 Utility and Maintenance Lights **SLS** .

Three portable, hand-held utility lights with coil cords are installed. Two lights are installed on the upper console; one each for the pilot and copilot. The third light is installed on the right side of the copilot's seat. A flexible light assembly is also installed on the lower console. All three utility lights provide blue or white lighting (Figures 9-1-8-16 and 9-1-8-17). System electrical power is supplied by a **70583-702374** BATT UTIL BUS **702375 - SUBQ** BATT BUS through the UTIL LTS CKPT circuit breaker on the lower console. The utility lights are attached to swivel-type detachable mountings and may be held to provide blue or white flood or spotlighting. Pressing the lock button on the casing assembly and

simultaneously turning the lens casing selects a light filter for blue or white lighting.

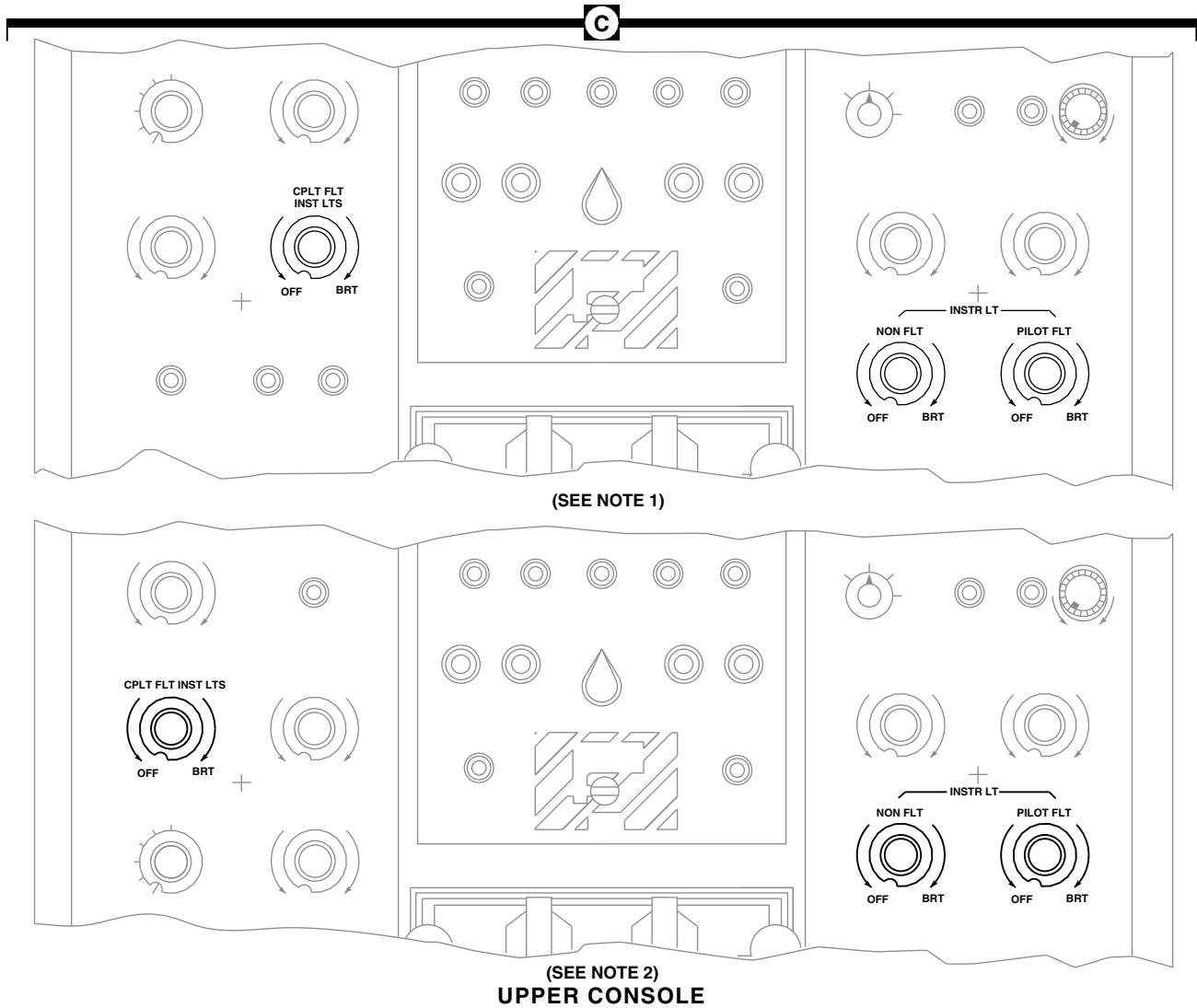
The maintenance light, stored in a stowage bag behind the pilot's seat, is a portable floodlight on a 20-foot cord. A DIM, OFF, and BRIGHT switch controls the brightness of the light. The cord of the light assembly can be connected to either of two maintenance light receptacles. One maintenance light receptacle is on the right side overhead in the forward cabin, while the other receptacle is on the bottom, outside of the tail cone aft of the tailwheel strut. Power to operate the maintenance light is provided from the **70583-702374** BATT UTIL BUS **702375 - SUBQ** BATT BUS through the UTIL LTS CKPT circuit breaker on the lower console circuit breaker panel.



NOTES	
1.	SLS
2.	W/O SLS

FB2219_1C
SA

FIGURE 9-1-8-1. INSTRUMENT PANEL LIGHTS LOCATION DIAGRAM (SHEET 1 OF 4)



FB2219_2A
SA

FIGURE 9-1-8-1. INSTRUMENT PANEL LIGHTS LOCATION DIAGRAM (SHEET 2 OF 4)

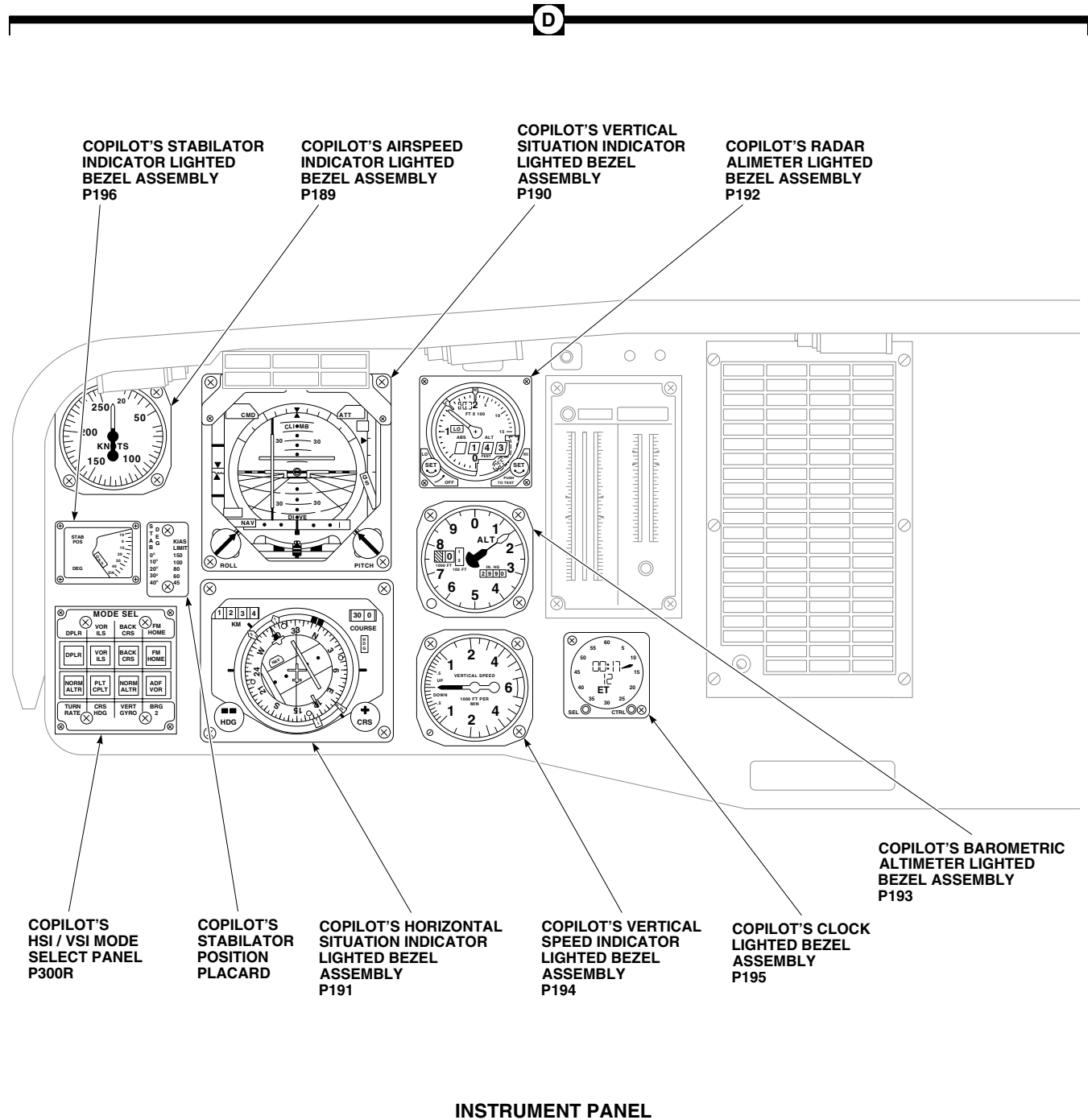


FIGURE 9-1-8-1. INSTRUMENT PANEL LIGHTS LOCATION DIAGRAM (SHEET 3 OF 4)

FB1582_5A
SA

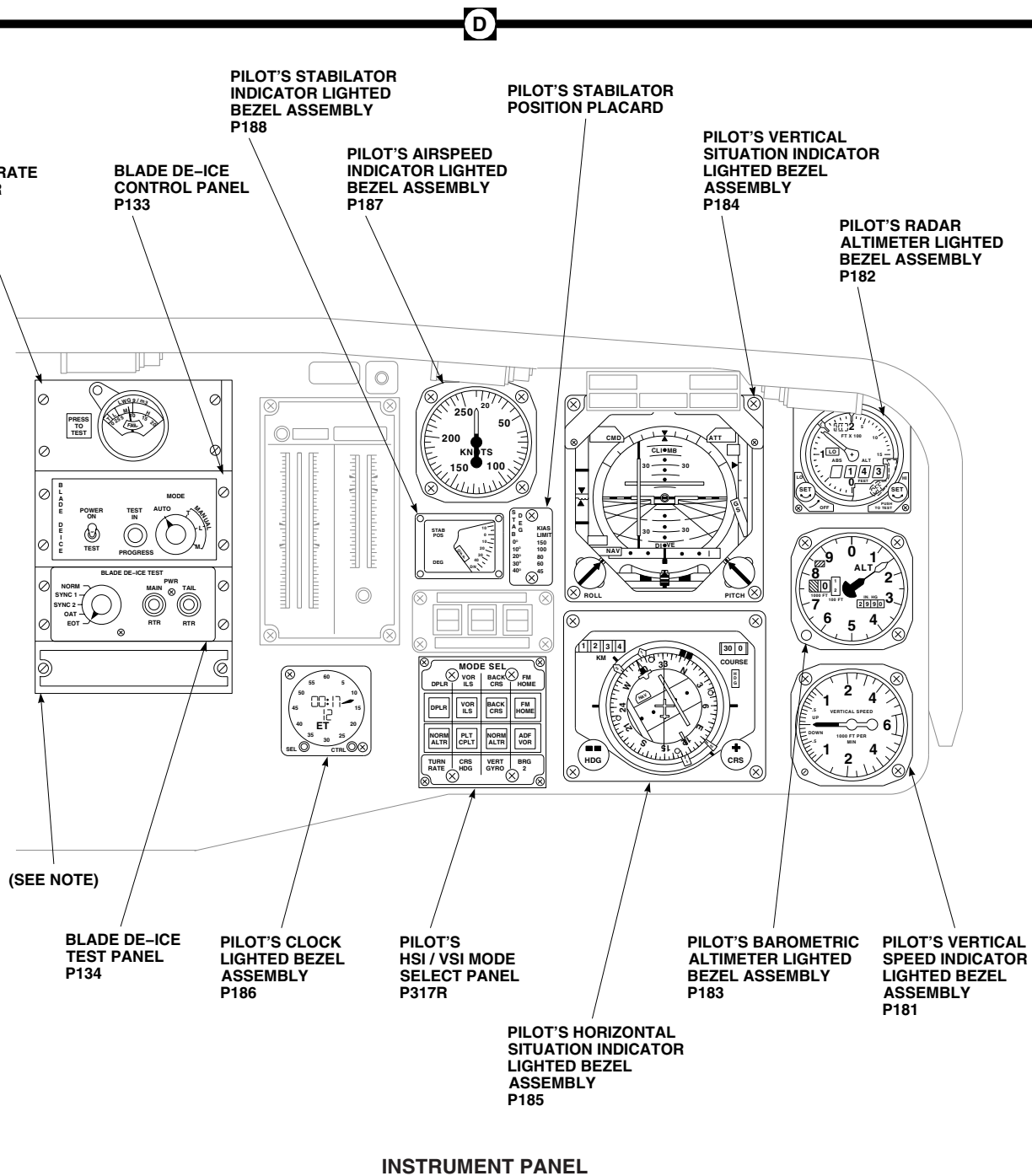
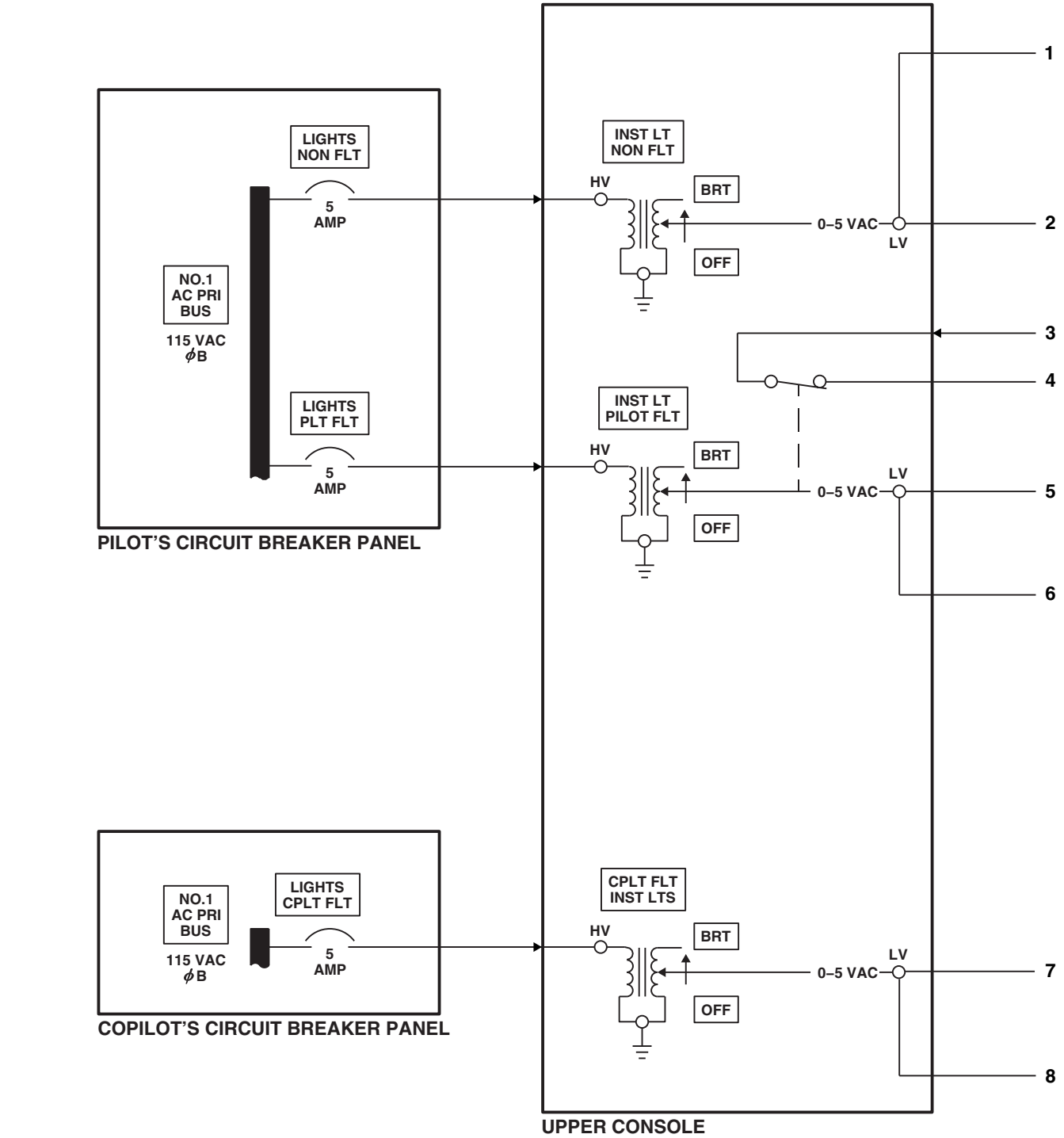
FB1582_6B
SA

FIGURE 9-1-8-1. INSTRUMENT PANEL LIGHTS LOCATION DIAGRAM (SHEET 4 OF 4)



EFFECTIVITY
SLS

FB2220_1C
SA

FIGURE 9-1-8-2. INSTRUMENT PANEL LIGHTS BLOCK DIAGRAM (SHEET 1 OF 2)

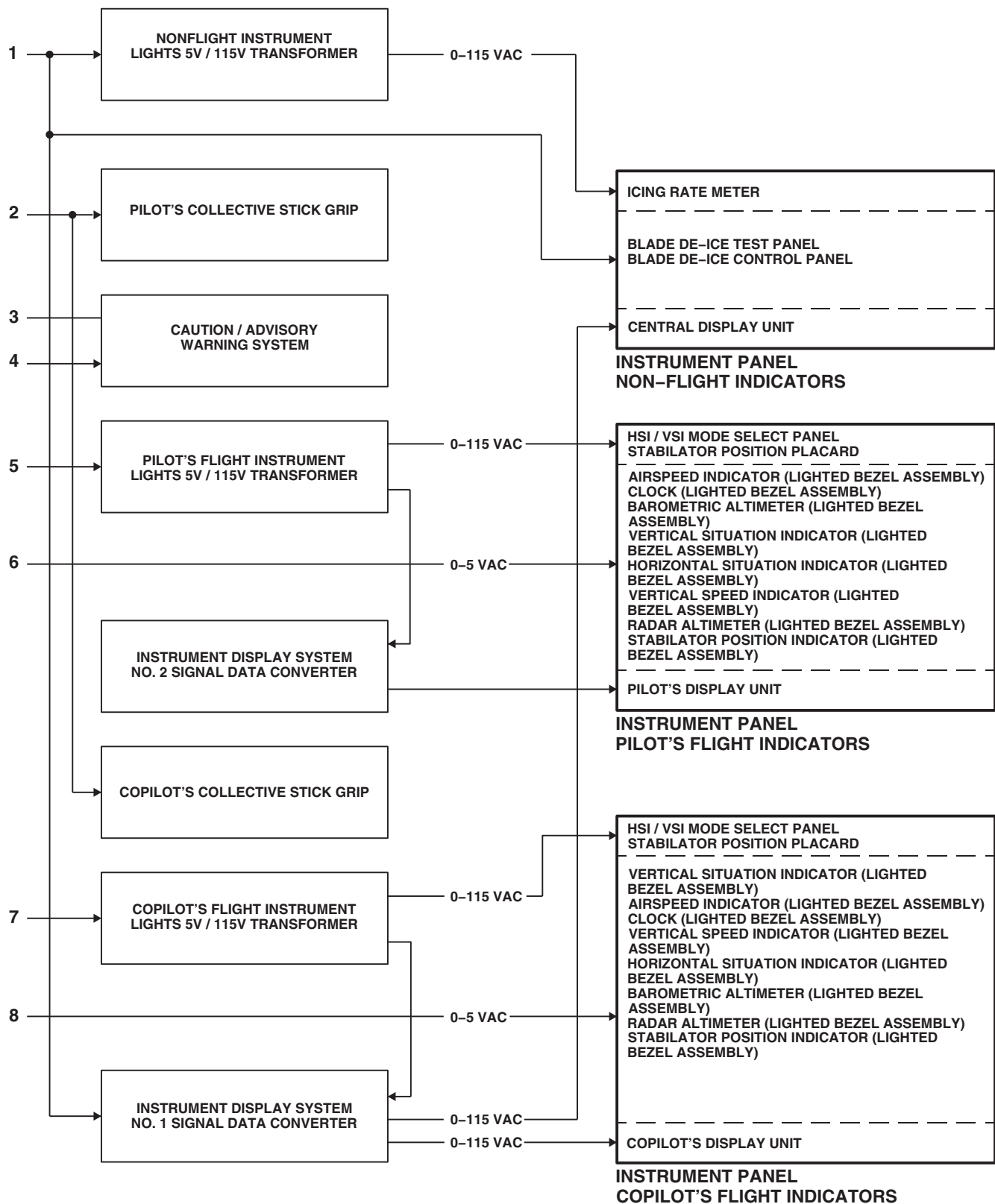
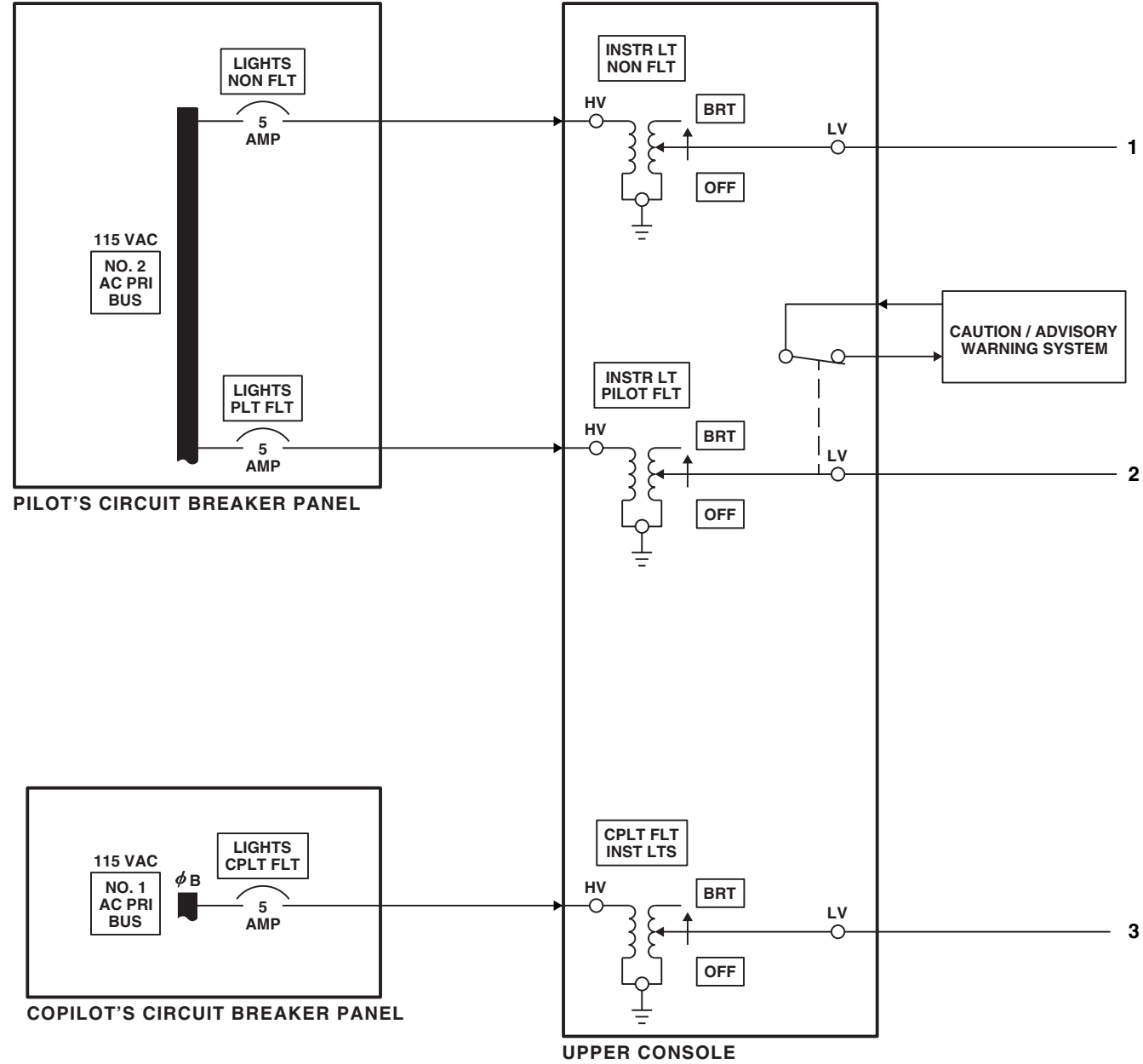
FB2220_2A
SA

FIGURE 9-1-8-2. INSTRUMENT PANEL LIGHTS BLOCK DIAGRAM (SHEET 2 OF 2)



EFFECTIVITY
W/O SLS

FB2220_3B
SA

FIGURE 9-1-8-3. INSTRUMENT PANEL LIGHTS BLOCK DIAGRAM (SHEET 1 OF 2)

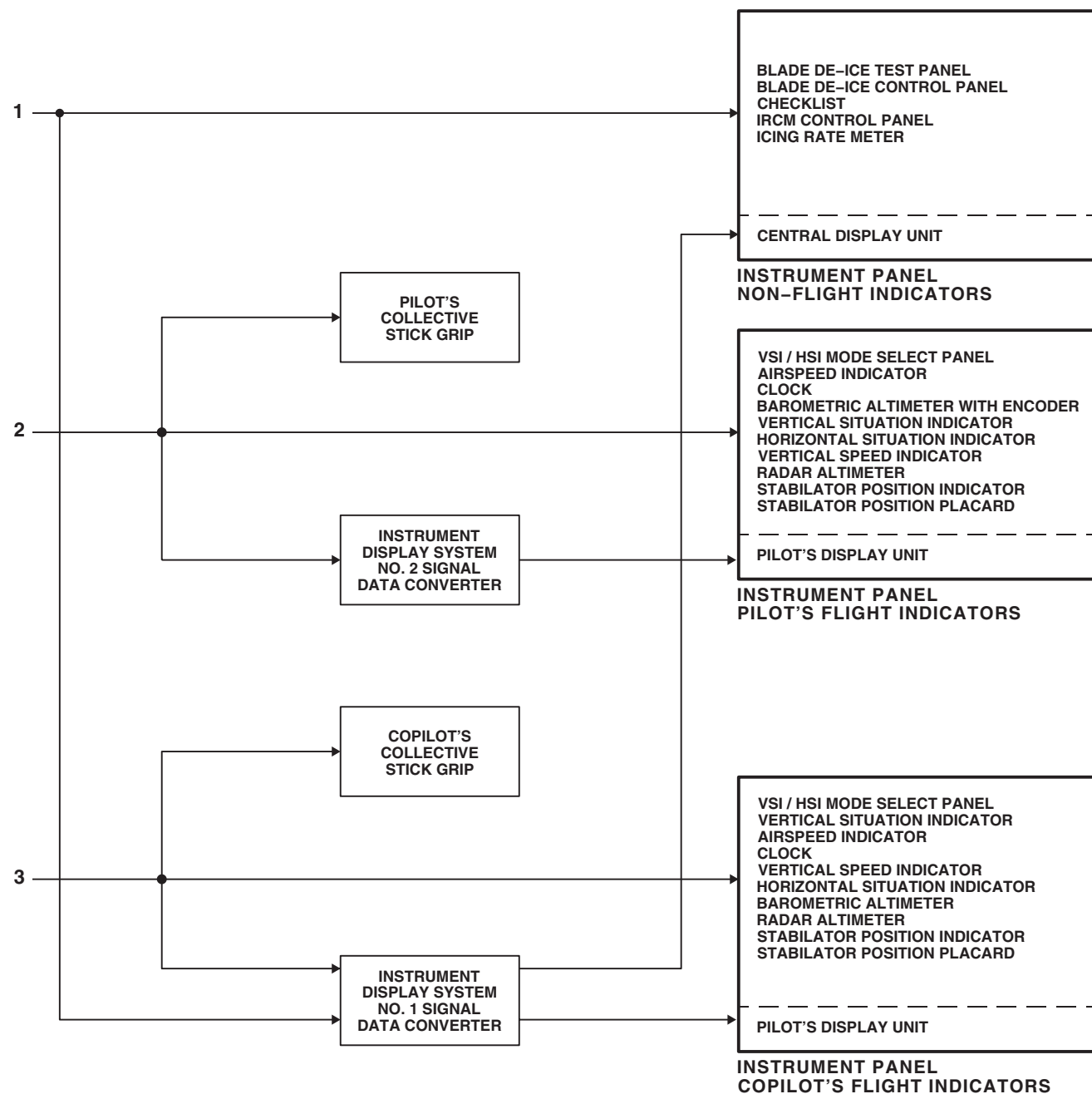
FB2220_4A
SA

FIGURE 9-1-8-3. INSTRUMENT PANEL LIGHTS BLOCK DIAGRAM (SHEET 2 OF 2)

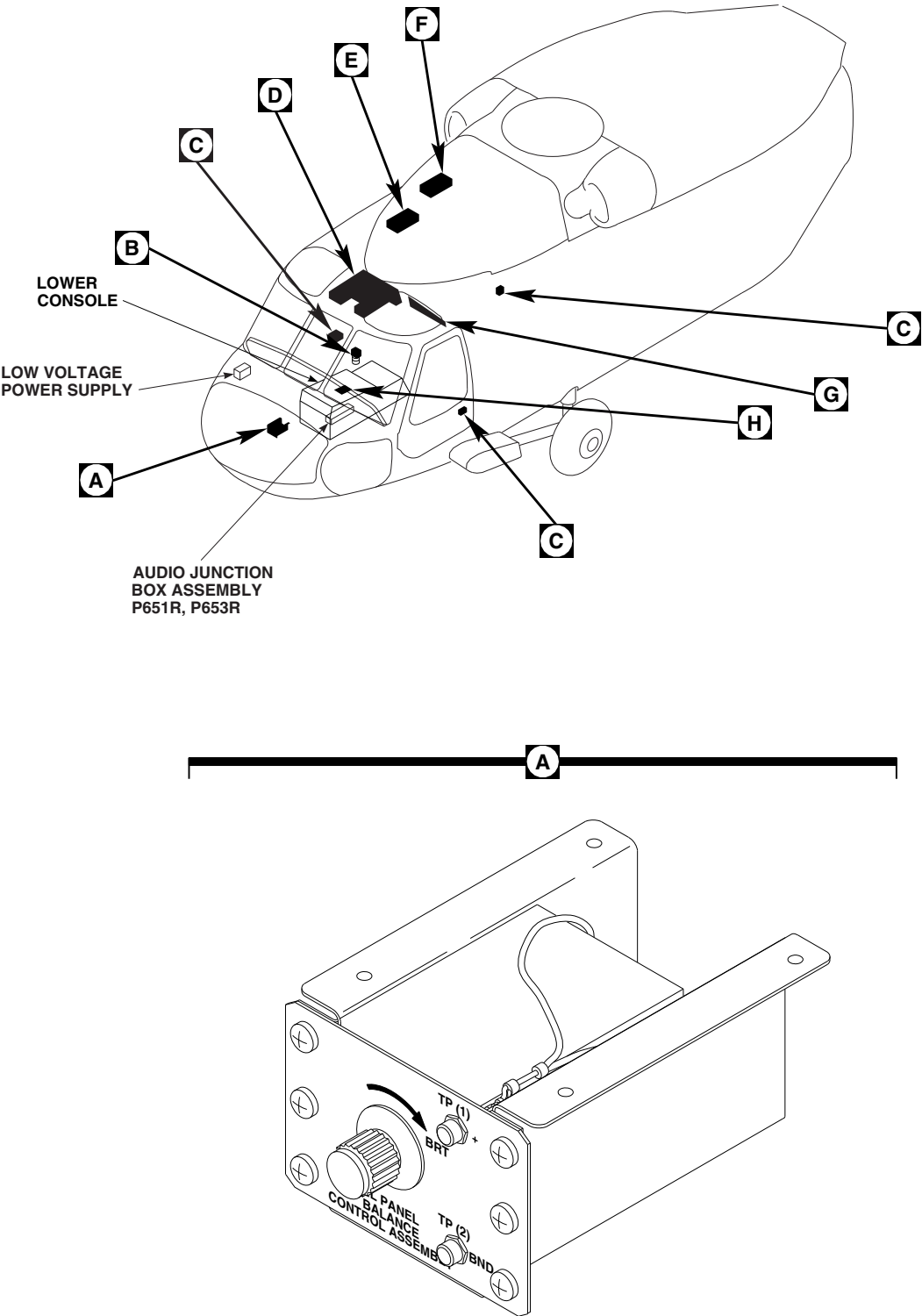
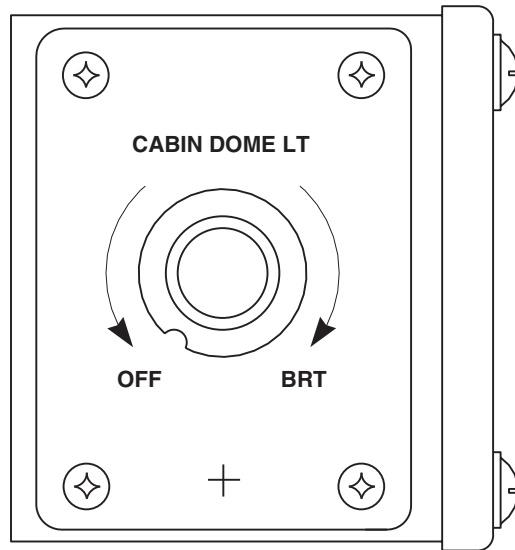


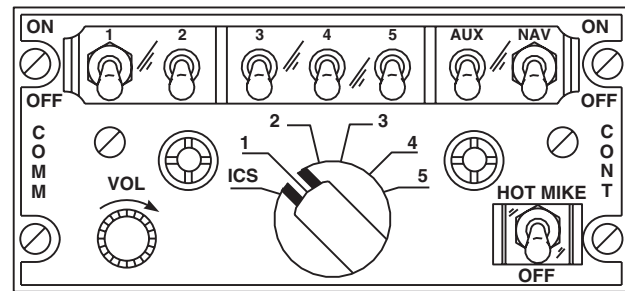
FIGURE 9-1-8-4. UPPER AND LOWER CONSOLE LIGHTS LOCATION DIAGRAM (SHEET 1 OF 3)

FB2221_1B
SA



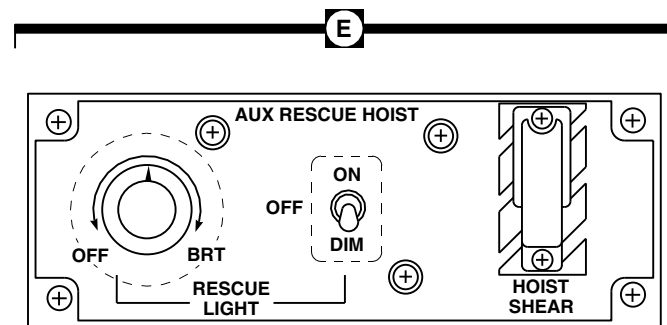
**CABIN DOME LIGHTS
DIMMER UNIT**

B



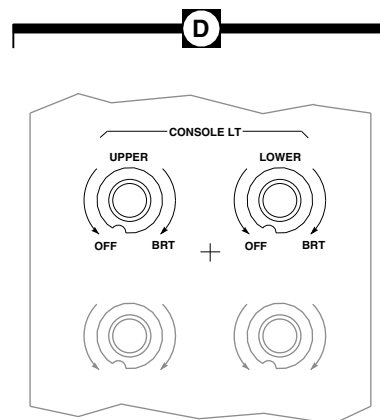
**INTERCOMMUNICATION SET (ICS)
CONTROL PANEL (TYPICAL)**

C



AUXILIARY RESCUE HOIST CONTROL PANEL

E



UPPER CONSOLE

D

FT1485_4
SA

**FIGURE 9-1-8-4. UPPER AND LOWER CONSOLE LIGHTS LOCATION DIAGRAM
(SHEET 2 OF 3)**

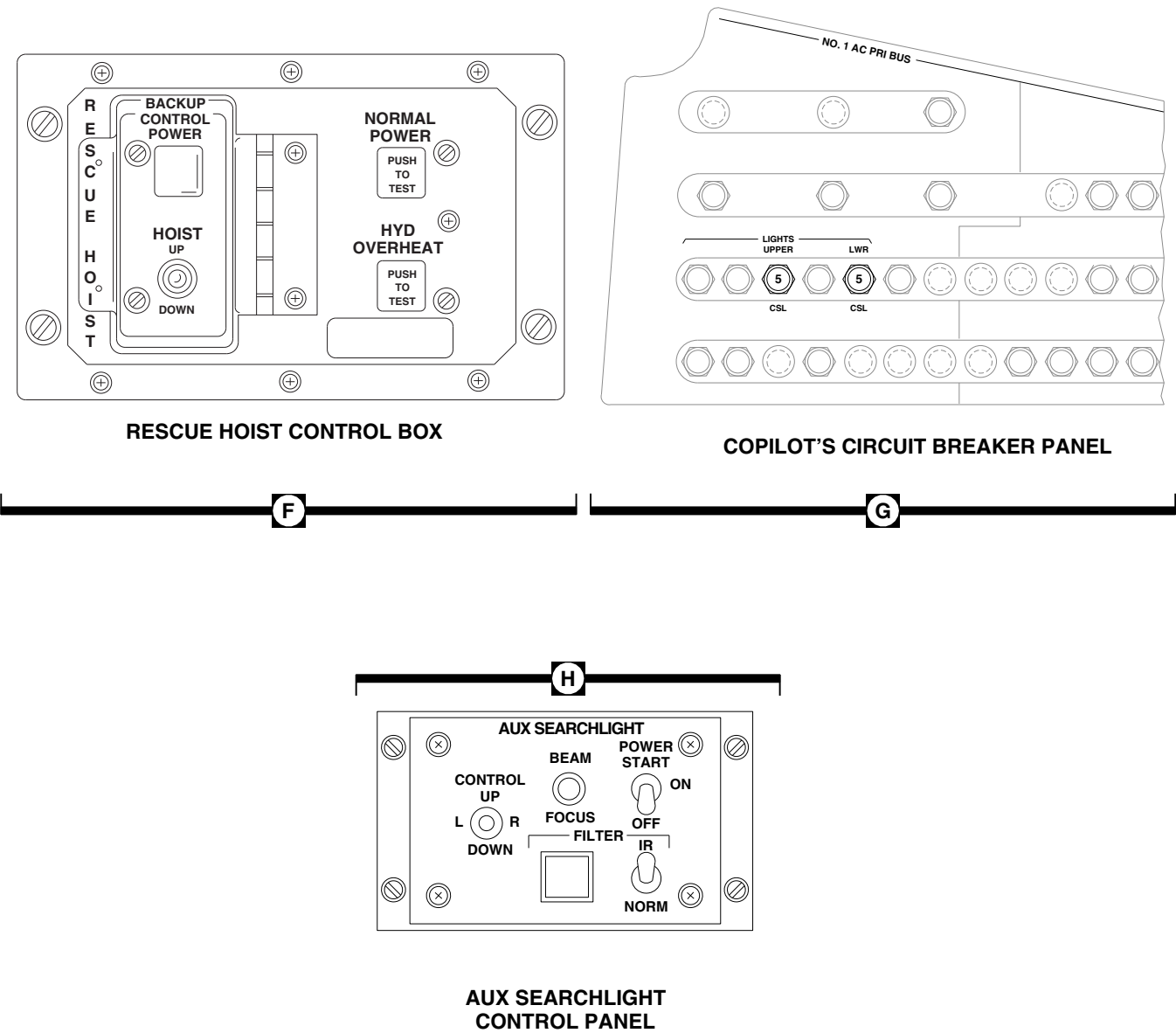


FIGURE 9-1-8-4. UPPER AND LOWER CONSOLE LIGHTS LOCATION DIAGRAM (SHEET 3 OF 3)

FB2221_3
SA

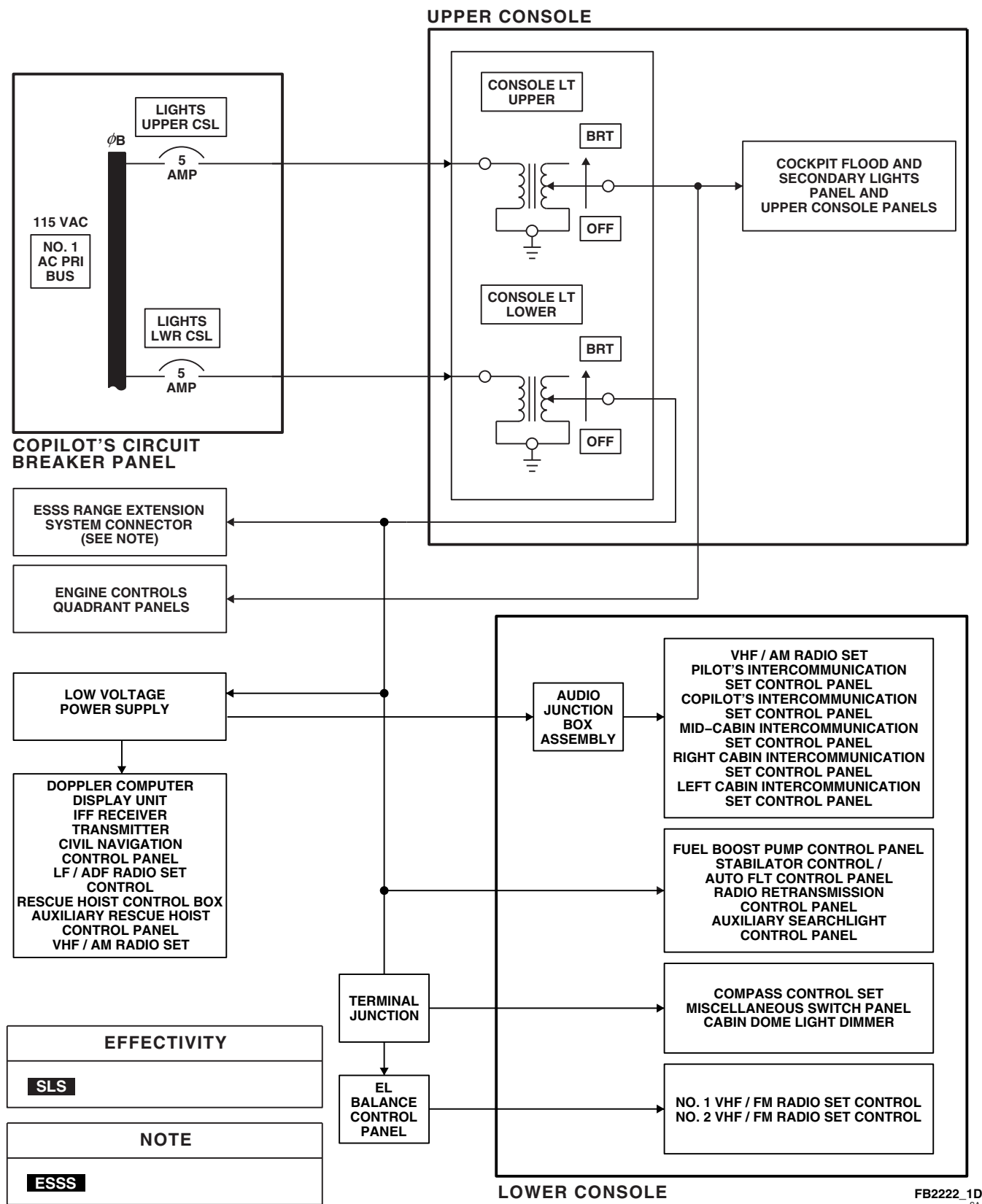


FIGURE 9-1-8-5. UPPER AND LOWER CONSOLE LIGHTS BLOCK DIAGRAM

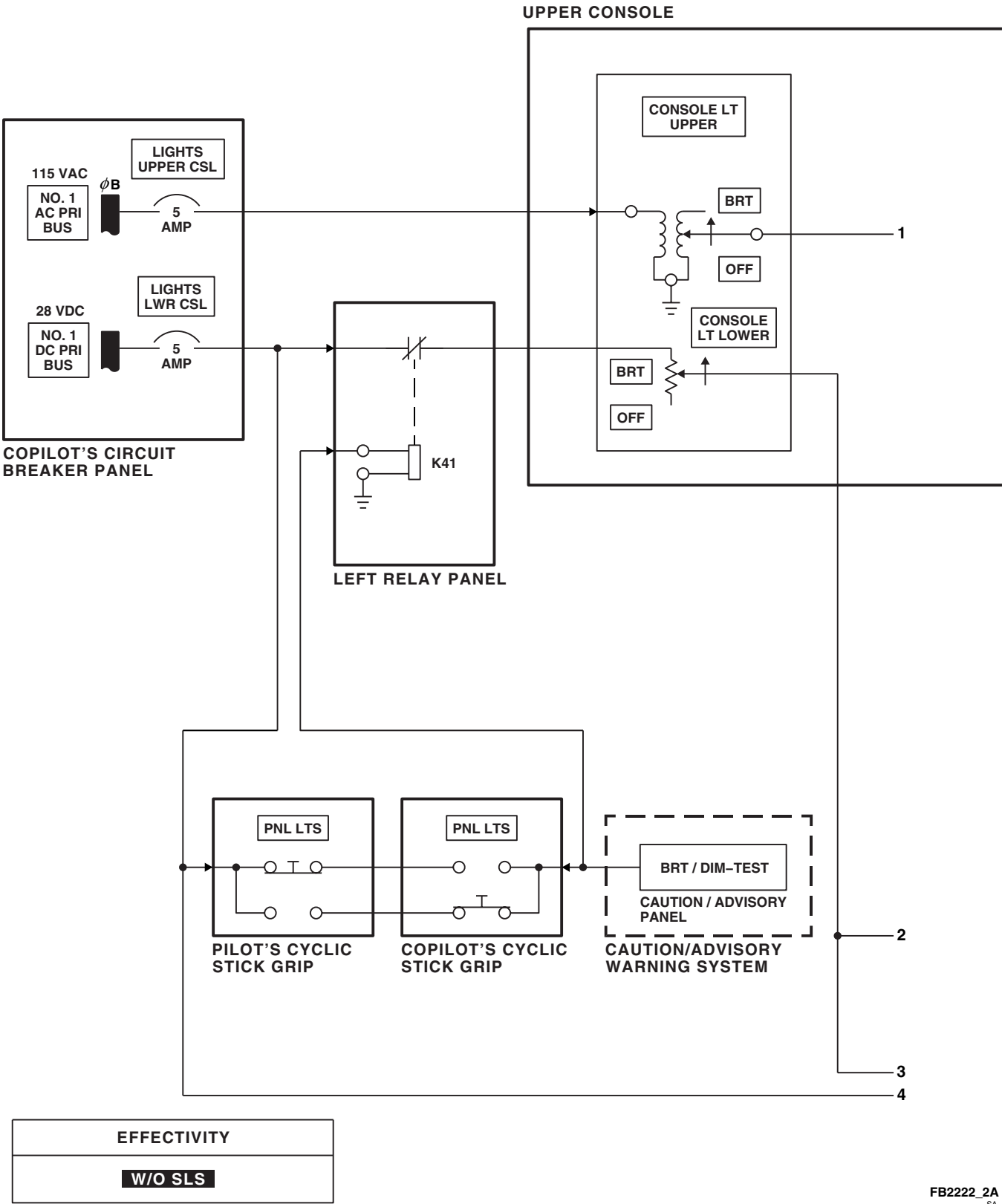


FIGURE 9-1-8-6. UPPER AND LOWER CONSOLE LIGHTS BLOCK DIAGRAM (SHEET 1 OF 2)

UPPER CONSOLE

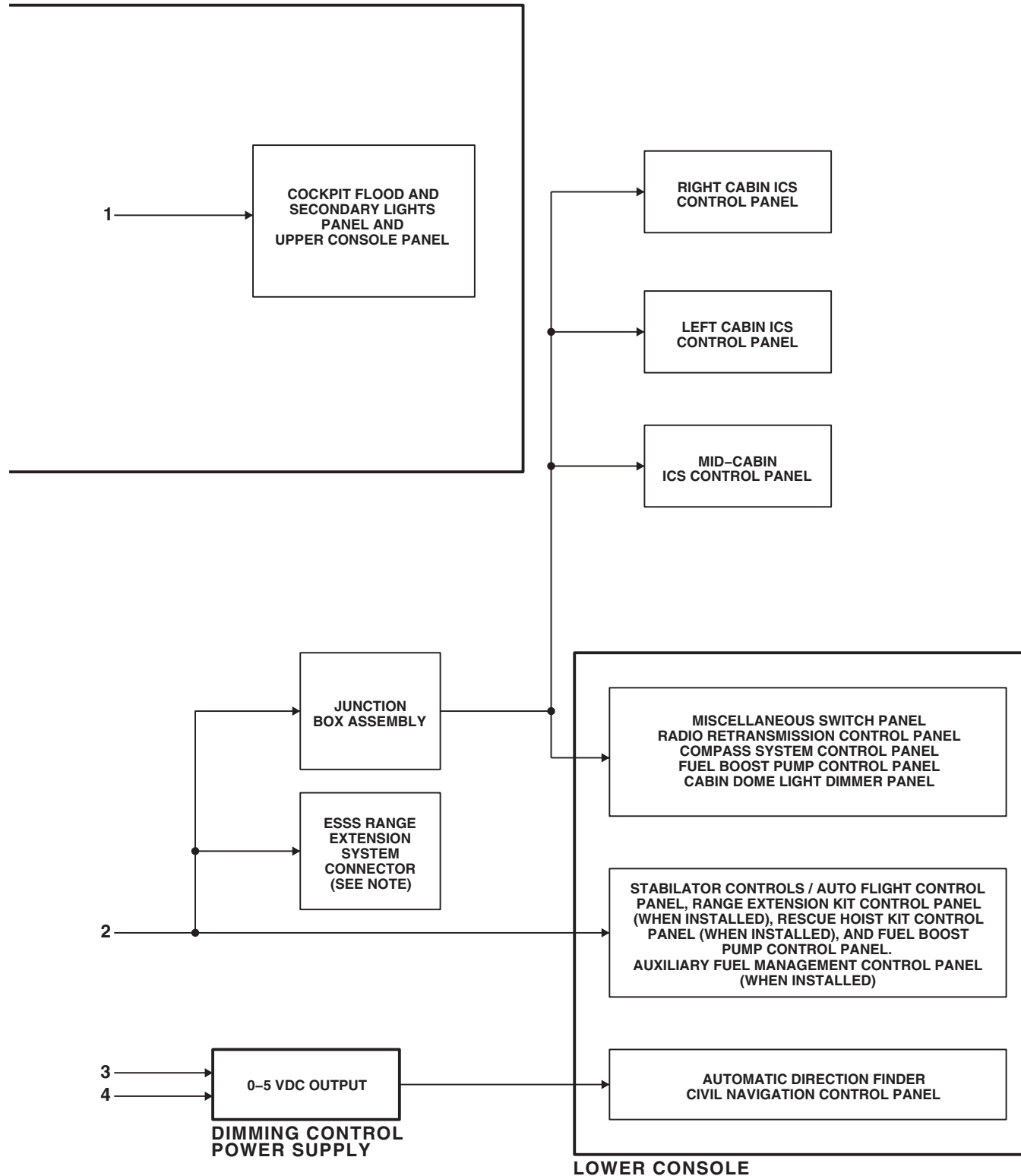
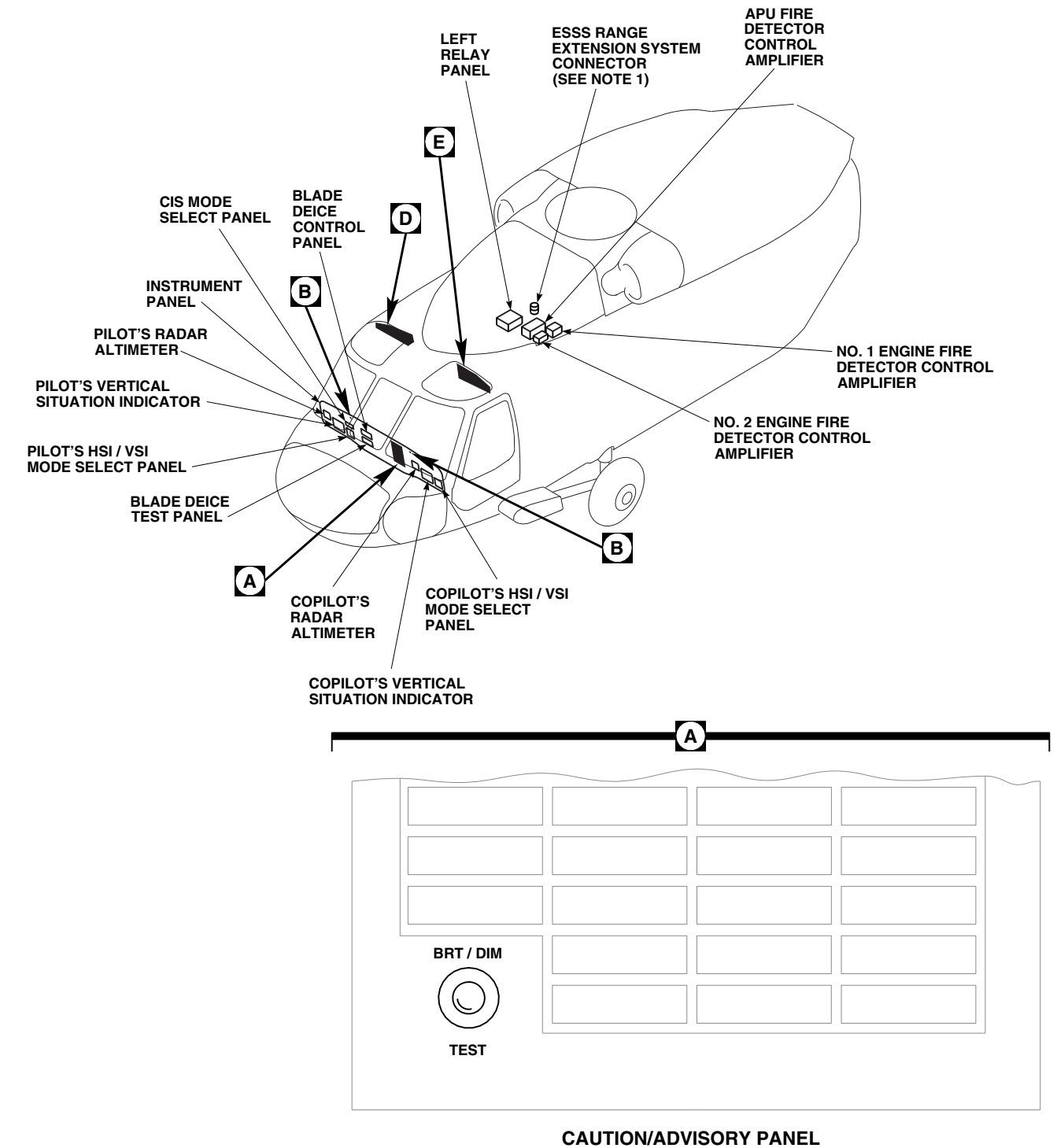
FB2222_3B
SA

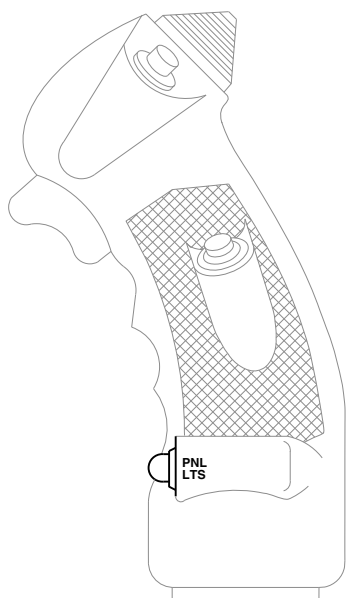
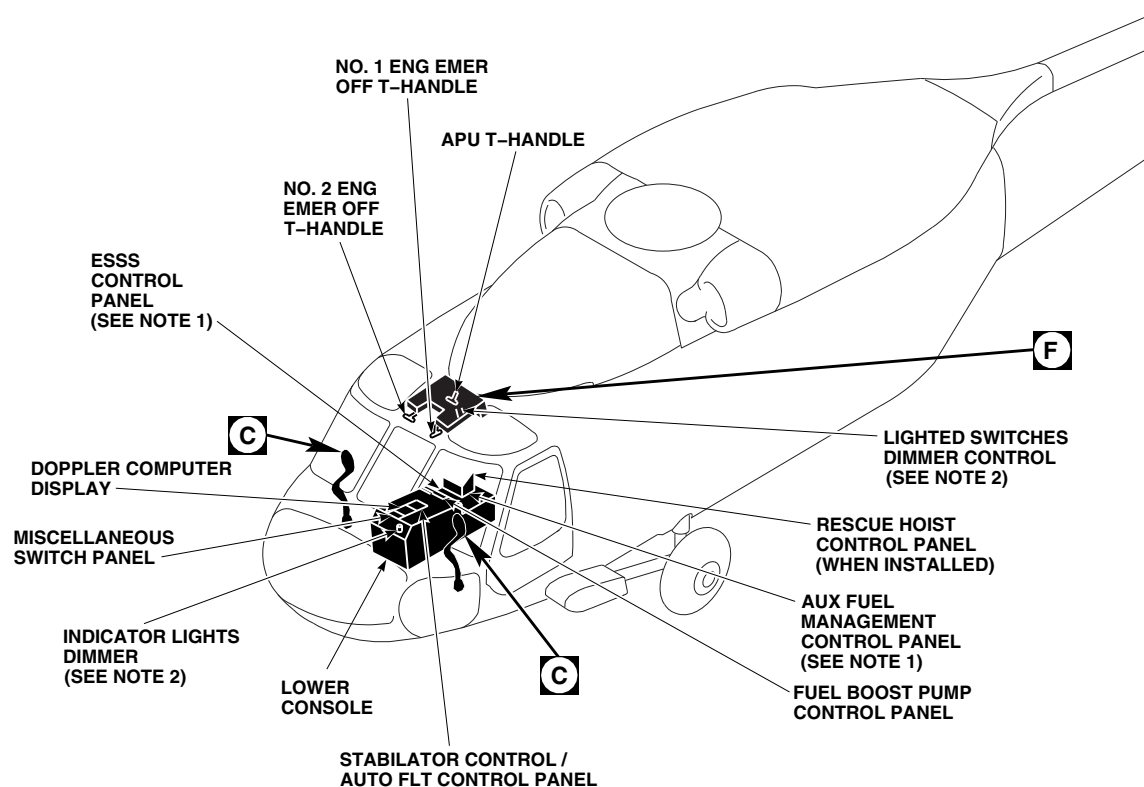
FIGURE 9-1-8-6. UPPER AND LOWER CONSOLE LIGHTS BLOCK DIAGRAM (SHEET 2 OF 2)



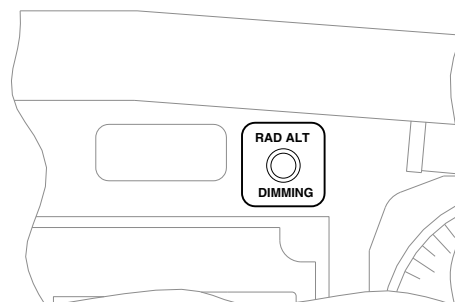
NOTES	
1.	ESSS
2.	SLS
3.	W/O SLS

FB2223_1D
SA

FIGURE 9-1-8-7. INSTRUMENT PANEL AND CONSOLES INDICATOR LIGHTS DIMMING LOCATION DIAGRAM (SHEET 1 OF 4)



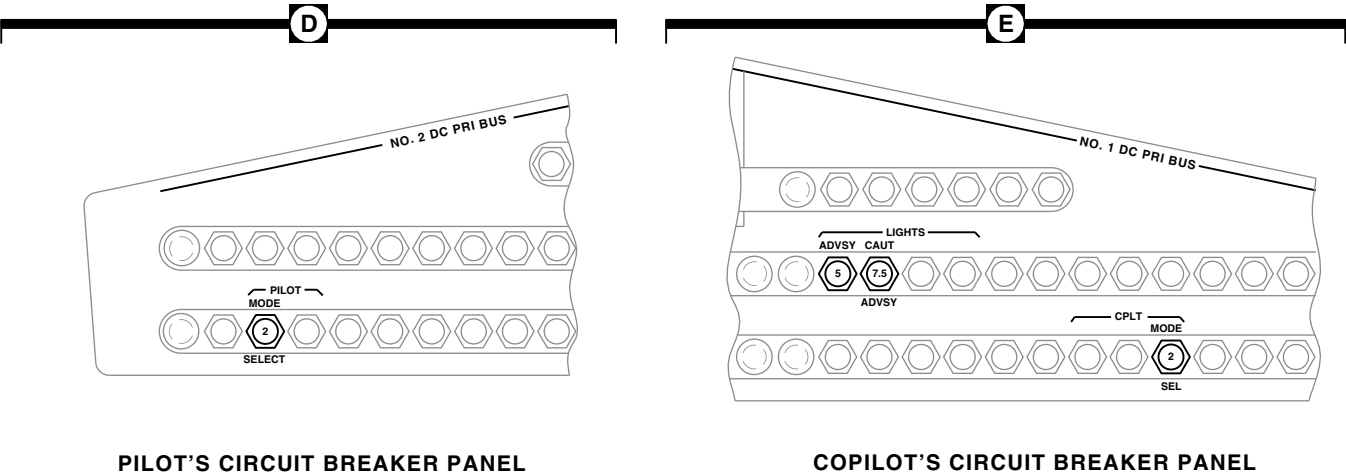
PILOT'S AND COPILOT'S
CYCLIC STICK GRIP



PILOT'S AND COPILOT'S RADAR
ALTIMETER DIMMING CONTROL

FB2223_2C
SA

FIGURE 9-1-8-7. INSTRUMENT PANEL AND CONSOLES INDICATOR LIGHTS DIMMING
LOCATION DIAGRAM (SHEET 2 OF 4)



FB2223 3
SA

FIGURE 9-1-8-7. INSTRUMENT PANEL AND CONSOLES INDICATOR LIGHTS DIMMING LOCATION DIAGRAM (SHEET 3 OF 4)

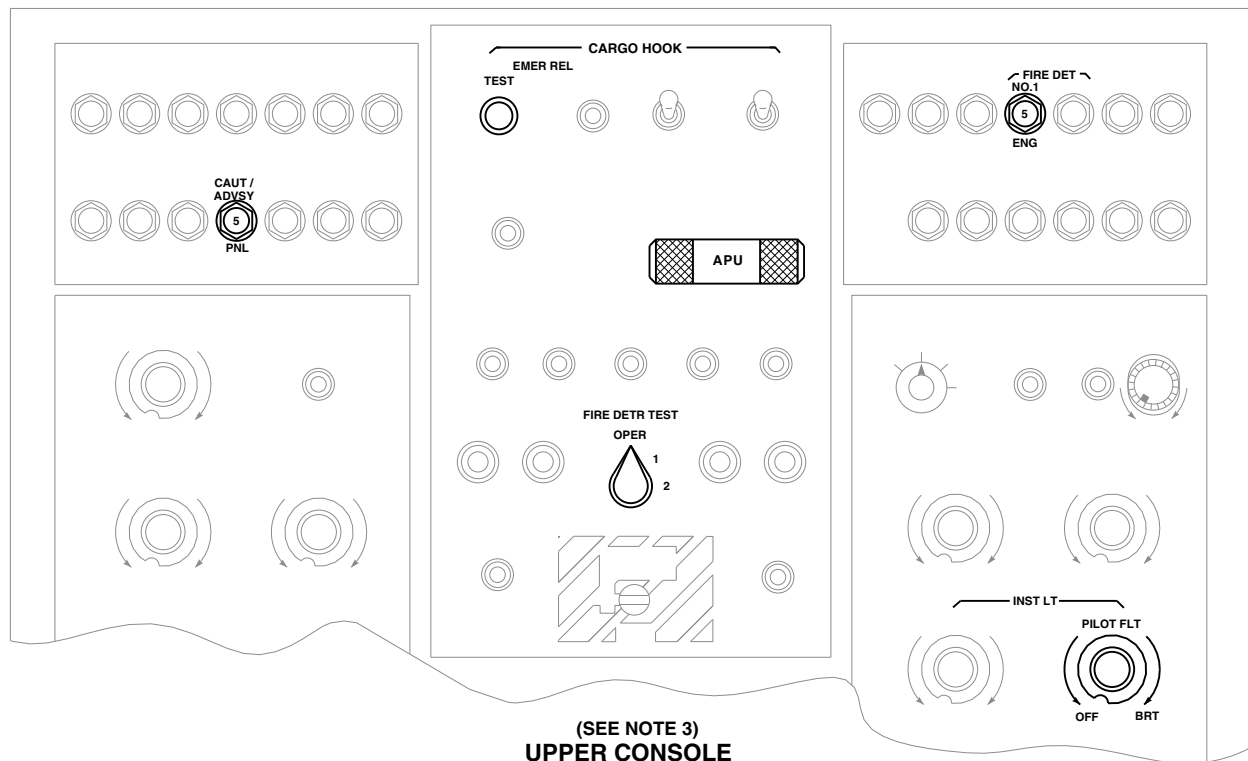
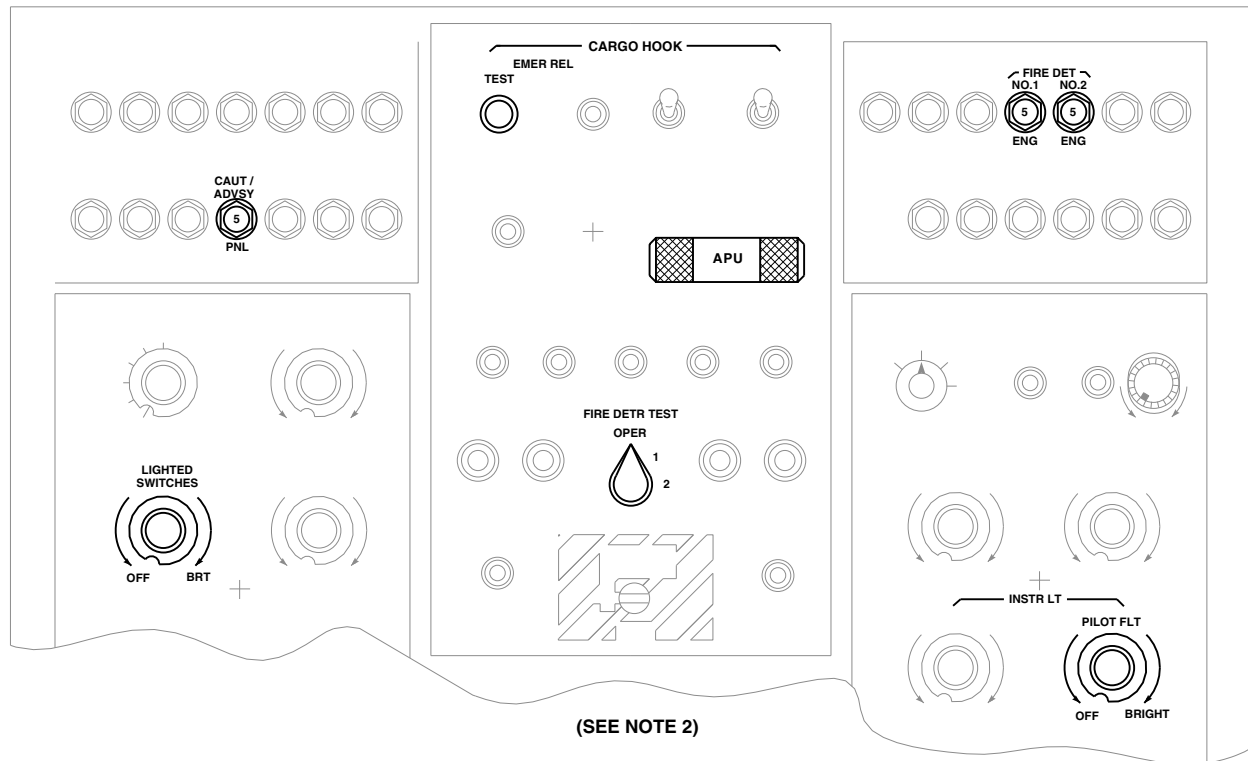
FFB2223_4B
SA

FIGURE 9-1-8-7. INSTRUMENT PANEL AND CONSOLES INDICATOR LIGHTS DIMMING LOCATION DIAGRAM (SHEET 4 OF 4)

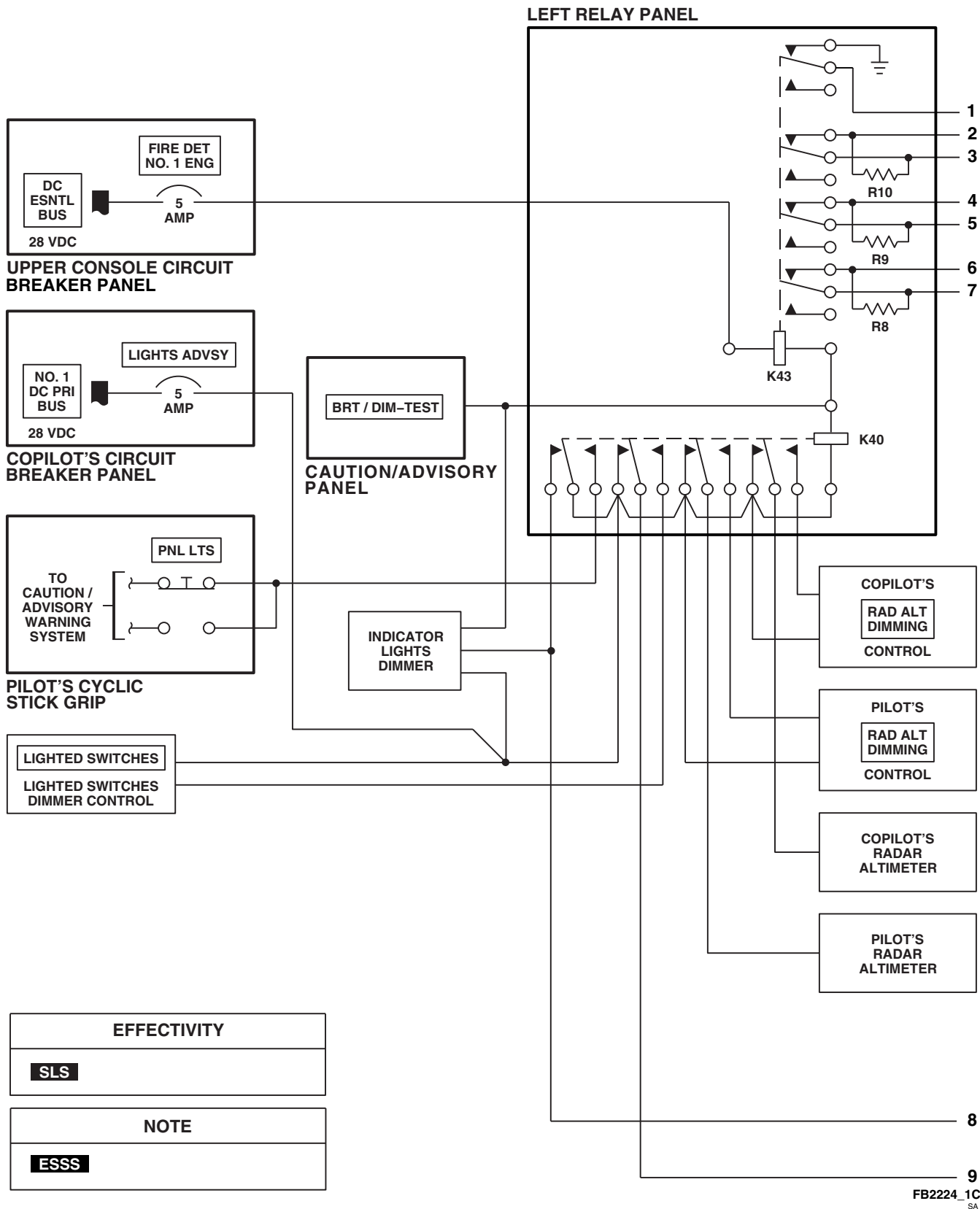


FIGURE 9-1-8-8. INSTRUMENT PANEL AND CONSOLES INDICATOR LIGHTS DIMMING
BLOCK DIAGRAM (SHEET 1 OF 2)

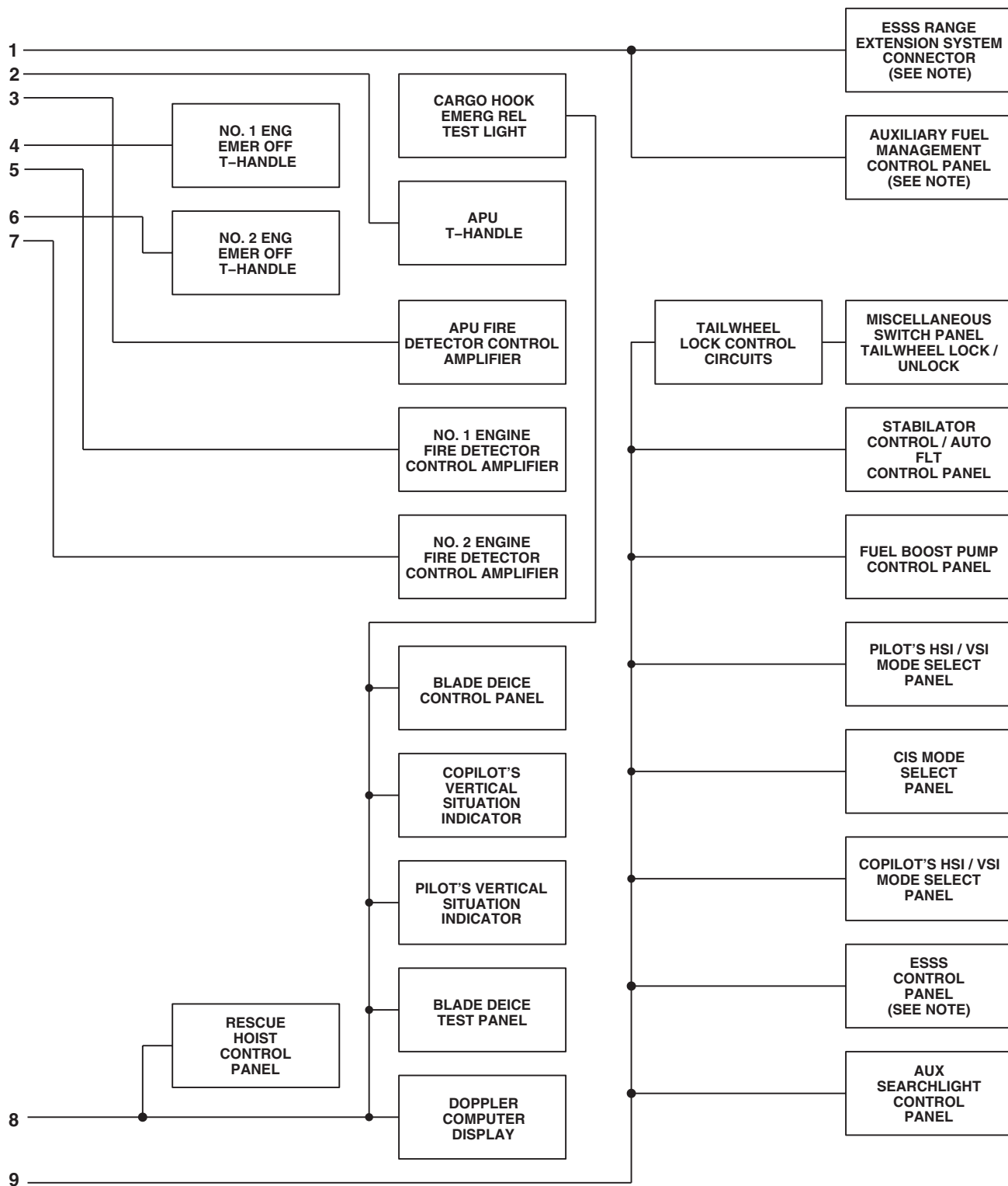
FB2224_2C
SA

FIGURE 9-1-8-8. INSTRUMENT PANEL AND CONSOLES INDICATOR LIGHTS DIMMING BLOCK DIAGRAM (SHEET 2 OF 2)

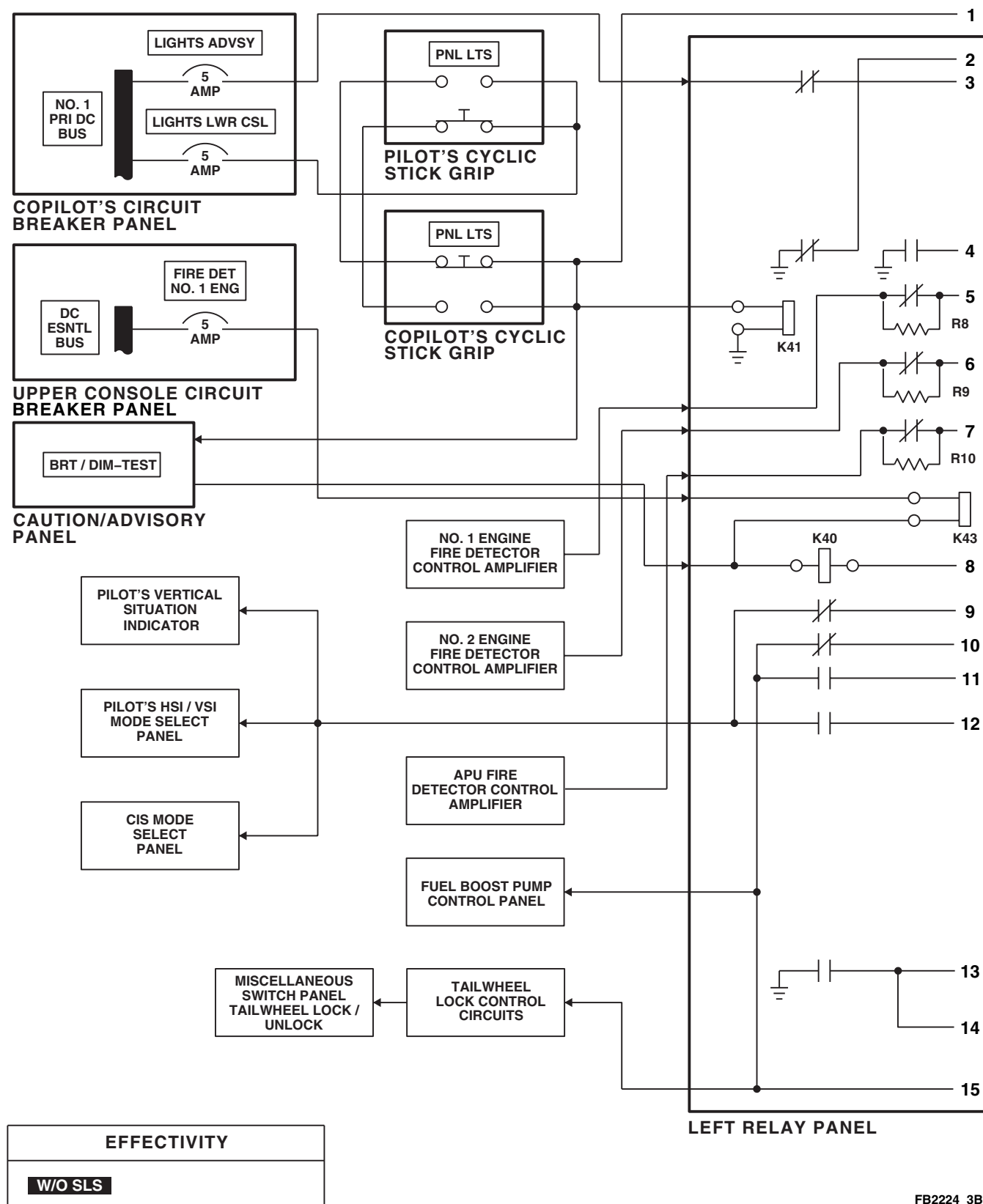
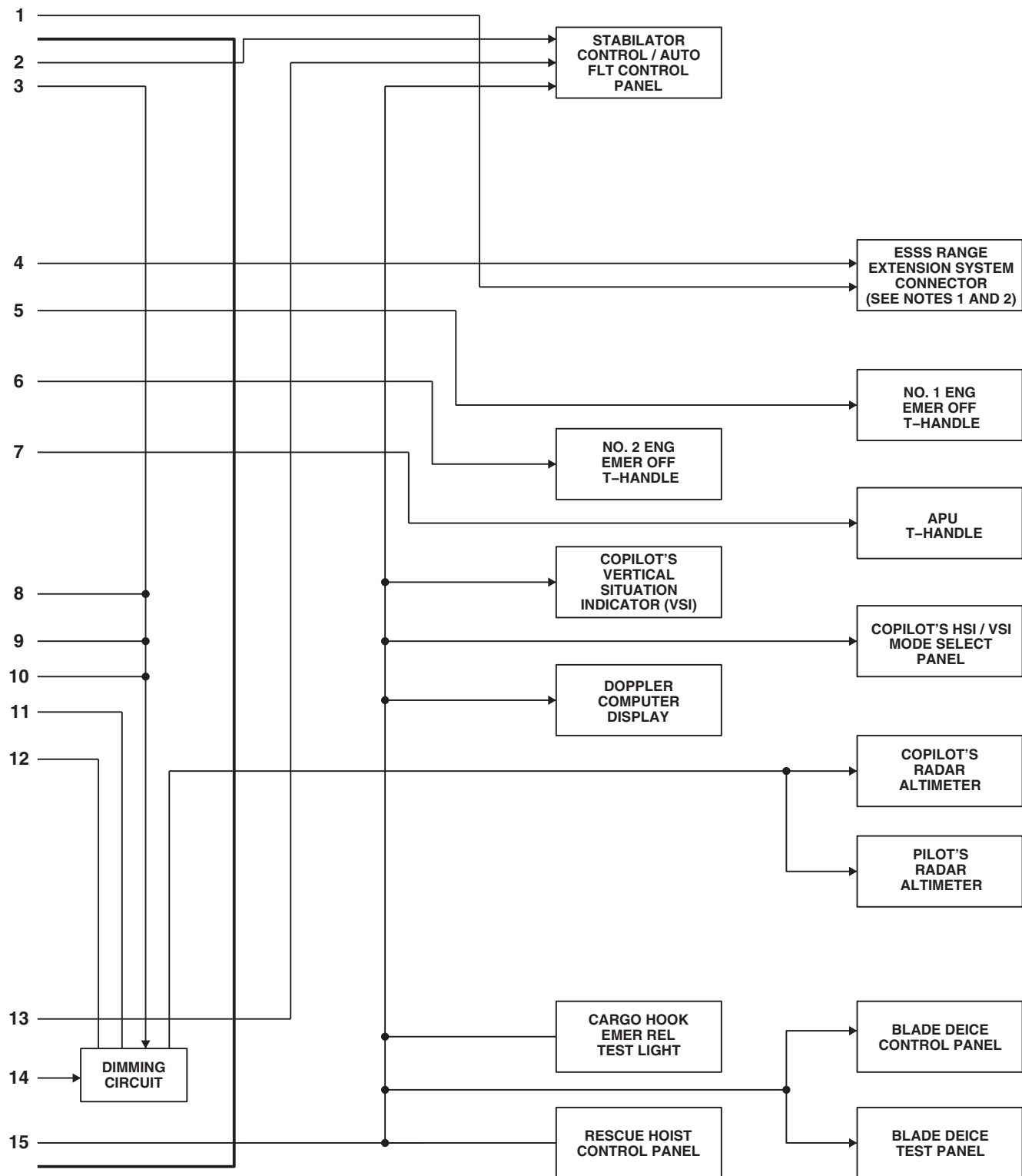
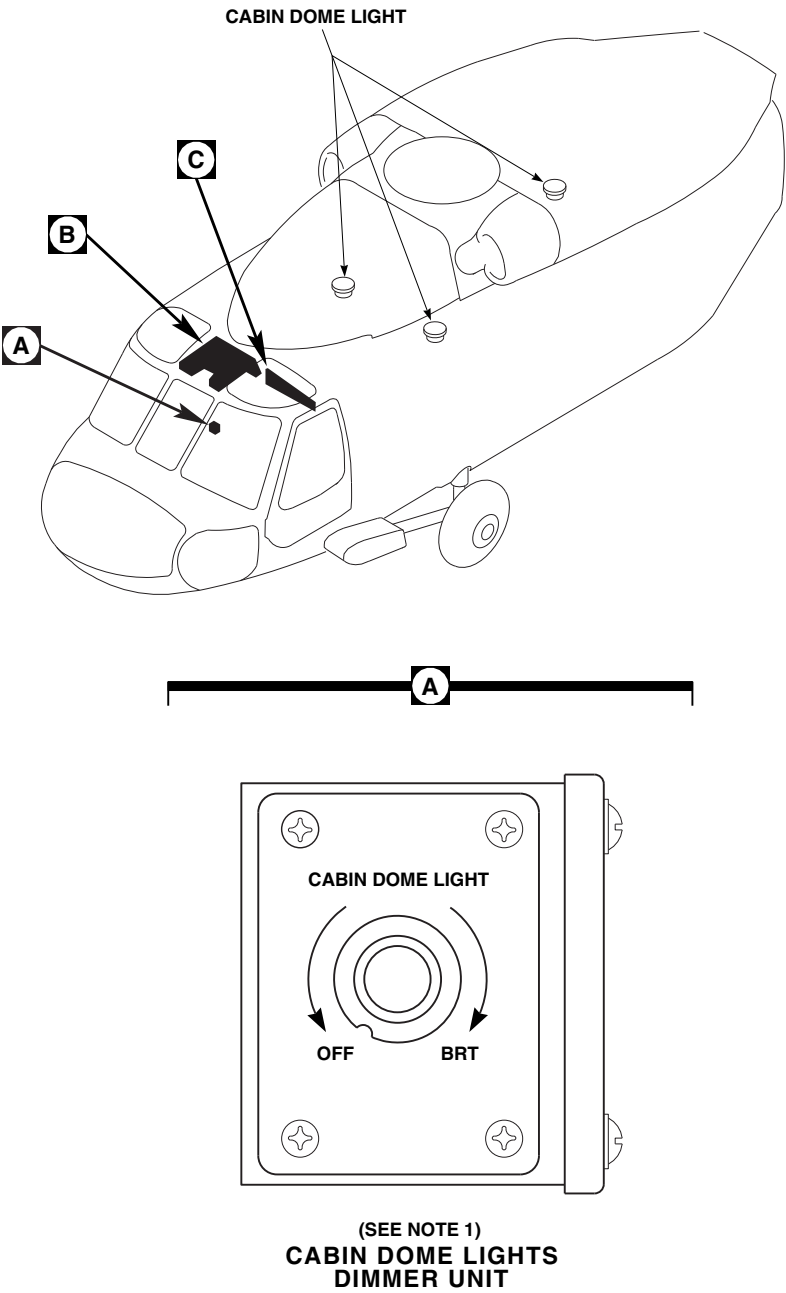


FIGURE 9-1-8-9. INSTRUMENT PANEL AND CONSOLES INDICATOR LIGHTS DIMMING BLOCK DIAGRAM (SHEET 1 OF 2)

FB2224_4C
SA

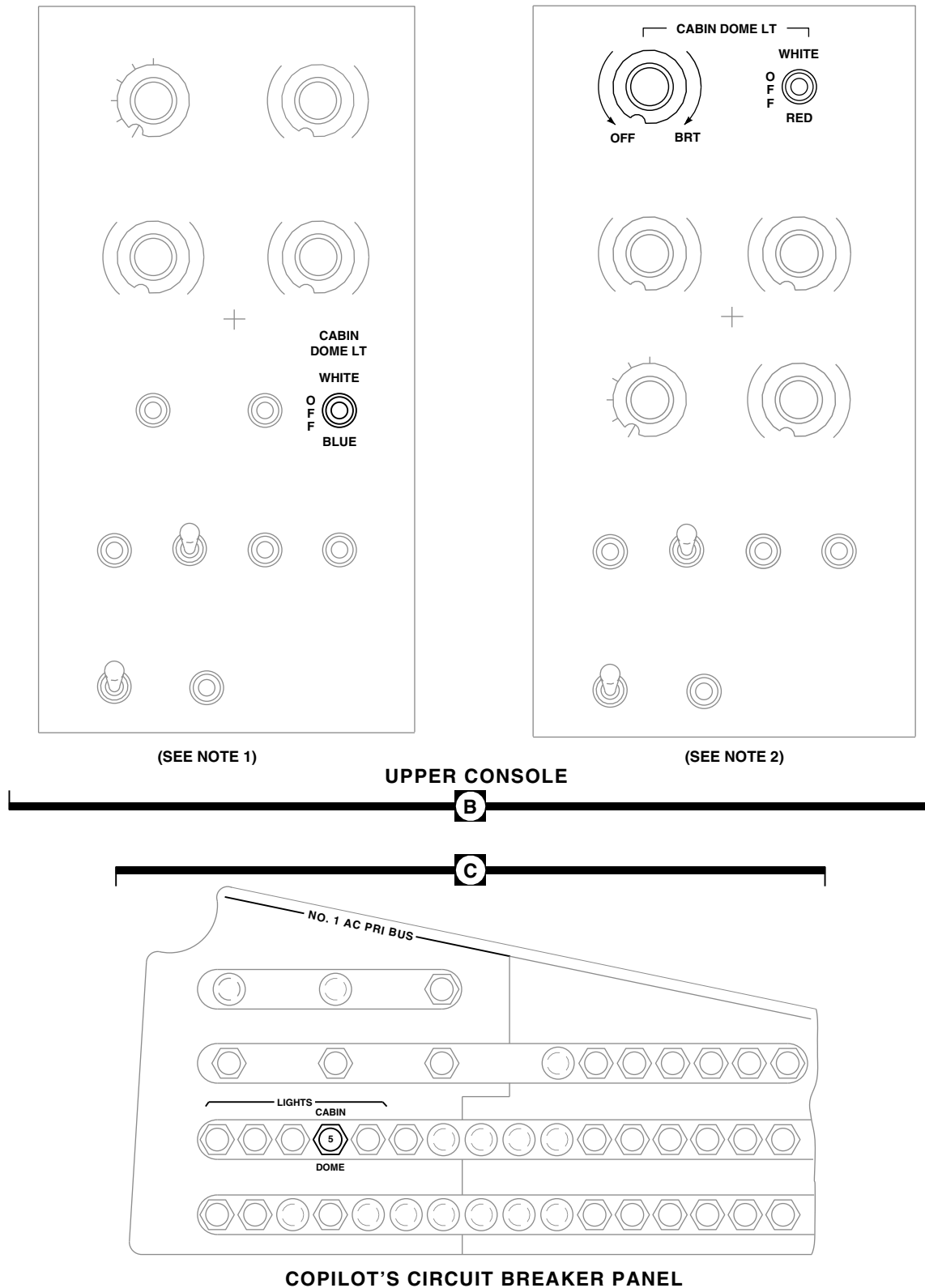
**FIGURE 9-1-8-9. INSTRUMENT PANEL AND CONSOLES INDICATOR LIGHTS DIMMING
BLOCK DIAGRAM (SHEET 2 OF 2)**



NOTES	
1.	SLS
2.	W/O SLS

FB2225_1A
 SA

FIGURE 9-1-8-10. CABIN DOME LIGHTS LOCATION DIAGRAM (SHEET 1 OF 2)



FB2225_2B
SA

FIGURE 9-1-8-10. CABIN DOME LIGHTS LOCATION DIAGRAM (SHEET 2 OF 2)

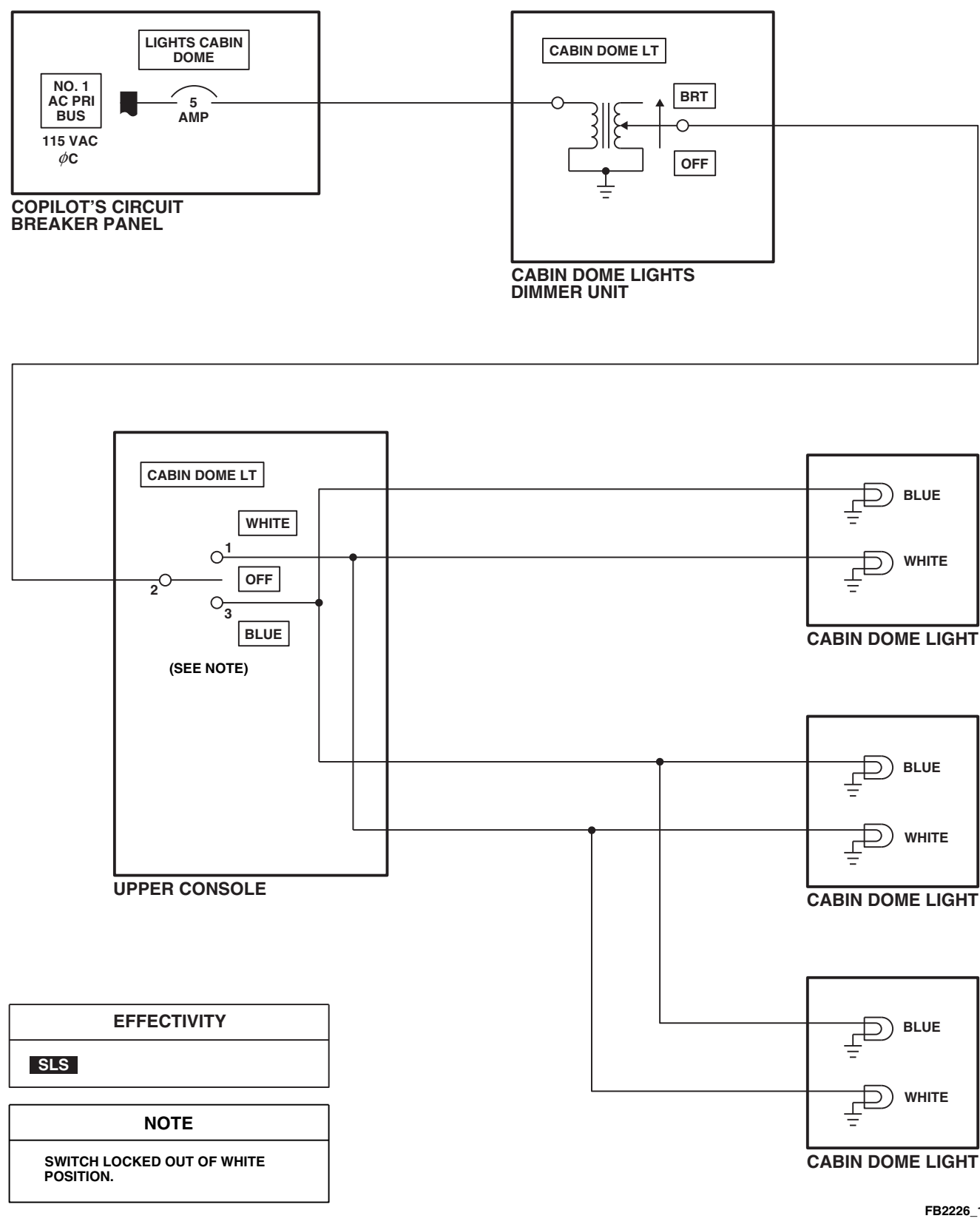
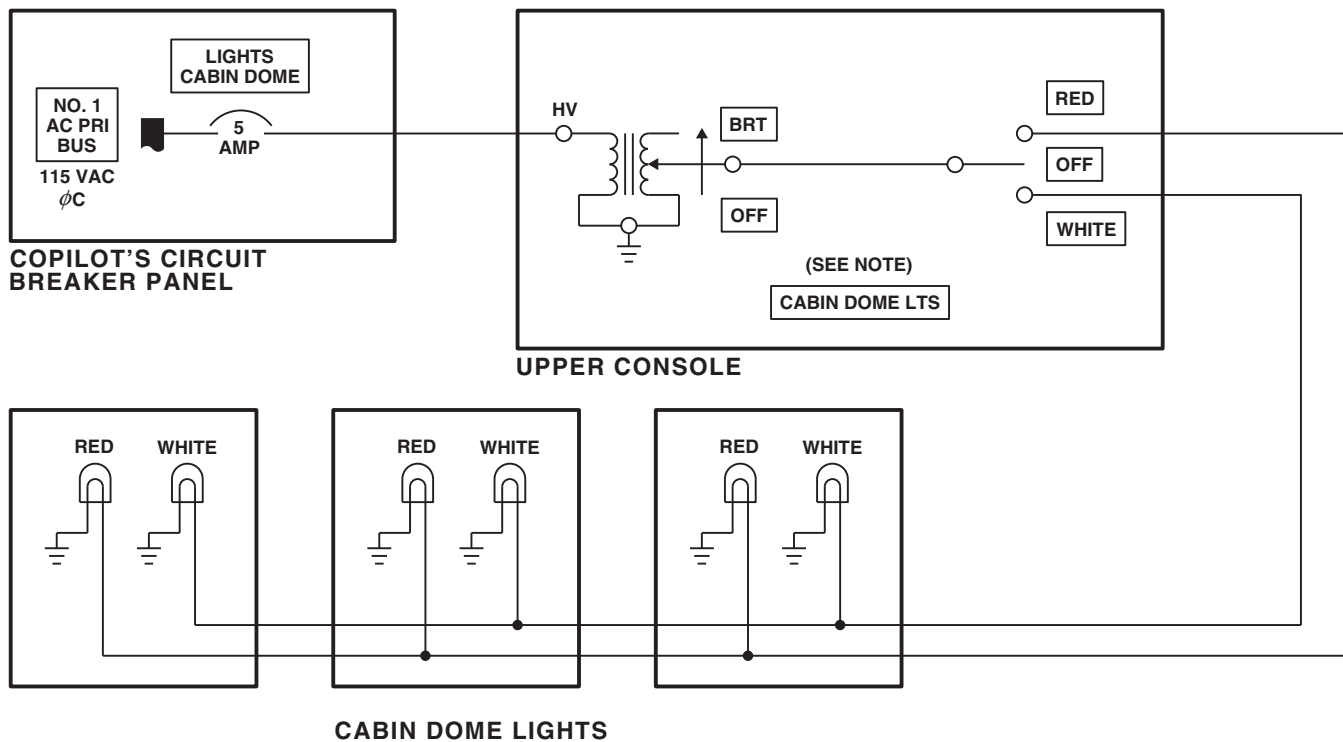


FIGURE 9-1-8-11. CABIN DOME LIGHTS BLOCK DIAGRAM



EFFECTIVITY
W/O SLS

FB2226_2A
SA

FIGURE 9-1-8-12. CABIN DOME LIGHTS BLOCK DIAGRAM

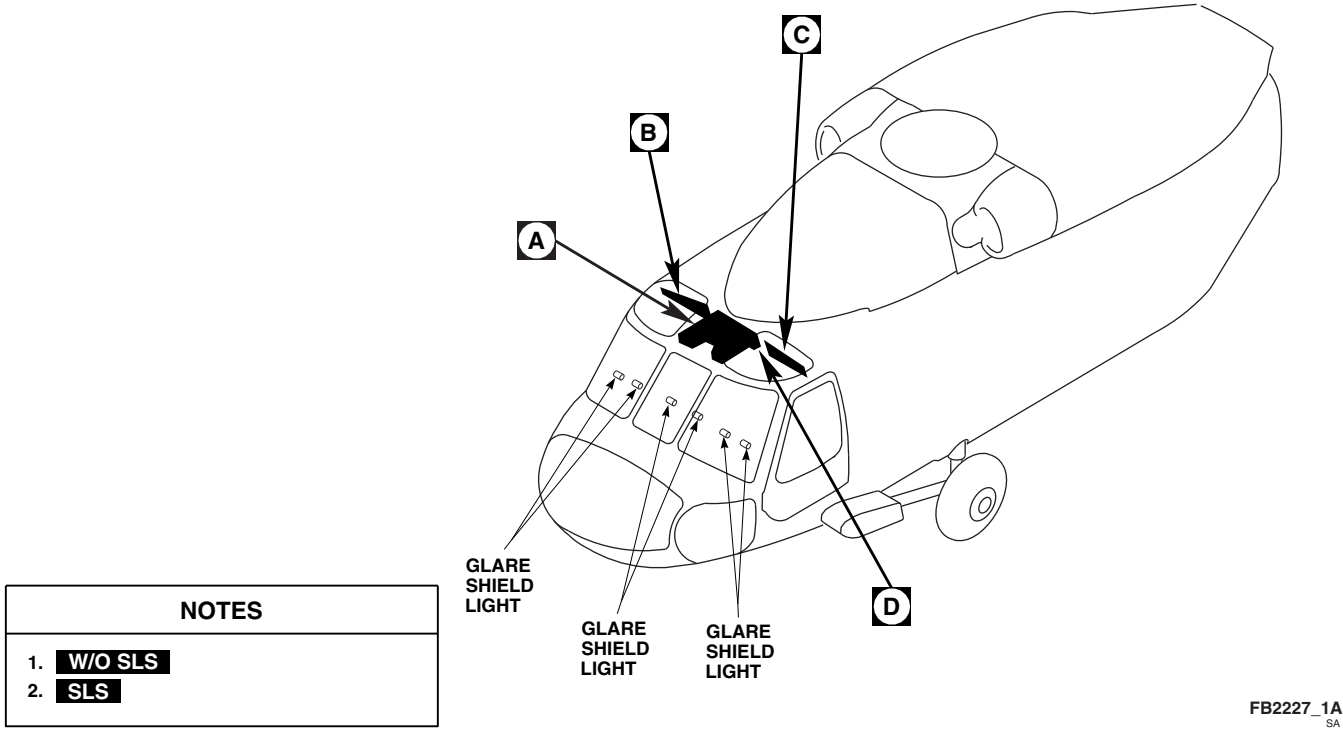
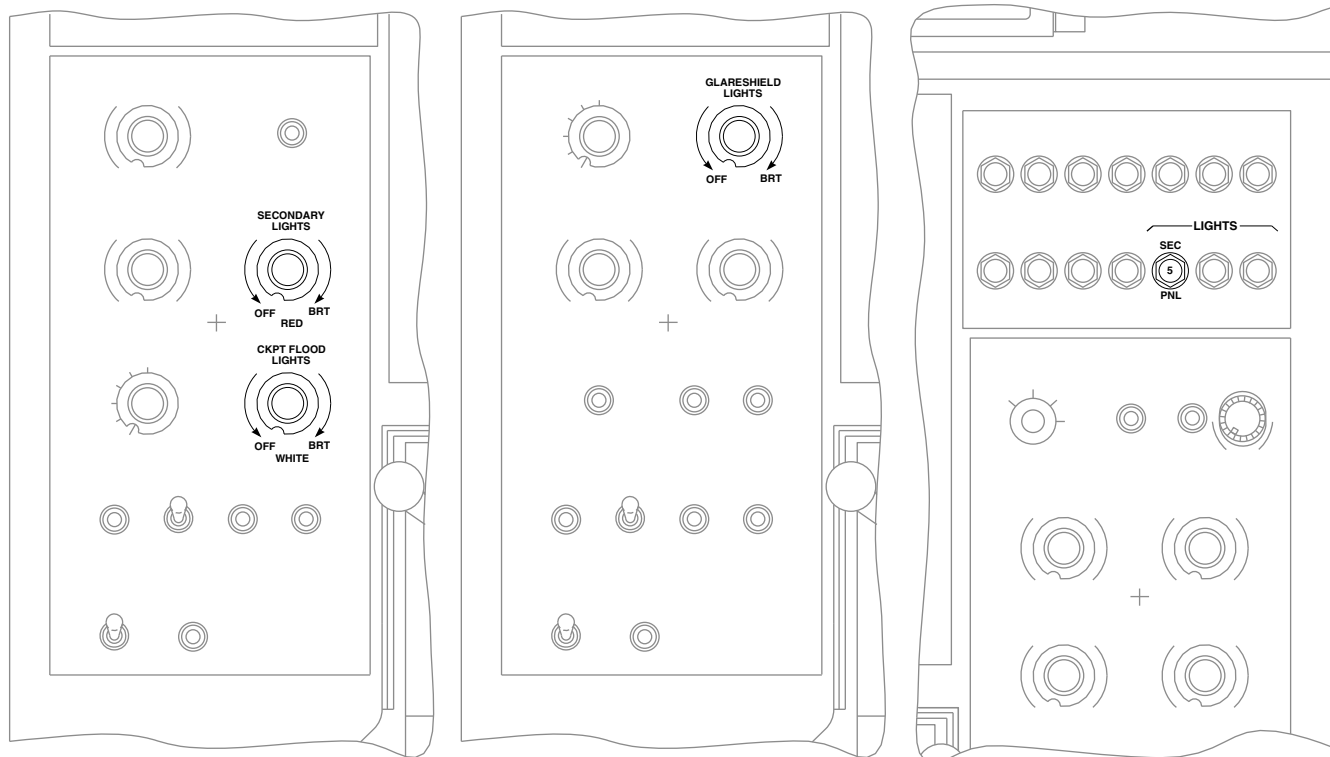


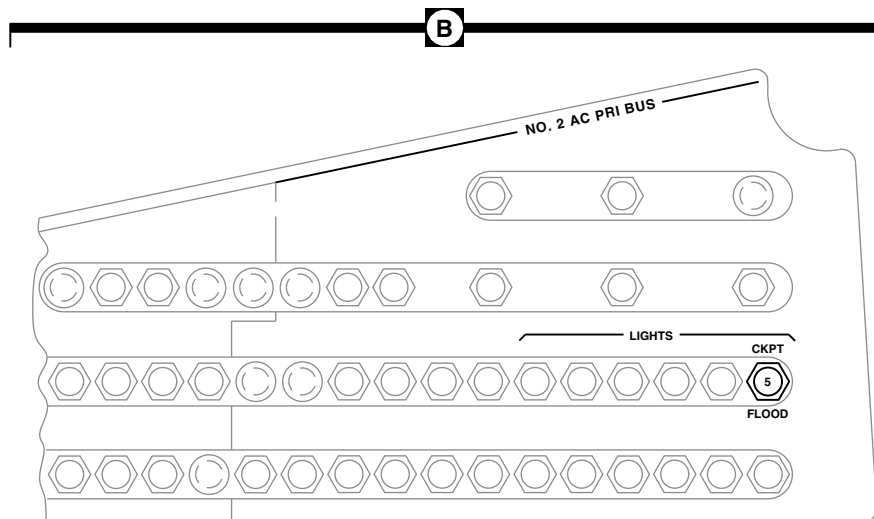
FIGURE 9-1-8-13. COCKPIT FLOOD AND SECONDARY LIGHTS LOCATION DIAGRAM
(SHEET 1 OF 4)



(SEE NOTE 1)

(SEE NOTE 2)

UPPER CONSOLE

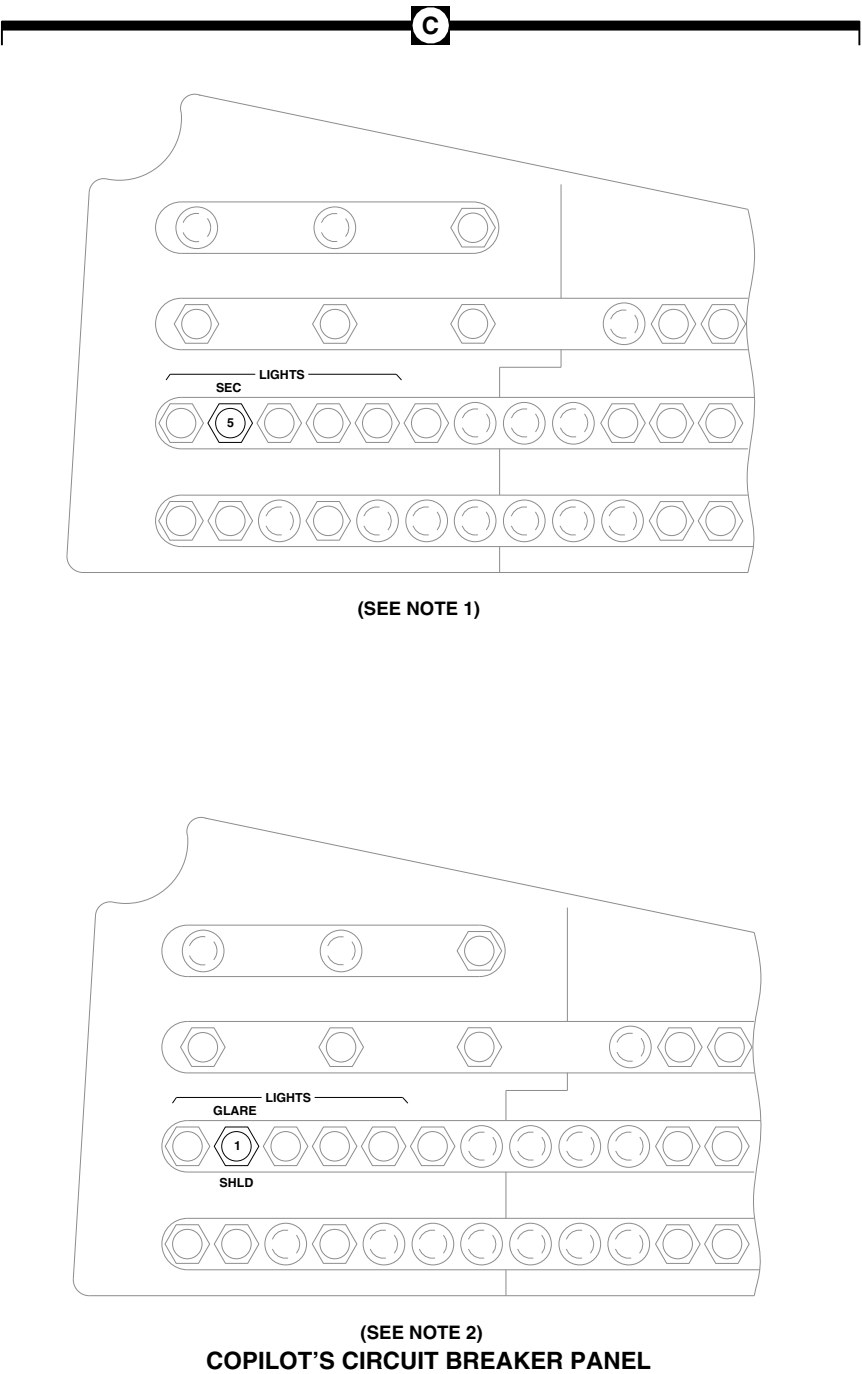


(SEE NOTE 1)

PILOT'S CIRCUIT BREAKER PANEL

FB2227_2B
SA

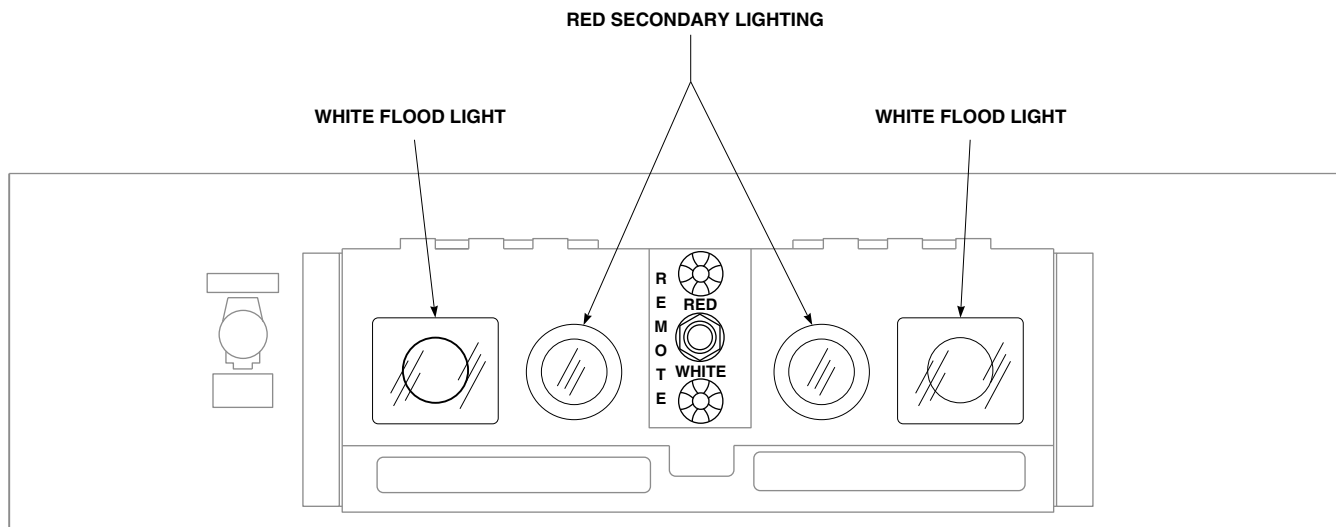
FIGURE 9-1-8-13. COCKPIT FLOOD AND SECONDARY LIGHTS LOCATION DIAGRAM
(SHEET 2 OF 4)



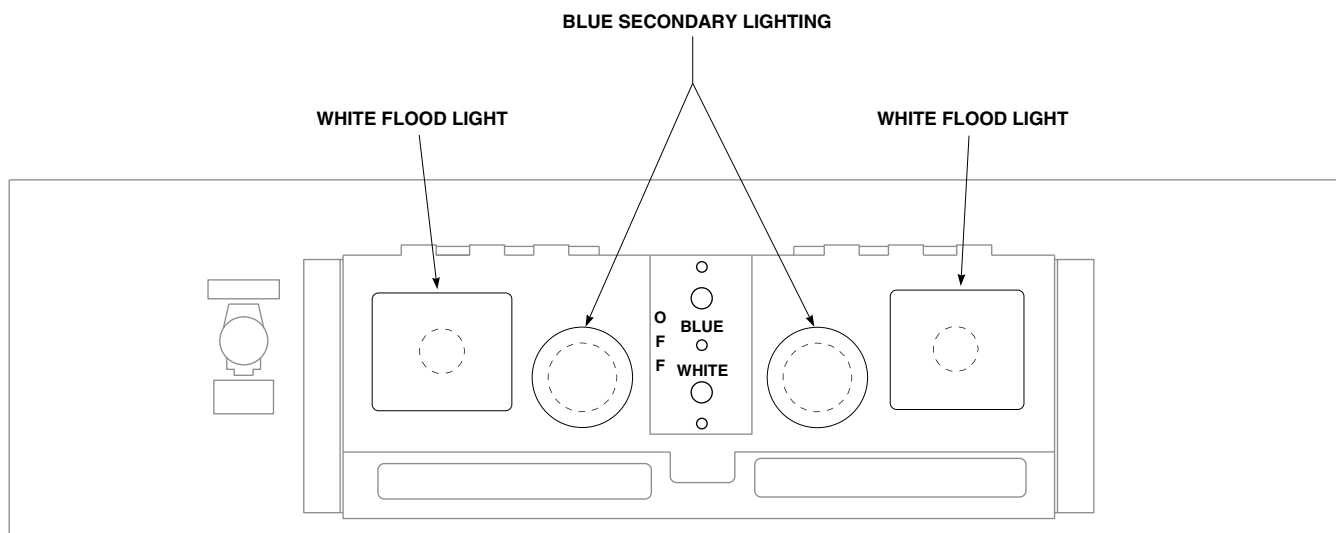
FB2227_3B
SA

**FIGURE 9-1-8-13. COCKPIT FLOOD AND SECONDARY LIGHTS LOCATION DIAGRAM
(SHEET 3 OF 4)**

D



(SEE NOTE 1)

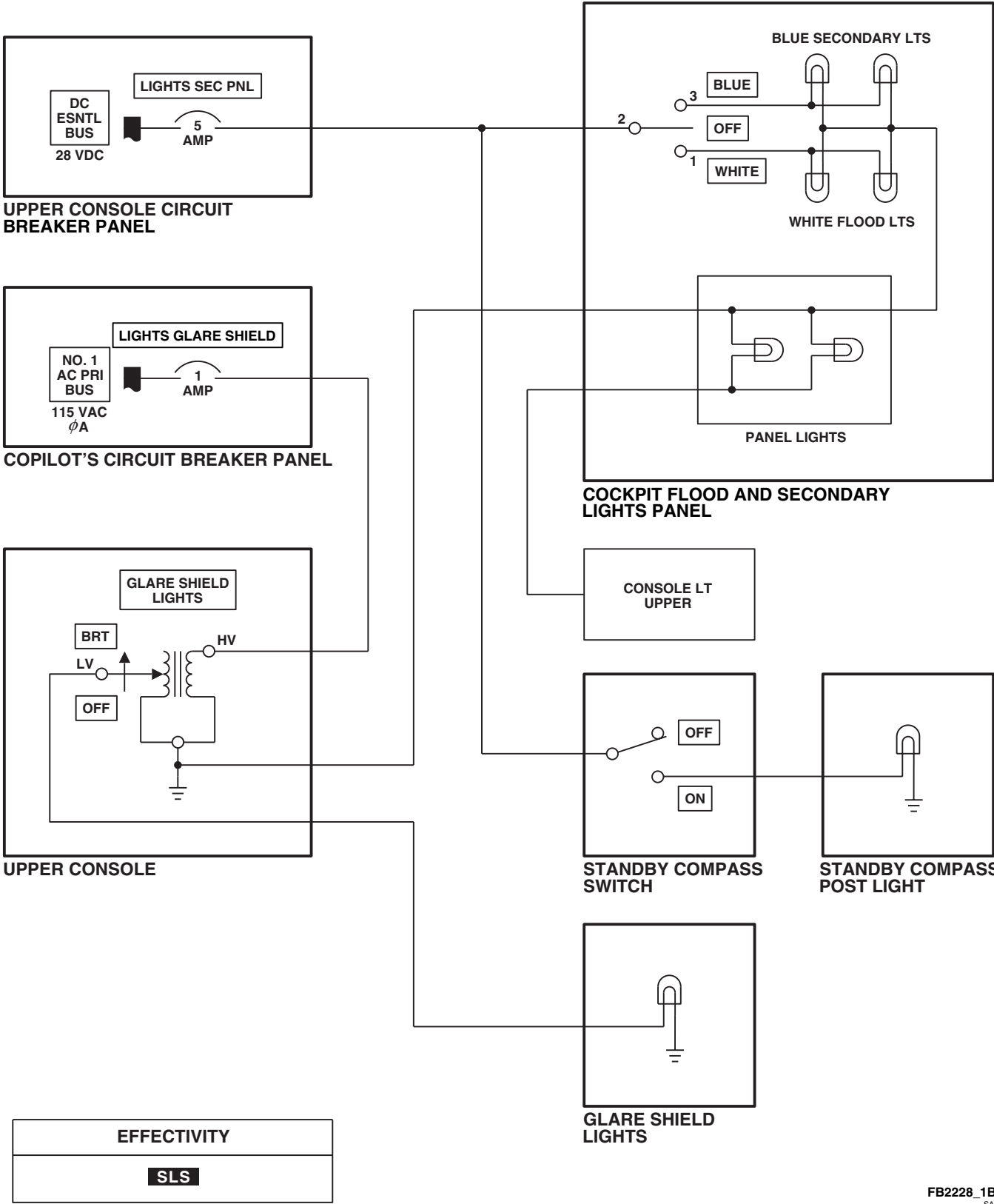


(SEE NOTE 2)

COCKPIT FLOOD AND SECONDARY LIGHT PANEL

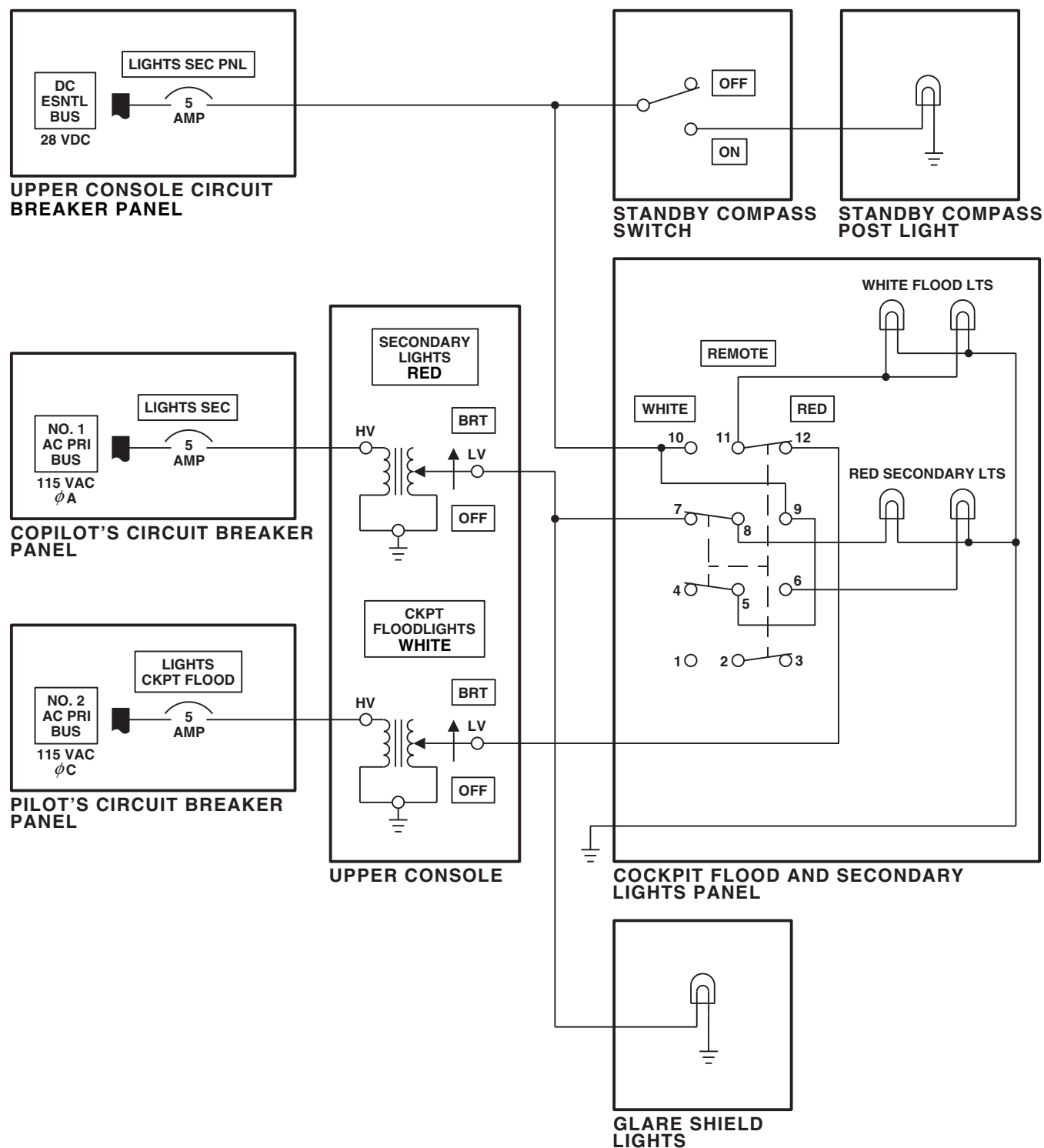
FB2227_4B
SB

**FIGURE 9-1-8-13. COCKPIT FLOOD AND SECONDARY LIGHTS LOCATION DIAGRAM
(SHEET 4 OF 4)**



FB2228_1B
SA

FIGURE 9-1-8-14. COCKPIT FLOOD AND SECONDARY LIGHTS BLOCK DIAGRAM



EFFECTIVITY
W/O SLS

FB2228 2B
SA

FIGURE 9-1-8-15. COCKPIT FLOOD AND SECONDARY LIGHTS BLOCK DIAGRAM

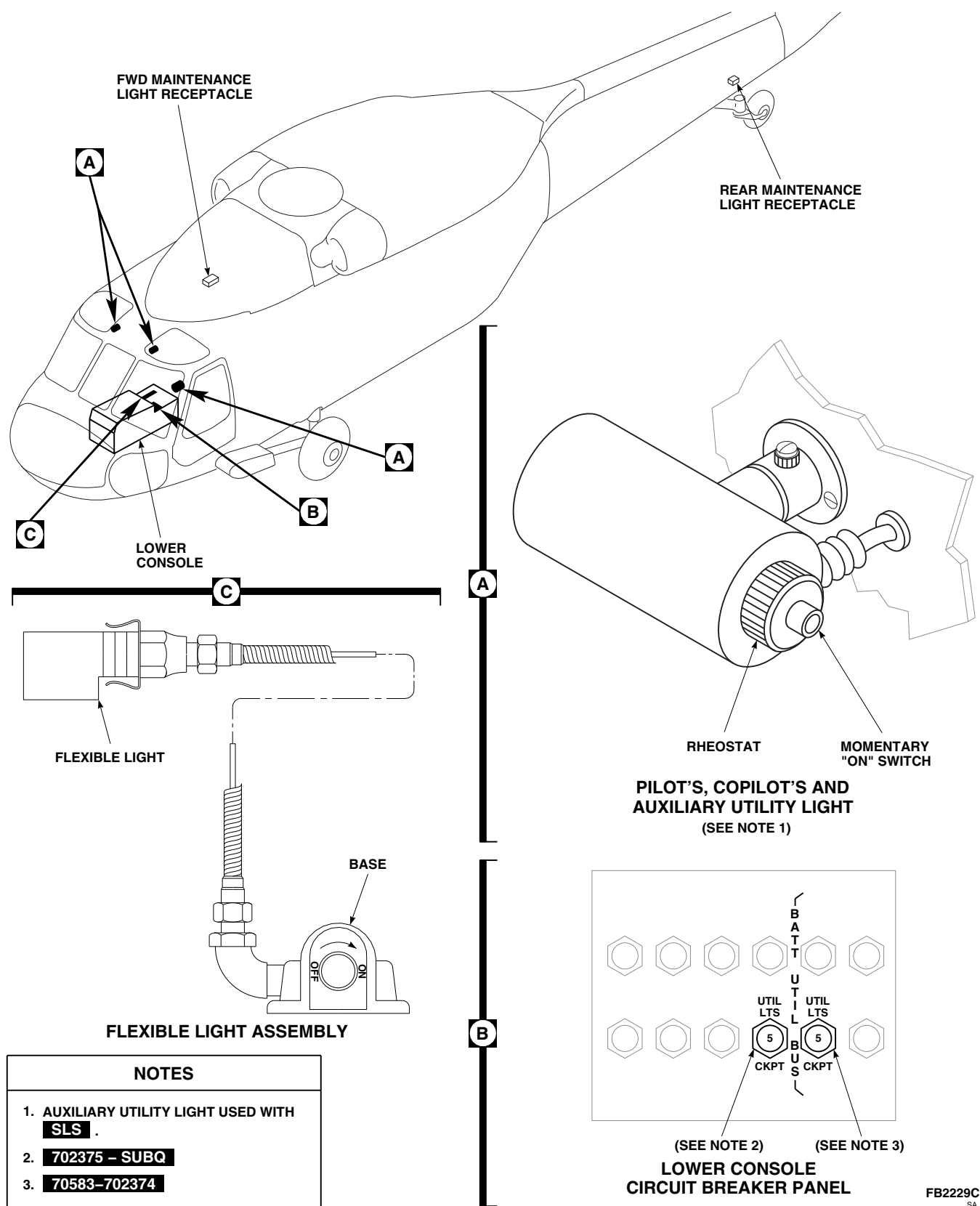


FIGURE 9-1-8-16. UTILITY AND MAINTENANCE LIGHTS LOCATION DIAGRAM

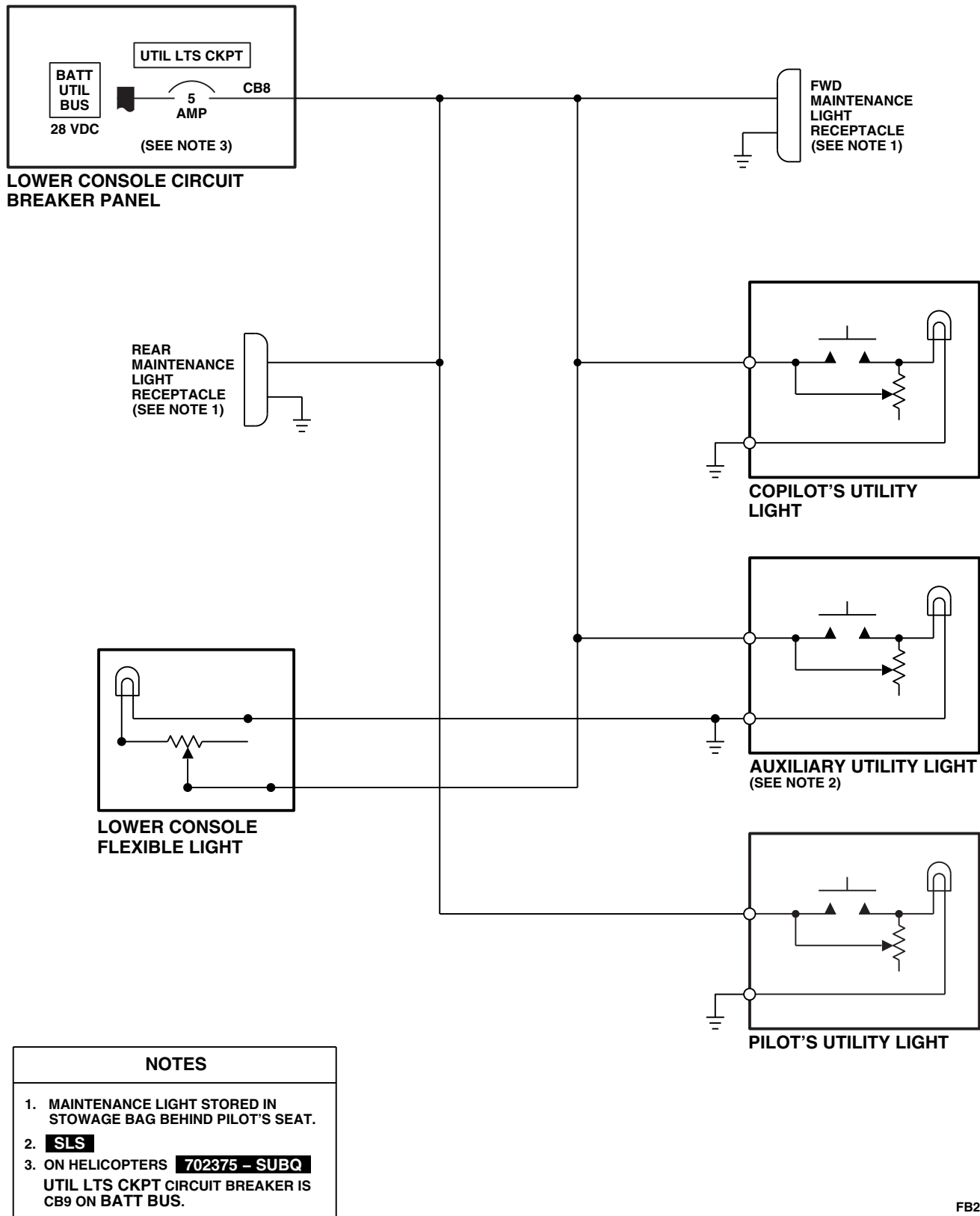


FIGURE 9-1-8-17. UTILITY AND MAINTENANCE LIGHTS BLOCK DIAGRAM

9-1-9 EXTERIOR LIGHTING DESCRIPTION AND DATA.

9-1-9.1 Exterior Lighting.

The Exterior Lighting Systems are used either as navigational aids or for lighting an area outside of the helicopter for search or landing operations (Figure 9-1-9-1). The Exterior Lighting Systems consist of formation lights, anticollision lights, position lights, a retractable landing light, controllable searchlight, auxiliary controllable searchlight, and cargo hook lights.

9-1-9.2 Formation Lights **W/O SLS** .

There are four formation lights; one each on the right and left horizontal stabilizers, one on the rear of the main rotor pylon, and one on the tail cone (Figure 9-1-9-3). Each formation light is an electro-luminescent light assembly that emits green light when an alternating electrical field is applied to the phosphorescent plate. Electrical power is routed from the NO. 2 AC PRI BUS through the LIGHTS FORM HV circuit breaker to the FORMATION LT control switch, and through the LIGHTS FORM LV circuit breaker to the left relay panel and the formation lights. The FORMATION LT control switch, on the upper console, has an OFF position, and positions marked from 1 through 5. The higher the control setting, the higher the output voltage from the FORMATION LT control and the brighter the formation lights.

9-1-9.3 Formation Lights **SLS** .

There are two sets of formation lights (Figure 9-1-9-2). One set is infrared, consisting of light-emitting diode assemblies, and one is incandescent, emitting a green light when an alternating electrical field is applied to the phosphorescent plate (NORM). The four formation lights include: one each on the right and left sides of the horizontal stabilator, one on the rear of the main rotor pylon, and one on top of the tail cone. With the NAV LTS switch at NORM, 115 VAC is routed from the NO. 2 AC PRI BUS through the LIGHTS FORM HV circuit breaker to the FORMATION LT control switch; the LIGHTS FORM LV circuit breaker feeds the voltage through the normally-closed contacts of relay K53 in the left relay panel to the formation lights.

With the NAV LTS switch at IR, 28 VDC is routed from the NO. 2 DC PRI BUS through the IR LTS circuit breaker to the NAV LTS switch to energize relay K53 in the left relay panel. Also, 28 VDC is routed to circuitry in the left relay panel, producing a 5 VDC output. This voltage is routed through the FORMATION LT control to the IR formation lights. The FORMATION LT control, on the upper console, has an OFF position and positions marked 1 through 5. With the NAV LTS switch positioned at NORM, the intensity of the electro-luminescent formation lights is controlled by rotating the FORMATION LT control clockwise from 1 to 5. Position 1 is dim and position 5 is bright. With the NAV LTS switch at IR, the electro-luminescent formation lights are disabled by energizing relay K53 in the left relay panel. With the FORMATION LT control at position 5, 5 VDC is applied to the clear (bright) side of the IR assemblies. At positions 1 through 4, 5 VDC is applied to the filtered (dim) side of the IR assemblies.

9-1-9.4 Anticollision Lights.

The helicopter has two anticollision light assemblies: one on the underside of the tail cone, and the other on the top of the tail rotor pylon (Figure 9-1-9-4). Each anticollision light assembly contains two strobe lamps. One lamp is enclosed in a clear lens and provides light for daylight operation. The second lamp is enclosed in a red lens and provides light for night operation. The mode of operation is controlled by the ANTI-COLLISION LIGHTS UPPER/BOTH/LOWER and DAY/OFF/NIGHT switches on the upper console. System electrical power is provided by the NO. 2 AC PRI BUS and routed through the LIGHTS ANTI COLL circuit breaker, on the pilot's circuit breaker panel, to the power supply. The power supply, located in the tail cone section, can power either one or both light assemblies in either the red or clear mode, as selected by the ANTI-COLLISION LIGHTS switches. With the ANTI-COLLISION LIGHTS DAY/OFF/NIGHT switch placed to NIGHT, and the ANTI-COLLISION LIGHTS UPPER/BOTH/LOWER switch placed to BOTH, the power supply alternately provides trigger voltages to the red lamps in both the upper and lower anticollision

light assemblies. With the ANTI-COLLISION LIGHTS DAY/OFF/NIGHT switch placed to DAY, the power supply alternately provides trigger voltages to the clear lamps in both the upper and lower anticollision light assemblies. With the ANTI-COLLISION LIGHTS UPPER/BOTH/LOWER switch placed to either UPPER or LOWER, the opposite light is disabled; the frequency of the trigger voltage to the selected lamps is 30 to 40 flashes per minute.

9-1-9.5 Position Lights **W/O SLS** .

The helicopter has either three or five position lights. On helicopters without External Stores Support System (ESSS) installed, the helicopter has three position lights: a white light on the rear of the tail rotor pylon, a red light on the left landing gear support fairing, and a green light on the right landing gear support fairing. On helicopters with ESSS installed, the helicopter has two additional position lights: a red light on the left support and a green light on the right support (Figure 9-1-9-6). On helicopters without External Stores Support System (ESSS), the ESSS disconnects are jumpered so the existing three lights are operational. On helicopters with ESSS installed, the position lights on the landing gear support fairings are disabled and the position lights on the HSS are operational. Therefore, three position lights are operational in either helicopter configuration. Power is applied to the lights from the NO. 2 DC PRI BUS through the POS LTS circuit breaker on the pilot's circuit breaker panel, and the POSITION LTS switches on the upper console. With the POSITION LIGHTS STEADY/FLASH switch placed in STEADY and DIM/OFF/BRT switch placed to either DIM or BRT, all three position lights illuminate accordingly. With the POSITION LIGHTS DIM/OFF/BRT switch placed to either DIM or BRT and STEADY/FLASH switch placed to FLASH, 28 VDC is applied to the flasher. Pulsing power is then routed from the flasher through the DIM/OFF/BRT switch to the lights, causing the three position lights to flash between 70 and 90 times per minute.

9-1-9.6 Position Lights **SLS** .

The helicopter has two sets of either three or five position lights. One set is infrared (IR) and one is

incandescent. On helicopters without External Stores Support System (ESSS), the helicopter has three position lights: a white light on the rear of the tail rotor pylon, a red light on the left landing gear support fairing, and a green light on the right landing gear support fairing. On helicopters with ESSS installed, the helicopter has two additional position lights: a red light on the left stores support and a green light on the right stores support (Figure 9-1-9-5). On helicopters without ESSS, the disconnects are jumpered so the existing three lights are operational. On helicopters with ESSS installed, the position lights on the landing gear support fairings are disabled and the position lights on the stores supports are operational. Therefore, three position lights are operational in either helicopter configuration. Power for the incandescent set of position lights is applied to the lights from the NO. 2 DC PRI BUS through the POS LTS circuit breaker, on the pilot's circuit breaker panel, and the POSITION LIGHTS switches on the upper console. With the POSITION LIGHTS STEADY/FLASH switch placed to STEADY and the DIM/OFF/BRT switch placed to either DIM or BRT, all three position lights illuminate accordingly. With the POSITION LIGHTS DIM/OFF/BRT switch placed to either DIM or BRT and the STEADY/FLASH switch placed to FLASH, 28 VDC is applied to the flasher. Pulsing power is then routed from the flasher through the DIM/OFF/BRT switch to the lights, causing the three position lights to flash between 70 and 90 times per minute.

Power for the IR position lights is applied when the NAV LTS switch is placed to the IR position. Power of 28 VDC is applied to the left relay panel from the NO. 2 DC PRI BUS through the IR LTS circuit breaker and the NAV LTS switch on the upper console. When the NAV LTS switch is placed to IR, the 28 VDC from the POSITION LIGHTS STEADY/FLASH switch is reduced to 5 VDC and applied through the POSITION LIGHTS DIM/OFF/BRT switch to the IR position lights. With the POSITION LIGHTS STEADY/FLASH switch placed to FLASH, 28 VDC is applied to the flasher and routed to the IR position lights, causing them to flash. This circuit is protected by a fuse mounted on the left relay panel.

9-1-9.7 Retractable Landing Light.

The retractable landing light assembly is on the left-forward lower nose area of the helicopter (Figure 9-1-9-7). Power of 28 VDC for the 600-watt quartz lamp is supplied by the NO. 1 DC PRI BUS through the LIGHTS RETR LDG PWR circuit breaker on the copilot's circuit breaker panel. The light assembly is controlled by a LDG LT combination pushbutton/toggle switch on both the pilot's and copilot's collective stick grips. Both LDG LT switches receive operating power from the NO. 1 DC PRI BUS through the LIGHTS RETR LDG CONT circuit breaker on the copilot's circuit breaker panel. When either LDG LT ON/OFF switch is pushed and released, 28 VDC control voltage is applied through relay latching circuitry in the right relay panel to the retractable landing light assembly, causing the lamp to light and the caution/advisory panel LDG LT ON caution capsule to light. Pressing the LDG LT ON/OFF switch again and releasing it causes the lamp and the LDG LT ON capsule to go out. When either LDG LT EXT/RETR switch is moved to EXT and held, 28 VDC power is applied to the retractable landing light assembly to extend the light. When either LDG LT EXT/RETR switch is moved to RETR, 28 VDC power is applied to the retractable landing light assembly to retract the light. The beam angle of the landing light can be controlled from straight down to straight ahead.

9-1-9.8 Controllable Searchlight **W/O SLS** .

The controllable searchlight assembly, on the forward lower fuselage, provides a directional beam of light that is manually controlled by switches on the pilot's and copilot's collective stick grips (Figure 9-1-9-9). The searchlight can be extended about 120° from the fully retracted position and can be rotated either left or right a full 360° when in any extended position. Power of 28 VDC for the 450-watt lamp is supplied by the DC ESNTL BUS through the LIGHTS CONTR SRCH PWR circuit breaker on the upper console. Power of 28 VDC to control the searchlight is supplied by the DC ESNTL BUS through the LIGHTS SRCH CONTR circuit breaker, on the upper console, to the SRCH LT ON/OFF/STOW switch and to a four-way thumb switch on the pilot's collective

stick grip. The SRCH LT ON/OFF/STOW switch and four-way thumb switch on the copilot's collective stick grip receive 28 VDC power and are functional only when the pilot's SRCH LT ON/OFF/STOW switch is OFF. Placing the SRCH LT ON/OFF/STOW switch ON causes the searchlight lamp to light while placing the switch to STOW causes the searchlight to turn off, retract, and stow from any extended position. Placing the four-way thumb switch to EXT causes the searchlight to extend. Placing the four-way switch to R or L causes the searchlight to turn right or left. Placing the four-way switch to RET cause the searchlight to retract, and also to stow if held at RET. The searchlight can be operated in the stow position without overheating.

9-1-9.9 Controllable Searchlight **SLS** .

The controllable searchlight assembly provides a directional beam of light that is manually controlled by switches on the pilot's or copilot's collective stick grips (Figure 9-1-9-8). The searchlight is mounted under the nose section of the helicopter. The searchlight can be extended about 120° from the fully-retracted position and can be rotated either left or right a full 360° when in any extended position. Electrical power of 28 VDC for the 150-watt lamp is supplied by the DC ESNTL BUS through the LIGHTS CONTR PWR circuit breaker on the upper console, to the dimming unit and the searchlight. Electrical power of 28 VDC to control the searchlight is supplied by the DC ESNTL BUS through the LIGHTS SRCH CONTR circuit breaker on the upper console to the left relay panel and to the SRCH LT PUSH ON-OFF/BRT-DIM switches and four-way thumb switches on the pilot's and copilot's collective stick grips. Pressing and releasing the SRCH LT PUSH ON-OFF/BRT-DIM switch causes the searchlight lamp to light and the SEARCH LT ON capsule on the caution/advisory panel to go on. If the controllable searchlight dimming unit, on the glareshield, OUTPUT NORM/BYPASS switch is in the NORM position, the lamp intensity is that of when it was last turned on and controllable by the SRCH LT PUSH ON-OFF/BRT-DIM switch BRT/DIM positions. If the controllable searchlight dimming unit OUTPUT NORM/BYPASS switch is in the BYPASS position, the lamp intensity is on full

bright and is not dimmable. Pressing and releasing the SRCH LT ON-OFF/BRT-DIM switch again turns the searchlight lamp off. The searchlight lamp has a removable infrared filter for Soft Lighting System (SLS) operation or incandescent lamp operation. Placing the four-way thumb switch to EXT on the collective stick grips causes the searchlight to extend. Placing the four-way thumb switch to R or L causes the searchlight to turn right or left. Placing the four-way thumb switch to RETR causes the searchlight to retract and also to stow if held at RETR.

9-1-9.10 Cargo Hook Lights **W/O SLS** .

Three lights are mounted in the cargo hook well area (Figure 9-1-9-10). Power for these lights is provided by the NO. 2 AC PRI BUS through the LIGHTS CARGO HOOK circuit breaker on the pilot's circuit breaker panel (Figure 9-1-9-12). From the circuit breaker, power is routed to the CARGO HOOK LT switch on the upper console. The power is then routed to the three cargo hook lights. The CARGO HOOK LT switch is a two-position toggle switch that provides on-off control of cargo hook lighting.

9-1-9.11 Cargo Hook Lights **SLS** .

Two lights are mounted in the cargo hook well area (Figure 9-1-9-11). Power for these lights is provided by the NO. 2 AC PRI BUS through the LIGHTS CARGO HOOK circuit breaker on the pilot's circuit breaker panel (Figure 9-1-9-13). From the circuit breaker, power is routed to the CARGO HOOK LT switch on the upper console and then to the two cargo hook lights through the cargo hook light dimming control panel. The CARGO HOOK LT switch is a two-position toggle switch that provides on-off control of cargo hook lighting.

9-1-9.12 Auxiliary Controllable Searchlight.

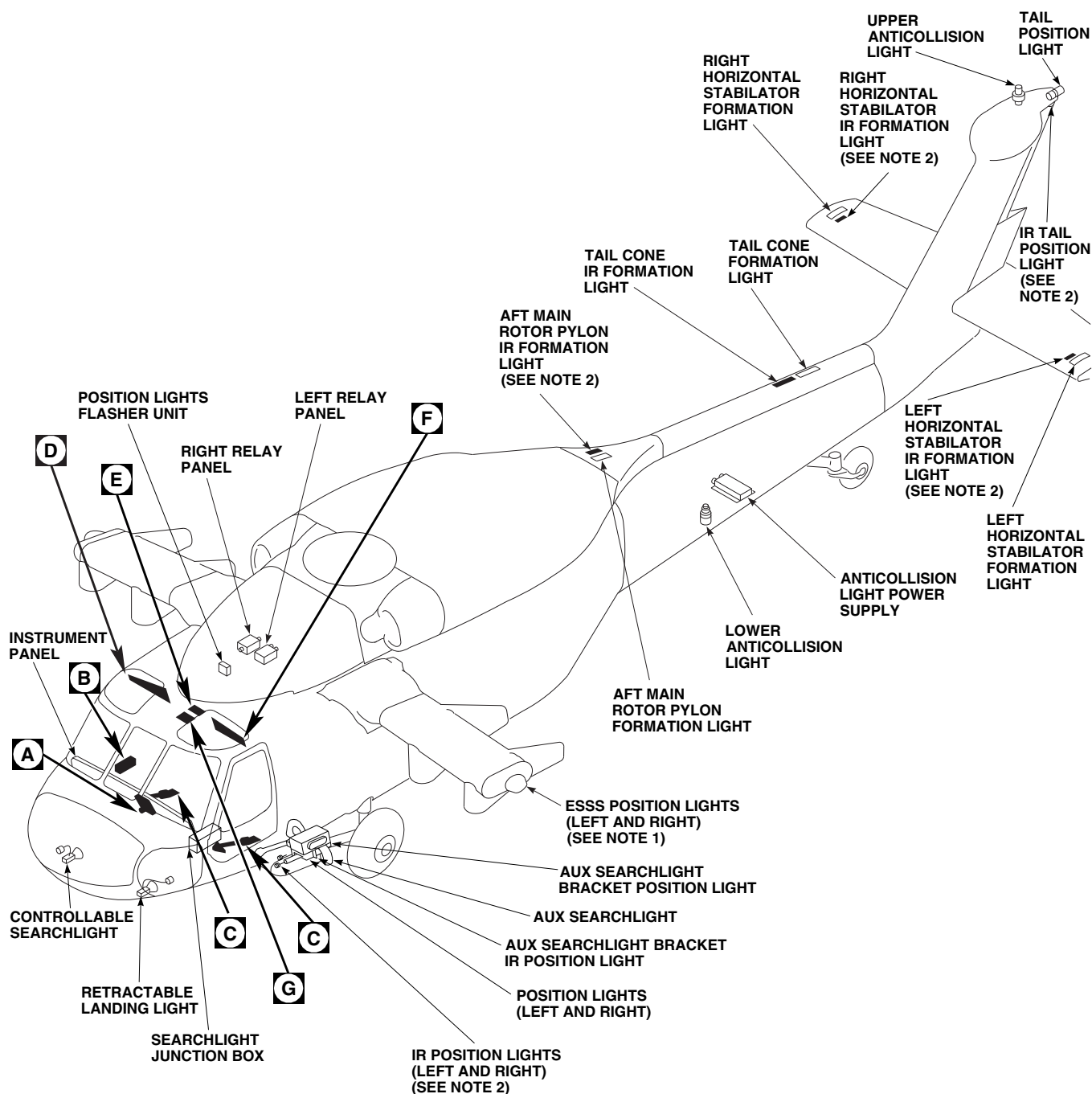
The auxiliary searchlight is a remote controlled, dual axis searchlight with the capability of focusing in-flight to a wide or narrow beam of white or IR light (Figures 9-1-9-14 and 9-1-9-15). The auxiliary searchlight is mounted on a bracket assembly attached to the end of the left hand stub wing. Electrical power of 28 VDC for the auxiliary

searchlight lamp and control functions is supplied by the NO. 2 DC PRI BUS through the AUX SEARCHLIGHT fuse in the No. 1 junction box to the searchlight junction box. The searchlight junction box supplies a fused 28 VDC through an internal relay to the searchlight lamp and the lower console auxiliary searchlight control panel. This fused voltage is interlocked through K15 DC BUS TIE CNTOR relay, requiring dual converter operation.

The POWER switch in the auxiliary searchlight control panel is a three-position switch with OFF and ON positions, and a momentary START position. Pressing and holding the POWER switch to the START position applies 28 VDC to energize the relay in the searchlight junction box. The 28 VDC is also applied to a booster assembly in the searchlight junction box, a blower in the searchlight assembly, and the AUX SRCHLT ON caution/advisory capsule. The energized relay in the searchlight junction box applies 28 VDC continuous operating power to the searchlight lamp. The power from the booster assembly causes the searchlight lamp to ignite within approximately five seconds of the POWER switch being held in the START position. Releasing the POWER switch to the ON position removes the boosted startup power to the lamp and continues to apply 28 VDC operating power to the searchlight lamp and blower to enable the auxiliary searchlight control functions. Setting the POWER switch to OFF disconnects the 28 VDC from the auxiliary searchlight lamp and blower and turns off the AUX SRCHLT ON caution/advisory capsule. Pressing and holding the momentary BEAM FOCUS switch varies the width of the auxiliary searchlight beam. Pressing and holding the momentary CONTROL switch in the UP, DOWN, L, or R position causes the gimbal assembly to move the auxiliary searchlight in the indicated direction. The FILTER switch selects either white light or IR operation. In the IR position, the IR window assembly is driven down over the auxiliary searchlight lamp lens, filtering all but the IR portion of the lamp beam spectrum. The auxiliary searchlight control panel FILTER indicator light goes on to indicate IR operation. With the FILTER switch in the NORM position, the IR window assembly is retracted and the FILTER indicator light goes off. The FILTER indicator light

can be tested by pressing the indicator lens. The intensity of the auxiliary searchlight control panel faceplate lighting can be varied along with other

lower console panel lighting by adjusting the CONSOLE LT LOWER dimming control on the upper console.



NOTES

1. **ESSS**
2. **SLS**
3. **W/O SLS**
4. **W/O SLS** SRCH LT SWITCH IS LABELLED ON/OFF/STOW.

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FIGURE 9-1-9-1. EXTERIOR LIGHTING SYSTEMS LOCATION DIAGRAM (SHEET 1 OF 4)

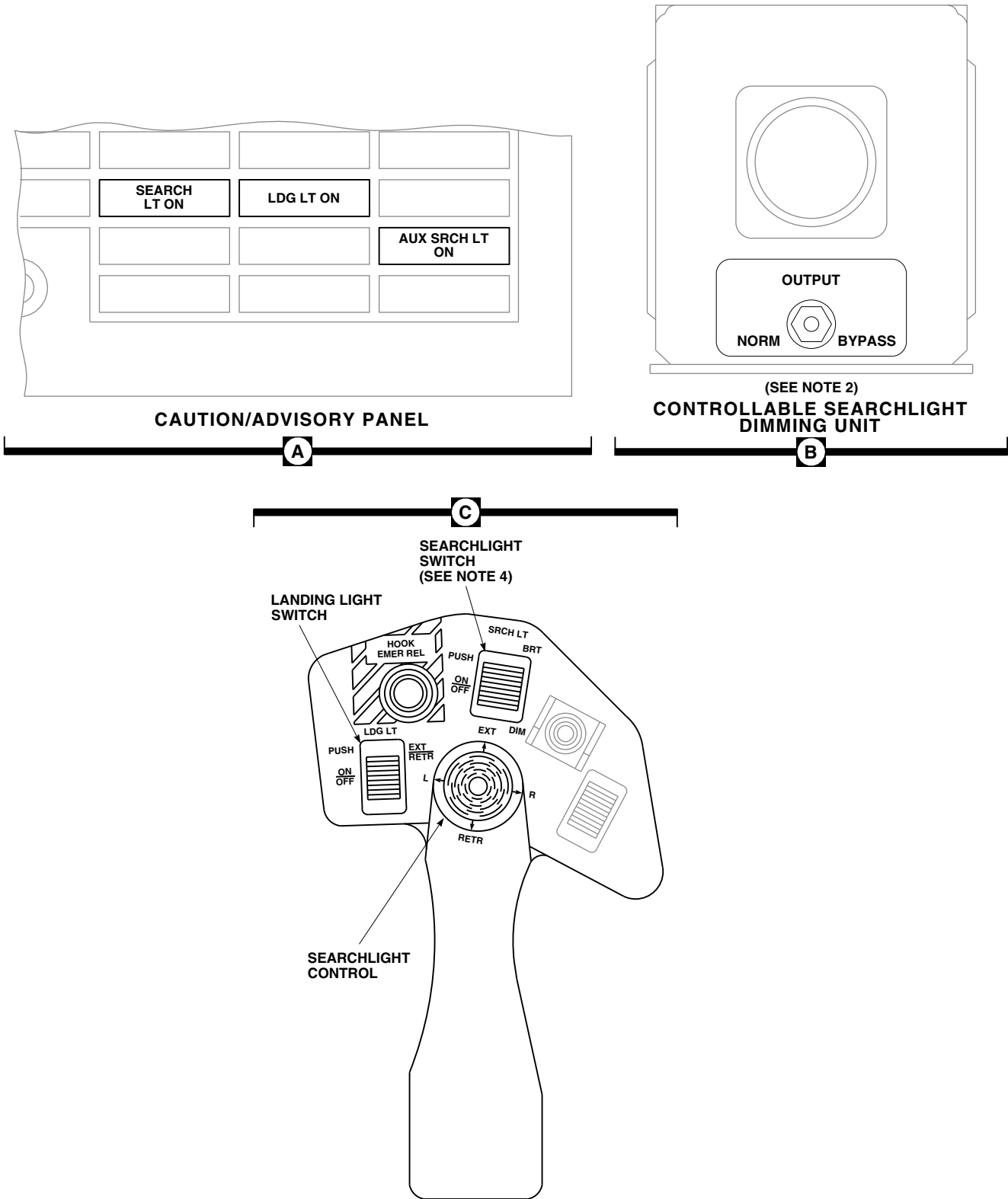
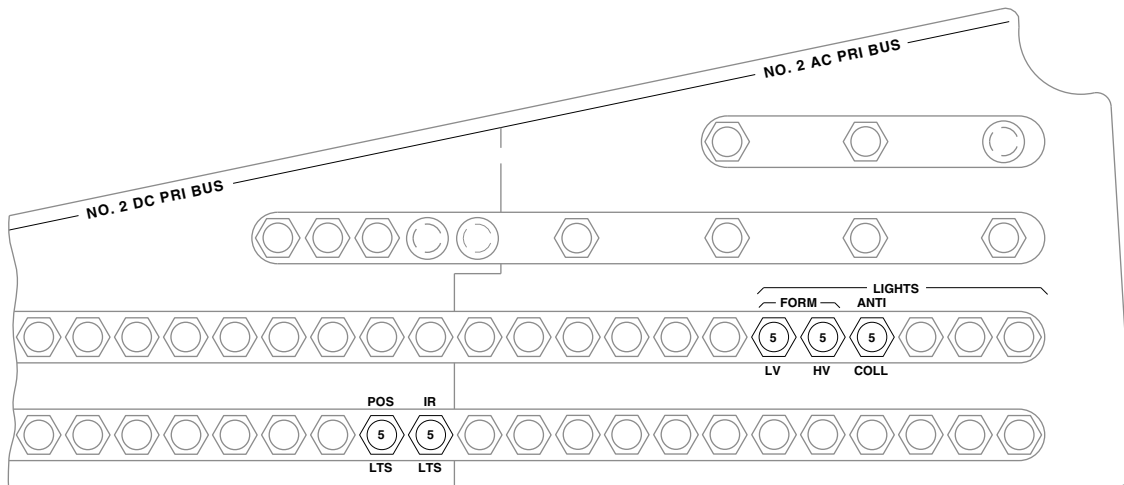
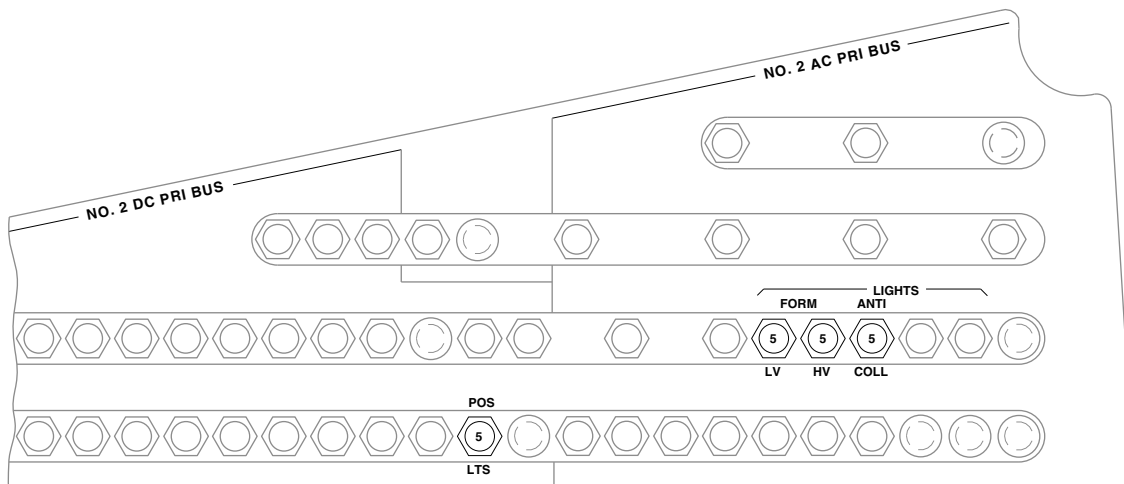


FIGURE 9-1-9-1. EXTERIOR LIGHTING SYSTEMS LOCATION DIAGRAM (SHEET 2 OF 4)



(SEE NOTE 2)



(SEE NOTE 3)

PILOT'S CIRCUIT BREAKER PANEL

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FIGURE 9-1-9-1. EXTERIOR LIGHTING SYSTEMS LOCATION DIAGRAM (SHEET 3 OF 4)

9-1-9-8 Change 7

This Document contains technical data subject to ITAR. See WARNING and classifications on first page.

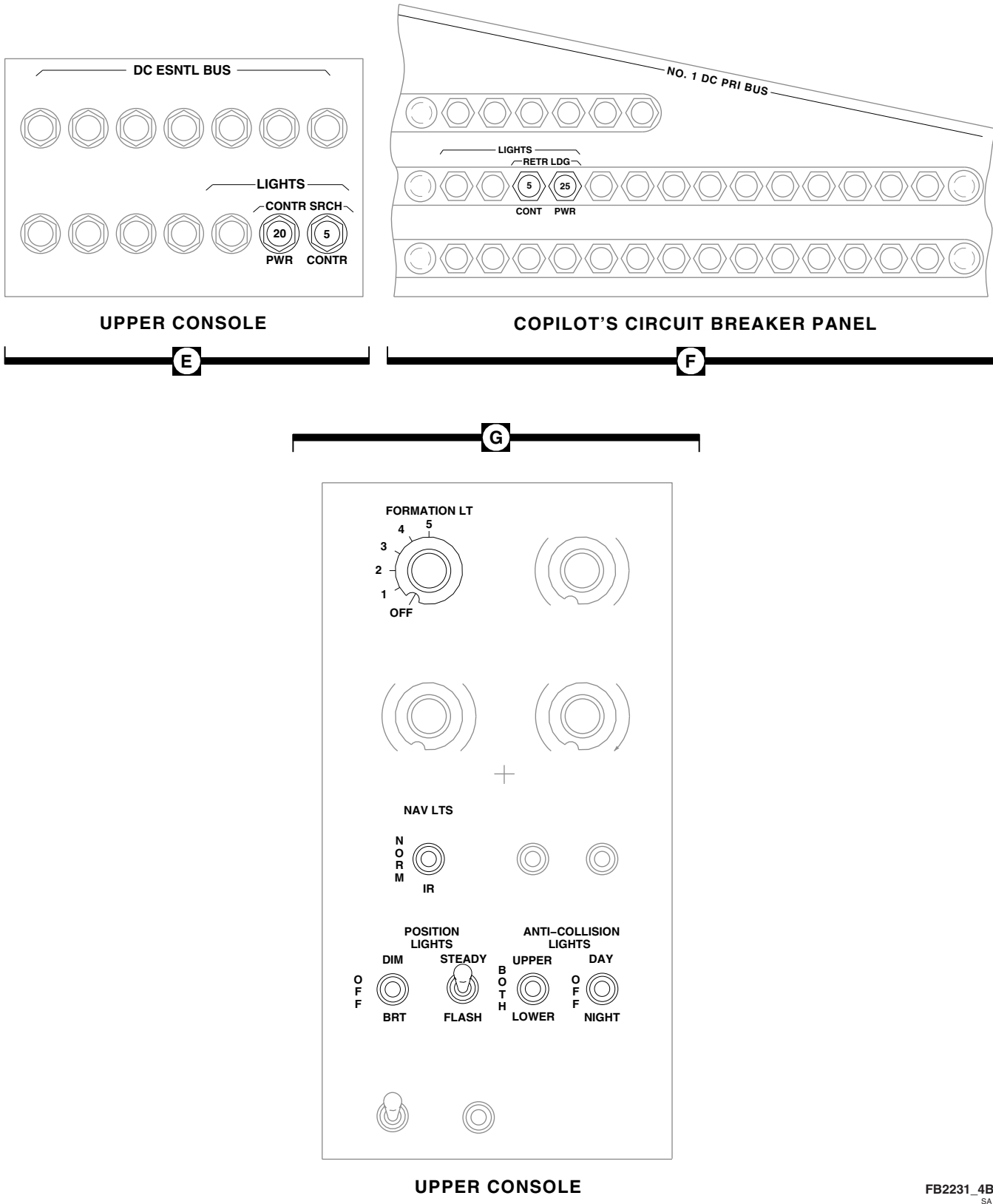
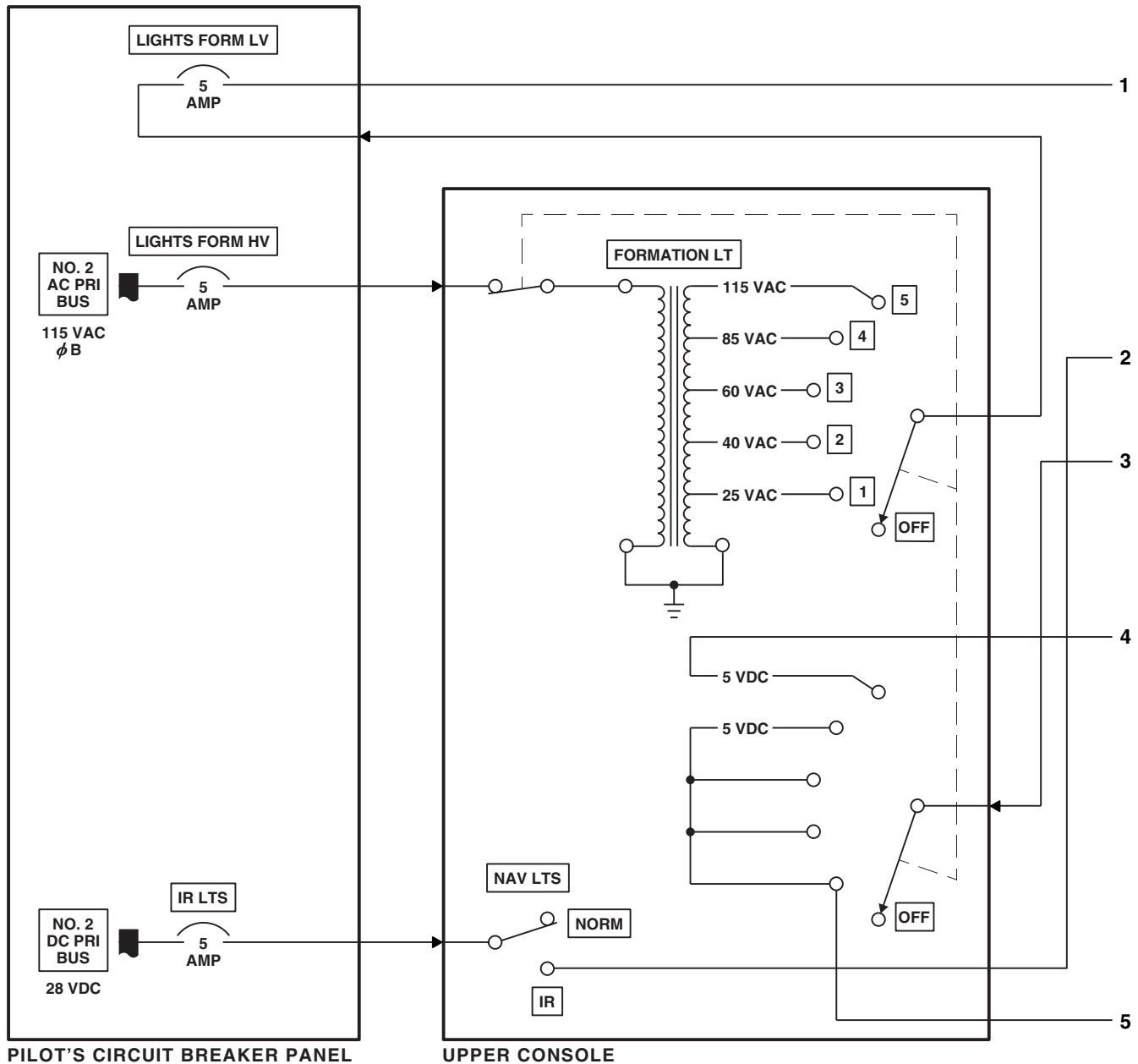


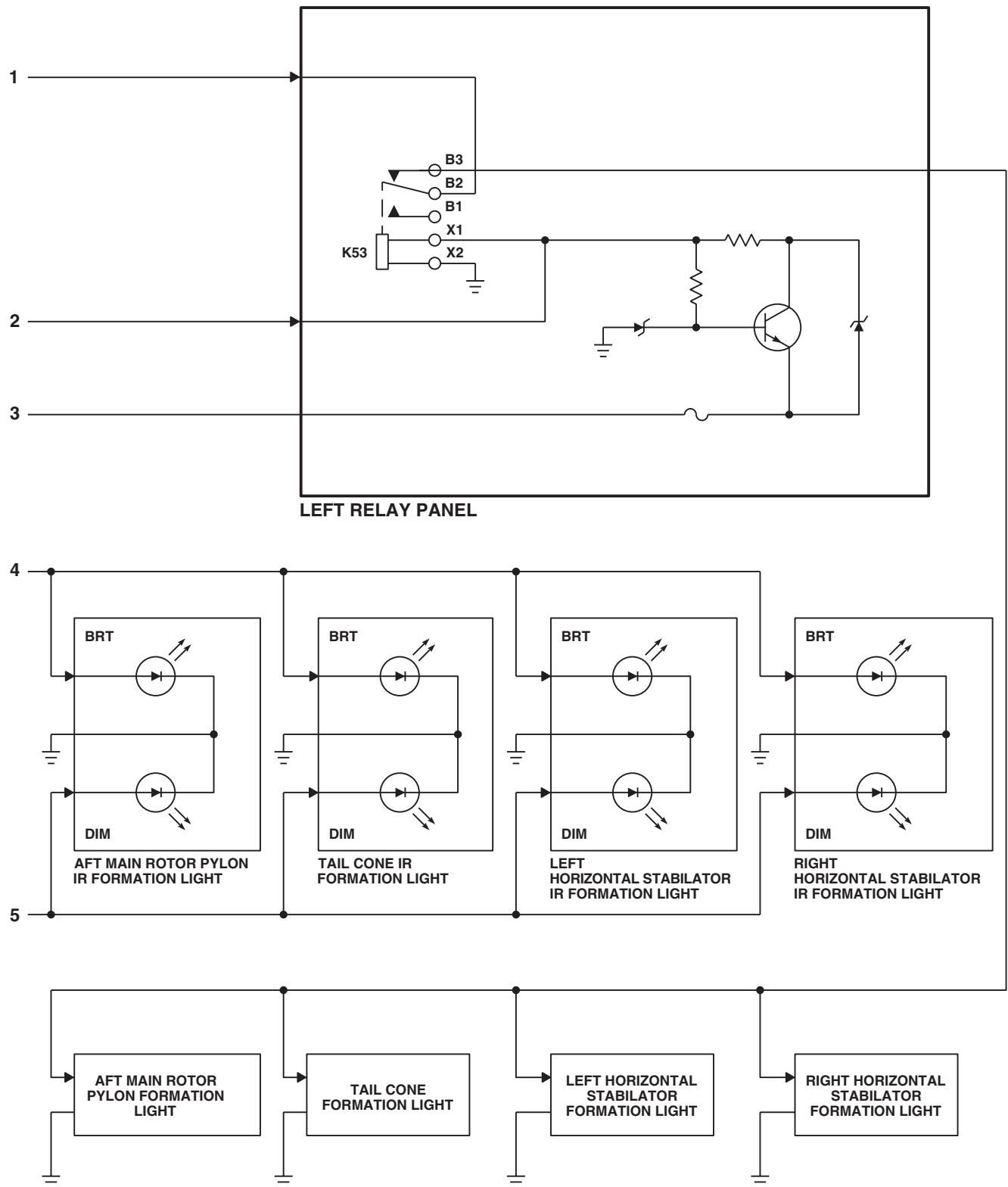
FIGURE 9-1-9-1. EXTERIOR LIGHTING SYSTEMS LOCATION DIAGRAM (SHEET 4 OF 4)



EFFECTIVITY
SLS

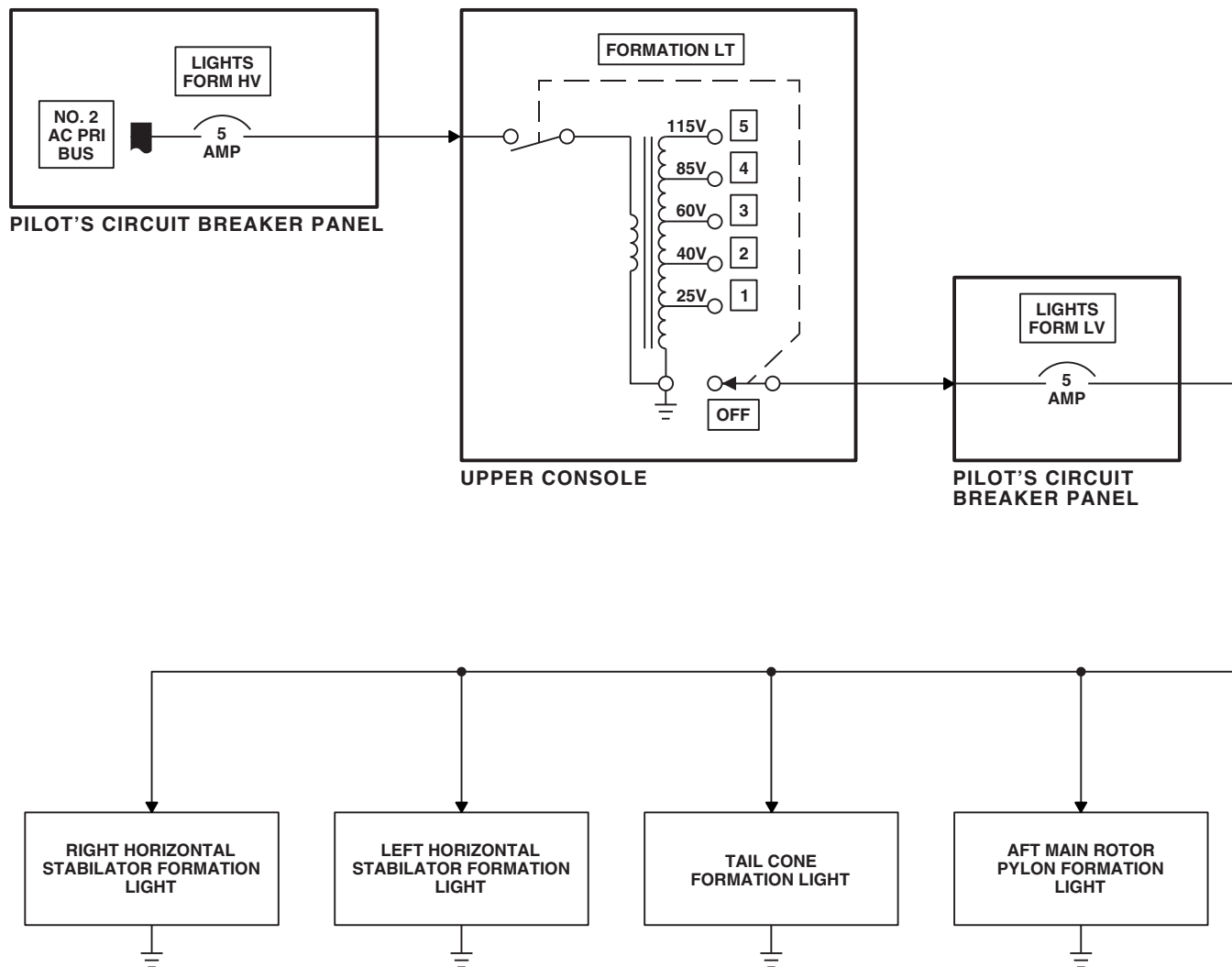
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FIGURE 9-1-9-2. FORMATION LIGHTS BLOCK DIAGRAM (SHEET 1 OF 2)



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SA

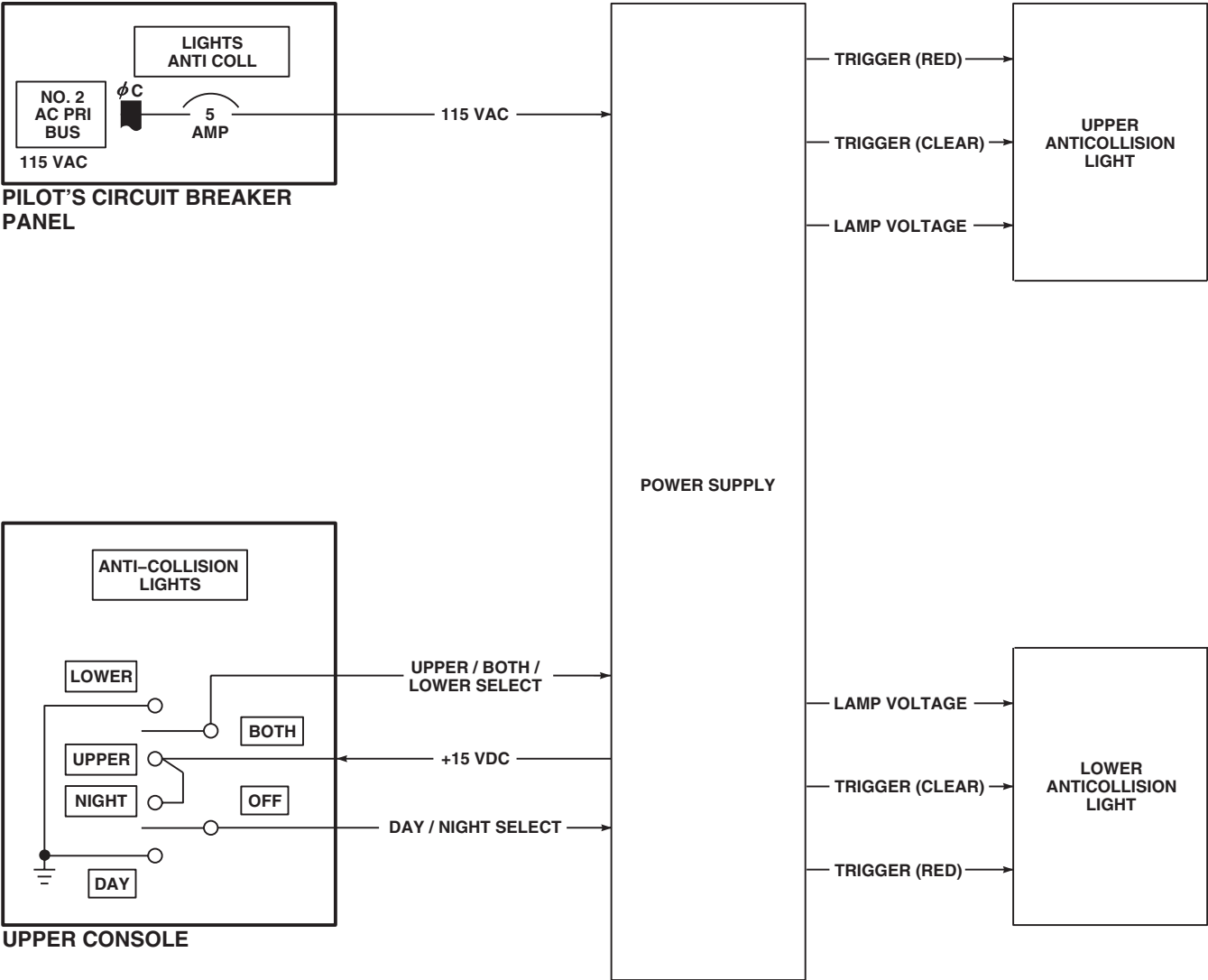
FIGURE 9-1-9-2. FORMATION LIGHTS BLOCK DIAGRAM (SHEET 2 OF 2)



EFFECTIVITY
W/O SLS

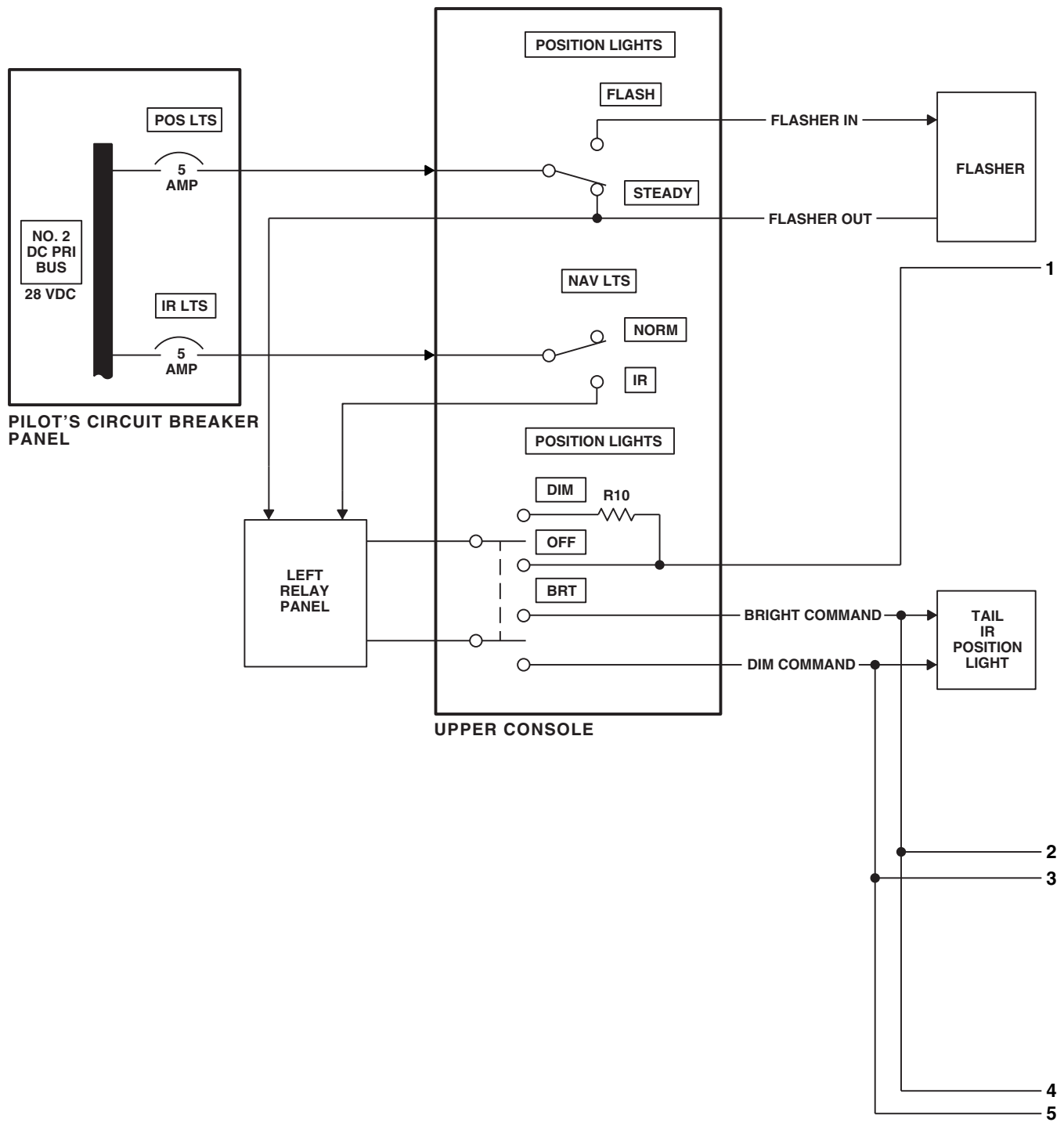
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FIGURE 9-1-9-3. FORMATION LIGHTS BLOCK DIAGRAM



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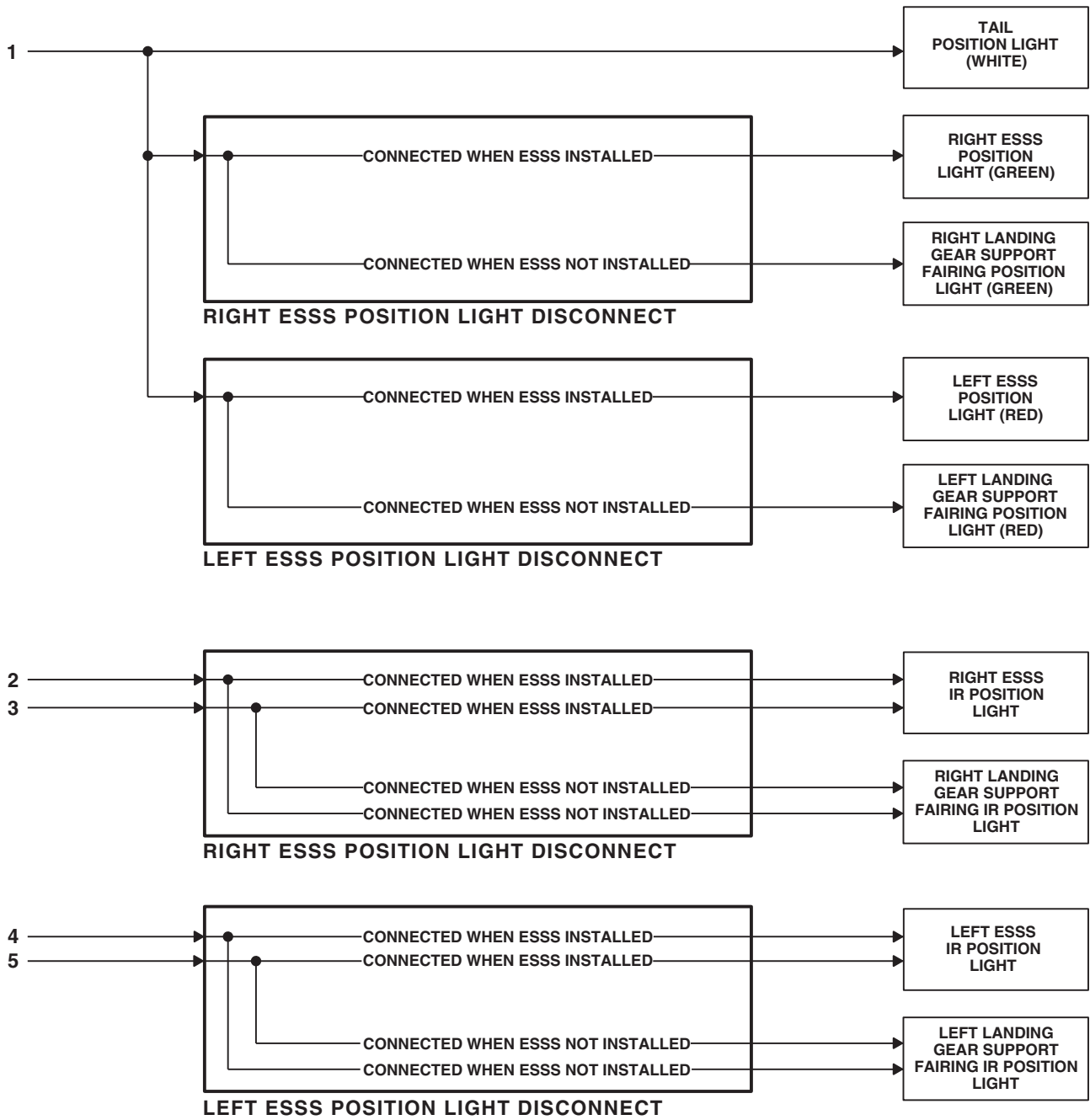
FIGURE 9-1-9-4. ANTICOLLISION LIGHTS BLOCK DIAGRAM



EFFECTIVITY
SLS

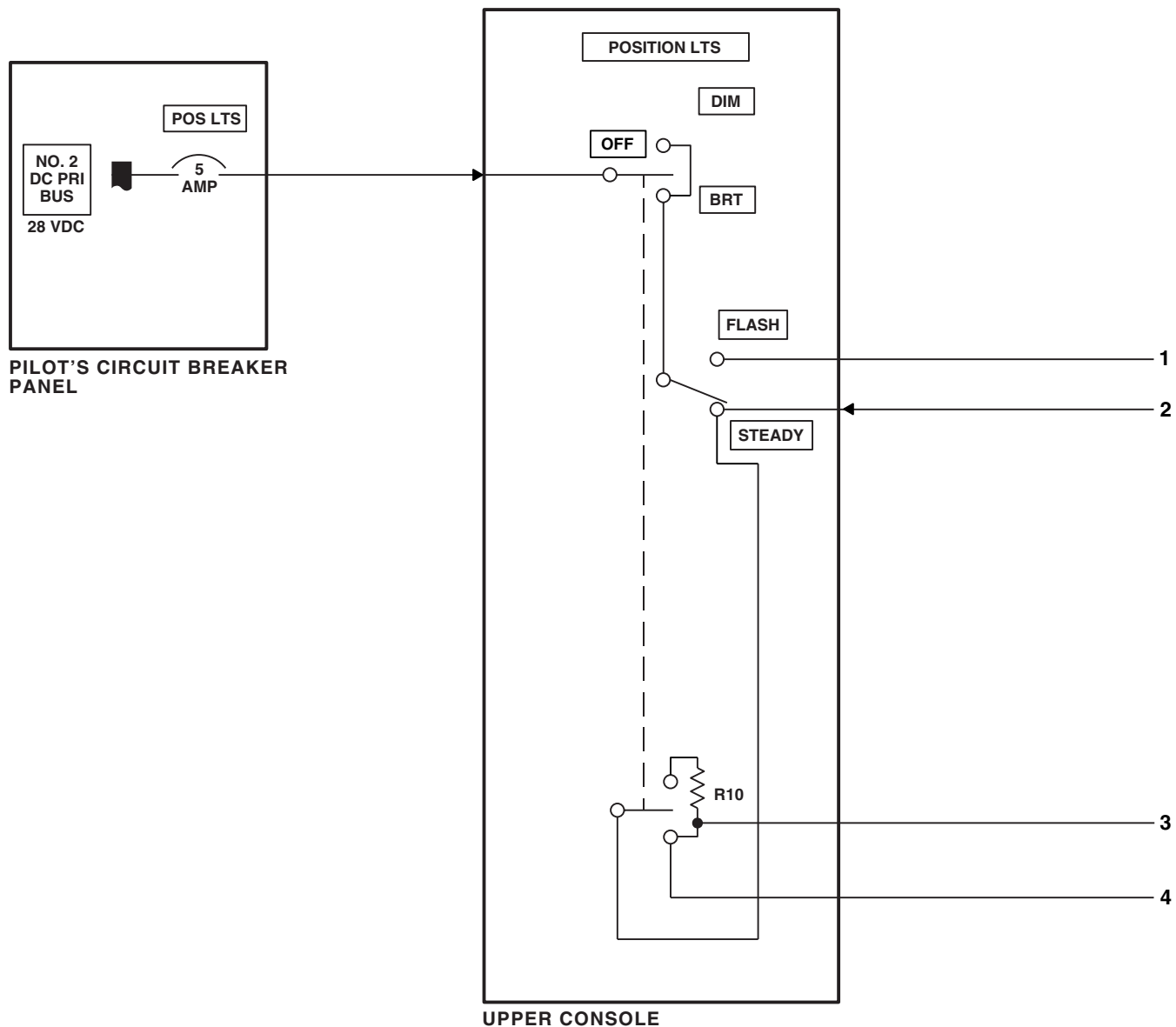
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FIGURE 9-1-9-5. POSITION LIGHTS BLOCK DIAGRAM (SHEET 1 OF 2)



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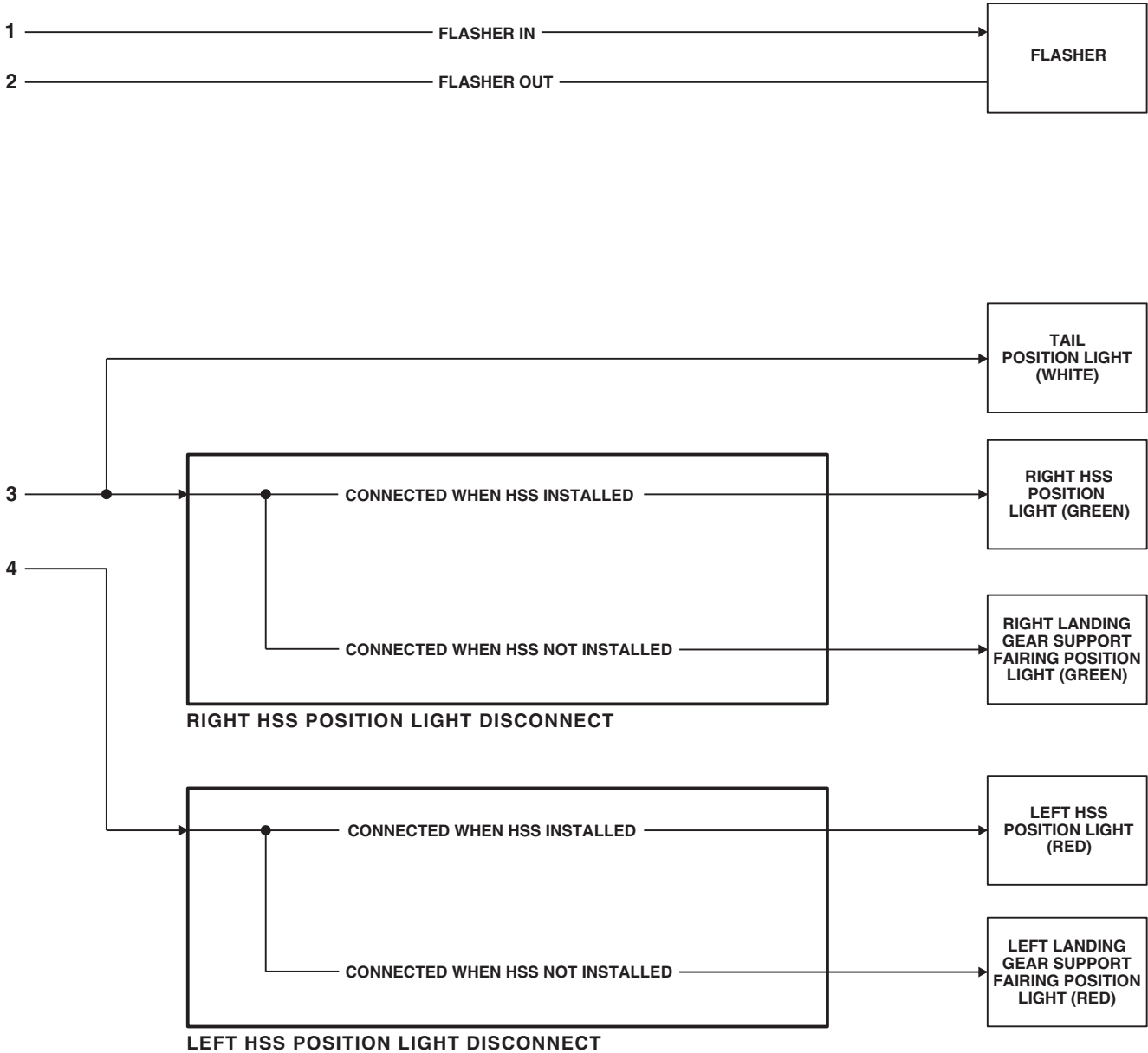
FIGURE 9-1-9-5. POSITION LIGHTS BLOCK DIAGRAM (SHEET 2 OF 2)



EFFECTIVITY
W / O SLS

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SA

FIGURE 9-1-9-6. POSITION LIGHTS BLOCK DIAGRAM (SHEET 1 OF 2)



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FIGURE 9-1-9-6. POSITION LIGHTS BLOCK DIAGRAM (SHEET 2 OF 2)

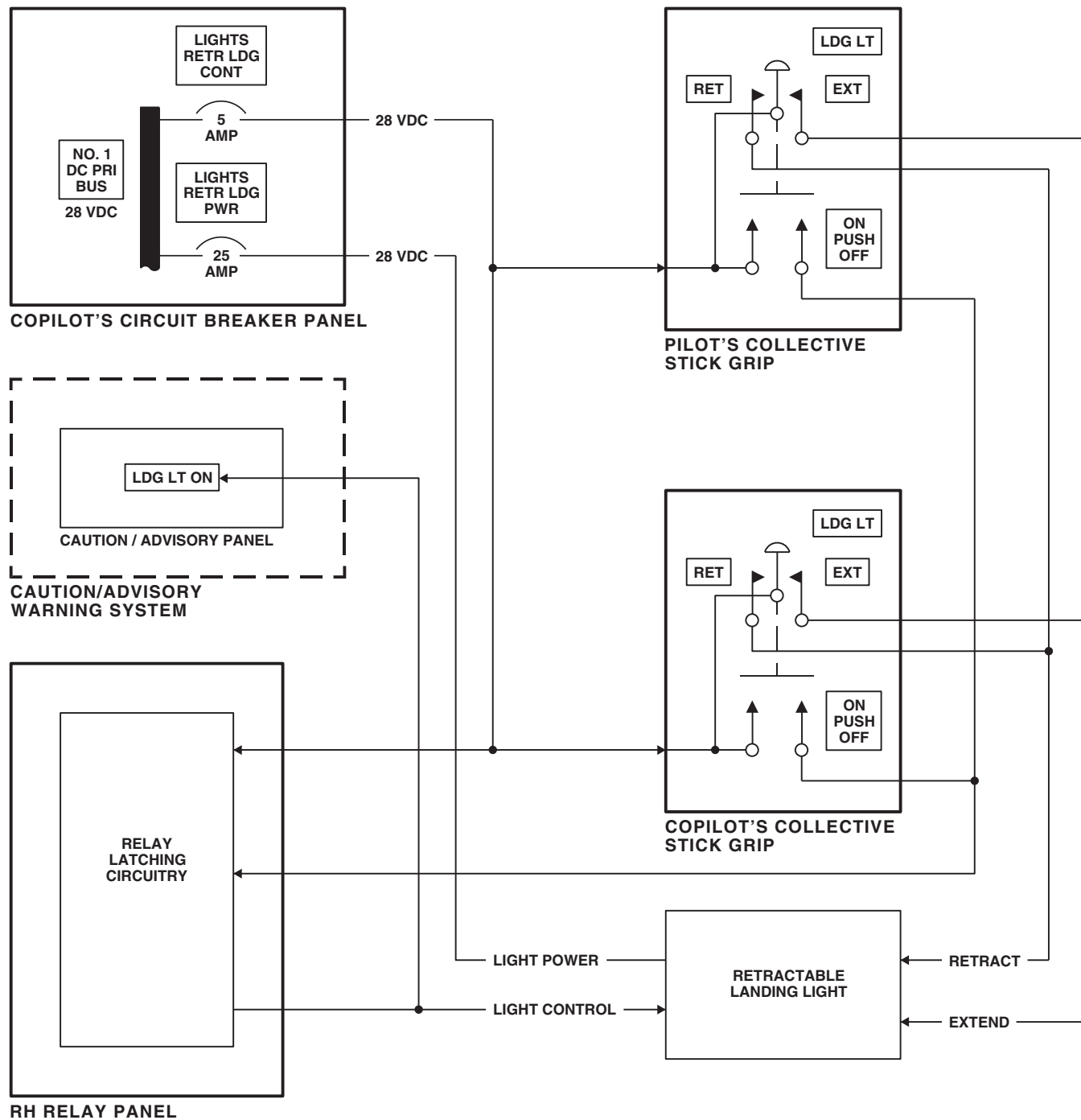
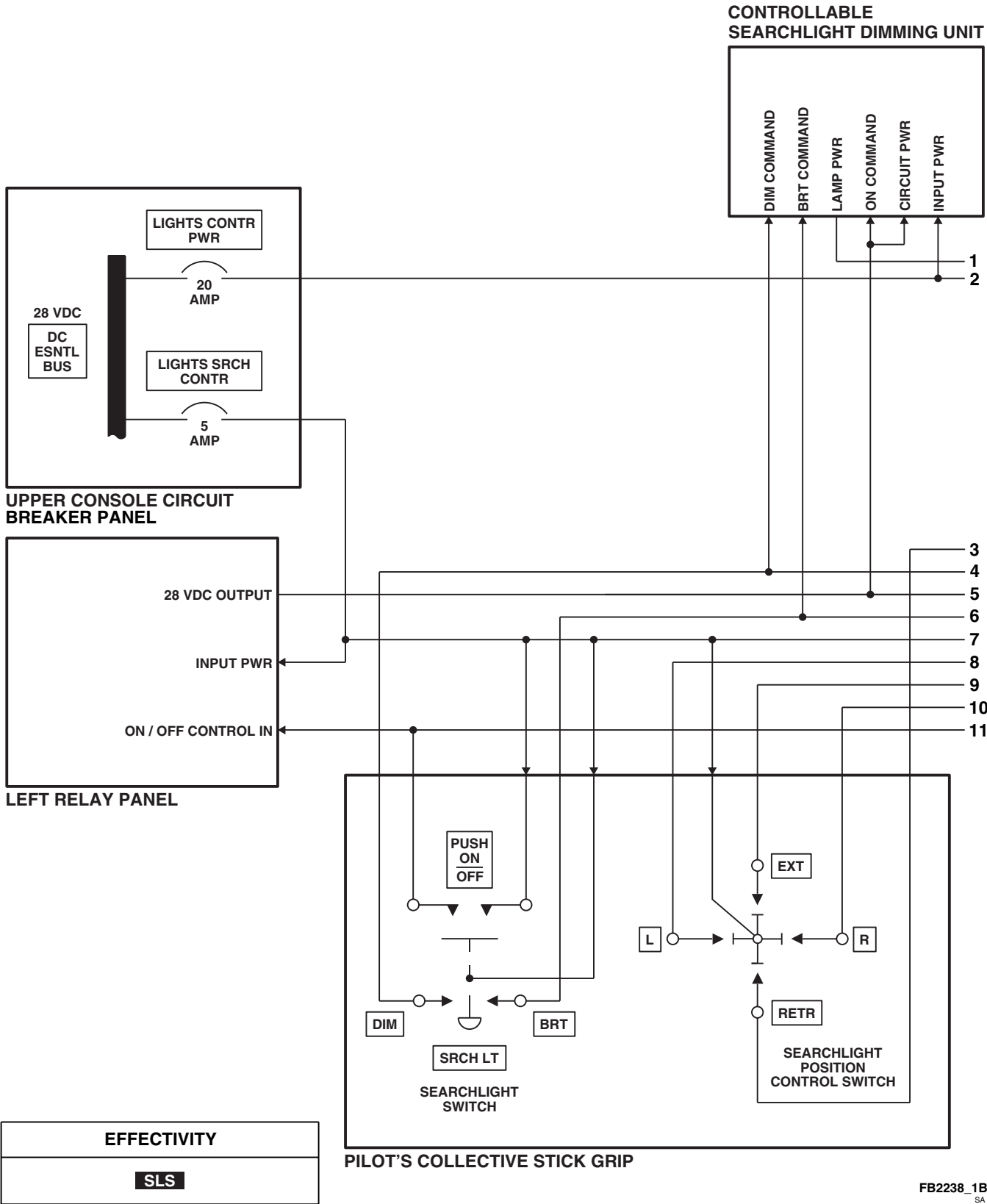
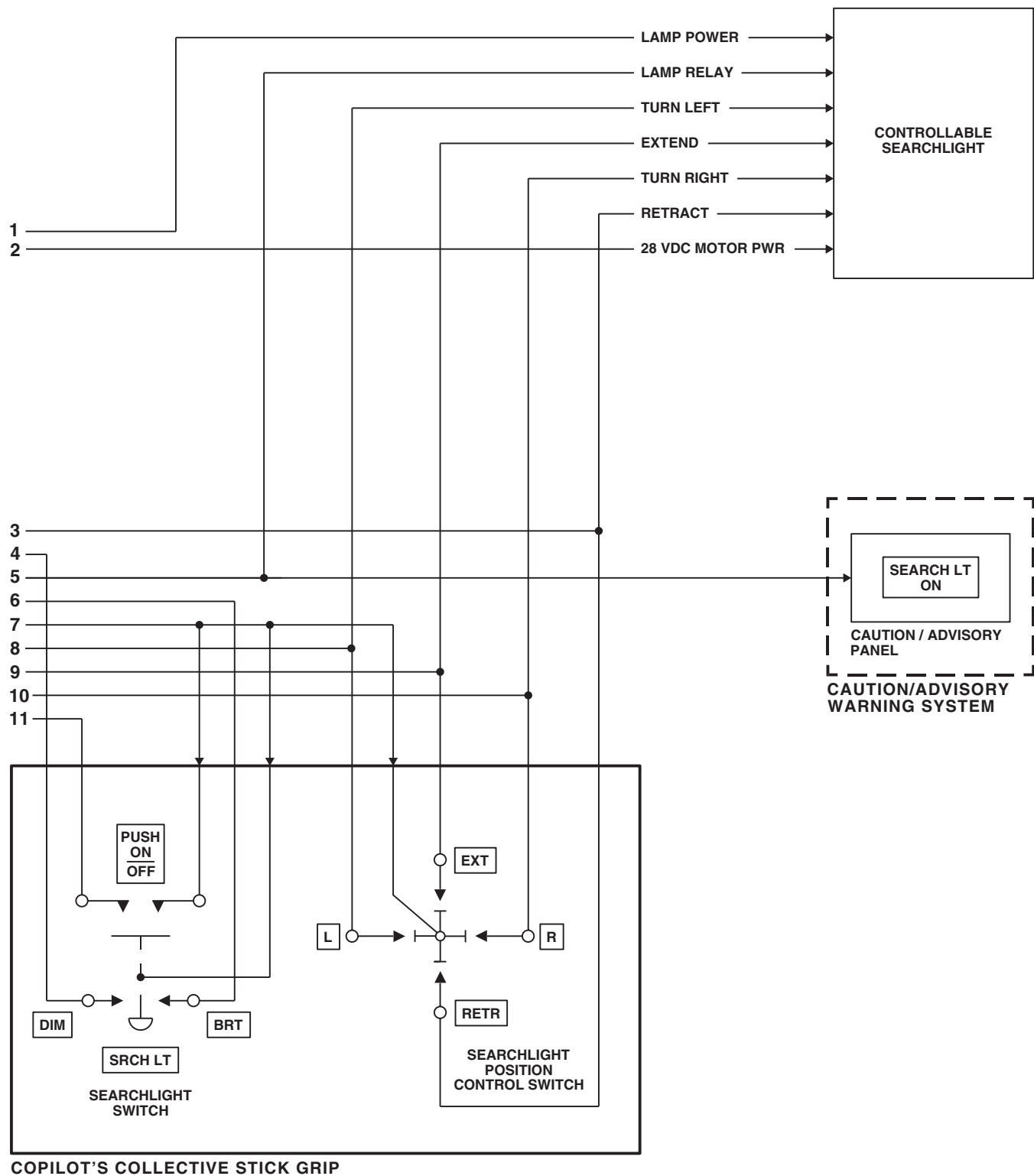
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FIGURE 9-1-9-7. RETRACTABLE LANDING LIGHT BLOCK DIAGRAM



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FIGURE 9-1-9-8. CONTROLLABLE SEARCHLIGHT BLOCK DIAGRAM (SHEET 1 OF 2)



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FIGURE 9-1-9-8. CONTROLLABLE SEARCHLIGHT BLOCK DIAGRAM (SHEET 2 OF 2)

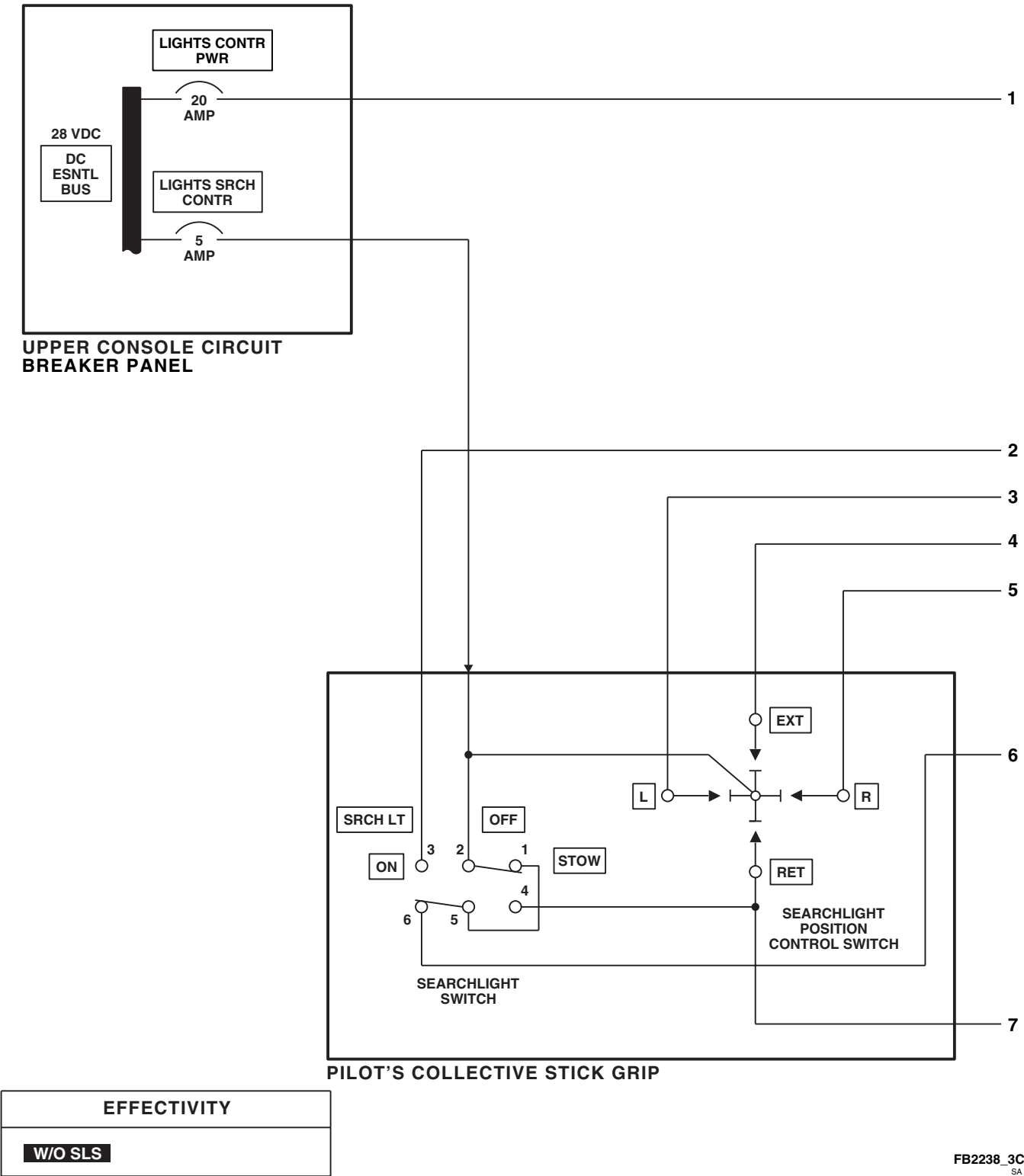
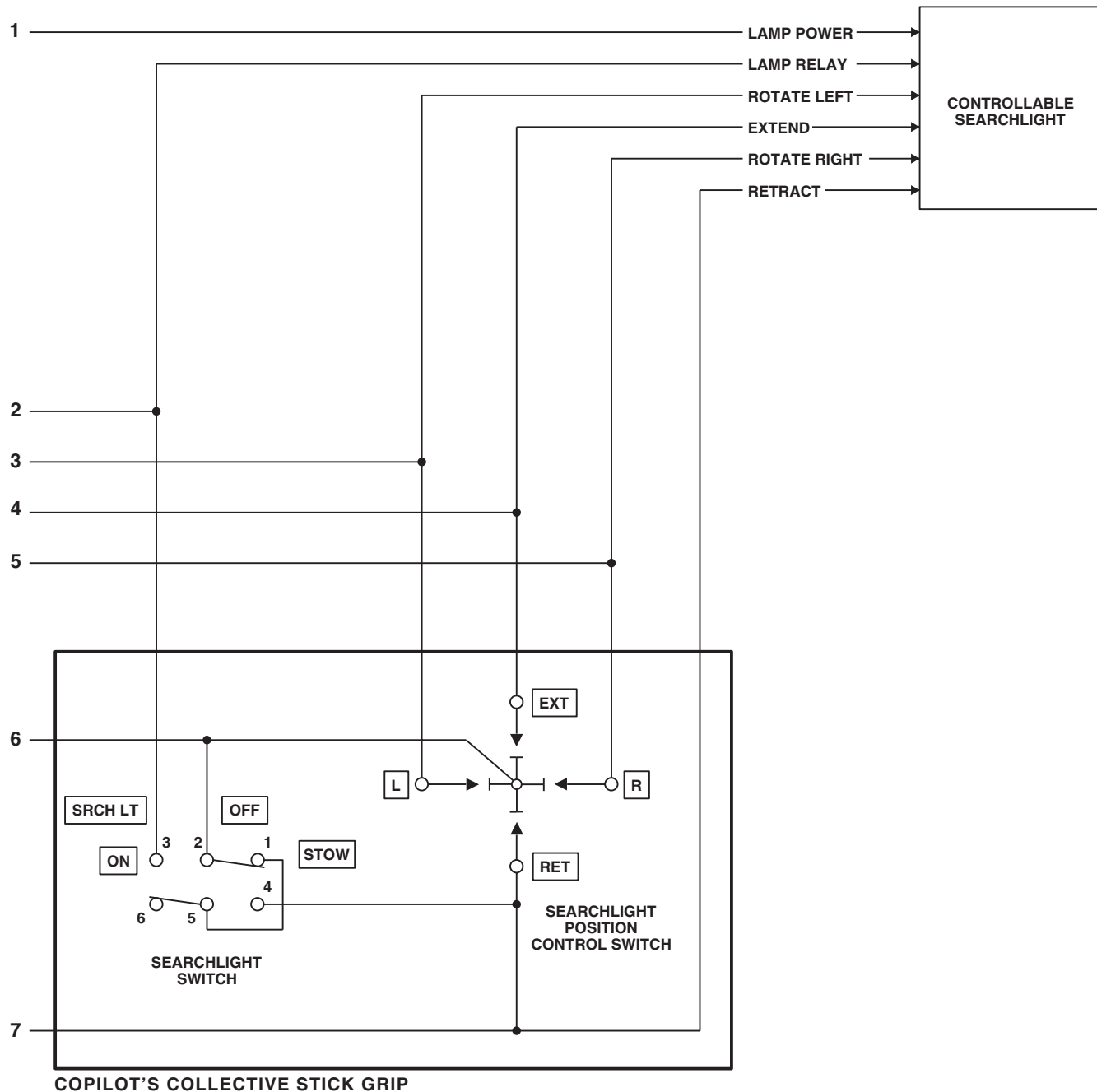
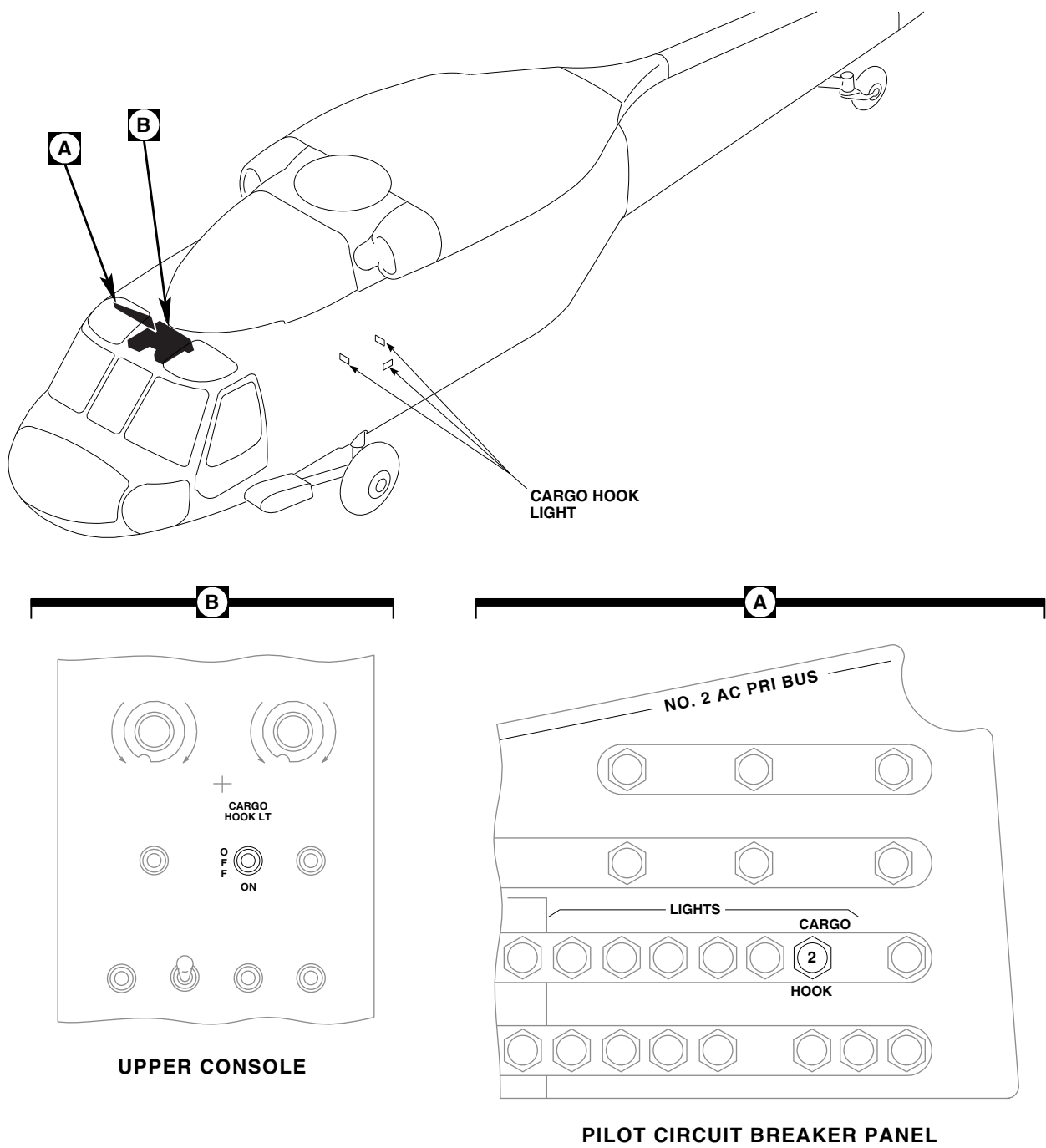


FIGURE 9-1-9-9. CONTROLLABLE SEARCHLIGHT BLOCK DIAGRAM (SHEET 1 OF 2)



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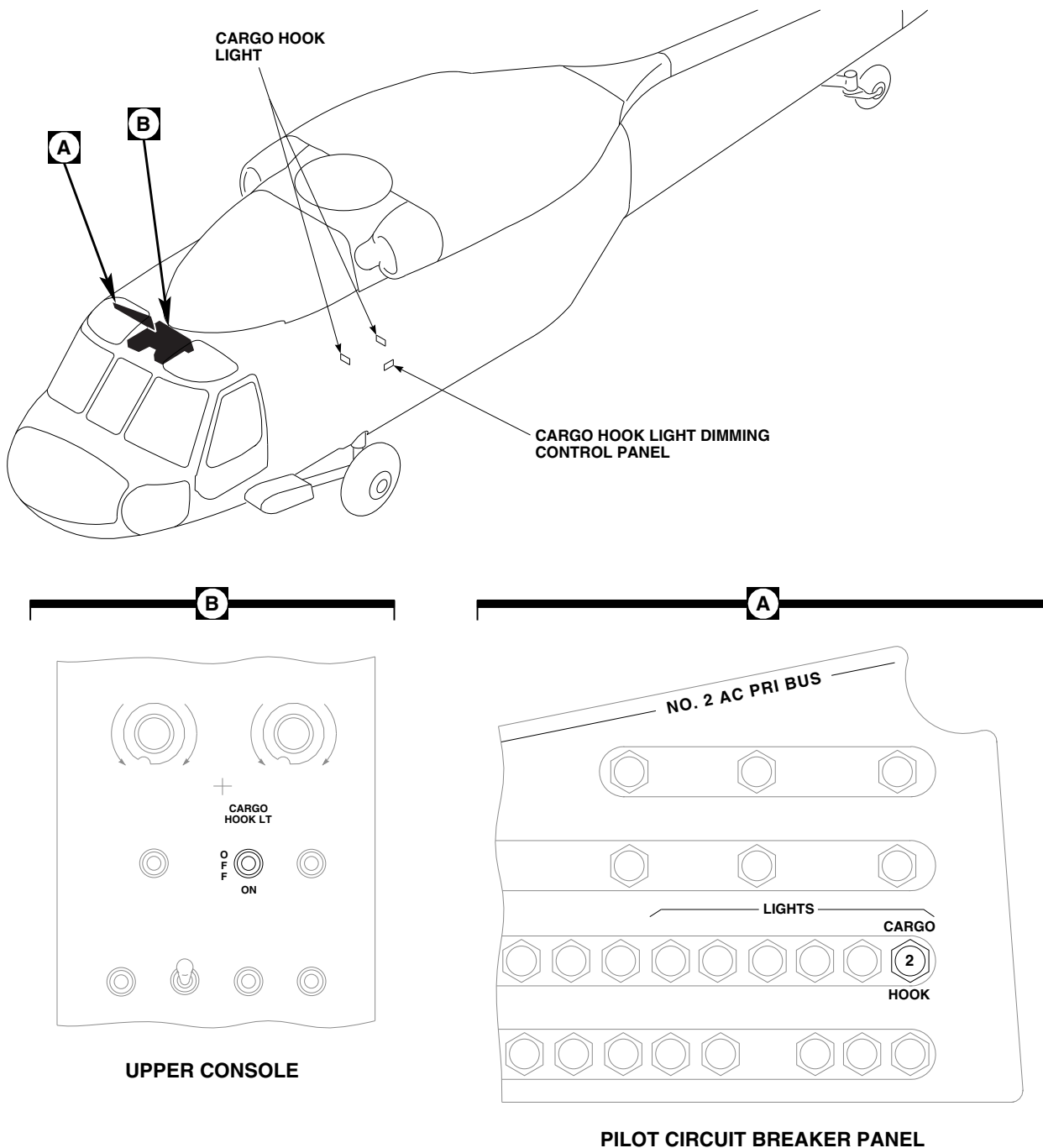
FIGURE 9-1-9-9. CONTROLLABLE SEARCHLIGHT BLOCK DIAGRAM (SHEET 2 OF 2)



EFFECTIVITY
W/O SLS

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SA

FIGURE 9-1-9-10. CARGO HOOK LIGHTS LOCATION DIAGRAM



EFFECTIVITY
SLS

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SA

FIGURE 9-1-9-11. CARGO HOOK LIGHTS LOCATION DIAGRAM

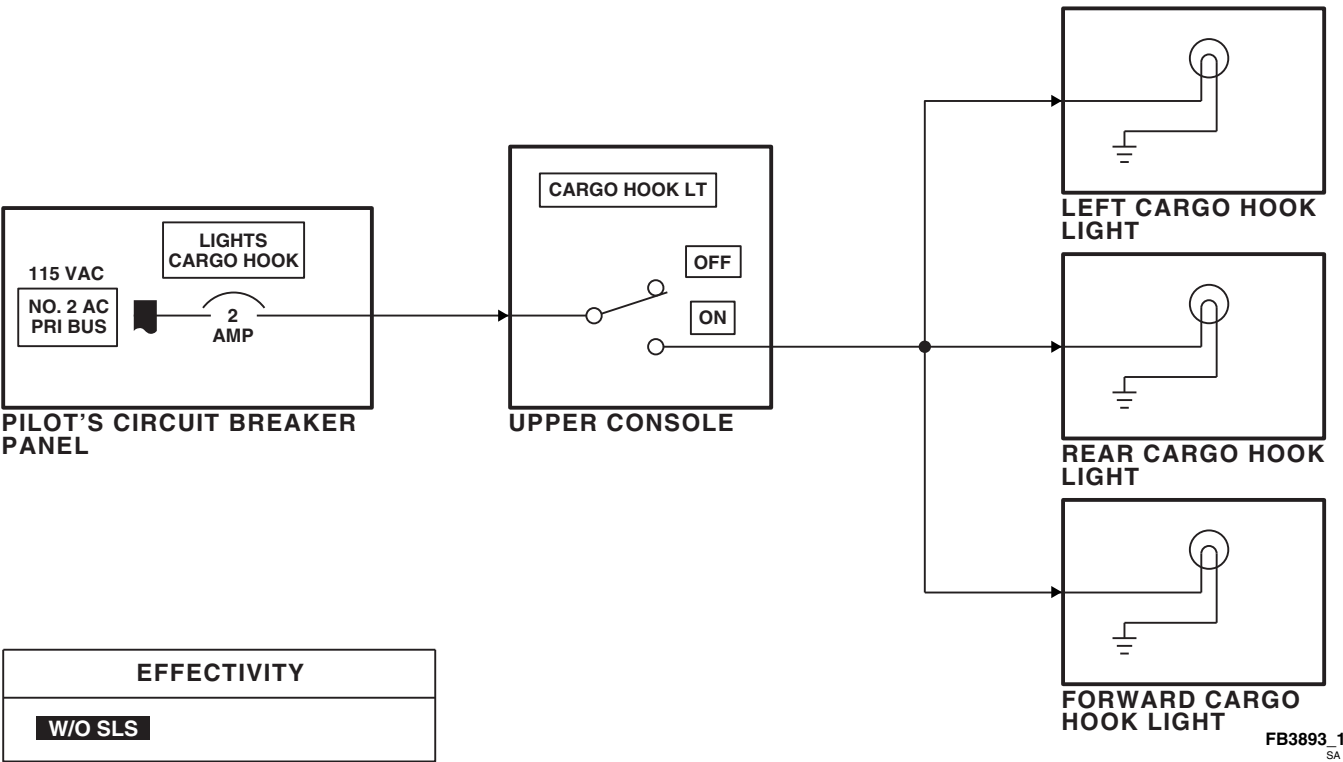


FIGURE 9-1-9-12. CARGO HOOK LIGHTS BLOCK DIAGRAM

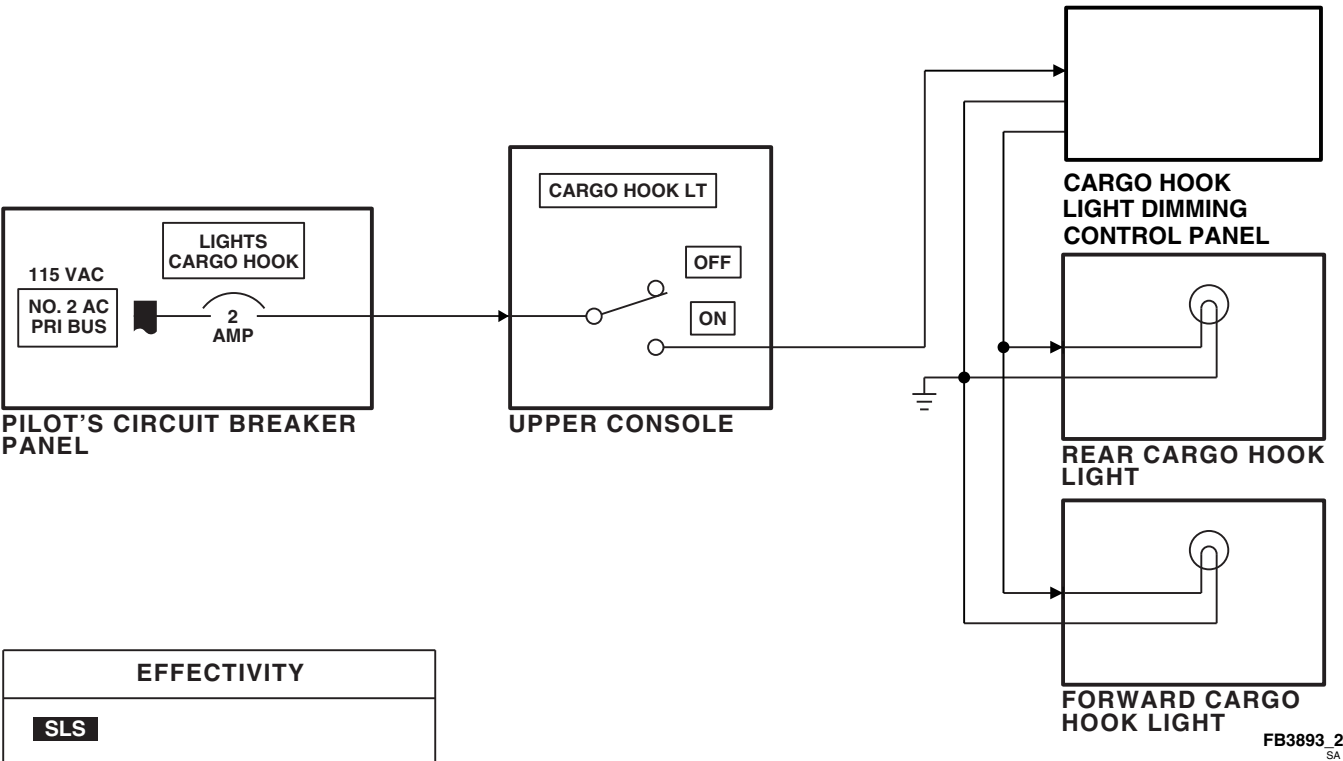
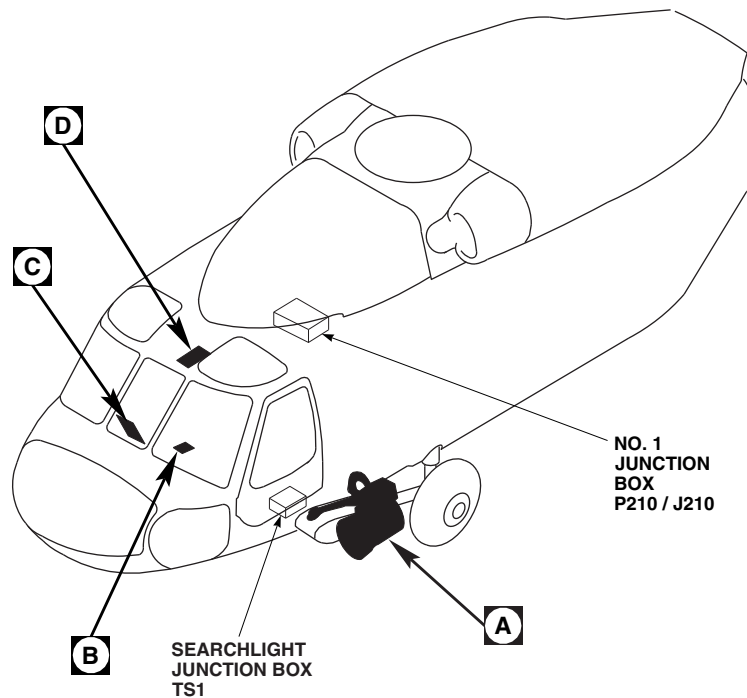


FIGURE 9-1-9-13. CARGO HOOK LIGHTS BLOCK DIAGRAM



TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
GND001	SEARCHLIGHT JUNCTION BOX
PWR1	SEARCHLIGHT JUNCTION BOX
PWR2	SEARCHLIGHT JUNCTION BOX
P118 / J118	CAUTION / ADVISORY PANEL
P142 / J142	LOWER CONSOLE DISCONNECT
P197 / J1	AUXILIARY SEARCHLIGHT CONTROL PANEL
P198 / J198	SEARCHLIGHT DISCONNECT
P210 / J210	NO. 1 JUNCTION BOX DISCONNECT
P240	DC BUS TIE CONTACTOR RELAY
P913 / J913	LOWER CONSOLE DISCONNECT
P914 / J914	LOWER CONSOLE DISCONNECT
TS1	SEARCHLIGHT JUNCTION BOX

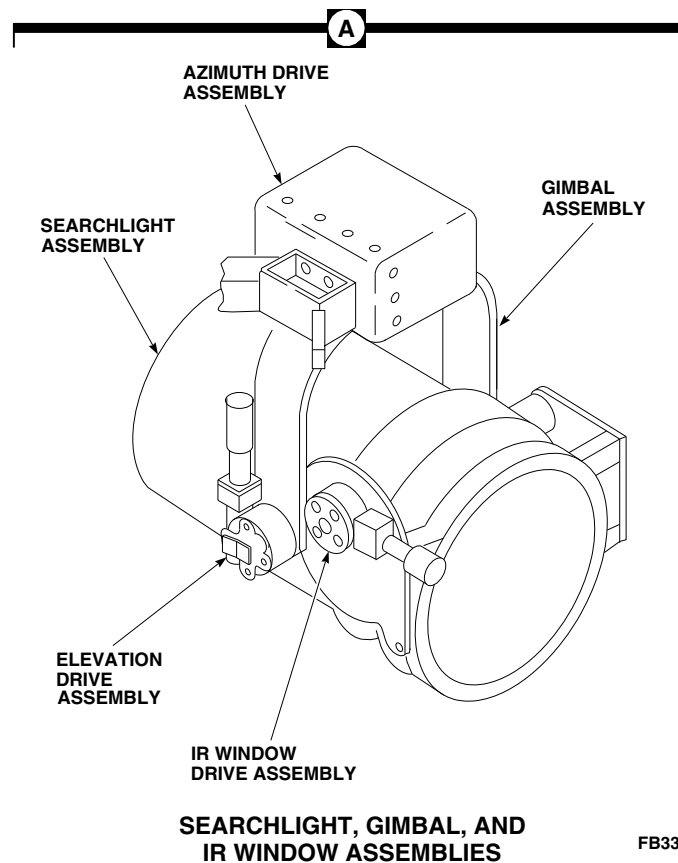
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SA

FIGURE 9-1-9-14. AUXILIARY CONTROLLABLE SEARCHLIGHT LOCATION DIAGRAM (SHEET 1 OF 2)

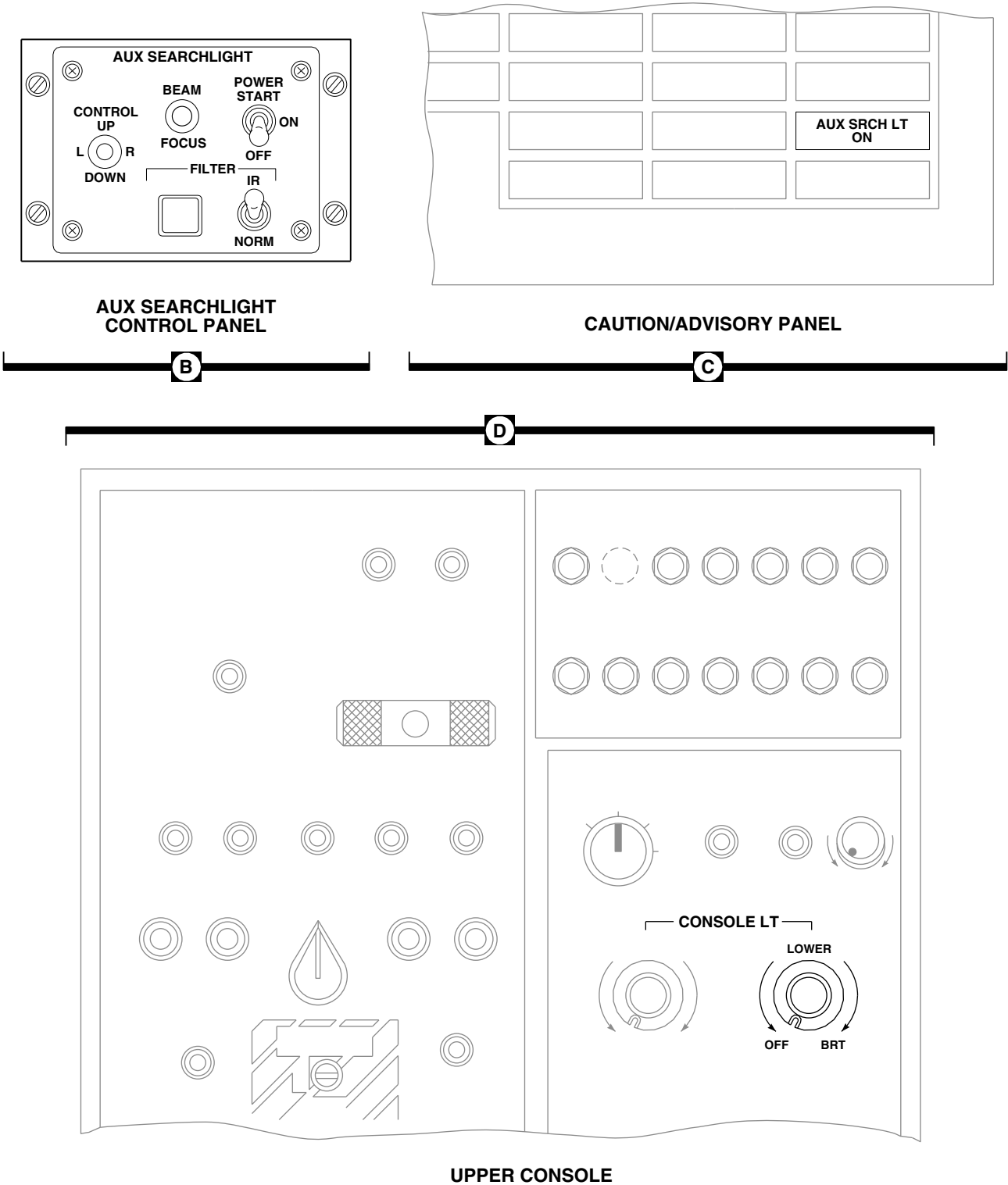


FIGURE 9-1-9-14. AUXILIARY CONTROLLABLE SEARCHLIGHT LOCATION DIAGRAM (SHEET 2 OF 2)

FT1162.2
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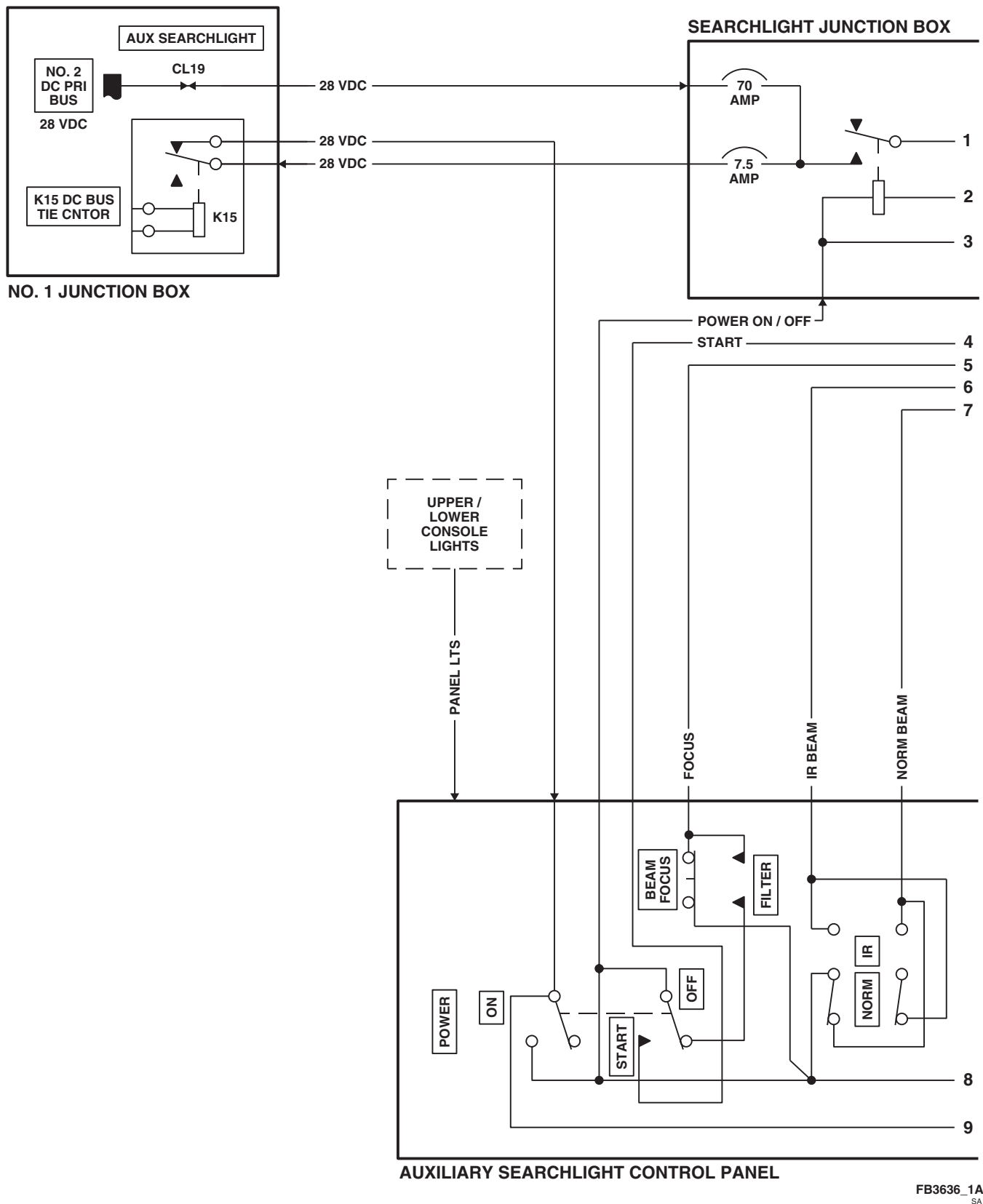


FIGURE 9-1-9-15. AUXILIARY CONTROLLABLE SEARCHLIGHT BLOCK DIAGRAM
(SHEET 1 OF 2)

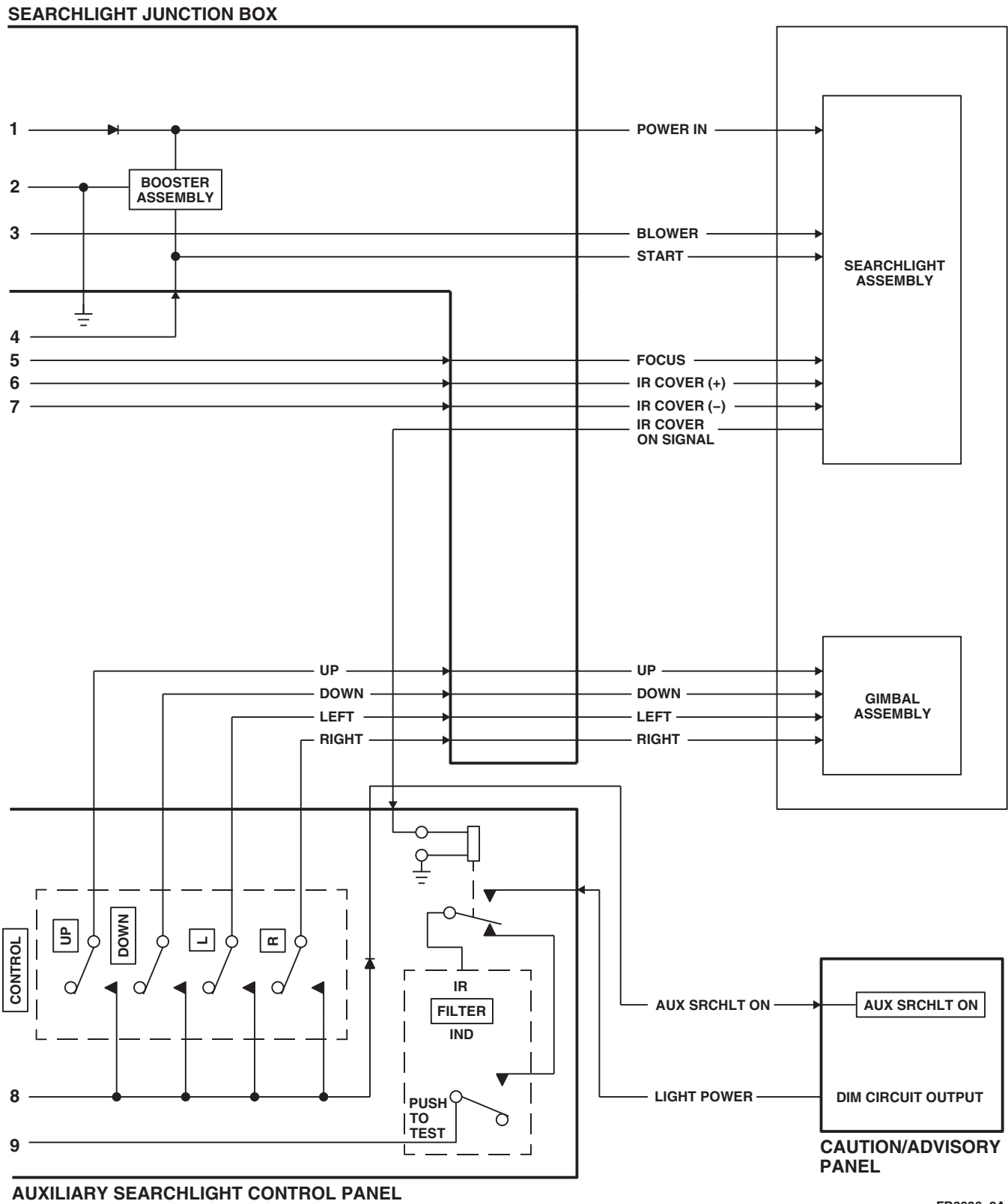


FIGURE 9-1-9-15. AUXILIARY CONTROLLABLE SEARCHLIGHT BLOCK DIAGRAM
(SHEET 2 OF 2)

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SECTION 9-2

TROUBLESHOOTING

SECTION OVERVIEW

PARAGRAPH	TITLE
9-2-1	AC Electrical System
9-2-2	DC Electrical System
9-2-3	System Circuit Breakers
9-2-4	Instrument Panel Lights
9-2-5	Upper and Lower Console Lights W/O SLS
9-2-6	Cabin Dome Lights
9-2-7	Cockpit Flood and Secondary Lights SLS
9-2-8	Cockpit Flood and Secondary Lights W/O SLS
9-2-9	Utility and Maintenance Lights
9-2-10	Instrument Panel and Consoles Indicator Lights Dimming SLS
9-2-11	Instrument Panel and Consoles Indicator Lights Dimming W/O SLS
9-2-12	Formation Lights
9-2-13	Controllable Searchlight
9-2-14	Position Lights
9-2-15	Retractable Landing Light
9-2-16	Anti-Collision Lights
9-2-17	Cargo Hook Lights W/O SLS
9-2-18	Fuel Boost Pump Control Panel (BENCH)
9-2-19	Right Relay Panel (BENCH)

PARAGRAPH	TITLE
9-2-20	Left Relay Panel (BENCH)
9-2-21	Auxiliary Controllable Searchlight
9-2-22	Nickel Cadmium Battery (BENCH)
9-2-23	Sealed Lead Acid Battery (BENCH)
9-2-24	Cargo Hook Lights SLS
9-2-25	Upper and Lower Console Lights SLS

9-2-1 AC ELECTRICAL SYSTEM.

INITIAL SETUP

Test Equipment

- Multimeter, Fluke 45
- Power System Analyzer, 60B63-5A

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer Toolkit
- Locally-Made Drag Beam Wedge, Figure H-167, Appendix H

References

- Appropriate Flight Manual
- PARA 1-6-16

Equipment Conditions

- External Electrical Power Available and APU Operational

CAUTION

In the following operational/troubleshooting procedures, to prevent overheating backup pump hydraulic system, use this chart for limits when running backup pump.

AIR TEMPERATURE F° (C°)	OPERATING TIME (MINUTES)	COOLDOWN TIME, (PUMP OFF) (MINUTES)
----------------------------	-----------------------------	---

32° (90°) max	Unlimited	-
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AIR TEMPERATURE F° (C°)	OPERATING TIME (MINUTES)	COOLDOWN TIME, (PUMP OFF) (MINUTES)
----------------------------	-----------------------------	---

91° - 100° (33° - 38°)	24	72
------------------------	----	----

101° - 125° (39° - 52°)	16	48
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CAUTION

Damage to the main and tail rotor blades will occur if PWR MAIN RTR or PWR TAIL RTR lamps go on for more than 10 seconds in this procedure. If lamps go on for more than 10 seconds, turn off electrical power and troubleshoot Blade De-Icing System (PARA 12-2-5).

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).

NOTE

When the upper console BATT switch is placed ON and no AC power is connected to the primary buses, the following capsules on the caution/advisory panel will go on from battery power:

AC ESS BUS OFF

#1 CONV

#2 CONV

Special Instructions

The AC Electrical System operational/troubleshooting procedure is subdivided as follows:

- | | |
|------------------------------|--|
| (1) APU Generator System | (5) 26 VAC Power System |
| (2) No. 1 Generator System | (6) UTIL RCPT 115 VAC 3ø 400 Hz Receptacle |
| (3) No. 2 Generator System | (7) 60 Hz Converter Receptacle |
| (4) AC External Power System | (8) Generator and GCU Troubleshooting |
-

9-2-1.1 APU Generator System.

- a. Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.

- b. Place upper console BATT switch to ON.

RESULT:

- (1) AC ESS BUS OFF capsule shall go on.
- (2) #1 CONV capsule shall go on.
- (3) #2 CONV capsule shall go on.

DC ESS BUS OFF capsule shall be off.

- If result (1) is not as specified, go to TABLE NO. 9-2-1-1.
- If result (2), (3), or (4) is not as specified, troubleshoot DC Electrical System (PARA 9-2-2).

- c. Make sure upper console GENERATORS NO. 1, GENERATORS NO. 2, and GENERATORS APU switches are in OFF/RESET.

- d. Start APU (Appropriate Flight Manual).

- e. Place upper console GENERATORS APU switch to ON.

RESULT:

- (1) APU GEN ON capsule shall go on.
- (2) #1 CONV capsule shall go off.

- (3) #2 CONV capsule shall go off.

- (4) AC ESS BUS OFF capsule shall be off.

- (5) #1 GEN and #2 GEN capsules shall go on.

- If results (1), (2), and (3) are not as specified, go to TABLE NO. 9-2-1-2.
- If result (1) is not as specified, go to TABLE NO. 9-2-1-3.
- If result (2) is not as specified, go to TABLE NO. 9-2-1-4.
- If result (3) is not as specified, check wiring from NO. 2 CONVERTER circuit breaker terminals A2, B2, and C2, to NO. 2 GEN CNTOR contactor terminals A2, B2, and C2.
 - (a) If continuity is present, replace NO. 2 GEN CNTOR (PARA 9-4-12).
 - (b) If continuity is not present, repair/replace wiring as required (Appendix F).

- If result (4) is not as specified, go to TABLE NO. 9-2-1-5.

- If result (5) is not as specified, perform No. 1 Generator System or No. 2 Generator System check, as required.

- f. Place upper console GENERATORS APU switch to OFF/RESET.

- g. Shut down APU (Appropriate Flight Manual).

- h. Place upper console BATT switch to OFF.

9-2-1-2 Change 7

9-2-1.2 No. 1 Generator System.

- a. Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.
- b. Place upper console BATT switch to ON.

RESULT: #1 GEN BRG capsule on caution/
advisory panel shall not go on.

- If result is not as specified, go to TABLE NO. 9-2-1-6.
- c. Make sure upper console GENERATORS NO. 1, GENERATORS NO. 2, and GENERATORS APU switches are placed to OFF/RESET.
- d. Insert locally-made drag beam wedge in right drag beam switch to simulate flight.
- e. Start APU (Appropriate Flight Manual).
- f. Place upper console GENERATORS APU switch to ON.

RESULT:

- (1) #1 GEN capsule shall go on.
- (2) #2 GEN capsule shall go on.
- If result (1) is not as specified, go to TABLE NO. 9-2-1-7.
- If result (2) is not as specified, perform No. 2 Generator System check.
- g. With qualified personnel at the controls, start No. 1 engine and operate with main rotor turning at 100% (Appropriate Flight Manual).
- h. Place upper console GENERATORS NO. 1 switch to ON.

RESULT:

- (1) #1 GEN capsule shall go off.
- (2) #2 GEN capsule shall remain on.

(3) #1 CONV capsule shall go off.

(4) #2 CONV capsule shall go off.

- If result (1) is not as specified, go to TABLE NO. 9-2-1-8.
- If result (2) is not as specified, perform No. 2 Generator System check.
- If result (3) is not as specified, go to TABLE NO. 9-2-1-9.
- If result (4) is not as specified, go to TABLE NO. 9-2-1-10.
- If No. 1 generator is disabled due to under-frequency, go to TABLE NO. 9-2-1-11.

- i. Place upper console GENERATORS NO. 1 switch to OFF/RESET.
- j. Remove locally-made drag beam wedge.
- k. Shut down No. 1 engine and APU (Appropriate Flight Manual).
- l. Place upper console BATT switch to OFF.

9-2-1.3 No. 2 Generator System.

- a. Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.
- b. Place upper console BATT switch to ON.

RESULT: #2 GEN BRG capsule on caution/
advisory panel shall not go on.

- If result is not as specified, go to TABLE NO. 9-2-1-6.
- c. Make sure upper console GENERATORS NO. 1, GENERATORS NO. 2, and GENERATORS APU switches are placed to OFF/RESET.
- d. Insert locally-made drag beam wedge in right drag beam switch to simulate flight.

- e. Start APU (Appropriate Flight Manual).
- f. Place upper console GENERATORS APU switch to ON.

RESULT:

- (1) #1 GEN capsule shall go on.
- (2) #2 GEN capsule shall go on.

- If result (1) is not as specified, perform No. 1 Generator System check.
- If result (2) is not as specified, go to TABLE NO. 9-2-1-12.

- g. With qualified personnel at the controls, start No. 2 engine and operate with main rotor turning at 100% (Appropriate Flight Manual).
- h. Place upper console GENERATORS NO. 2 switch to ON.

RESULT:

- (1) #2 GEN capsule shall go off.
- (2) #1 GEN capsule shall remain on.
- (3) #1 CONV and #2 CONV capsules shall go off.

- If result (1) is not as specified, go to TABLE NO. 9-2-1-13.
- If result (2) is not as specified, perform No. 1 Generator System check.
- If result (3) is not as specified, go to TABLE NO. 9-2-1-9 or TABLE NO. 9-2-1-10 as required.
- If No. 2 generator is disabled due to under-frequency, go to TABLE NO. 9-2-1-11.

- i. Place upper console GENERATORS NO. 2 switch to OFF/RESET.
- j. Remove locally-made drag beam wedge.

- k. Shut down No. 2 engine and APU (Appropriate Flight Manual).
- l. Place upper console BATT switch to OFF.

9-2-1.4 AC External Power System.

- a. Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.
- b. Place upper console BATT switch to ON.
- c. Make sure upper console GENERATORS NO. 1, GENERATORS NO. 2, and GENERATORS APU switches are in OFF/RESET.
- d. Turn on external electrical power (PARA 1-6-16).

RESULT:

- (1) EXT PWR CONNECTED capsule shall go on.
- (2) #1 CONV and #2 CONV capsules shall remain on.
- (3) AC ESS BUS OFF capsule shall remain on.

- If result (1) is not as specified, go to TABLE NO. 9-2-1-14.
- If result (2) is not as specified, troubleshoot DC Electrical System (PARA 9-2-2).
- If result (3) is not as specified, go to TABLE NO. 9-2-1-1.

- e. Momentarily place upper console EXT PWR switch to RESET and then to ON.

RESULT:

- (1) #1 CONV capsule shall go off.
- (2) #2 CONV capsule shall go off.
- (3) #1 GEN and #2 GEN capsules shall go on.

- If results (1) and (2) are not as specified, go to TABLE NO. 9-2-1-15.
- If result (1) is not as specified, go to TABLE NO. 9-2-1-4.
- If result (3) is not as specified, perform No. 1 Generator System or No. 2 Generator System check, as required.

f. Turn off external electrical power (PARA 1-6-16).

g. Place upper console BATT switch to OFF.

9-2-1.5 26 VAC Power System.

- a. Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.
- b. Make sure upper console GENERATORS NO. 1, GENERATORS NO. 2, and GENERATORS APU switches are set to OFF/RESET.
- c. Place upper console BATT switch to ON.
- d. Turn on electrical power (PARA 1-6-16).
- e. Check for 26 VAC between terminal 2 of STAB IND, INST, and DPLR circuit breakers (pilot's circuit breaker panel) and ground.

RESULT: Multimeter shall indicate 26 VAC.

- If result is not as specified, go to TABLE NO. 9-2-1-16.

f. Turn off electrical power (PARA 1-6-16).

g. Place upper console BATT switch to OFF.

9-2-1.6 UTIL RCPT 115 VAC 3Ø 400 Hz Receptacle.

- a. Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.
- b. Place upper console BATT switch to ON.

c. Turn on electrical power (PARA 1-6-16).

d. Check for 115 VAC between the following:

J257-A	and	ground
J257-B	and	ground
J257-C	and	ground

RESULT: Multimeter shall indicate 115 VAC.

- If result is not as specified, check for 115 VAC between terminals A2, B2, and C2 of UTIL RECP circuit breaker and ground.

(a) If voltage is as specified, repair/replace wiring between circuit breaker and J257 as required (Appendix F).

(b) If voltage is not as specified, replace circuit breaker (PARA 9-4-1).

e. Turn off electrical power (PARA 1-6-16).

f. Place upper console BATT switch to OFF.

9-2-1.7 60 Hz AC Converter Receptacle.

- a. Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.
- b. Place upper console BATT switch to ON.
- c. Turn on electrical power (PARA 1-6-16).
- d. Check for 115 VAC between the following:

J276-A	and	ground
J276-B	and	ground
J276-C	and	ground

RESULT: Multimeter shall indicate 115 VAC.

- If result is not as specified, check for 115 VAC between terminals A1, B1, and C1 of 60 HZ AC CONVERTER circuit breaker and ground.

- (a) If voltage is as specified, repair/replace wiring between circuit breaker and P276 as required (Appendix F).
- (b) If voltage is not as specified, replace circuit breaker (PARA 9-4-1).
- e. Turn off electrical power (PARA 1-6-16).
- f. Place upper console BATT switch to OFF.

- f. Disconnect P214, P215, or P216 from GCU and connect to P3 of analyzer adapter cable, as required.
- g. Connect P2 of analyzer adapter cable to GCU.
- h. Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.
- i. Start APU (Appropriate Flight Manual).
- j. With qualified personnel at controls, start No. 1 engine and operate with main rotor turning at 100% rpm (for No. 1 or No. 2 main generator) (Appropriate Flight Manual).

9-2-1.8 Generator and GCU Troubleshooting.

CAUTION

If generator control unit (GCU) is removed from its mounting at any time during system troubleshooting, the GCU case must be grounded to airframe ground.

- a. Plug ground cable (supplied with analyzer) into ground test jack on analyzer, and attach alligator clip to ground.
- b. Place upper console GENERATORS NO. 1, GENERATORS NO. 2, or GENERATORS APU switch to OFF/RESET, as required.
- c. Place power system analyzer GEN, CUR LIM, and FF toggle switches to OFF, and all other toggle switches down.
- d. Turn power system analyzer TEST SELECT switch to START TEST.
- e. Connect P1 of adapter cable (supplied with analyzer) to J1 AIRCRAFT connector on analyzer.

CAUTION

Always shut down APU or No. 1 engine (as applicable) before disconnecting GCU.

CAUTION

All tests with the system analyzer must be done with the GENERATORS NO. 1, GENERATORS NO. 2, or GENERATORS APU switch set to OFF/RESET to prevent applying a simulated fault to any helicopter load.

- k. Turn power system analyzer GEN/PMG VOLTS meter switch to A-B, B-C, and C-A (PMG 30V RANGE) and check PMG voltage on meter for each switch position.

RESULT: Each voltage indication shall be more than 20.5 volts.

- If result is not as specified, replace generator (PARA 9-4-16 or PARA 9-4-19).
- l. On power system analyzer, hold TEST SET switch at RESET and GEN switch to ON; then release TEST SET switch.
- m. Turn power system analyzer GEN/PMG VOLTS meter switch to T-1, T-2, and T-3 (GEN 140V RANGE) and check generator output voltage on meter for each switch position.

RESULT: Each voltage indication shall be between 112.5 and 117.5 VAC.

- If result is not as specified, replace generator (PARA 9-4-16 or PARA 9-4-19).

n. Check field current on power system analyzer FIELD AMPS/REG VOLTS meter.

RESULT: Field current shall be 1.0 ampere maximum.

- If result is not as specified, replace generator (PARA 9-4-16 or PARA 9-4-19).

o. Check power system analyzer FREQUENCY meter indication.

RESULT: FREQUENCY meter shall indicate between 380 and 420 Hz and ϕ SEQ T1-2-3 lamp shall be on.

- If result is not as specified, replace generator (PARA 9-4-16 or PARA 9-4-19).

p. Check power system analyzer GCR RESET, CCR RESET, CCR TRIP, SIM CONT, and TEST RELAY lamps.

RESULT: GCR RESET, CCR RESET, SIM CONT, and TEST RELAY lamps shall be on and CCR TRIP lamp shall be off.

- If result is not as specified, replace GCU (PARA 9-4-15).

q. Turn power system analyzer TEST SELECT switch to NL REG.

r. Place power system analyzer FIELD AMPS/REG VOLTS meter switch to READ REG BUS VOLTS and check voltage indication on meter.

RESULT: Voltage indication shall be between 21 and 27 volts.

- If result is not as specified, replace GCU (PARA 9-4-15).

s. Place power system analyzer FIELD AMPS/REG VOLTS meter switch to FLD CUR and GEN switch to OFF. Check GCR RESET, CCR RESET, CCR TRIP, SIM CONT, SIM WARN, and TEST RELAY lamps.

RESULT: CCR TRIP, SIM WARN, and TEST RELAY lamps shall be on and GCR RESET, CCR RESET, and SIM CONT lamps shall be off.

- If result is not as specified, replace GCU (PARA 9-4-15).

t. Hold power system analyzer TEST SET switch to RESET and hold GEN switch to TEST; then only release TEST SET switch.

u. Check power system analyzer GCR RESET, CCR RESET, CCR TRIP, SIM CONT, SIM WARN, and TEST RELAY lamps.

RESULT: GCR RESET and CCR RESET lamps shall be on and CCR TRIP, SIM CONT, SIM WARN, and TEST RELAY lamps shall be off.

- If result is not as specified, replace GCU (PARA 9-4-15).

v. Hold power system analyzer TEST SET switch to RESET and only release GEN switch to OFF, then place to ON; then only release TEST SET switch.

w. Turn power system analyzer GEN/PMG VOLTS meter switch to T-1, T-2, and T-3 (GEN 140V RANGE). **RECORD** generator output voltage on meter for each switch position.

RESULT: Each voltage indication shall be between 112.5 and 117.5 VAC.

- If result is not as specified, replace GCU (PARA 9-4-15).

x. Momentarily place power system analyzer FLYWHEEL DIODE switch to TEST.

RESULT: FLYWHEEL DIODE OPEN lamp shall be off.

- If result is not as specified, replace GCU (PARA 9-4-15).

y. Turn power system analyzer TEST SELECT switch to FL REG.

z. Turn power system analyzer GEN/PMG VOLTS meter switch to A-B, B-C, and C-A (PMG 30V RANGE) and check PMG voltage on meter for each switch position.

RESULT: Each voltage indication shall be at least 19 volts.

- If result is not as specified, replace generator (PARA 9-4-16 or PARA 9-4-19).

aa. Turn power system analyzer GEN/PMG VOLTS meter switch to T-1, T-2, and T-3 (GEN 140V RANGE); and check generator output voltage on meter for each switch position.

RESULT: Each voltage indication shall not change more than 2.5 VAC from indication **RECORDED** in step w.

- If result is not as specified, replace GCU (PARA 9-4-15).

ab. Turn power system analyzer GEN/PMG VOLTS meter switch to T-1 (GEN 140V RANGE).

ac. Place power system analyzer CUR LIM switch to 20 KVA.

NOTE

Voltage checks in step ad. must be done within 4 seconds of selection of each switch position to prevent an undervoltage trip. If an undervoltage trip occurs,

turn TEST SELECT switch to FL REG and place GEN switch OFF. Then with TEST SET switch held at RESET, place GEN switch ON and repeat voltage checks.

ad. Turn power system analyzer TEST SELECT switch to A $\emptyset$ CUR LIM, B $\emptyset$ CUR LIM, and C $\emptyset$ CUR LIM; and check GEN/PMG VOLTS meter T-1 indication for each switch position.

RESULT: Each voltage indication shall be between 90 and 110 volts.

- If result is not as specified, replace GCU (PARA 9-4-15).

ae. Place power system analyzer CUR LIM switch to 30 KVA and repeat step ad.

af. Place power system analyzer CUR LIM switch to OFF.

ag. Turn power system analyzer TEST SELECT switch to A $\emptyset$ GEN CT, press CT CUR switch to READ, and check CT TEST CURRENT meter indication.

NOTE

It is normal for the 60B63-5A Power System Analyzer to indicate a fault when it is switched to test current transformers and contactors.

RESULT: Current indication shall be between 0.70 and 1.70 mA.

- If result is not as specified, replace generator (PARA 9-4-16 or PARA 9-4-19).

ah. Repeat step ag. with power system analyzer TEST SELECT switch in B $\emptyset$ GEN CT and C $\emptyset$ GEN CT positions.

NOTE

Steps ai. through ap. are to be performed on helicopters with Blade De-Ice system installed. Otherwise continue to step aq.

- ai. Disconnect P333 from auxiliary AC junction box.
- aj. Turn power system analyzer TEST SELECT switch to A $\emptyset$ DP CT, press CT CUR switch to READ, and check CT TEST CURRENT meter indication.

RESULT: Current indication shall be between 0.6 and 2.0 mA.

- If result is not as specified, replace current transformer (PARA 9-4-14).
- ak. Repeat step aj. with power system analyzer TEST SELECT switch at B $\emptyset$ DP CT and C $\emptyset$ DP CT.
- al. Place power system analyzer GEN switch OFF and turn TEST SELECT switch to START TEST.
- am. Check continuity between the following:

J333-C	and	J333-D
J333-E	and	J333-F
J333-G	and	J333-H

RESULT: Continuity shall be present.

- If result is not as specified, replace auxiliary AC junction box (PARA 12-4-33).
- an. Disconnect P202 from No. 2 junction box.
- ao. Check continuity between the following:

P333-C	and	P333-D
P333-E	and	P333-F
P333-G	and	P333-H

RESULT: Continuity shall be present.

- If result is not as specified, replace T3 APU GEN CT (PARA 9-4-14).
- ap. Connect P202 to No. 2 junction box and P333 to auxiliary AC junction box.
- aq. Shut down APU and/or No. 1 engine, as applicable (Appropriate Flight Manual).
- ar. Disconnect P214, P215, or P216 (as applicable) from P3 of analyzer adapter cable and connect it to GCU.
- as. Disconnect adapter cable from analyzer.
- at. Disconnect ground cable from analyzer and ground.
- au. Shut down analyzer.

TABLE NO. 9-2-1-1. AC ESS BUS OFF Capsule Does Not Go On Without Power On AC ESNTL Bus.

TEST OR INSPECTION
CORRECTIVE ACTION
1. Check helicopter configuration. Step 1. EMEP Go to 2. Step 2. W/O EMEP Go to 4.

**TABLE NO. 9-2-1-1. AC ESS BUS OFF Capsule Does Not Go On Without Power On AC ESNTL Bus.
(Cont)**

	TEST OR INSPECTION CORRECTIVE ACTION
EMEP	2. Remove and check pin filter adapter from P117 (PARA 9-4-98).
	Step 1. If pin filter adapter is good, go to 3. Step 2. If pin filter adapter is not good, replace pin filter adapter (PARA 9-4-98). Go to 8.
EMEP	3. Install pin filter adapter (PARA 9-4-98). Go to 4.
	4. Check for 28 VDC between P117-20 and ground.
	Step 1. If voltage is as specified, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 8. Step 2. If voltage is not as specified, go to 5.
	5. Check for 28 VDC between K13 AC ESNTL BUS FAIL relay terminal A3 and ground.
	Step 1. If voltage is as specified, repair/replace wiring between K13 AC ESNTL BUS FAIL relay terminal A3 and P117-20 (Appendix F). Go to 8. Step 2. If voltage is not as specified, go to 6.
	6. Check for 28 VDC between K13 AC ESNTL BUS FAIL relay terminal A2 and ground.
	Step 1. If voltage is as specified, replace relay (PARA 9-4-22). Go to 8. Step 2. If voltage is not as specified, go to 7.
	7. Check for 28 VDC between terminal 1 (ESNTL BUS AC & CONV WARN circuit breaker, BATT BUS - lower console circuit breaker panel) and ground.
	Step 1. If voltage is as specified, repair/replace wiring between circuit breaker and relay terminal A2 (Appendix F). Go to 8. Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 8.
	8. Procedure completed.

TABLE NO. 9-2-1-2. With GENERATORS APU Switch ON, APU GEN ON Capsule Does Not Go On And #1 CONV And #2 CONV Capsules Remain On (Primary AC Buses Not Energized).

	TEST OR INSPECTION CORRECTIVE ACTION
	1. Check for 115 VAC between: J207-A and J207-D J207-B and J207-D J207-C and J207-D Step 1. If voltage is as specified, go to 2. Step 2. If voltage is not as specified, go to 25.
	2. Check helicopter configuration. Step 1. EMEP Go to 3. Step 2. W/O EMEP Go to 5.
EMEP	3. Remove and check pin filter adapters from P118 and P216 (PARA 9-4-98). Step 1. If pin filter adapters are good, go to 4. Step 2. If pin filter adapters are not good, replace pin filter adapters (PARA 9-4-98). Go to 26.
EMEP	4. Install pin filter adapters (PARA 9-4-98). Go to 5.
	5. Check for 28 VDC between GENERATORS APU switch terminal 2 and ground. Step 1. If voltage is as specified, go to 6. Step 2. If voltage is not as specified, go to 24.
	6. Check for 28 VDC between GENERATORS APU switch terminal 5 and ground. Step 1. If voltage is as specified, go to 7. Step 2. If voltage is not as specified, go to 23.
	7. Place GENERATORS APU switch to TEST and check that APU GEN ON capsule goes on. Step 1. If APU GEN ON capsule is on, go to 8. Step 2. If APU GEN ON capsule is not on, go to 21.
	8. Place GENERATORS APU switch to OFF/RESET and then ON. Go to 9.

TABLE NO. 9-2-1-2. With GENERATORS APU Switch ON, APU GEN ON Capsule Does Not Go On And #1 CONV And #2 CONV Capsules Remain On (Primary AC Buses Not Energized). (Cont)

**TEST OR INSPECTION
CORRECTIVE ACTION**

9. Check for 28 VDC between GENERATORS APU switch terminal 6 and ground.

Step 1. If voltage is as specified, go to **10**.

Step 2. If voltage is not as specified, replace GENERATORS APU switch (PARA 9-4-5). Go to **26**.

10. Check for 28 VDC between P205-P and P205-N.

Step 1. If voltage is as specified, go to **11**.

Step 2. If voltage is not as specified, go to **20**.

11. Check for 28 VDC between P206-C and ground.

Step 1. If voltage is as specified, go to **12**.

Step 2. If voltage is not as specified, replace K3 APU/EXT PWR CNTOR (PARA 9-4-17). Go to **26**.

12. Check for 28 VDC between P241-J and ground.

Step 1. If voltage is as specified, go to **13**.

Step 2. If voltage is not as specified, repair/replace wiring between P230-r and P241-J (Appendix F). Go to **26**.

13. Check for 28 VDC between P203-S and ground.

Step 1. If voltage is as specified, go to **14**.

Step 2. If voltage is not as specified, go to **19**.

14. Check for 28 VDC between P210-U and ground.

Step 1. If voltage is as specified, go to **15**.

Step 2. If voltage is not as specified, go to **18**.

15. Check for 28 VDC between P206-P and P206-N.

Step 1. If voltage is as specified, go to **16**.

Step 2. If voltage is not as specified, go to **17**.

TABLE NO. 9-2-1-2. With GENERATORS APU Switch ON, APU GEN ON Capsule Does Not Go On And #1 CONV And #2 CONV Capsules Remain On (Primary AC Buses Not Energized). (Cont)

TEST OR INSPECTION CORRECTIVE ACTION	
16. Check for 28 VDC between P118-E and ground.	
Step 1. If voltage is as specified, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 26 .	
Step 2. If voltage is not as specified, replace K4 AC BUS TIE CNTOR (PARA 9-4-21). Go to 26 .	
17. Check continuity between: P206-N and ground P206-P and P212-M	
Step 1. If continuity is present, replace K1 NO. 1 GEN CNTOR (PARA 9-4-11). Go to 26 .	
Step 2. If continuity is not present, repair/replace wiring as required (Appendix F). Go to 26 .	
18. Check continuity between P210-U and P203-P.	
Step 1. If continuity is present, replace K2 NO. 2 GEN CNTOR (PARA 9-4-12). Go to 26 .	
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 26 .	
19. Check continuity between P203-S and P241-P.	
Step 1. If continuity is present, replace right relay panel (PARA 9-4-2). Go to 26 .	
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 26 .	
20. Check continuity between P205-N and ground.	
Step 1. If continuity is present, repair/replace wiring between P205-P and GENERATORS APU switch terminal 6 (Appendix F). Go to 26 .	
Step 2. If continuity is not present, repair/replace wiring between P205-N and ground (Appendix F). Go to 26 .	
21. Check for 28 VDC between GENERATORS APU switch terminal 4 and ground.	
Step 1. If voltage is as specified, go to 22 .	
Step 2. If voltage is not as specified, replace GENERATORS APU switch (PARA 9-4-5). Go to 26 .	

TABLE NO. 9-2-1-2. With GENERATORS APU Switch ON, APU GEN ON Capsule Does Not Go On And #1 CONV And #2 CONV Capsules Remain On (Primary AC Buses Not Energized). (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

22. Check for 28 VDC between P118-E and ground.

- Step 1. If voltage is as specified, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to **26**.
- Step 2. If voltage is not as specified, repair/replace wiring between GENERATORS APU switch terminal 4 and P118-E (Appendix F). Go to **26**.

23. Check continuity between GENERATORS APU switch terminal 5 and P216-e.

- Step 1. If continuity is present, replace GCU (PARA 9-4-15). Go to **26**.
- Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **26**.

24. Check continuity between P230-a and P216-c.

- Step 1. If continuity is present, replace GCU (PARA 9-4-15). Go to **26**.
- Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **26**.

25. Check continuity between the following for APU GENERATOR:

Terminal T1 and J207-A
Terminal T2 and J207-B
Terminal T3 and J207-C
J207-D and ground

- Step 1. If continuity is present, perform Generator and GCU Troubleshooting procedure using system analyzer. Go to **26**.
- Step 2. If continuity is not present, repair/replace wiring as required (Appendix F). Go to **26**.

26. Procedure completed.

TABLE NO. 9-2-1-3. APU GEN ON Capsule Is Off With APU Generator Supplying Buses (#1 CONV And #2 CONV Capsules Off).

	TEST OR INSPECTION CORRECTIVE ACTION	
	1. Momentarily place caution/advisory panel BRT/DIM-TEST switch to TEST and check that APU GEN ON capsule goes on. Step 1. If APU GEN ON capsule is on, go to 2. Step 2. If APU GEN ON capsule is not on, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 10.	
	2. Check helicopter configuration. Step 1. EMEP Go to 3. Step 2. Go to 5.	
EMEP	3. Remove and check pin filter adapter from P118 (PARA 9-4-98). Step 1. If pin filter adapter is good, go to 4. Step 2. If pin filter adapter is not good, replace pin filter adapter (PARA 9-4-98). Go to 10.	
EMEP	4. Install pin filter adapter (PARA 9-4-98). Go to 5.	
	5. Check for 28 VDC between P118-E and ground. Step 1. If voltage is as specified, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 10. Step 2. If voltage is not as specified, go to 6.	
	6. Check for 28 VDC between P205-K and ground. Step 1. If voltage is as specified, go to 7. Step 2. If voltage is not as specified, repair/replace wiring between P205-K and J203-T (Appendix F). Go to 10.	
	7. Check for 28 VDC between P206-C and ground. Step 1. If voltage is as specified, go to 8. Step 2. If voltage is not as specified, go to 9.	

TABLE NO. 9-2-1-3. APU GEN ON Capsule Is Off With APU Generator Supplying Buses (#1 CONV And #2 CONV Capsules Off). (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
8. Check continuity between P118-E and P206-B.	Step 1. If continuity is present, replace K4 AC BUS TIE CNTOR (PARA 9-4-21). Go to 10. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 10.
9. Check continuity between P206-C and P205-L.	Step 1. If continuity is present, replace K3 APU/EXT PWR CNTOR (PARA 9-4-17). Go to 10. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 10.
10. Procedure completed.	

TABLE NO. 9-2-1-4. APU Generator Or AC External Power Does Not Supply No. 1 AC PRI Bus (#1 CONV Capsule On).

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for 115 VAC between the following for K1 NO. 1 GEN CNTOR: Terminal A2 and ground Terminal B2 and ground Terminal C2 and ground	Step 1. If voltage is as specified, repair/replace wiring between K1 NO. 1 GEN CNTOR and NO. 1 CONVERTER circuit breaker (Appendix F). Go to 4. Step 2. If voltage is not as specified, go to 2.
2. Check for 115 VAC between the following for K1 NO. 1 GEN CNTOR: Terminal A3 and ground Terminal B3 and ground Terminal C3 and ground	Step 1. If voltage is as specified, replace K1 NO. 1 GEN CNTOR (PARA 9-4-11). Go to 4. Step 2. If voltage is not as specified, go to 3.

TABLE NO. 9-2-1-4. APU Generator Or AC External Power Does Not Supply No. 1 AC PRI Bus (#1 CONV Capsule On). (Cont)

TEST OR INSPECTION CORRECTIVE ACTION
3. Check for open AC current limiters CL-1, CL-2, and CL-3. Step 1. If AC current limiters are open, replace them as required (PARA 9-4-13). Go to 4. Step 2. If AC current limiters are not open, repair/replace wiring between K1 NO. 1 GEN CNTOR and K2 NO. 2 GEN CNTOR as required (Appendix F). Go to 4. 4. Procedure completed.

TABLE NO. 9-2-1-5. AC ESS BUS OFF Capsule Does Not Go Off When No. 1 And No. 2 AC Primary Buses Are Energized.

TEST OR INSPECTION CORRECTIVE ACTION
1. Check for 115 VAC between K13 AC ESNTL BUS FAIL relay terminals X1 and X2. Step 1. If voltage is as specified, replace relay (PARA 9-4-22). Go to 15. Step 2. If voltage is not as specified, go to 2. 2. Check for 115 VAC between K13 AC ESNTL BUS FAIL relay terminal X1 and ground. Step 1. If voltage is as specified, repair/replace wiring between K13 AC ESNTL BUS FAIL relay terminal X2 and ground (Appendix F). Go to 15. Step 2. If voltage is not as specified, go to 3. 3. Check for 115 VAC between terminal 1 (AC ESNTL BUS WARN circuit breaker, AC ESNTL BUS - pilot's circuit breaker panel) and ground. Step 1. If voltage is as specified, repair/replace wiring between AC ESNTL BUS WARN circuit breaker terminal 1 and K13 AC ESNTL BUS FAIL relay terminal X1 (Appendix F). Go to 15. Step 2. If voltage is not as specified, replace AC ESNTL BUS WARN circuit breaker (PARA 9-4-1). Go to 15. Step 3. If trouble remains, go to 4.

TABLE NO. 9-2-1-5. AC ESS BUS OFF Capsule Does Not Go Off When No. 1 And No. 2 AC Primary Buses Are Energized. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
4. Check for 115 VAC between K8 AC ESNTL BUS XFER relay terminal A2 and ground.	Step 1. If voltage is as specified, go to 5. Step 2. If voltage is not as specified, go to 13.
5. Pull out AC ESNTL BUS SPLY circuit breaker (NO. 1 AC PRI BUS - copilot's circuit breaker panel). Go to 6.	
6. Check for 115 VAC between K8 AC ESNTL BUS XFER relay terminal A2 and ground.	Step 1. If voltage is as specified, go to 7. Step 2. If voltage is not as specified, go to 9.
7. Check for 115 VAC between K8 AC ESNTL BUS XFER relay terminal A3 and ground.	Step 1. If voltage is as specified, replace K8 AC ESNTL BUS XFER relay (PARA 9-4-22). Go to 15. Step 2. If voltage is not as specified, go to 8.
8. Check for 115 VAC between terminal 1 (AC ESNTL BUS SPLY circuit breaker, NO. 2 AC PRI BUS - pilot's circuit breaker panel) and ground.	Step 1. If voltage is as specified, repair/replace wiring between AC ESNTL BUS SPLY circuit breaker terminal 1 and K8 AC ESNTL BUS XFER relay terminal A3 (Appendix F). Go to 15. Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 15.
9. Push in AC ESNTL BUS SPLY circuit breaker (NO. 1 AC PRI BUS - copilot's circuit breaker panel). Go to 10.	
10. Check for 115 VAC between K8 AC ESNTL BUS XFER relay terminal A2 and ground.	Step 1. If voltage is as specified, repair/replace wiring between K8 AC ESNTL BUS XFER relay terminal A2 and AC ESNTL BUS WARN circuit breaker terminal 2 (Appendix F). Go to 15.

TABLE NO. 9-2-1-5. AC ESS BUS OFF Capsule Does Not Go Off When No. 1 And No. 2 AC Primary Buses Are Energized. (Cont)

TEST OR INSPECTION CORRECTIVE ACTION	
Step 2. If voltage is not as specified, go to 11 .	
11. Check for 115 VAC between K8 AC ESNTL BUS XFER relay terminal A1 and ground.	
Step 1. If voltage is as specified, repair/replace wiring between relay terminal X2 and ground (Appendix F). Go to 15 .	
Step 2. If voltage is not as specified, go to 12 .	
12. Check for 115 VAC between terminal 1 (AC ESNTL BUS SPLY circuit breaker, NO. 1 AC PRI BUS - copilot's circuit breaker panel) and ground.	
Step 1. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and K8 AC ESNTL BUS XFER relay terminal A1 (Appendix F). Go to 15 .	
Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 15 .	
13. Check for 115 VAC between K8 AC ESNTL BUS XFER relay terminal A1 and ground.	
Step 1. If voltage is as specified, repair/replace wiring between relay terminal X2 and ground (Appendix F). Go to 15 .	
Step 2. If voltage is not as specified, go to 14 .	
14. Check for 115 VAC between terminal 1 (AC ESNTL BUS SPLY circuit breaker, NO. 1 AC PRI BUS - copilot's circuit breaker panel) and ground.	
Step 1. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and K8 AC ESNTL BUS XFER relay terminal A1 (Appendix F). Go to 15 .	
Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 15 .	
15. Procedure completed.	

TABLE NO. 9-2-1-6. #1 GEN BRG And/Or #2 GEN BRG Capsule Goes On.

**TEST OR INSPECTION
CORRECTIVE ACTION**

1. Are both #1 GEN BRG and #2 GEN BRG capsules on?

Step 1. If both capsules are on, troubleshoot both #1 GEN BRG and #2 GEN BRG capsule circuits in the Caution/Advisory Warning System (PARA 8-2-4).

Go to **17**.

Step 2. If both capsules are not on, go to **2**.

2. Is #1 GEN BRG capsule on?

Step 1. If capsule is on, go to **3**.

Step 2. If capsule is not on, go to **10**.

3. Does #1 GEN BRG capsule go on steady for more than 1 minute?

Step 1. If capsule is steady, go to **4**.

Step 2. If capsule is not steady, no maintenance required. Go to **17**.

4. Check helicopter configuration.

Step 1. **EMEP** Go to **5**.

Step 2. **W/O EMEP** Go to **7**.

5. Remove and check pin filter adapter from P117 (PARA 9-4-98).

Step 1. If pin filter adapter is good, go to **6**.

Step 2. If pin filter adapter is not good, replace pin filter adapter (PARA 9-4-98).
Go to **17**.

6. Install pin filter adapter (PARA 9-4-98). Go to 7.**7. Pull out, then push in NO. 1 GEN WARN circuit breaker (NO. 1 DC PRI BUS - copilot's circuit breaker panel). Go to 8.****8. Does #1 GEN BRG capsule go off and remain off?**

Step 1. If capsule goes off and remains off, no maintenance required. Go to **17**.

Step 2. If capsule does not go off and remain off, go to **9**.

TABLE NO. 9-2-1-6. #1 GEN BRG And/Or #2 GEN BRG Capsule Goes On. (Cont)

	TEST OR INSPECTION CORRECTIVE ACTION
	9. Check for less than 50 ohms (Ω) between P117-18 and ground.
	Step 1. If resistance is as specified, replace No. 1 generator (PARA 9-4-16). Go to 17.
	Step 2. If resistance is not as specified, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 17.
	10. Does #2 GEN BRG capsule go on steady for more than 1 minute?
	Step 1. If capsule is steady, go to 11.
	Step 2. If capsule is not steady, no maintenance required. Go to 17.
	11. Check helicopter configuration.
	Step 1. EMEP Go to 12.
	Step 2. W/O EMEP Go to 14.
EMEP	12. Remove and check pin filter adapter from P117 (PARA 9-4-98).
	Step 1. If pin filter adapter is good, go to 13.
	Step 2. If pin filter adapter is not good, replace pin filter adapter (PARA 9-4-98). Go to 17.
EMEP	13. Install pin filter adapter (PARA 9-4-98). Go to 14.
	14. Pull out then push in NO. 2 GEN WARN circuit breaker (NO. 2 DC PRI BUS - pilot's circuit breaker panel). Go to 15.
	15. Does #2 GEN BRG capsule go off and remain off?
	Step 1. If capsule goes off and remains off, no maintenance required. Go to 17.
	Step 2. If capsule does not go off and remain off, go to 16.
	16. Check for less than 50 Ω between P117-34 and ground.
	Step 1. If resistance is as specified, replace No. 2 generator (PARA 9-4-16). Go to 17.
	Step 2. If resistance is not as specified, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 17.

TABLE NO. 9-2-1-6. #1 GEN BRG And/Or #2 GEN BRG Capsule Goes On. (Cont)

**TEST OR INSPECTION
CORRECTIVE ACTION**

17. Procedure completed.

TABLE NO. 9-2-1-7. #1 GEN Capsule Does Not Go On With No. 1 Generator Off.

**TEST OR INSPECTION
CORRECTIVE ACTION**

1. Momentarily place caution/advisory panel BRT/DIM-TEST switch to TEST and check that #1 GEN capsule goes on.

Step 1. If #1 GEN capsule goes on, go to **2**.

Step 2. If #1 GEN capsule does not go on, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to **12**.

2. Check helicopter configuration.

Step 1. **EMEP** Go to **3**.

Step 2. **W/O EMEP** Go to **5**.

EMEP

3. Remove and check pin filter adapters from P117 and P215 (PARA 9-4-98).

Step 1. If pin filter adapters are good, go to **4**.

Step 2. If pin filter adapters are not good, replace pin filter adapters (PARA 9-4-98). Go to **12**.

EMEP

4. Install pin filter adapters (PARA 9-4-98). Go to 5.

5. Check for 28 VDC between P117-17 and ground.

Step 1. If voltage is as specified, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to **12**.

Step 2. If voltage is not as specified, go to **6**.

6. Check for 28 VDC between P212-B and ground.

Step 1. If voltage is as specified, go to **7**.

Step 2. If voltage is not as specified, go to **8**.

TABLE NO. 9-2-1-7. #1 GEN Capsule Does Not Go On With No. 1 Generator Off. (Cont)

TEST OR INSPECTION CORRECTIVE ACTION	
7. Check continuity between P117-17 and P212-A.	
Step 1. If continuity is present, replace K1 NO. 1 GEN CNTOR (PARA 9-4-11). Go to 12.	
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 12.	
8. Check continuity between P212-B and P215-Z.	
Step 1. If continuity is present, go to 9.	
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 12.	
9. Check continuity between P212-A and P215-f.	
Step 1. If continuity is present, go to 10.	
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 12.	
10. Check for 28 VDC between P215-Y and ground.	
Step 1. If voltage is as specified, replace No. 1 GCU (PARA 9-4-15). Go to 12.	
Step 2. If voltage is not as specified, go to 11.	
11. Check for 28 VDC between terminal 1 (NO. 1 GEN WARN circuit breaker, NO. 1 DC PRI BUS - copilot's circuit breaker panel) and ground.	
Step 1. If voltage is as specified, repair/replace wiring between circuit breaker and P215-Y (Appendix F). Go to 12.	
Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 12.	
12. Procedure completed.	

TABLE NO. 9-2-1-8. #1 GEN Capsule Is On With GENERATORS NO. 1 Switch ON.

**TEST OR INSPECTION
CORRECTIVE ACTION**

- 1. Shut down No. 1 engine and APU (Appropriate Flight Manual). Go to 2.**
- 2. Inspect connector P251 for scorched sockets at the front and back ends of the connector.**
 - Step 1. If pins are scorched, clean or replace connector as required (Appendix F).
Go to **23**.
 - Step 2. If pins are not scorched, go to **3**.
- 3. With qualified personnel at the controls, start No. 1 engine and operate with main rotor turning at 100% (Appropriate Flight Manual). Go to 4.**
- 4. Check for 115 VAC between:**
 - J211-A and J211-D**
 - J211-B and J211-D**
 - J211-C and J211-D**
 - Step 1. If voltage is as specified, go to **5**.
 - Step 2. If voltage is not as specified, go to **22**.
- 5. Place and hold GENERATORS #1 switch to TEST. Go to 6.**
- 6. Check that #1 GEN capsule goes off.**
 - Step 1. If #1 GEN capsule goes off, go to **7**.
 - Step 2. If #1 GEN capsule does not go off, go to **11**.
- 7. Place GENERATORS NO. 1 switch to OFF/RESET and then to ON. Go to 8.**
- 8. Check for 28 VDC between P212-P and P212-N.**
 - Step 1. If voltage is as specified, replace K1 NO. 1 GEN CNTOR (PARA 9-4-11).
Go to **23**.
 - Step 2. If voltage is not as specified, go to **9**.
- 9. Check for 28 VDC between P212-P and ground.**
 - Step 1. If voltage is as specified, repair/replace wiring between P212-N and ground (Appendix F). Go to **23**.
 - Step 2. If voltage is not as specified, go to **10**.

TABLE NO. 9-2-1-8. #1 GEN Capsule Is On With GENERATORS NO. 1 Switch ON. (Cont)

	TEST OR INSPECTION CORRECTIVE ACTION	
	10. Check continuity between P212-P and GENERATORS NO. 1 switch terminal 6.	
	Step 1. If continuity is present, replace switch (PARA 9-4-5). Go to 23 .	
	Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 23 .	
	11. Check helicopter configuration.	
	Step 1. EMEP Go to 12 .	
	Step 2. W/O EMEP Go to 14 .	
EMEP	12. Remove and check pin filter adapter from P215 (PARA 9-4-98).	
	Step 1. If pin filter adapter is good, go to 13 .	
	Step 2. If pin filter adapter is not good, replace pin filter adapter (PARA 9-4-98). Go to 23 .	
EMEP	13. Install pin filter adapter (PARA 9-4-98). Go to 14.	
	14. Check for 28 VDC between GENERATORS NO. 1 switch terminal 5 and ground.	
	Step 1. If voltage is as specified, go to 15 .	
	Step 2. If voltage is not as specified, go to 17 .	
	15. With GENERATORS NO. 1 switch placed to TEST, check for 28 VDC between GENERATORS NO. 1 switch terminal 4 and ground.	
	Step 1. If voltage is as specified, go to 16 .	
	Step 2. If voltage is not as specified, replace switch (PARA 9-4-5). Go to 23 .	
	16. Check continuity between GENERATORS NO. 1 switch terminal 4 and P215-d.	
	Step 1. If continuity is present, replace No. 1 GCU (PARA 9-4-15). Go to 23 .	
	Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 23 .	
	17. Check continuity between GENERATORS NO. 1 switch terminal 5 and P215-e.	
	Step 1. If continuity is present, go to 18 .	
	Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 23 .	

TABLE NO. 9-2-1-8. #1 GEN Capsule Is On With GENERATORS NO. 1 Switch ON. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
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- 18. Check for 28 VDC between GENERATORS NO. 1 switch terminal 2 and ground.**

Step 1. If voltage is as specified, go to **19**.
 Step 2. If voltage is not as specified, go to **21**.

- 19. Check for 28 VDC between GENERATORS NO. 1 switch terminal 1 and ground.**

Step 1. If voltage is as specified, go to **20**.
 Step 2. If voltage is not as specified, replace switch (PARA 9-4-5). Go to **23**.

- 20. Check continuity between GENERATORS NO. 1 switch terminal 1 and P215-a.**

Step 1. If continuity is present, replace No. 1 GCU (PARA 9-4-15). Go to **23**.
 Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **23**.

- 21. Check continuity between GENERATORS NO. 1 switch terminal 2 and P215-c.**

Step 1. If continuity is present, replace No. 1 GCU (PARA 9-4-15). Go to **23**.
 Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **23**.

- 22. Check continuity between the following for No. 1 generator:**

Terminal T1 and J211-A
Terminal T2 and J211-B
Terminal T3 and J211-C
J211-D and ground

Step 1. If continuity is present, perform Generator and GCU Troubleshooting procedure using system analyzer. Go to **23**.
 Step 2. If continuity is not present, repair/replace wiring as required (Appendix F). Go to **23**.

- 23. Procedure completed.**
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TABLE NO. 9-2-1-9. #1 CONV Capsule Stays On When Either Generator Is Supplying Power To No. 1 AC Primary Bus.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for 115 VAC between the following for NO. 1 CONVERTER circuit breaker (NO. 1 AC PRI BUS - copilot's circuit breaker panel): Terminal A2 and ground Terminal B2 and ground Terminal C2 and ground	Step 1. If voltage is as specified, troubleshoot DC Electrical System (PARA 9-2-2). Go to 4. Step 2. If voltage is not as specified, go to 2.
2. Check for 115 VAC between the following for NO. 1 CONVERTER circuit breaker (NO. 1 AC PRI BUS - copilot's circuit breaker panel): Terminal A1 and ground Terminal B1 and ground Terminal C1 and ground	Step 1. If voltage is as specified, replace circuit breaker (PARA 9-4-1). Go to 4. Step 2. If voltage is not as specified, go to 3.
3. Check for 115 VAC between the following for K1 NO. 1 GEN CNTOR: Terminal A2 and ground Terminal B2 and ground Terminal C2 and ground	Step 1. If voltage is as specified, repair/replace wiring between K1 NO. 1 GEN CNTOR and NO. 1 CONVERTER circuit breaker (Appendix F). Go to 4. Step 2. If voltage is not as specified, replace K1 NO. 1 GEN CNTOR (PARA 9-4-11). Go to 4.
4. Procedure completed.	

TABLE NO. 9-2-1-10. #2 CONV Capsule Stays On When Either Generator Is Supplying Power To No. 2 AC Primary Bus.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for 115 VAC between the following for NO. 2 CONVERTER circuit breaker (NO. 2 AC PRI BUS - pilot's circuit breaker panel): Terminal A1 and ground Terminal B1 and ground Terminal C1 and ground	Step 1. If voltage is as specified, troubleshoot DC Electrical System (PARA 9-2-2). Go to 6. Step 2. If voltage is not as specified, go to 2.
2. Check for 115 VAC between the following for NO. 2 CONVERTER circuit breaker (NO. 2 AC PRI BUS - pilot's circuit breaker panel): Terminal A2 and ground Terminal B2 and ground Terminal C2 and ground	Step 1. If voltage is as specified, replace circuit breaker (PARA 9-4-1). Go to 6. Step 2. If voltage is not as specified, go to 3.
3. Check for 115 VAC between the following for K2 NO. 2 GEN CNTOR: Terminal A2 and ground Terminal B2 and ground Terminal C2 and ground	Step 1. If voltage is as specified, repair/replace wiring between K2 NO. 2 GEN CNTOR and NO. 2 CONVERTER circuit breaker (Appendix F). Go to 6. Step 2. If voltage is not as specified, go to 4.
4. Check for 115 VAC between the following for K1 NO. 1 GEN CNTOR: Terminal A2 and ground Terminal B2 and ground Terminal C2 and ground	Step 1. If voltage is as specified, go to 5. Step 2. If voltage is not as specified, replace K2 NO. 2 GEN CNTOR (PARA 9-4-12). Go to 6.
5. Check for open current limiters CL-4, CL-5, and CL-6.	Step 1. If AC current limiters are open, replace them as required (PARA 9-4-13). Go to 6.

TABLE NO. 9-2-1-10. #2 CONV Capsule Stays On When Either Generator Is Supplying Power To No. 2 AC Primary Bus. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 2. If AC current limiters are not open, repair/replace wiring between K1 NO. 1 GEN CNTOR and K2 NO. 2 GEN CNTOR (Appendix F). Go to 6 . Step 3. If trouble remains, replace K4 AC BUS TIE CNTOR (PARA 9-4-21). Go to 6 .
	6. Procedure completed.

TABLE NO. 9-2-1-11. No. 1 And/Or No. 2 Generator Disabled By Underfrequency Condition With Weight Off Wheels.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Are both generators disabled?	Step 1. If both are disabled, go to 2 . Step 2. If both are not disabled, go to 3 .
2. Is right drag beam switch in proper adjustment?	Step 1. If switch is in proper adjustment, replace switch (PARA 3-4-6). Go to 14 . Step 2. If switch is not is proper adjustment, adjust switch (PARA 3-4-6). Go to 14 .
3. Is No. 1 generator disabled?	Step 1. If No. 1 generator is disabled, go to 4 . Step 2. If No. 1 generator is not disabled, go to 9 .
4. Check continuity between P215-F and ground.	Step 1. If continuity is present, replace No. 1 GCU (PARA 9-4-15). Go to 14 . Step 2. If continuity is not present, go to 5 .
5. Check helicopter configuration.	Step 1. EMEP Go to 6 . Step 2. W/O EMEP Go to 8 .

TABLE NO. 9-2-1-11. No. 1 And/Or No. 2 Generator Disabled By Underfrequency Condition With Weight Off Wheels. (Cont)

	TEST OR INSPECTION CORRECTIVE ACTION
EMEP	6. Remove and check pin filter adapter from P215 (PARA 9-4-98).
	Step 1. If pin filter adapter is good, go to 7. Step 2. If pin filter adapter is not good, replace pin filter adapter (PARA 9-4-98). Go to 14.
EMEP	7. Install pin filter adapter on P215 (PARA 9-4-98). Go to 8.
	8. Repair/replace wiring between P215-F and J164-C (Appendix F). Go to 14.
	9. Check continuity between P214-F and ground.
	Step 1. If continuity is present, replace No. 2 GCU (PARA 9-4-15). Go to 14. Step 2. If continuity is not present, go to 10.
	10. Check helicopter configuration.
	Step 1. EMEP Go to 11.  Step 2. W/O EMEP Go to 13. 
EMEP	11. Remove and check pin filter adapter from P214 (PARA 9-4-98).
	Step 1. If pin filter adapter is good, go to 12. Step 2. If pin filter adapter is not good, replace pin filter adapter (PARA 9-4-98). Go to 14.
EMEP	12. Install pin filter adapter on P214 (PARA 9-4-98). Go to 13.
	13. Repair/replace wiring between P214-F and J164-C (Appendix F). Go to 14.
	14. Procedure completed.

TABLE NO. 9-2-1-12. #2 GEN Capsule Does Not Go On With No. 2 Generator Off.

	TEST OR INSPECTION CORRECTIVE ACTION
	1. Check helicopter configuration. Step 1. EMEP Go to 2. Step 2. W/O EMEP Go to 4.
EMEP	2. Remove and check pin filter adapters from P117 and P214 (PARA 9-4-98). Step 1. If pin filter adapters are good, go to 3. Step 2. If pin filter adapters are not good, replace pin filter adapters (PARA 9-4-98). Go to 13.
EMEP	3. Install pin filter adapters on P117 and P214 (PARA 9-4-98). Go to 4. 4. Momentarily place caution/advisory panel BRT/DIM-TEST switch to TEST and check that #2 GEN capsule goes on. Step 1. If capsule goes on, go to 5. Step 2. If capsule does not go on, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 13.
	5. Check for 28 VDC between P117-33 and ground. Step 1. If voltage is as specified, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 13. Step 2. If voltage is not as specified, go to 6.
	6. Check for 28 VDC between P204-B and ground. Step 1. If voltage is as specified, go to 7. Step 2. If voltage is not as specified, go to 8.
	7. Check continuity between P204-A and P117-33. Step 1. If continuity is present, replace K2 NO. 2 GEN CNTOR (PARA 9-4-12). Go to 13. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 13.

TABLE NO. 9-2-1-12. #2 GEN Capsule Does Not Go On With No. 2 Generator Off. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
8. Check continuity between P204-B and P214-Z.	Step 1. If continuity is present, go to 9. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 13.
9. Check continuity between P204-A and P214-f.	Step 1. If continuity is present, go to 10. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 13.
10. Check for 28 VDC between P214-Y and ground.	Step 1. If voltage is as specified, replace No. 2 GCU (PARA 9-4-15). Go to 13. Step 2. If voltage is not as specified, go to 11.
11. Check for 28 VDC between P117-66 and ground.	Step 1. If voltage is as specified, repair/replace wiring between NO. 2 GEN WARN circuit breaker and P214-Y (Appendix F). Go to 13. Step 2. If voltage is not as specified, go to 12.
12. Check for 28 VDC between terminal 1 (NO. 2 GEN WARN circuit breaker, NO. 2 DC PRI BUS - pilot's circuit breaker panel) and ground.	Step 1. If voltage is as specified, repair/replace wiring (Appendix F). Go to 13. Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 13.
13. Procedure completed.	

TABLE NO. 9-2-1-13. #2 GEN Capsule Is On With GENERATORS NO. 2 Switch ON.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Shut down No. 1 engine and APU (Appropriate Flight Manual). Go to 2.	
2. Inspect connector P250 for scorched sockets at the front and back ends of the connector.	

TABLE NO. 9-2-1-13. #2 GEN Capsule Is On With GENERATORS NO. 2 Switch ON. (Cont)

TEST OR INSPECTION CORRECTIVE ACTION	
Step 1. If pins are scorched, clean or replace connector as required (Appendix F). Go to 22. Step 2. If pins are not scorched, go to 3.	
3. With qualified personnel at the controls, start No. 1 engine and operate with main rotor turning at 100% (Appropriate Flight Manual). Go to 4.	
4. Check for 115 VAC between: J208-A and J208-D J208-B and J208-D J208-C and J208-D	
Step 1. If voltage is as specified, go to 5. Step 2. If voltage is not as specified, go to 21.	
5. Place and hold GENERATORS NO. 2 switch to TEST and check that #2 GEN capsule goes off.	
Step 1. If #2 GEN capsule goes off, go to 6. Step 2. If #2 GEN capsule does not go off, go to 10.	
6. Place GENERATORS NO. 2 switch to OFF/RESET and then to ON. Go to 7.	
7. Check for 28 VDC between P204-P and P204-N.	
Step 1. If voltage is as specified, replace K2 NO. 2 GEN CNTOR (PARA 9-4-12). Go to 22. Step 2. If voltage is not as specified, go to 8.	
8. Check for 28 VDC between P204-P and ground.	
Step 1. If voltage is as specified, repair/replace wiring between P204-N and ground (Appendix F). Go to 22. Step 2. If voltage is not as specified, go to 9.	
9. Check continuity between P204-P and GENERATORS NO. 2 switch terminal 6.	
Step 1. If continuity is present, replace switch (PARA 9-4-5). Go to 22. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 22.	

TABLE NO. 9-2-1-13. #2 GEN Capsule Is On With GENERATORS NO. 2 Switch ON. (Cont)

**TEST OR INSPECTION
CORRECTIVE ACTION**

10. Check helicopter configuration.

- Step 1. **EMEP** Go to **11.**
- Step 2. **W/O EMEP** Go to **13.**

11. Remove and check pin filter adapter from P214 (PARA 9-4-98).

- Step 1. If pin filter adapter is good, go to **12.**
- Step 2. If pin filter adapter is not good, replace pin filter adapter (PARA 9-4-98).
Go to **22.**

12. Install pin filter adapter on P214 (PARA 9-4-98). Go to 13.**13. Check for 28 VDC between GENERATORS NO. 2 switch terminal 5 and ground.**

- Step 1. If voltage is as specified, go to **14.**
- Step 2. If voltage is not as specified, go to **16.**

14. With GENERATORS NO. 2 switch placed at TEST, check for 28 VDC between GENERATORS NO. 2 switch terminal 4 and ground.

- Step 1. If voltage is as specified, go to **15.**
- Step 2. If voltage is not as specified, replace switch (PARA 9-4-5). Go to **22.**

15. Check continuity between GENERATORS NO. 2 switch terminal 4 and P214-d.

- Step 1. If continuity is present, replace switch (PARA 9-4-5). Go to **22.**
- Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **22.**

16. Check continuity between GENERATORS NO. 2 switch terminal 5 and P214-e.

- Step 1. If continuity is present, go to **17.**
- Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **22.**

17. Check for 28 VDC between GENERATORS NO. 2 switch terminal 2 and ground.

- Step 1. If voltage is as specified, go to **18.**
- Step 2. If voltage is not as specified, go to **20.**

TABLE NO. 9-2-1-13. #2 GEN Capsule Is On With GENERATORS NO. 2 Switch ON. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
18. Check for 28 VDC between GENERATORS NO. 2 switch terminal 1 and ground.	Step 1. If voltage is as specified, go to 19. Step 2. If voltage is not as specified, replace switch (PARA 9-4-5). Go to 22.
19. Check continuity between GENERATORS NO. 2 switch terminal 1 and P214-a.	Step 1. If continuity is present, replace No. 2 GCU (PARA 9-4-15). Go to 22. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 22.
20. Check continuity between GENERATORS NO. 2 switch terminal 2 and P214-c.	Step 1. If continuity is present, replace No. 2 GCU (PARA 9-4-15). Go to 22. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 22.
21. Check continuity between the following for No. 2 generator: Terminal T1 and J208-A Terminal T2 and J208-B Terminal T3 and J208-C J208-D and ground	Step 1. If continuity is present, perform Generator and GCU Troubleshooting procedure using system analyzer. Go to 22. Step 2. If continuity is not present, repair/replace wiring as required (Appendix F). Go to 22.
22. Procedure completed.	

TABLE NO. 9-2-1-14. EXT PWR CONNECTED Capsule Is Off With A Source Of External Power Connected.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check helicopter configuration.	Step 1. EMEP Go to 2. Step 2. W/O EMEP Go to 4.

TABLE NO. 9-2-1-14. EXT PWR CONNECTED Capsule Is Off With A Source Of External Power Connected. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
EMEP	2. Remove and check pin filter adapter from P118 (PARA 9-4-98). Step 1. If pin filter adapter is good, go to 3. Step 2. If pin filter adapter is not good, replace pin filter adapter (PARA 9-4-98). Go to 9.
EMEP	3. Install pin filter adapter on P118 (PARA 9-4-98). Go to 4. 4. Momentarily place caution/advisory panel BRT/DIM-TEST switch to TEST and check that EXT PWR CONNECTED capsule goes on. Step 1. If EXT PWR CONNECTED capsule goes on, go to 5. Step 2. If EXT PWR CONNECTED capsule does not go on, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 9. 5. Check for 28 VDC between J136-F and J136-N. Step 1. If voltage is as specified, go to 6. Step 2. If voltage is not as specified, go to 7. 6. Check for 28 VDC between P118-N and ground. Step 1. If voltage is as specified, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 9. Step 2. If voltage is not as specified, repair/replace wiring between J136-E and P118-N (Appendix F). Go to 9. 7. Check continuity between J136-N and ground. Step 1. If continuity is present, go to 8. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 9. 8. Check for 28 VDC between terminal 1 (BATT & ESNTL DC WARN EXT PWR CONTR circuit breaker, BATT BUS - lower console circuit breaker panel) and ground. Step 1. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and J136-F (Appendix F). Go to 9. Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 9.

TABLE NO. 9-2-1-14. EXT PWR CONNECTED Capsule Is Off With A Source Of External Power Connected. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION
9. Procedure completed.

TABLE NO. 9-2-1-15. External Power Does Not Supply AC Buses (#1 CONV And #2 CONV Capsules On).

TEST OR INSPECTION
CORRECTIVE ACTION
1. Check for 115 VAC between the following for K2 NO. 2 GEN CNTOR: Terminal A3 and ground Terminal B3 and ground Terminal C3 and ground Step 1. If voltage is as specified, replace K2 NO. 2 GEN CNTOR (PARA 9-4-12). Go to 18. Step 2. If voltage is not as specified, go to 2.
2. Check for 115 VAC between the following for K4 AC BUS TIE CNTOR: Terminal A2 and ground Terminal B2 and ground Terminal C2 and ground Step 1. If voltage is as specified, repair/replace wiring between K2 NO. 2 GEN CNTOR and K4 AC BUS TIE CNTOR as required (Appendix F). Go to 18. Step 2. If voltage is not as specified, go to 3.
3. Check for 115 VAC between the following for K4 AC BUS TIE CNTOR: Terminal A1 and ground Terminal B1 and ground Terminal C1 and ground Step 1. If voltage is as specified, go to 4. Step 2. If voltage is not as specified, go to 16.
4. Check for 28 VDC between: P206-P and ground P206-N and ground

TABLE NO. 9-2-1-15. External Power Does Not Supply AC Buses (#1 CONV And #2 CONV Capsules On). (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 1. If voltage is as specified, replace K4 AC BUS TIE CNTOR (PARA 9-4-21). Go to 18. Step 2. If voltage is not as specified, go to 5.
5. Check for 28 VDC between P206-P and ground.	Step 1. If voltage is as specified, repair/replace wiring between P206-N and ground (Appendix F). Go to 18. Step 2. If voltage is not as specified, go to 6.
6. Check for 28 VDC between P212-L and ground.	Step 1. If voltage is as specified, go to 7. Step 2. If voltage is not as specified, go to 8.
7. Check continuity between P212-M and P206-P.	Step 1. If continuity is present, replace K1 NO. 1 GEN CNTOR (PARA 9-4-11). Go to 18. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 18.
8. Check for 28 VDC between P204-M and ground.	Step 1. If voltage is as specified, go to 9. Step 2. If voltage is not as specified, go to 10.
9. Check continuity between P204-L and P212-L.	Step 1. If continuity is present, replace K2 NO. 2 GEN CNTOR (PARA 9-4-12). Go to 18. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 18.
10. Check for 28 VDC between P241-H and ground.	Step 1. If voltage is as specified, go to 11. Step 2. If voltage is not as specified, go to 12.

TABLE NO. 9-2-1-15. External Power Does Not Supply AC Buses (#1 CONV And #2 CONV Capsules On). (Cont)

TEST OR INSPECTION CORRECTIVE ACTION	
11. Check continuity between P241-P and P204-M.	
Step 1. If continuity is present, replace right relay panel (PARA 9-4-2). Go to 18 .	
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 18 .	
12. Check for 28 VDC between EXT PWR switch terminal 2 and ground.	
Step 1. If voltage is as specified, repair/replace wiring between switch terminal 2 and P241-H (Appendix F). Go to 18 .	
Step 2. If voltage is not as specified, go to 13 .	
13. Check for 28 VDC between EXT PWR switch terminal 3 and ground.	
Step 1. If voltage is as specified, replace switch (PARA 9-4-5). Go to 18 .	
Step 2. If voltage is not as specified, go to 14 .	
14. Check continuity between EXT PWR switch terminal 3 and P213-5.	
Step 1. If continuity is present, go to 15 .	
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 18 .	
15. Check continuity between the following for K3 APU/EXT PWR CNTOR:	
Terminal A3 and P213-3	
Terminal B3 and P213-2	
Terminal C3 and P213-1	
Step 1. If continuity is present, replace external power monitor panel (PARA 9-4-20). Go to 18 .	
Step 2. If continuity is not present, repair/replace wiring as required (Appendix F). Go to 18 .	
16. Check for 115 VAC between the following for K3 APU/EXT PWR CNTOR:	
Terminal A2 and ground	
Terminal B2 and ground	
Terminal C2 and ground	
Step 1. If voltage is as specified, repair/replace wiring between K3 APU/EXT PWR CNTOR and K4 AC BUS TIE CNTOR as required (Appendix F). Go to 18 .	
Step 2. If voltage is not as specified, go to 17 .	

TABLE NO. 9-2-1-15. External Power Does Not Supply AC Buses (#1 CONV And #2 CONV Capsules On). (Cont)

**TEST OR INSPECTION
CORRECTIVE ACTION**

- 17. Check for 115 VAC between the following for K3 APU/EXT PWR CNTOR:**
Terminal A3 and ground
Terminal B3 and ground
Terminal C3 and ground

Step 1. If voltage is as specified, replace K3 APU/EXT PWR CNTOR (PARA 9-4-17). Go to **18**.

Step 2. If voltage is not as specified, repair/replace wiring between K3 APU/EXT PWR CNTOR and external receptacle as required (Appendix F). Go to **18**.

- 18. Procedure completed.**
-

TABLE NO. 9-2-1-16. No 26 VAC Power On AC ESNTL Bus.

**TEST OR INSPECTION
CORRECTIVE ACTION**

- 1. Check for 115 VAC between T12 AUTO XFMR terminal HV and ground.**

Step 1. If voltage is as specified, go to **2**.

Step 2. If voltage is not as specified, go to **4**.

- 2. Check for 26 VAC between T12 AUTO XFMR terminal LV and ground.**

Step 1. If voltage is as specified, repair/replace wiring between transformer terminal LV and terminal 2 of AC ESNTL BUS 26 VAC INST circuit breaker (Appendix F). Go to **5**.

Step 2. If voltage is not as specified, go to **3**.

- 3. Check continuity between T12 AUTO XFMR terminal LV and ground.**

Step 1. If continuity is present, replace transformer (PARA 9-4-10). Go to **5**.

Step 2. If continuity is not present, repair/replace wiring between terminal LV and ground (Appendix F). Go to **5**.

TABLE NO. 9-2-1-16. No 26 VAC Power On AC ESNTL Bus. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
4. Check for 115 VAC between terminal 1 (AUTO XFMR circuit breaker, AC ESNTL BUS - pilot's circuit breaker panel) and ground.	Step 1. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and T12 transformer terminal HV (Appendix F). Go to 5. Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 5.
5. Procedure completed.	

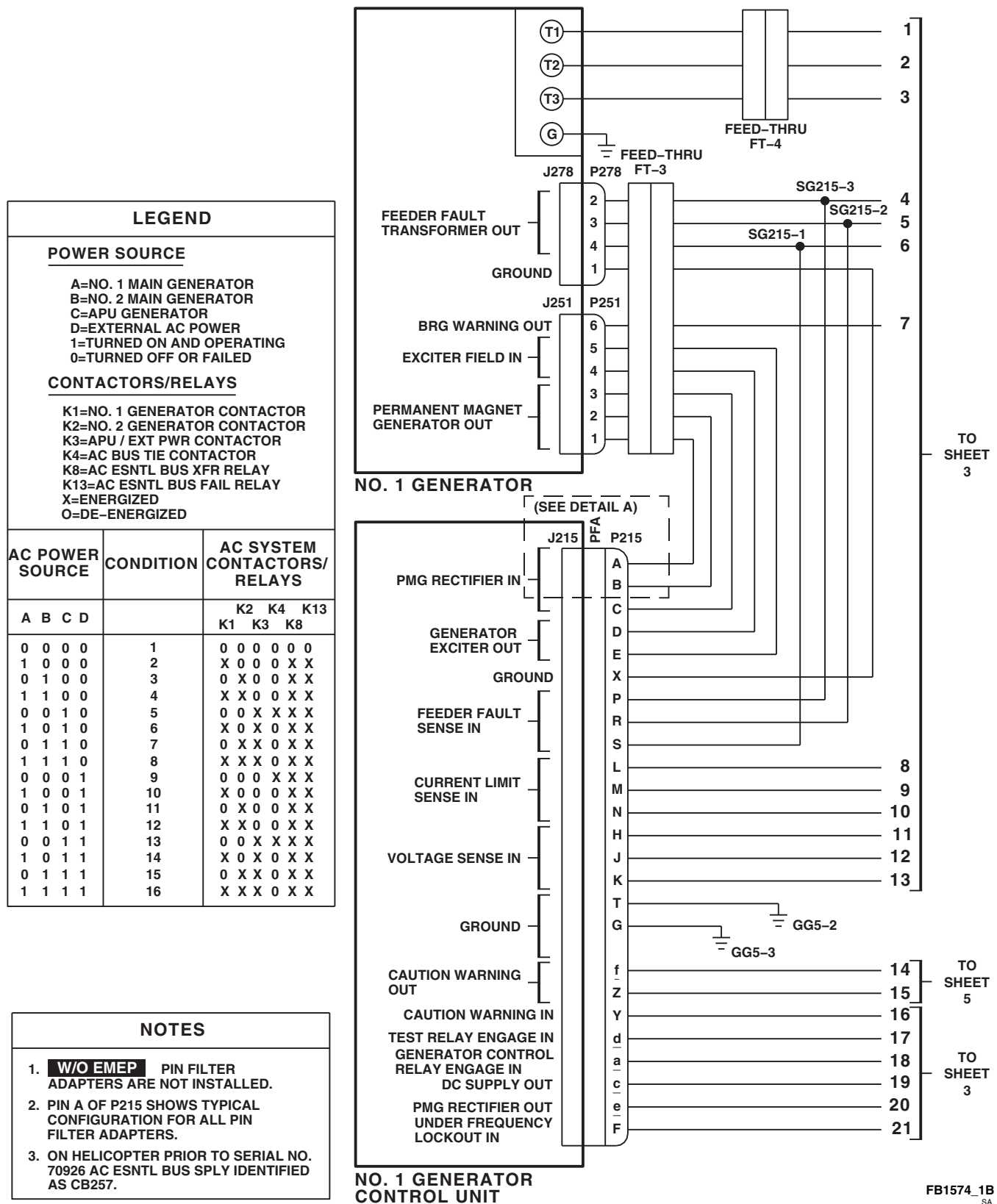


FIGURE 9-2-1-1. AC ELECTRICAL SYSTEM SCHEMATIC DIAGRAM (SHEET 1 OF 19)

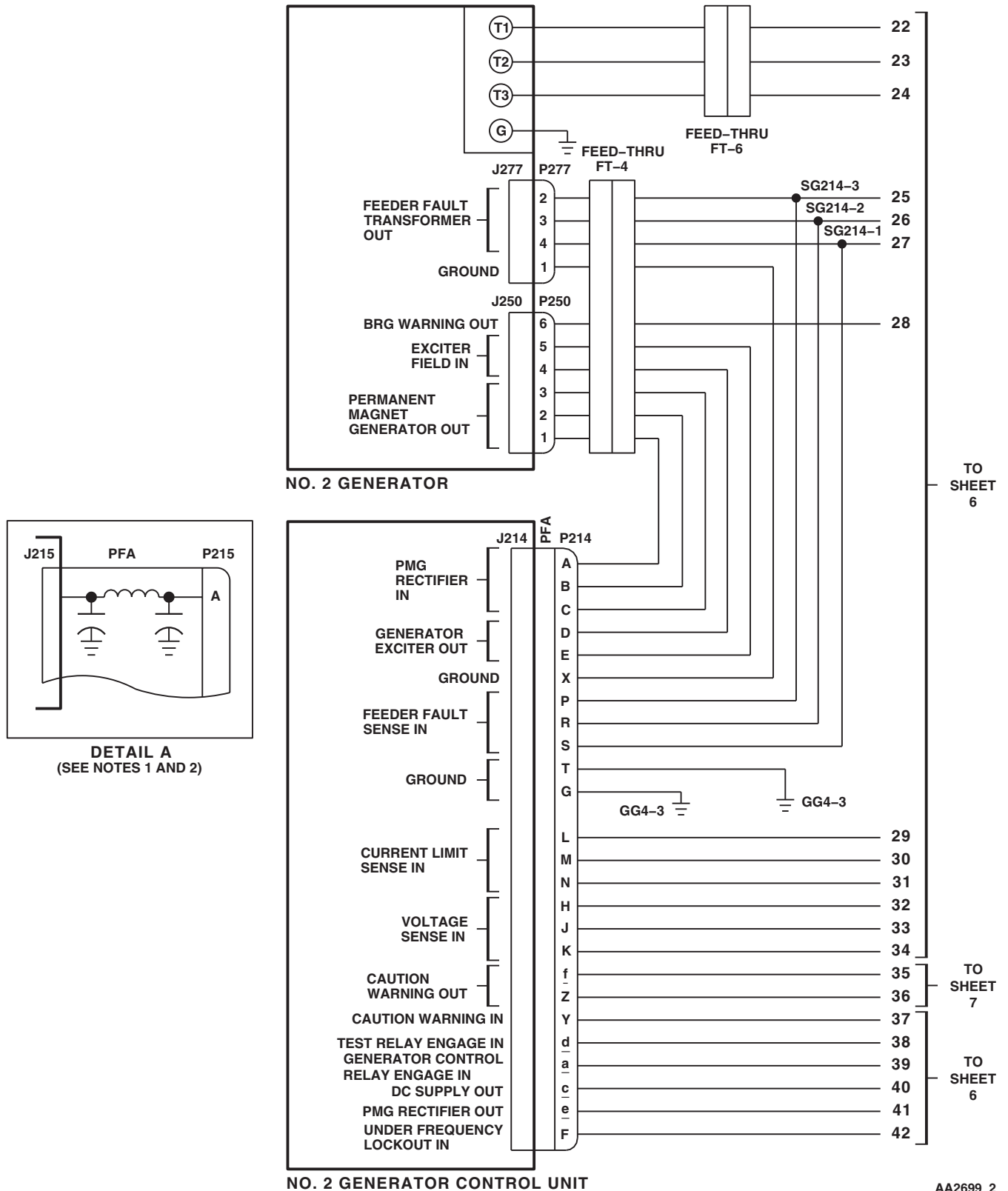


FIGURE 9-2-1-1. AC ELECTRICAL SYSTEM SCHEMATIC DIAGRAM (SHEET 2 OF 19)

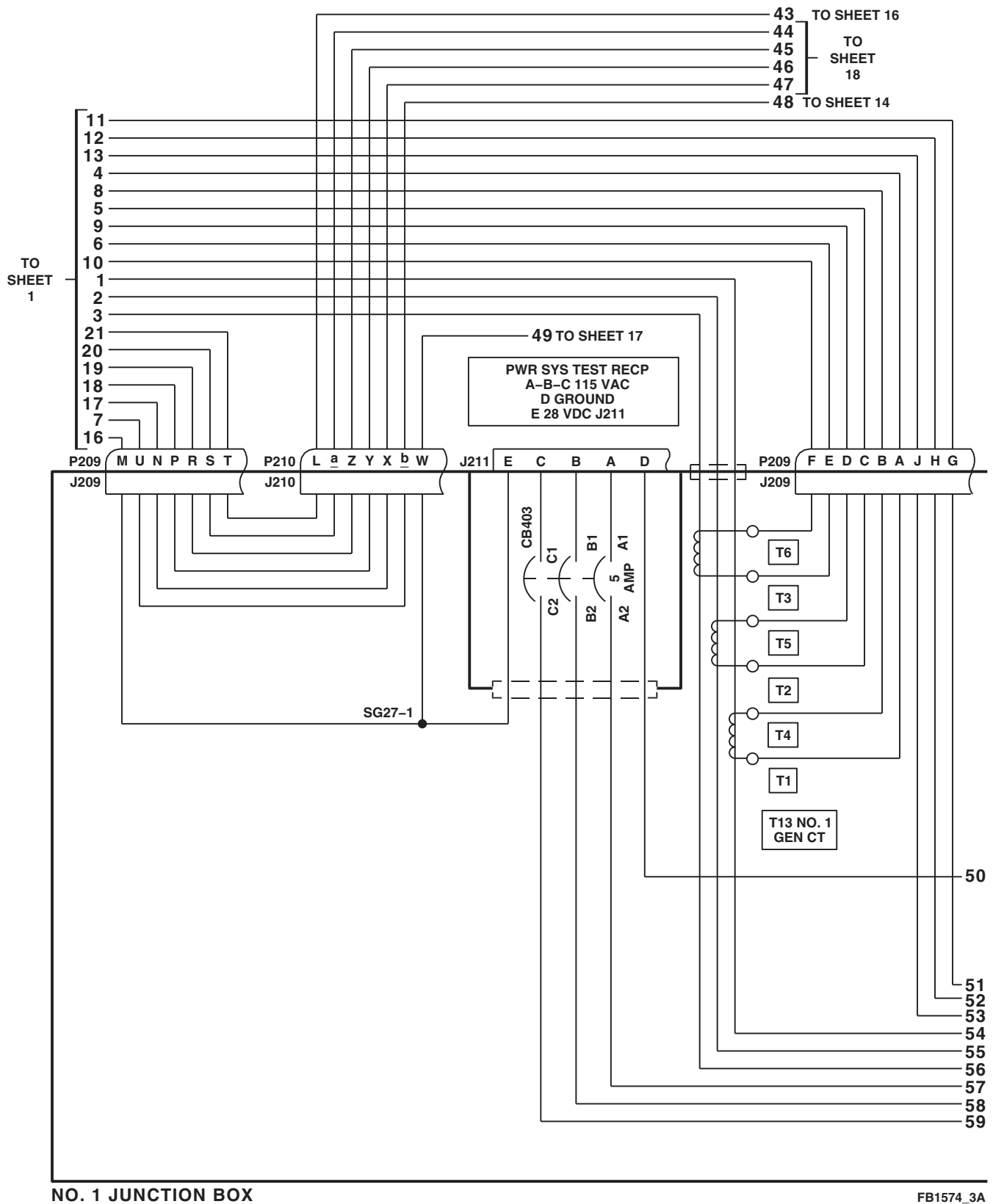
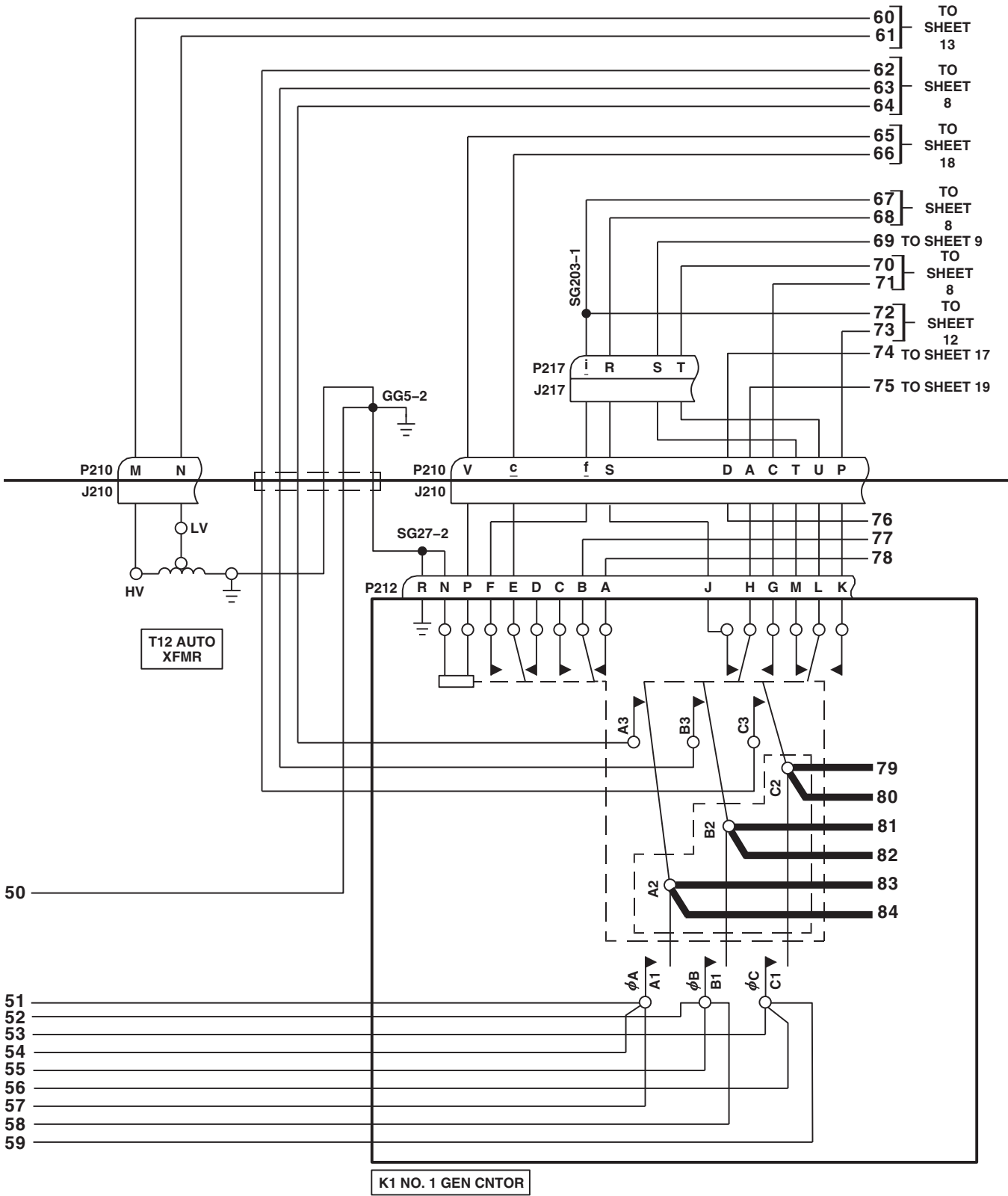


FIGURE 9-2-1-1. AC ELECTRICAL SYSTEM SCHEMATIC DIAGRAM (SHEET 3 OF 19)



NO. 1 JUNCTION BOX

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FIGURE 9-2-1-1. AC ELECTRICAL SYSTEM SCHEMATIC DIAGRAM (SHEET 4 OF 19)

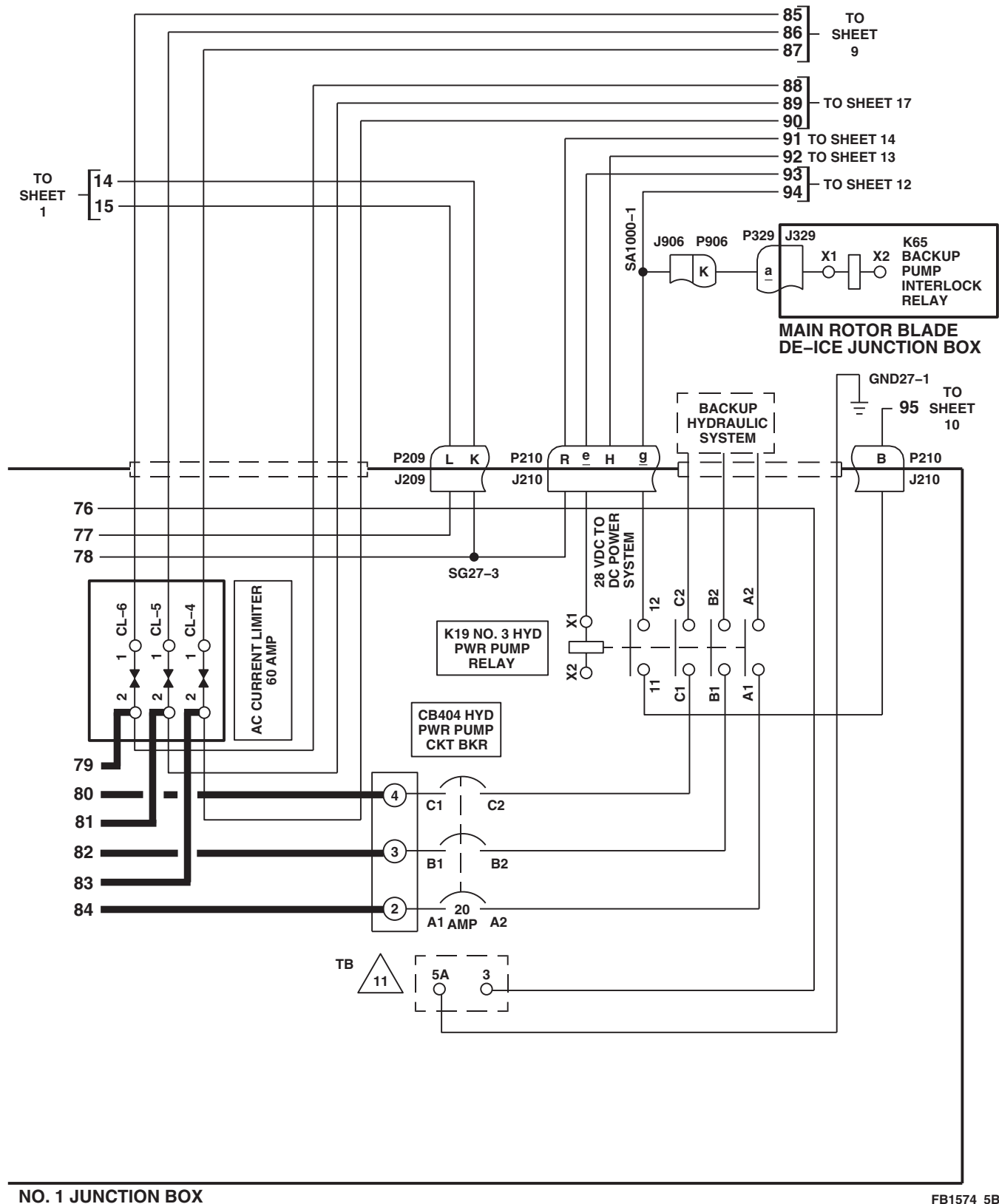


FIGURE 9-2-1-1. AC ELECTRICAL SYSTEM SCHEMATIC DIAGRAM (SHEET 5 OF 19)

9-2-1-46 Change 7

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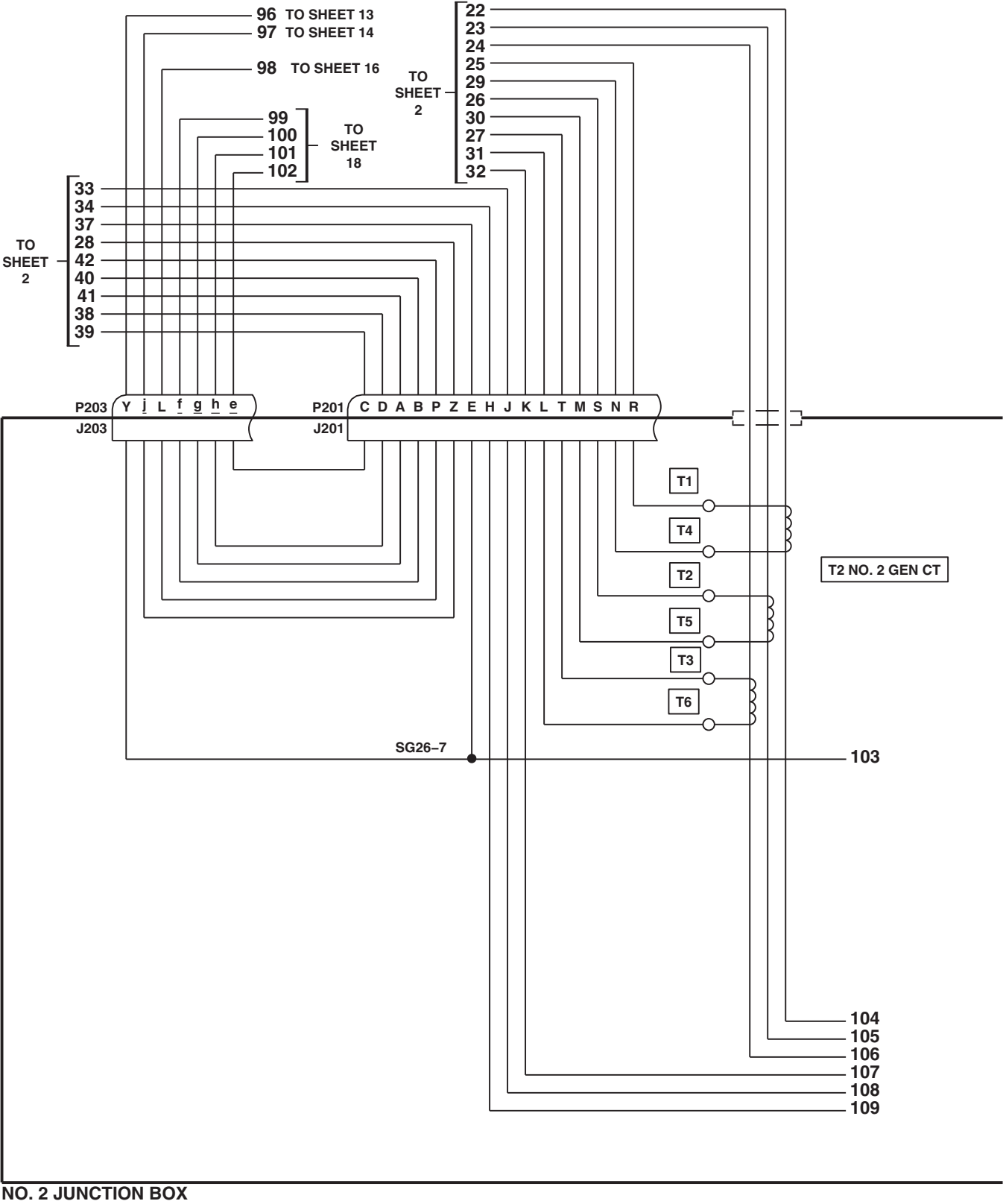


FIGURE 9-2-1-1. AC ELECTRICAL SYSTEM SCHEMATIC DIAGRAM (SHEET 6 OF 19)

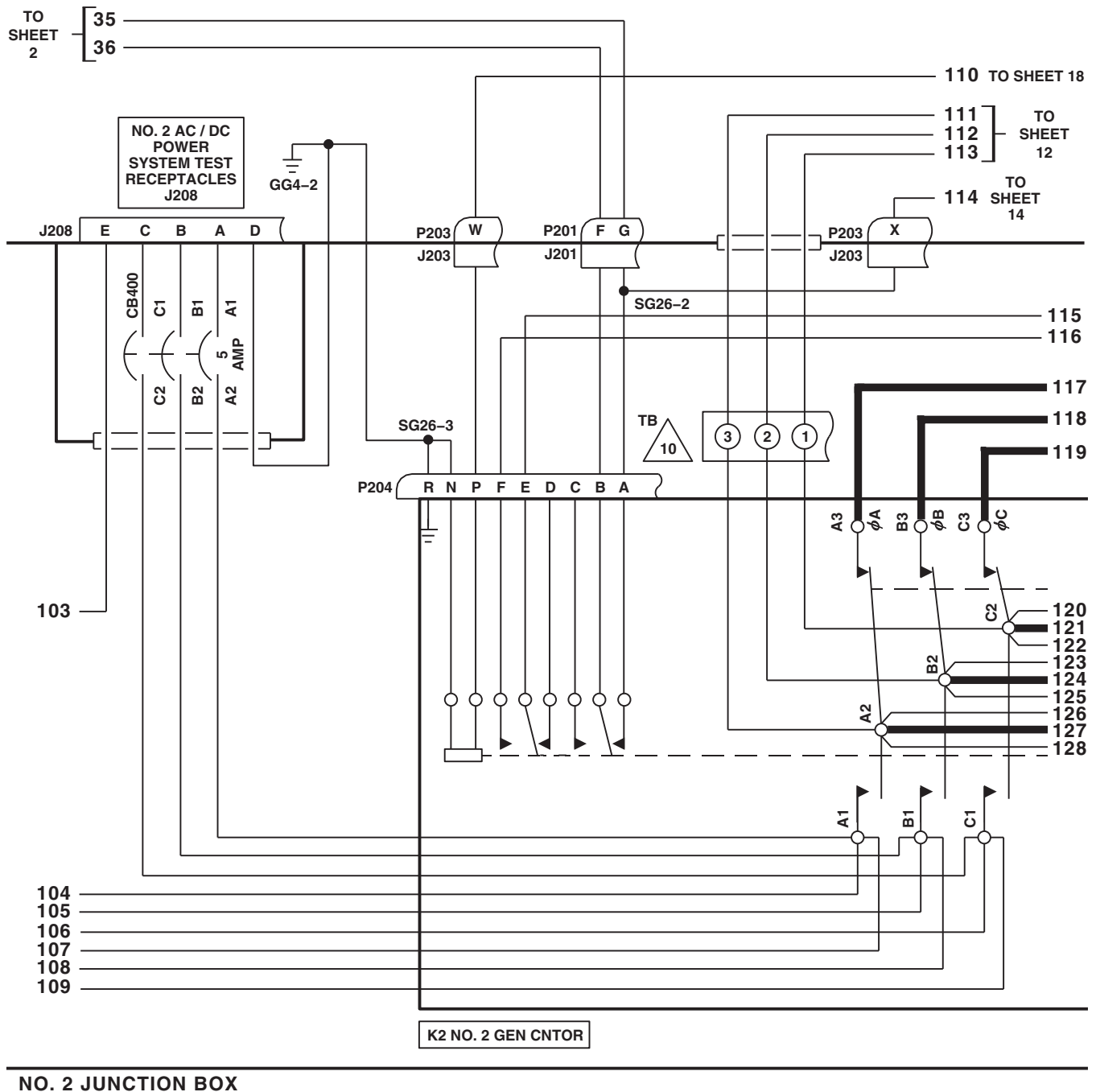
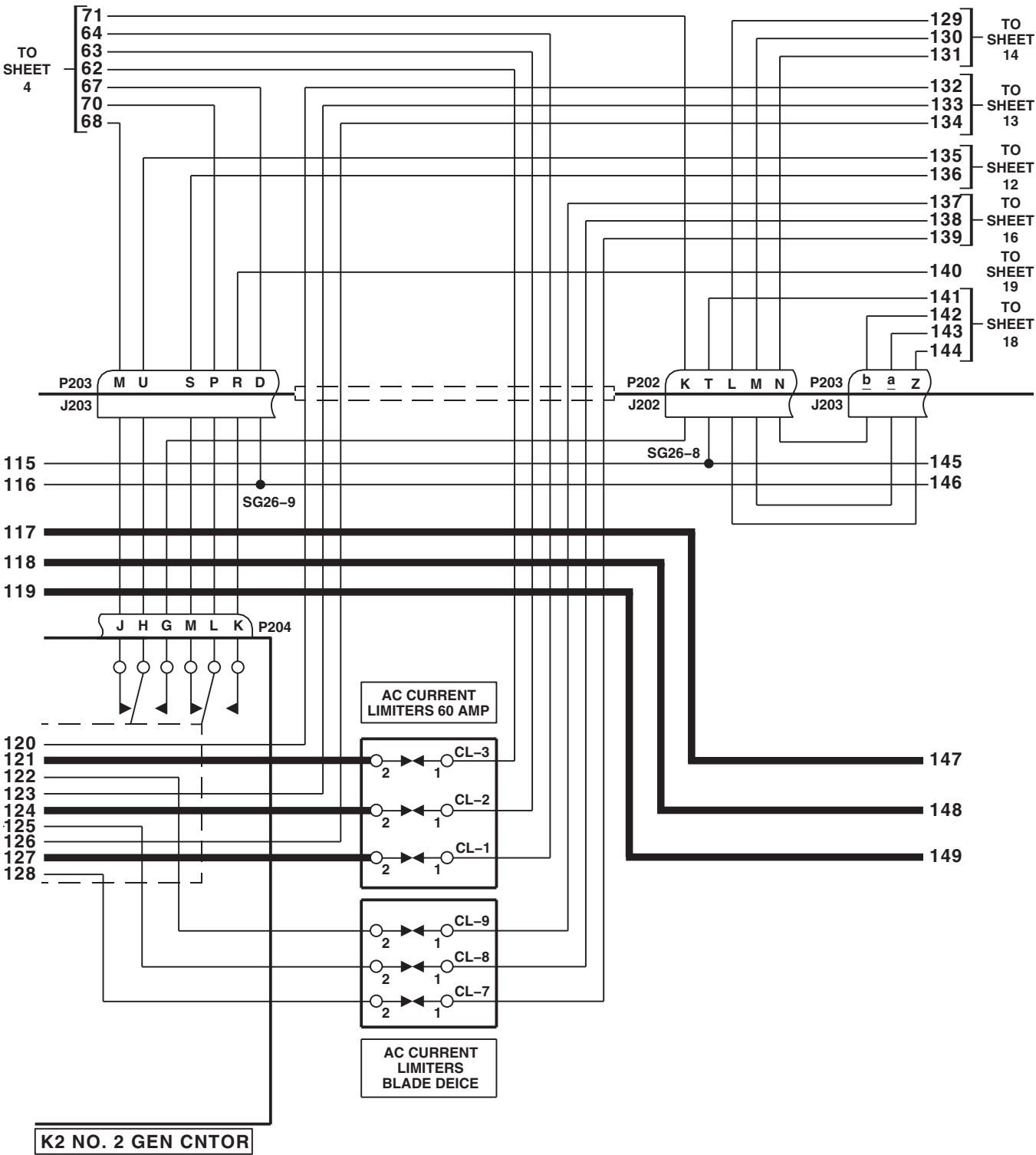
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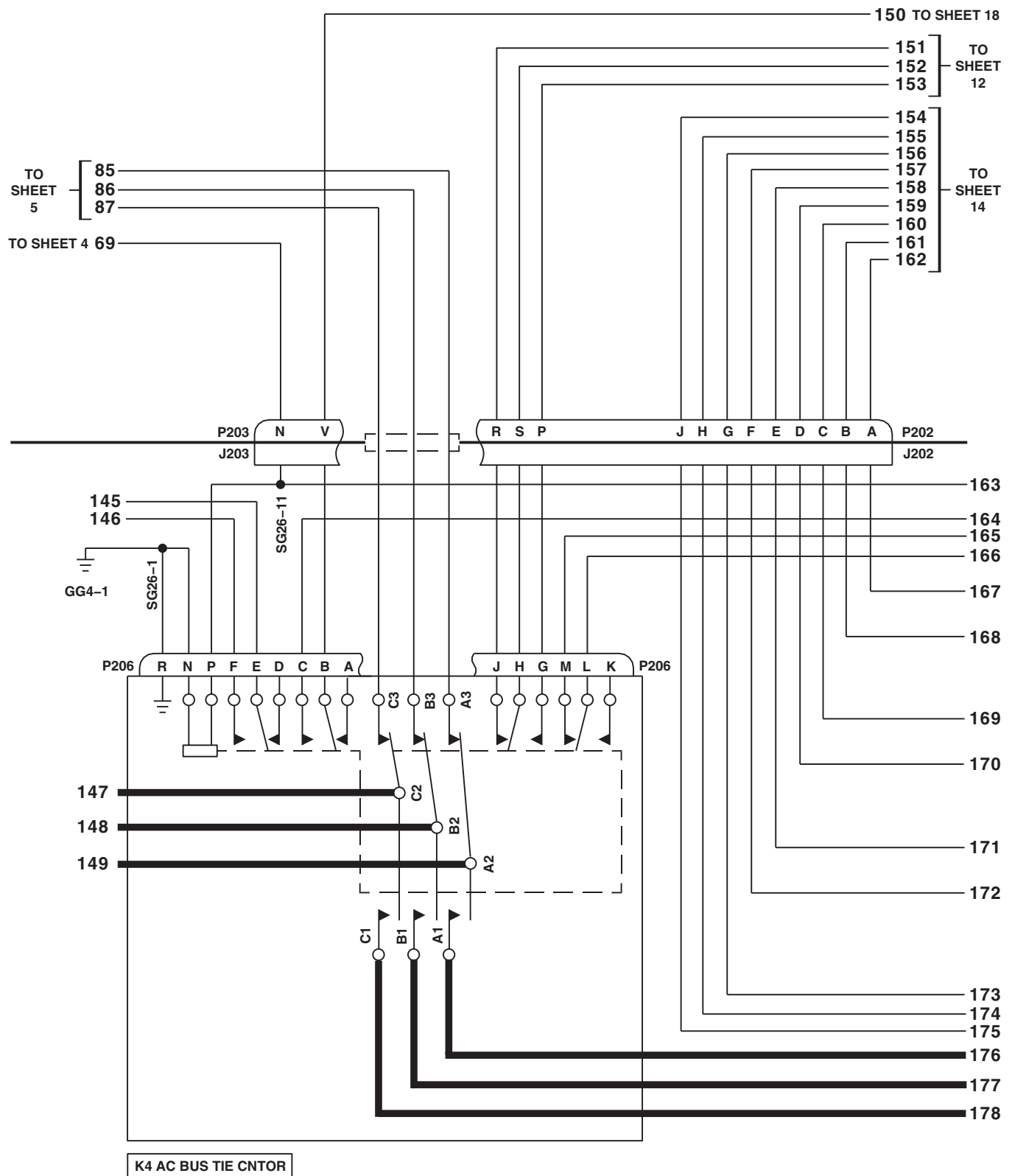
FIGURE 9-2-1-1. AC ELECTRICAL SYSTEM SCHEMATIC DIAGRAM (SHEET 7 OF 19)



NO. 2 JUNCTION BOX

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FIGURE 9-2-1-1. AC ELECTRICAL SYSTEM SCHEMATIC DIAGRAM (SHEET 8 OF 19)



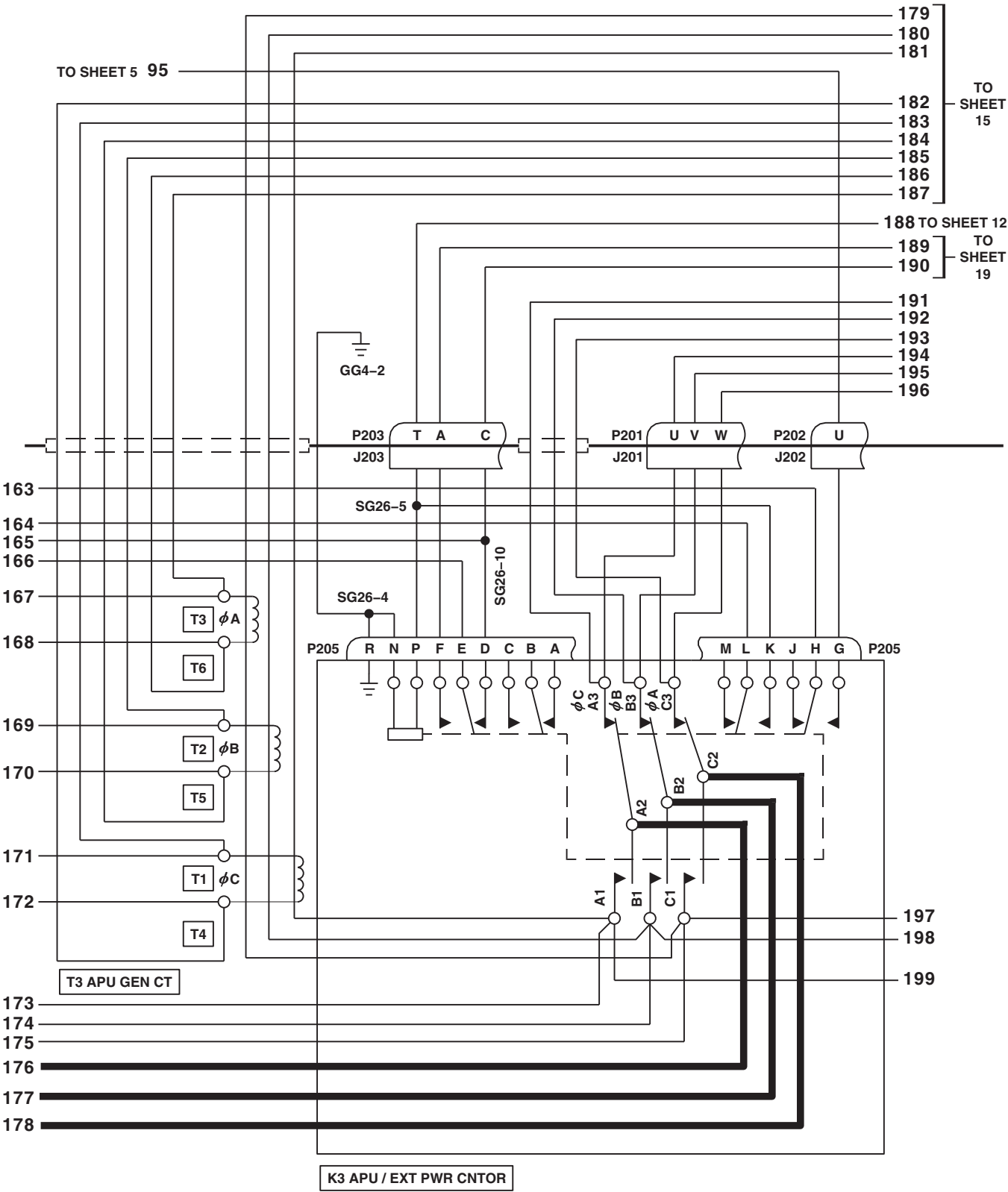
NO. 2 JUNCTION BOX

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FIGURE 9-2-1-1. AC ELECTRICAL SYSTEM SCHEMATIC DIAGRAM (SHEET 9 OF 19)

9-2-1-50

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NO. 2 JUNCTION BOX

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FIGURE 9-2-1-1. AC ELECTRICAL SYSTEM SCHEMATIC DIAGRAM (SHEET 10 OF 19)

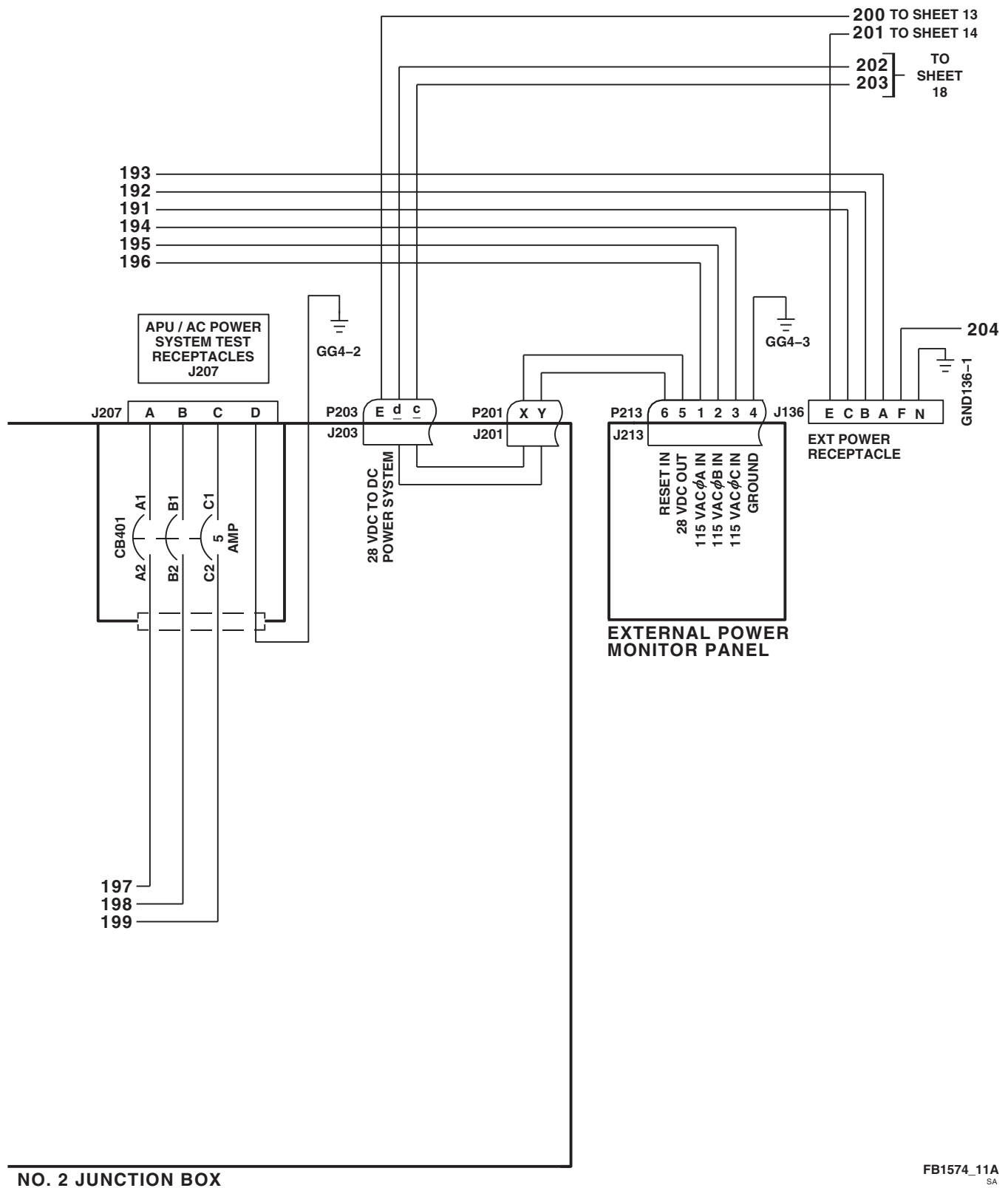
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FIGURE 9-2-1-1. AC ELECTRICAL SYSTEM SCHEMATIC DIAGRAM (SHEET 11 OF 19)

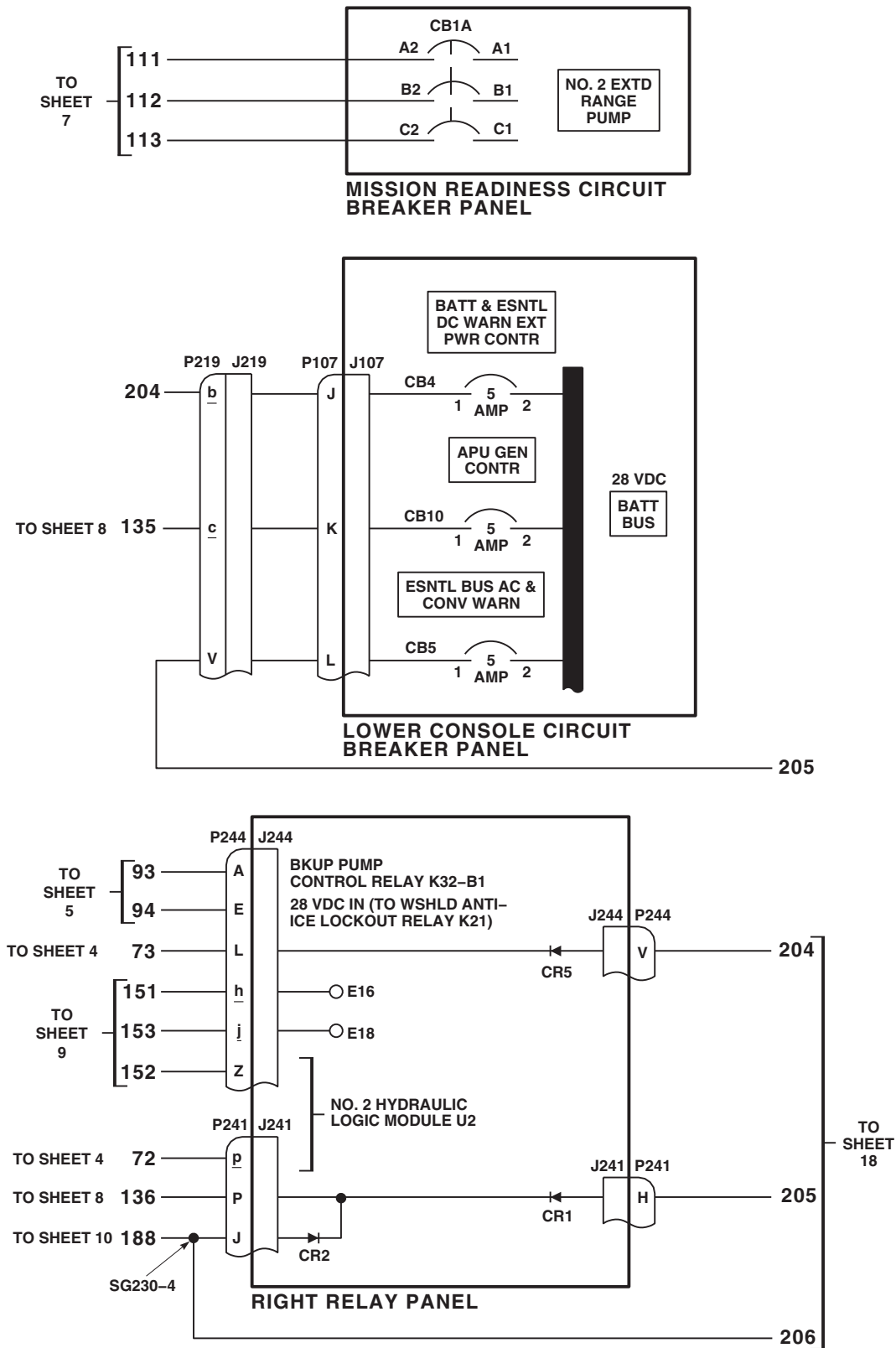


FIGURE 9-2-1-1. AC ELECTRICAL SYSTEM SCHEMATIC DIAGRAM (SHEET 12 OF 19)

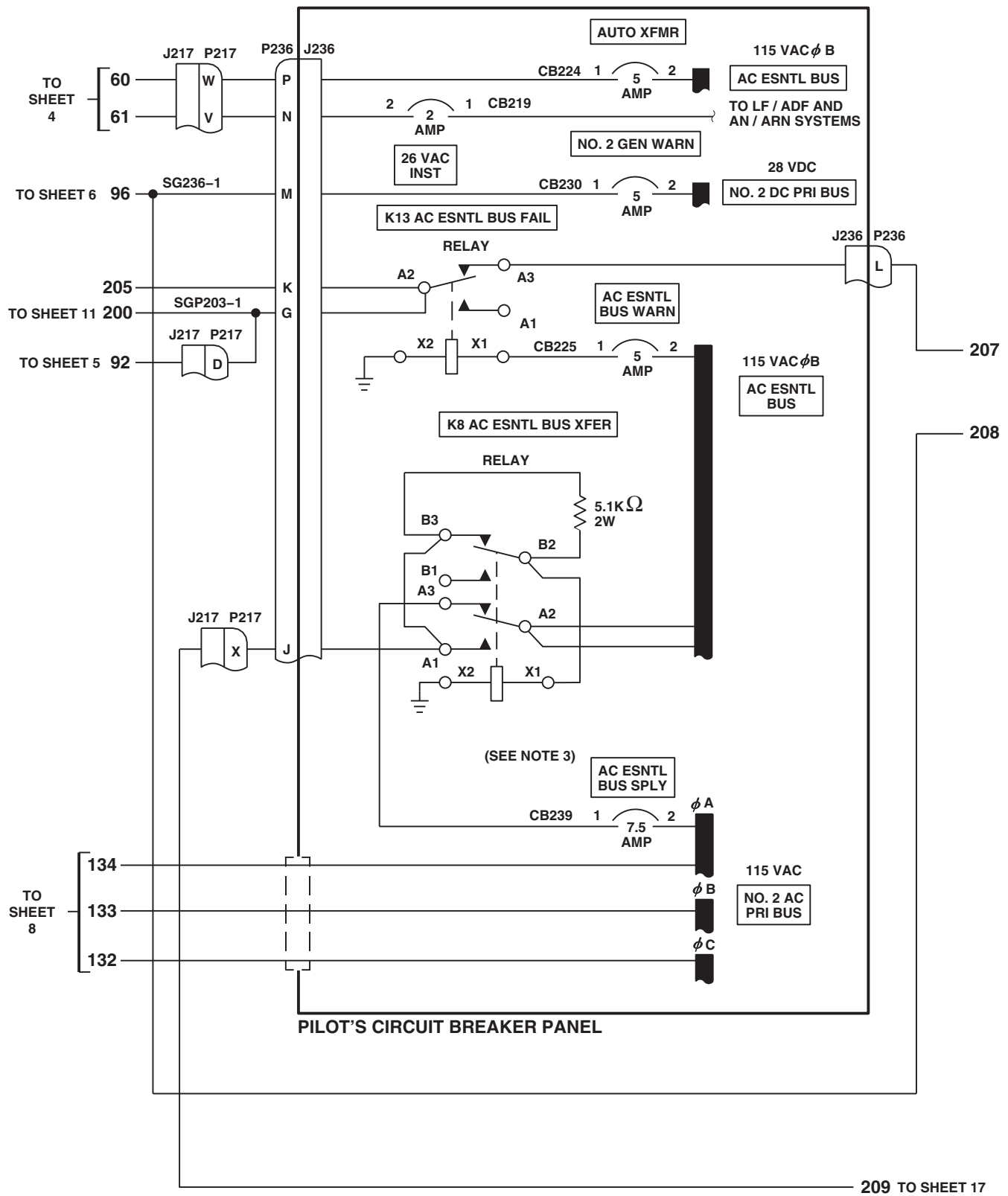


FIGURE 9-2-1-1. AC ELECTRICAL SYSTEM SCHEMATIC DIAGRAM (SHEET 13 OF 19)

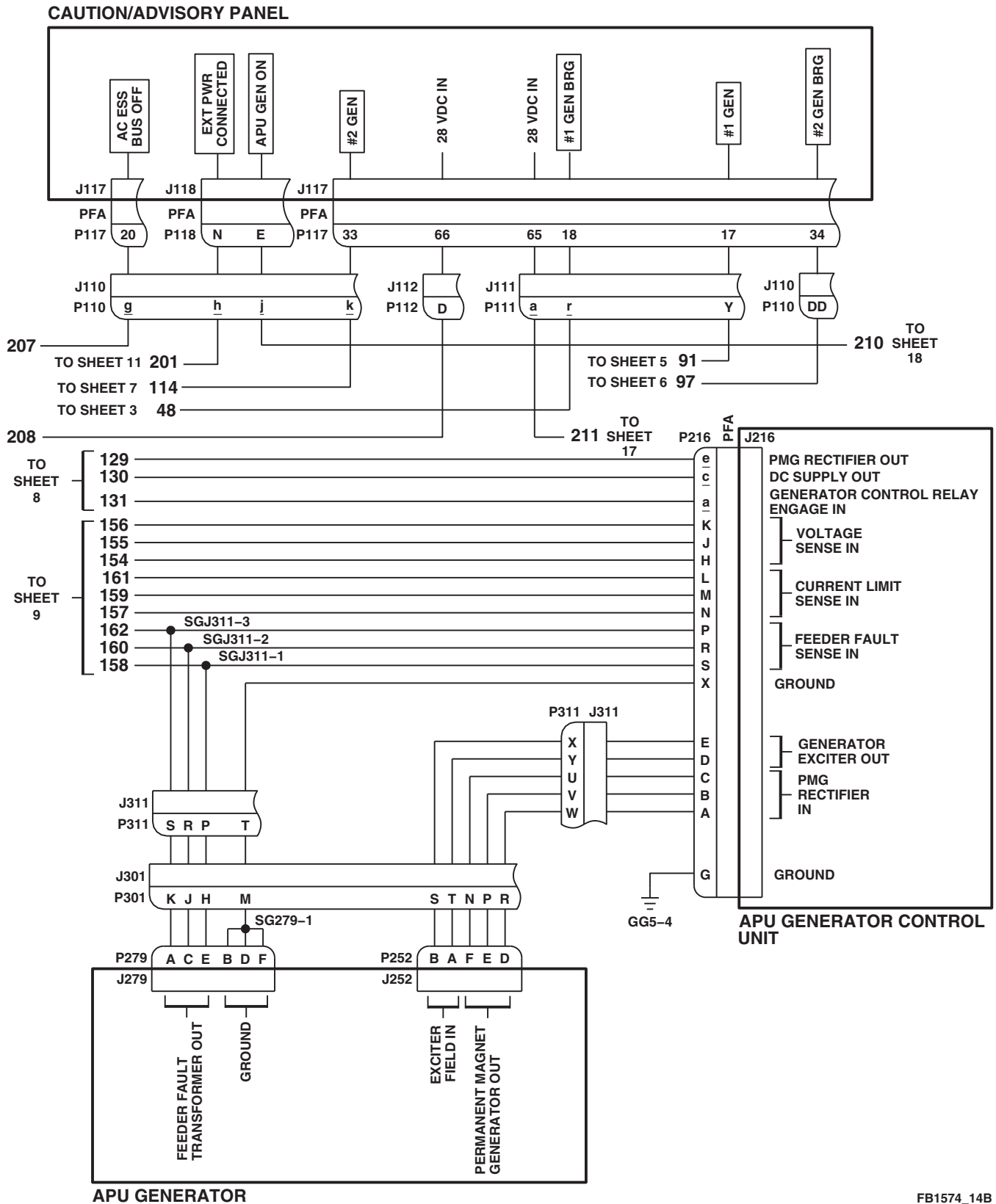


FIGURE 9-2-1-1. AC ELECTRICAL SYSTEM SCHEMATIC DIAGRAM (SHEET 14 OF 19)

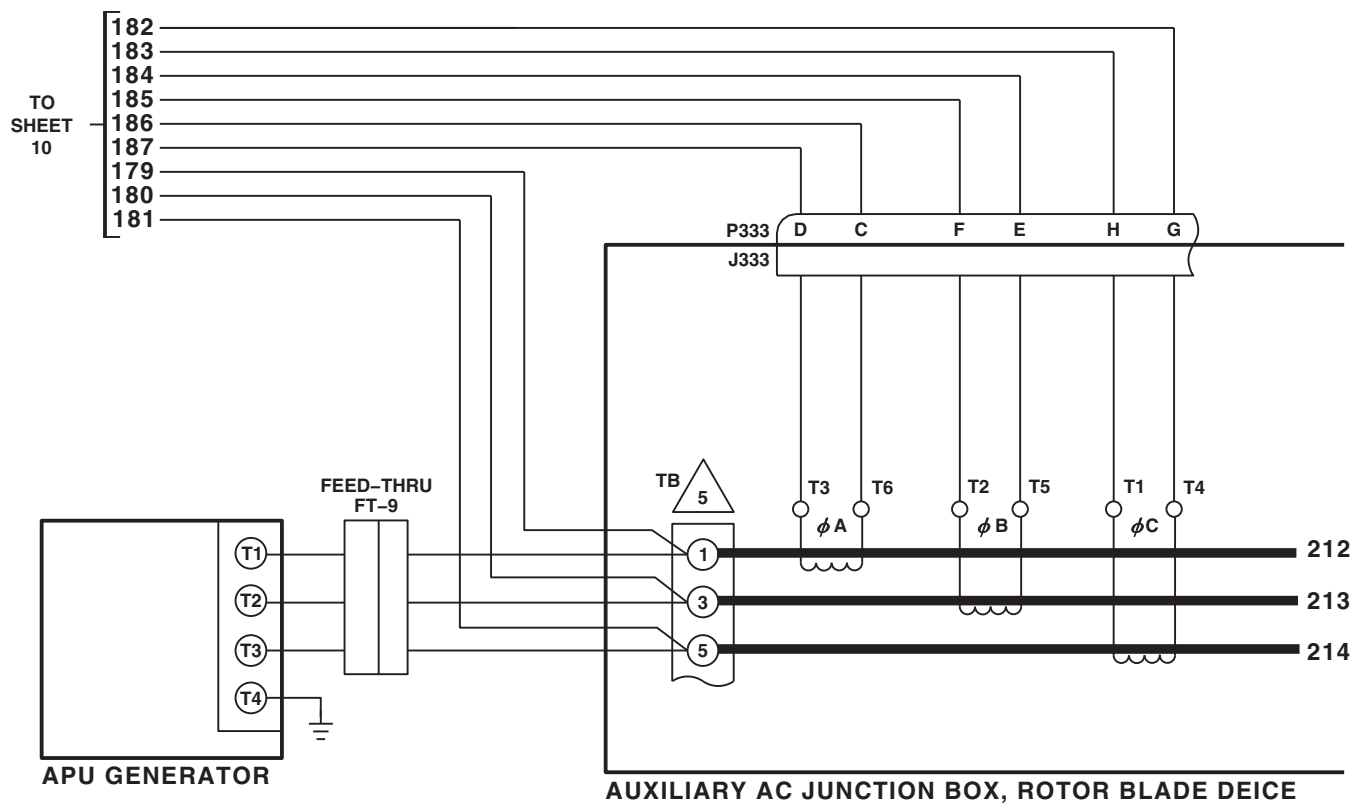
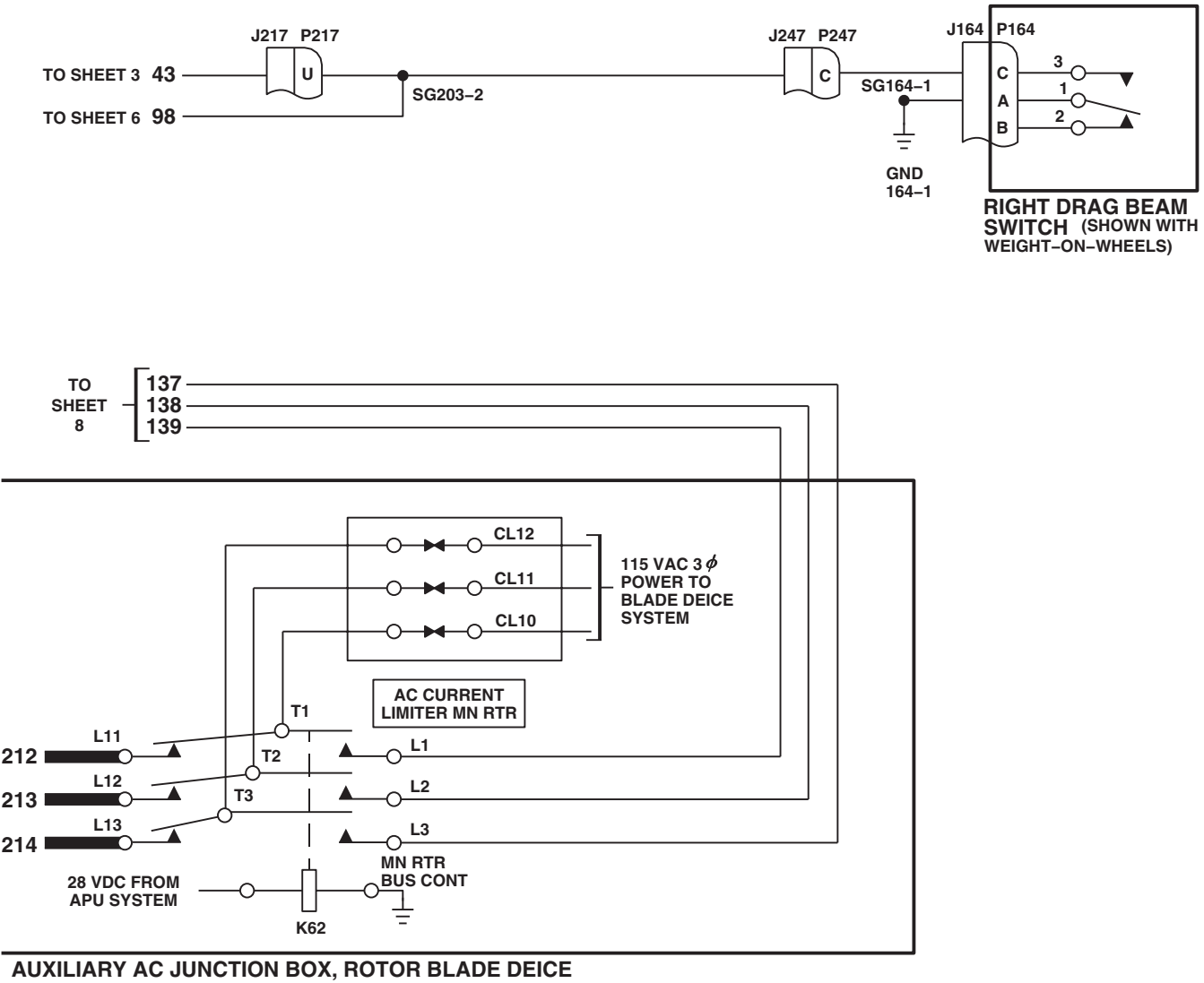
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FIGURE 9-2-1-1. AC ELECTRICAL SYSTEM SCHEMATIC DIAGRAM (SHEET 15 OF 19)



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FIGURE 9-2-1-1. AC ELECTRICAL SYSTEM SCHEMATIC DIAGRAM (SHEET 16 OF 19)

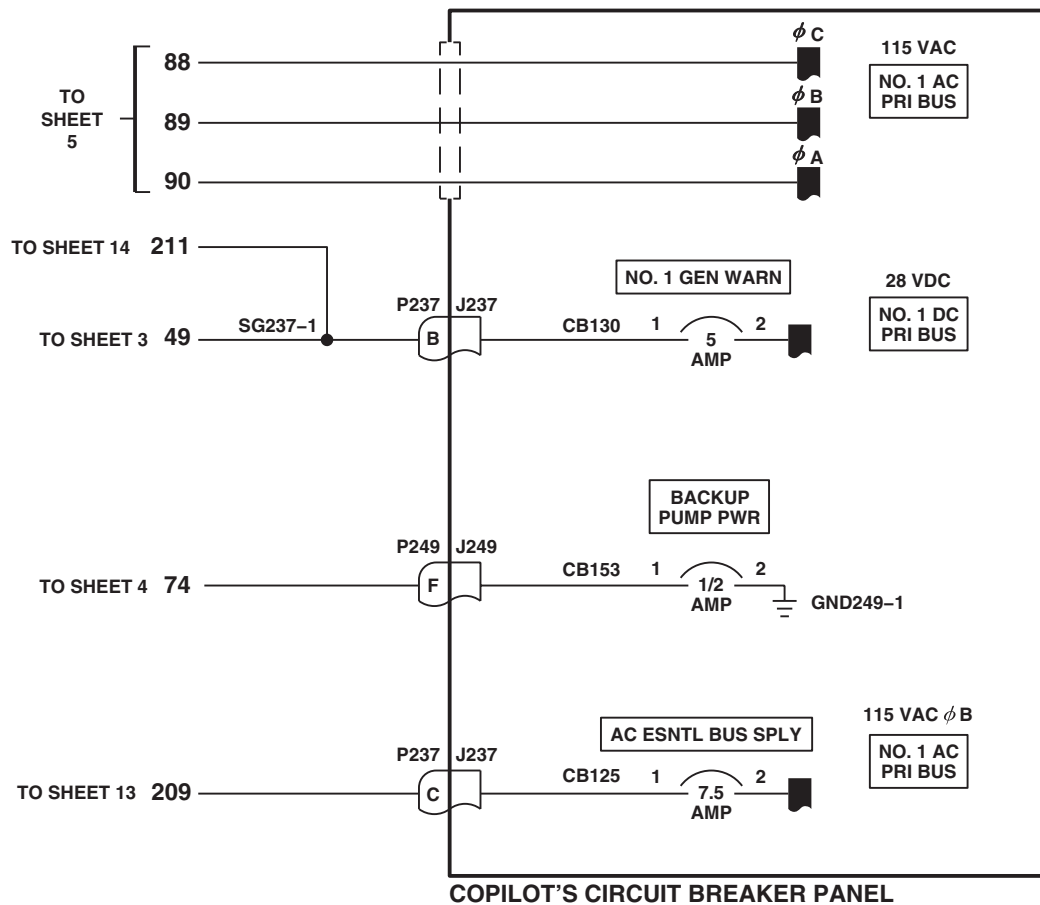
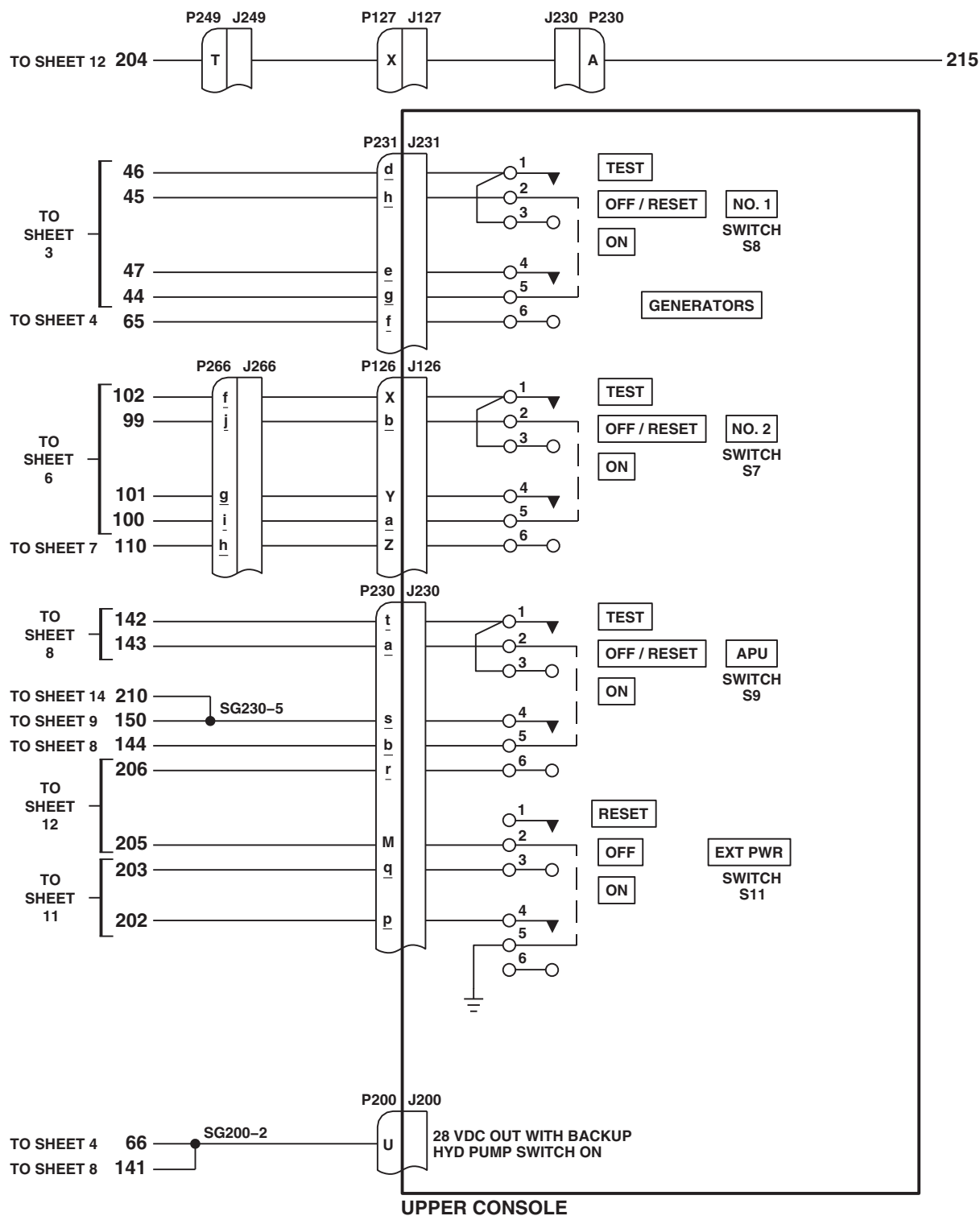
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FIGURE 9-2-1-1. AC ELECTRICAL SYSTEM SCHEMATIC DIAGRAM (SHEET 17 OF 19)

9-2-1-58 Change 7

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FIGURE 9-2-1-1. AC ELECTRICAL SYSTEM SCHEMATIC DIAGRAM (SHEET 18 OF 19)

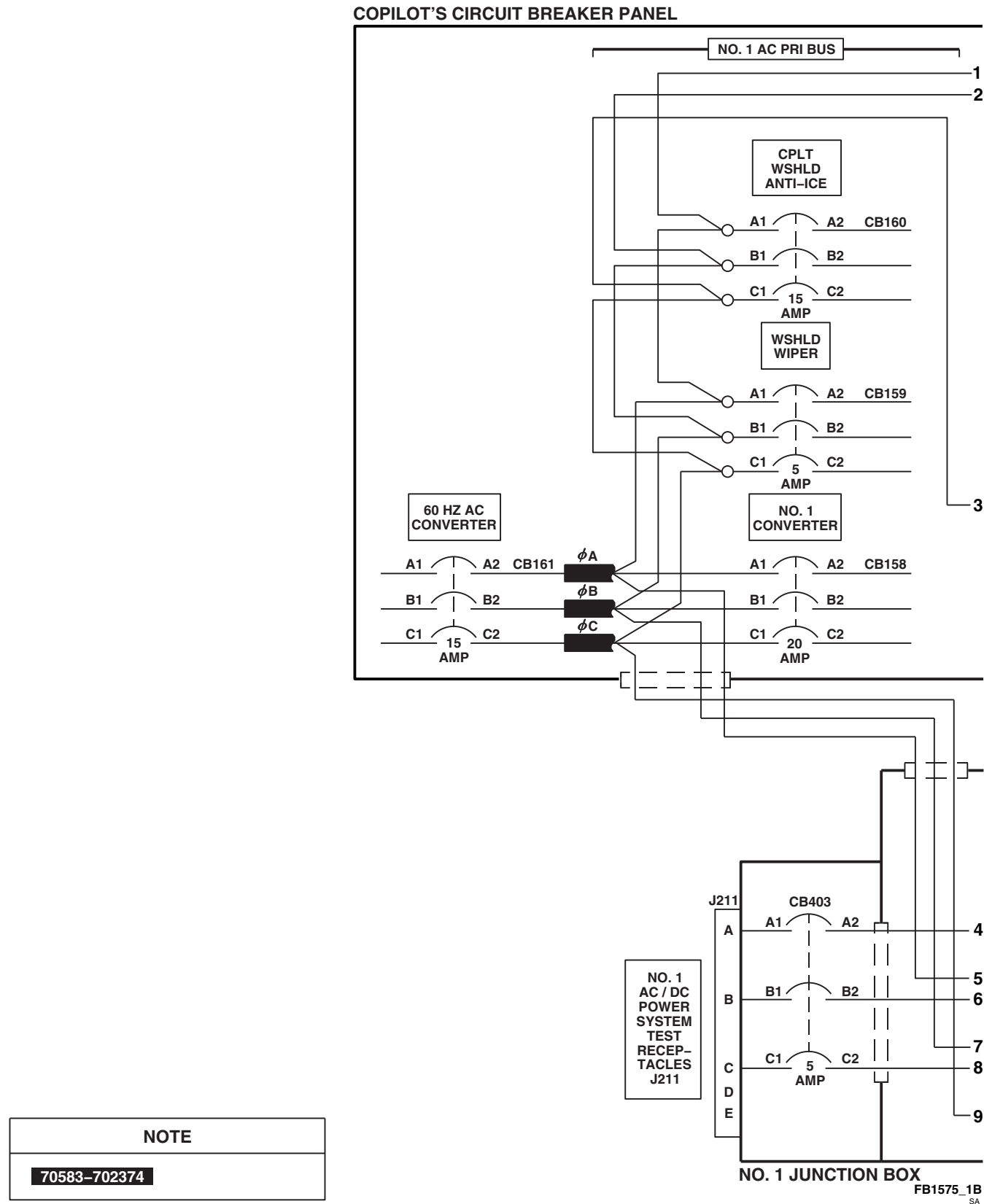


FIGURE 9-2-1-2. AC POWER DISTRIBUTION DIAGRAM (SHEET 1 OF 5)

COPILOT'S CIRCUIT BREAKER PANEL

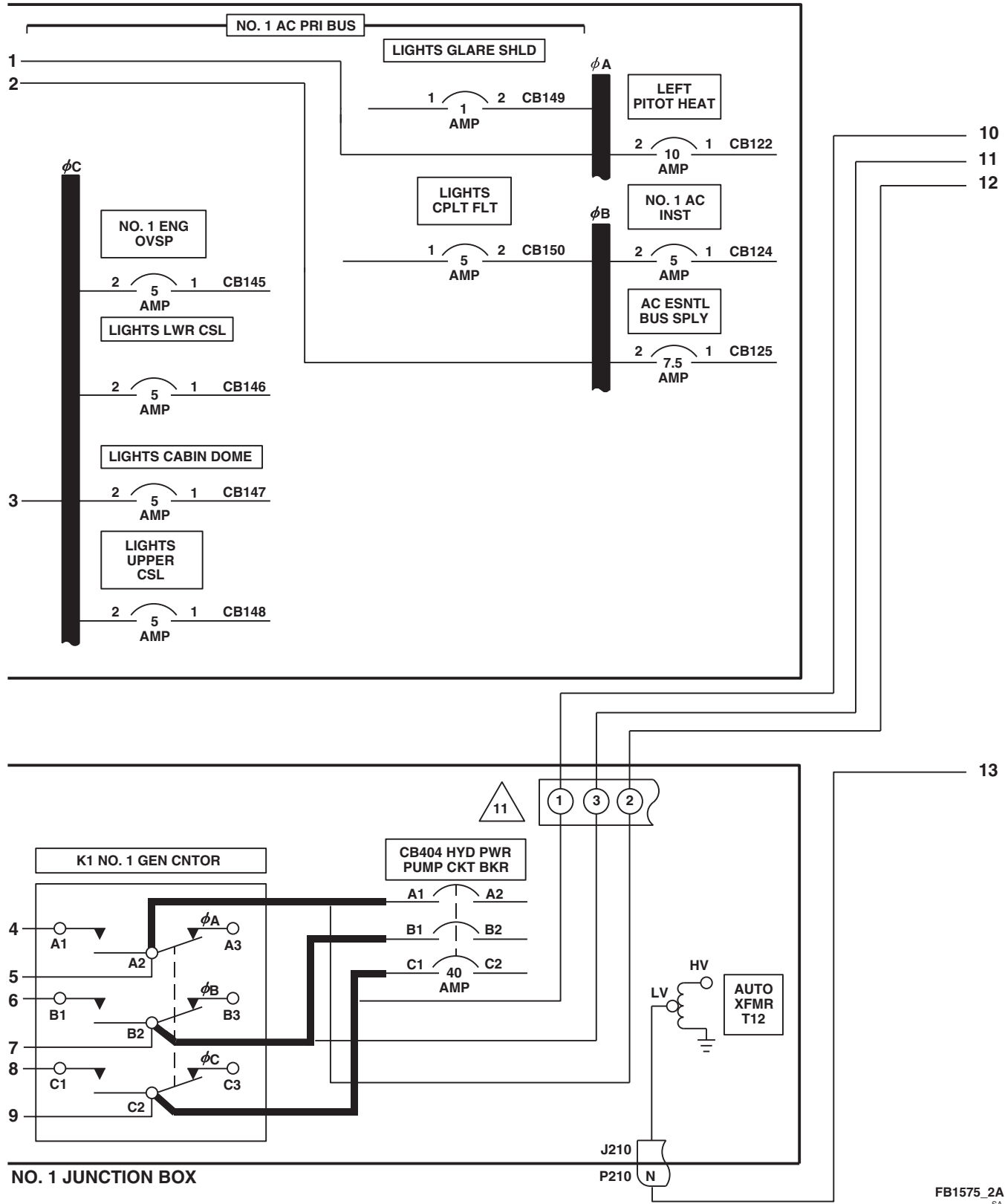
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FIGURE 9-2-1-2. AC POWER DISTRIBUTION DIAGRAM (SHEET 2 OF 5)

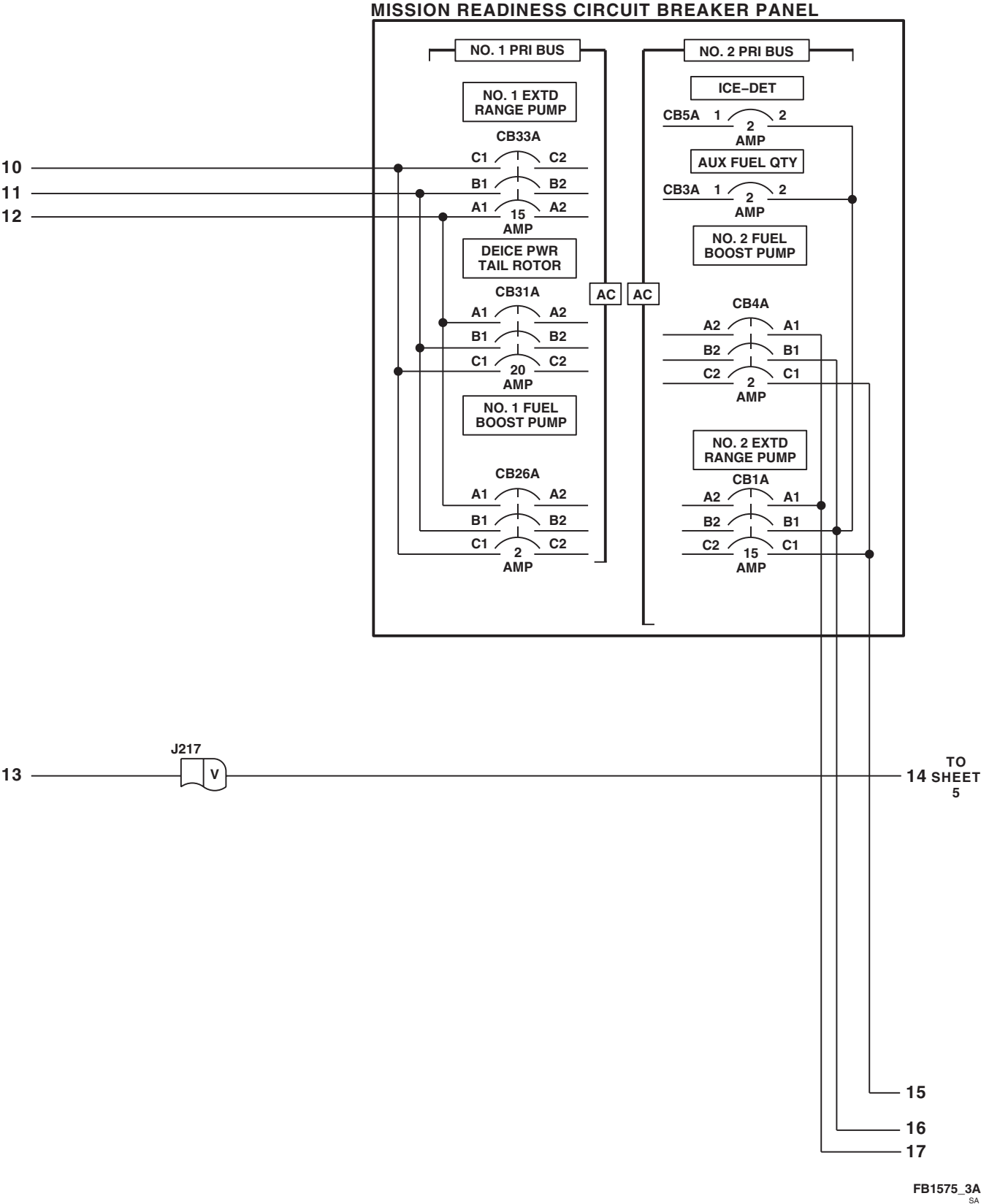
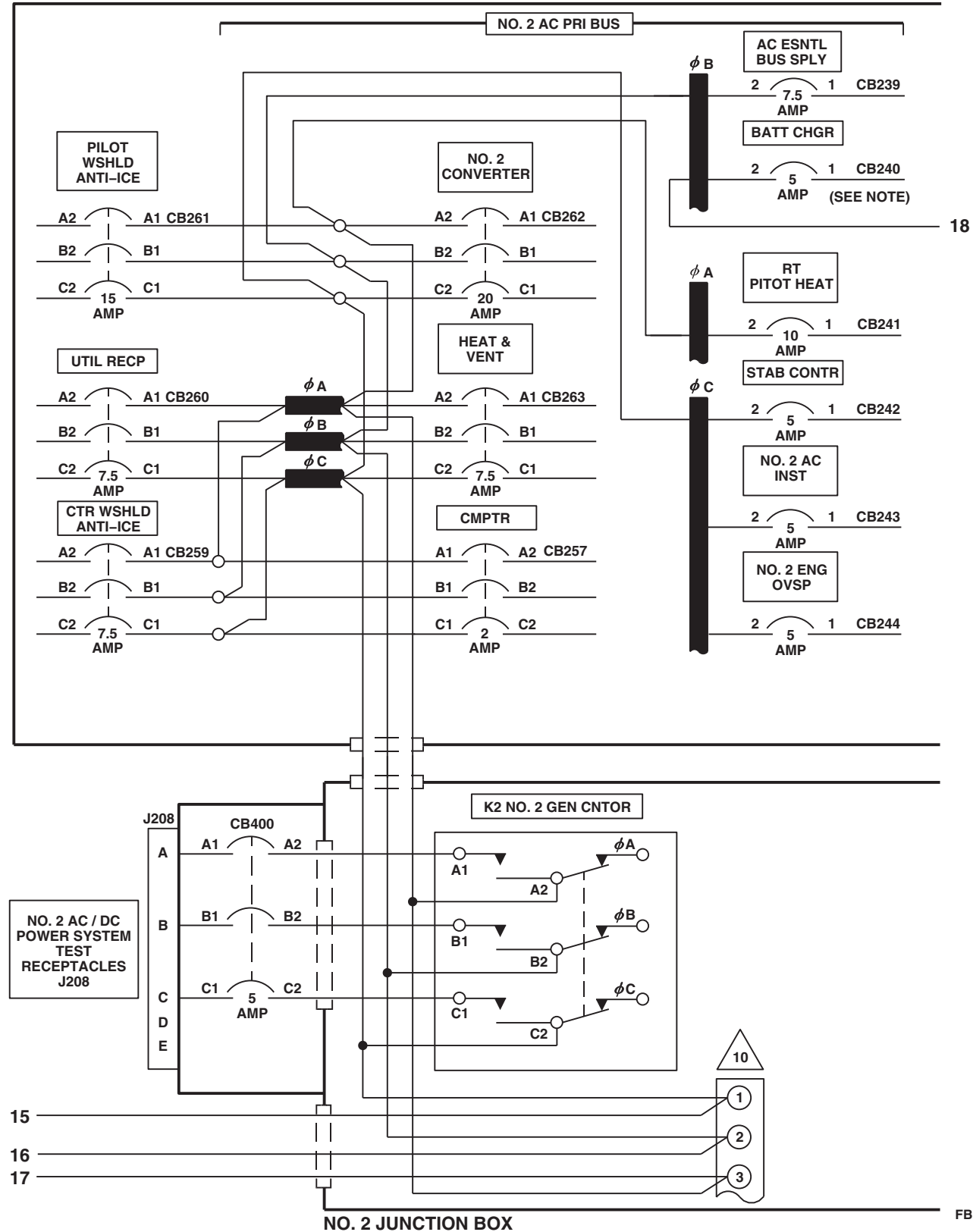


FIGURE 9-2-1-2. AC POWER DISTRIBUTION DIAGRAM (SHEET 3 OF 5)

PILOT'S CIRCUIT BREAKER PANEL



PILOT'S CIRCUIT BREAKER PANEL

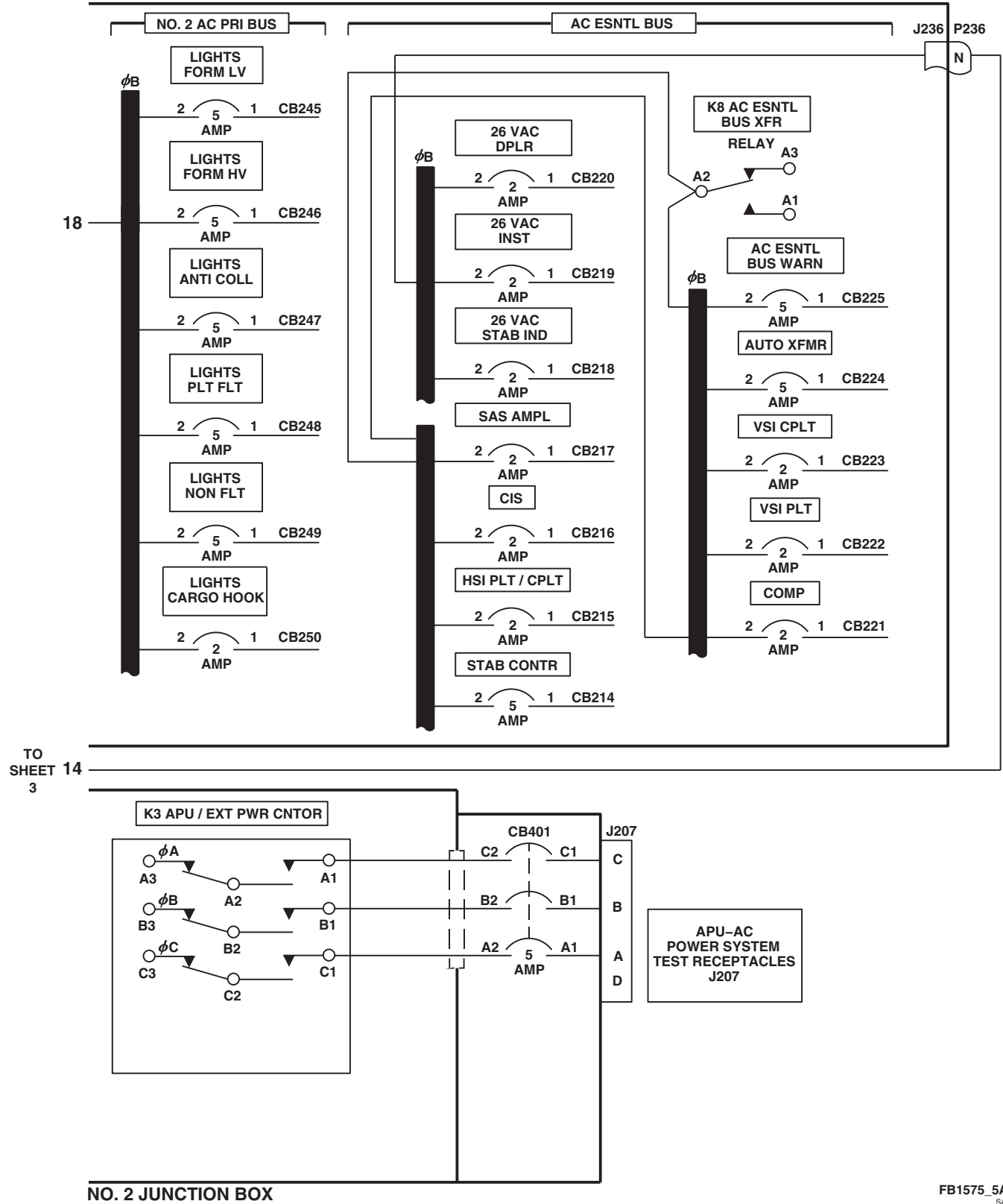


FIGURE 9-2-1-2. AC POWER DISTRIBUTION DIAGRAM (SHEET 5 OF 5)

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9-2-1-65/(9-2-1-66 Blank)

9-2-2 DC ELECTRICAL SYSTEM.

INITIAL SETUP

Test Equipment

- Multimeter, Fluke 45
- Battery Sensor Relay Tester, Figure H-255, Appendix H

Tools

- Electrical Repairer Toolkit

References

- PARA 1-6-16

Equipment Conditions

- External Power Available or APU Operational
- Battery Serviced and Fully Charged

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).

Special Instructions

The DC Electrical System operational/troubleshooting procedure is subdivided as follows:

- (1) Battery
- (2) No. 1 Converter
- (3) No. 2 Converter
- (4) Utility DC
- (5) Battery Low Sensing **702375 - SUBQ**

The battery sensor relay tester may be used at the bench level to check the battery low sensor relay. The following preset conditions must be met prior to applying power from a bench power supply:

- (1) S2 must be set to the on position on the battery sensor relay tester.
- (2) The rheostat R1 on the battery sensor relay tester must be set to approximately 5 ohms (Ω) to prevent damage to the zenner diode.

9-2-2.1 Battery.

NOTE

Do not start APU or connect external electrical power for battery system checkout.

- a. Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.
- b. Place BATT switch to ON.

RESULT:

- (1) BATT LOW CHARGE capsule shall be off.

- (2) **70583-702374** BATTERY FAULT capsule shall be off. **70583-702374**
- (3) DC ESS BUS OFF capsule shall be off.
- (4) SAS OFF, STABILATOR, AC ESS BUS OFF, and BOOST SERVO OFF capsules shall be on.
- (5) #1 CONV capsule shall be on.
- (6) #2 CONV capsule shall be on.

- If no caution/advisory capsules go on, go to **70583-702374** TABLE NO. 9-2-2-1 or **702375 - SUBQ** TABLE NO. 9-2-2-2.

- **70583-702374** ➤ If result (1) is not as specified, go to TABLE NO. 9-2-2-3.■

- **702375 - SUBQ** ➤ If result (1) is not as specified, perform Battery Low Sensing procedure.■

- **70583-702374** ➤ **EMEP** ➤ If result (2) is not as specified, place BATT switch to OFF, remove pin filter adapter (P123) (PARA 9-4-98) and check.■

- (a) If pin filter adapter is good, check continuity between P123 and P141.

- 1 If continuity is present, replace battery or battery conditioner/analyzer as required (PARA 9-4-34 or PARA 9-4-35).

- 2 If continuity is not present, repair/replace wiring as required (Appendix F).

- (b) If pin filter adapter is not good, replace it (PARA 9-4-98).

- (c) Install pin filter adapter and perform EME check (PARA 9-4-98).

- **70583-702374** ➤ **W/O EMEP** ➤ If result (2) is not as specified, place BATT switch to OFF and check continuity between P123 and P141.■

- (a) If continuity is present, replace battery or battery conditioner/analyzer as required (PARA 9-4-34 or PARA 9-4-35).

- (b) If continuity is not present, repair/replace wiring as required (Appendix F).

- If result (3) is not as specified, go to TABLE NO. 9-2-2-4.

- If results (5) and (6) are not as specified, go to TABLE NO. 9-2-2-5.

- If result (5) is not as specified, go to TABLE NO. 9-2-2-6.

- If result (6) is not as specified, go to TABLE NO. 9-2-2-7.

- c. Pull out ESNTL BUS DC SPLY circuit breaker (BATT BUS - lower console circuit breaker panel).

RESULT: DC ESS BUS OFF capsule shall go on.

- If result is not as specified, check helicopter configuration. **EMEP** ➤ Go to (a).■
W/O EMEP ➤ Go to (b).■

- (a) **EMEP** ➤ Remove pin filter adapter P117 (PARA 9-4-98) and check.■

- 1 If pin filter adapter is good, go to 3.

- 2 If pin filter adapter is not good, replace pin filter adapter (PARA 9-4-98).

- 3 Install pin filter adapter (PARA 9-4-98). Go to (b).

- (b) **W/O EMEP** ➤ Check for 28 VDC between P117-36 and ground.■

- 1 If voltage is as specified, troubleshoot Caution/Advisory Warning System (PARA 8-2-4).

- 2 If voltage is not as specified, check continuity between P117-36 and P900-G.

- a If continuity is present, replace right relay panel (PARA 9-4-2).

- b If continuity is not present, repair/replace wiring (Appendix F).

- d. Push in ESNTL BUS DC SPLY circuit breaker (lower console circuit breaker panel).
- e. Place upper console BATT switch to OFF.
- f. If no other check is to be done, turn off electrical power (PARA 1-6-16).

9-2-2.2 No. 1 Converter.

- a. Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.
- b. Turn on electrical power (PARA 1-6-16), if not already done.
- c. Make sure upper console BATT switch is OFF.
- d. Pull out NO. 2 CONVERTER circuit breaker (NO. 2 AC PRI BUS - pilot's circuit breaker panel).

RESULT:

- (1) #1 CONV capsule shall be off.
 - (2) #2 CONV capsule shall be on.
 - (3) DC ESS BUS OFF capsule shall be off.
- If result (1) is not as specified, go to TABLE NO. 9-2-2-8.
 - If result (2) is not as specified, go to TABLE NO. 9-2-2-7.
 - If result (3) is not as specified, go to TABLE NO. 9-2-2-4.
- e. Pull out ESNTL DC SENSE circuit breaker (DC ESNTL BUS - upper console circuit breaker panel).

RESULT: DC ESS BUS OFF capsule shall go on.

- If result is not as specified, go to TABLE NO. 9-2-2-9.

- f. Push in ESNTL DC SENSE circuit breaker (DC ESNTL BUS - upper console circuit breaker panel).
- g. Push in NO. 2 CONVERTER circuit breaker (NO. 2 AC PRI BUS - pilot's circuit breaker panel).
- h. If no other check is to be done, turn off electrical power (PARA 1-6-16).

9-2-2.3 No. 2 Converter.

- a. Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.
- b. Turn on electrical power (PARA 1-6-16), if not already done.
- c. Make sure upper console BATT switch is OFF.
- d. Pull out NO. 1 CONVERTER circuit breaker (NO. 1 AC PRI BUS - copilot's circuit breaker panel).

RESULT:

- (1) #2 CONV capsule shall be off.
 - (2) #1 CONV capsule shall be on.
 - (3) DC ESS BUS OFF capsule shall be off.
- If result (1) is not as specified, go to TABLE NO. 9-2-2-10.
 - If result (2) is not as specified, go to TABLE NO. 9-2-2-6.
 - If result (3) is not as specified, go to TABLE NO. 9-2-2-4.
- e. Pull out ESNTL DC SENSE circuit breaker (DC ESNTL BUS - upper console circuit breaker panel).

RESULT: DC ESS BUS OFF capsule shall go on.

- If result is not as specified, replace K9 NO. 2 DC ESNTL BUS SPLY relay (PARA 9-4-32).

- f. Push in ESNTL DC SENSE circuit breaker (DC ESNTL BUS - upper console circuit breaker panel).
- g. Push in NO. 1 CONVERTER circuit breaker (NO. 1 AC PRI BUS - copilot's circuit breaker panel).
- h. If no other check is to be done, turn off electrical power (PARA 1-6-16).

9-2-2.4 Utility DC.

- a. Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.
- b. Make sure upper console BATT switch is OFF.
- c. Turn on electrical power (PARA 1-6-16), if not already done.
- d. Check for 28 VDC between J258-B and J258-A.

RESULT: Multimeter shall indicate 28 VDC.

- If result is not as specified, check for 28 VDC between terminal 1 (UTIL RECP CABIN circuit breaker, NO. DC PRI BUS - copilot's circuit breaker panel) and ground.
 - (a) If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and J258-B, or between J258-A and ground, as required (Appendix F).
 - (b) If voltage is not as specified, replace circuit breaker (PARA 9-4-1).

- e. Check for 28 VDC between J265-B and J265-A.

RESULT: Multimeter shall indicate 28 VDC.

- If voltage is as specified, repair/replace wiring between terminal 1 (CMD CSL SET circuit breaker, NO. DC PRI BUS - copilot's

circuit breaker panel) and J265-B, or between J265-A and ground, as required (Appendix F).

- If voltage is not as specified, replace circuit breaker (PARA 9-4-1).

- f. If no other check is to be done, turn off electrical power (PARA 1-6-16).

9-2-2.5 Battery Low Sensing 702375 - SUBQ .

- a. Turn off electrical power (PARA 1-6-16).
- b. Make sure BATT switch on upper console is OFF.
- c. Disconnect electrical connector P140 from battery.
- d. Connect P140 to TP3 and TP4 of battery sensor relay tester as follows:
 - (1) P140(+) to red alligator clip.
 - (2) P140(-) to black alligator clip.
 - (3) Wire with red banana plug to TP3 on tester.
 - (4) Wire with black banana plug to TP4 on tester.
- e. Using battery connector P1 with pigtail wires, connect battery to tester as follows:
 - (1) Wire with black banana plug to TP2 of tester.
 - (2) Wire with red banana plug to TP1 of tester.
 - (3) Connector P1 to battery.
- f. Connect multimeter to TP3 and TP4 on tester.
- g. Place BATT switch on upper console to ON.
- h. On tester, push in circuit breaker CB1 and place switches S1 and S2 on.

- i. On tester, turn control R1 until multimeter indicates 24 VDC or higher.

RESULT: BATT LOW CHARGE capsule shall go on.

RESULT: BATT LOW CHARGE capsule shall be off.

- If result is not as specified, go to TABLE NO. 9-2-2-11A.

- If result is not as specified, go to TABLE NO. 9-2-2-11.

- j. On tester, turn R1 slowly clockwise until battery voltage decreases to 22 VDC.

- k. Place BATT switch on upper console to OFF.
- l. Disconnect all wiring from tester to electrical connector P140 and battery.
- m. Reconnect P140 to battery.

TABLE NO. 9-2-2-1. No Caution/Advisory Capsules Are On With BATT Switch ON.

70583-702374	TEST OR INSPECTION CORRECTIVE ACTION
	1. Check for greater than 23 VDC between K10-A3 and ground. Step 1. If voltage is as specified, repair/replace wiring between K10-A3 and DC ESNTL BUS (Appendix F). Go to 37. Step 2. If voltage is not as specified, go to 2. 2. Check for greater than 23 VDC between K10-A4 and ground. Step 1. If voltage is as specified, replace relay K10 (PARA 9-4-32). Go to 37. Step 2. If voltage is not as specified, go to 3. 3. Check for greater than 23 VDC between K9-A3 and ground. Step 1. If voltage is as specified, repair/replace wiring between K9-A3 and K10-A4 (Appendix F). Go to 37. Step 2. If voltage is not as specified, go to 4. 4. Check for greater than 23 VDC between K9-A1 and ground. Step 1. If voltage is as specified, repair/replace wiring between K9-A3 and K9-A1 (Appendix F). Go to 37. Step 2. If voltage is not as specified, go to 5. 5. Check for greater than 23 VDC between K9-A2 and ground. Step 1. If voltage is as specified, go to 6. Step 2. If voltage is not as specified, go to 20.

TABLE NO. 9-2-2-1. No Caution/Advisory Capsules Are On With BATT Switch ON. (Cont)

70583-702374

**TEST OR INSPECTION
CORRECTIVE ACTION**

6. Check for greater than 23 VDC between K9-X1 and ground.

Step 1. If voltage is as specified, go to 7.

Step 2. If voltage is not as specified, go to 8.

7. Check continuity between K9-X2 and ground.

Step 1. If continuity is present, replace NO. 2 DC ESNTL BUS SPLY relay K9 (PARA 9-4-32). Go to 37.

Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 37.

8. Check continuity between P123-N and K9-X1.

Step 1. If continuity is present, go to 9.

Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 37.

9. Check for greater than 23 VDC between P123-M and ground.

Step 1. If voltage is as specified, go to 10.

Step 2. If voltage is not as specified, repair/replace wiring between P123-M and P238-G (Appendix F). Go to 37.

10. Check for greater than 23 VDC between P123-R and ground.

Step 1. If voltage is as specified, go to 11.

Step 2. If voltage is not as specified, go to 18.

11. Check continuity between P123-P and ground.

Step 1. If continuity is present, go to 12.

Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 37.

12. Check continuity between P123-D and ground.

Step 1. If continuity is present, go to 13.

Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 37.

13. Check for less than 0.2 VDC between P123-C and P123-V.

Step 1. If voltage is as specified, go to 14.

Step 2. If voltage is not as specified, go to 15.

TABLE NO. 9-2-2-1. No Caution/Advisory Capsules Are On With BATT Switch ON. (Cont)

70583-702374	TEST OR INSPECTION CORRECTIVE ACTION
14. Check for less than 0.2 VDC between J141-C and J141-H.	Step 1. If voltage is as specified, replace battery (PARA 9-4-34). Go to 37. Step 2. If voltage is not as specified, repair/replace wiring, as required, between (Appendix F), then go to 37: P123-V and P141-H P123-C and P141-C
15. Check for 0.2 to 1.03 VDC between P123-C and P123-V.	Step 1. If voltage is as specified, go to 16. Step 2. If voltage is not as specified, check for less than 23 VDC between P117-21 and ground. Step 3. If condition exists, replace battery conditioner/analyzer (PARA 9-4-35). Go to 37.
16. Check for greater than 23 VDC between P117-21 and ground.	Step 1. If voltage is as specified, replace battery (PARA 9-4-34). Go to 37. Step 2. If voltage is not as specified, go to 17.
17. Check continuity between P123-K and P117-21.	Step 1. If continuity is present, replace battery conditioner/analyzer (PARA 9-4-35). Go to 37. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 37.
18. Check for greater than 23 VDC between BATT switch terminal 6 and ground.	Step 1. If voltage is as specified, repair/replace wiring between P123-R and BATT switch terminal 6 (Appendix F), then go to 37. Step 2. If voltage is not as specified, go to 19.
19. Check for greater than 23 VDC between BATT switch terminal 5 and ground.	Step 1. If voltage is as specified, replace BATT switch S10 (PARA 9-4-5). Go to 37. Step 2. If voltage is not as specified, repair/replace wiring between BATT switch terminal 5 and terminal 1 (BATT BUS CONTR circuit breaker, BATT UTIL BUS - lower console circuit breaker panel) (Appendix F). Go to 37.

TABLE NO. 9-2-2-1. No Caution/Advisory Capsules Are On With BATT Switch ON. (Cont)

70583-702374

TEST OR INSPECTION
CORRECTIVE ACTION

- 20. Check for greater than 23 VDC between terminal 1 (ESNTL BUS DC SPLY circuit breaker, BATT BUS - lower console circuit breaker) and ground.**

Step 1. If voltage is as specified, repair/replace wiring between circuit breaker and K9-A2 (Appendix F). Go to **37**.

Step 2. If voltage is not as specified, go to **21**.

- 21. Check for greater than 23 VDC between terminal 2 (ESNTL BUS DC SPLY circuit breaker, BATT BUS - lower console circuit breaker) and ground.**

Step 1. If voltage is as specified, replace circuit breaker (PARA 9-4-1). Go to **37**.

Step 2. If voltage is not as specified, go to **22**.

- 22. Check for greater than 23 VDC between lower console circuit breaker panel terminal E1 and ground.**

Step 1. If voltage is as specified, repair/replace wiring between terminal E1 and BATT bus (Appendix F). Go to **37**.

Step 2. If voltage is not as specified, go to **23**.

- 23. Check for greater than 23 VDC between K7-A1 and ground.**

Step 1. If voltage is as specified, repair/replace wiring between relay K7 and terminal E1 (Appendix F). Go to **37**.

Step 2. If voltage is not as specified, go to **24**.

- 24. Check for greater than 23 VDC between K7-A2 and ground.**

Step 1. If voltage is as specified, go to **25**.

Step 2. If voltage is not as specified, go to **35**.

- 25. Check for greater than 23 VDC between K7-X1 and K7-X2.**

Step 1. If voltage is as specified, replace battery relay K7 (PARA 9-4-33). Go to **37**.

Step 2. If voltage is not as specified, go to **26**.

- 26. Check continuity between K7-X2 and ground.**

Step 1. If continuity is present, go to **27**.

Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **37**.

TABLE NO. 9-2-2-1. No Caution/Advisory Capsules Are On With BATT Switch ON. (Cont)

70583-702374	TEST OR INSPECTION CORRECTIVE ACTION
27. Check for greater than 23 VDC between BATT switch terminal 3 and ground.	Step 1. If voltage is as specified, repair/replace wiring between BATT switch terminal 3 and K7-X1 (Appendix F). Go to 37. Step 2. If voltage is not as specified, go to 28.
28. Check for greater than 23 VDC between BATT switch terminal 2 and ground.	Step 1. If voltage is as specified, replace BATT switch (PARA 9-4-5). Go to 37. Step 2. If voltage is not as specified, go to 29.
29. Check continuity between BATT switch terminal 2 and P238-G.	Step 1. If continuity is present, go to 30. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 37.
30. Check for greater than 23 VDC between P238-H and ground.	Step 1. If voltage is as specified, replace K6 NO. 2 DC PRI BUS CNTOR (PARA 9-4-30). Go to 37. Step 2. If voltage is not as specified, go to 31.
31. Check continuity between P238-H and P239-H.	Step 1. If continuity is present, go to 32. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 37.
32. Check for greater than 23 VDC between P239-G and ground.	Step 1. If voltage is as specified, replace K16 NO. 1 DC PRI BUS CNTOR (PARA 9-4-30). Go to 37. Step 2. If voltage is not as specified, go to 33.
33. Check for greater than 23 VDC between terminal 1 (BATT BUS CONTR circuit breaker, BATT UTIL BUS - lower console circuit breaker panel) and ground.	Step 1. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P239-G (Appendix F). Go to 37. Step 2. If voltage is not as specified, go to 34.

TABLE NO. 9-2-2-1. No Caution/Advisory Capsules Are On With BATT Switch ON. (Cont)

70583-702374	TEST OR INSPECTION CORRECTIVE ACTION
	34. Check for greater than 23 VDC between terminal 2 (BATT BUS CONTR circuit breaker, BATT UTIL BUS - lower console circuit breaker panel) and ground. Step 1. If voltage is as specified, replace circuit breaker (PARA 9-4-1). Go to 37. Step 2. If voltage is not as specified, repair/replace wiring between circuit breaker terminal 2 and K7-A2 (Appendix F). Go to 37. 35. Check continuity between: K7-A2 and P140 (+) K7-A2 and terminal E2 on lower console circuit breaker panel Step 1. If continuity is present, go to 36. Step 2. If continuity is not present, repair/replace wiring as required (Appendix F). Go to 37. 36. Check continuity between P140 (-) and ground. Step 1. If continuity is present, replace battery (PARA 9-4-34). Go to 37. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 37. 37. Procedure completed.

TABLE NO. 9-2-2-2. No Caution/Advisory Capsules Are On With BATT Switch ON.

702375 - SUBQ	TEST OR INSPECTION CORRECTIVE ACTION
	1. Check for greater than 23 VDC between K10-A3 and ground. Step 1. If voltage is as specified, repair/replace wiring between K10-A3 and DC ESNTL BUS (Appendix F). Go to 27. Step 2. If voltage is not as specified, go to 2. 2. Check for greater than 23 VDC between K10-A4 and ground. Step 1. If voltage is as specified, replace relay K10 (PARA 9-4-32). Go to 27. Step 2. If voltage is not as specified, go to 3.

TABLE NO. 9-2-2-2. No Caution/Advisory Capsules Are On With BATT Switch ON. (Cont)

702375 - SUBQ	TEST OR INSPECTION CORRECTIVE ACTION
3. Check for greater than 23 VDC between K9-A3 and ground.	Step 1. If voltage is as specified, repair/replace wiring between K9-A3 and K10-A4 (Appendix F). Go to 27. Step 2. If voltage is not as specified, go to 4.
4. Check for greater than 23 VDC between K9-A1 and ground.	Step 1. If voltage is as specified, repair/replace wiring between K9-A3 and K9-A1 (Appendix F). Go to 27. Step 2. If voltage is not as specified, go to 5.
5. Check for greater than 23 VDC between K9-A2 and ground.	Step 1. If voltage is as specified, go to 6. Step 2. If voltage is not as specified, go to 8.
6. Check for greater than 23 VDC between K9-X1 and ground.	Step 1. If voltage is as specified, go to 7. Step 2. If voltage is not as specified, repair/replace wiring between K9-X1 and P230-h (Appendix F). Go to 27.
7. Check continuity between K9-X2 and ground.	Step 1. If continuity is present, replace NO. 2 DC ESNTL BUS SPLY relay K9 (PARA 9-4-32). Go to 27. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 27.
8. Check for greater than 23 VDC between terminal 1 (ESNTL BUS DC SPLY circuit breaker, BATT BUS - lower console circuit breaker panel) and ground.	Step 1. If voltage is as specified, repair/replace wiring between circuit breaker and K9-A2 (Appendix F). Go to 27. Step 2. If voltage is not as specified, go to 9.
9. Check for greater than 23 VDC between terminal 2 (ESNTL BUS DC SPLY circuit breaker, BATT BUS - lower console circuit breaker panel) and ground.	Step 1. If voltage is as specified, replace circuit breaker (PARA 9-4-1). Go to 27. Step 2. If voltage is not as specified, go to 10.

TABLE NO. 9-2-2-2. No Caution/Advisory Capsules Are On With BATT Switch ON. (Cont)

702375 – SUBQ	TEST OR INSPECTION CORRECTIVE ACTION
	10. Check for greater than 23 VDC between lower console circuit breaker panel terminal E1 and ground. Step 1. If voltage is as specified, repair/replace wiring between terminal E1 and BATT bus (Appendix F). Go to 27. Step 2. If voltage is not as specified, go to 11.
	11. Check for greater than 23 VDC between K7-A1 and ground. Step 1. If voltage is as specified, repair/replace wiring between relay K7 and terminal E1 (Appendix F). Go to 27. Step 2. If voltage is not as specified, go to 12.
	12. Check for greater than 23 VDC between K7-A2 and ground. Step 1. If voltage is as specified, replace battery relay K7 (PARA 9-4-33). Go to 27. Step 2. If voltage is not as specified, go to 13.
	13. Check for greater than 23 VDC between K7-X1 and ground. Step 1. If voltage is as specified, repair/replace wiring between K9-X1 and K7-X1 (Appendix F). Go to 27. Step 2. If voltage is not as specified, go to 14.
	14. Check for greater than 23 VDC between BATT switch terminal 3 and ground. Step 1. If voltage is as specified, repair/replace wiring between BATT switch terminal 3 and K7-X1 (Appendix F). Go to 27. Step 2. If voltage is not as specified, go to 15.
	15. Check for greater than 23 VDC between BATT switch terminal 2 and ground. Step 1. If voltage is as specified, replace BATT switch (PARA 9-4-5). Go to 27. Step 2. If voltage is not as specified, go to 16.
	16. Check continuity between BATT switch terminal 2 and P238-G. Step 1. If continuity is present, go to 17. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 27.

TABLE NO. 9-2-2-2. No Caution/Advisory Capsules Are On With BATT Switch ON. (Cont)

702375 - SUBQ	TEST OR INSPECTION CORRECTIVE ACTION
17. Check for greater than 23 VDC between P238-H and ground.	Step 1. If voltage is as specified, replace K6 NO. 2 DC PRI BUS CNTOR (PARA 9-4-30). Go to 27. Step 2. If voltage is not as specified, go to 18.
18. Check continuity between P238-H and P239-H.	Step 1. If continuity is present, go to 19. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 27.
19. Check for greater than 23 VDC between P239-G and ground.	Step 1. If voltage is as specified, replace K16 NO. 1 DC PRI BUS CNTOR (PARA 9-4-30). Go to 27. Step 2. If voltage is not as specified, go to 20.
20. Check for greater than 23 VDC between terminal 1 (BATT BUS CONTR circuit breaker, BATT UTIL BUS - lower console circuit breaker panel) and ground.	Step 1. If voltage is as specified, go to 21. Step 2. If voltage is not as specified, go to 24.
21. Check for greater than 23 VDC between BATT switch terminal 5 and ground.	Step 1. If voltage is as specified, go to 22. Step 2. If voltage is not as specified, repair/replace wiring between BATT switch terminal 5 and P219-N (Appendix F). Go to 27.
22. Check for greater than 23 VDC between BATT switch terminal 6 and ground.	Step 1. If voltage is as specified, go to 23. Step 2. If voltage is not as specified, replace BATT switch (PARA 9-4-5). Go to 27.
23. Check continuity between: K200-X1 and BATT switch terminal 6 K200-X2 and ground	Step 1. If continuity is present, replace relay K200 (PARA 9-4-105). Go to 27. Step 2. If continuity is not present, repair/replace wiring as required (Appendix F). Go to 27.

TABLE NO. 9-2-2-2. No Caution/Advisory Capsules Are On With BATT Switch ON. (Cont)

702375 - SUBQ	TEST OR INSPECTION CORRECTIVE ACTION
24. Check for greater than 23 VDC between terminal 1 (BATT BUS CONTR circuit breaker, BATT UTIL BUS - lower console circuit breaker panel) and ground.	Step 1. If voltage is as specified, replace circuit breaker (PARA 9-4-1). Go to 27. Step 2. If voltage is not as specified, go to 25.
25. Check continuity between BATT UTIL BUS and K7-A2.	Step 1. If continuity is present, go to 26. Step 2. If continuity is not present, repair/replace wiring as required (Appendix F). Go to 27.
26. Check continuity between: CL25-1 and K200-A1 K200-A2 (forward through CR96) and TLTB24-1 TLTB24-1 and P140 (+) K7-A2 and P140 (+) P140 (-) and ground	Step 1. If continuity is present, replace battery (PARA 9-4-106). Go to 27. Step 2. If trouble remains, replace battery charging relay K200 (PARA 9-4-105). Go to 27. Step 3. If continuity is not present, repair/replace wiring as required (Appendix F). Go to 27.
27. Procedure completed.	

TABLE NO. 9-2-2-3. BATT LOW CHARGE Capsule Is On.

70583-702374	TEST OR INSPECTION CORRECTIVE ACTION
1. Turn on electrical power (PARA 1-6-16). Go to 2.	
2. Place BATT switch to OFF. Go to 3.	

TABLE NO. 9-2-2-3. BATT LOW CHARGE Capsule Is On. (Cont)

70583-702374	TEST OR INSPECTION CORRECTIVE ACTION	
	3. Check helicopter configuration.	
	Step 1. EMEP Go to 4.	
	Step 2. W/O EMEP Go to 6.	
EMEP	4. Remove and check pin filter adapter from P123 (PARA 9-4-98).	
	Step 1. If pin filter adapter is good, go to 5.	
	Step 2. If pin filter adapter is not good, replace pin filter adapter (PARA 9-4-98). Go to 14.	
EMEP	5. Install pin filter adapter (PARA 9-4-98). Go to 6.	
	6. Check for 28 VDC between P123-S and ground.	
	Step 1. If voltage is as specified, go to 7.	
	Step 2. If voltage is not as specified, go to 11.	
	7. Check for 115 VAC between P123-H and ground.	
	Step 1. If voltage is as specified, go to 8.	
	Step 2. If voltage is not as specified, go to 10.	
	8. Check for 28 VDC between P123-J and ground.	
	Step 1. If voltage is as specified, replace battery conditioner/analyzer (PARA 9-4-35). Go to 14.	
	Step 2. If voltage is not as specified, go to 9.	
	9. Check continuity between P123-J and terminal 1 (BATT CHGR circuit breaker, NO. 2 DC PRI BUS - pilot's circuit breaker panel).	
	Step 1. If continuity is present, replace circuit breaker (PARA 9-4-1). Go to 14.	
	Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 14.	
	10. Check for 115 VAC between terminal 1 (BATT CHGR circuit breaker, NO. 2 AC PRI BUS - pilot's circuit breaker panel) and ground.	
	Step 1. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P123-H (Appendix F). Go to 14.	

TABLE NO. 9-2-2-3. BATT LOW CHARGE Capsule Is On. (Cont)

70583-702374

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 14.
11. Check for 28 VDC between K7-X1 and K7-X2.	Step 1. If voltage is as specified, replace BATT switch S10 (PARA 9-4-5). Go to 14. Step 2. If voltage is not as specified, go to 12.
12. Check continuity between P123-S and K7-13.	Step 1. If continuity is present, go to 13. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 14.
13. Check continuity between K7-12 and K7-A2.	Step 1. If continuity is present, replace battery relay K7 (PARA 9-4-33). Go to 14. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 14.
14. Procedure completed.	

TABLE NO. 9-2-2-4. DC ESS BUS OFF Capsule Is On With BATT Switch ON.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for greater than 23 VDC between P900-J and ground.	Step 1. If voltage is as specified, replace right relay panel (PARA 9-4-2). Go to 9. Step 2. If voltage is not as specified, go to 2.
2. Check for greater than 23 VDC between terminal 1 (ESNTL DC SENSE circuit breaker, DC ESNTL BUS - upper console circuit breaker panel) and ground.	Step 1. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P900-J (Appendix F). Go to 9. Step 2. If voltage is not as specified, go to 3.

TABLE NO. 9-2-2-4. DC ESS BUS OFF Capsule Is On With BATT Switch ON. (Cont)

TEST OR INSPECTION CORRECTIVE ACTION	
3. Check for greater than 23 VDC between terminal 2 (ESNTL DC SENSE circuit breaker, DC ESNTL BUS - upper console circuit breaker panel) and ground.	
Step 1. If voltage is as specified, replace circuit breaker (PARA 9-4-1). Go to 9. Step 2. If voltage is not as specified, go to 4.	
4. Check for greater than 23 VDC between K10-A3 and ground.	
Step 1. If voltage is as specified, repair/replace wiring between K10-A3 and DC ESNTL BUS (Appendix F). Go to 9. Step 2. If voltage is not as specified, go to 5.	
5. Check for greater than 23 VDC between K10-A4 and ground.	
Step 1. If voltage is as specified, replace relay K10 (PARA 9-4-32). Go to 9. Step 2. If voltage is not as specified, go to 6.	
6. Check continuity between K9-A3 and K10-A4.	
Step 1. If continuity is present, go to 7. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 9.	
7. Check for greater than 23 VDC between K9-A2 and ground.	
Step 1. If voltage is as specified, replace relay K9 (PARA 9-4-32). Go to 9. Step 2. If voltage is not as specified, go to 8.	
8. Check continuity between K9-A2 and terminal 1 (ESNTL BUS DC SPLY circuit breaker, BATT BUS - lower console circuit breaker panel).	
Step 1. If continuity is present, replace ESNTL BUS DC SPLY circuit breaker (PARA 9-4-1). Go to 9. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 9.	
9. Procedure completed.	

TABLE NO. 9-2-2-5. #1 And #2 CONV Capsules Are Not On.

**TEST OR INSPECTION
CORRECTIVE ACTION**

1. Check for 28 VDC between K13 AC ESNTL BUS FAIL relay terminal A2 and ground.

Step 1. If voltage is as specified, repair/replace wiring between terminal A2 and SGP203-1 (Appendix F). Go to 3.

Step 2. If voltage is not as specified, go to 2.

2. Check for 28 VDC between terminal 1 (ESNTL BUS AC & CONV WARN circuit breaker, BATT BUS - lower console circuit breaker panel) and ground.

Step 1. If voltage is as specified, repair/replace wiring between terminal 1 and K13 AC ESNTL BUS FAIL relay terminal A2 (Appendix F). Go to 3.

Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 3.

3. Procedure completed.
-

TABLE NO. 9-2-2-6. #1 CONV Capsule Is Not On.

**TEST OR INSPECTION
CORRECTIVE ACTION**

1. Place and hold caution/advisory panel BRT/DIM-TEST switch to TEST.

Step 1. If #1 CONV capsule goes on, go to 2.

Step 2. If #1 CONV capsule does not go on, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 8.

2. Check helicopter configuration.

Step 1. **EMEP** Go to 3.◀

Step 2. **W/O EMEP** Go to 6.◀

3. Remove and check pin filter adapter from P117 (PARA 9-4-98).

Step 1. If pin filter adapter is good, go to 4.

Step 2. If pin filter adapter is not good, replace pin filter adapter (PARA 9-4-98). Go to 8.

4. Install pin filter adapter (PARA 9-4-98). Go to 5.

TABLE NO. 9-2-2-6. #1 CONV Capsule Is Not On. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
5. Check for 28 VDC between P117-19 and ground.	Step 1. If voltage is as specified, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 8. Step 2. If voltage is not as specified, go to 6.
6. Check continuity between P117-19 and P239-E.	Step 1. If continuity is present, go to 7. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 8.
7. Check for 28 VDC between P239-J and ground.	Step 1. If voltage is as specified, replace K16 NO. 1 DC PRI BUS CNTOR (PARA 9-4-30). Go to 8. Step 2. If voltage is not as specified, repair/replace wiring between K13 AC ESNTL BUS FAIL relay terminal A2 and P239-J (Appendix F). Go to 8.
8. Procedure completed.	

TABLE NO. 9-2-2-7. #2 CONV Capsule Is Not On.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Place and hold caution/advisory panel BRT/DIM-TEST switch to TEST.	Step 1. If #2 CONV capsule goes on, go to 2. Step 2. If #2 CONV capsule does not go on, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 8.
2. Check helicopter configuration.	Step 1. EMEP Go to 3. Step 2. W/O EMEP Go to 5.
EMEP 3. Remove and check pin filter adapter from P117 (PARA 9-4-98).	Step 1. If pin filter adapter is good, go to 4.

TABLE NO. 9-2-2-7. #2 CONV Capsule Is Not On. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 2. If pin filter adapter is not good, replace pin filter adapter (PARA 9-4-98). Go to 8.
EMEP	4. Install pin filter adapter (PARA 9-4-98). Go to 5.
	5. Check for 28 VDC between P117-35 and ground.
	Step 1. If voltage is as specified, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 8. Step 2. If voltage is not as specified, go to 6.
	6. Check continuity between P117-35 and P238-C. Step 1. If continuity is present, go to 7. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 8.
	7. Check for 28 VDC between P238-K and ground. Step 1. If voltage is as specified, replace K6 NO. 2 DC PRI BUS CNTOR (PARA 9-4-30). Go to 8. Step 2. If voltage is not as specified, repair/replace wiring between K13 AC ESNTL BUS FAIL terminal A2 and P238-K (Appendix F). Go to 7.
	8. Procedure completed.

TABLE NO. 9-2-2-8. #1 CONV Capsule Is On.

TEST OR INSPECTION	CORRECTIVE ACTION
	1. Check for 28 VDC between P239-A and P239-B. Step 1. If voltage is as specified, replace K16 NO. 1 DC PRI BUS CNTOR (PARA 9-4-30). Go to 8. Step 2. If voltage is not as specified, go to 2.

TABLE NO. 9-2-2-8. #1 CONV Capsule Is On. (Cont)

TEST OR INSPECTION CORRECTIVE ACTION	
2. Check for 28 VDC between P239-A and ground.	
Step 1. If voltage is as specified, repair/replace wiring between P239-B and ground (Appendix F). Go to 8. Step 2. If voltage is not as specified, go to 3.	
3. Check for 28 VDC between K16 NO. 1 DC PRI BUS CNTOR terminal A1 and ground.	
Step 1. If voltage is as specified, repair/replace wiring between contactor terminal A1 and P239-A (Appendix F). Go to 8. Step 2. If voltage is not as specified, go to 4.	
4. Check continuity between: No. 1 converter (+) and K16-A1 No. 1 converter (-) and ground	
Step 1. If continuity is present, go to 5. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 8.	
5. Check for 115 VAC between: P903-A and P903-D P903-B and P903-D P903-C and P903-D	
Step 1. If voltage is as specified, replace No. 1 converter (PARA 9-4-36). Go to 8. Step 2. If voltage is not as specified, go to 6.	
6. Check continuity between P903-D and ground.	
Step 1. If continuity is present, go to 7. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 8.	
7. Check for 115 VAC between the following for NO. 1 CONVERTER circuit breaker (NO. 1 AC PRI BUS - copilot's circuit breaker panel): Terminal A1 and ground Terminal B1 and ground Terminal C1 and ground	
Step 1. If voltage is as specified, repair/replace wiring between circuit breaker and No. 1 converter (Appendix F). Go to 8. Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 8.	

TABLE NO. 9-2-2-8. #1 CONV Capsule Is On. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	8. Procedure completed.

TABLE NO. 9-2-2-9. DC ESS BUS OFF Capsule Is Not On With ESNTL DC SENSE Circuit Breaker Pulled.

TEST OR INSPECTION	CORRECTIVE ACTION
	1. Check helicopter configuration.
	Step 1. EMEP Go to 2. Step 2. W/O EMEP Go to 4.
EMEP	2. Remove and check pin filter adapter from P117 (PARA 9-4-98).
	Step 1. If pin filter adapter is good, go to 3. Step 2. If pin filter adapter is not good, replace pin filter adapter (PARA 9-4-98). Go to 15.
EMEP	3. Install pin filter adapter (PARA 9-4-98). Go to 4.
	4. Check for 28 VDC between P117-36 and ground.
	Step 1. If voltage is as specified, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 15. Step 2. If voltage is not as specified, go to 5.
	5. Check for 28 VDC between P900-H and ground.
	Step 1. If voltage is as specified, replace right relay panel (PARA 9-4-2). Go to 15. Step 2. If trouble remains, repair/replace wiring between P900-G and P117-36 (Appendix F). Go to 15. Step 3. If voltage is not as specified, go to 6.

TABLE NO. 9-2-2-9. DC ESS BUS OFF Capsule Is Not On With ESNTL DC SENSE Circuit Breaker Pulled. (Cont)

TEST OR INSPECTION CORRECTIVE ACTION	
6. Check for 28 VDC between terminal 1 (BATT & ESNTL DC WARN EXT PWR CONTR circuit breaker, BATT BUS - lower console circuit breaker panel) and ground.	
Step 1. If voltage is as specified, repair/replace wiring between P900-H and circuit breaker terminal 1 (Appendix F). Go to 15 .	
Step 2. If voltage is not as specified, go to 7 .	
7. Check for 28 VDC between terminal 2 (BATT & ESNTL DC WARN EXT PWR CONTR circuit breaker, BATT BUS - lower console circuit breaker panel) and ground.	
Step 1. If voltage is as specified, replace circuit breaker (PARA 9-4-1). Go to 15 .	
Step 2. If voltage is not as specified, go to 8 .	
8. Check for 28 VDC between K7-A1 and ground.	
Step 1. If voltage is as specified, repair/replace wiring between K7-A1 and BATT BUS (Appendix F). Go to 15 .	
Step 2. If voltage is not as specified, go to 9 .	
9. Check for 28 VDC between terminal 1 (BATT BUS SPLY circuit breaker, DC ESNTL BUS - upper console circuit breaker panel) and ground.	
Step 1. If voltage is as specified, go to 10 .	
Step 2. If voltage is not as specified, go to 11 .	
10. Check continuity between K7-A4 and terminal 1 (BATT BUS SPLY circuit breaker, DC ESNTL BUS - upper console circuit breaker panel).	
Step 1. If continuity is present, replace battery relay K7 (PARA 9-4-33). Go to 15 .	
Step 2. If continuity is not present, repair/replace wiring between K7-A4 and circuit breaker terminal 1 (Appendix F). Go to 15 .	
11. Check for 28 VDC between terminal 2 (BATT BUS SPLY circuit breaker, DC ESNTL BUS - upper console circuit breaker panel) and ground.	
Step 1. If voltage is as specified, replace circuit breaker (PARA 9-4-1). Go to 15 .	
Step 2. If voltage is not as specified, go to 12 .	

TABLE NO. 9-2-2-9. DC ESS BUS OFF Capsule Is Not On With ESNTL DC SENSE Circuit Breaker Pulled. (Cont)

	TEST OR INSPECTION CORRECTIVE ACTION
	12. Check for 28 VDC between K10 NO. 1 DC ESNTL BUS SPLY relay terminal A2 and ground. Step 1. If voltage is as specified, go to 13. Step 2. If voltage is not as specified, go to 14.
	13. Check for 28 VDC between K10 NO. 1 DC ESNTL BUS SPLY relay terminal A3 and ground. Step 1. If voltage is as specified, repair/replace wiring between relay terminal A3 and DC ESNTL BUS (Appendix F). Go to 15. Step 2. If voltage is not as specified, replace relay K10 (PARA 9-4-32). Go to 15.
	14. Check for 28 VDC between terminal 1 (DC ESNTL BUS SPLY circuit breaker, NO. 1 DC PRI BUS - copilot's circuit breaker panel) and ground. Step 1. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and K10 NO. 1 DC ESNTL BUS SPLY relay terminal A2 (Appendix F). Go to 15. Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 15.
	15. Procedure completed.

TABLE NO. 9-2-2-10. #2 CONV Capsule Is On.

	TEST OR INSPECTION CORRECTIVE ACTION
	1. Check for 28 VDC between P238-A and P238-B. Step 1. If voltage is as specified, replace K6 NO. 2 DC PRI BUS CNTOR (PARA 9-4-30). Go to 8. Step 2. If voltage is not as specified, go to 2.

TABLE NO. 9-2-2-10. #2 CONV Capsule Is On. (Cont)

TEST OR INSPECTION CORRECTIVE ACTION	
2. Check for 28 VDC between P238-A and ground.	
Step 1. If voltage is as specified, repair/replace wiring between P238-B and ground (Appendix F). Go to 8. Step 2. If voltage is not as specified, go to 3.	
3. Check for 28 VDC between K6 NO. 2 DC PRI BUS CNTOR terminal A2 and ground.	
Step 1. If voltage is as specified, repair/replace wiring between K6 NO. 2 DC PRI BUS CNTOR terminal A2 and P238-A (Appendix F). Go to 8. Step 2. If voltage is not as specified, go to 4.	
4. Check continuity between: No. 2 converter (+) and K6-A2 No. 2 converter (-) and ground	
Step 1. If continuity is present, go to 5. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 8.	
5. Check for 115 VAC between: P904-A and P904-D P904-B and P904-D P904-C and P904-D	
Step 1. If voltage is as specified, replace converter (PARA 9-4-36). Go to 8. Step 2. If voltage is not as specified, go to 6.	
6. Check continuity between P904-D and ground.	
Step 1. If continuity is present, go to 7. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 8.	
7. Check for 115 VAC between the following for NO. 2 CONVERTER circuit breaker (NO. 2 AC PRI BUS - pilot's circuit breaker panel): Terminal A1 and ground Terminal B1 and ground Terminal C1 and ground	
Step 1. If voltage is as specified, repair/replace wiring between circuit breaker and No. 2 converter (Appendix F). Go to 8. Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 8.	

TABLE NO. 9-2-2-10. #2 CONV Capsule Is On. (Cont)

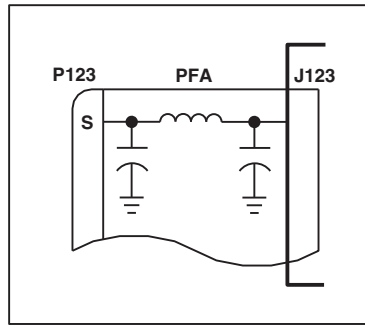
TEST OR INSPECTION	CORRECTIVE ACTION
	8. Procedure completed.

TABLE NO. 9-2-2-11. BATT LOW CHARGE Capsule Is Off When Battery Voltage Is 22.7 Vdc Or Less.

TEST OR INSPECTION	CORRECTIVE ACTION
	1. Remove and check pin filter adapter from P117 (PARA 9-4-98).
	Step 1. If pin filter adapter is good, go to 2.
	Step 2. If pin filter adapter is not good, replace pin filter adapter (PARA 9-4-98). Go to 10.
	2. Install pin filter adapter (PARA 9-4-98). Go to 3.
	3. Check for less than 23 VDC between P117-21 and ground.
	Step 1. If voltage is as specified, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 10.
	Step 2. If voltage is not as specified, go to 4.
	4. Check continuity between K201-B3 and P117-36.
	Step 1. If continuity is present, go to 5.
	Step 2. If continuity is not present, repair/replace wiring as required (Appendix F). Go to 10.
	5. Check continuity between K201-B3 and K201-B2.
	Step 1. If continuity is present, go to 6.
	Step 2. If continuity is not present, replace relay K201 (PARA 9-4-105). Go to 10.
	6. With BATT switch ON, check for 23 VDC or less between BATT switch terminal 5 and ground.
	Step 1. If voltage is as specified, go to 7.
	Step 2. If voltage is not as specified, go to 8.

**TABLE NO. 9-2-2-11. BATT LOW CHARGE Capsule Is Off When Battery Voltage Is 22.7 Vdc Or Less.
(Cont)**

TEST OR INSPECTION CORRECTIVE ACTION	
7. Check for 23 VDC or less between BATT switch terminal 6 and ground.	
Step 1. If voltage is as specified, repair/replace wiring between BATT switch terminal 6 and K201-X1 (Appendix F), then go to 10 .	
Step 2. If voltage is not as specified, replace BATT switch (PARA 9-4-5). Go to 10 .	
8. Check continuity between: S10-5 and BATT BUS CONTR circuit breaker terminal 1 P140 (+) and terminal E2 on lower console circuit breaker panel P140 (-) and ground	
Step 1. If continuity is present, go to 9 .	
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 10 .	
9. Check continuity between K7-A2 and BATT UTIL BUS.	
Step 1. If continuity is present, replace BATT BUS CONTR circuit breaker (PARA 9-4-1). Go to 10 .	
Step 2. If trouble remains, replace battery (PARA 9-4-106). Go to 10 .	
Step 3. If continuity is not present, repair/replace wiring (Appendix F). Go to 10 .	
10. Procedure completed.	



DETAIL A
(SEE NOTE 2)

EFFECTIVITY

70853 - 702374

LEGEND

POWER SOURCE

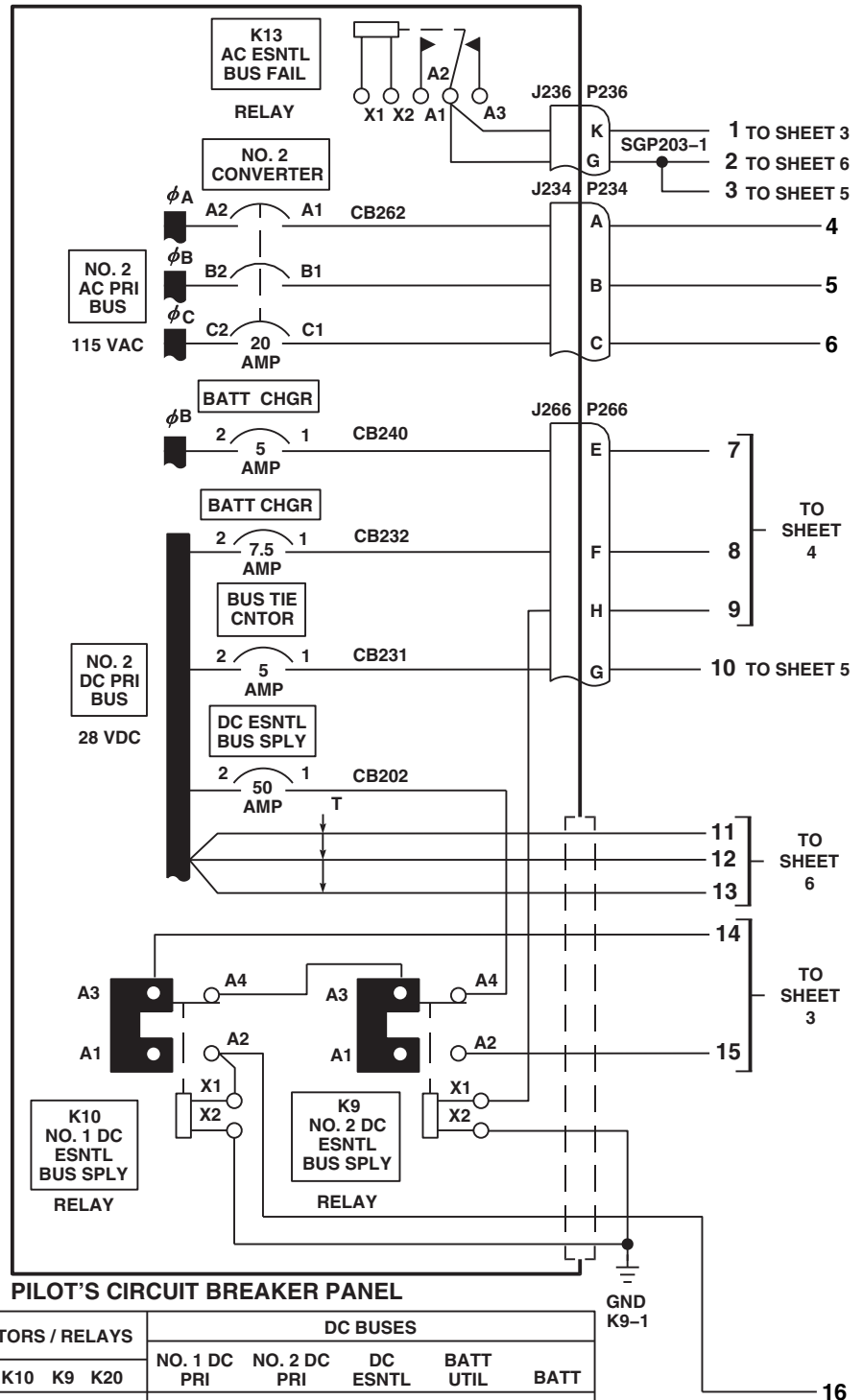
A = NO. 1 CONVERTER
B = NO. 2 CONVERTER
C = BATTERY (GREATER THAN 35% CHARGED)
1 = TURNED ON AND OPERATING
0 = TURNED OFF OR FAILED

CONTACTORS/RELAYS/BUSES

K7 = BATTERY RELAY
K16 = NO. 1 DC PRI BUS CONTACTOR
K6 = NO. 2 DC PRI BUS CONTACTOR
K15 = DC BUS TIE CONTACTOR
K10 = NO. 1 DC ESNTL BUS SPLY RELAY
K9 = NO. 2 ESNTL BUS SPLY RELAY
K20 = DC ESNTL BUS FAIL RELAY
X = ENERGIZED
O = DE-ENERGIZED

NOTES

- W/O EMEP**
PIN FILTER ADAPTERS (PFA) ARE NOT INSTALLED.
- PIN S OF P123 SHOWS TYPICAL CONFIGURATION FOR ALL PIN FILTER ADAPTERS.



DC POWER SOURCE			CONDITION	DC SYSTEM CONTACTORS / RELAYS								DC BUSES				
												NO. 1 DC PRI	NO. 2 DC PRI	DC ESNTL	BATT UTIL	BATT
A	B	C		K7	K16	K6	K15	K10	K9	K20						
0	0	0	1	O	O	O	O	O	O	O		O	O	O	O	O
1	0	0	2	O	X	O	X	X	O	X		X	X	X	X	X
0	1	0	3	O	O	X	X	X	O	X		X	X	X	X	X
1	1	0	4	O	X	X	O	X	O	X		X	X	X	X	X
0	0	1	5	X	O	O	O	O	X	X		O	O	X	X	X
1	0	1	6	O	X	O	X	X	O	X		X	X	X	X	X
0	1	1	7	O	O	X	X	X	O	X		X	X	X	X	X
1	1	1	8	O	X	X	O	X	O	X		X	X	X	X	X

FIGURE 9-2-2-1. DC ELECTRICAL SYSTEM SCHEMATIC DIAGRAM (SHEET 1 OF 6)

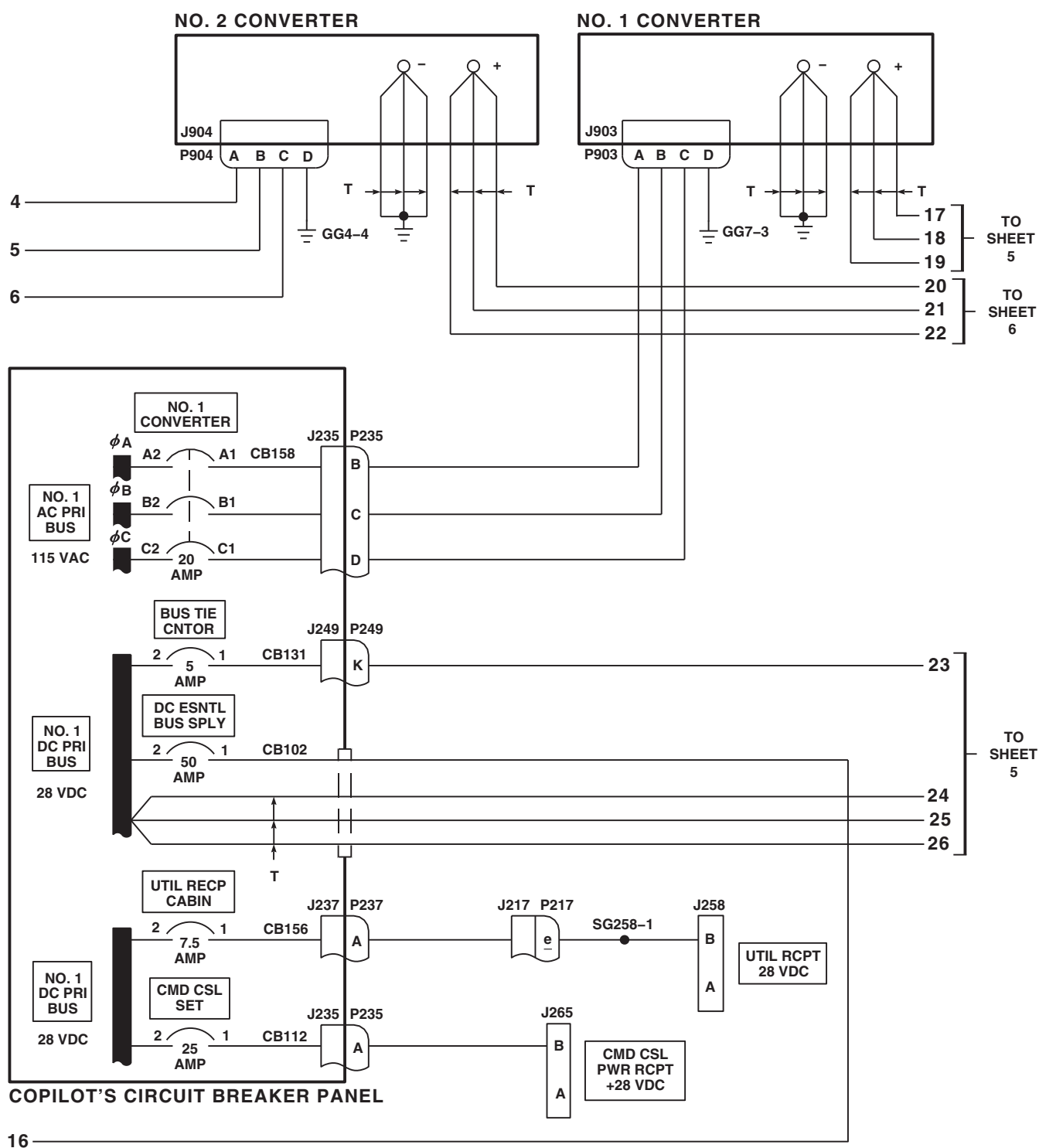


FIGURE 9-2-2-1. DC ELECTRICAL SYSTEM SCHEMATIC DIAGRAM (SHEET 2 OF 6)

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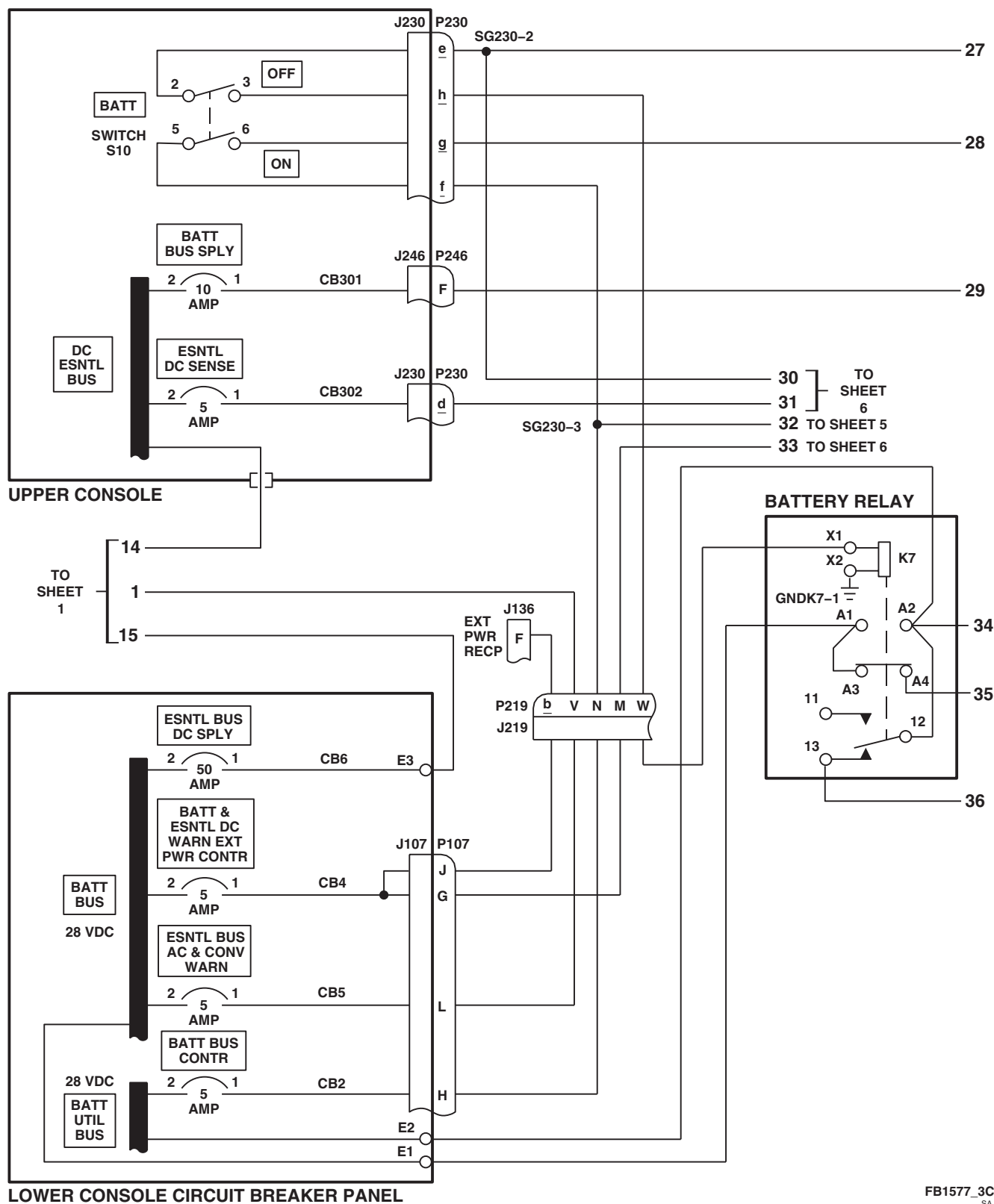
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FIGURE 9-2-2-1. DC ELECTRICAL SYSTEM SCHEMATIC DIAGRAM (SHEET 3 OF 6)

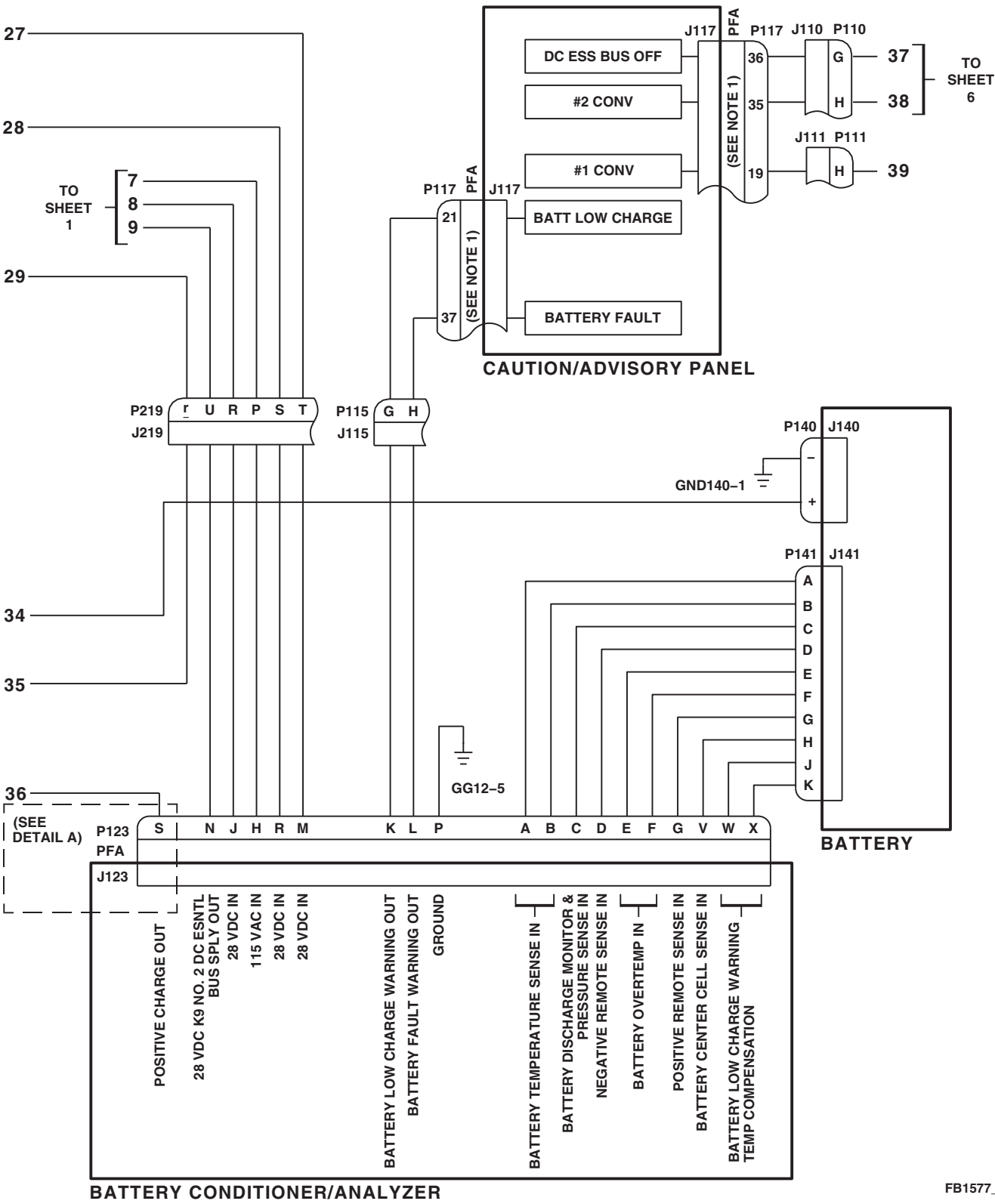


FIGURE 9-2-2-1. DC ELECTRICAL SYSTEM SCHEMATIC DIAGRAM (SHEET 4 OF 6)

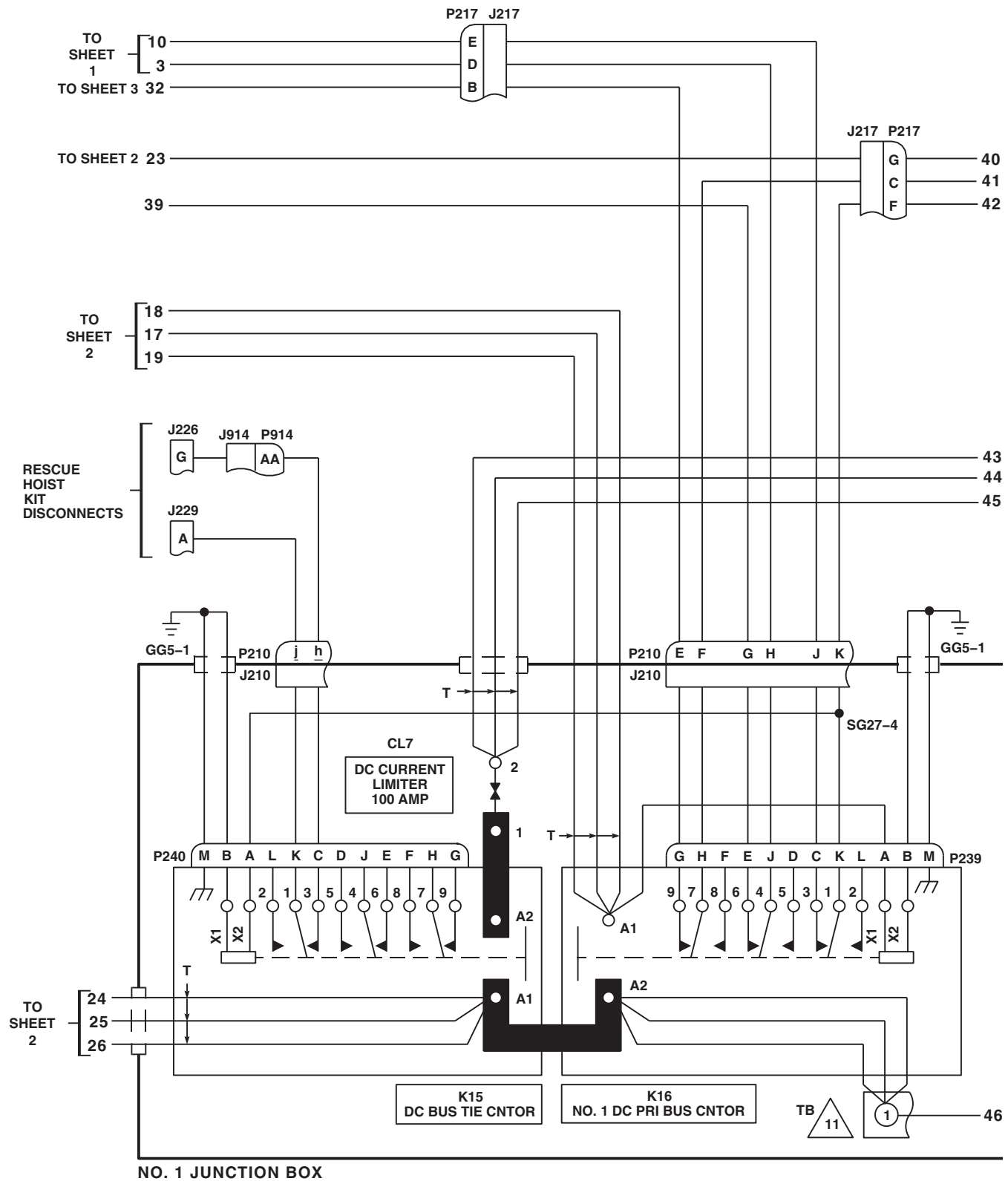
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FIGURE 9-2-2-1. DC ELECTRICAL SYSTEM SCHEMATIC DIAGRAM (SHEET 5 OF 6)

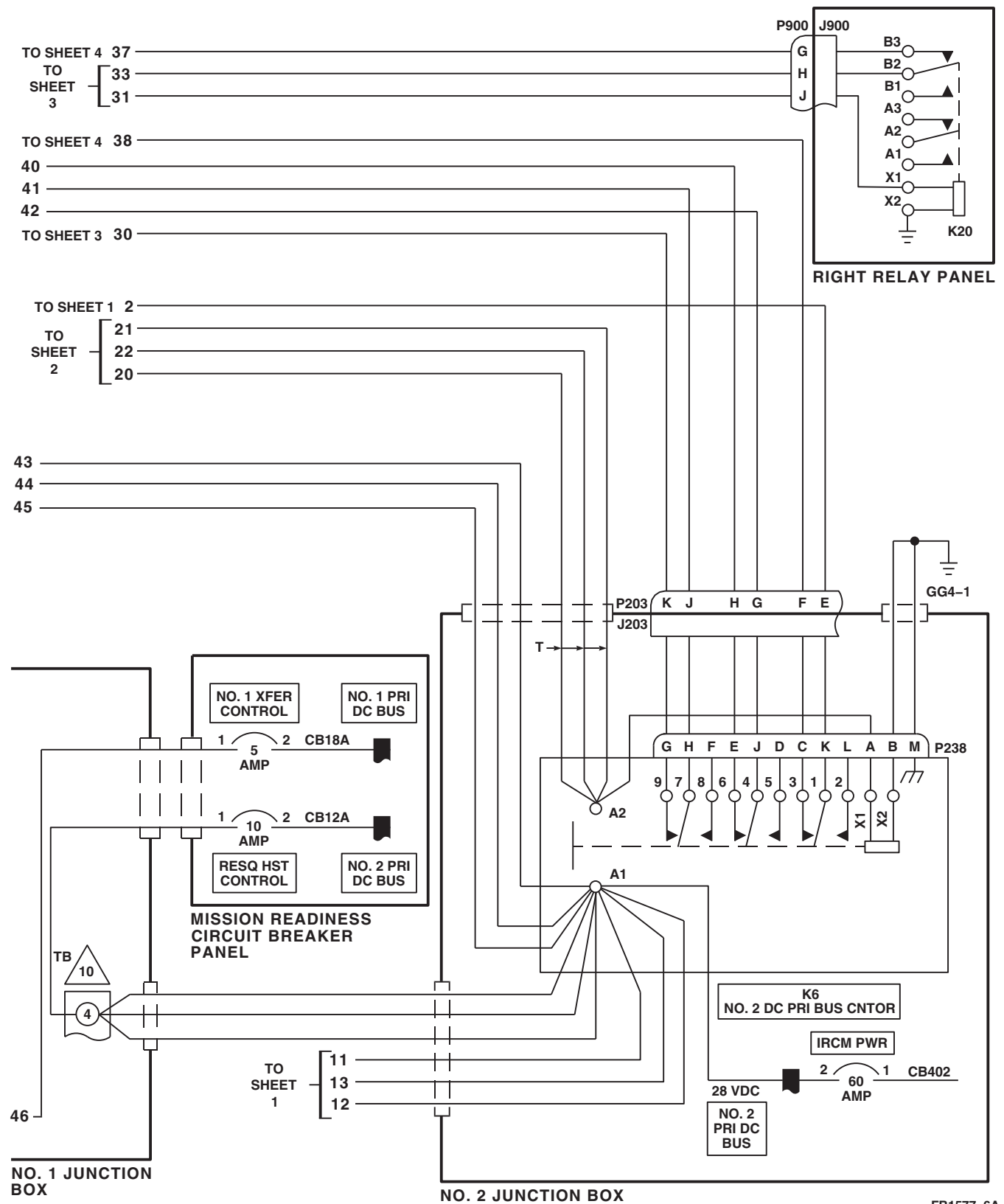


FIGURE 9-2-2-1. DC ELECTRICAL SYSTEM SCHEMATIC DIAGRAM (SHEET 6 OF 6)

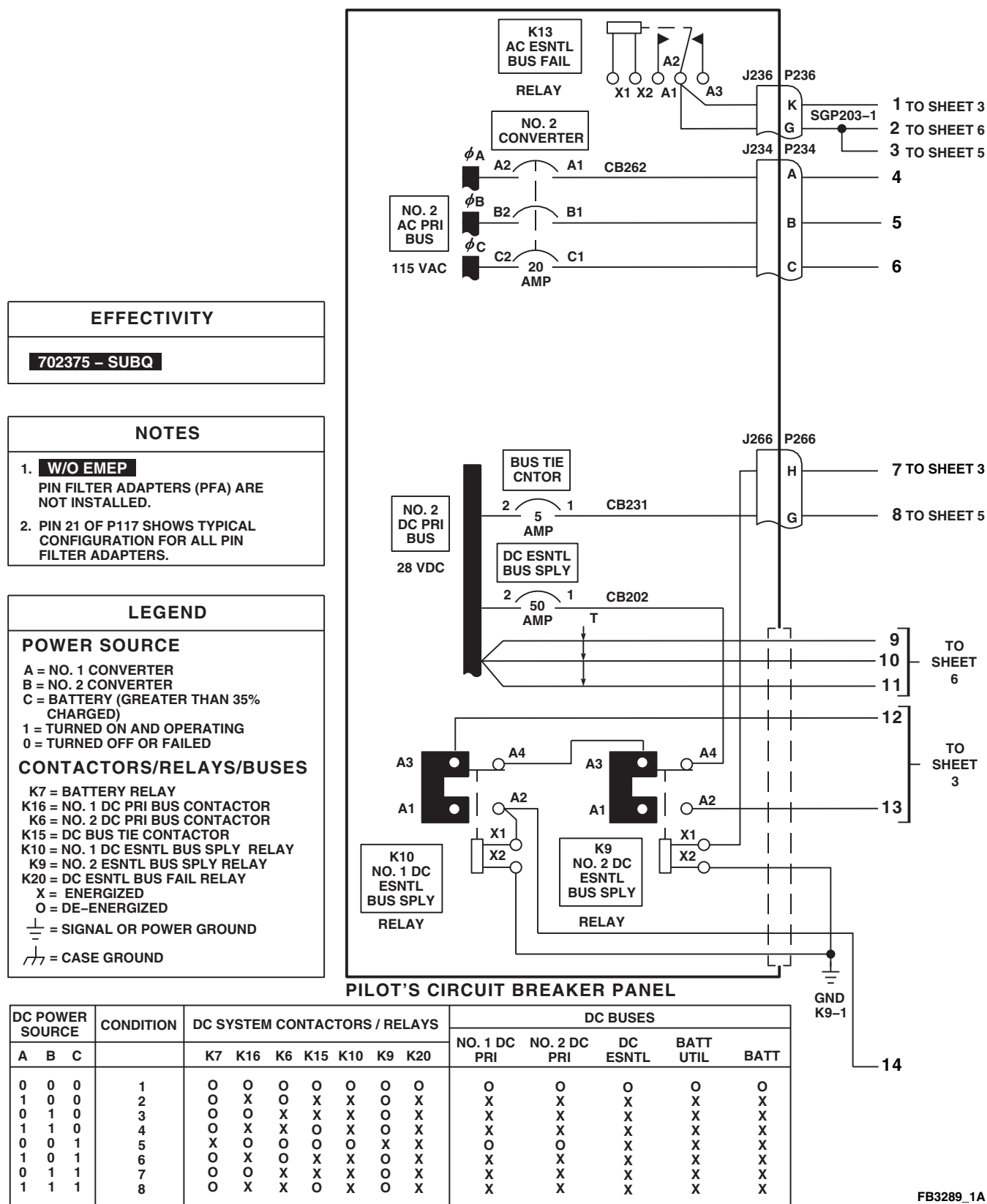


FIGURE 9-2-2-2. DC ELECTRICAL SYSTEM SCHEMATIC DIAGRAM (SHEET 1 OF 6)

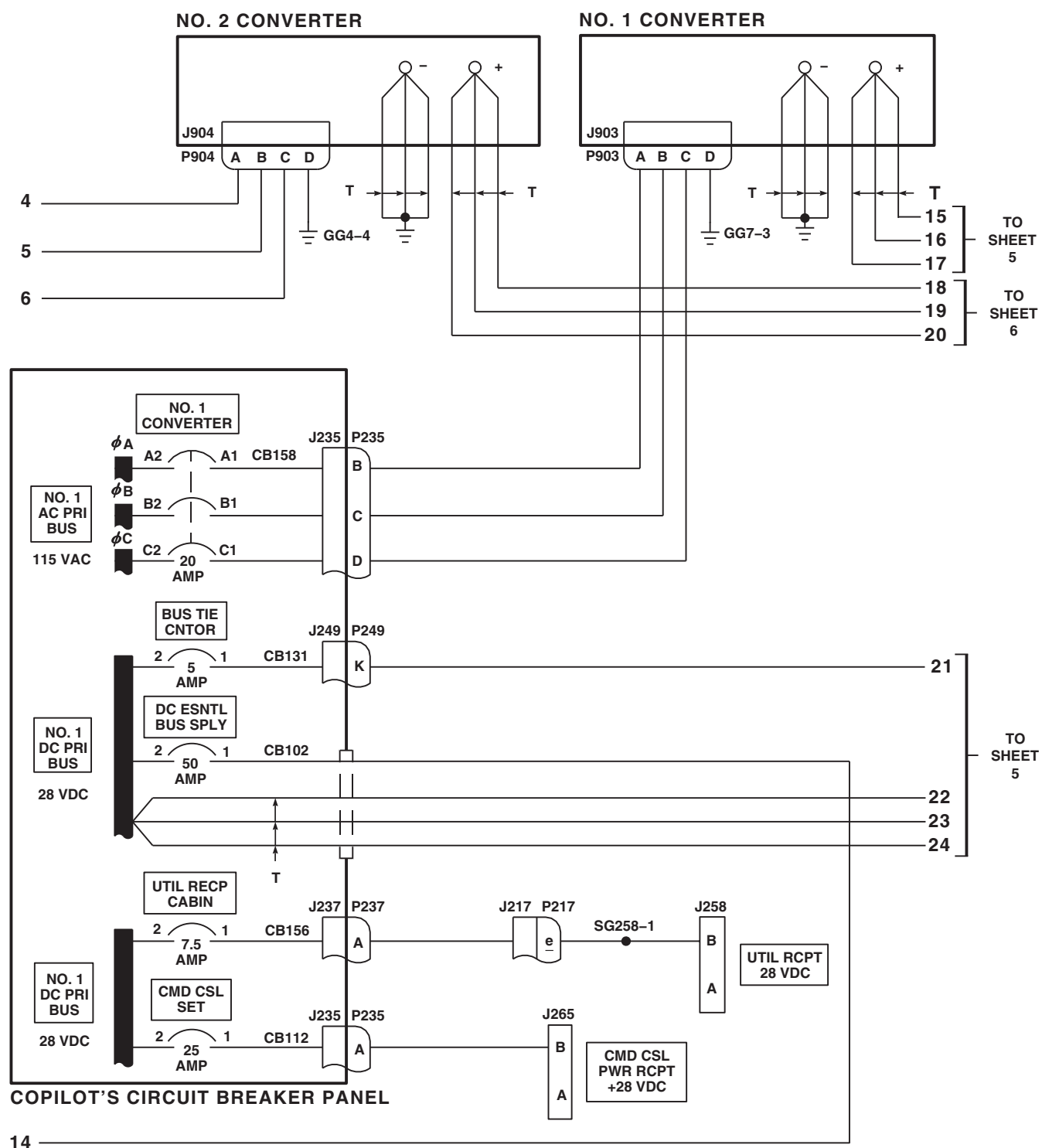


FIGURE 9-2-2-2. DC ELECTRICAL SYSTEM SCHEMATIC DIAGRAM (SHEET 2 OF 6)

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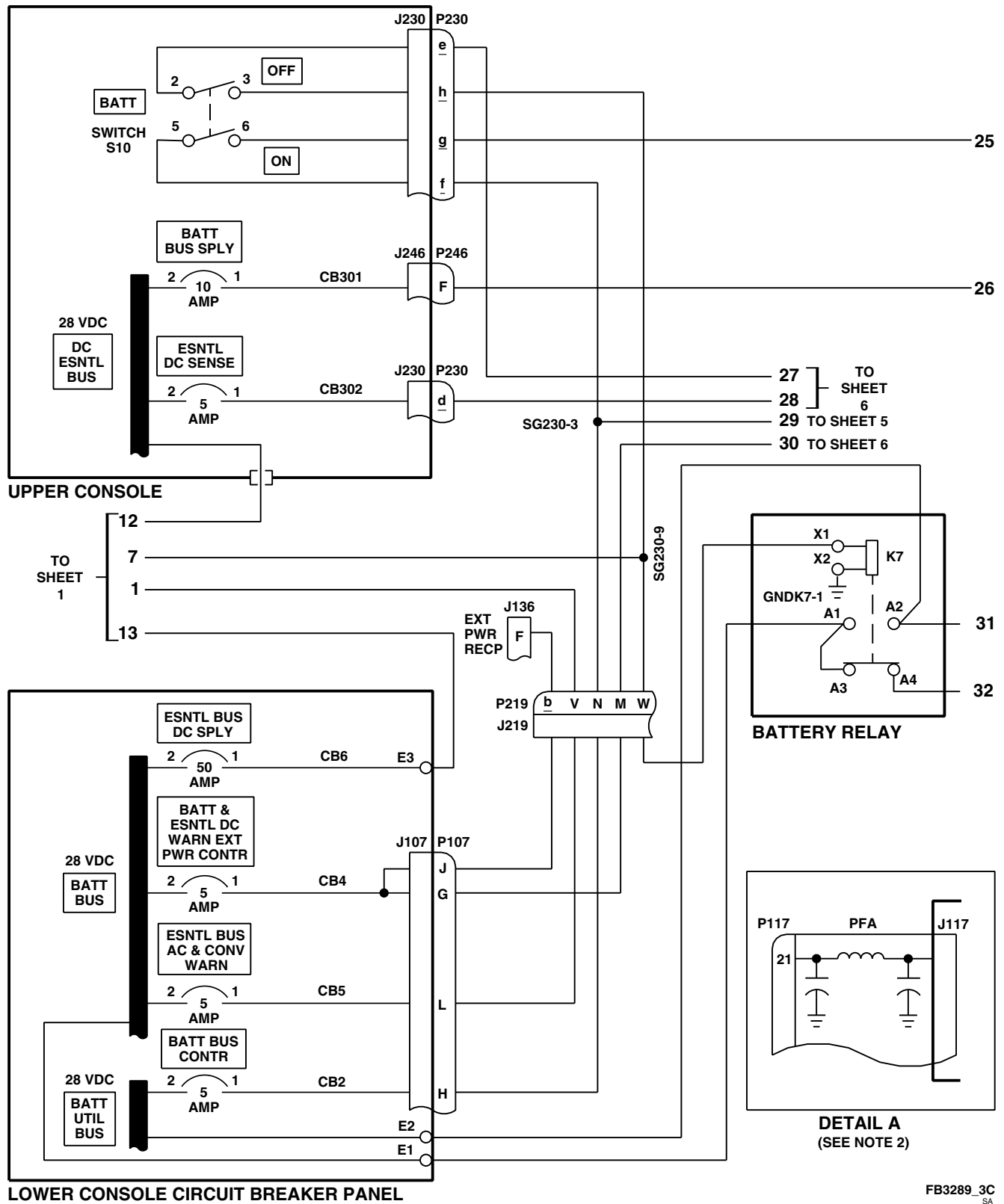


FIGURE 9-2-2-2. DC ELECTRICAL SYSTEM SCHEMATIC DIAGRAM (SHEET 3 OF 6)

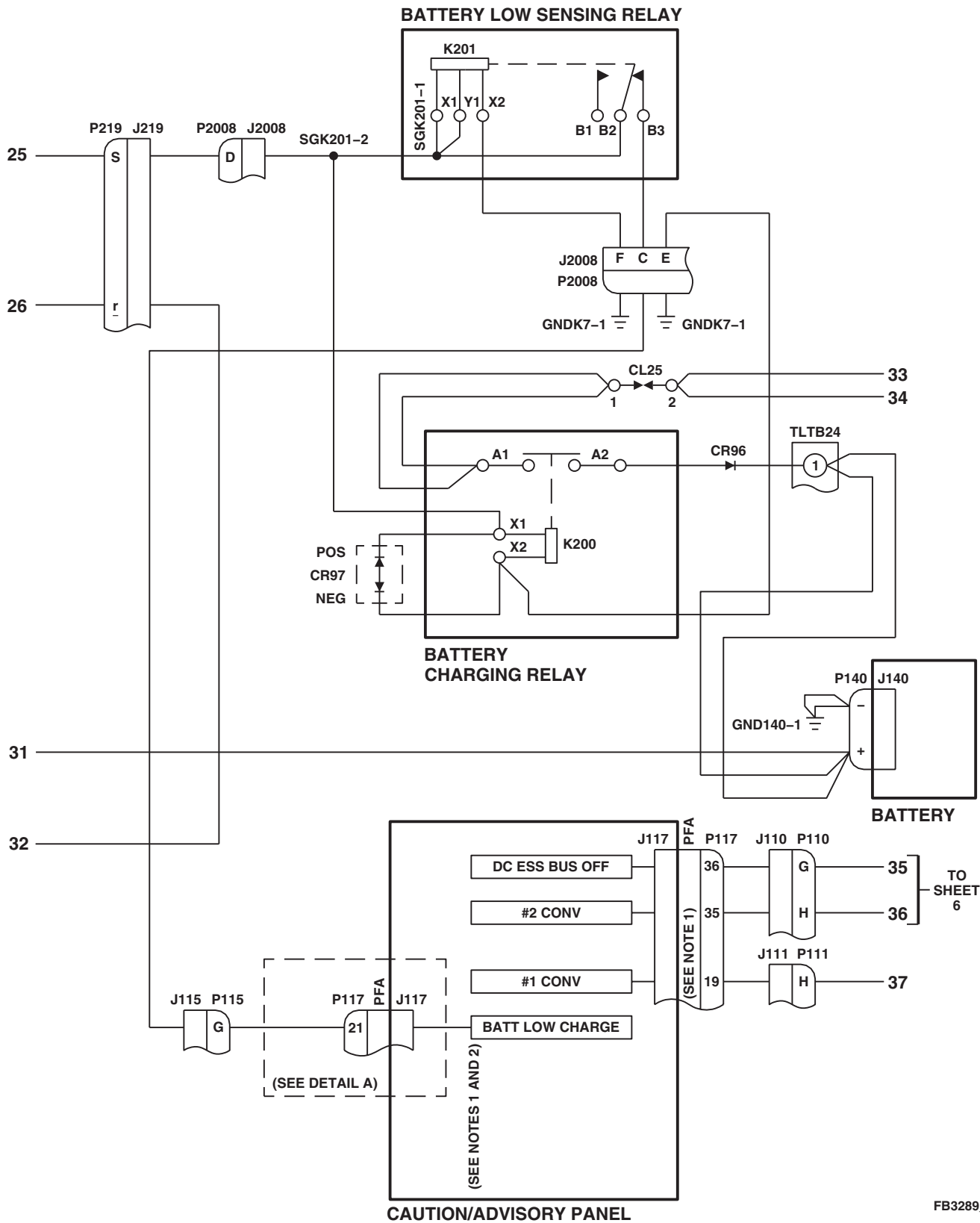


FIGURE 9-2-2-2. DC ELECTRICAL SYSTEM SCHEMATIC DIAGRAM (SHEET 4 OF 6)

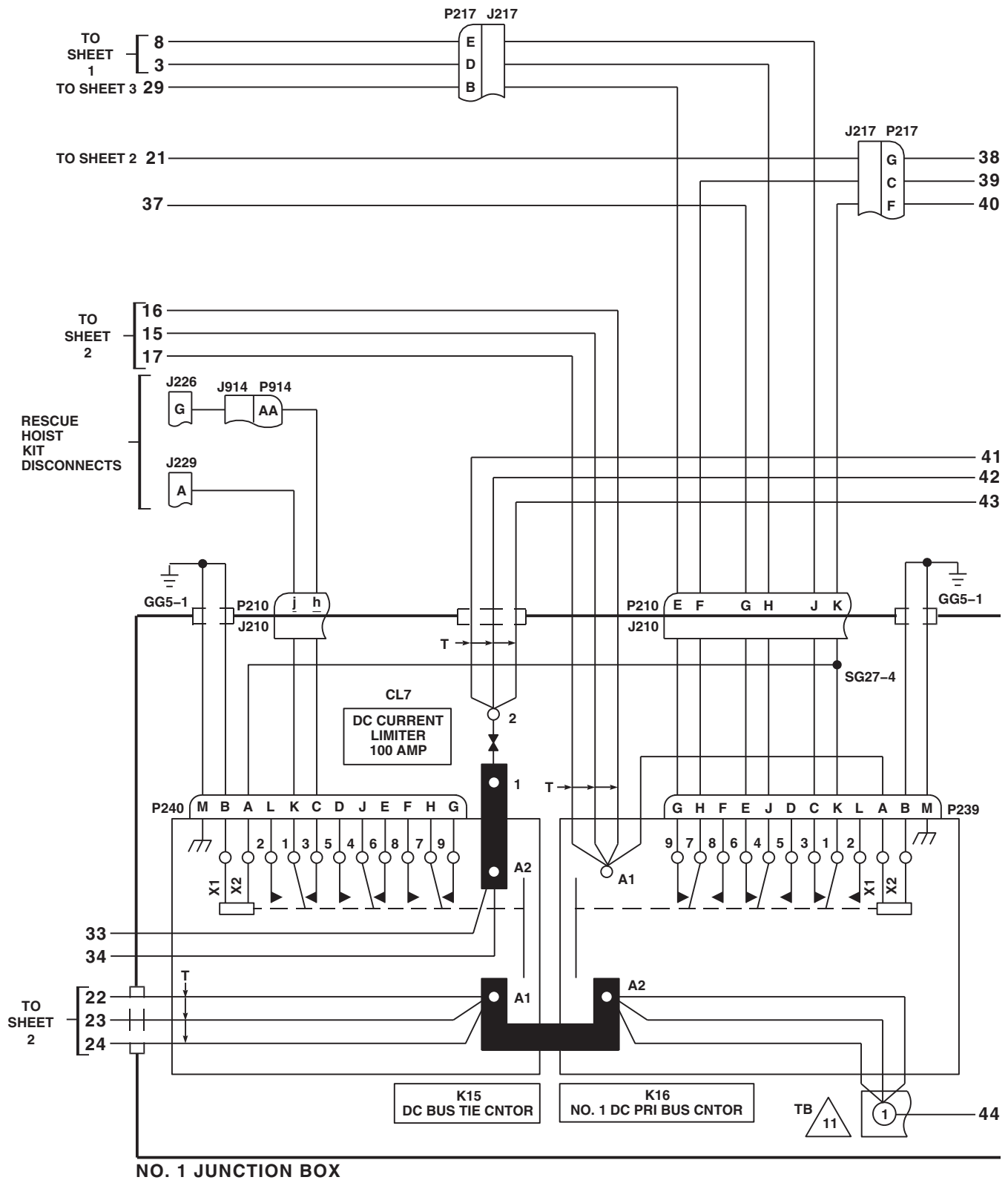
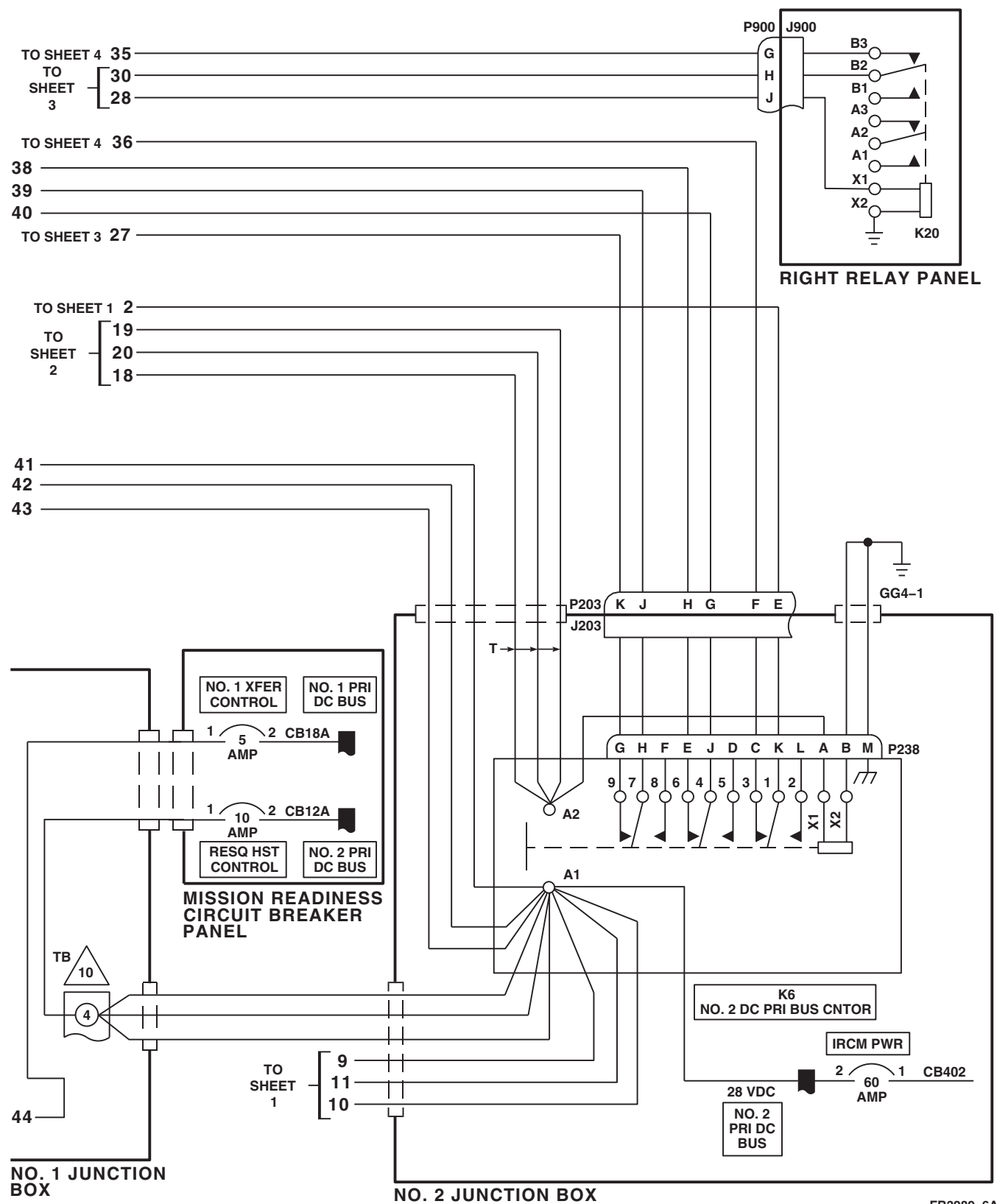
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FIGURE 9-2-2-2. DC ELECTRICAL SYSTEM SCHEMATIC DIAGRAM (SHEET 5 OF 6)



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FIGURE 9-2-2-2. DC ELECTRICAL SYSTEM SCHEMATIC DIAGRAM (SHEET 6 OF 6)

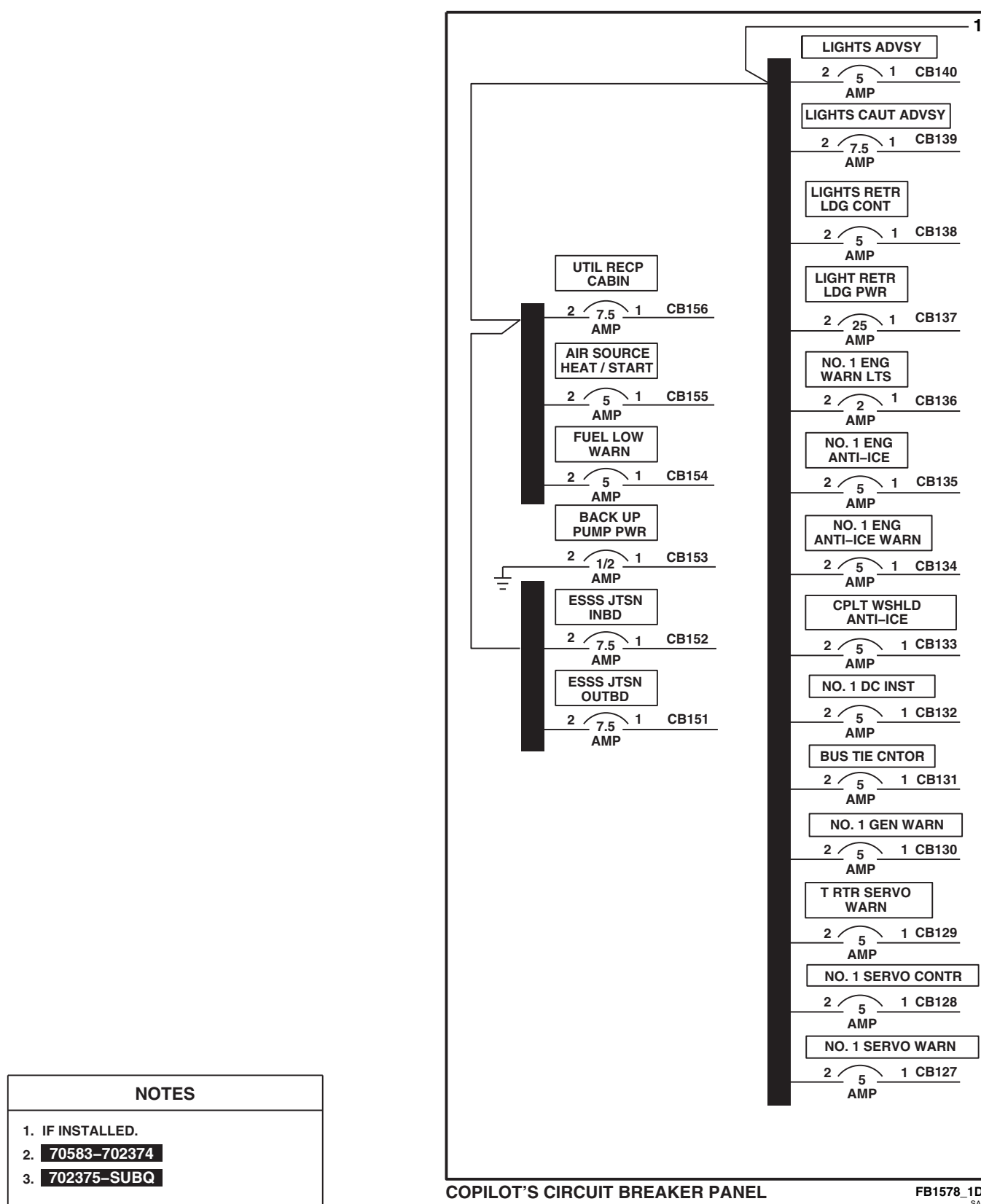


FIGURE 9-2-2-3. DC POWER DISTRIBUTION DIAGRAM (SHEET 1 OF 7)

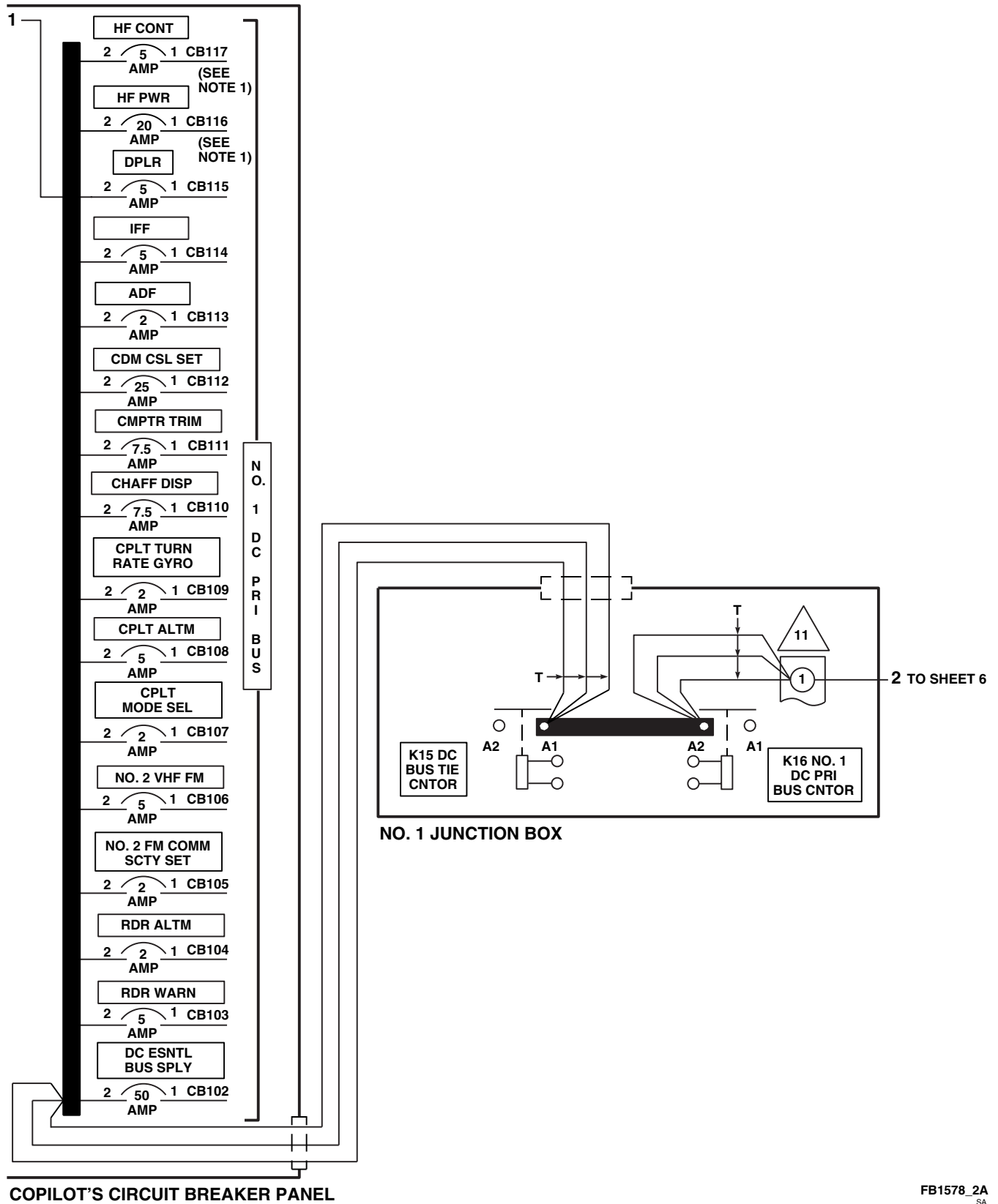
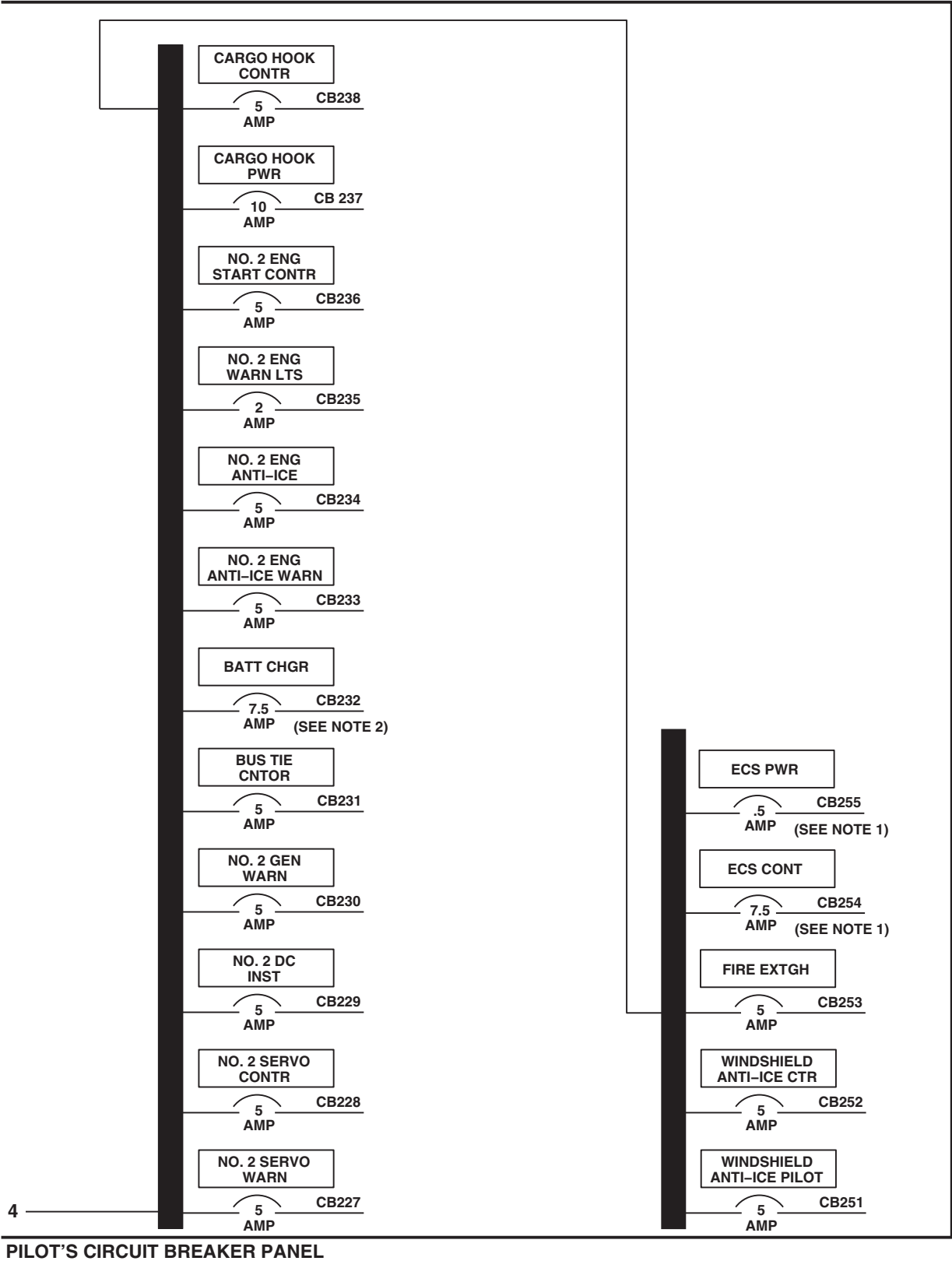


FIGURE 9-2-2-3. DC POWER DISTRIBUTION DIAGRAM (SHEET 2 OF 7)

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SA



FB1578_4E
SA

FIGURE 9-2-2-3. DC POWER DISTRIBUTION DIAGRAM (SHEET 4 OF 7)

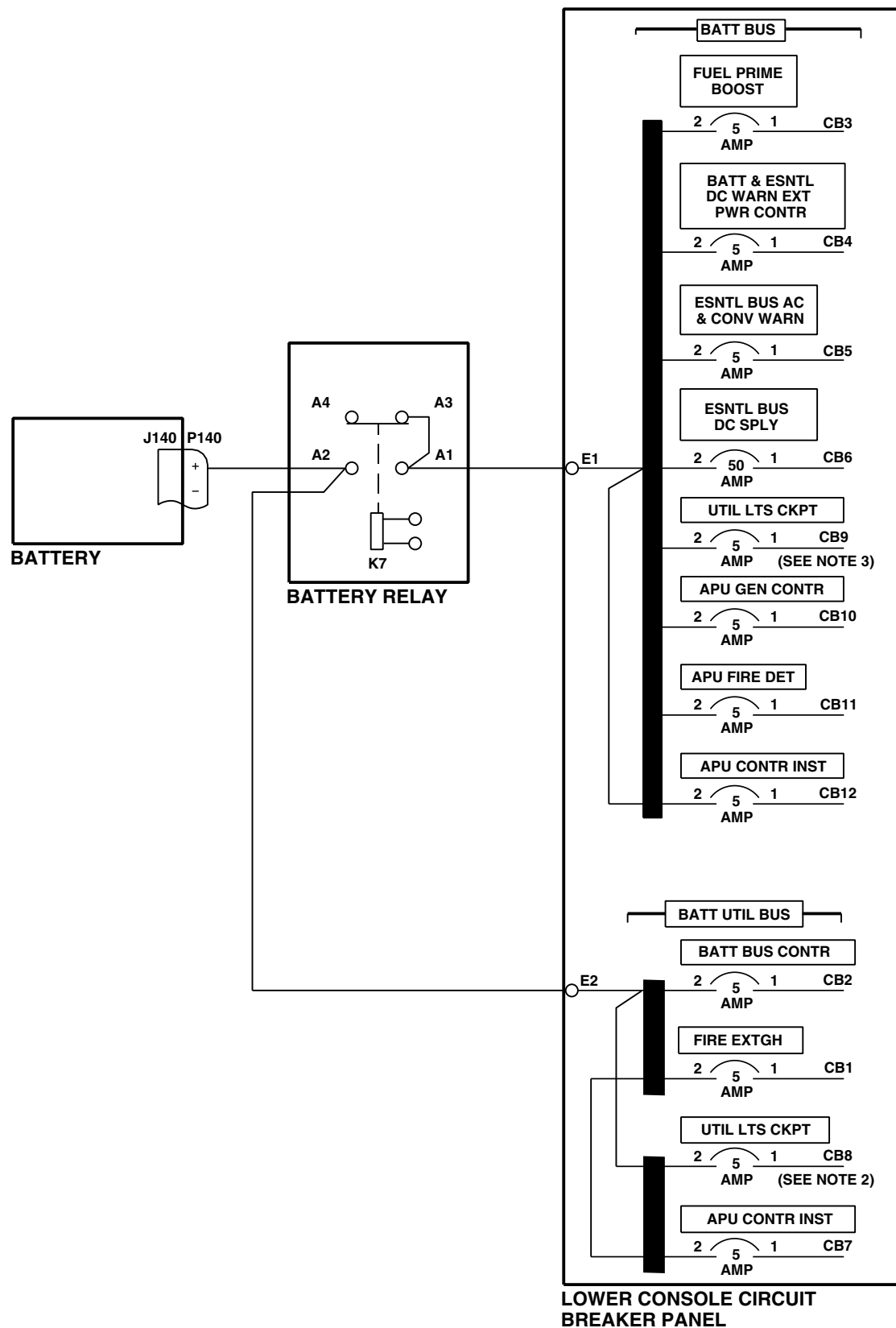
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FIGURE 9-2-2-3. DC POWER DISTRIBUTION DIAGRAM (SHEET 5 OF 7)

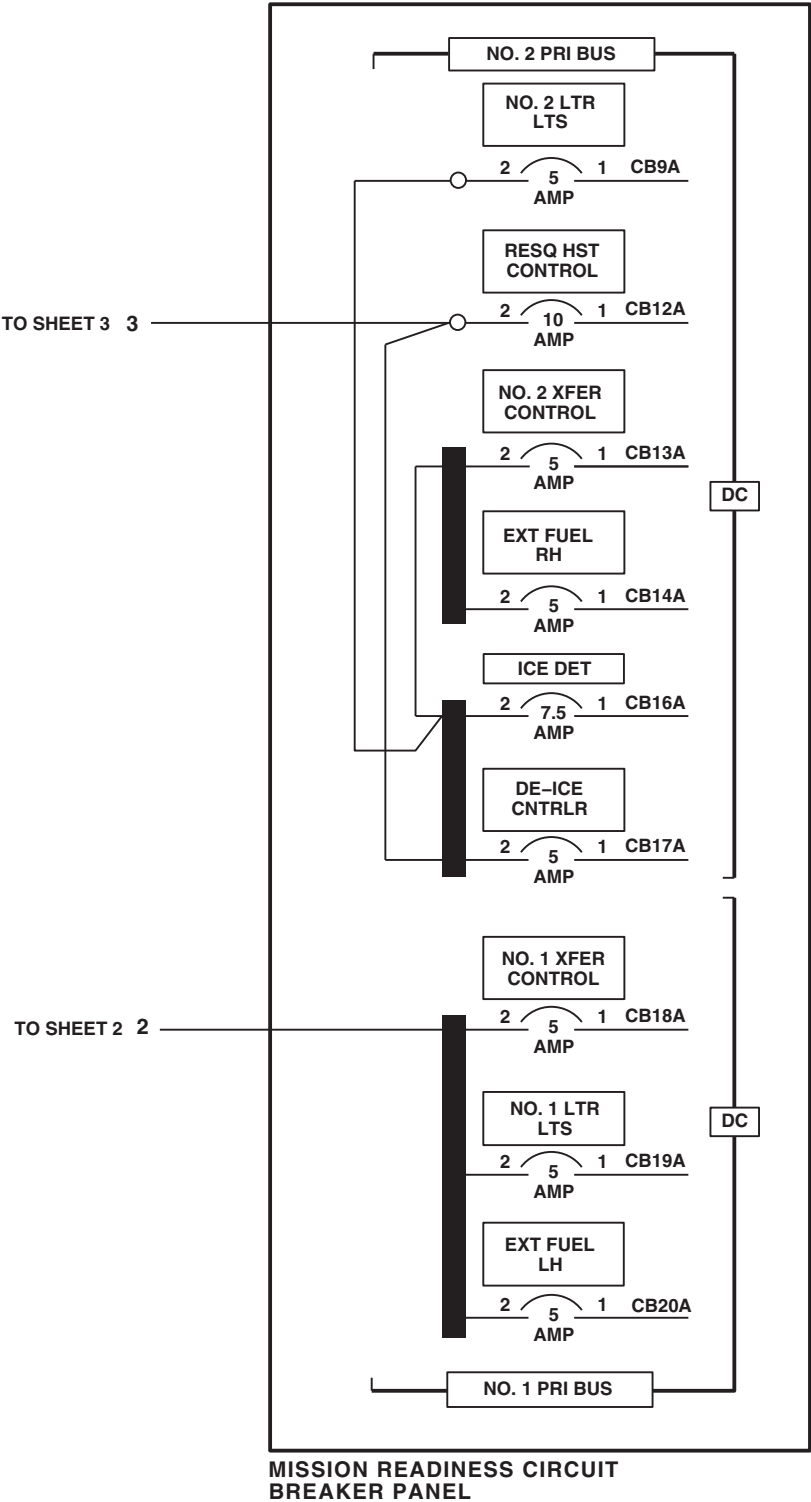


FIGURE 9-2-2-3. DC POWER DISTRIBUTION DIAGRAM (SHEET 6 OF 7)

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SA

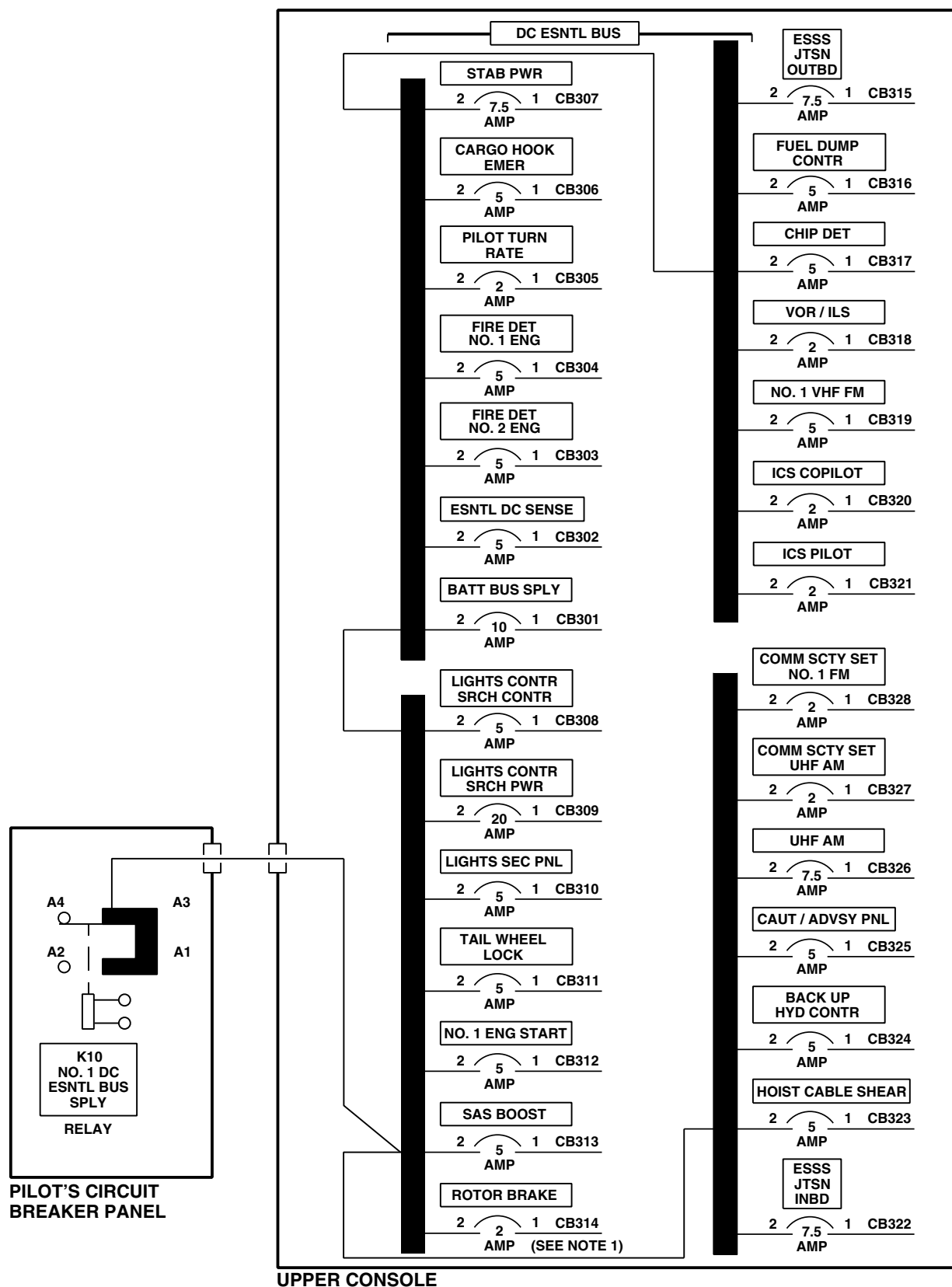
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FIGURE 9-2-2-3. DC POWER DISTRIBUTION DIAGRAM (SHEET 7 OF 7)

9-2-3 SYSTEM CIRCUIT BREAKERS.

INITIAL SETUP

Test Equipment

- Multimeter, Fluke 27/AN
- Multimeter, Fluke 45

Tools

- Electrical Repairer Toolkit

Equipment Conditions

- 28 VDC or 115 VAC (as applicable), Available at System Circuit Breaker

9-2-3.1 Operational/Troubleshooting Procedure.

NOTE

The following applies to any circuit breaker which pops out during system operation.

- Disconnect electrical connectors from all system components and check for signs of overheating, bent connector pins, or improper connection. Leave connectors off.
- Push in circuit breaker.
 - If circuit breaker stays in, component malfunction is indicated. Isolate malfunctioning component as follows:

- Pull out circuit breaker.
 - Sequentially connect all components serviced by circuit breaker, one component at a time.
 - Push in circuit breaker after each connection. When circuit breaker pops out, last component connected is then known to be malfunctioning.
- If circuit breaker pops, short circuit in helicopter wiring is indicated. Check for short circuit between helicopter ground and system wiring using multimeter. Repair/replace wiring as required (Appendix F).

9-2-4 INSTRUMENT PANEL LIGHTS.

INITIAL SETUP

Test Equipment

- Multimeter, Fluke 45

Tools

- Electrical Repairer Toolkit

References

- PARA 1-6-16

Equipment Conditions

- External Electrical Power Available or APU Operational

CAUTION

To prevent damage to lighting, do not measure voltage or continuity across transformers T5, T6, and T7.

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).

9-2-4.1 Operational/Troubleshooting Procedure.

- Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.
- Turn on electrical power (PARA 1-6-16).
- Turn CPLT FLT INST LTS control to OFF.
- Turn INST LT PILOT FLT control to OFF, and then turn slightly clockwise to get out of detent.
- Turn INST LT NON FLT control to BRT.

RESULT:

- All non-flight instrument panel lights shall go on.
 - Pilot's and copilot's collective stick grip lights shall go on.
- If results (1) and (2) are not as specified, go to TABLE NO. 9-2-4-1.

- If only icing rate meter light does not go on, go to TABLE NO. 9-2-4-2.
- If result (2) is not as specified, go to TABLE NO. 9-2-4-3.
- If lights on individual non-flight instruments do not go on, check continuity between P280-L and affected instrument.
 - If continuity is present, replace instrument or lighted bezel assembly (PARA 8-4-6).
 - If continuity is not present, repair/replace wiring (Appendix F).
- Turn INST LT NON FLT control to OFF.

RESULT: All non-flight instrument panel lights and pilot's and copilot's collective stick grip lights shall dim and go off.

- If result is not as specified, replace INST LT NON FLT dimmer control T7 (PARA 9-4-49).
- Turn INST LT PILOT FLT control to BRT.

RESULT: All pilot flight instrument panel lights shall go on.

- If result is not as specified, go to TABLE NO. 9-2-4-4.
- If lights on pilot's stabilator position placard and pilot's HSI/VSI mode select panel do not go on, go to TABLE NO. 9-2-4-5.
- If individual pilot flight instrument lights do not go on, check continuity between instrument panel connector J112 and affected instrument light.
 - (a) If continuity is present, replace lighted bezel assembly (PARA 8-4-6).
 - (b) If continuity is not present, repair/replace wiring (Appendix F).

h. Turn INST LT PILOT FLT control to OFF.

RESULT: All pilot flight instrument panel lights shall dim and go off.

- If result is not as specified, replace INST LT PILOT FLT dimmer control T6 (PARA 9-4-49).
- i. Turn CPLT FLT INST LTS control to BRT.

RESULT: All copilot flight instrument panel lights shall go on.

- If result is not as specified, go to TABLE NO. 9-2-4-6.
- If lights on copilot's stabilator position placard and copilot's HSI/VSI mode select panel do not go on, go to TABLE NO. 9-2-4-7.
- If lights on individual copilot flight instruments do not go on, check continuity between instrument panel connector J113 and affected instrument light.
 - (a) If continuity is present, replace lighted bezel assembly.
 - (b) If continuity is not present, repair/replace wiring (Appendix F).

j. Turn CPLT FLT INST LTS control to OFF.

RESULT: All copilot flight instrument panel lights shall dim and go off.

- If result is not as specified, replace CPLT FLT INST LTS dimmer control T5 (PARA 9-4-49).
- k. Turn off electrical power (PARA 1-6-16).

TABLE NO. 9-2-4-1. Non-flight Instrument Panel Lights And Pilot's/Copilot's Collective Stick Grip Lights Do Not Go On.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Do all non-flight instrument lights fail to go on?	Step 1. If all non-flight instrument lights do not go on, go to 2. Step 2. If some non-flight instrument lights go on, go to 5.
2. Check continuity between T7-G and ground.	Step 1. If continuity is present, go to 3. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 11.

TABLE NO. 9-2-4-1. Non-flight Instrument Panel Lights And Pilot's/Copilot's Collective Stick Grip Lights Do Not Go On. (Cont)

TEST OR INSPECTION CORRECTIVE ACTION
3. Check for 115 VAC between T7-HV and T7-G. Step 1. If voltage is as specified, replace T7 (PARA 9-4-49). Go to 11. Step 2. If voltage is not as specified, go to 4.
4. Check for 115 VAC between terminal 1 (LIGHTS NON FLT circuit breaker, NO. 2 AC PRI BUS - pilot's circuit breaker panel) and ground. Step 1. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and T7-HV (Appendix F). Go to 11. Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 11.
5. Do pilot's or copilot's collective stick grip lights go on? Step 1. If collective stick grip lights go on, go to 6. Step 2. If collective stick grip lights do not go on, repair/replace wiring, as required (Appendix F) between the following, then go to 11: J103-T and T7-LV J104-T and T7-LV
6. Do pilot's collective stick grip lights go on? Step 1. If lights go on, go to 7. Step 2. If lights do not go on, go to 9.
7. Check continuity between J103-V and ground. Step 1. If continuity is present, go to 8. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 11.
8. Check continuity between J103-T and J104-T. Step 1. If continuity is present, troubleshoot copilot's collective stick (PARA 11-2-2). Go to 11. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 11.
9. Check continuity between J104-V and ground.

TABLE NO. 9-2-4-1. Non-flight Instrument Panel Lights And Pilot's/Copilot's Collective Stick Grip Lights Do Not Go On. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 1. If continuity is present, go to 10. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 11.
10. Check continuity between J103-T and J104-T.	Step 1. If continuity is present, troubleshoot pilot's collective stick (PARA 11-2-2). Go to 11. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 11.
11. Procedure completed.	

TABLE NO. 9-2-4-2. Icing Rate Meter Light Does Not Come On.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for 115 VAC between P388-C and ground.	Step 1. If voltage is as specified, go to 2. Step 2. If voltage is not as specified, repair/replace wiring between P280-L and P388-C (Appendix F). Go to 4.
2. Check for 5 VAC between P132-M and ground.	Step 1. If voltage is as specified, replace icing rate meter (PARA 12-4-27). Go to 4. Step 2. If voltage is not as specified, go to 3.
3. Check continuity between P132-M and P388-A.	Step 1. If continuity is present, replace non-flight instrument lights 5V/115V transformer (PARA 9-4-55). Go to 4. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 4.
4. Procedure completed.	

TABLE NO. 9-2-4-3. Pilot's And/Or Copilot's Collective Stick Grip Lighting Does Not Go On.

TEST OR INSPECTION CORRECTIVE ACTION	
1. Does either pilot's or copilot's collective stick grip lighting go on?	
Step 1. If lighting goes on, go to 2 . Step 2. If lighting does not go on, go to 5 .	
2. Does pilot's collective stick grip lighting go on?	
Step 1. If lighting goes on, go to 3 . Step 2. If lighting does not go on, go to 4 .	
3. Check for 115 VAC between J103-T and J103-V.	
Step 1. If voltage is as specified, troubleshoot copilot's collective stick (PARA 11-2-2). Go to 6 . Step 2. If voltage is not as specified, repair/replace wiring, as required (Appendix F) between the following, then go to 6 : J103-T and P231-U J103-V and ground	
4. Check for 115 VAC between J104-T and J104-V.	
Step 1. If voltage is as specified, troubleshoot pilot's collective stick (PARA 11-2-2). Go to 6 . Step 2. If voltage is not as specified, repair/replace wiring, as required (Appendix F) between the following, then go to 6 : J104-T and P231-U J104-V and ground	
5. Check for 115 VAC between J231-U and ground.	
Step 1. If voltage is as specified, repair/replace wiring, as required (Appendix F) between the following, then go to 6 : P231-U and J103-T P231-U and J104-T Step 2. If voltage is not as specified, repair/replace wiring between J231-U and T7-LV (Appendix F). Go to 6 .	
6. Procedure completed.	

TABLE NO. 9-2-4-4. No Pilot's Flight Instrument Panel Lights Go On.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check continuity between T6-G and ground.	Step 1. If continuity is present, go to 2. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 4.
2. Check for 115 VAC between T6-HV and T6-G.	Step 1. If voltage is as specified, replace T6 (PARA 9-4-49). Go to 4. Step 2. If voltage is not as specified, go to 3.
3. Check for 115 VAC between terminal 1 (LIGHTS PLT FLT circuit breaker, NO. 2 AC PRI BUS - pilot's circuit breaker panel) and ground.	Step 1. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and T6-HV (Appendix F). Go to 4. Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 4.
4. Procedure completed.	

TABLE NO. 9-2-4-5. Lights On Pilot's Stabilator Position Placard And Pilot's HSI/VSI Mode Select Panel Do Not Go On.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for 5 VAC between J246-a and ground.	Step 1. If voltage is as specified, go to 2. Step 2. If voltage is not as specified, repair/replace wiring between J246-a and T6-LV (Appendix F). Go to 4.
2. Check for 5 VAC between: P386-A and P386-B P386-A and P386-D P386-A and P386-E	Step 1. If voltage is as specified, go to 3. Step 2. If voltage is not as specified, repair/replace wiring, as required (Appendix F) between the following, then go to 4:

TABLE NO. 9-2-4-5. Lights On Pilot’s Stabilator Position Placard And Pilot’s HSI/VSI Mode Select Panel Do Not Go On. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	P386-A and P246-a P386-B and ground P386-D and ground P386-E and ground
3.	Check continuity between P386-C and P112-B. Step 1. If continuity is present, replace pilot’s flight instrument lights 5V/115V transformer (PARA 9-4-55). Go to 4. Step 2. If continuity is not present, repair/replace wiring, as required (Appendix F) between, then go to 4: P317R-36 and P386-C Pilot’s stabilator position placard terminal PWR and P386-C
4.	Procedure completed.

TABLE NO. 9-2-4-6. No Copilot’s Flight Instrument Panel Lights Go On.

TEST OR INSPECTION	CORRECTIVE ACTION
1.	Check continuity between T5-G and ground. Step 1. If continuity is present, go to 2. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 4.
2.	Check for 115 VAC between T5-HV and T5-G. Step 1. If voltage is as specified, replace T5 (PARA 9-4-49). Go to 4. Step 2. If voltage is not as specified, go to 3.
3.	Check for 115 VAC between terminal 1 (LIGHTS CPLT FLT circuit breaker, NO. 1 AC PRI BUS - copilot’s circuit breaker panel) and ground. Step 1. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and T5-HV (Appendix F). Go to 4. Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 4.

TABLE NO. 9-2-4-6. No Copilot's Flight Instrument Panel Lights Go On. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
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4. Procedure completed.

TABLE NO. 9-2-4-7. Lights On Copilot's Stabilator Position Placard And Copilot's HSI/VSI Mode Select Panel Do Not Go On.

TEST OR INSPECTION	CORRECTIVE ACTION
--------------------	-------------------

1. Check for 5 VAC between J246-g and ground.

Step 1. If voltage is as specified, go to **2**.

Step 2. If voltage is not as specified, repair/replace wiring between J246-g and T5-LV (Appendix F). Go to **4**.

2. Check for 5 VAC between:

P387-A and P387-B

P387-A and P387-D

P387-A and P387-E

Step 1. If voltage is as specified, go to **3**.

Step 2. If voltage is not as specified, repair/replace wiring between P387-A and P246-g (Appendix F). Go to **4**.

3. Check continuity between P387-C and P300R-36.

Step 1. If continuity is present, replace copilot's flight instrument lights 5V/115V transformer (PARA 9-4-55). Go to **4**.

Step 2. If continuity is not present, repair/replace wiring, as required (Appendix F) between the following, then go to **4**:

P300R-36 and P387-C

Copilot's stabilator position placard terminal PWR and P387-C

4. Procedure completed.

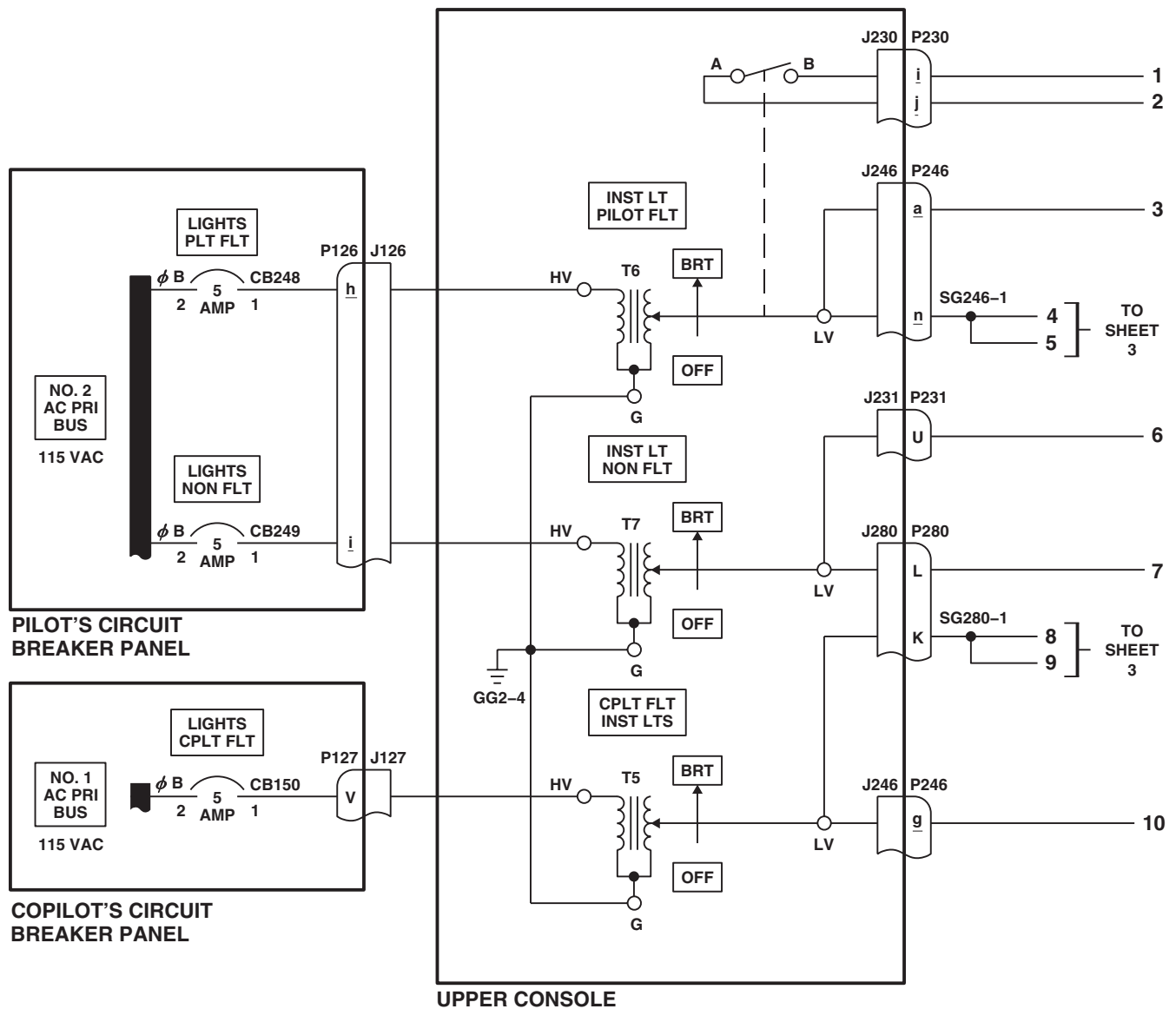
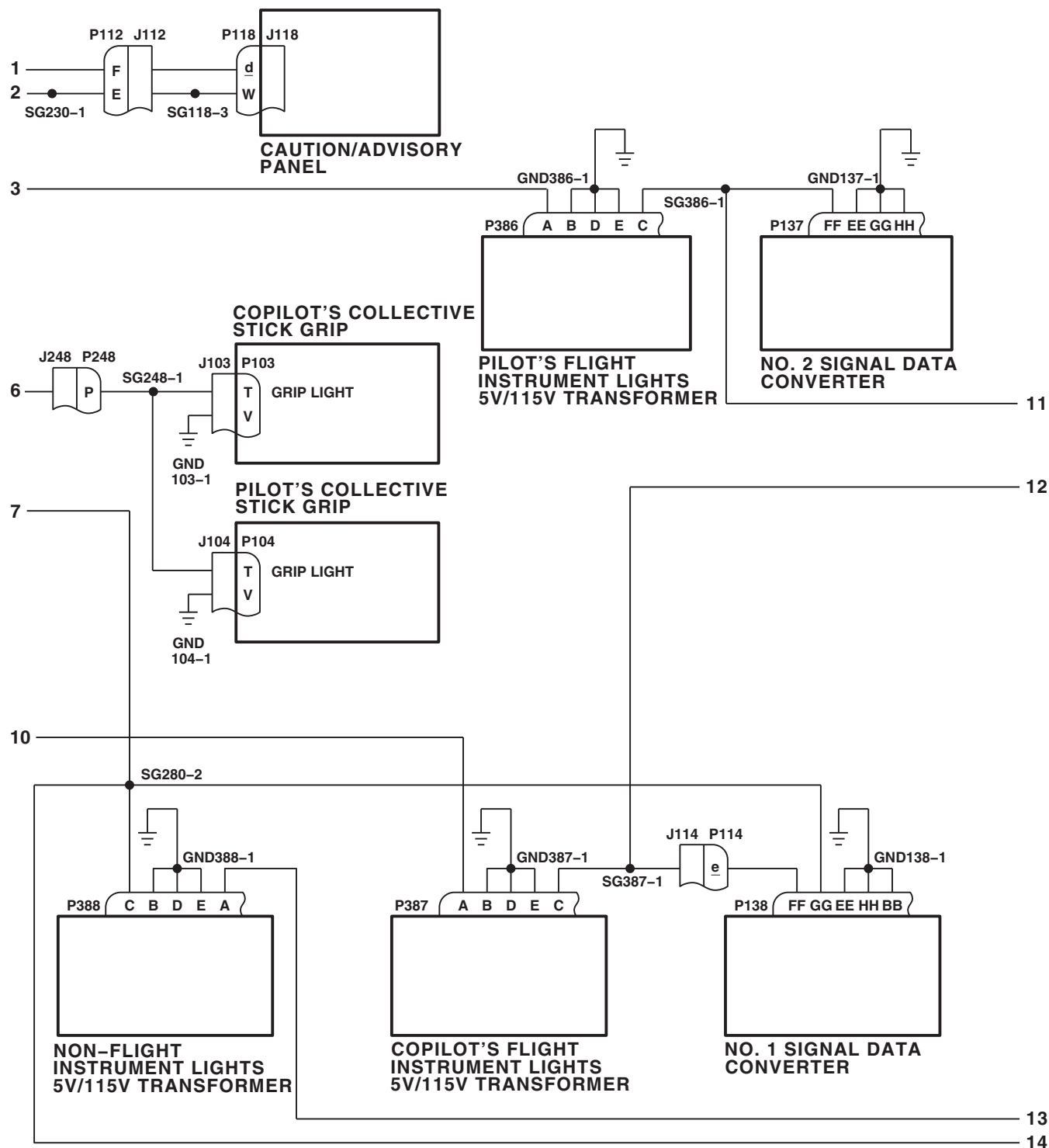


FIGURE 9-2-4-1. INSTRUMENT PANEL LIGHTS SCHEMATIC DIAGRAM (SHEET 1 OF 5)

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FB1583_2A
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FIGURE 9-2-4-1. INSTRUMENT PANEL LIGHTS SCHEMATIC DIAGRAM (SHEET 2 OF 5)

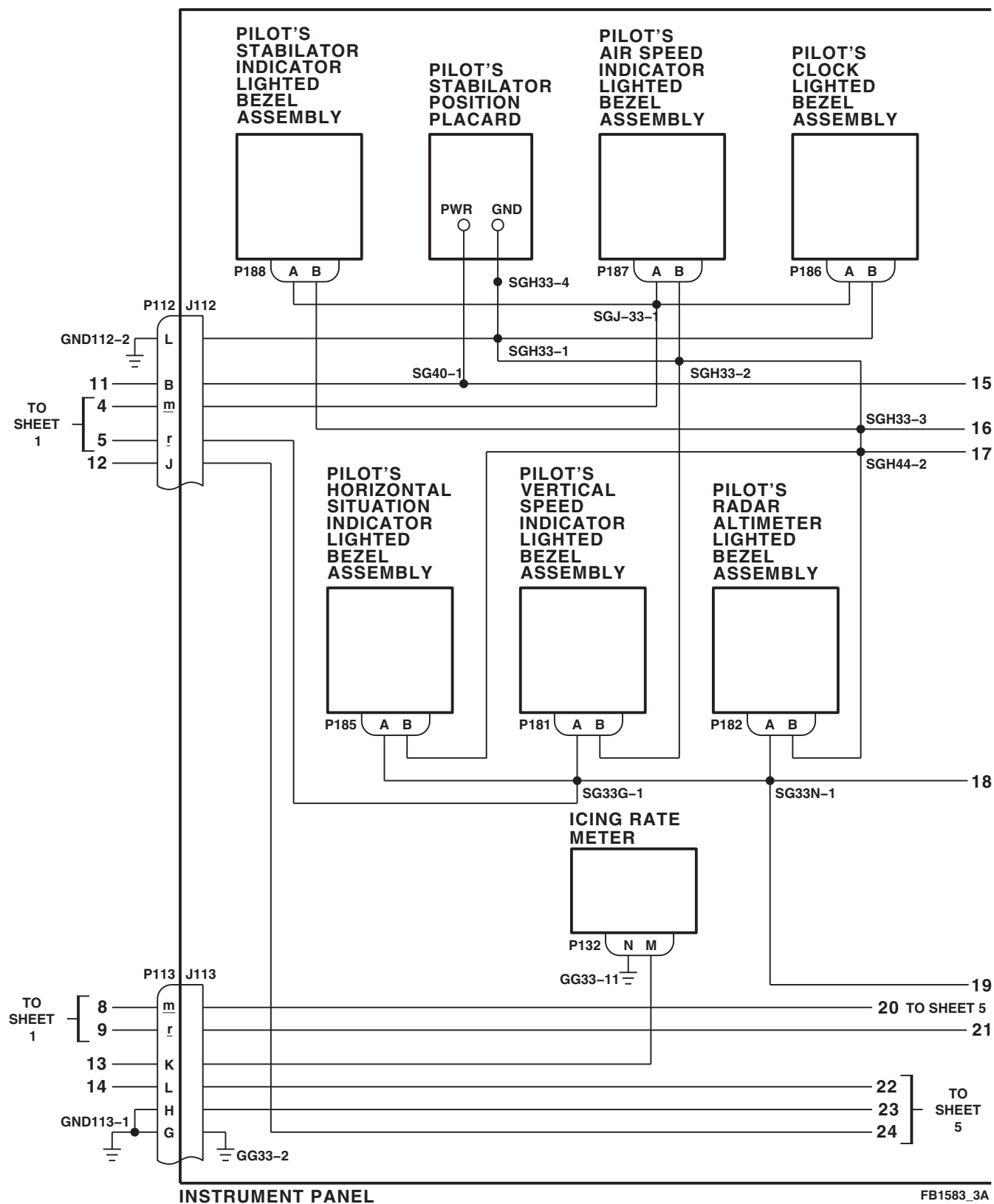
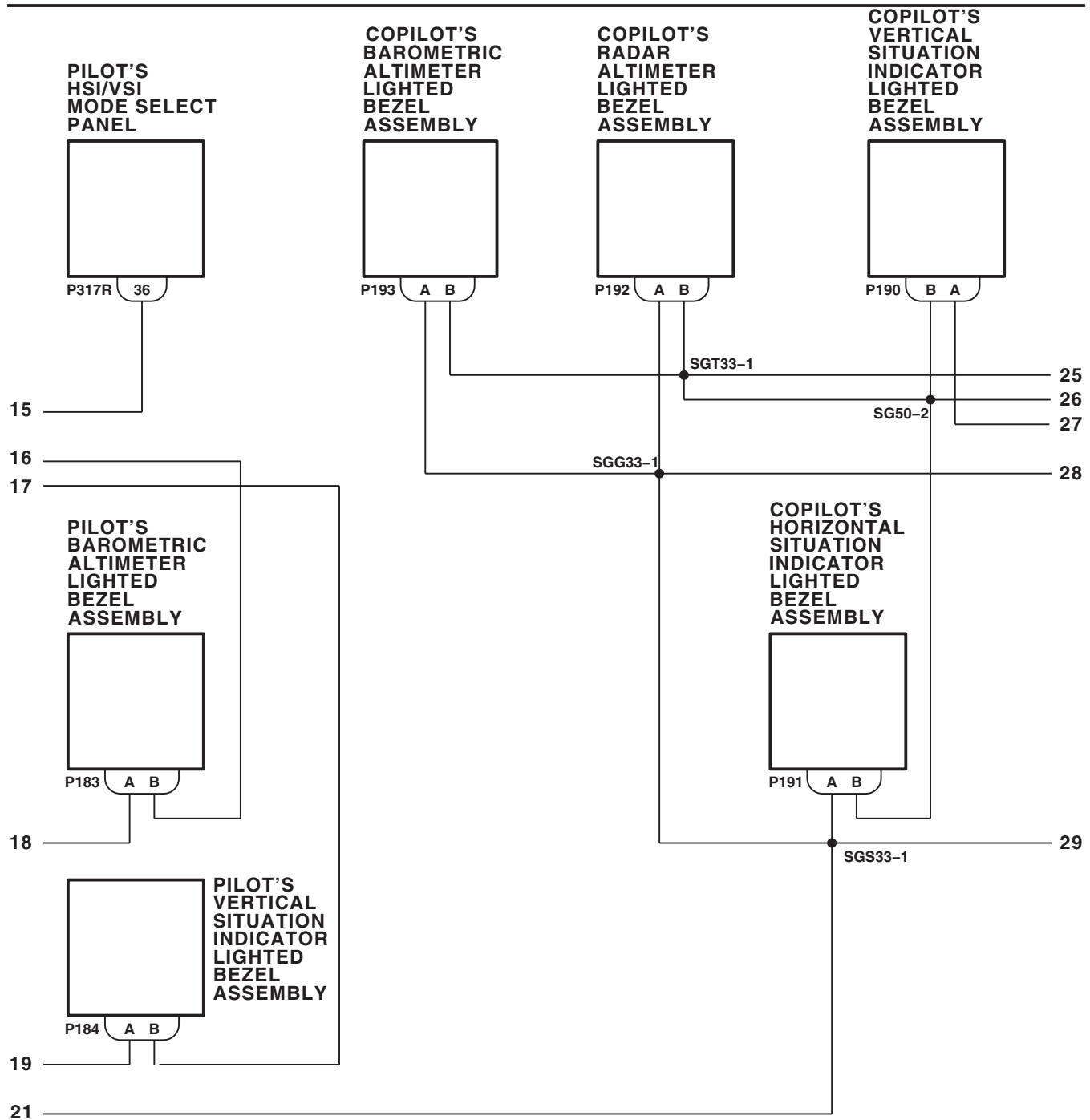


FIGURE 9-2-4-1. INSTRUMENT PANEL LIGHTS SCHEMATIC DIAGRAM (SHEET 3 OF 5)

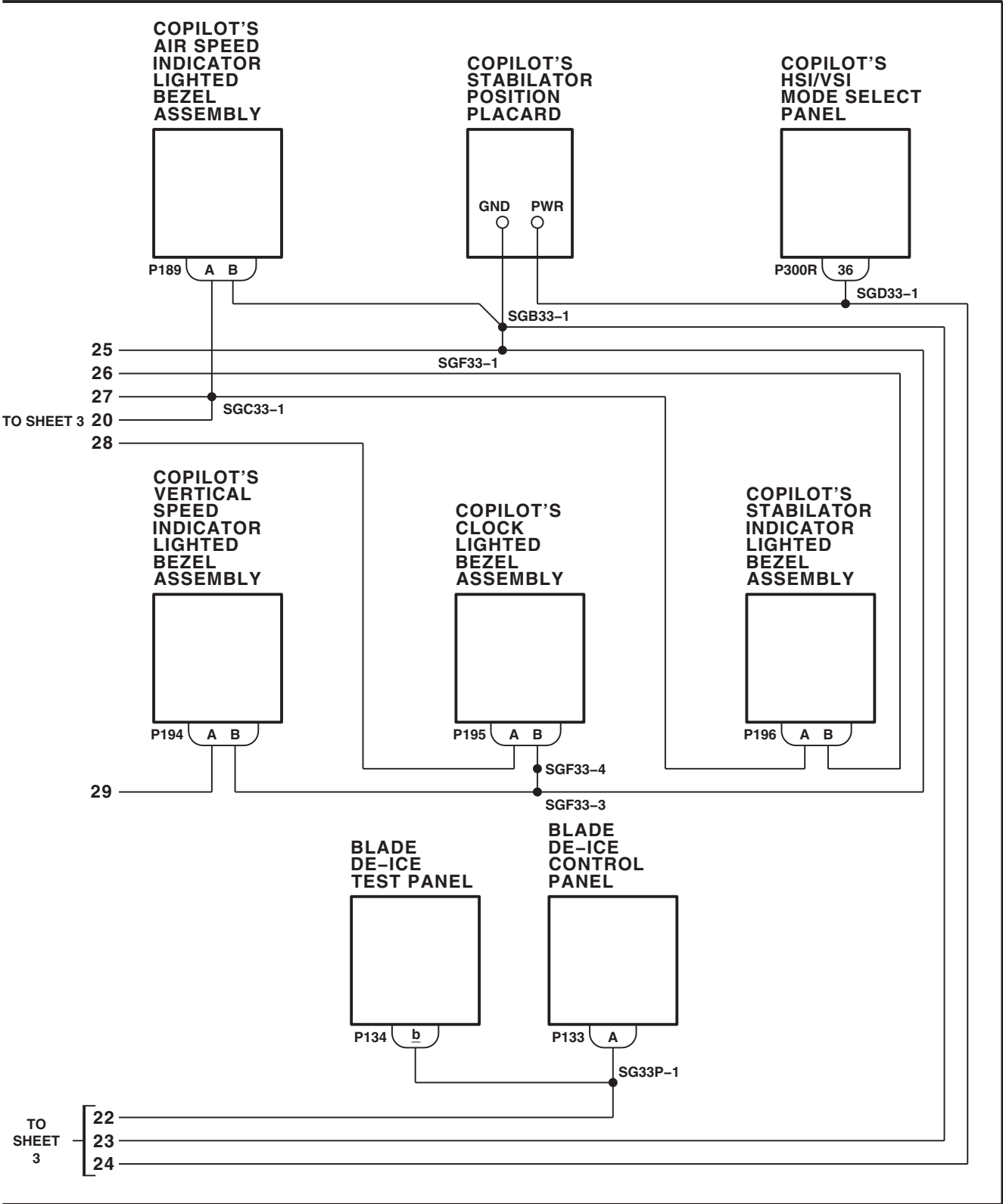


INSTRUMENT PANEL

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FIGURE 9-2-4-1. INSTRUMENT PANEL LIGHTS SCHEMATIC DIAGRAM (SHEET 4 OF 5)

9-2-4-12 Change 7



INSTRUMENT PANEL

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FIGURE 9-2-4-1. INSTRUMENT PANEL LIGHTS SCHEMATIC DIAGRAM (SHEET 5 OF 5)

9-2-5 UPPER AND LOWER CONSOLE LIGHTS **W/O SLS** .

INITIAL SETUP

Test Equipment

- Multimeter, Fluke 45

Tools

- Electrical Repairer Toolkit

References

- PARA 1-6-16

Equipment Conditions

- External Electrical Power Available or APU Operational

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).

9-2-5.1 Operational/Troubleshooting Procedure.

- Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.
- Turn on electrical power (PARA 1-6-16).
- On upper console, turn CONSOLE LT LOWER control to BRT.

RESULT: All lower console lights and ICS control panel floodlights shall go from dim to bright as control is turned clockwise.

- If result is not as specified, go to TABLE NO. 9-2-5-1.
- If any individual 0-115 VAC light does not go on, go to CHART NO. 9-2-5-2.
- If the lower console lights flicker when control is turned clockwise, replace CONSOLE LT LOWER dimming control T11 (PARA 9-4-49).
- If ICS control panel floodlights do not go on, check for 115 VAC between P657R-Z and ground with dimming control in BRT position.

- If voltage is as specified, replace audio junction box assembly (TM 11-70-23).

- If voltage is not as specified, repair/replace wiring between P657R-Z and J219-m (Appendix F).

- If VOR/ILS control panel and LF/ADF set control panel lights do not go on, check continuity between P102-A and P138R-18.

- If continuity is present, replace lower console lights 5V/115V transformer. █

- If continuity is not present, repair/replace wiring between P102-A and P138R-18 (Appendix F).

- On upper console, turn CONSOLE LT LOWER control to OFF.

RESULT: Lower console lights shall go from bright to dim, then off as control is turned counterclockwise.

- If result is not as specified, replace CONSOLE LT LOWER dimming control (PARA 9-4-49).

- On upper console, turn CONSOLE LT UPPER control to BRT.

RESULT: The following lights shall go on bright as control is turned clockwise:

- (1) Upper console controls
- (2) Secondary lights panel
- (3) Engine controls quadrant
- If result is not as specified, go to TABLE NO. 9-2-5-3.
- If the upper console lights flicker when control is turned clockwise, replace CONSOLE LT UPPER dimming control T3 (PARA 9-4-49).
- If any individual upper console light does not go on, go to CHART NO. 9-2-5-4.

f. On upper console, turn CONSOLE LT UPPER control to OFF.

RESULT: The following lights shall go from bright to dim, then off as control is turned counterclockwise:

- (1) Upper console controls
 - (2) Cockpit flood and secondary lights panel
 - (3) Engine controls quadrant
 - If result is not as specified, replace CONSOLE LT UPPER dimming control T3 (PARA 9-4-49).
- g. Turn off electrical power (PARA 1-6-16).

TABLE NO. 9-2-5-1. Lower Console Lights Do Not Go From Dim To Bright.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for 115 VAC between T11-HV and ground.	Step 1. If voltage is as specified, go to 2. Step 2. If voltage is not as specified, go to 3.
2. Check continuity between CONSOLE LT LOWER dimming control T11-G and ground.	Step 1. If continuity is present, go to 4. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 5.
3. Check for 115 VAC between terminal 1 (LIGHTS LWR CSL circuit breaker, NO. 1 AC PRI BUS - copilot's circuit breaker panel) and ground.	Step 1. If voltage is as specified, repair/replace wiring between T11-HV and circuit breaker terminal 1 (Appendix F). Go to 5. Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 5.
4. With CONSOLE LT LOWER control set to BRT, check for 115 VAC between T11-LV and ground.	

1. Check for 115 VAC between T11-HV and ground.

Step 1. If voltage is as specified, go to 2.
 Step 2. If voltage is not as specified, go to 3.

2. Check continuity between CONSOLE LT LOWER dimming control T11-G and ground.

Step 1. If continuity is present, go to 4.
 Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 5.

3. Check for 115 VAC between terminal 1 (LIGHTS LWR CSL circuit breaker, NO. 1 AC PRI BUS - copilot's circuit breaker panel) and ground.

Step 1. If voltage is as specified, repair/replace wiring between T11-HV and circuit breaker terminal 1 (Appendix F). Go to 5.
 Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 5.

4. With CONSOLE LT LOWER control set to BRT, check for 115 VAC between T11-LV and ground.

TABLE NO. 9-2-5-1. Lower Console Lights Do Not Go From Dim To Bright. (Cont)

**TEST OR INSPECTION
CORRECTIVE ACTION**

Step 1. If voltage is as specified, repair/replace wiring between T11-LV and P219-m (Appendix F). Go to **5**.

Step 2. If voltage is not as specified, replace dimming control T11 (PARA 9-4-49). Go to **5**.

5. Procedure completed.

CHART NO. 9-2-5-2. Any Individual 0-115 VAC Light Does Not Go On.

COMPONENT	CHECK CONTINUITY BETWEEN	IF YES	IF NO
COMPASS SET CONTROL	P134R-U and T11-LV	Replace panel (TM 11-70-23).	Repair/replace wiring (Appendix F).
CABIN DOME LIGHT DIMMER PANEL	P455-D and T11-LV	Replace panel (PARA 9-4-51).	Repair/replace wiring (Appendix F).
MISCELLANEOUS SWITCH PANEL	P659R-G and T11-LV	Replace panel (PARA 9-4-7).	Repair/replace wiring (Appendix F).
STABILATOR CONTROL/ AUTO FLT CONTROL PANEL	P4R-e and T11-LV	Replace panel (TM 11-70-23).	Repair/replace wiring (Appendix F).
FUEL BOOST PUMP CONTROL PANEL	P139-L and T11-LV	Replace panel (PARA 9-4-40).	Repair/replace wiring (Appendix F).
RADIO RETRANSMISSION CONTROL PANEL	P116R-J and T11-LV	Replace panel (TM 11-70-23).	Repair/replace wiring (Appendix F).
RESCUE HOIST KIT CONTROL PANEL	P228-K and T11-LV	Replace panel (PARA 14-4-25).	Repair/replace wiring (Appendix F).
AUXILIARY FUEL MANAGE- MENT CONTROL PANEL	P360-T and T11-LV	Replace panel (PARA 16-4-38).	Repair/replace wiring (Appendix F).

TABLE NO. 9-2-5-3. No Upper Console Lights Go On.

TEST OR INSPECTION CORRECTIVE ACTION
1. Check for 115 VAC between T3-HV and ground.
Step 1. If voltage is as specified, go to 2 . Step 2. If voltage is not as specified, go to 3 .
2. Check continuity between T3-G and ground.
Step 1. If continuity is present, replace T3 (PARA 9-4-49). Go to 4 . Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 4 .
3. Check for 115 VAC between terminal 1 (LIGHTS UPPER CSL circuit breaker, NO. 1 AC PRI BUS - copilot's circuit breaker panel) and ground.
Step 1. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and T3-HV (Appendix F). Go to 4 . Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 4 .
4. Procedure completed.

CHART NO. 9-2-5-4. Any Individual Upper Console Light Does Not Go On.

COMPONENT	CHECK CONTINUITY BETWEEN	IF YES	IF NO
COCKPIT FLOOD AND SECONDARY LIGHTS PANEL	P288-E and T3-LV	Replace panel (PARA 9-4-69).	Repair/replace wiring (Appendix F).
LEFT SIDE CONTROL PANEL	1J2 and T3-LV	Replace panel (PARA 9-4-6).	Repair/replace wiring (Appendix F).
CENTER CONTROL PANEL	1J1 and T3-LV	Replace panel (PARA 9-4-6).	Repair/replace wiring (Appendix F).
RIGHT SIDE CONTROL PANEL	1J3 and T3-LV	Replace panel (PARA 9-4-6).	Repair/replace wiring (Appendix F).
ENGINE CONTROL LEFT QUADRANT PANEL	J394-1 and T3-LV	Replace panel (PARA 4-4-21).	Repair/replace wiring (Appendix F).
ENGINE CONTROL CENTER QUADRANT PANEL	J395-1 and T3-LV	Replace panel (PARA 4-4-21).	Repair/replace wiring (Appendix F).
ENGINE CONTROL RIGHT QUADRANT PANEL	J396-1 and T3-LV	Replace panel (PARA 4-4-21).	Repair/replace wiring (Appendix F).

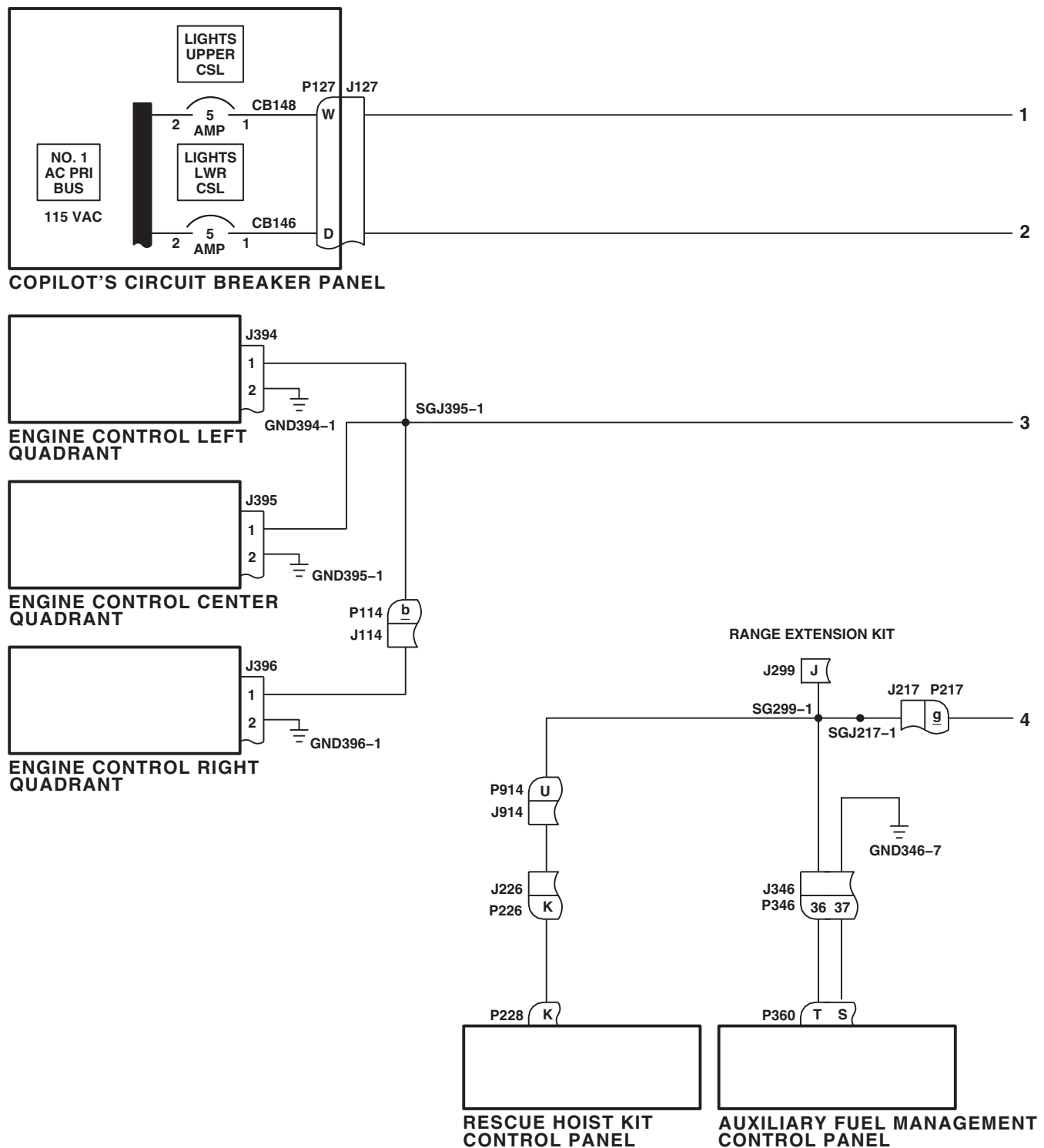
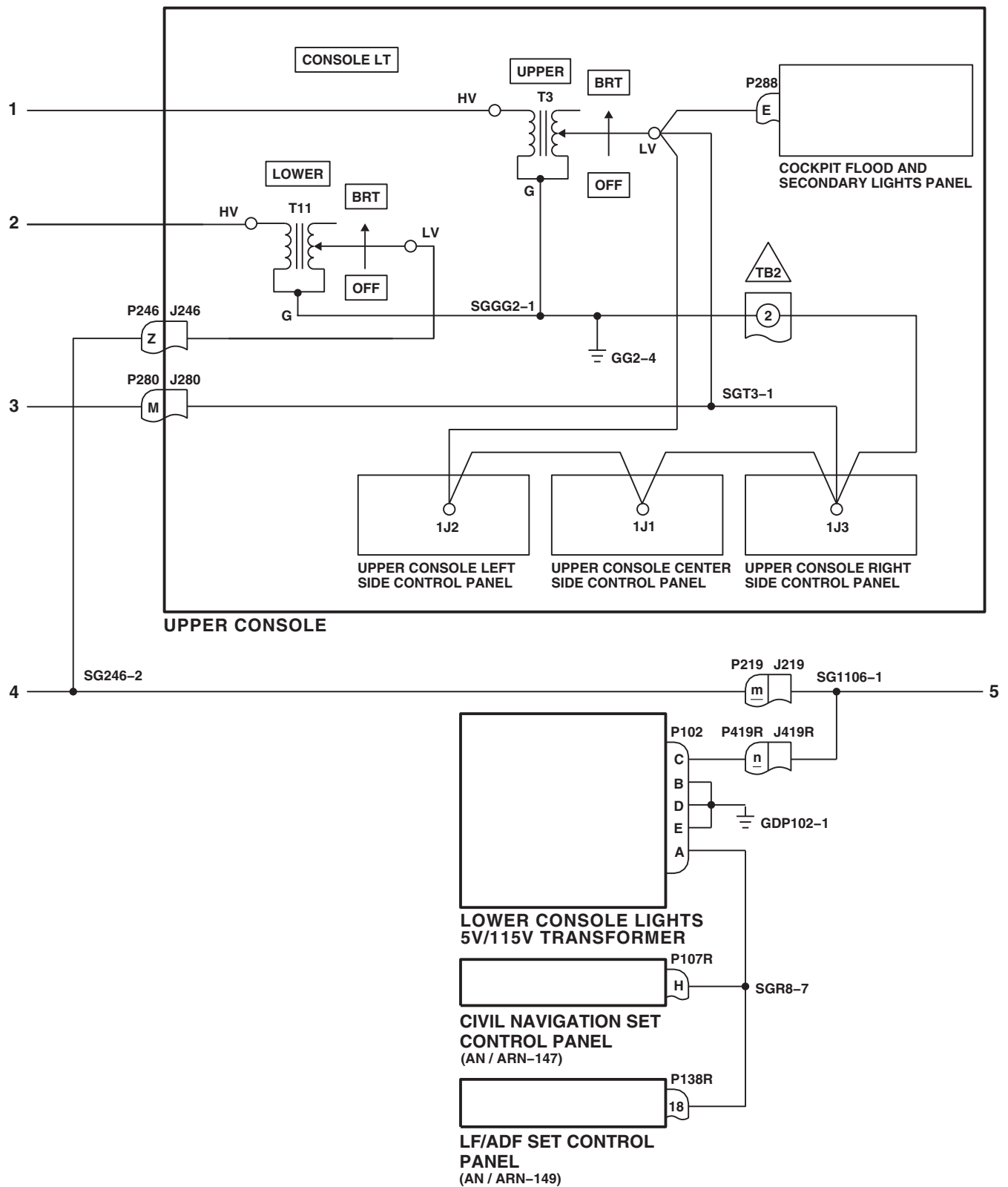
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FIGURE 9-2-5-1. UPPER AND LOWER CONSOLE LIGHTS SCHEMATIC DIAGRAM
(SHEET 1 OF 4)



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FIGURE 9-2-5-1. UPPER AND LOWER CONSOLE LIGHTS SCHEMATIC DIAGRAM (SHEET 2 OF 4)

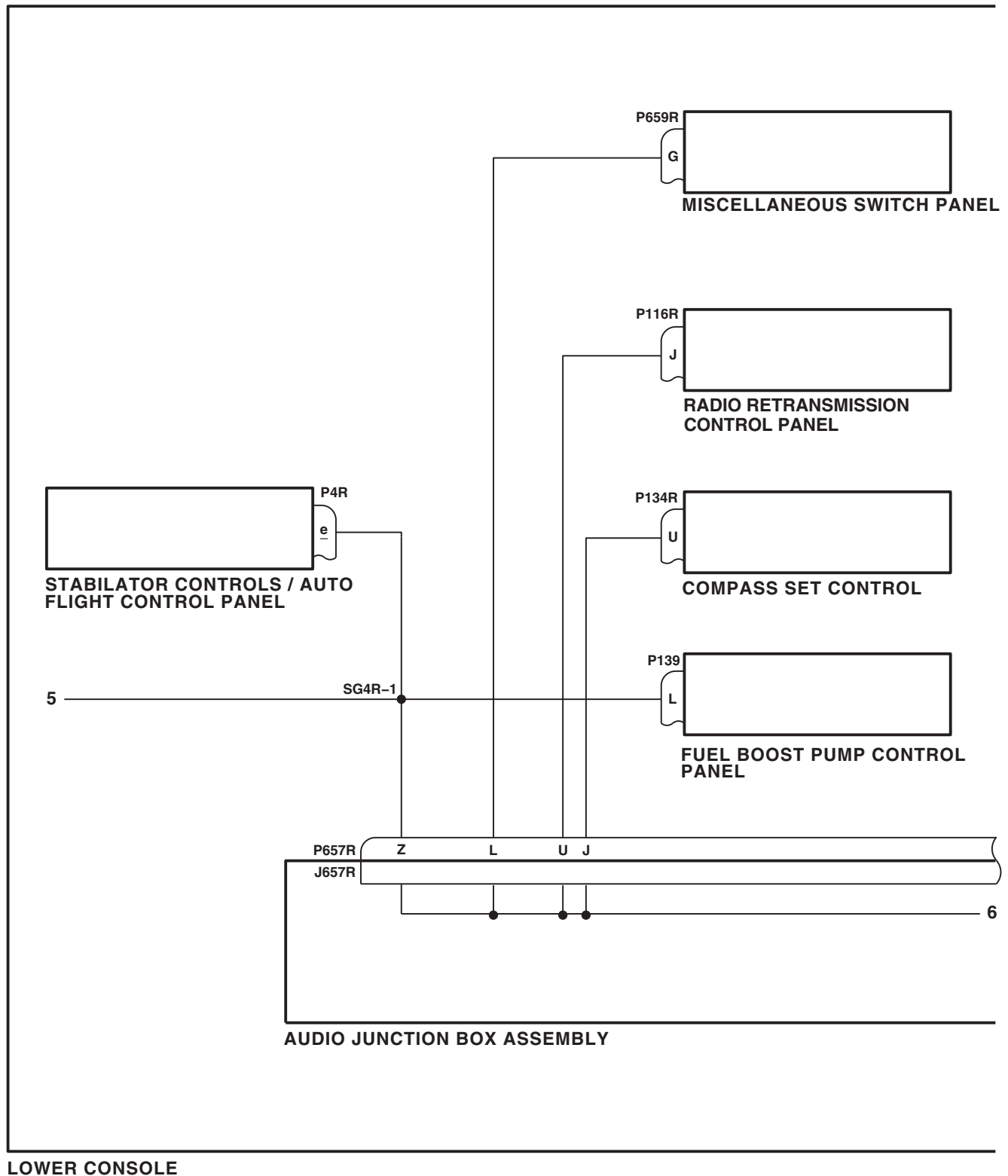
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FIGURE 9-2-5-1. UPPER AND LOWER CONSOLE LIGHTS SCHEMATIC DIAGRAM
(SHEET 3 OF 4)

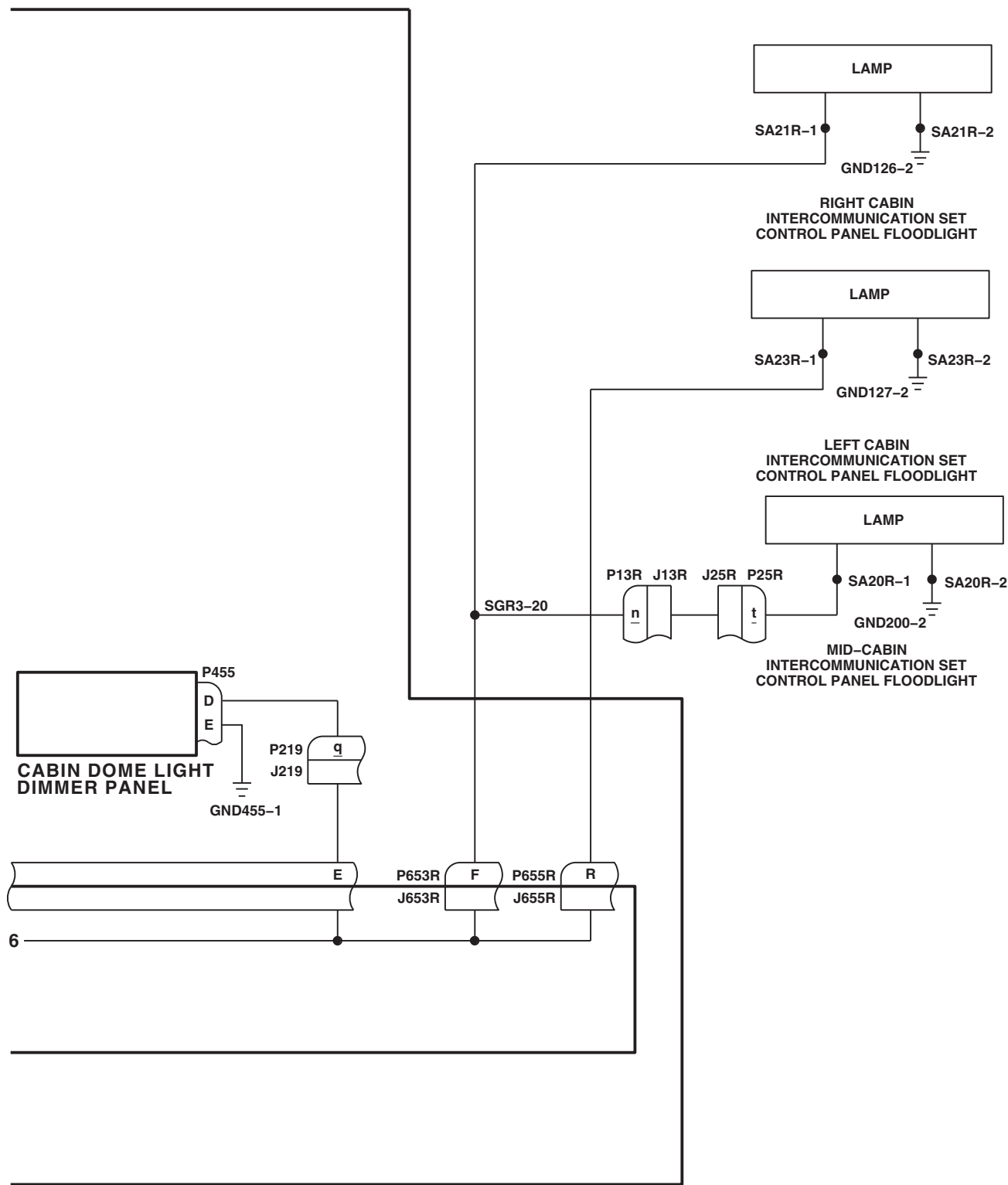


FIGURE 9-2-5-1. UPPER AND LOWER CONSOLE LIGHTS SCHEMATIC DIAGRAM
 (SHEET 4 OF 4)

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9-2-6 CABIN DOME LIGHTS.

INITIAL SETUP

Test Equipment

- Multimeter, Fluke 45

Tools

- Electrical Repairer Toolkit

References

- PARA 1-6-16

Equipment Conditions

- External Electrical Power Available or APU Operational

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).

9-2-6.1 Operational/Troubleshooting Procedure.

- Make sure all circuit breakers are pushed in, except those as noted being pulled for maintenance or safety.
- Turn on electrical power (PARA 1-6-16).
- Turn CABIN DOME LT control to BRT.
- Place upper console CABIN DOME LT switch to WHITE.

RESULT: All white cabin dome lights shall go on.

- If result is not as specified, go to TABLE NO. 9-2-6-1.
- If one white cabin dome light does not go on, check lamp.
 - If lamp is not good, replace lamp (PARA 9-4-60).
 - If lamp is good, repair/replace wiring between CABIN DOME LT switch terminal 1 and the following, as required (Appendix F):

- J262-A

- J263-A
- J264-A
- e. Place upper console CABIN DOME LT switch to BLUE.

RESULT: All white cabin dome lights shall go off and all blue cabin dome lights shall go on.

- If result is not as specified, replace CABIN DOME LT switch (PARA 9-4-5).
- If one blue cabin dome light does not go on, check lamp.
 - If lamp is not good, replace lamp (PARA 9-4-60).
 - If lamp is good, repair/replace wiring between CABIN DOME LT switch terminal 3 and the following, as required (Appendix F):

- J262-B
- J263-B
- J264-B

- Turn CABIN DOME LT control to OFF.

RESULT: Lights shall dim and go off.

g. Turn off electrical power (PARA 1-6-16).

- If result is not as specified, replace CABIN DOME LT dimming control T1 (PARA 9-4-51).

TABLE NO. 9-2-6-1. No White Cabin Dome Lights Go On.

TEST OR INSPECTION CORRECTIVE ACTION
1. Check for 115 VAC between CABIN DOME LT dimming control T1-HV and ground. Step 1. If voltage is as specified, go to 2 . Step 2. If voltage is not as specified, go to 4 .
2. Check for 28 VAC between T1-LV and ground. Step 1. If voltage is as specified, go to 3 . Step 2. If voltage is not as specified, replace dimming control T1 (PARA 9-4-51). Go to 5 .
3. Check for 28 VAC between CABIN DOME LT switch S12-2 and ground. Step 1. If voltage is as specified, replace switch S12 (PARA 9-4-5). Go to 5 . Step 2. If voltage is not as specified, repair/replace wiring between S12-2 and T1-LV (Appendix F). Go to 5 .
4. Check for 115 VAC between terminal 1 (LIGHTS CABIN DOME circuit breaker, NO. 1 AC PRI BUS - copilot's circuit breaker panel) and ground. Step 1. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and T1-HV (Appendix F). Go to 5 . Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 5 .
5. Procedure completed.

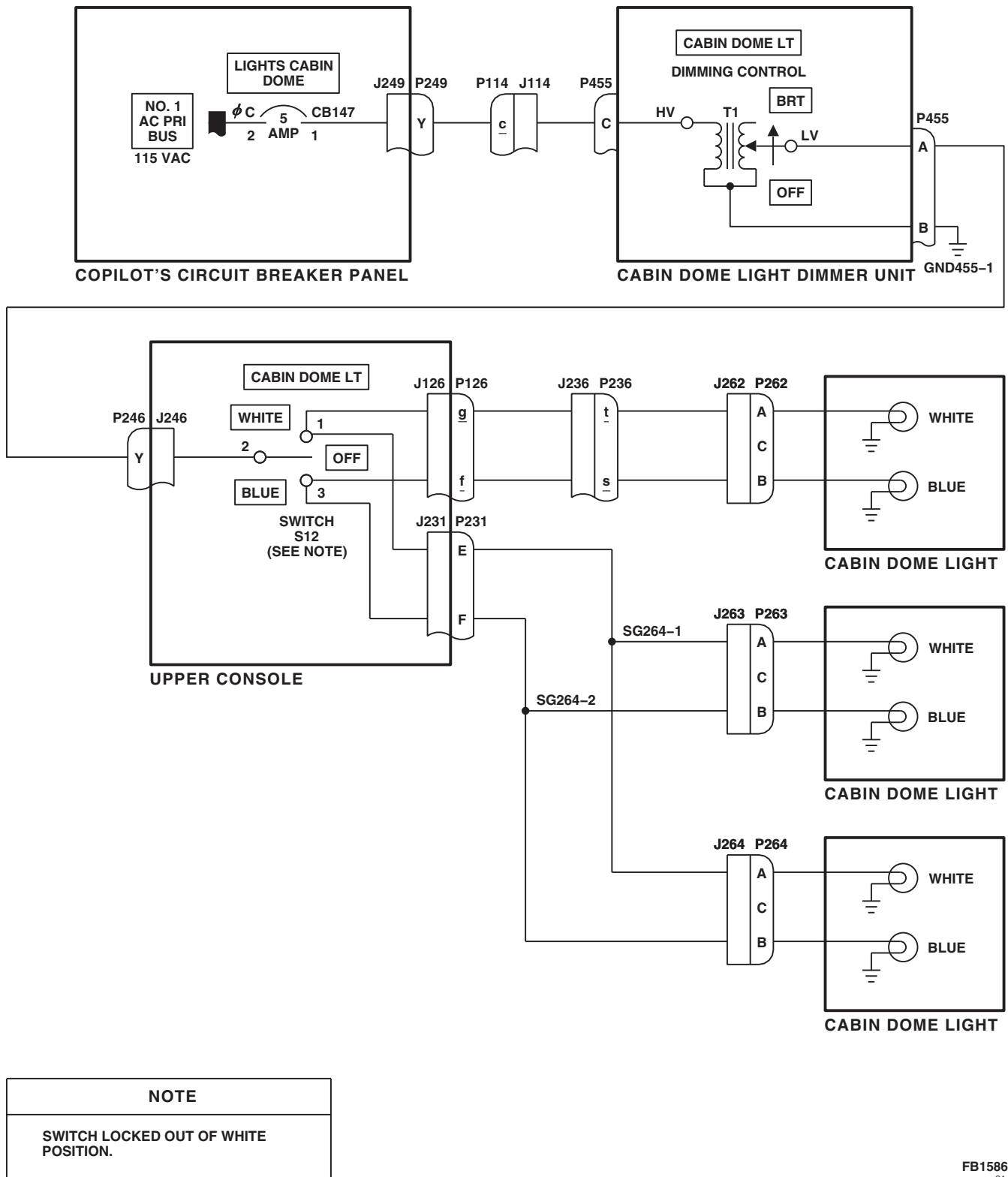


FIGURE 9-2-6-1. CABIN DOME LIGHTS SCHEMATIC DIAGRAM

9-2-7 COCKPIT FLOOD AND SECONDARY LIGHTS.

INITIAL SETUP

Test Equipment

- Multimeter, Fluke 45

Tools

- Electrical Repairer Toolkit

References

- PARA 1-6-16

Equipment Conditions

- External Electrical Power Available or APU Operational

NOTE

If a circuit breaker pops out during the Cockpit Flood And Secondary Lights Operational/Troubleshooting Procedure, a short circuit is indicated (PARA 9-2-3).

9-2-7.1 Operational/Troubleshooting Procedure

- Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.
- Make sure GLARESHIELD LIGHTS dimming control is in OFF position.
- Turn on electrical power (PARA 1-6-16).
- Place cockpit flood and secondary lights panel BLUE/OFF/WHITE switch to OFF.
- Turn GLARESHIELD LIGHTS control to BRT.

RESULT: Instrument panel glare shield lights shall go on and brighten.

- If result is not as specified, go to TABLE NO. 9-2-7-1.
- If one glare shield light does not go on, replace lamp or socket (PARA 9-4-65).
- f. Turn GLARESHIELD LIGHTS control to OFF.

RESULT: Instrument panel glare shield lights shall dim and go off.

- If result is not as specified, replace GLARESHIELD LIGHTS dimming control T4 (PARA 9-4-49).
- g. Place cockpit flood and secondary lights panel BLUE/OFF/WHITE switch to WHITE.

RESULT: Cockpit upper console flood and secondary lights panel white flood lights shall go on.

- If result is not as specified, go to TABLE NO. 9-2-7-2.
- If one cockpit white flood light does not go on, replace flood light lamp (PARA 9-4-67).
- h. Place upper console cockpit flood and secondary lights panel BLUE/OFF/WHITE switch to BLUE.

RESULT: Both white flood lights shall go off and both blue lights shall go on.

- If result is not as specified, replace cockpit flood and secondary lights panel (PARA 9-4-69).
- If one blue secondary light does not go on, replace secondary light lamp (PARA 9-4-67).

- i. Place upper console cockpit flood and secondary lights panel BLUE/OFF/WHITE switch to OFF.

RESULT: All secondary overhead lights shall go off.

- If result is not as specified, replace cockpit flood and secondary lights panel (PARA 9-4-69).
- j. Place standby magnetic compass ON/OFF light switch to ON.

RESULT: Standby magnetic compass post light shall go on.

- If result is not as specified, go to TABLE NO. 9-2-7-3.

- k. Place standby magnetic compass ON/OFF light switch to OFF.

- l. Turn off electrical power (PARA 1-6-16).

TABLE NO. 9-2-7-1. Glare Shield Lights Do Not Go On With GLARESHIELD LIGHTS Control At BRT.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for 115 VAC between P110-E and ground. Step 1. If voltage is as specified, go to 2. Step 2. If voltage is not as specified, go to 3. 2. Check continuity between: J110-E and J162-A J110-E and J162-C J162-B and ground J162-D and ground Step 1. If continuity is present, replace glare shield harness (PARA 9-4-66). Go to 6. Step 2. If continuity is not present, repair/replace wiring (Appendix F) as required. Go to 6. 3. Check continuity between: P110-E and T4-LV T4-G and ground Step 1. If continuity is present, go to 4. Step 2. If continuity is not present, repair/replace wiring (Appendix F) as required. Go to 6. 4. Check for 115 VAC between dimming control T4-HV and ground. Step 1. If voltage is as specified, replace dimming control T4 (PARA 9-4-49). Go to 6.	

TABLE NO. 9-2-7-1. Glare Shield Lights Do Not Go On With GLARESHIELD LIGHTS Control At BRT. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 2. If voltage is not as specified, go to 5 .
5. Check for 115 VAC between terminal 1 (LIGHTS GLARE SHLD circuit breaker, NO. 1 AC PRI BUS - copilot's circuit breaker panel) and ground.	
	Step 1. If voltage is as specified, repair/replace wiring between terminal 1 and T4 (Appendix F), then go to 6 .
	Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 6 .
6. Procedure completed.	

TABLE NO. 9-2-7-2. Cockpit White Flood Lights Do Not Go On.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for 28 VDC between P288-C and ground.	
	Step 1. If voltage is as specified, go to 3 .
	Step 2. If voltage is not as specified, go to 2 .
2. Check continuity between P288-C and terminal 1 (LIGHTS SEC PNL circuit breaker, DC ESNTL BUS - pilot's side upper console circuit breaker panel).	
	Step 1. If continuity is present, replace circuit breaker (PARA 9-4-1). Go to 4 .
	Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 4 .
3. Check continuity between P288-D and ground.	
	Step 1. If continuity is present, replace cockpit flood and secondary lights panel (PARA 9-4-69). Go to 4 .
	Step 2. If continuity is not present, repair/replace wiring (Appendix F) as required. Go to 4 .
4. Procedure completed.	

TABLE NO. 9-2-7-3. Standby Magnetic Compass Post Light Does Not Go On.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for 28 VDC between compass post light and ground.	Step 1. If voltage is as specified, go to 3. Step 2. If voltage is not as specified, go to 2.
2. Check continuity between: S39-2 and terminal 1 (LIGHTS SEC PNL circuit breaker, DC ESNTL BUS - pilot's side upper console circuit breaker panel) S39-3 and compass post light	Step 1. If continuity is present, replace standby compass switch (PARA 8-4-9). Go to 4. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 4.
3. Check continuity between post light and ground.	Step 1. If continuity is present, replace post light (PARA 8-4-8). Go to 4. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 4.
4. Procedure completed.	

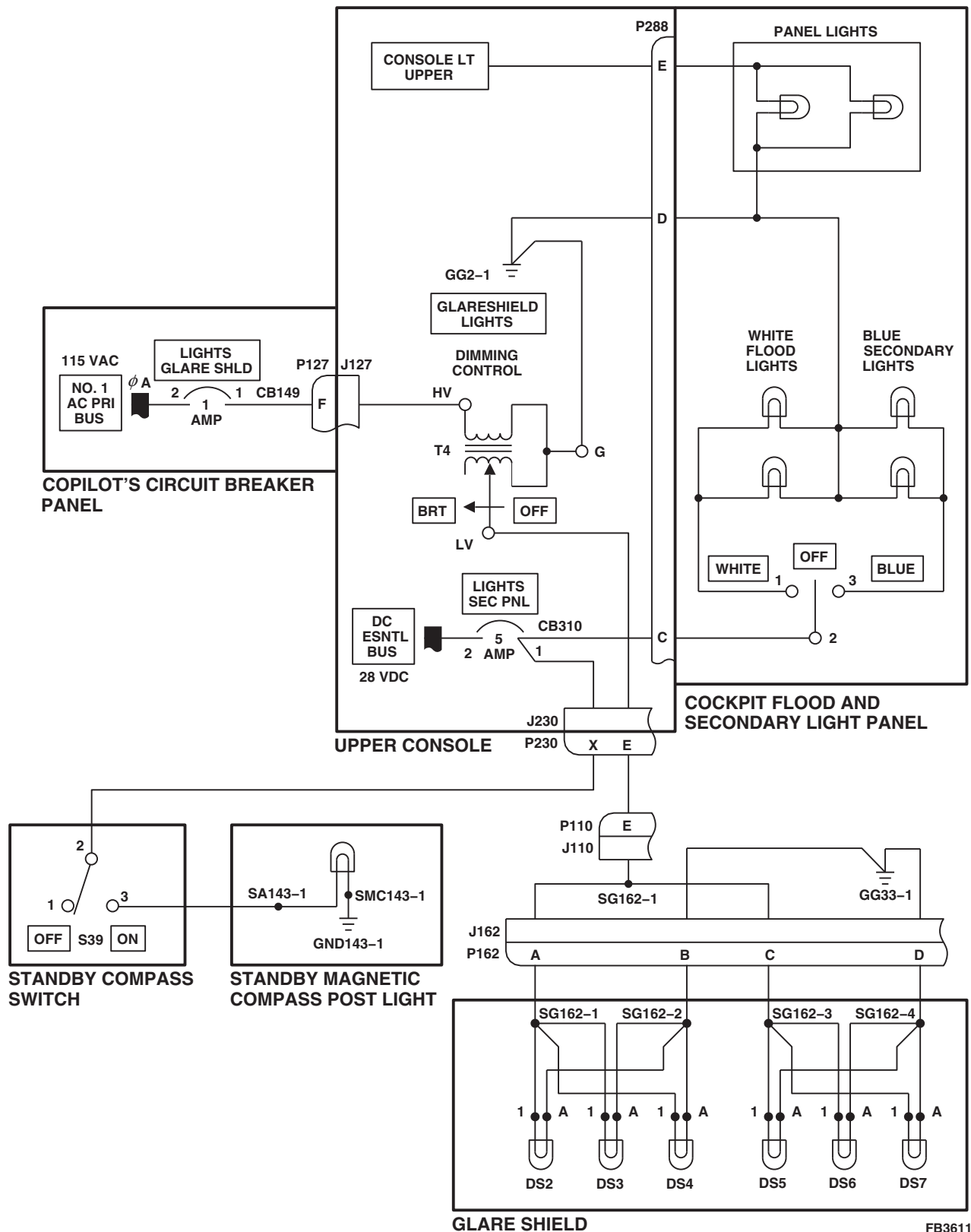


FIGURE 9-2-7-1. COCKPIT FLOOD AND SECONDARY LIGHTS SCHEMATIC DIAGRAM

9-2-8 COCKPIT FLOOD AND SECONDARY LIGHTS **W/O SLS** .

INITIAL SETUP

Test Equipment

- Multimeter, Fluke 45

Tools

- Electrical Repairer Toolkit

References

- PARA 1-6-16

Equipment Conditions

- External Electrical Power Available or APU Operational

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).

9-2-8.1 Operational/Troubleshooting Procedure.

- Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.
- Turn on electrical power (PARA 1-6-16).
- Place cockpit flood and secondary lights panel RED/REMOTE/WHITE switch to REMOTE.
- Turn upper console CKPT FLOOD LIGHTS WHITE control to BRT.

RESULT: Both cockpit white floodlights shall go on.

- If result is not as specified, go to TABLE NO. 9-2-8-1.

- Turn upper console CKPT FLOOD LIGHTS WHITE control to OFF.

RESULT: Both cockpit white floodlights shall dim and go off.

- If result is not as specified, replace CKPT FLOOD LIGHTS WHITE dimmer control T8 (PARA 9-4-49).

- Turn upper console SECONDARY LIGHTS RED control to BRT.

RESULT:

- Red secondary lights on cockpit flood and secondary lights panel shall go on.
- Red lights on glareshield shall go on.

- If result (1) is not as specified, go to TABLE NO. 9-2-8-2.
- If result (2) is not as specified, go to TABLE NO. 9-2-8-3.
- If results (1) and (2) are not as specified, go to TABLE NO. 9-2-8-4.

- Turn upper console SECONDARY LIGHTS RED control to OFF.

RESULT:

- Red secondary lights on cockpit flood and secondary lights panel shall go off.
- Red lights on glareshield shall go off.

- If results are not as specified, replace SECONDARY LIGHTS RED dimmer control T4 (PARA 9-4-49).

- Place cockpit flood and secondary lights panel RED/REMOTE/WHITE switch to WHITE.

RESULT: Both cockpit white floodlights shall go on.

- If result is not as specified, go to TABLE NO. 9-2-8-5.

If trouble remains, replace floodlight lamp (PARA 9-4-67).

- i. Place cockpit flood and secondary lights panel RED/REMOTE/WHITE switch to RED.

RESULT: Both secondary lights on cockpit flood and secondary lights panel shall go on.

- If result is not as specified, replace cockpit flood and secondary lights panel (PARA 9-4-69).

- (a) If trouble remains, replace red secondary light lamp (PARA 9-4-68).

- j. Place standby magnetic compass ON/OFF light switch to ON.

RESULT: Standby magnetic compass post light shall go on.

- If result is not as specified, go to TABLE NO. 9-2-8-6.

- k. Place standby magnetic compass ON/OFF light switch to OFF.

- l. Turn off electrical power (PARA 1-6-16).

TABLE NO. 9-2-8-1. Cockpit White Flood Lights Do Not Go On.

TEST OR INSPECTION	CORRECTIVE ACTION
	1. Check for 28 VAC between dimming control T8-LV and ground. Step 1. If voltage is as specified, go to 2. Step 2. If voltage is not as specified, go to 3.
	2. Check for 28 VAC between P288-A and ground. Step 1. If voltage is as specified, go to 4. Step 2. If voltage is not as specified, repair/replace wiring between P288-A and T8-LV (Appendix F). Go to 6.
	3. Check for 115 VAC between dimming control T8-HV and ground. Step 1. If voltage is as specified, replace dimming control T8 (PARA 9-4-49). Go to 6. Step 2. If voltage is not as specified, go to 5.
	4. Check continuity between P288-D and ground. Step 1. If continuity is present, replace cockpit flood and secondary lights panel (PARA 9-4-69). Go to 6.

TABLE NO. 9-2-8-1. Cockpit White Flood Lights Do Not Go On. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 2. If continuity is not present, repair/replace wiring between P288-D and GG2-1 (Appendix F). Go to 6 .
5. Check for 115 VAC between terminal 1 (LIGHTS CKPT FLOOD circuit breaker, NO. 2 AC PRI BUS - pilot's circuit breaker panel) and ground.	
	Step 1. If voltage is as specified, repair/replace wiring between terminal 1 and T8-HV (Appendix F). Go to 6 .
	Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 6 .
6. Procedure completed.	

TABLE NO. 9-2-8-2. Red Secondary Lights On Panel Do Not Go On.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for 28 VAC between dimming control T4-LV and ground.	
	Step 1. If voltage is as specified, go to 2 .
	Step 2. If voltage is not as specified, go to 3 .
2. Check for 28 VAC between P288-B and ground.	
	Step 1. If voltage is as specified, replace cockpit flood and secondary lights panel (PARA 9-4-69). Go to 5 .
	Step 2. If voltage is not as specified, repair/replace wiring between P288-B and T4-LV (Appendix F). Go to 5 .
3. Check for 115 VAC between dimming control T4-HV and ground.	
	Step 1. If voltage is as specified, replace dimming control T4 (PARA 9-4-49). Go to 5 .
	Step 2. If voltage is not as specified, go to 4 .
4. Check for 115 VAC between terminal 1 (LIGHTS SEC circuit breaker, NO. 1 AC PRI BUS - copilot's circuit breaker panel) and ground.	

TABLE NO. 9-2-8-2. Red Secondary Lights On Panel Do Not Go On. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 1. If voltage is as specified, repair/replace wiring between terminal 1 and T4-HV (Appendix F). Go to 5. Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 5.
5. Procedure completed.	

TABLE NO. 9-2-8-3. Red Lights On Glareshield Do Not Go On.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for 28 VAC between dimming control T4-LV and ground.	Step 1. If voltage is as specified, go to 2. Step 2. If voltage is not as specified, go to 3.
2. Check for 28 VAC between J162-A and ground.	Step 1. If voltage is as specified, replace red glareshield lamps (PARA 9-4-65). Go to 5. Step 2. If voltage is not as specified, repair/replace wiring between J162-A and T4-LV (Appendix F). Go to 5.
3. Check for 115 VAC between dimming control T4-HV and ground.	Step 1. If voltage is as specified, replace dimming control T4 (PARA 9-4-49). Go to 5. Step 2. If voltage is not as specified, go to 4.
4. Check for 115 VAC between terminal 1 (LIGHTS SEC circuit breaker, NO. 1 AC PRI BUS - copilot's circuit breaker panel) and ground.	Step 1. If voltage is as specified, repair/replace wiring between terminal 1 and T4-HV (Appendix F). Go to 5. Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 5.
5. Procedure completed.	

TABLE NO. 9-2-8-4. No Red Lights Go On.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for 28 VAC between dimming control T4-LV and ground.	Step 1. If voltage is as specified, repair/replace wiring between T4-LV and S43-7 (Appendix F). Go to 4. Step 2. If voltage is not as specified, go to 2.
2. Check for 115 VAC between dimming control T4-HV and ground.	Step 1. If voltage is as specified, replace dimming control T4 (PARA 9-4-49). Go to 4. Step 2. If voltage is not as specified, go to 3.
3. Check for 115 VAC between terminal 1 (LIGHTS SEC circuit breaker, NO. 1 AC PRI BUS - copilot's circuit breaker panel) and ground.	Step 1. If voltage is as specified, repair/replace wiring between terminal 1 and T4-HV (Appendix F). Go to 4. Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 4.
4. Procedure completed.	

TABLE NO. 9-2-8-5. Cockpit White Flood Lights Do Not Go On.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for 28 VDC between P288-C and ground.	Step 1. If voltage is as specified, replace cockpit flood and secondary lights panel (PARA 9-4-69). Go to 3. Step 2. If voltage is not as specified, go to 2.
2. Check continuity between P288-C and terminal 1 (LIGHTS SEC PNL circuit breaker, DC ESNTL BUS - upper console circuit breaker panel).	Step 1. If continuity is present, replace circuit breaker (PARA 9-4-1). Go to 3.

TABLE NO. 9-2-8-5. Cockpit White Flood Lights Do Not Go On. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 3 .
	3. Procedure completed.

TABLE NO. 9-2-8-6. Pilot's Standby Magnetic Compass Post Light Does Not Go On.

TEST OR INSPECTION	CORRECTIVE ACTION
	1. Check for 28 VDC between P143-1 and P143-2.
	Step 1. If voltage is as specified, go to 2 .
	Step 2. If voltage is not as specified, go to 5 .
	2. Check continuity between P143-2 and ground.
	Step 1. If continuity is present, go to 3 .
	Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 6 .
	3. Check continuity between P143-1 and standby magnetic compass switch S39-3.
	Step 1. If continuity is present, go to 4 .
	Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 6 .
	4. Check continuity between S39-2 and S39-3.
	Step 1. If continuity is present, replace standby magnetic compass (PARA 8-4-7). Go to 6 .
	Step 2. If continuity is not present, replace S39 (PARA 8-4-9). Go to 6 .
	5. Check for 28 VDC between terminal 1 (LIGHTS SEC PNL circuit breaker, DC ESNTL BUS - upper console circuit breaker panel) and ground.
	Step 1. If voltage is as specified, repair/replace wiring between terminal 1 and S39-2 (Appendix F). Go to 6 .
	Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 6 .
	6. Procedure completed.

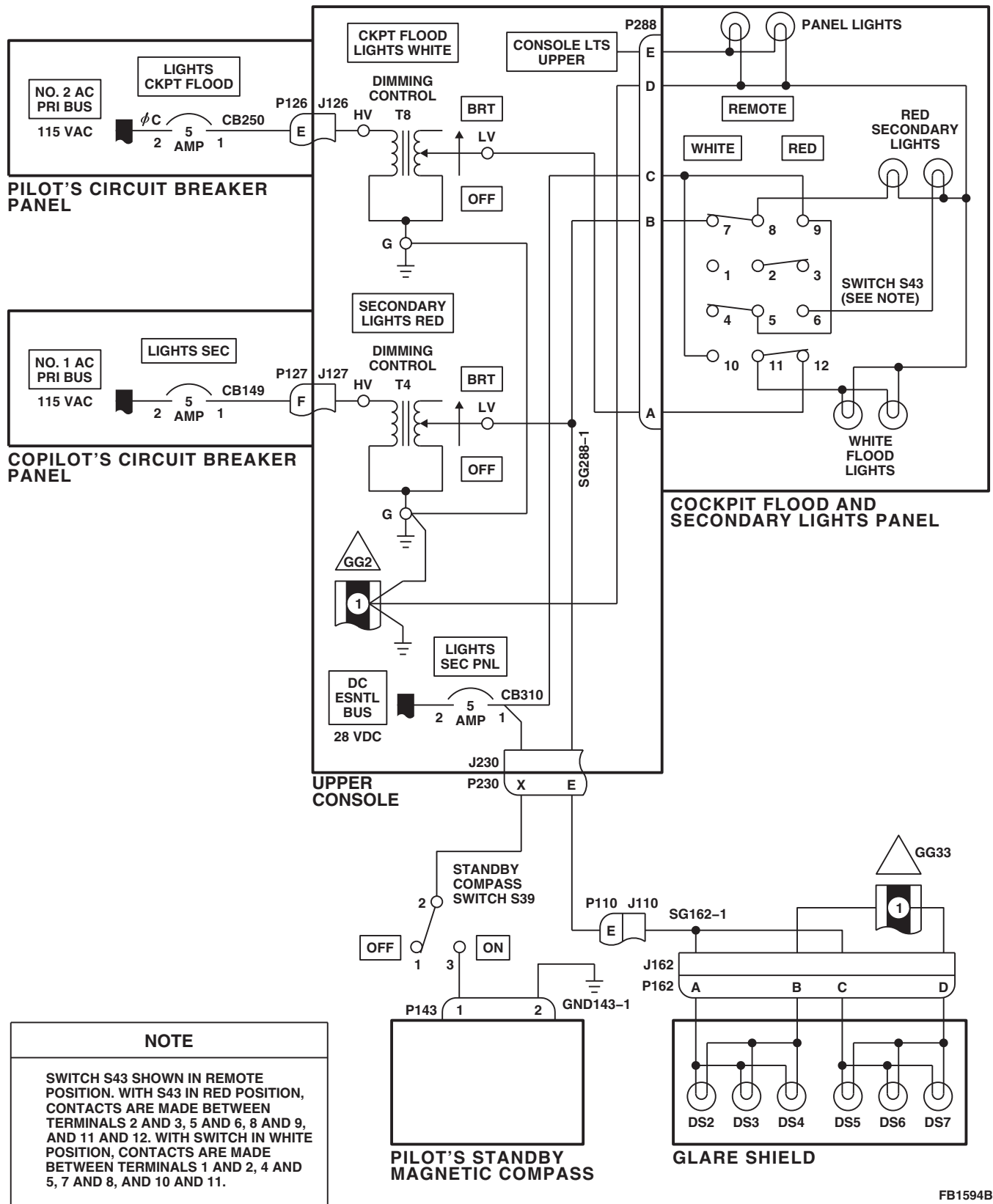


FIGURE 9-2-8-1. COCKPIT FLOOD AND SECONDARY LIGHTS SCHEMATIC DIAGRAM

9-2-9 UTILITY AND MAINTENANCE LIGHTS.

NOTE

INITIAL SETUP

Test Equipment

- Multimeter, Fluke 45

Tools

- Electrical Repairer Toolkit

Equipment Conditions

- Battery at Least 35% Charged

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).

Special Instructions

Repeat steps c. through j. for pilot's, copilot's, and auxiliary utility lights. Then do steps k. and l. for forward and rear maintenance light receptacles.

9-2-9.1 Operational/Troubleshooting Procedure.

- Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.

- Place BATT switch to ON.

- Press and hold button in center of rheostat knob on back of each pilot's, copilot's, or auxiliary utility light assembly.

RESULT: Utility light shall go on to full brightness.

- If result is not as specified, go to TABLE NO. 9-2-9-1.

- Release button in center of rheostat knob.

RESULT: Utility light shall go off.

- If result is not as specified, replace utility light assembly (PARA 9-4-63).

- Turn rheostat on back of pilot's, copilot's, or auxiliary utility light assembly fully clockwise.

RESULT: Utility light shall go to full brightness.

- If result is not as specified, replace utility light assembly (PARA 9-4-63).

- Pull collar on light assembly forward, over lens.

- Press lock button and turn collar so index marks on collar and light assembly line up.

RESULT: Utility light output shall be a bright, white, spotlight beam.

- If result is not as specified, replace utility light assembly (PARA 9-4-63).

- Push collar backwards, away from lens.

- Press lock button and turn collar to obtain blue lens.

RESULT: Light output shall be bright blue.

- If result is not as specified, replace utility light assembly (PARA 9-4-63).

- Turn rheostat on back of utility light assembly fully counterclockwise to OFF.

RESULT: Light shall dim and go off.

- If result is not as specified, replace utility light assembly (PARA 9-4-63).

- Connect maintenance light assembly to rear maintenance light receptacle J509.

RESULT: Maintenance light shall go on.

- If result is not as specified, go to TABLE NO. 9-2-9-2.

- l. Connect maintenance light assembly to forward maintenance light receptacle J259.

RESULT: Maintenance light shall go on.

- If result is not as specified, go to TABLE NO. 9-2-9-3.

- m. Rotate rheostat clockwise on lower console flexible light to ON position.

RESULT: Lower console flexible light shall go on.

- If result is not as specified, go to TABLE NO. 9-2-9-4.

- n. Rotate rheostat counterclockwise on lower console flexible light to OFF position.

RESULT: Lower console flexible light shall go off.

- If result is not as specified, replace lower console flexible light (PARA 9-4-63).

TABLE NO. 9-2-9-1. Utility Light Does Not Go On.

**TEST OR INSPECTION
CORRECTIVE ACTION**

1. Does any utility light go on?

Step 1. If any utility light is on, go to **2**.

Step 2. If any utility light is not on, go to **10**.

2. Does pilot's utility light go on?

Step 1. If pilot's light is on, go to **5**.

Step 2. If pilot's light is not on, go to **3**.

3. Check lamp.

Step 1. If lamp is good, go to **4**.

Step 2. If lamp is not good, replace lamp (PARA 9-4-62). Go to **11**.

4. Check for 28 VDC between TB2-1 and TB2-2.

Step 1. If voltage is as specified, replace pilot's utility light assembly (PARA 9-4-63). Go to **11**.

Step 2. If voltage is not as specified, repair/replace wiring to TB2-1 or TB2-2 (Appendix F) as required. Go to **11**.

5. Does copilot's utility light go on?

Step 1. If copilot's light is on, go to **8**.

TABLE NO. 9-2-9-1. Utility Light Does Not Go On. (Cont)

TEST OR INSPECTION CORRECTIVE ACTION	
Step 2. If copilot's light is not on, go to 6 .	
6. Check lamp.	
Step 1. If lamp is good, go to 7 .	
Step 2. If lamp is not good, replace lamp (PARA 9-4-62). Go to 11 .	
7. Check for 28 VDC between TB1-1 and TB1-2.	
Step 1. If voltage is as specified, replace copilot's utility light assembly (PARA 9-4-63). Go to 11 .	
Step 2. If voltage is not as specified, repair/replace wiring to TB1-1 or TB1-2 (Appendix F) as required. Go to 11 .	
8. Check auxiliary utility light lamp.	
Step 1. If lamp is good, go to 9 .	
Step 2. If lamp is not good, replace lamp (PARA 9-4-62). Go to 11 .	
9. Check for 28 VDC between TB6-1 and TB6-2.	
Step 1. If voltage is as specified, replace auxiliary utility light assembly (PARA 9-4-63). Go to 11 .	
Step 2. If voltage is not as specified, repair/replace wiring to TB6-1 or TB6-2 (Appendix F). Go to 11 .	
10. Check for 28 VDC between terminal 1 (UTIL LTS CKPT circuit breaker, BATT UTIL BUS - lower console circuit breaker panel) and ground.	
Step 1. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and TB1-1 (Appendix F). Go to 11 .	
Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 11 .	
11. Procedure completed.	

TABLE NO. 9-2-9-2. Maintenance Light Does Not Go On When Connected To Rear Maintenance Light Receptacle.

	TEST OR INSPECTION CORRECTIVE ACTION
I	1. Check for 28 VDC between J509-A and ground. Step 1. If voltage is as specified, replace maintenance light assembly (PARA 9-4-64). Go to 5 . Step 2. If voltage is not as specified, go to 2 .
	2. Check continuity between J509-B and ground. Step 1. If continuity is present, go to 3 . Step 2. If continuity is not present, repair/replace wiring between J509-B and ground (Appendix F). Go to 5 .
I	3. Check for 28 VDC between J107-E and ground. Step 1. If voltage is as specified, repair/replace wiring between P107-E and J509-A (Appendix F). Go to 5 . Step 2. If voltage is not as specified, go to 4 .
	4. Check for 28 VDC between terminal 1 (UTIL LTS CKPT circuit breaker, BATT UTIL BUS - lower console circuit breaker panel) and ground. Step 1. If voltage is as specified, repair/replace wiring between J107-E and UTIL LTS CKPT circuit breaker terminal 1 (Appendix F). Go to 5 . Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 5 .
I	5. Procedure completed.

TABLE NO. 9-2-9-3. Maintenance Light Does Not Go On When Connected To Forward Maintenance Light Receptacle.

	TEST OR INSPECTION CORRECTIVE ACTION
I	1. Check for 28 VDC between J259-A and ground. Step 1. If voltage is as specified, replace maintenance light assembly (PARA 9-4-64). Go to 5 . Step 2. If voltage is not as specified, go to 2 .

TABLE NO. 9-2-9-3. Maintenance Light Does Not Go On When Connected To Forward Maintenance Light Receptacle. (Cont)

TEST OR INSPECTION CORRECTIVE ACTION
2. Check continuity between J259-B and ground.
Step 1. If continuity is present, go to 3. Step 2. If continuity is not present, repair/replace wiring between J259-B and ground (Appendix F). Go to 5.
3. Check for 28 VDC between J107-E and ground.
Step 1. If voltage is as specified, repair/replace wiring between P107-E and J259-A (Appendix F). Go to 5. Step 2. If voltage is not as specified, go to 4.
4. Check for 28 VDC between terminal 1 (UTIL LTS CKPT circuit breaker, BATT UTIL BUS - lower console circuit breaker panel) and ground.
Step 1. If voltage is as specified, repair/replace wiring between J107-E and circuit breaker terminal 1 (Appendix F). Go to 5. Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 5.
5. Procedure completed.

TABLE NO. 9-2-9-4. Lower Console Flexible Light Does Not Go On.

TEST OR INSPECTION CORRECTIVE ACTION
1. Check for 28 VDC between TB6-1 and TB6-2.
Step 1. If voltage is as specified, go to 2. Step 2. If voltage is not as specified, go to 3.
2. Disconnect lower console flexible light splices and check continuity between TB6-1 and TB6-2 and disconnected wiring.
Step 1. If continuity is present, replace lower console flexible light assembly (PARA 9-4-63). Go to 5. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 5.

TABLE NO. 9-2-9-4. Lower Console Flexible Light Does Not Go On. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
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3.	Check for 28 VDC between J107-E and ground.
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	Step 1. If voltage is as specified, repair/replace wiring (Appendix F). Go to 5.
--	--

	Step 2. If voltage is not as specified, go to 4.
--	--

4.	Check for 28 VDC between terminal 1 (UTIL LTS CKPT circuit breaker, BATT UTIL BUS - lower console circuit breaker panel) and ground.
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	Step 1. If voltage is as specified, repair/replace wiring (Appendix F). Go to 5.
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	Step 2. If voltage is not as specified, replace UTIL LTS CKPT circuit breaker (PARA 9-4-1). Go to 5.
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5.	Procedure completed.
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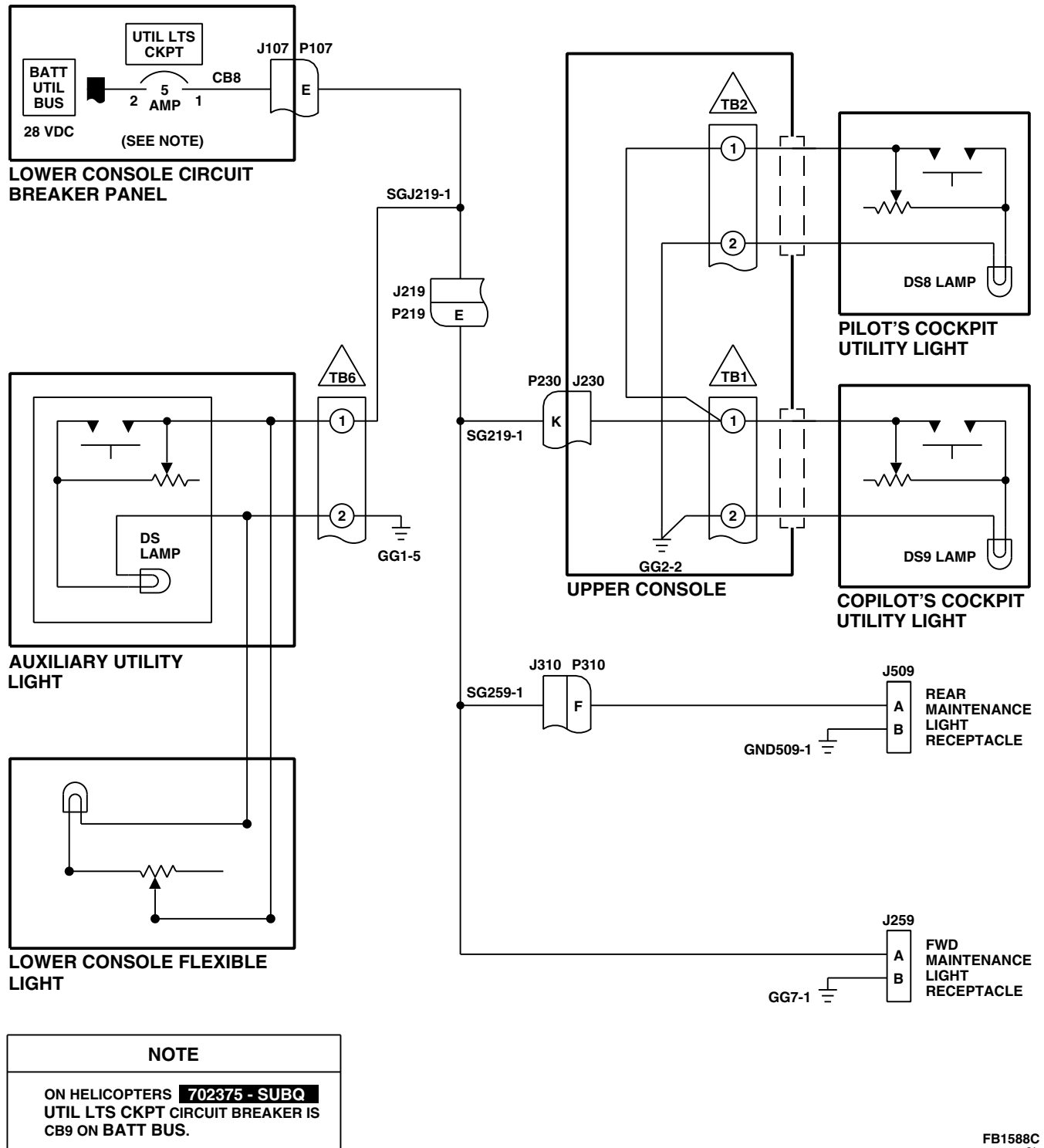


FIGURE 9-2-9-1. UTILITY AND MAINTENANCE LIGHTS SCHEMATIC DIAGRAM

9-2-10 INSTRUMENT PANEL AND CONSOLES INDICATOR LIGHTS DIMMING SLS .

INITIAL SETUP

Test Equipment

- Multimeter, Fluke 27/AN

Tools

- Electrical Repairer Toolkit

References

- PARA 1-6-16

Equipment Conditions

- External Electrical Power Available or APU Operational

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).

9-2-10.1 Operational/Troubleshooting Procedure.

- Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.
- Make sure the INST LT PILOT FLT and LIGHTED SWITCHES controls are turned OFF.
- Make sure the pilot's and copilot's RAD ALT DIMMING controls are turned fully counterclockwise.
- Turn on electrical power (PARA 1-6-16).
- Test the following press-to-test indicators:
 - Blade de-ice control panel TEST IN PROGRESS lamp.
 - Blade de-ice test panel PWR MAIN RTR and PWR TAIL RTR lamps.
 - Rescue hoist control panel SQUIB IND lamp (when installed).
 - Upper console CARGO HOOK EMERG REL TEST lamp.
 - Fuel boost pump control panel NO. 1 PUMP and NO. 2 PUMP lamps.

RESULT: When pressed, each indicator shall light brightly.

- If result is not as specified, go to TABLE NO. 9-2-10-1.
- If one press-to-test indicator does not go on, replace applicable lamp (PARA 9-4-42).
 - If trouble remains, check continuity of affected indicator wiring.
 - If continuity is present, replace indicator (PARA 9-5-18).
 - If continuity is not present, repair/replace wiring (Appendix F) as required.
 - If trouble remains, replace left relay panel (PARA 9-4-2).
- f. Place and hold the caution/advisory panel BRT/DIM-TEST switch to TEST.

RESULT:

- All capsules on caution/advisory panel and pilot's and copilot's master warning panels shall go on bright.
- Master warning panel LOW ROTOR RPM and #1 and #2 FUEL LOW capsules shall flash.

- (3) All switch legends and indicator lamps on instrument panel and upper and lower console panels and indicators shall go on bright.

- If result (1) or (2) is not as specified, troubleshoot Caution/Advisory Warning System (PARA 8-2-4).
- If result (3) is not as specified, check continuity of applicable wiring to panel containing lamps or indicators not lit.

(a) If continuity is present, replace panel (TM 11-70-23).

(b) If continuity is not present, repair/replace wiring (Appendix F) as required.

g. Release BRT/DIM-TEST switch.

h. Place Radar Altimeter System on by turning pilot's and copilot's radar altimeter LO/SET/OFF control from OFF.

RESULT:

- (1) Pilot's and copilot's OFF flags shall go out of view.
- (2) Pilot's and copilot's digital displays shall light brightly.

- If result (1) is not as specified, troubleshoot Radar Altimeter (AN/APN-209) System (TM 11-70-23).
- If result (2) is not as specified, go to TABLE NO. 9-2-10-2.

i. Press and hold miscellaneous switch panel TAIL WHEEL switch. Observe TAIL WHEEL switch legends.

RESULT: When pressed, TAIL WHEEL switch shall indicate LOCK or UNLK depending upon position of tail wheel lockpin.

- If result is not as specified, replace lamp (PARA 9-4-8).

(a) If trouble remains, troubleshoot Tail-wheel Lock System (PARA 3-2-2).

(b) If trouble still remains, replace miscellaneous switch panel (PARA 9-4-7).

j. Place Doppler computer display MODE switch to LAMP TEST.

RESULT: MEM and MAL indicators shall go on bright.

- If result is not as specified, repair/replace wiring between J111-k and P687R-J (Appendix F) or replace Doppler computer display (TM 11-70-23), as required.

k. Turn upper console INST LT PILOT FLT control one quarter turn from OFF.

l. Place caution/advisory panel BRT/DIM-TEST switch to BRT/DIM to select dim mode and release.

RESULT:

- (1) Any lit caution/advisory panel and master warning panel capsules shall dim.
- (2) Switch legends on pilot's and copilot's HSI/VSI mode select panels and miscellaneous switch panel TAIL WHEEL LOCK switch shall dim.
- (3) Digital displays on pilot's and copilot's radar altimeters shall dim.
- (4) Rescue hoist SQUIB IND lamp and CARGO HOOK EMERG REL TEST lamp shall light with decreased intensity when pressed.
- (5) Indicator lights on blade de-ice test and control panels shall dim.
- (6) Indicator lights on Doppler computer display control panel, CIS mode select panel, and pilot's and copilot's VSIs shall dim.

- If all results are not as specified, troubleshoot caution/advisory warning system (PARA 8-2-4).
 - (a) If trouble remains, replace left relay panel (PARA 9-4-2).
- If results (4), (5), and (6) are not as specified, go to TABLE NO. 9-2-10-3.
- m. Place Doppler computer display MODE switch to OFF.
- n. Turn upper console LIGHTED SWITCHES control from OFF to BRT.

RESULT: Selected legends on miscellaneous switch panel TAIL WHEEL LOCK switch and bottom row of pilot's and copilot's HSI/VSI mode select panels shall vary smoothly from off to bright.

- If result is not as specified, replace LIGHTED SWITCHES dimmer control (PARA 9-4-50).
- If lights do not go on, go to TABLE NO. 9-2-10-4.
- o. Turn pilot's and copilot's instrument panel RAD ALT DIMMING control fully counterclockwise and then fully clockwise.

RESULT: Digital displays on pilot's and copilot's radar altimeters shall vary smoothly from dim to bright as associated RAD ALT DIMMING control is turned clockwise.

- If result is not as specified, replace associated RAD ALT DIMMING control (TM 11-70-23).

- If either digital display does not go on, go to TABLE NO. 9-2-10-5.
- p. Place caution/advisory panel BRT/DIM-TEST switch to BRT/DIM to select bright mode and release.

RESULT: All indicators and switch legends previously dimmed shall increase in brightness.

- q. Place FIRE DETR TEST switch to position 1 and hold, then release.

RESULT: APU, #1 ENG EMER OFF, and #2 ENG EMER OFF T-handles shall go on bright.

- If result is not as specified, troubleshoot Engine Fire Detection System (PARA 12-2-1).

- (a) If trouble remains, replace left relay panel (PARA 9-4-2).

- r. Place caution/advisory panel BRT/DIM-TEST switch to BRT/DIM to select dim mode and release.

RESULT: APU, #1 ENG EMER OFF, and #2 ENG EMER OFF T-handles shall go on dimly.

- If result is not as specified, go to TABLE NO. 9-2-10-6.

- s. Turn off electrical power (PARA 1-6-16).

TABLE NO. 9-2-10-1. Instrument Panel And Consoles Indicator Press-To-Test Lights Do Not Go On Bright.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for 28 VDC between P902-V and ground.	Step 1. If voltage is as specified, replace left relay panel (PARA 9-4-2). Go to 3. Step 2. If voltage is not as specified, go to 2.
2. Check for 28 VDC between terminal 1 (LIGHTS ADVSY circuit breaker, NO. 1 DC PRI BUS - copilot's circuit breaker panel) and ground.	Step 1. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P902-V (Appendix F). Go to 3. Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 3.
3. Procedure completed.	

TABLE NO. 9-2-10-2. Pilot's And Copilot's Radar Altimeter Digital Displays Do Not Go On Bright.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for 28 VDC between: P675R-7 and ground P676R-6 and ground	Step 1. If voltage is as specified, replace malfunctioning radar altimeter (TM 11-70-23). Go to 3. Step 2. If voltage is not as specified, go to 2.
2. Check continuity between: P675R-7 and P902-Y P676R-6 and P902-n	Step 1. If continuity is present, replace left relay panel (PARA 9-4-2). Go to 3. Step 2. If continuity is not present, repair/replace wiring (Appendix F) as required. Go to 3.

TABLE NO. 9-2-10-2. Pilot’s And Copilot’s Radar Altimeter Digital Displays Do Not Go On Bright.
 (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION
3. Procedure completed.

TABLE NO. 9-2-10-3. Indicator Lights Do Not Dim.

TEST OR INSPECTION
CORRECTIVE ACTION
1. Check fuse on indicator lights dimmer.
Step 1. If fuse is good, go to 2. Step 2. If fuse is not good, replace fuse (PARA 9-4-53). Go to 9.
2. Check for 28 VDC between P393-B and ground.
Step 1. If voltage is as specified, go to 3. Step 2. If voltage is not as specified, go to 4.
3. Check continuity between P393-C and ground.
Step 1. If continuity is present, replace indicator lights dimmer (PARA 9-4-52). Go to 9. Step 2. If continuity is not present, go to 5.
4. Check for 28 VDC between terminal 1 (LIGHTS ADVSY circuit breaker, NO. 1 DC PRI BUS - copilot’s circuit breaker panel) and ground.
Step 1. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P393-B (Appendix F). Go to 9. Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 9.
5. Remove and check pin filter adapter from P118 (PARA 9-4-98).
Step 1. If pin filter adapter is good, go to 6. Step 2. If pin filter adapter is not good, replace pin filter adapter (PARA 9-4-98). Go to 9.

TABLE NO. 9-2-10-3. Indicator Lights Do Not Dim. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	6. Install pin filter adapter on P118 (PARA 9-4-98). Go to 7. 7. Check continuity between P393-C and P118-T. Step 1. If continuity is present, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 9. Step 2. If trouble remains, go to 8. Step 3. If continuity is not present, repair/replace wiring (Appendix F). Go to 9. 8. Check continuity between P118-T and P902-q. Step 1. If continuity is present, replace left relay panel (PARA 9-4-2). Go to 9. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 9. 9. Procedure completed.

TABLE NO. 9-2-10-4. Pilot's Or Copilot's Radar Altimeter Digital Display Lighting Does Not Go On.

TEST OR INSPECTION	CORRECTIVE ACTION
	1. With affected altimeter's RAD ALT dimming control turned fully clockwise, check for 28 VDC between: R4-2 and R4-3 R5-2 and R5-3 Step 1. If voltage is as specified, replace left relay panel (PARA 9-4-2). Go to 4. Step 2. If trouble remains, repair/replace wiring between (Appendix F), then go to 4: R4-2 and P902-T R5-2 and P902-p Step 3. If voltage is not as specified, go to 2. 2. Check for 28 VDC between: R4-1 and ground R5-1 and ground Step 1. If voltage is as specified, replace dimming control (TM 11-70-23). Go to 4. Step 2. If voltage is not as specified, go to 3.

TABLE NO. 9-2-10-4. Pilot’s Or Copilot’s Radar Altimeter Digital Display Lighting Does Not Go On.
(Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
3. Check continuity between: R4-1 and P902-W R5-1 and P902-f	Step 1. If continuity is present, replace left relay panel (PARA 9-4-2). Go to 4. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 4.
4. Procedure completed.	

TABLE NO. 9-2-10-5. T-Handles Do Not Dim.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for 28 VDC between P902-P and ground.	Step 1. If voltage is as specified, replace left relay panel (PARA 9-4-2). Go to 3. Step 2. If voltage is not as specified, go to 2.
2. Check continuity between P902-P and terminal 1 (FIRE DET NO. 1 ENG circuit breaker, DC ESNTL BUS - upper console circuit breaker panel).	Step 1. If continuity is present, replace circuit breaker (PARA 9-4-1). Go to 3. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 3.
3. Procedure completed.	

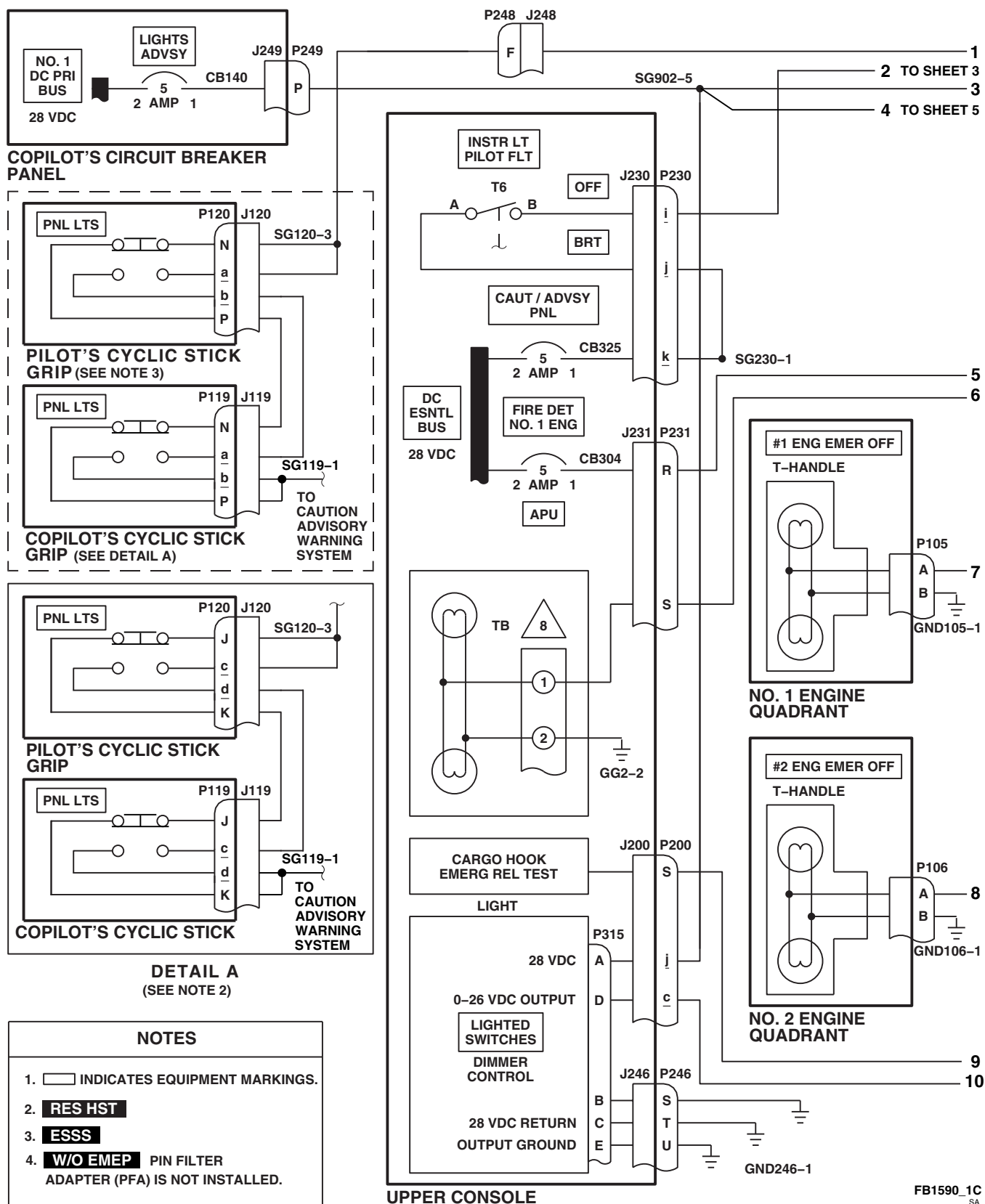


FIGURE 9-2-10-1. INSTRUMENT PANEL AND CONSOLES INDICATOR LIGHTS DIMMING SCHEMATIC DIAGRAM (SHEET 1 OF 6)

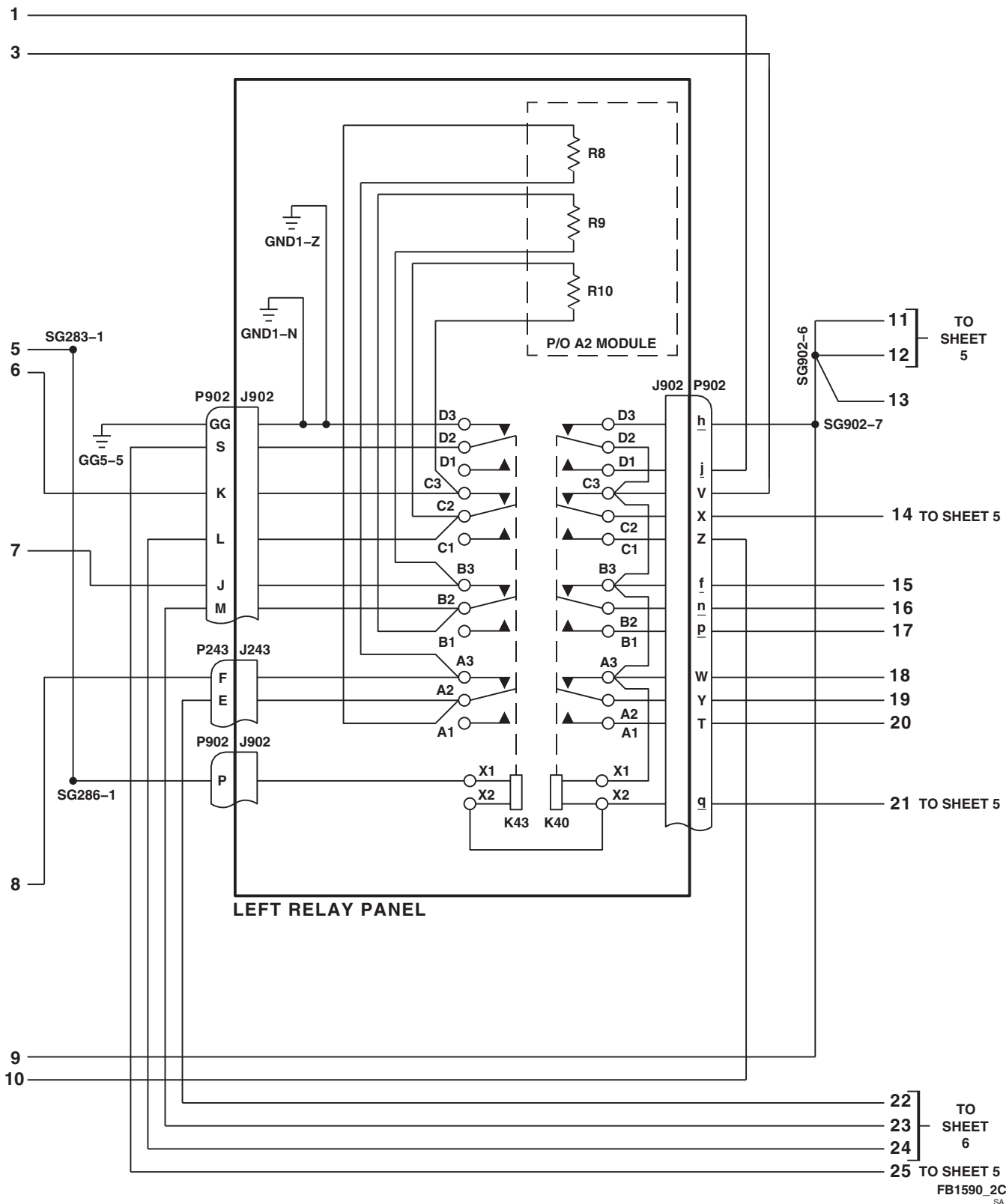


FIGURE 9-2-10-1. INSTRUMENT PANEL AND CONSOLES INDICATOR LIGHTS DIMMING SCHEMATIC DIAGRAM (SHEET 2 OF 6)

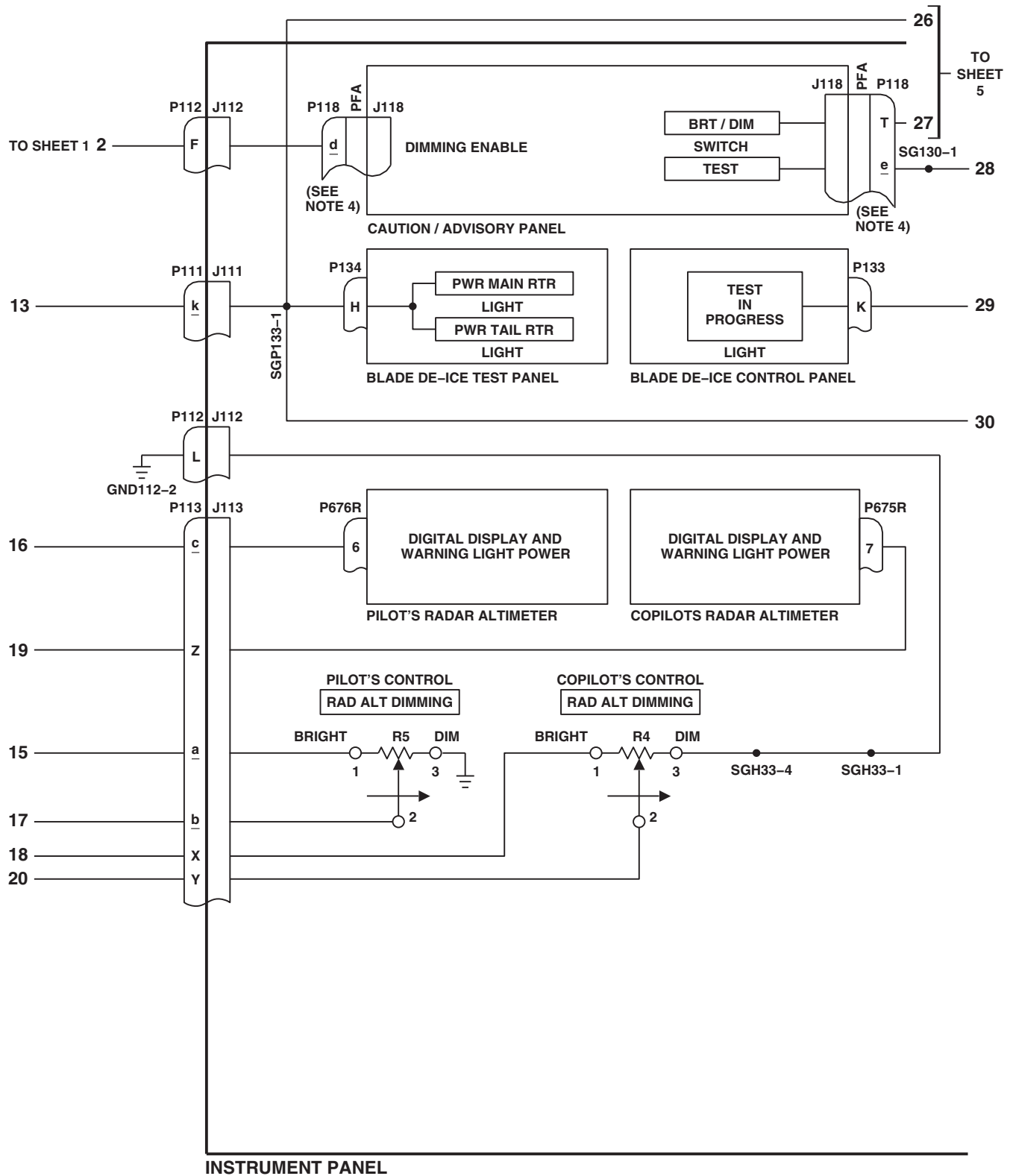


FIGURE 9-2-10-1. INSTRUMENT PANEL AND CONSOLES INDICATOR LIGHTS DIMMING SCHEMATIC DIAGRAM (SHEET 3 OF 6)

9-2-10-10

Change 7

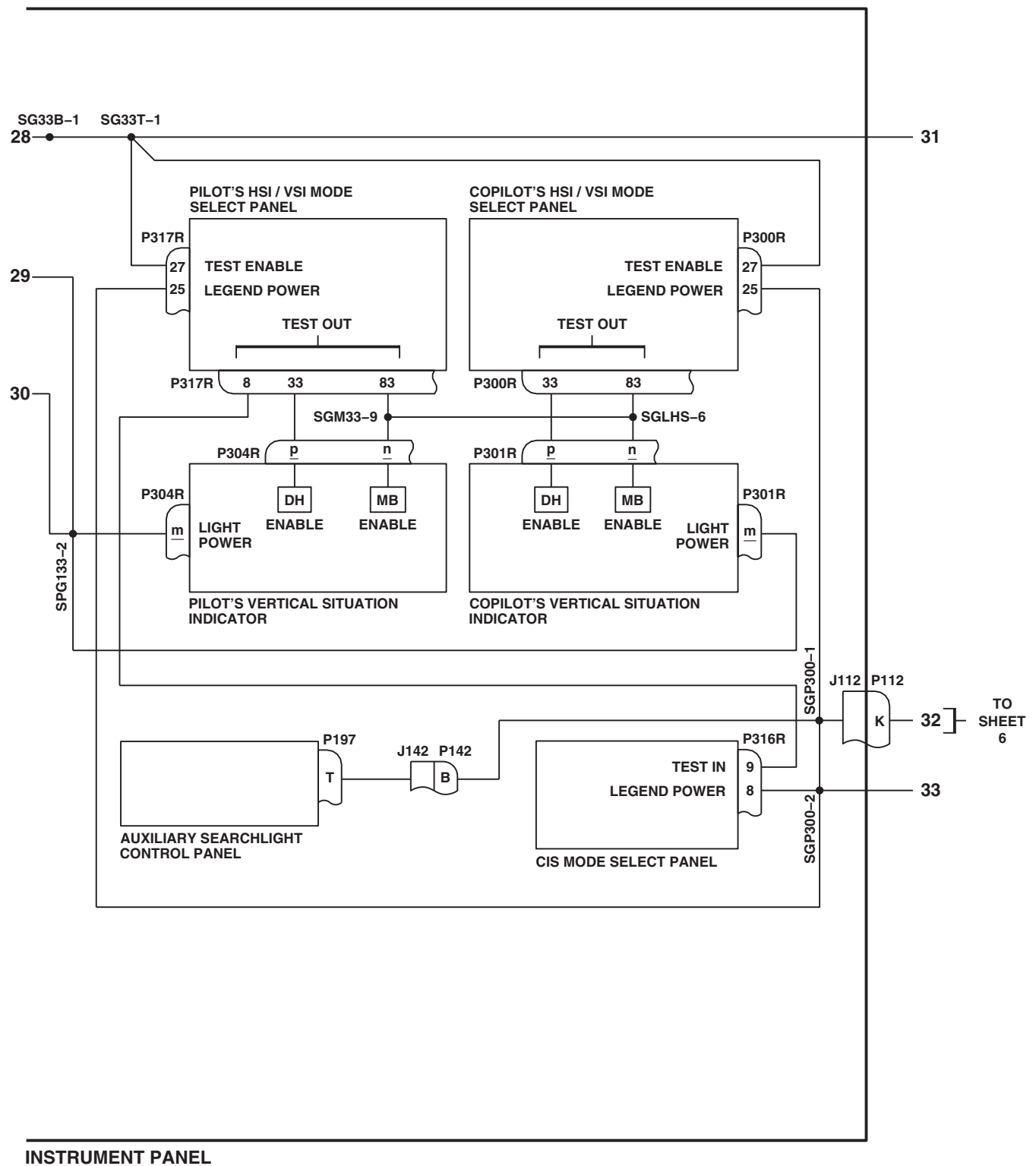


FIGURE 9-2-10-1. INSTRUMENT PANEL AND CONSOLES INDICATOR LIGHTS DIMMING SCHEMATIC DIAGRAM (SHEET 4 OF 6)

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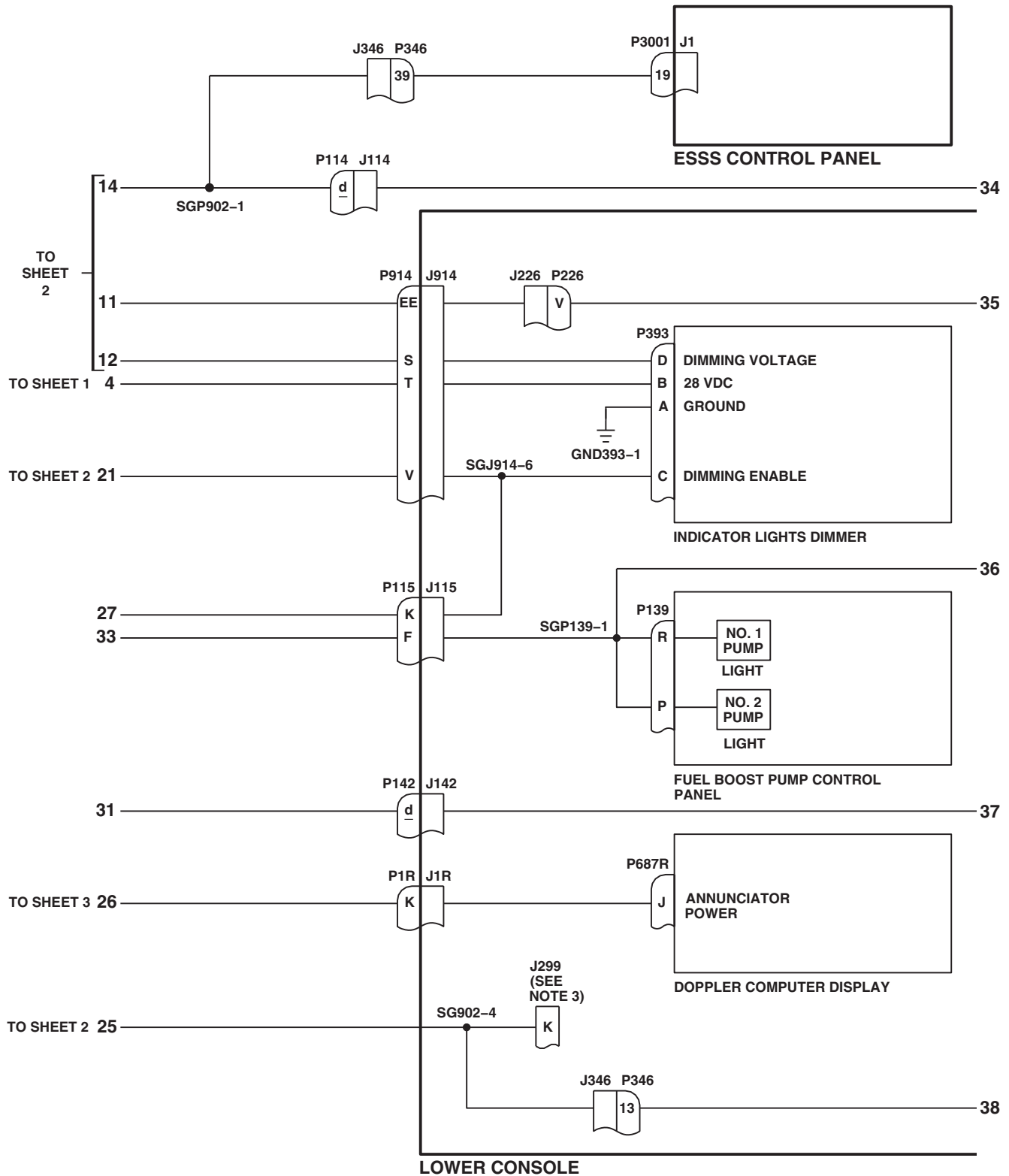
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FIGURE 9-2-10-1. INSTRUMENT PANEL AND CONSOLES INDICATOR LIGHTS DIMMING SCHEMATIC DIAGRAM (SHEET 5 OF 6)

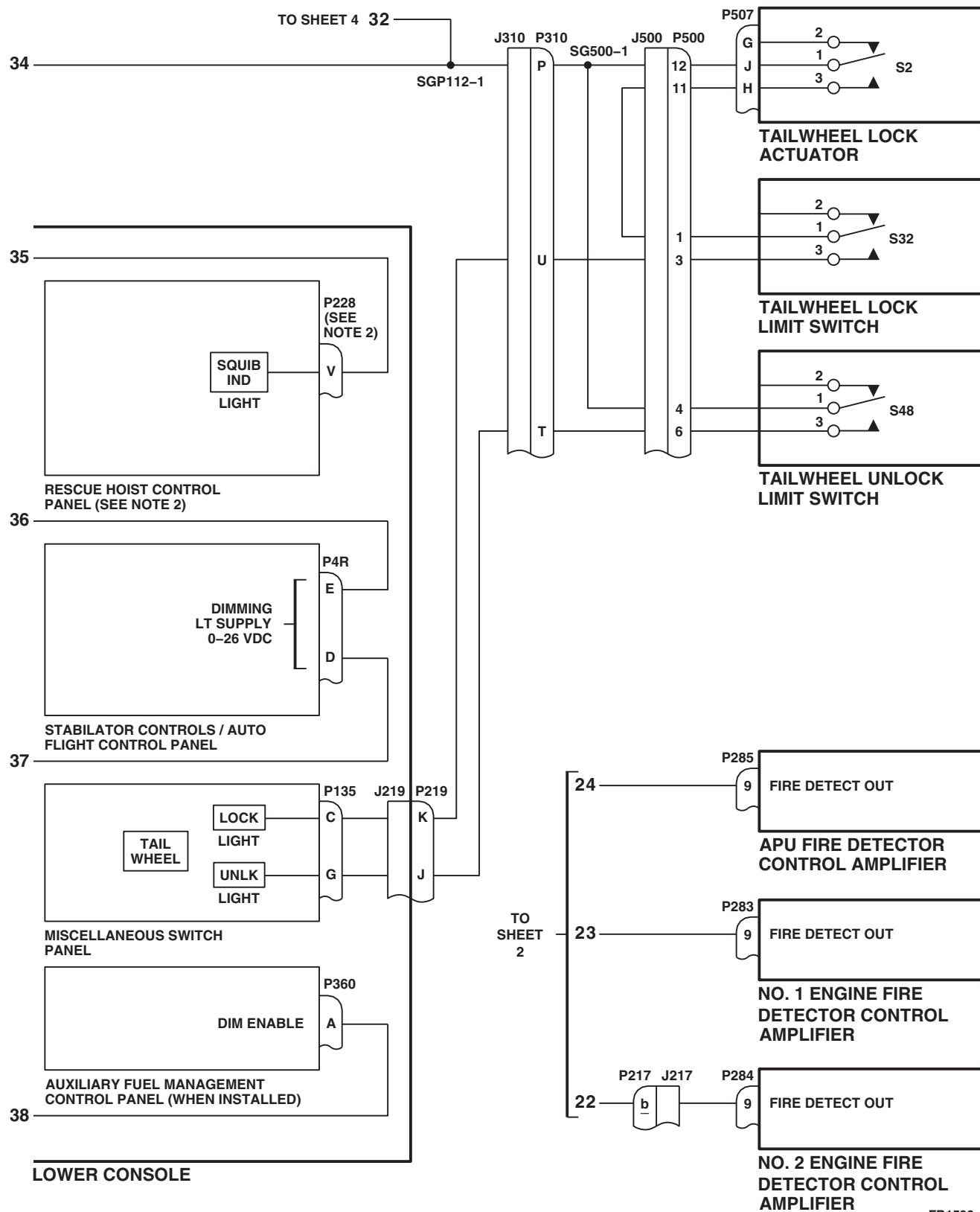


FIGURE 9-2-10-1. INSTRUMENT PANEL AND CONSOLES INDICATOR LIGHTS DIMMING SCHEMATIC DIAGRAM (SHEET 6 OF 6)

9-2-11 INSTRUMENT PANEL AND CONSOLES INDICATOR LIGHTS DIMMING W/O SLS .

INITIAL SETUP

Test Equipment

- Multimeter, Fluke 27/AN

Tools

- Electrical Repairer Toolkit

References

- PARA 1-6-16

Equipment Conditions

- External Electrical Power Available and/or APU Operational

NOTE

If a circuit breakers pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).

9-2-11.1 Operational/Troubleshooting Procedure.

- Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.
- Turn on electrical power (PARA 1-6-16).
- Turn upper console INST LTS PILOT FLT control to OFF.
- If upper portions of lower four switches on pilot's and copilot's HSI/VSI mode select panels are off, press pilot's cyclic stick PNL LTS switch.

RESULT: Upper portions of switches shall go on bright.

- If result is not as specified, go to TABLE NO. 9-2-11-1.
- If upper portion of switches go on but not bright, check continuity between P902-q and ground.
 - If continuity is present, troubleshoot Caution/Advisory Warning System (PARA 8-2-4).
 - If continuity is not present, replace left relay panel (PARA 9-4-1).

- Place caution/advisory BRT/DIM-TEST switch to BRT/DIM.

RESULT: Upper portions of lower four switches on pilot's and copilot's HSI/VSI mode select panels shall be on with less brightness than that noted in step d.

- If result is not as specified, go to TABLE NO. 9-2-11-2.

- Press pilot's cyclic stick PNL LTS switch.

RESULT: All switches on pilot's and copilot's HSI/VSI mode select panels shall go off.

- If result is not as specified, go to TABLE NO. 9-2-11-3.

- Press pilot's cyclic stick PNL LTS switch.

- Place and hold upper console FIRE DET TEST switch to position 1.

RESULT: APU, #1 ENG EMER OFF, and #2 ENG EMER OFF T-handles on engine control quadrant shall go on bright.

- If result is not as specified, troubleshoot Fire Detection System (PARA 12-2-1).

- If T-handles go on dimly, replace left relay panel (PARA 9-4-2).
 - i. Place caution/advisory panel BRT/DIM-TEST switch to BRT/DIM.
- RESULT: APU, #1 ENG EMER OFF, and #2 ENG EMER OFF T-handles on engine control quadrant shall go on dimly.
- If result is not as specified, go to TABLE NO. 9-2-11-4.
 - j. Turn off electrical power (PARA 1-6-16).

TABLE NO. 9-2-11-1. Instrument Panel And Consoles Indicator Lights Are Off.

TEST OR INSPECTION	CORRECTIVE ACTION
--------------------	-------------------

1. Check for 28 VDC between P243-j and ground.	
--	--

Step 1. If voltage is as specified, troubleshoot upper and lower console lights (PARA 9-2-5). Go to 4. Step 2. If voltage is not as specified, go to 2.	
---	--

2. Check for 28 VDC between P902-k and ground.	
--	--

Step 1. If voltage is as specified, replace left relay panel (PARA 9-4-2). Go to 4. Step 2. If voltage is not as specified, go to 3.	
--	--

3. Check for 28 VDC between terminal 1 (LIGHTS ADVSY circuit breaker, NO. 1 DC PRI BUS - copilot's circuit breaker panel) and ground.	
---	--

Step 1. If voltage is as specified, repair/replace wiring between terminal 1 and P902-k (Appendix F). Go to 4. Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 4.	
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4. Procedure completed.	
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TABLE NO. 9-2-11-2. Instrument Panel And/Or Console Indicator Lights Do Not Dim.

TEST OR INSPECTION CORRECTIVE ACTION	
1. Do any of the following indicator lights dim: pilot's HSI/VSI mode select panel, pilot's vertical situation indicator, or CIS mode select panel.	
Step 1. If any indicator light dims, go to 2 .	
Step 2. If no indicator light dims, go to 4 .	
2. Do any of the following indicator lights dim: rescue hoist control panel, upper console flight control panel, Doppler computer display, blade deice control, copilot's HSI/VSI mode select panel, or copilot's vertical situation indicator.	
Step 1. If any indicator lights dim, repair/replace wiring to panel on which indicator lights do not dim (Appendix F). Go to 8 .	
Step 2. If no indicator lights dim, go to 3 .	
3. Check fuse F2-1A continuity.	
Step 1. If continuity is present, replace left relay panel (PARA 9-4-2). Go to 8 .	
Step 2. If continuity is not present, replace fuse (PARA 9-4-3). Go to 8 .	
Step 3. If trouble remains, replace left relay panel (PARA 9-4-2). Go to 8 .	
4. Do any of the following indicator lights dim: upper console, flight control panel, Doppler computer display, blade deice control panel, copilot's HSI/VSI mode select panel, or copilot's vertical situation indicator.	
Step 1. If any indicator lights dim, go to 5 .	
Step 2. If no indicator lights dim, go to 6 .	
5. Check fuse F1-1A continuity.	
Step 1. If continuity is present, replace left relay panel (PARA 9-4-2). Go to 8 .	
Step 2. If continuity is not present, replace fuse (PARA 9-4-3). Go to 8 .	
Step 3. If trouble remains, replace left relay panel (PARA 9-4-2). Go to 8 .	
6. Check continuity between P902-q and ground.	
Step 1. If continuity is present, replace left relay panel (PARA 9-4-2). Go to 8 .	
Step 2. If continuity is not present, go to 7 .	

TABLE NO. 9-2-11-2. Instrument Panel And/Or Console Indicator Lights Do Not Dim. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
7. Check continuity between P902-q and P118-T.	Step 1. If continuity is present, troubleshoot caution/advisory warning system (PARA 8-2-4). Go to 8. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 8.
8. Procedure completed.	

TABLE NO. 9-2-11-3. Instrument Panel And Console Indicator Lights Do Not Go Off With PNL LTS Switch Pressed.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for 28 VDC between the following: J120-N and ground J120-a and ground	Step 1. If voltage is as specified, go to 2. Step 2. If voltage is not as specified, go to 8.
2. Check continuity between P120-N and P120-P while pressing pilot's cyclic stick PNL LTS switch. Reading shall alternate between continuity and infinity.	Step 1. If reading alternates, go to 3. Step 2. If reading does not alternate, troubleshoot pilot's cyclic stick (PARA 11-2-1). Go to 9.
3. Check continuity between P120-a and P120-b while pressing pilot's cyclic stick PNL LTS switch. Reading shall alternate between continuity and infinity.	Step 1. If reading alternates, go to 4. Step 2. If reading does not alternate, troubleshoot pilot's cyclic stick (PARA 11-2-1). Go to 9.
4. Check continuity between the following: J120-b and J119-a J120-P and J119-N	

TABLE NO. 9-2-11-3. Instrument Panel And Console Indicator Lights Do Not Go Off With PNL LTS Switch Pressed. (Cont)

TEST OR INSPECTION CORRECTIVE ACTION	
Step 1. If continuity is present, go to 5. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 9.	
5. Check continuity between P119-N and P119-P while pressing copilot's cyclic stick PNL LTS switch. Reading shall alternate between continuity and infinity.	
Step 1. If reading alternates, go to 6. Step 2. If reading does not alternate, troubleshoot copilot's cyclic stick (PARA 11-2-1). Go to 9.	
6. Check continuity between P119-a and P119-b while pressing pilot's cyclic stick PNL LTS switch. Reading shall alternate between continuity and infinity.	
Step 1. If reading alternates, go to 7. Step 2. If reading does not alternate, troubleshoot copilot's cyclic stick (PARA 11-2-1). Go to 9.	
7. Check for 28 VDC between P243-j and ground.	
Step 1. If voltage is as specified, replace left relay panel (PARA 9-4-2). Go to 9. Step 2. If voltage is not as specified, repair/replace wiring, as required, between (Appendix F), then go to 9: P243-j and J119-b P243-j and J119-P	
8. Check for 28 VDC between terminal 1 (LIGHTS LWR CSL circuit breaker, NO. 1 DC PRI BUS - copilot's circuit breaker panel) and ground.	
Step 1. If voltage is as specified, repair/replace wiring between J120-N and terminal 1 (Appendix F). Go to 9. Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 9.	
9. Procedure completed.	

TABLE NO. 9-2-11-4. T-Handles Remain Bright.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Do caution/advisory capsules go on dim?	Step 1. If capsules go on dim, go to 2. Step 2. If capsules do not go on dim, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 6.
2. Do instrument panel and lower console lights go on dim?	Step 1. If lights go on dim, go to 3. Step 2. If lights do not go on dim, go to 5.
3. Check for 28 VDC between P902-P and ground.	Step 1. If voltage is as specified, replace left relay panel (PARA 9-4-2). Go to 6. Step 2. If voltage is not as specified, go to 4.
4. Check continuity between P902-P and P231-R.	Step 1. If continuity is present, replace FIRE DET NO. 1 ENG circuit breaker (PARA 9-4-1). Go to 6. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 6.
5. Check continuity between P902-q and P118-T.	Step 1. If continuity is present, replace left relay panel (PARA 9-4-2). Go to 6. Step 2. If trouble remains, replace caution/advisory panel (PARA 8-4-19). Go to 6. Step 3. If continuity is not present, repair/replace wiring (Appendix F). Go to 6.
6. Procedure completed.	

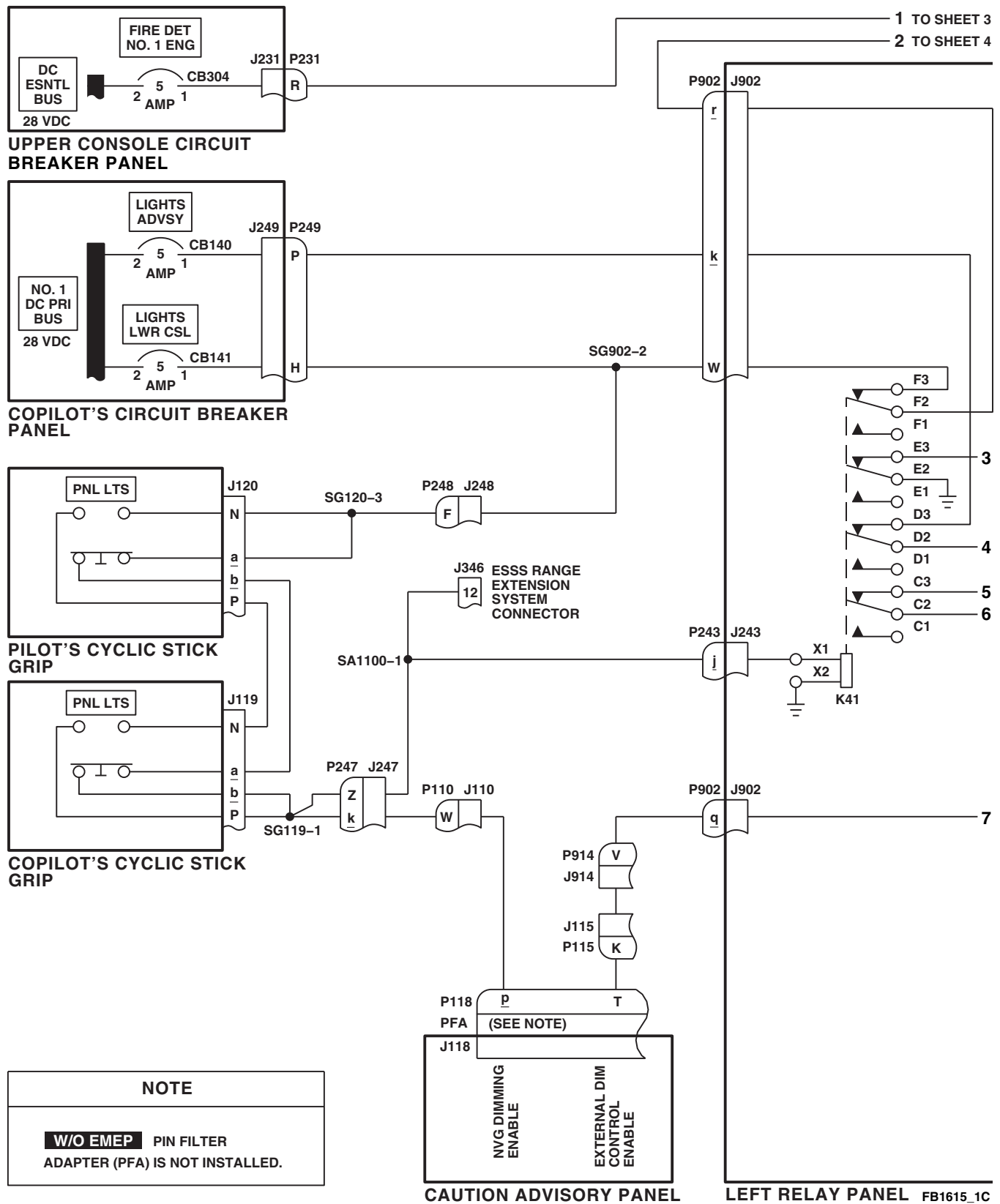
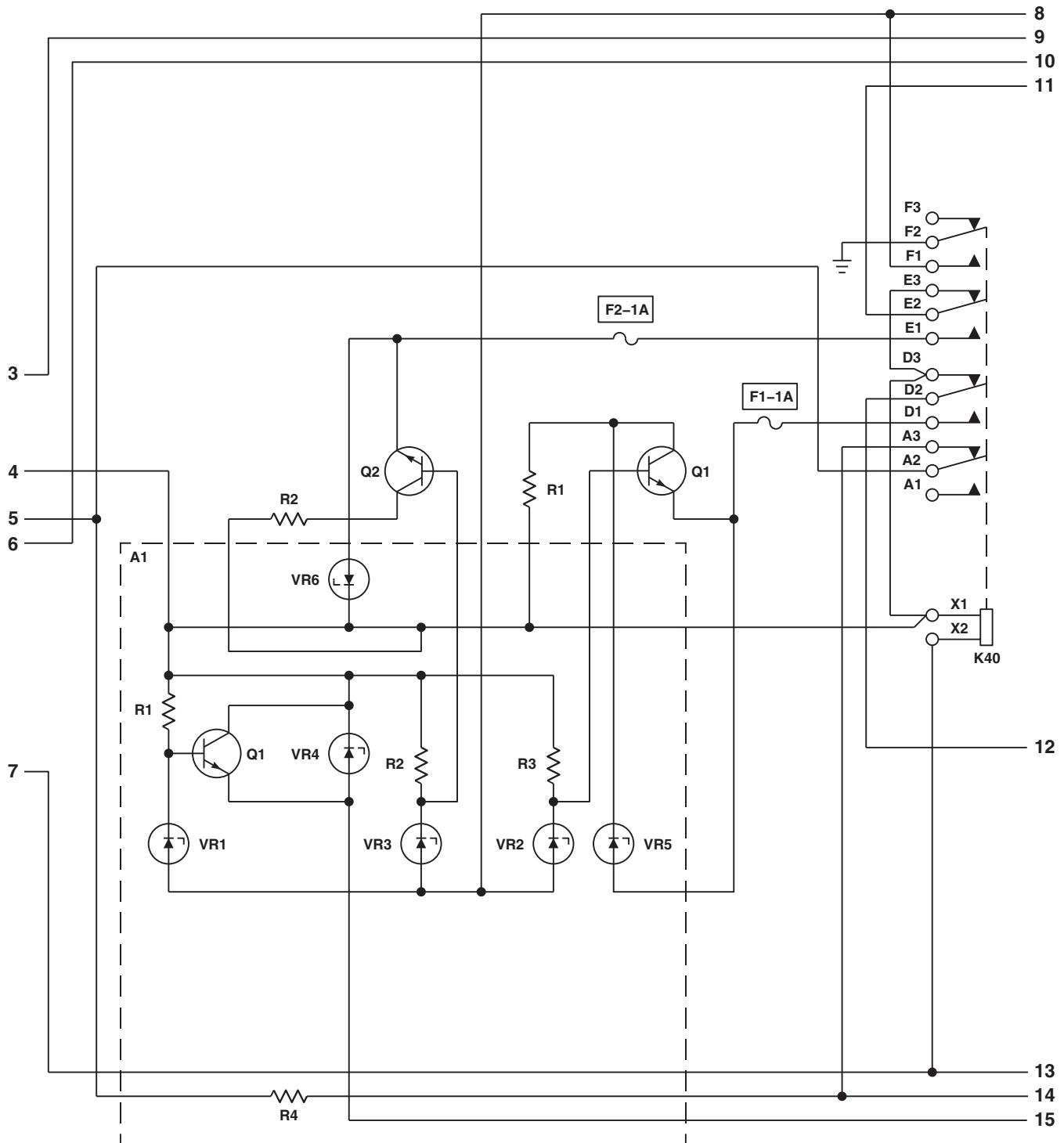


FIGURE 9-2-11-1. INSTRUMENT PANEL AND CONSOLE INDICATOR LIGHTS DIMMING SCHEMATIC DIAGRAM (SHEET 1 OF 6)



LEFT RELAY PANEL

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FIGURE 9-2-11-1. INSTRUMENT PANEL AND CONSOLE INDICATOR LIGHTS DIMMING SCHEMATIC DIAGRAM (SHEET 2 OF 6)

9-2-11-8 Change 7

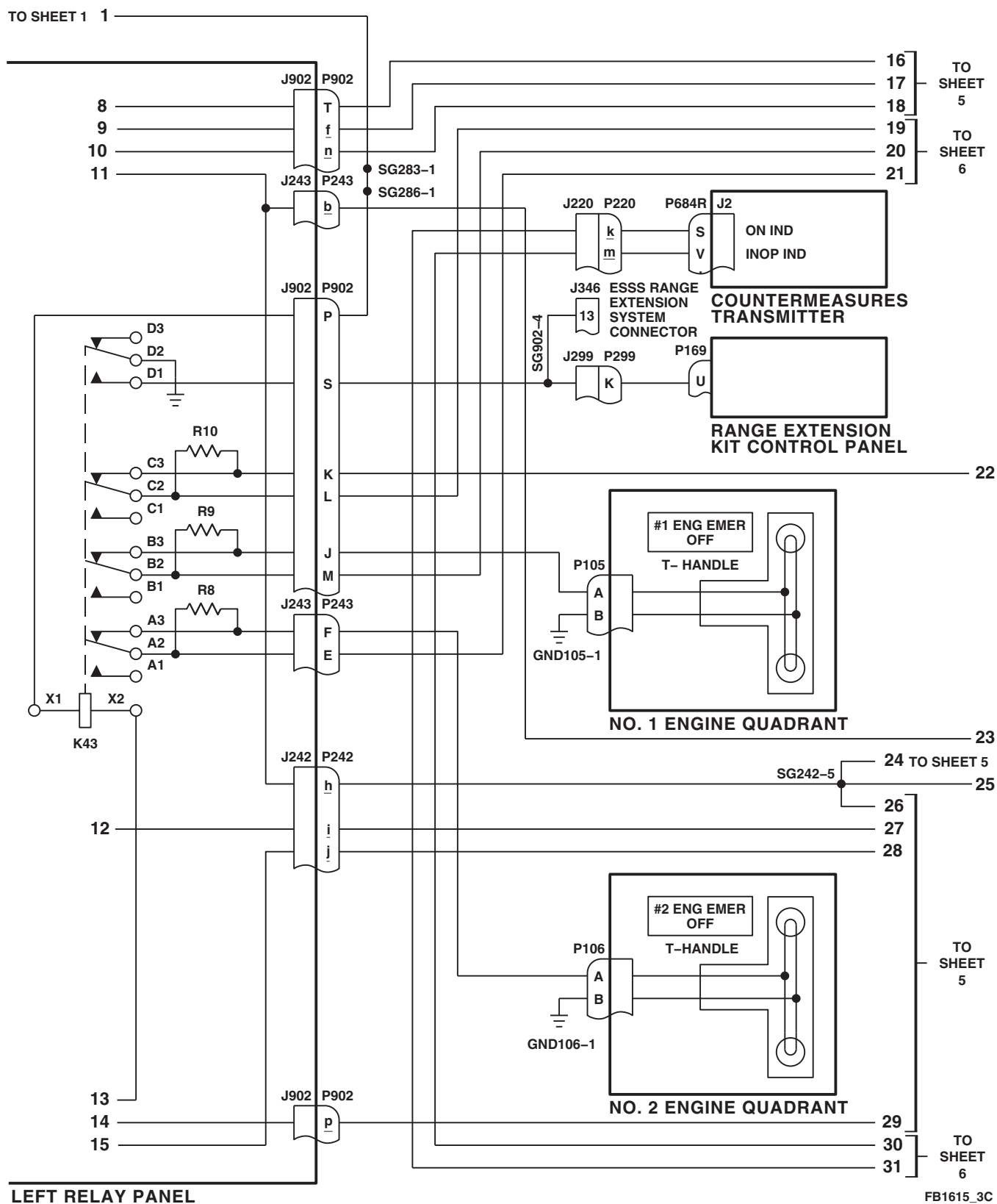


FIGURE 9-2-11-1. INSTRUMENT PANEL AND CONSOLE INDICATOR LIGHTS DIMMING SCHEMATIC DIAGRAM (SHEET 3 OF 6)

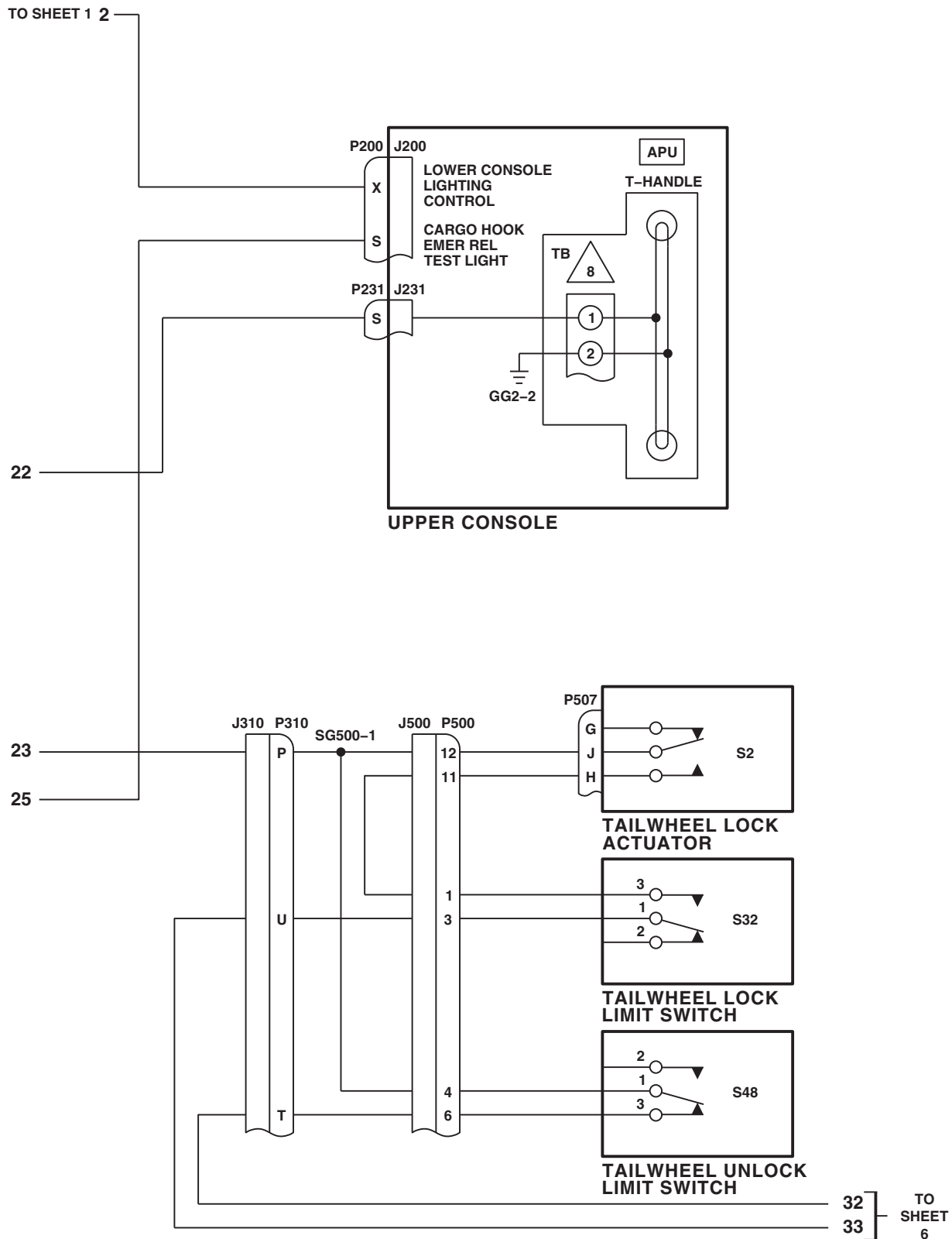
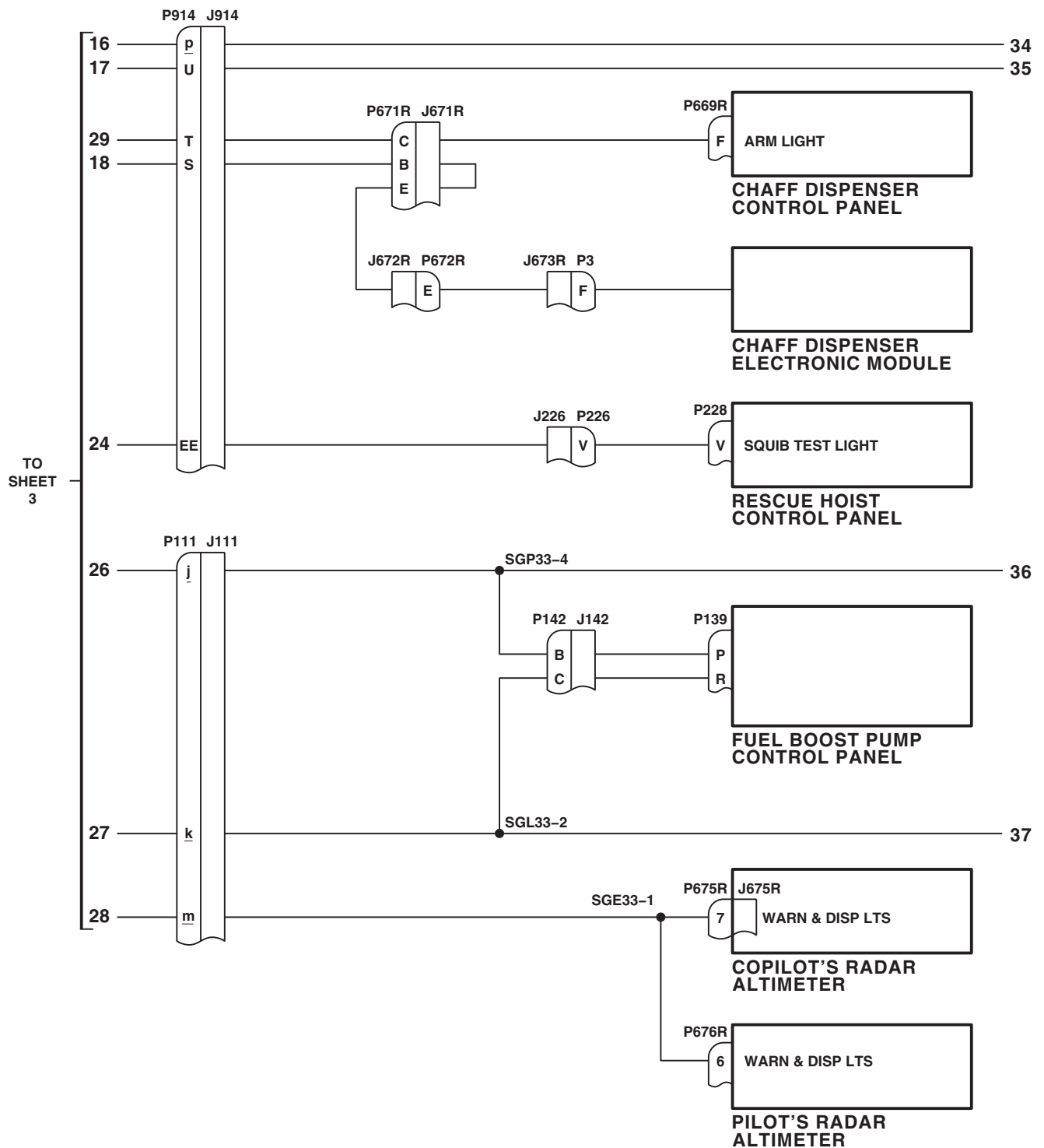


FIGURE 9-2-11-1. INSTRUMENT PANEL AND CONSOLE INDICATOR LIGHTS DIMMING SCHEMATIC DIAGRAM (SHEET 4 OF 6)

9-2-11-10

Change 7



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FIGURE 9-2-11-1. INSTRUMENT PANEL AND CONSOLE INDICATOR LIGHTS DIMMING SCHEMATIC DIAGRAM (SHEET 5 OF 6)

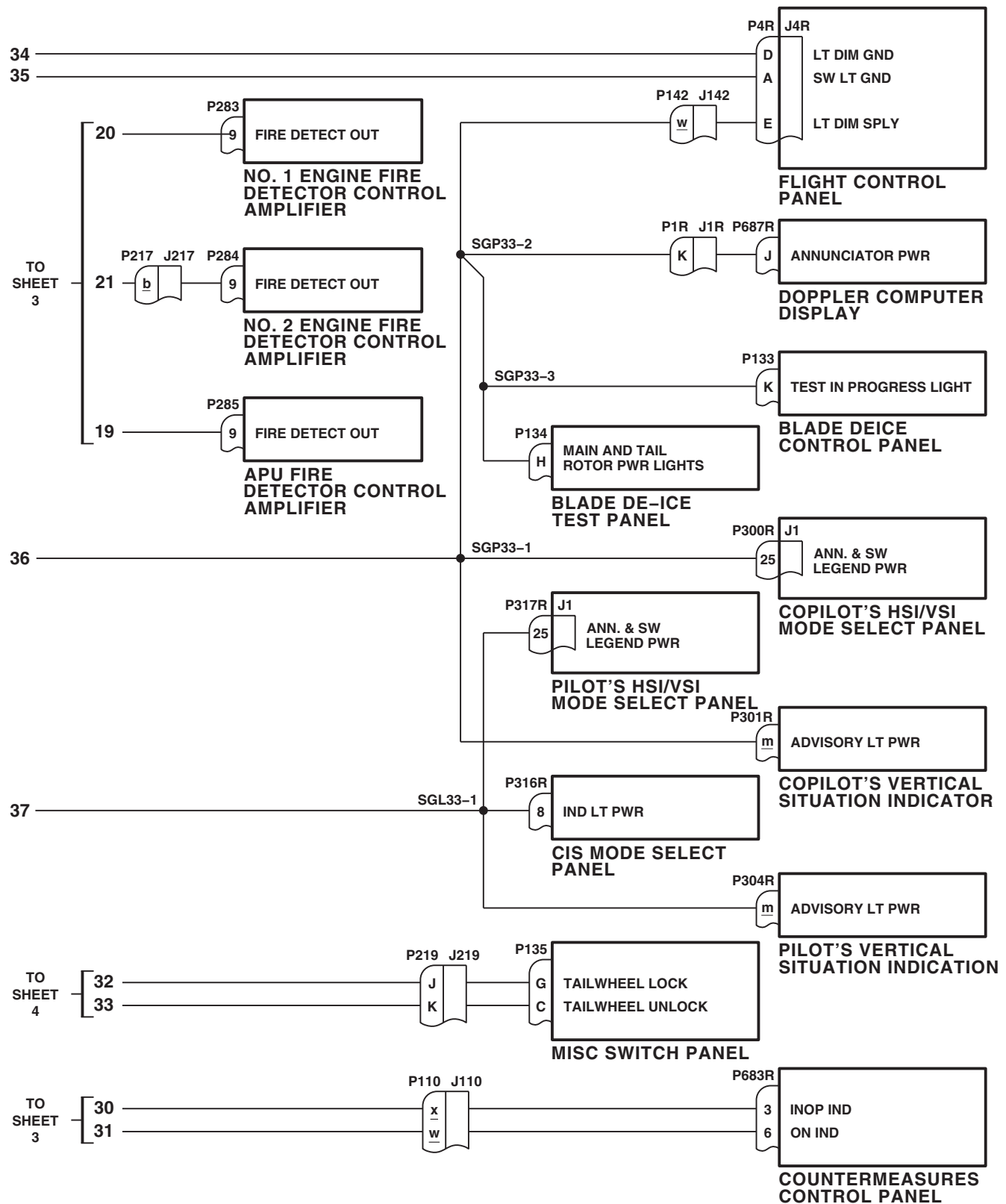
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FIGURE 9-2-11-1. INSTRUMENT PANEL AND CONSOLE INDICATOR LIGHTS DIMMING SCHEMATIC DIAGRAM (SHEET 6 OF 6)

9-2-12 FORMATION LIGHTS.

INITIAL SETUP

Test Equipment

- Locally-Made IR Light Test Set, Figure H-149, Appendix H
- Multimeter, Fluke 45

Tools

- Electrical Repairer Toolkit

References

- PARA 1-6-16

Equipment Conditions

- External Electrical Power Available or APU Operational

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).

9-2-12.1 Operational/Troubleshooting Procedure.

- Check all four formation light assemblies for cracked or broken lenses.

RESULT: No formation light lenses shall be cracked or broken.

- If result is not as specified, replace formation light assembly (PARA 9-4-70, PARA 9-4-72, or PARA 9-4-74).
- Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.
 - Turn on electrical power (PARA 1-6-16).
 - Make sure NAV LTS switch is set to NORM.
 - Turn FORMATION LT control to 5.

RESULT: All four formation lights shall go on.

- If no formation lights go on, go to TABLE NO. 9-2-12-1.
- If one or more formation light does not go on, go to TABLE NO. 9-2-12-2.

- Turn FORMATION LT control through positions 4, 3, 2, 1, and OFF.

RESULT: Formation lights shall gradually dim and go off when control is OFF.

- If result is not as specified, replace FORMATION LT control (PARA 9-4-5).
- Check all four IR formation light assemblies for cracked or broken lenses.

RESULT: No IR formation light lenses shall be cracked or broken.

- If result is not as specified, replace IR formation light assembly (PARA 9-4-71, PARA 9-4-73, or PARA 9-4-75).
- Place locally-made IR light test set (test set) power switch to ON.

RESULT: Test set POWER ON lamp shall come on.

- If result is not as specified, troubleshoot test set.
- Place FORMATION LT control to position 5.
 - Place NAV LTS switch to IR.

- k. Place test set IR detector lead over clear lens of each IR formation light.

RESULT: Test set IR LIGHT ON lamp shall go on for each IR formation light.

- If lamp does not go on for all IR formation lights, go to TABLE NO. 9-2-12-3.
- If lamp does not go one for one or more IR formation lights, go to TABLE NO. 9-2-12-4.

1. Turn FORMATION LT control through position 1 through 4. At each position, place IR detector lead over dark lens of each IR formation light.

RESULT: Test set IR LIGHT ON lamp shall go on at each IR formation light for each switch position.

- If lamp does not go on for all IR formation lights, check for 5 VDC between terminals C and G of FORMATION LT control.

- If lamp does not go on for one or more IR formation lights, check for 5 VDC between the following, as required:

J454-3	and	J 4 5 4 - 2 (main rotor pylon)
J547-3	and	J 5 4 7 - 2 (tail cone)
J711-3	and	J 7 1 1 - 2 (left stabilator)
J712-3	and	J 7 1 2 - 2 (right stabilator)

- (a) If voltage is as specified, replace light assembly (PARA 9-4-71, PARA 9-4-73, or PARA 9-4-75) as required.

- (b) If voltage is not as specified, repair/replace wiring between FORMATION LT control terminal C and the following (Appendix F) as required:

- (a) If voltage is as specified, repair/replace wiring between FORMATION LT control terminal C and J922-R (Appendix F).

J454-3	(main rotor pylon)
J547-3	(tail cone)
J711-3	(left stabilator)
J712-3	(right stabilator)

- (b) If voltage is not as specified, replace FORMATION LT control (PARA 9-4-5).

- m. Turn off electrical power (PARA 1-6-16).

TABLE NO. 9-2-12-1. No Formation Lights Go On.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for 115 VAC between P243-N and ground.	
Step 1. If voltage is as specified, go to 2.	
Step 2. If voltage is not as specified, go to 3.	

TABLE NO. 9-2-12-1. No Formation Lights Go On. (Cont)

TEST OR INSPECTION CORRECTIVE ACTION	
2. Check continuity between J701-A and P243-L.	
Step 1. If continuity is present, replace left relay panel (PARA 9-4-2). Go to 10 . Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 10 .	
3. Check for 115 VAC between terminal 1 (LIGHTS FORM LV circuit breaker, pilot's circuit breaker panel) and ground.	
Step 1. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P243-N (Appendix F). Go to 10 . Step 2. If voltage is not as specified, go to 4 .	
4. Check for 115 VAC between terminal 2 (LIGHTS FORM LV circuit breaker, pilot's circuit breaker panel) and ground.	
Step 1. If voltage is as specified, replace circuit breaker (PARA 9-4-1). Go to 10 . Step 2. If voltage is not as specified, go to 5 .	
5. Check for 115 VAC between FORMATION LT control T9-LV and ground.	
Step 1. If voltage is as specified, repair/replace wiring between T9-LV and LIGHTS FORM LV circuit breaker terminal 2 (Appendix F). Go to 10 . Step 2. If voltage is not as specified, go to 6 .	
6. Check for 115 VAC between T9-5 and T9-G.	
Step 1. If voltage is as specified, replace FORMATION LT control (PARA 9-4-5). Go to 10 . Step 2. If voltage is not as specified, go to 7 .	
7. Check continuity between T9-G and ground.	
Step 1. If continuity is present, go to 8 . Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 10 .	
8. Check for 115 VAC between T9-HV and ground.	
Step 1. If voltage is as specified, replace FORMATION LT control (PARA 9-4-5). Go to 10 . Step 2. If voltage is not as specified, go to 9 .	

TABLE NO. 9-2-12-1. No Formation Lights Go On. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
9. Check for 115 VAC between terminal 1 (LIGHTS FORM HV circuit breaker, NO. 2 AC PRI BUS, pilot's circuit breaker panel) and ground.	Step 1. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and T9-HV (Appendix F). Go to 10. Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 10.
10. Procedure completed.	

TABLE NO. 9-2-12-2. One Or More Formation Lights Does Not Go On.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for 115 VAC between: J402-A and J402-B (main rotor pylon) J506-A and J506-B (tail cone) J700-A and J700-B (right stabilator) J701-A and J701-B (left stabilator)	Step 1. If voltage is as specified, replace light assembly (PARA 9-4-70, PARA 9-4-72, or PARA 9-4-74) as required. Go to 3. Step 2. If voltage is not as specified, go to 2.
2. Check continuity between: J402-B and ground (main rotor pylon) J506-B and ground (tail cone) J700-B and ground (right stabilator) J701-B and ground (left stabilator)	Step 1. If continuity is present, repair/replace wiring between the following (Appendix F) as required, then go to 3: P243-L and J402-A P243-L and J506-A P243-L and J700-A P243-L and J701-A Step 2. If continuity is not present, repair/replace wiring (Appendix F) as required. Go to 3.

TABLE NO. 9-2-12-2. One Or More Formation Lights Does Not Go On. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
3. Procedure completed.	

TABLE NO. 9-2-12-3. No IR Formation Lights (Clear Lens) Go On With FORMATION LT Control In Position 5.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for 28 VDC between P243-j and ground.	I
Step 1. If voltage is as specified, go to 2 .	
Step 2. If voltage is not as specified, go to 5 .	
2. Check for 5 VDC between FORMATION LT control T9-B and T9-G.	I
Step 1. If voltage is as specified, go to 3 .	
Step 2. If voltage is not as specified, go to 4 .	
3. Check for 5 VDC between T9-A and T9-G.	I
Step 1. If voltage is as specified, repair/replace wiring between T9-A and J922-P (Appendix F). Go to 8 .	
Step 2. If voltage is not as specified, replace FORMATION LT control (PARA 9-4-5). Go to 8 .	
4. Check continuity between T9-B and P243-g.	
Step 1. If continuity is present, replace left relay panel (PARA 9-4-2). Go to 8 .	
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 8 .	
5. Check for 28 VDC between NAV LTS switch S69-3 and ground.	I
Step 1. If voltage is as specified, repair/replace wiring between S69-3 and P243-j (Appendix F). Go to 8 .	
Step 2. If voltage is not as specified, go to 6 .	
6. Check for 28 VDC between S69-2 and ground.	I
Step 1. If voltage is as specified, replace NAV LTS switch (PARA 9-4-5). Go to 8 .	

TABLE NO. 9-2-12-3. No IR Formation Lights (Clear Lens) Go On With FORMATION LT Control In Position 5. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 2. If voltage is not as specified, go to 7.
7. Check for 28 VDC between terminal 1 (IR LTS circuit breaker, NO. 2 DC PRI BUS - pilot's circuit breaker panel) and ground.	Step 1. If voltage is as specified, repair/replace wiring between S69-2 and terminal 1 of circuit breaker (Appendix F). Go to 8. Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 8.
8. Procedure completed.	

TABLE NO. 9-2-12-4. One Or More IR Formation Lights (Clear Lens) Does Not Go On.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for 5 VDC between:	J454-1 and J454-2 (main rotor pylon) J547-1 and J547-2 (tail cone) J711-1 and J711-2 (left stabilator) J712-1 and J712-2 (right stabilator)
	Step 1. If voltage is as specified, replace light assembly (PARA 9-4-71, PARA 9-4-73, or PARA 9-4-75) as required. Go to 3. Step 2. If voltage is not as specified, go to 2.
2. Check continuity between:	J454-2 and ground (main rotor pylon) J547-2 and ground (tail cone) J711-2 and ground (left stabilator) J712-3 and ground (right stabilator)
	Step 1. If continuity is present, repair/replace wiring (Appendix F) as required between the following, then go to 3: J454-1 and T9-A J547-1 and T9-A J711-1 and T9-A J712-1 and T9-A

TABLE NO. 9-2-12-4. One Or More IR Formation Lights (Clear Lens) Does Not Go On. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 2. If continuity is not present, repair/replace wiring (Appendix F) as required. Go to 3.
	3. Procedure completed.

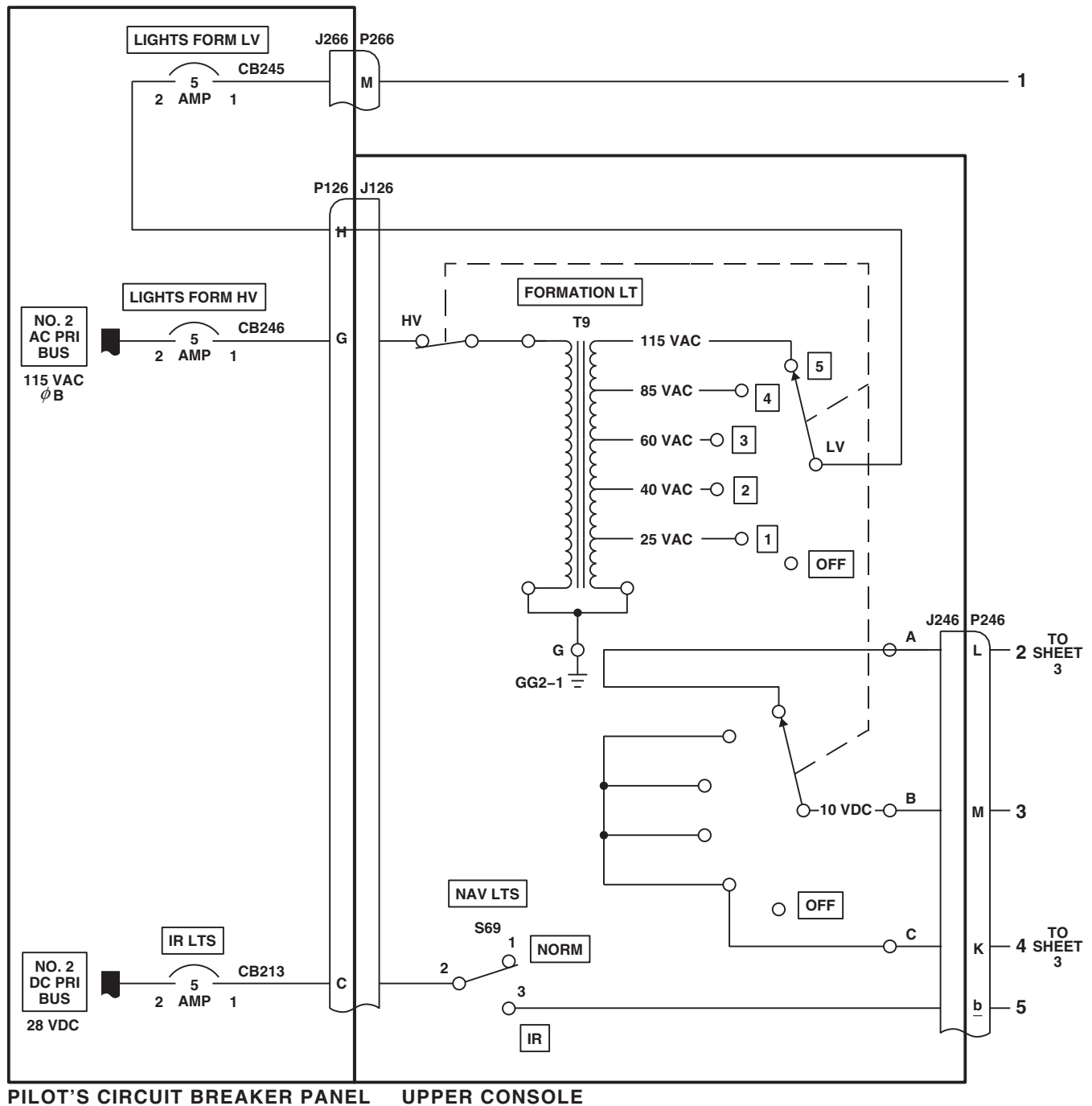


FIGURE 9-2-12-1. FORMATION LIGHTS SCHEMATIC DIAGRAM (SHEET 1 OF 4)

FB1592_1B
SA

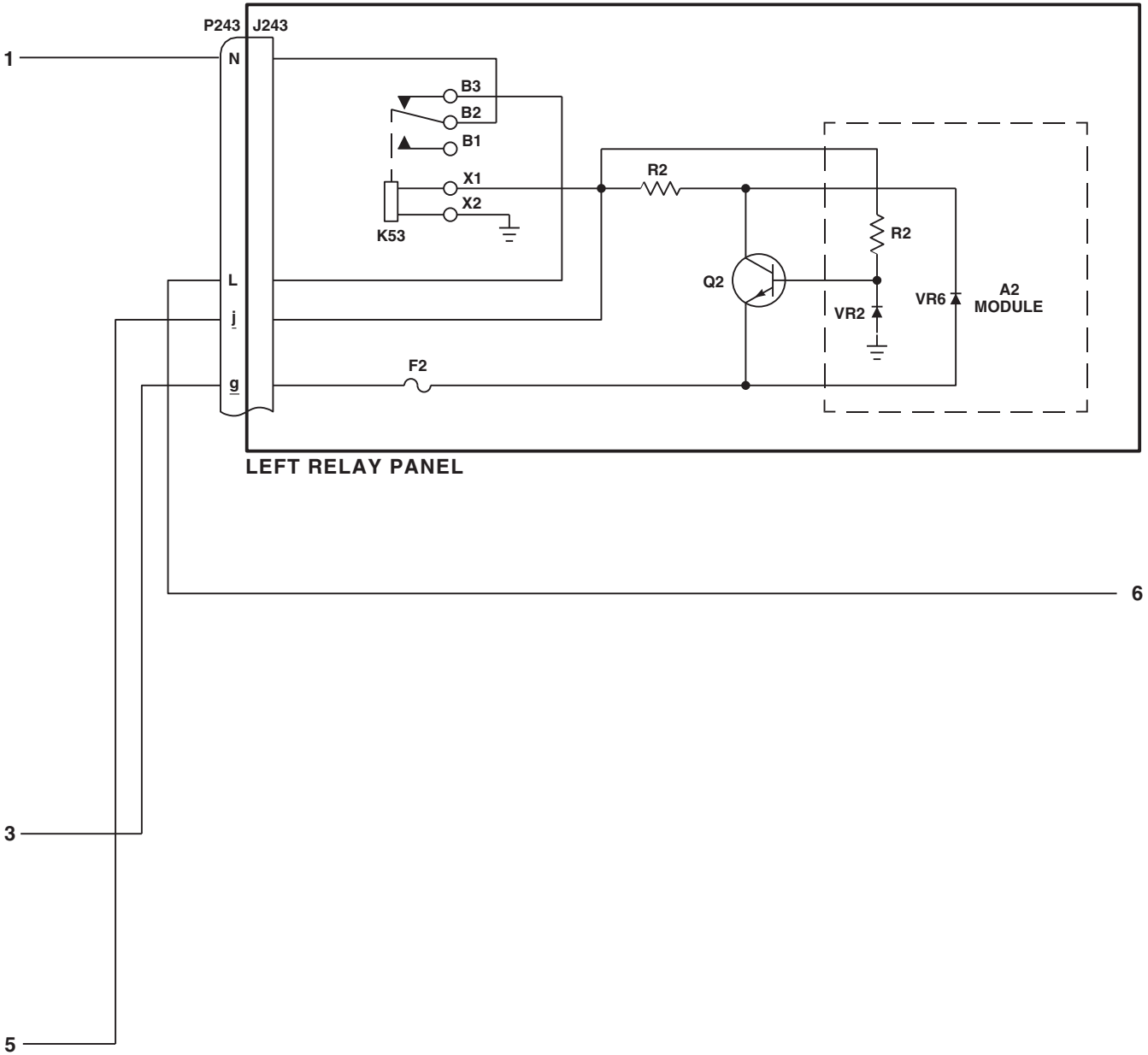


FIGURE 9-2-12-1. FORMATION LIGHTS SCHEMATIC DIAGRAM (SHEET 2 OF 4)

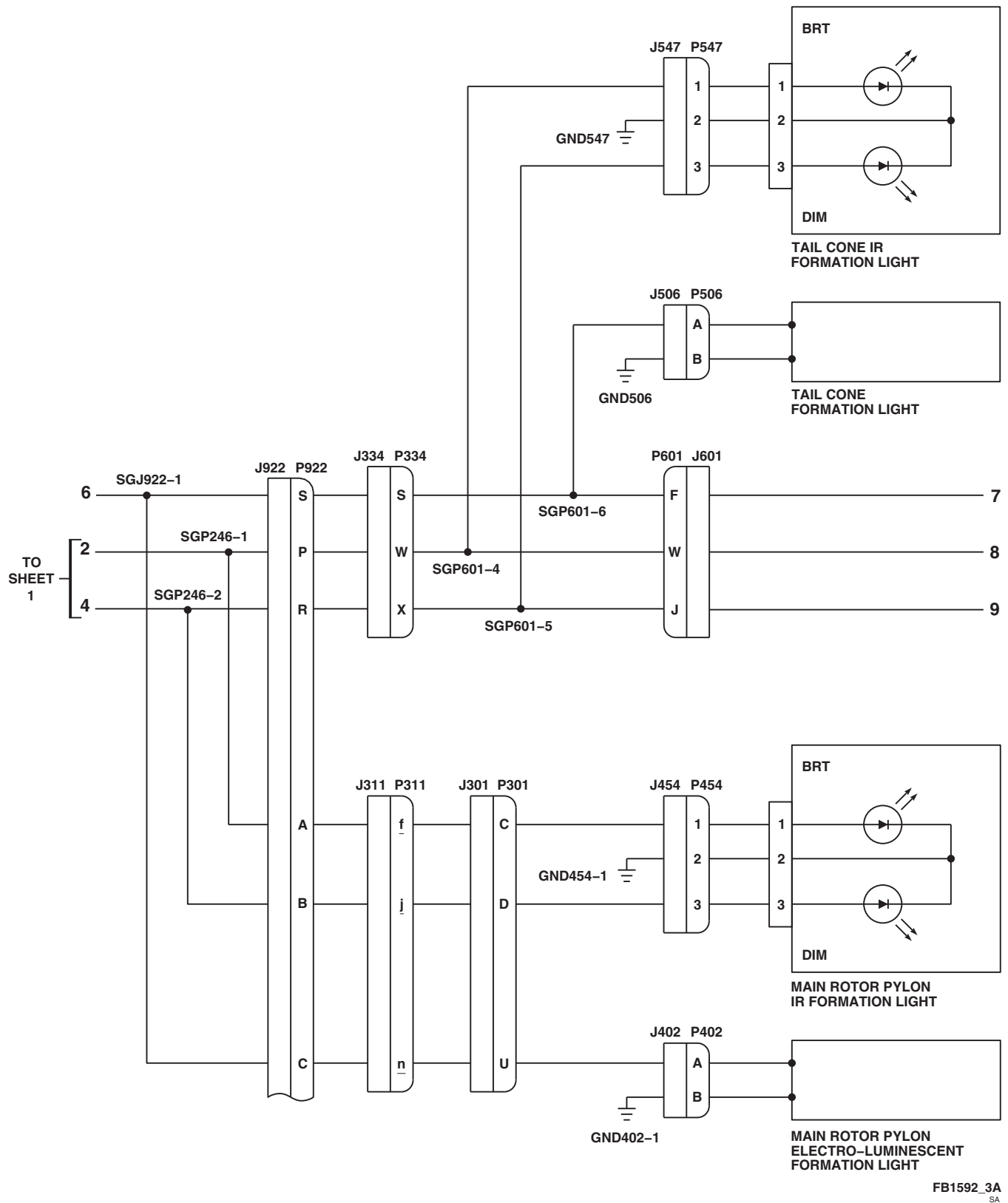


FIGURE 9-2-12-1. FORMATION LIGHTS SCHEMATIC DIAGRAM (SHEET 3 OF 4)

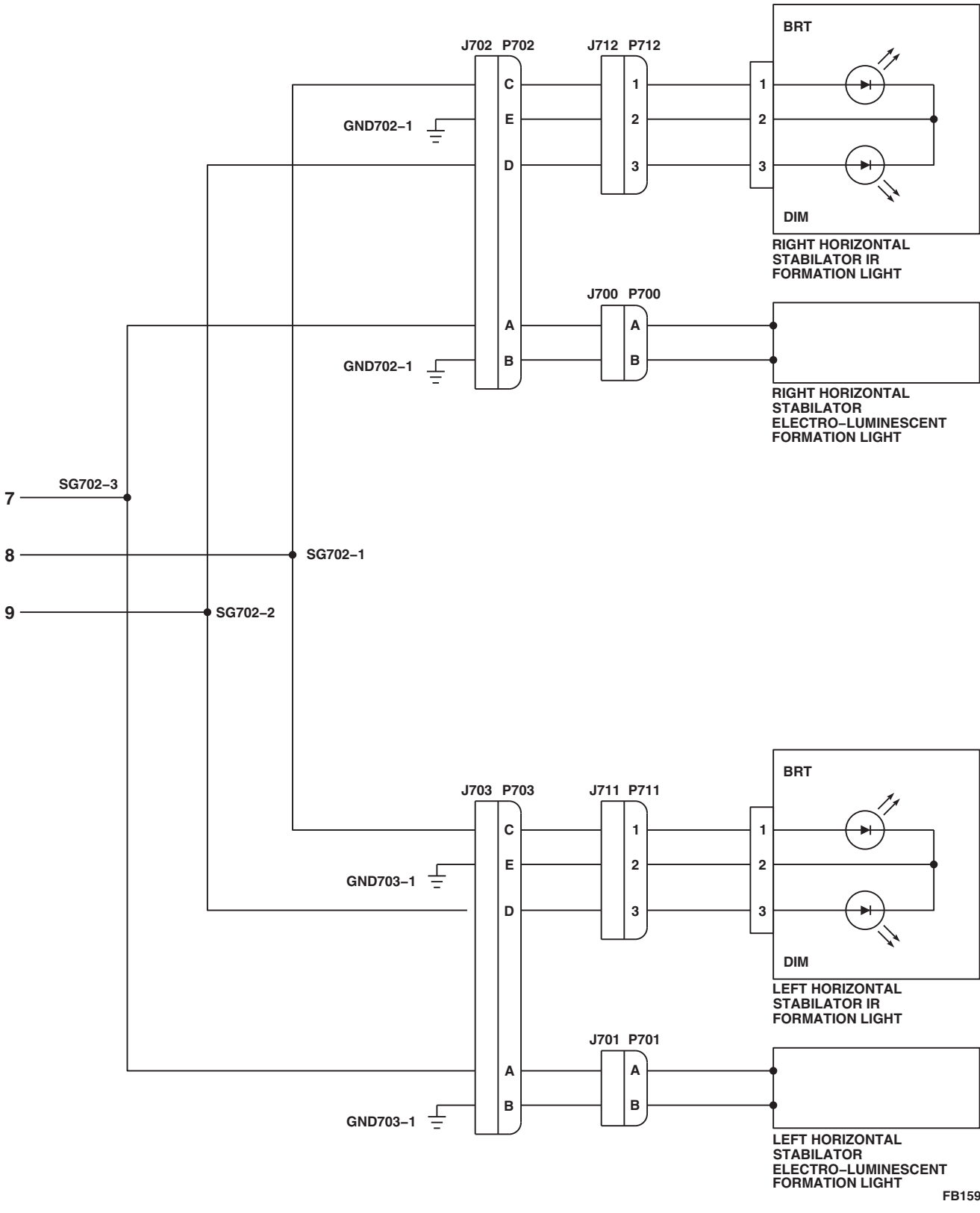


FIGURE 9-2-12-1. FORMATION LIGHTS SCHEMATIC DIAGRAM (SHEET 4 OF 4)

9-2-13 CONTROLLABLE SEARCHLIGHT.

INITIAL SETUP

Test Equipment

- Multimeter, Fluke 45

Tools

- Electrical Repairer Toolkit

References

- PARA 1-6-16
- PARA 9-4-91

Equipment Conditions

- External Electrical Power Available or APU Operational



To prevent personal injury, do not look directly into searchlight when it is turned on.



To avoid damage to the searchlight dimming unit, do not operate OUTPUT switch with the searchlight on.

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).

9-2-13.1 Operational/Troubleshooting Procedure.

- Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.
- Turn on electrical power (PARA 1-6-16).
- Press pilot's collective grip searchlight control to EXT and hold for 5 seconds.

RESULT: Searchlight shall extend from stored position.

- If result is not as specified, go to TABLE NO. 9-2-13-1.
- d. Press pilot's collective grip searchlight control to R and release.

RESULT: Controllable searchlight shall turn to right.

- If result is not as specified, go to TABLE NO. 9-2-13-2.

- Press pilot's collective grip searchlight control to L and release.

RESULT: Controllable searchlight shall turn to left.

- If result is not as specified, go to TABLE NO. 9-2-13-3.

- Press pilot's collective grip searchlight control to RETR and hold for 10 seconds.

RESULT: Controllable searchlight shall retract.

- If result is not as specified, go to TABLE NO. 9-2-13-4.

- Press copilot's collective grip searchlight control to EXT and hold for 5 seconds.

RESULT: Controllable searchlight shall extend from stored position.

- If result is not as specified, go to TABLE NO. 9-2-13-5.

- h. Press copilot's collective grip searchlight control to R.

RESULT: Controllable searchlight shall turn to right.

- If result is not as specified, go to TABLE NO. 9-2-13-6.

- i. Press copilot's collective grip searchlight control to L.

RESULT: Controllable searchlight shall turn to left.

- If result is not as specified, go to TABLE NO. 9-2-13-7.

- j. Press copilot's collective grip searchlight control to RETR and hold for 10 seconds.

RESULT: Controllable searchlight shall retract.

- If result is not as specified, go to TABLE NO. 9-2-13-8.

- k. If installed, remove infrared filter from the searchlight (PARA 9-4-91).

- l. Make sure searchlight dimming unit OUTPUT switch is in NORM position.

- m. Press pilot's collective grip SRCH LT PUSH ON/OFF switch and release.

RESULT: Searchlight shall go on, and SEARCH LT ON capsule on the caution/ advisory panel shall go on.

- If result is not as specified, go to TABLE NO. 9-2-13-9.

- n. Press pilot's collective grip SRCH LT switch to DIM and hold.

RESULT: Searchlight shall decrease in brightness and go off.

- If result is not as specified, go to TABLE NO. 9-2-13-10.

- o. Press pilot's collective grip SRCH LT switch to BRT and hold.

RESULT: Searchlight shall increase in brightness.

- If result is not as specified, go to TABLE NO. 9-2-13-11.

- p. Press pilot's collective grip SRCH LT PUSH ON/OFF switch and release.

RESULT: Searchlight shall go off, and SEARCH LT ON capsule on caution/ advisory panel shall go off.

- If result is not as specified, replace left relay panel (PARA 9-4-2).

- q. Set searchlight dimming unit OUTPUT switch to BYPASS.

- r. Press pilot's collective grip SRCH LT PUSH ON/OFF switch and release.

- s. Press pilot's collective grip SRCH LT switch to DIM and hold.

RESULT: Searchlight shall not decrease in brightness.

- If result is not as specified, replace controllable searchlight dimming unit (PARA 9-4-56).

- t. Press pilot's collective grip SRCH LT PUSH ON/OFF switch and release.

- u. Place dimming unit OUTPUT switch to NORM.

- v. Press copilot's collective grip SRCH LT PUSH ON/OFF switch and release.

RESULT: Searchlight shall go on, and SEARCH LT ON capsule on the caution/ advisory panel shall go on.

- If result is not as specified, go to TABLE NO. 9-2-13-12.

- w. Press copilot’s collective grip SRCH LT PUSH ON/OFF switch to DIM and hold.

RESULT: Searchlight shall decrease in brightness and go off.

 - If result is not as specified, go to TABLE NO. 9-2-13-13.

x. Press copilot’s collective grip SRCH LT PUSH ON/OFF switch to BRT and hold.

RESULT: Searchlight shall increase in brightness.

 - If result is not as specified, go to TABLE NO. 9-2-13-14.
- y. Press copilot’s collective grip SRCH LT PUSH ON/OFF switch and release.

RESULT: Searchlight shall go off, and SEARCH LT ON capsule on caution/advisory panel shall go off.

 - If result is not as specified, replace left relay panel (PARA 9-4-2).

z. If removed, install infrared filter on the controllable searchlight (PARA 9-4-91).

aa. Turn off electrical power (PARA 1-6-16).

TABLE NO. 9-2-13-1. Searchlight Does Not Extend With Pilot’s Searchlight Control Pressed To EXT.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for 28 VDC between J104-C and ground.	Step 1. If voltage is as specified, go to 2. Step 2. If voltage is not as specified, go to 7.
2. Press and hold pilot’s searchlight control to EXT. Go to 3.	
3. Check continuity between:	W/O MOD MOD P104-C and P104-E P104-C and P104-e Step 1. If continuity is present, go to 4. Step 2. If continuity is not present, troubleshoot pilot’s collective stick (PARA 11-2-2). Go to 8.
4. Check for 28 VDC between J109-A and J109-I.	Step 1. If voltage is as specified, replace controllable searchlight (PARA 9-4-92). Go to 8. Step 2. If voltage is not as specified, go to 5.
5. Release pilot’s searchlight control. Go to 6.	

**TABLE NO. 9-2-13-1. Searchlight Does Not Extend With Pilot's Searchlight Control Pressed To EXT.
(Cont)**

TEST OR INSPECTION	CORRECTIVE ACTION
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6. Check continuity between J109-I and ground.

Step 1. If continuity is present, repair/replace wiring between the following (Appendix F). Go to **8**.

W/O MOD J109-A and J104-E

MOD J109-A and J104-e

Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **8**.

7. Check for 28 VDC between terminal 1 (LIGHTS SRCH CONTR circuit breaker, DC ESNTL BUS - upper console circuit breaker panel) and ground.

Step 1. If voltage is as specified, repair/replace wiring between circuit breaker and J104-C (Appendix F). Go to **8**.

Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to **8**.

8. Procedure completed.

TABLE NO. 9-2-13-2. Searchlight Does Not Turn Right With Pilot's Searchlight Control Pressed To R.

TEST OR INSPECTION	CORRECTIVE ACTION
--------------------	-------------------

1. Press and hold pilot's searchlight control to R. Go to 2.

2. Check for 28 VDC between J109-C and ground.

Step 1. If voltage is as specified, replace controllable searchlight (PARA 9-4-92).
Go to **4**.

Step 2. If voltage is not as specified, go to **3**.

3. Check continuity between P104-C and P104-D.

Step 1. If continuity is present, repair/replace wiring between J104-D and J109-C (Appendix F). Go to **4**.

Step 2. If continuity is not present, troubleshoot pilot's collective stick (PARA 11-2-2). Go to **4**.

TABLE NO. 9-2-13-2. Searchlight Does Not Turn Right With Pilot’s Searchlight Control Pressed To R.
 (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	4. Procedure completed.

TABLE NO. 9-2-13-3. Searchlight Does Not Turn Left With Pilot’s Searchlight Control Pressed To L.

TEST OR INSPECTION	CORRECTIVE ACTION
	1. Press and hold pilot’s searchlight control to L. Go to 2.
	2. Check for 28 VDC between J109-D and ground. Step 1. If voltage is as specified, replace controllable searchlight (PARA 9-4-92). Go to 4. Step 2. If voltage is not as specified, go to 3.
	3. Check continuity between P104-A and P104-C. Step 1. If continuity is present, repair/replace wiring between J104-A and J109-D (Appendix F). Go to 4. Step 2. If continuity is not present, troubleshoot pilot’s collective stick (PARA 11-2-2). Go to 4.
	4. Procedure completed.

TABLE NO. 9-2-13-4. Searchlight Does Not Retract With Pilot’s Searchlight Control Pressed To RETR.

TEST OR INSPECTION	CORRECTIVE ACTION
	1. Press and hold pilot’s searchlight control to RETR. Go to 2.
	2. Check for 28 VDC between J109-E and ground. Step 1. If voltage is as specified, replace controllable searchlight (PARA 9-4-92). Go to 4.

**TABLE NO. 9-2-13-4. Searchlight Does Not Retract With Pilot's Searchlight Control Pressed To RETR.
(Cont)**

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 2. If voltage is not as specified, go to 3 .
3. Check continuity between P104-B and P104-C.	Step 1. If continuity is present, repair/replace wiring between J104-B and J109-E (Appendix F). Go to 4. Step 2. If continuity is not present, troubleshoot pilot's collective stick (PARA 11-2-2). Go to 4.
4. Procedure completed.	

TABLE NO. 9-2-13-5. Searchlight Does Not Extend With Copilot's Searchlight Control Pressed To EXT.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for 28 VDC between J103-C and ground.	Step 1. If voltage is as specified, go to 2. Step 2. If voltage is not as specified, go to 5.
2. Press and hold copilot's searchlight control to EXT. Go to 3.	
3. Check continuity between:	W/O MOD P103-C and P104-E MOD P103-C and P104-e Step 1. If continuity is present, go to 4. Step 2. If continuity is not present, troubleshoot copilot's collective stick (PARA 11-2-2). Go to 6.
4. Check for 28 VDC between J109-A and ground.	Step 1. If voltage is as specified, replace controllable searchlight (PARA 9-4-92). Go to 6. Step 2. If voltage is not as specified, repair/replace wiring between the following (Appendix F). Go to 6.

TABLE NO. 9-2-13-5. Searchlight Does Not Extend With Copilot’s Searchlight Control Pressed To EXT.
 (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	W/O MOD J109-A and J104-E MOD J109-A and J104-e
5.	Check for 28 VDC between terminal 1 (LIGHTS SRCH CONTR circuit breaker, DC ESNTL BUS - upper console circuit breaker panel) and ground. Step 1. If voltage is as specified, repair/replace wiring between circuit breaker and J103-C (Appendix F). Go to 6. Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 6.
6.	Procedure completed.

TABLE NO. 9-2-13-6. Searchlight Does Not Turn Right With Copilot’s Searchlight Control Pressed To R.

TEST OR INSPECTION	CORRECTIVE ACTION
1.	Press and hold copilot’s searchlight control to R. Go to 2.
2.	Check for 28 VDC between J109-C and ground. Step 1. If voltage is as specified, replace controllable searchlight (PARA 9-4-92). Go to 4. Step 2. If voltage is not as specified, go to 3.
3.	Check continuity between P103-C and P103-D. Step 1. If continuity is present, repair/replace wiring between J103-D and J109-C (Appendix F). Go to 4. Step 2. If continuity is not present, troubleshoot copilot’s collective stick (PARA 11-2-2). Go to 4.
4.	Procedure completed.

TABLE NO. 9-2-13-7. Searchlight Does Not Turn Left With Copilot's Searchlight Control Pressed To L.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Press and hold copilot's searchlight control to L. Go to 2.	
2. Check for 28 VDC between J109-D and ground.	Step 1. If voltage is as specified, replace controllable searchlight (PARA 9-4-92). Go to 4. Step 2. If voltage is not as specified, go to 3.
3. Check continuity between P103-A and P103-C.	Step 1. If continuity is present, repair/replace wiring between J103-A and J109-D (Appendix F). Go to 4. Step 2. If continuity is not present, troubleshoot copilot's collective stick (PARA 11-2-2). Go to 4.
4. Procedure completed.	

TABLE NO. 9-2-13-8. Searchlight Does Not Retract With Copilot's Searchlight Control Pressed To RETR.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Press and hold copilot's searchlight control to RETR. Go to 2.	
2. Check for 28 VDC between J109-E and ground.	Step 1. If voltage is as specified, replace controllable searchlight (PARA 9-4-92). Go to 4. Step 2. If voltage is not as specified, go to 3.
3. Check continuity between P103-B and P103-C.	Step 1. If continuity is present, repair/replace wiring between J103-B and J109-E (Appendix F). Go to 4. Step 2. If continuity is not present, troubleshoot copilot's collective stick (PARA 11-2-2). Go to 4.

TABLE NO. 9-2-13-8. Searchlight Does Not Retract With Copilot’s Searchlight Control Pressed To RETR. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION
4. Procedure completed.

TABLE NO. 9-2-13-9. Searchlight Does Not Light With Pilot’s SRCH LT Switch ON.

TEST OR INSPECTION
CORRECTIVE ACTION
1. Check for 28 VDC between J104-K and ground.
Step 1. If voltage is as specified, go to 2. Step 2. If voltage is not as specified, go to 16.
2. Check for 28 VDC between J109-G and ground.
Step 1. If voltage is as specified, go to 3. Step 2. If voltage is not as specified, go to 4.
3. Check continuity between J109-I and ground.
Step 1. If continuity is present, replace controllable searchlight (PARA 9-4-92). Go to 17. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 17.
4. Press and hold pilot’s SRCH LT ON/OFF switch. Go to 5.
5. Check for 28 VDC between P243-Y and ground.
Step 1. If voltage is as specified, go to 6. Step 2. If voltage is not as specified, go to 15.
6. Release pilot’s SRCH LT ON/OFF switch. Go to 7.
7. Check continuity between P243-W and P230-L.
Step 1. If continuity is present, go to 8. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 17.

TABLE NO. 9-2-13-9. Searchlight Does Not Light With Pilot's SRCH LT Switch ON. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
8. Press and release pilot's SRCH LT ON/OFF switch. Go to 9.	
9. Is SEARCH LT ON advisory capsule on?	Step 1. If SEARCH LT ON capsule is on, go to 10. Step 2. If SEARCH LT ON capsule is not on, replace left relay panel (PARA 9-4-2). Go to 17.
10. Check continuity between J109-G and P243-V.	Step 1. If continuity is present, go to 11. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 17.
11. Check for 28 VDC between J109-F and ground.	Step 1. If voltage is as specified, replace controllable searchlight (PARA 9-4-92). Go to 17. Step 2. If voltage is not as specified, go to 12.
12. Check for 28 VDC between terminal 1 (LIGHTS CONTR PWR circuit breaker, DC ESNTL BUS - upper console circuit breaker panel) and ground.	Step 1. If voltage is as specified, go to 13. Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 17.
13. Check continuity between terminal 1 (LIGHTS CONTR PWR circuit breaker, DC ESNTL BUS - upper console circuit breaker panel) and P167-B.	Step 1. If continuity is present, go to 14. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 17.
14. Check continuity between J109-F and P167-I.	Step 1. If continuity is present, replace dimming unit (PARA 9-4-56). Go to 17. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 17.
15. Check continuity between P104-K and P104-P.	Step 1. If continuity is present, repair/replace wiring between J104-P and P243-Y (Appendix F). Go to 17.

TABLE NO. 9-2-13-9. Searchlight Does Not Light With Pilot's SRCH LT Switch ON. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 2. If continuity is not present, troubleshoot pilot's collective stick (PARA 11-2-2). Go to 17 .
16. Check for 28 VDC between terminal 1 (LIGHTS SRCH CONTR circuit breaker, DC ESNTL BUS - upper console circuit breaker pane) and ground.	
	Step 1. If voltage is as specified, repair/replace wiring between J104-K and circuit breaker terminal 1 (Appendix F). Go to 17 .
	Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 17 .
17. Procedure completed.	

TABLE NO. 9-2-13-10. Searchlight Does Not Decrease In Brightness And Go Off With Pilot's SRCH LT Switch Held To DIM.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Make sure that the OUTPUT switch on the searchlight dimming unit is positioned to NORM. Go to 2.	
2. With SRCH LT switch at DIM, check for 28 VDC between P167-D and ground.	
	Step 1. If voltage is as specified, replace dimming unit (PARA 9-4-56). Go to 5 .
	Step 2. If voltage is not as specified, go to 3 .
3. Check for 28 VDC between J104-M and ground.	
	Step 1. If voltage is as specified, go to 4 .
	Step 2. If voltage is not as specified, repair/replace wiring between J104-M and P247-F (Appendix F). Go to 5 .
4. With SRCH LT switch at DIM, check continuity between P104-M and P104-N.	
	Step 1. If continuity is present, repair/replace wiring between J104-N and P167-D (Appendix F). Go to 5 .
	Step 2. If continuity is not present, troubleshoot pilot's collective stick (PARA 11-2-2). Go to 5 .

TABLE NO. 9-2-13-10. Searchlight Does Not Decrease In Brightness And Go Off With Pilot's SRCH LT Switch Held To DIM. (Cont)

TEST OR INSPECTION CORRECTIVE ACTION

5. Procedure completed.

TABLE NO. 9-2-13-11. Searchlight Does Not Increase In Brightness With Pilot's SRCH LT Switch Held To BRT.

TEST OR INSPECTION CORRECTIVE ACTION

1. Make sure that the OUTPUT switch on the searchlight dimming unit is positioned to NORM. Go to 2.

2. With SRCH LT switch at BRT, check for 28 VDC between P167-C and ground.
--

Step 1. If voltage is as specified, replace controllable searchlight dimming unit (PARA 9-4-56). Go to 4.

Step 2. If voltage is not as specified, go to 3.
--

3. With SRCH LT switch at BRT, check continuity between P104-L and P104-M.
--

Step 1. If continuity is present, repair/replace wiring between J104-L and P167-C (Appendix F). Go to 4.
--

Step 2. If continuity is not present, troubleshoot pilot's collective stick (PARA 11-2-2). Go to 4.

4. Procedure completed.

TABLE NO. 9-2-13-12. Searchlight Does Not Light With Copilot's SRCH LT Switch ON.

TEST OR INSPECTION CORRECTIVE ACTION

1. Check for 28 VDC between J103-K and ground.
--

Step 1. If voltage is as specified, go to 2.
--

TABLE NO. 9-2-13-12. Searchlight Does Not Light With Copilot's SRCH LT Switch ON. (Cont)

TEST OR INSPECTION CORRECTIVE ACTION
Step 2. If voltage is not as specified, repair/replace wiring between J103-K and P247-F (Appendix F). Go to 3. 2. With copilot's SRCH LT ON/OFF switch pressed, check continuity between P103-K and P103-P. Step 1. If continuity is present, repair/replace wiring between J103-P and P247-A (Appendix F). Go to 3. Step 2. If continuity is not present, troubleshoot copilot's collective stick (PARA 11-2-2). Go to 3. 3. Procedure completed.

TABLE NO. 9-2-13-13. Searchlight Does Not Decrease In Brightness And Go Off With Copilot's SRCH LT Switch At DIM.

TEST OR INSPECTION CORRECTIVE ACTION
1. Make sure that the OUTPUT switch on the searchlight dimming unit is positioned to NORM. Go to 2. 2. Check for 28 VDC between J103-M and ground. Step 1. If voltage is as specified, go to 3. Step 2. If voltage is not as specified, repair/replace wiring between J103-M and P247-F (Appendix F). Go to 4. 3. With SRCH LT switch at DIM, check continuity between P103-M and P103-N. Step 1. If continuity is present, repair/replace wiring between J103-N and P167-D (Appendix F). Go to 4. Step 2. If continuity is not present, troubleshoot copilot's collective stick (PARA 11-2-2). Go to 4. 4. Procedure completed.

TABLE NO. 9-2-13-14. Searchlight Does Not Increase In Brightness With Copilot's SRCH LT Switch At BRT.

TEST OR INSPECTION CORRECTIVE ACTION
1. Make sure that the OUTPUT switch on the searchlight dimming unit is positioned to NORM. Go to 2. 2. With SRCH LT switch at BRT, check continuity between P103-L and P103-M. Step 1. If continuity is present, repair/replace wiring between J103-L and P167-C (Appendix F). Go to 3. Step 2. If continuity is not present, troubleshoot copilot's collective stick (PARA 11-2-2). Go to 3. 3. Procedure completed.

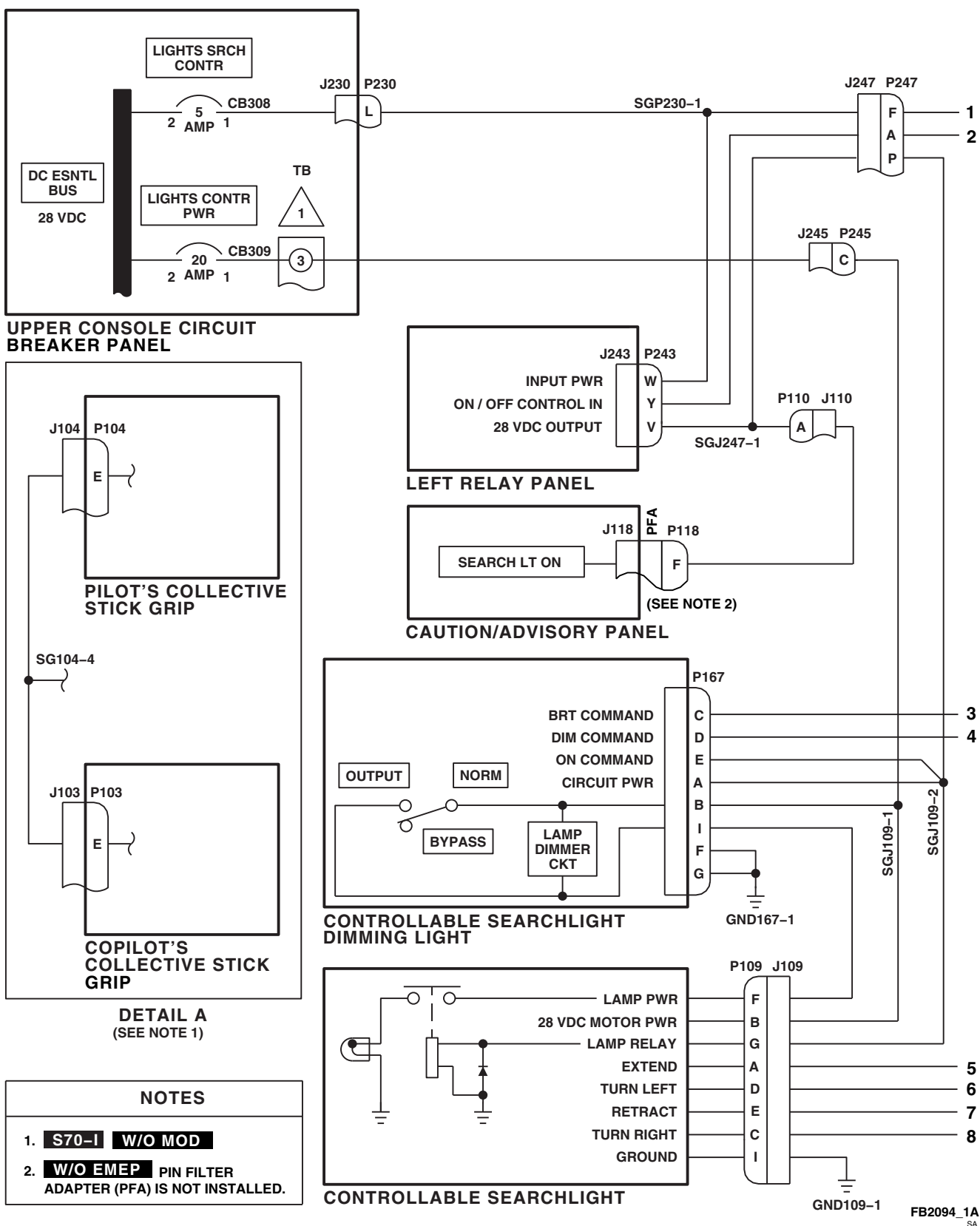


FIGURE 9-2-13-1. CONTROLLABLE SEARCHLIGHT SCHEMATIC DIAGRAM (SHEET 1 OF 2)

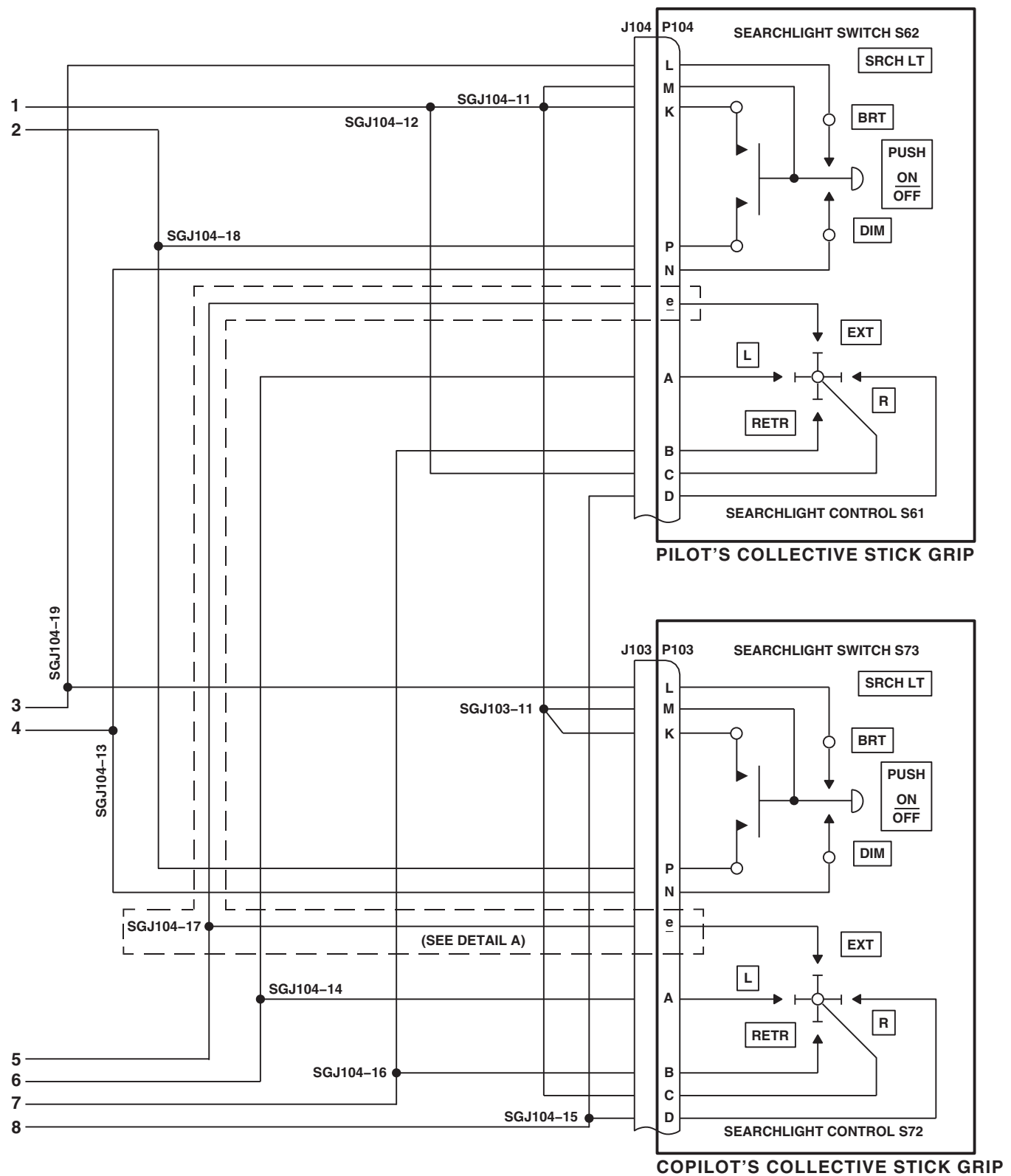
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FIGURE 9-2-13-1. CONTROLLABLE SEARCHLIGHT SCHEMATIC DIAGRAM (SHEET 2 OF 2)

9-2-14 POSITION LIGHTS.

INITIAL SETUP

Test Equipment

- Locally-Made IR Light Test Set, Figure H-149, Appendix H
- Multimeter, Fluke 45

Tools

- Electrical Repairer Toolkit

References

- PARA 1-6-16

Equipment Conditions

- External Electrical Power Available or APU Operational

NOTE

W/O ESSS Disconnects (P681 and P682) are jumpered so the existing three lights are operational.

NOTE

ESSS Position lights on the landing gear support fairings are disabled and the position lights on the external stores support are operational. Therefore, three position lights are operational in either helicopter configuration.

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).

9-2-14.1 Operational/Troubleshooting Procedure.

- Check position light assemblies for cracked or broken lenses.

 RESULT: No position light lenses shall be cracked or broken.
 - If result is not as specified, replace lens (PARA 9-4-77 through PARA 9-4-84).
- Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.
- Turn on electrical power (PARA 1-6-16).
- Place NAV LTS switch to NORM.
- Place POSITION LIGHTS DIM/OFF/BRT switch to BRT.

- Place POSITION LIGHTS STEADY/FLASH switch to STEADY.

RESULT: Three position lights shall go on.

- If result is not as specified, go to TABLE NO. 9-2-14-1.
- If one position light does not go on, go to TABLE NO. 9-2-14-2.

- Place POSITION LIGHTS DIM/OFF/BRT switch to DIM.

RESULT: Position lights shall remain on but at reduced intensity.

- If result is not as specified, go to TABLE NO. 9-2-14-3.

- Place POSITION LIGHTS STEADY/FLASH switch to FLASH.

RESULT: Position lights shall flash.

- If result is not as specified, go to TABLE NO. 9-2-14-4.
- i. Place POSITION LIGHTS DIM/OFF/BRT switch to OFF.

RESULT: Position lights shall go off.

- If result is not as specified, replace POSITION LIGHTS DIM/OFF/BRT switch S29 (PARA 9-4-5).
- j. Place NAV LIGHTS switch to IR.
- k. Place locally-made IR light test set POWER switch to ON.

RESULT: POWER ON lamp shall go on.

- If result is not as specified, troubleshoot test set as required.

NOTE

- **W/O ESSS** Perform steps l. through t. for each of the three IR position lights (tail rotor pylon, and left and right landing gear support fairings).■
- **ESSS** Perform steps l. through t. for each of the three IR position lights (tail rotor pylon, and left and right external stores support system beams).■

- l. Place POSITION LIGHTS DIM/OFF/BRT switch to DIM.
- m. Place POSITION LIGHTS STEADY/FLASH switch to STEADY.
- n. Place test set IR detector lead over each of the IR position light dark lenses.

RESULT: Test set IR LIGHT ON lamp shall go on steady for all IR position lights.

- If result is not as specified, go to TABLE NO. 9-2-14-5.

- If one IR position light does not go on, go to TABLE NO. 9-2-14-6.

- o. Place POSITION LIGHTS DIM/OFF/BRT switch to BRT.
- p. Place test set IR detector lead over each of the IR position light clear lenses.

RESULT: Test set IR LIGHT ON lamp shall go on for all IR position lights.

- If result is not as specified, check fuse F1 on left relay panel.

(a) If fuse is good, go to TABLE NO. 9-2-14-5.

(b) If fuse is not good, replace fuse.

- If one IR position light does not go on, go to TABLE NO. 9-2-14-6.

- q. Place POSITION LIGHTS STEADY/FLASH switch to FLASH.

- r. Place IR detector lead over each of the IR position light clear lenses.

RESULT: Test set IR LIGHT ON lamp shall flash on and off for all IR position lights.

- If result is not as specified, go to TABLE NO. 9-2-14-4.

- s. Place POSITION LIGHTS DIM/OFF/BRT switch to OFF.

- t. Place test set IR detector lead over each IR position light clear lenses.

RESULT: Test set IR LIGHT ON lamp shall not go on for all IR position lights.

- If result is not as specified, replace POSITION LIGHTS DIM/OFF/BRT switch S29 (PARA 9-4-5).

- u. Place test set POWER switch to OFF.

- v. Turn off electrical power (PARA 1-6-16).

TABLE NO. 9-2-14-1. Incandescent Position Lights Do Not Go On.

TEST OR INSPECTION CORRECTIVE ACTION	
1. Check for 28 VDC between S29-6 and ground.	
Step 1. If voltage is as specified, repair/replace wiring between S29-6 and TB2-4 (Appendix F). Go to 8 . Step 2. If voltage is not as specified, go to 2 .	
2. Check for 28 VDC between S29-5 and ground.	
Step 1. If voltage is as specified, replace switch (PARA 9-4-5). Go to 8 . Step 2. If voltage is not as specified, go to 3 .	
3. Check continuity between S29-5 and P243-R.	
Step 1. If continuity is present, go to 4 . Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 8 .	
4. Check for 28 VDC between P243-T and ground.	
Step 1. If voltage is as specified, replace left relay panel (PARA 9-4-2). Go to 8 . Step 2. If voltage is not as specified, go to 5 .	
5. Check for 28 VDC between S28-1 and ground.	
Step 1. If voltage is as specified, repair/replace wiring between S28-1 and P243-T (Appendix F). Go to 8 . Step 2. If voltage is not as specified, go to 6 .	
6. Check for 28 VDC between S28-2 and ground.	
Step 1. If voltage is as specified, replace switch (PARA 9-4-5). Go to 8 . Step 2. If voltage is not as specified, go to 7 .	
7. Check for 28 VDC between terminal 1 (POS LTS circuit breaker, NO. 2 DC PRI BUS - pilot's circuit breaker panel) and ground.	
Step 1. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and S28-2 (Appendix F). Go to 8 . Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 8 .	

TABLE NO. 9-2-14-1. Incandescent Position Lights Do Not Go On. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
8. Procedure completed.	

TABLE NO. 9-2-14-2. One Incandescent Position Light Does Not Go On.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check light lamp.	
Step 1. If lamp is good, go to 2.	
Step 2. If lamp is not good, replace lamp (PARA 9-4-77 through PARA 9-4-84) as required. Go to 4.	
2. Check for bad ground on light assembly mounting screw.	
Step 1. If bad ground is present, repair ground connection (Appendix F). Go to 4.	
Step 2. If bad ground is not present, go to 3.	
3. Check wiring between S29-6 and malfunctioning light.	
Step 1. If wiring is good, replace position light assembly (PARA 9-4-77 through PARA 9-4-84) as required. Go to 4.	
Step 2. If wiring is not good, repair/replace wiring (Appendix F). Go to 4.	
4. Procedure completed.	

TABLE NO. 9-2-14-3. Incandescent Position Lights Do Not Light At A Reduced Intensity With POSITION LIGHTS DIM/OFF/BRT Switch At DIM.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for open dimming resistor R10.	Step 1. If dimming resistor is open, replace dimming resistor R10 (PARA 9-4-109). Go to 3. Step 2. If dimming resistor is not open, go to 2.
2. Check for 28 VDC between S29-4 and ground.	Step 1. If voltage is as specified, repair/replace wiring between S29-4 and TB2-3 (Appendix F). Go to 3. Step 2. If voltage is not as specified, replace switch (PARA 9-4-5). Go to 3.
3. Procedure completed.	

TABLE NO. 9-2-14-4. Position Lights Do Not Flash.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for 28 VDC between P232-B and ground.	Step 1. If voltage is as specified, go to 2. Step 2. If voltage is not as specified, go to 3.
2. Check continuity between: S28-1 and P232-C S28-1 and P232-D P232-E and ground	Step 1. If continuity is present, replace flasher (PARA 9-4-76). Go to 4. Step 2. If continuity is not present, repair/replace wiring (Appendix F) as required. Go to 4.
3. Check continuity between switch S28-3 and P232-B.	Step 1. If continuity is present, replace switch (PARA 9-4-5). Go to 4.

TABLE NO. 9-2-14-4. Position Lights Do Not Flash. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 4. 4. Procedure completed.

TABLE NO. 9-2-14-5. IR Position Lights Do Not Go On.

TEST OR INSPECTION	CORRECTIVE ACTION
1. With NAV LTS switch at IR, check for 10 VDC between S29-2 and ground.	Step 1. If voltage is as specified, replace switch (PARA 9-4-5). Go to 7. Step 2. If voltage is not as specified, go to 2.
2. Check for 28 VDC between P243-j and ground.	Step 1. If voltage is as specified, go to 3. Step 2. If voltage is not as specified, go to 4.
3. Check continuity between P243-f and S29-2.	Step 1. If continuity is present, replace left relay panel (PARA 9-4-2). Go to 7. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 7.
4. Check for 28 VDC between S69-3 and ground.	Step 1. If voltage is as specified, repair/replace wiring between P243-j and S69-3 (Appendix F). Go to 7. Step 2. If voltage is not as specified, go to 5.
5. Check for 28 VDC between S69-2 and ground.	Step 1. If voltage is as specified, replace switch (PARA 9-4-5). Go to 7. Step 2. If voltage is not as specified, go to 6.
6. Check for 28 VDC between terminal 1 (IR LTS circuit breaker, NO. 2 DC PRI BUS - pilot's circuit breaker panel) and ground.	

TABLE NO. 9-2-14-5. IR Position Lights Do Not Go On. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 1. If voltage is as specified, repair/replace wiring between S69-2 and circuit breaker terminal 1 (Appendix F). Go to 7. Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 7.
7. Procedure completed.	

TABLE NO. 9-2-14-6. One IR Position Light Does Not Go On.

TEST OR INSPECTION	CORRECTIVE ACTION
	1. Check continuity between: P340-2 and ground (left IR position light) P341-2 and ground (right IR position light) P616-2 and ground (tail IR position light) P719-2 and ground (left ESSS IR position light) P720-2 and ground (right ESSS IR position light) Step 1. If continuity is present, go to 2. Step 2. If continuity is not present, repair/replace wiring (Appendix F) as required. Go to 3.
	2. Check wiring between malfunctioning IR light and P200 or P246 as applicable. Step 1. If wiring is good, replace malfunctioning IR position light (PARA 9-4-77 through PARA 9-4-84). Go to 3. Step 2. If wiring is not good, repair/replace wiring (Appendix F). Go to 3.
3. Procedure completed.	

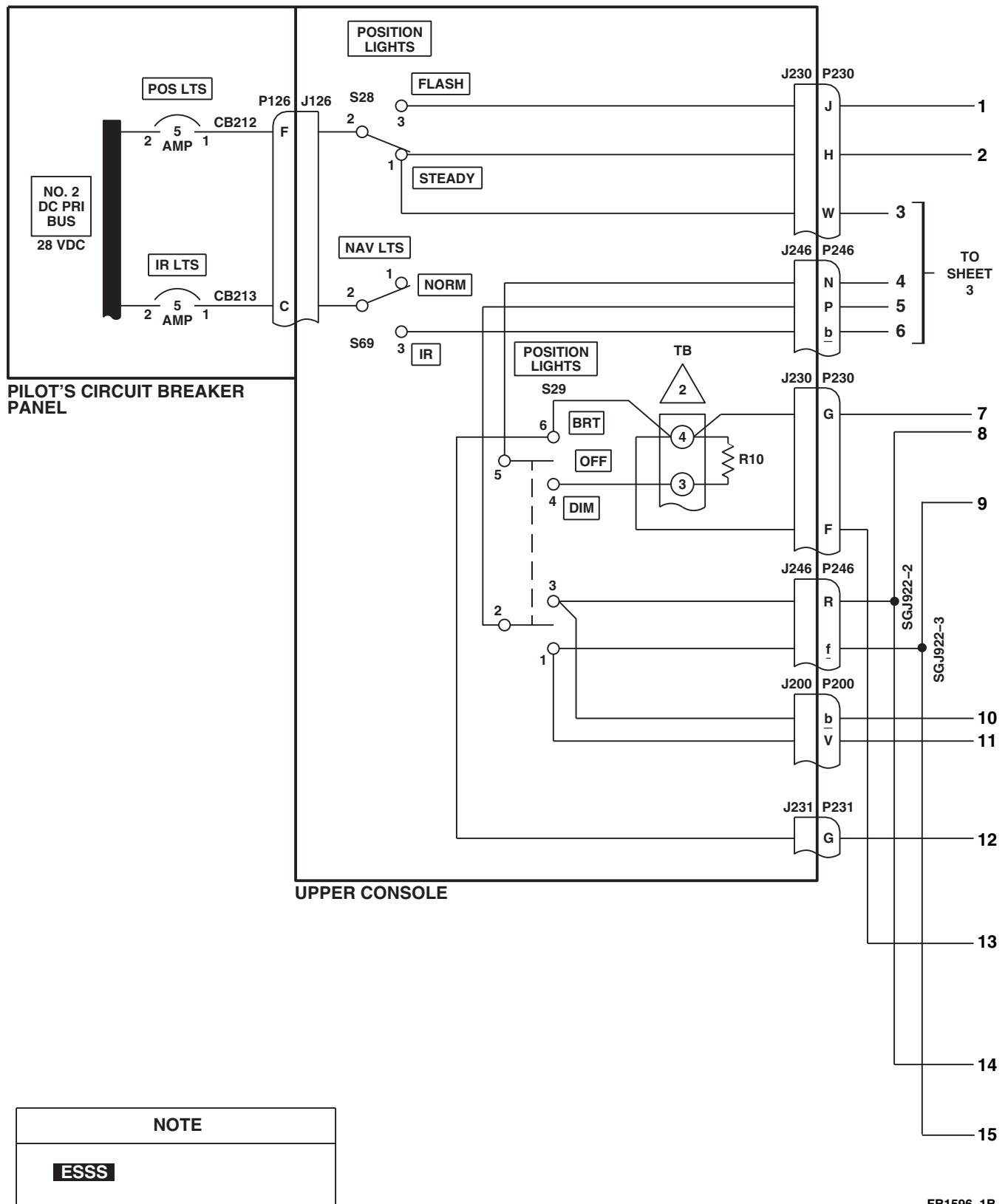


FIGURE 9-2-14-1. POSITION LIGHTS SCHEMATIC DIAGRAM (SHEET 1 OF 4)

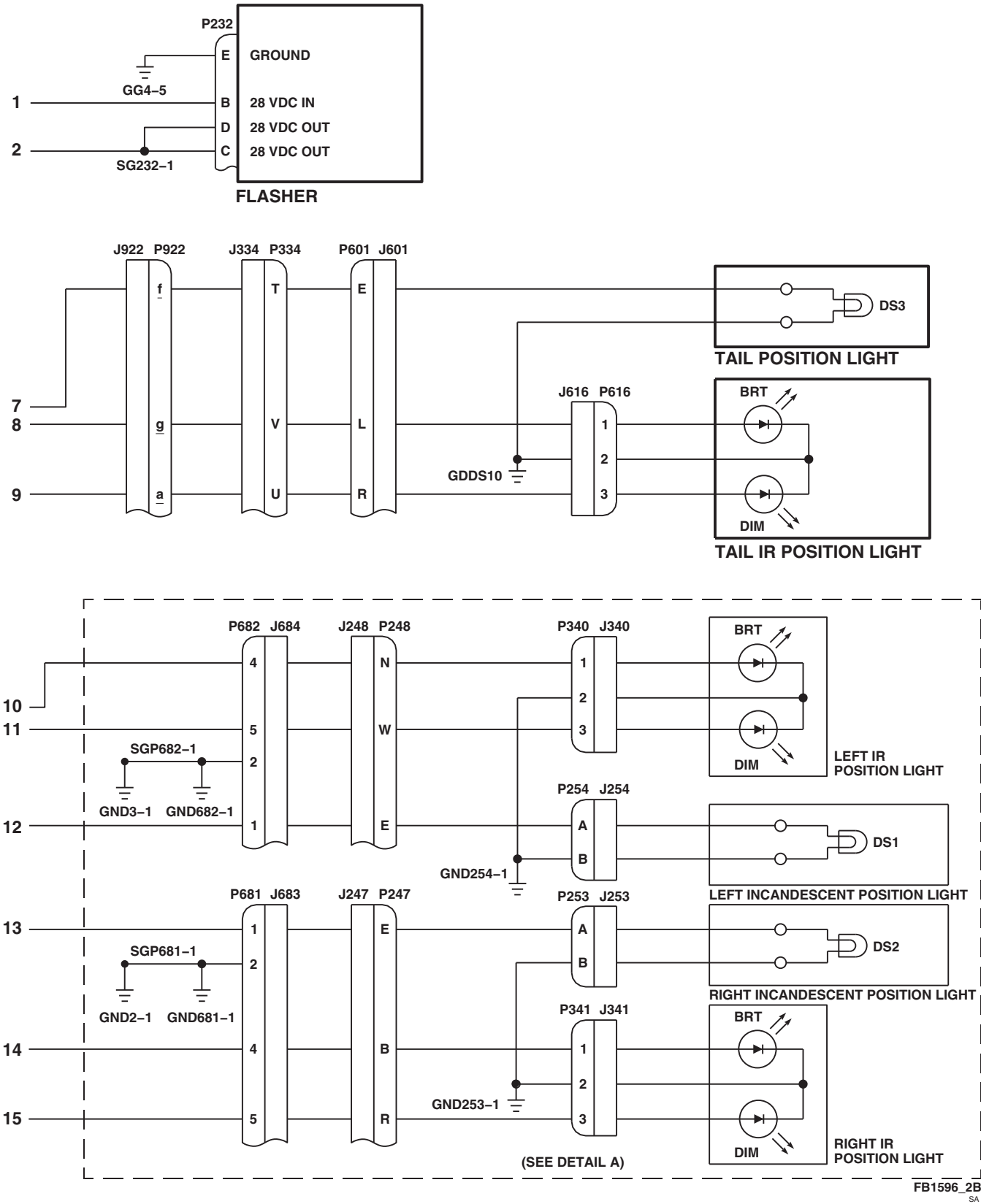


FIGURE 9-2-14-1. POSITION LIGHTS SCHEMATIC DIAGRAM (SHEET 2 OF 4)

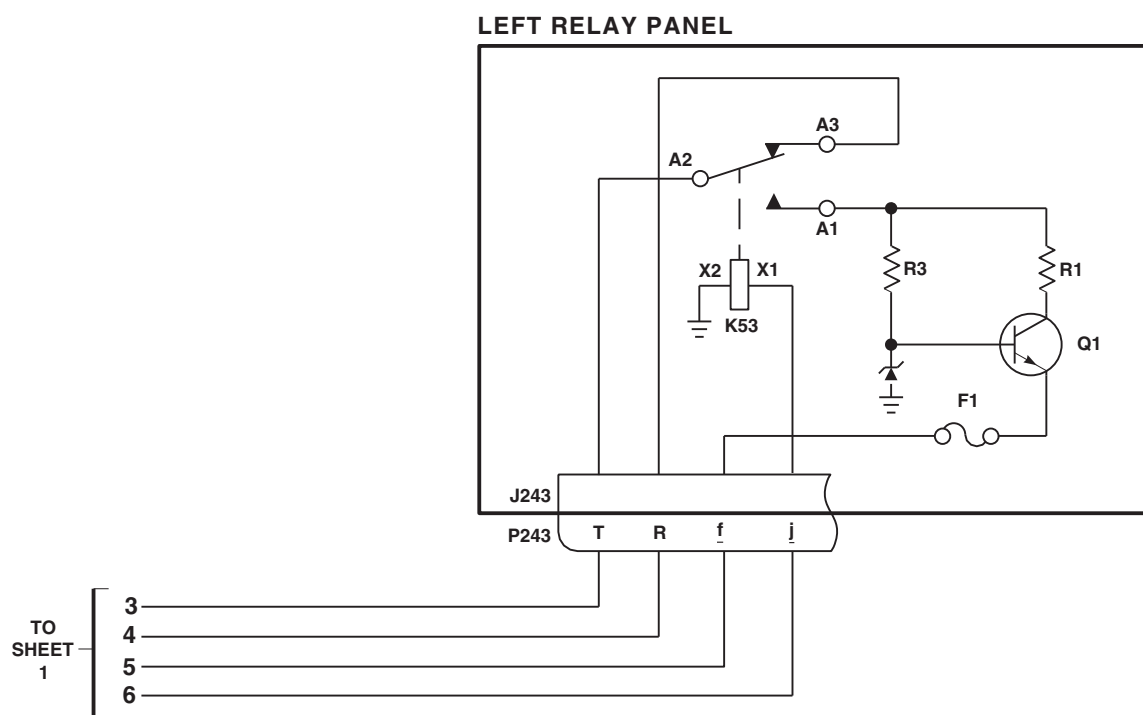
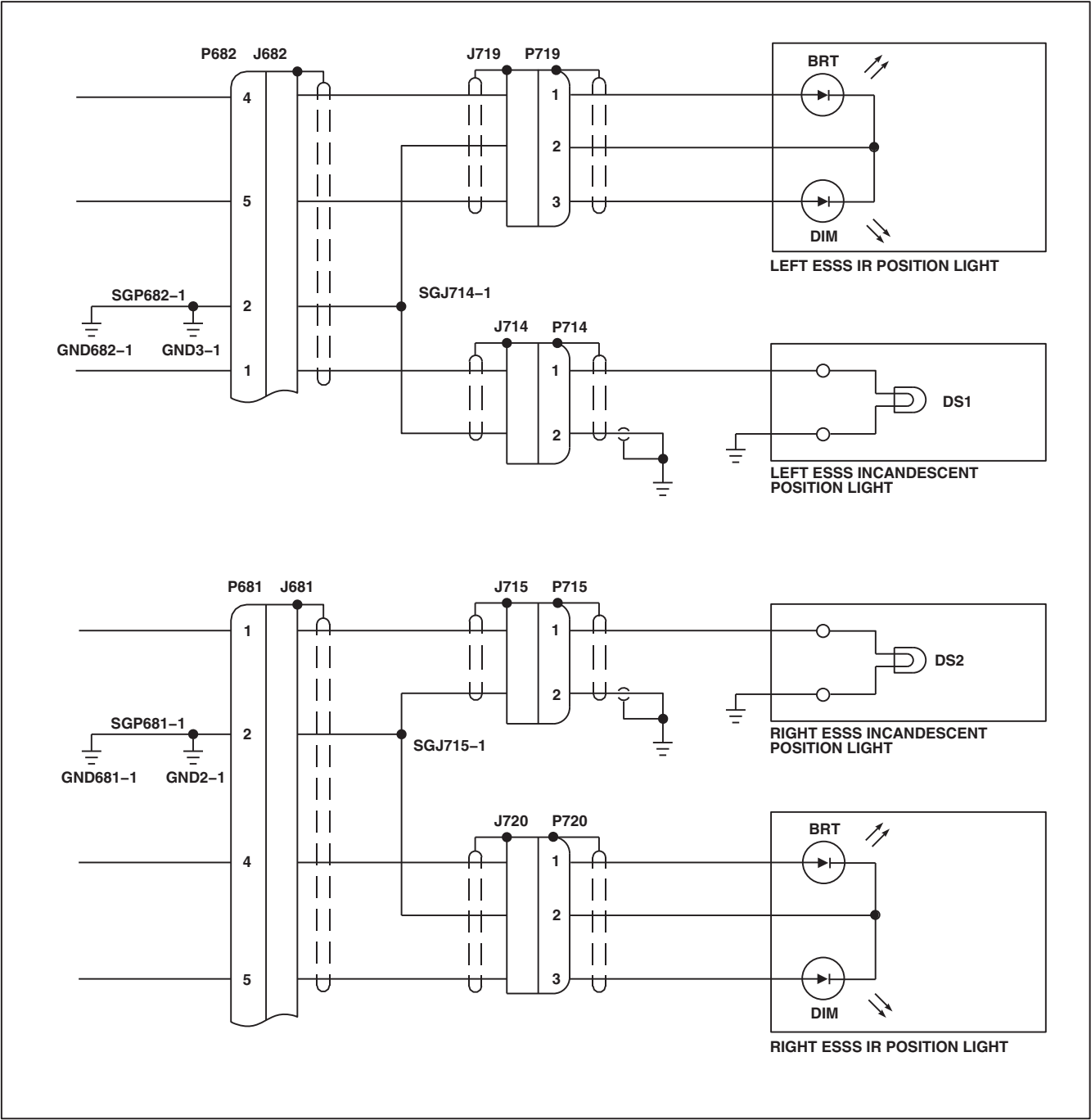
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FIGURE 9-2-14-1. POSITION LIGHTS SCHEMATIC DIAGRAM (SHEET 3 OF 4)



DETAIL A
 (SEE NOTE)

FIGURE 9-2-14-1. POSITION LIGHTS SCHEMATIC DIAGRAM (SHEET 4 OF 4)

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9-2-15 RETRACTABLE LANDING LIGHT.

INITIAL SETUP

Test Equipment

- Multimeter, Fluke 45

Tools

- Electrical Repairer Toolkit

References

- PARA 1-6-16

Equipment Conditions

- External Electrical Power Available or APU Operational

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).

9-2-15.1 Operational/Troubleshooting Procedure.

- Slide pilot's and copilot's collective stick grip LDG LT switches to center position.
- Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.
- Turn on electrical power (PARA 1-6-16).
- Slide pilot's collective stick grip LDG LT switch to EXT and hold.

RESULT: Landing light shall move out to full extension.

- If result is not as specified, go to TABLE NO. 9-2-15-1.
- If landing light moves but stops short of fully extended position, replace light assembly (PARA 9-4-90).
- e. Momentarily slide pilot's collective stick grip LDG LT switch to RETR and release.

RESULT: Landing light shall retract while switch is moved to RETR. Switch shall return to center when released and landing light shall stop.

- If result is not as specified, troubleshoot pilot's collective stick (PARA 11-2-2).

- If landing light does not move, go to TABLE NO. 9-2-15-2.

- Slide pilot's collective stick grip LDG LT switch to RETR and hold until landing light is fully retracted.

RESULT: Landing light shall retract.

- If result is not as specified, replace light assembly (PARA 9-4-90).
- g. Slide copilot's collective stick grip LDG LT switch to EXT and hold.

RESULT: Landing light shall move out to full extension.

- If landing light does not move, go to TABLE NO. 9-2-15-3.
- h. Momentarily slide copilot's collective stick grip LDG LT switch to RETR.

RESULT: Landing light shall retract while switch is held to RETR. Switch shall return to center when released and landing light shall stop.

- If result is not as specified, with copilot's LDG LT switch held to RETR, check continuity between P103-S and P103-U.

- (a) If continuity is present, repair/replace wiring between J103-S and J128-E (Appendix F).
- (b) If continuity is not present, troubleshoot copilot's collective stick (PARA 11-2-2).

i. Momentarily press pilot's collective stick grip LDG LT switch.

RESULT:

- (1) Landing light shall go on.
- (2) LDG LT ON caution/advisory capsule shall go on.
- If result (1) is not as specified, go to TABLE NO. 9-2-15-4.
- If result (2) is not as specified, check helicopter configuration.

EMEP go to 1, **W/O EMEP** go to 2. 

1 **EMEP** Remove pin filter adapter P118 (PARA 9-4-98) and check. 

- (a) If pin filter adapter is good, go to (c).
- (b) If pin filter adapter is not good, replace pin filter adapter (PARA 9-4-98).
- (c) Install pin filter adapter (PARA 9-4-98). Go to 2.

2 **W/O EMEP** Check for 28 VDC between P118-L and ground. 

(a) If voltage is as specified, troubleshoot Caution/Advisory Warning System (PARA 8-2-4).

(b) If voltage is not as specified, repair/replace wiring between P118-L and P900-p (Appendix F).

j. Momentarily press pilot's collective stick grip LDG LT switch.

RESULT: Landing light shall go off.

- If result is not as specified, replace right relay panel (PARA 9-4-2).

k. Momentarily press copilot's collective stick grip LDG LT switch.

RESULT: Landing light and LDG LT ON caution/advisory capsule shall go on.

- If result is not as specified, go to TABLE NO. 9-2-15-5.

l. Momentarily press copilot's collective stick grip LDG LT switch.

RESULT: Landing light shall go off.

- If result is not as specified, replace right relay panel (PARA 9-4-2).

m. Slide copilot's collective stick grip landing light switch to RETR until landing light is fully retracted.

n. Turn off electrical power (PARA 1-6-16).

TABLE NO. 9-2-15-1. Landing Light Does Not Move With Pilot's LDG LT Switch Moved To EXT.

TEST OR INSPECTION CORRECTIVE ACTION	
1. While holding pilot's LDG LT switch to EXT, check for 28 VDC between J128-F and J128-D.	
Step 1. If voltage is as specified, replace retractable landing light (PARA 9-4-90). Go to 6 .	
Step 2. If voltage is not as specified, go to 2 .	
2. While holding pilot's LDG LT switch to EXT, check for 28 VDC between J128-F and ground.	
Step 1. If voltage is as specified, repair/replace wiring between J128-D and ground (Appendix F). Go to 6 .	
Step 2. If voltage is not as specified, go to 3 .	
3. Check for 28 VDC between J104-U and ground.	
Step 1. If voltage is as specified, go to 4 .	
Step 2. If voltage is not as specified, go to 5 .	
4. With pilot's LDG LT switch held at EXT, check continuity between P104-U and P104-W.	
Step 1. If continuity is present, repair/replace wiring between J104-W and J128-F (Appendix F). Go to 6 .	
Step 2. If continuity is not present, troubleshoot pilot's collective stick (PARA 11-2-2). Go to 6 .	
5. Check for 28 VDC between terminal 1 (LIGHTS RETR LDG CONT circuit breaker, NO. 1 DC PRI BUS - copilot's circuit breaker panel) and ground.	
Step 1. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and J104-U (Appendix F). Go to 6 .	
Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 6 .	
6. Procedure completed.	

TABLE NO. 9-2-15-2. Landing Light Does Not Move With Pilot's LDG LT Switch Moved To RETR.

	TEST OR INSPECTION
	CORRECTIVE ACTION

- 1. While holding pilot's LDG LT switch to RETR, check for 28 VDC between J128-E and ground.**

Step 1. If voltage is as specified, replace retractable landing light (PARA 9-4-90).

Go to **3.**

Step 2. If voltage is not as specified, go to **2.**

- 2. With pilot's LDG LT switch held at RETR, check continuity between P104-U and P104-S.**

Step 1. If continuity is present, repair/replace wiring between J104-S and J128-E (Appendix F). Go to **3.**

Step 2. If continuity is not present, troubleshoot pilot's collective stick (PARA 11-2-2). Go to **3.**

- 3. Procedure completed.**
-

TABLE NO. 9-2-15-3. Landing Light Does Not Move With Copilot's LDG LT Switch Moved To EXT.

	TEST OR INSPECTION
	CORRECTIVE ACTION

- 1. Check for 28 VDC between J103-U and ground.**

Step 1. If voltage is as specified, go to **2.**

Step 2. If voltage is not as specified, repair/replace wiring between J103-U and terminal 1 (LIGHTS RETR LDG CONT circuit breaker, NO. 1 DC PRI BUS - copilot's circuit breaker panel) (Appendix F). Go to **3.**

- 2. With copilot's LDG LT switch held at EXT, check continuity between P103-U and P103-W.**

Step 1. If continuity is present, repair/replace wiring between J103-W and J128-F (Appendix F). Go to **3.**

Step 2. If continuity is not present, troubleshoot copilot's collective stick (PARA 11-2-2). Go to **3.**

- 3. Procedure completed.**
-

TABLE NO. 9-2-15-4. Landing Light And LDG LT ON Capsule Do Not Go On When Pilot's LDG LT Switch Is Pressed And Released.

TEST OR INSPECTION CORRECTIVE ACTION	
1. Check for 28 VDC between J128-A and ground.	
Step 1. If voltage is as specified, go to 2 .	
Step 2. If voltage is not as specified, go to 9 .	
2. With pilot's LDG LT switch pushed in and held, check for 28 VDC between J128-B and ground.	
Step 1. If voltage is as specified, go to 3 .	
Step 2. If voltage is not as specified, go to 4 .	
3. Check if lamp is good.	
Step 1. If lamp is good, replace landing light (PARA 9-4-90). Go to 10 .	
Step 2. If lamp is not good, replace lamp (PARA 9-4-89). Go to 10 .	
4. Check for 28 VDC between P900-n and ground.	
Step 1. If voltage is as specified, go to 5 .	
Step 2. If voltage is not as specified, go to 8 .	
5. With pilot's LDG LT switch pushed in and held, check for 28 VDC between P900-k and ground.	
Step 1. If voltage is as specified, replace right relay panel (PARA 9-4-2). Go to 10 .	
Step 2. If voltage is not as specified, go to 6 .	
6. Check continuity between J104-X and P900-k.	
Step 1. If continuity is present, go to 7 .	
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 10 .	
7. Check for 28 VDC between J104-Y and ground.	
Step 1. If voltage is as specified, troubleshoot pilot's collective stick (PARA 11-2-2). Go to 10 .	

TABLE NO. 9-2-15-4. Landing Light And LDG LT ON Capsule Do Not Go On When Pilot's LDG LT Switch Is Pressed And Released. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 2. If voltage is not as specified, repair/replace wiring between J104-Y and LIGHTS RETR LDG CONT circuit breaker terminal 1 (Appendix F). Go to 10.
8. Check continuity between P900-n and terminal 1 (LIGHTS RETR LDG CONT circuit breaker, NO. 1 DC PRI BUS - copilot's circuit breaker panel).	Step 1. If continuity is present, replace circuit breaker (PARA 9-4-1). Go to 10. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 10.
9. Check continuity between J128-A and terminal 1 (LIGHTS RETR LDG PWR circuit breaker, NO. 1 DC PRI BUS - copilot's circuit breaker panel).	Step 1. If continuity is present, replace circuit breaker (PARA 9-4-1). Go to 10. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 10.
10. Procedure completed.	

TABLE NO. 9-2-15-5. Landing Light And LDG LT ON Capsule Do Not Go On When Copilot's LDG LT Switch Is Pressed And Released.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for 28 VDC between J103-Y and ground.	Step 1. If voltage is as specified, go to 2. Step 2. If voltage is not as specified, repair/replace wiring between J103-Y and P248-D (Appendix F). Go to 3.
2. With copilot's LDG LT switch pushed in and held, check continuity between P103-X and P103-Y.	Step 1. If continuity is present, repair/replace wiring between J103-X and P900-k (Appendix F). Go to 3. Step 2. If continuity is not present, troubleshoot copilot's collective stick (PARA 11-2-2). Go to 3.

TABLE NO. 9-2-15-5. Landing Light And LDG LT ON Capsule Do Not Go On When Copilot’s LDG LT Switch Is Pressed And Released. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION
3. Procedure completed.

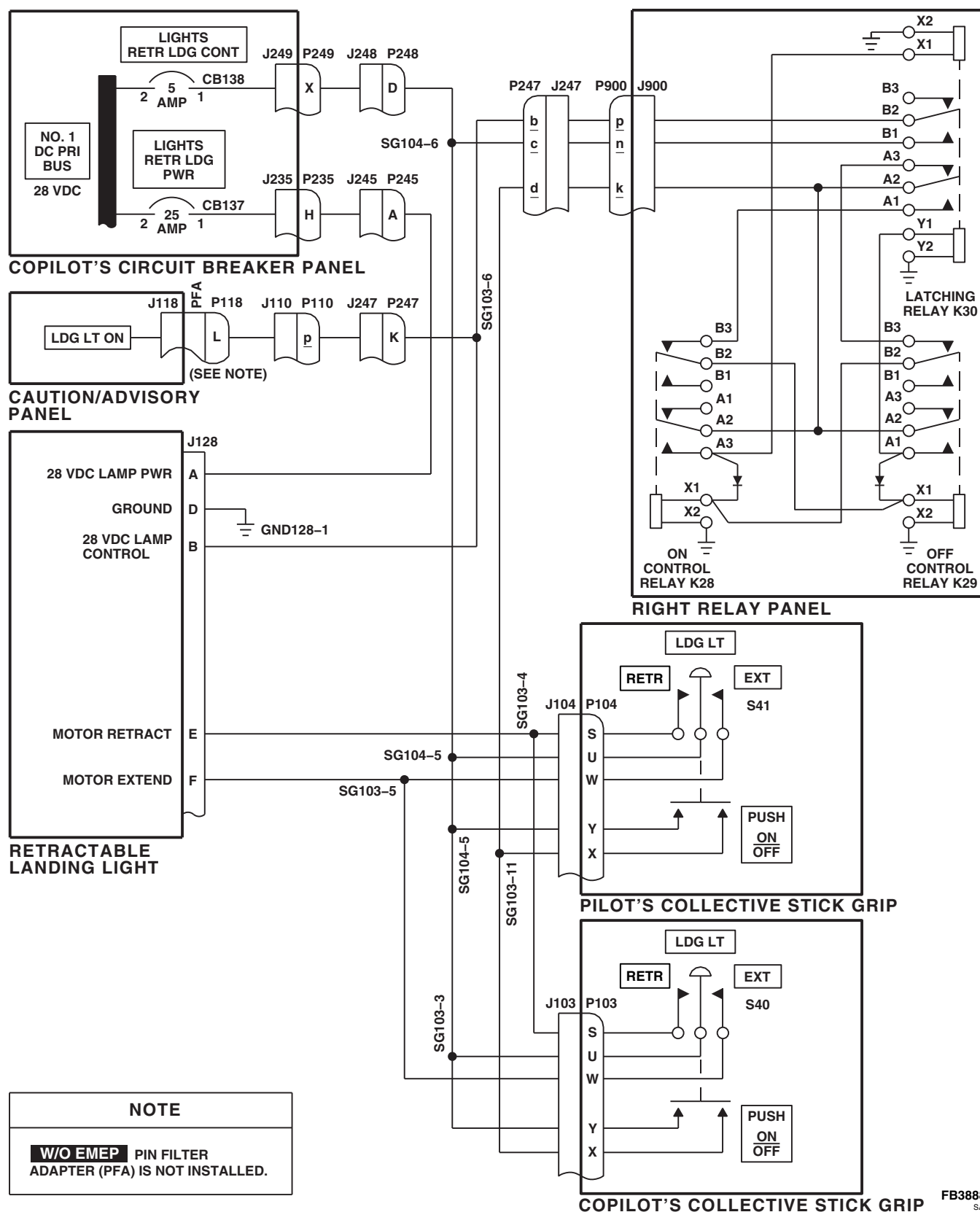


FIGURE 9-2-15-1. RETRACTABLE LANDING LIGHT SCHEMATIC DIAGRAM

9-2-16 ANTICOLLISION LIGHTS.

INITIAL SETUP

Test Equipment

- Multimeter, Fluke 45

Tools

- Electrical Repairer Toolkit

References

- PARA 1-6-16

Equipment Conditions

- External Electrical Power Available or APU Operational

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).

9-2-16.1 Operational/Troubleshooting Procedure.

- Check anticollision lights for cracked or broken lenses.

RESULT: Anticollision light lenses shall not be cracked or broken.

- If result is not as specified, replace lens (PARA 9-4-87).
- Make sure that all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.
 - Turn on electrical power (PARA 1-6-16).
 - Place upper console ANTI-COLLISION LIGHTS UPPER/BOTH/LOWER switch to BOTH.
 - Place upper console ANTI-COLLISION LIGHTS DAY/OFF/NIGHT switch to DAY.

RESULT: Upper and lower anticollision lights shall alternately flash white (clear).

- If result is not as specified, go to TABLE NO. 9-2-16-1.
- If one light does not flash white, go to TABLE NO. 9-2-16-2.

- If both day (white) and night (red) anticollision lights are flashing simultaneously, replace power supply (PARA 9-4-88).
- Place upper console ANTI-COLLISION LIGHTS DAY/OFF/NIGHT switch to NIGHT.

RESULT: Upper and lower anticollision lights shall alternately flash red.

- If result is not as specified, go to TABLE NO. 9-2-16-3.
 - If one light does not flash red, go to TABLE NO. 9-2-16-4.
 - If both day (white) and night (red) anticollision lights are flashing simultaneously, replace power supply (PARA 9-4-88).
- Place upper console ANTI-COLLISION LIGHTS UPPER/BOTH/LOWER switch to UPPER.

RESULT:

- Upper anticollision light shall flash red.
- Lower anticollision light shall be off.

- If result (1) is not as specified, go to TABLE NO. 9-2-16-5.

- If result (2) is not as specified, replace ANTI-COLLISION LIGHTS UPPER/BOTH/LOWER light switch S27 (PARA 9-4-5).
- h. Place upper console ANTI-COLLISION LIGHTS UPPER/BOTH/LOWER switch to LOWER.

RESULT:

- (1) Lower anticollision light shall flash red.
 - (2) Upper anticollision light shall be off.
- If result (1) is not as specified, check continuity between terminals 2 and 3 of ANTI-COLLISION LIGHTS UPPER/BOTH/LOWER switch.
 - (a) If continuity is present, repair/replace wiring to switch (Appendix F).

- (b) If continuity is not present, replace switch (PARA 9-4-5).

- If result (2) is not as specified, replace ANTI-COLLISION LIGHTS UPPER/BOTH/LOWER switch S27 (PARA 9-4-5).

- i. Place upper console ANTI-COLLISION UPPER/BOTH/LOWER switch to BOTH and DAY/OFF/NIGHT switch to OFF.

RESULT: Upper and lower anticollision lights shall go off.

- If result is not as specified, replace ANTI-COLLISION LIGHTS DAY/OFF/NIGHT switch S26 (PARA 9-4-5).

- j. Turn off electrical power (PARA 1-6-16).

TABLE NO. 9-2-16-1. Neither Anticollision Light Flashes White (Clear).

TEST OR INSPECTION	CORRECTIVE ACTION
1. Place ANTI-COLLISION LIGHTS DAY/OFF/NIGHT switch to NIGHT.	Go to 2.
2. Check that upper and lower anticollision lights alternately flash red.	Step 1. If lights alternately flash red, go to 3. Step 2. If lights do not alternately flash red, go to 4.
3. Check continuity between S26-1 and ground.	Step 1. If continuity is present, replace switch S26 (PARA 9-4-5). Go to 9. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 9.
4. Check for 115 VAC between P503-A and P503-C.	Step 1. If voltage is as specified, go to 5. Step 2. If voltage is not as specified, go to 7.

TABLE NO. 9-2-16-1. Neither Anticollision Light Flashes White (Clear). (Cont)

TEST OR INSPECTION CORRECTIVE ACTION
5. Check continuity between P503-E and P503-B. Step 1. If continuity is present, replace power supply (PARA 9-4-88). Go to 9. Step 2. If continuity is not present, go to 6. 6. Check continuity between P503-E and S26-2. Step 1. If continuity is present, replace switch S26 (PARA 9-4-5). Go to 9. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 9. 7. Check continuity between P503-C and ground. Step 1. If continuity is present, go to 8. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 9. 8. Check for 115 VAC between terminal 1 (LIGHTS ANTI COLL circuit breaker, NO. AC PRI BUS - pilot's circuit breaker panel) and ground. Step 1. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P503-A (Appendix F). Go to 9. Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 9. 9. Procedure completed.

WARNING

Hazardous voltage exists on power supply connectors even with power removed. Use extreme care when disconnecting P502 and P504.

TABLE NO. 9-2-16-2. One Anticollision Light Does Not Flash White (Clear) With ANTI-COLLISION LIGHTS Switch Set To BOTH.

TEST OR INSPECTION CORRECTIVE ACTION
1. Is upper anticollision light operating? Step 1. If light is operating, go to 2. Step 2. If light is not operating, go to 9. 2. Check continuity between P503-B and P503-D. Step 1. If continuity is present, replace switch S27 (PARA 9-4-5). Go to 16. Step 2. If continuity is not present, go to 3. 3. Swap P502 and P504. Go to 4. 4. Is lower anticollision light operating? Step 1. If light is operating, go to 5. Step 2. If light is not operating, go to 7. 5. Swap P502 and P504 to original connections. Go to 6. 6. Check continuity between: P502-A and P505-A P502-B and P505-B P502-C and P505-C P502-E and P505-E Step 1. If continuity is present, replace power supply (PARA 9-4-88). Go to 16. Step 2. If continuity is not present, repair/replace wiring (Appendix F) as required. Go to 16. 7. Swap P502 and P504 to original connections. Go to 8. 8. Replace lower anticollision light lamp (PARA 9-4-87). Go to 16.

TABLE NO. 9-2-16-2. One Anticollision Light Does Not Flash White (Clear) With ANTI-COLLISION LIGHTS Switch Set To BOTH. (Cont)

TEST OR INSPECTION CORRECTIVE ACTION
9. Check continuity between P503-D and ground.
Step 1. If continuity is present, replace switch S27 (PARA 9-4-5). Go to 16. Step 2. If continuity is not present, go to 10.
10. Swap P502 and P504. Go to 11.
11. Is upper anticollision light operating?
Step 1. If light is operating, go to 12. Step 2. If light is not operating, go to 14.
12. Swap P502 and P504 to original connections. Go to 13.
13. Check continuity between: P504-A and P602-A P504-B and P602-B P504-C and P602-C P504-E and P602-E
Step 1. If continuity is present, replace power supply (PARA 9-4-88). Go to 16. Step 2. If continuity is not present, repair/replace wiring (Appendix F) as required. Go to 16.
14. Swap P502 and P504 to original connections. Go to 15.
15. Replace upper anticollision light lamp (PARA 9-4-87). Go to 16.
16. Procedure completed.

TABLE NO. 9-2-16-3. Neither Anticollision Light Flashes Red.

TEST OR INSPECTION	CORRECTIVE ACTION
--------------------	-------------------

1. Check continuity between P503-B and P503-E.	Step 1. If continuity is present, replace power supply (PARA 9-4-88). Go to 4. Step 2. If continuity is not present, go to 2.
--	---

2. Check continuity between P503-B and S27-1.	Step 1. If continuity is present, go to 3. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 4.
---	---

3. Check continuity between S27-1 and S26-3.	Step 1. If continuity is present, replace switch S26 (PARA 9-4-5). Go to 4. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 4.
--	--

4. Procedure completed.	
-------------------------	--

WARNING

Hazardous voltages exist on power supply connectors even with power removed. Use extreme care when disconnecting P502 and P504.

TABLE NO. 9-2-16-4. One Anticollision Light Does Not Flash Red With Switch Set To BOTH.

TEST OR INSPECTION	CORRECTIVE ACTION
--------------------	-------------------

1. Is upper anticollision light operating?	Step 1. If light is operating, go to 2. Step 2. If light is not operating, go to 8.
--	---

2. Swap P502 and P504. Go to 3.	
---------------------------------	--

3. Is lower anticollision light operating?	
--	--

TABLE NO. 9-2-16-4. One Anticollision Light Does Not Flash Red With Switch Set To BOTH. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
Step 1. If light is operating, go to 4. Step 2. If light is not operating, go to 6.	4. Swap P502 and P504 to original connections. Go to 5.
5. Check continuity between P502-D and P505-D.	Step 1. If continuity is present, replace power supply (PARA 9-4-88). Go to 14. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 14.
6. Swap P502 and P504 to original connections. Go to 7.	7. Replace lower anticollision light lamp (PARA 9-4-87). Go to 14.
8. Swap P502 and P504. Go to 9.	9. Is upper anticollision light operating?
Step 1. If light is operating, go to 10. Step 2. If light is not operating, go to 12.	10. Swap P502 and P504 to original connections. Go to 11.
11. Check continuity between P504-D and P602-D.	Step 1. If continuity is present, replace power supply (PARA 9-4-88). Go to 14. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 14.
12. Swap P502 and P504 to original connections. Go to 13.	13. Replace upper anticollision light lamp (PARA 9-4-87). Go to 14.
14. Procedure completed.	

**TABLE NO. 9-2-16-5. Upper Anticollision Light Does Not Flash With ANTI-COLLISION LIGHTS
Switch Set To UPPER.**

TEST OR INSPECTION	CORRECTIVE ACTION
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1. Check continuity between P503-B and P503-D.

Step 1. If continuity is present, replace power supply (PARA 9-4-88). Go to **3.**

Step 2. If continuity is not present, go to **2.**

2. Check continuity between P503-D and S27-2.

Step 1. If continuity is present, replace switch S27 (PARA 9-4-5). Go to **3.**

Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **3.**

3. Procedure completed.

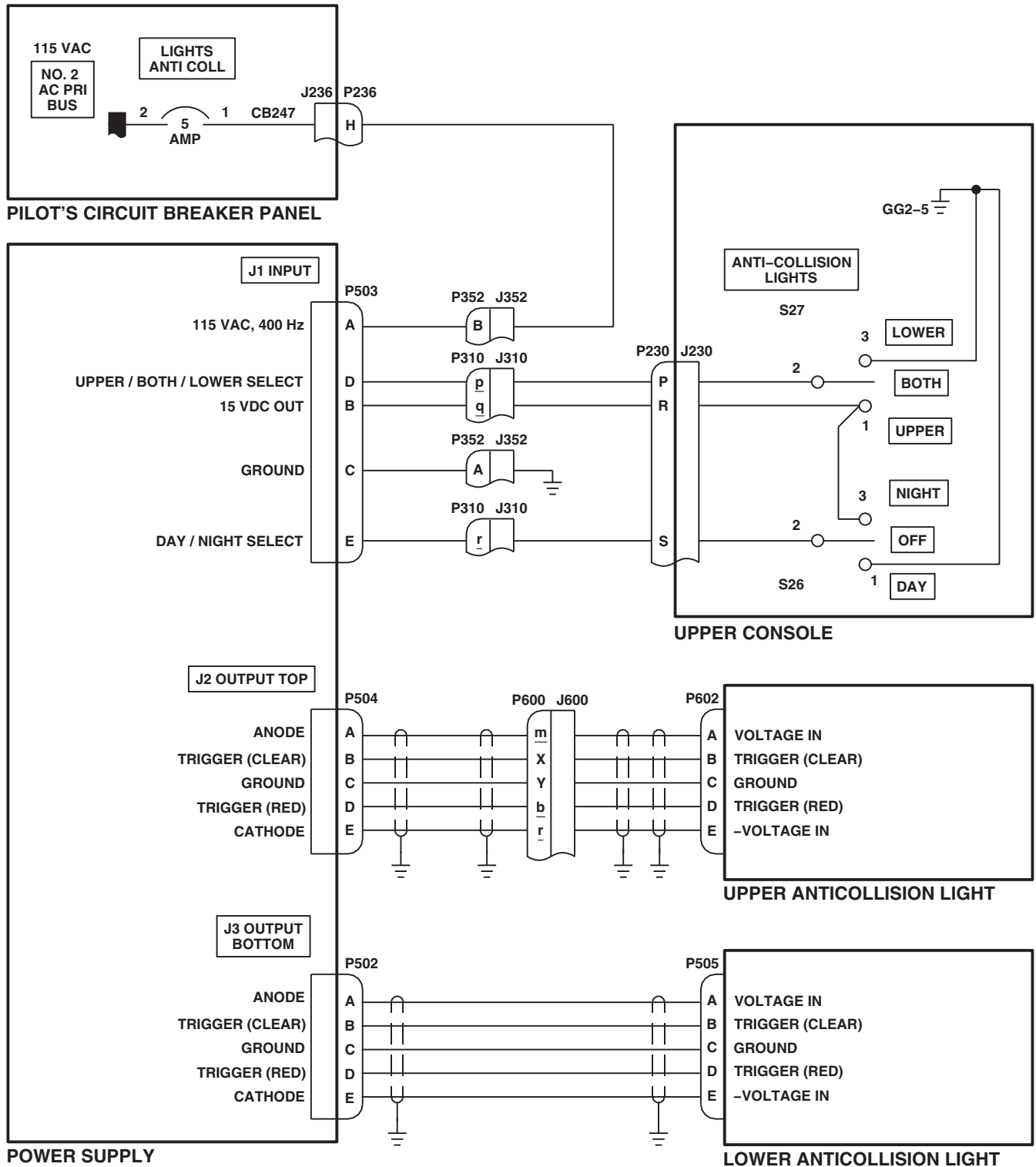


FIGURE 9-2-16-1. ANTICOLLISION LIGHTS SCHEMATIC DIAGRAM

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9-2-17 CARGO HOOK LIGHTS **W/O SLS** .

INITIAL SETUP

Test Equipment

- Multimeter, Fluke 45

Tools

- Electrical Repairer Toolkit

References

- PARA 1-6-16

Equipment Conditions

- External Electrical Power Available or APU Operational

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).

9-2-17.1 Operational/Troubleshooting Procedure.

- Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.
- Turn on electrical power (PARA 1-6-16).
- Place upper console CARGO HOOK LT switch to ON.

RESULT: Cargo hook lights shall go on.

- If result is not as specified, go to TABLE NO. 9-2-17-1.
- If one cargo hook light does not go on, check for 115 VAC between the following:

J389-1 and ground

J390-1	and	ground
J391-1	and	ground

- If voltage is not as specified, repair/replace wiring as required (Appendix F).
- If voltage is as specified, replace light assembly (PARA 9-4-93).

- Place upper console CARGO HOOK LT switch to OFF.

RESULT: Cargo hook lights shall go off.

- If result is not as specified, replace CARGO HOOK LT switch (PARA 9-4-5).
- Turn off electrical power (PARA 1-6-16).

TABLE NO. 9-2-17-1. No Cargo Hook Lights Go On.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for 115 VAC between CARGO HOOK LT switch terminal 2 and ground.	Step 1. If voltage is as specified, go to 2. Step 2. If voltage is not as specified, go to 6.
2. Check for 115 VAC between CARGO HOOK LT switch terminal 3 and ground.	Step 1. If voltage is as specified, go to 3. Step 2. If voltage is not as specified, replace switch (PARA 9-4-5). Go to 7.
3. Check for 115 VAC between J392-1 and J392-2.	Step 1. If voltage is as specified, go to 4. Step 2. If voltage is not as specified, go to 5.
4. Check continuity between:	P392-1 and J389-1 P392-1 and J390-1 P392-1 and J391-1
	Step 1. If continuity is present, repair/replace wiring, as required, between the following (Appendix F), then go to 7: P392-2 and J389-2 P392-2 and J390-2 P392-2 and J391-2 Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 7.
5. Check for 115 VAC between J392-1 and ground.	Step 1. If voltage is as specified, repair/replace wiring between J392-2 and ground (Appendix F). Go to 7. Step 2. If voltage is not as specified, repair/replace wiring between J392-1 and CARGO HOOK LT switch terminal 3 (Appendix F). Go to 7.

1. Check for 115 VAC between CARGO HOOK LT switch terminal 2 and ground.

Step 1. If voltage is as specified, go to 2.

Step 2. If voltage is not as specified, go to 6.

2. Check for 115 VAC between CARGO HOOK LT switch terminal 3 and ground.

Step 1. If voltage is as specified, go to 3.

Step 2. If voltage is not as specified, replace switch (PARA 9-4-5). Go to 7.

3. Check for 115 VAC between J392-1 and J392-2.

Step 1. If voltage is as specified, go to 4.

Step 2. If voltage is not as specified, go to 5.

4. Check continuity between:

P392-1 and J389-1

P392-1 and J390-1

P392-1 and J391-1

Step 1. If continuity is present, repair/replace wiring, as required, between the following (Appendix F), then go to 7:

P392-2 and J389-2

P392-2 and J390-2

P392-2 and J391-2

Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 7.

5. Check for 115 VAC between J392-1 and ground.

Step 1. If voltage is as specified, repair/replace wiring between J392-2 and ground (Appendix F). Go to 7.

Step 2. If voltage is not as specified, repair/replace wiring between J392-1 and CARGO HOOK LT switch terminal 3 (Appendix F). Go to 7.

TABLE NO. 9-2-17-1. No Cargo Hook Lights Go On. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
6. Check for 115 VAC between terminal 1 (LIGHTS CARGO HOOK circuit breaker, NO. 2 AC PRI BUS - pilot's circuit breaker panel) and ground.	Step 1. If voltage is as specified, replace circuit breaker (PARA 9-4-1). Go to 7. Step 2. If voltage is not as specified, repair/replace wiring between circuit breaker terminal 1 and CARGO HOOK LT switch terminal 2 (Appendix F). Go to 7.
7. Procedure completed.	

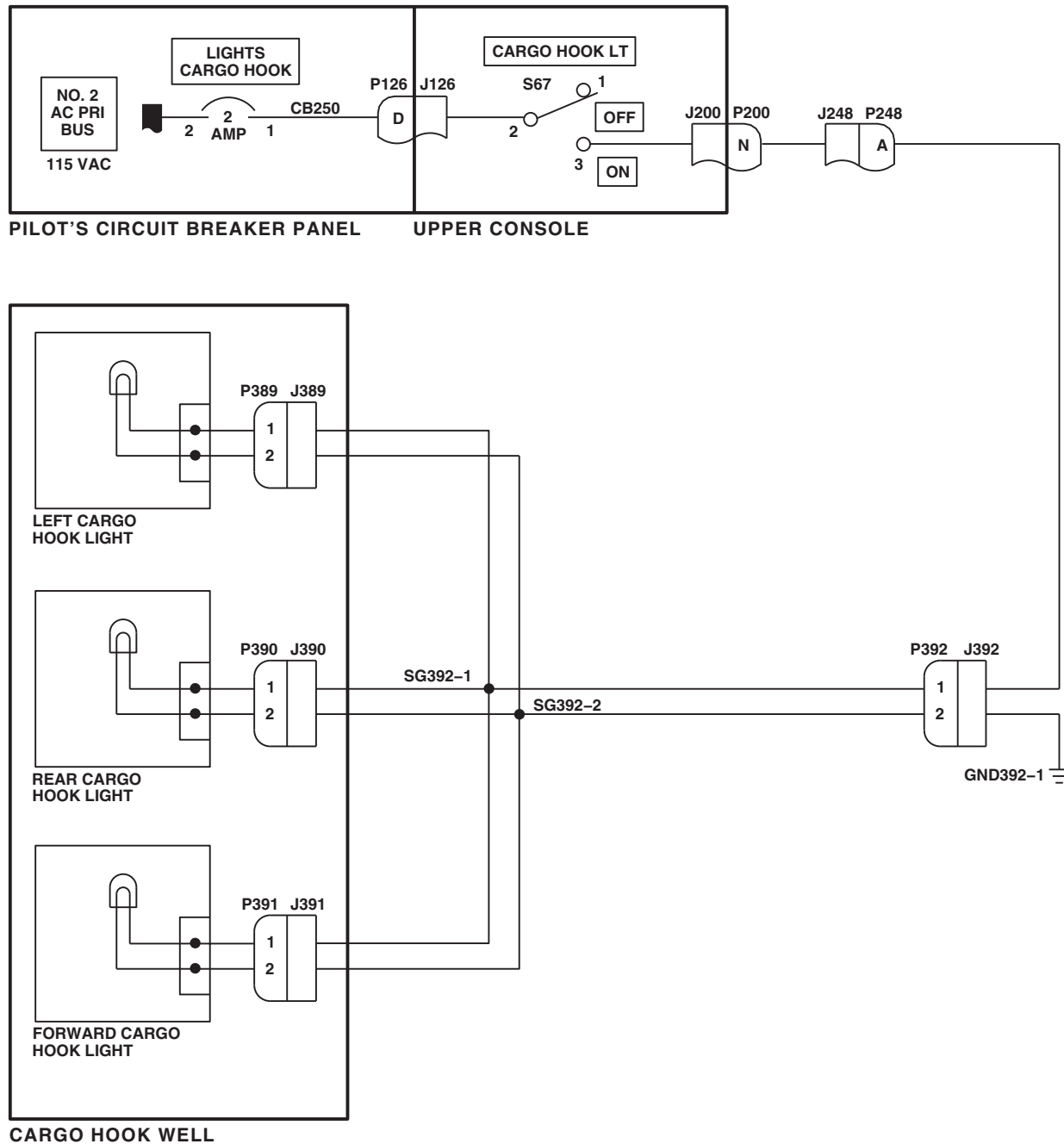
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FIGURE 9-2-17-1. CARGO HOOK LIGHTS SCHEMATIC DIAGRAM

9-2-17-4 Change 7

9-2-18 FUEL BOOST PUMP CONTROL PANEL (BENCH).

INITIAL SETUP

Test Equipment

- Multimeter, Fluke 45

Tools

- Electrical Repairer Toolkit

- Locally-Made Fuel Boost Pump Control Panel Test Harness, Figure H-153, Appendix H

Equipment Conditions

- Between 27 and 28 VDC, and 115 VAC Bench Power Available

9-2-18.1 Operational/Troubleshooting Procedure.

- Remove cover from control panel and inspect for broken or burnt wires, security of components, and general condition.

RESULT: Wires shall not be broken or burnt, and components shall be secure and in good condition.

- If result is not as specified, repair/replace as required.
- Connect locally-made test harness P1 to fuel boost pump control panel J1.
 - With control panel NO. 1 PUMP switch placed ON, check continuity between the following jacks on test harness:

T	and	J
U	and	H
V	and	G

RESULT: Meter shall indicate continuity.

- If result is not as specified, check continuity between the following NO. 1 PUMP switch terminals:

4	and	5
7	and	8

10	and	11
----	-----	----

- If continuity is present, repair/replace wiring.
- If continuity is not present, replace NO. 1 PUMP switch (PARA 9-5-17).

- Place NO. 1 PUMP switch to OFF.
- Check continuity between the following jacks on test harness:

T	and	J
U	and	H
V	and	G

RESULT: Meter shall indicate no continuity.

- If result is not as specified, check continuity between the following NO. 1 PUMP switch terminals:

4	and	5
7	and	8
10	and	11

- If continuity is present, replace NO. 1 PUMP switch (PARA 9-5-17).

- (b) If continuity is not present, repair/replace wiring.

- f. With control panel NO. 2 PUMP switch placed ON, check continuity between the following jacks on test harness:

A	and	F
B	and	E
C	and	D

RESULT: Meter shall indicate continuity.

- If result is not as specified, check continuity between the following NO. 2 PUMP switch terminals:

1	and	2
4	and	5
7	and	8

- (a) If continuity is present, repair/replace wiring.
- (b) If continuity is not present, replace NO. 2 PUMP switch (PARA 9-5-17).

- g. Place NO. 2 PUMP switch to OFF.

- h. Check continuity between the following jacks on test harness:

A	and	F
B	and	E
C	and	D

RESULT: Meter shall indicate no continuity.

- If result is not as specified, check continuity between the following NO. 2 PUMP switch terminals:

1	and	2
4	and	5
7	and	8

- (a) If continuity is present, replace NO. 2 PUMP switch (PARA 9-5-17).
- (b) If continuity is not present, repair/replace wiring.

- i. Disconnect meter.



To prevent possible damage to fuel boost pump control panel, verify if fuel boost pump control panel information plate uses 5 VAC or 115 VAC.

- j. Apply 5 VAC or 115 VAC across jacks L and Y on test harness.

RESULT: Information plate legend shall light evenly.

- If result is not as specified, check information plate lamps and replace as required (PARA 9-5-19).

- (a) If trouble remains, check continuity between J1-L and J2-1, and J1-Y and J2-2.

1 If continuity is present, replace information plate (PARA 9-4-41).

2 If continuity is not present, repair/replace wiring.

- k. Disconnect 5 VAC or 115 VAC from test harness.

- l. Apply 28 VDC between the following jacks on test harness:

P and Z
R and Z

- m. On control panel, momentarily press NO. 1 PUMP press-to-test indicator.

RESULT: NO. 1 PUMP indicator shall light when pressed.

- If result is not as specified, check indicator lamp and replace if burnt-out (PARA 9-5-18).

- (a) If trouble remains, check continuity between J1-R and NO. 1 PUMP indicator terminal 1, and J1-Z and NO. 1 PUMP indicator terminal 3.

1 If continuity is present, replace NO. 1 PUMP indicator (PARA 9-5-18).

2 If continuity is not present, repair/replace wiring.

- n. On control panel, momentarily press NO. 2 PUMP press-to-test indicator.

RESULT: NO. 2 PUMP indicator shall light when pressed.

- If result is not as specified, check indicator lamp and replace if burnt-out (PARA 9-5-18).

- (a) If trouble remains, check continuity between J1-P and NO. 2 PUMP indicator terminal 1, and J1-Z and NO. 2 PUMP indicator terminal 3.

1 If continuity is present, replace NO. 2 PUMP indicator (PARA 9-5-18).

2 If continuity is not present, repair/replace wiring.

- o. Connect negative lead of 28 VDC power source to M test jack.

RESULT: NO. 1 PUMP indicator shall light.

- If result is not as specified, check continuity between J1-M and NO. 1 PUMP indicator terminal 2.

(a) If continuity is present, replace NO. 1 PUMP indicator (PARA 9-5-18).

(b) If continuity is not present, repair/replace wiring.

- p. Connect negative lead of 28 VDC power source to N test jack.

RESULT: NO. 2 PUMP indicator shall light.

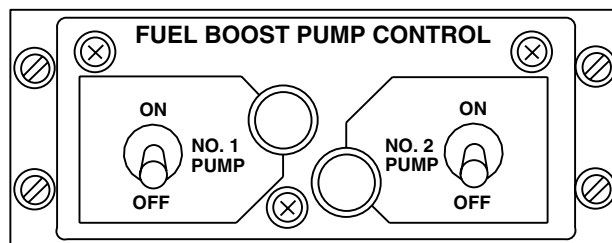
- If result is not as specified, check continuity between J1-N and NO. 2 PUMP indicator terminal 2.

(a) If continuity is present, replace NO. 2 PUMP indicator (PARA 9-5-18).

(b) If continuity is not present, repair/replace wiring.

- q. Disconnect 28 VDC from test harness.

- r. Disconnect test harness from control panel.



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FIGURE 9-2-18-1. FUEL BOOST PUMP CONTROL PANEL

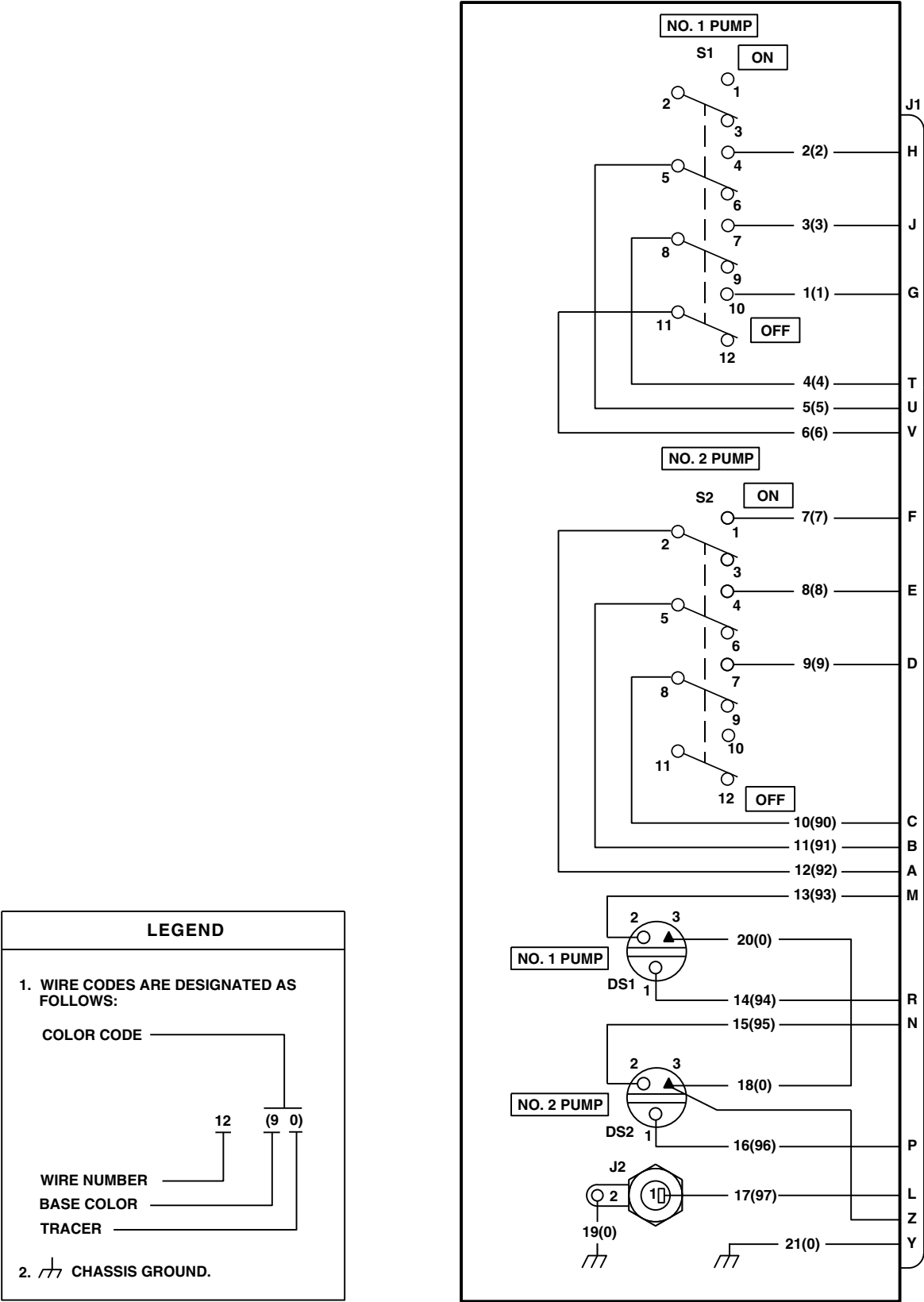


FIGURE 9-2-18-2. FUEL BOOST PUMP CONTROL PANEL SCHEMATIC DIAGRAM

9-2-19 RIGHT RELAY PANEL (BENCH).

INITIAL SETUP

Tools

- Decade Resistor, ZM57U
- Electrical Repairer Toolkit
- Locally-Made Test Lamp, Figure H-150, Appendix H
- Multimeter, Fluke 27/AN

Equipment Conditions

- Bench Power Available, 28 VDC and 115 VAC, 400 Hz

Special Instructions

The right relay panel operational/troubleshooting procedure is subdivided as follows:

- (1) T1 Current Sensor
- (2) K20 DC Essential Bus Fail Relay
- (3) K21 Windshield Anti-ice Lockout Relay
- (4) K26 No. 2 Engine Start Relay
- (5) K32 Hydraulic Accumulator Relay
- (6) K22 APU Surge Relay
- (7) Shutdown

9-2-19.1 T1 Current Sensor.

NOTE

See Figure 9-2-19-2 for test setup.

- a. Set decade resistors R1 and R2 to 200 ohms (Ω).
- b. Apply 115 VAC 400 Hz to J241-m.
- c. Apply 28 VDC to J241-d.
- d. Decrease resistance of R1 and R2 until each ammeter reads a minimum of 0.83 amps. Record resistance.

RESULT: Test lamp shall go off.

- If result is not as specified, go to TABLE NO. 9-2-19-1.
- e. Increase resistance of R1 while holding R2 constant until R1 ammeter reads less than 0.67 amps.

RESULT: Test lamp shall go on.

- If result is not as specified, go to TABLE NO. 9-2-19-2.
- f. Decrease resistance of R1 until it returns to original resistance measured in step d.
- g. Increase resistance of R2 while holding R1 constant until R2 ammeter reads less than 0.67 amps.

RESULT: Test lamp shall go on.

- If result is not as specified, go to TABLE NO. 9-2-19-2.
- h. If no other test is to be done, go to Shutdown.

9-2-19.2 K20 DC Essential Bus Fail Relay.

- a. Check continuity between J900-G and J900-H.

RESULT: Continuity shall be present.

- If result is not as specified, repair/replace wiring.

- (a) If trouble remains, replace K20 relay (PARA 9-5-8).

- b. Connect 28 VDC between J900-J and J900-M.
- c. Check for no continuity between J900-G and J900-H.

RESULT: No continuity shall be present.

- If result is not as specified, repair/replace wiring.

- (a) If trouble remains, replace K20 relay (PARA 9-5-8).

- d. If no other check is to be done, go to Shutdown.

9-2-19.3 K21 Windshield Anti-ice Lockout Relay.

- a. Check continuity between:
J244-F and J244-G
J900-E and J900-F
J900-C and J900-D

RESULT: Continuity shall be present.

- If continuity is not present, go to TABLE NO. 9-2-19-3.

- b. For right relay panels 70550-02102-049, -051, and -056 only, check continuity between J900-C and J900-D.

RESULT: Continuity shall be present.

- If results are not as specified, go to TABLE NO. 9-2-19-3.

- c. Connect 28 VDC between J244-E and J900-M.

- d. Check for no continuity between:
J244-F and J244-G
J900-E and J900-F
J900-C and J900-D

RESULT: No continuity shall be present.

- If results are not as specified, go to TABLE NO. 9-2-19-3.

- e. For right relay panels 70550-02102-049, -051, and -056, check for open circuit between J900-C and J900-D.

RESULT: No continuity shall be present.

- If results are not as specified, go to TABLE NO. 9-2-19-3.

- f. If no other check is to be done, go to Shutdown.

9-2-19.4 K26 No. 2 Engine Start Relay.

- a. Check continuity between the following:
J241-F and J244-J
J241-G and J244-K
J241-X and J241-Y
J900-L and J900-K

RESULT: Continuity shall be present.

- If results are not as specified, go to TABLE NO. 9-2-19-4.

- b. Connect 28 VDC between J241-R and J900-M.

RESULT: 28 VDC shall be measured between J241-U and J900-M.

- If result is not as specified, go to TABLE NO. 9-2-19-4.

- c. Check continuity between:
J241-a and J241-Z
J241-a and J241-b

RESULT: Continuity shall be present.

- If results are not as specified, go to TABLE NO. 9-2-19-4.

- d. Check for no continuity between:
J241-F and J244-J

J241-G and J244-K
 J241-X and J241-Y
 J900-L and J900-K

RESULT: No continuity shall be present.

- If results are not as specified, go to TABLE NO. 9-2-19-4.
- e. If no other check is to be done, go to Shutdown.

9-2-19.5 K32 Hydraulic Accumulator Relay.

- a. Connect 28 VDC between J241-p and J900-M.
- b. Check continuity between J244-A and J244-C.

RESULT: Continuity shall be present.

- If result is not as specified, go to TABLE NO. 9-2-19-5.
- c. Connect 28 VDC between J244-a and J900-M.
- d. Disconnect 28 VDC from J241-p.
- e. Check continuity between J244-A and J244-C.

RESULT: Continuity shall be present.

- If result is not as specified, go to TABLE NO. 9-2-19-5.
- f. Disconnect 28 VDC from J244-a.
- g. Check for no continuity between J244-A and J244-C.

RESULT: No continuity shall be present.

- If result is not as specified, go to TABLE NO. 9-2-19-5.
- h. If no other check is to be done, go to Shutdown.

9-2-19.6 K22 APU Surge Relay.

- a. Connect 28 VDC between J244-T and J244-U.
- b. Check resistance between J244-M and J244-S.

RESULT: Resistance between J244-M and J244-S shall be 10X greater in one direction than in the other direction.

- If result is not as specified, go to TABLE NO. 9-2-19-6.
- c. Disconnect 28 VDC from between J244-T and J244-U.
- d. Check for no continuity between J244-M and J244-S.

RESULT: No continuity shall be present.

- If result is not as specified, go to TABLE NO. 9-2-19-6.
- e. If no other check is to be done, go to Shutdown.

9-2-19.7 Shutdown.

- a. Turn off and disconnect electrical power.

TABLE NO. 9-2-19-1. Test Lamp Stays On.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check electrical connections.	
Step 1. If electrical connections are good, go to 2.	

TABLE NO. 9-2-19-1. Test Lamp Stays On. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 2. If electrical connections are not good, reconnect test equipment and repeat operational test. Go to 3. 2. Check continuity between: J241-c and T1-X2 J241-d and T1-A1 J241-f and T1-A2 J241-m and T1-X1 J241-r and T1-X4 T1-G and ground Step 1. If continuity is present, replace T1 (PARA 9-5-7). Go to 3. Step 2. If continuity is not present, repair/replace wiring. Go to 3. 3. Procedure completed.

TABLE NO. 9-2-19-2. Test Lamp Stays Off.

TEST OR INSPECTION	CORRECTIVE ACTION
	1. Check electrical connections. Step 1. If electrical connections are good, go to 2. Step 2. If electrical connections are not good, reconnect test equipment and repeat operational test. Go to 3. 2. Check continuity between: J241-c and T1-X2 J241-d and T1-A1 J241-f and T1-A2 J241-m and T1-X1 J241-r and T1-X4 T1-G and ground Step 1. If continuity is present, replace T1 (PARA 9-5-7). Go to 3. Step 2. If continuity is not present, repair/replace wiring. Go to 3.

TABLE NO. 9-2-19-2. Test Lamp Stays Off. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION
3. Procedure completed.

TABLE NO. 9-2-19-3. K21 Windshield Anti-ice Lockout Relay Test Fails.

TEST OR INSPECTION
CORRECTIVE ACTION
1. Check helicopter configuration.
Step 1. If right relay panel is 70550-02102-043, -045, or -047, go to 2. Step 2. If right relay panel is 70550-02102-049, -051, or -056, go to 3.
2. Check continuity between:
J244-E and K21-X1 GND 1-B and K21-X2 J244-F and K21-A3 J244-G and K21-A2 J900-E and K21-B3 J900-F and K21-B2
Step 1. If continuity is present, replace relay K21 (PARA 9-5-9). Go to 4. Step 2. If continuity is not present, repair/replace wiring. Go to 4.
3. Check continuity between:
J244-E and K21-X1 K21-X2 and ground J244-F and K21-A3 J244-G and K21-A2 J900-C and K21-C2 J900-D and K21-C3 J900-E and K21-B3 J900-F and K21-B2
Step 1. If continuity is present, replace relay K21 (PARA 9-5-9). Go to 4. Step 2. If continuity is not present, repair/replace wiring. Go to 4
4. Procedure completed.

TABLE NO. 9-2-19-4. K26 No. 2 Engine Start Relay Test Fails.

TEST OR INSPECTION	CORRECTIVE ACTION
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1. Check continuity between:

J241-F and K26-E2
 J241-G and K26-F2
 J241-R and K26-X1
 J241-U and K26-A2
 J241-X and K26-D2
 J241-Y and K26-D3
 J241-Z and K26-C2
 J241-a and K26-C1
 J241-b and K26-C2
 J241-e and ground
 J244-J and K26-E3
 J244-K and K26-F3
 J900-K and K26-B3
 J900-L and K26-B2
 J900-g and K26-B1
 K26-A1 and K26-X1
 K26-X2 and ground

Step 1. If continuity is present, replace relay K26 (PARA 9-5-11). Go to **2**.

Step 2. If continuity is not present, repair/replace wiring. Go to **2**.

2. Procedure completed.

TABLE NO. 9-2-19-5. K32 Hydraulic Accumulator Relay Test Fails.

TEST OR INSPECTION	CORRECTIVE ACTION
--------------------	-------------------

1. Check helicopter configuration.

Step 1. If right relay panel is 70550-02102-043, -045, -047, -049, or -051, go to **2**.

Step 2. If right relay panel is 70550-02102-056, go to **3**.

2. Check continuity between:

TABLE NO. 9-2-19-5. K32 Hydraulic Accumulator Relay Test Fails. (Cont)

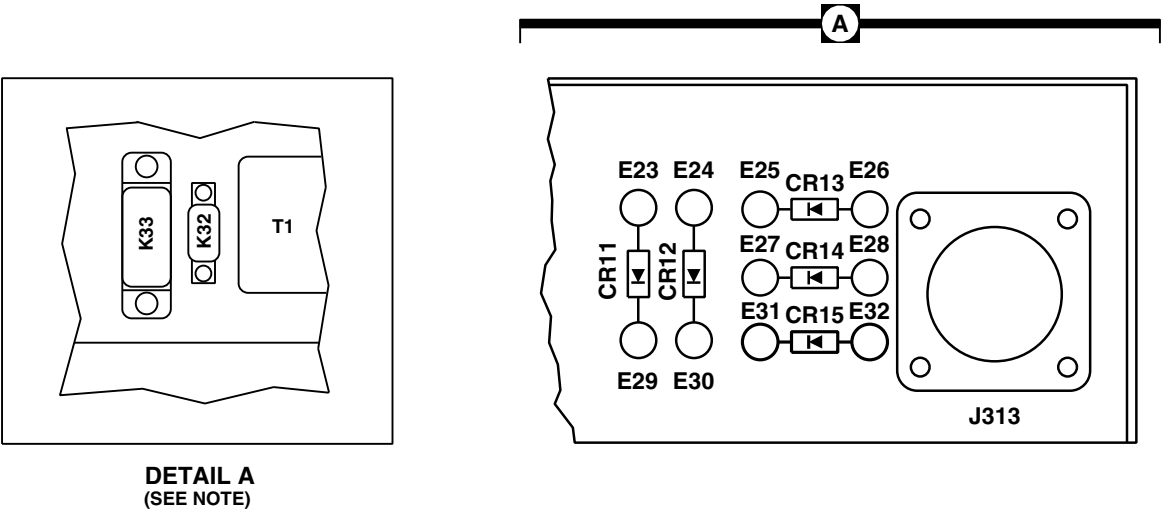
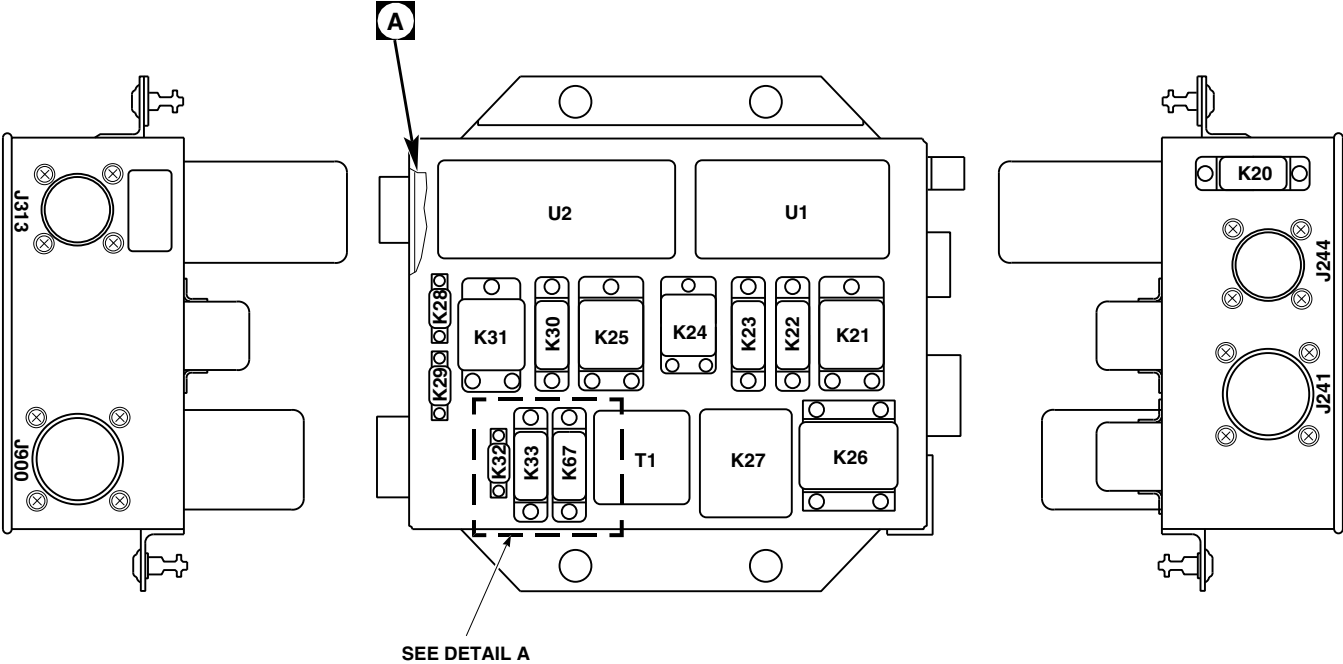
TEST OR INSPECTION CORRECTIVE ACTION	
J241-p and U2-18 J244-a and U2-7 E26 and U2-7 K31-A3 and U2-18 J244-A and K31-X1 J244-C and K33-B2 K31-A2 and K32-A1 K31-A3 and K32-X1 K31-X1 and K32-B1 K32-A2 and K33-B1 K32-B2 and K33-B2 K32-X2 and ground E25 and K33-B1	Step 1. If continuity is present, go to 4. Step 2. If continuity is not present, repair/replace wiring. Go to 5.
3. Check continuity between: J241-p and K31-A3 J244-a and E26 J244-A and K31-X1 J244-C and K33-B2 K31-A2 and K32-A1 K31-A3 and K32-X1 K31-B3 and E24 K31-X1 and K32-B1 K32-A2 and K33-B1 K32-B2 and K33-B2 K32-X2 and ground E25 and K33-B1	Step 1. If continuity is present, go to 4. Step 2. If continuity is not present, repair/replace wiring. Go to 5.
4. Check that resistance across diode CR13 is 10X greater in one direction than in the other direction.	Step 1. If measured resistance is as specified, replace relay K32 (PARA 9-5-12). Go to 5. Step 2. If measured resistance is not as specified, replace diode CR13 (PARA 9-4-25). Go to 5.

TABLE NO. 9-2-19-5. K32 Hydraulic Accumulator Relay Test Fails. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
5. Procedure completed.	

TABLE NO. 9-2-19-6. K22 APU Surge Relay Test Fails.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check continuity between: K22-A1 and E5 K22-A2 and J244-S K22-X1 and J244-T K22-X2 and J244-U J241-W and E4 J244-M and E6	Step 1. If continuity is present, go to 2. Step 2. If continuity is not present, repair/replace wiring. Go to 4.
2. Check that resistance across diodes CR3 and CR4 is 10X greater in one direction than in the other direction.	Step 1. If measured resistance is as specified, go to 3. Step 2. If measured resistance is not as specified, replace CR3 or CR4. Go to 4.
3. With 28 VDC removed from J244-T, check for no continuity between E5 and J244-S.	Step 1. If continuity is present, replace relay K22 (PARA 9-5-10). Go to 4. Step 2. If continuity is not present, repair/replace wiring. Go to 4.
4. Procedure completed.	



NOTE
DETAIL A IS FOR RELAY PANEL PART
NUMBERS 70550-02102-043 AND
70550-02102-045.

AA9147
SA

FIGURE 9-2-19-1. RIGHT RELAY PANEL LOCATION DIAGRAM

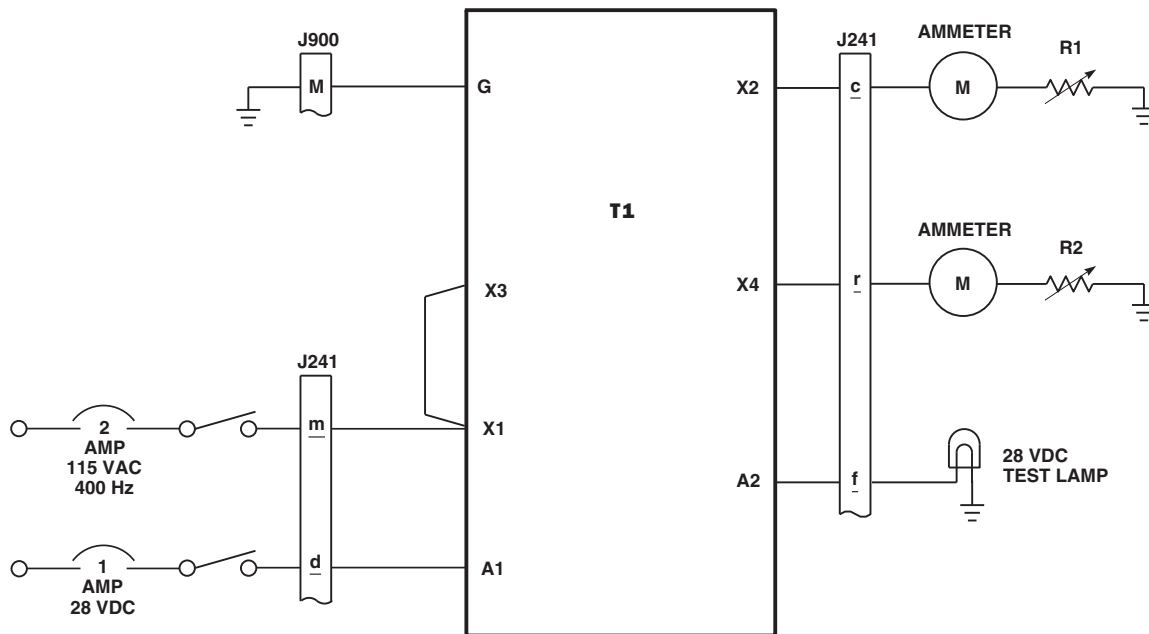
AA9146
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FIGURE 9-2-19-2. T1 CURRENT SENSOR TEST SETUP

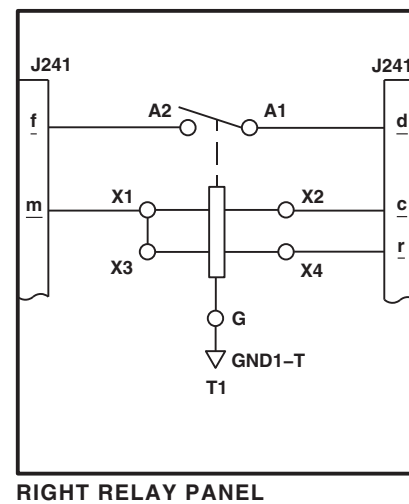
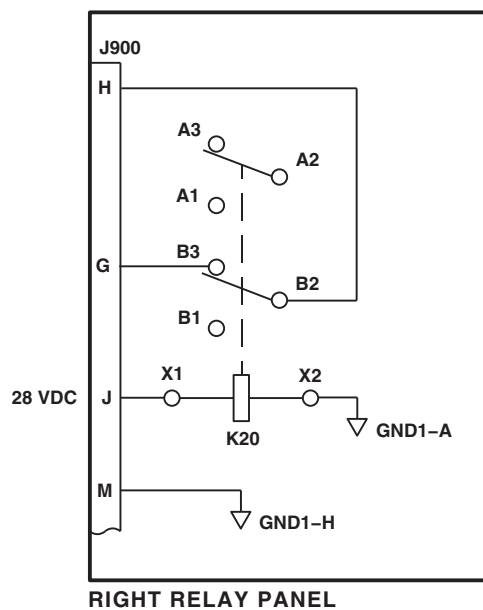
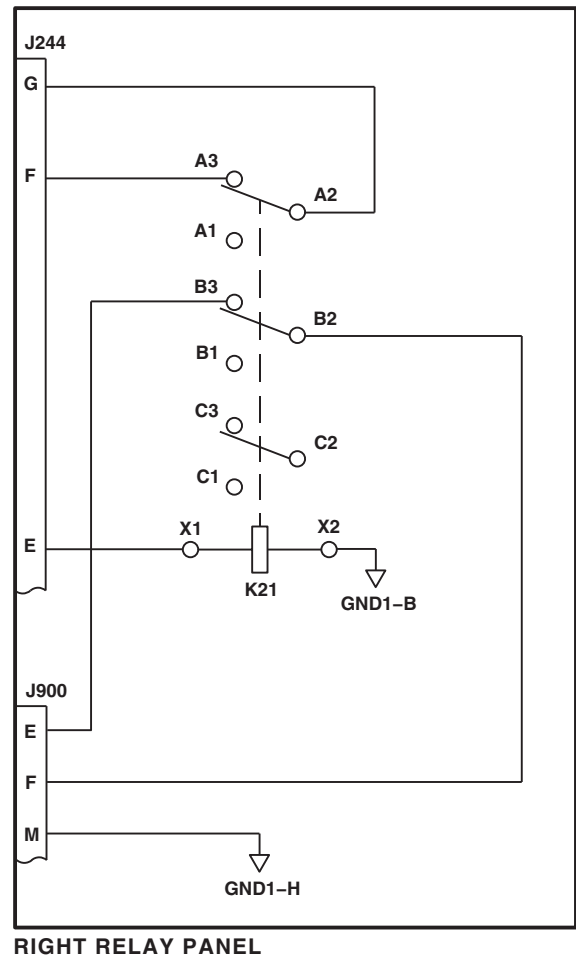
AA9148
SA

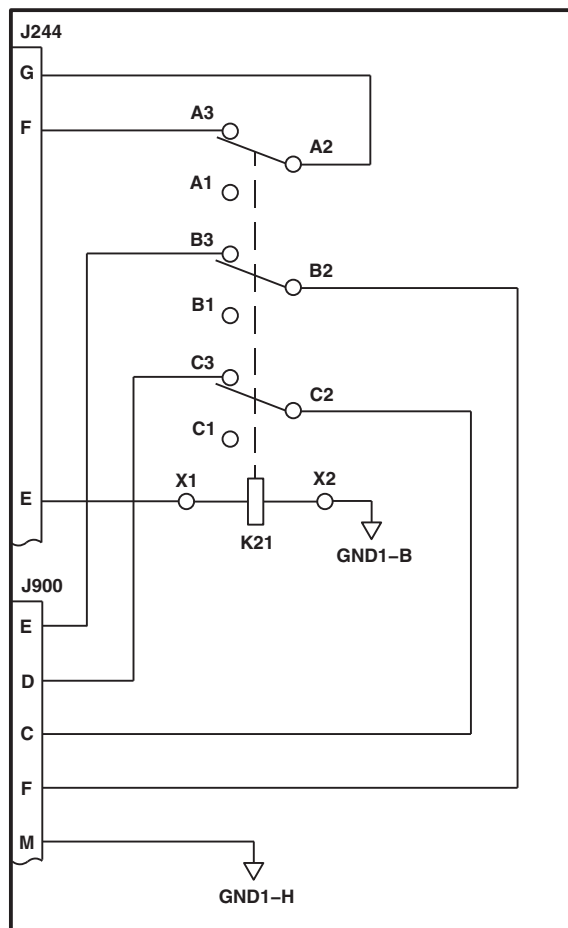
FIGURE 9-2-19-3. RIGHT RELAY PANEL SCHEMATIC DIAGRAM (SHEET 1 OF 6)



EFFECTIVITY
RELAY PANEL 70550-02102-043, -045, AND -047.

AA9150
 SA

FIGURE 9-2-19-3. RIGHT RELAY PANEL SCHEMATIC DIAGRAM (SHEET 2 OF 6)

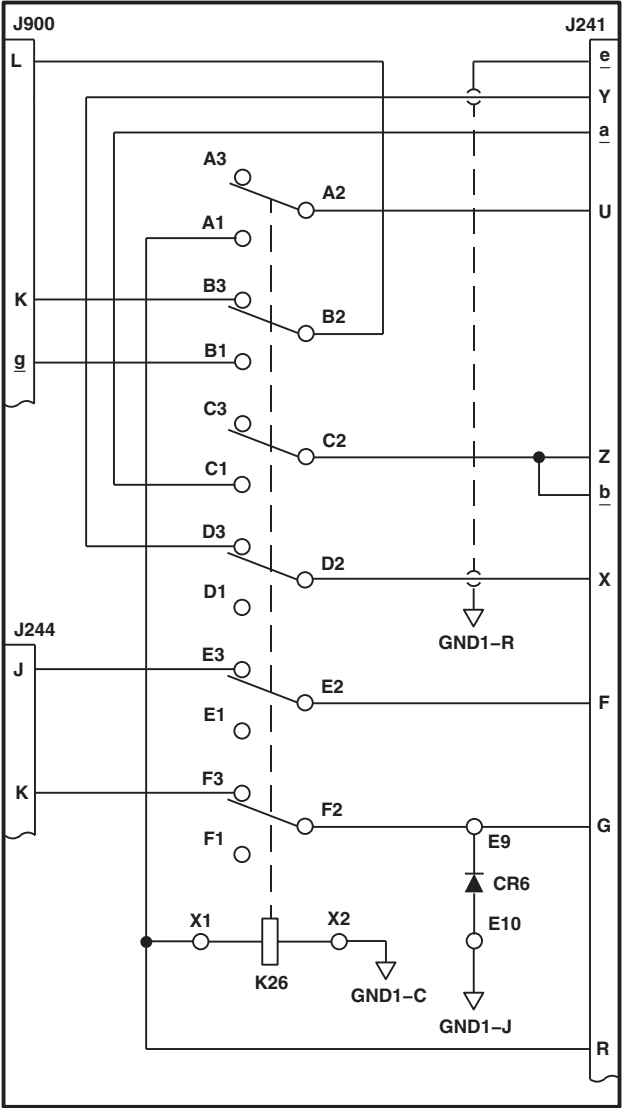


RIGHT RELAY PANEL

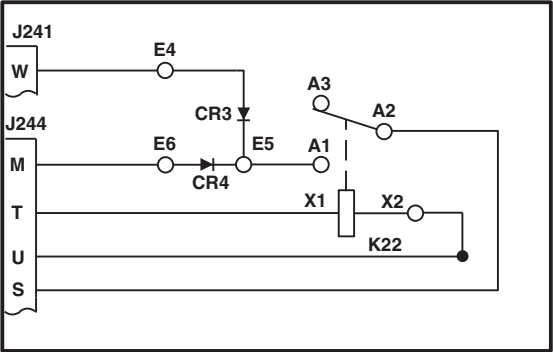
EFFECTIVITY
RELAY PANEL 70550-02102-049, -051, AND -056.

AA9149
SA

FIGURE 9-2-19-3. RIGHT RELAY PANEL SCHEMATIC DIAGRAM (SHEET 3 OF 6)



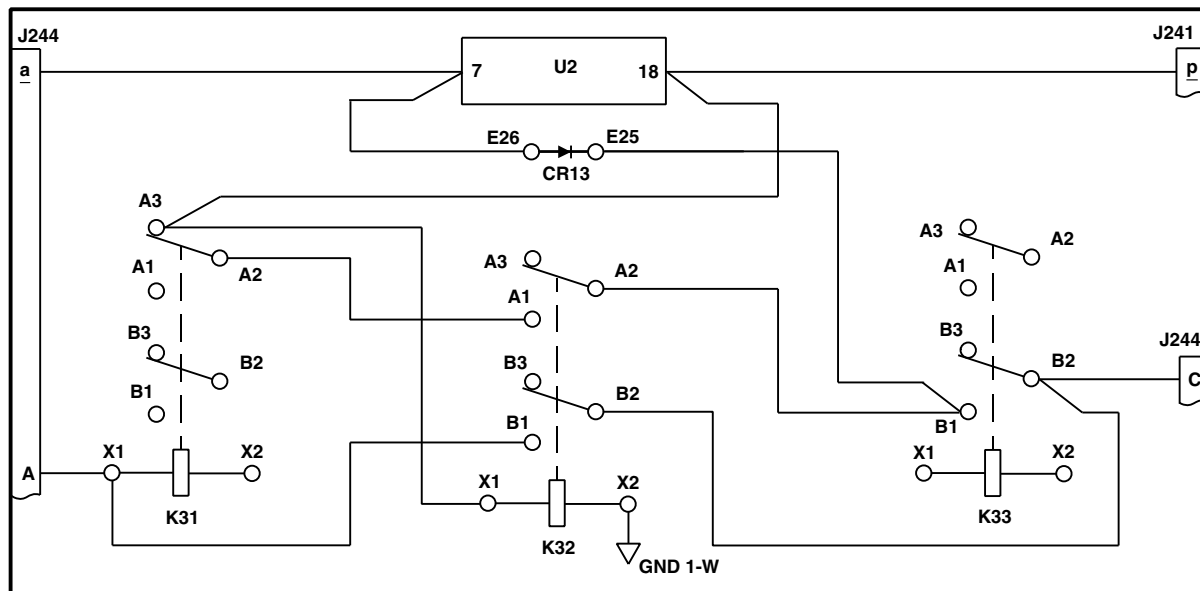
RIGHT RELAY PANEL



RIGHT RELAY PANEL

FIGURE 9-2-19-3. RIGHT RELAY PANEL SCHEMATIC DIAGRAM (SHEET 4 OF 6)

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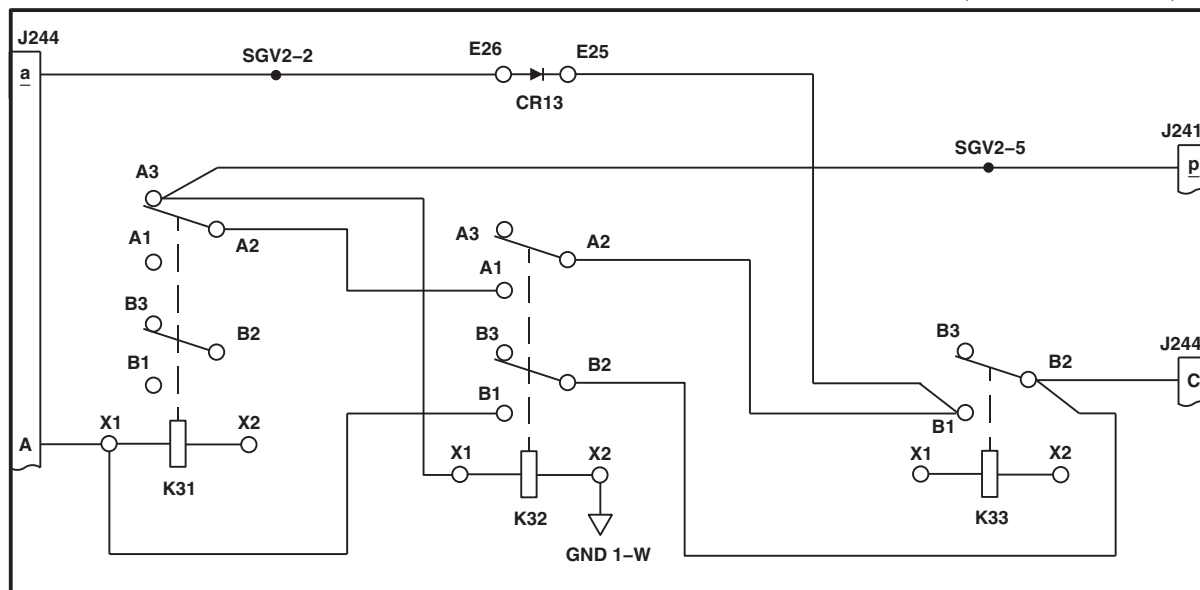


RIGHT RELAY PANEL

EFFECTIVITY
RELAY PANEL 70550-02102-043, -045, -047, -049, AND -051.

FB3612
SA

FIGURE 9-2-19-3. RIGHT RELAY PANEL SCHEMATIC DIAGRAM (SHEET 5 OF 6)



RIGHT RELAY PANEL

EFFECTIVITY
RELAY PANEL 70550-02102-056.

AA9152
SA

FIGURE 9-2-19-3. RIGHT RELAY PANEL SCHEMATIC DIAGRAM (SHEET 6 OF 6)

9-2-20 LEFT RELAY PANEL (BENCH).

INITIAL SETUP

Tools

- Decade Resistor, ZM57U
- Electrical Repairer Toolkit
- Locally-Made Test Lamp, Figure H-150, Appendix H
- Multimeter, Fluke 27/AN

Equipment Conditions

- Bench Power Available, 28 VDC and 115 VAC, 400 Hz

Special Instructions

The left relay panel operational/troubleshooting procedure is subdivided as follows:

- (1) T1 Current Sensor
- (2) K41 Searchlight Relay, 70550-02105-048 and -050
- (3) K41 Search Light Relay, 70550-02105-051, -054, -057, and -058
- (4) K45 No. 1 Engine Start Relay
- (5) Shutdown

9-2-20.1 T1 Current Sensor.

NOTE

See Figure 9-2-20-2 for test setup.

- a. Set decade resistors R1 and R2 to 200 ohms (Ω).
- b. Apply 115 VAC 400 Hz to J242-r.
- c. Apply 28 VDC to J243-X.
- d. Decrease resistance of R1 and R2 until each ammeter reads a minimum of 0.83 amps. Record resistance.

RESULT: Test lamp shall go off.

- If result is not as specified, go to TABLE NO. 9-2-20-1.
- e. Increase resistance of R1 while holding R2 constant until ammeter reads less than 0.67 amps.

RESULT: Test lamp shall go on.

- If result is not as specified, go to TABLE NO. 9-2-20-2.

- f. Return R1 to original resistance measured in step d.
- g. Increase resistance of R2 while holding R1 constant until ammeter reads less than 0.67 amps.

RESULT: Test lamp shall go on.

- If result is not as specified, go to TABLE NO. 9-2-20-2.
- h. If no other check is to be done, go to Shutdown.

9-2-20.2 K41 Searchlight Relay, 70550-02105-048 and -050.

- a. Check continuity between:
J243-d and J243-f
J243-e and J243-g
P902-f and P243-P
P902-k and J242-i
J902-n and J902-p
P902-W and J902-Y

RESULT: Continuity shall be present.

- If results are not as specified, go to TABLE NO. 9-2-20-3.

- b. Connect 28 VDC between P243-i and P243-P.
- c. Check continuity between:
 - J243-d and J243-f
 - J243-e and J243-g
 - P902-f and P243-P
 - P902-k and J242-i
 - J902-n and J902-p
 - P902-W and J902-Y

RESULT: No continuity shall be present.

- If results are not as specified, go to TABLE NO. 9-2-20-3.

- d. If no other check is to be done, go to Shutdown.

9-2-20.3 K41 Searchlight Relay, 70550-02105-051, -054, -057, and -058.

NOTE

K41 searchlight relay is a two-coil latching relay. The de-energized state of the relay depends upon which coil was last energized.

- a. Apply and remove 28 VDC between J243-Y and J243-P.
- b. Measure state (continuity/no continuity) between J243-V and J243-W.
- c. Apply and remove 28 VDC between J243-Y and J243-P.

RESULT: Measured state (continuity/no continuity) between J243-V and J243-W shall cycle upon each application of 28 VDC.

- If results are not as specified, go to TABLE NO. 9-2-20-4.
- d. If no other check is to be done, go to Shutdown.

9-2-20.4 K45 No. 1 Engine Start Relay.

- a. Check continuity between:
 - J242-R and J242-S

J242-W and J242-X
J902-a and J902-c
J902-b and J902-d

RESULT: Continuity shall be present.

- If continuity is not present, go to TABLE NO. 9-2-20-5.

- b. Check continuity between:
 - J242-T and J242-V
 - J242-P and J242-S
 - J242-Y and J242-Z
 - J242-Y and J242-U

RESULT: No continuity shall be present.

- If continuity is present, go to TABLE NO. 9-2-20-5.

- c. Connect 28 VDC lamp between J242-V and J902-GG.

- d. Connect 28 VDC between J242-T and J243-P.

RESULT: 28 VDC lamp shall light.

- If results are not as specified, go to TABLE NO. 9-2-20-5.

- e. Check continuity between:
 - J242-P and J242-S
 - J242-Y and J242-Z

RESULT: Continuity shall be present.

- If results are not as specified, go to TABLE NO. 9-2-20-5.

- f. Check for open circuit between:
 - J242-R and J242-S
 - J242-W and J242-X
 - J902-a and J902-c
 - J902-b and J902-d

RESULT: No continuity shall be present.

- If results are not as specified, go to TABLE NO. 9-2-20-5.

- g. If no other check is to be done, go to Shutdown.

9-2-20.5 Shutdown.

- a. Turn off and disconnect electrical power.

TABLE NO. 9-2-20-1. Test Lamp Stays On.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check electrical connections.	Step 1. If electrical connections are good, go to 2. Step 2. If electrical connections are not good, reconnect test equipment and repeat operational test. Go to 3.
2. Check continuity between:	J242-K and T1-X2 J243-X and T1-A1 J242-b and T1-A2 J242-r and T1-X1 J242-m and T1-X4 T1-G and ground T1-X1 and T1-X3 Step 1. If continuity is present, replace T1 (PARA 9-5-1). Go to 3. Step 2. If continuity is not present, repair/replace wiring. Go to 3.
3. Procedure completed.	

TABLE NO. 9-2-20-2. Test Lamp Stays Off.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check electrical connections.	Step 1. If electrical connections are good, go to 2. Step 2. If electrical connections are not good, reconnect test equipment and repeat operational test. Go to 3.
2. Check continuity between:	

TABLE NO. 9-2-20-2. Test Lamp Stays Off. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	J242-K and T1-X2 J243-X and T1-A1 J242-b and T1-A2 J242-r and T1-X1 J242-m and T1-X4 T1-G and ground T1-X1 and T1-X3
	Step 1. If continuity is present, replace T1 (PARA 9-5-1). Go to 3 . Step 2. If continuity is not present, repair/replace wiring. Go to 3 .
	3. Procedure completed.

TABLE NO. 9-2-20-3. K41 Search Light Relay Test Fails.

TEST OR INSPECTION	CORRECTIVE ACTION
	1. Check continuity between: K41-A2 and K41-A3 K41-B2 and K41-B3 K41-C2 and K41-C3 K41-D2 and K41-D3 K41-E2 and K41-E3 K41-F2 and K41-F3
	Step 1. If continuity is present, go to 2 . Step 2. If continuity is not present, replace relay K41 (PARA 9-5-2). Go to 8 .
	2. Check continuity between: K41-A1 and K41-A2 K41-B1 and K41-B2 K41-C1 and K41-C2 K41-D1 and K41-D2 K41-E1 and K41-E2 K41-F1 and K41-F2
	Step 1. If continuity is present, replace relay K41 (PARA 9-5-2). Go to 8 . Step 2. If continuity is not present, go to 3 .

TABLE NO. 9-2-20-3. K41 Search Light Relay Test Fails. (Cont)

TEST OR INSPECTION CORRECTIVE ACTION	
3. Connect 28 VDC between K41-X1 and K41-X2. Check continuity between: K41-A1 and K41-A2 K41-B1 and K41-B2 K41-C1 and K41-C2 K41-D1 and K41-D2 K41-E1 and K41-E2 K41-F1 and K41-F2 Step 1. If continuity is present, go to 4. Step 2. If continuity is not present, replace relay K41 (PARA 9-5-2). Go to 8.	
4. Check continuity between: K41-A2 and K41-A3 K41-B2 and K41-B3 K41-C2 and K41-C3 K41-D2 and K41-D3 K41-E2 and K41-E3 K41-F2 and K41-F3 Step 1. If continuity is present, replace relay K41 (PARA 9-5-2). Go to 8. Step 2. If continuity is not present, go to 5.	
5. Check continuity between: K41-X2 and GND 1-A K41-X1 and P243-i K41-A2 and J243-g K41-A3 and K40-C2 K41-B2 and J243-f K41-B3 and K40-B2 K41-C2 and J902-n K41-C3 and J1-13 K41-C3 and K40-A2 K41-D2 and K40-D3 K41-F2 and J902-Y K41-F3 and P902-W J243-P and GND 1-M K41-D3 and P902-k K41-E2 and GND 1-B K41-E3 and P902-f Step 1. If continuity is present, go to 6. Step 2. If continuity is not present, repair/replace wiring. Go to 8.	

TABLE NO. 9-2-20-3. K41 Search Light Relay Test Fails. (Cont)

**TEST OR INSPECTION
CORRECTIVE ACTION**

6. Check continuity between:

**K40-A2 and K40-A3
K40-B2 and K40-B3
K40-C2 and K40-C3
K40-D2 and K40-D3
K40-E2 and K40-E3
K40-F2 and K40-F3**

Step 1. If continuity is present, go to **7**.

Step 2. If continuity is not present, replace relay K40 (PARA 9-5-6). Go to **8**.

7. Check continuity between:

**K40-A2 and K40-A1
K40-B2 and K40-B1
K40-C2 and K40-C1
K40-D2 and K40-D1
K40-E2 and K40-E1
K40-F2 and K40-F1**

Step 1. If continuity is present, replace relay K40 (PARA 9-5-6). Go to **8**.

Step 2. If continuity is not present, go to **8**.

8. Procedure completed.

TABLE NO. 9-2-20-4. K41 Searchlight Relay Test Fails.

**TEST OR INSPECTION
CORRECTIVE ACTION**

1. Apply and remove 28 VDC between K41-X1 and K41-X2. Check continuity between:

**K41-A1 and K41-A2
K41-B1 and K41-B2**

Step 1. If continuity is present, go to **2**.

Step 2. If continuity is not present, replace relay K41 (PARA 9-5-2). Go to **17**.

2. Check continuity between:

TABLE NO. 9-2-20-4. K41 Searchlight Relay Test Fails. (Cont)

TEST OR INSPECTION CORRECTIVE ACTION	
K41-A2 and K41-A3 K41-B2 and K41-B3	
Step 1. If continuity is present, replace relay K41 (PARA 9-5-2). Go to 17 . Step 2. If continuity is not present, go to 3 .	
3. Apply and remove 28 VDC from K41-X1 and K41-X2 and connect 28 VDC between K41-Y1 and K41-Y2. Check continuity between: K41-A2 and K41-A3 K41-B2 and K41-B3	
Step 1. If continuity is present, go to 4 . Step 2. If continuity is not present, replace relay K41 (PARA 9-5-2). Go to 17 .	
4. Check continuity between: K41-A1 and K41-A2 K41-B1 and K41-B2	
Step 1. If continuity is present, replace relay K41 (PARA 9-5-2). Go to 17 . Step 2. If continuity is not present, go to 5 .	
5. Check continuity between: K41-X2 and ground K41-X1 and K51-A1 K41-A1 and K51-B3 K41-A2 and J243-Y K41-A2 and K52-A2 K52-A2 and K51-A2 K41-B2 and J243-V K41-B1 and J243-W K41-Y1 and K52-A1 K41-Y2 and ground	
Step 1. If continuity is present, go to 6 . Step 2. If continuity is not present, repair/replace wiring as required. Go to 17 .	
6. Check continuity between: K52-A2 and K52-A3 K52-B2 and K52-B3	
Step 1. If continuity is present, go to 7 . Step 2. If continuity is not present, replace relay K52 (PARA 9-5-4). Go to 17 .	

TABLE NO. 9-2-20-4. K41 Searchlight Relay Test Fails. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
7. Check continuity between: K52-A2 and K52-A1 K52-B2 and K52-B1	Step 1. If continuity is present, replace relay K52 (PARA 9-5-4). Go to 17. Step 2. If continuity is not present, go to 8.
8. Connect 28 VDC between K52-X1 and K52-X2. Check continuity between: K52-A2 and K52-A1 K52-B2 and K52-B1	Step 1. If continuity is present, go to 9. Step 2. If continuity is not present, replace relay K52 (PARA 9-5-4). Go to 17.
9. Check continuity between: K52-A2 and K52-A3 K52-B2 and K52-B3	Step 1. If continuity is present, go to 10. Step 2. If continuity is not present, replace relay K52 (PARA 9-5-4). Go to 17.
10. Disconnect 28 VDC. Check continuity between: K52-A1 and K41-Y1 K52-A2 and K41-A2 K52-A2 and K51-A2 K52-B2 and K51-X1 K52-B3 and K41-A3 K52-X1 and K51-B2 K52-X2 and ground	Step 1. If continuity is present, go to 11. Step 2. If continuity is not present, replace relay K52 (PARA 9-5-4). Go to 17.
11. Check continuity between: K51-A2 and K51-A3 K51-B2 and K51-B3	Step 1. If continuity is present, go to 12. Step 2. If continuity is not present, replace relay K51 (PARA 9-5-3). Go to 17.
12. Check continuity between:	

7. Check continuity between:

K52-A2 and K52-A1

K52-B2 and K52-B1

Step 1. If continuity is present, replace relay K52 (PARA 9-5-4). Go to 17.

Step 2. If continuity is not present, go to 8.

8. Connect 28 VDC between K52-X1 and K52-X2. Check continuity between:

K52-A2 and K52-A1

K52-B2 and K52-B1

Step 1. If continuity is present, go to 9.

Step 2. If continuity is not present, replace relay K52 (PARA 9-5-4). Go to 17.

9. Check continuity between:

K52-A2 and K52-A3

K52-B2 and K52-B3

Step 1. If continuity is present, go to 10.

Step 2. If continuity is not present, replace relay K52 (PARA 9-5-4). Go to 17.

10. Disconnect 28 VDC. Check continuity between:

K52-A1 and K41-Y1

K52-A2 and K41-A2

K52-A2 and K51-A2

K52-B2 and K51-X1

K52-B3 and K41-A3

K52-X1 and K51-B2

K52-X2 and ground

Step 1. If continuity is present, go to 11.

Step 2. If continuity is not present, replace relay K52 (PARA 9-5-4). Go to 17.

11. Check continuity between:

K51-A2 and K51-A3

K51-B2 and K51-B3

Step 1. If continuity is present, go to 12.

Step 2. If continuity is not present, replace relay K51 (PARA 9-5-3). Go to 17.

12. Check continuity between:

TABLE NO. 9-2-20-4. K41 Searchlight Relay Test Fails. (Cont)

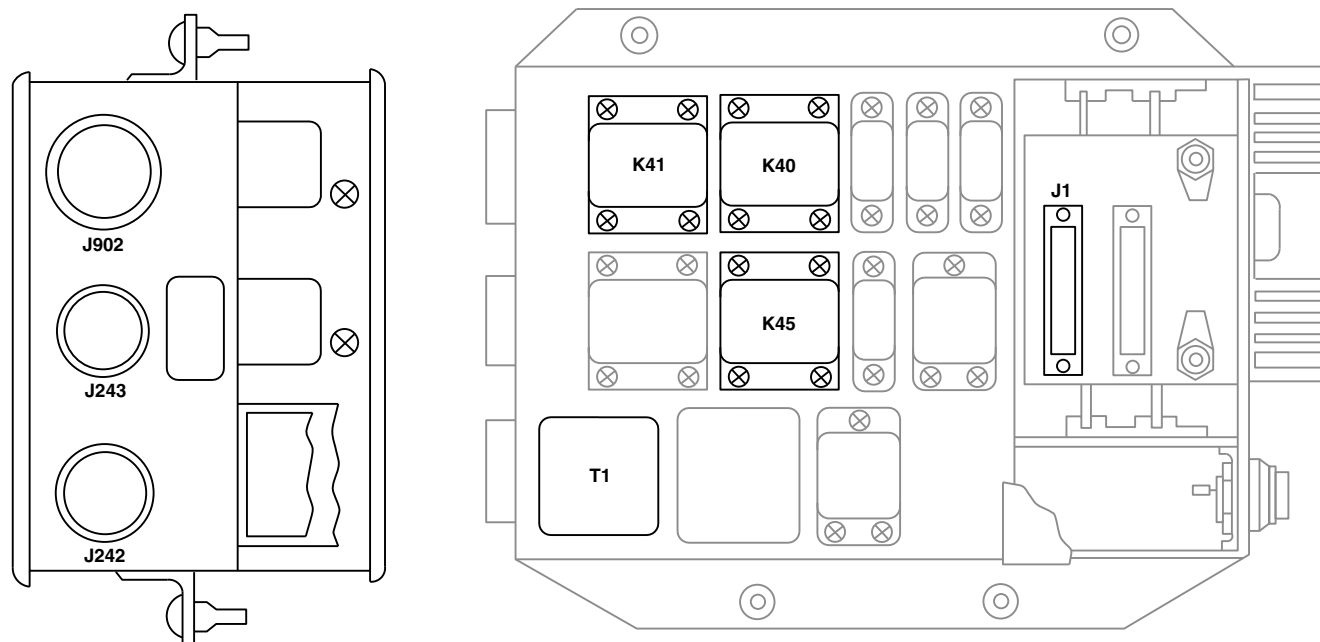
TEST OR INSPECTION CORRECTIVE ACTION	
K51-A2 and K51-A1 K51-B2 and K51-B1	
Step 1. If continuity is present, replace relay K51 (PARA 9-5-3). Go to 17 . Step 2. If continuity is not present, go to 13 .	
13. Connect 28 VDC between K51-X1 and K51-X2. Check continuity between: K51-A2 and K51-A1 K51-B2 and K51-B1	
Step 1. If continuity is present, go to 14 . Step 2. If continuity is not present, replace relay K51 (PARA 9-5-3). Go to 17 .	
14. Check continuity between: K51-B2 and K51-B3 K51-A2 and K51-A3	
Step 1. If continuity is present, replace relay K51 (PARA 9-5-3). Go to 17 . Step 2. If continuity is not present, go to 15 .	
15. Disconnect 28 VDC from K51. Check continuity between: K51-X2 and ground K51-X1 and K52-B2 K51-A1 and K41-X1 K51-A2 and K52-A2 K51-B3 and K41-A1	
Step 1. If continuity is present, go to 16 . Step 2. If continuity is not present, repair/replace wiring as required. Go to 17 .	
16. Check continuity between: K51-A1 and J1-11 K51-X1 and J1-5 K52-A1 and J1-12 K52-B2 and J1-5 K52-X1 and J1-7	
Step 1. If continuity is present, continue operational checkout. Go to 17 . Step 2. If continuity is not present, repair/replace wiring as required. Go to 17 .	
17. Procedure completed.	

TABLE NO. 9-2-20-5. K45 No. 1 Engine Start Relay Test Fails.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check continuity between: K45-A2 and K45-A3 K45-B2 and K45-B3 K45-C2 and K45-C3 K45-D2 and K45-D3 K45-E2 and K45-E3 K45-F2 and K45-F3	Step 1. If continuity is present, go to 2. Step 2. If continuity is not present, replace relay K45 (PARA 9-5-5). Go to 6.
2. Check continuity between: K45-A2 and K45-A1 K45-B2 and K45-B1 K45-C2 and K45-C1 K45-D2 and K45-D1 K45-E2 and K45-E1 K45-F2 and K45-F1	Step 1. If continuity is present, replace relay K45 (PARA 9-5-5). Go to 6. Step 2. If continuity is not present, go to 3.
3. Connect 28 VDC between K45-X1 and K45-X2. Check continuity between: K45-A2 and K45-A1 K45-B2 and K45-B1 K45-C2 and K45-C1 K45-D2 and K45-D1 K45-E2 and K45-E1 K45-F2 and K45-F1	Step 1. If continuity is present, go to 4. Step 2. If continuity is not present, replace relay K45 (PARA 9-5-5). Go to 6.
4. Check continuity between: K45-A2 and K45-A3 K45-B2 and K45-B3 K45-C2 and K45-C3 K45-D2 and K45-D3 K45-E2 and K45-E3 K45-F2 and K45-F3	

TABLE NO. 9-2-20-5. K45 No. 1 Engine Start Relay Test Fails. (Cont)

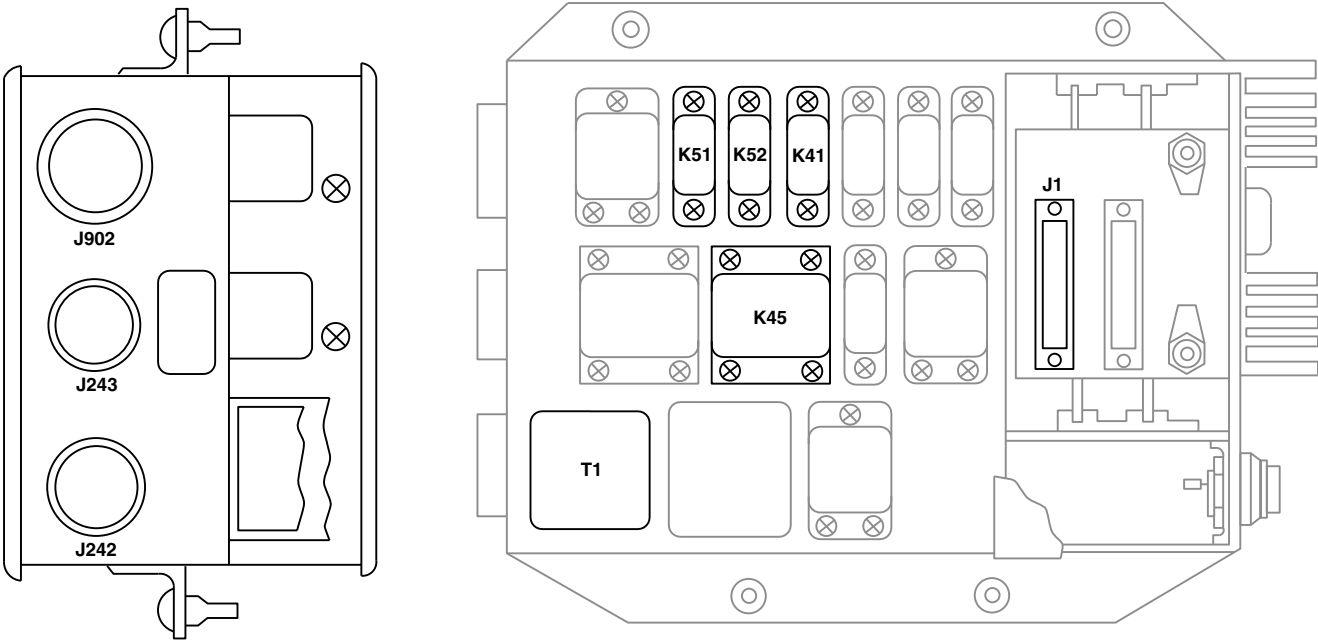
TEST OR INSPECTION	CORRECTIVE ACTION
	Step 1. If continuity is present, replace relay K45 (PARA 9-5-5). Go to 6. Step 2. If continuity is not present, go to 5.
5. Disconnect 28 VDC from K45. Check continuity between: K45-X2 and ground K45-X1 and K45-A1 K45-A1 and J242-T K45-A2 and J242-V K45-B2 and J242-S K45-B1 and J242-P K45-B3 and J242-R K45-B3 and J1-9 K45-C2 and J242-Z K45-C2 and J242-U K45-D2 and J242-X K45-D3 and J242-W K45-E2 and J902-<u>c</u> K45-E3 and J902-<u>a</u> K45-F2 and J902-<u>d</u> K45-F3 and J902-<u>b</u>	Step 1. If continuity is present, continue operational checkout. Go to 6. Step 2. If continuity is not present, repair/replace wiring as required. Go to 6.
6. Procedure completed.	

**EFFECTIVITY**

RELAY PANEL PART NUMBERS
70550-02105-048 AND -050.

AA9168_1
SA

FIGURE 9-2-20-1. LEFT RELAY PANEL LOCATION DIAGRAM (SHEET 1 OF 2)

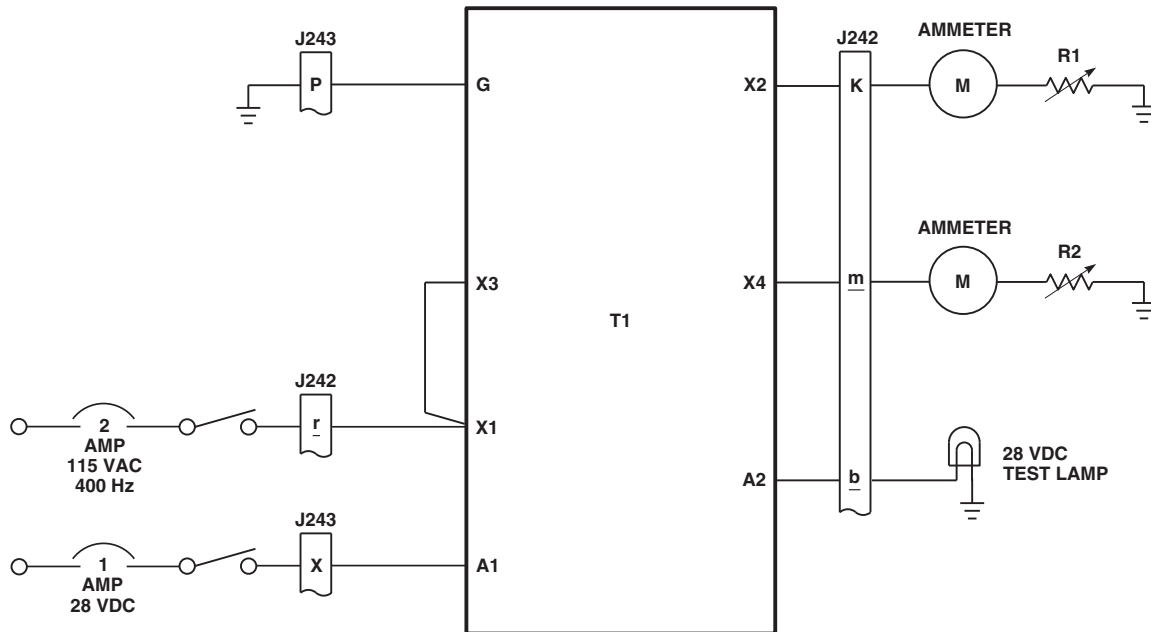


EFFECTIVITY

RELAY PANEL PART NUMBERS
 70550-02105-051, -054, AND -057.

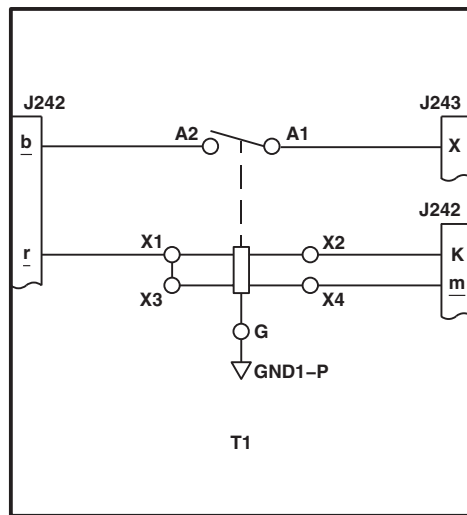
AA9168_2
 SA

FIGURE 9-2-20-1. LEFT RELAY PANEL LOCATION DIAGRAM (SHEET 2 OF 2)



AA9169
SA

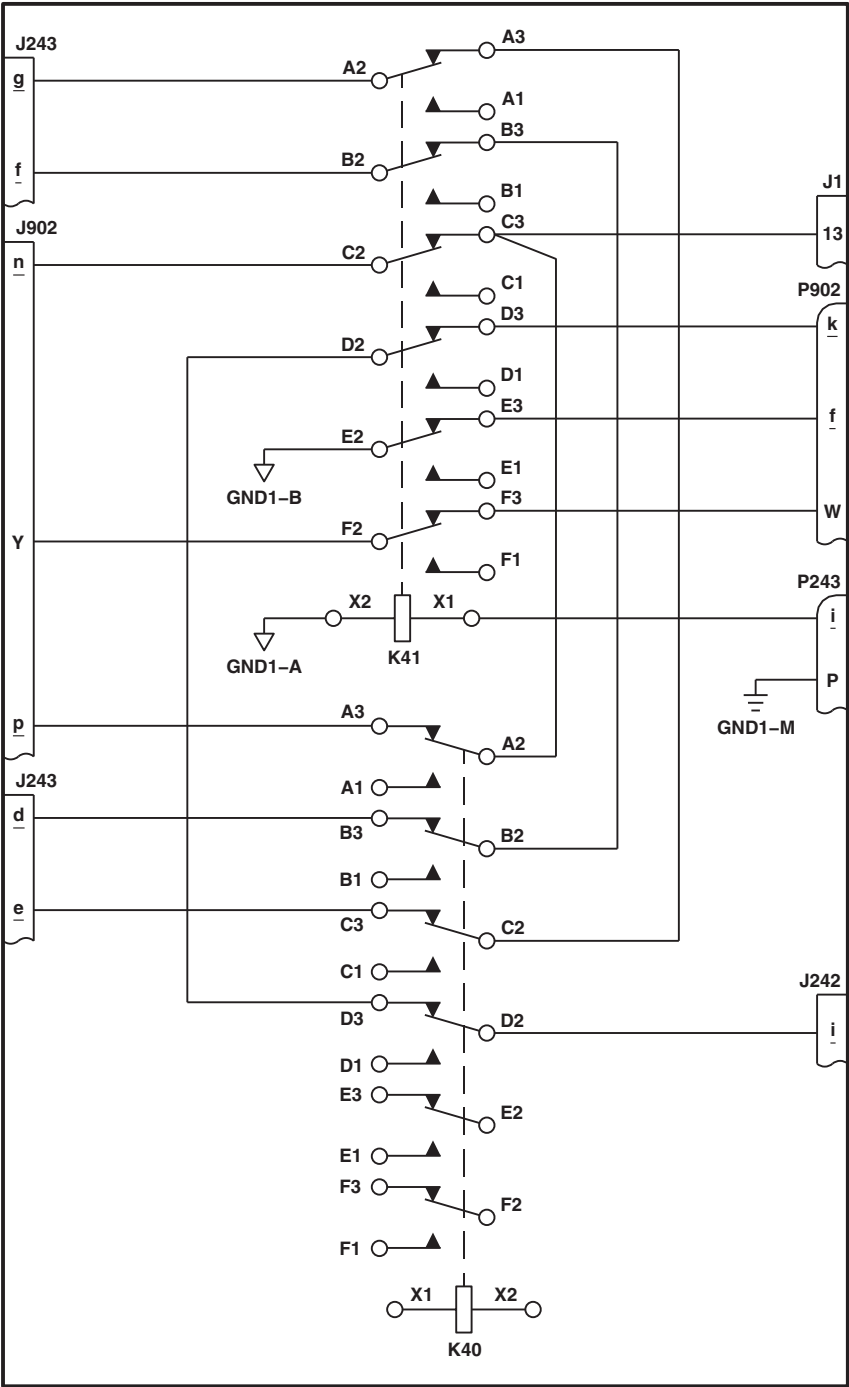
FIGURE 9-2-20-2. T1 CURRENT SENSOR TEST SETUP



LEFT RELAY PANEL

AA9170
SA

FIGURE 9-2-20-3. LEFT RELAY PANEL SCHEMATIC DIAGRAM (SHEET 1 OF 4)

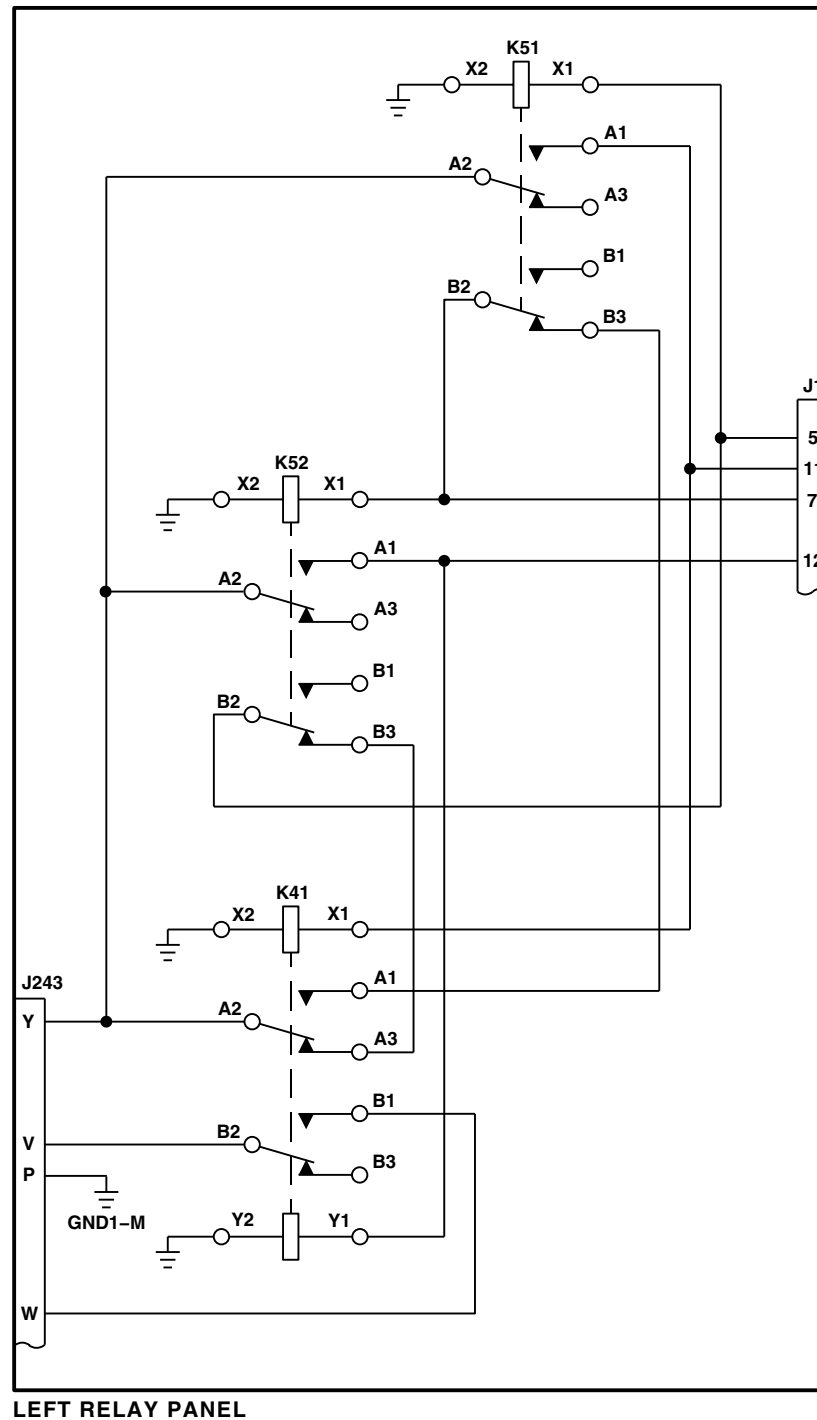


LEFT RELAY PANEL

EFFECTIVITY
RELAY PANEL PART NUMBERS 70550-02105-048 AND -050.

FIGURE 9-2-20-3. LEFT RELAY PANEL SCHEMATIC DIAGRAM (SHEET 2 OF 4)

AA9172
SA



LEFT RELAY PANEL

EFFECTIVITY

RELAY PANEL PART NUMBERS
70550-02105-051, -054, -057,
AND -058.

AA9171
SA

FIGURE 9-2-20-3. LEFT RELAY PANEL SCHEMATIC DIAGRAM (SHEET 3 OF 4)

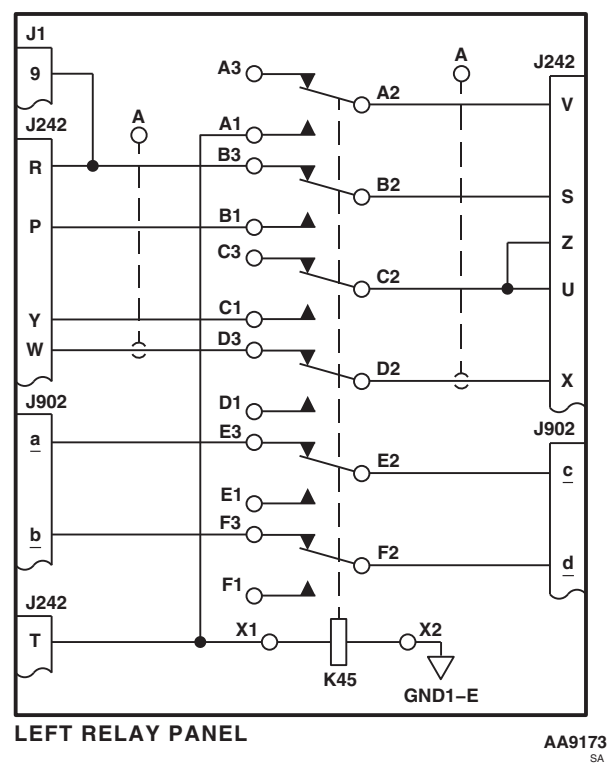


FIGURE 9-2-20-3. LEFT RELAY PANEL SCHEMATIC DIAGRAM (SHEET 4 OF 4)

9-2-21 AUXILIARY CONTROLLABLE SEARCHLIGHT.

INITIAL SETUP

Test Equipment

- Multimeter, Fluke 45

Tools

- Electrical Repairer Toolkit

References

- Appropriate Flight Manual

Equipment Conditions

- External Electrical Power Available or APU Operational



To prevent serious injury to personnel, do not look directly into searchlight when it is turned on.



Avoid operation of the searchlight below 10 feet to prevent the ignition of flammable or dry materials.



To prevent injury to personnel or damage to equipment, discharge booster capacitor in junction box with an insulated 6 gauge shorting jumper before performing any troubleshooting.



Holding the auxiliary searchlight control panel POWER switch in the START position after lamp ignites or during a series of strobes will seriously damage the equipment.

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).

9-2-21.1 Operational/Troubleshooting Procedure.

- a. Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.
- b. Set FILTER switch on auxiliary searchlight control panel to NORM position.
- c. Set BRT/DIM-TEST switch on caution/advisory panel to BRT position.
- d. Start APU (Appropriate Flight Manual).

- e. Set auxiliary searchlight control panel POWER switch to the ON position.

RESULT:

- (1) The AUX SRCHLT ON caution/advisory capsule shall go on.
- (2) The searchlight assembly blower motor shall be running.
- If result (1) is not as specified, go to TABLE NO. 9-2-21-1.

- If result (2) is not as specified, go to TABLE NO. 9-2-21-2.

f. Set auxiliary searchlight control panel FILTER switch to the infrared (IR) position.

RESULT:

- (1) Auxiliary searchlight infrared window assembly shall close.
 - (2) The microswitch on bottom of IR filter shall activate and IR filter drive motor shall stop.
 - (3) The IR light on the auxiliary searchlight control panel shall go on.
- If infrared window assembly does not close, go to TABLE NO. 9-2-21-3.
 - If microswitch does not activate, loosen adjusting bracket set screws and adjust microswitch until it actuates.
 - If IR light does not go on, go to TABLE NO. 9-2-21-4.
- g. Set auxiliary searchlight control panel FILTER switch to NORM.

RESULT:

- (1) The infrared window assembly shall open, and IR light shall go off.
 - (2) Microswitch on the upper outboard IR lens shall activate and IR drive motor shall stop when IR lens is in full up position.
- If infrared window assembly does not open, replace auxiliary searchlight control panel (PARA 9-4-100).
 - If microswitch does not activate, adjust microswitch position.
 - If trouble still remains, replace IR window assembly (PARA 9-4-104).

- h. Press auxiliary searchlight control panel CONTROL switch to UP position. Release switch when auxiliary searchlight stops.

RESULT: Auxiliary searchlight beam shall move to the 0° (pitch up) position.

- If result is not as specified, go to TABLE NO. 9-2-21-5.

- i. Press auxiliary searchlight control panel CONTROL switch to the DOWN position. Release switch when searchlight stops.

RESULT: Auxiliary searchlight shall move from 0° pitch up to 52° pitch down.

- If result is not as specified, go to TABLE NO. 9-2-21-6.

- j. Press auxiliary searchlight control panel CONTROL switch to the L position. Release switch when searchlight stops.

RESULT: Auxiliary searchlight shall move from 0° to 90° laterally.

- If result is not as specified, go to TABLE NO. 9-2-21-7.

CAUTION

Ensure that the searchlight does not travel more than 15° right in order to prevent the searchlight from being aimed at the airframe.

- k. Press auxiliary searchlight control panel CONTROL switch to the R position. Release switch when searchlight stops.

RESULT: Auxiliary searchlight shall move from 90° left to 15° right.

- If result is not as specified, go to TABLE NO. 9-2-21-8.

- l. Turn CONSOLE LT LOWER control, on upper console, from BRT to OFF.

RESULT: Auxiliary searchlight control panel faceplate light shall decrease in brightness.

- If result is not as specified, troubleshoot upper and lower console lighting (PARA 9-2-25).

WARNING

The auxiliary searchlight beam is 30 million candlepower as it leaves the searchlight lens. Viewing the lens directly during start-up or continuous operation can cause serious injury.

NOTE

Raise auxiliary searchlight to the horizontal position before igniting auxiliary searchlight.

- m. Press auxiliary searchlight control panel power switch to the START position and hold

for approximately 5 seconds, or until auxiliary searchlight ignites.

RESULT: Auxiliary searchlight lamp shall illuminate.

- If result is not as specified, go to TABLE NO. 9-2-21-9.

- n. Press and hold auxiliary searchlight control panel BEAM/FOCUS switch.

RESULT: Auxiliary searchlight focus motor shall drive the focus mechanism, changing the light beam from a narrow to wide beam.

- If result is not as specified, go to TABLE NO. 9-2-21-10.

- o. Release auxiliary searchlight control panel BEAM/FOCUS switch.

- p. Set auxiliary searchlight control panel POWER switch to OFF.

- q. Shut down APU (Appropriate Flight Manual).

TABLE NO. 9-2-21-1. Caution/Advisory Panel AUX SRCHLT ON Capsule Does Not Go On.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for 28 VDC between P118-Y and ground.	
	Step 1. If voltage is as specified, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 9. Step 2. If voltage is not as specified, go to 2.
2. Check for 28 VDC between P197-A and ground.	
	Step 1. If voltage is as specified, go to 3. Step 2. If voltage is not as specified, go to 4.
3. Check continuity between J197-A and J197-H.	
	Step 1. If continuity is present, repair/replace wiring between P197-H and P118-Y (Appendix F). Go to 9.

TABLE NO. 9-2-21-1. Caution/Advisory Panel AUX SRCHLT ON Capsule Does Not Go On. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 2. If continuity is not present, replace auxiliary searchlight control panel (PARA 9-4-100). Go to 9 .
4. Check for 28 VDC between P210-j and ground.	Step 1. If voltage is as specified, repair/replace wiring between P210-h and P197-A (Appendix F). Go to 9. Step 2. If voltage is not as specified, go to 5.
5. Check continuity between J210-h and J210-j.	Step 1. If continuity is present, go to 6. Step 2. If continuity is not present, troubleshoot dc electrical system (PARA 9-2-2). Go to 9.
6. Check for 28 VDC between TS1 terminal 1 and ground.	Step 1. If voltage is as specified, repair/replace wiring between TS1-1 and P210-j (Appendix F). Go to 9. Step 2. If voltage is not as specified, go to 7.
7. Check for 28 VDC between searchlight junction box power stud PWR1 and ground.	Step 1. If voltage is as specified, replace searchlight junction box (PARA 9-4-101). Go to 9. Step 2. If voltage is not as specified, go to 8.
8. Check for 28 VDC between terminal 1 of AUX SEARCHLIGHT current limiter CL19 and ground.	Step 1. If voltage is as specified, repair/replace wiring between power stud PWR1 and terminal 1 of CL19 (Appendix F). Go to 9. Step 2. If voltage is not as specified, replace CL19. Go to 9.
9. Procedure completed.	

TABLE NO. 9-2-21-2. Blower Motor Is Not Running.

TEST OR INSPECTION CORRECTIVE ACTION	
1. Check for 28 VDC between P1-E and P1-C.	
Step 1. If voltage is as specified, replace auxiliary searchlight assembly (PARA 9-4-103). Go to 11. Step 2. If voltage is not as specified, go to 2.	
2. Check for 28 VDC between TS1-6 and GND001.	
Step 1. If voltage is as specified, replace auxiliary searchlight assembly (PARA 9-4-103). Go to 11. Step 2. If voltage is not as specified, go to 3.	
3. Check continuity between: GND001 (ground stud) and ground GND001 (ground stud) and terminal TS1-3	
Step 1. If continuity is present, go to 4. Step 2. If continuity is not present, replace searchlight junction box assembly (PARA 9-4-101). Go to 11.	
4. Check for 28 VDC between TS1-6 and TS1-3.	
Step 1. If voltage is as specified, replace searchlight junction box assembly (PARA 9-4-101). Go to 11. Step 2. If voltage is not as specified, go to 5.	
5. Check for 28 VDC between P197-A and ground.	
Step 1. If voltage is as specified, replace auxiliary searchlight control panel (PARA 9-4-100). Go to 11. Step 2. If voltage is not as specified, go to 6.	
6. Check for 28 VDC between P210-j and ground.	
Step 1. If voltage is as specified, go to 7. Step 2. If voltage is not as specified, go to 8.	
7. Check continuity between J210-h and J210-j.	
Step 1. If continuity is present, repair/replace wiring between P210-h and P197-A (Appendix F). Go to 11.	

TABLE NO. 9-2-21-2. Blower Motor Is Not Running. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 2. If continuity is not present, troubleshoot DC Electrical System (PARA 9-2-2). Go to 11 .
8. Check for 28 VDC between TS1 terminal 1 and ground.	Step 1. If voltage is as specified, repair/replace wiring between TS1-1 and P210-j (Appendix F). Go to 11 . Step 2. If voltage is not as specified, go to 9 .
9. Check for 28 VDC between searchlight junction box power stud PWR1 and ground.	Step 1. If voltage is as specified, replace searchlight junction box assembly (PARA 9-4-101). Go to 11 . Step 2. If voltage is not as specified, go to 10 .
10. Check for 28 VDC between terminal 1 of AUX SEARCHLIGHT current limiter CL19 and ground.	Step 1. If voltage is as specified, repair/replace wiring between power stud PWR1 and terminal 1 of CL19. (Appendix F). Go to 11 . Step 2. If voltage is not as specified, replace CL19. Go to 11 .
11. Procedure completed.	

TABLE NO. 9-2-21-3. Infrared Window Assembly Does Not Close.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Connect multimeter between P2-A and P2-D. Go to 2.	
2. Place auxiliary searchlight control panel NORM/IR switch to NORM position. Go to 3.	
3. Observe that 28 VDC is present and that polarity of P2-A is negative.	Step 1. If voltage and polarity are as specified, go to 4 . Step 2. If voltage is not as specified, go to 6 .

TABLE NO. 9-2-21-3. Infrared Window Assembly Does Not Close. (Cont)

TEST OR INSPECTION CORRECTIVE ACTION	
Step 3. If polarity is not as specified, replace auxiliary searchlight control panel (PARA 9-4-100). Go to 7.	
4. Place auxiliary searchlight control panel NORM/IR switch to IR position. Go to 5.	
5. Observe that 28 VDC is present and that polarity of P2-A is positive.	I
Step 1. If voltage and polarity are as specified, replace auxiliary searchlight assembly (PARA 9-4-103). Go to 7.	
Step 2. If voltage is not as specified, go to 6.	
Step 3. If polarity is not as specified, replace auxiliary searchlight control panel (PARA 9-4-100). Go to 7.	
6. Check continuity between: P2-A and P197-F P2-B and P197-G P2-D and P197-J	
Step 1. If continuity is present, replace auxiliary searchlight control panel (PARA 9-4-100). Go to 7.	
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 7.	
7. Procedure completed.	

TABLE NO. 9-2-21-4. IR Light Does Not Go On.

TEST OR INSPECTION CORRECTIVE ACTION	
1. Press FILTER light lens on searchlight control panel and observe that IR light goes on.	
Step 1. If IR light goes on, go to 2.	
Step 2. If IR light does not go on, go to 5.	
2. Check for 28 VDC between P142-B and ground.	I
Step 1. If voltage is as specified, go to 3.	

TABLE NO. 9-2-21-4. IR Light Does Not Go On. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 2. If voltage is not as specified, troubleshoot instrument panel and console indicator lights dimming (PARA 9-2-10). Go to 6. 3. Check continuity between J142-B and P197-T. Step 1. If continuity is present, go to 4. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 6. 4. Check continuity between J197-T and K1-A1. Step 1. If continuity is present, go to 5. Step 2. If continuity is not present, replace auxiliary searchlight control panel (PARA 9-4-100). Go to 6. 5. Press FILTER push-to-test switch and check continuity between J197-A and K1-A2. Step 1. If continuity is present, replace burnt out lamps (PARA 9-4-99). Go to 6. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 6. 6. Procedure completed.

TABLE NO. 9-2-21-5. Auxiliary Searchlight Beam Does Not Go Up.

TEST OR INSPECTION	CORRECTIVE ACTION
	1. With control switch in UP position, check for 28 VDC between P3-A and P3-B. Step 1. If voltage is as specified, go to 2. Step 2. If voltage is not as specified, go to 3. 2. With control switch in UP position, check continuity between P3-B and ground. Step 1. If continuity is as specified, replace gimbal assembly (PARA 9-4-102). Go to 5. Step 2. If continuity is not as specified, go to 4.

TABLE NO. 9-2-21-5. Auxiliary Searchlight Beam Does Not Go Up. (Cont)

TEST OR INSPECTION CORRECTIVE ACTION
3. Check continuity between P197-K and P3-A. Step 1. If continuity is present, replace auxiliary searchlight control panel (PARA 9-4-100). Go to 5. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 5. 4. Check continuity between P197-L and P3-B. Step 1. If continuity is present, replace auxiliary searchlight control panel (PARA 9-4-100). Go to 5. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 5. 5. Procedure completed.

TABLE NO. 9-2-21-6. Auxiliary Searchlight Beam Does Not Go Down.

TEST OR INSPECTION CORRECTIVE ACTION
1. With control switch in DOWN position, check for 28 VDC between P3-B and P3-A. Step 1. If voltage is as specified, replace gimbal assembly (PARA 9-4-102). Go to 2. Step 2. If voltage is not as specified, replace auxiliary searchlight control panel (PARA 9-4-100). Go to 2. 2. Procedure completed.

TABLE NO. 9-2-21-7. Auxiliary Searchlight Beam Does Not Go Left.

TEST OR INSPECTION	CORRECTIVE ACTION
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1. With control switch in L position, check for 28 VDC between P3-C and P3-D.	
---	--

Step 1. If voltage is as specified, go to 2.	
--	--

Step 2. If voltage is not as specified, go to 3.	
--	--

2. With control switch in L position, check continuity between P3-D and ground.	
---	--

Step 1. If continuity is as specified, replace gimbal assembly (PARA 9-4-102).	
--	--

Go to 5.	
----------	--

Step 2. If continuity is not as specified, go to 4.	
---	--

3. Check continuity between P197-M and P3-C.	
--	--

Step 1. If continuity is present, replace auxiliary searchlight control panel (PARA 9-4-100). Go to 5.	
--	--

Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 5.	
--	--

4. Check continuity between P197-N and P3-D.	
--	--

Step 1. If continuity is present, replace auxiliary searchlight control panel (PARA 9-4-100). Go to 5.	
--	--

Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 5.	
--	--

5. Procedure completed.	
-------------------------	--

TABLE NO. 9-2-21-8. Auxiliary Searchlight Beam Does Not Go Right.

TEST OR INSPECTION	CORRECTIVE ACTION
--------------------	-------------------

1. With control switch in R position, check for 28 VDC between P3-D (+) and P3-C (-).	
---	--

Step 1. If voltage is as specified, replace gimbal assembly (PARA 9-4-102).	
---	--

Go to 2.	
----------	--

Step 2. If voltage is not as specified, replace auxiliary searchlight control panel (PARA 9-4-100). Go to 2.	
--	--

TABLE NO. 9-2-21-8. Auxiliary Searchlight Beam Does Not Go Right. (Cont)

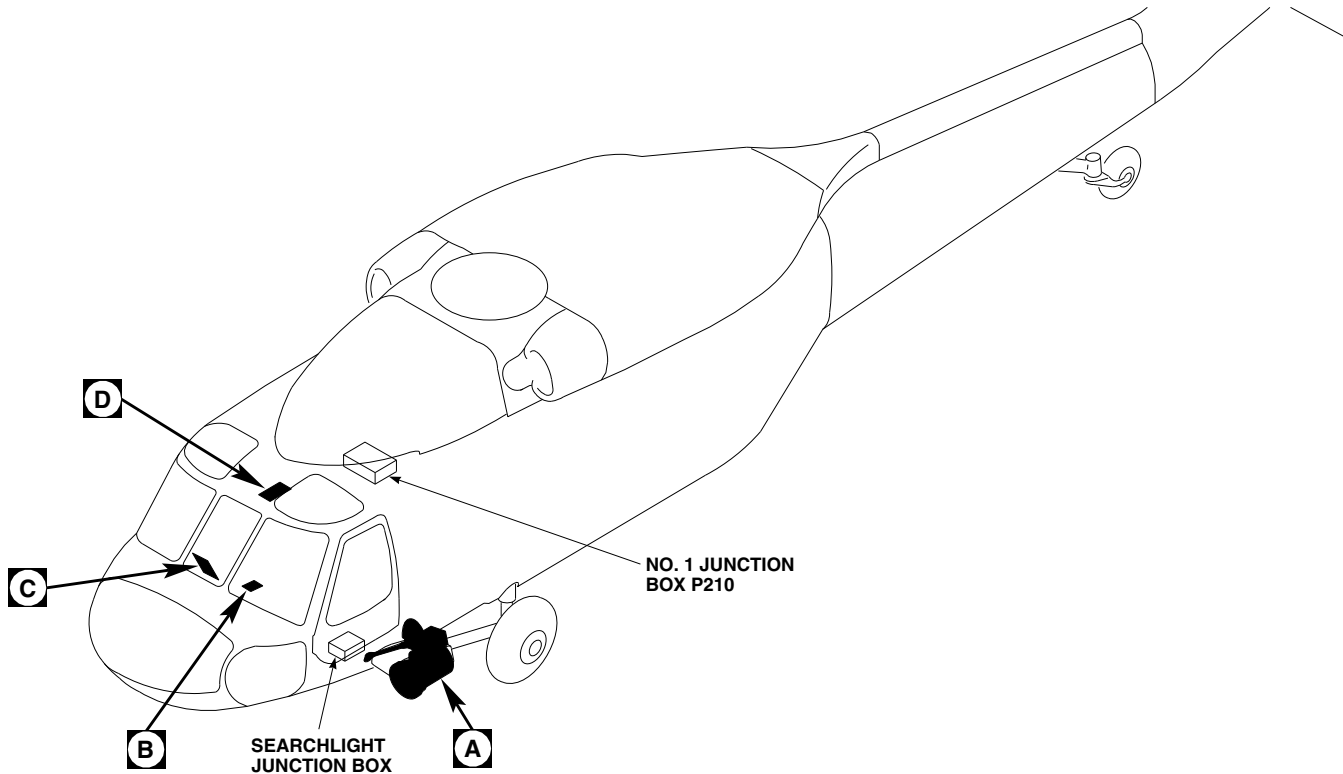
TEST OR INSPECTION	CORRECTIVE ACTION
2. Procedure completed.	

TABLE NO. 9-2-21-9. Auxiliary Searchlight Lamp Does Not Illuminate.

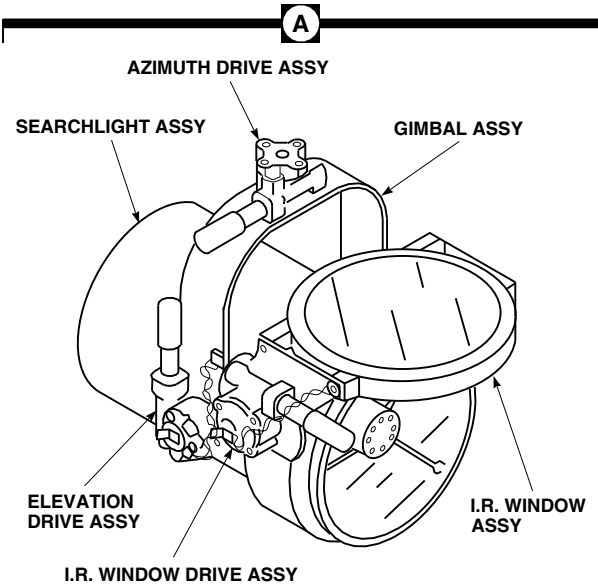
TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for 28 VDC between P1-A and P1-C.	Step 1. If voltage is as specified, go to 2. Step 2. If voltage is not as specified, go to 4.
2. With POWER switch held to START position, check for 28 VDC between P1-B and P1-C.	Step 1. If voltage is as specified, replace auxiliary searchlight assembly (PARA 9-4-103). Go to 5. Step 2. If voltage is not as specified, go to 3.
3. Check continuity between P1-B and P197-B.	Step 1. If continuity is present, replace auxiliary searchlight control panel (PARA 9-4-100). Go to 5. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 5.
4. Check for 28 VDC between power stud PWR2 and ground.	Step 1. If voltage is as specified, repair/replace wiring between power stud PWR2 and P1-A (Appendix F). Go to 5. Step 2. If voltage is not as specified, replace auxiliary searchlight junction box (PARA 9-4-101). Go to 5.
5. Procedure completed.	

TABLE NO. 9-2-21-10. Auxiliary Searchlight Beam Focus Switch Does Not Change Size Of Beam.

TEST OR INSPECTION	CORRECTIVE ACTION
1. With BEAM/FOCUS switch pressed, check for 28 VDC between P1-D and P1-C.	Step 1. If voltage is as specified, replace auxiliary searchlight assembly (PARA 9-4-103). Go to 3. Step 2. If voltage is not as specified, go to 2.
2. Check continuity between P197-D and P1-D.	Step 1. If continuity is as specified, replace auxiliary searchlight control panel (PARA 9-4-100). Go to 3. Step 2. If continuity is not as specified, repair/replace wiring (Appendix F). Go to 3.
3. Procedure completed.	



TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
GND001	SEARCHLIGHT JUNCTION BOX
P118 / J118	CAUTION / ADVISORY PANEL
P142 / J142	LOWER CONSOLE DISCONNECT
P197 / J197	AUX SEARCHLIGHT CONTROL PANEL
P198 / J198	SEARCHLIGHT DISCONNECT
P210 / J210	NO. 1 JUNCTION BOX DISCONNECT
P217 / J217	LH MAIN DISCONNECT
P240	DC BUS TIE CONTACTOR RELAY
P246 / J246	UPPER CONSOLE DISCONNECT
P913 / J913	LOWER CONSOLE DISCONNECT
P914 / J914	LOWER CONSOLE DISCONNECT
PWR1	SEARCHLIGHT JUNCTION BOX
PWR2	SEARCHLIGHT JUNCTION BOX
TS1	SEARCHLIGHT JUNCTION BOX



SEARCHLIGHT, GIMBAL, AND
I.R. WINDOW ASSEMBLIES

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FIGURE 9-2-21-1. AUXILIARY CONTROLLABLE SEARCHLIGHT LOCATION DIAGRAM
(SHEET 1 OF 2)

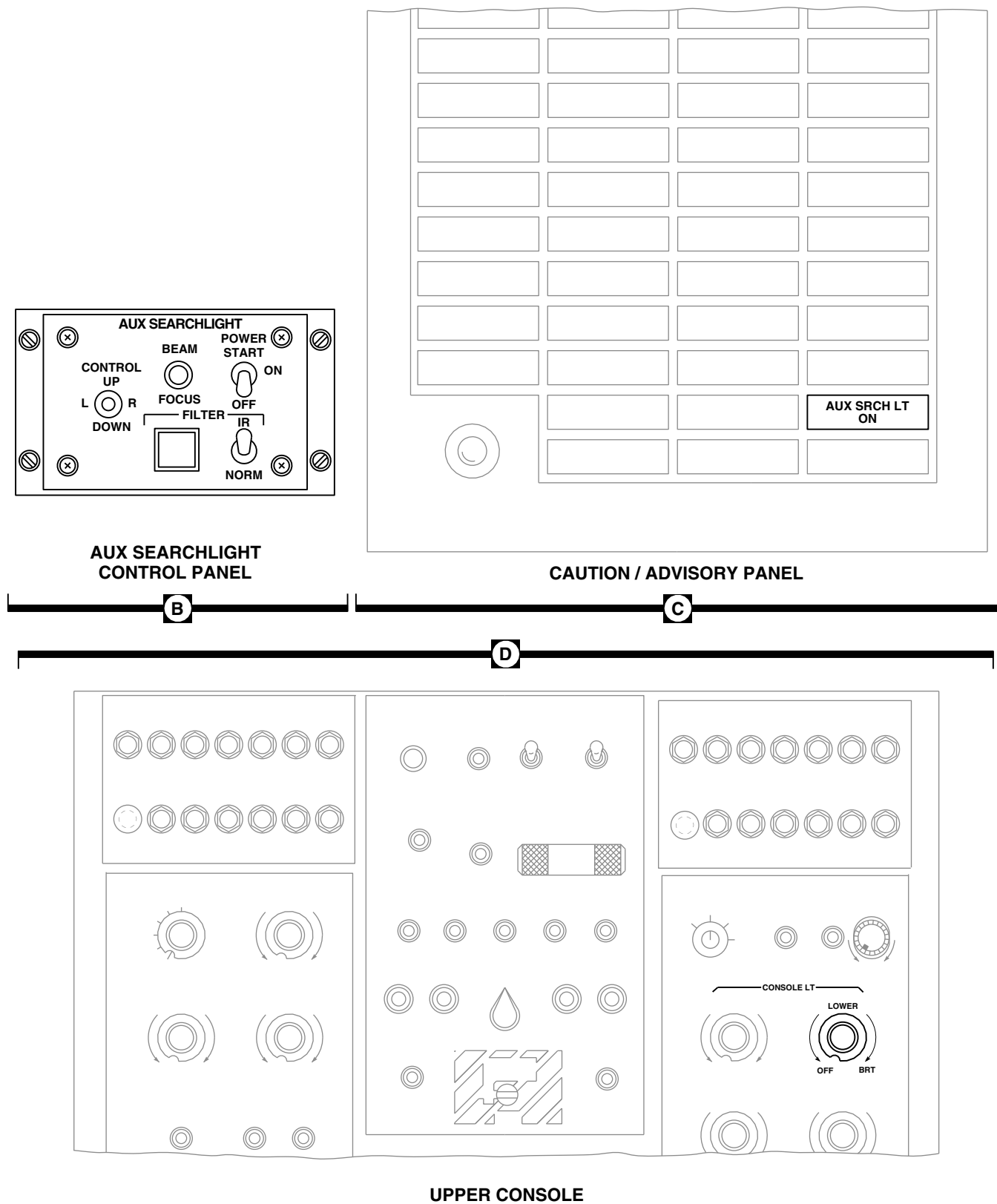


FIGURE 9-2-21-1. AUXILIARY CONTROLLABLE SEARCHLIGHT LOCATION DIAGRAM (SHEET 2 OF 2)

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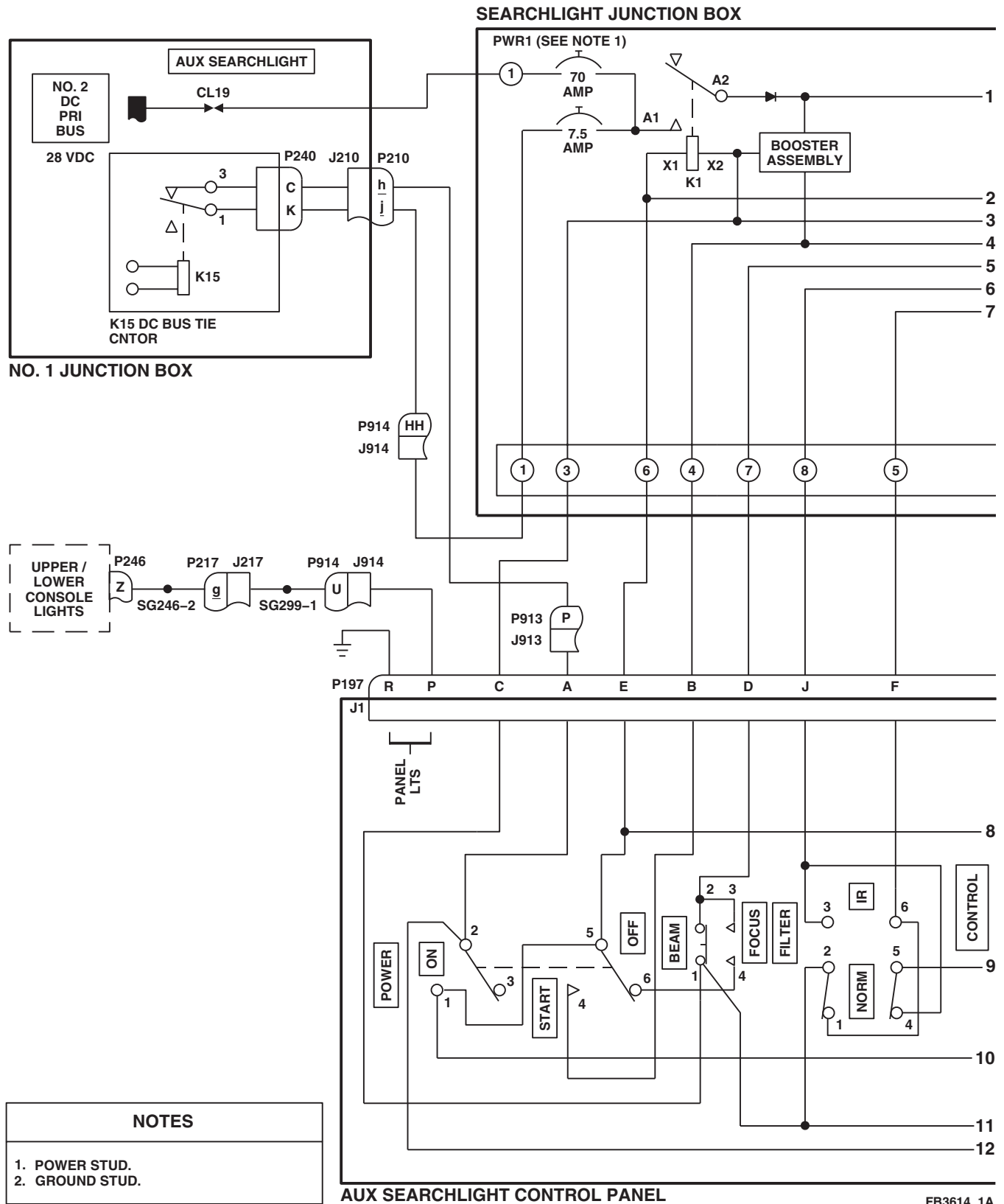


FIGURE 9-2-21-2. AUXILIARY CONTROLLABLE SEARCHLIGHT SCHEMATIC DIAGRAM (SHEET 1 OF 2)

SEARCHLIGHT JUNCTION BOX

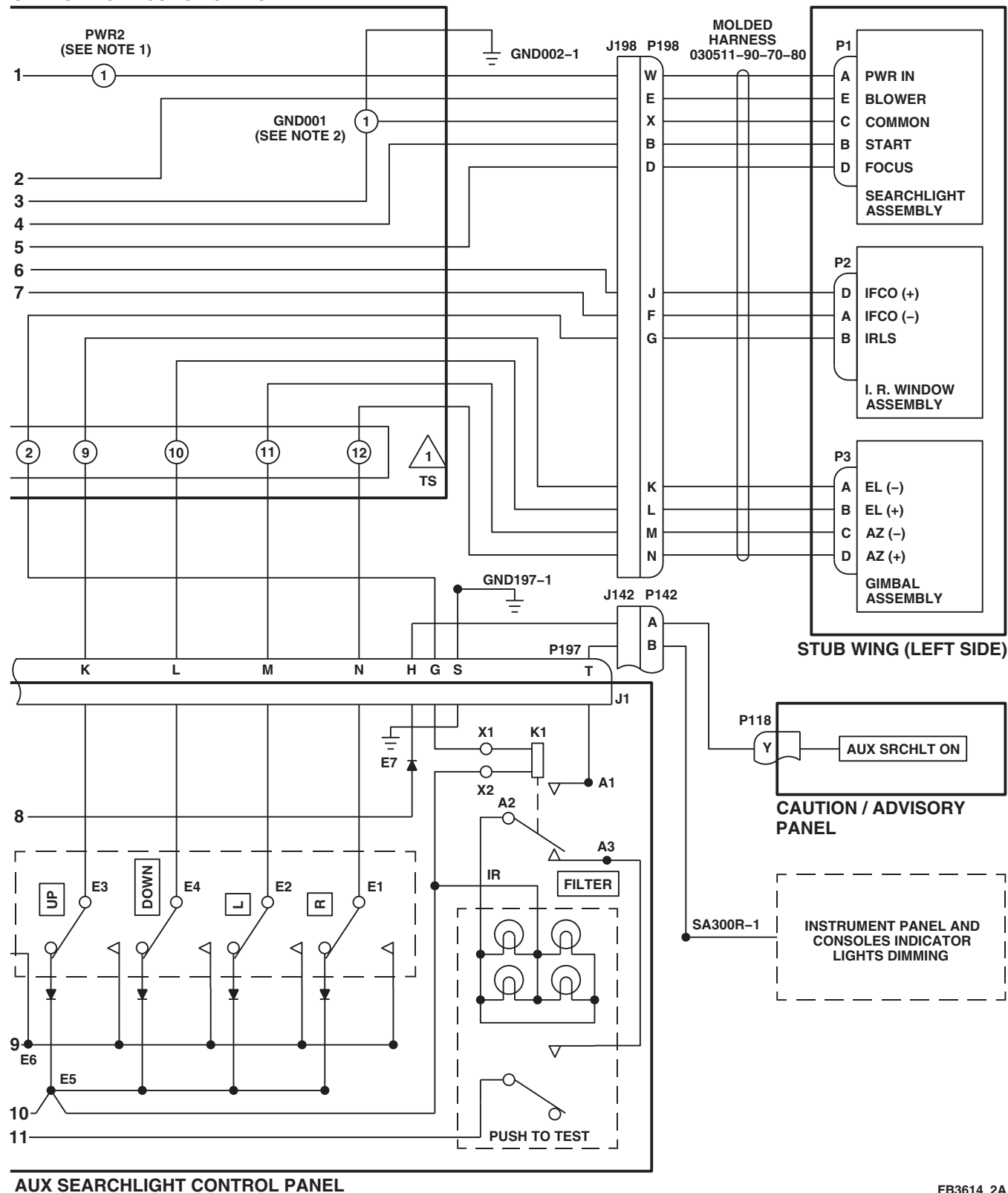
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FIGURE 9-2-21-2. AUXILIARY CONTROLLABLE SEARCHLIGHT SCHEMATIC DIAGRAM
(SHEET 2 OF 2)

9-2-22 NICKEL CADMIUM BATTERY (BENCH).

INITIAL SETUP

Test Equipment

- Battery Analyzer/Charger, Cristie Model RF80-K, 121630-001
- Battery Analyzer/Charger, PCA-131 50/60
- Multimeter, Fluke 45 (3 required)

Tools

- Apron, Impermeable, Battery Worker's
- Electrolyte Level Checker, Locally-Made, Figure H-242, Appendix H
- Electronic Equipment Toolkit
- Face Shield, Industrial
- Knife Switch, 60 Amp, 250 VAC, A-1142
- Shorting Jumpers, Locally-Made (15 Required), Figure H-246, Appendix H
- Shorting Jumpers, 1-ohm, 2 watts, Locally-Made (5 Required), Figure H-246, Appendix H
- Torque Wrench, GGG-W-00686
- Vent Cap Wrench, VCW-001
- Variable Resistor, AA66

Materials/Parts

- Cheesecloth, Item 90, Appendix D
- Corrosion Preventive Compound, Item 114, Appendix D
- Cotton Swabs, Item 119, Appendix D
- Distilled Water, Item 123, Appendix D

- Dry-Cleaning Solvent, Item 124, Appendix D
- Gloves, Rubber, Item 265, Appendix D
- Nylon Bristle Brush, Item 87, Appendix D

Equipment Conditions

- Bench Power Available, 115 VAC, 50/60 Hz, Single Phase
- Battery Removed From Helicopter

Special Instructions

Battery servicing is subdivided into 50 HOUR INSPECTION (Paragraphs 1 through 6) and 100 HOUR INSPECTION (Paragraphs 1 through 12) as follows:

- (1) Inspection
- (2) Cleaning
- (3) Electrolyte Level Check/Adjustment
- (4) Current Leakage Check
- (5) Topping Charge (Using PCA-131 50/60 Battery Analyzer/Charger)
- (6) Topping Charge (Using RF80-K Battery Analyzer/Charger)
- (7) Battery Harness Check
- (8) Clamping Diode Check
- (9) Discharge-Primary Method
- (10) Discharge-Alternate Method
- (11) Normal Charge (Using PCA-131 50/60 Battery Analyzer/Charger)
- (12) Normal Charge (Using RF80-K Battery Analyzer/Charger)

9-2-22.1 Inspection.

WARNING

The electrolyte in the battery is very caustic and will cause serious burns. Wear protective clothing when working around batteries. In case of accident, flush with water. Do not wear jewelry that may contact cell terminals.

CAUTION

Do not perform maintenance on a nickel-cadmium battery in a lead acid shop. Never use tools or material which have been used on lead-acid batteries.

- a. Remove battery case cover.
- b. Inspect cells for correct polarity installation.

RESULT: Cell shall have correct polarity installation.

- If cell is improperly installed, reinstall cell and perform battery reconditioning (PARA 9-5-31).
- c. Inspect cell for swelling or physical damage.

RESULT: Cell shall have no swelling or physical damage.

- If result is not as specified, replace cell (PARA 9-5-26).

NOTE

If five or more cells are found to be defective, replace battery.

- d. Inspect cells and area around cells for leakage.

RESULT: There shall be no leakage.

- If result is not as specified, clean as required (PARA 9-2-22.2).

WARNING

Use extreme care when tightening cell terminal nuts. Serious injury and damage to equipment may result if torque wrench short circuits cell terminal nuts.

- e. Inspect vent caps for corrosion or foreign matter.

RESULT: There shall be no corrosion or foreign matter.

- If result is not as specified, clean or replace (PARA 9-2-22.2).
- f. Inspect intercell connections for corrosion, burn marks, excessive spewage, and discoloration.

RESULT: There shall be no corrosion, burn marks, excessive spewage, or discoloration.

- If result is not as specified, clean intercell connection (PARA 9-2-22.2).
- g. Apply a light coat of corrosion preventive compound on cell terminals and intercell connectors (PARA 9-2-22.2).
- h. **TORQUE CELL TERMINAL NUTS TO 15 - 18 INCH-POUNDS.**
- i. **TORQUE VENT CAPS TO 8 INCH-POUNDS.**
- j. Inspect battery harness for white deposits.

RESULT: There shall be no white deposits on battery harness.

- If result is not as specified, clean battery harness (PARA 9-2-22.2).
- k. Inspect battery harness for fraying or broken insulation.

RESULT: There shall be no fraying or broken insulation on battery harness.

- If result is not as specified, replace battery harness (PARA 9-5-21).
- l. Inspect external electrical receptacles for corrosion, burn marks, and condition of pins.

RESULT: There shall be no corrosion, burn marks, or damaged pins.

- If result is not as specified, replace receptacles (PARA 9-5-22).
- m. Inspect cover, cover latches, and cover gaskets for cleanness.

RESULT: The cover, cover latches, and cover gaskets shall be clean.

- If result is not as specified, clean as required (PARA 9-2-22.2).
- n. Inspect cover, cover latches, and cover gaskets for serviceability.

RESULT: Cover, cover latches, and cover gaskets shall be serviceable.

- If result is not as specified, replace cover, cover latches, or cover gaskets (PARA 9-5-29).

9-2-22.2 Cleaning.



- Do not use wire brush to clean any part of battery.
 - Do not rub protective coating off internal hardware.
- a. **TORQUE VENT CAPS TO 8 INCH-POUNDS.**
 - b. Loosen white deposits from cell tops using stiff, nylon bristle brush.



The water used to wash the battery may become caustic; avoid contact with skin.



Do not allow water to enter the cell when washing, or electrolyte will be contaminated.

- c. Tilting battery at a 45° angle, flush white deposits from cell tops using distilled or tap water.
- d. Flush vent cap tops with tap water.
- e. Using cheesecloth moistened with distilled water, remove white deposits from battery harness.
- f. Dry cleaned areas with cheesecloth and/or compressed air at less than 30 psi.
- g. After cleaning, apply a light protective coat of corrosion preventive compound, to all exposed metal on cell tops.

9-2-22.3 Electrolyte Level Check/Adjustment.



The electrolyte in the battery is very caustic and will cause serious burns. Wear protective clothing when working around batteries. In case of accident, flush with water. Do not wear jewelry that may contact cell terminals.

CAUTION

- Do not perform any maintenance on a nickel cadmium battery in a lead acid shop. Never use tools or materials which have been used on lead acid batteries.
- Electrolyte level adjustment must be accomplished within 1 hour after battery is fully charged or after helicopter flight. Do not add water when battery is in a discharged state (any cell below 1.28 VDC open circuit).

NOTE

Remove one vent cap at a time and replace immediately after electrolyte level is checked/adjusted.

- a. Using vent cap wrench, remove one vent cap assembly at a time.
- b. Inspect vent cap and O-ring for damage.

RESULT: Vent cap and O-ring shall not be damaged.

- If result is not as specified, replace O-ring or vent cap (PARA 9-5-27).
- c. Clean vent cap assembly by sloshing in warm water.
- d. Submerge vent cap in water and apply 3 - 10 psi of oil free compressed air to the bottom of the cap until bubbles are observed emerging from beneath the rubber collar vent slot.
- e. Gently insert locally-made electrolyte level checking tube into cell until it rests on top of plates.

NOTE

Use distilled water only. Do not overfill or add potassium hydroxide (electrolyte).

- f. Place a finger over open end of tube and remove it from cell. Note height of electrolyte in tube.

RESULT:

- (1) Electrolyte level for blacktop reference cell (No. 10) shall be 15/16 to 1-1/16 inch.
- (2) Electrolyte level for regular cells shall be 5/16 to 7/16 inch.
- If electrolyte level is too low in any cell, add distilled water as required.
- If electrolyte level is too high in any cell, remove excess with hollow tube.

- g. Install vent cap assembly.

- h. **TORQUE VENT CAP TO 8 INCH-POUNDS.**

9-2-22.4 Current Leakage Check.

- a. Measure current leakage between positive (+) battery terminal and battery case latch, then between negative (-) battery terminal and battery case latch.

RESULT: Current leakage shall not be greater than 50 mA.

- If result is not as specified, perform the following steps:
 - (1) Clean battery cell tops and intercell connectors.
 - (2) Recheck current leakage.
 - (3) If trouble remains, go to step b.
 - (4) If result is as specified, check completed.

- b. Set multimeter function switch to read VDC.

- c. Place one meter probe on battery latch and the other probe on one of the battery terminals. If multimeter indicates negative polarity, reverse probes.

- d. Keeping meter probe on battery latch, move other probe counterclockwise from cell terminal to cell terminal, beginning with the cell closest to receptacle J2 positive terminal.

RESULT: Voltage shall slowly decrease to 0 VDC.

- If voltage goes negative before reaching the last cell, replace the leaking cell at the changeover point (PARA 9-5-26).
- If trouble remains, repair/clean battery case and cover.

9-2-22.5 Topping Charge (Using PCA-131 50/60 Battery Analyzer/Charger).

NOTE

- Perform steps a. through j. if part of 50 HOUR INSPECTION.
 - If 100 HOUR INSPECTION is being performed, go to Battery Harness Check.
 - Battery analyzer/charger is set up to operate with 115 VAC, 50-60 Hertz, single phase. A 12 amp fuse must also be used.
- a. For 50 cycle (Hz) operation, set TIMER control to S2 and connect power cord of battery analyzer to electrical source.
 - b. Connect power cord of battery analyzer/charger to electrical source.
 - c. Turn SELECTOR switch to MA-500H.
 - d. Connect P2B battery cable between battery and battery analyzer/charger.
 - e. Set AC POWER switch to ON.

RESULT: AC POWER light goes on.

- If result is not as specified, check ac power fuse.

- (1) If fuse is not good, replace fuse.
- (2) If fuse is good, turn battery analyzer/charger in for repair.

- f. Set AUTOMATIC CYCLE-MANUAL DISCHARGE switch to AUTOMATIC CYCLE.
- g. Turn TIMER control to TOPPING charge and record start time.
- h. Periodically monitor total cell voltage reading on battery analyzer/charger.

RESULT: Charging shall be complete when battery voltage reaches 30 VDC, or 30 minutes elapse.

- If battery does not reach 30 VDC within 30 minutes, perform battery deep cycle (PARA 9-5-32).
- i. Install battery case cover.
 - j. Return battery to service.

9-2-22.6 Topping Charge (Using RF80-K Battery Analyzer/Charger).

NOTE

- Perform steps a. through i. if part of 50 HOUR INSPECTION.
 - If 100 HOUR INSPECTION is being performed, go to Battery Harness Check.
- a. Connect power cord to battery analyzer/charger and then to electrical source, if not already done.
 - b. Adjust battery analyzer/charger CHARGE CURRENT to 1.0 ampere.
 - c. Set battery analyzer/charger REFLEX - CELLS NICKEL CADMIUM to 19.
 - d. Connect P2B battery cable between battery and battery analyzer/charger.

- e. Set battery analyzer/charger SET CHARGE TIME for 1 hour.
- f. Turn battery analyzer/charger AC POWER switch to ON.

RESULT: Charging shall be complete when battery analyzer/charger CYCLE COMPLETE light goes on.

- g. Turn battery analyzer/charger AC POWER switch to OFF and disconnect battery cable.
- h. Connect digital multimeter to battery and measure for voltage.

RESULT: Digital multimeter shall read between 30.95 and 31.05 VDC. If result is not as specified, replace battery.

- i. Install battery case cover.
- j. Return battery to service.

9-2-22.7 Battery Harness Check.

NOTE

A 50 HOUR INSPECTION (PARA 9-2-22.1 thru 9-2-22.6) must be completed before performing a 100 HOUR INSPECTION (PARA 9-2-22.1 thru 9-2-22.12).

- a. Check continuity between J1-E and J1-F.

RESULT: Continuity shall be present.

- If result is not as specified, replace battery harness (PARA 9-5-21).

- b. Check resistance between J1-E and each of intercell connectors with overtemperature switches mounted.

RESULT: Resistance shall be greater than 1 megohm.

- If result is not as specified, replace battery harness (PARA 9-5-21).

- c. Check resistance between J1-A and J1-B, and J1-J and J1-K.

RESULT: Resistance shall be within limits referenced in CHART NO. 9-2-22-1.

- If result is not as specified, replace temperature sensor (PARA 9-5-23).
- If trouble remains, replace battery harness (PARA 9-5-21).

- d. With No. 1 temperature sensor mounted, check resistance between J1-A and intercell connector.

RESULT: Resistance shall be greater than 1 megohm.

- If result is not as specified, replace battery harness (PARA 9-5-21).

- e. With No. 2 temperature sensor mounted, check resistance between J1-J and intercell connector.

RESULT: Resistance shall be greater than 1 megohm.

- If result is not as specified, replace battery harness (PARA 9-5-21).

CAUTION

Touch only pins specified in steps f. and g. Touching the sides of receptacle J1 or any other pins may cause battery harness to short out.

- f. Check continuity between receptacle J1-G and positive (+) terminal of receptacle J2.

RESULT: Continuity shall be present.

- If result is not as specified, replace battery harness (PARA 9-5-21).

- g. Check continuity between receptacle J1-D and negative (-) terminal of receptacle J2.

RESULT: Continuity shall be present.

- If result is not as specified, replace battery harness (PARA 9-5-21).
- h. Check continuity between receptacle J1-C and positive (+) terminal of blacktop reference cell (No. 10).

RESULT: Continuity shall be present.

- If result is not as specified, replace battery harness (PARA 9-5-21).
- i. Check continuity between receptacle J1-H and negative (-) terminal of blacktop reference cell (No. 10).

RESULT: Continuity shall be present.

- If result is not as specified, replace battery harness (PARA 9-5-21).

9-2-22.8 Clamping Diode Check.

- a. Remove cell terminal nut and Belleville washer attaching clamping diode cathode terminal wire (red) to cell (No. 9) negative (-) terminal.
- b. Check resistance in both directions between cathode wire terminal lug and intercell connecting/diode mounting link.

RESULT: Resistance in one direction shall be 10 times greater than in other direction.

- If result is not as specified, replace clamping diode (PARA 9-5-25).

NOTE

Install Belleville washer with concave side down.

- c. Place cathode terminal wire lug on cell (No. 9) negative terminal and install Belleville washer and cell terminal nut.
- d. **TORQUE CELL TERMINAL NUT TO 15-18 INCH-POUNDS.**

9-2-22.9 Discharge-Primary Method.

WARNING

The electrolyte in the battery is very caustic and will cause serious burns. Wear protective clothing when working around batteries. In case of accident, flush with water. Do not wear jewelry that may contact cell terminals.

CAUTION

- Do not loosen or remove vent caps during discharge. Open cells may cause the formation of potassium carbonate.
- When discharging battery, cells with voltages of 0.5 VDC or less should be shorted to prevent cell polarity reversal.
- Do not discharge battery unless each individual cell has at least 1/16 inch of electrolyte above plates. Add distilled water, only, if necessary.
- Never discharge batteries that have cells connected in parallel.

NOTE

Battery analyzer/charger PCA-131 50/60 is set up to operate with 115 VAC, 50-60 Hertz, single phase. A 12 amp fuse must also be used.

- a. For 50 cycle (Hz) operation, set TIMER control to S2 and connect power cord of battery analyzer to electrical source.
- b. Turn SELECTOR switch to MA-500H.
- c. Connect P2B battery cable between battery and battery analyzer/charger.
- d. Set AC POWER switch to ON.

RESULT: AC POWER light goes on.

- If result is not as specified, check ac power fuse.

(1) If fuse is not good, replace fuse.

(2) If fuse is good, turn battery analyzer/charger in for repair.

e. Set AUTOMATIC CYCLE-MANUAL DISCHARGE switch to AUTOMATIC CYCLE.

f. Turn TIMER control to DISCHARGE.

RESULT: Yellow END OF CYCLE light shall go off.

- If result is not as specified, turn in battery analyzer/charger for repair.

NOTE

- Battery shall discharge for approximately 2 hours and TIMER control will automatically advance clockwise.
- If step g. is not followed, the battery analyzer/charger will automatically start the charging cycle.

g. When BATTERY LOW light goes on, automatic cycle will stop. Set AUTOMATIC CYCLE-MANUAL DISCHARGE switch to MANUAL DISCHARGE.

h. Continue to discharge battery until reading drops below 10 VDC.

NOTE

If it is necessary to discharge battery to extinction, perform steps i. through n. Otherwise, go to step o.

i. When the battery is discharged, keep the AUTOMATIC CYCLE-MANUAL DISCHARGE switch in MANUAL DISCHARGE position.

NOTE

The blacktop reference cell (No. 10) will discharge at a faster rate than regular cells. When monitoring cell voltages, check reference cell voltage often. As reference cell voltage approaches 0.3 VDC, closely monitor voltage.

j. Using multimeter, monitor individual cell voltages.

k. When reference cell reaches 0.3 VDC, connect a locally-made shorting jumper across cell terminals.

l. When individual regular cells reach 0.5 VDC, connect locally-made shorting jumper across cell terminals.

m. Discharge cells until 75% (15) of cells are shorted with jumpers. Place a 1 ohm, 2 watt shorting jumper across terminals of each of the remaining cells.

CAUTION

With battery completely discharged, electrolyte is absorbed into cell plates and battery appears dry. Do not add water.

n. Turn off and disconnect battery analyzer/charger and allow battery to remain shorted for a minimum of 3 hours.

o. Perform battery charge (PARA 9-2-22.11 or PARA 9-2-22.12).

9-2-22.10 Discharge-Alternate Method.

WARNING

The electrolyte in the battery is very caustic and will cause serious burns. Wear protective clothing when working around batteries. In case of accident, flush with water. Do not wear jewelry that may contact cell terminals.

CAUTION

- Never discharge batteries that have cells connected in parallel.
 - When discharging battery, cells with voltages of 0.5 VDC or less should be shorted to prevent cell polarity reversal.
 - Do not loosen or remove vent caps during discharge. Open cells may cause the formation of potassium carbonate.
 - Do not discharge battery unless each individual cell has at least 1/16 inch of electrolyte above plates. Add distilled water, only, if necessary.
- a. Connect positive probe of multimeter (M1) to positive (+) terminal of regular cell and negative probe of multimeter to negative (-) terminal of regular cell.
 - b. Connect second multimeter (M2) as follows: connect positive probe of multimeter to positive (+) terminal of reference cell and negative probe of multimeter to negative (-) terminal of blacktop reference cell (No. 10).
 - c. Connect one lead of variable resistor (R1) to negative (-) terminal of reference cell, and other lead of variable resistor (R1) to negative (-) terminal of a third multimeter (M3).
 - d. Connect positive (+) terminal of reference cell to one pole of knife switch. Connect other pole of knife switch to positive (+) terminal of third multimeter (M3).

- e. Turn variable resistor (R1) to about mid-scale.

NOTE

To hold a constant 6.0 amp discharge rate, variable resistor (R1) will require periodic adjustment as battery voltage level decreases.

- f. Close knife switch and adjust variable resistor (R1) for a 6.0 amp discharge current rate as indicated on multimeter (M3).

NOTE

The blacktop reference cell (No. 10) will discharge at a faster rate than the regular cells. Check reference cell voltage often when monitoring regular cell voltages in steps g. and h. As reference cell voltage approaches 0.3 VDC, closely monitor voltage.

- g. Using multimeter (M2), monitor reference cell voltage and connect locally-made shorting jumper across cell when voltage is between -0.6 VDC and 0.3 VDC.
- h. Using multimeter (M1), monitor regular cell voltage and connect locally-made shorting jumper across each cell when voltage approaches 0.5 VDC.

CAUTION

With battery completely discharged, electrolyte is absorbed into cell plates and battery appears dry. Do not add water.

NOTE

Cells shall remain shorted together until battery is prepared for either charging or storage.

- i. When all battery cells have been shorted, disconnect test equipment, and let shorted battery stand for a minimum of 3 hours.

9-2-22.11 Normal Charge (Using PCA-131 50/60 Battery Analyzer/Charger).

WARNING

Explosive gases may be released during charging. Make sure that charging area is well ventilated. Never use matches or open flame. Avoid short circuit arcing which may cause explosion. Never disconnect charging cable from battery until battery analyzer/charger is off.

CAUTION

- Battery case should never be more than slightly warm to the touch (100°F/ 37.8°C) during charging. If battery exhibits overtemperature condition during charging, stop charging, discharge battery, and restart charging.
 - Charge battery at ambient temperature between 40°F - 90°F (4.4°C - 32.2°C).
 - Do not loosen or remove vent caps during discharge. Open cells may cause the formation of potassium carbonate.
 - Never charge batteries with cells connected in parallel.
- a. Disconnect all shorting jumpers from battery cells (if used).

NOTE

Battery analyzer/charger PCA-131 50/60 is set up to operate with 115 VAC, 50 - 60 Hertz, single phase. A 12 amp fuse must also be used.

- b. For 50 cycle (Hz) operation, set TIMER control to S2 and connect power cord of battery analyzer to electrical source.
- c. Turn selector switch to MA-500H.

- d. Connect red/white cable lead of battery analyzer/charger battery cable P2B to positive (+) terminal riser on back of battery J2 connector, and black cable lead of battery analyzer/charger battery cable P2B to negative (-) terminal riser.

- e. **TORQUE CELL TERMINAL NUTS TO 15 - 18 INCH-POUNDS.**

- f. Set AC POWER switch to ON.

RESULT: AC POWER light shall go on.

- If result is not as specified, check ac power fuse.

(1) If fuse is not good, replace fuse.

(2) If fuse is good, turn in battery analyzer/charger for repair.

- g. Set AUTOMATIC CYCLE-MANUAL DISCHARGE switch to AUTOMATIC CYCLE.

- h. With battery analyzer/charger off, connect a shorting jumper between cell No. 9 negative (-) terminal and cell No. 11 positive (+) terminal.

NOTE

Automatic operation will not start until battery voltage charges up to 19 VDC (within 5 minutes). At that time the BATTERY LOW light will go off and automatic cycle will begin. The TIMER control will automatically advance through the charging cycle. MAIN charging takes approximately 2 hours, then TOPPING charge will begin, and continue for 2 hours.

- i. Turn TIMER control to MAIN CHARGE (S2). Record start time.

RESULT: BATTERY LOW light will go off within 5 minutes.

- If result is not as specified, turn in battery to designated overhaul facility for repair.

- j. After 2 minutes, quickly check each cell voltage (except cell No. 10) with digital multimeter.

NOTE

If five or more cells are found to be defective, replace battery.

RESULT: Cell voltage shall be between 0.00 and 1.75 VDC.

- If any cell has a voltage over 1.75 VDC, replace cell (PARA 9-5-26).
 - If any cell except cell No. 10 has a voltage of 0.0 VDC, replace cell (PARA 9-5-26).
- k. After 1 hour of MAIN charging, turn battery analyzer/charger off. Remove shorting jumper between cell No. 9 negative (-) terminal and cell No. 11 positive (+) terminal.
- l. Restart battery analyzer/charger MAIN charge. MAIN charge will continue for an additional 1 hour. After 1 hour, switch to TOPPING charge.
- m. After 1 hour and 55 minutes have elapsed on timer during TOPPING charge, check each cell voltage with a digital multimeter.

RESULT:

- (1) Blacktop reference cell (No. 10) voltage shall be between 1.53 and 1.75 VDC.
 - (2) Regular cell voltage shall be between 1.55 and 1.75 VDC.
- If results are not as specified, continue topping charge an additional 30 minutes and repeat voltage check. If voltages are not as specified, discharge and replace cell (PARA 9-5-26).
- n. At the end of the TOPPING charge, with the battery analyzer/charger still on charge,

measure voltage across 10 cells on one side of battery and across 10 cells on opposite side of battery.

RESULT: Voltage difference shall be 0.2 VDC or less.

- If result is not as specified, perform reconditioning (PARA 9-5-31).
- o. Turn off battery analyzer/charger and disconnect battery cable P2B from battery.
- p. Perform Electrolyte Level Check/Adjustment.
- q. After 1 hour, perform Battery Inspection (PARA 9-2-22.1).
- r. Install and latch battery case cover.

9-2-22.12 Normal Charge (Using RF80-K Battery Analyzer/Charger).

WARNING

Explosive gases may be released during charging. Make sure that charging area is well ventilated. Never use matches or open flame. Avoid short circuit arcing which may cause explosion. Never disconnect charging cable from battery until battery analyzer/charger is off.

CAUTION

- Battery case should never be more than slightly warm to the touch (100°F/ 37.8°C) during charging. If battery exhibits overtemperature condition during charging, stop charging, discharge battery, and restart charging.
- Charge battery at ambient temperature between 40°F - 90°F (4.4°C - 32.2°C).
- Do not loosen or remove vent caps during discharge. Open cells may cause the formation of potassium carbonate.

- Never charge batteries with cells connected in parallel.
 - When using battery analyzer/charger RF80H, use the CONSTANT CURRENT mode only.
 - Charge battery in an upright position to avoid spillage and allow proper venting.
- a. Disconnect all shorting jumpers from battery cells (if used).
 - b. Connect power cord to battery analyzer/charger and then to electrical source.
 - c. Connect red/white cable lead of battery analyzer/charger battery cable P2B to positive (+) terminal riser on back of battery J2 connector, and black cable lead of battery analyzer/charger battery cable P2B to negative (-) terminal riser.
 - d. **TORQUE CELL TERMINAL NUTS TO 15 - 18 INCH-POUNDS.**

NOTE

DigiFLEX LED displays relative state of charge and battery condition.

- e. Set Charge Method switch to REFLEX-CELLS NICKEL CADMIUM to 19.
- f. Turn battery analyzer/charger switch to CELL VOLTS and connect leads to CELL VOLTS jacks.
- g. Turn battery analyzer/charger MODE switch to CHARGE.
- h. With battery analyzer/charger off, connect a shorting jumper between cell No. 9 negative (-) terminal and cell No. 11 positive (+) terminal.
- i. Set battery analyzer/charger SET CHARGE TIME for 2 hours.

NOTE

Automatic operation will not start until battery voltage charges up to 19 VDC (within 5 minutes). MAIN charging takes approximately 2 hours, then TOPPING charge will begin, and continue for 2 hours.

- j. Adjust battery analyzer/charger CHARGE CURRENT to 1.0 ampere, and then turn AC POWER ON.
- k. After 2 minutes, quickly check each cell voltage (except cell No. 10) with CELL VOLTS leads.

NOTE

If five or more cells are found to be defective, replace battery.

RESULT: Cell voltage shall be between 0.00 and 1.75 VDC.

- If any cell has a voltage over 1.75 VDC, replace cell (PARA 9-5-26).
- If any cell, except cell No. 10, has a voltage of 0.0 VDC, replace cell (PARA 9-5-26).

- l. Adjust battery analyzer/charger CHARGE CURRENT to 3.0 ampere.
- m. Restart battery analyzer/charger MAIN charge. MAIN charge will continue for an additional 1 hour. After 1 hour, switch to TOPPING charge.
- n. After 1 hour of MAIN charging, turn battery analyzer/charger off. Remove shorting jumper between cell No. 9 negative (-) terminal and cell No. 11 positive (+) terminal.
- o. During final 5 minutes of TOPPING charge, check each cell voltage with the CELL VOLTS leads.

RESULT:

- (1) Blacktop reference cell (No. 10) voltage shall be between 1.60 and 1.75 VDC.

(2) Regular cell voltage shall be between 1.55 and 1.75 VDC.

- If results are not as specified, continue topping charge an additional 30 minutes and repeat voltage check. If voltages are not as specified, discharge and replace cell (PARA 9-5-26).

p. At the end of the TOPPING charge, with the battery analyzer/charger still on charge, measure voltage across 10 cells on one side of battery and across 10 cells on opposite side of battery.

RESULT: Voltage difference shall be 0.2 VDC or less.

- If result is not as specified, perform reconditioning (PARA 9-5-31).

q. Turn off battery analyzer/charger and disconnect battery cable P2B from battery.

r. Perform Electrolyte Level Check/Adjustment (PARA 9-2-22.3).

s. After 1 hour, perform battery Inspection (PARA 9-2-22.1).

t. Install and latch battery case cover.

CHART NO. 9-2-22-1. TEMPERATURE/RESISTANCE

DEGREES	OHMS	DEGREES	OHMS
F°(C°)	±10%	F°(C°)	±10%
50.0(10.0)	1990.3	77.0(25.0)	1000.0
51.8(11.0)	1897.2	78.8(26.0)	957.1
53.6(12.0)	1809.0	80.6(27.0)	916.4
55.4(13.0)	1725.5	82.4(28.0)	877.6
57.2(14.0)	1645.4	84.2(29.0)	840.7
59.0(15.0)	1571.4	86.0(30.0)	805.6
60.8(16.0)	1500.0	87.8(31.0)	772.0
62.6(17.0)	1432.3	89.6(32.0)	740.1
64.4(18.0)	1368.1	91.4(33.0)	709.6
66.2(19.0)	1307.1	93.2(34.0)	680.6
68.0(20.0)	1249.3	95.0(35.0)	653.0
69.8(21.0)	1194.2	96.8(36.0)	626.6
71.6(22.0)	1141.8	98.6(37.0)	601.4
73.4(23.0)	1092.1	100.4(38.0)	577.4
75.2(24.0)	1044.9	102.2(39.0)	554.6

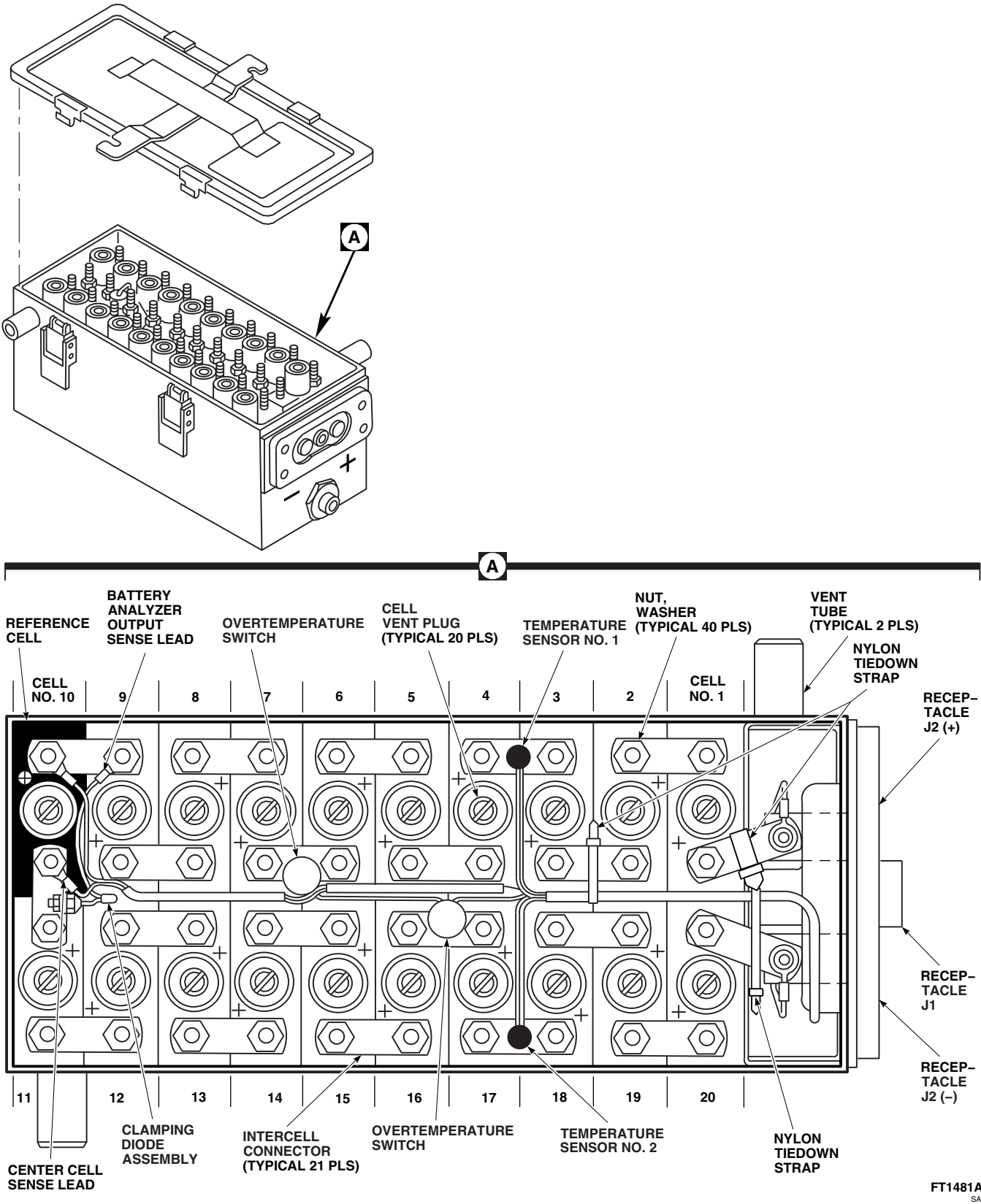


FIGURE 9-2-22-1. NICKEL CADMIUM BATTERY LOCATION DIAGRAM

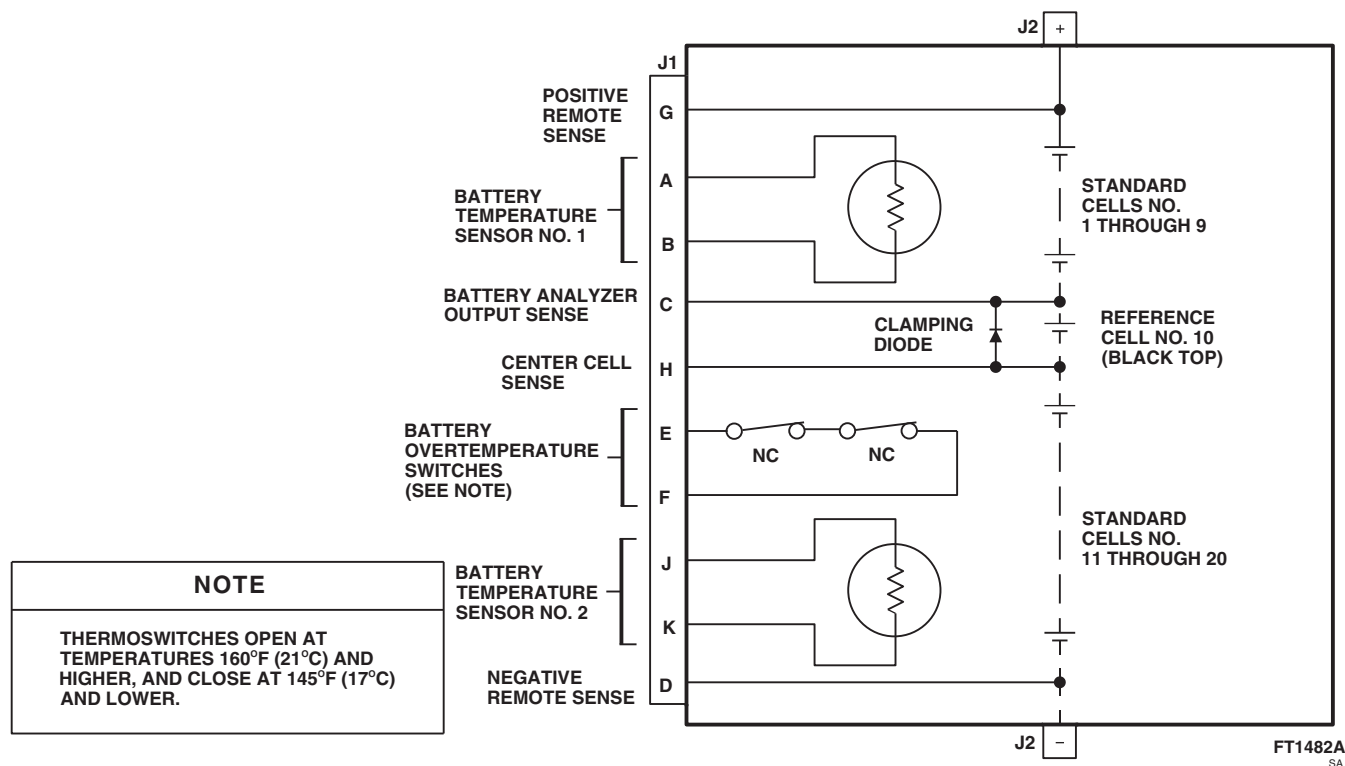


FIGURE 9-2-22-2. NICKEL CADMIUM BATTERY SCHEMATIC DIAGRAM

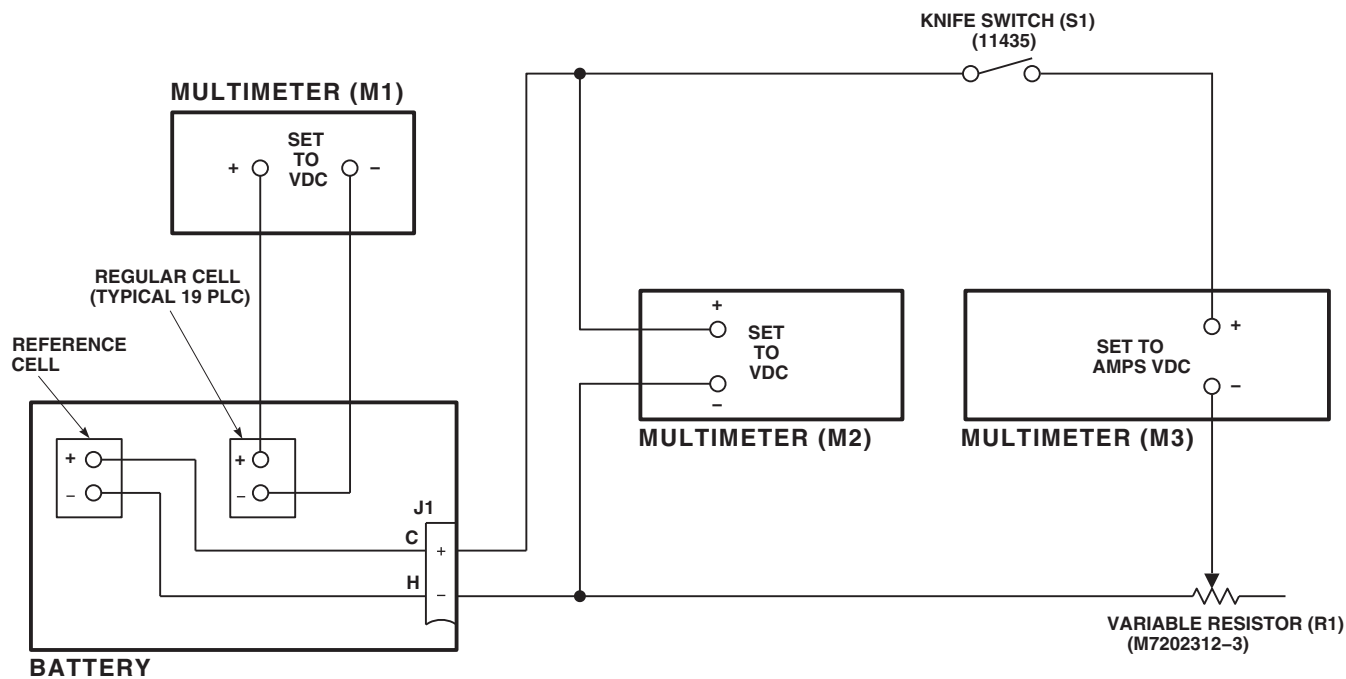


FIGURE 9-2-22-3. NICKEL CADMIUM BATTERY DISCHARGE TEST SETUP

9-2-23 SEALED LEAD ACID BATTERY (BENCH).

INITIAL SETUP

Test Equipment

- Battery Analyzer/Charger, 4159
- Multimeter, Fluke 45

Tools

- Apron, Impermeable, Battery Worker’s
- Electronic Equipment Toolkit
- Face Shield, Industrial
- Torque Wrench, GGG-W-00686

Materials/Parts

- Cheesecloth, Item 90, Appendix D
- Corrosion Preventive Compound, Item 111, Appendix D
- Cotton Swabs, Item 119, Appendix D
- Distilled Water, Item 123, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D

- Gloves, Rubber, Item 265, Appendix D
- Nylon Bristle Brush, Item 87, Appendix D

Equipment Conditions

- Bench Power Available, 115 VAC, 50/60 Hz, Single Phase
- Battery Removed From Helicopter

Special Instructions

This procedure is to be used with older battery charger/analyzer model number 4159. To avoid confusion refer to figure -2.

Battery servicing is subdivided into 50 HOUR INSPECTION (Paragraphs 1 and 2) and 100 HOUR INSPECTION (Paragraphs 1 through 5) as follows:

- (1) Inspection
- (2) Cleaning
- (3) Discharge
- (4) Normal Charge
- (5) Conditioning Charge

9-2-23.1 Inspection.

- a. Inspect battery for swelling or physical damage.

RESULT: Battery shall have no swelling or physical damage.

- If result is not as specified, replace battery.

- b. Inspect case for evidence of leakage.

RESULT: There shall be no evidence of leakage.

- If result is not as specified, replace battery.

- c. Inspect case for evidence of corrosion.

RESULT: There shall be no evidence of corrosion.

- If result is not as specified, clean or replace battery (PARA 9-2-23.2).

- d. Inspect external electrical receptacles for corrosion, burn marks, and condition of pins.

RESULT: There shall be no corrosion, burn marks, or damaged pins.

- If result is not as specified, replace receptacles.

- e. Inspect battery case for cleanness.

RESULT: The battery case shall be clean.

- If result is not as specified, clean as required (PARA 9-2-23.2).

- f. Inspect battery case for serviceability.

RESULT: Battery case shall be serviceable.

- If result is not as specified, replace battery.

9-2-23.2 Cleaning.

- Loosen corrosion from battery case using stiff, nylon bristle brush.
- Tilting battery at a 45° angle, flush corrosion from battery using distilled or tap water.
- Dry cleaned areas with cheesecloth and/or compressed air at less than 30 psi.
- After cleaning, apply a light protective coat of corrosion preventive compound, to all exposed metal on battery.

9-2-23.3 Discharge.

WARNING

Ensure that charging/discharging current is turned OFF before disconnecting or connecting battery terminal connector. High current arcing may cause hydrogen gas generated in the cells to explode. Wear protective clothing when working around batteries. Do not wear jewelry that may contact cell terminals.

CAUTION

- Do not allow battery to remain in discharged state for more than 24 hours.

- Do not perform maintenance on lead acid battery in nickel cadmium battery service shop. Never use tools or material which have been used on nickel cadmium batteries.

NOTE

- Overtemperature indicator lamp will not be operational during discharge cycle. Battery temperature may normally exceed 130°F (54°C). Electrolyte will not normally be consumed during this discharge cycle.
 - If not already completed, perform normal charge procedures to prevent premature rejection of a battery in a low charge state before starting this procedure.
 - Set battery analyzer/charger selector switch to appropriate setting for power source.
- Connect battery analyzer/charger power cord to electrical power source.
 - Connect battery analyzer/charger cable between battery and analyzer/charger.
 - Turn battery analyzer/charger rotary switch to D (discharge) position.
 - Set battery analyzer/charger timer to 60 minutes.
 - Adjust the DISCHARGE CURRENT knob to minimum (CCW) position.
 - Set battery analyzer/charger ON/OFF switch to ON.

RESULT:

- (1) DISCHARGE light shall go on.
- (2) Both the accept and reject lights illuminate.

NOTE

- Audible signal will sound until START button is pressed.
 - VOLTS and AMPS meters shall read all 6's, 8's, or 9's.
- g. Press and hold START button for 2 to 3 seconds. When VOLTS meter reads open circuit voltage and AMPS meter reads at or near 0, slowly adjust DISCHARGE CURRENT knob to obtain 8 amps.

RESULT: After one minute the timer LED display will come on and count upward.

NOTE

Audible signal will sound and all meter readings will freeze and lock at any time when ACCEPT/REJECT parameters are reached.

- h. Monitor voltage (VDC) and discharge current (-A).

RESULT:

- (1) If red (REJECT) LED comes ON indicating voltage of less than 18.0 VDC before 45 minutes of discharge cycle has been completed, discard battery.
- (2) If red (REJECT) LED comes ON indicating voltage of less than 18.0 VDC before TIMER times out, proceed as follows:
 - (a) Allow battery to cool to ambient temperature.
 - (b) Perform normal charge (PARA 9-2-23.4).
 - (c) Wait two hours, then repeat discharge (PARA 9-2-23.3).
 - (d) If battery charger/analyzer indicates REJECT again, replace battery.

- (3) When TIMER times out, if green (ACCEPT) LED comes ON indicating voltage of 18.0 VDC or greater, proceed as follows:

- (a) Allow battery to cool to ambient temperature.
- (b) Perform normal charge (PARA 9-2-23.4).
- (c) Return battery to normal service.

9-2-23.4 Normal Charge.

WARNING

Ensure that charging/discharging current is turned OFF before disconnecting or connecting battery terminal connector. High current arcing may cause hydrogen gas generated in the cells to explode. Wear protective clothing when working around batteries. Do not wear jewelry that may contact cell terminals.

CAUTION

- Do not allow battery to remain in discharged state for more than 24 hours.
- Do not perform maintenance on lead acid battery in nickel cadmium battery service shop. Never use tools or material which have been used on nickel cadmium batteries.

NOTE

- Normal charging is recommended before and after discharge testing.
- Set battery analyzer/charger selector switch to appropriate setting for power source.

- a. Connect battery analyzer/charger power cord to electrical power source.
- b. Connect battery analyzer/charger cable between battery and analyzer/charger.
- c. Turn battery analyzer/charger rotary switch to CP (Constant Potential).
- d. Set battery analyzer/charger timer to 4 hours.
- e. Adjust the CHARGE CURRENT knob to minimum (fully CCW).
- f. Set battery analyzer/charger ON/OFF switch to ON.

RESULT:

- (1) CHARGE light shall go on.
- (2) VOLTS meter shall indicate open circuit voltage.

- g. Press START button on battery analyzer/charger.

RESULT: After one minute the timer LED display will come on and count upward.

- h. Adjust the CHARGE CURRENT knob to maximum (fully CW).
- i. Monitor voltage (VDC) and charging current (A).

RESULT: As charging progresses, if the battery voltage reaches approximately 28.5 volts and the current reduces to a value less than 2 amps, charging is complete and the battery is ready for installation or discharge testing.

- If result is not as specified, perform discharge testing.

NOTE

Monitor the voltage and amperage readings. The voltmeter shall indicate the voltage increase and the ammeter

the charge current reading. This reading could be as high as 16.5 amps unless the battery is partially charged.

- j. Turn off battery analyzer/charger and disconnect battery analyzer/charger from battery.

9-2-23.5 Conditioning Charge.**WARNING**

Ensure that charging/discharging current is turned OFF before disconnecting or connecting battery terminal connector. High current arcing may cause hydrogen gas generated in the cells to explode. Wear protective clothing when working around batteries. Do not wear jewelry that may contact cell terminals.

CAUTION

- Do not allow battery to remain in discharged state for more than 24 hours.
- Do not perform maintenance on lead acid battery in nickel cadmium battery service shop. Never use tools or material which have been used on nickel cadmium batteries.

NOTE

- Set battery analyzer/charger selector switch to appropriate setting for power source.
 - A conditioning charge is used when a battery has not been charged for 90 days or more, left discharged for 24 hours, or is sulfated.
 - A conditioning charge requires 12-24 hours to complete.
- a. Connect analyzer/charger cable between battery and battery analyzer/charger.

- b. Turn battery analyzer/charger rotary switch to CI (3.5 amperes).
- c. Set battery analyzer/charger timer for 30 minutes.
- d. Adjust the CHARGE CURRENT knob to minimum (CCW).

RESULT:

- (1) VOLTS meter will read battery open circuit voltage.
- (2) AMPS meter will read zero charge current.
- e. Turn battery analyzer/charger ON/OFF switch to ON.
- f. Press START button on battery analyzer/charger.

RESULT:

- (1) CHARGE light shall go on indicating charge has started.
- (2) Counter will count up from zero to time set.
- g. Monitor voltage and amperage on battery analyzer/charger VOLTS and AMPS meter. Adjust the CHARGE CURRENT control knob to between 2 to 3 amps charge current. Allow the charge to continue for the full 30 minutes.

RESULT: If voltage does not reach at least 28.5 volts, reset by turning off the power toggle switch and continuing the charge until the voltage reaches 28.8 to 29.5 volts. Amperes will be held constant at the adjusted level.

- h. Turn battery analyzer/charger ON/OFF switch to OFF and disconnect battery cable.
- i. Return battery to service.

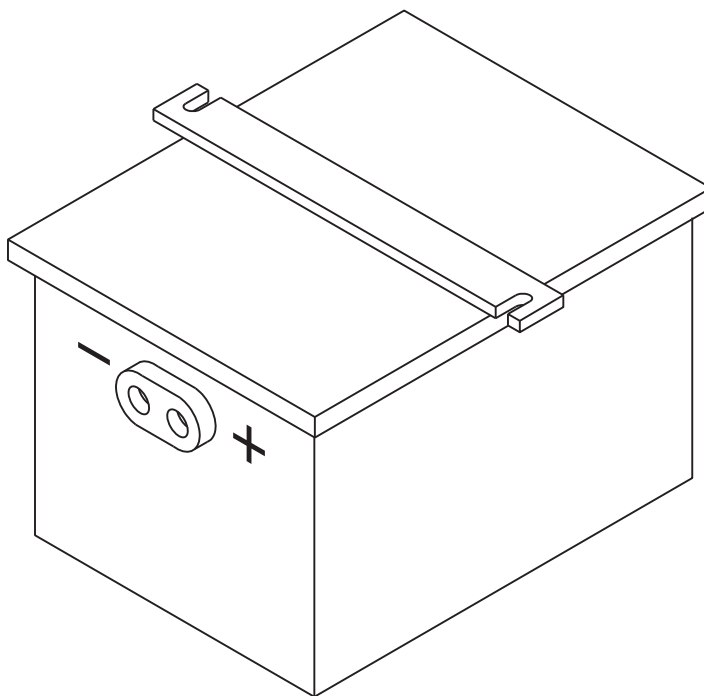
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FIGURE 9-2-23-1. BATTERY COMPONENT LOCATION DIAGRAM

9-2-24 CARGO HOOK LIGHTS **SLS** .

INITIAL SETUP

Test Equipment

- Multimeter, Fluke 45

Tools

- Electrical Repairer Toolkit

References

- PARA 1-6-16

Equipment Conditions

- External Electrical Power Available or APU Operational

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).

9-2-24.1 Operational/Troubleshooting Procedure.

- Make sure LIGHTS CARGO HOOK circuit breaker on pilot’s circuit breaker panel is pushed in.
- Turn on electrical power (PARA 1-6-16).
- Rotate HOOK LIGHT DIMMING CONTROL PANEL knob fully clockwise to BRT.
- Place upper console CARGO HOOK LT switch to ON.

RESULT: Cargo hook lights shall go on bright.

- If result is not as specified, go to TABLE NO. 9-2-24-1. If one cargo hook light does not go on, check for 115 VAC between the following:

J390-1 and ground

J391-1 and ground

- If voltage is not as specified, repair/replace wiring as required (Appendix F).
- If voltage is as specified, replace light assembly (PARA 9-4-93).
- Rotate HOOK LIGHT DIMMING CONTROL PANEL knob fully counterclockwise.

RESULT: Cargo hook lights shall dim.

- If result is not as specified, replace control panel (PARA 9-4-59).
- Place upper console CARGO HOOK LT switch to OFF.

RESULT: Cargo hook lights shall go off.

- If result is not as specified, replace CARGO HOOK LT switch (PARA 9-4-5).
- Turn off electrical power (PARA 1-6-16).

TABLE NO. 9-2-24-1. No Cargo Hook Lights Go On With HOOK LIGHT Control Panel Knob Fully Clockwise.

	TEST OR INSPECTION CORRECTIVE ACTION
	1. Check for 115 VAC between CARGO HOOK LT switch terminal 2 and ground. Step 1. If voltage is as specified, go to 2. Step 2. If voltage is not as specified, go to 7.
	2. Check for 115 VAC between CARGO HOOK LT switch terminal 3 and ground. Step 1. If voltage is as specified, go to 3. Step 2. If voltage is not as specified, replace switch (PARA 9-4-5). Go to 9.
	3. Check for 115 VAC between J392-1 and J392-2. Step 1. If voltage is as specified, go to 4. Step 2. If voltage is not as specified, go to 6. 4. Check continuity between: J389-1 and P392-2 J390-2 and P392-2 J391-2 and P392-2 Step 1. If continuity is present, go to 5. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 9.
	5. Check for 115 VAC between J389-3 and ground. Step 1. If voltage is as specified, go to 8. Step 2. If voltage is not as specified, repair/replace wiring (Appendix F). Go to 9.
	6. Check for 115 VAC between J392-1 and ground. Step 1. If voltage is as specified, repair/replace wiring between J392-2 and ground (Appendix F). Go to 9. Step 2. If voltage is not as specified, repair/replace wiring between J392-1 and CARGO HOOK LT switch terminal 3 (Appendix F). Go to 9.
	7. Check for 115 VAC between terminal 1 (LIGHTS CARGO HOOK circuit breaker, NO. 2 AC PRI BUS - pilot's circuit breaker panel) and ground. Step 1. If voltage is as specified, replace circuit breaker (PARA 9-4-1). Go to 9.

TABLE NO. 9-2-24-1. No Cargo Hook Lights Go On With HOOK LIGHT Control Panel Knob Fully Clockwise. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 2. If voltage is not as specified, repair/replace wiring between circuit breaker terminal 1 and CARGO HOOK LT switch terminal 2 (Appendix F). Go to 9 .
8. Check for 115 VAC between: J390-1 and ground J391-1 and ground	Step 1. If voltage is as specified, replace cargo hook light (PARA 9-4-93). Go to 9 . Step 2. If voltage is not as specified, replace HOOK LIGHT DIMMING control panel (PARA 9-4-59). Go to 9 .
9. Procedure completed.	

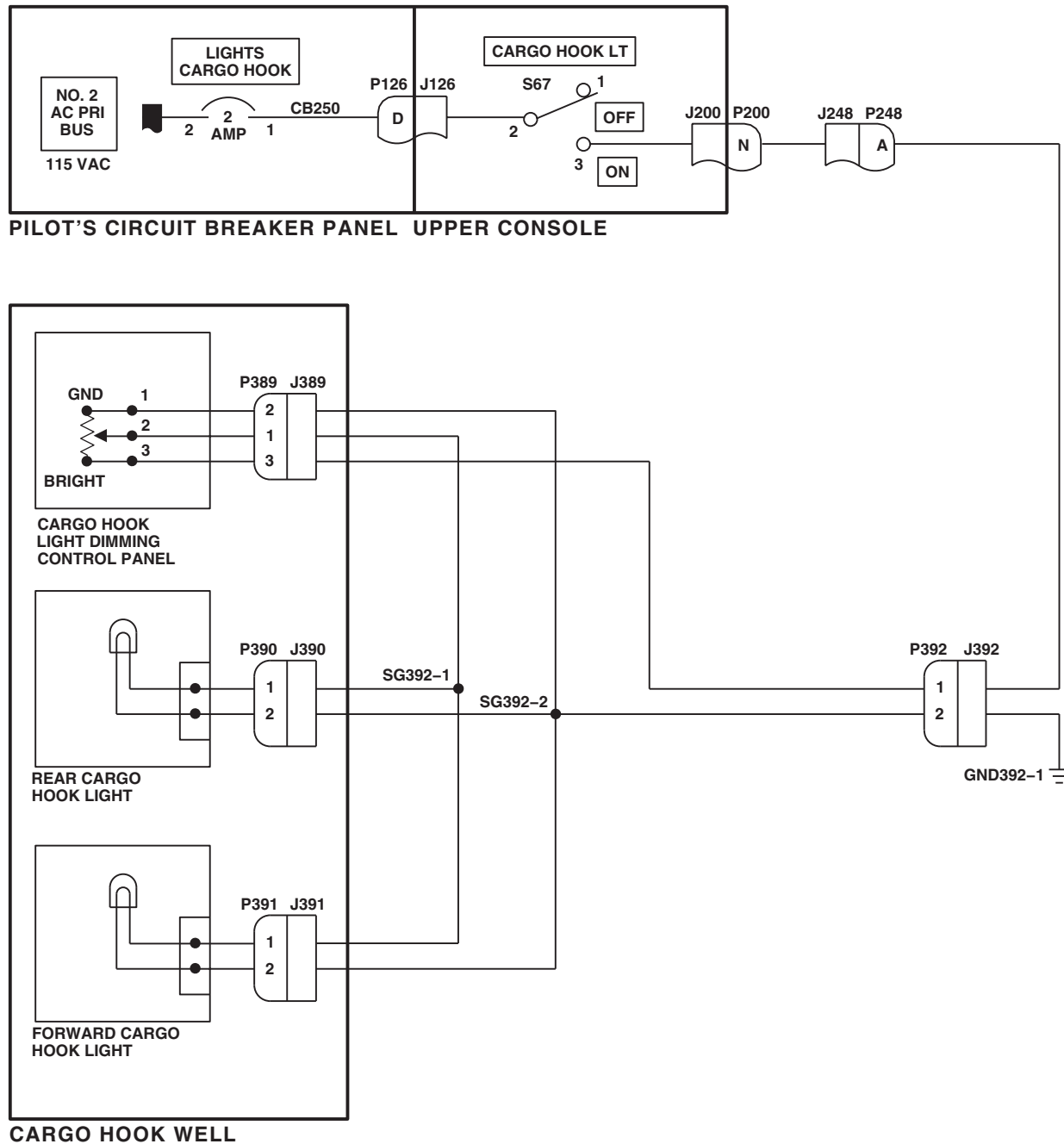
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FIGURE 9-2-24-1. CARGO HOOK LIGHTS SCHEMATIC DIAGRAM

9-2-25 UPPER AND LOWER CONSOLE LIGHTS **SLS** .

INITIAL SETUP

Test Equipment

- Multimeter, Fluke 45

Tools

- Electrical Repairer Toolkit

References

- PARA 1-6-16

Equipment Conditions

- External Electrical Power Available or APU Operational

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).

9-2-25.1 Operational/Troubleshooting Procedure.

- Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.
- Turn on electrical power (PARA 1-6-16).
- On upper console, turn CONSOLE LT LOWER control to BRT.

RESULT: All lower console lights shall go from dim to bright as control is turned clockwise.

- If result is not as specified, go to TABLE NO. 9-2-25-1.

- If any individual 0-115 VAC light does not go on, go to TABLE NO. 9-2-25-2.

- If no 0-28 VDC lights go on, go to TABLE NO. 9-2-25-3.

- If any individual 0-28 VDC light does not go on, go to TABLE NO. 9-2-25-4.

- If no 0-5 VAC lights go on, go to TABLE NO. 9-2-25-5.

- If any individual 0-5 VAC light does not go on, go to TABLE NO. 9-2-25-6.

- If the lower console lights flicker when control is turned clockwise, replace CONSOLE LT LOWER dimming control T11 (PARA 9-4-49).

- On upper console, turn CONSOLE LT LOWER control to OFF.

RESULT: Lower console lights shall go from bright to dim, then off as control is turned counterclockwise.

- If result is not as specified, replace CONSOLE LT LOWER dimming control T11 (PARA 9-4-49).

- On upper console, turn CONSOLE LT UPPER control to BRT.

RESULT: The following lights shall go on bright as control is turned clockwise:

- Upper console controls
- Cockpit flood and secondary lights panel
- Engine controls quadrant

- If result is not as specified, go to TABLE NO. 9-2-25-7.

- If any individual upper console light does not go on, go to TABLE NO. 9-2-25-8.

- If the upper console lights flicker when control is turned clockwise, replace CONSOLE LT UPPER dimming control T3 (PARA 9-4-49).
 - f. On upper console, turn CONSOLE LT UPPER control to OFF.
- RESULT: The following lights shall go from bright to dim, then off as control is turned counterclockwise:
- (1) Upper console controls
 - (2) Cockpit flood and secondary lights panel
 - (3) Engine controls quadrant
 - If result is not as specified, replace CONSOLE LT UPPER dimming control T3 (PARA 9-4-49).
 - g. Turn off electrical power (PARA 1-6-16).

TABLE NO. 9-2-25-1. Lower Console Lights Do Not Go From Dim To Bright.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for 115 VAC between T11-HV and ground.	Step 1. If voltage is as specified, go to 2.
	Step 2. If voltage is not as specified, go to 3.
2. Check continuity between CONSOLE LT LOWER dimming control T11-G and ground.	Step 1. If continuity is present, go to 4.
	Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 5.
3. Check for 115 VAC between terminal 1 (LIGHTS LWR CSL circuit breaker, NO. 1 AC PRI BUS - copilot's circuit breaker panel) and ground.	Step 1. If voltage is as specified, repair/replace wiring between T11-HV and circuit breaker terminal 1 (Appendix F). Go to 5.
	Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 5.
4. With CONSOLE LT LOWER control set to BRT, check for 115 VAC between T11-LV and ground.	Step 1. If voltage is as specified, repair/replace wiring between T11-LV and P219-m (Appendix F). Go to 5.
	Step 2. If voltage is not as specified, replace dimming control T11 (PARA 9-4-49). Go to 5.
5. Procedure completed.	

CHART NO. 9-2-25-2. Any Individual 0-115 Vac Light Does Not Go On.

COMPONENT	CHECK CONTINUITY BETWEEN	IF YES	IF NO
COMPASS SET CONTROL	P134R-U and TJ1-F	Replace panel (TM 11-70-23). If trouble remains, replace terminal board.	Repair/replace wiring (Appendix F).
CABIN DOME LIGHTS DIMMER PANEL	P455-D and TJ1-D	Replace panel (PARA 9-4-51). If trouble remains, replace terminal board.	Repair/replace wiring (Appendix F).
MISCELLANEOUS SWITCH PANEL	P659R-G and TJ1-B	Replace panel (PARA 9-4-7). If trouble remains, replace terminal board.	Repair/replace wiring (Appendix F).
STABILATOR CONTROL/ AUTO FLT CONTROL PANEL	P4R- <u>e</u> and T11-LV	Replace panel (TM 11-70-23).	Repair/replace wiring (Appendix F).
FUEL BOOST PUMP CONTROL PANEL	P139-L and T11-LV	Replace panel (PARA 9-4-40).	Repair/replace wiring (Appendix F).
RADIO RETRANSMISSION CONTROL PANEL	P116R-J and TJ1-C	Replace panel (TM 11-70-23). If trouble remains, replace terminal board.	Repair/replace wiring (Appendix F).
NO. 1 VHF/FM RADIO SET CONTROL PANEL	P820R- <u>a</u> and T110-LV	Replace panel (TM 11-70-23). If trouble remains, replace EL balance control panel assembly.	Repair/replace wiring (Appendix F).
NO. 2 VHF/FM RADIO SET CONTROL PANEL	P817R- <u>a</u> and T110-LV	Replace panel (TM 11-70-23). If trouble remains, replace EL balance control panel assembly.	Repair/replace wiring (Appendix F).
AUXILIARY SEARCHLIGHT CONTROL PANEL	P197-P and T110-LV	Replace panel (PARA 9-4-103).	Repair/replace wiring (Appendix F).

TABLE NO. 9-2-25-3. No 0-28 Vdc Lights Go On.

**TEST OR INSPECTION
CORRECTIVE ACTION**

1. Check for 28 VAC between P657R-Z and ground.

Step 1. If voltage is as specified, replace audio junction box (TM 11-70-23).

Step 2. If voltage is not as specified, go to **2**.

2. Check continuity between P168-D and P657R-Z.

Step 1. If continuity is present, replace low voltage power supply (PARA 9-4-58).

Go to **3**.

Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **3**.

3. Procedure completed.

CHART NO. 9-2-25-4. Any Individual 0-28 Vdc Light Does Not Go On.

COMPONENT	CHECK CONTINUITY BETWEEN	IF YES	IF NO
MID CABIN-INTER-COMMUNICATION SET (ICS) CONTROL PANEL	P200R-U and P653R-F	Replace panel (TM 11-70-23). If trouble remains, replace audio junction box (TM 11-70-23).	Repair/replace wiring (Appendix F).
RIGHT CABIN-INTER-COMMUNICATION SET (ICS) CONTROL PANEL	P126R-U and P653R-F	Replace panel (TM 11-70-23). If trouble remains, replace audio junction box (TM 11-70-23).	Repair/replace wiring (Appendix F).
LEFT CABIN-INTER-COMMUNICATION SET (ICS) CONTROL PANEL	P127R-U and P655R-R	Replace panel (TM 11-70-23). If trouble remains, replace audio junction box (TM 11-70-23).	Repair/replace wiring (Appendix F).
PILOT'S INTER-COMMUNICATION SET (ICS) CONTROL PANEL	P124R-U and P651R-AA	Replace panel (TM 11-70-23). If trouble remains, replace audio junction box (TM 11-70-23).	Repair/replace wiring (Appendix F).

CHART NO. 9-2-25-4. Any Individual 0-28 Vdc Light Does Not Go On. (Cont)

COMPONENT	CHECK CONTINUITY BETWEEN	IF YES	IF NO
COPILOT'S INTER-COMMUNICATION SET (ICS) CONTROL PANEL	P125R-U and P651R-AA	Replace panel (TM 11-70-23). If trouble remains, replace audio junction box (TM 11-70-23).	Repair/replace wiring (Appendix F).
VHF/AM RADIO SET	P118R-C and P657R-X	Replace panel (TM 11-70-23). If trouble remains, replace audio junction box (TM 11-70-23).	Repair/replace wiring (Appendix F).

TABLE NO. 9-2-25-5. No 0-5 Vac Lights Go On.

**TEST OR INSPECTION
CORRECTIVE ACTION**

1. Check for 115 VAC between P168-F and ground.

Step 1. If voltage is as specified, go to **2**.

Step 2. If voltage is not as specified, repair/replace wiring between T11-LV and P168-F (Appendix F). Go to **3**.

2. Check continuity between:

P168-K and ground

P168-L and ground

P168-E and ground

Step 1. If continuity is present, replace low voltage power supply (PARA 9-4-58). Go to **3**.

Step 2. If continuity is not present, repair/replace wiring, as required (Appendix F). Go to **3**.

3. Procedure completed.

CHART NO. 9-2-25-6. Any Individual 0-5 Vac Light Does Not Go On.

COMPONENT	CHECK CONTINUITY BETWEEN	IF YES	IF NO
DOPPLER COMPUTER DISPLAY UNIT	P687R-K and P168-J	Replace panel (TM 11-70-23). If trouble remains, replace low voltage power supply (PARA 9-4-58).	Repair/replace wiring (Appendix F).
IFF RECEIVER TRANSMITTER	P664R-15 and P168-J	Replace panel (TM 11-70-23). If trouble remains, replace low voltage power supply (PARA 9-4-58).	Repair/replace wiring (Appendix F).
CIVIL NAVIGATION CONTROL PANEL	P107R-H and P168-M	Replace panel (TM 11-70-23). If trouble remains, replace low voltage power supply (PARA 9-4-58).	Repair/replace wiring (Appendix F).
LF/ADF CONTROL PANEL	P138R-18 and P168-M	Replace panel (TM 11-70-23). If trouble remains, replace low voltage power supply (PARA 9-4-58).	Repair/replace wiring (Appendix F).
VHF/AM RADIO SET CONTROL PANEL	P814R-a and P168-X	Replace panel (TM 11-70-23). If trouble remains, replace low voltage power supply (PARA 9-4-58).	Repair/replace wiring (Appendix F).
UHF/AM RADIO SET CONTROL PANEL	P120R-C and P168-N	Replace panel (TM 11-70-23). If trouble remains, replace low voltage power supply (PARA 9-4-58).	Repair/replace wiring (Appendix F).
RESCUE HOIST CONTROL BOX	P965-14 and P168-A	Replace panel (PARA 14-4-12). If trouble remains, replace low voltage power supply (PARA 9-4-58).	Repair/replace wiring (Appendix F).
AUXILIARY RESCUE HOIST CONTROL PANEL	J352-E and P168-H	Replace panel (PARA 14-4-25). If trouble remains, replace low voltage power supply (PARA 9-4-58).	Repair/replace wiring (Appendix F).

TABLE NO. 9-2-25-7. No Upper Console Lights Go On.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for 115 VAC between T3-HV and ground.	Step 1. If voltage is as specified, go to 2. Step 2. If voltage is not as specified, go to 3.
2. Check continuity between T3-G and ground.	Step 1. If continuity is present, replace T3 (PARA 9-4-49). Go to 4. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 4.
3. Check for 115 VAC between terminal 1 (LIGHTS UPPER CSL circuit breaker, NO. 1 AC PRI BUS - copilot's circuit breaker panel) and ground.	Step 1. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and T3-HV (Appendix F). Go to 4. Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 4.
4. Procedure completed.	

CHART NO. 9-2-25-8. Any Individual Upper Console Light Does Not Go On.

COMPONENT	CHECK CONTINUITY BETWEEN	IF YES	IF NO
COCKPIT FLOOD AND SECONDARY LIGHTS PANEL	P288-E and T3-LV	Replace information plate (PARA 9-4-69).	Repair/replace wiring (Appendix F).
LEFT SIDE CONTROL PANEL	1J2 and T3-LV	Replace panel (PARA 9-4-6).	Repair/replace wiring (Appendix F).
CENTER CONTROL PANEL	1J1 and T3-LV	Replace panel (PARA 9-4-6).	Repair/replace wiring (Appendix F).
RIGHT SIDE CONTROL PANEL	1J3 and T3-LV	Replace panel (PARA 9-4-6).	Repair/replace wiring (Appendix F).
ENGINE CONTROL LEFT QUADRANT PANEL	J394-1 and T3-LV	Replace panel (PARA 4-4-21).	Repair/replace wiring (Appendix F).
ENGINE CONTROL CENTER QUADRANT PANEL	J395-1 and T3-LV	Replace panel (PARA 4-4-21).	Repair/replace wiring (Appendix F).
ENGINE CONTROL RIGHT QUADRANT PANEL	J396-1 and T3-LV	Replace panel (PARA 4-4-21).	Repair/replace wiring (Appendix F).

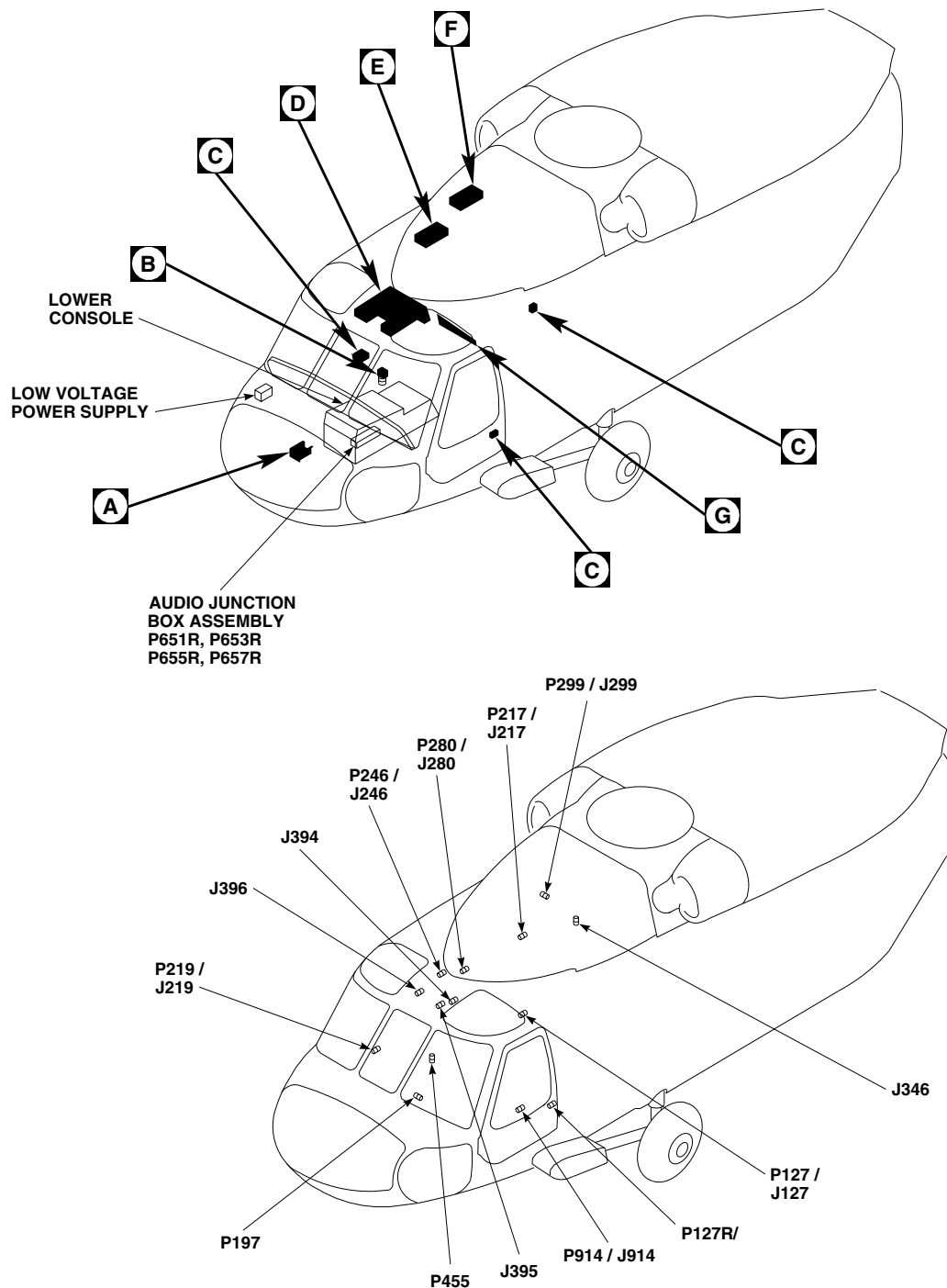


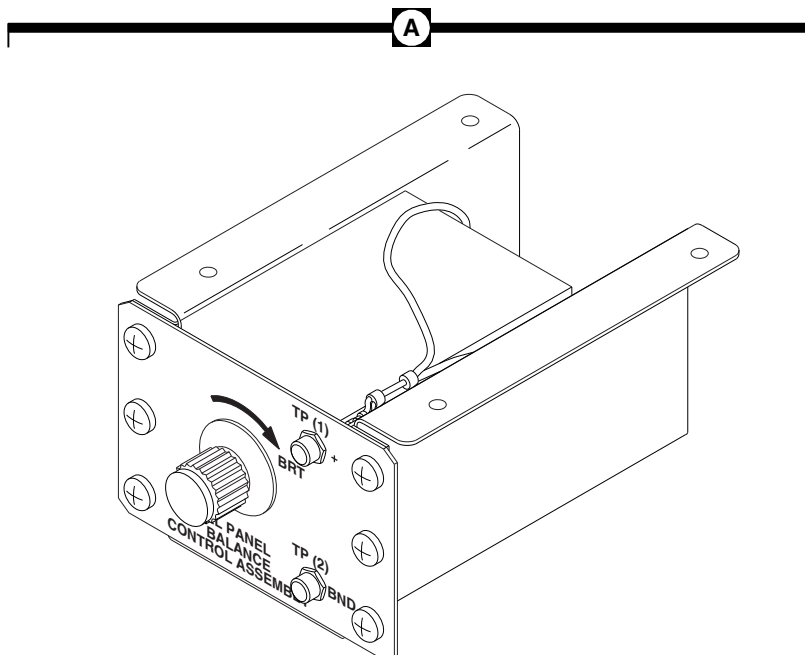
FIGURE 9-2-25-1. UPPER AND LOWER CONSOLE LIGHTS LOCATION DIAGRAM
(SHEET 1 OF 5)

FB4470_1B
SA

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
J346	CABIN CEILING, BL 18 LH, STA 301
J352	AUX RESCUE HOIST CONTROL PANEL
J352	AUXILIARY RESCUE HOIST CONTROL PANEL
J394	COCKPIT CEILING BL 9 LH, STA 233
J395	COCKPIT CEILING BL 4 LH, STA 229
J396	COCKPIT CEILING BL 4 RH, STA 229
P4R	STABILATOR CONTROLS / AUTO FLIGHT CONTROL PANEL
P13R / J13R	CABIN BL 20 LH, STA 247
P25R / J25R	CABIN BL 40 RH, STA 280
P107R	CIVIL NAVIGATION SYSTEM RECEIVER CONTROL PANEL
P114 / J114	COCKPIT MAIN DISCONNECT
P116R / J1	RADIO RETRANSMISSION CONTROL PANEL
P118R	VHF / AM RADIO SET
P120R	UHF / AM RADIO SET CONTROL PANEL
P124R	PILOT'S ICS CONTROL PANEL
P125R	COPILOT'S ICS CONTROL PANEL
P126R	RIGHT CABIN ICS CONTROL PANEL
P127 / J127	BEHIND COPILOT'S CIRCUIT BREAKER PANEL BL 28 LH, STA 243
P127R	LEFT CABIN ICS CONTROL PANEL
P134R	COMPASS SET CONTROL
P138R	LF / ADF RADIO SET CONTROL
P139	FUEL BOOST PUMP CONTROL PANEL
P168	LOW VOLTAGE POWER SUPPLY

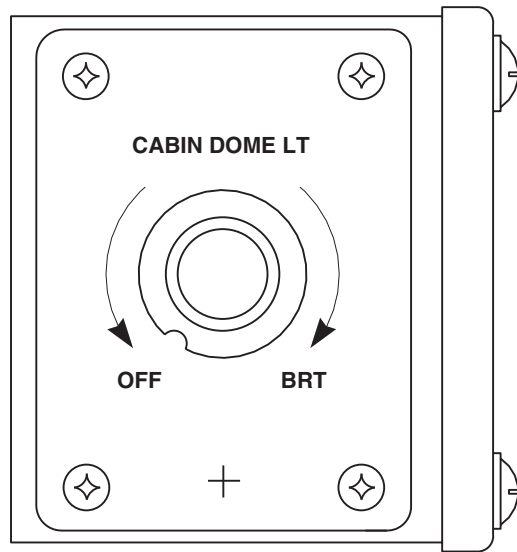
TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P197	AUXILIARY SEARCHLIGHT CONTROL PANEL
P200R	MID CABIN ICS CONTROL PANEL
P217 / J217	MAIN ROTOR PYLON DECK BL 0, STA 284
P219 / J219	CABIN FLOOR BL 25 RH, STA 247
P246 / J246	CABIN CEILING BL 5 RH, STA 247
P280 / J280	CABIN CEILING BL 6 LH, STA 247
P288	COCKPIT FLOOD AND SECONDARY LIGHTS PANEL
P299 / J299	CABIN CEILING, BL 10 LH, STA 302
P346 / J346	ESSS DISCONNECT
P455	CABIN DOME LT DIMMER BEHIND PILOT'S SEAT BL 11 RH, STA 245
P651R	AUDIO JUNCTION BOX
P653R	AUDIO JUNCTION BOX
P655R	AUDIO JUNCTION BOX
P657R	AUDIO JUNCTION BOX
P659R	MISCELLANEOUS SWITCH PANEL
P664R	IFF RECEIVER- TRANSMITTER
P687R	DOPPLER COMPUTER DISPLAY UNIT
P814R	VHF / AM RADIO SET CONTROL PANEL
P817R	NO. 2 VHF / FM RADIO SET CONTROL PANEL
P820R	NO. 1 VHF / FM RADIO SET CONTROL PANEL
P914 / J914	COCKPIT, BL 24 LH, STA 247
P965	RESCUE HOIST CONTROL BOX

FB4470_2B
SAFIGURE 9-2-25-1. UPPER AND LOWER CONSOLE LIGHTS LOCATION DIAGRAM
(SHEET 2 OF 5)



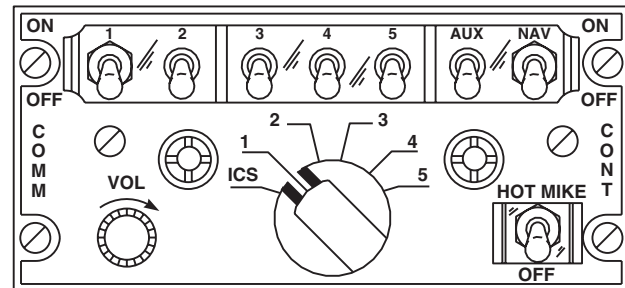
FB4470_3
SA

FIGURE 9-2-25-1. UPPER AND LOWER CONSOLE LIGHTS LOCATION DIAGRAM
(SHEET 3 OF 5)



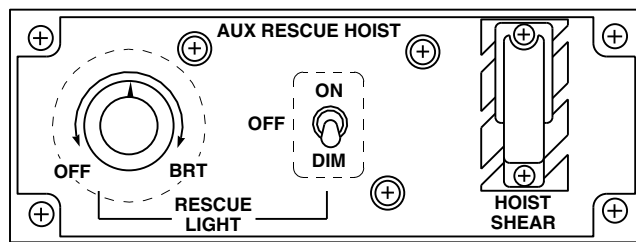
**CABIN DOME LIGHTS
DIMMER UNIT**

B



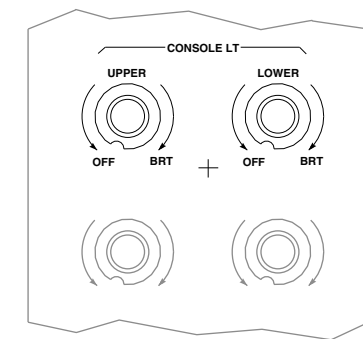
**INTERCOMMUNICATION SET (ICS)
CONTROL PANEL (TYPICAL)**

C



AUXILIARY RESCUE HOIST CONTROL PANEL

E



UPPER CONSOLE

D

FIGURE 9-2-25-1. UPPER AND LOWER CONSOLE LIGHTS LOCATION DIAGRAM
(SHEET 4 OF 5)

FT1485_4
SA

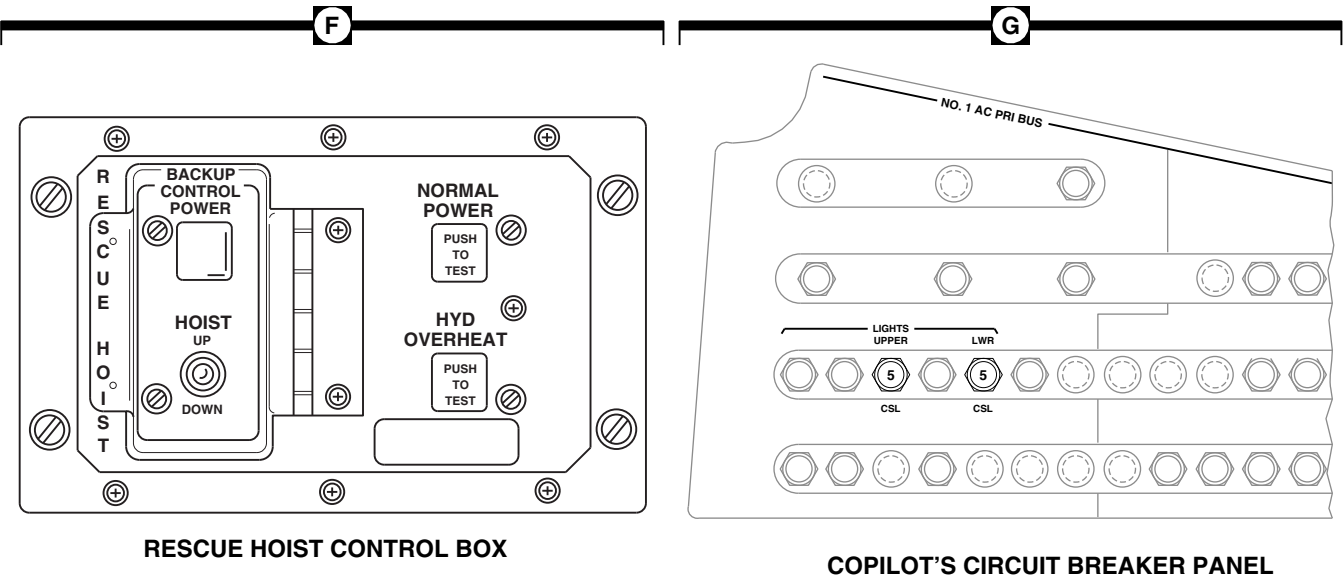
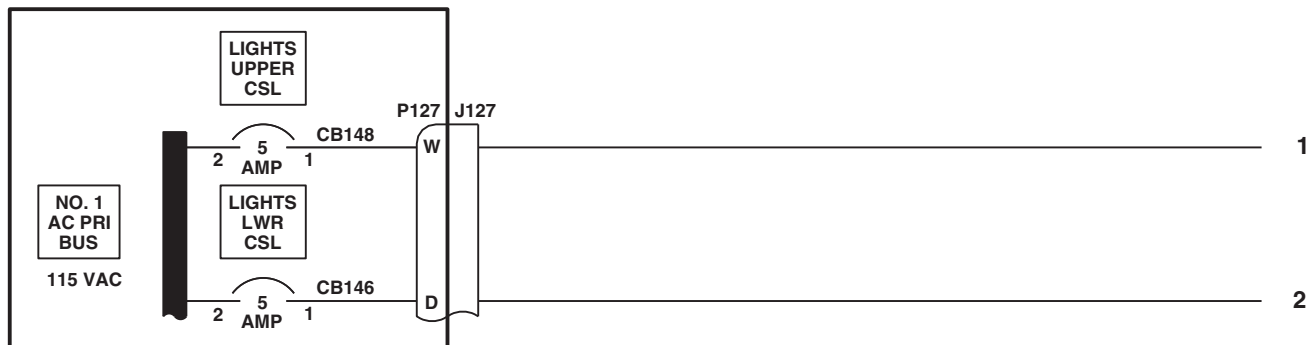
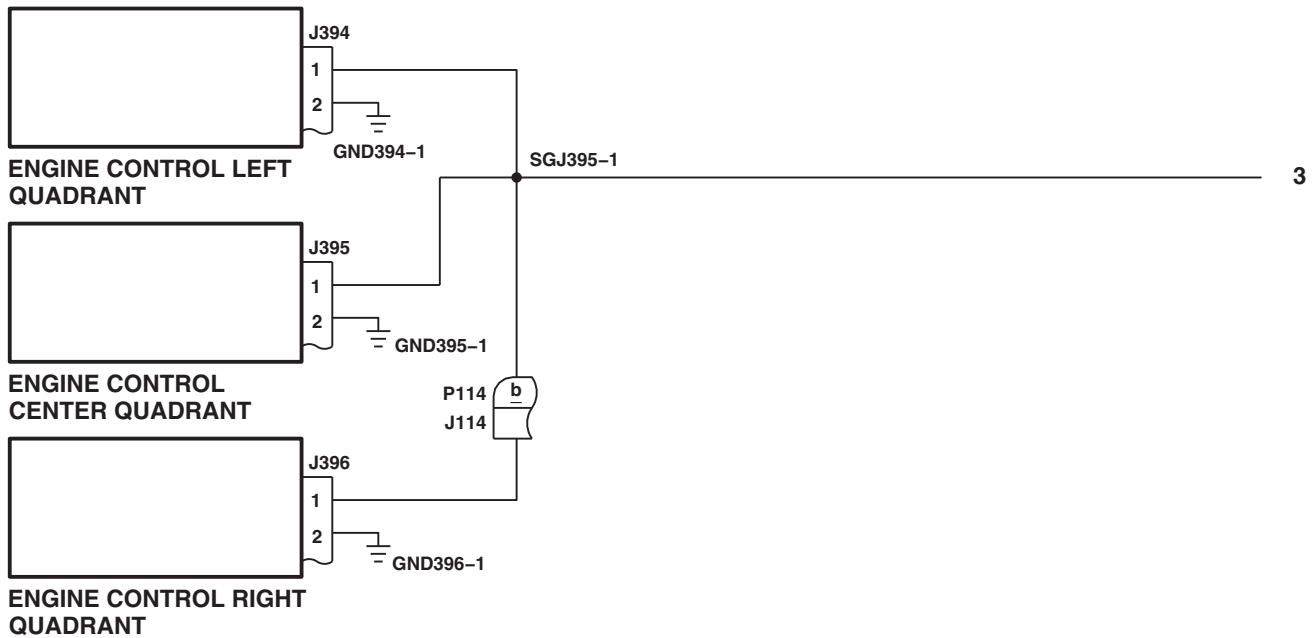


FIGURE 9-2-25-1. UPPER AND LOWER CONSOLE LIGHTS LOCATION DIAGRAM (SHEET 5 OF 5)

FT1485_5
SA



COPILOT'S CIRCUIT BREAKER PANEL

FT1486_1
SAFIGURE 9-2-25-2. UPPER AND LOWER CONSOLE LIGHTS SCHEMATIC DIAGRAM
(SHEET 1 OF 5)

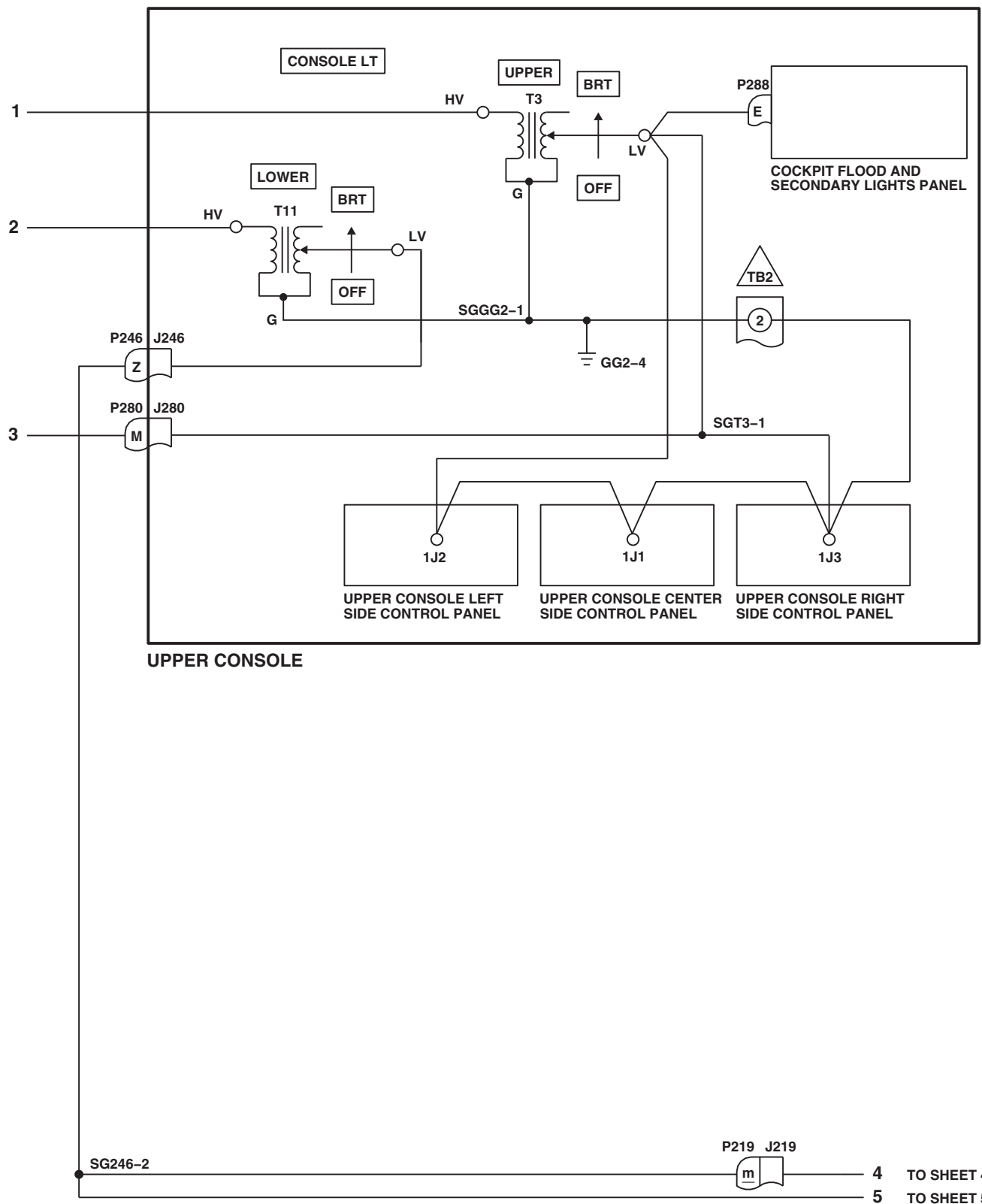


FIGURE 9-2-25-2. UPPER AND LOWER CONSOLE LIGHTS SCHEMATIC DIAGRAM
(SHEET 2 OF 5)

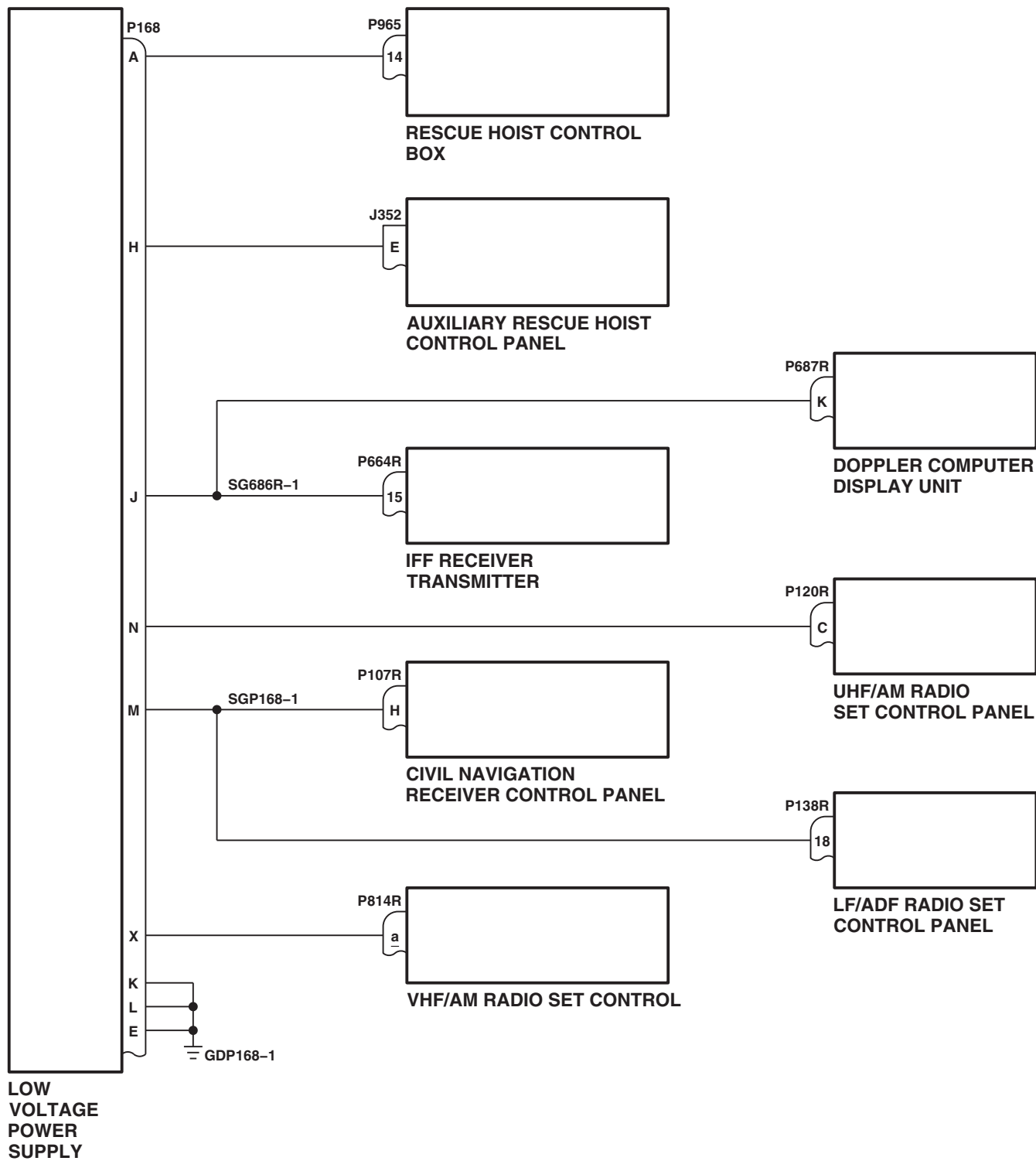
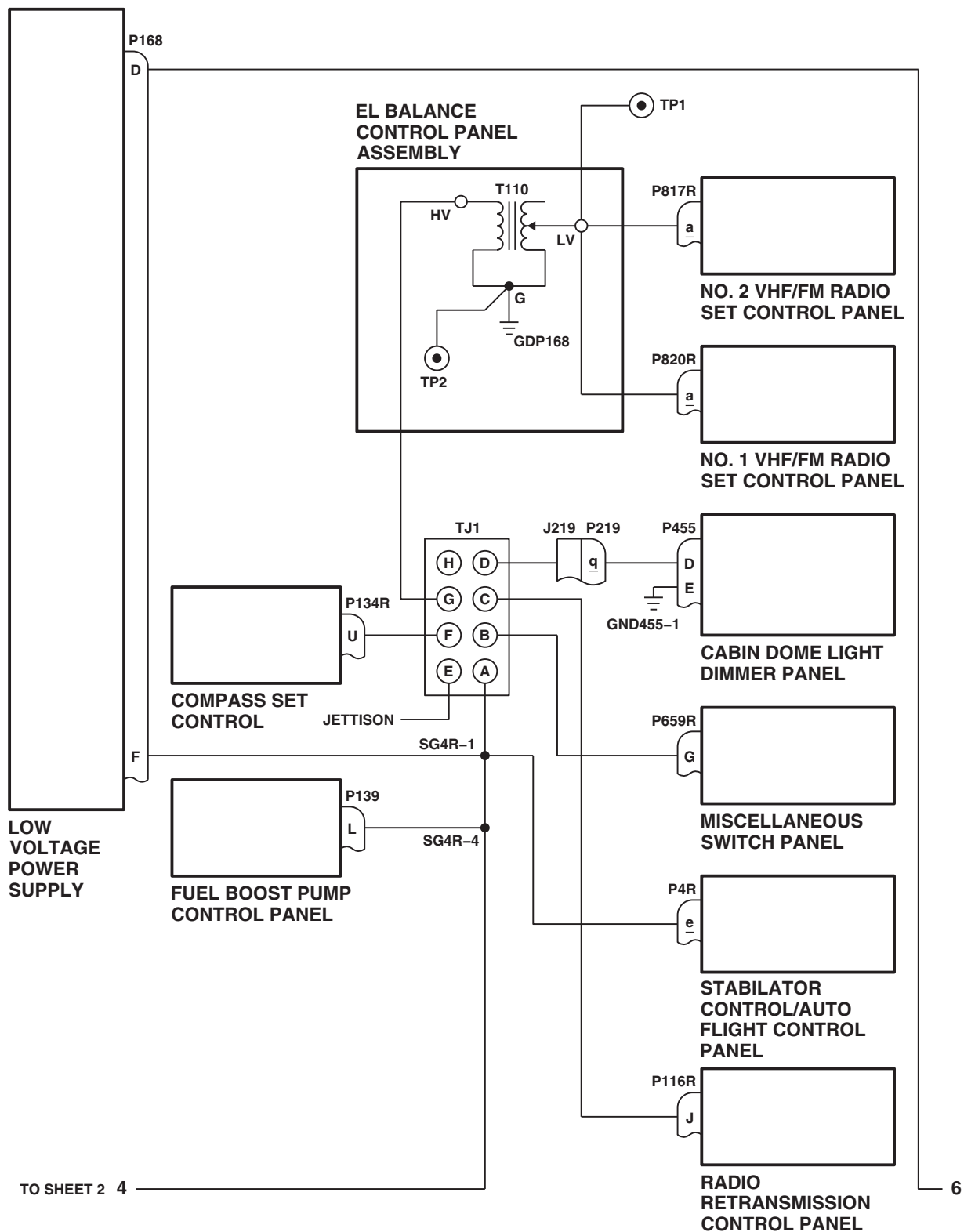
FB4471_3B
SA

FIGURE 9-2-25-2. UPPER AND LOWER CONSOLE LIGHTS SCHEMATIC DIAGRAM
(SHEET 3 OF 5)



FB4471_4C
SA

FIGURE 9-2-25-2. UPPER AND LOWER CONSOLE LIGHTS SCHEMATIC DIAGRAM (SHEET 4 OF 5)

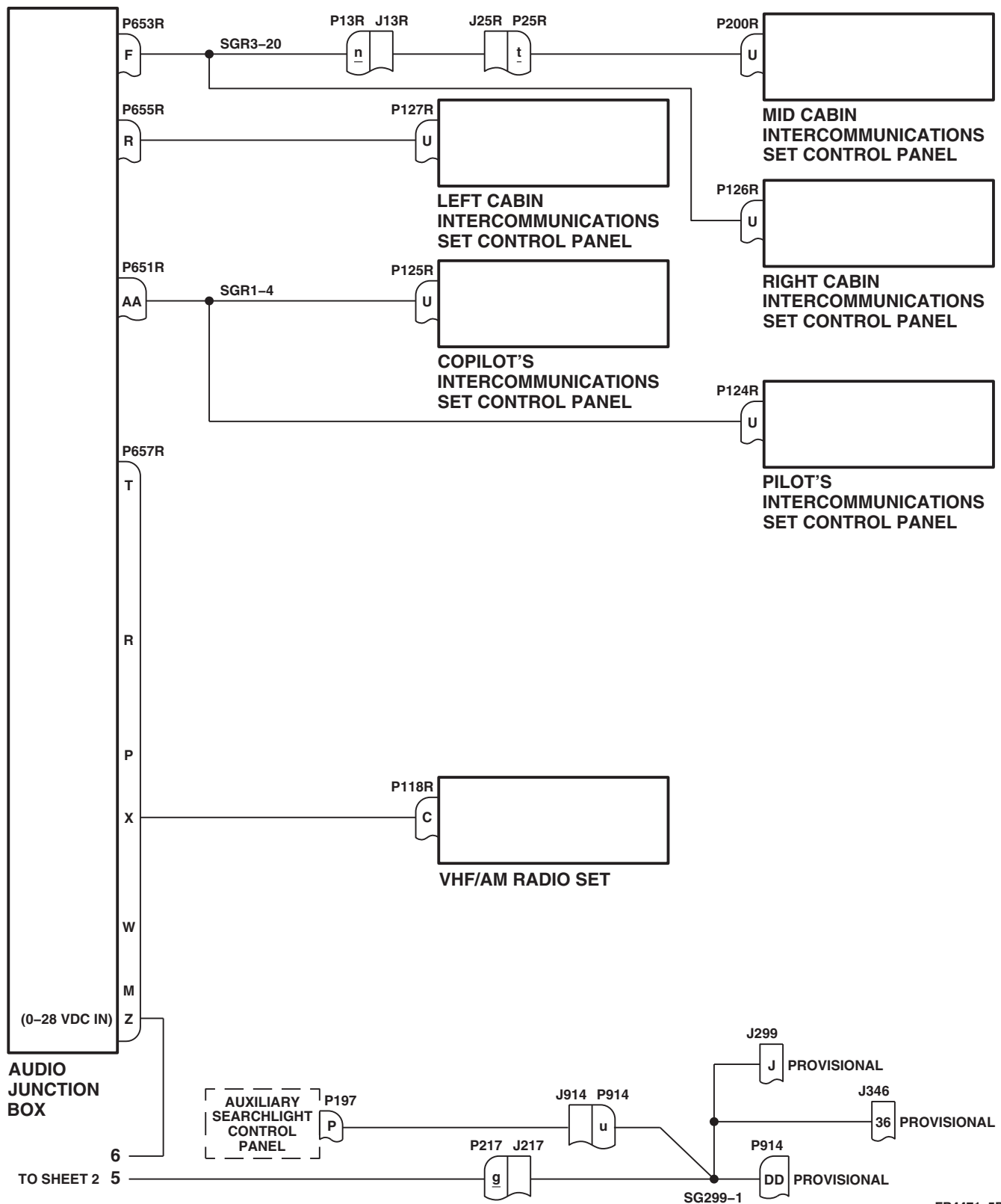


FIGURE 9-2-25-2. UPPER AND LOWER CONSOLE LIGHTS SCHEMATIC DIAGRAM
(SHEET 5 OF 5)

9-2-25-18

Change 7

This Document contains technical data subject to ITAR. See WARNING and classifications on first page.

SECTION 9-3
ON AIRCRAFT INSPECTIONS
NOT APPLICABLE

SECTION 9-4

MAINTENANCE PROCEDURES

SECTION OVERVIEW

PARAGRAPH	TITLE
9-4-1	Circuit Breakers
9-4-2	Left/Right Relay Panels
9-4-3	Left Relay Panel Fuses
9-4-4	No. 3 Relay Panel
9-4-5	Upper Console Switches
9-4-6	Upper Console Information Plates
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9-4-8	Miscellaneous Switch Panel Pushbutton Lamps
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9-4-15	Generator Control Units
9-4-16	Main Generators
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PARAGRAPH	TITLE
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9-4-24	AC Secondary Bus Contactor, K11
9-4-25	Diodes, CR13 and CR14
9-4-26	Diodes, CR17, CR18, CR19, and CR20
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9-4-30	DC Primary Bus Contactors, K6 and K16
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9-4-32	DC Essential Bus Supply Relays, K9 and K10
9-4-33	Battery Relay, K7
9-4-34	Nickel-Cadmium Battery 70583-702374
9-4-35	Battery Conditioner/Analyzer 70583-702374
9-4-36	Converters
9-4-37	DC Current Limiter
9-4-38	DC Monitor Bus Supply Relay, K5
9-4-39	APU Electronic Sequence Unit
9-4-40	Fuel Boost Pump Control Panel
9-4-41	Fuel Boost Pump Control Panel Information Plate
9-4-42	Fuel Boost Pump Control Panel Indicator Light Lamps
9-4-43	Engine Prime Boost Diode Board
9-4-44	Backup Hydraulic Pump Motor
9-4-45	Backup Hydraulic Pump Motor Hood

PARAGRAPH	TITLE
9-4-46	Backup Hydraulic Pump Motor Fan
9-4-47	Backup Hydraulic Pump Relay, K19
9-4-48	Remote Circuit Breaker
9-4-49	Upper Console Dimming Controls
9-4-50	Lighted Switches Dimming Control
9-4-51	Cabin Dome Dimming Control
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9-4-53	Indicator Lights Dimmer Box Fuse
9-4-54	Cargo Hook Emergency Release System Resistor, R4
9-4-55	Transformer Boxes
9-4-56	Controllable Searchlight Dimming Control
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9-4-60	Cabin Dome Light Lamps
9-4-61	Cabin Dome Lights
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9-4-64	Portable Maintenance Light Lamp
9-4-65	Glare Shield Lamps
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9-4-67	Secondary Light Enclosure Floodlight Lamps
9-4-68	Secondary Light Enclosure Information Plate Lamps
9-4-69	Secondary Light Enclosure
9-4-70	Main Rotor Pylon Electro-Luminescent Formation Light

PARAGRAPH

TITLE

9-4-71	Main Rotor Pylon Infrared Formation Light
9-4-72	Tail Cone Electro-Luminescent Formation Light
9-4-73	Tail Cone Infrared Formation Light
9-4-74	Stabilator Electro-Luminescent Formation Lights
9-4-75	Stabilator Infrared Formation Lights
9-4-76	Position Lights Flasher Unit
9-4-77	ESSS Horizontal Stores Support (HSS) Incandescent Position Lights Lamps and Lenses ESSS
9-4-78	Drag Beam Support Fairing Incandescent Position Lights
9-4-79	Drag Beam Support Fairing Infrared Position Lights
9-4-80	ESSS Horizontal Stores Support (HSS) Incandescent Position Lights ESSS
9-4-81	ESSS Horizontal Stores Support (HSS) Infrared Position Lights ESSS
9-4-82	Tail Incandescent Position Light Lamp and Lens
9-4-83	Tail Incandescent Position Light
9-4-84	Tail Infrared Position Light
9-4-85	Anticollision Light Red and White Subassemblies
9-4-86	Anticollision Lights
9-4-87	Anticollision Light Base Subassemblies
9-4-88	Anticollision Lights Power Supply Unit
9-4-89	Retractable Landing Light Lamp
9-4-90	Retractable Landing Light
9-4-91	Controllable Searchlight Lamp
9-4-92	Controllable Searchlight
9-4-93	Cargo Hook Lights

PARAGRAPH	TITLE
9-4-94	Cabin Intercommunication System Floodlight Control Panel Light
9-4-95	Passenger's Intercommunication System Floodlight Control Panel Light
9-4-96	Engine Ignition Switch
9-4-97	Engine Overspeed Relays
9-4-98	Pin Filter Adapters EMEP
9-4-99	Instrument Panel Pushbutton Lamps
9-4-100	Auxiliary Searchlight Control Panel
9-4-101	Auxiliary Searchlight Junction Box
9-4-102	Auxiliary Searchlight Gimbal Assembly
9-4-103	Auxiliary Searchlight Assembly
9-4-104	Auxiliary Searchlight Infrared Filter Lens
9-4-105	Sealed Lead-Acid Battery Junction Box Relays, K200 and K201 702375 - SUBQ
9-4-106	Sealed Lead-Acid Battery 702375 - SUBQ
9-4-107	Sealed Lead-Acid Battery Junction Box 702375 - SUBQ
9-4-108	No. 7 Relay Panel
9-4-109	Dimming Resistor, R10
9-4-110	Auxiliary Searchlight Bracket Incandescent Position Light Lamps and Lenses
9-4-111	Auxiliary Searchlight Bracket Incandescent Position Light
9-4-112	Auxiliary Searchlight Bracket Infrared Position Light

9-4-1 CIRCUIT BREAKERS.

PARAGRAPH BREAKDOWN

	PAGE
9-4-1.1 Replace Circuit Breakers	9-4-1-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

Materials/Parts

- Tags

References

- PARA 1-6-16
- PARA 9-2-3
- PARA 9-4-34
- PARA 9-4-106

9-4-1.1 Replace Circuit Breakers.

NOTE

Replacement of all circuit breakers on pilot’s and copilot’s circuit breaker panels is the same.

9-4-1.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Loosen fasteners and carefully lower circuit breaker panel (Figure 9-4-1-1, Detail A).
- Tag wires. Remove screws, lockwashers, and if present, washers securing wires to circuit breaker (Figure 9-4-1-1, Detail A and Detail B).
- If required, remove bus bars from circuit breaker being replaced.

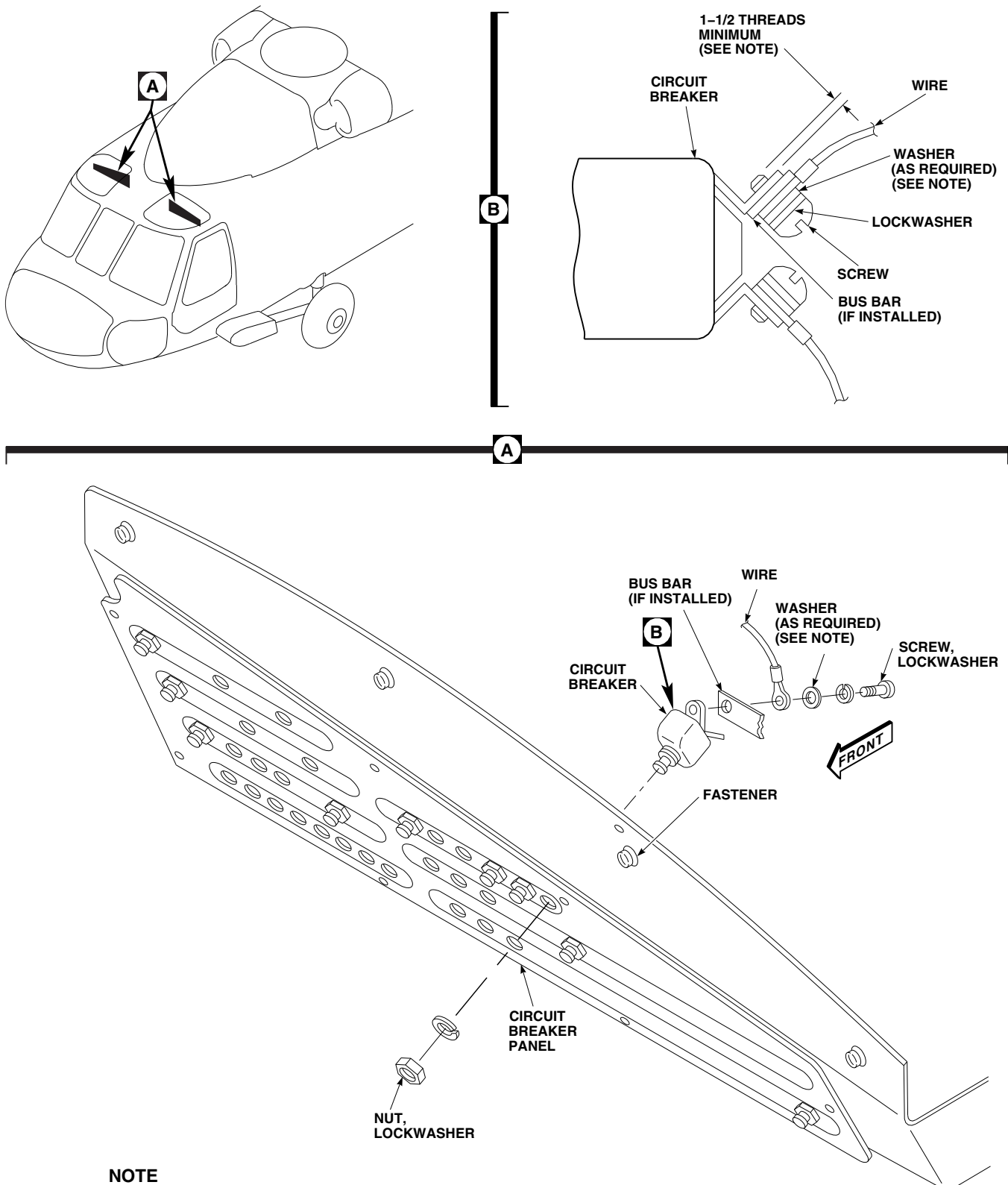
- Remove nut and lockwasher from front of circuit breaker panel (Figure 9-4-1-1, Detail A).
- Remove circuit breaker from back of circuit breaker panel.

9-4-1.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).
- Install replacement circuit breaker into back of circuit breaker panel and secure with lockwasher and nut on front of circuit breaker panel (Figure 9-4-1-1, Detail A).
- If removed, connect bus bars to replacement circuit breaker.



Damage to body of circuit breaker will result if screw is too long. Add washer(s) under lockwasher if necessary.



NOTE

ADD WASHER(S) UNDER LOCKWASHER, IF NECESSARY.

TYPICAL

FF0336G
SA

FIGURE 9-4-1-1. REPLACE CIRCUIT BREAKERS

NOTE

Stackup of wire terminals may require use of longer screw.

- d. Install wires on circuit breaker and secure with screws, lockwashers, and, if required, washers (Figure 9-4-1-1, Detail A and Detail B). Remove tags.
- e. Make sure area is clean and free of foreign material.
- f. Carefully swing circuit breaker panel into closed position and secure with fasteners.
- g. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- h. Perform operational check of system circuit breakers (PARA 9-2-3).

9-4-2 LEFT/RIGHT RELAY PANELS.

PARAGRAPH BREAKDOWN

	PAGE
9-4-2.1 Replace Left/Right Relay Panels	9-4-2-2

INITIAL SETUP

- PARA 4-2-7

Tools

- PARA 4-2-8

- Aircraft Mechanic’s Toolkit
- Digital Low-Resistance Ohmmeter
- Electrical Repairer’s Toolkit

- PARA 7-2-1
- PARA 8-2-2
- PARA 8-2-4

Materials/Parts

- PARA 9-2-10

- Cheesecloth, Item 90, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Protective Caps and Plugs
- Tags
- Tiedown Straps

- PARA 9-2-13
- PARA 9-2-15
- PARA 9-4-34
- PARA 9-4-106
- PARA 12-2-1

References

- PARA 12-2-2

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-69
- PARA 2-6-5

- PARA 12-2-4
- PARA 12-2-6
- PARA 13-2-1
- PARA 15-2-2
- TM 11-70-23

9-4-2.1 Replace Left/Right Relay Panels.**NOTE**

Replacement of left and right relay panels is the same, except as noted.

9-4-2.1.1 Remove.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- c. Remove soundproofing panel (PARA 2-4-69).
- d. Unlock 1/4-turn fasteners securing side of relay panel to overhead support (Figure 9-4-2-1, Detail A or Detail B).
- e. Remove screws, lockwashers, and washers on opposite side securing relay panel to overhead support. Tilt relay panel and carefully lower it to gain access to electrical connectors.

NOTE

When winterization kit is installed, electrical connector, P244B, is connected to electrical connector, J244. When winterization kit is not installed, electrical connector, P244, is connected to electrical connector, J244.

- f. Tag and disconnect electrical connectors from relay panel as follows (Figure 9-4-2-1, Detail A or Detail B):

Right Relay Panel	Left Relay Panel
P241 from J241	P242 from J242
P244 or P244B from J244	P243 from J243
P313 from J313	P902 from J902
P900 from J900	

- g. Install protective caps and plugs on electrical connectors.

- h. Remove relay panel.

9-4-2.1.2 Install.

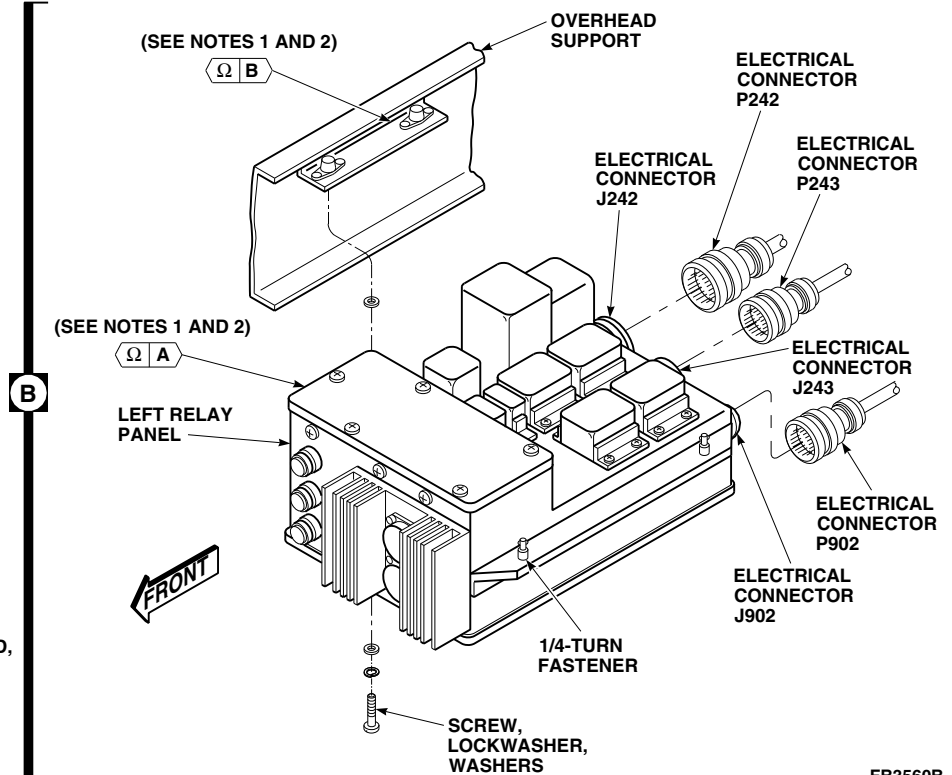
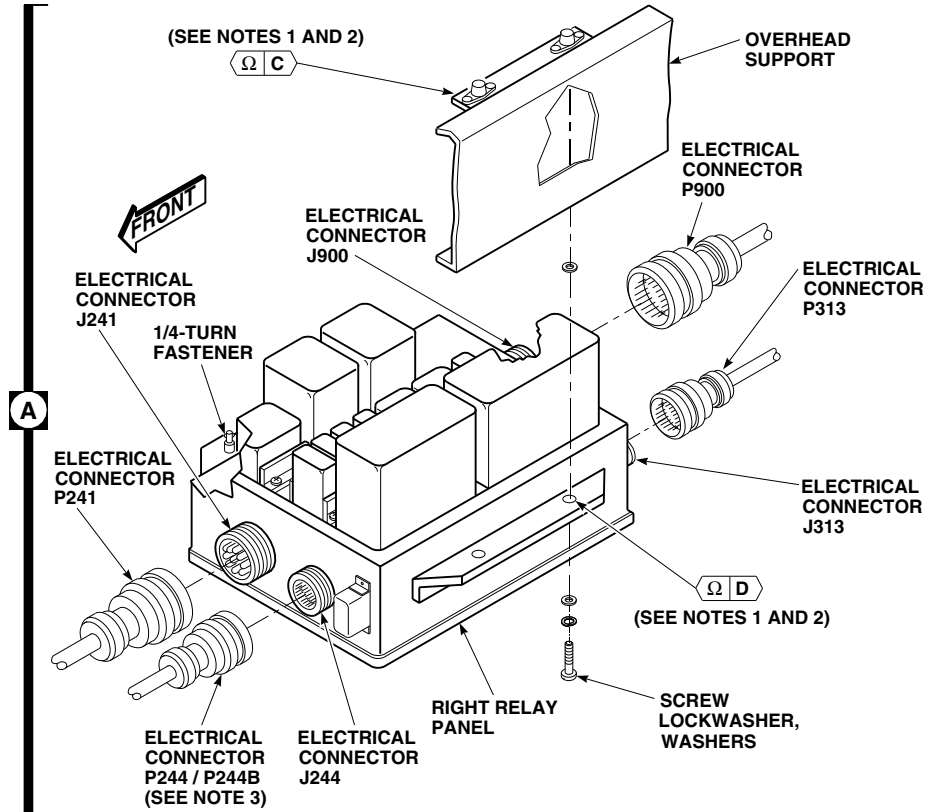
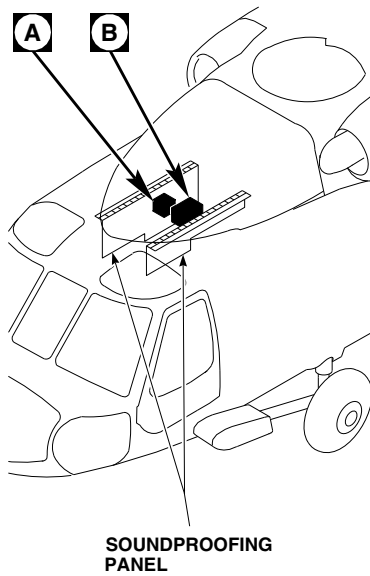
- a. Turn off all electrical power (PARA 1-6-16).
- b. **EMEP** Inspect chemical conversion coating as follows:
 - (1) Using cheesecloth, Item 90, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D, clean chemical conversion coated surfaces. Using clean, dry cheesecloth, Item 90, Appendix D, dry surfaces.
 - (2) Inspect chemical conversion coated surfaces for breaks and scratches. **NONE ALLOWED.** If coated surfaces have breaks or scratches, apply chemical conversion coating (PARA 2-6-5).
- c. Remove protective caps and plugs from electrical connectors.
- d. Inspect and treat electrical connectors (PARA 1-7-20).

NOTE

When winterization kit is installed, electrical connector, P244B, is connected to electrical connector, J244. When winterization kit is not installed, electrical connector, P244, is connected to electrical connector, J244.

- e. Connect electrical connectors to relay panel as follows (Figure 9-4-2-1, Detail A or Detail B), then remove tags:

Right Relay Panel	Left Relay Panel
P241 to J241	P242 to J242
P244 or P244B to J244	P243 to J243
P313 to J313	P902 to J902
P900 to J900	



NOTES

1. EMEP

POINTS FROM	TO	RESISTANCE SHALL BE
A	B	20.0 MILLIOHMS OR LESS
C	D	20.0 MILLIOHMS OR LESS

3. WHEN WINTERIZATION KIT IS INSTALLED, ELECTRICAL CONNECTOR P244B IS CONNECTED TO ELECTRICAL CONNECTOR J244. WHEN WINTERIZATION KIT IS NOT INSTALLED, ELECTRICAL CONNECTOR P244 IS CONNECTED TO ELECTRICAL CONNECTOR J244.

FIGURE 9-4-2-1. REPLACE LEFT/RIGHT RELAY PANELS

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SA

- f. Install relay panel to overhead support and secure with screws, lockwashers, and washers (Figure 9-4-2-1, Detail A or Detail B). Complete installation by locking 1/4-turn fasteners.

WARNING

No slack, especially retermination slack, should be located above relay panels. Excess slack in this area may cause chafing and possible arcing, causing damage to wires, relay panels, or the hydraulic logic modules.

- g. Inspect/adjust wire harness routing and clamping above relay panels at STA 265.5 and STA 281.0 as follows (Figure 9-4-2-2):
- (1) Visually inspect for proper clamping and harness ties.
 - (2) Check for adequate clearances and possible chafing points between wires and top of relay panels.
 - (3) Adjust clamping and secure with tiedown straps as required.
- h. **EMEP** Perform EME resistance check of relay panel as follows (Figure 9-4-2-1, Detail A and Detail B):■

NOTE

Resistance measurements must be taken at least three times.

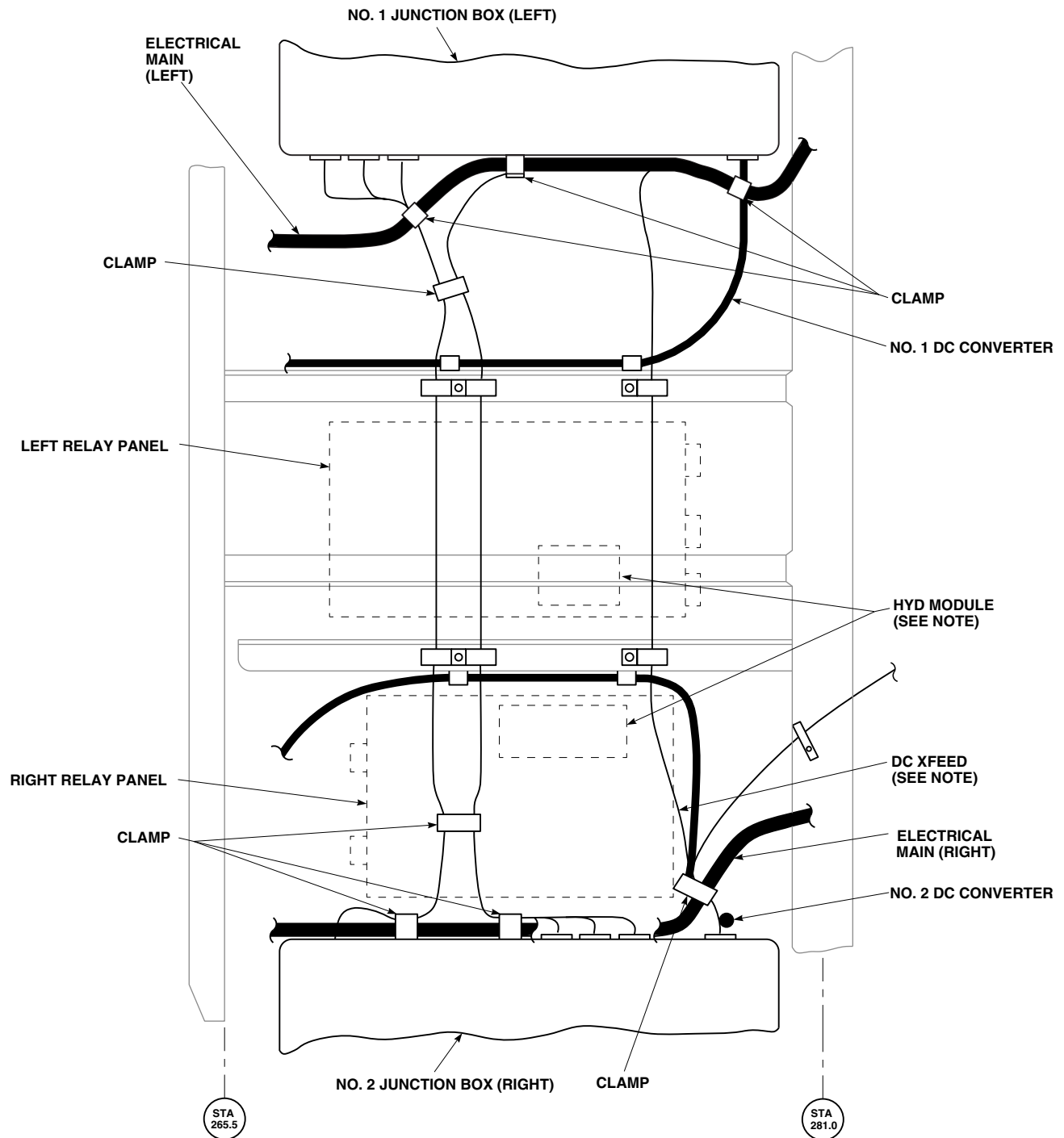
- (1) Using digital low-resistance ohmmeter, measure resistance between each electrical connector of relay panel and overhead support. **RESISTANCE SHALL BE 20.0 MILLIOHMS OR LESS.**
- (2) If resistance is greater than 20.0 milliohms, remove relay panel and apply chemical conversion coating (PARA 2-6-5).

- i. Make sure area is clean and free of foreign material.
- j. Touch up paint, as required (PARA 2-6-5).
- k. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).

NOTE

When relay panel is removed, replaced, or repaired, only affected system(s) require operational checks.

- l. If left relay panel is replaced, perform operational checks of the following as required:
 - (1) Heating and Ventilating System (PARA 13-2-1).
 - (2) Engine Start and Ignition System (PARA 4-2-8).
 - (3) Engine Anti-Ice System (PARA 4-2-7).
 - (4) Fire Detection System (PARA 12-2-1).
 - (5) Fire Extinguishing System (PARA 12-2-2).
 - (6) Auxiliary Power Unit System (PARA 15-2-2).
 - (7) Hydraulic System (PARA 7-2-1).
 - (8) Instrument Panel and Consoles Indicator Lights Dimming System (PARA 9-2-10).
 - (9) Pitot Heater System (PARA 8-2-2).
 - (10) Caution/Advisory Warning System (PARA 8-2-4).
 - (11) DAFCS (TM 11-70-23).
 - (12) Controllable Searchlight (PARA 9-2-13).
 - (13) Countermeasure Set (TM 11-70-23).
 - (14) Cargo Hook System (PARA 12-2-6).



NOTE

CLAMPING AND ROUTING OF HARNESS, 70552-02100, TO AVOID CONTACT WITH THE HYDRAULIC MODULES MOUNTED ON TOP OF THE LEFT / RIGHT RELAY PANELS.

FA1514
SA

FIGURE 9-4-2-2 LEFT/RIGHT RELAY PANEL HARNESS CLAMPING AND ROUTING

- m. If right relay panel is replaced, perform operational checks of the following as required:
 - (1) Heating and Ventilating System (PARA 13-2-1).
 - (2) Windshield Anti-Ice System (PARA 12-2-4).
 - (3) Engine Start and Ignition System (PARA 4-2-8).
 - (4) Engine Anti-Ice System (PARA 4-2-7).
 - (5) Auxiliary Power Unit System (PARA 15-2-2).
 - (6) Hydraulic System (PARA 7-2-1).
 - (7) Pitot Heater System (PARA 8-2-2).
 - (8) Retractable Landing Light System (PARA 9-2-15).
 - (9) Fire Extinguishing System (PARA 12-2-2).
 - (10) Cargo Hook System (PARA 12-2-6).
- n. Make sure area is clean and free of foreign material.
- o. Install soundproofing panel (PARA 2-4-69).

9-4-3 LEFT RELAY PANEL FUSES.

PARAGRAPH BREAKDOWN

		PAGE
9-4-3.1	Replace Left Relay Panel Fuses	9-4-3-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

References

- PARA 1-6-16
- PARA 2-4-69

- PARA 9-2-12
- PARA 9-4-34
- PARA 9-4-106

9-4-3.1 Replace Left Relay Panel Fuses.

NOTE

Replacement of left relay panel fuses, F1-1A and F2-1A, is the same.

9-4-3.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Remove soundproofing panel (PARA 2-4-69).
- Push and turn fuse holder cap counterclockwise and remove fuse holder cap (Figure 9-4-3-1, Detail A).
- Remove fuse from fuse holder.

9-4-3.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Install fuse in fuse holder (Figure 9-4-3-1, Detail A).
- Install fuse holder cap on fuse holder by pushing fuse holder cap and turning it clockwise.
- Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Perform operational check of position and formation lights (PARA 9-2-12).
- Make sure area is clean and free of foreign material.
- Install soundproofing panel (PARA 2-4-69).

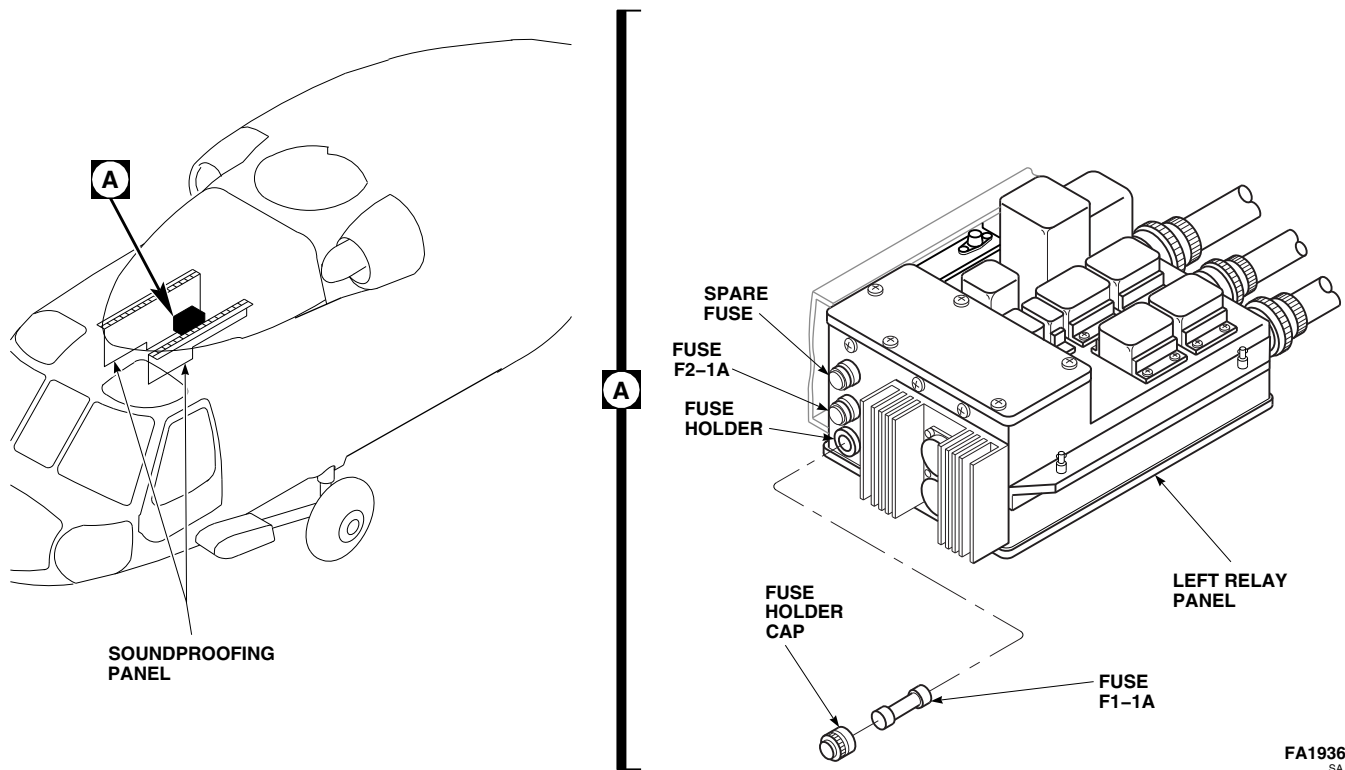


FIGURE 9-4-3-1. REPLACE LEFT RELAY PANEL FUSES

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9-4-4 NO. 3 RELAY PANEL.

PARAGRAPH BREAKDOWN

	PAGE
9-4-4.1 Replace No. 3 Relay Panel	9-4-4-1

INITIAL SETUP	• PARA 1-7-20
Tools	• PARA 2-4-69
• Aircraft Mechanic’s Toolkit • Electrical Repairer’s Toolkit	• PARA 9-2-1
Materials/Parts	• PARA 9-2-10
• Cheesecloth, Item 90, Appendix D • Dry-Cleaning Solvent, Item 124, Appendix D • Protective Caps and Plugs	• PARA 9-2-11 • PARA 12-2-6 • PARA 13-4-22
References	• PARA 15-2-2
• Appendix D • PARA 1-6-16	• PARA 16-2-1

9-4-4.1 Replace No. 3 Relay Panel.

9-4-4.1.1 Remove.

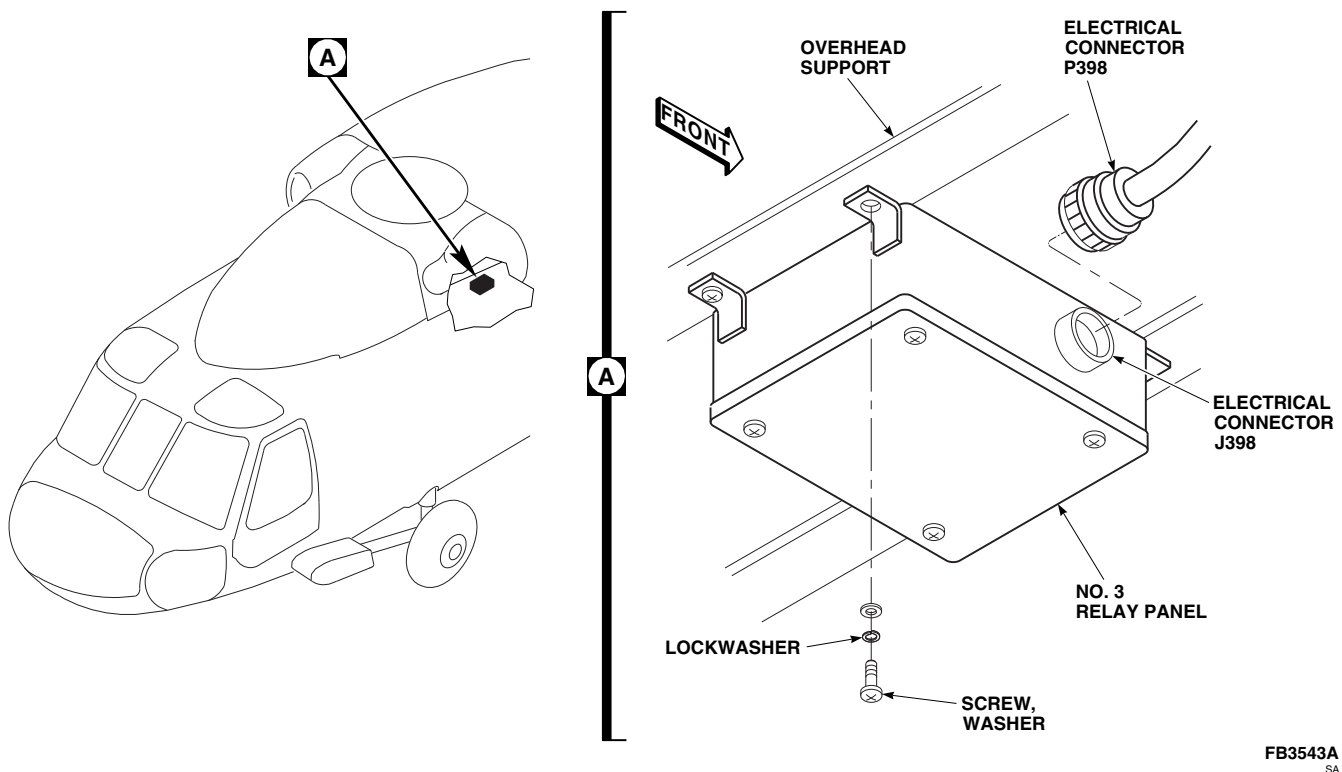
- Turn off all electrical power (PARA 1-6-16).
- Remove soundproofing panel (PARA 2-4-69).
- Remove air conditioning duct section to access No. 3 relay panel (PARA 13-4-22).
- Remove screws, lockwashers, and washers securing No. 3 relay panel to overhead support, and carefully lower to gain access to electrical connector (Figure 9-4-4-1, Detail A).
- Disconnect electrical connector, P398 from J398.

- Remove No. 3 relay panel.

- Install protective caps and plugs on electrical connectors.

9-4-4.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).
- Using cheesecloth, Item 90, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D, clean surface around all mounting holes. Using clean, dry cheesecloth, Item 90, Appendix D, dry surface.
- Remove protective caps and plugs from electrical connectors.

FB3543A
SA**FIGURE 9-4-4-1. REPLACE NO. 3 RELAY PANEL**

- d. Inspect and treat electrical connectors (PARA 1-7-20).
- e. Support No. 3 relay panel and connect electrical connector, P398 to J398, to relay panel (Figure 9-4-4-1, Detail A).
- f. Install No. 3 relay panel to overhead supports and secure with screws, lockwashers, and washers.
- g. Perform operational check of the following:
 - (1) Cargo Hook Assembly (PARA 12-2-6).
 - (2) Instrument Panel and Consoles Indicator Lights Dimming (PARA 9-2-10 or PARA 9-2-11).
 - (3) Auxiliary Power Unit System (PARA 15-2-2).
 - (4) AC Electrical System (PARA 9-2-1).
 - (5) ESSS Range Extension System (PARA 16-2-1).
- h. Make sure area is clean and free of foreign material.
- i. Install air conditioning duct section (PARA 13-4-22).
- j. Install soundproofing panel (PARA 2-4-69).

9-4-5 UPPER CONSOLE SWITCHES.

PARAGRAPH BREAKDOWN

		PAGE
9-4-5.1	Replace Upper Console Switches	9-4-5-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

Materials/Parts

- Tags

References

- PARA 1-6-16
- PARA 9-2-5
- PARA 9-4-6
- PARA 9-4-34
- PARA 9-4-106

9-4-5.1 Replace Upper Console Switches.

NOTE

Replacement of all upper console switches is the same.

9-4-5.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- If required, loosen setscrew securing knob to shaft and remove knob from affected switch (Figure 9-4-5-1, Detail C).
- Remove information plate from upper console (PARA 9-4-6).
- Loosen fasteners securing upper console and carefully lower (Figure 9-4-5-1, Detail A).

- Tag wires. Remove screws securing wires to terminals on switch (Figure 9-4-5-1, Detail B).
- Remove nut and lockwasher securing switch to upper console. Remove switch.

9-4-5.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).
- Install switch from rear of upper console, aligning switch key in slot, and secure with nut and lockwasher (Figure 9-4-5-1, Detail B).
- Install wires to terminals on switch and secure with screws. Remove tags.
- Make sure area is clean and free of foreign material.
- Carefully raise upper console and secure with fasteners (Figure 9-4-5-1, Detail A).
- Install information plate on upper console (PARA 9-4-6).

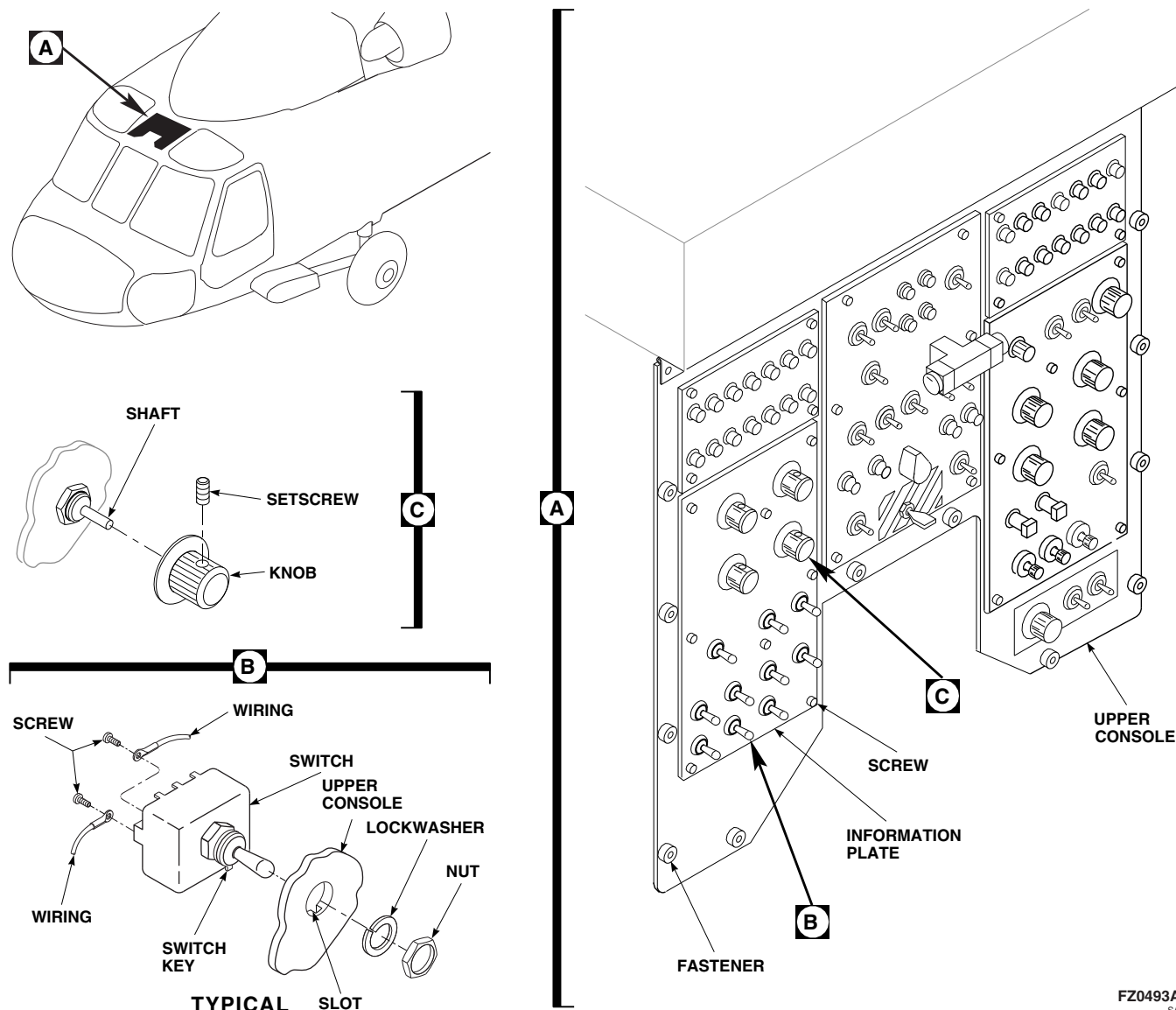


FIGURE 9-4-5-1. REPLACE UPPER CONSOLE SWITCHES

- g. If required, install knob on shaft and secure with setscrew (Figure 9-4-5-1, Detail C).
- h. Make sure area is clean and free of foreign material.
- i. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- j. Perform operational check of upper console lights (PARA 9-2-5).

9-4-6 UPPER CONSOLE INFORMATION PLATES.

PARAGRAPH BREAKDOWN

		PAGE
9-4-6.1	Replace Upper Console Information Plates	9-4-6-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

References

- PARA 1-6-16
- PARA 9-2-5
- PARA 12-4-7

9-4-6.1 Replace Upper Console Information Plates.

NOTE

NOTE

There are three information plates on the upper console. Replacement of all three information plates is the same, except as noted.

Lamp will remain with CARGO HOOK EMERG REL TEST light lens cap when removed.

- (2) Unscrew CARGO HOOK EMERG REL TEST light lens cap, with lamp, and remove from upper console.
- (3) Loosen setscrew securing knob to FIRE DETR TEST switch and remove knob.

9-4-6.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Place toggle switches to up position on information plate to be removed (Figure 9-4-6-1, Detail A).
- Turn control knobs fully counterclockwise.
- Loosen setscrews and remove knobs from shafts on information plate.
- If removing center information plate, perform the following:
 - (1) Remove APU T-handle (PARA 12-4-7).

- Remove screws securing information plate to upper console.
- Remove information plate.

9-4-6.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Install information plate on upper console and secure with screws (Figure 9-4-6-1, Detail A).
- Turn control shafts fully counterclockwise.
- Install knobs on control shafts with index marks lined up with OFF markings and secure with setscrews.

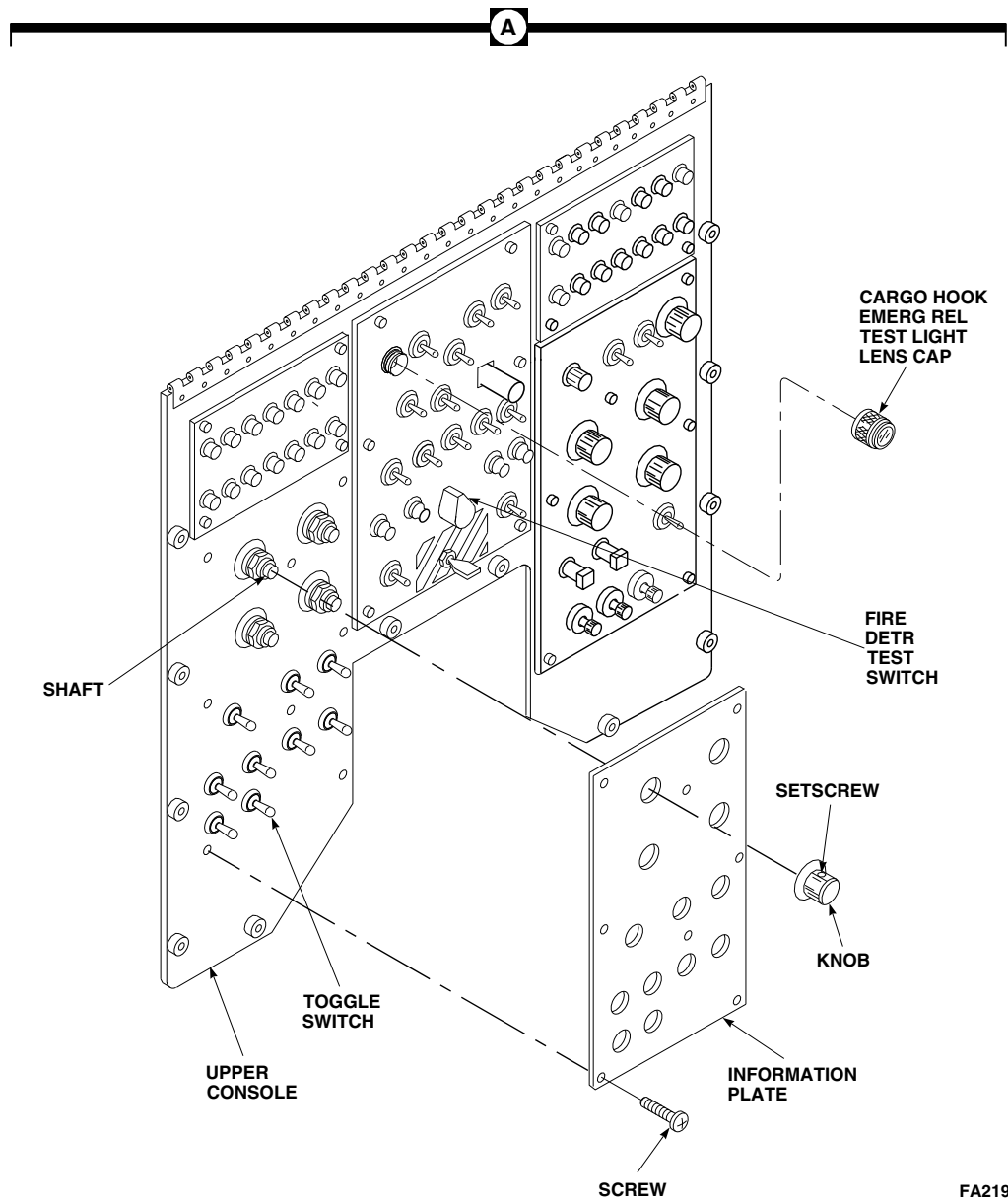
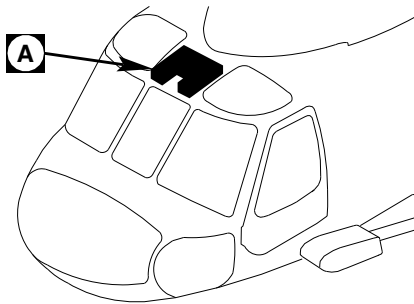
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FIGURE 9-4-6-1. REPLACE UPPER CONSOLE PANEL INFORMATION PLATES

9-4-6-2

- e. If installing center information plate, perform the following:
 - (1) Install APU T-handle (PARA 12-4-7).
 - (2) Screw CARGO HOOK EMERG REL TEST light lens cap, with lamp, in and secure in upper console.
 - (3) Install knob on FIRE DETR TEST switch, line up index mark on knob with OPER position on information plate, and secure with setscrew.
- f. Perform operational check of upper console lighting (PARA 9-2-5).

9-4-7 MISCELLANEOUS SWITCH PANEL.

PARAGRAPH BREAKDOWN

	PAGE
9-4-7.1 Replace Miscellaneous Switch Panel	9-4-7-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Digital Low-Resistance Ohmmeter
- Electrical Repairer’s Toolkit

Materials/Parts

- Cheesecloth, Item 90, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Protective Caps and Plugs
- Tags

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-6-5
- PARA 3-2-2
- PARA 7-2-1
- PARA 9-2-5
- PARA 9-4-34
- PARA 9-4-106
- PARA 10-2-1
- TM 11-70-23

9-4-7.1 Replace Miscellaneous Switch Panel.

9-4-7.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Loosen turnlock fasteners securing miscellaneous switch panel to lower console (Figure 9-4-7-1, Detail A).

- Remove miscellaneous switch panel to access electrical connectors.
- Tag electrical connectors. Disconnect electrical connectors, P659R from J659R, and, P135 from J135, from miscellaneous switch panel.
- Install protective caps and plugs on electrical connectors.

9-4-7.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).

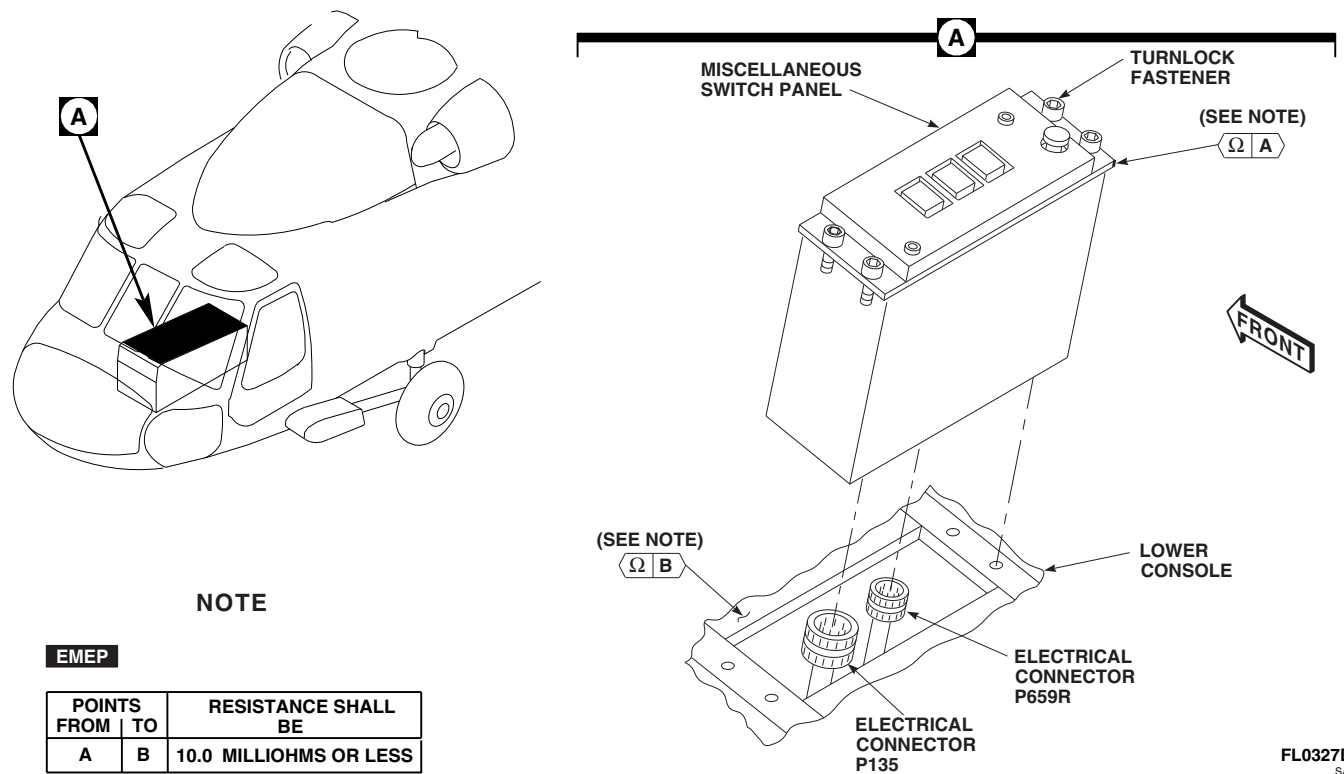


FIGURE 9-4-7-1. REPLACE MISCELLANEOUS SWITCH PANEL

- b. **EMEP** Inspect chemical conversion coating as follows:
 - (1) Using cheesecloth, Item 90, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D, clean surface around all mounting holes. Using clean, dry cheesecloth, Item 90, Appendix D, dry surface.
 - (2) Inspect chemical conversion coated surfaces for breaks and scratches. **NONE ALLOWED.** If coated surfaces have breaks or scratches, apply chemical conversion coating (PARA 2-6-5).
- c. Remove protective caps and plugs from electrical connectors.
- d. Inspect and treat electrical connectors (PARA 1-7-20).
- e. Connect electrical connectors, P135 to J135, and P659R to J659R, to miscellaneous switch panel (Figure 9-4-7-1, Detail A). Remove tags.
- f. Make sure area is clean and free of foreign material.
- g. Carefully install miscellaneous switch panel in lower console and secure with turnlock fasteners.
- h. **EMEP** Perform EME resistance check of miscellaneous switch panel as follows:
 - (1) Using digital low-resistance ohmmeter, check resistance between faceplate of miscellaneous switch panel and lower console. **RESISTANCE SHALL BE 10.0 MILLIOHMS OR LESS.**
 - (2) If resistance is greater than 10.0 milliohms, remove miscellaneous switch panel and apply chemical conversion coating (PARA 2-6-5).

NOTE

Resistance measurements must be taken at least three times.

- i. Touch up paint, as required (PARA 2-6-5).
- j. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- k. Perform operational check of the following:
 - (1) Fuel Quantity System (PARA 10-2-1).
 - (2) Tailwheel Lock System (PARA 3-2-2).
 - (3) Attitude Indicating System (TM 11-70-23).
 - (4) Tail Rotor Servo in No. 1 Hydraulic and Backup System (PARA 7-2-1).
 - (5) Upper and Lower Console Lights (PARA 9-2-5).

9-4-8 MISCELLANEOUS SWITCH PANEL PUSHBUTTON LAMPS.

PARAGRAPH BREAKDOWN

		PAGE
9-4-8.1	Replace Miscellaneous Switch Panel Pushbutton Lamps	9-4-8-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit

References

- PARA 1-6-16
- PARA 3-2-2

- PARA 7-2-1
- PARA 9-2-5
- PARA 10-2-1
- TM 11-70-23

9-4-8.1 Replace Miscellaneous Switch Panel Pushbutton Lamps.



NOTE

Miscellaneous switch panel has three pushbutton lamps. Replacement of all three miscellaneous switch panel push-button lamps is the same.

If soft lighting system is installed, to prevent damage to lamps, make sure soft lighting system compatible lamps are installed.

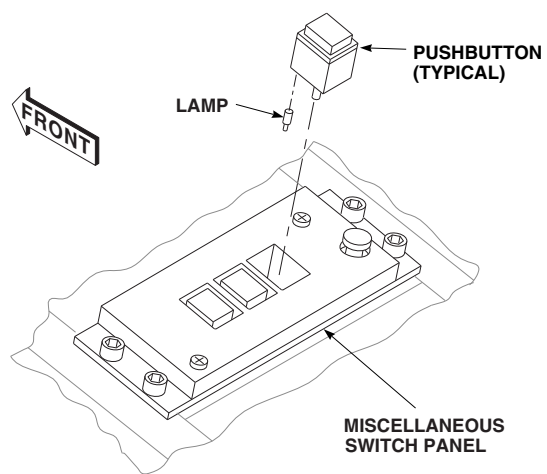
9-4-8.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Pull and remove pushbutton from miscellaneous switch panel (Figure 9-4-8-1).
- Twist and remove lamp from pushbutton.

9-4-8.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).

- Twist lamp into pushbutton (Figure 9-4-8-1).
- Install pushbutton into miscellaneous switch panel.
- Perform operational check of the following:
 - Fuel Quantity System (PARA 10-2-1).
 - Tailwheel Lock System (PARA 3-2-2).
 - Attitude Indicating System (TM 11-70-23).
 - Tail Rotor Servo in No. 1 Hydraulic and Backup System (PARA 7-2-1).

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SA**FIGURE 9-4-8-1. REPLACE MISCELLANEOUS SWITCH PANEL PUSHBUTTON LAMPS**

- (5) Upper and Lower Console Lights (PARA 9-2-5).

9-4-9 MISCELLANEOUS SWITCH PANEL INFORMATION PLATE AND LAMPS.

PARAGRAPH BREAKDOWN

		PAGE
9-4-9.1	Replace Miscellaneous Switch Panel Information Plate and Lamps	9-4-9-1

INITIAL SETUP

- PARA 9-2-25

Tools

- Electrical Repairer’s Toolkit

References

- PARA 1-6-16
- PARA 9-2-5

9-4-9.1 Replace Miscellaneous Switch Panel Information Plate and Lamps.

9-4-9.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Remove screws securing information plate to miscellaneous switch panel and remove information plate (Figure 9-4-9-1).
- Remove screws securing lamp to back of information plate.
- Remove lamp.

9-4-9.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).

- Install lamp in back of information plate and secure with screws (Figure 9-4-9-1).
- Install information plate on miscellaneous switch panel and secure with screws.
- On copilot’s circuit breaker panel, make sure LIGHTS LWR CNSL circuit breaker is pushed in.
- On upper console, turn CONSOLE LT LOWER OFF-BRT control to BRT.
- Make sure area is clean and free of foreign material.
- Perform operational check of upper and lower console lights (PARA 9-2-5 or PARA 9-2-25).

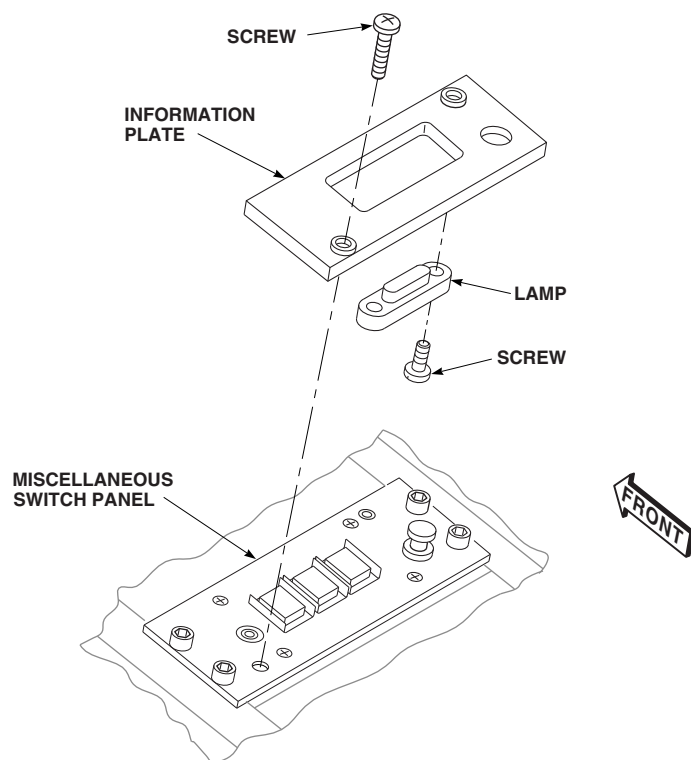
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FIGURE 9-4-9-1. REPLACE MISCELLANEOUS SWITCH PANEL INFORMATION PLATE AND LAMPS

9-4-10 AUTOTRANSFORMER, T12.

PARAGRAPH BREAKDOWN

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9-4-10.1 Replace Autotransformer, T12	9-4-10-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

Materials/Parts

- Tags

References

- PARA 1-6-16
- PARA 2-4-69
- PARA 9-2-1
- PARA 9-4-34
- PARA 9-4-106

9-4-10.1 Replace Autotransformer, T12.

9-4-10.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Remove soundproofing panel (PARA 2-4-69).
- Loosen fasteners securing cover on No. 1 junction box and let cover hang by chain (Figure 9-4-10-1, Detail A).
- Tag wires. Remove screws, lockwashers, and washers securing wiring to terminals of autotransformer, T12.
- Remove screws and washers securing autotransformer, T12 to No. 1 junction box.
- Remove autotransformer, T12.

9-4-10.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Install autotransformer, T12, in No. 1 junction box and secure with screws and washers (Figure 9-4-10-1, Detail A).
- Install wiring to terminals of autotransformer, T12, and secure with screws, lockwashers, and washers. Remove tags.
- Install cover on No. 1 junction box and secure with fasteners.
- Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Perform operational check of ac electrical system (PARA 9-2-1).
- Make sure area is clean and free of foreign material.
- Install soundproofing panel (PARA 2-4-69).

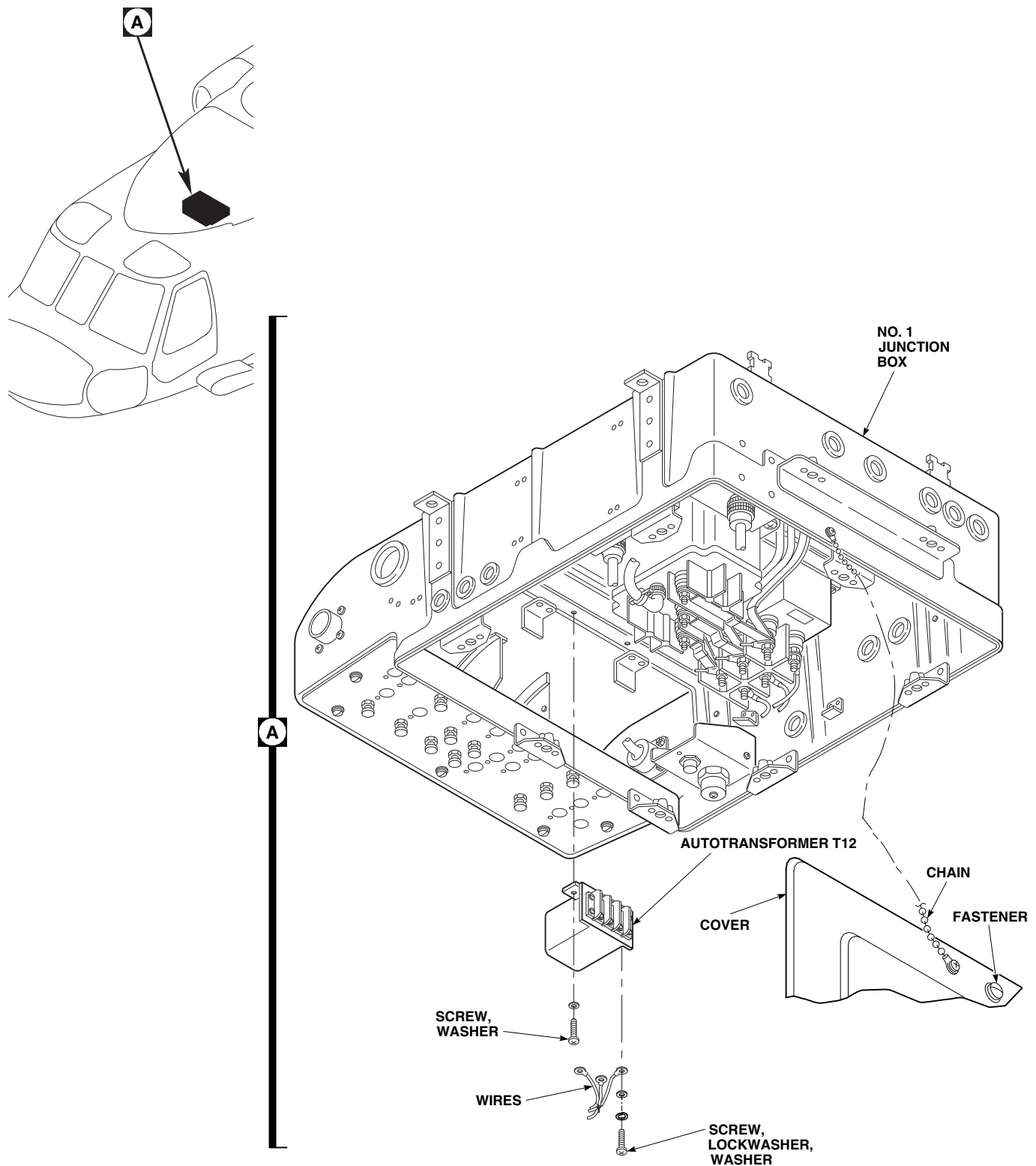
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FIGURE 9-4-10-1. REPLACE AUTOTRANSFORMER, T12

9-4-10-2

9-4-11 NO. 1 AC GENERATOR CONTACTOR, K1.

PARAGRAPH BREAKDOWN

	PAGE
9-4-11.1 Replace No. 1 AC Generator Contactor, K1	9-4-11-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

Materials/Parts

- Protective Caps and Plugs
- Tags

References

- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-69
- PARA 9-2-1
- PARA 9-4-34
- PARA 9-4-106

9-4-11.1 Replace No. 1 AC Generator Con-
tactor, K1.

9-4-11.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Remove soundproofing panel (PARA 2-4-69).
- Loosen fasteners securing cover on No. 1 junction box and let cover hang by chain (Figure 9-4-11-1, Detail A).
- Remove screws and washers securing terminal shield to No. 1 ac generator contactor, K1.
- Disconnect electrical connector, P212 from J212, on No. 1 ac generator contactor, K1.

- Install protective caps and plugs on electrical connectors.
- Remove nuts, lockwashers, and washers securing bus bars to No. 1 ac generator contactor, K1.
- Tag wires. Remove nuts, lockwashers, and washers, securing bus bars and wiring to CB404 HYD PWR PUMP CKT BKR.
- Remove screws and washers securing No. 1 ac generator contactor, K1, to No. 1 junction box.
- Remove No. 1 ac generator contactor, K1.

9-4-11.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Install No. 1 ac generator contactor in No. 1 junction box and secure with screws and washers (Figure 9-4-11-1, Detail A).

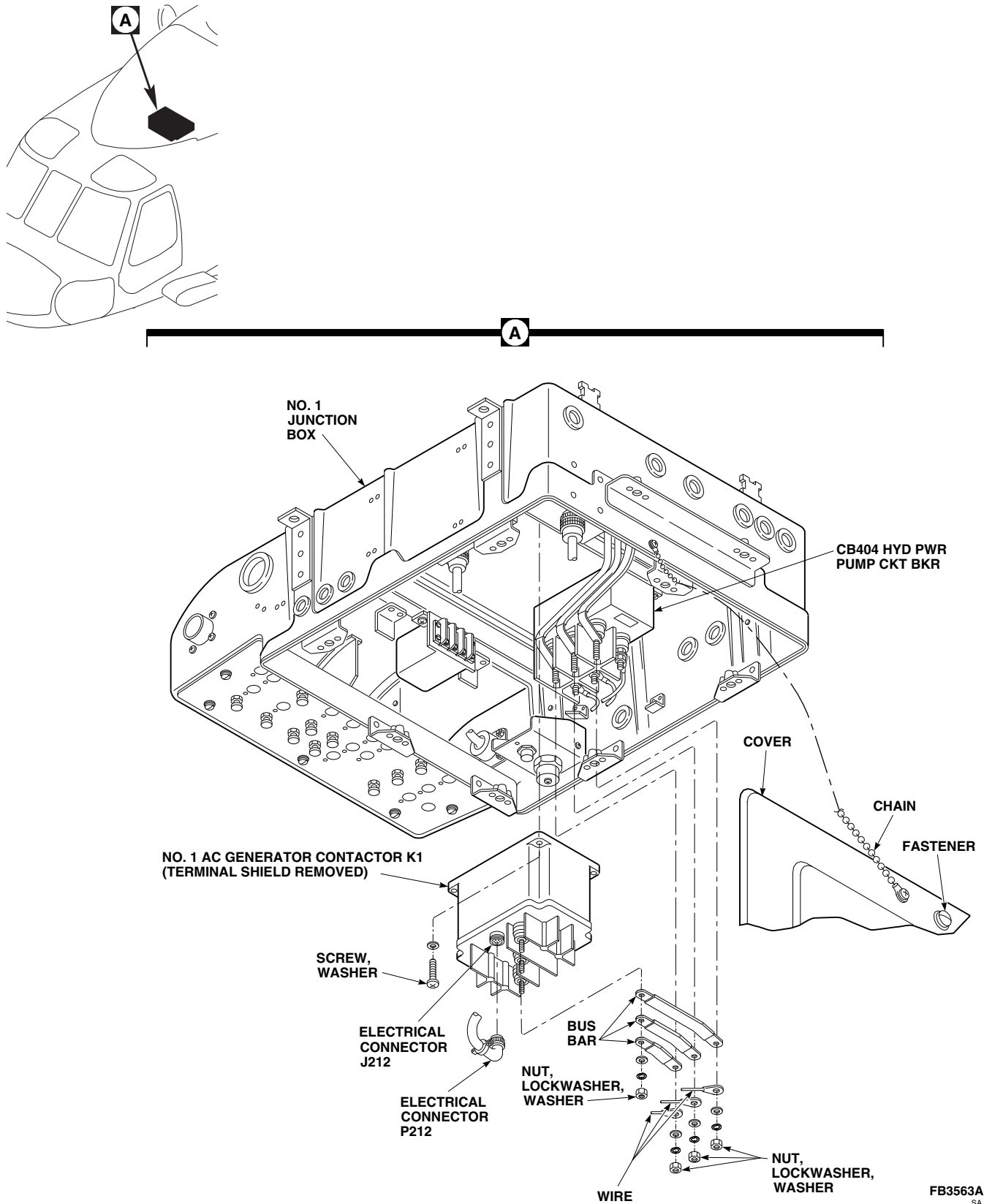


FIGURE 9-4-11-1. REPLACE NO. 1 AC GENERATOR CONTACTOR, K1

9-4-11-2

- c. Install bus bars on No. 1 ac generator contactor, K1, and CB404 HYD PWR PUMP CKT BKR. Secure bus bars to No. 1 ac generator contactor, K1, with nuts, lockwashers, and washers.
- d. Install wiring over bus bars on CB404 HYD PWR PUMP CKT BKR. Secure wiring and bus bars with nuts, lockwashers, and washers. Remove tags.
- e. Remove protective caps and plugs from electrical connectors.
- f. Inspect and treat electrical connectors (PARA 1-7-20).
- g. Connect electrical connector, P212 to J212, on No. 1 ac generator contactor, K1.
- h. Install terminal shield on No. 1 ac generator contactor, K1, and secure with screws and washers.
- i. Install cover on No. 1 junction box and secure with fasteners.
- j. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- k. Perform operational check of ac electrical system (PARA 9-2-1).
- l. Make sure area is clean and free of foreign material.
- m. Install soundproofing panel (PARA 2-4-69).

9-4-12 NO. 2 AC GENERATOR CONTACTOR, K2.

PARAGRAPH BREAKDOWN

	PAGE
9-4-12.1 Replace No. 2 AC Generator Contactor, K2	9-4-12-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

Materials/Parts

- Protective Caps and Plugs
- Tags

References

- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-69
- PARA 9-2-1
- PARA 9-4-34
- PARA 9-4-106

9-4-12.1 Replace No. 2 AC Generator Con-
tactor, K2.

9-4-12.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Remove soundproofing panel (PARA 2-4-69).
- Loosen fasteners securing cover on No. 2 junction box and let cover hang by chain (Figure 9-4-12-1, Detail A).
- Remove screws and washers securing terminal shields to No. 2 ac generator contactor, K2, and ac bus tie contactor, K4.
- Disconnect electrical connector, P204 from J204, on No. 2 ac generator contactor, K2.

- Install protective caps and plugs on electrical connectors.
- Tag wires. Remove nuts, washers, and lock-washers securing bus bars and wiring to No. 2 ac generator contactor, K2.
- Remove screws, lockwashers and washers securing bus bars to mounting block.
- Remove nuts, washers, and lockwashers secur-ing bus bars to No. 2 ac generator contactor, K2, and ac bus tie contactor, K4.
- Remove screws and washers securing No. 2 ac generator contactor, K2 to No. 2 junction box.
- Remove No. 2 ac generator contactor, K2.

9-4-12.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).

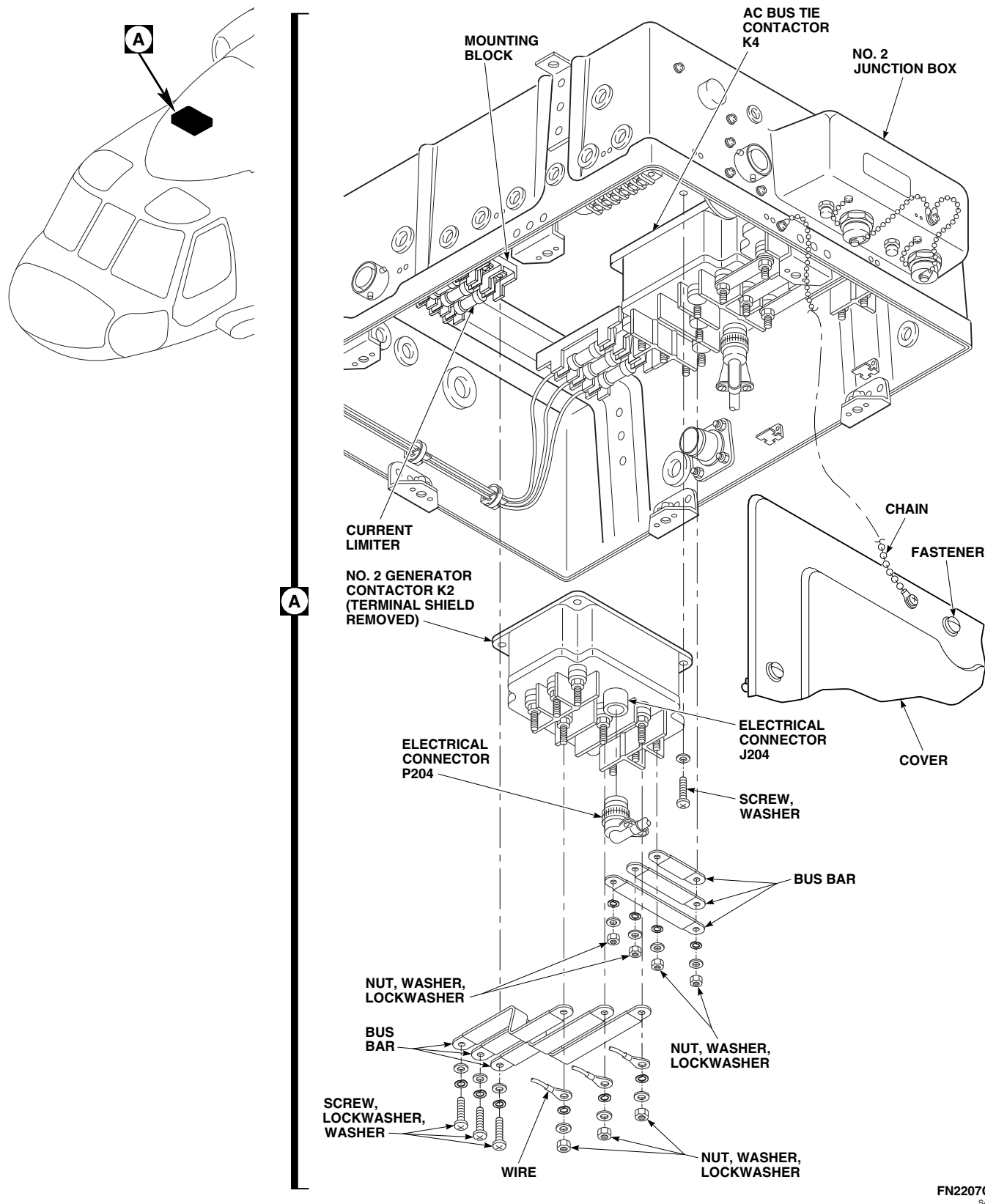


FIGURE 9-4-12-1. REPLACE NO. 2 AC GENERATOR CONTACTOR, K2

9-4-12-2

- b. Install No. 2 ac generator contactor, K2, in No. 2 junction box and secure with screws and washers (Figure 9-4-12-1, Detail A).
- c. Install bus bars on No. 2 ac generator contactor, K2, and ac bus tie contactor, K4. Secure with nuts, washers, and lockwashers.
- d. Install bus bars on No. 2 ac generator contactor, K2, and mounting block. Secure bus bars to mounting block with screws, lockwashers, and washers.
- e. Install wiring over bus bars on No. 2 ac generator contactor, K2. Secure wiring and bus bars with nuts, washers, and lockwashers. Remove tags.
- f. Remove protective caps and plugs from electrical connectors.
- g. Inspect and treat electrical connectors (PARA 1-7-20).
- h. Connect electrical connector, P204 to J204, on No. 2 ac generator contactor, K2.
- i. Install terminal shields on No. 2 ac generator contactor, K2, and ac bus tie contactor, K4, and secure with screws and washers.
- j. Install cover on No. 2 junction box and secure with fasteners.
- k. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- l. Perform operational check of ac electrical system (PARA 9-2-1).
- m. Make sure area is clean and free of foreign material.
- n. Install soundproofing panel (PARA 2-4-69).

9-4-13 CURRENT LIMITERS, CL1 THROUGH CL9.

PARAGRAPH BREAKDOWN

	PAGE
9-4-13.1 Replace Current Limiters, CL1 through CL9	9-4-13-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

References

- PARA 1-6-16
- PARA 2-4-69

- PARA 9-2-1
- PARA 9-4-34
- PARA 9-4-106

9-4-13.1 Replace Current Limiters, CL1 through CL9.

NOTE

Replacement of current limiters, CL1 through CL9, is the same.

9-4-13.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Remove soundproofing panels as required (PARA 2-4-69).
- Loosen fasteners securing cover on No. 1 or No. 2 junction box and let cover hang by chain (Figure 9-4-13-1, Detail A and Detail B).
- Pull and remove current limiter from its mounting block.

9-4-13.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Position current limiter over its mounting block and press it into mounting block (Figure 9-4-13-1, Detail A and Detail B).
- Install cover on No. 1 or No. 2 junction box and secure with fasteners.
- Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Perform operational check of ac electrical system (PARA 9-2-1).
- Make sure area is clean and free of foreign material.
- Install soundproofing panels (PARA 2-4-69).

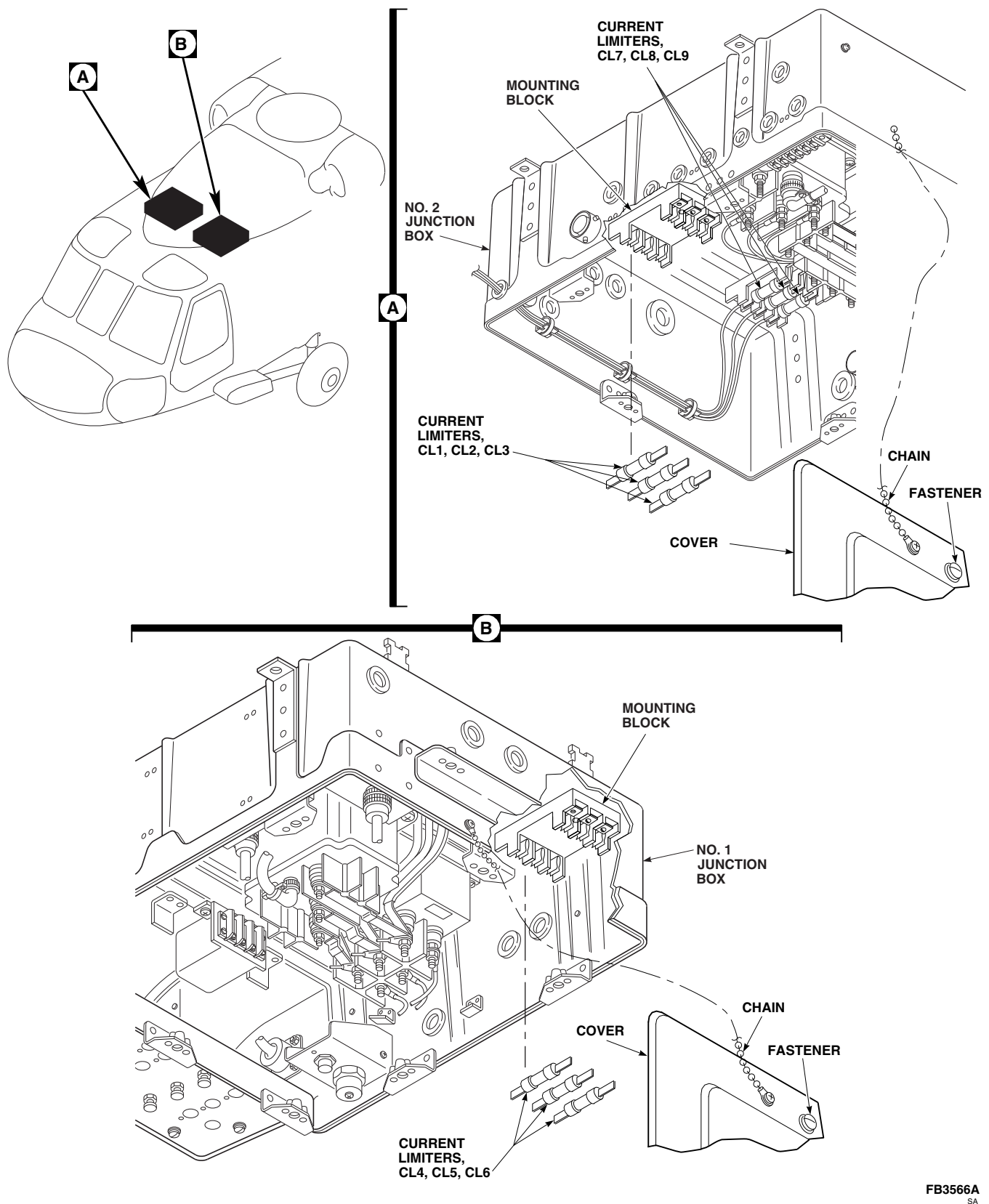
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FIGURE 9-4-13-1. REPLACE CURRENT LIMITERS, CL1 THROUGH CL9

9-4-13-2

9-4-14 GENERATOR AND APU CURRENT TRANSFORMERS, T2, T3, AND T13.

PARAGRAPH BREAKDOWN

	PAGE
9-4-14.1 Replace Generator and APU Current Transformers, T2, T3, and T13	9-4-14-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

Materials/Parts

- Tags

References

- PARA 1-6-16
- PARA 2-4-69
- PARA 9-2-1
- PARA 9-4-34
- PARA 9-4-106

9-4-14.1 Replace Generator and APU Current Transformers, T2, T3, and T13.

NOTE

Replacement of No. 1 generator (T13), No. 2 generator (T2), or APU (T3) current transformers is the same.

9-4-14.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Remove soundproofing panel (PARA 2-4-69).
- Loosen fasteners securing cover on No. 1 or No. 2 junction box and let cover hang by chain (Figure 9-4-14-1, Detail A or Detail B).

- Tag wires. Remove screws and washers securing wiring to current transformer.
- Remove screws and washers securing terminal shield to No. 1 or No. 2 ac generator contactor, K1 or K2, or APU/external power contactor, K3, as applicable.
- Tag wires. Remove screws and washers securing wiring to terminals, A1, B1, and C1, of No. 1 or No. 2 ac generator contactor.
- Remove screws and washers securing current transformer to junction box, carefully passing wires through holes in current transformer.
- Remove current transformer.

9-4-14.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Carefully pass wires through proper holes in current transformer.

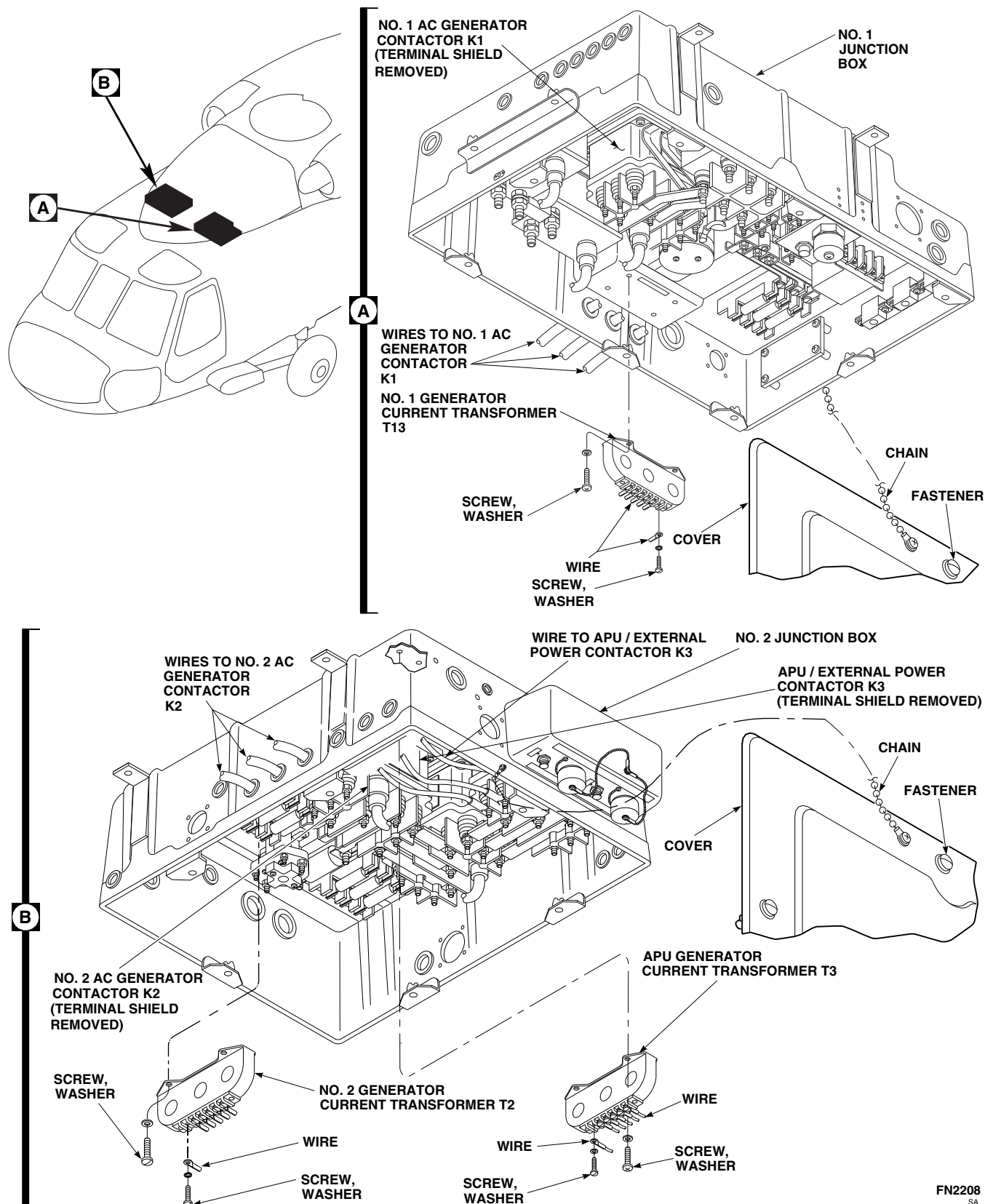


FIGURE 9-4-14-1. REPLACE GENERATOR AND APU CURRENT TRANSFORMERS, T2, T3, AND T13

- c. Install current transformer in junction box and secure with screws and washers (Figure 9-4-14-1, Detail A or Detail B).
- d. Install wiring to terminals, A1, B1, and C1, of No. 1 or No. 2 ac generator contactor, APU/external power contactor, as applicable, with screws and washers. Remove tags.
- e. Install terminal shield on No. 1 or No. 2 generator contactor, K1 or K2, or APU/external power contactor, K3, and secure with screws and washers.
- f. Install wiring to current transformer and secure with screws and washers. Remove tags.
- g. Install cover on No. 1 or No. 2 junction box and secure with fasteners.
- h. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- i. Perform operational check of ac electrical system (PARA 9-2-1).
- j. Make sure area is clean and free of foreign material.
- k. Install soundproofing panel (PARA 2-4-69).

9-4-15 GENERATOR CONTROL UNITS.

PARAGRAPH BREAKDOWN

	PAGE
9-4-15.1 Replace Generator Control Units	9-4-15-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Digital Low-Resistance Ohmmeter
- Electrical Repairer’s Toolkit

Materials/Parts

- Cheesecloth, Item 90, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Protective Caps and Plugs
- Tags

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-69
- PARA 2-6-5
- PARA 9-2-1
- PARA 9-4-34
- PARA 9-4-106

9-4-15.1 Replace Generator Control Units.

NOTE

NOTE

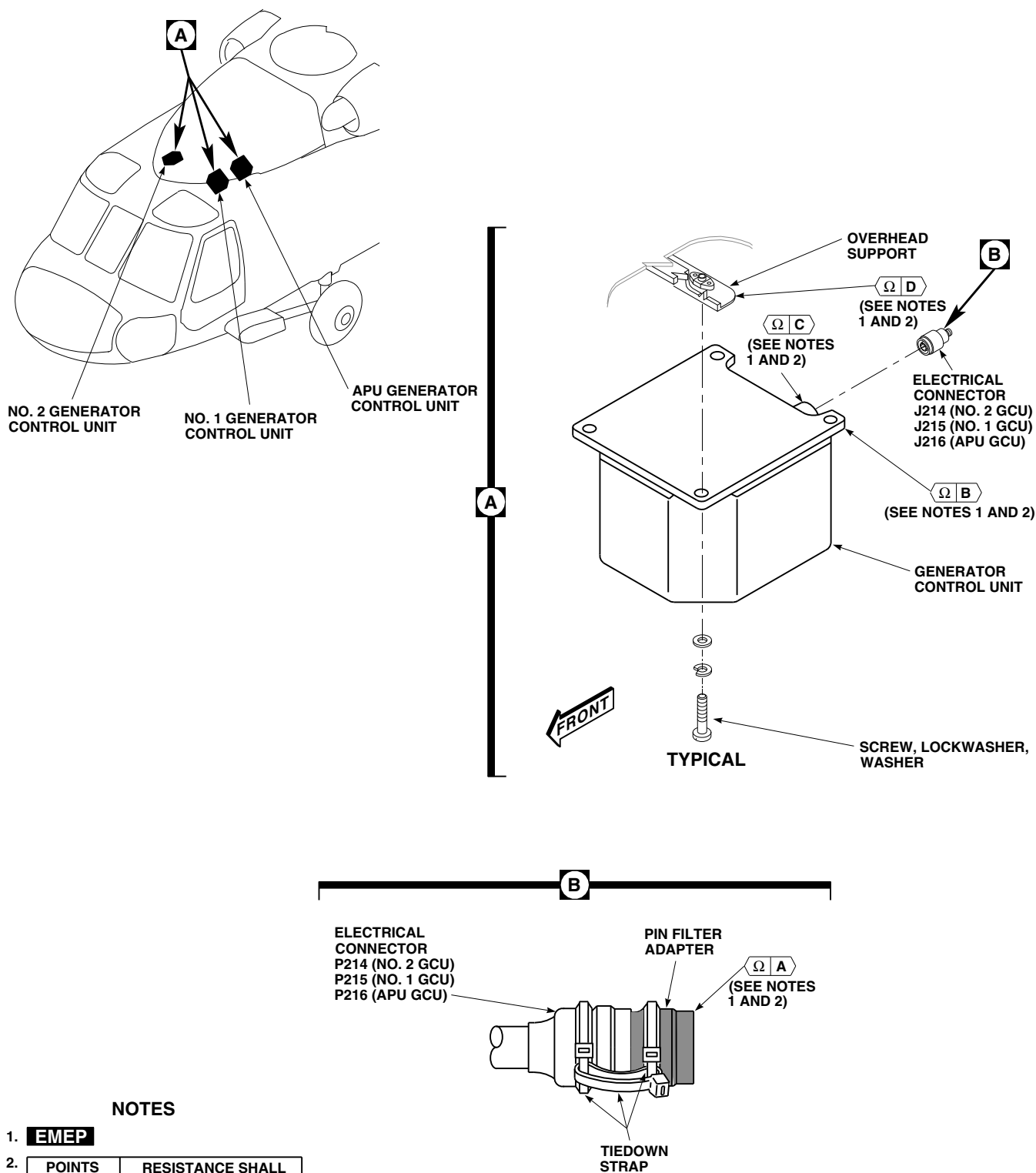
Replacement of No. 1, No. 2, and APU generator control units (GCU) is the same, except as noted.

EMEP Pin filter adapters are considered part of harness assembly. Do not remove pin filter adapter from electrical connectors, P214, P215, and P216 (Figure 9-4-15-1, Detail B).

9-4-15.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Remove soundproofing panels as required (PARA 2-4-69).

- Tag electrical connectors. Disconnect electrical connector from GCU as follows (Figure 9-4-15-1, Detail A):
 - P214 from J214 (No. 2 GCU).
 - P215 from J215 (No. 1 GCU).
 - P216 from J216 (APU GCU).



NOTES

1. **EMEP**

POINTS FROM	TO	RESISTANCE SHALL BE
A	B	5.0 MILLIOHMS OR LESS
A	C	5.0 MILLIOHMS OR LESS
B	D	2.5 MILLIOHMS OR LESS

FIGURE 9-4-15-1. REPLACE GENERATOR CONTROL UNITS

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- e. Install protective caps and plugs on electrical connectors.
- f. Remove screws, lockwashers, and washers securing GCU to overhead support.
- g. Remove GCU.

9-4-15.1.2 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. **EMEP** Inspect chemical conversion coating as follows:
 - (1) Using cheesecloth, Item 90, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D, clean surface around all mounting surface. Using clean, dry cheesecloth, Item 90, Appendix D, dry surface.
 - (2) Inspect chemical conversion coated surfaces for breaks and scratches. **NONE ALLOWED.** If coated surfaces have breaks or scratches, apply chemical conversion coating (PARA 2-6-5).
- c. Install GCU to overhead support and secure with screws, lockwashers, and washers (Figure 9-4-15-1, Detail A).
- d. Remove protective caps and plugs from electrical connectors.
- e. Inspect and treat electrical connectors (PARA 1-7-20).
- f. Connect electrical connector to GCU as follows:
 - (1) P214 to J214 (No. 2 GCU).
 - (2) P215 to J215 (No. 1 GCU).
 - (3) P216 to J216 (APU GCU).
- g. Remove tags.
- h. **EMEP** Perform pin filter adapter EME resistance check as follows:

NOTE

Resistance measurements must be taken at least three times.

- (1) Using digital low-resistance ohmmeter, measure resistance between shell of pin filter adapter and GCU. **RESISTANCE SHALL BE 5.0 MILLIOHMS OR LESS.**
- (2) If resistance is greater than 5.0 milliohms, measure resistance between shell of pin filter adapter and electrical connector on GCU. **RESISTANCE SHALL BE 5.0 MILLIOHMS OR LESS.**
- (3) If resistance is still beyond limits, clean surface between pin filter adapter and GCU.

- i. **EMEP** Perform EME resistance check of GCU as follows:

NOTE

Resistance measurements must be taken at least three times.

- (1) Using digital low-resistance ohmmeter, check resistance between GCU and overhead support. **RESISTANCE SHALL BE 2.5 MILLIOHMS OR LESS.**
- (2) If resistance is greater than 2.5 milliohms, remove generator unit and apply chemical conversion coating (PARA 2-6-5).
- j. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- k. Perform operational check of ac electrical system (PARA 9-2-1).
- l. Make sure area is clean and free of foreign material.
- m. Install soundproofing panels (PARA 2-4-69).

9-4-16 MAIN GENERATORS.

PARAGRAPH BREAKDOWN

	PAGE
9-4-16.1 Replace Main Generators	9-4-16-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Container, 1-Quart
- Electrical Repairer’s Toolkit
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Lubricating Oil, Item 205, Appendix D
- Lubricating Oil, Item 206, Appendix D
- Lubricating Oil, Item 208, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D

- Protective Caps and Plugs
- Sealing Compound, Item 292, Appendix D
- Spiral Wrap, Item 315, Appendix D
- Tags
- Tiedown Strap, Item 334, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 6-4-15
- PARA 6-4-29
- PARA 9-2-1
- PARA 9-4-34
- PARA 9-4-106

9-4-16.1 Replace Main Generators.

NOTE

Replacement of No. 1 and No. 2 main generators is the same, except as noted.

9-4-16.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).

- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Open main rotor pylon sliding cover.
- If installed, remove screws and washers and slide terminal block cover forward on electrical leads of main generator, or pull rubber terminal covers back from terminals, as required, to gain

access to electrical leads (Figure 9-4-16-1, Sheet 2, Detail D or Detail F).

- e. Tag electrical leads. Remove nuts, lockwashers, and washers securing electrical leads to terminals.

NOTE

Terminal block cover will not be reused.

- f. If terminal block cover was installed, pull electrical leads through terminal block cover and remove terminal block cover.
- g. Remove screw, washer, and clamp securing electrical leads to end of main generator.
- h. Tag electrical connectors. Disconnect electrical connectors as follows (Figure 9-4-16-1, Sheet 1, Detail A):

No. 1 Main Generator	No. 2 Main Generator
P251 from J251	P250 from J250
P278 from J278	P277 from J277

- i. Install protective caps and plugs on electrical connectors.
- j. Using nonmetallic scraper, remove sealing compound from nuts.
- k. Position 1-quart container to catch any leaking oil.

CAUTION

- Damage to main generator will result if rotor shaft is turned when main generator is removed from accessory module. Do not turn main generator rotor shaft with main generator removed.
- Damage to accessory module will result if main generator is not carefully

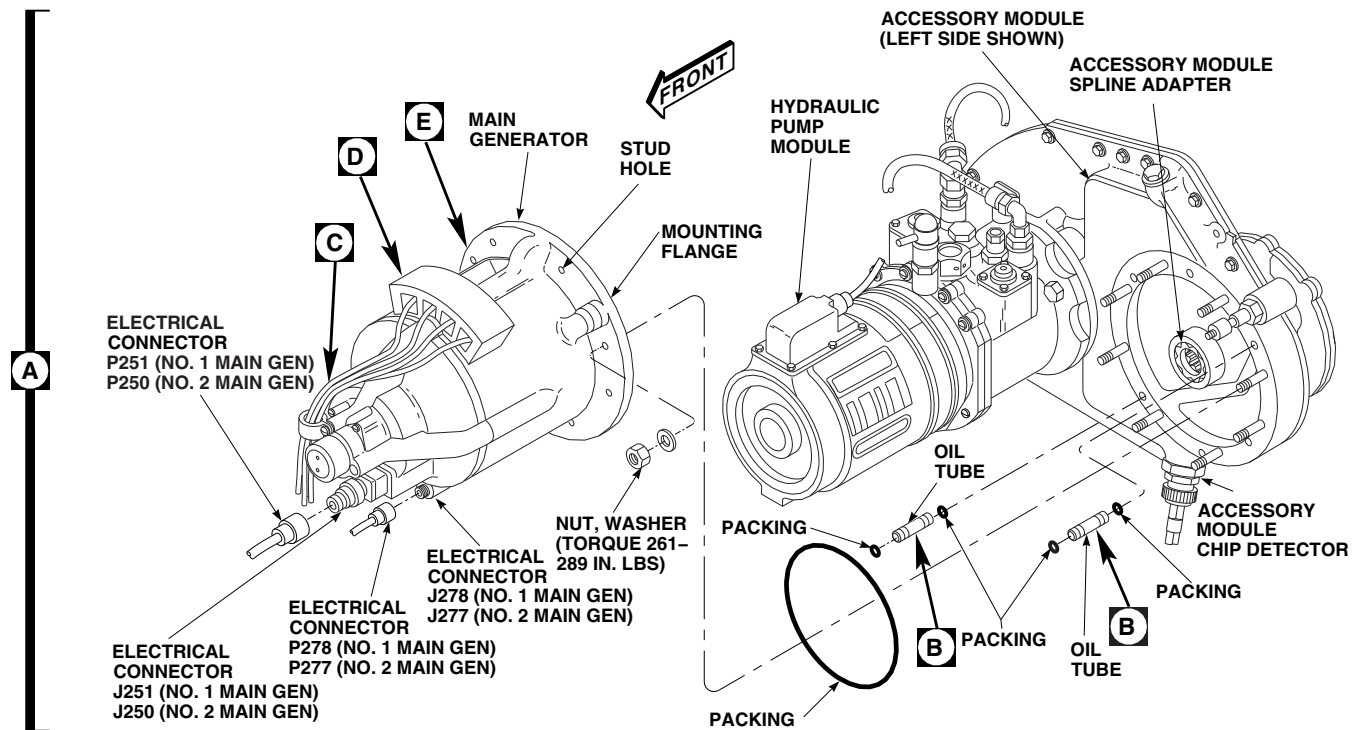
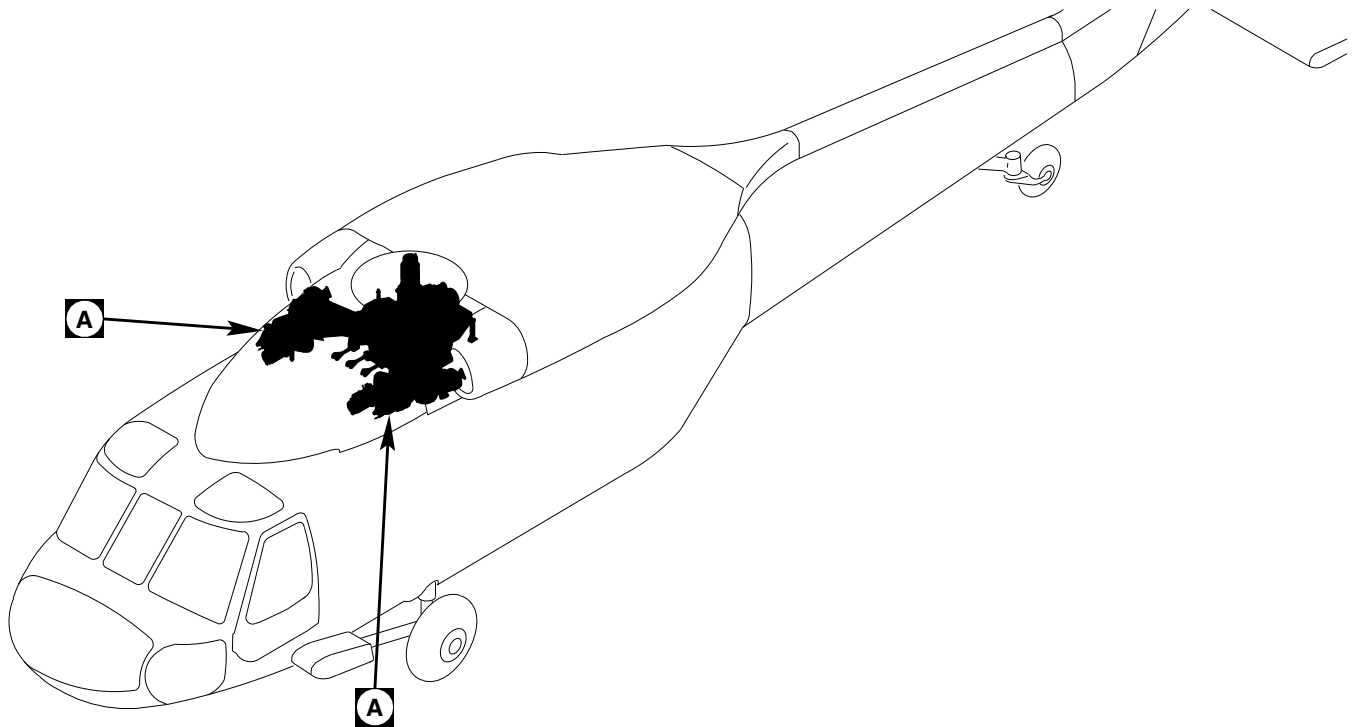
removed. Use care when removing main generator to prevent surface damage and subsequent corrosion.

NOTE

- Main generator weighs approximately 30 pounds.
 - To prevent oil spillage, remove main generator with front end down.
- Remove nuts and washers securing main generator to accessory module and remove main generator and packing. Discard packing.
 - Using nonmetallic scraper, remove sealing compound from mating surfaces and all mounting nuts.
 - If necessary, using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, remove remaining sealing compound.
 - Remove oil tubes and packings from main generator. Keep oil tubes with accessory module and discard packings.

9-4-16.1.2 Inspect.

- Inspect accessory module spline adapter for wear, cracks, or heat damage (PARA 6-4-29). **NONE ALLOWED.** If removal is required to complete inspection, make sure no broken pieces enter input module through hole in main generator gear.
- Inspect cover/screen flathead screws at back of main generator (Figure 9-4-16-1, Sheet 2, Detail E). If any of the six flathead screws are missing, inspect accessory module chip detector (PARA 6-4-15).
- Inspect electrical leads in main generator harness for chafing. **NONE ALLOWED.** If chafed, repair electrical leads.
- If installed, inspect rubber terminal covers for cracks. **NONE ALLOWED.** If cracked, replace rubber terminal covers.



NOTE

NO. 1 MAIN GENERATOR.

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FIGURE 9-4-16-1. REPLACE MAIN GENERATORS (SHEET 1 OF 2)

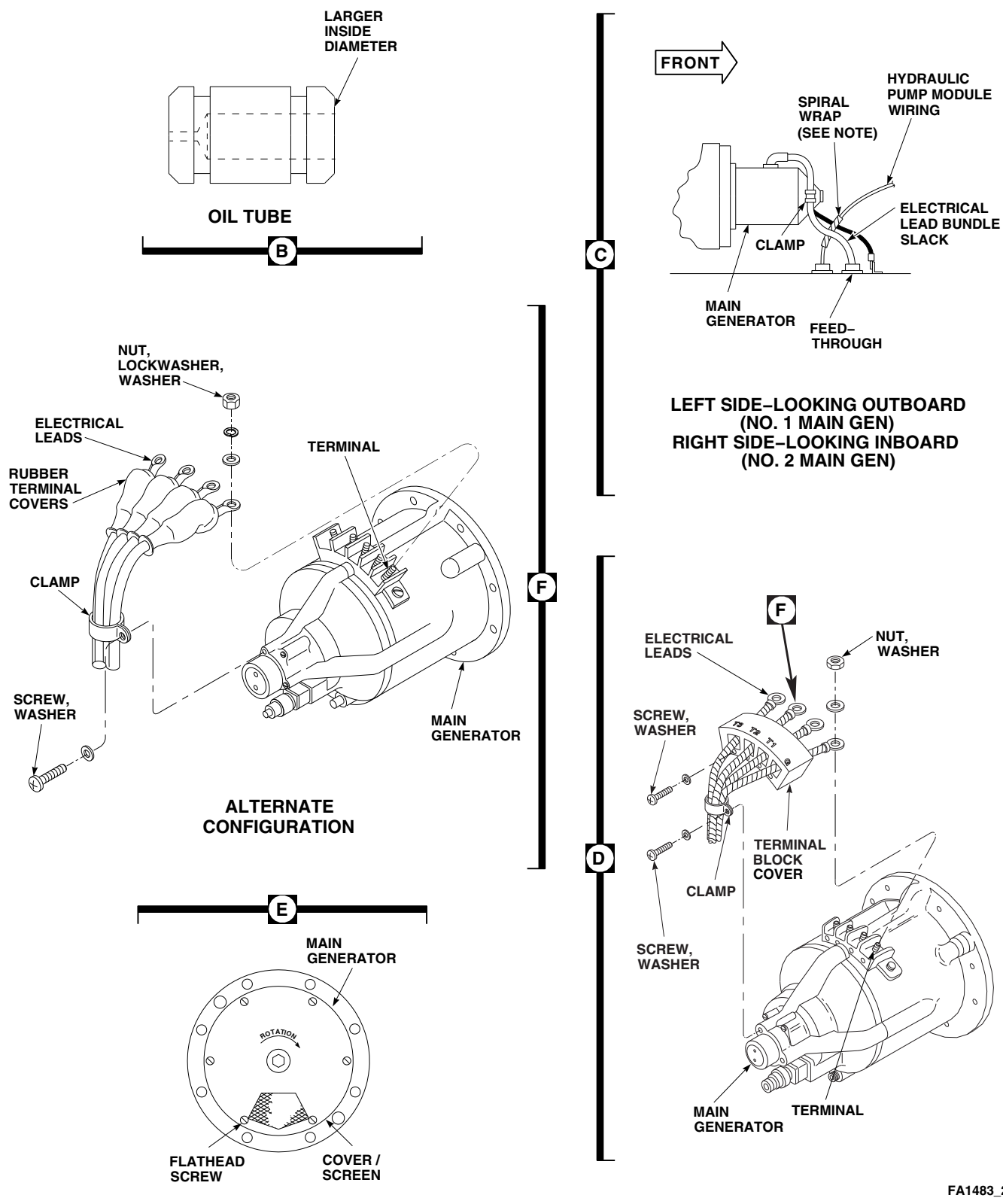


FIGURE 9-4-16-1. REPLACE MAIN GENERATORS (SHEET 2 OF 2)

- e. Inspect area around mounting flange and stud holes for corrosion. **NONE ALLOWED.**

9-4-16.1.3 Install.

NOTE

If new main generator is being installed, new accessory module spline adapter must be installed.

- a. If necessary, install new accessory module spline adapter (PARA 6-4-29).
- b. Turn off all electrical power (PARA 1-6-16).
- c. Using lubricating oil, Item 205, Item 206, or Item 208, Appendix D, lubricate packings and install on oil tubes (Figure 9-4-16-1, Sheet 1, Detail A).

CAUTION

Damage to main generator will result if oil tubes are not properly installed. Oil will not flow properly to main generator. Make sure larger inside diameter end of oil tube is installed in accessory module.

NOTE

Make sure main generator oil tubes and packings are installed to prevent leakage from accessory module.

- d. Install oil tubes with larger inside diameter end in accessory module ports (Figure 9-4-16-1, Sheet 1, Detail A and Figure 9-4-16-1, Sheet 2, Detail B).
- e. Using lubricating oil, Item 205, Item 206, or Item 208, Appendix D, lubricate packing and install on main generator (Figure 9-4-16-1, Sheet 1, Detail A).

CAUTION

Damage to main generator will result if rotor shaft is turned when main genera-

tor is installed in accessory module. Do not turn main generator rotor shaft with main generator removed.

NOTE

Main generator weighs approximately 30 pounds.

- f. Line up main generator oil ports with accessory module oil tubes and place main generator on accessory module flange.
- g. Install main generator on accessory module and secure with nuts and washers. **TORQUE NUTS TO 261 - 289 INCH-POUNDS IN A CRISSCROSS PATTERN.**
- h. Remove protective caps and plugs from electrical connectors.
- i. Inspect and treat electrical connectors (PARA 1-7-20).
- j. Connect electrical connectors as follows:

No. 1 Main Generator	No. 2 Main Generator
P251 to J251	P250 to J250
P278 to J278	P277 to J277

- k. Remove tags from electrical connectors.

NOTE

Do not reinstall terminal block cover. Terminal block cover will be replaced with rubber terminal covers.

- l. If required, install rubber terminal covers on electrical leads (Figure 9-4-16-1, Sheet 2, Detail F).
- m. If installing No. 1 main generator, install electrical leads to terminals and secure with washers, lockwashers, and nuts as follows:

Terminal	Electrical Lead No.
T-1	X101A6A
T-2	X102A6B
T-3	X103A6C
G	X104A6N

- n. If installing No. 2 main generator, install electrical leads to terminals and secure with washers, lockwashers, and nuts as follows:

Terminal	Electrical Lead No.
T-1	X1A6A
T-2	X2A6B
T-3	X3A6C
G	X4A6N

- o. Remove tags.
- p. Push rubber terminal covers firmly over terminal connections.
- q. Install electrical leads with clamp to end of main generator and secure with screw and washer.
- r. Arrange electrical lead bundle with most of slack between clamp and feed-through (Figure 9-4-16-1, Sheet 2, Detail C).
- s. If not already done, using tiedown straps, Item 334, Appendix D, secure electrical lead bundle below clamp at 4 to 6-inch intervals.
- t. For No. 1 main generator only, if not already done, using spiral wrap, Item 315, Appendix D, wrap backup hydraulic pump module wiring from 2 or 3-inches above to 2 or 3-inches below point adjacent to main generator electrical lead bundle. Using tiedown straps, Item 334, Appendix D, secure spiral wrap.
- u. Apply watertight bead of sealing compound, Item 292, Appendix D, to all nuts securing main generator and to main generator/accessory module seam.
- v. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- w. Perform operational check of AC electrical system (PARA 9-2-1).
- x. Make sure area is clean and free of foreign material.
- y. Close main rotor pylon sliding cover.

9-4-17 APU/EXTERNAL POWER CONTACTOR, K3.

PARAGRAPH BREAKDOWN

	PAGE
9-4-17.1 Replace APU/External Power Contactor, K3	9-4-17-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

Materials/Parts

- Protective Caps and Plugs
- Tags

References

- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-69
- PARA 9-2-1
- PARA 9-4-34
- PARA 9-4-106

9-4-17.1 Replace APU/External Power Contactor, K3.

9-4-17.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Remove soundproofing panel (PARA 2-4-69).
- Loosen fasteners securing cover on No. 2 junction box and let cover hang by chain (Figure 9-4-17-1, Detail A).
- Remove screws and washers securing terminal shields to APU/external power contactor, K3, and ac bus tie contactor, K4.
- Disconnect electrical connector, P205 from J205, on APU/external power contactor, K3.

- Install protective caps and plugs on electrical connectors.
- Remove nuts, lockwashers, and washers securing bus bars to APU/external power contactor, K3, and ac bus tie contactor, K4.
- Tag wires. Remove nuts, lockwashers, and washers securing wiring to APU/external power contactor, K3.
- Remove screws and washers securing APU/external power contactor, K3, from No. 2 junction box.
- Remove APU/external power contactor, K3.

9-4-17.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Position APU/external power contactor, K3, in No. 2 junction box and secure with screws and washers (Figure 9-4-17-1, Detail A).

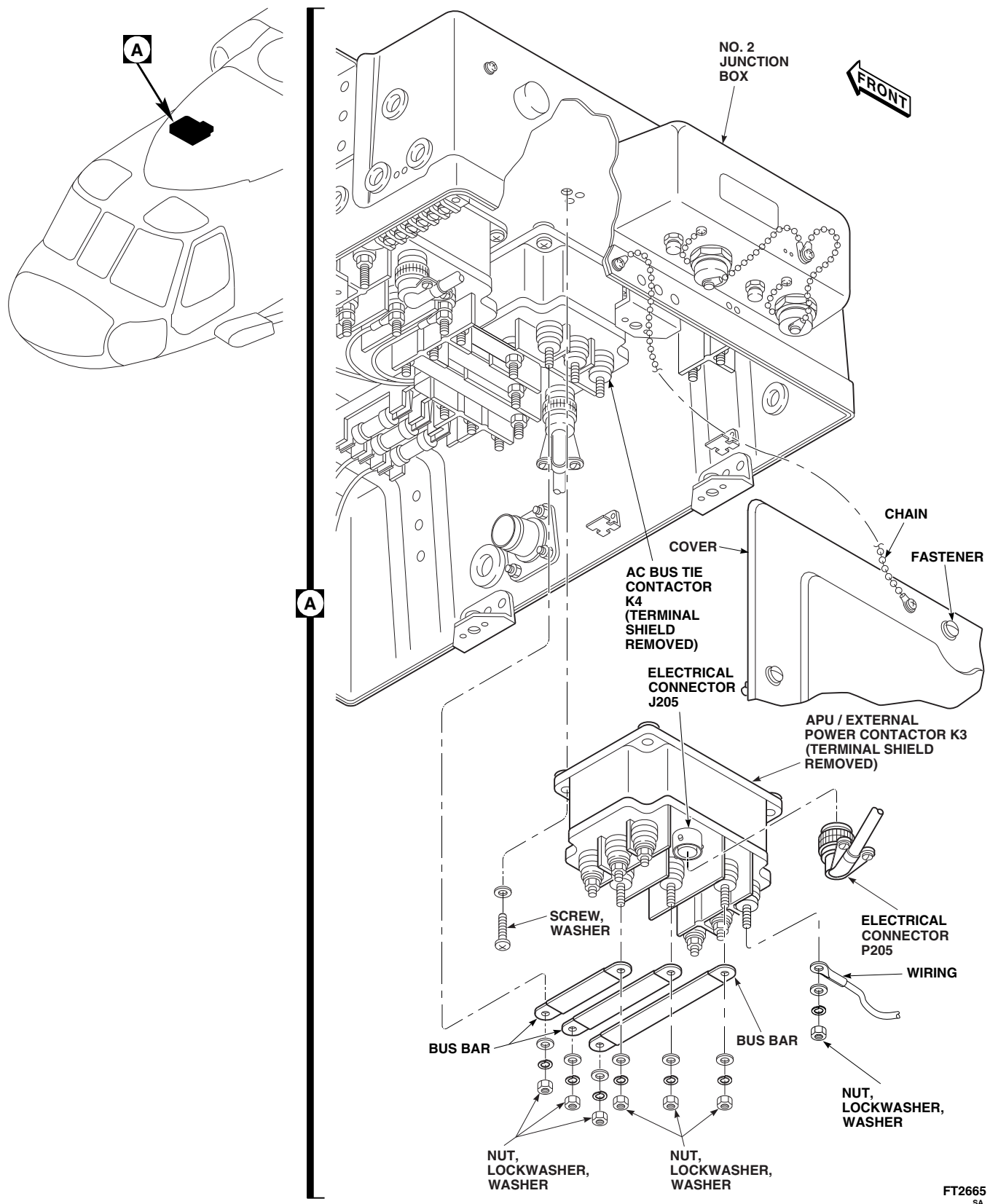
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FIGURE 9-4-17-1. REPLACE APU/EXTERNAL POWER CONTACTOR, K3

9-4-17-2

- c. Install wiring to APU/external power contactor, K3, and secure with nuts, washers, and lockwashers. Remove tags.
- d. Install bus bars on APU/external power contactor, K3, and ac bus tie contactor, K4, and secure with nuts, lockwashers, and washers.
- e. Remove protective caps and plugs from electrical connectors.
- f. Inspect and treat electrical connectors (PARA 1-7-20).
- g. Connect electrical connector, P205 to J205, on APU/external power contactor, K3.
- h. Install terminal shields on APU/external power contactor, K3, and ac bus tie contactor, K4, and secure with screws and washers.
- i. Install cover on No. 2 junction box and secure with fasteners.
- j. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- k. Perform operational check of ac electrical system (PARA 9-2-1).
- l. Make sure area is clean and free of foreign material.
- m. Install soundproofing panel (PARA 2-4-69).

9-4-18 EXTERNAL POWER RECEPTACLE.

PARAGRAPH BREAKDOWN

	PAGE
9-4-18.1 Replace External Power Receptacle	9-4-18-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit
- Nonmetallic Scraper, Locally-Made

Materials/Parts

- Adhesive, Item 37, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D

- Tags

References

- Appendix D
- PARA 1-6-16
- PARA 9-2-1
- PARA 9-4-34
- PARA 9-4-106

9-4-18.1 Replace External Power Receptacle.

9-4-18.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Open external power receptacle access door.
- Open right crew chief’s window to gain access to external power receptacle.
- Remove nuts and washers securing terminal cover to mounting box (Figure 9-4-18-1, Detail A).
- Tag wires. Remove nuts, lockwashers, and washers securing wiring to terminal posts.

- Remove screws, washers, and nuts securing external power receptacle to mounting box by pulling out through external power receptacle access opening.

9-4-18.1.2 Inspect.

Inspect gasket on mounting box for damage. If damaged, using nonmetallic scraper, remove gasket.

9-4-18.1.3 Install.

- Turn off all electrical power (PARA 1-6-16).
- Make sure mounting box is free of foreign objects before installing external power receptacle.
- If replacing gasket, perform the following (Figure 9-4-18-1, Detail A):

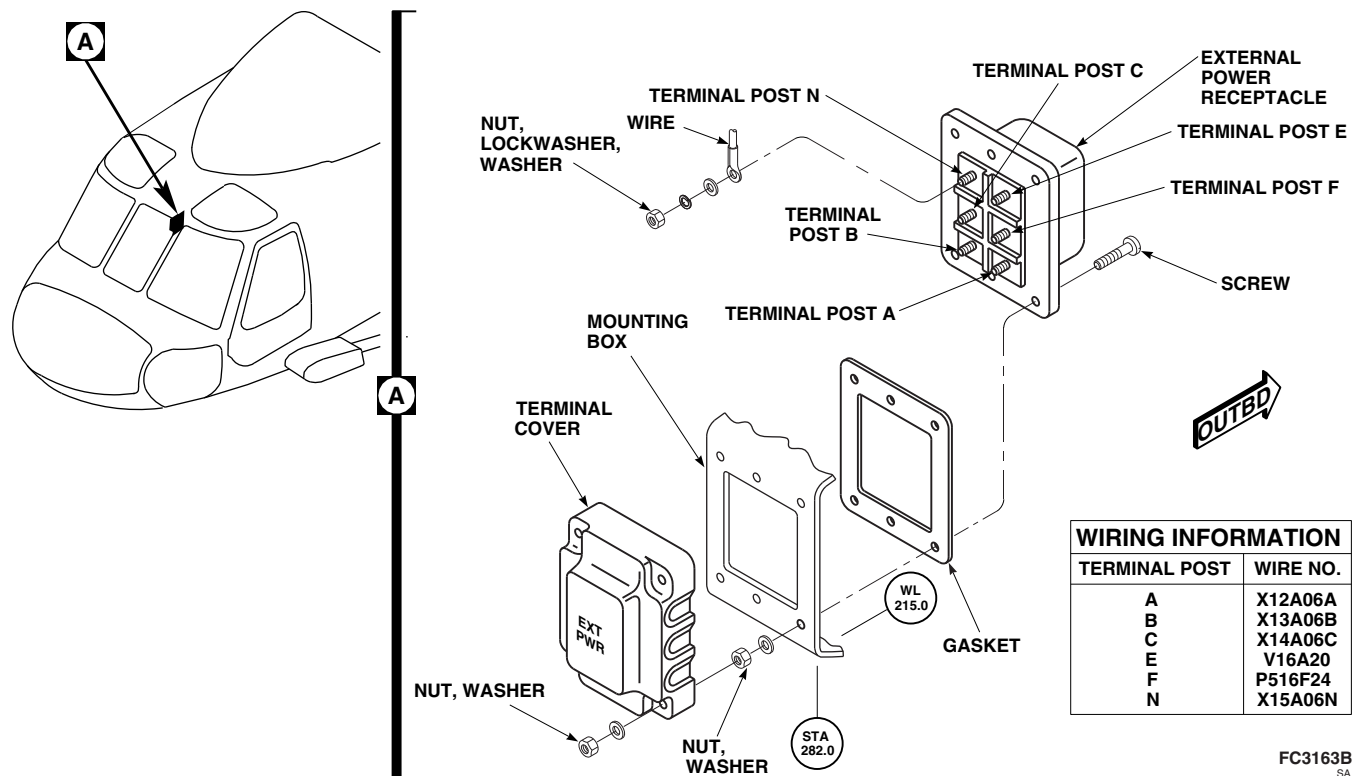


FIGURE 9-4-18-1. REPLACE EXTERNAL POWER RECEPTACLE

- (1) Using dry-cleaning solvent, Item 124, Appendix D, clean mating surface on mounting box.
 - (2) Using adhesive, Item 37, Appendix D, install new gasket in mounting box.
- d. Install external power receptacle in mounting box and secure with screws, washers, and nuts.
- e. Connect wiring to terminal posts with nuts, lockwashers, and washers. Remove tags.
- f. Install terminal cover and secure with washers and nuts.
- g. Make sure area is clean and free of foreign material.
- h. Close right crew chief's window.
- i. Close external power receptacle access door.
- j. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- k. Perform operational check of AC electrical system (PARA 9-2-1).

9-4-19 APU GENERATOR.

PARAGRAPH BREAKDOWN

	PAGE
9-4-19.1 Replace APU Generator	9-4-19-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Container, 1-Gallon
- Electrical Repairer’s Toolkit
- Mallet
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Antiseize Compound, Item 74, Appendix D
- Lockwire, Item 196, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Protective Caps and Plugs

- Sealing Compound, Item 292, Appendix D
- Silicone Compound, Item 309, Appendix D
- Tags

References

- Appendix D
- PARA 1-3-10
- PARA 1-6-16
- PARA 1-7-20
- PARA 9-2-1
- PARA 9-4-34
- PARA 9-4-106
- PARA 15-4-1
- PARA 15-4-7
- PARA 15-4-8

9-4-19.1 Replace APU Generator.

9-4-19.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).

- Manually release auxiliary power unit (APU) accumulator charge (PARA 1-3-10).
- Open APU access doors.
- If APU generator, AGH958-1, is being removed, remove APU starter motor (PARA 15-4-7).

- f. Position 1-gallon container to catch any leaking fuel.
- g. Disconnect fuel inlet hose from fitting on fuel control enclosure and fuel drain hose from fuel drain port (Figure 9-4-19-1, Detail A).
- h. Disconnect lubrication line and starter motor drain tube from fittings on APU compartment deck.
- i. Install protective plugs in hoses, line, and tube, and caps on fittings and fuel drain port.
- j. Tag electrical connectors. Disconnect electrical connectors, P252 from J252, and P279 from J279, on APU generator.
- k. Install protective caps and plugs on electrical connectors.

NOTE

APU generator weighs about 30 pounds.

- l. Loosen bolt on quick attach-disconnect (QAD) mount clamp and remove APU generator and QAD mount clamp from APU.
- m. Place APU generator on APU compartment deck.

NOTE

Metal terminal cover will not be reused.

- n. If installed, remove screws and washers securing metal terminal cover, or pull rubber terminal covers back from terminals, as required, to gain access to electrical leads (Figure 9-4-19-1, Detail A and Detail B).
- o. Tag electrical leads. Remove nuts, lockwashers, and washers securing electrical leads to terminals on APU generator.
- p. Using nonmetallic scraper, remove sealing compound from QAD mounting flange.
- q. Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean remaining sealing compound.

9-4-19.1.2 Inspect.

If installed, inspect rubber terminal covers for cracks. **NONE ALLOWED.** If cracked, replace rubber terminal covers.

9-4-19.1.3 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Inspect auxiliary power unit (APU) spline adapter (PARA 15-4-8).
- c. If APU generator, AGH958-1, is being installed, remove APU starter motor (PARA 15-4-7).
- d. Place APU generator in compartment.

NOTE

Do not reinstall metal terminal cover. Metal terminal cover will be replaced with rubber terminal covers.

- e. If required, install rubber terminal covers on each electrical lead (Figure 9-4-19-1, Detail B).
- f. Install electrical leads to terminals and secure with washers, lockwashers, and nuts as follows:

Terminal	Electrical Lead No.
T-1	X8E6A
T-2	X9E6B
T-3	X10E6C
T-4	X11A6N

- g. Remove tags.
- h. Push rubber terminal cover firmly over terminal connections.
- i. Using light amount of silicone compound, Item 309, Appendix D, coat groove in quick attach-disconnect (QAD) mount.

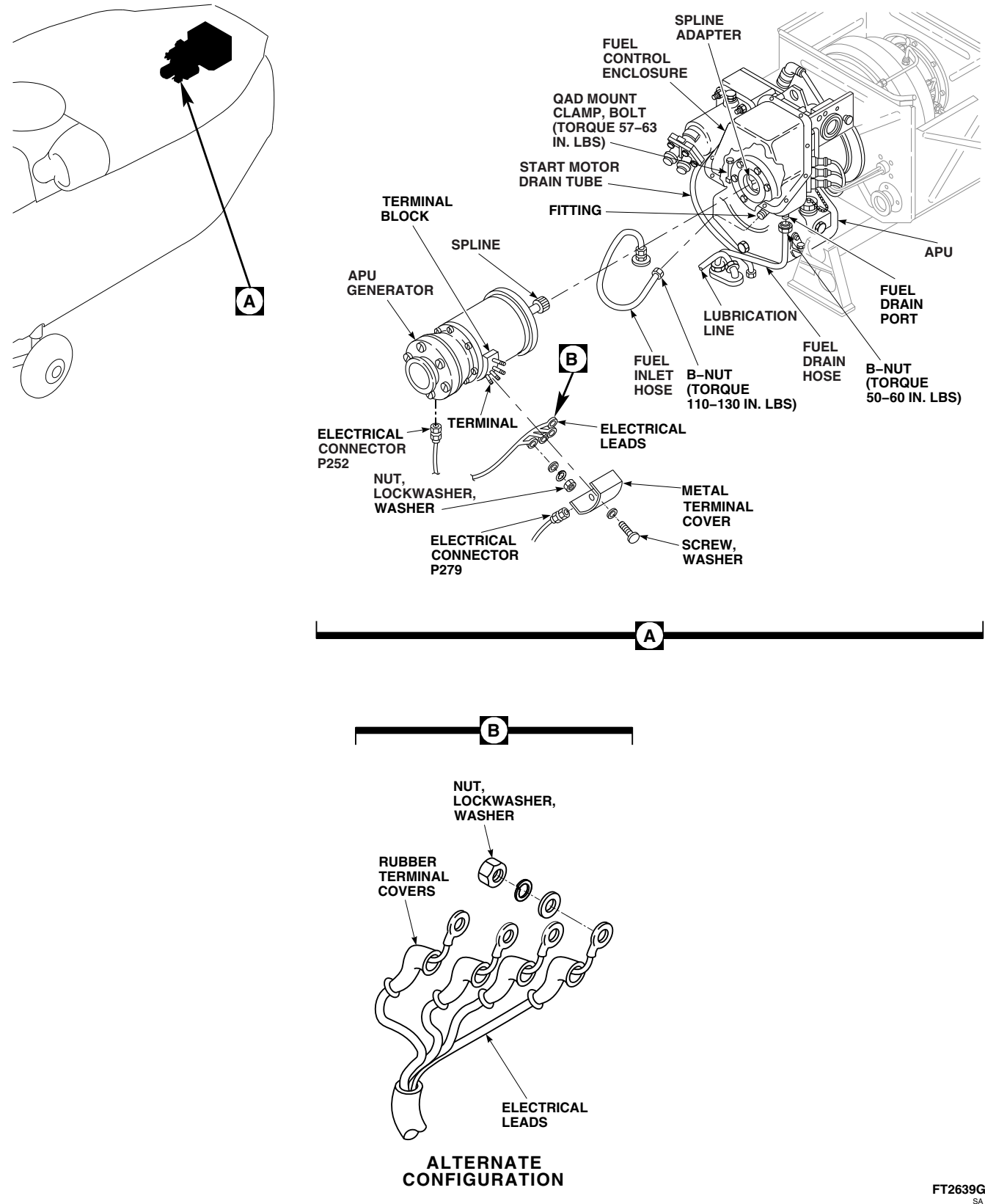


FIGURE 9-4-19-1. REPLACE APU GENERATOR

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CAUTION

- Damage to APU generator housing and flange will result if QAD mount clamp is not completely around APU generator flange. Make sure APU generator is flush against QAD mount while tightening clamp bolt.
- Damage to APU will result if spline on APU generator does not properly engage with spline adapter on APU. Make sure spline on APU generator properly engages spline adapter of APU to prevent pushing spline adapter out of drive spline and into gear box.

NOTE

If spline on APU generator and spline adapter on APU do not mesh easily, remove APU generator and slightly turn APU generator shaft until splines mesh easily.

- j. Place APU generator flat against QAD mount face so spline on APU generator meshes with spline adapter on APU (Figure 9-4-19-1, Detail A). Position APU generator with terminal block facing toward left side of APU and slightly down.

CAUTION

Clamping action could cause cracks in APU generator housing and damage to flange. To make sure APU generator mounting flange mates all around with groove in QAD mount clamp, hold APU generator flat against QAD mount clamp, with its front end supported, while tightening QAD mount clamp.

- k. Before installing QAD mount clamp on APU generator, using antiseize compound, Item 74, Appendix D, coat inner surface of QAD mount clamp.

- l. Check QAD mount clamp to make sure groove in QAD mount clamp mates all around with APU generator and QAD mounting flanges.

CAUTION

Damage to APU generator housing and flange will result if QAD mount clamp is not properly torqued. To make sure APU generator mounting flange mates all around with groove in QAD mount clamp, hold APU generator flat against QAD mount clamp, with its front end supported, while tightening QAD mount clamp.

- m. Hold APU generator flat against QAD mount clamp. **TORQUE CLAMP BOLT TO 57 - 63 INCH-POUNDS.**
- n. Using mallet, tap lightly around QAD mount clamp. **RETORQUE CLAMP BOLT TO 57 - 63 INCH-POUNDS.**
- o. Reach through exhaust and turn turbine wheel to make sure there is engagement between turbine and gear box. If turbine wheel spins freely, ring separation may have occurred. Replace APU (PARA 15-4-1).
- p. Continue tapping and retorquing until there is no loss in torque two times in a row.
- q. Using lockwire, Item 196, Appendix D, lockwire bolt in QAD mount clamp.
- r. Remove protective caps and plugs from electrical connectors.
- s. Inspect and treat electrical connectors (PARA 1-7-20).
- t. Connect electrical connectors, P252 to J252, and P279 to J279, to APU generator. Remove tags.
- u. Remove protective plugs from line and tube, and caps from fittings on APU compartment deck.

- v. Connect lubrication line and starter motor drain tube to fittings on APU compartment deck.
- w. Apply watertight bead of sealing compound, Item 292, Appendix D, to lubrication line and starter motor drain tube fittings, and to APU generator/QAD mount clamp flange.
- x. Remove protective plug from fuel inlet hose and cap from fitting on fuel control enclosure.
- y. Connect fuel inlet hose to fitting. **TORQUE B-NUT TO 110 - 130 INCH-POUNDS.**
- z. Remove plugs from fuel drain hose and cap from fuel drain port.
- aa. Connect fuel drain hose to fuel drain port. **TORQUE B-NUT TO 50 - 60 INCH-POUNDS.**
- ab. If APU generator, AGH958-1, is installed, install APU starter motor (PARA 15-4-7).
- ac. Recharge APU accumulator (PARA 1-3-10).
- ad. Make sure area is clean and free of foreign material.
- ae. Close APU access doors.
- af. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- ag. Perform operational check of AC electrical system (PARA 9-2-1).

9-4-20 EXTERNAL POWER MONITOR PANEL.

PARAGRAPH BREAKDOWN

	PAGE
9-4-20.1 Replace External Power Monitor Panel	9-4-20-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

Materials/Parts

- Cheesecloth, Item 90, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-69
- PARA 9-2-1
- PARA 9-4-34
- PARA 9-4-106

9-4-20.1 Replace External Power Monitor Panel.

9-4-20.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Remove soundproofing panel (PARA 2-4-69).
- Disconnect electrical connector, P213 from J213, on external power monitor panel (Figure 9-4-20-1, Detail A).
- Install protective caps and plugs on electrical connectors.

- Remove screws, lockwashers, and washers securing external power monitor panel to support.
- Remove external power monitor panel.

9-4-20.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Using cheesecloth, Item 90, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D, clean surface around all mounting holes.
- Install external power monitor panel on support and secure with screws, lockwashers, and washers (Figure 9-4-20-1, Detail A).
- Remove protective caps and plugs from electrical connectors.

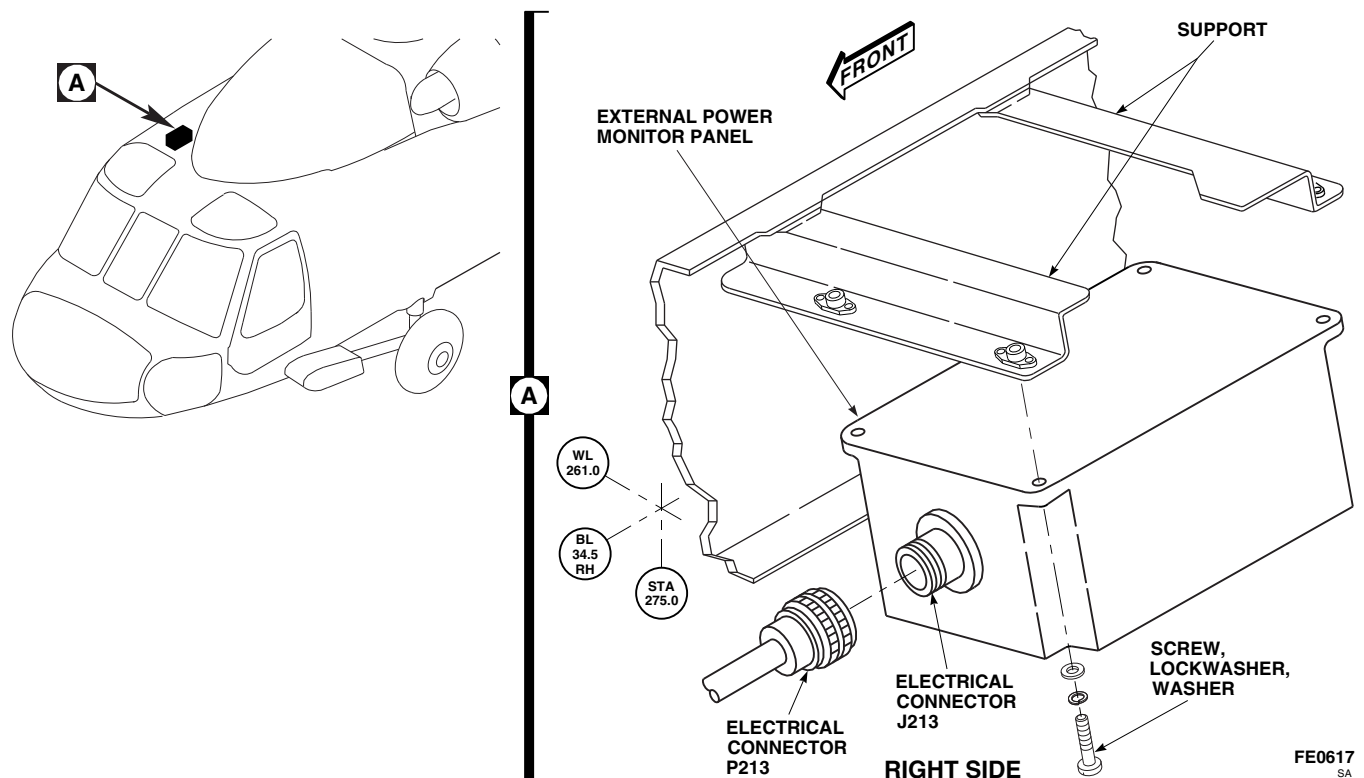


FIGURE 9-4-20-1. REPLACE EXTERNAL POWER MONITOR PANEL

- | | |
|--|--|
| <p>e. Inspect and treat electrical connectors (PARA 1-7-20).</p> <p>f. Connect electrical connector, P213 to J213, on external power monitor panel.</p> <p>g. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).</p> | <p>h. Perform operational check of ac electrical system (PARA 9-2-1).</p> <p>i. Make sure area is clean and free of foreign material.</p> <p>j. Install soundproofing panel (PARA 2-4-69).</p> |
|--|--|

9-4-21 AC BUS TIE CONTACTOR, K4.

PARAGRAPH BREAKDOWN

	PAGE
9-4-21.1 Replace AC Bus Tie Con- tactor, K4	9-4-21-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

Materials/Parts

- Protective Caps and Plugs
- Tags

References

- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-69
- PARA 9-2-1
- PARA 9-4-34
- PARA 9-4-106

9-4-21.1 Replace AC Bus Tie Contactor, K4.

9-4-21.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Remove soundproofing panel (PARA 2-4-69).
- Loosen fasteners securing cover on No. 2 junction box and let cover hang by chain (Figure 9-4-21-1, Detail A).
- Remove screws and washers securing terminal shields from ac bus tie contactor, K4, No. 2 ac generator contactor, K2, and APU/external power contactor, K3.
- Disconnect electrical connector, P206 from J206, on ac bus tie contactor, K4.

- Install protective caps and plugs on electrical connectors.
- Remove nuts, lockwashers, and washers securing bus bars to ac bus tie contactor, K4, and No. 2 ac generator contactor, K2.
- Remove nuts, lockwashers, and washers securing bus bars to ac bus tie contactor, K4, and APU/external power contactor, K3.
- Tag wires. Remove nuts, lockwashers, and washers securing wiring to ac bus tie contactor, K4.
- Remove screws and washers securing ac bus tie contactor, K4, from No. 2 junction box.
- Remove ac bus tie contactor, K4.

9-4-21.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).

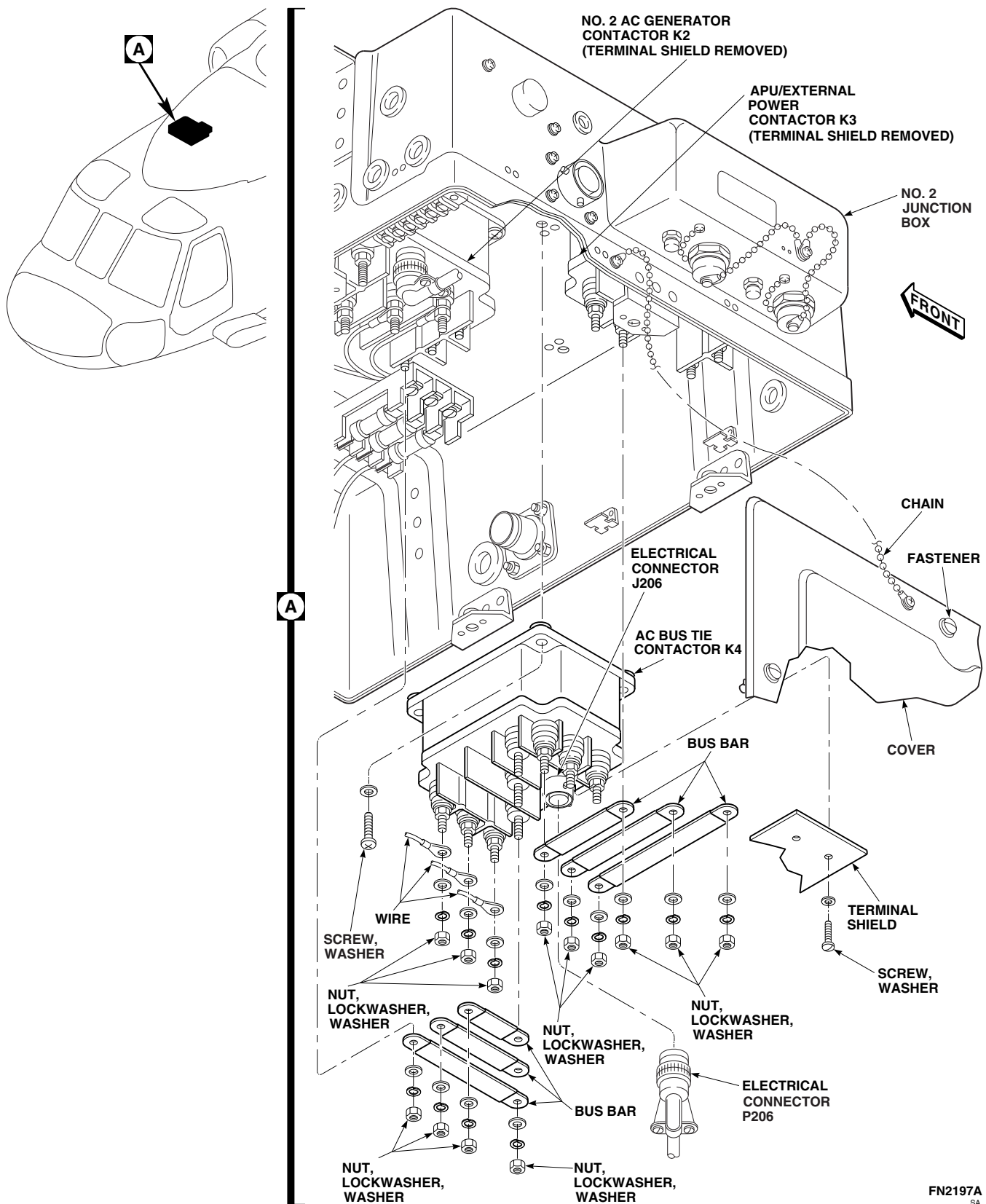


FIGURE 9-4-21-1. REPLACE AC BUS TIE CONTACTOR, K4

9-4-21-2

- b. Install ac bus tie contactor, K4, in No. 2 junction box and secure with screws and washers (Figure 9-4-21-1, Detail A).
- c. Install wiring on ac bus tie contactor, K4, and secure with nuts, lockwashers, and washers. Remove tags.
- d. Install bus bars on ac bus tie contactor, K4, and No. 2 ac generator contactor, K2, and secure with nuts, lockwashers, and washers.
- e. Install bus bars on ac bus tie contactor, K4, and APU/external power contactor, K3, and secure with nuts, lockwashers, and washers.
- f. Remove protective caps and plugs from electrical connectors.
- g. Inspect and treat electrical connectors (PARA 1-7-20).
- h. Connect electrical connector, P206 to J206, on ac bus tie contactor, K4.
- i. Install terminal shields on ac bus tie contactor, K4, No. 2 ac generator contactor, K2, and APU/external power contactor, K3, and secure with screws and washers.
- j. Install cover on No. 2 junction box and secure with fasteners.
- k. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- l. Perform operational check of ac electrical system (PARA 9-2-1).
- m. Make sure area is clean and free of foreign material.
- n. Install soundproofing panel (PARA 2-4-69).

9-4-22 AC ESSENTIAL BUS RELAYS, K8 AND K13.

PARAGRAPH BREAKDOWN

	PAGE
9-4-22.1 Replace AC Essential Bus Relays, K8 and K13	9-4-22-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit
- Heat Gun
- Protection, Eye and Skin
- Soldering Iron

Materials/Parts

- Insulation Sleeving, Heat-Shrinkable, Item 162, Appendix D

- Solder, Item 313, Appendix D

- Tags

References

- Appendix D
- PARA 1-6-16
- PARA 9-2-1
- PARA 9-4-34
- PARA 9-4-106

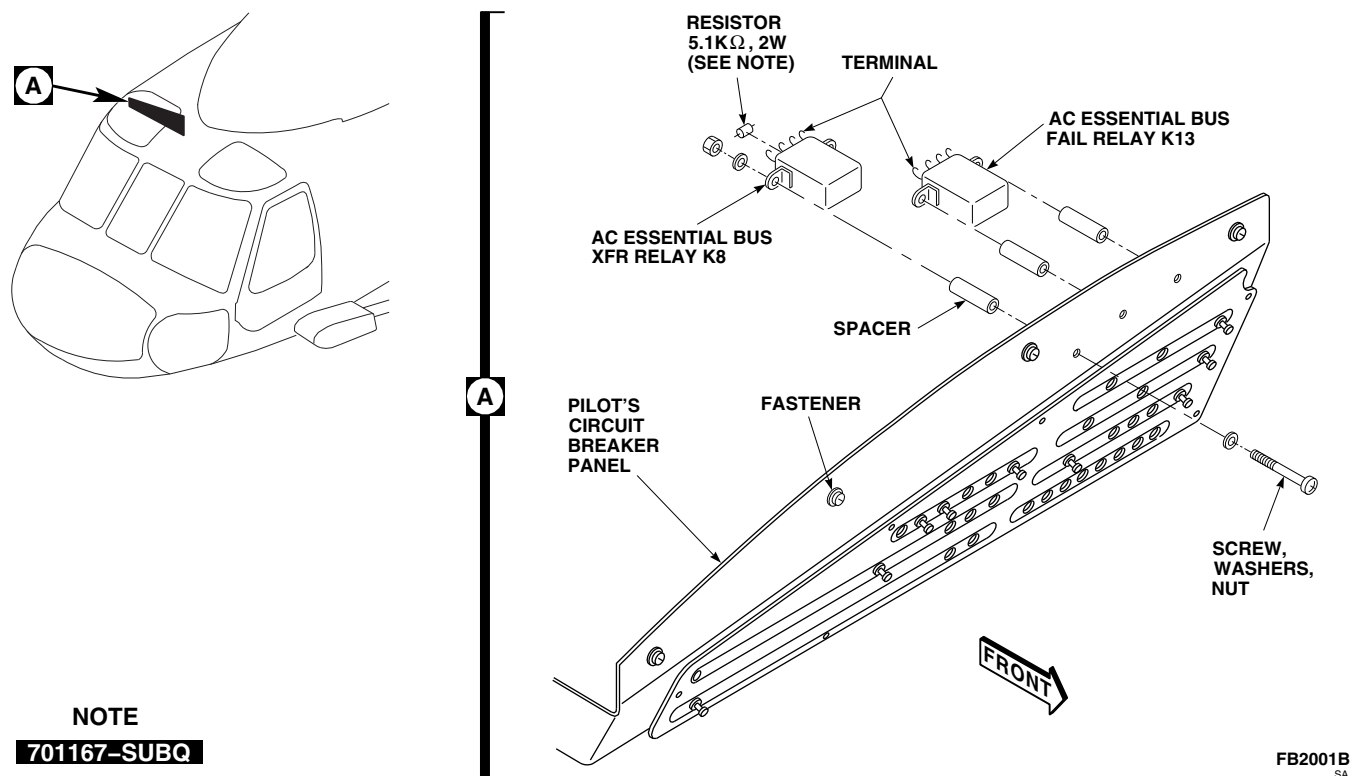
9-4-22.1 Replace AC Essential Bus Relays, K8 and K13.

NOTE

- AC essential bus XFR relay, K8, and ac essential bus fail relay, K13 are located on the pilot’s circuit breaker panel.
- Replacement of ac essential bus XFR relay, K8, and ac essential bus fail relay, K13 is the same, except as noted.
- AC essential bus XFR relay, K8, has a 5.1K ohm, 2W resistor connected between terminals, B2 and B3.

9-4-22.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Loosen fasteners and carefully lower pilot’s circuit breaker panel (Figure 9-4-22-1, Detail A).
- Slide heat-shrinkable insulation sleeving back from terminals. Inspect heat-shrinkable insulation sleeving for damage. If damaged, remove heat-shrinkable insulation sleeving.



NOTE

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FIGURE 9-4-22-1. REPLACE AC ESSENTIAL BUS RELAYS, K8 AND K13

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- e. Tag wires. Using soldering iron, unsolder wiring from terminals.
- f. If required, on ac essential bus XFR relay, K8, using soldering iron, unsolder and remove resistor.

- g. Remove screws, washers, spacers, and nuts securing ac essential bus relays to pilot's circuit breaker panel.

- h. Remove ac essential bus relays.

9-4-22.1.2. Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Install ac essential bus relay and spacers on pilot's circuit breaker panel and secure with screws, washers, and nuts (Figure 9-4-22-1, Detail A).
- c. If necessary, install heat-shrinkable insulation sleeving, Item 162, Appendix D, on wire.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- d. Using soldering iron and solder, Item 313, Appendix D, solder wiring to terminals. Remove tags.
- e. If required, on ac essential bus XFR relay, K8, using soldering iron and solder, Item 313, Appendix D, solder resistor to terminals, B2 and B3.
- f. Slide heat-shrinkable insulation sleeving over solder terminals.
- g. Using heat gun, shrink heat-shrinkable insulation sleeving.
- h. Make sure area is clean and free of foreign material.
- i. Carefully swing pilot's circuit breaker panel into closed position and secure with fasteners.
- j. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- k. Perform operational check of ac electrical system (PARA 9-2-1).

9-4-23 SECONDARY BUS CURRENT LIMITERS, CL16 THROUGH CL18.

PARAGRAPH BREAKDOWN

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9-4-23.1 Replace Secondary Bus Current Limiters, CL16 through CL18	9-4-23-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

References

- PARA 1-6-16
- PARA 2-4-69

- PARA 9-2-1
- PARA 9-4-34
- PARA 9-4-106

9-4-23.1 Replace Secondary Bus Current Limiters, CL16 through CL18.

NOTE

Replacement of secondary bus current limiters, CL16 through CL18, is the same, except as noted.

9-4-23.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Remove soundproofing panel (PARA 2-4-69).
- Loosen fasteners securing cover on No. 1 junction box and let cover hang by chain (Figure 9-4-23-1, Detail A).
- Pull and remove secondary bus current limiter from its mounting block.

9-4-23.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Position secondary bus current limiter over its mounting block and press it into mounting block (Figure 9-4-23-1, Detail A).
- Install cover on No. 1 junction box and secure with fasteners.
- Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Perform operational check of ac electrical system (PARA 9-2-1).
- Make sure area is clean and free of foreign material.
- Install soundproofing panel (PARA 2-4-69).

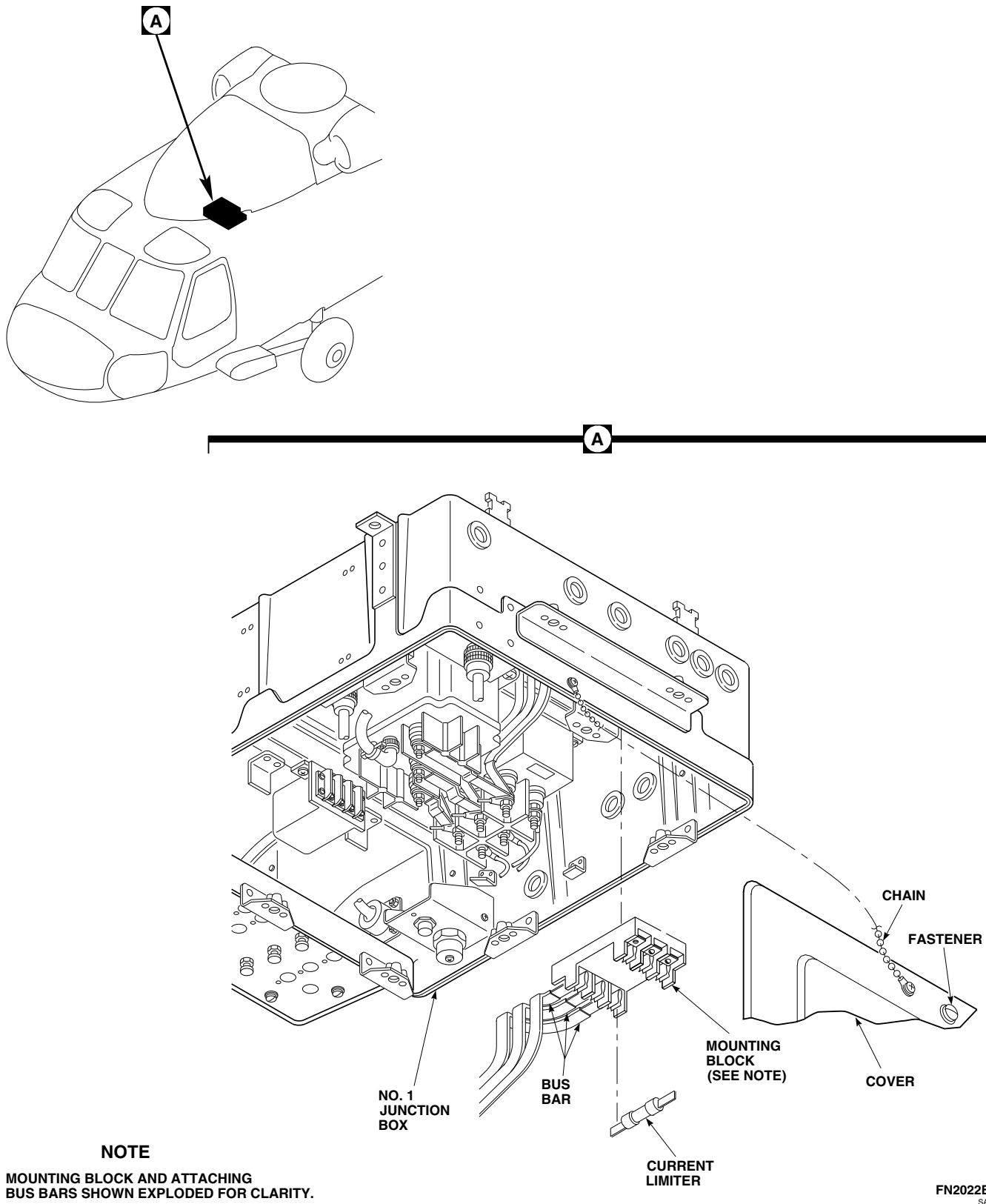


FIGURE 9-4-23-1. REPLACE SECONDARY BUS CURRENT LIMITERS, CL16 THROUGH CL18

9-4-24 AC SECONDARY BUS CONTACTOR, K11.

PARAGRAPH BREAKDOWN

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9-4-24.1 Replace AC Secondary Bus Contactor, K11	9-4-24-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

Materials/Parts

- Tags

References

- PARA 1-6-16
- PARA 2-4-69
- PARA 9-2-1
- PARA 9-4-34
- PARA 9-4-106

9-4-24.1 Replace AC Secondary Bus Contactor, K11.

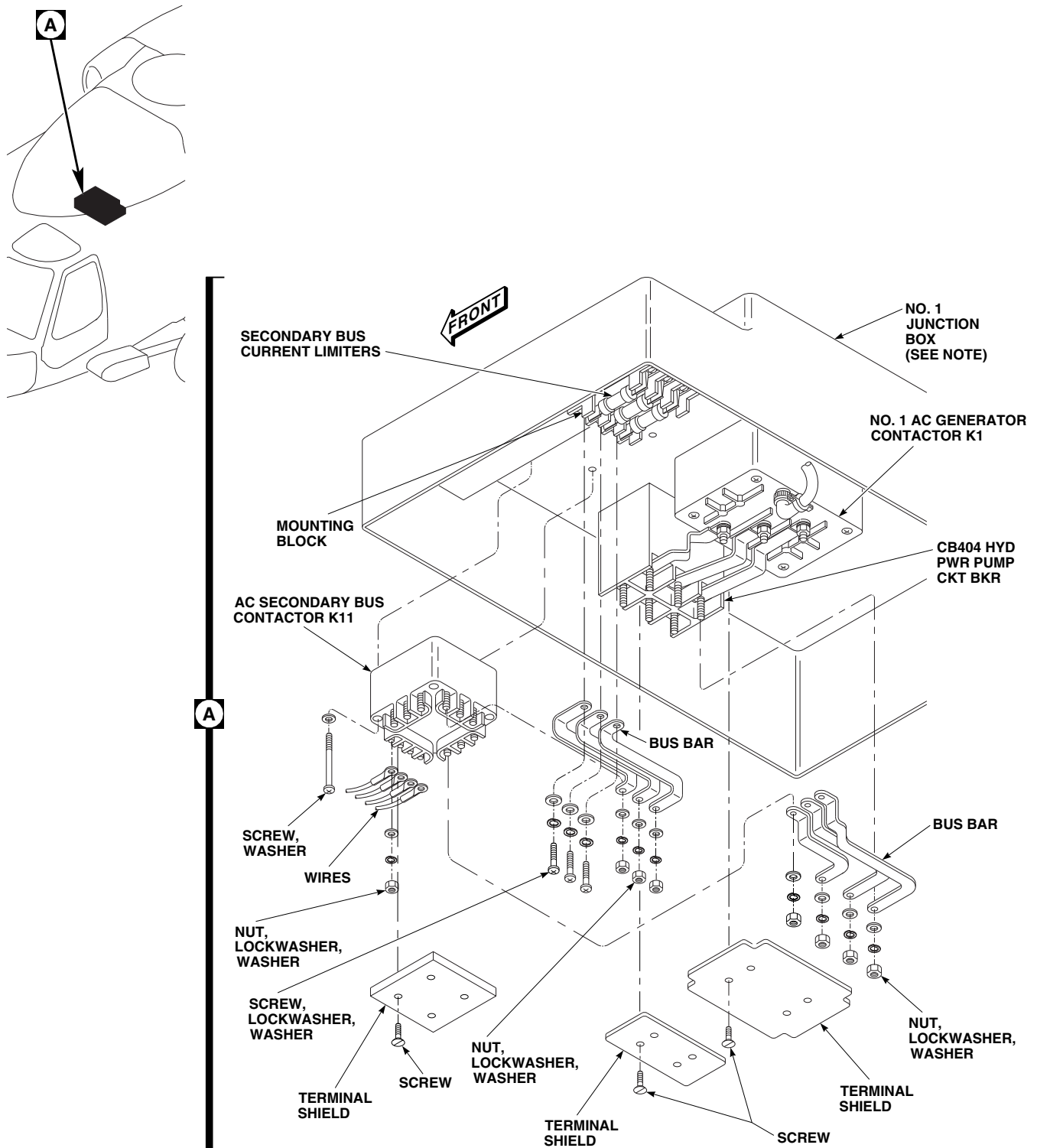
9-4-24.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Remove soundproofing panel (PARA 2-4-69).
- Loosen fasteners securing cover on No. 1 junction box and let cover hang by chain.
- Remove screws securing terminal shields to ac secondary bus contactor, K11, CB404 HYD PWR PUMP CKT BKR, and No. 1 ac generator contactor, K1 (Figure 9-4-24-1, Detail A).
- Tag wires. Remove nuts, lockwashers, and washers securing wires to ac secondary bus contactor, K11.

- Remove screws, nuts, lockwashers, and washers securing bus bars connected to secondary bus current limiters and mounting block of ac secondary bus contactor, K11.
- Remove nuts, lockwashers, and washers securing bus bars connected to CB404 HYD PWR PUMP CKT BKR and ac secondary bus contactor, K11.
- Remove screws and washers securing ac secondary bus contactor, K11, to No. 1 junction box.
- Remove ac secondary bus contactor, K11.

9-4-24.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Install ac secondary bus contactor, K11, in No. 1 junction box and secure with screws and washers (Figure 9-4-24-1, Detail A).
- Install bus bars on ac secondary bus contactor, K11, and mounting block of secondary bus

**NOTE**

NO. 1 JUNCTION BOX COVER NOT SHOWN.

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FIGURE 9-4-24-1. REPLACE AC SECONDARY BUS CONTACTOR, K11

9-4-24-2

- current limiters and secure with screws, nuts, lockwashers, and washers.
- d. Install bus bars on CB404 HYD PWR PUMP CKT BKR and ac secondary bus contactor, K11, and secure with nuts, lockwashers, and washers.
- e. Install wires on ac secondary bus contactor, K11, and secure with nuts, lockwashers, and washers. Remove tags.
- f. Install terminal shields on ac secondary bus contactor, K11, CB404 HYD PWR PUMP CKT BKR, and No. 1 ac generator contactor, K1, and secure with screws.
- g. Install cover on No. 1 junction box and secure with fasteners.
- h. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- i. Perform operational check of ac electrical system (PARA 9-2-1).
- j. Make sure area is clean and free of foreign material.
- k. Install soundproofing panel (PARA 2-4-69).

9-4-25 DIODES, CR13 AND CR14.

PARAGRAPH BREAKDOWN

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9-4-25.1 Replace Diodes, CR13 and CR14	9-4-25-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit
- Protection, Eye and Skin
- Soldering Iron

Materials/Parts

- Solder, Item 313, Appendix D
- Tags

References

- Appendix D
- PARA 1-6-16
- PARA 9-2-1
- PARA 9-4-2

9-4-25.1 Replace Diodes, CR13 and CR14.

NOTE

Replacement of diodes, CR13 and CR14, is the same.

9-4-25.1.1 Remove.

- Remove right relay panel (PARA 9-4-2).
- Remove screws and washers securing bottom cover to right relay panel (Figure 9-4-25-1).

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and

injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- Tag diode leads. Using soldering iron, unsolder diode leads from terminals and remove diode (Figure 9-4-25-1, Detail A).

9-4-25.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).
- Install diode between terminals with diode leads positioned as shown on diode board (Figure 9-4-25-1, Detail A).

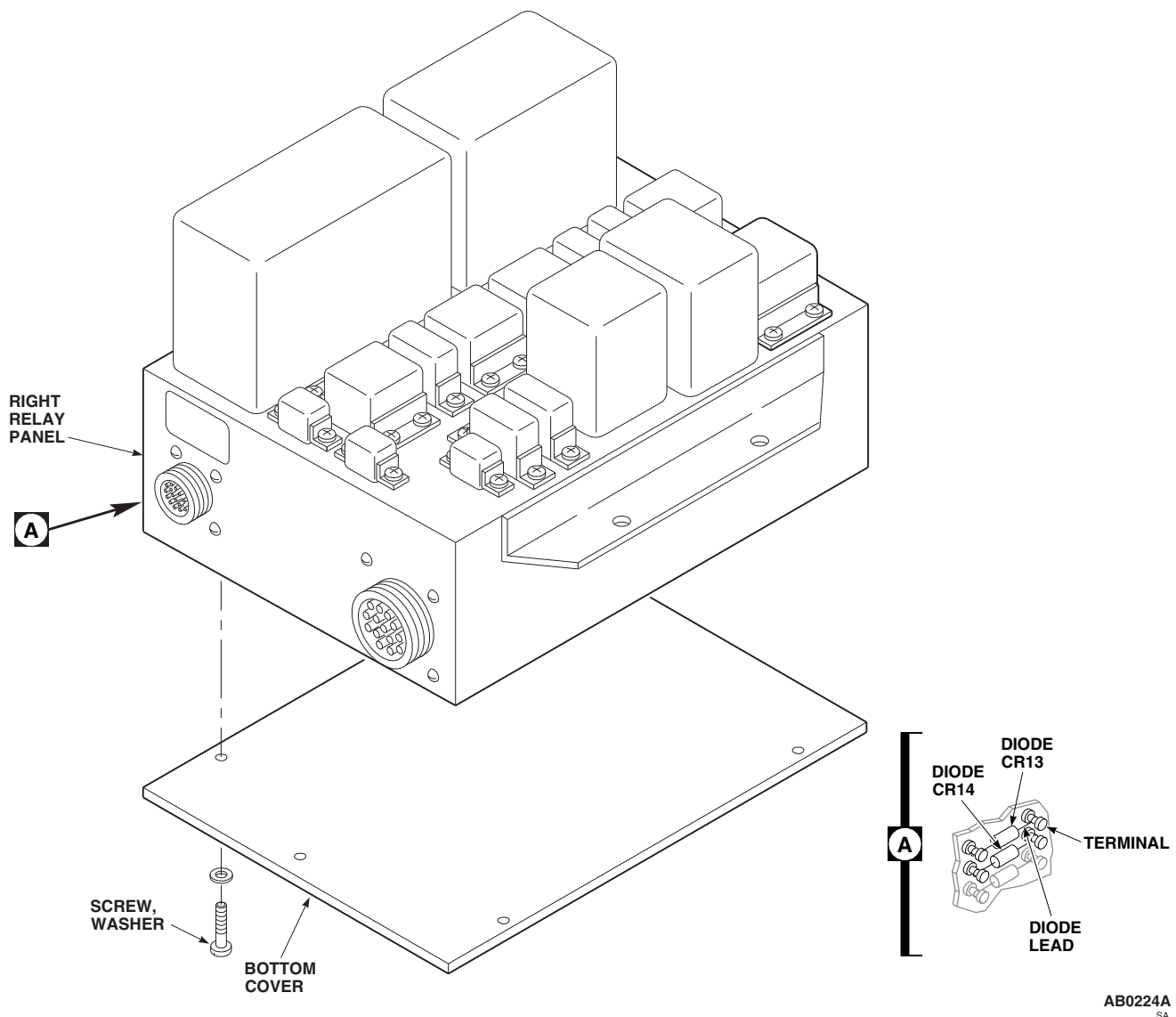


FIGURE 9-4-25-1. REPLACE DIODES, CR13 AND CR14

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- c. Using soldering iron and solder, Item 313, Appendix D, solder diode leads to terminals. Remove tags.
- d. Install bottom cover on right relay panel and secure with screws and washers (Figure 9-4-25-1).
- e. Install right relay panel (PARA 9-4-2).
- f. Perform operational check of ac electrical system (PARA 9-2-1).

9-4-26 DIODES, CR17, CR18, CR19, AND CR20.

PARAGRAPH BREAKDOWN

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9-4-26.1 Replace Diodes, CR17, CR18, CR19, and CR20	9-4-26-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit
- Protection, Eye and Skin
- Soldering Iron

Materials/Parts

- Solder, Item 313, Appendix D
- Tags

References

- Appendix D
- PARA 1-6-16
- PARA 2-4-69
- PARA 9-2-1
- PARA 9-4-34
- PARA 9-4-106

9-4-26.1 Replace Diodes, CR17, CR18, CR19, and CR20.

NOTE

Replacement of diodes, CR17, CR18, CR19, and CR20, is the same.

9-4-26.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Remove soundproofing panel (PARA 2-4-69).
- Loosen fasteners securing cover on No. 1 junction box and let cover hang by chain.



Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- Tag diode leads. Using soldering iron, unsolder diode leads from terminals and remove diode (Figure 9-4-26-1, Detail B).
- Remove diode.

9-4-26.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).

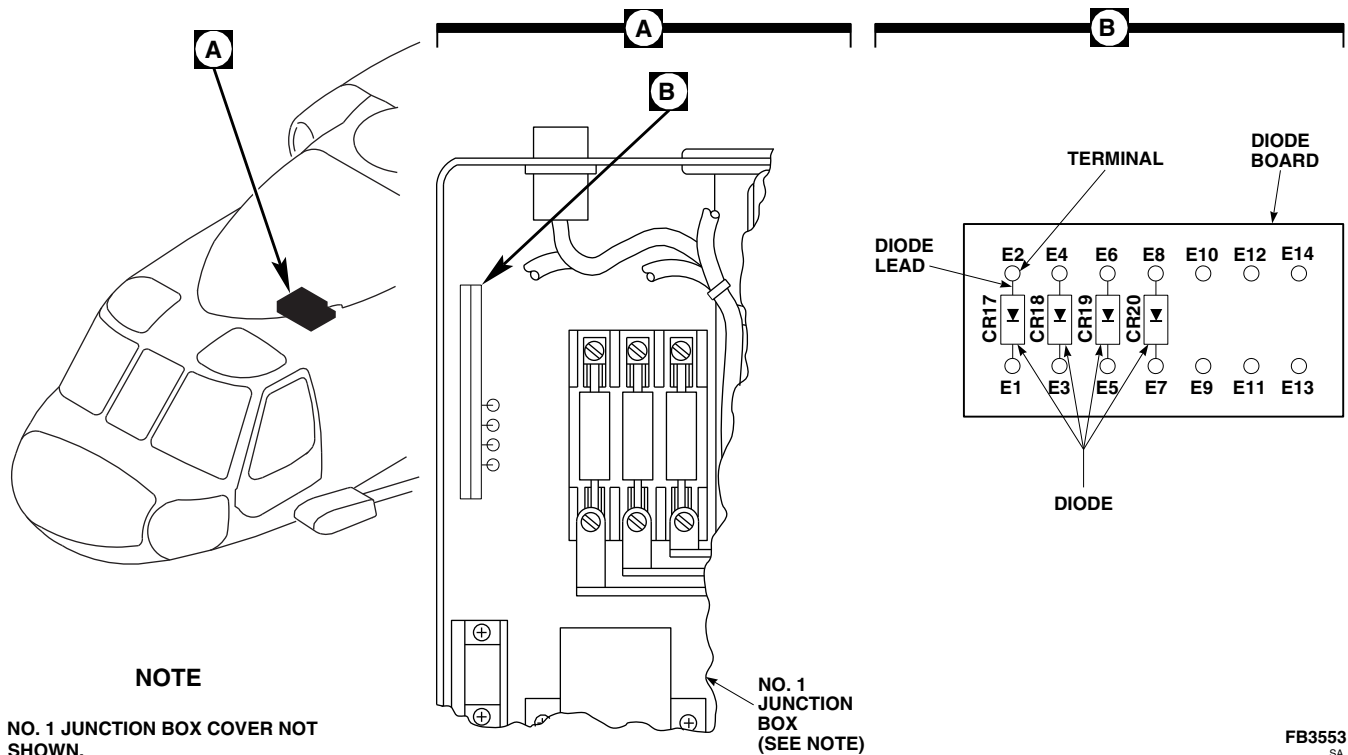


FIGURE 9-4-26-1. REPLACE DIODES CR17, CR18, CR19, AND CR20

- b. Install diode between terminals with diode leads positioned as shown on diode board (Figure 9-4-26-1, Detail B).

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- c. Using soldering iron and solder, Item 313, Appendix D, solder diode leads to terminals. Remove tags.

- d. Install cover on No. 1 junction box and secure with fasteners.
- e. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- f. Perform operational check of ac electrical system (PARA 9-2-1).
- g. Make sure area is clean and free of foreign material.
- h. Install soundproofing panel (PARA 2-4-69).

9-4-27 DIODES, CR21, CR22, AND CR23.

PARAGRAPH BREAKDOWN

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9-4-27.1 Replace Diodes, CR21, CR22, and CR23	9-4-27-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit
- Protection, Eye and Skin
- Soldering Iron

Materials/Parts

- Solder, Item 313, Appendix D
- Tags

References

- Appendix D
- PARA 1-6-16
- PARA 2-4-69
- PARA 9-2-1
- PARA 9-4-34
- PARA 9-4-106

9-4-27.1 Replace Diodes, CR21, CR22, and CR23.

NOTE

Replacement of diodes, CR21, CR22, and CR23, is the same.

9-4-27.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Remove soundproofing panel (PARA 2-4-69).
- Loosen fasteners securing cover on No. 2 junction box and let cover hang by chain.

- Remove screws, washers, and spacers securing diode board to No. 2 junction box (Figure 9-4-27-1, Detail A).

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- Tag diode leads. Using soldering iron, unsolder diode leads from terminals and remove diode.

9-4-27.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).

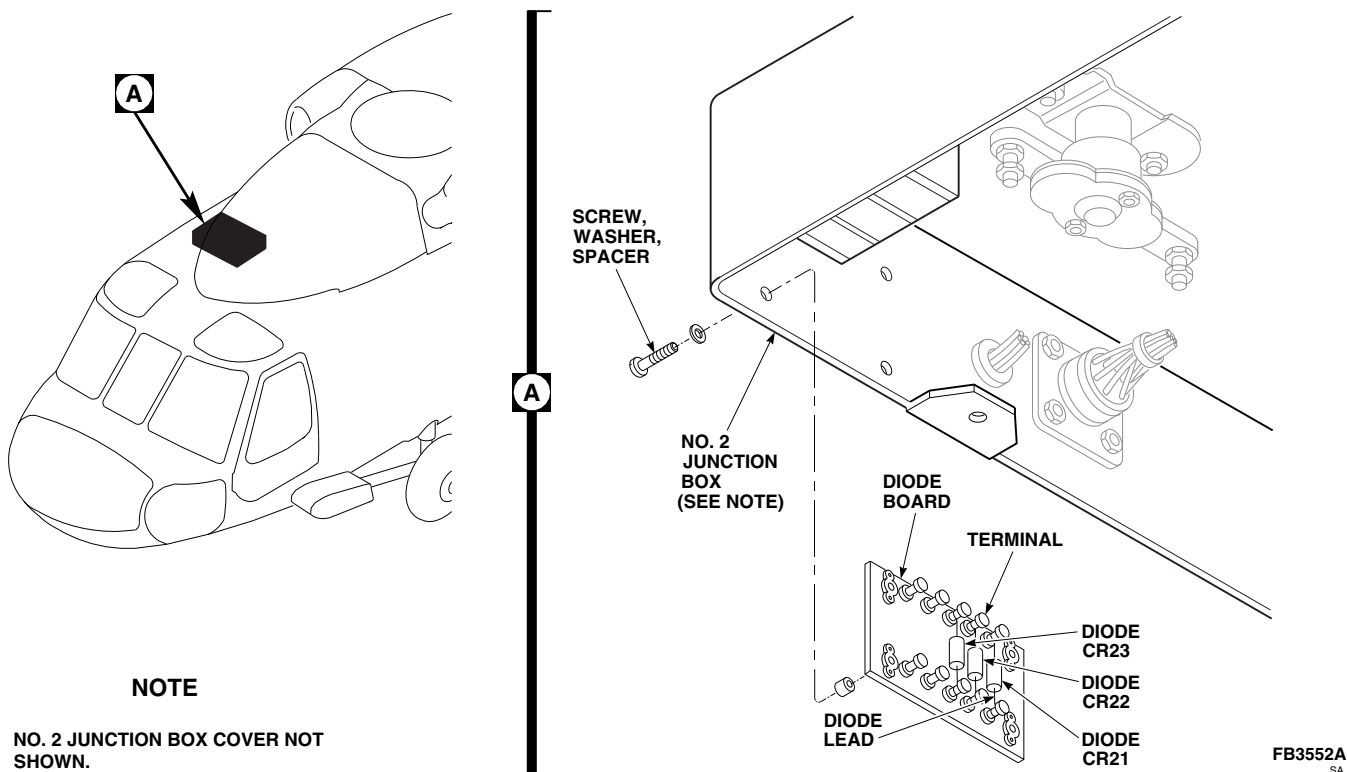


FIGURE 9-4-27-1. REPLACE DIODES, CR21, CR22, AND CR23

- b. Install diode between terminals with diode leads positioned as shown on diode board (Figure 9-4-27-1, Detail A).

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- c. Using soldering iron and solder, Item 313, Appendix D, solder diode leads to terminals. Remove tags.

- d. Install diode board in No. 2 junction box and secure with screws, washers, and spacers.
- e. Install cover on No. 2 junction box and secure with fasteners.
- f. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- g. Perform operational check of ac electrical system (PARA 9-2-1).
- h. Make sure area is clean and free of foreign material.
- i. Install soundproofing panel (PARA 2-4-69).

9-4-28 JUNCTION BOX RELAYS, K80, K81, AND K82.

PARAGRAPH BREAKDOWN

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9-4-28.1 Replace Junction Box Relays, K80, K81, and K82	9-4-28-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Heat Gun
- Protection, Eye and Skin
- Soldering Iron

Materials/Parts

- Insulation Sleeving, Heat-Shrinkable, Item 162, Appendix D
- Solder, Item 313, Appendix D

- Tags

References

- Appendix D
- PARA 1-6-16
- PARA 2-4-69
- PARA 9-2-1
- PARA 9-4-34
- PARA 9-4-106

9-4-28.1 Replace Junction Box Relays, K80, K81, and K82.

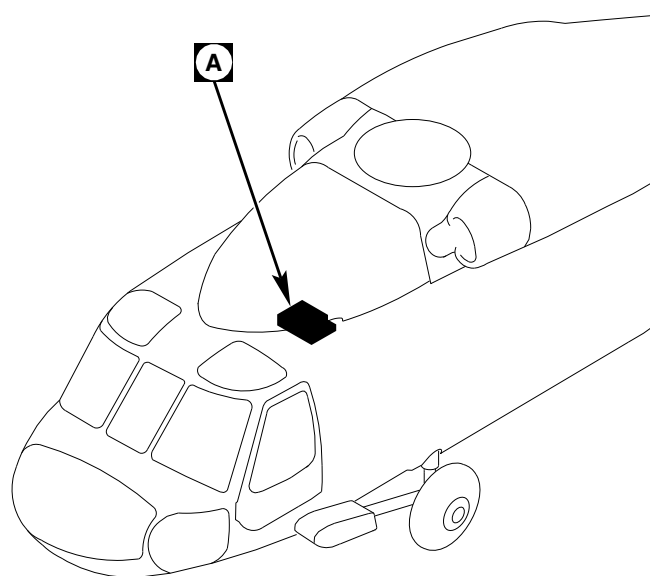
NOTE

Replacement of junction box relays, K80, K81, and K82, is the same.

9-4-28.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).

- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Remove soundproofing panel (PARA 2-4-69).
- Loosen fasteners securing cover on No. 1 junction box and let cover hang by chain.
- Slide heat-shrinkable insulation sleeving back from terminals (Figure 9-4-28-1, Detail A).



NOTE
NO. 1 JUNCTION BOX COVER
NOT SHOWN.

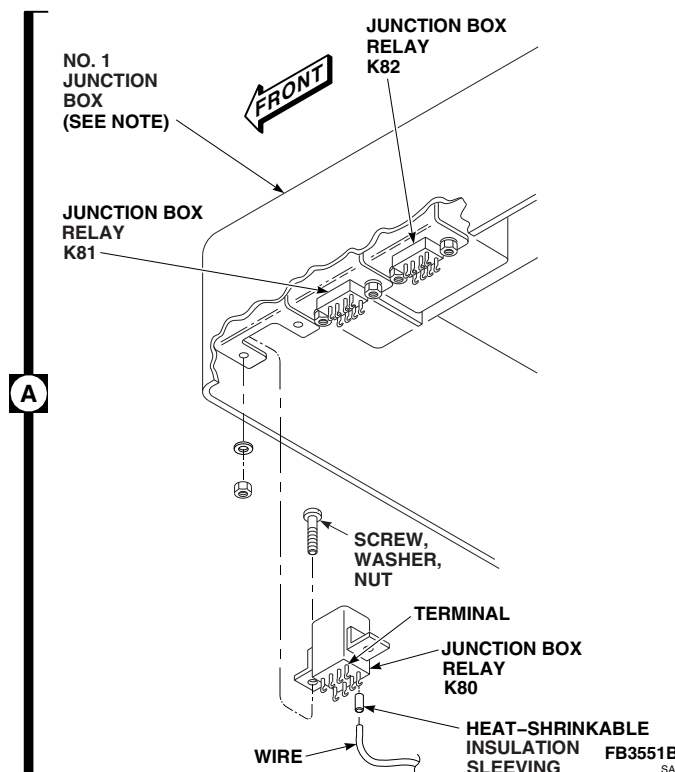


FIGURE 9-4-28-1. REPLACE JUNCTION BOX RELAYS, K80, K81, AND K82

- f. Inspect heat-shrinkable insulation sleeving for damage. **NONE ALLOWED.** If damaged, remove heat-shrinkable insulation sleeving.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection, while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- g. Tag wires. Using soldering iron, unsolder wiring from terminals.
- h. Remove screws, washers, and nuts securing junction box relay to No. 1 junction box.
- i. Remove junction box relay.

9-4-28.1.2 Install.

- a. Turn off all electrical power (PARA 1-6-16).

- b. Install junction box relay in No. 1 junction box and secure with screws, washers, and nuts (Figure 9-4-28-1, Detail A).
- c. If necessary, install heat-shrinkable insulation sleeving, Item 162, Appendix D, on wire.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection, while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- d. Using soldering iron and solder, Item 313, Appendix D, solder wiring to terminals. Remove tags.
- e. Slide heat-shrinkable insulation sleeving over solder terminals.

- f. Using heat gun, shrink heat-shrinkable insulation sleeving.
- g. Install cover on No. 1 junction box and secure with fasteners.
- h. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- i. Perform operational check of AC electrical system (PARA 9-2-1).
- j. Make sure area is clean and free of foreign material.
- k. Install soundproofing panel (PARA 2-4-69).

9-4-29 TRANSFORMER, T11 (26 VAC).

PARAGRAPH BREAKDOWN

	PAGE
9-4-29.1 Replace Transformer, T11 (26 vac)	9-4-29-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

Materials/Parts

- Tags

References

- PARA 1-6-16
- PARA 2-4-69
- PARA 9-2-1
- PARA 9-4-34
- PARA 9-4-106

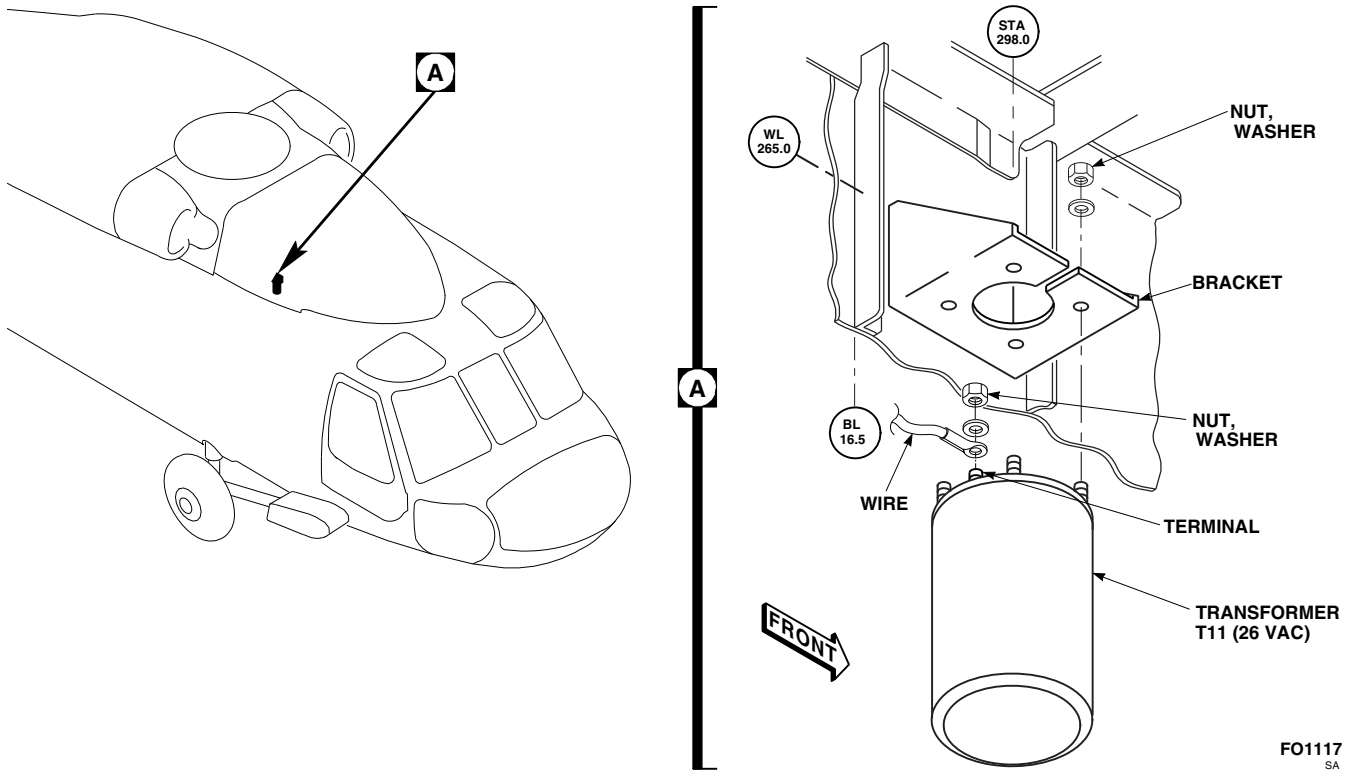
9-4-29.1 Replace Transformer, T11 (26 vac).

9-4-29.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Remove soundproofing panel (PARA 2-4-69).
- Remove nuts and washers securing transformer, T11 (26 vac), to bracket (Figure 9-4-29-1, Detail A).
- Tag wires. Remove nuts and washers securing wires to terminals of transformer, T11 (26 vac).
- Remove transformer, T11 (26 vac).

9-4-29.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).
- Install wiring to terminals and secure with nuts and washers (Figure 9-4-29-1, Detail A). Remove tags.
- Install transformer, T11 (26 vac), to bracket and secure with washers and nuts.
- Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Perform operational check of ac electrical system (PARA 9-2-1).
- Make sure area is clean and free of foreign material.
- Install soundproofing panel (PARA 2-4-69).



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FIGURE 9-4-29-1. REPLACE TRANSFORMER, T11 (26 VAC)

9-4-30 DC PRIMARY BUS CONTACTORS, K6 AND K16.

PARAGRAPH BREAKDOWN

	PAGE
9-4-30.1 Replace DC Primary Bus Contactors, K6 and K16	9-4-30-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

Materials/Parts

- Cheesecloth, Item 90, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Protective Caps and Plugs
- Tags

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-69
- PARA 9-2-2
- PARA 9-4-34
- PARA 9-4-106

9-4-30.1 Replace DC Primary Bus Contactors, K6 and K16.

NOTE

Replacement of dc primary bus contactors, K16 (No. 1) and K6 (No. 2), is the same, except as noted.

9-4-30.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Remove soundproofing panel (PARA 2-4-69).

- Loosen fasteners securing cover on No. 1 or No. 2 junction box and let cover hang by chain (Figure 9-4-30-1, Detail A or Detail B).
- Disconnect electrical connector, P238 from J238 (K6), or P239 from J239 (K16), from dc primary bus contactor.
- Install protective caps and plugs on electrical connectors.
- Tag wires. Remove nuts, lockwashers, and washers securing wiring to dc primary bus contactor, K6 or K16.
- For dc primary bus contactor, K16, remove nut, lockwasher, and washer securing bus bar to dc bus tie contactor, K15. Remove bus bar (Figure 9-4-30-1, Detail A).

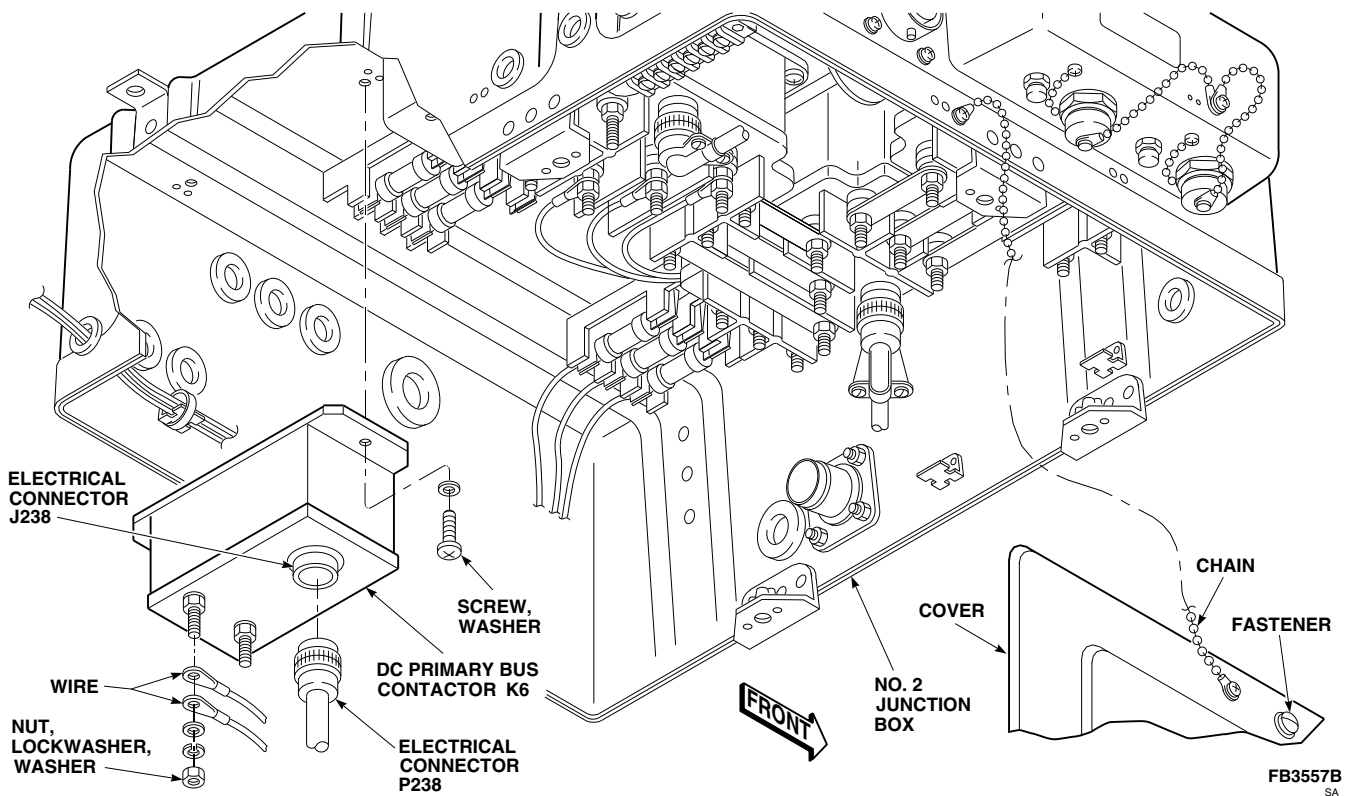
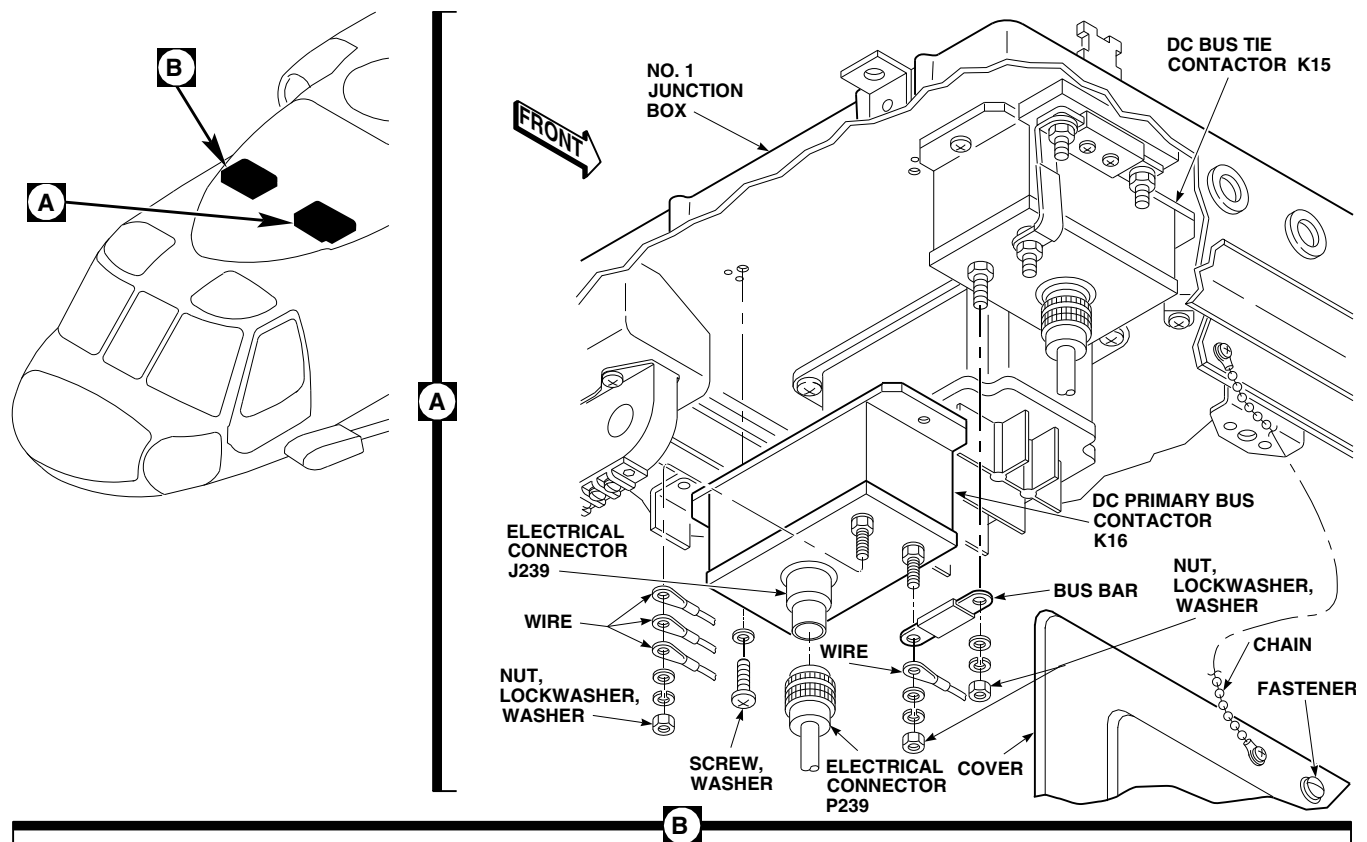


FIGURE 9-4-30-1. REPLACE DC PRIMARY BUS CONTACTORS, K6 AND K16

9-4-30-2

- i. Remove screws and washers securing dc primary bus contactor, K6 (No. 2 junction box), or K16 (No. 1 junction box), to junction box (Figure 9-4-30-1, Detail A or Detail B).
- j. Remove dc primary bus contactor.
- e. Install wiring on dc primary bus contactor, K6 or K16, and secure with nuts, lockwashers, and washers (Figure 9-4-30-1, Detail A or Detail B). Remove tags.
- f. Remove protective caps and plugs from electrical connectors.

9-4-30.1.2. Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Using cheesecloth, Item 90, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D, clean surface around all mounting holes. Using clean, dry cheesecloth, Item 90, Appendix D, dry surface.
- c. Install dc primary bus contactor, K6 (No. 2 junction box), or K16 (No. 1 junction box) to junction box and secure with screws and washers (Figure 9-4-30-1, Detail A or Detail B).
- d. For dc primary bus contactor, K16, install bus bar to dc primary bus contactor, K16, and dc bus tie contactor, K15. Secure bus bar to dc bus tie contactor, K15, with nut, lockwasher, and washer (Figure 9-4-30-1, Detail A).
- g. Inspect and treat electrical connectors (PARA 1-7-20).
- h. Connect electrical connector, P239 to J239 (K16), or P238 to J238 (K6).
- i. Install cover on No. 1 or No. 2 junction box and secure with fasteners.
- j. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- k. Perform operational check of dc electrical system (PARA 9-2-2).
- l. Make sure area is clean and free of foreign material.
- m. Install soundproofing panel (PARA 2-4-69).

9-4-31 DC BUS TIE CONTACTOR, K15.

PARAGRAPH BREAKDOWN

	PAGE
9-4-31.1 Replace DC Bus Tie Con- tactor, K15	9-4-31-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

Materials/Parts

- Cheesecloth, Item 90, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-69
- PARA 9-2-2
- PARA 9-4-34
- PARA 9-4-106

9-4-31.1 Replace DC Bus Tie Contactor, K15.

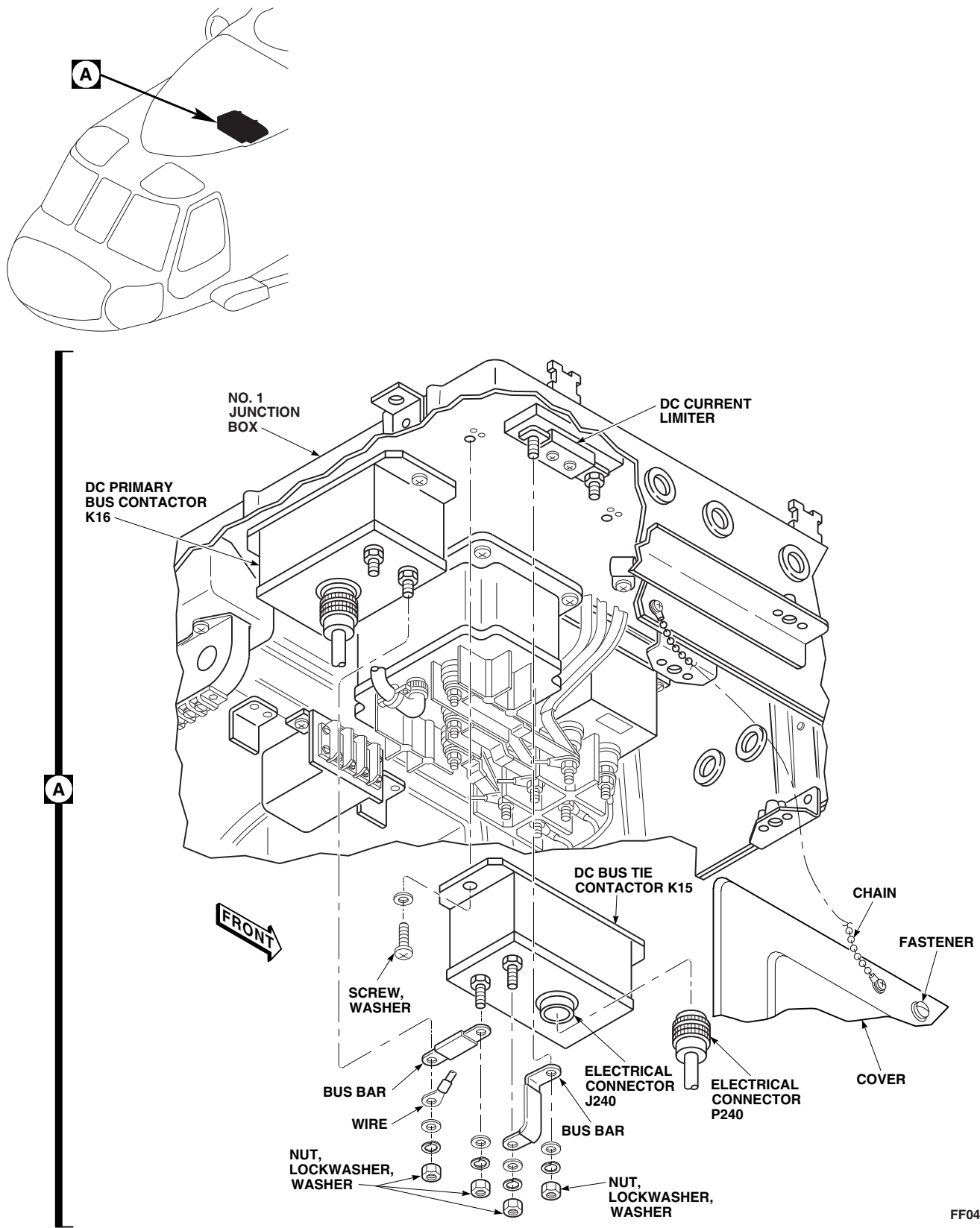
9-4-31.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Remove soundproofing panel (PARA 2-4-69).
- Loosen fasteners securing cover on No. 1 junction box and let cover hang by chain (Figure 9-4-31-1, Detail A).
- Disconnect electrical connector, P240 from J240, on dc bus tie contactor, K15.
- Install protective caps and plugs on electrical connectors.

- Remove nuts, lockwashers, and washers securing wiring and bus bar to dc bus tie contactor, K15, and dc primary bus contactor, K16. Remove bus bar.
- Remove nuts, lockwashers, and washers securing bus bar to dc bus tie contactor, K15, and dc current limiter.
- Remove screws and washers securing dc bus tie contactor, K15, to No. 1 junction box.
- Remove dc bus tie contactor, K15.

9-4-31.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Using cheesecloth, Item 90, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D, clean surface around all mounting



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FIGURE 9-4-31-1. REPLACE DC BUS TIE CONTACTOR, K15

9-4-31-2

- holes. Using clean, dry cheesecloth, Item 90, Appendix D, dry surface.
- c. Install dc bus tie contactor, K15, in No. 1 junction box and secure with screws and washers (Figure 9-4-31-1, Detail A).
- d. Install bus bar on dc bus tie contactor, K15, and dc current limiter and secure with nuts, lockwashers, and washers.
- e. Install bus bar on dc bus tie contactor, K15, and dc primary bus contactor, K16. Secure bus bar to dc bus tie contactor, K15, with washer, lockwasher, and nut.
- f. Install wiring over bus bar on dc primary bus contactor, K16. Secure wiring and bus bar with nut, lockwasher, and washer.
- g. Remove protective caps and plugs from electrical connectors.
- h. Inspect and treat electrical connectors (PARA 1-7-20).
- i. Connect electrical connector, P240 to J240, on dc bus tie contactor, K15.
- j. Install cover on No. 1 junction box and secure with fasteners.
- k. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- l. Perform operational check of dc electrical system (PARA 9-2-2).
- m. Make sure area is clean and free of foreign material.
- n. Install soundproofing panel (PARA 2-4-69).

9-4-32 DC ESSENTIAL BUS SUPPLY RELAYS, K9 AND K10.

PARAGRAPH BREAKDOWN

	PAGE
9-4-32.1 Replace DC Essential Bus Supply Relays, K9 and K10	9-4-32-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

Materials/Parts

- Cheesecloth, Item 90, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D

- Tags

References

- Appendix D
- PARA 1-6-16
- PARA 9-2-2
- PARA 9-4-34
- PARA 9-4-106

9-4-32.1 Replace DC Essential Bus Supply Relays, K9 and K10.

NOTE

Replacement of dc essential bus supply relays, K10 (No. 1), and K9 (No. 2), is the same.

9-4-32.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Loosen fasteners and carefully lower pilot’s circuit breaker panel.
- Remove screw securing clamp to support (Figure 9-4-32-1, Detail A).

- Remove screws, lockwashers, and washers securing terminal shield to dc essential bus supply relay.
- Tag wires. Remove nuts, lockwashers, and washers securing bus bar and wiring to dc essential bus supply relay.
- Remove screws, washers, and lockwashers securing dc essential bus supply relay to support.
- Remove dc essential bus supply relay.

9-4-32.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Using cheesecloth, Item 90, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D, clean surface around all mounting holes. Using clean, dry cheesecloth, Item 90, Appendix D, dry surface.

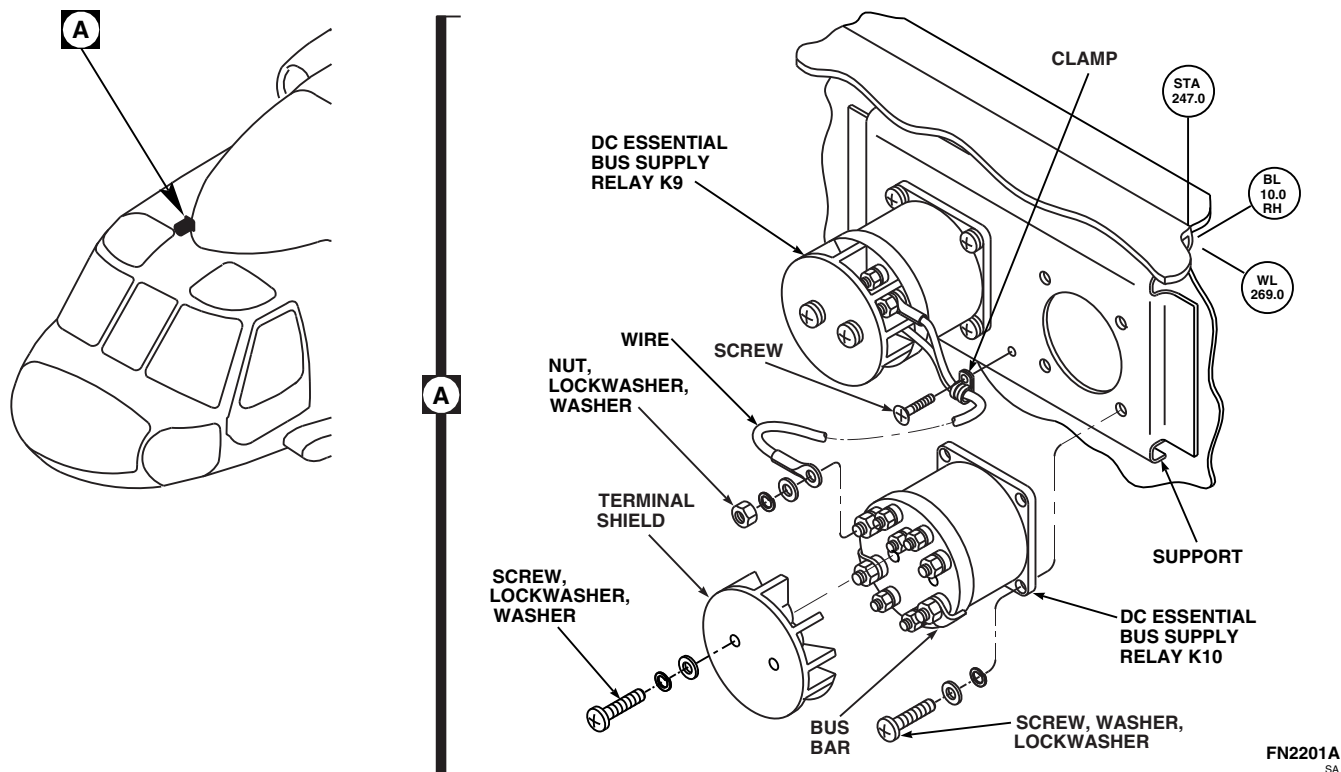


FIGURE 9-4-32-1. REPLACE DC ESSENTIAL BUS SUPPLY RELAYS, K9 AND K10

- c. Install dc essential bus supply relay on support and secure with screws, washers, and lockwashers (Figure 9-4-32-1, Detail A).
- d. Install wiring and bus bar to dc essential bus supply relay and secure with nuts, washers, and lockwashers. Remove tags.
- e. Install terminal shield on dc essential bus supply relay and secure with screws, lockwashers, and washers.
- f. Install clamp to support with screw.
- g. Make sure area is clean and free of foreign material.
- h. Carefully swing pilot's circuit breaker panel into closed position and secure with fasteners.
- i. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- j. Perform operational check of dc electrical system (PARA 9-2-2).

9-4-33 BATTERY RELAY, K7.

PARAGRAPH BREAKDOWN

	PAGE
9-4-33.1 Replace Battery Relay, K7	9-4-33-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

Materials/Parts

- Cheesecloth, Item 90, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D

- Tags

References

- Appendix D
- PARA 1-6-16
- PARA 9-2-2
- PARA 9-4-34
- PARA 9-4-106

9-4-33.1 Replace Battery Relay, K7.

9-4-33.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).

WARNING

Do not remove battery relay junction box while external power is connected, while either engine or APU is operating, or while battery is connected. Serious fires can result.

- Remove screws and washers securing battery relay junction box, with battery relay, K7, to structure near cabin floor (Figure 9-4-33-1, Detail A).

- Remove screws, lockwashers, and washers securing terminal shield to battery relay, K7 (Figure 9-4-33-1, Detail B).
- Tag wires. Remove nuts, lockwashers, and washers securing wiring to battery relay, K7.
- Remove screws, washers, and nuts securing battery relay, K7, to battery relay junction box (Figure 9-4-33-1, Detail A).
- Remove battery relay, K7.

9-4-33.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Using cheesecloth, Item 90, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D, clean surface around all mounting holes. Using clean, dry cheesecloth, Item 90, Appendix D, dry surface.
- Install battery relay, K7, in battery relay junction box and secure with screws, washers, and

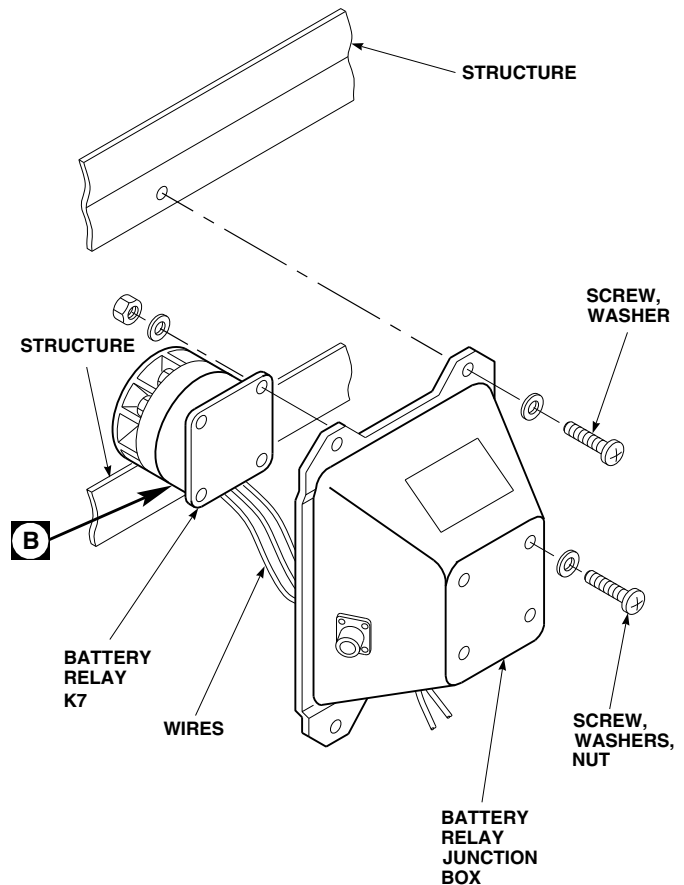
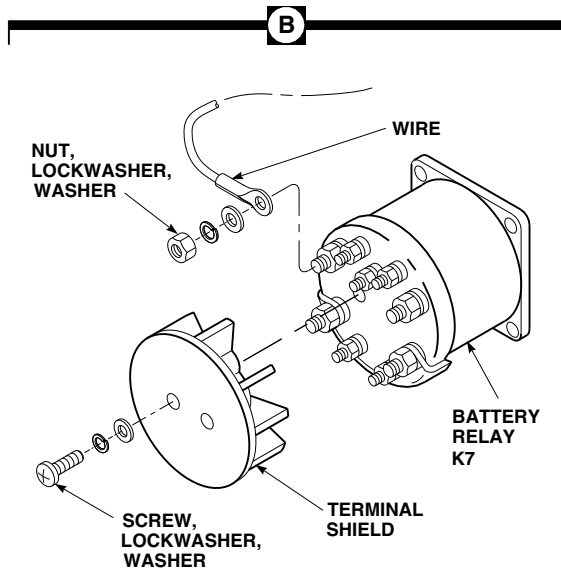
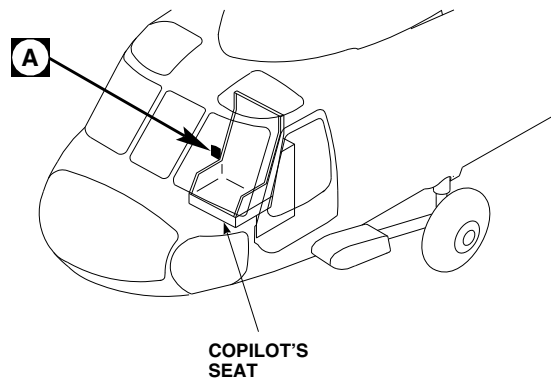


FIGURE 9-4-33-1. REPLACE BATTERY RELAY, K7

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9-4-33-2

- nuts, with screwheads on outside (Figure 9-4-33-1, Detail A).
- d. Install wiring to battery relay, K7, and secure with nuts, lockwashers, and washers. Remove tags (Figure 9-4-33-1, Detail B).
- e. Install terminal shield on battery relay, K7, and secure with screws, lockwashers, and washers.
- f. Install battery relay junction box on structure and secure with screws and washers (Figure 9-4-33-1, Detail A).
- g. Make sure area is clean and free of foreign material.
- h. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- i. Perform operational check of dc electrical system (PARA 9-2-2).

9-4-34 NICKEL-CADMIUM BATTERY 70583–702374 .

PARAGRAPH BREAKDOWN

		PAGE
9-4-34.1	Replace Nickel-Cadmium Battery	9-4-34-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

Materials/Parts

- Cheesecloth, Item 90, Appendix D
- Distilled Water, Item 123, Appendix D
- Lockwire, Item 195, Appendix D

- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 9-2-2
- PARA 9-5-33

9-4-34.1 Replace Nickel-Cadmium Battery.

9-4-34.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Loosen fasteners and remove battery cover from battery support (Figure 9-4-34-1, Detail A).
- Disconnect electrical connector, P140, from battery.
- Install protective caps and plugs on electrical connectors.
- Disconnect electrical connector, P141, from battery and connect to dummy electrical connector on battery relay junction box.

- If present, carefully pull vent tubes from battery.
- Loosen wingnuts on clamp rods and remove clamp rods from slots on battery clamp.
- Remove battery from battery support.

9-4-34.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Using cheesecloth, Item 90, Appendix D, dampened with distilled water, Item 123, Appendix D, clean battery, battery cover, and battery support. Using clean, dry cheesecloth, Item 90, Appendix D, dry surface.
- Make sure that battery support is free of foreign material.

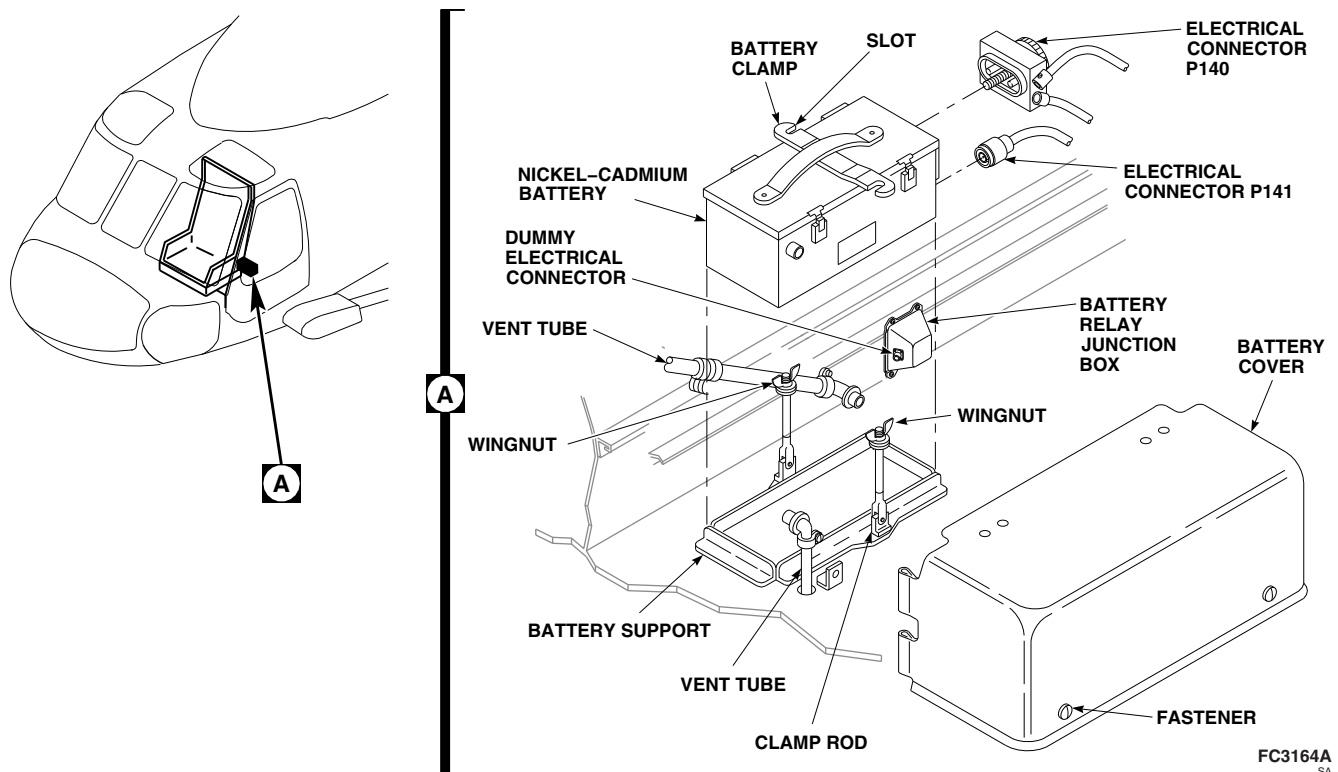


FIGURE 9-4-34-1. REPLACE NICKEL-CADMIUM BATTERY

NOTE

Make sure that replacement battery has been serviced prior to installation.

- d. If necessary, service replacement battery (PARA 9-5-33).
- e. Install battery in battery support between clamp rods (Figure 9-4-34-1, Detail A).
- f. Install clamp rods in slots in battery clamp.
- g. **TIGHTEN WINGNUTS.**
- h. Using lockwire, Item 195, Appendix D, lockwire wingnuts.
- i. If required, install vent tubes on battery.
- j. Disconnect electrical connector, P141, from dummy electrical connector on battery relay junction box.
- k. Inspect and treat electrical connectors (PARA 1-7-20).
- l. Connect electrical connector, P141, to battery.
- m. Remove protective caps and plugs from electrical connectors.
- n. Inspect and treat electrical connectors (PARA 1-7-20).
- o. Connect electrical connector, P140, to battery.
- p. Make sure area is clean and free of foreign material.
- q. Perform operational check of dc electrical system (PARA 9-2-2).
- r. Install battery cover to battery support and secure with fasteners.

9-4-35 BATTERY CONDITIONER/ANALYZER 70583–702374 .

PARAGRAPH BREAKDOWN

	PAGE
9-4-35.1 Replace Battery Conditioner/Analyzer	9-4-35-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Digital Low-Resistance Ohmmeter
- Electrical Repairer’s Toolkit

Materials/Parts

- Cheesecloth, Item 90, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D

- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-6-5
- PARA 9-2-2
- PARA 9-4-34

9-4-35.1 Replace Battery Conditioner/
Analyzer.

9-4-35.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34).
- Loosen fasteners and open access panel at rear of lower console (Figure 9-4-35-1).

NOTE

EMEP Pin filter adapters are considered part of harness assembly. Do not remove pin filter adapter from electrical connector, P123 (Figure 9-4-35-1, Detail B).

- Disconnect electrical connector, P123 from J123, from battery conditioner/analyzer (Figure 9-4-35-1, Detail A).
- Install protective caps and plugs on electrical connectors.
- Remove screws, washers, and lockwashers securing bracket assembly to outside of lower console and battery conditioner/analyzer to inside of lower console.
- Remove battery conditioner/analyzer from inside lower console.

9-4-35.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- EMEP** Inspect chemical conversion coating as follows:

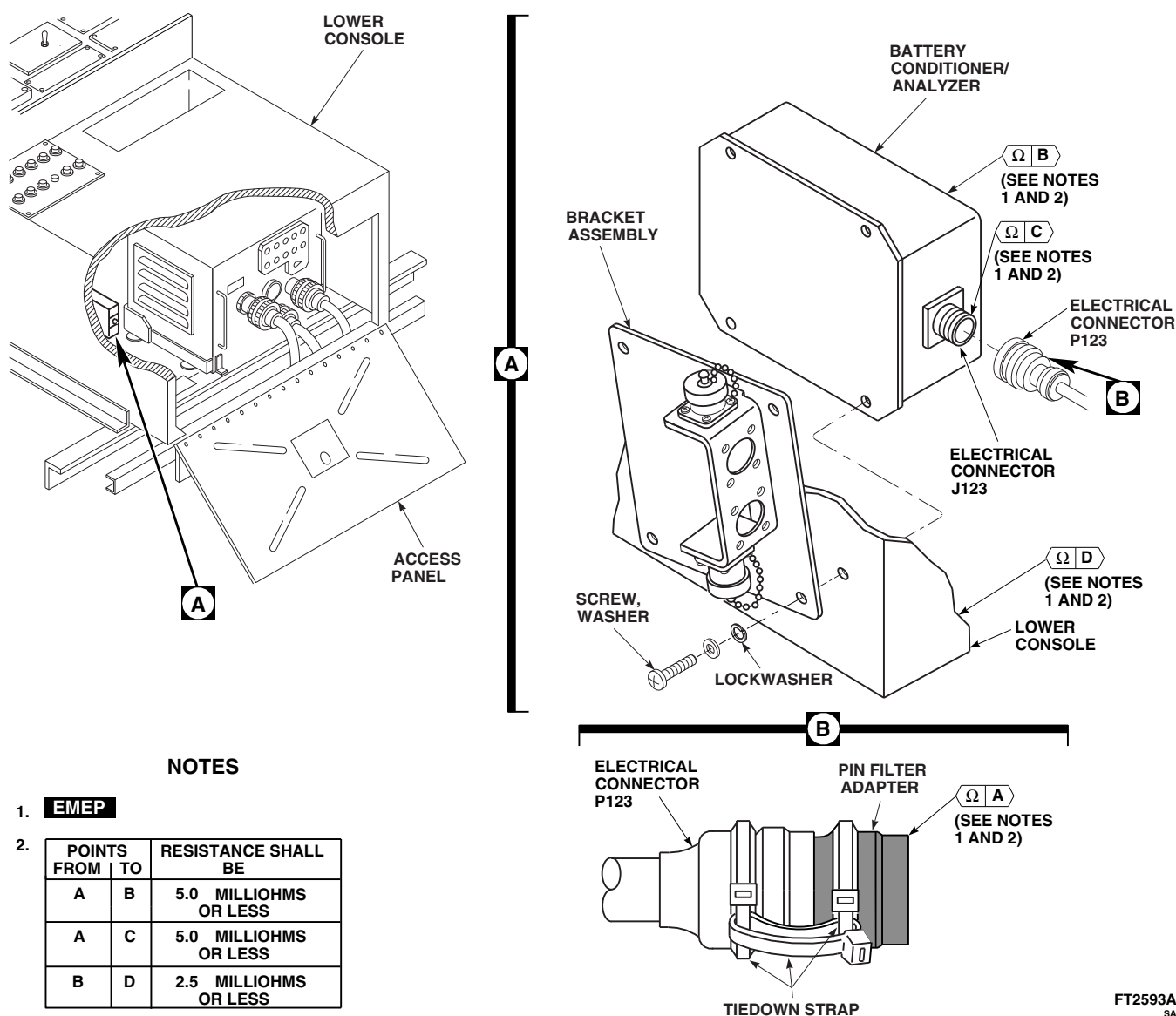


FIGURE 9-4-35-1. REPLACE BATTERY CONDITIONER/ANALYZER

- Using cheesecloth, Item 90, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D, clean surface around all mounting holes. Using clean, dry cheesecloth, Item 90, Appendix D, dry surface.
- Inspect chemical conversion coated surfaces for breaks and scratches. **NO BREAKS AND SCRATCHES ALLOWED.** If coated surfaces have breaks or scratches, apply chemical conversion coating (PARA 2-6-5).
- Position battery conditioner/analyzer on inside of lower console, with electrical connector, J123, at rear (Figure 9-4-35-1, Detail A).
- Install bracket assembly on outside of lower console, and secure bracket assembly and battery conditioner/analyzer to lower console with screws, washers, and lockwashers.
- Remove protective caps and plugs from electrical connectors.
- Inspect and treat electrical connectors (PARA 1-7-20).

- g. Connect electrical connector, P123 to J123, on battery conditioner/analyzer.
- h. **EMEP** Perform pin filter adapter EME resistance check as follows:■

NOTE

Resistance measurements must be taken at least three times.

- (1) Using digital low-resistance ohmmeter, measure resistance between shell of pin filter adapter and battery conditioner/analyzer. **RESISTANCE SHALL BE 5.0 MILLIOHMS OR LESS.**
- (2) If resistance is greater than 5.0 milliohms, measure resistance between shell of pin filter adapter and electrical connector on battery conditioner/analyzer. **RESISTANCE SHALL BE 5.0 MILLIOHMS OR LESS.**
- (3) If resistance is still beyond limits, clean surface between pin filter adapter and battery conditioner/analyzer.
- i. **EMEP** Perform resistance check of battery conditioner/analyzer as follows:■

NOTE

Resistance measurements must be taken at least three times.

- (1) Using digital low-resistance ohmmeter, check resistance between battery conditioner/analyzer and lower console. **RESISTANCE SHALL BE 2.5 MILLIOHMS OR LESS.**
- (2) If resistance is greater than 2.5 milliohms, remove battery conditioner/analyzer and apply chemical conversion coating (PARA 2-6-5).
- j. Touch up paint, as required (PARA 2-6-5).
- k. Make sure area is clean and free of foreign material.
- l. Close access panel on lower console and secure with fasteners (Figure 9-4-35-1).
- m. Connect electrical connector, P140, to battery (PARA 9-4-34).
- n. Perform operational check of dc electrical system (PARA 9-2-2).

9-4-36 CONVERTERS.

PARAGRAPH BREAKDOWN

	PAGE
9-4-36.1 Replace Converters	9-4-36-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

Materials/Parts

- Cheesecloth, Item 90, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Protective Caps and Plugs
- Tags

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-69
- PARA 9-2-2
- PARA 9-4-34
- PARA 9-4-106

9-4-36.1 Replace Converters.

NOTE

Replacement of No. 1 and No. 2 converter is the same, except as noted.

9-4-36.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Remove soundproofing panel (PARA 2-4-69).
- Disconnect electrical connector, P903 from J903 (No. 1), or P904 from J904 (No. 2) from converter (Figure 9-4-36-1, Detail A).

- Install protective caps and plugs on electrical connectors.
- Loosen fasteners on terminal board cover and remove.
- Tag wires. Remove nuts, lockwashers, and washers securing wiring to terminal board.
- Remove bolts, washers, and lockwashers from two nonslotted mounting holes securing converter to overhead support.
- Loosen bolts in two slotted mounting holes and slide converter to remove from bolts.

9-4-36.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Using cheesecloth, Item 90, Appendix D, dampened with dry-cleaning solvent, Item 124,

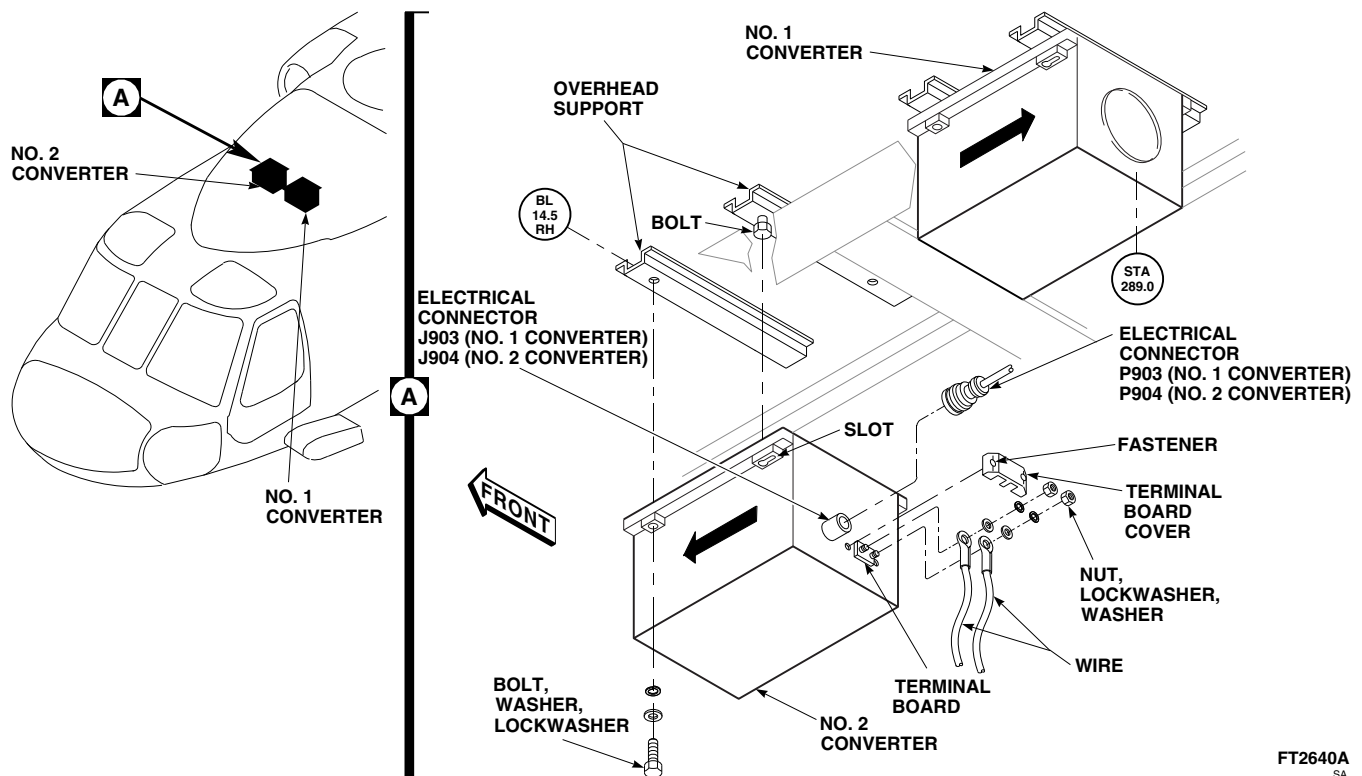


FIGURE 9-4-36-1. REPLACE CONVERTERS

- Appendix D, clean surface around all mounting holes. Using clean, dry cheesecloth, Item 90, Appendix D, dry surface.
- c. Install converter on overhead support and secure with mounting bolts in slotted mounting holes (Figure 9-4-36-1, Detail A).
 - d. Slide converter to line up nonslotted mounting holes and install bolts, washers, and lockwashers.
 - e. **TIGHTEN ALL BOLTS.**
 - f. Install wiring to terminal board and secure with nuts, lockwashers, and washers. Remove tags.
 - g. Install terminal board cover to terminal board and secure with fasteners.
 - h. Remove protective caps and plugs from electrical connectors.
 - i. Inspect and treat electrical connectors (PARA 1-7-20).
 - j. Connect electrical connector, P903 to J903 (No. 1), or P904 to J904 (No. 2), on converter.
 - k. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
 - l. Perform operational check of dc electrical system (PARA 9-2-2).
 - m. Make sure area is clean and free of foreign material.
 - n. Install soundproofing panel (PARA 2-4-69).

9-4-37 DC CURRENT LIMITER.

PARAGRAPH BREAKDOWN

	PAGE
9-4-37.1 Replace DC Current Limiter	9-4-37-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

References

- PARA 1-6-16
- PARA 2-4-69

- PARA 9-2-2
- PARA 9-4-34
- PARA 9-4-106

9-4-37.1 Replace DC Current Limiter.

9-4-37.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Remove soundproofing panel (PARA 2-4-69).
- Loosen fasteners securing cover on No. 1 junction box and let cover hang by chain (Figure 9-4-37-1, Detail A).
- Loosen nuts securing dc current limiter to fuse holder.
- Push slotted side of dc current limiter out from under nut and then pull dc current limiter to release slotted end from under nut.

9-4-37.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).

- Insert front side slotted end of DC current limiter between washer and fuse holder. Push until firmly seated (Figure 9-4-37-1, Detail A).
- Push rear side slotted side of dc current limiter between bus bar and fuse holder.
- TIGHTEN NUTS SECURING DC CURRENT LIMITER TO FUSE HOLDER.**
- Install cover on No. 1 junction box and secure with fasteners.
- Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Perform operational check of dc electrical system (PARA 9-2-2).
- Make sure area is clean and free of foreign material.
- Install soundproofing panel (PARA 2-4-69).

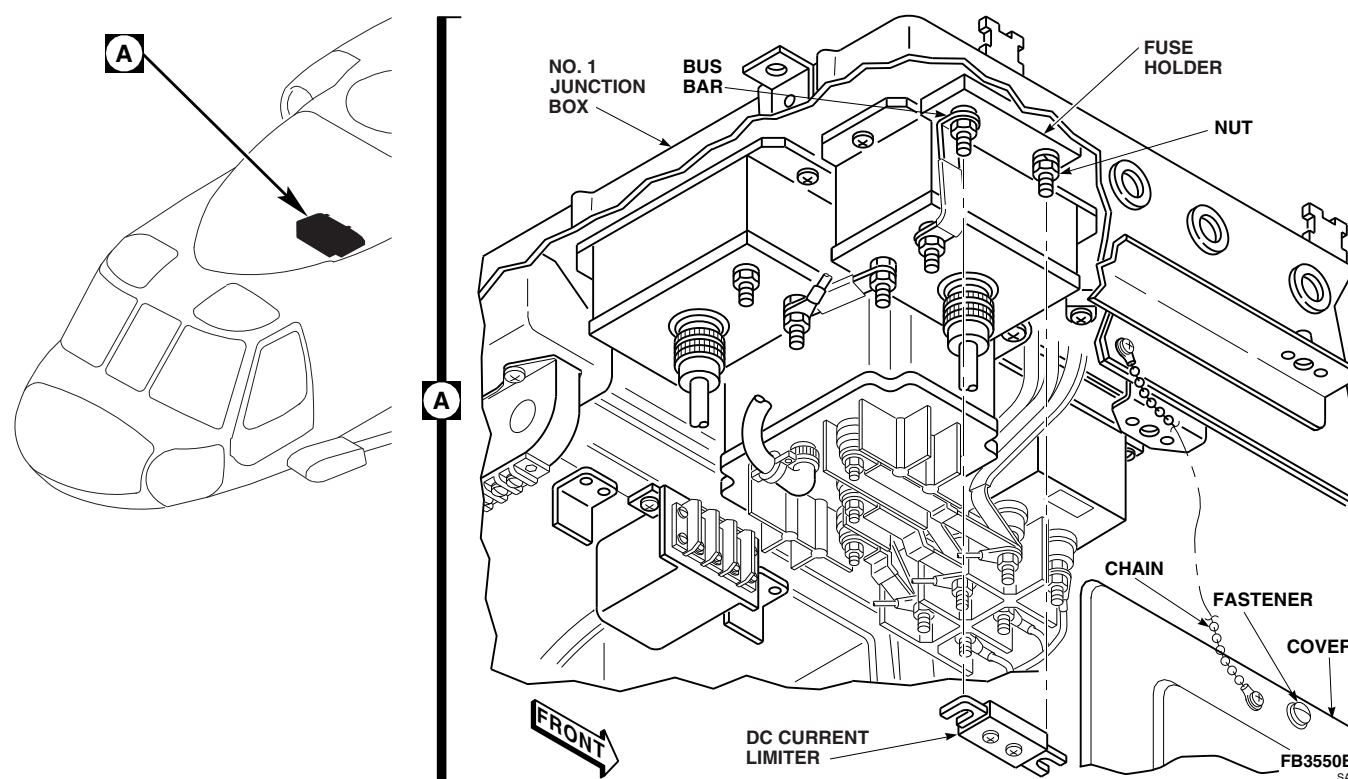


FIGURE 9-4-37-1. REPLACE DC CURRENT LIMITER

9-4-38 DC MONITOR BUS SUPPLY RELAY, K5.

PARAGRAPH BREAKDOWN

	PAGE
9-4-38.1 Replace DC Monitor Bus Supply Relay, K5	9-4-38-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit
- Heat Gun
- Protection, Eye and Skin
- Soldering Iron

Materials/Parts

- Insulation Sleeving, Heat-Shrinkable, Item 162, Appendix D

- Solder, Item 313, Appendix D
- Tags

References

- Appendix D
- PARA 1-6-16
- PARA 2-4-69
- PARA 9-2-2
- PARA 9-4-34
- PARA 9-4-106

9-4-38.1 Replace DC Monitor Bus Supply Relay, K5.

9-4-38.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Remove soundproofing panel (PARA 2-4-69).
- Loosen fasteners securing cover on No. 2 junction box and let cover hang by chain.
- Slide heat-shrinkable insulation sleeving back from terminals (Figure 9-4-38-1, Detail A).

Inspect heat-shrinkable insulation sleeving for damage. If damaged, remove heat-shrinkable insulation sleeving.



- Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.
- Tag wires. Using soldering iron, unsolder and remove wiring from terminals.

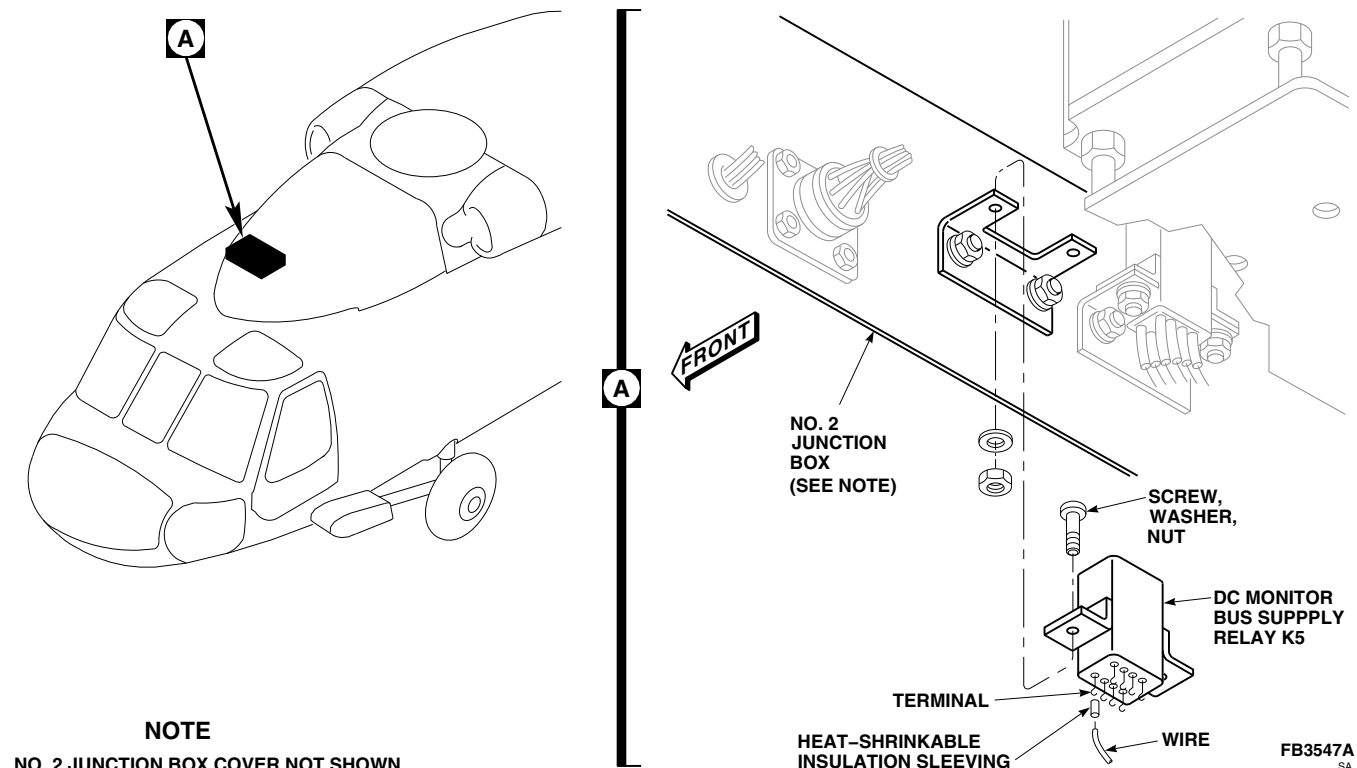


FIGURE 9-4-38-1. REPLACE DC MONITOR BUS SUPPLY RELAY, K5

g. Remove screws, washers, and nuts securing dc monitor bus supply relay, K5, from No. 2 junction box.

h. Remove dc monitor bus supply relay, K5.

9-4-38.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Install dc monitor bus supply relay, K5, in No. 2 junction box and secure with screws, washers, and nuts (Figure 9-4-38-1, Detail A).
- If necessary, install heat-shrinkable insulation sleeving, Item 162, Appendix D, on wire.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while

soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- Using soldering iron and solder, Item 313, Appendix D, solder wiring to terminals of dc monitor bus supply relay, K5. Remove tags.
- Slide heat-shrinkable insulation sleeving over solder terminals.
- Using heat gun, shrink heat-shrinkable insulation sleeving.
- Install cover on No. 2 junction box and secure with fasteners.
- Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Perform operational check of dc electrical system (PARA 9-2-2).
- Make sure area is clean and free of foreign material.

- k. Install soundproofing panel (PARA 2-4-69).

9-4-39 APU ELECTRONIC SEQUENCE UNIT.

PARAGRAPH BREAKDOWN

	PAGE
9-4-39.1 Replace APU Electronic Sequence Unit	9-4-39-1

INITIAL SETUP	• PARA 2-4-69
Tools	• PARA 9-4-34
• Aircraft Mechanic’s Toolkit • Electrical Repairer’s Toolkit	• PARA 9-4-106
Materials/Parts	• PARA 15-2-1
• Protective Caps and Plugs • Tags	• PARA 15-2-2 • PARA 15-2-3
References	
• PARA 1-6-16 • PARA 1-7-20	

9-4-39.1 Replace APU Electronic Sequence Unit.

NOTE

- Digital electronic sequence unit (DESU) is a suitable replacement for electronic sequence unit (ESU). Helicopters may be equipped with either ESU or DESU.
- Replacement of ESU or DESU is the same.

9-4-39.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).

- Remove soundproofing panel (PARA 2-4-69).
- Tag electrical connectors. Disconnect electrical connectors, P260 from J2, and P261 from J1, from APU electronic sequence unit (ESU) (Figure 9-4-39-1, Detail A).
- Install protective caps and plugs on electrical connectors.
- Remove screws, lockwashers, and washers securing ESU to support.
- Remove ESU.

9-4-39.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).

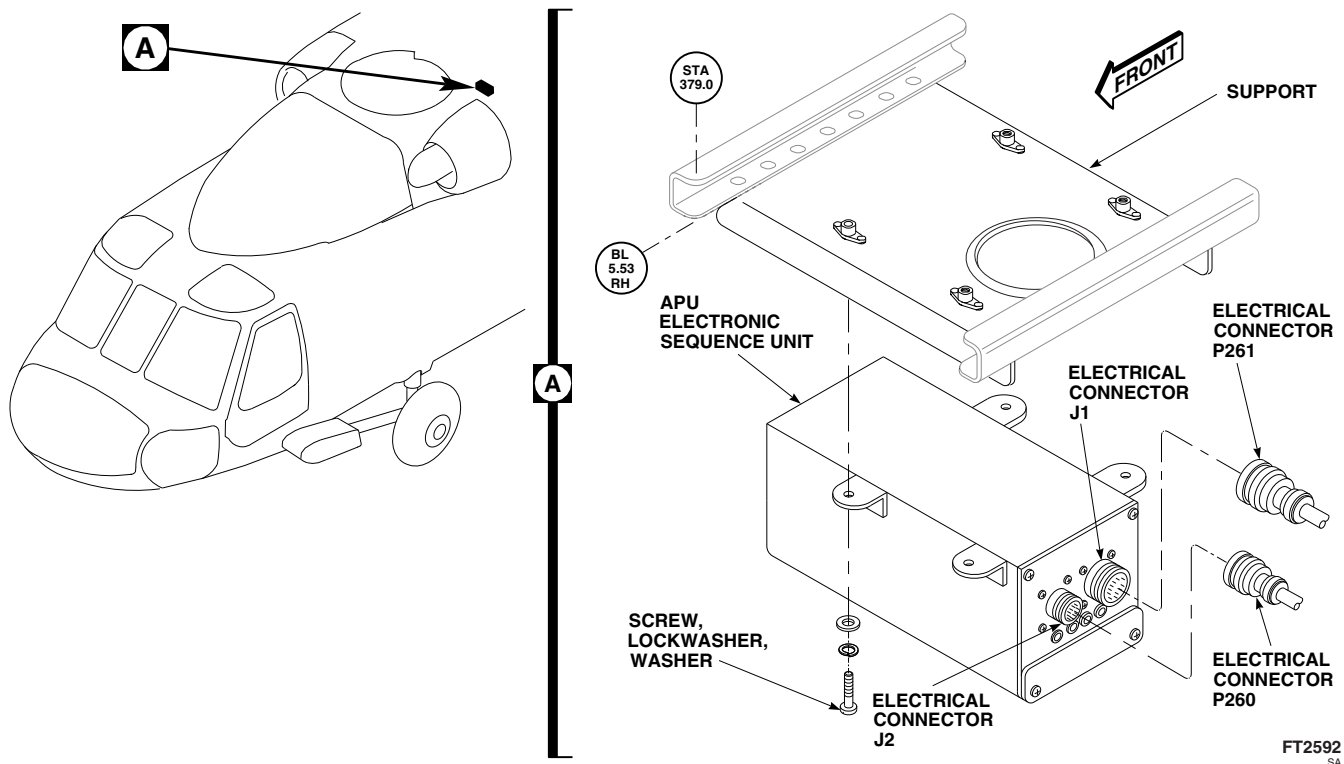


FIGURE 9-4-39-1. REPLACE APU ELECTRONIC SEQUENCE UNIT

- b. Install APU electronic sequence unit (ESU) to support and secure with screws, lockwashers, and washers (Figure 9-4-39-1, Detail A).
- c. Remove protective caps and plugs from electrical connectors.
- d. Inspect and treat electrical connectors (PARA 1-7-20).
- e. Connect electrical connectors, P260 to J2, and P261 to J1, on ESU. Remove tags.
- f. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- g. Perform operational check of auxiliary power unit (PARA 15-2-1, PARA 15-2-2, or PARA 15-2-3).
- h. Make sure area is clean and free of foreign material.
- i. Install soundproofing panel (PARA 2-4-69).

9-4-40 FUEL BOOST PUMP CONTROL PANEL.

PARAGRAPH BREAKDOWN

	PAGE
9-4-40.1 Replace Fuel Boost Pump Control Panel	9-4-40-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit

Materials/Parts

- Protective Caps and Plugs

References

- PARA 1-6-16
- PARA 1-7-20

- PARA 9-4-34
- PARA 9-4-106
- PARA 10-2-2

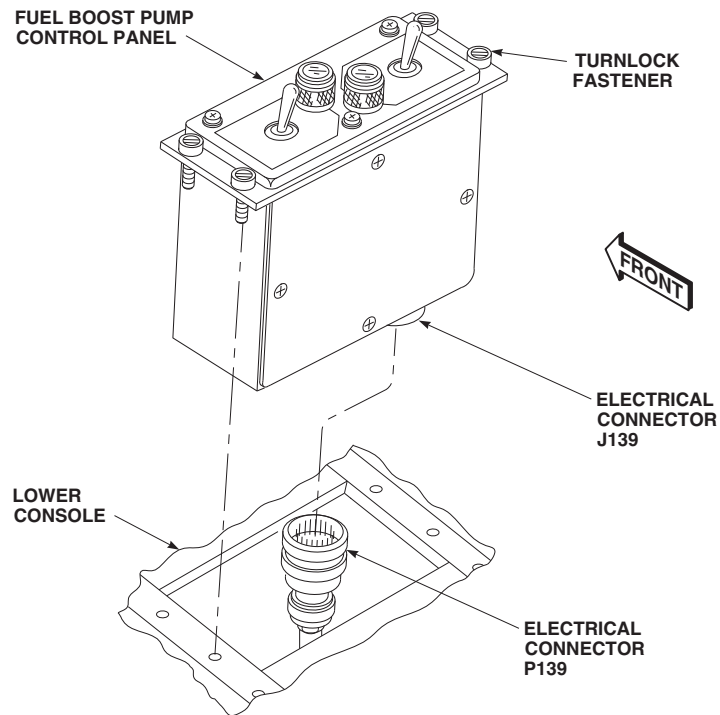
9-4-40.1 Replace Fuel Boost Pump Control Panel.

9-4-40.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Loosen turnlock fasteners securing fuel boost pump control panel to lower console and carefully lift out to gain access to electrical connector (Figure 9-4-40-1).
- Disconnect electrical connector, P139 from J139, and remove fuel boost pump control panel.
- Install protective caps and plugs on electrical connectors.

9-4-40.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Remove protective caps and plugs from electrical connectors.
- Inspect and treat electrical connectors (PARA 1-7-20).
- Connect electrical connector, P139 to J139, to fuel boost pump control panel (Figure 9-4-40-1).
- Make sure area is clean and free of foreign material.
- Install fuel boost pump control panel in lower console and secure with turnlock fasteners.
- Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).

FF0420
SA**FIGURE 9-4-40-1. REPLACE FUEL BOOST PUMP CONTROL PANEL**

- h. Perform operational check of submerged fuel boost pump system (PARA 10-2-2).

9-4-41 FUEL BOOST PUMP CONTROL PANEL INFORMATION PLATE.

PARAGRAPH BREAKDOWN

		PAGE
9-4-41.1	Replace Fuel Boost Pump Control Panel Information Plate	9-4-41-1

INITIAL SETUP

- PARA 9-2-25

Tools

- Electrical Repairer’s Toolkit

References

- PARA 1-6-16
- PARA 9-2-5

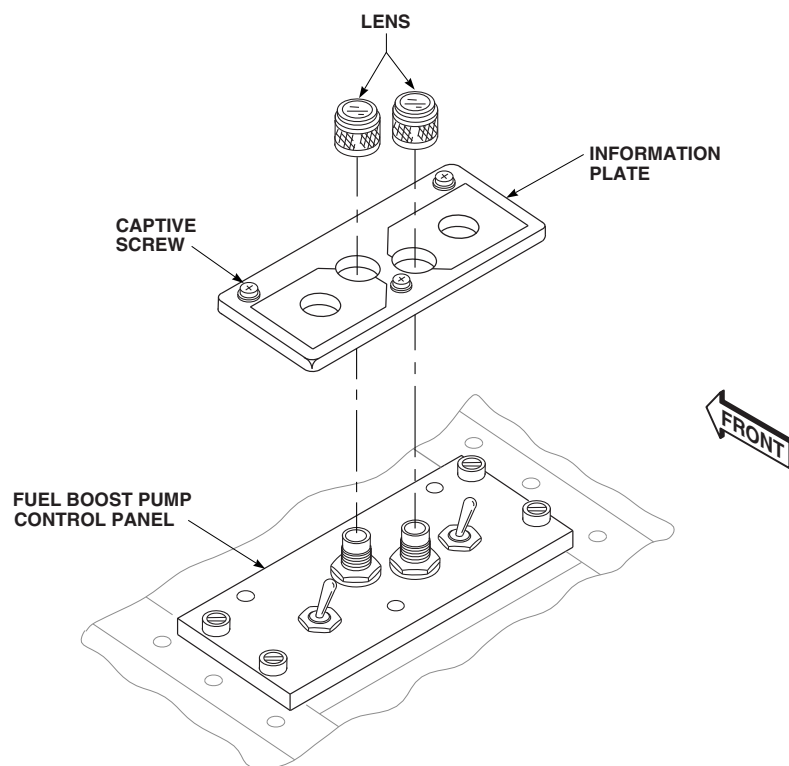
9-4-41.1 Replace Fuel Boost Pump Control Panel Information Plate.

9-4-41.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Unscrew lenses from fuel boost pump control panel (Figure 9-4-41-1).
- Loosen captive screws securing information plate to fuel boost pump control panel.
- Remove information plate.

9-4-41.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Install information plate on fuel boost pump control panel and secure with captive screws (Figure 9-4-41-1).
- Screw lenses on fuel boost pump control panel.
- Make sure area is clean and free of foreign material.
- Perform operational check of upper and lower console lights (PARA 9-2-5 or PARA 9-2-25).

FF0338A
SA**FIGURE 9-4-41-1. REPLACE FUEL BOOST PUMP CONTROL PANEL INFORMATION PLATE**

9-4-42 FUEL BOOST PUMP CONTROL PANEL INDICATOR LIGHT LAMPS.

PARAGRAPH BREAKDOWN

	PAGE
9-4-42.1 Replace Fuel Boost Pump Control Panel Indicator Light Lamps	9-4-42-1

INITIAL SETUP

- PARA 9-2-11

Tools

- PARA 10-2-2

- Electrical Repairer’s Toolkit

References

- PARA 1-6-16
- PARA 9-2-10

9-4-42.1 Replace Fuel Boost Pump Control Panel Indicator Light Lamps.

NOTE

Fuel boost pump control panel has two indicator lights. Replacement of both fuel boost pump control panel indicator light lamps is the same.

9-4-42.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Unscrew and remove lens from fuel boost pump control panel (Figure 9-4-42-1).

- Remove lamp from lens.

9-4-42.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Install lamp in lens (Figure 9-4-42-1).
- Screw lens onto fuel boost pump control panel.
- Perform operational check of submerged fuel boost pump system (PARA 10-2-2).
- Perform operational check of instrument panel and consoles indicator lights dimming (PARA 9-2-10 or PARA 9-2-11).

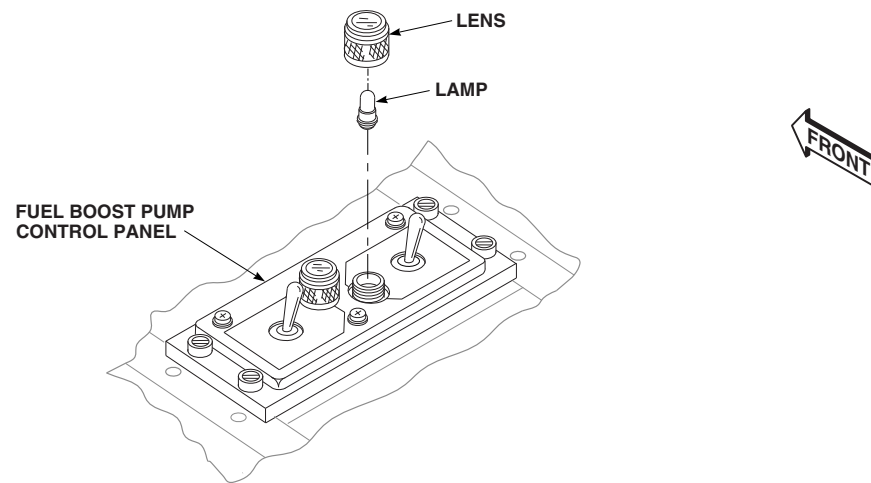
FL0329A
SA

FIGURE 9-4-42-1. REPLACE FUEL BOOST PUMP CONTROL PANEL INDICATOR LIGHT LAMPS

9-4-43 ENGINE PRIME BOOST DIODE BOARD.

PARAGRAPH BREAKDOWN

	PAGE
9-4-43.1 Replace Engine Prime Boost Diode Board	9-4-43-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit

Materials/Parts

- Tags

References

- PARA 1-6-16
- PARA 9-4-34

- PARA 9-4-106
- PARA 10-2-6
- PARA 10-2-5

9-4-43.1 Replace Engine Prime Boost Diode Board.

9-4-43.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Loosen fasteners securing upper console and carefully lower.
- Tag wires. Remove nuts, lockwashers, and washers securing wires to terminal boards, TB1 and TB2 (Figure 9-4-43-1, Detail A).
- Remove screws, washers, and nuts securing terminal boards, TB1 and TB2, to supports.

- Remove screws, washers, and spacers securing engine prime boost diode board to support.
- Remove engine prime boost diode board.

9-4-43.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Install engine prime boost diode board on support with screws, spacers, and washers (Figure 9-4-43-1, Detail A).
- Install terminal boards, TB1 and TB2, to supports and secure with screws, washers, and nuts.
- Install wires to terminal boards, TB1 and TB2, and secure with nuts, lockwashers, and washers as follows:

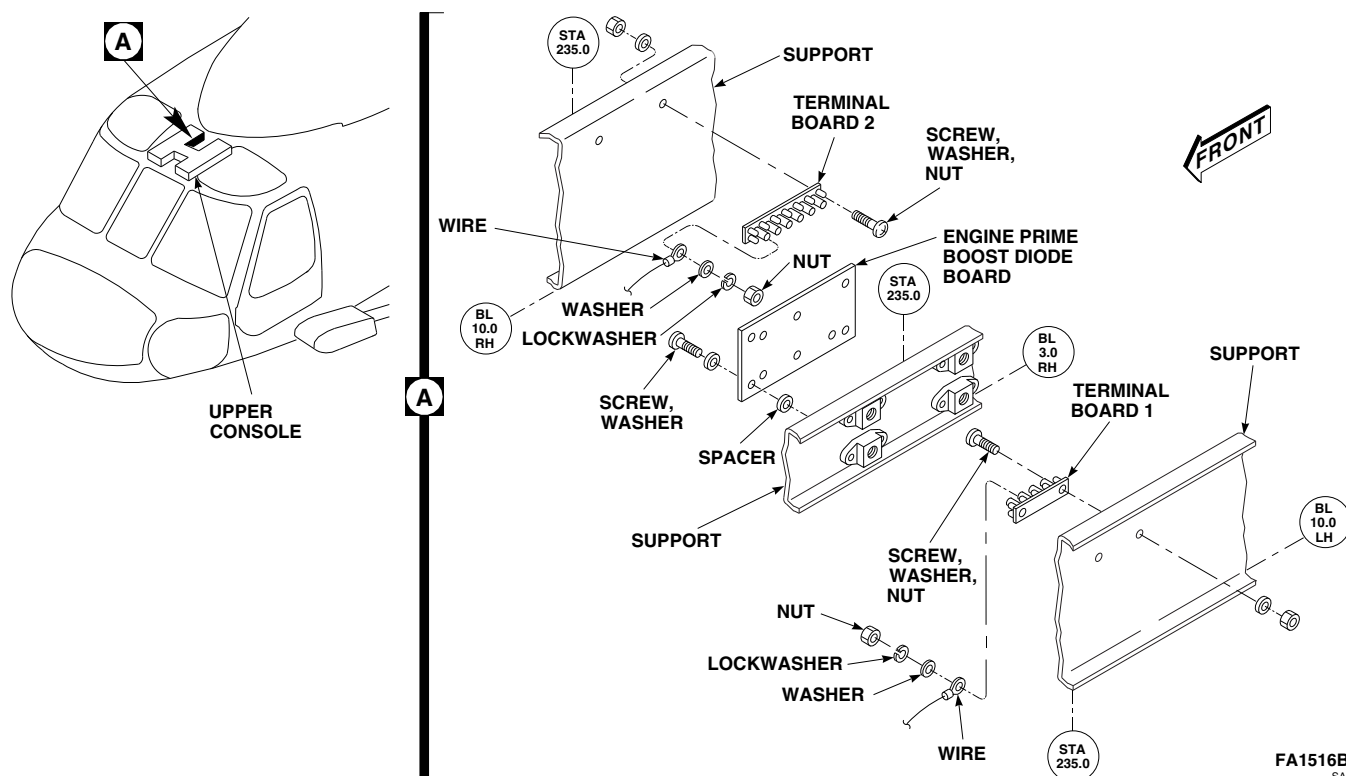


FIGURE 9-4-43-1. REPLACE ENGINE PRIME BOOST DIODE BOARD

WIRE NO.	TERMINAL BOARD	TERMINAL
Q3211C20	2	7
Q3213A20	2	8
Q3212A20	2	9
Q3204C20	1	8
Q2314C20	1	7

e. Remove tags.

- f. Make sure area is clean and free of foreign material.
- g. Carefully raise upper console and secure with fasteners.
- h. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- i. Perform operational check of fuel quantity system (PARA 10-2-5 or PARA 10-2-6).

9-4-44 **BACKUP HYDRAULIC PUMP MOTOR.**

PARAGRAPH BREAKDOWN

	PAGE
9-4-44.1 Replace Backup Hydraulic Pump Motor	9-4-44-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs
- Wood Blocks (2-inches × 4-inches × 6-inches)

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Lockwire, Item 193, Appendix D

- Protective Caps and Plugs
- Tags

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 1-7-20
- PARA 7-2-1
- PARA 9-4-34
- PARA 9-4-106

9-4-44.1 **Replace Backup Hydraulic Pump Motor.**

J423, from backup hydraulic pump motor (Figure 9-4-44-1, Detail A).

9-4-44.1.1. **Remove.**

- | | |
|---|---|
| <ul style="list-style-type: none"> a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15). b. Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)). c. Open main rotor pylon sliding cover. d. Tag electrical connectors. Disconnect electrical connectors, P422 from J422, and P423 from | <ul style="list-style-type: none"> e. Install protective caps and plugs on electrical connectors. f. Loosen nuts securing backup hydraulic pump motor to backup hydraulic pump module. g. Place wood blocks, as required, under backup hydraulic pump module to support backup hydraulic pump module when backup hydraulic pump motor is removed. h. Turn backup hydraulic pump module counterclockwise to remove from backup |
|---|---|

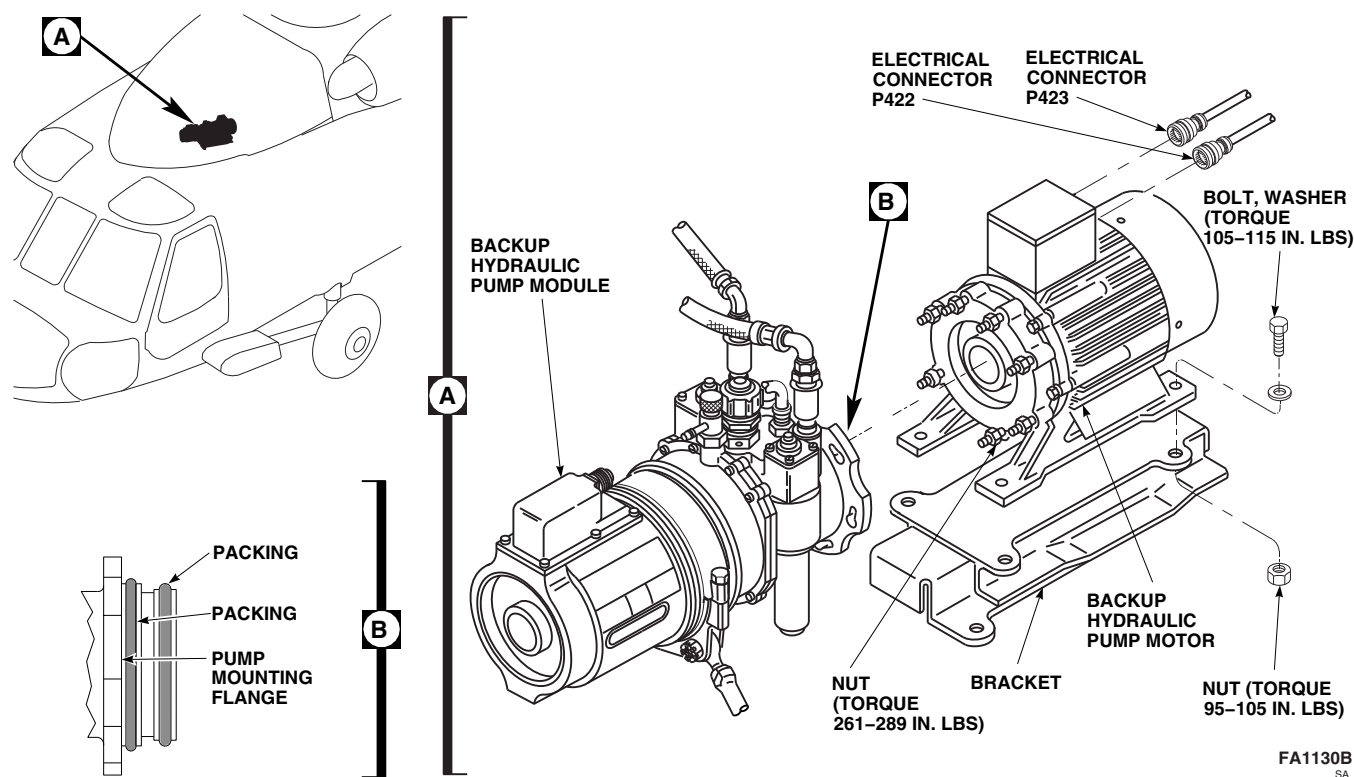


FIGURE 9-4-44-1. REPLACE BACKUP HYDRAULIC PUMP MOTOR

hydraulic pump motor and support backup hydraulic pump module on wood blocks.

NOTE

Backup hydraulic pump motor may be secured to bracket by either a bolt/nutplate or bolt/nut combination.

- i. Remove bolts, washers, and, if present, nuts securing backup hydraulic pump motor to bracket.
- j. Remove packings from backup hydraulic pump module (Figure 9-4-44-1, Detail B). Discard packings.

9-4-44.1.2. Install.

- a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

NOTE

Backup hydraulic pump motor may be secured to bracket by either a bolt/nutplate or bolt/nut combination.

- b. Install backup hydraulic pump motor on bracket and secure with bolts, washers, and, if required, nuts (Figure 9-4-44-1, Detail A).
- c. **TORQUE NUTS TO 95 - 105 INCH-POUNDS, OR TORQUE BOLTS TO 105 - 115 INCH-POUNDS.**
- d. Lubricate new packings with hydraulic fluid, Item 154 or Item 155, Appendix D, and install on backup hydraulic pump module (Figure 9-4-44-1, Detail B).
- e. Install backup hydraulic pump module on backup hydraulic pump motor. Turn backup hydraulic pump module clockwise to engage mount studs on backup hydraulic pump motor in narrow slots of pump mounting flange. Secure with washers and nuts (Figure 9-4-44-1, Detail A).
- f. **TORQUE NUTS ON STUDS TO 261 - 289 INCH-POUNDS.**
- g. Remove protective caps and plugs from electrical connectors.

9-4-44-2

- h. Inspect and treat electrical connectors (PARA 1-7-20).
- i. Connect electrical connectors, P422 to J422, and P423 to J423, to backup hydraulic pump motor. Remove tags.
- j. Lockwire, Item 193, Appendix D, electrical connector, P423.
- k. Remove wood blocks from under backup hydraulic pump module.
- l. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- m. Perform operational check of hydraulic systems (PARA 7-2-1).
- n. Make sure area is clean and free of foreign material.
- o. Close main rotor pylon sliding cover.

9-4-45 **BACKUP HYDRAULIC PUMP MOTOR HOOD.**

PARAGRAPH BREAKDOWN

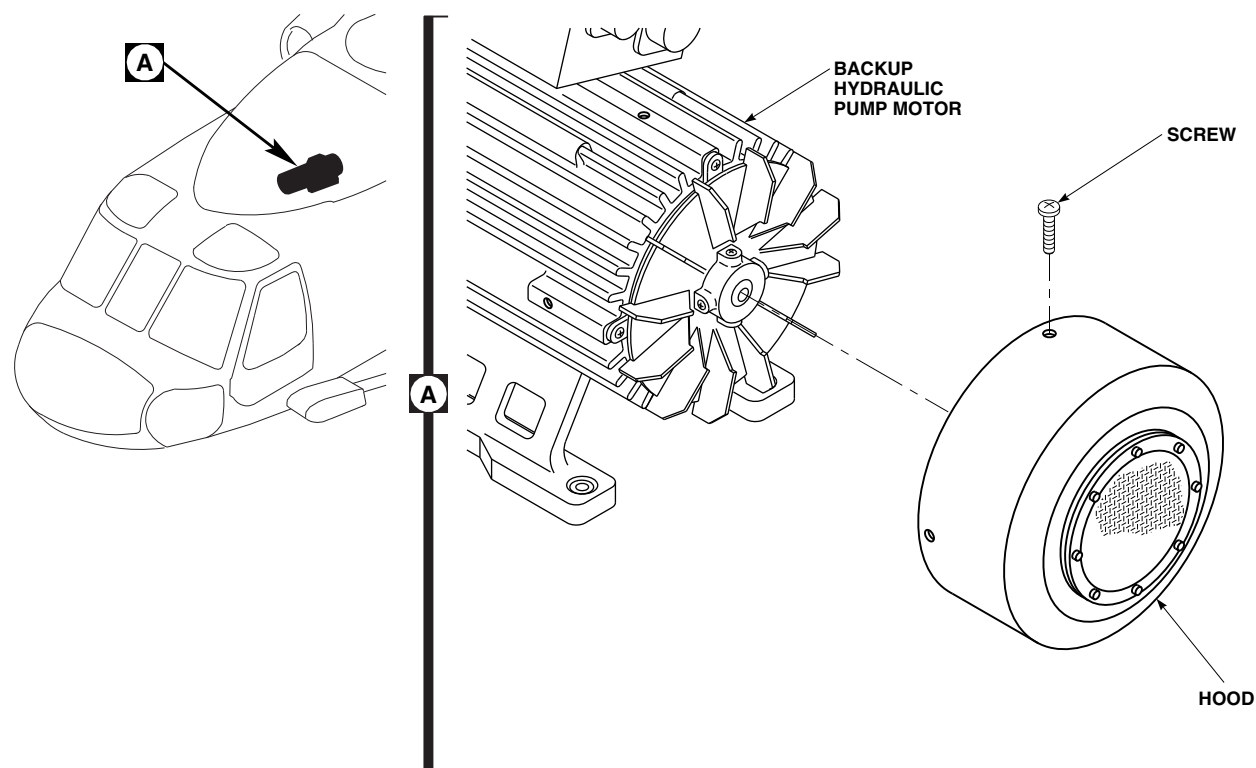
		PAGE
9-4-45.1	Replace Backup Hydraulic Pump Motor Hood	9-4-45-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit

9-4-45.1 Replace Backup Hydraulic Pump Motor Hood.	9-4-45.1.2. Install.
9-4-45.1.1. Remove.	
a. Open main rotor pylon sliding cover.	a. Install hood on hydraulic pump motor and secure with screws (Figure 9-4-45-1, Detail A).
b. Remove screws securing hood to hydraulic pump motor and remove hood (Figure 9-4-45-1, Detail A).	b. Close main rotor pylon sliding cover.

FT2602
SA**FIGURE 9-4-45-1. REPLACE BACKUP HYDRAULIC PUMP MOTOR HOOD**

9-4-46 **BACKUP HYDRAULIC PUMP MOTOR FAN.**

PARAGRAPH BREAKDOWN

	PAGE
9-4-46.1 Replace Backup Hydraulic Pump Motor Fan	9-4-46-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Abrasive Cloth, Item 10, Appendix D

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 9-4-34
- PARA 9-4-44
- PARA 9-4-45
- PARA 9-4-106

9-4-46.1 **Replace Backup Hydraulic Pump Motor Fan.**

9-4-46.1.1. **Remove.**

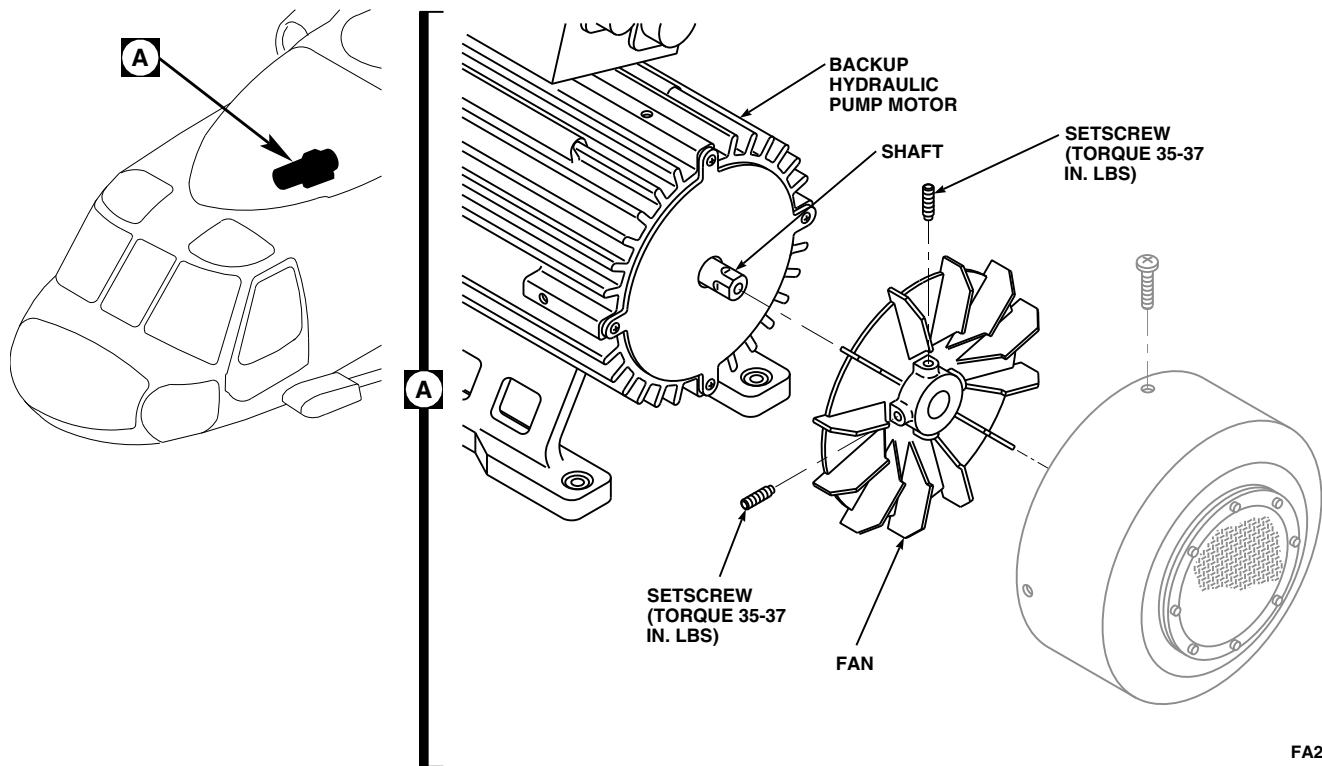
- Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Remove backup hydraulic pump motor hood (PARA 9-4-45).
- Loosen setscrews securing fan to shaft (Figure 9-4-46-1, Detail A).
- Remove fan from shaft.

9-4-46.1.2. **Inspect/Repair.**

Inspect shaft for scratches. **SCRATCHES SHALL NOT EXCEED 0.030-INCH DEEP.** If scratches are less than 0.030-inch, using abrasive cloth, Item 10, Appendix D, remove scratches. If scratches are deeper than 0.030-inch, replace backup hydraulic pump motor (PARA 9-4-44).

9-4-46.1.3. **Install.**

- Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).
- Install fan onto shaft. Line up flat in shaft with setscrew hole in fan (Figure 9-4-46-1, Detail A).
- Install setscrews. **TIGHTEN SETSCREWS, MAKING SURE AT LEAST ONE ENTERS**

FA2062
SA**FIGURE 9-4-46-1. REPLACE BACKUP HYDRAULIC PUMP MOTOR FAN**

HOLE IN SHAFT. TORQUE SETSCREWS TO 35 - 37 INCH-POUNDS.

- d. Install backup hydraulic pump motor hood (PARA 9-4-45).
- e. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).

9-4-47 **BACKUP HYDRAULIC PUMP RELAY, K19.**

PARAGRAPH BREAKDOWN

	PAGE
9-4-47.1 Replace Backup Hydraulic Pump Relay, K19	9-4-47-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

Materials/Parts

- Tags

References

- PARA 1-6-15
- PARA 1-6-16

- PARA 2-4-69
- PARA 7-2-1
- PARA 9-4-34
- PARA 9-4-106

9-4-47.1 **Replace Backup Hydraulic Pump Relay, K19.**

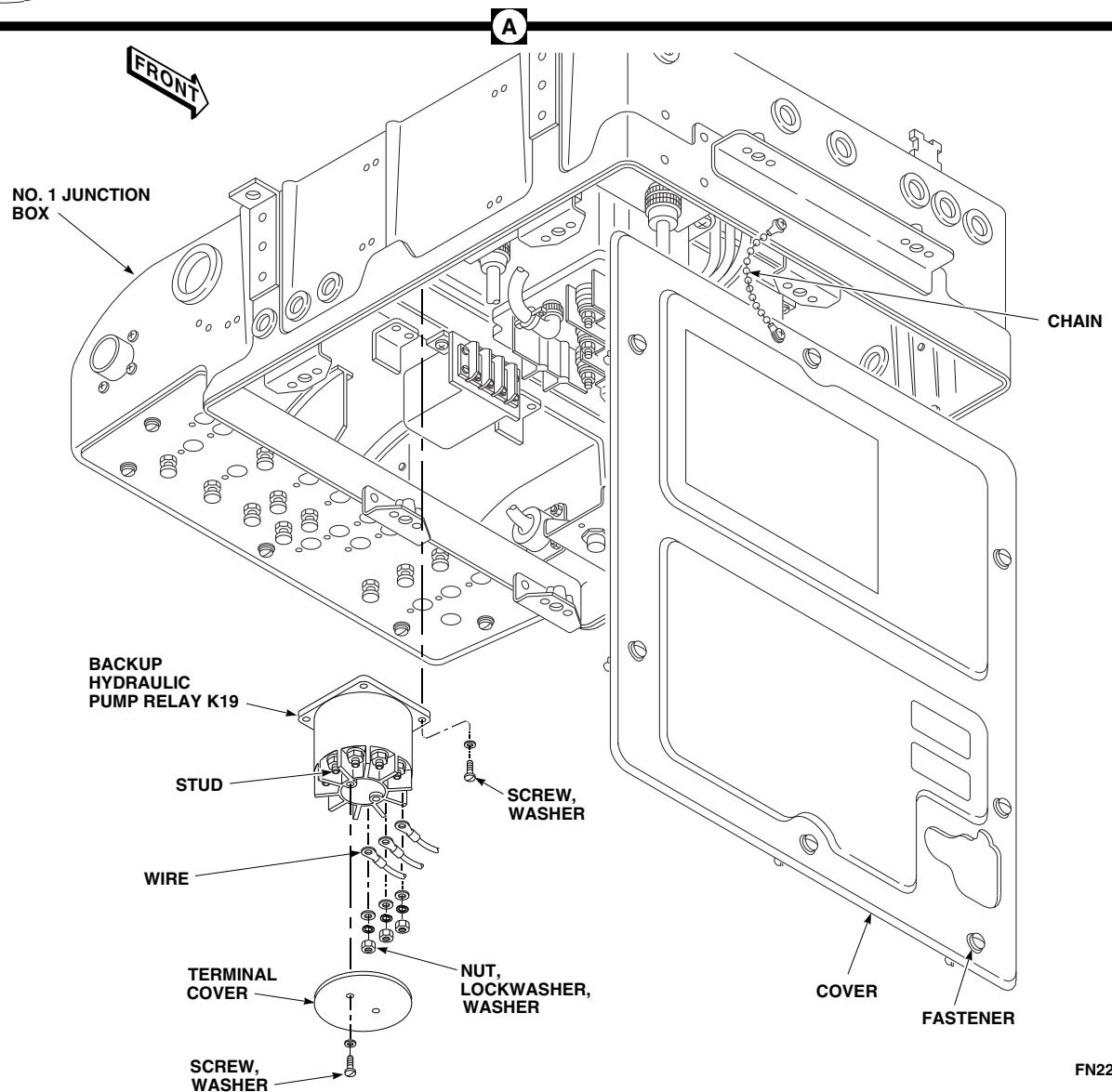
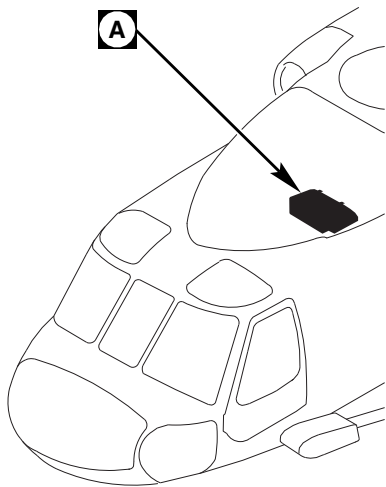
9-4-47.1.1. **Remove.**

- Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Remove soundproofing panel (PARA 2-4-69).
- Loosen fasteners securing cover on No. 1 junction box and let cover hang by chain (Figure 9-4-47-1, Detail A).
- Remove screws and washers securing terminal cover to backup hydraulic pump relay, K19.

- Tag wires. Remove nuts, lockwashers, and washers securing wires to studs on backup hydraulic pump relay, K19.
- Remove screws and washers securing backup hydraulic pump relay, K19 to No. 1 junction box.
- Remove backup hydraulic pump relay, K19.

9-4-47.1.2. **Install.**

- Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).
- Install backup hydraulic pump relay, K19, in No. 1 junction box and secure with screws and washers (Figure 9-4-47-1, Detail A).



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FIGURE 9-4-47-1. REPLACE BACKUP HYDRAULIC PUMP RELAY, K19

9-4-47-2

- c. Install wires to studs on backup hydraulic pump relay, K19, and secure with nuts, lock-washers, and washers. Remove tags.
- d. Install terminal cover on backup hydraulic pump relay, K19, and secure with screws and washers.
- e. Install cover on No. 1 junction box and secure with fasteners.
- f. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- g. Perform operational check of hydraulic systems (PARA 7-2-1).
- h. Make sure area is clean and free of foreign material.
- i. Install soundproofing panel (PARA 2-4-69).

9-4-48 REMOTE CIRCUIT BREAKER.

PARAGRAPH BREAKDOWN

	PAGE
9-4-48.1 Replace Remote Circuit Breaker	9-4-48-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit
- Insert/Extract Tool

Materials/Parts

- Tags

References

- PARA 1-6-16
- PARA 2-4-69
- PARA 7-2-1
- PARA 9-4-34
- PARA 9-4-106

9-4-48.1 Replace Remote Circuit Breaker.

9-4-48.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Remove soundproofing panel (PARA 2-4-69).
- Loosen fasteners securing cover on No. 1 junction box and let cover hang by chain (Figure 9-4-48-1, Detail A).

NOTE

Remote circuit breaker is labeled CB404 HYD PWR PUMP CKT BKR.

- Remove screws securing terminal shields to No. 1 ac generator contactor, K1, and remote circuit breaker.

- Tag wires. Remove nuts, lockwashers, and washers securing wires to remote circuit breaker.
- Tag wires. Remove nuts, lockwashers, and washers securing bus bars and wires to remote circuit breaker.
- Remove nuts, lockwashers, and washers securing bus bars to No. 1 ac generator contactor, K1.
- Remove screws and washers securing bus bars to mounting block of current limiters, CL4, CL5, and CL6.
- Using insert/extract tool, remove wires, 3413A20N and 3414C20, from terminal junction block on remote circuit breaker (Figure 9-4-48-1, Detail B).
- Remove screws and washers securing remote circuit breaker to No. 1 junction box.
- Remove remote circuit breaker.

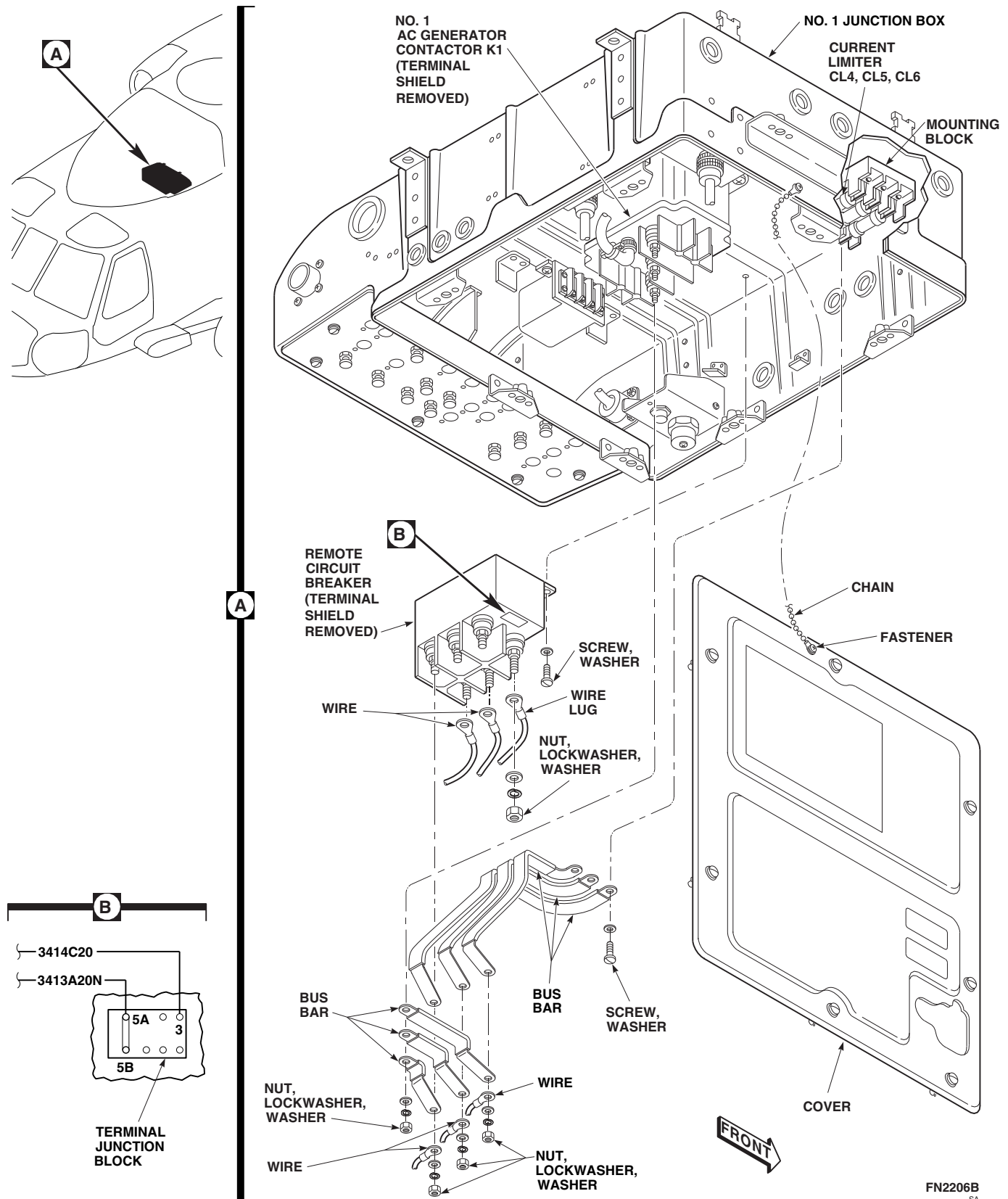


FIGURE 9-4-48-1. REPLACE REMOTE CIRCUIT BREAKER

9-4-48-2

9-4-48.1.2. Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Install remote circuit breaker in No. 1 junction box and secure with screws and washers (Figure 9-4-48-1, Detail A).
- c. Using insert/extract tool, insert wire number, 3413A20N, onto terminal, 5A, and wire number, 3414C20, onto terminal, 3 of terminal junction block (Figure 9-4-48-1, Detail B).
- d. Install bus bars on remote circuit breaker and mounting block of current limiters, CL4, CL5, and CL6. Secure bus bars to mounting block with screws and washers (Figure 9-4-48-1, Detail A).
- e. Install bus bars on remote circuit breaker and No. 1 ac generator contactor, K1. Secure bus bars to No. 1 ac generator contactor, K1, with nuts, lockwashers, and washers.
- f. Install wires over bus bars on remote circuit breaker and secure with nuts, lockwashers, and washers. Remove tags.
- g. Install wires on remote circuit breaker and secure with nuts, lockwashers, and washers. Remove tags.
- h. Install terminal shields on No. 1 ac generator contactor, K1, and remote circuit breaker and secure with screws.
- i. Install cover on No. 1 junction box and secure with fasteners.
- j. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- k. Perform operational check of hydraulic systems (PARA 7-2-1).
- l. Make sure area is clean and free of foreign material.
- m. Install soundproofing panel (PARA 2-4-69).

9-4-49 UPPER CONSOLE DIMMING CONTROLS.

PARAGRAPH BREAKDOWN

	PAGE
9-4-49.1 Replace Upper Console Dimming Controls	9-4-49-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit
- Torque Wrench, 0 - 30 in. lbs

Materials/Parts

- Tags

References

- PARA 1-6-16
- PARA 9-2-10

- PARA 9-2-11
- PARA 9-4-6
- PARA 9-4-34
- PARA 9-4-50
- PARA 9-4-51
- PARA 9-4-106

9-4-49.1 Replace Upper Console Dimming Controls.

NOTE

Replacement of rheostat-type dimming controls located on the upper console is the same, except lighted switches and cabin dome dimming control. To replace the lighted switches dimming control, refer to PARA 9-4-50. To replace the cabin dome dimming control, refer to PARA 9-4-51.

9-4-49.1.1 Remove.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- c. Remove information plate (PARA 9-4-6).
- d. Loosen fasteners securing upper console and carefully lower (Figure 9-4-49-1, Detail A).
- e. Tag wires. Remove nuts, lockwashers, and washers securing wires to dimming control.
- f. Remove nut and lockwasher securing dimming control to upper console. Remove dimming control.
- g. Remove washer from dimming control.

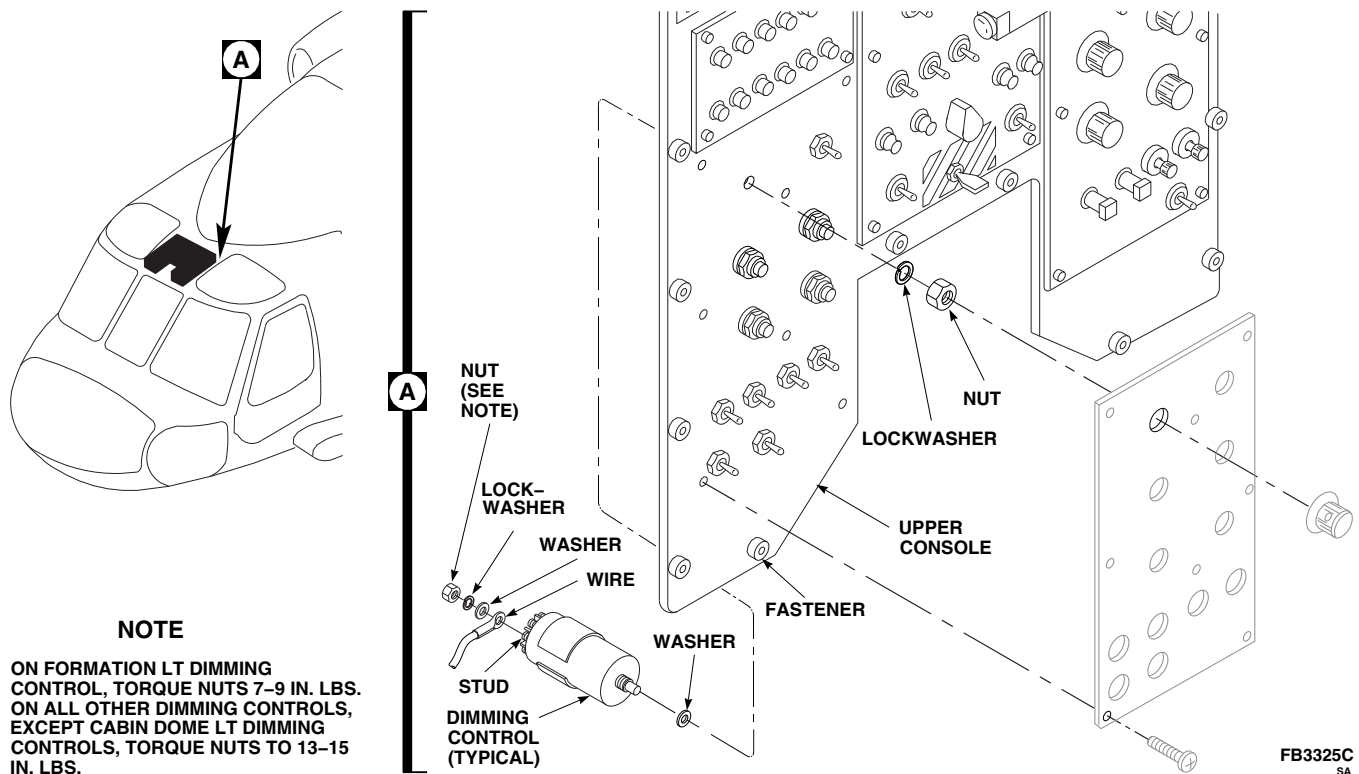


FIGURE 9-4-49-1. REPLACE UPPER CONSOLE DIMMING CONTROLS

9-4-49.1.2 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Install washer on dimming control (Figure 9-4-49-1, Detail A).
- c. Install dimming control from rear of upper console, with locking key in hole.
- d. Install nut and lockwasher on dimming control and secure. Make sure threaded portion of dimming control is at least flush with bottom of nut.
- e. Install wires on dimming control and secure with nuts, lockwashers, and washers. Remove tags.
- f. **ON FORMATION LT DIMMING CONTROL, TORQUE NUTS TO 7-9 INCH-**

POUNDS. ON ALL OTHER DIMMING CONTROLS, EXCEPT LIGHTED SWITCHES AND CABIN DOME LT DIMMING CONTROLS, TORQUE NUTS TO 13 - 15 INCH-POUNDS.

- g. Make sure area is clean and free of foreign material.
- h. Carefully raise upper console and secure with fasteners.
- i. Install information plate (PARA 9-4-6).
- j. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- k. Perform operational check of instrument panel and consoles indicator lights dimming (PARA 9-2-10 or PARA 9-2-11).

9-4-50 **LIGHTED SWITCHES DIMMING CONTROL.**

PARAGRAPH BREAKDOWN

	PAGE
9-4-50.1 Replace Lighted Switches Dimming Control	9-4-50-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

Materials/Parts

- Protective Caps and Plugs

References

- PARA 1-6-16
- PARA 1-7-20

- PARA 9-2-10

- PARA 9-2-11

- PARA 9-4-6

- PARA 9-4-34

- PARA 9-4-106

9-4-50.1 **Replace Lighted Switches Dimming Control.**

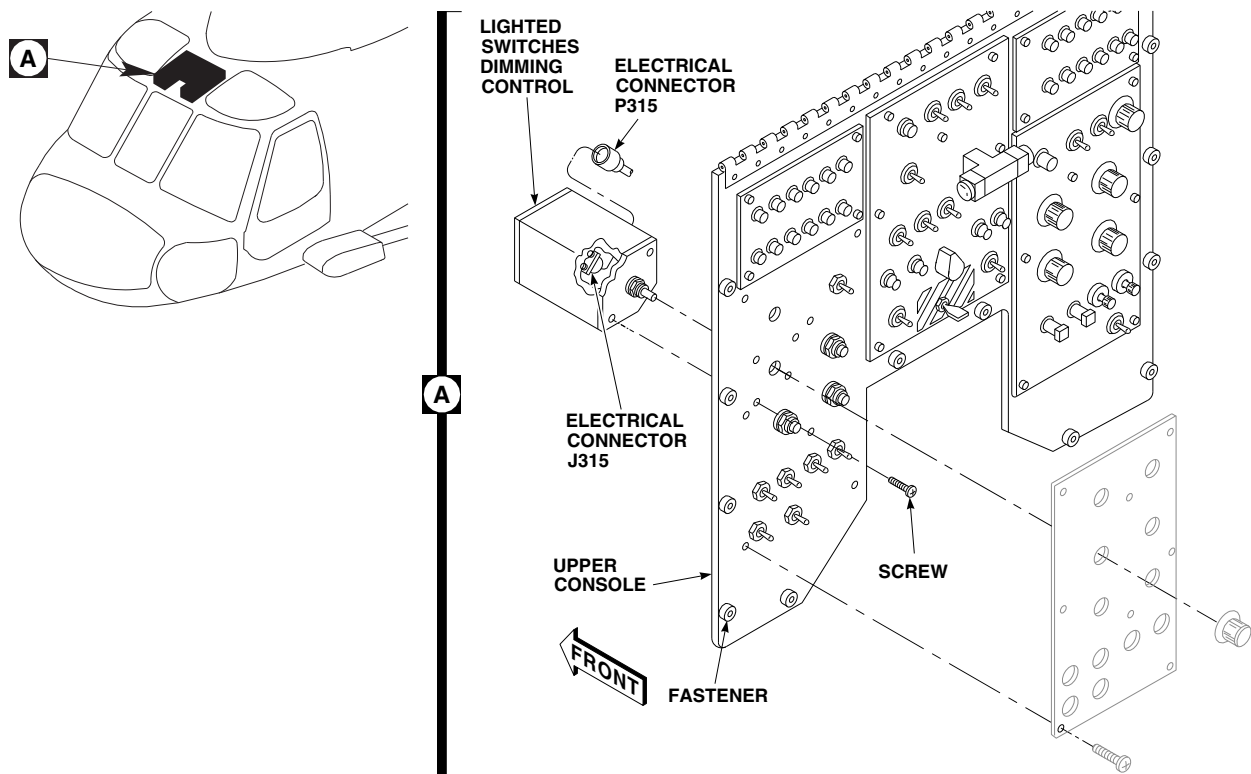
9-4-50.1.1 **Remove.**

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Remove left information plate from upper console (PARA 9-4-6).
- Loosen fasteners securing upper console and carefully lower console (Figure 9-4-50-1, Detail A).
- Disconnect electrical connector, P315 from J315, on lighted switches dimming control.
- Install protective caps and plugs on electrical connectors.

- Remove screws securing lighted switches dimming control to upper console. Remove lighted switches dimming control.

9-4-50.1.2 **Install.**

- Turn off all electrical power (PARA 1-6-16).
- Install lighted switches dimming control on rear of upper console and secure with screws (Figure 9-4-50-1, Detail A).
- Remove protective caps and plugs from electrical connectors.
- Inspect and treat electrical connectors (PARA 1-7-20).
- Connect electrical connector, P315 to J315, to lighted switches dimming control.
- Make sure area is clean and free of foreign material.

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SA**FIGURE 9-4-50-1. REPLACE LIGHTED SWITCHES DIMMING CONTROL**

- g. Carefully raise upper console and secure with fasteners.
- h. Install left information plate on upper console (PARA 9-4-6).
- i. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- j. Perform operational check of instrument panel and consoles indicator lights dimming (PARA 9-2-10 or PARA 9-2-11).

9-4-51 CABIN DOME DIMMING CONTROL.

PARAGRAPH BREAKDOWN

	PAGE
9-4-51.1 Replace Cabin Dome Dimming Control	9-4-51-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit
- Torque Wrench, 0 - 30 in. lbs

Materials/Parts

- Protective Caps and Plugs
- Tags

References

- PARA 1-6-16
- PARA 1-7-20
- PARA 9-2-10
- PARA 9-4-34
- PARA 9-4-106

9-4-51.1 Replace Cabin Dome Dimming Control.

9-4-51.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Disconnect electrical connector, P455 from J455, on cabin dome dimming control (Figure 9-4-51-1, Detail A).
- Install protective caps and plugs on electrical connectors.
- Remove screws and washers securing cover to enclosure assembly.
- Remove screws and washers securing enclosure assembly to pilot’s seat.

9-4-51.1.2. Disassemble.

- Loosen setscrew and remove knob from cabin dome dimming control (Figure 9-4-51-1, Detail A).
- Remove screws securing information plate to enclosure assembly.
- Remove nut and lockwasher securing cabin dome dimming control to enclosure assembly.
- Tag wires. Remove nuts, lockwashers, and washers securing wires to cabin dome dimming control.

9-4-51.1.3. Assemble.

- Install wires on terminals of cabin dome dimming control and secure with nuts, lockwashers, and washers (Figure 9-4-51-1, Detail A). Remove tags.

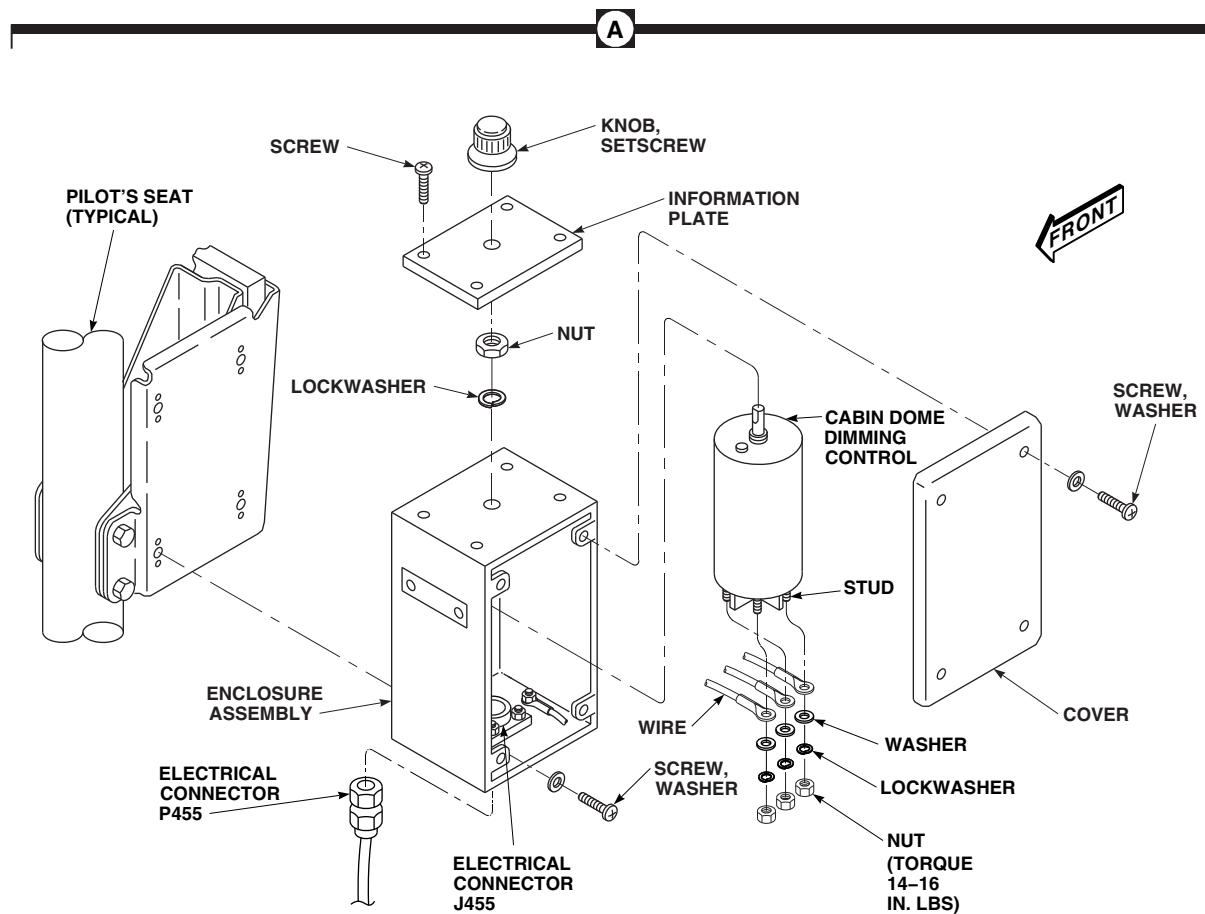
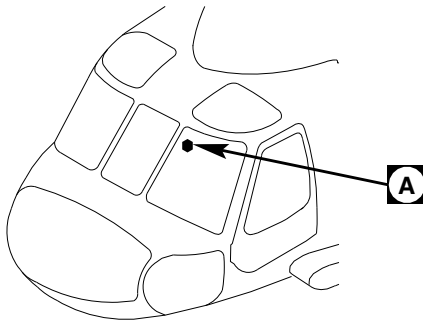
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FIGURE 9-4-51-1. REPLACE CABIN DOME DIMMING CONTROL

9-4-51-2

- b. **TORQUE NUTS ON STUDS TO 14 - 16 INCH-POUNDS.**
- c. Install cabin dome dimming control in enclosure assembly and secure with nut and lockwasher.
- d. Install information plate on enclosure assembly and secure with screws.
- e. Turn shaft of cabin dome dimming control fully counterclockwise.
- f. Install knob on shaft, lining up knob index mark with OFF marking on information plate.
- g. **TIGHTEN SETSCREW.**

9-4-51.1.4. Install.

- a. Turn off all electrical power (PARA 1-6-16).

- b. Install enclosure assembly on pilot's seat and secure with screws and washers (Figure 9-4-51-1, Detail A).
- c. Install cover on enclosure assembly and secure with screws and washers.
- d. Remove protective caps and plugs from electrical connectors.
- e. Inspect and treat electrical connectors (PARA 1-7-20).
- f. Connect electrical connector, P455 to J455, on cabin dome dimming control.
- g. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- h. Perform operational check of dimming control (PARA 9-2-10).

9-4-52 INDICATOR LIGHTS DIMMER BOX.

PARAGRAPH BREAKDOWN

	PAGE
9-4-52.1 Replace Indicator Lights Dimmer Box	9-4-52-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

Materials/Parts

- Alcohol, Denatured, Item 70, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 9-2-10
- PARA 9-4-34
- PARA 9-4-106

9-4-52.1 Replace Indicator Lights Dimmer Box.

9-4-52.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Open nose door.
- Disconnect electrical connector, P393 from J393, on indicator lights dimmer box (Figure 9-4-52-1, Detail A or Detail B).
- Install protective caps and plugs on electrical connectors.

NOTE

Spacers are installed in alternate locations.

- Remove screws, washers, lockwasher, and spacers securing indicator lights dimmer box to support on lower console.
- Remove indicator lights dimmer box.

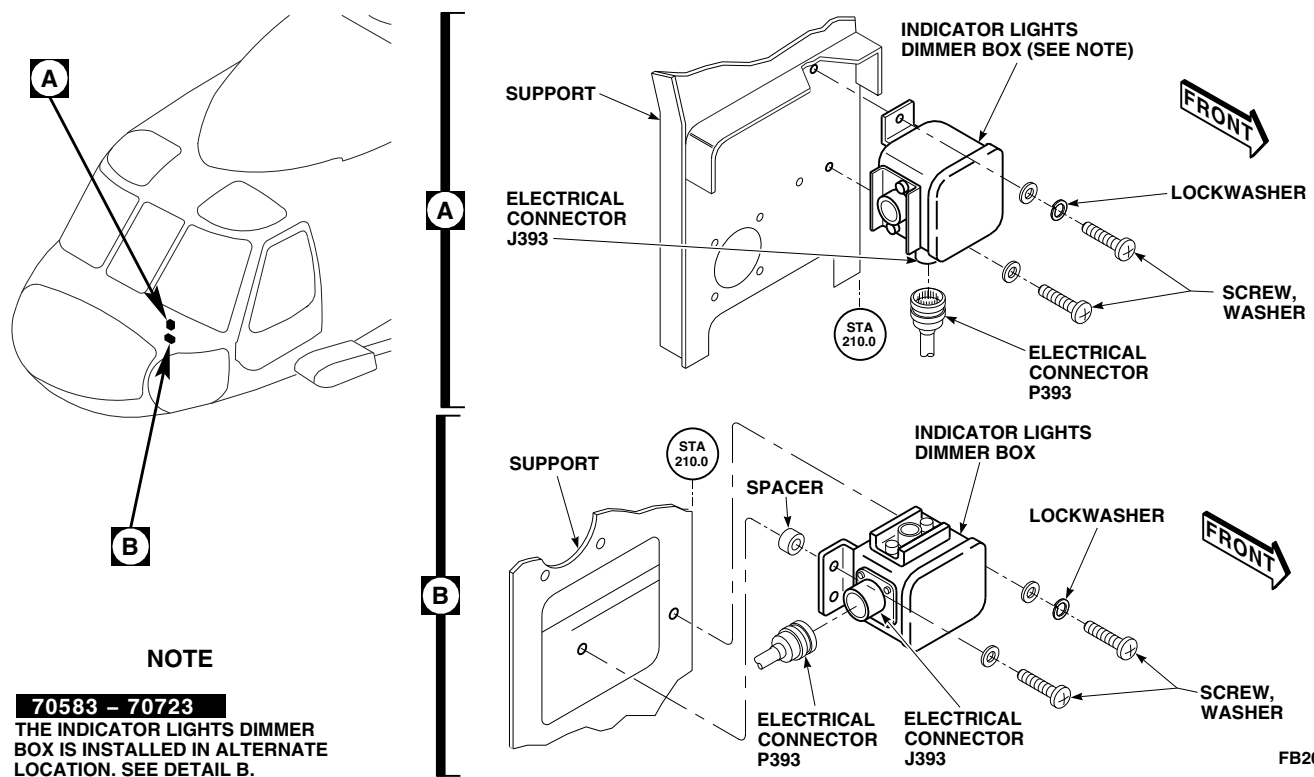
9-4-52.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).
- Using denatured alcohol, Item 70, Appendix D, clean mounting surface.

NOTE

Spacers are installed in alternate location.

- Install indicator lights dimmer box to support on lower console and secure with screws, washers, lockwasher, and spacers (Figure 9-4-52-1, Detail A or Detail B).

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- d. Remove protective caps and plugs from electrical connectors.
- e. Inspect and treat electrical connectors (PARA 1-7-20).
- f. Connect electrical connector, P393 to J393, on indicator lights dimmer box.
- g. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- h. Perform operational check of instrument panel and consoles indicator lights dimming system (PARA 9-2-10).
- i. Make sure area is clean and free of foreign material.
- j. Close nose door.

9-4-53 INDICATOR LIGHTS DIMMER BOX FUSE.

PARAGRAPH BREAKDOWN

	PAGE
9-4-53.1 Replace Indicator Lights Dimmer Box Fuse	9-4-53-1

INITIAL SETUP

- PARA 9-2-11

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

- PARA 9-4-34

- PARA 9-4-106

References

- PARA 1-6-16
- PARA 9-2-10

9-4-53.1 Replace Indicator Lights Dimmer Box Fuse.

9-4-53.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Open nose door.

NOTE

If indicator light dimmer box is installed in alternate location, do not perform step d.

- Loosen screws on indicator lights dimmer box and carefully move indicator lights dimmer box away from support to gain access to fuse holder (Figure 9-4-53-1, Detail A).
- Remove fuse holder and fuse (Figure 9-4-53-1, Detail A or Detail B). Remove fuse and discard.

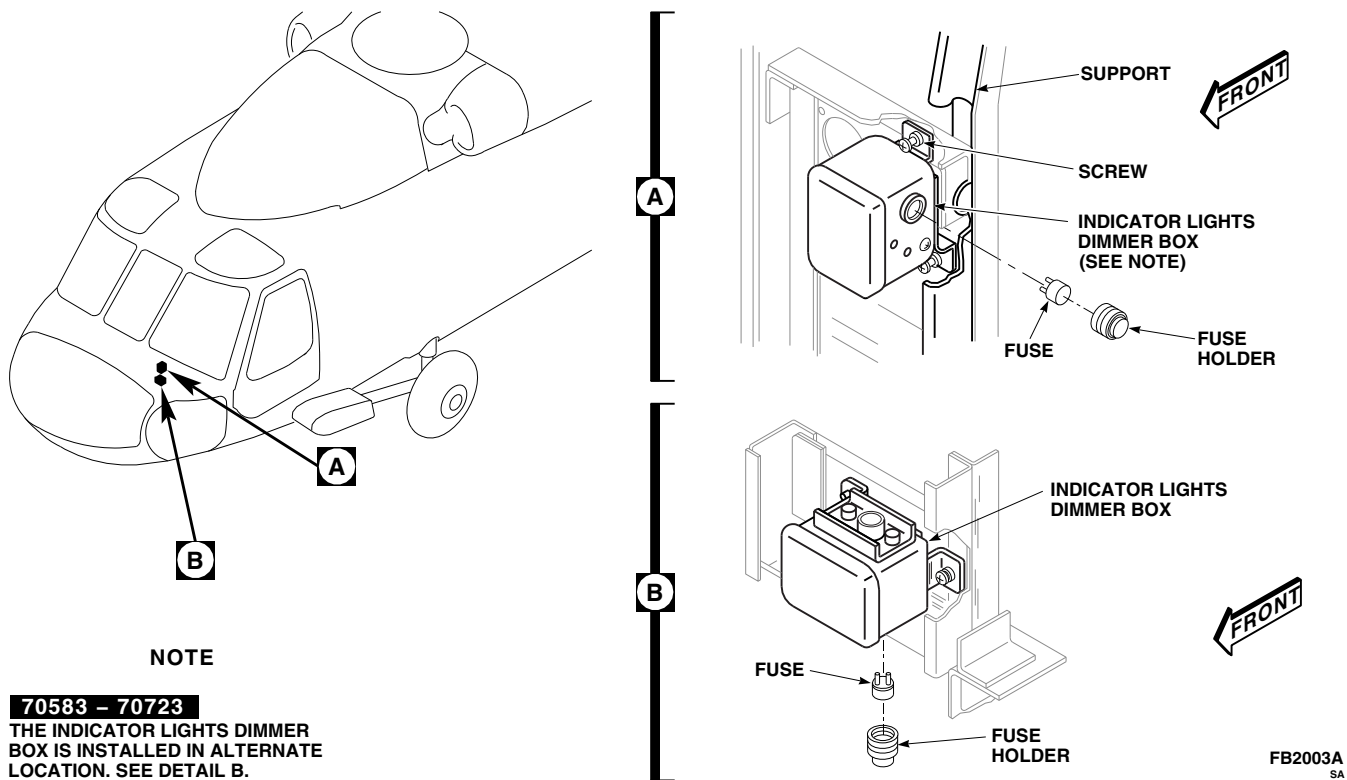
9-4-53.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).
- Install fuse into fuse holder (Figure 9-4-53-1, Detail A or Detail B).
- Install fuse holder into indicator lights dimmer box.

NOTE

If indicator lights dimmer box is installed in alternate location, do not perform step d.

- Secure indicator lights dimmer box to support with screws.
- Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Perform operational check of instrument panel and consoles indicator lights dimming (PARA 9-2-10 and PARA 9-2-11).



g. Close nose door.

9-4-54 CARGO HOOK EMERGENCY RELEASE SYSTEM RESISTOR, R4.

PARAGRAPH BREAKDOWN

	PAGE
9-4-54.1 Replace Cargo Hook Emergency Release System Resistor, R4	9-4-54-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit

Materials/Parts

- Tags
- Tiedown Strap, Item 329, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 9-4-34
- PARA 9-4-106
- PARA 12-2-6

9-4-54.1 Replace Cargo Hook Emergency Release System Resistor, R4.

9-4-54.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Loosen fasteners securing upper console and carefully lower.
- Tag wires. Remove nuts, lockwashers, and washers securing wires to terminal board 1 (Figure 9-4-54-1, Detail A).
- Remove tiedown strap securing cargo hook emergency release system resistor, R4, wires.
- Remove screws, spacers, washers, and nuts securing cargo hook emergency release system resistor, R4, to support.

- Remove cargo hook emergency release system resistor, R4.

9-4-54.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).



- To prevent damage to cargo hook emergency release system resistor, R4, do not overtighten screws.
- Install cargo hook emergency release system resistor, R4, on support and secure with screws, spacers, washers, and nuts (Figure 9-4-54-1, Detail A).
 - Install tiedown strap, Item 329, Appendix D, to secure wires.

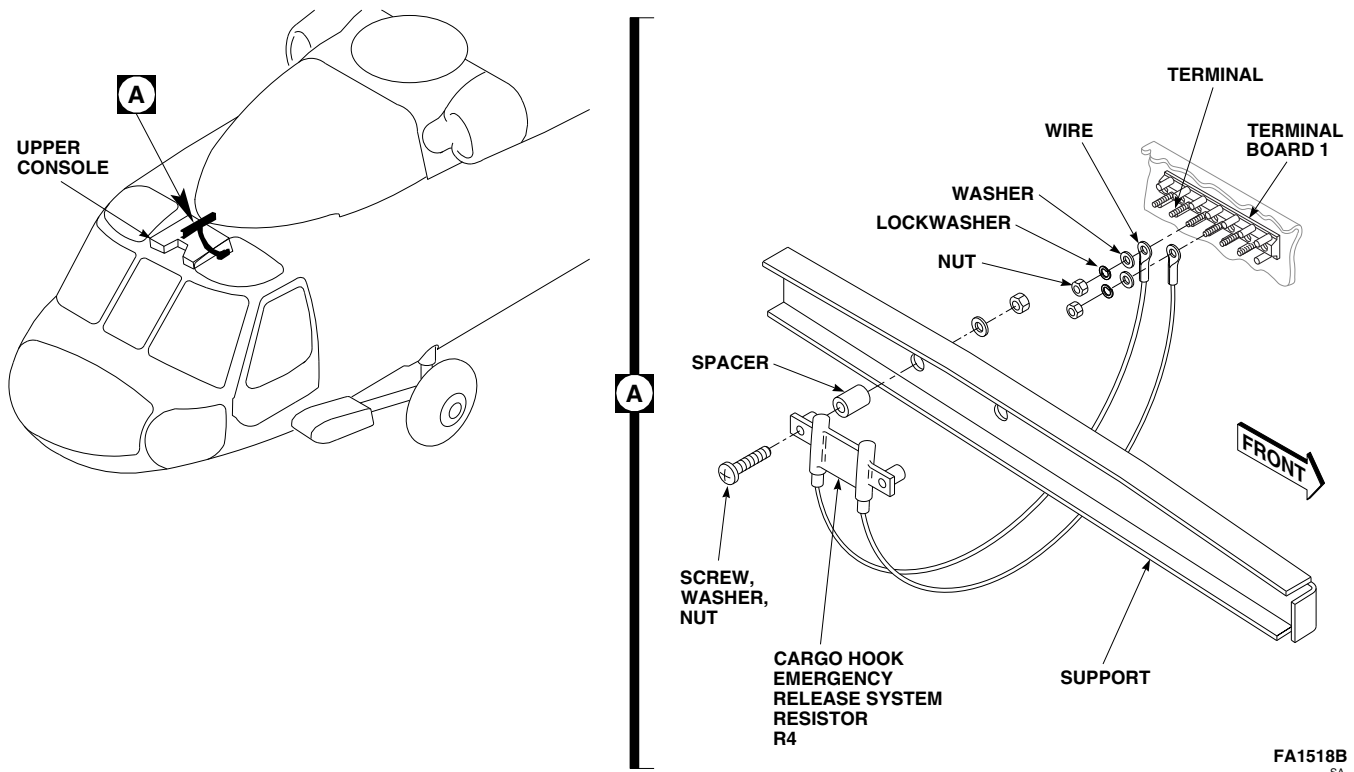


FIGURE 9-4-54-1. REPLACE CARGO HOOK EMERGENCY RELEASE SYSTEM RESISTOR, R4

- d. Install wires to terminal board 1, on terminals 4 and 5, and secure with nuts, lockwashers, and washers. Remove tags.
- e. Make sure area is clean and free of foreign material.
- f. Carefully raise upper console and secure with fasteners.
- g. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- h. Perform operational check of cargo hook system (PARA 12-2-6).

9-4-55 TRANSFORMER BOXES.

PARAGRAPH BREAKDOWN

		PAGE
9-4-55.1	Replace Transformer Boxes	9-4-55-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit

Materials/Parts

- Dry-Cleaning Solvent, Item 124, Appendix D
- Protective Caps and Plugs
- Tags

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 9-2-4
- PARA 9-4-34
- PARA 9-4-106

9-4-55.1 Replace Transformer Boxes.

NOTE

Replacement of all transformer boxes is the same, except as noted.

9-4-55.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Open nose door.
- Tag electrical connector. Disconnect electrical connector from applicable transformer box(es) as follows (Figure 9-4-55-1, Detail A):
 - P386 from J386 (top right side).
 - P387 from J387 (bottom right side).

(3) P388 from J388 (left side).

- Install protective caps and plugs on electrical connectors.
- Remove screws, lockwashers, and washers securing transformer box to helicopter.
- Remove transformer box.

9-4-55.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).
- Using dry-cleaning solvent, Item 124, Appendix D, remove paint from mounting surfaces and clean.
- Install transformer box to helicopter with screws, lockwashers, and washers (Figure 9-4-55-1, Detail A).
- Remove protective caps and plugs from electrical connectors.

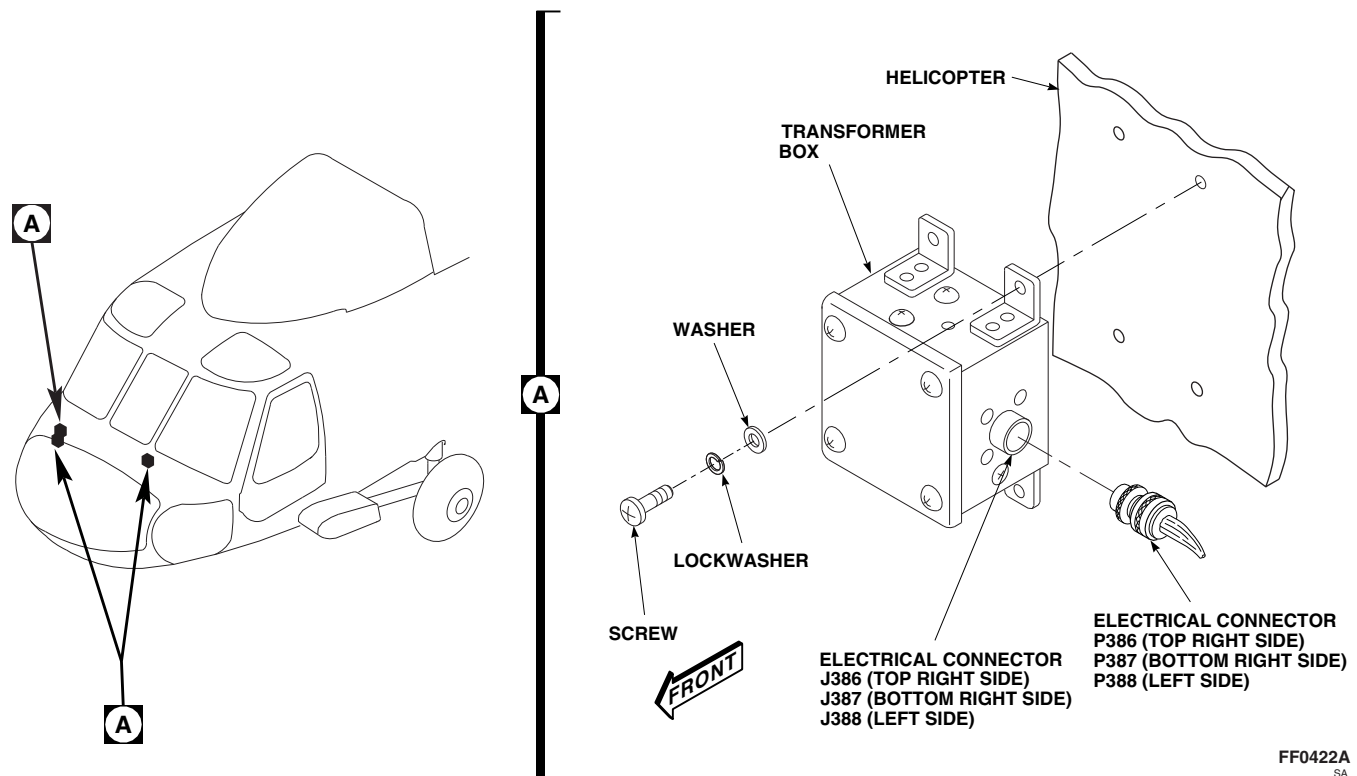


FIGURE 9-4-55-1. REPLACE TRANSFORMER BOXES

- e. Inspect and treat electrical connectors (PARA 1-7-20).
- f. Connect electrical connector to applicable transformer box(es) as follows, then remove tag:
 - (1) P386 to J386 (top right side).
 - (2) P387 to J387 (bottom right side).
 - (3) P388 to J388 (left side).
- g. Make sure area is clean and free of foreign material.
- h. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- i. Close nose door.
- j. Perform operational check of instrument panel lights (PARA 9-2-4).

9-4-56 CONTROLLABLE SEARCHLIGHT DIMMING CONTROL.

PARAGRAPH BREAKDOWN

		PAGE
9-4-56.1	Replace Controllable Searchlight Dimming Control	9-4-56-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit

Materials/Parts

- Cheesecloth, Item 90, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 9-2-13
- PARA 9-4-34
- PARA 9-4-106

9-4-56.1 Replace Controllable Searchlight Dimming Control.

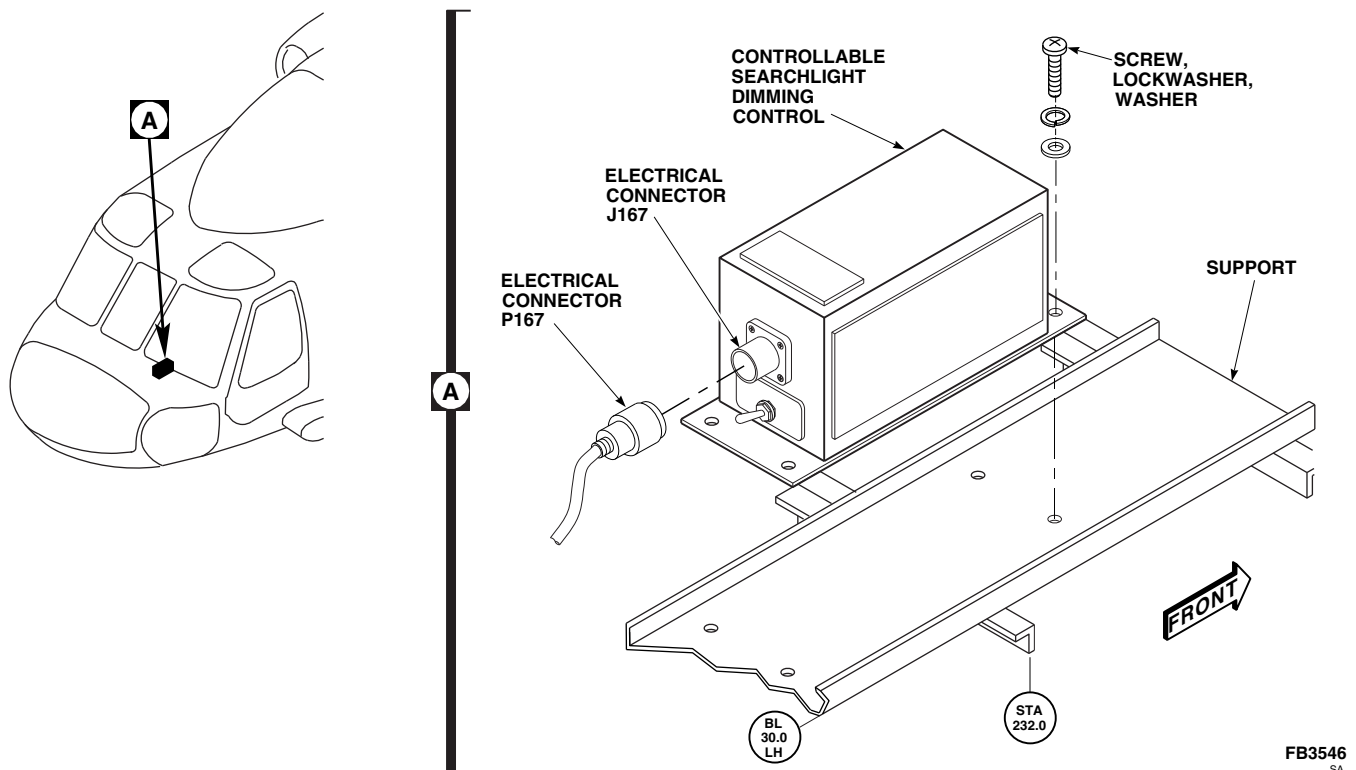
9-4-56.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Pull pilot’s seat EMER TILT REL HDL and push seat back.
- Remove cover over pilot’s seatwell by pulling back on fabric cover.
- Disconnect electrical connector, P167 from J167, on controllable searchlight dimming control (Figure 9-4-56-1, Detail A).
- Install protective caps and plugs on electrical connectors.

- Remove screws, lockwashers, and washers securing controllable searchlight dimming control to support.
- Remove controllable searchlight dimming control.

9-4-56.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Using cheesecloth, Item 90, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D, clean surface around all mounting holes. Using clean, dry cheesecloth, Item 90, Appendix D, dry surface.
- Install controllable searchlight dimming control to support and secure with screws, lockwashers, and washers (Figure 9-4-56-1, Detail A).

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SA**FIGURE 9-4-56-1. REPLACE CONTROLLABLE SEARCHLIGHT DIMMING CONTROL**

- d. Remove protective caps and plugs from electrical connectors.
- e. Inspect and treat electrical connectors (PARA 1-7-20).
- f. Connect electrical connector, P167 to J167, on controllable searchlight dimming control.
- g. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- h. Perform operational check of controllable searchlight (PARA 9-2-13).
- i. Make sure area is clean and free of foreign material.
- j. Install cover over pilot's seatwell.
- k. Return pilot's seat to normal position.

9-4-57 RADAR ALTIMETER DIMMING CONTROL.

PARAGRAPH BREAKDOWN

		PAGE
9-4-57.1	Replace Radar Altimeter Dimming Control	9-4-57-1

INITIAL SETUP

References

- TM 11-70-23

9-4-57.1 Replace Radar Altimeter Dimming Control.

Replace radar altimeter dimming control (TM 11-70-23).

9-4-58 DIMMING CONTROL POWER SUPPLY.

PARAGRAPH BREAKDOWN

	PAGE
9-4-58.1 Replace Dimming Control Power Supply	9-4-58-1

INITIAL SETUP

Tools

- Digital Low-Resistance Ohmmeter
- Electrical Repairer’s Toolkit

Materials/Parts

- Cheesecloth, Item 90, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-6-5
- PARA 9-2-25
- PARA 9-4-34
- PARA 9-4-106

9-4-58.1 Replace Dimming Control Power Supply.

9-4-58.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Open nose door.
- Disconnect electrical connector, P168 from J168, on dimming control power supply (Figure 9-4-58-1, Detail A).
- Install protective caps and plugs on electrical connectors.
- Have assistant in nose compartment hold dimming control power supply while screws, lock-

washers, and washers securing dimming control power supply to canted bulkhead are removed from cockpit.

- Remove dimming control power supply.

9-4-58.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).
- Inspect chemical conversion coating as follows:
 - Clean surface around all mounting surface with cheesecloth, Item 90, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D. Dry surface with clean, dry cheesecloth, Item 90, Appendix D.
 - Inspect chemical conversion coated surfaces for breaks and scratches. **NONE**

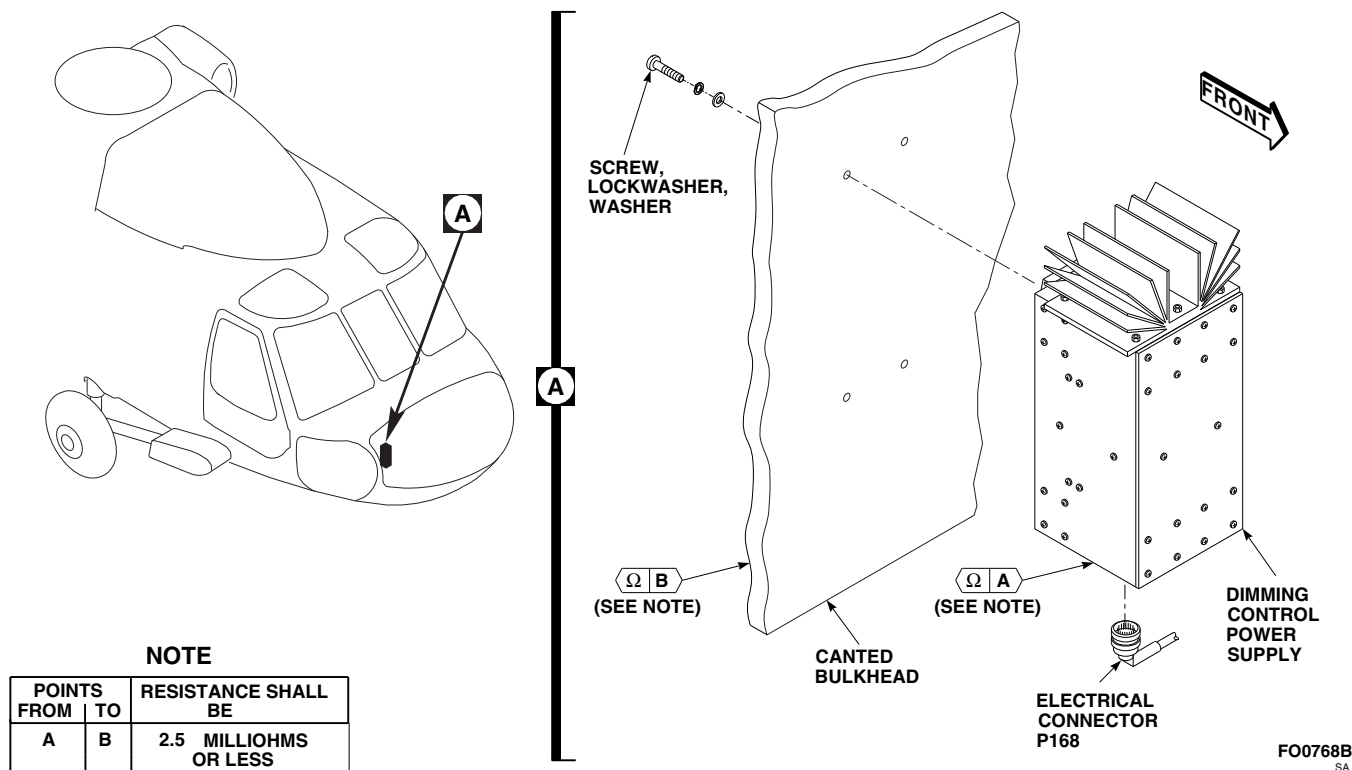


FIGURE 9-4-58-1. REPLACE DIMMING CONTROL POWER SUPPLY

ALLOWED. If coated surfaces have breaks or scratches, apply chemical conversion coating (PARA 2-6-5).

- c. Have assistant in nose compartment position dimming control power supply on canted bulkhead while screws, lockwashers, and washers are used to secure from cockpit.
- d. Remove protective caps and plugs from electrical connectors.
- e. Inspect and treat electrical connectors (PARA 1-7-20).
- f. Connect electrical connector, P168 to J168, on dimming control power supply.
- g. Perform EME resistance check as follows:

NOTE

Resistance measurements must be taken at least three times.

- (1) Using digital low-resistance ohmmeter, check resistance between dimming control power supply and canted bulkhead. **RESISTANCE SHALL BE 2.5 MILLIOHMS OR LESS.**

- (2) If resistance is greater than 2.5 milliohms, remove dimming control power supply and apply chemical conversion coating (PARA 2-6-5).

- h. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- i. Perform operational check of instrument panel and consoles indicator lights dimming system (PARA 9-2-25).
- j. Make sure area is clean and free of foreign material.
- k. Close nose door.

9-4-59 CARGO HOOK LIGHT DIMMING CONTROL PANEL.

PARAGRAPH BREAKDOWN

	PAGE
9-4-59.1 Replace Cargo Hook Light Dimming Control Panel	9-4-59-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

Materials/Parts

- Cheesecloth, Item 90, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-45
- PARA 9-2-10
- PARA 9-4-34
- PARA 9-4-106

9-4-59.1 Replace Cargo Hook Light Dimming Control Panel.

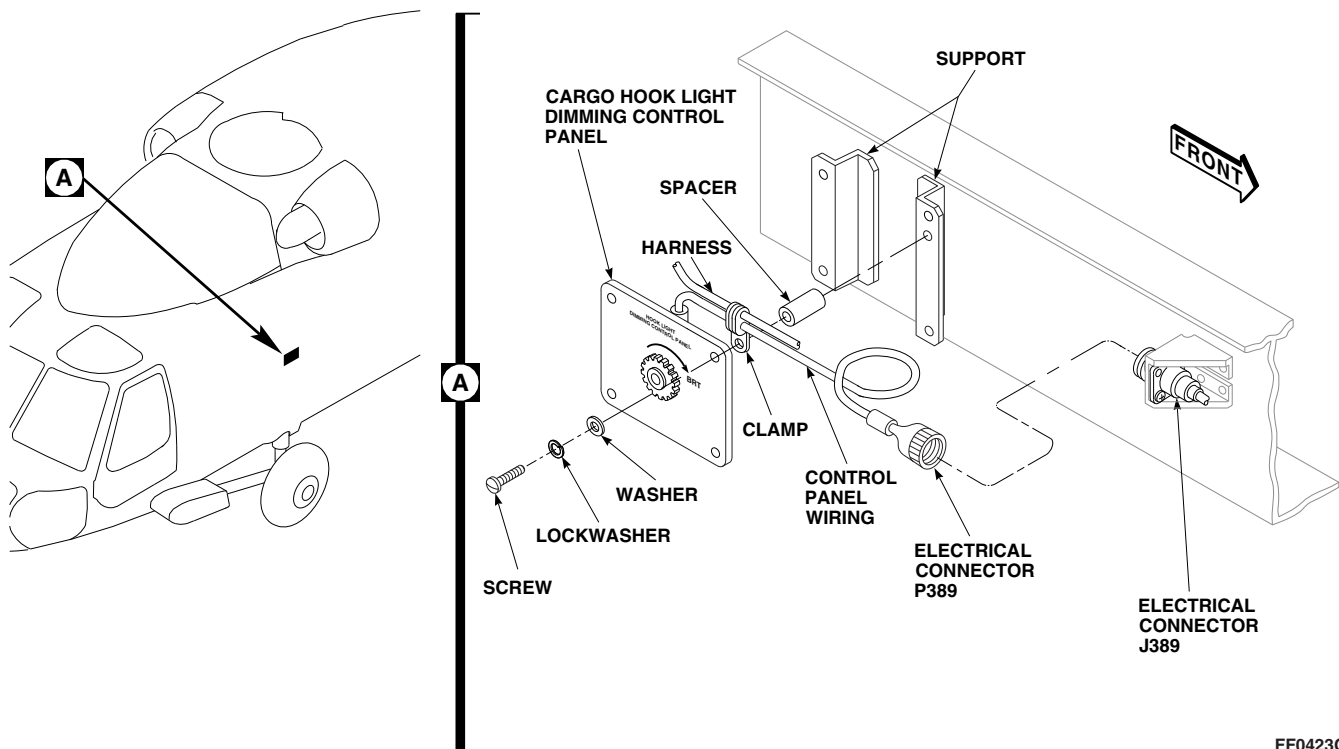
9-4-59.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Remove passenger’s seats as required (PARA 2-4-45).
- Open cargo hook door.
- Disconnect electrical connector, P389 from J389 (Figure 9-4-59-1, Detail A).

- Install protective caps and plugs on electrical connectors.
- Remove screws, lockwashers, washers, and spacers securing clamp and cargo hook light dimming control panel to support.
- Remove harness from clamp and remove cargo hook light dimming control panel.

9-4-59.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Using cheesecloth, Item 90, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D, clean surface around all mounting holes. Using clean, dry cheesecloth, Item 90, Appendix D, dry surface.

FF0423C
SA**FIGURE 9-4-59-1. REPLACE CARGO HOOK LIGHT DIMMING CONTROL PANEL****NOTE**

Do not install hardware in location that requires clamp at this time.

- c. Install cargo hook light dimming control panel to support, and secure with screws, lockwashers, washers, and spacers in locations that do not require clamp (Figure 9-4-59-1, Detail A).
- d. Install harness in clamp next to control panel wiring. Position clamp on control panel and secure to support with screw, lockwasher, washer, and spacer.
- e. Remove protective caps and plugs from electrical connectors.
- f. Inspect and treat electrical connectors (PARA 1-7-20).
- g. Connect electrical connector, P389 to J389, to cargo hook light dimming control panel.
- h. Make sure area is clean and free of foreign material.
- i. Close cargo hook door.
- j. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- k. Perform operational check of cargo hook light dimming control panel (PARA 9-2-10).
- l. Install passenger's seats (PARA 2-4-45).

9-4-60 CABIN DOME LIGHT LAMPS.

PARAGRAPH BREAKDOWN

	PAGE
9-4-60.1 Replace Cabin Dome Light Lamps	9-4-60-1

INITIAL SETUP

- PARA 9-4-34

Tools

- Electrical Repairer’s Toolkit

- PARA 9-4-106

References

- PARA 1-6-16
- PARA 9-2-6

9-4-60.1 Replace Cabin Dome Light Lamps.

- d. Remove lamp as required.

NOTE

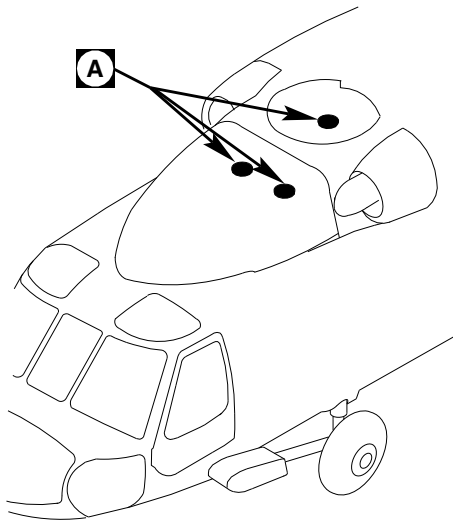
- Replacement of each pair is the same for all cabin dome light lamps.
- **70967 – SUBQ** A blue lens is installed over a white lamp.◀

9-4-60.1.2. Install.

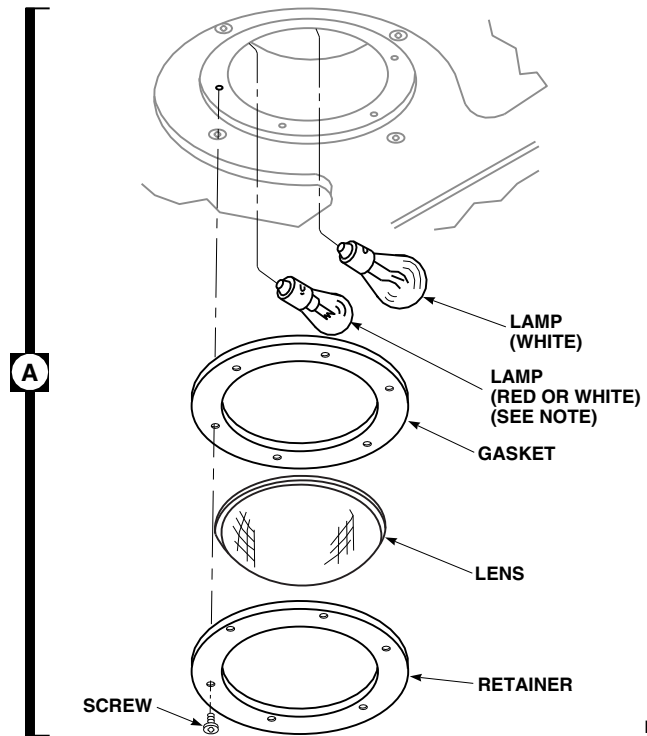
9-4-60.1.1. Remove.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- c. Remove screws, retainer, lens, and gasket (Figure 9-4-60-1, Detail A).

- a. Turn off all electrical power (PARA 1-6-16).
- b. Install lamp as required (Figure 9-4-60-1, Detail A).
- c. Install gasket, lens, and retainer and secure with screws.
- d. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- e. Perform operational check of cabin dome lights (PARA 9-2-6).

**NOTE**

RED ON HELICOPTERS PRIOR TO SERIAL NO. 70967. WHITE ON HELICOPTERS SERIAL NOS. 70967 AND SUBSEQUENT.



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SA

FIGURE 9-4-60-1. REPLACE CABIN DOME LIGHT LAMPS

9-4-61 CABIN DOME LIGHTS.

PARAGRAPH BREAKDOWN

	PAGE
9-4-61.1 Replace Cabin Dome Lights	9-4-61-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

Materials/Parts

- Dry-Cleaning Solvent, Item 124, Appendix D
- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-69
- PARA 9-2-6
- PARA 9-4-34
- PARA 9-4-106

9-4-61.1 Replace Cabin Dome Lights.

NOTE

Replacement of all cabin dome lights is the same, except as noted.

9-4-61.1.1 Remove.

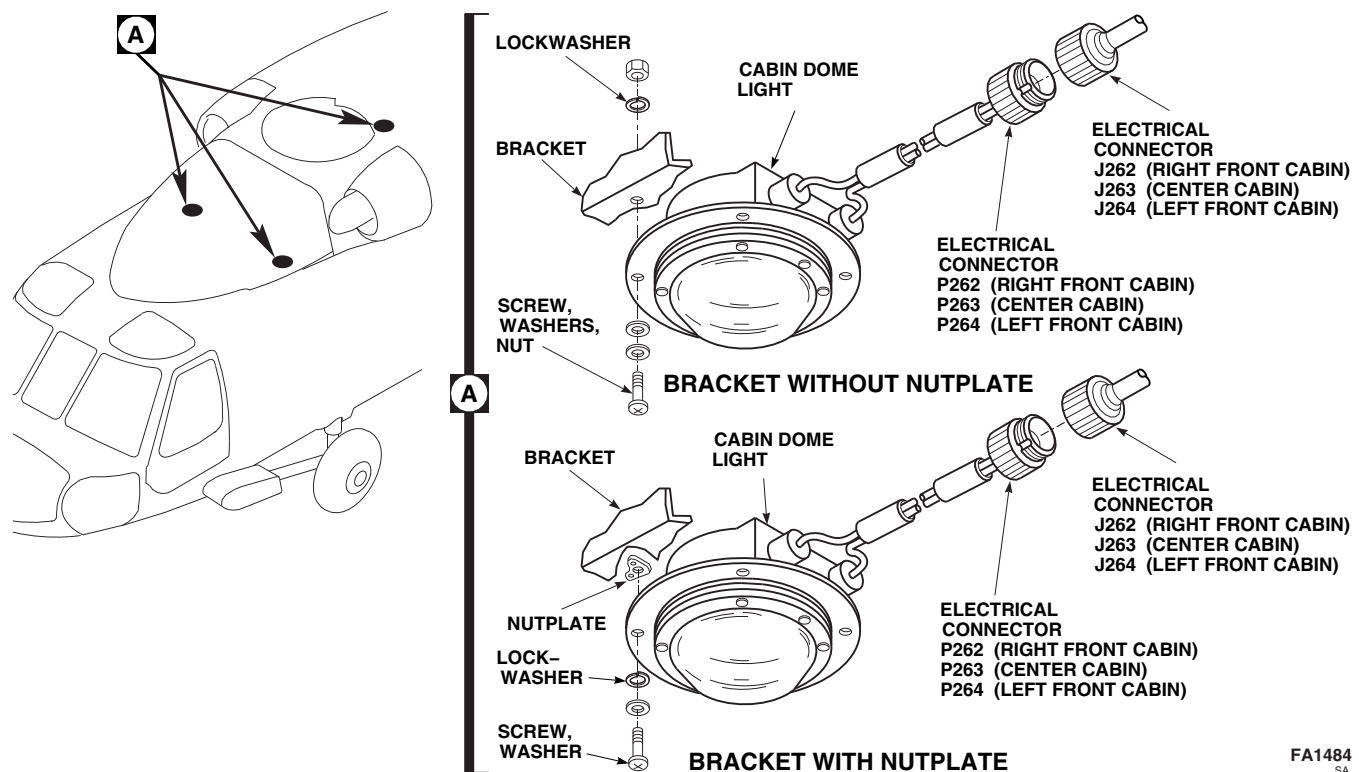
- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Remove soundproofing panel (PARA 2-4-69).
- Remove cabin dome light from bracket as follows (Figure 9-4-61-1, Detail A):
 - If nutplates are installed, remove screws, washers, and lockwashers securing cabin dome light to bracket.

- If nutplates are not installed, remove screws, washers, lockwashers, and nuts securing cabin dome light to bracket.

- Disconnect applicable electrical connector(s) from cabin dome light as follows:
 - P262 from J262 (right front cabin).
 - P263 from J263 (center cabin).
 - P264 from J264 (left front cabin).
- Remove cabin dome light.
- Install protective caps and plugs on electrical connectors.

9-4-61.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).

FA1484
SA**FIGURE 9-4-61-1. REPLACE CABIN DOME LIGHTS**

- b. Using machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D, clean surfaces around mounting holes. Using clean, dry machinery towel, Item 211, Appendix D, dry surface.
- c. Remove protective caps and plugs from electrical connectors.
- d. Inspect and treat electrical connectors (PARA 1-7-20).
- e. Connect applicable electrical connector(s) to cabin dome light as follows (Figure 9-4-61-1, Detail A):
 - (1) P262 to J262 (right front cabin).
 - (2) P263 to J263 (center cabin).
 - (3) P264 to J264 (left front cabin).
- f. Install cabin dome light in bracket as follows (Figure 9-4-61-1, Detail A):
 - (1) If nutplates are installed, secure cabin dome light in bracket with screws, washers, and lockwashers.
 - (2) If nutplates are not installed, secure cabin dome light in bracket with screws, washers, lockwashers, and nuts.
- g. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- h. Perform operational check of cabin lights (PARA 9-2-6).
- i. Make sure area is clean and free of foreign material.
- j. Install soundproofing panel (PARA 2-4-69).

9-4-62 PILOT’S AND COPILOT’S UTILITY LIGHT LAMPS.

PARAGRAPH BREAKDOWN

		PAGE
9-4-62.1	Replace Pilot’s and Copilot’s Utility Light Lamps	9-4-62-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

References

- PARA 1-6-16
- PARA 9-2-9

9-4-62.1 Replace Pilot’s and Copilot’s Utility Light Lamps.

NOTE

There are three utility lights; one on the pilot’s side, and two on the copilot’s side. Replacement of all three utility light lamps is the same.

9-4-62.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- On BATT BUS circuit breaker panel on lower console, pull out UTIL LTS CKPT circuit breaker.

- Remove snap-in retainer and lens from utility light (Figure 9-4-62-1, Detail A).
- Remove lamp.

9-4-62.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).
- Install lamp (Figure 9-4-62-1, Detail A).
- Install lens and snap-in retainer in utility light.
- On BATT BUS circuit breaker panel on lower console, push in UTIL LTS CKPT circuit breaker.
- Perform operational check of utility and maintenance lights (PARA 9-2-9).

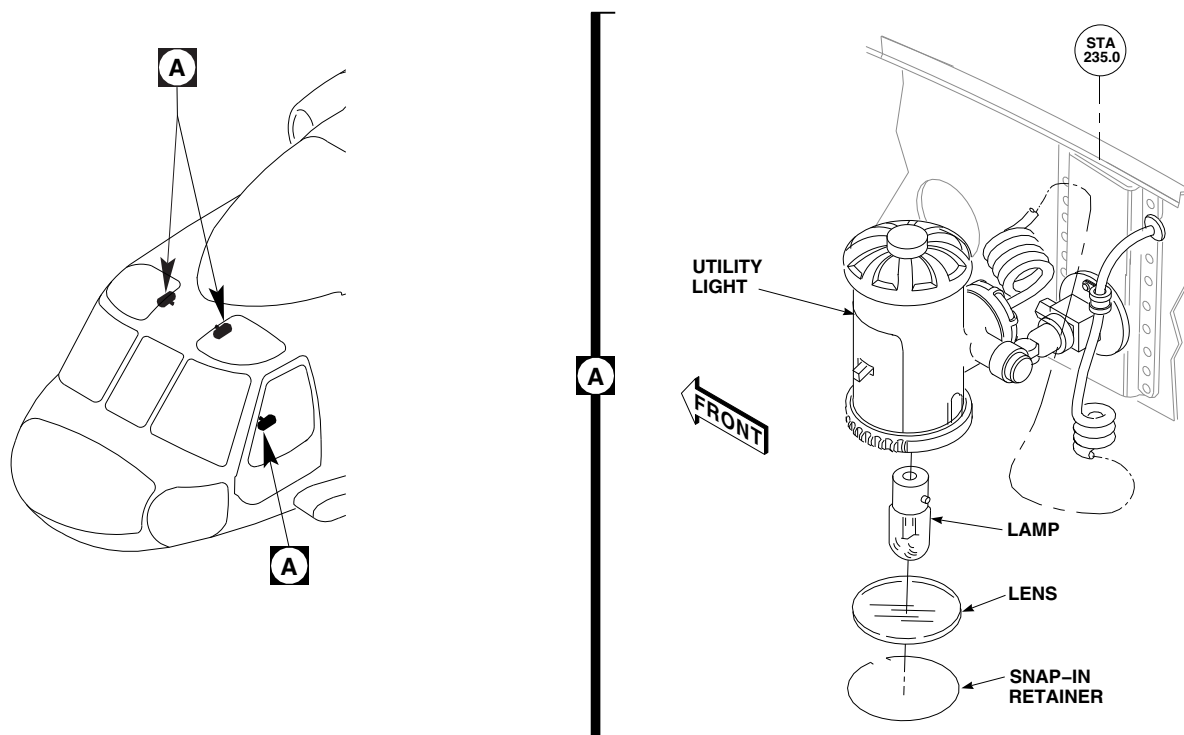
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FIGURE 9-4-62-1. REPLACE PILOT'S AND COPILOT'S UTILITY LIGHT LAMPS

9-4-63 PILOT’S AND COPILOT’S UTILITY LIGHTS.

PARAGRAPH BREAKDOWN

	PAGE
9-4-63.1 Replace Pilot’s and Copilot’s Utility Lights	9-4-63-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

Materials/Parts

- Tags

References

- PARA 1-6-16
- PARA 9-2-9
- PARA 9-4-34
- PARA 9-4-106

9-4-63.1 Replace Pilot’s and Copilot’s Utility Lights.

NOTE

There are three utility lights; one on the pilot’s side, and two on the copilot’s side. Replacement of all three utility lights is the same, except as noted.

9-4-63.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- On BATT BUS circuit breaker panel on lower console, pull out UTIL LTS CKPT circuit breaker.
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Loosen fasteners securing upper console and carefully lower.
- Tag wires. Remove nuts securing wiring to terminal board, TB1 (copilot’s overhead), or TB2

(pilot’s overhead), terminals 1 and 2 (Figure 9-4-63-1, Detail A).

- If replacing third utility light on copilot’s seat, tag wires and remove terminal cover. Remove nuts securing wiring to terminal board, TB6 (copilot’s seat), terminals 1 and 2 (Figure 9-4-63-1, Detail B).
- Loosen screw securing strain-relief clamp in place (Figure 9-4-63-1, Detail A or Detail B).
- Slowly pull coil cord through grommet and strain-relief clamp.
- Loosen knob on mounting base and remove utility light.

9-4-63.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).
- Install utility light into mounting base. **TIGHTEN KNOB** (Figure 9-4-63-1, Detail A or Detail B).
- Slide coil cord through strain-relief clamp on mounting base and through grommet.

- d. Install white wire to terminal 1 and black wire to terminal 2 of terminal board, TB1 (copilot's overhead), TB2 (pilot's overhead), or TB6 (copilot's seat) and secure with nuts. Remove tags.
- e. Install terminal cover on terminal board, TB6 (copilot's seat) (Figure 9-4-63-1, Detail B).
- f. **TIGHTEN SCREWS SECURING STRAIN-RELIEF CLAMP IN PLACE** (Figure 9-4-63-1, Detail A or Detail B).
- g. Make sure area is clean and free of foreign material.
- h. Carefully raise upper console and secure with fasteners.
- i. On BATT BUS circuit breaker panel on lower console, push in UTIL LTS CKPT circuit breaker.
- j. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- k. Perform operational check of utility and maintenance lights (PARA 9-2-9).

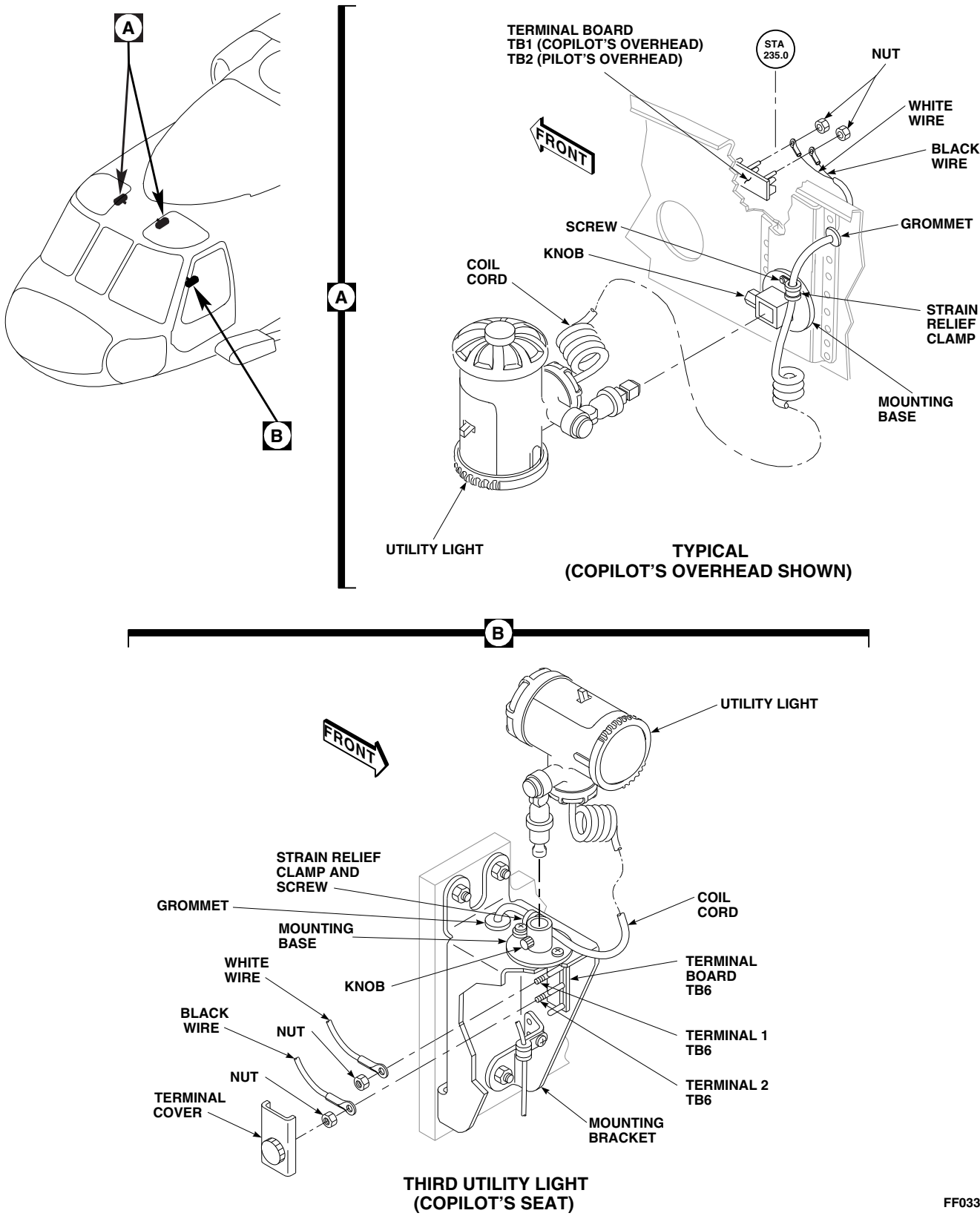


FIGURE 9-4-63-1. REPLACE PILOT'S AND COPILOT'S UTILITY LIGHTS

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9-4-64 PORTABLE MAINTENANCE LIGHT LAMP.

PARAGRAPH BREAKDOWN

	PAGE
9-4-64.1 Replace Portable Maintenance Light Lamp	9-4-64-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

References

- PARA 1-6-16
- PARA 9-2-9

9-4-64.1 Replace Portable Maintenance Light Lamp.

NOTE

Portable maintenance light may be in either the stowed, front, or rear location.

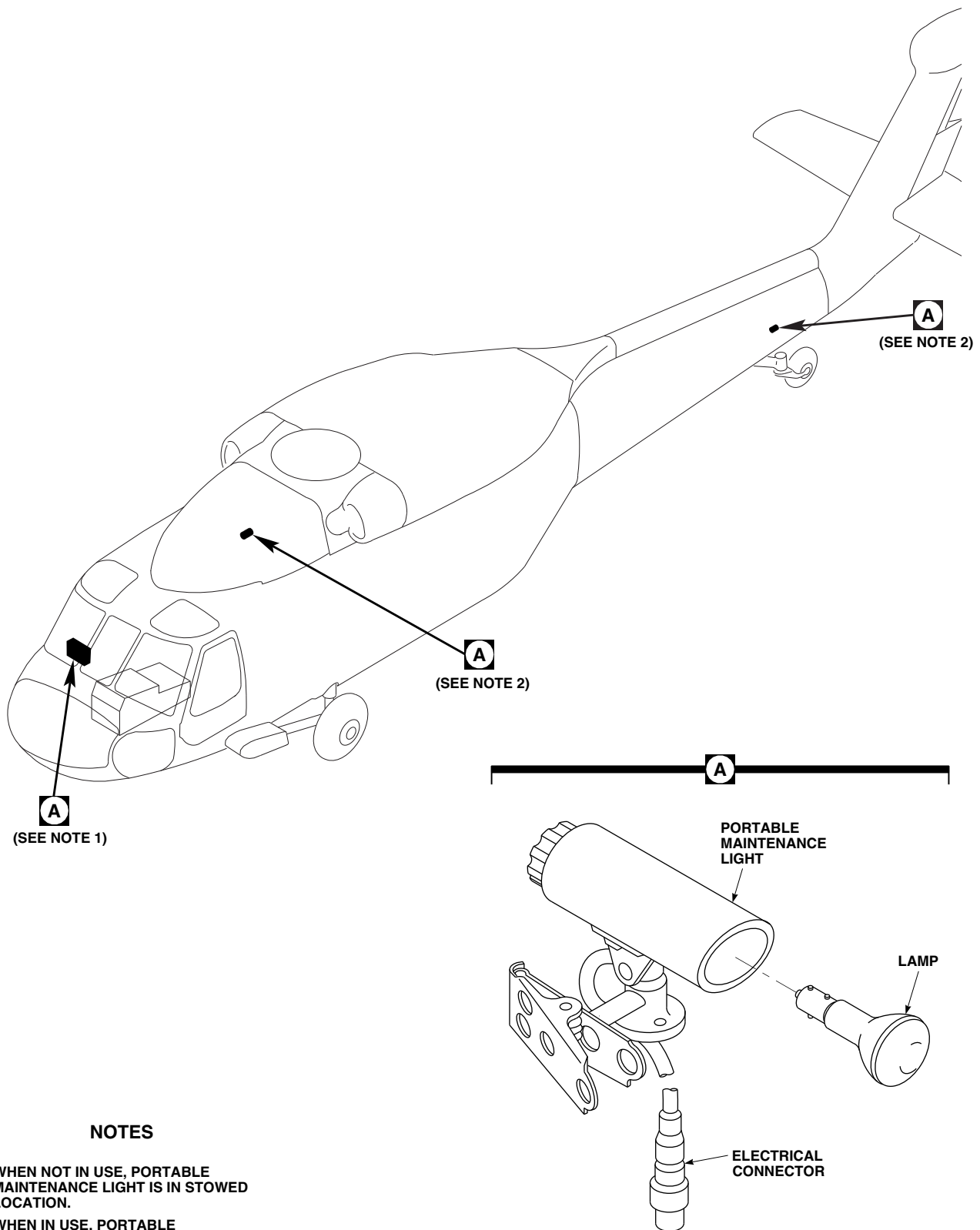
9-4-64.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- If required, disconnect portable maintenance light from electrical connector in either front or rear location (Figure 9-4-64-1).

- Push in on lamp and turn counterclockwise to remove it from portable maintenance light (Figure 9-4-64-1, Detail A).

9-4-64.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Push lamp into portable maintenance light and turn it clockwise (Figure 9-4-64-1, Detail A).
- Connect portable maintenance light to electrical connector in either front or rear location (Figure 9-4-64-1).
- Perform operational check of utility and maintenance lights (PARA 9-2-9).



NOTES

1. WHEN NOT IN USE, PORTABLE MAINTENANCE LIGHT IS IN STOWED LOCATION.
2. WHEN IN USE, PORTABLE MAINTENANCE LIGHT MAY BE IN EITHER FRONT OR REAR LOCATION.

FIGURE 9-4-64-1. REPLACE PORTABLE MAINTENANCE LIGHT LAMP

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9-4-64-2

9-4-65 GLARE SHIELD LAMPS.

PARAGRAPH BREAKDOWN

		PAGE
9-4-65.1	Replace Glare Shield Lamps	9-4-65-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit

References

- PARA 1-6-16
- PARA 9-2-7

- PARA 9-2-8
- PARA 9-4-34
- PARA 9-4-106

9-4-65.1 Replace Glare Shield Lamps.

NOTE

Replacement of all glare shield lamps is the same.

9-4-65.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Hold base of glare shield light and pull to remove hood (Figure 9-4-65-1, Detail A).
- Push down on lamp and turn counterclockwise to remove lamp from glare shield light.

9-4-65.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Push lamp into glare shield light socket and turn lamp clockwise (Figure 9-4-65-1, Detail A).
- Install hood over lamp and line up hood with positioning pin. Press hood onto glare shield light to secure.
- Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Perform operational check of cockpit flood and secondary lights (PARA 9-2-7 or PARA 9-2-8).

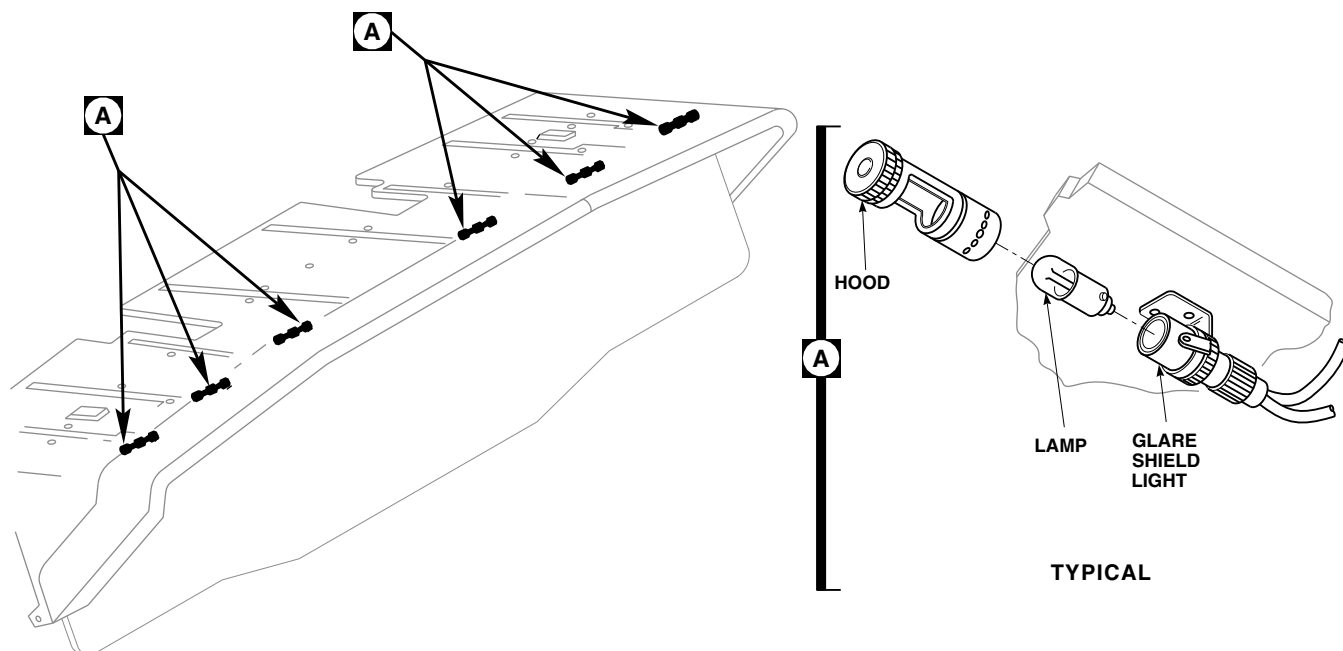


FIGURE 9-4-65-1. REPLACE GLARE SHIELD LAMPS

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9-4-66 GLARE SHIELD HARNESS.

PARAGRAPH BREAKDOWN

	PAGE
9-4-66.1 Replace Glare Shield Har-	9-4-66-1
ness	
INITIAL SETUP	• PARA 1-7-20
Tools	• PARA 9-2-7
• Electrical Repairer’s Toolkit	• PARA 9-2-8
Materials/Parts	
• Protective Caps and Plugs	• PARA 9-4-34
• Tiedown Strap, Item 334, Appendix D	• PARA 9-4-65
References	
• Appendix D	• PARA 9-4-106
• PARA 1-6-16	• TM 11-70-23

9-4-66.1 Replace Glare Shield Harness.

9-4-66.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Remove copilot’s radar altimeter (TM 11-70-23).
- Reach through altimeter mounting hole and disconnect electrical connector, P162 from J162 (Figure 9-4-66-1, Detail A).
- Install protective caps and plugs on electrical connectors.
- Remove hoods and glare shield lamps from glare shield lights (PARA 9-4-65).

- Remove screws, washers, and nuts securing glare shield lights and ground wires to glare shield (Figure 9-4-66-1, Detail B).
- Remove five left-most screws on top of instrument panel, and screws on left side of glare shield.
- Remove tiedown straps securing glare shield harness to glare shield (Figure 9-4-66-1, Detail A).
- Remove glare shield harness from instrument panel by flexing glare shield around feed-through connector.

9-4-66.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).
- Install glare shield harness by flexing glare shield around feed-through connector and pass-

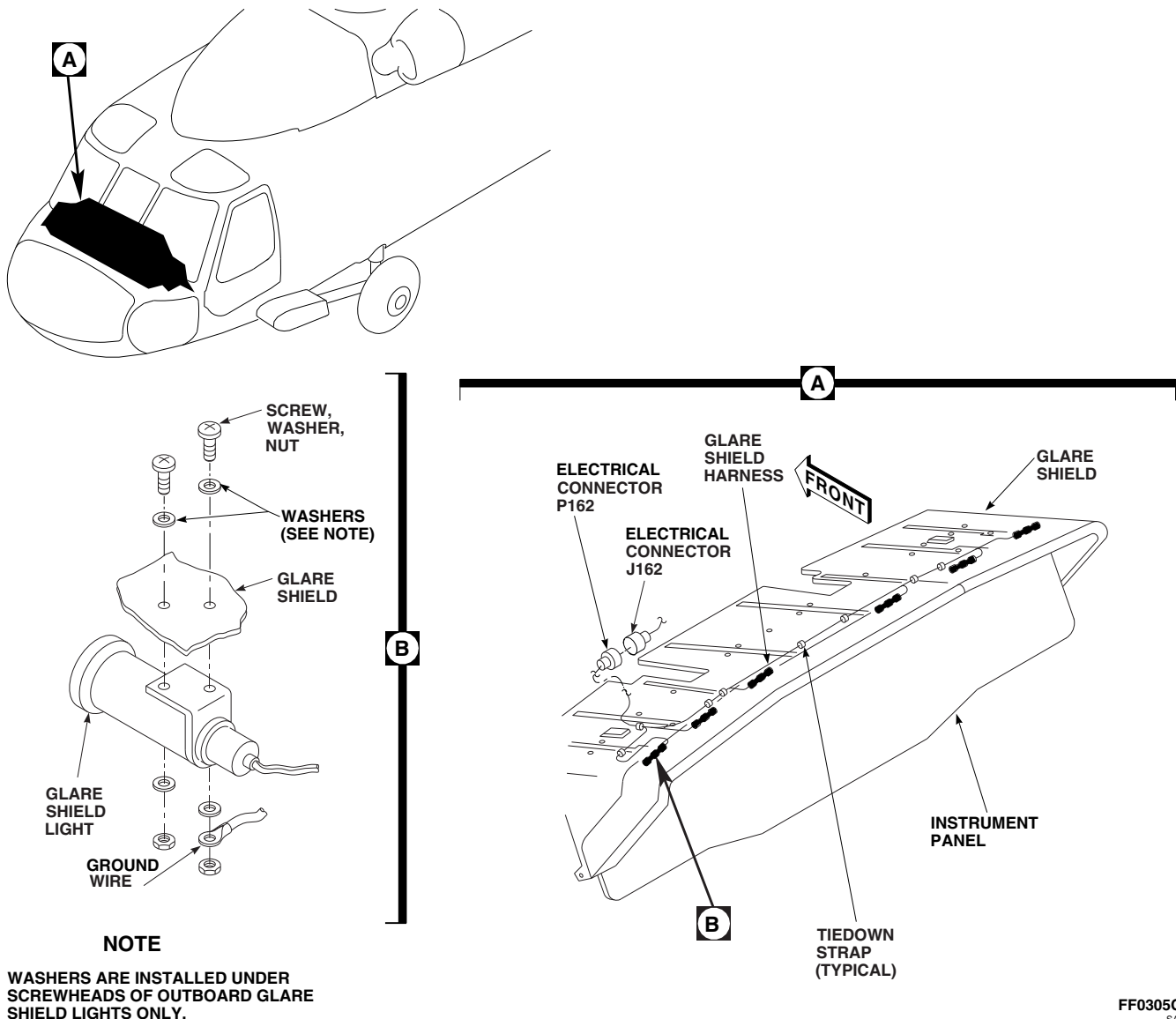


FIGURE 9-4-66-1. REPLACE GLARE SHIELD HARNESS

ing through electrical connector, P162 (Figure 9-4-66-1, Detail A).

- c. Remove protective caps and plugs from electrical connectors.
- d. Inspect and treat electrical connectors (PARA 1-7-20).
- e. Connect electrical connector, P162 to J162.
- f. **INSTALL AND TIGHTEN FIVE LEFT-MOST SCREWS ON TOP OF INSTRUMENT PANEL AND SCREWS ON LEFT SIDE OF GLARE SHIELD.**

NOTE

Make sure washers are installed under screwheads of outboard lights.

- g. Install glare shield lights and ground wires on glare shield and secure with screws, washers, and nuts (Figure 9-4-66-1, Detail B).
- h. Secure glare shield harness to glare shield with tiedown straps, Item 334, Appendix D (Figure 9-4-66-1, Detail A).
- i. Install glare shield lamps and hoods on glare shield lights (PARA 9-4-65).

- j. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- k. Perform operational check of cockpit flood and secondary lights (PARA 9-2-7 or PARA 9-2-8).
- l. Make sure area is clean and free of foreign material.
- m. Install copilot's radar altimeter (TM 11-70-23).

9-4-67 **SECONDARY LIGHT ENCLOSURE FLOODLIGHT LAMPS.**

PARAGRAPH BREAKDOWN

	PAGE
9-4-67.1 Replace Secondary Light Enclosure Floodlight Lamps	9-4-67-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit

References

- PARA 1-6-16
- PARA 9-2-7

- PARA 9-2-8
- PARA 9-4-34
- PARA 9-4-106

9-4-67.1 **Replace Secondary Light Enclosure Floodlight Lamps.**

NOTE

Replacement of all secondary light enclosure floodlight lamps is the same.

9-4-67.1.1. **Remove.**

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Loosen fasteners and carefully lower cover of secondary light enclosure (Figure 9-4-67-1, Detail A).

NOTE

Record floodlight lamp type and position for correct installation.

- Remove floodlight lamp by pressing down and twisting counterclockwise to release it from socket.

9-4-67.1.2. **Install.**

- Turn off all electrical power (PARA 1-6-16).

NOTE

Make sure correct type floodlight lamp is installed in correct position in secondary light enclosure.

- Insert floodlight lamp in inboard or outboard socket by pressing down while twisting lamp clockwise until seated (Figure 9-4-67-1, Detail A).
- Make sure area is clean and free of foreign material.
- Carefully swing cover of secondary light enclosure into closed position and secure with fasteners.

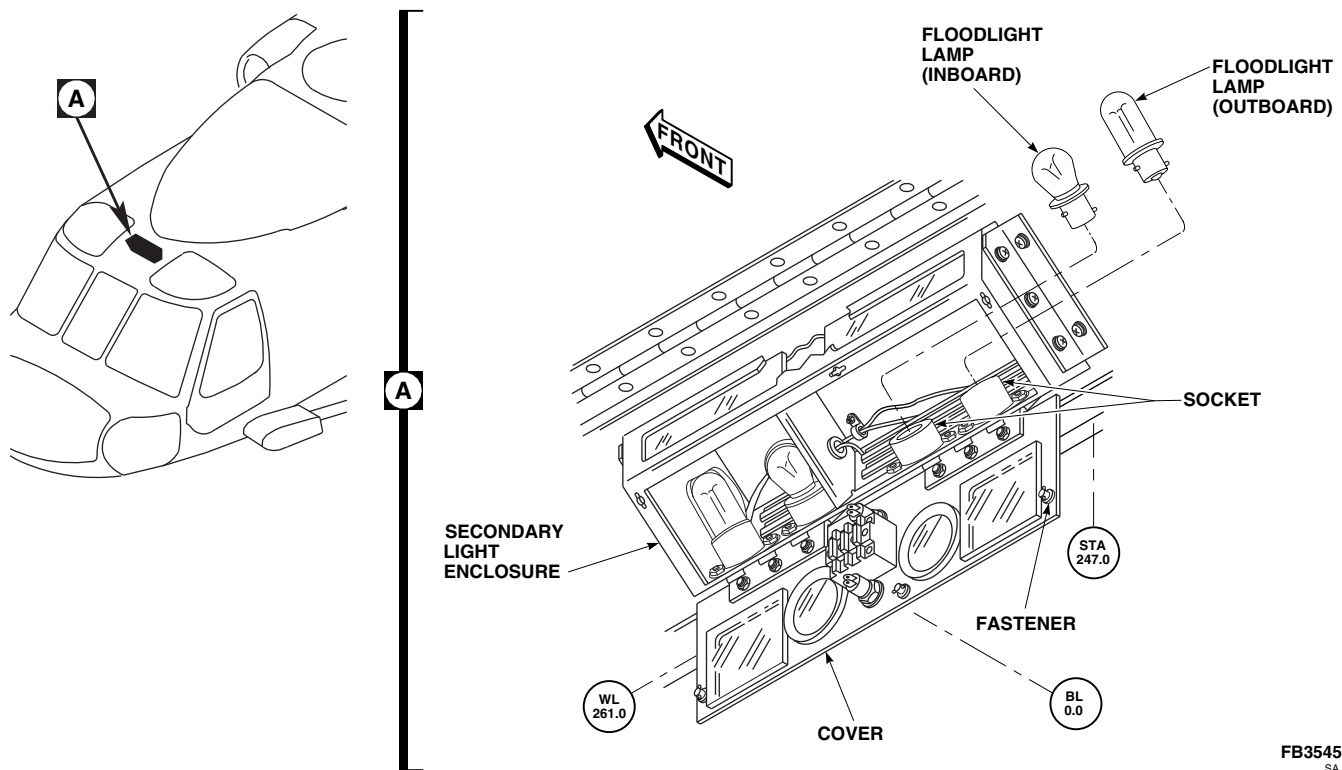


FIGURE 9-4-67-1. REPLACE SECONDARY LIGHT ENCLOSURE FLOODLIGHT LAMPS

- e. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- f. Perform operational check of cockpit flood and secondary lights (PARA 9-2-7 or PARA 9-2-8).

9-4-68 SECONDARY LIGHT ENCLOSURE INFORMATION PLATE LAMPS.

PARAGRAPH BREAKDOWN

	PAGE
9-4-68.1 Replace Secondary Light Enclosure Information Plate Lamps	9-4-68-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit

References

- PARA 1-6-16
- PARA 9-2-7

- PARA 9-2-8
- PARA 9-4-34
- PARA 9-4-106

9-4-68.1 Replace Secondary Light Enclosure Information Plate Lamps.

NOTE

Replacement of all secondary light enclosure information plate lamps is the same.

9-4-68.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Loosen fasteners and carefully lower cover of secondary light enclosure (Figure 9-4-67-1, Detail A).
- Remove screws securing information plate to secondary light enclosure.
- Remove nut and lockwasher securing information plate to toggle switch.

- Remove moisture seal from back of information plate and remove disc lamp by turning counterclockwise.

9-4-68.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Install disc lamp in information plate, turn clockwise to secure, and install moisture seal over disc lamp (Figure 9-4-68-1, Detail A).
- Install information plate on secondary light enclosure and secure with screws.
- Secure panel to toggle switch using nut and lockwasher.
- Make sure area is clean and free of foreign material.
- Carefully swing cover of secondary light enclosure into closed position and secure with fasteners.

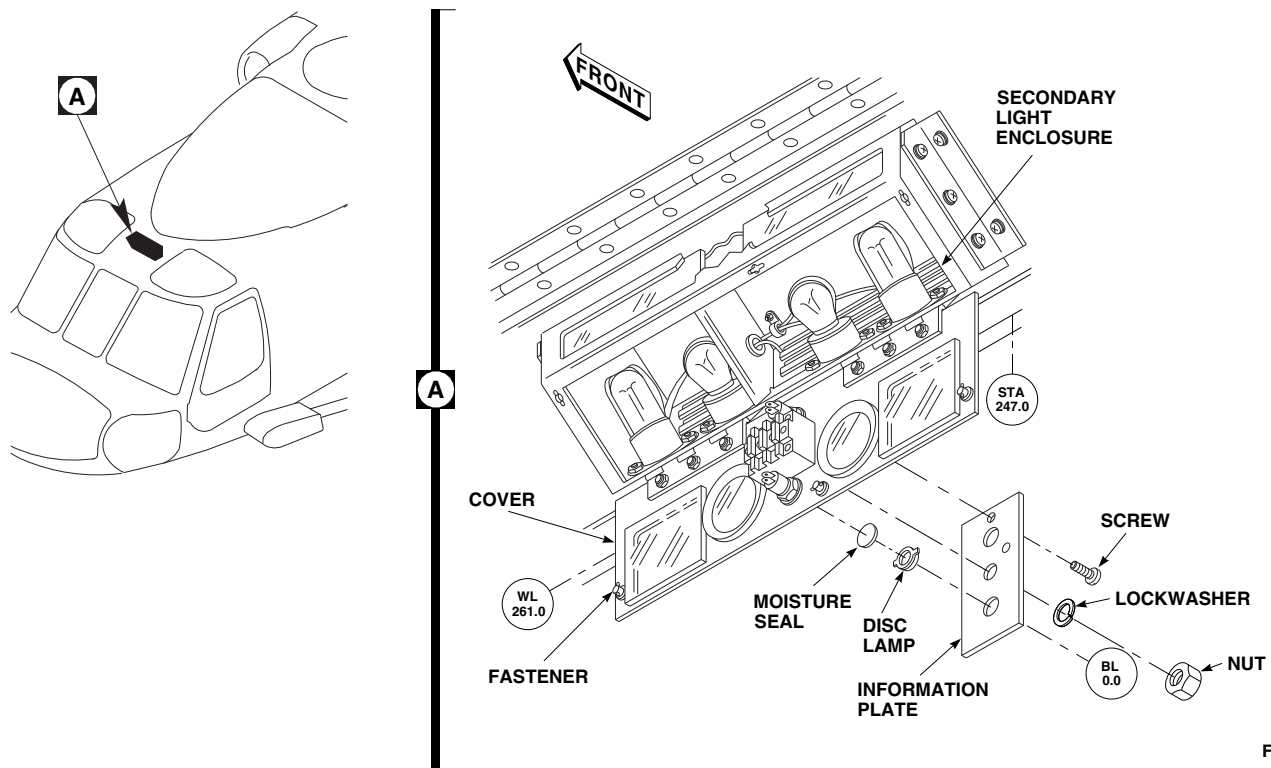
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FIGURE 9-4-68-1. REPLACE SECONDARY LIGHT ENCLOSURE INFORMATION PLATE LAMPS

- g. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- h. Perform operational check of cockpit flood and secondary lights (PARA 9-2-7 or PARA 9-2-8).

9-4-69 SECONDARY LIGHT ENCLOSURE.

PARAGRAPH BREAKDOWN

	PAGE
9-4-69.1 Replace Secondary Light Enclosure	9-4-69-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit

Materials/Parts

- Protective Caps and Plugs

References

- PARA 1-6-16
- PARA 1-7-20

- PARA 9-2-7
- PARA 9-2-8
- PARA 9-4-34
- PARA 9-4-106

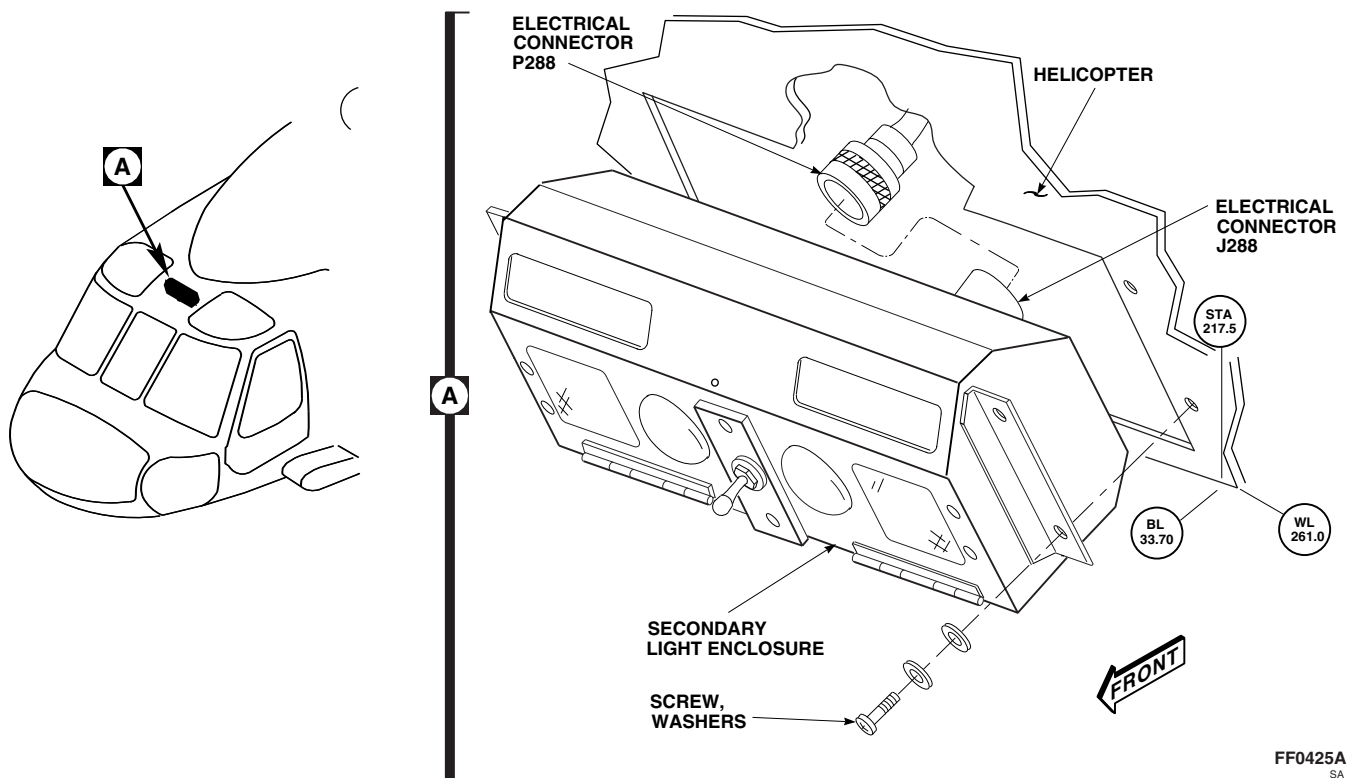
9-4-69.1 Replace Secondary Light Enclosure.

9-4-69.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Remove screws and washers securing secondary light enclosure to helicopter and carefully lower to gain access to electrical connector (Figure 9-4-69-1, Detail A).
- Disconnect electrical connector, P288 from J288, on secondary light enclosure.
- Remove secondary light enclosure.
- Install protective caps and plugs on electrical connectors.

9-4-69.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Remove protective caps and plugs from electrical connectors.
- Inspect and treat electrical connectors (PARA 1-7-20).
- Connect electrical connector, P288 to J288, on secondary light enclosure (Figure 9-4-69-1, Detail A).
- Install secondary light enclosure on helicopter and secure with screws and washers.
- Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Perform operational check of cockpit flood and secondary lights (PARA 9-2-7 or PARA 9-2-8).



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FIGURE 9-4-69-1. REPLACE SECONDARY LIGHT ENCLOSURE

9-4-70 MAIN ROTOR PYLON ELECTRO-LUMINESCENT FORMATION LIGHT.

PARAGRAPH BREAKDOWN

		PAGE
9-4-70.1	Replace Main Rotor Pylon Electro-Luminescent Formation Light	9-4-70-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

Materials/Parts

- Protective Caps and Plugs

References

- PARA 1-6-16
- PARA 1-7-20
- PARA 9-2-12
- PARA 9-4-34
- PARA 9-4-106

9-4-70.1 Replace Main Rotor Pylon Electro-Luminescent Formation Light.

9-4-70.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Open APU access doors.
- Disconnect electrical connector, P402 from J402 (Figure 9-4-70-1, Detail A).
- Install protective caps and plugs on electrical connectors.
- Remove screws and washers securing main rotor pylon electro-luminescent formation light to main rotor pylon rear fairing.

- Remove main rotor pylon electro-luminescent formation light.

9-4-70.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).
- Remove protective caps and plugs from electrical connectors.
- Inspect and treat electrical connectors (PARA 1-7-20).
- Connect electrical connector, P402 to J402 (Figure 9-4-70-1, Detail A).
- Install main rotor pylon electro-luminescent formation light to main rotor pylon rear fairing and secure with screws and washers.
- Make sure area is clean and free of foreign material.

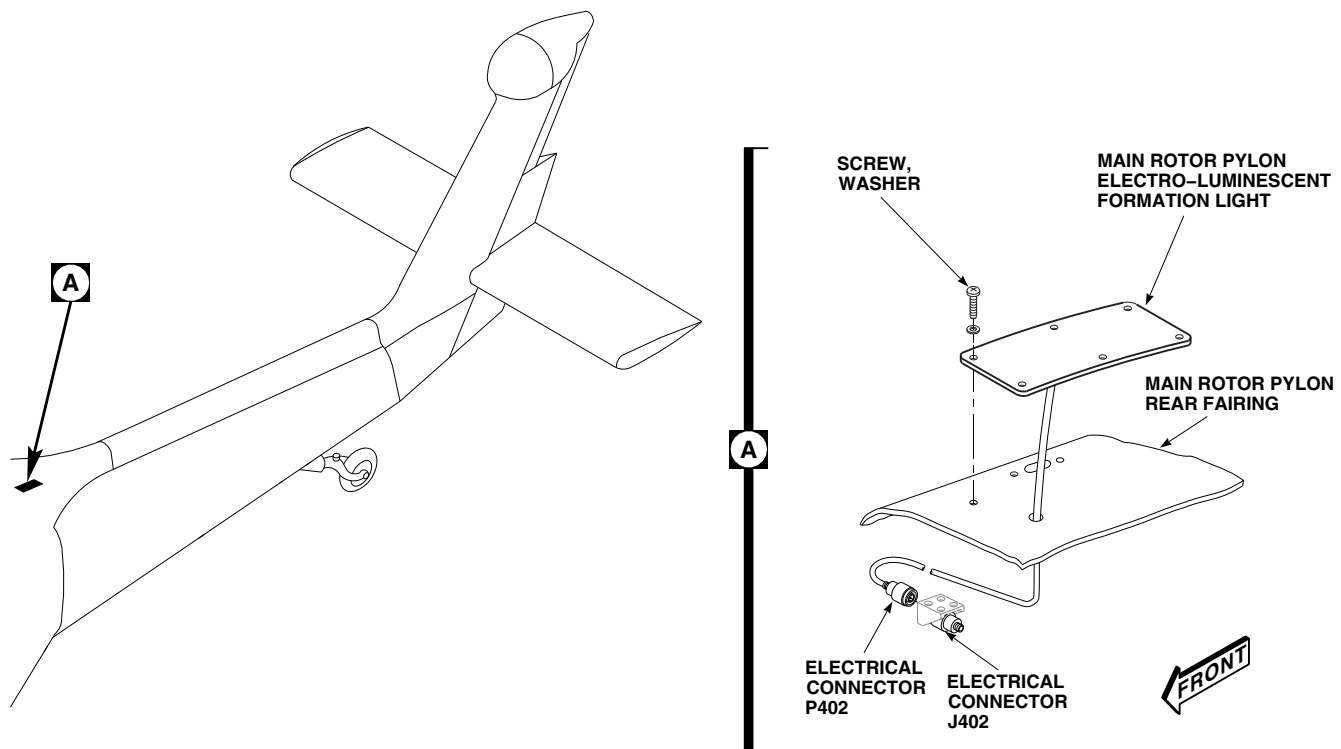
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FIGURE 9-4-70-1. REPLACE MAIN ROTOR PYLON ELECTRO-LUMINESCENT FORMATION LIGHT

- g. Close APU access doors.
- h. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- i. Perform operational check of formation lights (PARA 9-2-12).

9-4-71 MAIN ROTOR PYLON INFRARED FORMATION LIGHT.

PARAGRAPH BREAKDOWN

	PAGE
9-4-71.1 Replace Main Rotor Pylon Infrared Formation Light	9-4-71-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

Materials/Parts

- Protective Caps and Plugs

References

- PARA 1-6-16
- PARA 1-7-20
- PARA 9-2-12
- PARA 9-4-34
- PARA 9-4-106

9-4-71.1 Replace Main Rotor Pylon Infrared Formation Light.

9-4-71.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Open APU access doors.
- Disconnect electrical connector, P454 from J454 (Figure 9-4-71-1, Detail A).
- Install protective caps and plugs on electrical connectors.
- Remove screws and washers securing main rotor pylon infrared formation light to main rotor pylon rear fairing.
- Remove main rotor pylon infrared formation light.

9-4-71.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).
- Install main rotor pylon infrared formation light to main rotor pylon rear fairing and secure with screws and washers (Figure 9-4-71-1, Detail A).
- Remove protective caps and plugs from electrical connectors.
- Inspect and treat electrical connectors (PARA 1-7-20).
- Connect electrical connector, P454 to J454.
- Make sure area is clean and free of foreign material.
- Close APU access doors.
- Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).

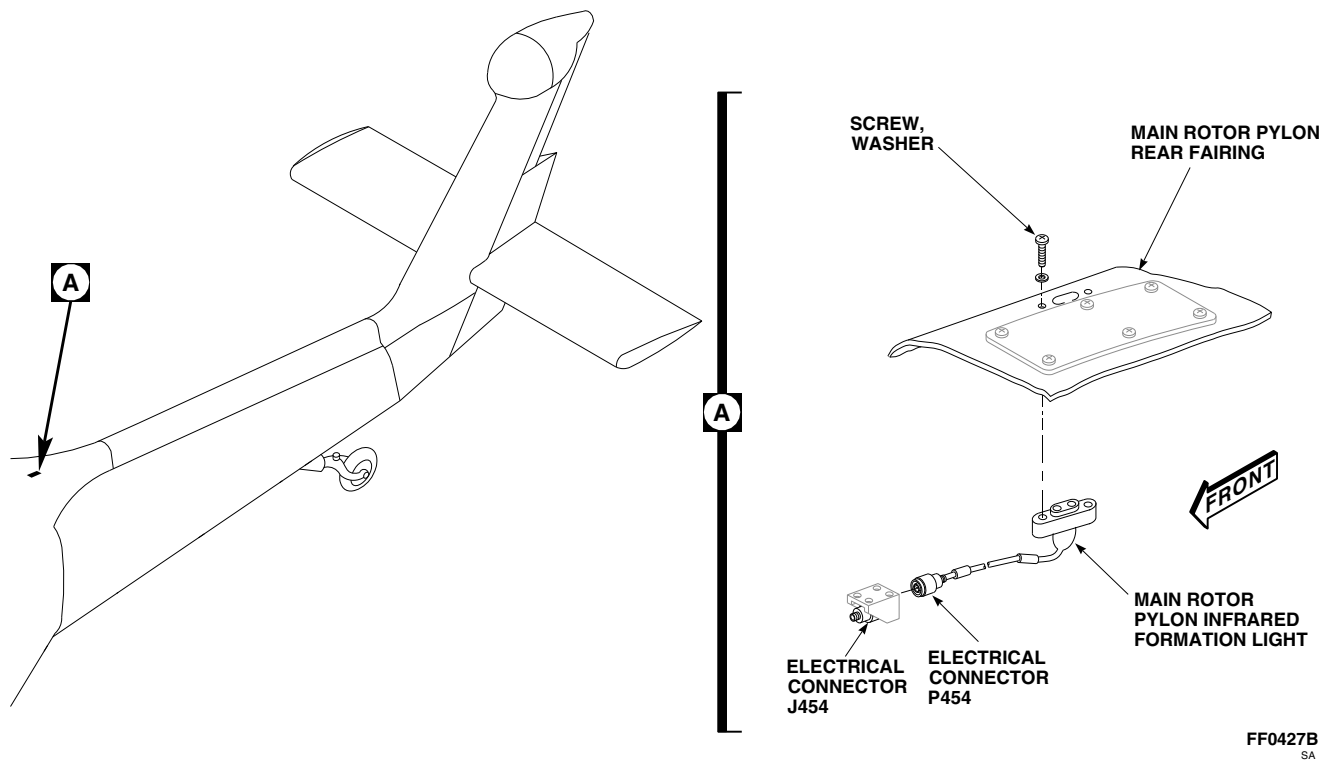


FIGURE 9-4-71-1. REPLACE MAIN ROTOR PYLON INFRARED FORMATION LIGHT

- i. Perform operational check of formation lights (PARA 9-2-12).

9-4-72 TAIL CONE ELECTRO-LUMINESCENT FORMATION LIGHT.

PARAGRAPH BREAKDOWN

	PAGE
9-4-72.1 Replace Tail Cone Electro-Luminescent Formation Light	9-4-72-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

Materials/Parts

- Protective Caps and Plugs

References

- PARA 1-6-16
- PARA 1-7-20
- PARA 9-2-12
- PARA 9-4-34
- PARA 9-4-106

9-4-72.1 Replace Tail Cone Electro-Luminescent Formation Light.

9-4-72.1.1. Remove.

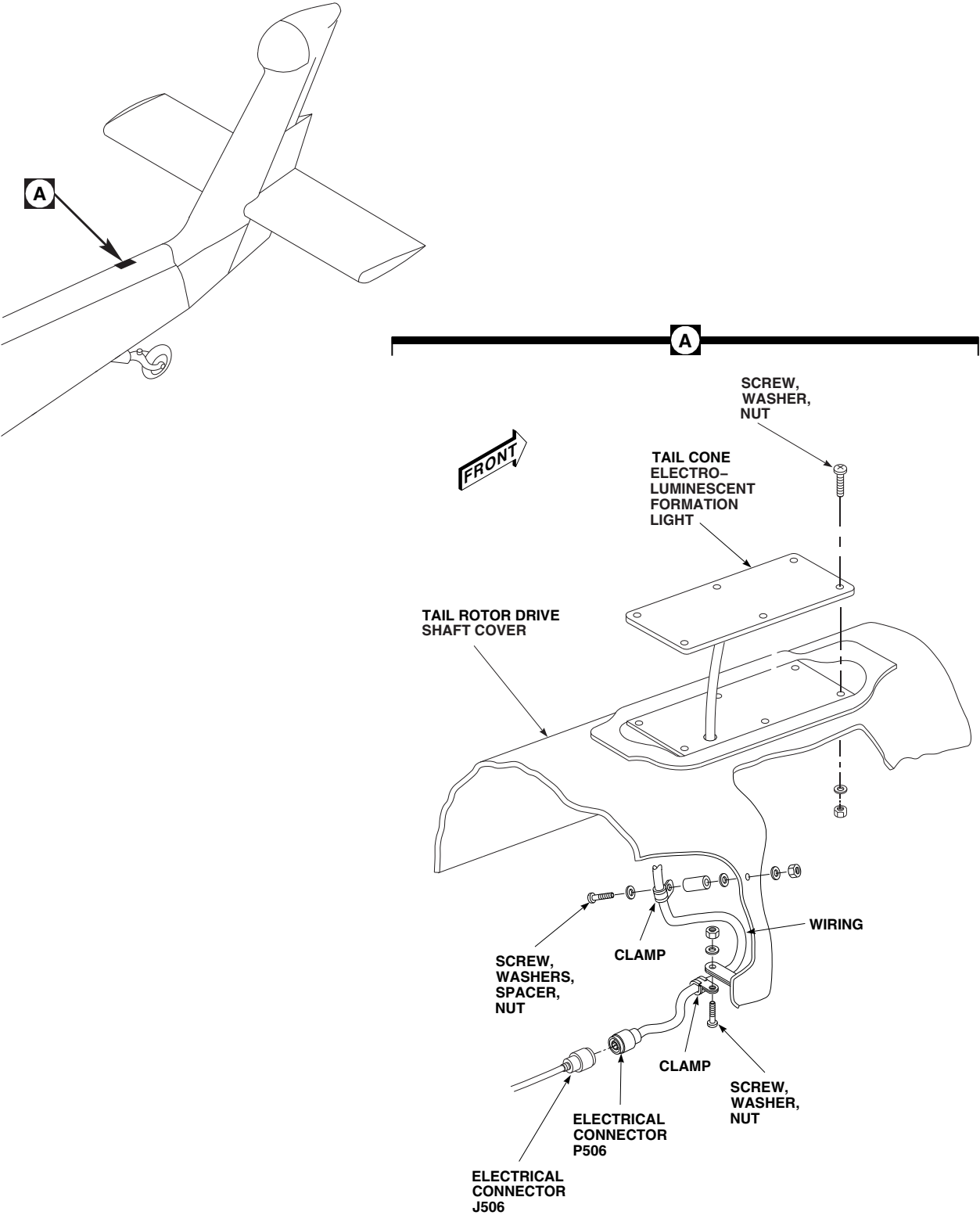
- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Open rear tail rotor drive shaft cover.
- Remove screws, washers, spacer, clamps, and nuts securing wiring to tail rotor drive shaft cover (Figure 9-4-72-1, Detail A).
- Disconnect electrical connector, P506 from J506.
- Install protective caps and plugs on electrical connectors.
- Remove screws, washers, and nuts securing tail cone electro-luminescent formation light to tail rotor drive shaft cover.

- Remove tail cone electro-luminescent formation light.

9-4-72.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Remove protective caps and plugs from electrical connectors.
- Inspect and treat electrical connectors (PARA 1-7-20).
- Connect electrical connector, P506 to J506 (Figure 9-4-72-1, Detail A).
- Install tail cone electro-luminescent formation light to tail rotor drive shaft cover and secure with screws, washers, and nuts.
- Install screws, washers, spacer, clamps, and nuts securing wiring to tail rotor drive shaft cover.

- g. Make sure area is clean and free of foreign material.
- h. Close rear tail rotor drive shaft cover.
- i. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- j. Perform operational check of formation lights (PARA 9-2-12).



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FIGURE 9-4-72-1. REPLACE TAIL CONE ELECTRO-LUMINESCENT FORMATION LIGHT

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9-4-73 TAIL CONE INFRARED FORMATION LIGHT.

PARAGRAPH BREAKDOWN

	PAGE
9-4-73.1 Replace Tail Cone Infrared Formation Light	9-4-73-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

Materials/Parts

- Protective Caps and Plugs
- Tiedown Strap, Item 329, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 9-2-12
- PARA 9-4-34
- PARA 9-4-106

9-4-73.1 Replace Tail Cone Infrared Formation Light.

9-4-73.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Open rear tail rotor drive shaft cover.
- Cut tiedown straps, as necessary, to remove tail cone infrared formation light.
- Disconnect electrical connector, P547 from J547 (Figure 9-4-73-1, Detail A).
- Install protective caps and plugs on electrical connectors.
- Remove screws, washers, spacers, nuts, and clamps securing wiring harness to tail rotor drive shaft cover.

- Remove screws and washers securing tail cone infrared formation light to tail rotor drive shaft cover.
- Remove tail cone infrared formation light.

9-4-73.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).
- Fasten tail cone infrared formation light to tail rotor drive shaft cover with screws and washers (Figure 9-4-73-1, Detail A).
- Remove protective caps and plugs from electrical connectors.
- Inspect and treat electrical connectors (PARA 1-7-20).
- Connect electrical connector, P547 to J547.
- Install wiring harness to tail rotor drive shaft cover and secure with screws, washers, spacers, clamps, and nuts.

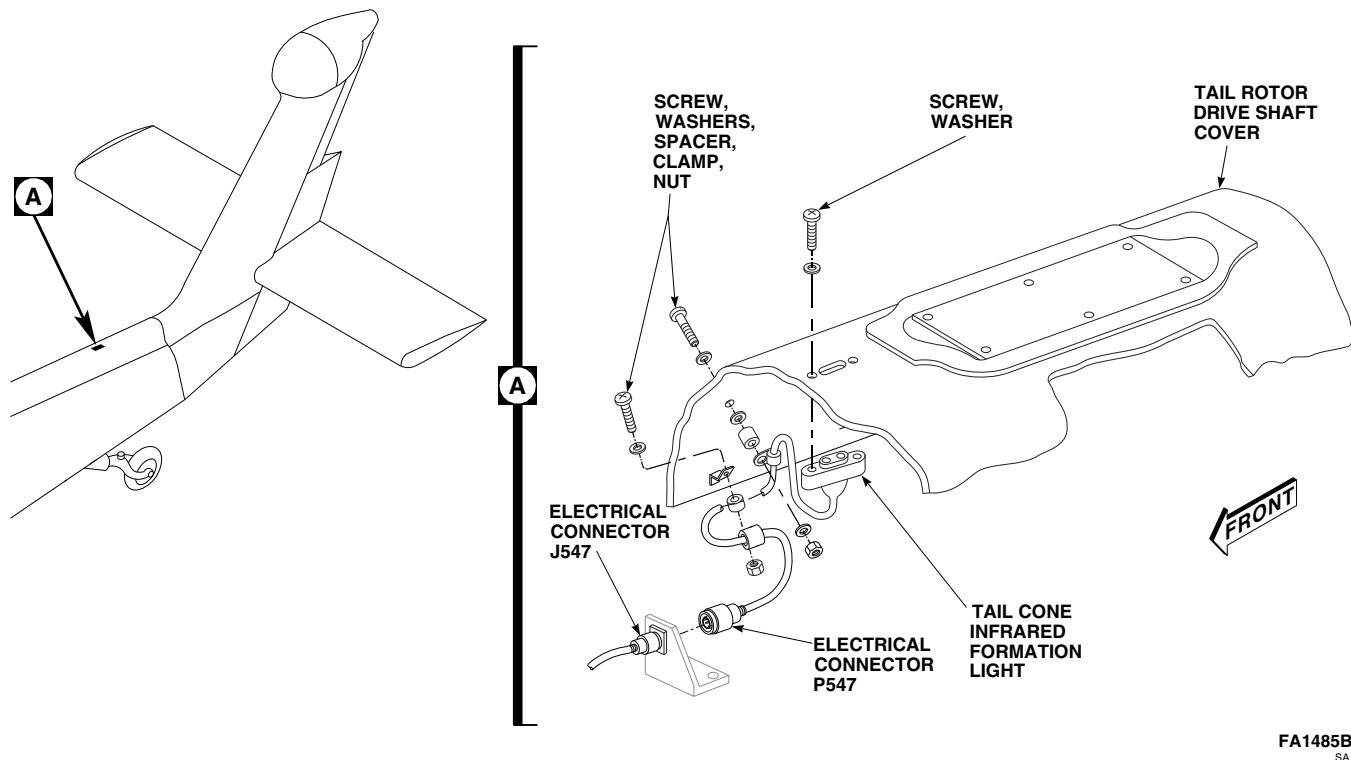
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SA

FIGURE 9-4-73-1. REPLACE TAIL CONE INFRARED FORMATION LIGHT

WARNING

Injury to personnel or damage to equipment will result if wiring harness is not properly secured. Wiring harness may chafe against tail rotor drive shaft. Make sure wiring harness is properly secured.

- g. Install tiedown straps, Item 329, Appendix D, where necessary to prevent chafing of wiring harness with tail rotor drive shaft.
- h. Make sure area is clean and free of foreign material.
- i. Close rear tail rotor drive shaft cover.
- j. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- k. Perform operational check of formation lights (PARA 9-2-12).

9-4-74 STABILATOR ELECTRO-LUMINESCENT FORMATION LIGHTS.

PARAGRAPH BREAKDOWN

	PAGE
9-4-74.1 Replace Stabilator Electro-Luminescent Formation Lights	9-4-74-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit
- Nonmetallic Scraper, Locally-Made

Materials/Parts

- Protective Caps and Plugs
- Sealing Compound, Item 292, Appendix D
- Tiedown Strap, Item 329, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-6-5
- PARA 9-2-12
- PARA 9-4-34
- PARA 9-4-106

9-4-74.1 Replace Stabilator Electro-Luminescent Formation Lights.

NOTE

Replacement of left and right stabilator electro-luminescent formation lights is the same, except as noted.

9-4-74.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).

- Using nonmetallic scraper, remove sealing compound from edges of stabilator electro-luminescent formation light.
- Remove screws and washers securing formation light to stabilator and carefully lift to gain access to wiring harness (Figure 9-4-74-1, Detail A).
- Cut and remove tiedown strap securing wiring harness to mounting plate.
- Disconnect electrical connector, P700 from J700 (right), or P701 from J701 (left), and remove stabilator electro-luminescent formation light.

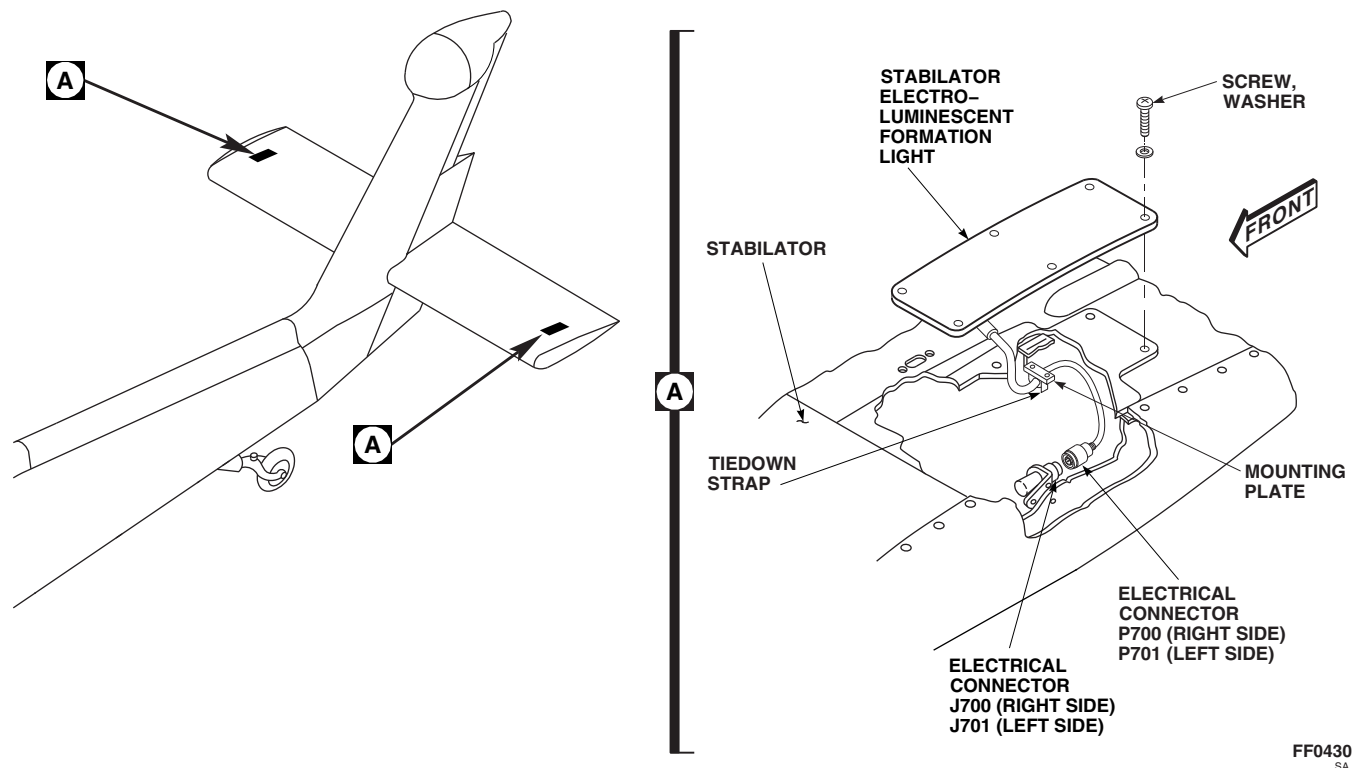


FIGURE 9-4-74-1. REPLACE STABILATOR ELECTRO-LUMINESCENT FORMATION LIGHTS

- g. Install protective caps and plugs on electrical connectors.

9-4-74.1.2. Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Make sure area is clean and free of foreign material.
- c. Remove protective caps and plugs from electrical connectors.
- d. Inspect and treat electrical connectors (PARA 1-7-20).
- e. Connect electrical connector, P700 to J700 (right), or P701 to J701 (left) (Figure 9-4-74-1, Detail A).

- f. Install tiedown strap, Item 329, Appendix D, to secure wiring harness to mounting plate.

- g. Install stabilator electro-luminescent formation light to stabilator and secure with screws and washers.
- h. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- i. Perform operational check of formation lights (PARA 9-2-12).
- j. Apply sealing compound, Item 292, Appendix D, around exterior edge of stabilator electro-luminescent formation light.
- k. Touch up paint (PARA 2-6-5).

9-4-75 STABILATOR INFRARED FORMATION LIGHTS.

PARAGRAPH BREAKDOWN

	PAGE
9-4-75.1 Replace Stabilator Infrared Formation Lights	9-4-75-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

Materials/Parts

- Protective Caps and Plugs
- Tiedown Strap, Item 329, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 9-4-74

9-4-75.1 Replace Stabilator Infrared Formation Lights.

NOTE

Replacement of left and right stabilator infrared formation lights is the same, except as noted.

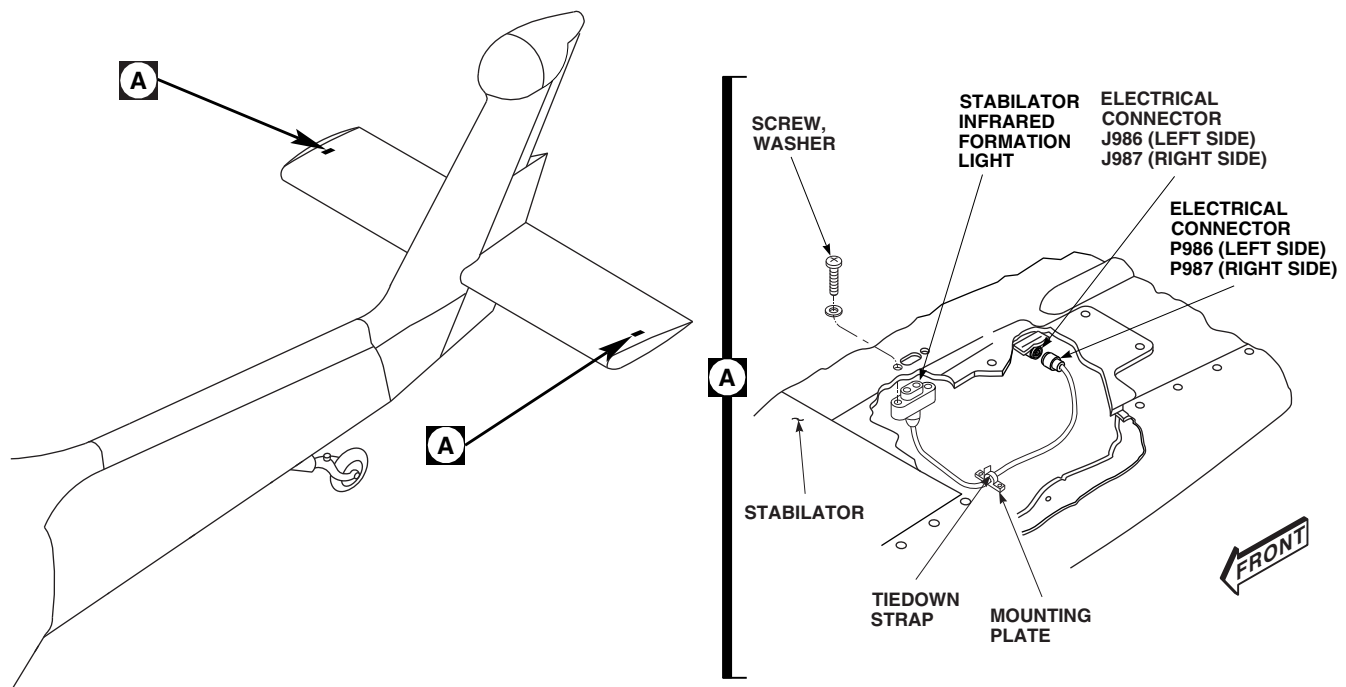
9-4-75.1.1 Remove.

- Remove stabilator electro-luminescent formation light (PARA 9-4-74).
- Disconnect electrical connector, P987 from J987 (right), or P986 from J986 (left) (Figure 9-4-75-1, Detail A).
- Install protective caps and plugs on electrical connectors.
- Cut and remove tiedown strap securing wiring harness to mounting plate.
- Remove screws and washers securing stabilator infrared formation light to stabilator.

- Remove stabilator infrared formation light.

9-4-75.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).
- Install stabilator infrared formation light to stabilator and secure with screws and washers (Figure 9-4-75-1, Detail A).
- Remove protective caps and plugs from electrical connectors.
- Inspect and treat electrical connectors (PARA 1-7-20).
- Connect electrical connector, P987 to J987 (right), or P986 to J986 (left).
- Install tiedown strap, Item 329, Appendix D, to secure wiring harness to mounting plate.
- Make sure area is clean and free of foreign material.

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SA**FIGURE 9-4-75-1. REPLACE STABILATOR INFRARED FORMATION LIGHTS**

- h. Install stabilator electro-luminescent formation lights (PARA 9-4-74).

9-4-76 POSITION LIGHTS FLASHER UNIT.

PARAGRAPH BREAKDOWN

	PAGE
9-4-76.1 Replace Position Lights Flasher Unit	9-4-76-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

Materials/Parts

- Protective Caps and Plugs

References

- PARA 1-6-16
- PARA 1-7-20

- PARA 2-4-69
- PARA 9-2-14
- PARA 9-4-34
- PARA 9-4-106

9-4-76.1 Replace Position Lights Flasher Unit.

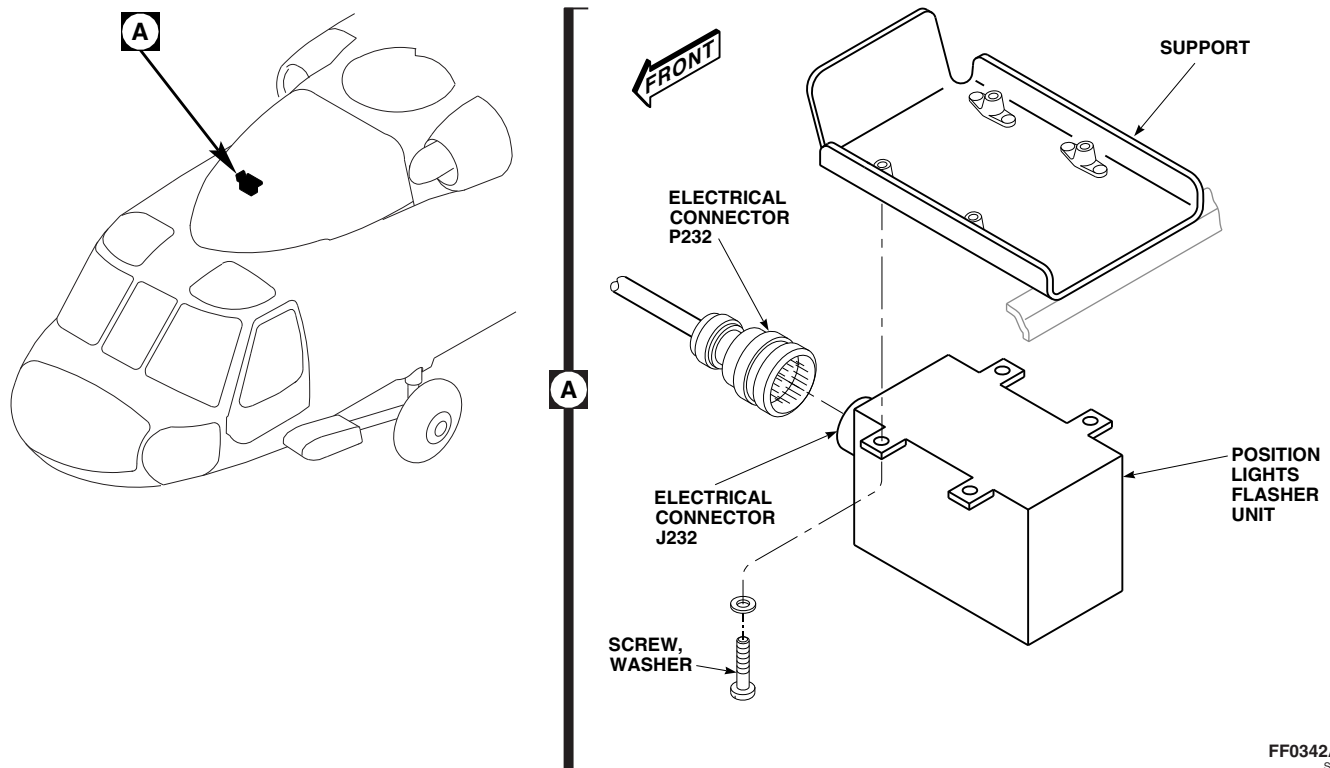
9-4-76.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Remove soundproofing panel (PARA 2-4-69).
- Disconnect electrical connector, P232 from J232, on position lights flasher unit at STA 285.0 (Figure 9-4-76-1, Detail A).
- Install protective caps and plugs on electrical connectors.
- Remove screws and washers securing position lights flasher unit to support.

- Remove position lights flasher unit.

9-4-76.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Install position lights flasher unit to support, at STA 285.0, and secure with screws and washers (Figure 9-4-76-1, Detail A).
- Remove protective caps and plugs from electrical connectors.
- Inspect and treat electrical connectors (PARA 1-7-20).
- Connect electrical connector, P232 to J232, to position lights flasher unit.
- Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).

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SA**FIGURE 9-4-76-1. REPLACE POSITION LIGHTS FLASHER UNIT**

- g. Perform operational check of position lights (PARA 9-2-14).
- h. Make sure area is clean and free of foreign material.
- i. Install soundproofing panel (PARA 2-4-69).

9-4-77 ESSS HORIZONTAL STORES SUPPORT (HSS) INCANDESCENT POSITION LIGHTS LAMPS AND LENSES **ESSS** .

PARAGRAPH BREAKDOWN

	PAGE
9-4-77.1 Replace ESSS Horizontal Stores Support (HSS) Incandescent Position Lights Lamps and Lenses	9-4-77-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit
- Maintenance Platform

References

- PARA 1-6-16
- PARA 9-2-14
- PARA 9-4-34
- PARA 9-4-106

9-4-77.1 Replace ESSS Horizontal Stores Support (HSS) Incandescent Position Lights Lamps and Lenses.

NOTE

Replacement of left and right horizontal stores support incandescent position light lamp and lens is the same.

9-4-77.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Position suitable maintenance platform next to incandescent position light.
- Remove screw and rubber washer securing lens to incandescent position light (Figure 9-4-77-1, Detail A).

- Remove lamp from incandescent position light.
- Remove and discard gasket from lens.

9-4-77.1.2 Install.

- Install lamp in incandescent position light (Figure 9-4-77-1, Detail A).
- Install gasket on lens.

NOTE

Right side lens is green. Left side lens is red.

- Install lens on incandescent position light and secure with rubber washer and screw.
- Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Perform operational check of position lights (PARA 9-2-14).

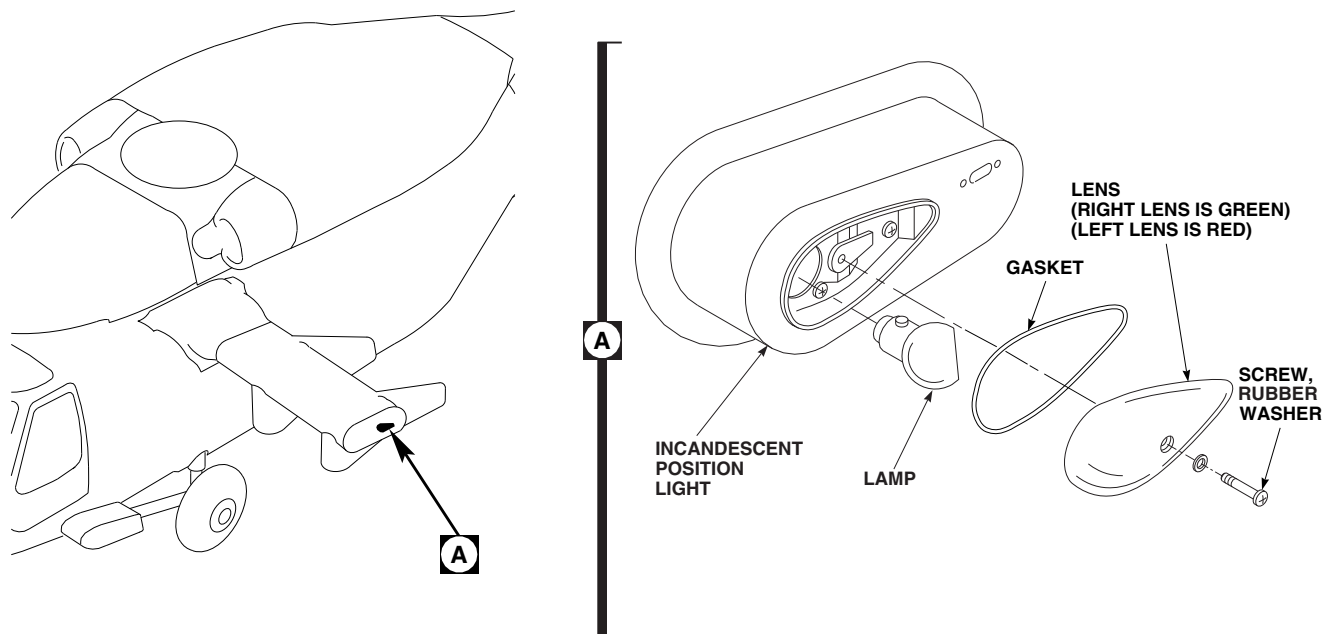
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FIGURE 9-4-77-1. REPLACE ESSS HORIZONTAL STORES SUPPORT (HSS) INCANDESCENT POSITION LIGHTS LAMPS AND LENSES

9-4-78 DRAG BEAM SUPPORT FAIRING INCANDESCENT POSITION LIGHTS.

PARAGRAPH BREAKDOWN

	PAGE
9-4-78.1 Replace Drag Beam Support Fairing Incandescent Position Lights	9-4-78-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

Materials/Parts

- Protective Caps and Plugs
- Sealing Compound, Item 292, Appendix D
- Tape, Item 317C, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-94
- PARA 9-2-14
- PARA 9-4-34
- PARA 9-4-106

9-4-78.1 Replace Drag Beam Support Fairing Incandescent Position Lights.

NOTE

Replacement of left or right drag beam support fairing incandescent position light is the same, except as noted.

9-4-78.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Remove drag beam support fairing (PARA 2-4-94).

- Remove screws, washers, and nuts securing electrical connector, J253 (right), or J254 (left), to bracket on inside of drag beam support fairing (Figure 9-4-78-1, Detail A).
- Install protective caps and plugs on electrical connectors.
- Remove tape securing wiring and incandescent position light (Figure 9-4-78-1, Detail B).
- Remove screws, washers, spacer, nuts, and clamps securing wiring to brackets (Figure 9-4-78-1, Detail A).
- Remove screw and rubber washer securing lens to incandescent position light.
- Remove lamp from incandescent position light.

- j. Remove screw, washers, lockwasher, and nut securing ground wire and incandescent position light to drag beam support fairing.
- k. Remove screw, washer, and nut securing incandescent position light to drag beam support fairing.
- l. Remove incandescent position light.
- m. Remove and discard gasket from lens.

9-4-78.1.2. Install.

- a. Turn off all electrical power (PARA 1-6-16).
 - b. Install incandescent position light to drag beam support fairing and secure with screw, washer, and nut (Figure 9-4-78-1, Detail A).
 - c. Apply sealing compound, Item 292, Appendix D, around exterior edge of incandescent position light and drag beam support fairing.
 - d. Install ground wire and secure ground wire and incandescent position light to drag beam support fairing with screw, washers, lockwasher, and nut.
 - e. Install lamp in incandescent position light.
 - f. Install gasket on lens.
- NOTE**
- Right side lens is green. Left side lens is red.
- g. Install lens on incandescent position light and secure with rubber washer and screw.
 - h. Remove protective caps and plugs from electrical connectors.
 - i. Inspect and treat electrical connectors (PARA 1-7-20).
 - j. Install electrical connector, J254 (left), or J253 (right), to bracket on inside of drag beam support fairing with screws, washers, and nuts.
 - k. Secure wiring to brackets with screws, washers, spacer, nuts, and clamps.
 - l. Using tape, Item 317C, Appendix D, wrap around body of lamp. Secure mounting bracket to bottom of lamp as required. Secure wiring with tape as required. **AFTER INSTALLATION, WIRE MUST NOT BE ALLOWED TO EXCEED 0.25-INCH** (Figure 9-4-78-1, Detail B).
 - m. Install drag beam support fairing (PARA 2-4-94).
 - n. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
 - o. Perform operational check of position lights (PARA 9-2-14).

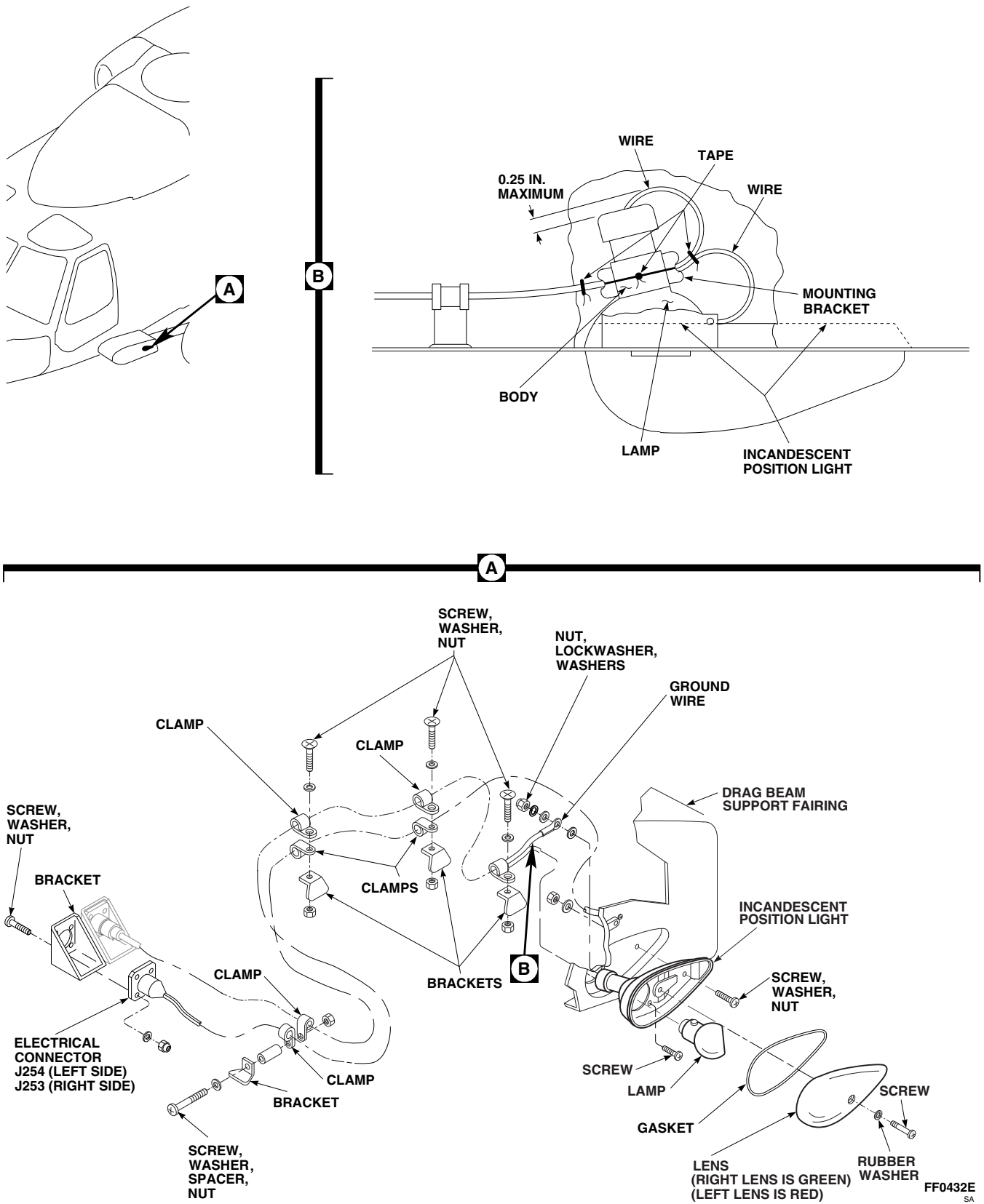


FIGURE 9-4-78-1. REPLACE DRAG BEAM SUPPORT FAIRING INCANDESCENT POSITION LIGHTS

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9-4-79 DRAG BEAM SUPPORT FAIRING INFRARED POSITION LIGHTS.

PARAGRAPH BREAKDOWN

	PAGE
9-4-79.1 Replace Drag Beam Support Fairing Infrared Position Lights	9-4-79-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

Materials/Parts

- Protective Caps and Plugs
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-94
- PARA 9-2-14
- PARA 9-4-34
- PARA 9-4-106

9-4-79.1 Replace Drag Beam Support Fairing Infrared Position Lights.

NOTE

Replacement of left or right drag beam support fairing infrared position light is the same, except as noted.

9-4-79.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Remove drag beam support fairing (PARA 2-4-94).
- Remove screws and nuts securing electrical connector, J340 (left), or J341 (right), to

bracket on inside of drag beam support fairing (Figure 9-4-79-1, Detail A).

- Install protective caps and plugs on electrical connectors.
- Remove screws, washers, spacer, nuts, and clamps securing wiring to brackets.
- Remove screws and washers securing infrared position light to drag beam support fairing.
- Remove infrared position light.

9-4-79.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Install infrared position light to drag beam support fairing and secure with screws and washers (Figure 9-4-79-1, Detail A).



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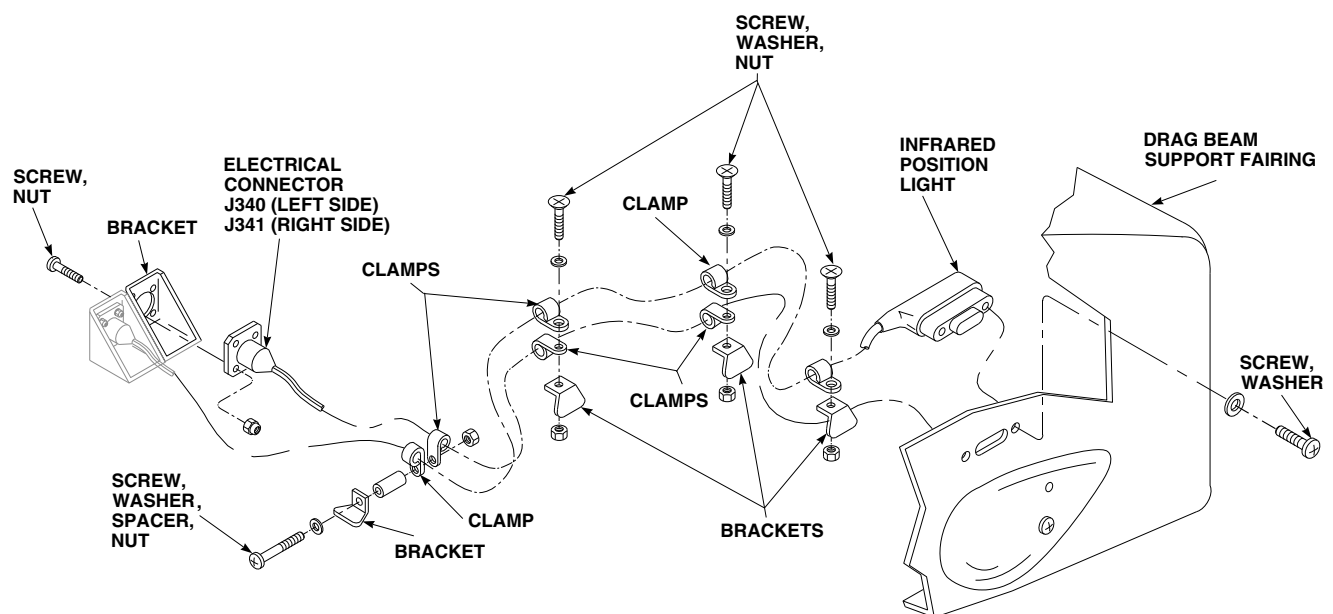
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FIGURE 9-4-79-1. REPLACE DRAG BEAM SUPPORT FAIRING INFRARED POSITION LIGHTS

9-4-79-2

- c. Apply sealing compound, Item 292, Appendix D, around exterior edge of infrared position light and drag beam support fairing.
- d. Remove protective caps and plugs from electrical connectors.
- e. Inspect and treat electrical connectors (PARA 1-7-20).
- f. Install electrical connector, J340 (left), or J341 (right), to bracket on inside of drag beam support fairing with screws and nuts.
- g. Secure wiring to brackets with screws, washers, spacer, nuts, and clamps.
- h. Install drag beam support fairing (PARA 2-4-94).
- i. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- j. Perform operational check of position lights (PARA 9-2-14).

9-4-80 **ESSS HORIZONTAL STORES SUPPORT (HSS) INCANDESCENT POSITION LIGHTS** **ESSS** .

PARAGRAPH BREAKDOWN

		PAGE
9-4-80.1	Replace ESSS Horizontal Stores Support (HSS) Incandescent Position Lights	9-4-80-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit
- Maintenance Platform
- Nonmetallic Scraper, Locally-Made

Materials/Parts

- Methyl Ethyl Ketone, Item 217, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 9-2-14
- PARA 9-4-34
- PARA 9-4-77
- PARA 9-4-106
- PARA 16-4-21

9-4-80.1 **Replace ESSS Horizontal Stores Support (HSS) Incandescent Position Lights.**

NOTE

Replacement of left and right horizontal stores support incandescent position lights is the same.

9-4-80.1.1 **Remove.**

- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Position suitable maintenance platform next to incandescent position light.

- Remove horizontal stores support tip cap (PARA 16-4-21).
- Remove horizontal stores support incandescent position light lamp and lens (PARA 9-4-77).
- Using nonmetallic scraper, remove sealing compound from edges of incandescent position light.
- Remove screws, washers, lockwasher, and nuts securing ground wires and incandescent position light to horizontal stores support tip cap (Figure 9-4-80-1, Detail A).
- Remove incandescent position light.

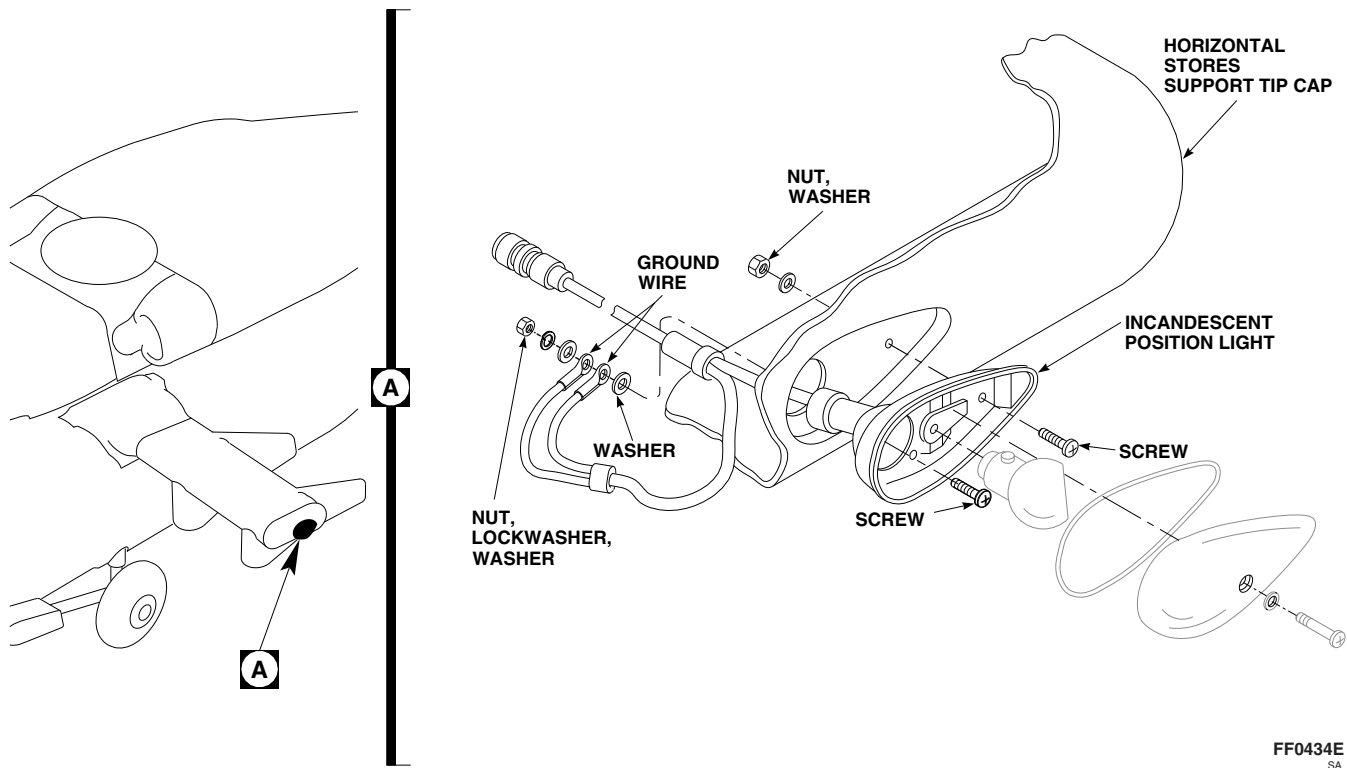
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FIGURE 9-4-80-1. REPLACE ESSS HORIZONTAL STORES SUPPORT (HSS) INCANDESCENT POSITION LIGHTS

- h. Using methyl ethyl ketone, Item 217, Appendix D, remove sealing compound.

9-4-80.1.2 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Make sure area is clean and free of foreign material.
- c. Apply sealing compound, Item 292, Appendix D, around exterior edge of incandescent position light and horizontal stores support tip cap.
- d. Install ground wires and incandescent position light to horizontal stores support tip cap with screws, washers, lockwasher, and nuts (Figure 9-4-80-1, Detail A).
- e. Install horizontal stores support incandescent position light lamp and lens (PARA 9-4-77).
- f. Make sure area is clean and free of foreign material.
- g. Install horizontal stores support tip cap (PARA 16-4-21).
- h. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- i. Perform operational check of position lights (PARA 9-2-14).

9-4-81 **ESSS HORIZONTAL STORES SUPPORT (HSS) INFRARED POSITION LIGHTS** **ESSS** .

PARAGRAPH BREAKDOWN

		PAGE
9-4-81.1	Replace ESSS Horizontal Stores Support (HSS) Infrared Position Lights	9-4-81-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit
- Maintenance Platform
- Nonmetallic Scraper, Locally-Made

Materials/Parts

- Methyl Ethyl Ketone, Item 217, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 9-2-14
- PARA 9-4-34
- PARA 9-4-106
- PARA 16-4-21

9-4-81.1 **Replace ESSS Horizontal Stores Support (HSS) Infrared Position Lights.**

NOTE

Replacement of left and right horizontal stores support infrared position lights is the same, except as noted.

9-4-81.1.1 **Remove.**

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Position suitable maintenance platform next to horizontal stores support position light.

- Remove horizontal stores support tip cap (PARA 16-4-21).
- Using nonmetallic scraper, remove sealing compound from edges of infrared position light.
- Remove screws, washers, and nuts securing infrared position light to horizontal stores support tip cap (Figure 9-4-81-1, Detail A).
- Remove infrared position light.
- Using methyl ethyl ketone, Item 217, Appendix D, remove sealing compound.

9-4-81.1.2 **Install.**

- Turn off all electrical power (PARA 1-6-16).

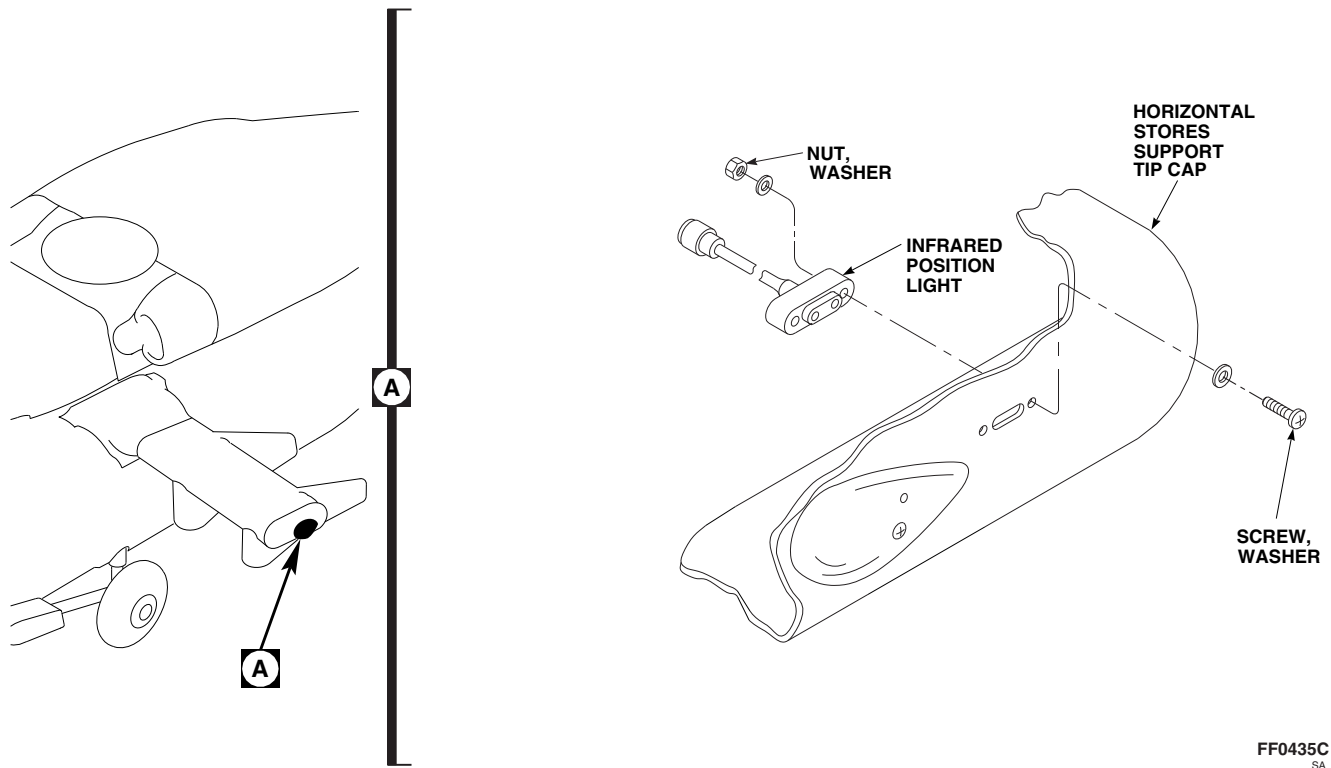


FIGURE 9-4-81-1. REPLACE ESSS HORIZONTAL STORES SUPPORT (HSS) INFRARED POSITION LIGHTS

- b. Install infrared position light to horizontal stores support tip cap and secure with screws, washers, and nuts (Figure 9-4-81-1, Detail A).
- c. Apply sealing compound, Item 292, Appendix D, around exterior edge of infrared position light and horizontal stores support tip cap.
- d. Make sure area is clean and free of foreign material.
- e. Install ESSS horizontal stores support (HSS) tip cap (PARA 16-4-21).
- f. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- g. Perform operational check of position lights (PARA 9-2-14).

9-4-82 TAIL INCANDESCENT POSITION LIGHT LAMP AND LENS.

PARAGRAPH BREAKDOWN

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9-4-82.1 Replace Tail Incandescent Position Light Lamp and Lens	9-4-82-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit
- Maintenance Platform

References

- PARA 1-6-16
- PARA 9-2-14
- PARA 9-4-34
- PARA 9-4-106

9-4-82.1 Replace Tail Incandescent Position Light Lamp and Lens.

9-4-82.1.1. Remove.

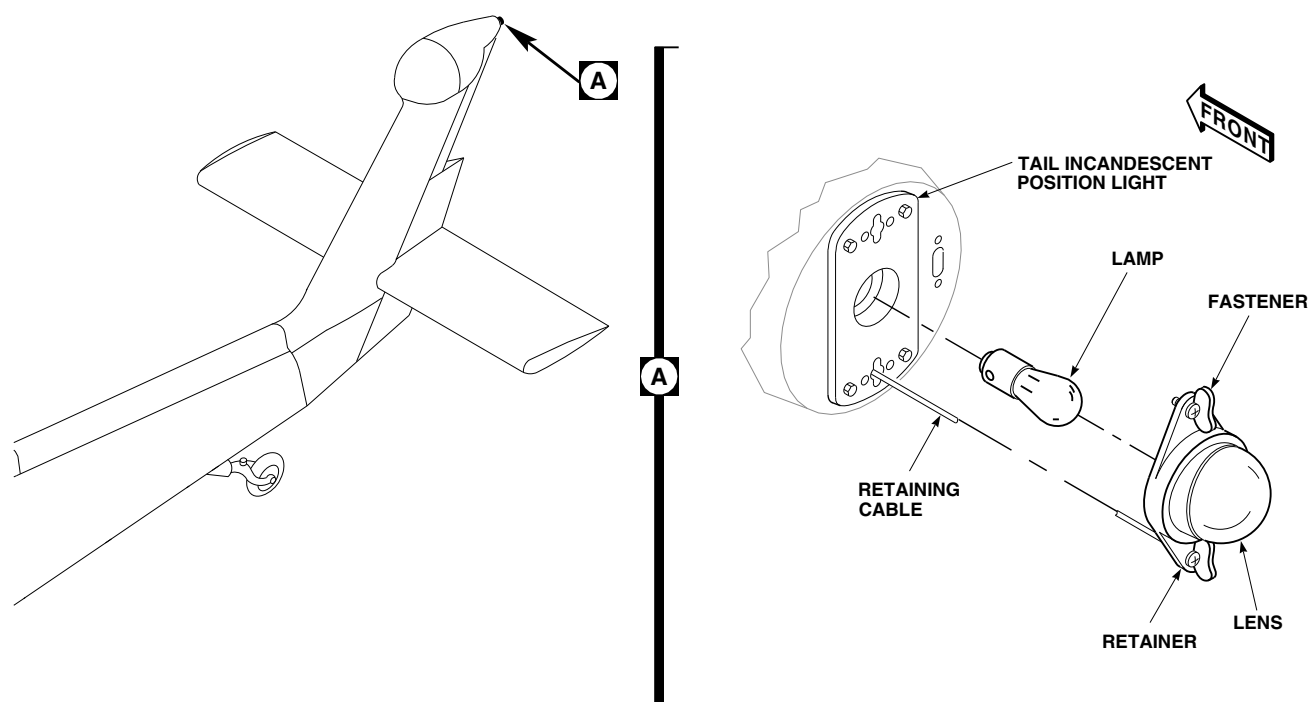
- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Position suitable maintenance platform next to tail incandescent position light.
- Loosen fasteners securing lens and retainer to tail incandescent position light (Figure 9-4-82-

1, Detail A). Let retaining cable support retainer.

- Remove lamp.

9-4-82.1.2. Install.

- Install lamp (Figure 9-4-82-1, Detail A).
- Install lens and retainer on tail incandescent position light and secure with fasteners.
- Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Perform operational check of position lights (PARA 9-2-14).

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SA**FIGURE 9-4-82-1. REPLACE TAIL INCANDESCENT POSITION LIGHT LAMP AND LENS**

9-4-83 TAIL INCANDESCENT POSITION LIGHT.

PARAGRAPH BREAKDOWN

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9-4-83.1 Replace Tail Incandescent Position Light	9-4-83-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit
- Maintenance Platform

Materials/Parts

- Tags

References

- PARA 1-6-16
- PARA 9-2-14
- PARA 9-4-34
- PARA 9-4-106

9-4-83.1 Replace Tail Incandescent Position Light.

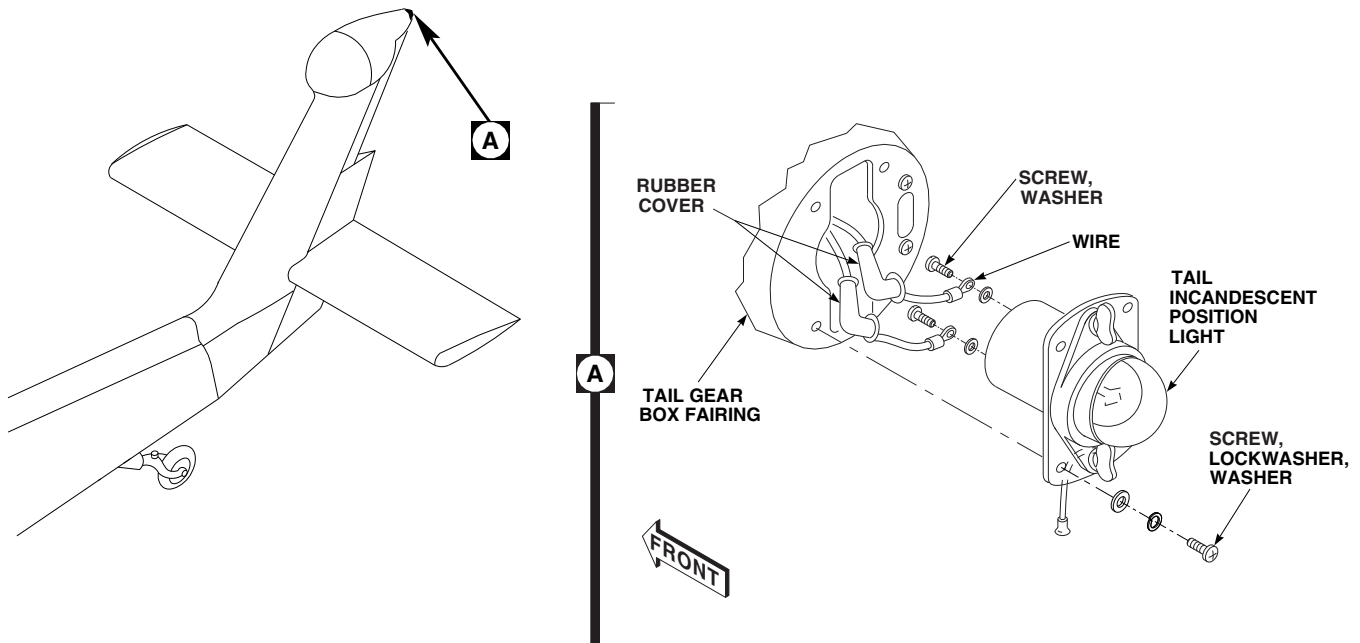
9-4-83.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Position suitable maintenance platform next to tail incandescent position light.
- Remove screws, lockwashers, and washers securing tail incandescent position light to tail gear box fairing (Figure 9-4-83-1, Detail A).
- Carefully pull tail incandescent position light out and slide rubber covers back to gain access to wire connections.
- Tag wires. Remove screws and washers securing wires to tail incandescent position light.

- Remove tail incandescent position light.

9-4-83.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Install wires to tail incandescent position light and secure with screws and washers. Remove tags (Figure 9-4-83-1, Detail A).
- Slide rubber covers securely over wire connections.
- Make sure area is clean and free of foreign material.
- Install tail incandescent position light to tail gear box fairing and secure with screws, lockwashers, and washers.
- Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Perform operational check of position lights (PARA 9-2-14).



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FIGURE 9-4-83-1. REPLACE TAIL INCANDESCENT POSITION LIGHT

9-4-84 TAIL INFRARED POSITION LIGHT.

PARAGRAPH BREAKDOWN

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9-4-84.1	Replace Tail Infrared Position Light	9-4-84-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit
- Maintenance Platform

Materials/Parts

- Protective Caps and Plugs

References

- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-132
- PARA 9-2-14
- PARA 9-4-34
- PARA 9-4-106

9-4-84.1 Replace Tail Infrared Position Light.

9-4-84.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Position suitable maintenance platform next to tail incandescent position light.
- Remove screws and washers securing tail infrared position light to tail gear box fairing (Figure 9-4-84-1, Detail A).
- Remove front tail gear box fairing (PARA 2-4-132).
- Disconnect electrical connector, P616 from J616.

- Install protective caps and plugs on electrical connectors.

- Remove tail infrared position light.

9-4-84.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Install tail infrared position light to tail gear box fairing and secure with screws and washers (Figure 9-4-84-1, Detail A).
- Remove protective caps and plugs from electrical connectors.
- Inspect and treat electrical connectors (PARA 1-7-20).
- Connect electrical connector, P616 to J616.
- Install front tail gear box fairing (PARA 2-4-132).

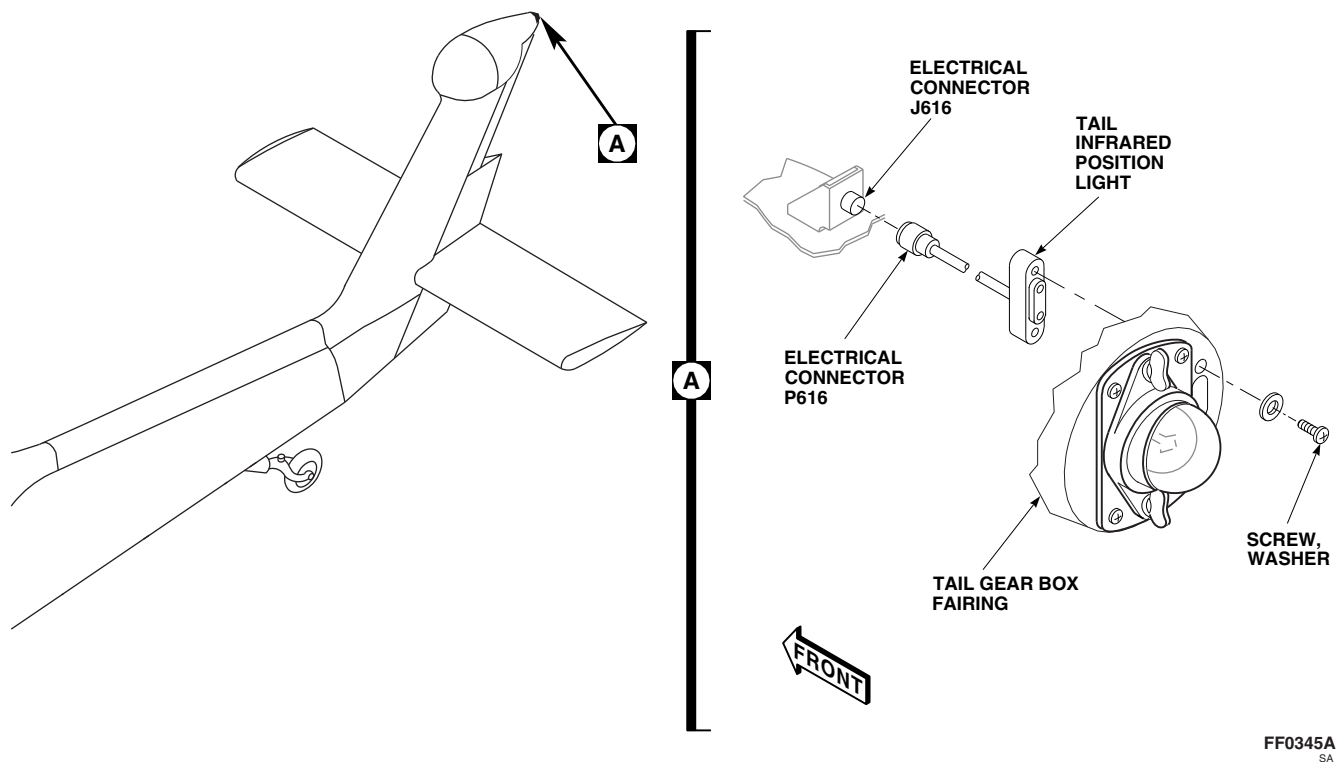


FIGURE 9-4-84-1. REPLACE TAIL INFRARED POSITION LIGHT

- g. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- h. Perform operational check of position lights (PARA 9-2-14).

9-4-85 ANTICOLLISION LIGHT RED AND WHITE SUBASSEMBLIES.

PARAGRAPH BREAKDOWN

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9-4-85.1 Replace Anticollision Light Red and White Subassemblies	9-4-85-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit
- Maintenance Platform

Materials/Parts

- Lockwire, Item 193, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 9-2-16
- PARA 9-4-34
- PARA 9-4-106

9-4-85.1 Replace Anticollision Light Red
and White Subassemblies.

NOTE

Replacement of red or white subas-
sembly on tail gear box fairing and tail
cone anticollision lights is the same,
except as noted.

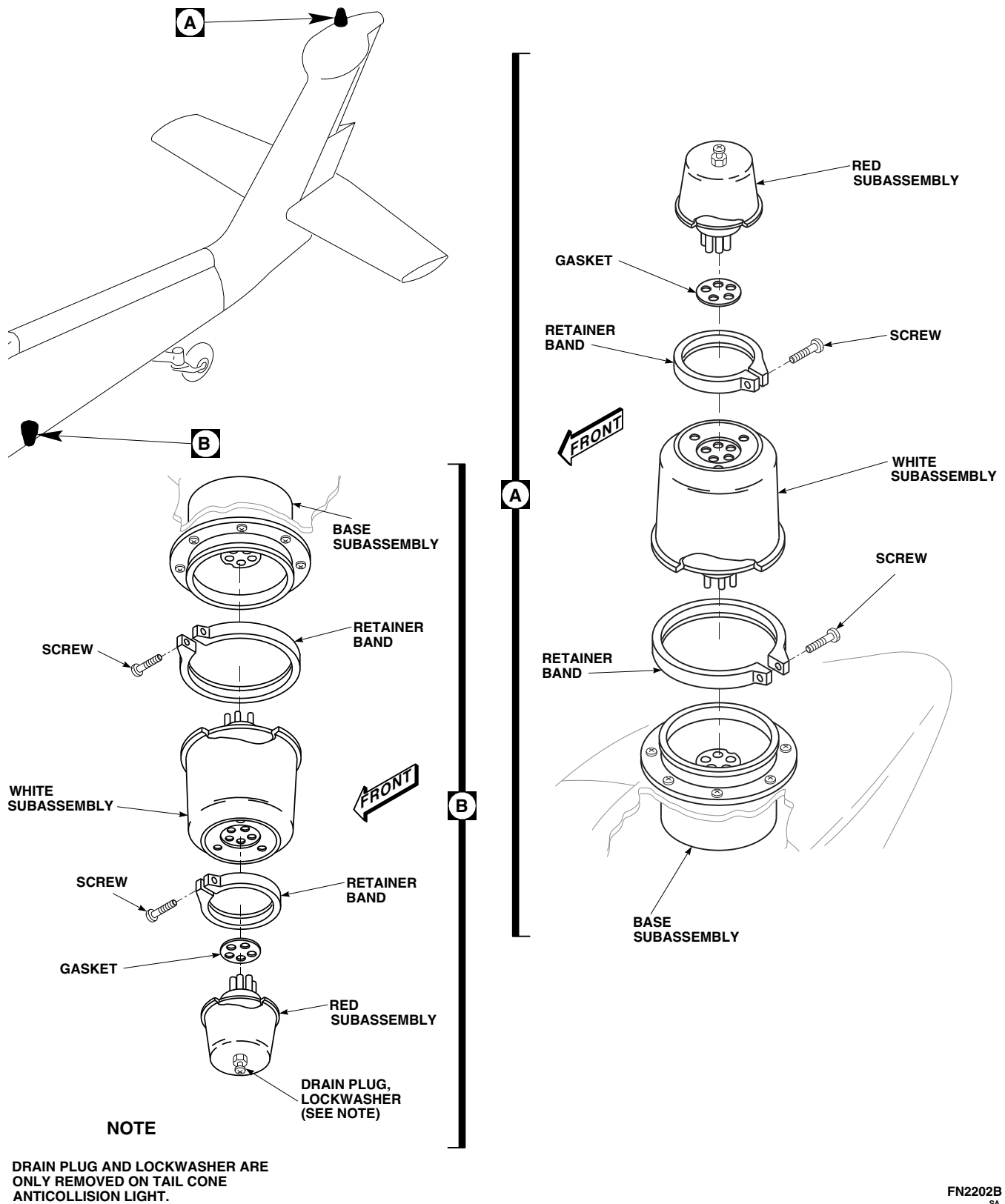
9-4-85.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- If replacing red and/or white subassemblies on tail pylon, position suitable maintenance platform next to anticollision light.
- Remove screw and retainer band securing red subassembly to white subassembly (Figure 9-4-85-1, Detail A or Detail B).

- Remove red subassembly from white subassembly.
- Remove and inspect gasket beneath red subassembly. If damaged, replace gasket.
- Remove screw and retainer band securing white subassembly to base subassembly.
- Remove white subassembly from base subassembly.

9-4-85.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Install white subassembly to base subassembly and secure with retainer band and screw (Figure 9-4-85-1, Detail A or Detail B).
- Lockwire, Item 193, Appendix D, screw.
- For anticollision light located on tail gear box fairing, install gasket on top of white subassembly.

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- sembly. For anticollision light located on tail cone, install gasket on red subassembly.
- e. Prior to installing red subassembly on tail cone anticollision light, perform the following:
 - (1) Remove drain plug and lockwasher from red subassembly (Figure 9-4-85-1, Detail B).
 - (2) Drain water, if any.
 - (3) Install drain plug and lockwasher in red subassembly.
 - f. Install red subassembly on white subassembly and secure with retainer band and screw (Figure 9-4-85-1, Detail A and Detail B).
 - g. Lockwire, Item 193, Appendix D, screw.
 - h. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
 - i. Perform operational check of anticollision lights (PARA 9-2-16).

9-4-86 **ANTICOLLISION LIGHTS.**

PARAGRAPH BREAKDOWN

		PAGE
9-4-86.1	Replace Anticollision Lights	9-4-86-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit
- Maintenance Platform

Materials/Parts

- Dry-Cleaning Solvent, Item 124, Appendix D
- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs

- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 9-2-16
- PARA 9-4-34
- PARA 9-4-106

9-4-86.1 **Replace Anticollision Lights.**

NOTE

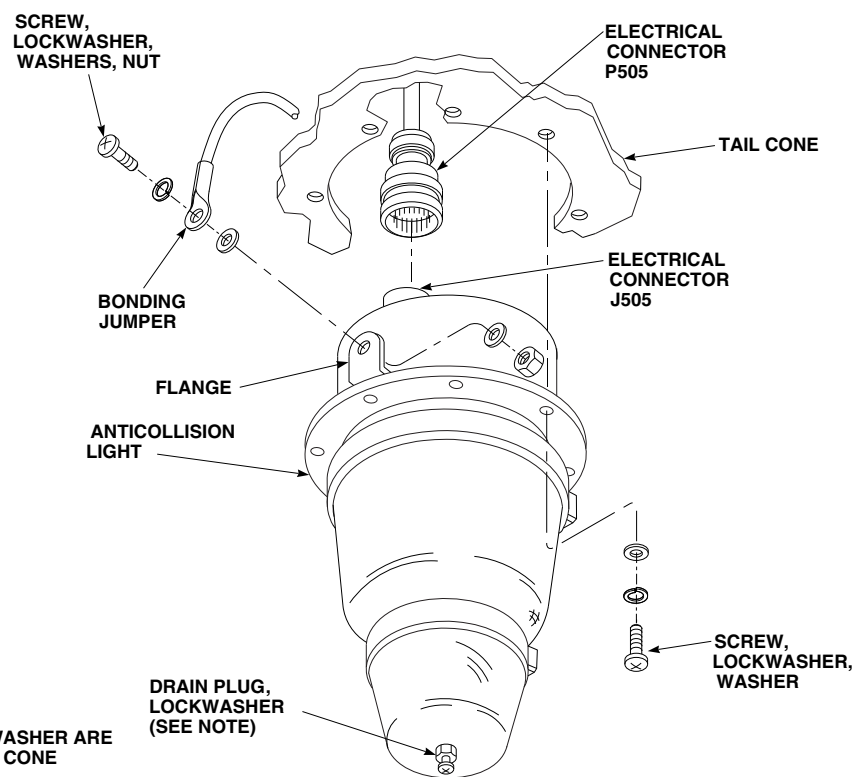
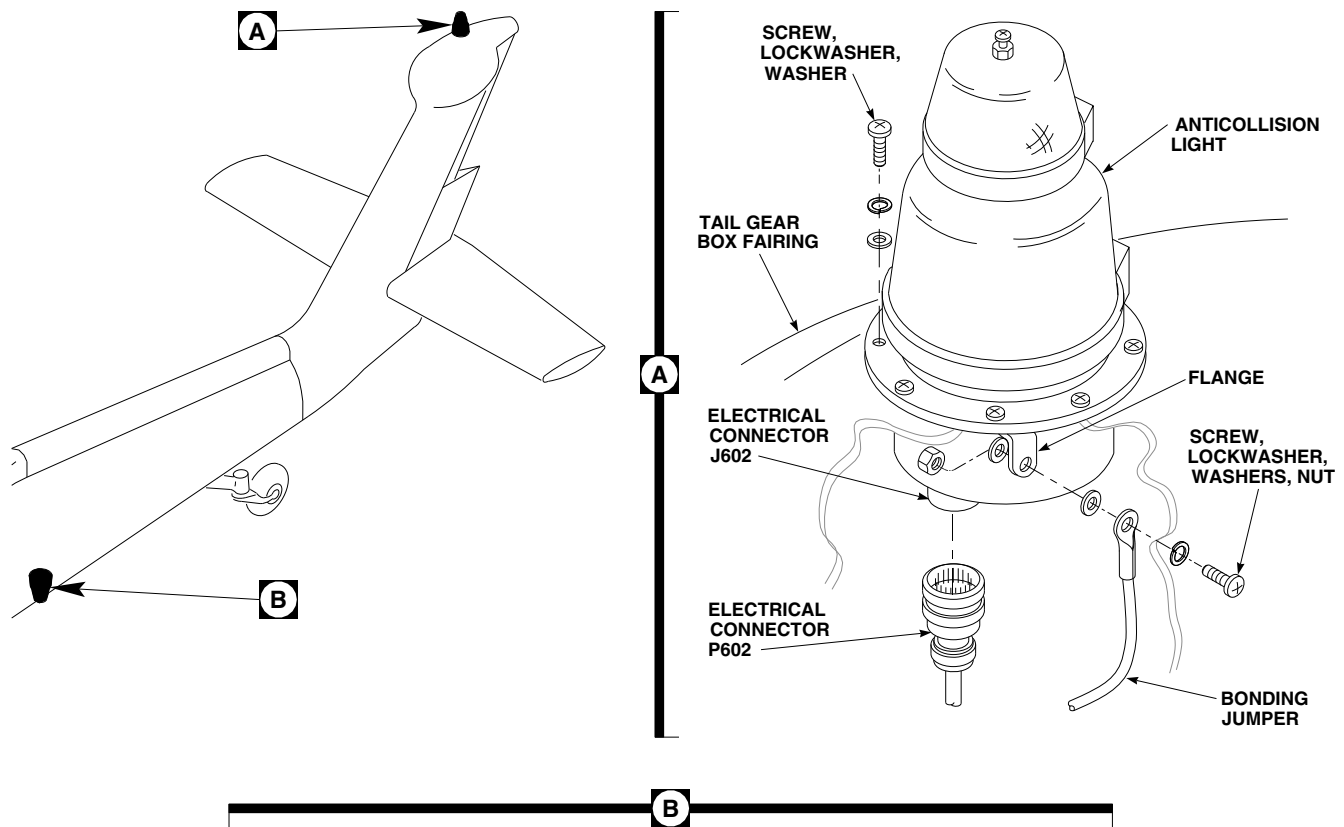
Replacement of both anticollision lights is the same, except as noted.

9-4-86.1.1. **Remove.**

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- If replacing anticollision light on tail gear box fairing, position suitable maintenance platform next to anticollision light.
- Remove screws, lockwashers, and washers securing anticollision light to tail gear box fair-

ing or tail cone (Figure 9-4-86-1, Detail A or Detail B).

- Carefully raise or lower anticollision light to gain access to bonding jumper and electrical connector.
- Remove screw, lockwasher, washers, and nut securing bonding jumper to flange.
- Disconnect electrical connector, P602 from J602 (tail gear box fairing), or P505 from J505 (tail cone), from anticollision light.
- Install protective caps and plugs on electrical connectors.
- Remove anticollision light.

**NOTE**

DRAIN PLUG AND LOCKWASHER ARE ONLY REMOVED ON TAIL CONE ANTICOLLISION LIGHT.

FIGURE 9-4-86-1. REPLACE ANTICOLLISION LIGHTS

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9-4-86.1.2. Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Using machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D, clean surface around all mounting holes and flange. Using clean, dry machinery towel, Item 211, Appendix D, dry surface.
- c. Install bonding jumper to flange and secure with screw, lockwasher, washers, and nut (Figure 9-4-86-1, Detail A or Detail B).
- d. Seal bonded connection with sealing compound, Item 292, Appendix D.
- e. Remove protective caps and plugs from electrical connectors.
- f. Inspect and treat electrical connectors (PARA 1-7-20).
- g. Connect electrical connector, P602 to J602 (tail gear box fairing), and P505 to J505 (tail cone), to anticollision light.
- h. Make sure area is clean and free of foreign material.
- i. Prior to installing red subassembly on tail cone anticollision light, perform the following:
 - (1) Remove drain plug and lockwasher from red subassembly (Figure 9-4-86-1, Detail B).
 - (2) Drain water, if any.
 - (3) Install drain plug and lockwasher in red subassembly.
- j. Line up mounting holes and install anticollision light and secure with screws, lockwashers, and washers (Figure 9-4-86-1, Detail A or Detail B).
- k. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- l. Perform operational check of anticollision lights (PARA 9-2-16).

9-4-87 ANTICOLLISION LIGHT BASE SUBASSEMBLIES.

PARAGRAPH BREAKDOWN

		PAGE
9-4-87.1	Replace Anticollision Light Base Subassemblies	9-4-87-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit
- Maintenance Platform

Materials/Parts

- Dry-Cleaning Solvent, Item 124, Appendix D
- Lockwire, Item 193, Appendix D
- Machinery Towel, Item 211, Appendix D

- Protective Caps and Plugs
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 9-2-16
- PARA 9-4-34
- PARA 9-4-106

9-4-87.1 Replace Anticollision Light Base Subassemblies.

NOTE

Replacement of both anticollision light base subassemblies is the same, except as noted.

9-4-87.1.1. Remove.

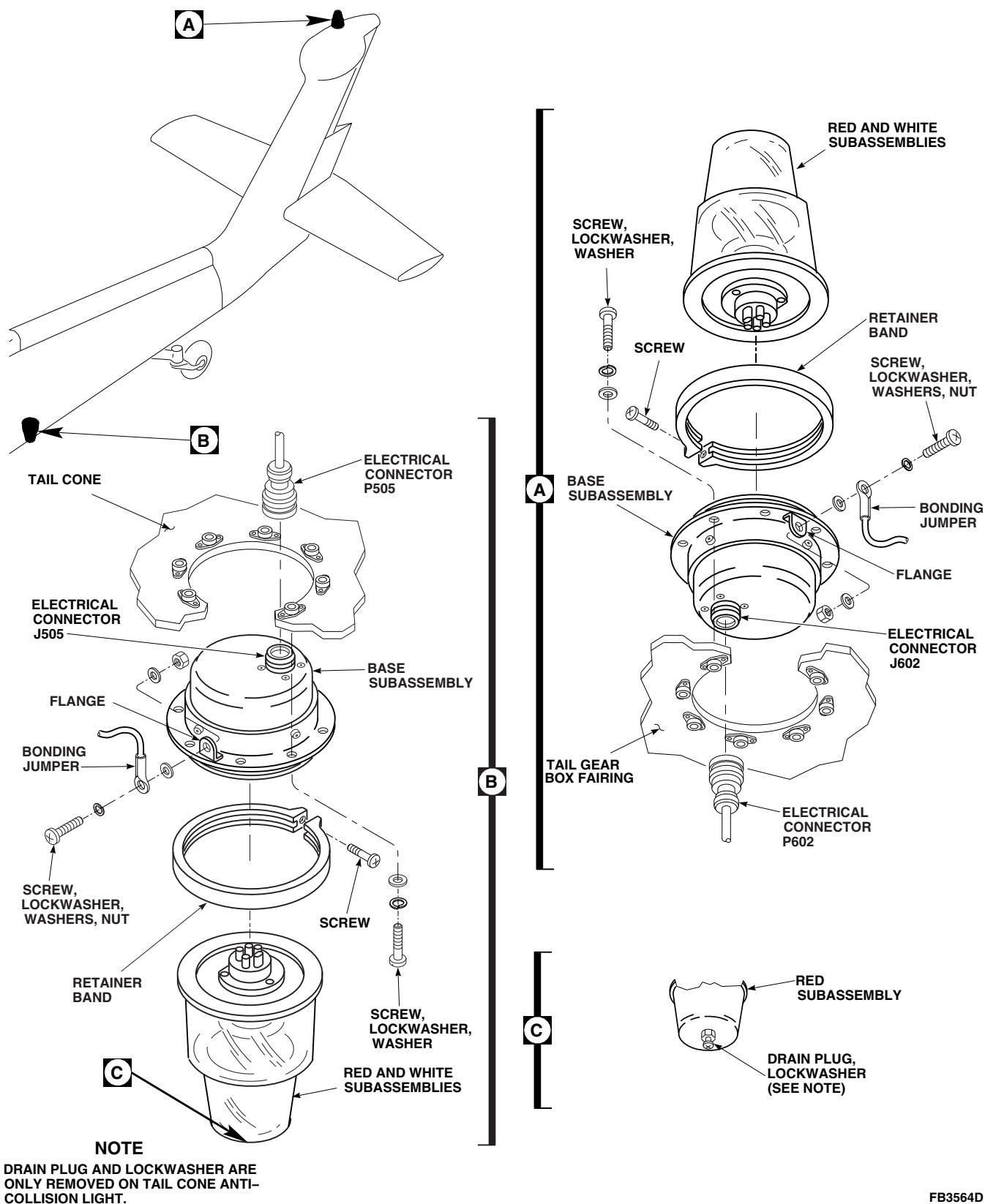
- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- If replacing base subassembly on the tail gear box fairing, position suitable maintenance platform next to anticollision light.

- Remove screw and retainer band securing red and white subassemblies to base subassembly. Remove red and white subassemblies (Figure 9-4-87-1, Detail A or Detail B).
- Remove screws, lockwashers, and washers securing base subassembly to tail gear box fairing or tail cone.
- Carefully raise or lower base subassemblies to gain access to bonding jumper and electrical connector.
- Remove screw, lockwasher, washers, and nut securing bonding jumper to flange.
- Disconnect electrical connector, P602 from J602 (tail gear box fairing), or P505 from J505 (tail cone), from base subassembly.

- i. Remove base subassembly.
- j. Install protective caps and plugs on electrical connectors.

9-4-87.1.2. Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Using machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D, clean surface around all mounting holes and flange. Using clean, dry machinery towel, Item 211, Appendix D, dry surface.
- c. Install bonding jumper to flange and secure with screw, lockwasher, washers, and nut (Figure 9-4-87-1, Detail A or Detail B).
- d. Seal bonded connection with sealing compound, Item 292, Appendix D.
- e. Remove protective caps and plugs from electrical connectors.
- f. Inspect and treat electrical connectors (PARA 1-7-20).
- g. Connect electrical connector, P602 to J602 (tail gear box fairing), and P505 to J505 (tail cone), to base subassembly.
- h. Make sure area is clean and free of foreign material.
- i. Line up mounting holes and install base subassembly on tail gear box fairing or tail cone and secure with screws, lockwashers, and washers.
- j. Prior to installing red and white subassemblies on tail cone anticollision light, perform the following:
 - (1) Remove drain plug and lockwasher from red subassembly (Figure 9-4-87-1, Detail C).
 - (2) Drain water, if any.
 - (3) Install drain plug and lockwasher in red subassembly.
- k. Install red and white subassemblies on base subassembly and secure with retainer band and screw (Figure 9-4-87-1, Detail A or Detail B).
- l. Lockwire, Item 193, Appendix D, screw.
- m. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- n. Perform operational check of anticollision light (PARA 9-2-16).



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FIGURE 9-4-87-1. REPLACE ANTICOLLISION LIGHT BASE SUBASSEMBLIES

9-4-88 ANTICOLLISION LIGHTS POWER SUPPLY UNIT.

PARAGRAPH BREAKDOWN

	PAGE
9-4-88.1 Replace Anticollision Lights Power Supply Unit	9-4-88-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

Materials/Parts

- Protective Caps and Plugs
- Tags

References

- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-45
- PARA 2-4-69
- PARA 9-2-16
- PARA 9-4-34
- PARA 9-4-106

9-4-88.1 Replace Anticollision Lights Power Supply Unit.

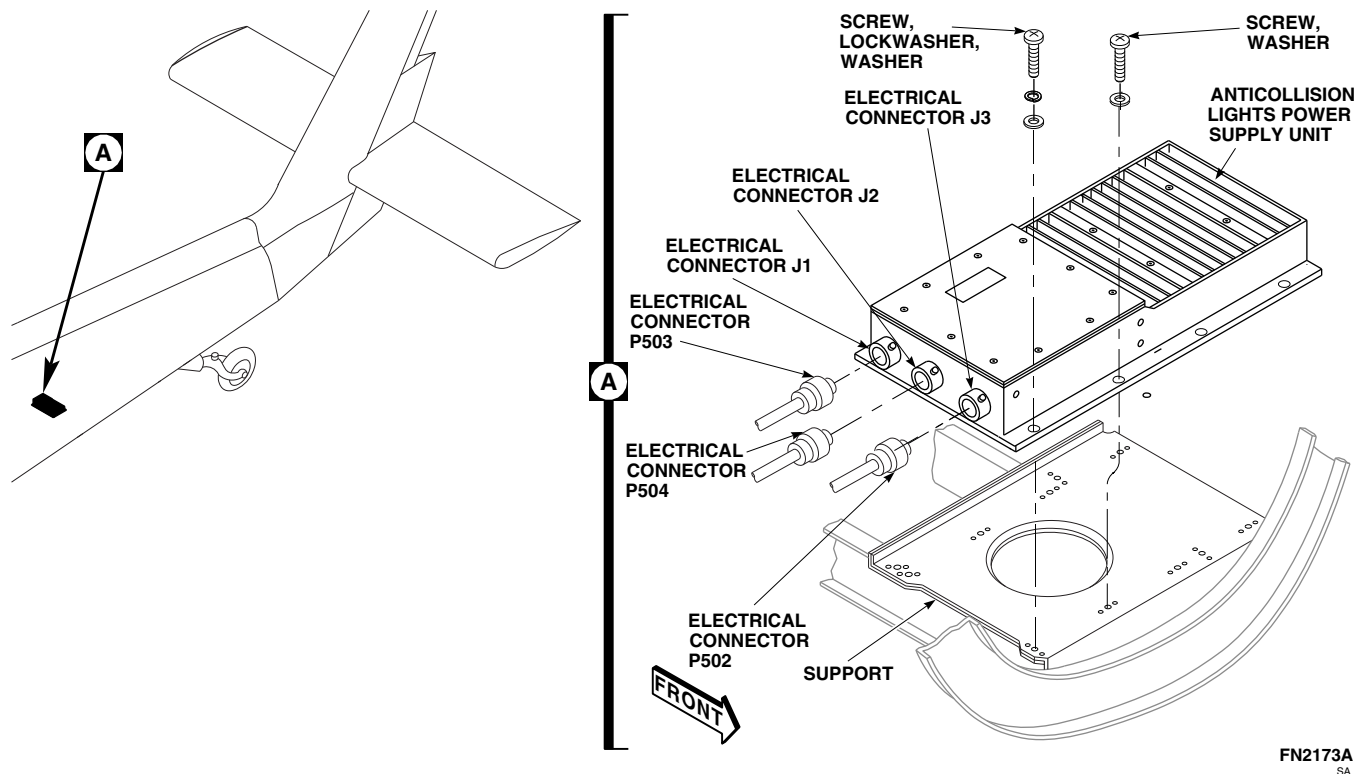
9-4-88.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Remove rear passenger’s seats as required (PARA 2-4-45).
- Remove soundproofing panel from rear bulkhead covering left or right stowage compartment (PARA 2-4-69).
- Remove cargo netting from stowage compartment.

- Tag electrical connectors. Disconnect electrical connectors from anticollision lights power supply unit as follows (Figure 9-4-88-1, Detail A):
 - P502 from J3.
 - P503 from J1.
 - P504 from J2.
- Install protective caps and plugs on electrical connectors.
- Remove screws, lockwashers, and washers securing anticollision lights power supply unit to support.
- Remove anticollision lights power supply unit.

9-4-88.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).

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SA**FIGURE 9-4-88-1. REPLACE ANTICOLLISION LIGHTS POWER SUPPLY UNIT**

- b. Install anticollision lights power supply unit on support and secure with screws, lockwashers, and washers (Figure 9-4-88-1, Detail A).
- c. Remove protective caps and plugs from electrical connectors.
- d. Inspect and treat electrical connectors (PARA 1-7-20).
- e. Connect electrical connectors to anticollision lights power supply unit as follows, then remove tags:
 - (1) P502 to J3.
 - (2) P503 to J1.
 - (3) P504 to J2.
- f. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- g. Perform operational check of anticollision lights (PARA 9-2-16).
- h. Make sure area is clean and free of foreign material.
- i. Install cargo netting.
- j. Install soundproofing panel (PARA 2-4-69).
- k. Install rear passenger's seats (PARA 2-4-45).

9-4-89 RETRACTABLE LANDING LIGHT LAMP.

PARAGRAPH BREAKDOWN

	PAGE
9-4-89.1 Replace Retractable Land- ing Light Lamp	9-4-89-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

Materials/Parts

- Tags

References

- PARA 1-6-16
- PARA 9-2-15
- PARA 9-4-34
- PARA 9-4-106

9-4-89.1 Replace Retractable Landing Light Lamp.

9-4-89.1.1 Remove.

- Turn on all electrical power (PARA 1-6-16).
- Make sure LIGHTS RETR LDG CONT circuit breaker (NO. 1 DC PRI BUS - copilot’s circuit breaker panel) is pushed in.
- Set pilot’s collective stick grip LDG LT switch to EXT, to fully extend retractable landing light.

WARNING

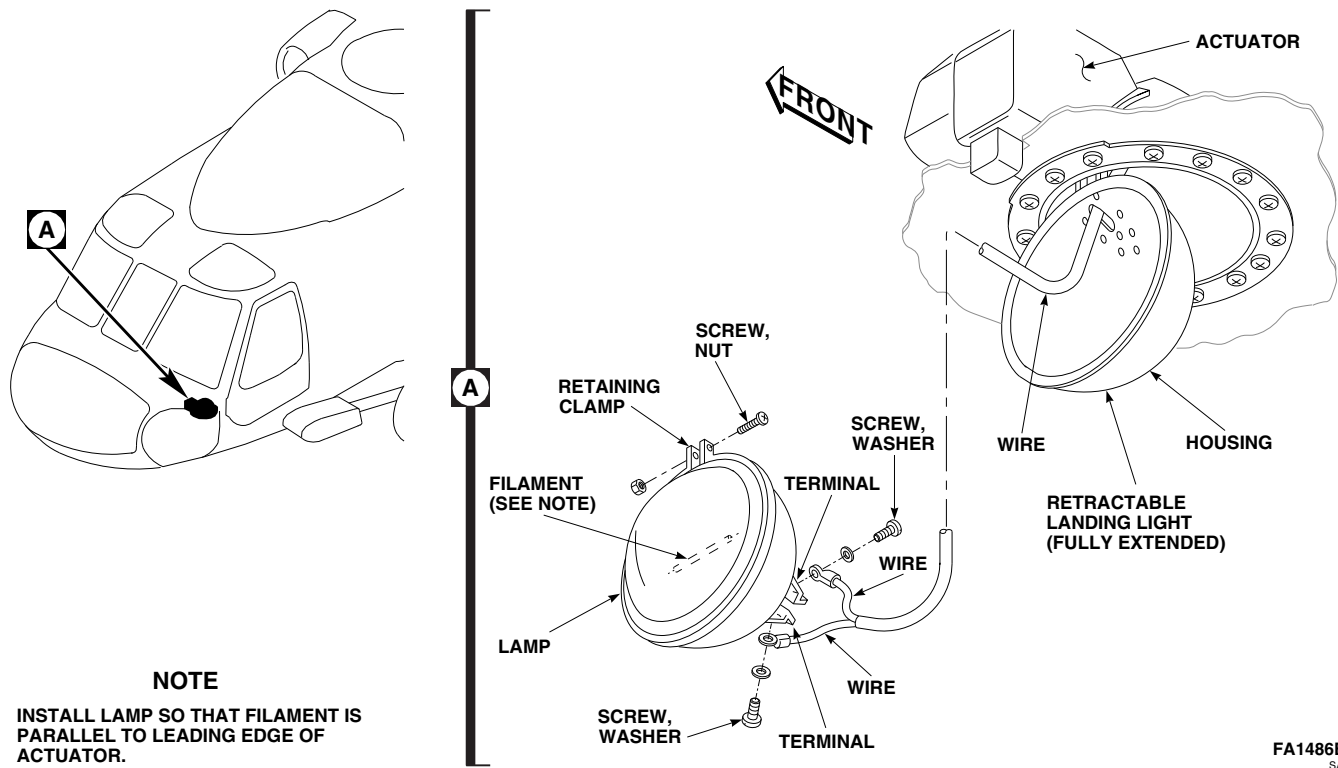
Lamp is hot when on. Never handle a lamp that may have power connected. Make sure lamp is cool before handling.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).

- Remove screw and nut securing retaining clamp to lamp (Figure 9-4-89-1, Detail A).
- Carefully lower lamp from housing to gain access to wires.
- Tag wires. Remove screws and washers securing wires to terminals on lamp.
- Remove lamp.

9-4-89.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).
- Install wires to terminal on lamp and secure with screws and washers (Figure 9-4-89-1, Detail A). Remove tags.
- Position lamp in housing with filament parallel to leading edge of actuator.
- Install retaining clamp on lamp and secure with screw and nut.
- Make sure area is clean and free of foreign material.

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f. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).

g. Perform operational check of retractable landing light (PARA 9-2-15).

9-4-90 RETRACTABLE LANDING LIGHT.

PARAGRAPH BREAKDOWN

	PAGE
9-4-90.1 Replace Retractable Land- ing Light	9-4-90-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

Materials/Parts

- Protective Caps and Plugs
- Sealing Compound, Item 301A, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 9-2-15
- PARA 9-4-34
- PARA 9-4-89
- PARA 9-4-106

9-4-90.1 Replace Retractable Landing Light.

9-4-90.1.1 Remove.

- Turn on all electrical power (PARA 1-6-16).
- On copilot’s circuit breaker panel, make sure LIGHTS RETR LDG CONT circuit breaker is pushed in.
- Set pilot’s collective stick grip LDG LT switch to EXT, to fully extend retractable landing light.

WARNING

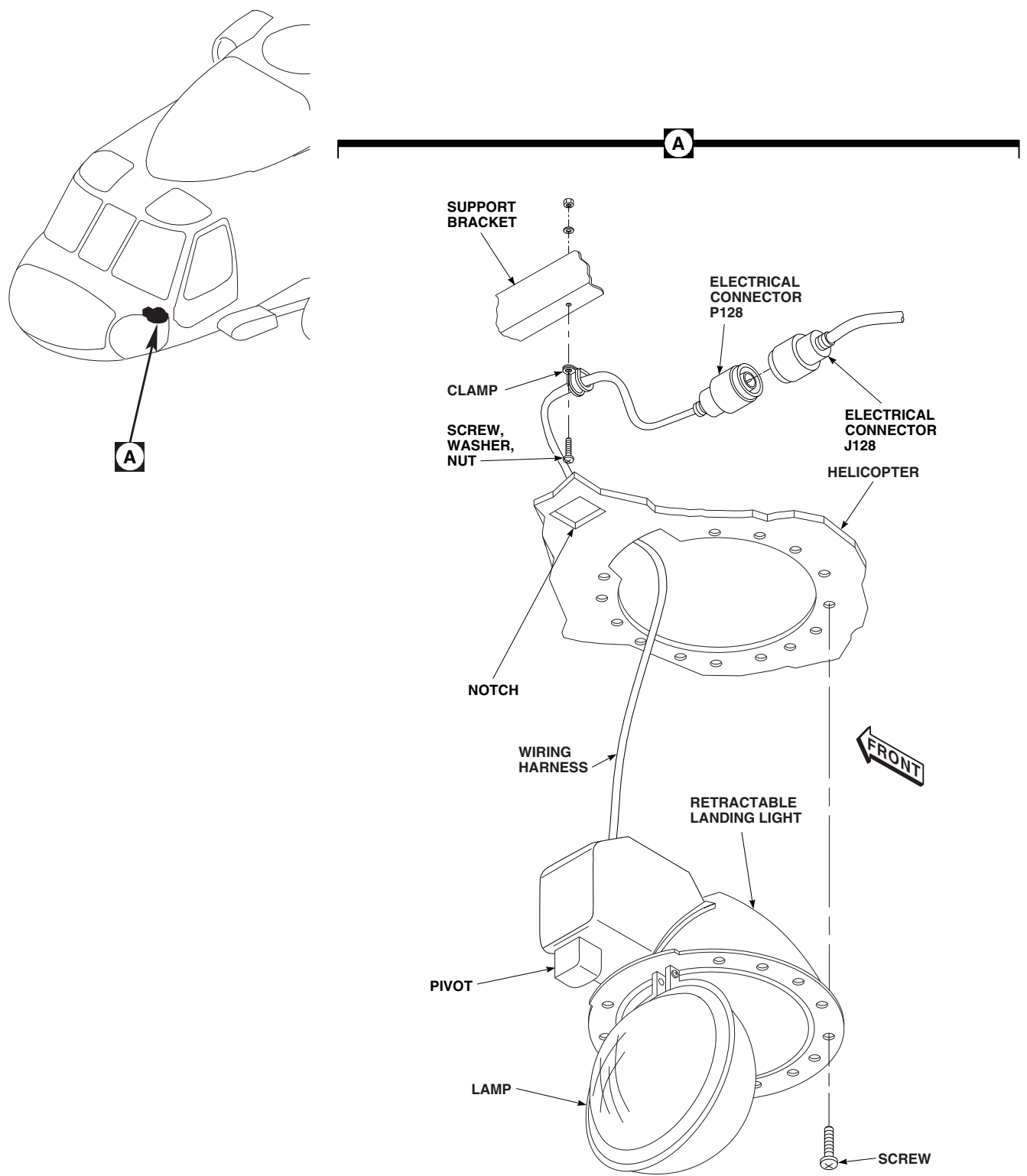
Lamp is hot when on. Never handle a lamp that may have power connected. Make sure lamp is cool before handling.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Support retractable landing light and remove screws securing retractable landing light to helicopter (Figure 9-4-90-1, Detail A).
- Slide pivot of retractable landing light toward center of opening in helicopter and carefully lower retractable landing light.
- Disconnect electrical connector, P128 from J128, from inside helicopter.
- Install protective caps and plugs on electrical connectors.
- Remove sealing compound from screw, washer, nut, and clamp.

- k. Remove screw, washer, and nut securing clamp and wiring harness to support bracket.
- l. Remove retractable landing light.
- m. Remove retractable landing light lamp (PARA 9-4-89). Retain lamp for installation.

9-4-90.1.2 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Install retractable landing light lamp (PARA 9-4-89).
- c. Install clamp and wiring harness to support bracket and secure with screw, washer, and nut wet with sealing compound, Item 301A, Appendix D (Figure 9-4-90-1, Detail A).
- d. Remove protective caps and plugs from electrical connectors.
- e. Inspect and treat electrical connectors (PARA 1-7-20).
- f. Connect electrical connector, P128 to J128, inside of helicopter.
- g. Make sure area is clean and free of foreign material.
- h. Install retractable landing light through opening in helicopter. Position pivot of retractable landing light so it is in notch of square opening.
- i. Position retractable landing light with mounting holes in helicopter and secure with screws.
- j. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- k. Perform operational check of retractable landing light (PARA 9-2-15).



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FIGURE 9-4-90-1. REPLACE RETRACTABLE LANDING LIGHT

9-4-91 CONTROLLABLE SEARCHLIGHT LAMP.

PARAGRAPH BREAKDOWN

	PAGE
9-4-91.1 Replace Controllable Searchlight Lamp	9-4-91-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit

Materials/Parts

- Tags

References

- PARA 1-6-16
- PARA 9-2-13
- PARA 9-4-34
- PARA 9-4-106

9-4-91.1 Replace Controllable Searchlight Lamp.

9-4-91.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Loosen captive screws securing lamp retaining ring and infrared filter (if installed) and remove from housing (Figure 9-4-91-1, Detail A).
- Carefully lower lamp from housing to gain access to wires.
- Tag wires. Remove screws and washers securing wires to lamp.
- Remove lamp.

9-4-91.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Install wires to lamp and secure with screws and washers. Remove tags (Figure 9-4-91-1, Detail A).
- Position lamp in housing with locating lug on lamp in locating lug receptacle in housing.
- Install infrared filter (if removed) and lamp retaining ring over lamp and secure to housing with captive screws.
- Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Perform operational check of controllable searchlight (PARA 9-2-13).

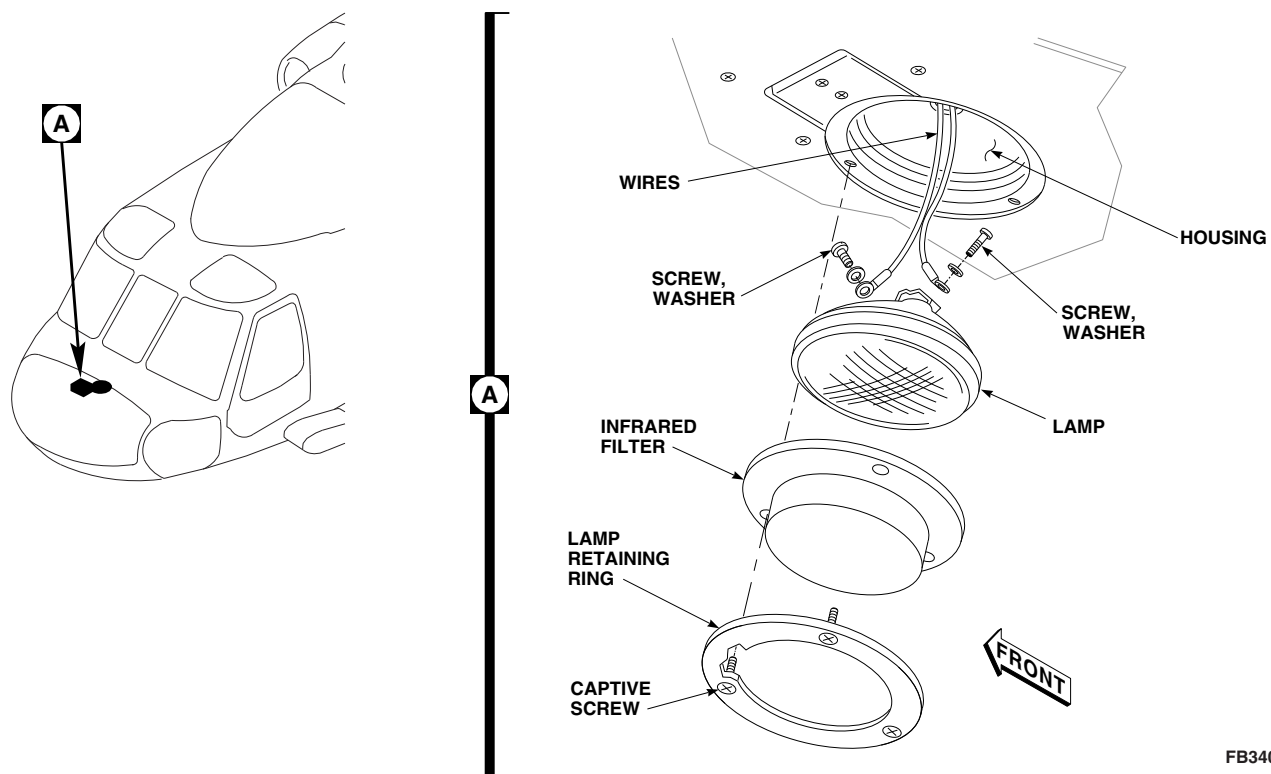
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FIGURE 9-4-91-1. REPLACE CONTROLLABLE SEARCHLIGHT LAMP

9-4-92 CONTROLLABLE SEARCHLIGHT.

PARAGRAPH BREAKDOWN

		PAGE
9-4-92.1	Replace Controllable Searchlight	9-4-92-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

Materials/Parts

- Dry-Cleaning Solvent, Item 124, Appendix D
- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-6-5
- PARA 9-2-13
- PARA 9-4-34
- PARA 9-4-106

9-4-92.1 Replace Controllable Searchlight.

9-4-92.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Remove screws, lockwashers, and washers securing controllable searchlight to helicopter (Figure 9-4-92-1, Detail A).
- Carefully lower controllable searchlight to gain access to electrical connector.
- Disconnect electrical connector, P109 from J109, from inside of helicopter.
- Install protective caps and plugs on electrical connectors.

- Remove controllable searchlight.

9-4-92.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- EMEP

Inspect chemical conversion coating as follows:

 - Using machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D, clean surface around all mounting holes. Using clean, dry machinery towel, Item 211, Appendix D, dry surface.
 - Inspect chemical conversion coated surfaces for breaks and scratches. **NONE ALLOWED.** If coated surfaces have breaks or scratches, apply chemical conversion coating (PARA 2-6-5).

- c. Remove protective caps and plugs from electrical connectors.
- d. Inspect and treat electrical connectors (PARA 1-7-20).
- e. Connect electrical connector, P109 to J109, to inside of helicopter (Figure 9-4-92-1, Detail A).
- f. Make sure area is clean and free of foreign material.
- g. Install controllable searchlight and secure with screws, lockwashers, and washers.
- h. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- i. Perform operational check of controllable searchlight (PARA 9-2-13).

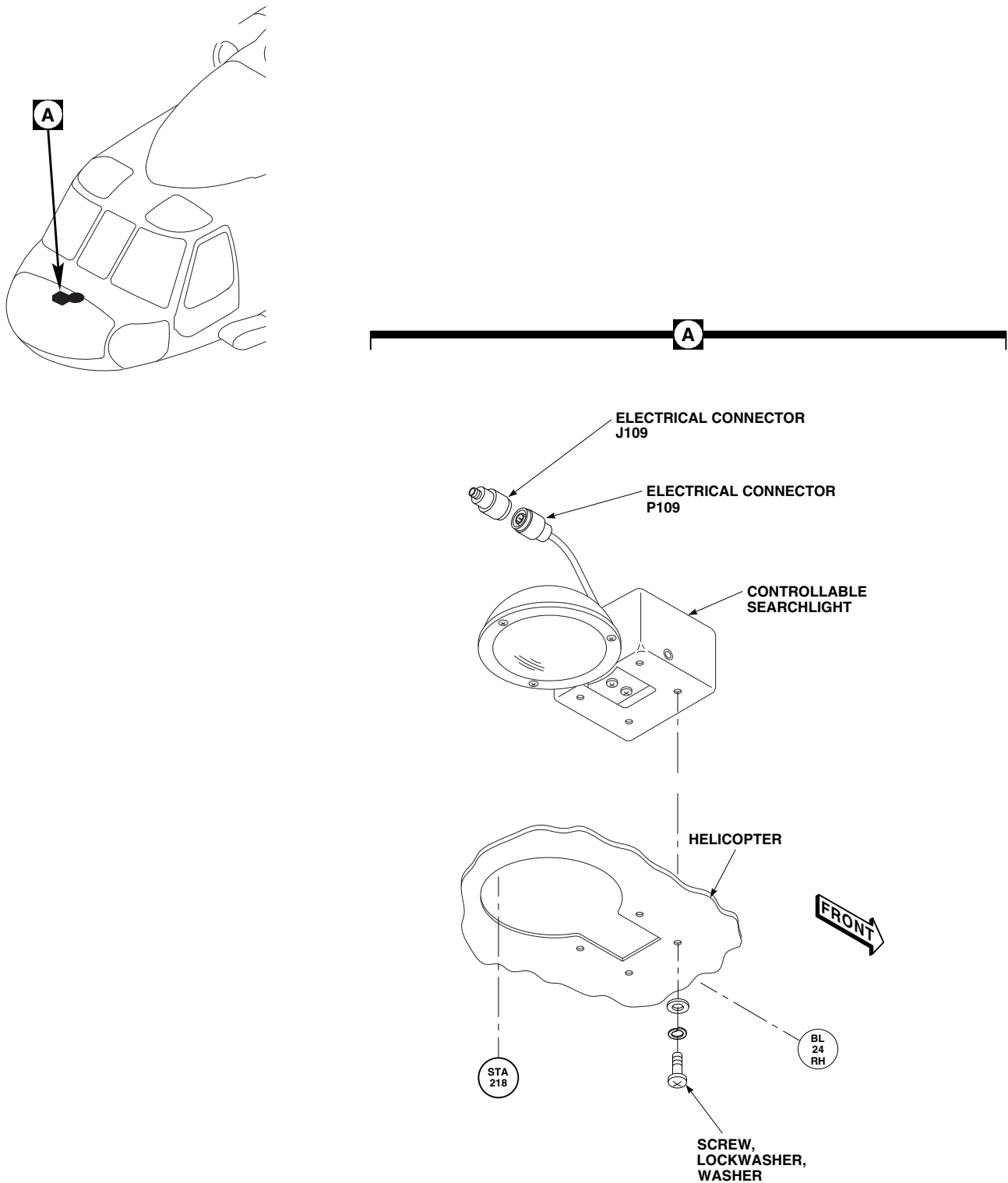


FIGURE 9-4-92-1. REPLACE CONTROLLABLE SEARCHLIGHT

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9-4-93 CARGO HOOK LIGHTS.

PARAGRAPH BREAKDOWN

	PAGE
9-4-93.1 Replace Cargo Hook Lights	9-4-93-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

Materials/Parts

- Protective Caps and Plugs

References

- PARA 1-6-16
- PARA 1-7-20

- PARA 2-4-45
- PARA 9-2-17
- PARA 9-4-34
- PARA 9-4-106

9-4-93.1 Replace Cargo Hook Lights.

NOTE

Replacement of all cargo hook lights is the same, except as noted.

9-4-93.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Remove passenger’s seats as required (PARA 2-4-45).
- Open cargo hook door.
- Disconnect applicable electrical connector(s) as follows (Figure 9-4-93-1, Detail A):

- P390 from J390 (rear).
 - P391 from J391 (front).
 - P389 from J389 (left).
- Install protective caps and plugs on electrical connectors.
 - Remove screws, washers, and nuts securing clamps and harness to supports.
 - Remove screws securing cargo hook light to supports.
 - Remove cargo hook light.

9-4-93.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).
- Install cargo hook light to support and secure with screws (Figure 9-4-93-1, Detail A).

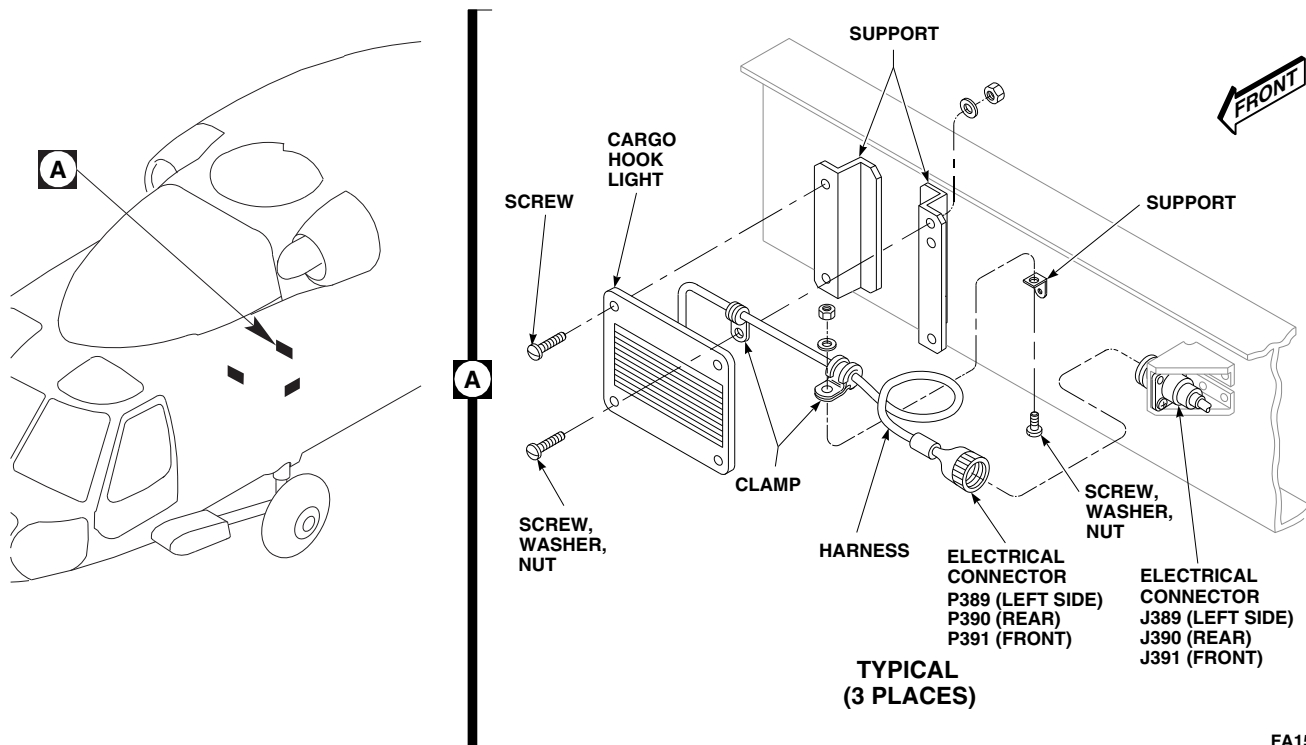


FIGURE 9-4-93-1. REPLACE CARGO HOOK LIGHTS

- c. Install harness and clamps to supports and secure with screws, washers, and nuts.
- d. Remove protective caps and plugs from electrical connectors.
- e. Inspect and treat electrical connectors (PARA 1-7-20).
- f. Connect applicable electrical connector(s) as follows:
 - (1) P390 to J390 (rear).
 - (2) P391 to J391 (front).
 - (3) P389 to J389 (left).
- g. Make sure area is clean and free of foreign material.
- h. Close cargo hook door.
- i. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- j. Perform operational check of cargo hook lights (PARA 9-2-17).
- k. Install passenger's seats (PARA 2-4-45).

9-4-94 CABIN INTERCOMMUNICATION SYSTEM FLOODLIGHT CONTROL PANEL LIGHT.

PARAGRAPH BREAKDOWN

	PAGE
9-4-94.1 Replace Cabin Intercommunication System Floodlight Control Panel Light	9-4-94-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Crimping Tool
- Electrical Repairer’s Toolkit
- Insert/Extract Tool

Materials/Parts

- Adhesive, Item 35, Appendix D
- Alcohol, Isopropyl, Item 72, Appendix D

- Cheesecloth, Item 90, Appendix D
- Tags

References

- Appendix D
- PARA 1-6-16
- PARA 9-2-5
- PARA 9-2-25
- PARA 9-4-34
- PARA 9-4-106

9-4-94.1 Replace Cabin Intercommunication System Floodlight Control Panel Light.

NOTE

Replacement of both cabin intercommunication floodlight control panel lights is the same.

9-4-94.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).

- Tag wires. Using insert/extract tool, disconnect pins from electrical connectors (Figure 9-4-94-1, Detail A).
- Pull wire through grommet on support box.
- Remove screws, washers, and clamp securing cabin ICS floodlight control panel light to support box.
- Remove cabin ICS floodlight control panel light.

9-4-94.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).

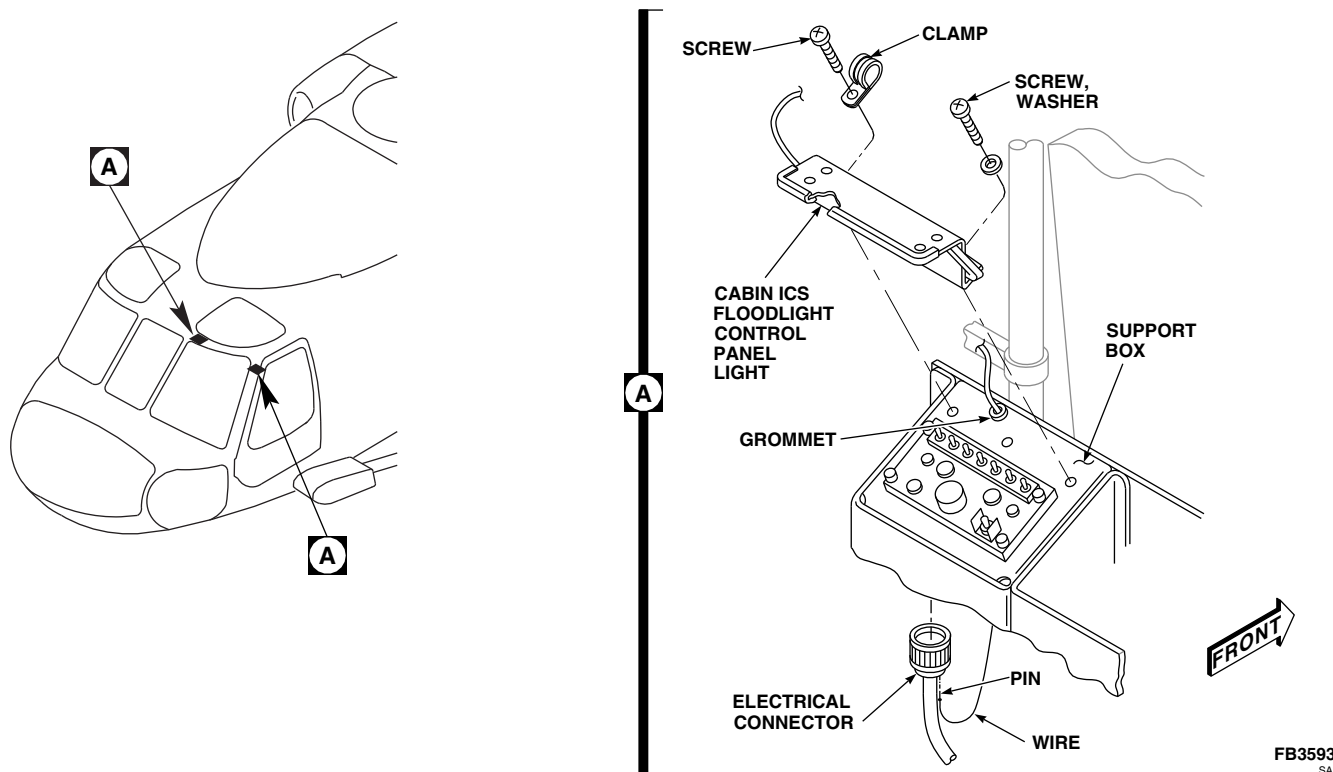


FIGURE 9-4-94-1. REPLACE CABIN INTERCOMMUNICATION SYSTEM FLOODLIGHT CONTROL PANEL LIGHT

- b. Install identification plate on intercommunication system (ICS) floodlight control panel light as follows:
 - (1) Using isopropyl alcohol, Item 72, Appendix D, and cheesecloth, Item 90, Appendix D, clean back of identification plate and ICS floodlight control panel light.
 - (2) Apply adhesive, Item 35, Appendix D, to bonding area and install identification plate.
- c. Install cabin ICS floodlight control panel light on support box and secure with screws, washers, and clamp (Figure 9-4-94-1, Detail A).
- d. Route wires through grommet, behind support box, and connect wires along wiring harness to electrical connector as follows:
 - (1) Using crimping tool, crimp pins supplied with electrical connector to wires.
 - (2) Using insert/extract tool, insert pins into body of electrical connector. Remove tags.
- e. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- f. Make sure area is clean and free of foreign material.
- g. Perform operational check of upper and lower console lights (PARA 9-2-5 or PARA 9-2-25).

9-4-95 PASSENGER’S INTERCOMMUNICATION SYSTEM FLOODLIGHT CONTROL PANEL LIGHT.

PARAGRAPH BREAKDOWN

		PAGE
9-4-95.1	Replace Passenger’s Intercommunication System Floodlight Control Panel Light	9-4-95-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Crimping Tool
- Electrical Repairer’s Toolkit
- Insert/Extract Tool
- Torque Wrench, 0 - 30 in. lbs

Materials/Parts

- Tags

References

- PARA 1-6-16
- PARA 2-4-69
- PARA 9-2-5
- PARA 9-2-25
- PARA 9-4-34
- PARA 9-4-106

9-4-95.1 Replace Passenger’s Intercommunication System Floodlight Control Panel Light.

9-4-95.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Remove soundproofing panel (PARA 2-4-69).
- Tag wires. Using insert/extract tool, disconnect pins from electrical connectors.
- Pull wires through adapter in support box (Figure 9-4-95-1, Detail A).

- Remove washers and nuts securing passenger’s intercommunication system (ICS) floodlight control panel light to support box.

- Remove passenger’s ICS floodlight control panel light.

9-4-95.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Install passenger’s intercommunication system (ICS) floodlight control panel light to support box and secure with washers and nuts (Figure 9-4-95-1, Detail A).
- TORQUE NUTS TO 15 - 17 INCH-POUNDS.

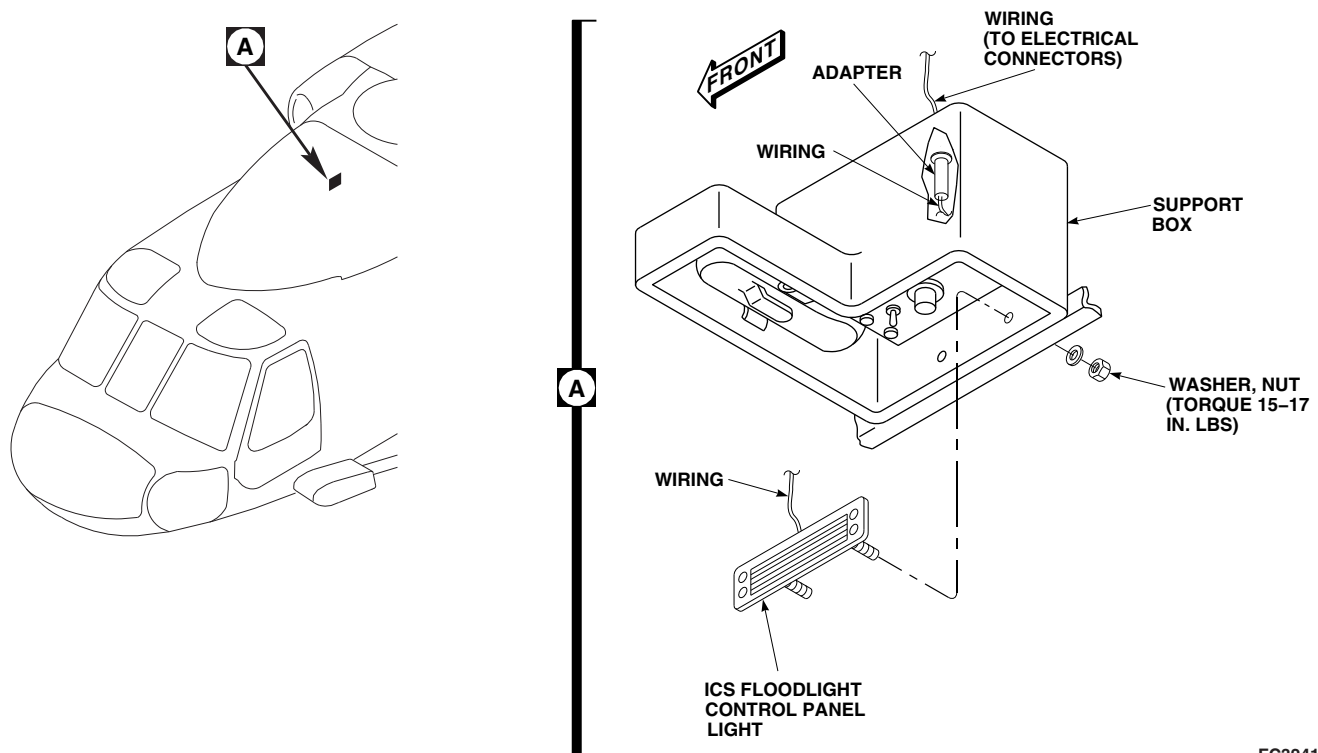
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FIGURE 9-4-95-1. REPLACE PASSENGER'S INTERCOMMUNICATION SYSTEM FLOODLIGHT CONTROL PANEL LIGHT

- d. Route wires through adapter and connect wires along wiring harness to electrical connector as follows:
 - (1) Using crimping tool, crimp pins supplied with electrical connector to wires.
 - (2) Using insert/extract tool, insert pins into body of electrical connector. Remove tags.
- e. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- f. Perform operational check of upper and lower console lights (PARA 9-2-5 or PARA 9-2-25).
- g. Make sure area is clean and free of foreign material.
- h. Install soundproofing panel (PARA 2-4-69).

9-4-96 ENGINE IGNITION SWITCH.

PARAGRAPH BREAKDOWN

		PAGE
9-4-96.1	Replace Engine Ignition Switch	9-4-96-1

INITIAL SETUP

Tools

- Digital Multimeter
- Electrical Repairer’s Toolkit
- Protection, Eye and Skin
- Soldering Iron

Materials/Parts

- Solder, Item 313, Appendix D
- Tags

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 9-4-34
- PARA 9-4-106

9-4-96.1 Replace Engine Ignition Switch.

9-4-96.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Remove knurled nut and lockwasher from rear of engine ignition switch (Figure 9-4-96-1, Detail A).
- Slide knurled nut and lockwasher away from engine ignition switch.
- Pull engine ignition switch out of instrument panel to gain access to contacts.



Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- Tag wires. Using soldering iron, unsolder wires from engine ignition switch.
- Remove engine ignition switch, knurled nut, and lockwasher.

9-4-96.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).

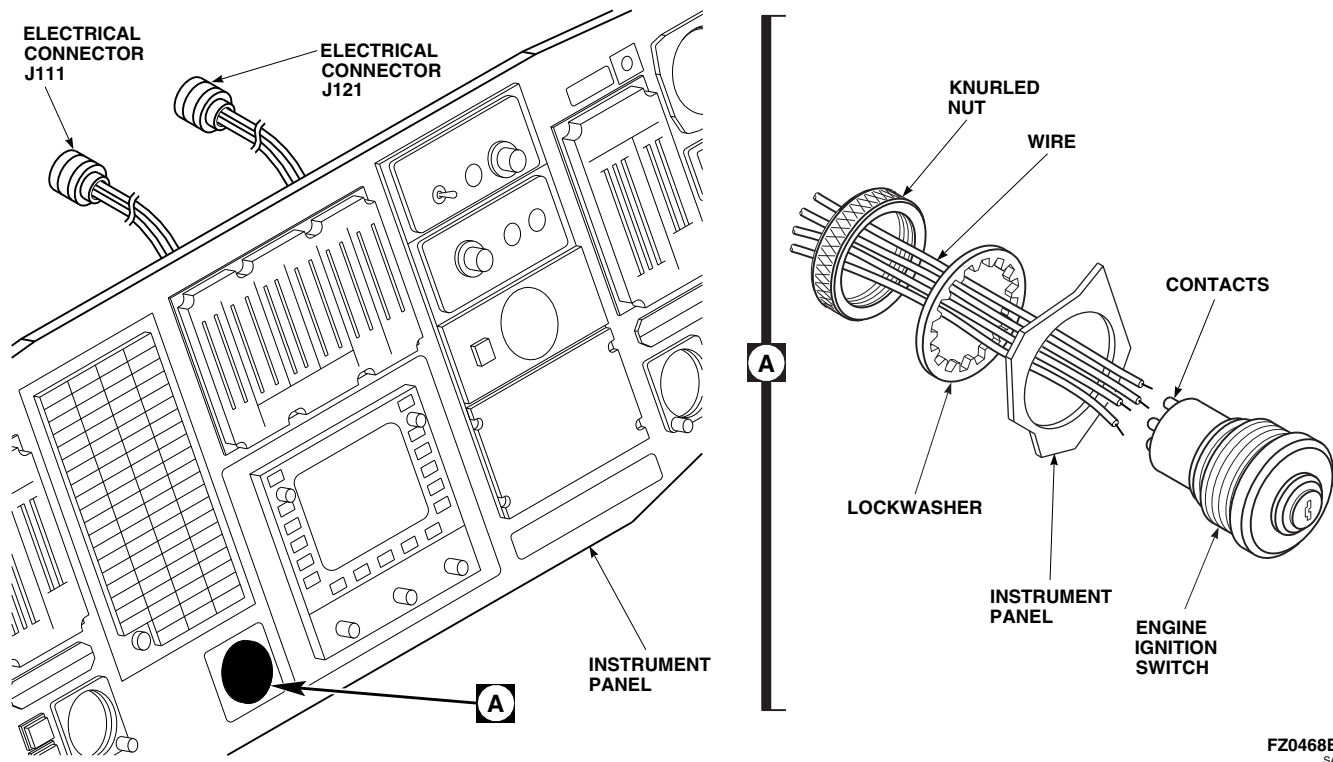
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FIGURE 9-4-96-1. REPLACE ENGINE IGNITION SWITCH

- b. Install knurled nut and lockwasher over wires (Figure 9-4-96-1, Detail A).
- c. Slide wires through hole in instrument panel.
- d. Using soldering iron and solder, Item 313, Appendix D, solder wires to contacts on engine ignition switch. Remove tags.
- e. With engine ignition switch and instrument panel slots lined up, install engine ignition switch in instrument panel and secure with lockwasher and knurled nut.
- f. Make sure area is clean and free of foreign material.
- g. Disconnect electrical connectors, J111 and J121 (Figure 9-4-96-1).
- h. With ENGINE IGNITION switch OFF, using digital multimeter, check for continuity between pins b and d of electrical connector, J121, and between pins CC and BB of electrical connector, J111.
- i. With ENGINE IGNITION switch ON, check for open circuit between pins b and d of electrical connector, J121, and between pins CC and BB of electrical connector, J111.
- j. Inspect and treat electrical connectors (PARA 1-7-20).
- k. Connect electrical connectors, J111 and J121.
- l. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

9-4-97 ENGINE OVERSPEED RELAYS.

PARAGRAPH BREAKDOWN

	PAGE
9-4-97.1 Replace Engine Over-speed Relays	9-4-97-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

Materials/Parts

- Cheesecloth, Item 90, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-69
- PARA 4-2-4
- PARA 9-4-34
- PARA 9-4-106

9-4-97.1 Replace Engine Overspeed Relays.

NOTE

Replacement of No. 1 and No. 2 engine overspeed relays is the same, except as noted.

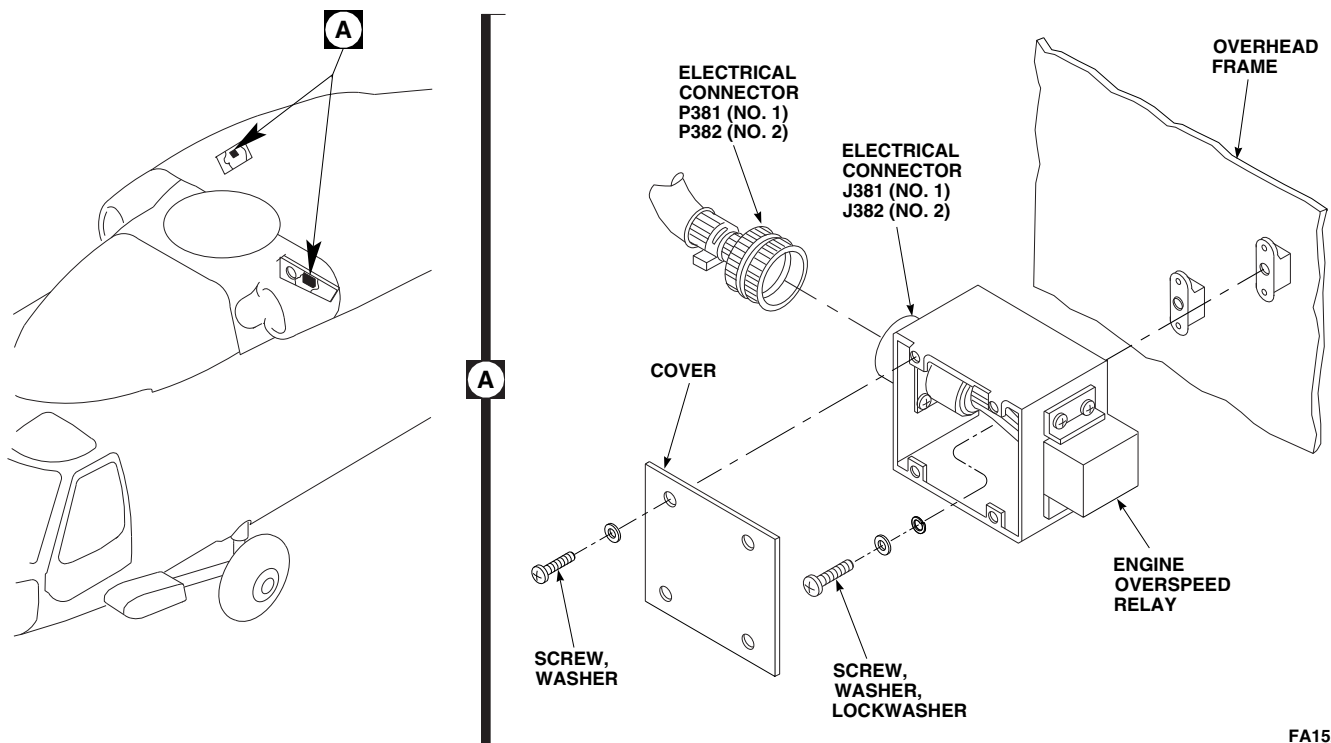
9-4-97.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Remove soundproofing panel (PARA 2-4-69).
- Remove screws and washers securing cover to engine overspeed relay (Figure 9-4-97-1, Detail A).

- Remove screws, washers, and lockwashers securing engine overspeed relay to overhead frame.
- Disconnect applicable electrical connector(s) from engine overspeed relay as follows:
 - P381 from J381 (No. 1 engine overspeed relay).
 - P382 from J382 (No. 2 engine overspeed relay).
- Install protective caps and plugs on electrical connectors.

9-4-97.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Using cheesecloth, Item 90, Appendix D, dampened with dry-cleaning solvent, Item 124,

FA1527
SA**FIGURE 9-4-97-1. REPLACE ENGINE OVERSPEED RELAYS**

Appendix D, clean surface around all mounting holes. Using clean, dry cheesecloth, Item 90, Appendix D, dry surface.

- c. Remove protective caps and plugs from electrical connectors.
- d. Inspect and treat electrical connectors (PARA 1-7-20).
- e. Connect applicable electrical connector(s) to engine overspeed relay as follows (Figure 9-4-97-1, Detail A):
 - (1) P381 to J381 (No. 1 engine overspeed relay).
 - (2) P382 to J382 (No. 2 engine overspeed relay).
- f. Install engine overspeed relay to overhead frame and secure with screws, washers, and lockwashers.
- g. Install cover on engine overspeed relay and secure with washers and screws.
- h. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- i. Perform operational check of engine overspeed protection system (PARA 4-2-4).
- j. Make sure area is clean and free of foreign material.
- k. Install soundproofing panel (PARA 2-4-69).

9-4-98 PIN FILTER ADAPTERS **EMEP** .

PARAGRAPH BREAKDOWN

	PAGE
9-4-98.1 Replace Pin Filter Adapt- ers	9-4-98-2

INITIAL SETUP

Tools

- Digital Multimeter
- Electrical Repairer’s Toolkit

Materials/Parts

- Cheesecloth, Item 90, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D

- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs
- Tiedown Strap, Item 331, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20

9-4-98.1 Replace Pin Filter Adapters.**NOTE**

- Replacement of all pin filter adapters is the same.
- The following list identifies those components having replaceable pin filter adapters:
 - Blade Deice Test Panel.
 - Blade Deice Controller.
 - Pilot's HSI.
 - Copilot's HSI.
 - SAS/FPS Computer.
 - Rate Gyro.
 - No. 1 Generator Control Unit.
 - No. 2 Generator Control Unit.
 - APU Generator Control Unit.
 - Pilot's VSI.
 - Copilot's VSI.
 - Caution/Advisory Panel.
 - No. 1 Lateral Accelerometer.
 - No. 2 Lateral Accelerometer.
 -  **70583-702374** Battery Conditioner/Analyzer.
 - Pilot's HSI/VSI Mode Select Panel.
 - Copilot's HSI/VSI Mode Select Panel.
 - Radar Altimeter Receiver Transmitter.
 - Airspeed Transducer.

- No. 1 Signal Data Converter.
- No. 2 Signal Data Converter.
- No. 1 Stabilator Actuator.
- No. 2 Stabilator Actuator.
- Roll Trim Servo.
- Yaw Trim Servo.
- No. 1 Stabilator Amplifier.
- No. 2 Stabilator Amplifier.
- VOR/ILS Receiver.
- CIS Processor.
- Air Data Transducer.
- Compass Control Panel.
- Doppler SDC.
- Doppler Computer Display Unit.
- Pilot's Master Warning Panel.
- Copilot's Master Warning Panel.

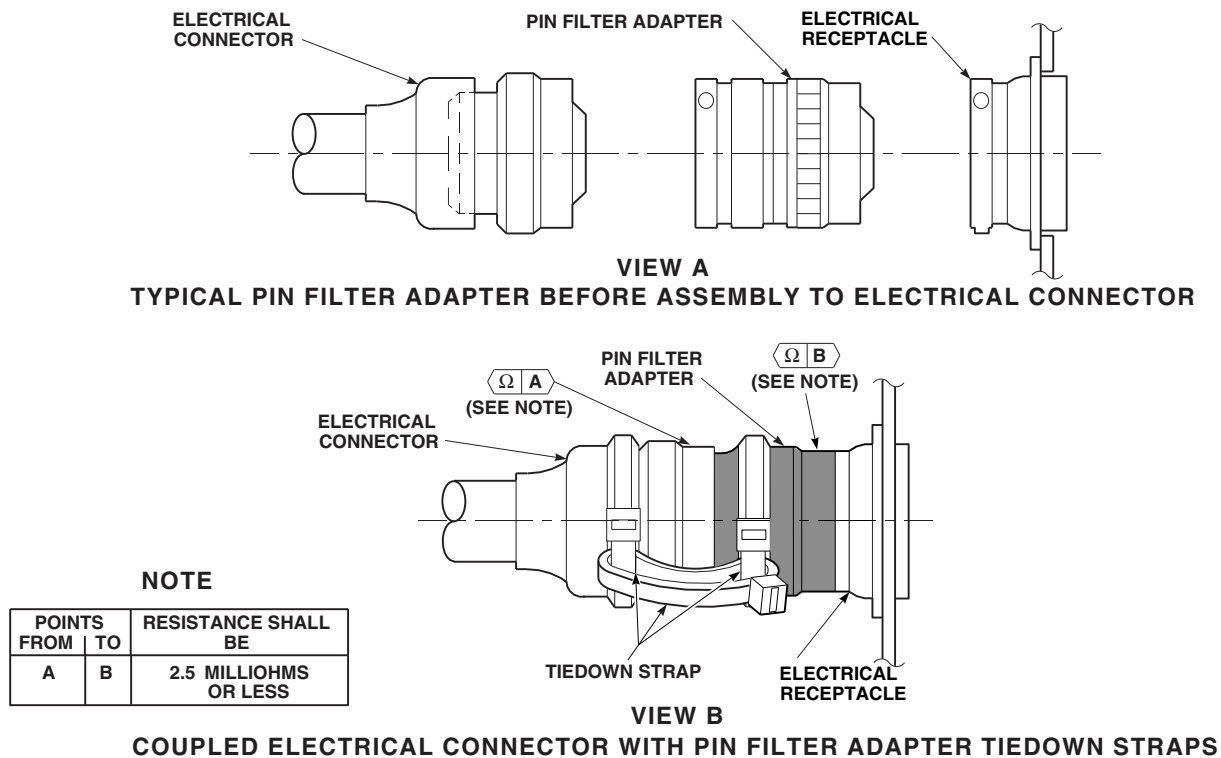
9-4-98.1.1. Remove.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Disconnect pin filter adapter from electrical receptacle on component (Figure 9-4-98-1).
- c. Install protective caps and plugs on electrical connectors.
- d. Remove tiedown straps and pin filter adapter.

9-4-98.1.2. Test.

- a. Using digital multimeter, check pin to pin continuity as follows:
 - (1) Check continuity of each pin of pin filter adapter. **CONTINUITY SHALL BE PRESENT.**

9-4-98-2 Change 2



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FIGURE 9-4-98-1. REPLACE PIN FILTER ADAPTERS

- (2) If continuity is not present, replace pin filter adapter.
- b. Using digital multimeter, check pin to shell continuity as follows:
 - (1) Check continuity from each pin to shell of pin filter adapter. **CONTINUITY SHALL NOT BE PRESENT.**
 - (2) Check each pin to surrounding pins.
 - (3) If continuity is present, replace pin filter adapter.
- c. Remove protective caps and plugs from electrical connectors.
- d. Inspect and treat electrical connectors (PARA 1-7-20).
- e. Connect pin filter adapter to electrical connector (Figure 9-4-98-1).
- f. Using digital multimeter, take resistance measurement between pin filter adapter and electrical connector. **RESISTANCE SHALL BE 2.5 MILLIOHMS OR LESS.**
- g. If resistance is not as specified, repeat step b. through step f.
- h. Install tiedown straps, Item 331, Appendix D.
- i. Connect pin filter adapter from electrical receptacle to component.

9-4-98.1.3. Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Using cheesecloth, Item 90, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D, clean threads of pin filter adapter

9-4-99 INSTRUMENT PANEL PUSHBUTTON LAMPS.

PARAGRAPH BREAKDOWN

		PAGE
9-4-99.1	Replace Instrument Panel Pushbutton Lamps	9-4-99-1

INITIAL SETUP

- PARA 9-2-4

References

- PARA 1-6-16

9-4-99.1 Replace Instrument Panel Pushbutton Lamps.

NOTE

Replacement of all instrument panel pushbutton lamps is the same.

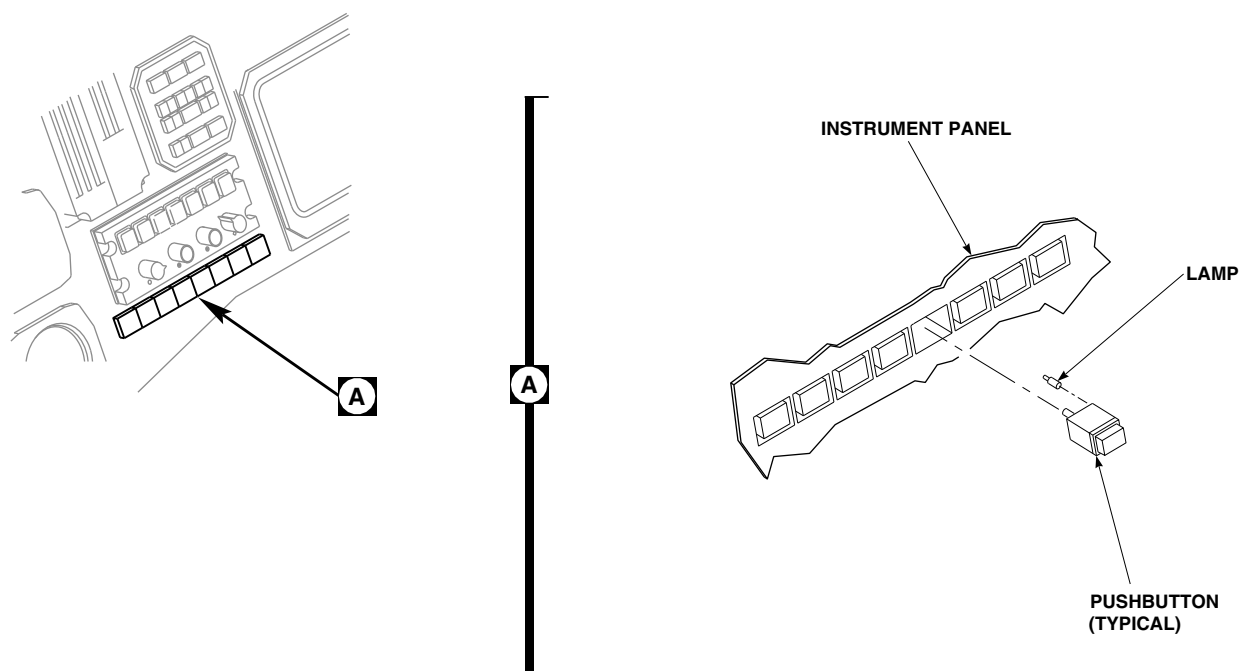
9-4-99.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Pull pushbutton from instrument panel (Figure 9-4-99-1, Detail A).

- Twist and remove lamp from pushbutton.

9-4-99.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Twist lamp into pushbutton (Figure 9-4-99-1, Detail A).
- Install pushbutton into instrument panel.
- Perform operational check of affected lamp (PARA 9-2-4).

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SA**FIGURE 9-4-99-1. REPLACE INSTRUMENT PANEL PUSHBUTTON LAMPS**

9-4-100 AUXILIARY SEARCHLIGHT CONTROL PANEL.

PARAGRAPH BREAKDOWN

	PAGE
9-4-100.1 Replace Auxiliary Searchlight Control Panel	9-4-100-1

INITIAL SETUP

References

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

- PARA 1-6-16
- PARA 1-7-20
- PARA 9-2-21

Materials/Parts

- Protective Caps and Plugs

- PARA 9-4-34
- PARA 9-4-106

9-4-100.1 Replace Auxiliary Searchlight Control Panel.

- b. Remove protective caps and plugs from electrical connectors.

9-4-100.1.1. Remove.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- c. Loosen turnlock fasteners securing auxiliary searchlight control panel to lower console and carefully lift out to gain access to electrical connector (Figure 9-4-100-1, Detail A).
- d. Disconnect electrical connector, P197 from J197, on auxiliary searchlight control panel.
- e. Remove auxiliary searchlight control panel.
- f. Install protective caps and plugs on electrical connectors.

- c. Inspect and treat electrical connectors (PARA 1-7-20).
- d. Connect electrical connector, P197 to J197, on auxiliary searchlight control panel (Figure 9-4-100-1, Detail A).
- e. Carefully install auxiliary searchlight control panel into lower console and secure with turnlock fasteners.
- f. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- g. Make sure area is clean and free of foreign material.

9-4-100.1.2. Install.

- a. Turn off all electrical power (PARA 1-6-16).

- h. Perform operational check of auxiliary searchlight (PARA 9-2-21).

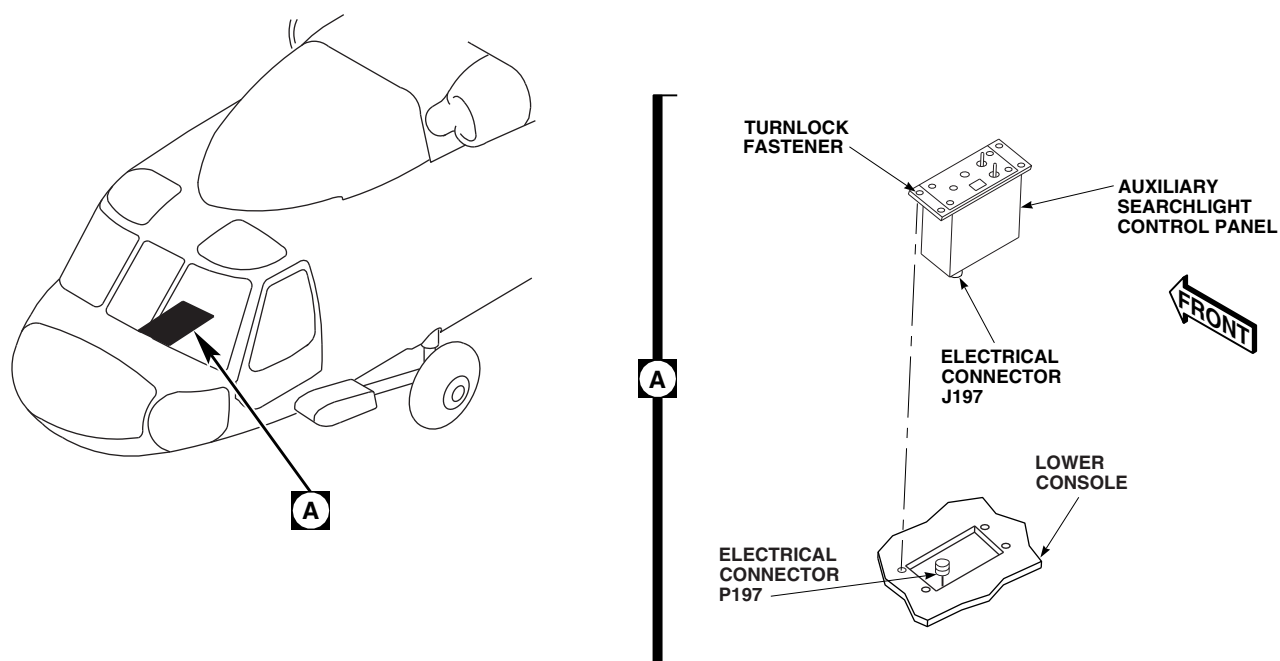
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FIGURE 9-4-100-1. REPLACE AUXILIARY SEARCHLIGHT CONTROL PANEL

9-4-101 AUXILIARY SEARCHLIGHT JUNCTION BOX.

PARAGRAPH BREAKDOWN

	PAGE
9-4-101.1 Replace Auxiliary Searchlight Junction Box	9-4-101-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

Materials/Parts

- Tags

References

- PARA 1-6-16
- PARA 2-4-44
- PARA 9-2-21
- PARA 9-4-34
- PARA 9-4-106

9-4-101.1 Replace Auxiliary Searchlight Junction Box.

9-4-101.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Remove copilot’s seat (PARA 2-4-44).
- Remove screws and washers securing cover to auxiliary searchlight junction box and remove cover (Figure 9-4-101-1, Detail A).
- For the wiring marked CONTROL on outside of auxiliary searchlight junction box, tag wires and remove screws securing wires to terminal board (TB1) (Figure 9-4-101-1, Detail B).
- For the wiring marked 28 V.D.C. on outside of auxiliary searchlight junction box, tag wires and remove nut securing one wire to ground

stud. Remove screw and washer securing other wire to circuit breaker (CB2).

- For the wiring marked GIMBAL on outside of auxiliary searchlight junction box, tag wire and remove screw securing wire to capacitor.
- For the wiring marked LIGHT on outside of auxiliary searchlight junction box, tag wires and remove screw securing wire to capacitor. Remove wire from ground stud.
- Carefully remove wiring from auxiliary searchlight junction box by pulling wiring through grommets.
- Remove screws, lockwashers, and washers securing auxiliary searchlight junction box to nose compartment (Figure 9-4-101-1, Detail A).
- Remove auxiliary searchlight junction box.

9-4-101.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).

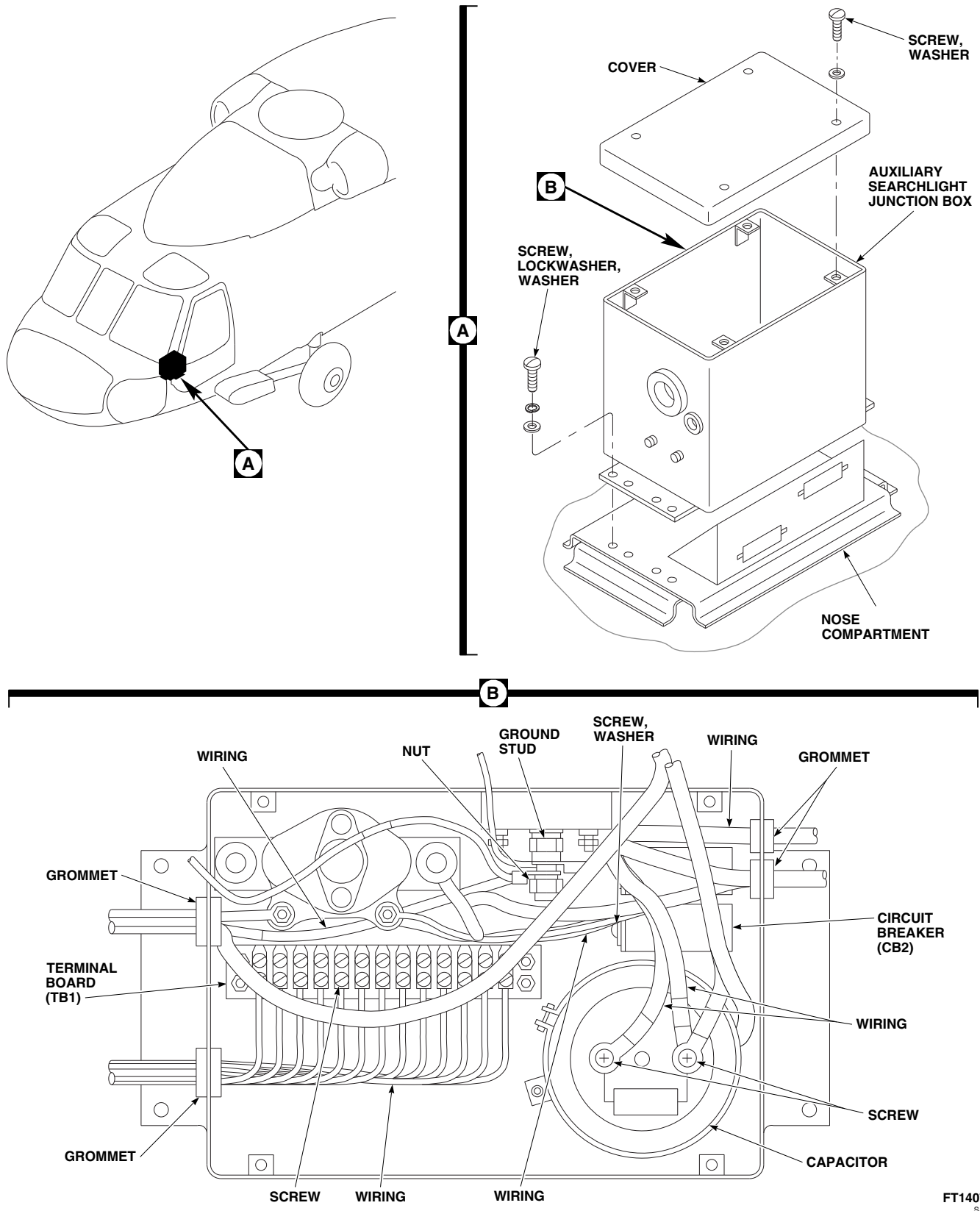


FIGURE 9-4-101-1. REPLACE AUXILIARY SEARCHLIGHT JUNCTION BOX

9-4-101-2

- b. Install auxiliary searchlight junction box into nose compartment and secure with screws, lockwashers, and washers (Figure 9-4-101-1, Detail A).
- c. Carefully pass wiring through grommets.
- d. For the wiring marked CONTROL on outside of auxiliary searchlight junction box, install wires to terminal board (TB1) and secure with screws. Remove tags (Figure 9-4-101-1, Detail B).
- e. For the wiring marked LIGHT on outside of auxiliary searchlight junction box, install wire on capacitor and secure with screw. Install other wire on ground stud. Remove tags.
- f. For the wiring marked 28 V.D.C. on outside of auxiliary searchlight junction box, install wire to circuit breaker (CB2) and secure with screw and washer. Install other wire on ground stud and secure with nut. Remove tags.
- g. For the wiring marked GIMBAL on outside of auxiliary searchlight junction box, install wire on capacitor and secure with screw. Remove tags.
- h. Install cover on auxiliary searchlight junction box and secure with screws and washers (Figure 9-4-101-1, Detail A).
- i. Install copilot's seat (PARA 2-4-44).
- j. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- k. Make sure area is clean and free of foreign material.
- l. Perform operational check of auxiliary searchlight (PARA 9-2-21).

9-4-102 AUXILIARY SEARCHLIGHT GIMBAL ASSEMBLY.

PARAGRAPH BREAKDOWN

	PAGE
9-4-102.1 Replace Auxiliary Searchlight Gimbal Assembly	9-4-102-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Protective Caps and Plugs
- Tags

References

- PARA 1-6-16
- PARA 1-7-20
- PARA 9-2-21
- PARA 9-4-34
- PARA 9-4-103
- PARA 9-4-106

9-4-102.1 Replace Auxiliary Searchlight Gimbal Assembly.

9-4-102.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Remove auxiliary searchlight assembly (PARA 9-4-103).
- Tag and disconnect electrical connector, P198 from J198 (Figure 9-4-102-1, Detail A).
- Tag and disconnect electrical connector, P3 from J3, on auxiliary searchlight gimbal assembly.

- Install protective caps and plugs on electrical connectors.
- Remove screws, washers, and clamps securing wiring harnesses to support assembly.
- Remove screw and washer securing safety wire to position light junction box.
- Remove bolts, lockwashers, and washers securing auxiliary searchlight gimbal assembly to support assembly.
- Remove auxiliary searchlight gimbal assembly.

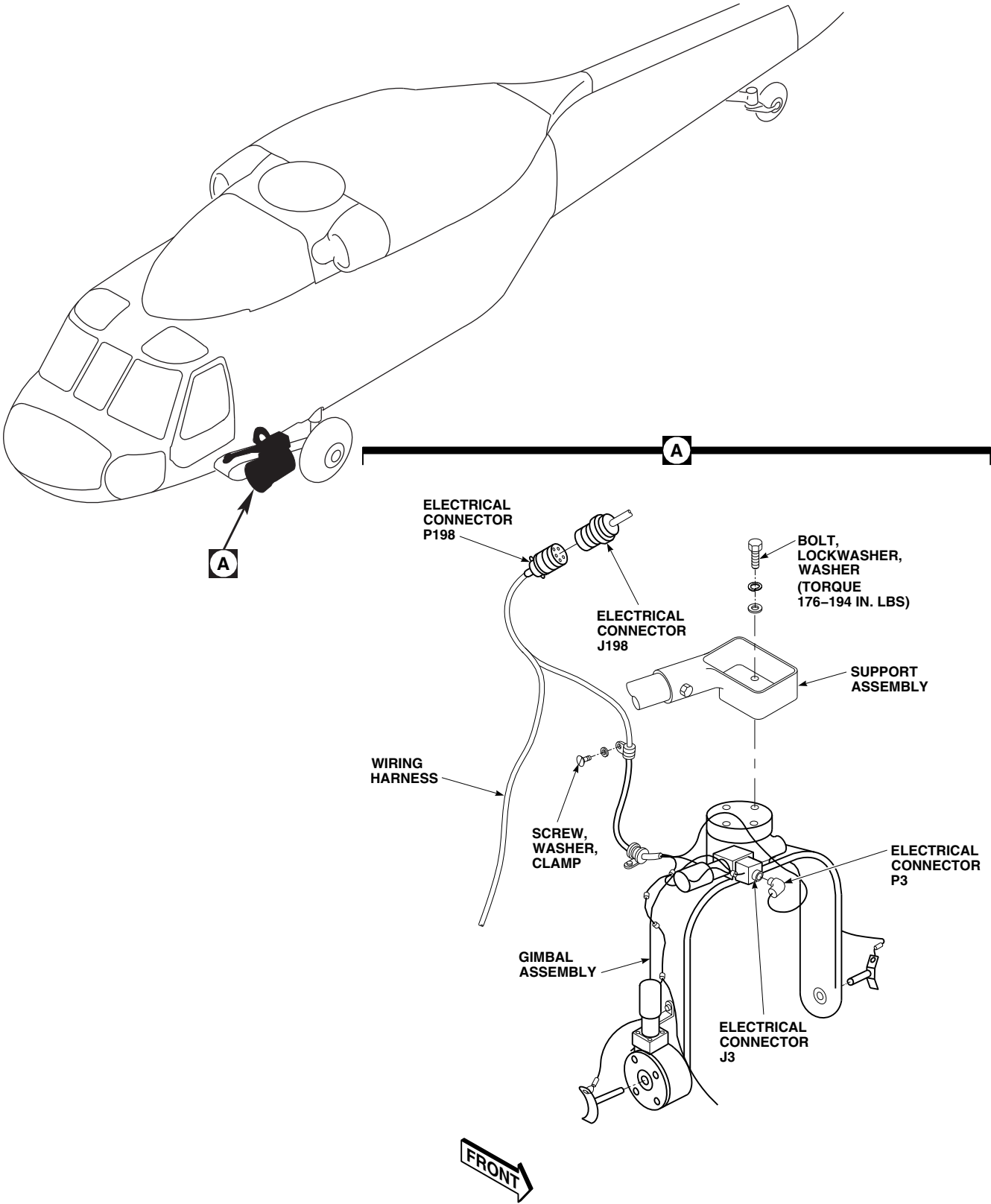
9-4-102.1.2. Inspect.

- Inspect entire auxiliary searchlight gimbal assembly for the following; replace or repair as necessary:

- (1) Loose mountings.
- (2) Freedom of movement.
- (3) Worn bearings.
- (4) Clutch wear.
- (5) Securing of fasteners.
- (6) Stops/rollpins for cracks.
- (7) Dust or dirt.
- (8) General condition.

9-4-102.1.3. Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Install auxiliary searchlight gimbal assembly on support assembly and secure with bolts, lockwashers, and washers (Figure 9-4-102-1, Detail A).
- c. **TORQUE BOLTS TO 176 - 194 INCH-POUNDS.**
- d. Install safety wire to position junction box using screw and washer.
- e. Remove protective caps and plugs from electrical connectors as they are connected.
- f. Inspect and treat electrical connectors (PARA 1-7-20).
- g. Connect electrical connector, P3 to J3, on auxiliary searchlight gimbal assembly.
- h. Connect electrical connector, P198 to J198.
- i. Install wiring harness to support assembly and secure with screws, washers, and clamps.
- j. Install auxiliary searchlight assembly (PARA 9-4-103).
- k. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- l. Make sure area is clean and free of foreign material.
- m. Perform operational check of auxiliary searchlight (PARA 9-2-21).



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FIGURE 9-4-102-1. REPLACE AUXILIARY SEARCHLIGHT GIMBAL ASSEMBLY

9-4-102-3/(9-4-102-4 Blank)

9-4-103 AUXILIARY SEARCHLIGHT ASSEMBLY.

PARAGRAPH BREAKDOWN

	PAGE
9-4-103.1 Replace Auxiliary Searchlight Assembly	9-4-103-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

Materials/Parts

- Protective Caps and Plugs
- Tags

References

- PARA 1-6-16
- PARA 1-7-20
- PARA 9-2-21
- PARA 9-4-34
- PARA 9-4-102
- PARA 9-4-106

9-4-103.1 Replace Auxiliary Searchlight Assembly.

9-4-103.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Tag and disconnect electrical connector, P1 from J1, on auxiliary searchlight assembly (Figure 9-4-103-1, Detail A).
- Tag and disconnect electrical connector, P2 from J2, on microswitch.
- Install protective caps and plugs on electrical connectors.
- Remove bolt securing safety cable to auxiliary searchlight assembly. Remove safety cable.

NOTE

- It may be necessary to remove wing bolt lanyard in order to fully remove wing bolt without damaging wing bolt lanyard.
- If necessary, remove screw and clamp securing wing bolt lanyard to auxiliary searchlight gimbal assembly.

NOTE

- Due to the size and weight of the auxiliary searchlight assembly, it is recommended that two people remove the auxiliary searchlight assembly.
- Support auxiliary searchlight assembly and remove wing bolts.
 - Remove auxiliary searchlight assembly.

- j. Inspect auxiliary searchlight gimbal assembly (PARA 9-4-102).

9-4-103.1.2. Install.

- a. Turn off all electrical power (PARA 1-6-16).

NOTE

Due to the size and weight of the auxiliary searchlight assembly, it is recommended that two people install the auxiliary searchlight assembly.

- b. Install auxiliary searchlight assembly onto auxiliary searchlight gimbal assembly and secure with wing bolts (Figure 9-4-103-1, Detail A).
- c. If removed, install wing bolt lanyard to auxiliary searchlight gimbal assembly and secure with screw and clamp.
- d. Install safety cable to auxiliary searchlight assembly and secure with bolt.
- e. Remove protective caps and plugs from electrical connectors.
- f. Inspect and treat electrical connectors (PARA 1-7-20).
- g. Connect electrical connector, P1 to J1, on auxiliary searchlight assembly. Remove tag.
- h. Connect electrical connector, P2 to J2, on microswitch. Remove tag.
- i. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- j. Make sure area is clean and free of foreign material.
- k. Perform operational check of auxiliary searchlight (PARA 9-2-21).

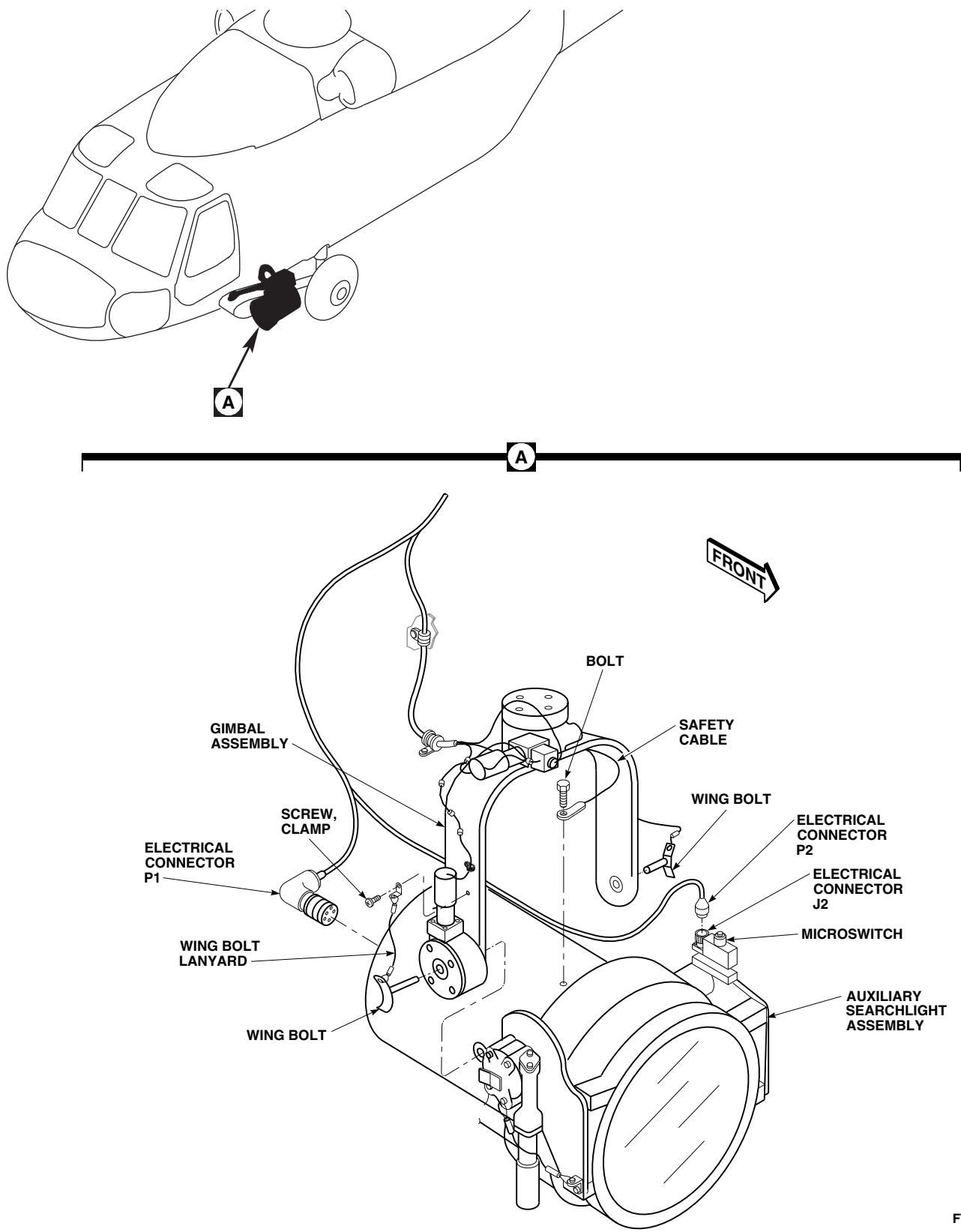


FIGURE 9-4-103-1. REPLACE AUXILIARY SEARCHLIGHT ASSEMBLY

9-4-103-3/(9-4-103-4 Blank)

9-4-104 AUXILIARY SEARCHLIGHT INFRARED FILTER LENS.

PARAGRAPH BREAKDOWN

	PAGE
9-4-104.1 Adjust Auxiliary Searchlight Infrared Filter Lens	9-4-104-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Lockwire, Item 193, Appendix D

References

- Appendix D

9-4-104.1 Adjust Auxiliary Searchlight Infrared Filter Lens.

9-4-104.1.1 Adjust.

- Actuate infrared (IR) filter lens to full down position. Make sure IR filter lens rubber seal fully seats around auxiliary searchlight assembly shroud.
- If rubber seal is not fully seated with limit switch actuated, adjust IR filter lens as follows:
 - Loosen mounting bolts on IR filter frame (Figure 9-4-104-1, Detail A).
 - Adjust IR filter frame to fully seat against auxiliary searchlight assembly shroud.
 - TORQUE MOUNTING BOLTS TO 53 - 57 INCH-POUNDS.**
 - Actuate IR filter lens to up position then to full down position. If motor stops and rubber seal does not fully seat against auxiliary searchlight assembly shroud,

loosen setscrews on adjusting bracket and adjust bracket to stop motor when rubber seal fully seats. **TIGHTEN SETSCREWS ON ADJUSTING BRACKET AND REPEAT STEP b. (4)** (Figure 9-4-104-1, Detail B).

- Actuate IR filter lens to full up position. IR filter frame must be perpendicular to clear window lens within $\pm 5^{\circ}$ (near horizontal and clear of light beam).
 - If required, loosen bolt through spacer.
 - Rotate spacer to adjust lower limit switch until IR filter frame is perpendicular to clear window lens within $\pm 5^{\circ}$ (near horizontal and clear of light beam).
 - TORQUE MOUNTING BOLTS TO 53 - 57 INCH-POUNDS.**
- Using lockwire, Item 193, Appendix D, lock-wire mounting bolts on IR filter frame.
- Make sure area is clean and free of foreign material.

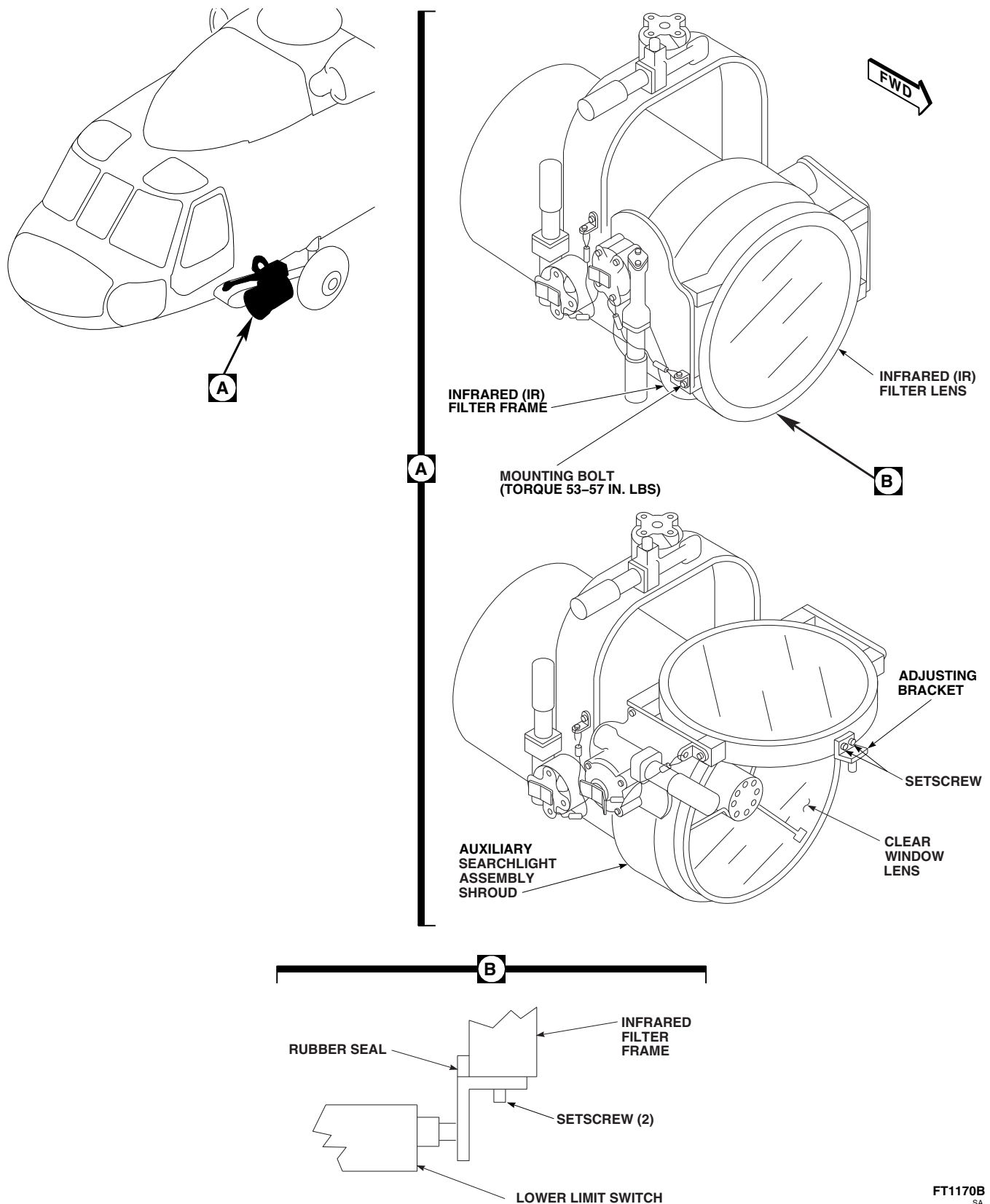


FIGURE 9-4-104-1. ADJUST AUXILIARY SEARCHLIGHT INFRARED FILTER LENS

9-4-105 SEALED LEAD-ACID BATTERY JUNCTION BOX RELAYS, K200 AND K201 **702375 – SUBQ** .

PARAGRAPH BREAKDOWN

	PAGE
9-4-105.1 Replace Sealed Lead-Acid Battery Junction Box Relays, K200 and K201	9-4-105-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Tags

References

- PARA 1-6-16
- PARA 9-4-107

9-4-105.1 Replace Sealed Lead-Acid Battery Junction Box Relays, K200 and K201.

9-4-105.1.1 Remove.

- Remove sealed lead-acid battery (SLAB) junction box (PARA 9-4-107).
- Loosen captive screws securing cover to SLAB junction box. Remove cover (Figure 9-4-105-1).
- To remove relay, K200, perform the following:
 - Tag wires. Remove nuts and washers securing wires and diode to terminals of relay, K200.
 - Remove bolts and washers securing relay, K200, to SLAB junction box and remove relay, K200.
 - Remove relay, K200, from SLAB junction box.
- To remove relay, K201, perform the following:

- Remove screws, lockwashers, and washers securing relay, K201, to SLAB junction box.
- Remove relay, K201, from SLAB junction box.

9-4-105.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).
- Install relay, K201, in (SLAB) junction box and secure with screws, lockwashers, and washers (Figure 9-4-105-1).
- To install relay, K200, perform the following:
 - Install relay, K200, in SLAB junction box and secure with bolts and washers.
 - TORQUE BOLTS TO 49 - 55 INCH-POUNDS.**
 - Install diode and wires on terminals and secure with nuts and washers. Remove tags.

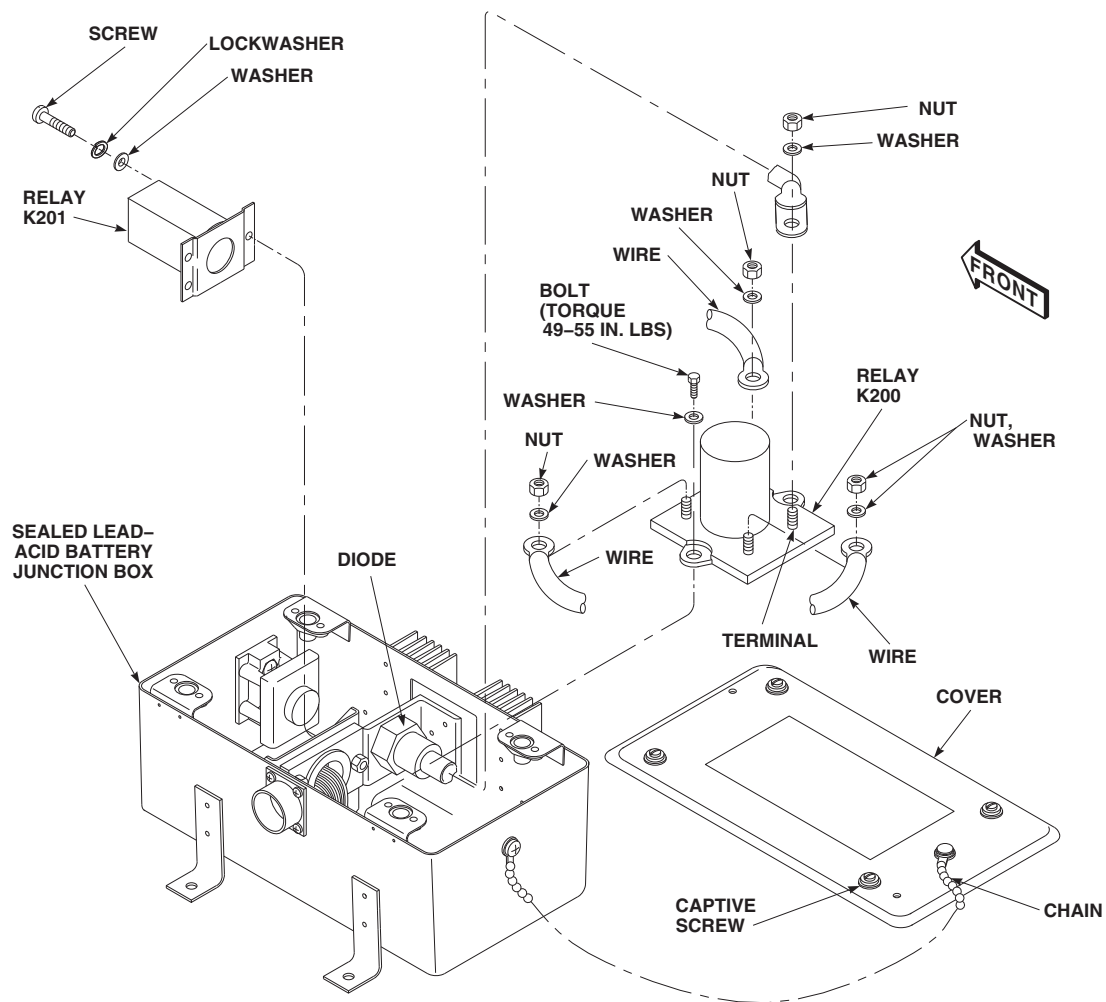
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FIGURE 9-4-105-1. REPLACE SEALED LEAD-ACID BATTERY JUNCTION BOX RELAYS, K200 AND K201

- d. Make sure area is clean and free of foreign material.
- e. Install cover on SLAB junction box.
- f. **TIGHTEN CAPTIVE SCREWS.**
- g. Install SLAB junction box (PARA 9-4-107).

9-4-106 SEALED LEAD-ACID BATTERY 702375 – SUBQ

PARAGRAPH BREAKDOWN

	PAGE
9-4-106.1 Replace Sealed Lead-Acid Battery	9-4-106-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

Materials/Parts

- Cheesecloth, Item 90, Appendix D
- Distilled Water, Item 123, Appendix D

- Lockwire, Item 195, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 9-2-2

9-4-106.1 Replace Sealed Lead-Acid Battery.

9-4-106.1.1. Remove.

WARNING

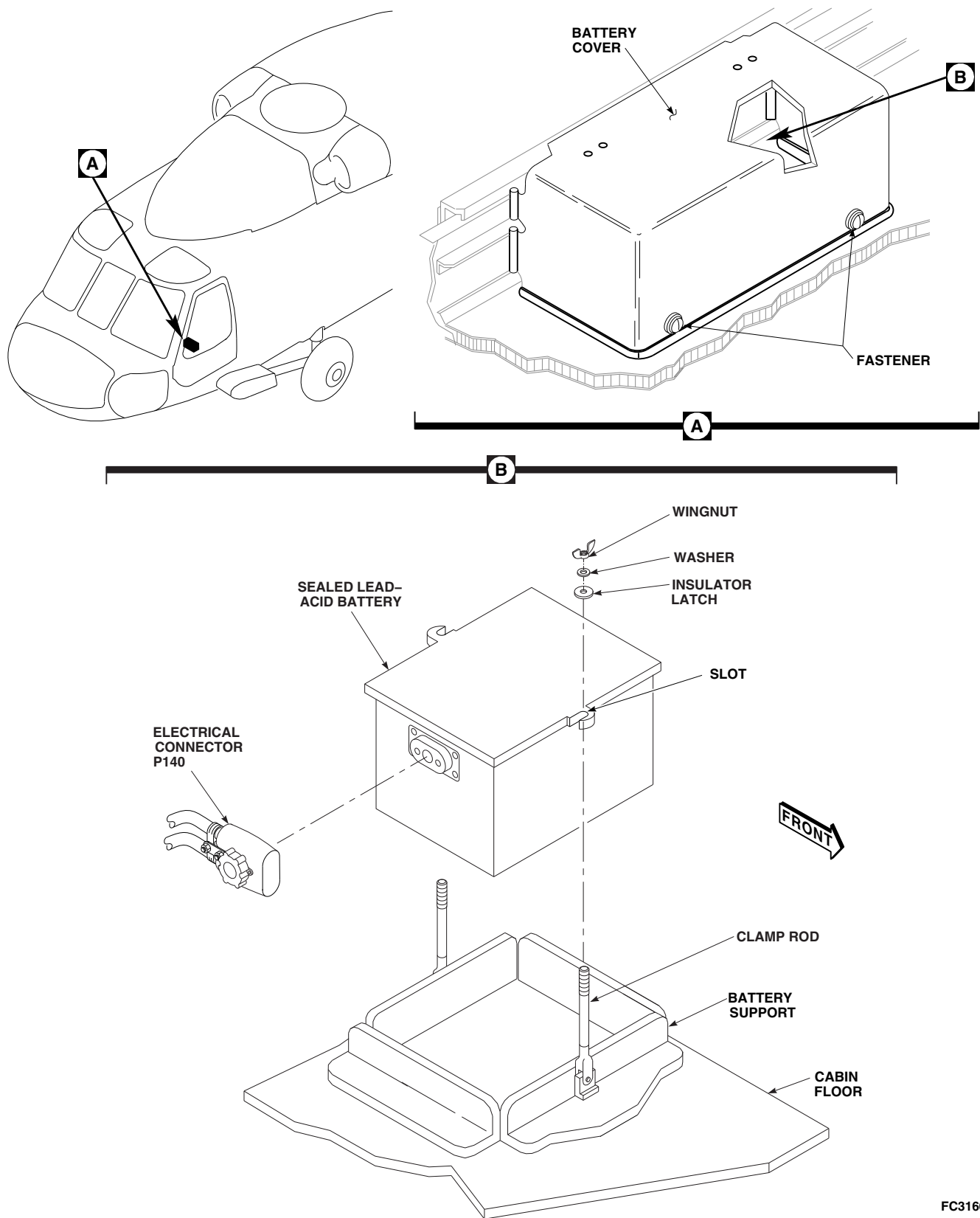
Do not remove sealed lead-acid battery (SLAB) while external power is connected, or while either engine or APU is operating. Serious fires can result.

- Turn off all electrical power (PARA 1-6-16).
- Loosen fasteners and remove battery cover from battery support (Figure 9-4-106-1, Detail A).
- Disconnect electrical connector, P140, from battery (Figure 9-4-106-1, Detail B).
- Install protective caps and plugs on electrical connectors.

- Remove wingnuts, washers, and insulator latches from clamp rods.
- Remove SLAB from battery support and cabin floor.

9-4-106.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Using cheesecloth, Item 90, Appendix D, dampened with distilled water, Item 123, Appendix D, clean sealed lead-acid battery (SLAB), battery cover, and battery support. Using clean, dry cheesecloth, Item 90, Appendix D, dry surface.
- Make sure that battery support is free of foreign material.
- Before installing SLAB, make sure replacement SLAB has been serviced.
- Place SLAB in battery support on cabin floor, between clamp rods. Position clamp rods

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- through slots in SLAB (Figure 9-4-106-1, Detail B).
- f. Install insulator latches, washers, and wingnuts on clamp rods securing SLAB.
- g. **TIGHTEN WINGNUTS.**
- h. Lockwire, Item 195, Appendix D, wingnuts.
- i. Remove protective caps and plugs from electrical connectors.
- j. Inspect and treat electrical connectors (PARA 1-7-20).
- k. Connect electrical connector, P140, to SLAB.
- l. Lockwire, Item 195, Appendix D, electrical connector.
- m. Make sure area is clean and free of foreign material.
- n. Install battery cover on battery support and secure with fasteners (Figure 9-4-106-1, Detail A).
- o. Perform operational check of dc electrical system (PARA 9-2-2).

9-4-107 SEALED LEAD-ACID BATTERY JUNCTION BOX
 702375 – SUBQ

PARAGRAPH BREAKDOWN

	PAGE
9-4-107.1 Replace Sealed Lead-Acid Battery Junction Box	9-4-107-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

Materials/Parts

- Cheesecloth, Item 90, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Protective Caps and Plugs

- Tags

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 9-2-2
- PARA 9-4-106

9-4-107.1 Replace Sealed Lead-Acid Battery Junction Box.

9-4-107.1.1 Remove.

WARNING

Do not remove sealed lead-acid battery (SLAB) junction box while external power is connected, while either engine or APU is operating, or while SLAB is connected. Serious fires can result.

- Turn off all electrical power (PARA 1-6-16).
- Remove bolts and washers securing lower panel to cabin avionics rack and remove lower panel.

- Disconnect electrical connector, P140, from battery (PARA 9-4-106).
- Disconnect electrical connector, P2008, from SLAB junction box (Figure 9-4-107-1, Detail A).
- Install protective caps and plugs on electrical connectors.
- Loosen fasteners and remove battery cover from battery support.
- Unscrew and remove cover from terminal board, TB24.
- Tag wires. Remove nuts, lockwashers, and washers securing wires to terminal board, TB24.

- i. Tag wire. Remove nut and washer securing wire to relay, K200, stud, A1.
- j. Remove grommet and slide down wiring away from SLAB junction box.
- k. Carefully remove wiring by passing through access hole in SLAB junction box.
- l. Remove screws and washers securing SLAB junction box to helicopter.
- m. Remove SLAB junction box from helicopter.

9-4-107.1.2 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Using cheesecloth, Item 90, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D, clean surface around all mounting holes. Using clean, dry cheesecloth, Item 90, Appendix D, dry surface.
- c. Install sealed lead-acid battery (SLAB) junction box on helicopter and secure with screws and washers (Figure 9-4-107-1, Detail A).
- d. Carefully pass wiring through access hole and install grommet.
- e. Install wire on relay, K200, stud, A1, and secure with nut and washer. Remove tag.
- f. Install wires on terminal board, TB24, and secure with nuts, lockwashers, and washers. Remove tags.
- g. Install cover on terminal board, TB24.
- h. Install battery cover to battery support and secure with fasteners.
- i. Remove protective cap and plug from electrical connectors.
- j. Inspect and treat electrical connectors (PARA 1-7-20).
- k. Connect electrical connector, P2008, to SLAB junction box.
- l. Connect electrical connector, P140, to battery (PARA 9-4-106).
- m. Make sure area is clean and free of foreign material.
- n. Perform operational check of DC electrical system (PARA 9-2-2).
- o. Install lower panel on cabin avionics rack with bolts and washers.

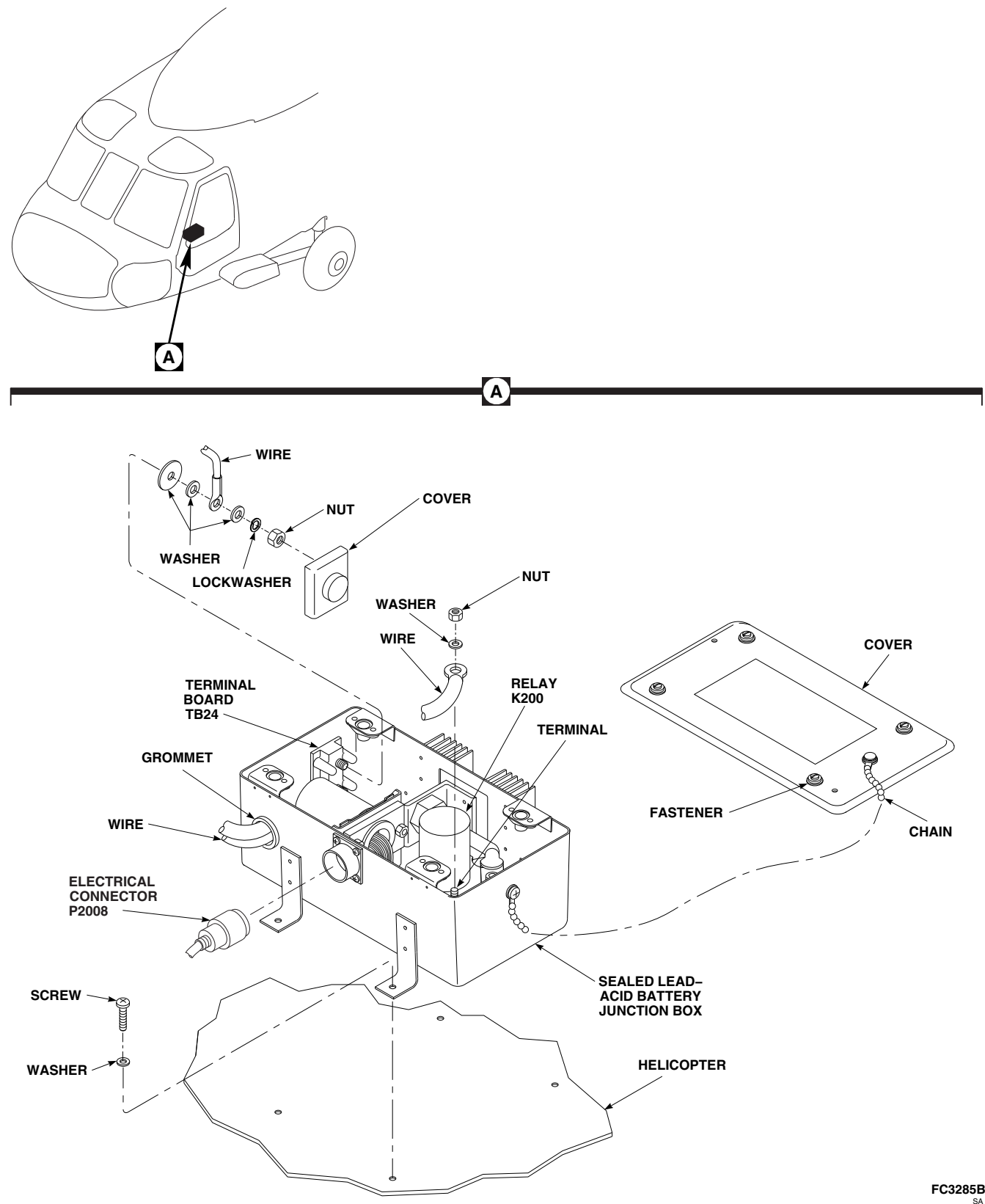


FIGURE 9-4-107-1. REPLACE SEALED LEAD-ACID BATTERY JUNCTION BOX

9-4-108 NO. 7 RELAY PANEL.

PARAGRAPH BREAKDOWN

	PAGE
9-4-108.1 Replace No. 7 Relay Panel	9-4-108-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

Materials/Parts

- Dry-Cleaning Solvent, Item 124, Appendix D
- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs

- Tags

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-69
- PARA 14-2-1

9-4-108.1 Replace No. 7 Relay Panel.

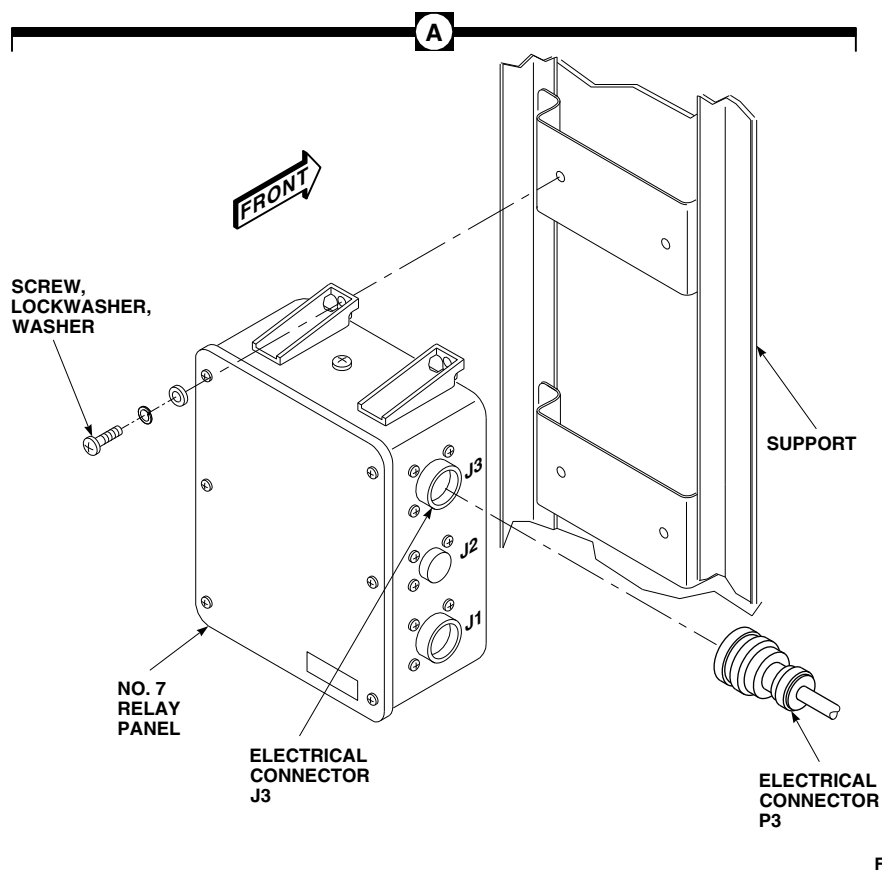
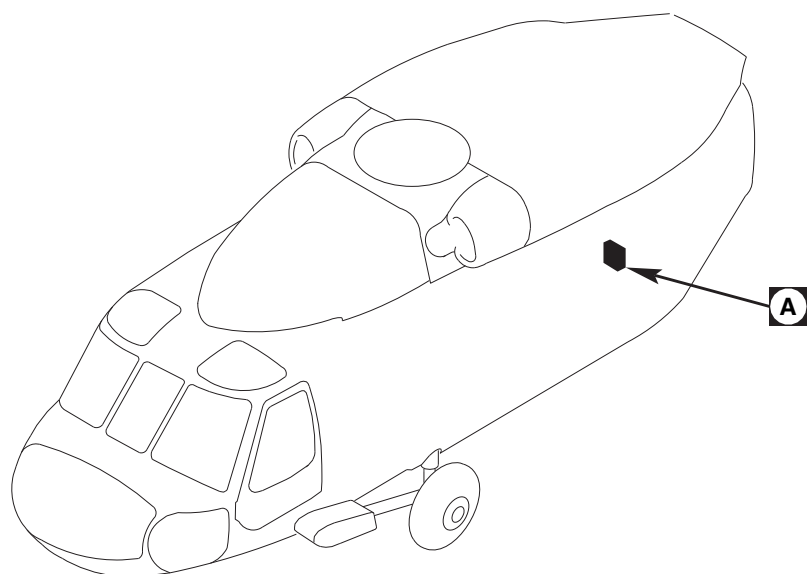
9-4-108.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Remove soundproofing panel (PARA 2-4-69).
- Tag electrical connectors. Disconnect electrical connector, P3 from J3, on No. 7 relay panel (Figure 9-4-108-1, Detail A).
- Install protective caps and plugs on electrical connectors.
- Remove screws, lockwashers, and washers securing No. 7 relay panel to support.
- Remove No. 7 relay panel.

9-4-108.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).

- Using machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D, clean surfaces around mounting holes for good electrical ground contact.
- Install No. 7 relay panel on support with screws, lockwashers, and washers (Figure 9-4-108-1, Detail A).
- Remove protective caps and plugs from electrical connectors.
- Inspect and treat electrical connectors (PARA 1-7-20).
- Connect electrical connector, P3 to J3, to No. 7 relay panel. Remove tags.
- Perform operational check of the rescue hoist system (PARA 14-2-1).
- Make sure area is clean and free of foreign material.



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FIGURE 9-4-108-1. REPLACE NO. 7 RELAY PANEL

9-4-108-2

- i. Install soundproofing panel (PARA 2-4-69).

9-4-109 DIMMING RESISTOR, R10.

PARAGRAPH BREAKDOWN

	PAGE
9-4-109.1 Replace Dimming Resistor, R10	9-4-109-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit

Materials/Parts

- Tags
- Tiedown Strap, Item 329, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 9-2-14

9-4-109.1 Replace Dimming Resistor, R10.

9-4-109.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- On upper console, place BATT switch to OFF.
- Loosen fasteners securing upper console and carefully lower.
- Remove screws, washers, spacers, and nuts securing dimming resistor, R10, to support (Figure 9-4-109-1, Detail A).
- Remove tiedown straps securing dimming resistor, R10, wires.
- Tag wires. Remove nuts, lockwashers, and washers securing terminal lugs to terminal board, TB2.
- Remove dimming resistor, R10.

9-4-109.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Install wires to terminal board, TB2, on terminals 3 and 4, and secure with nuts, lockwashers, and washers (Figure 9-4-109-1, Detail A). Remove tags.
- Install tiedown straps, Item 329, Appendix D, to secure wires.
- Secure dimming resistor, R10, to support with screws, washers, spacers, and nuts.
- Make sure area is clean and free of foreign material.
- Carefully raise upper console and secure with fasteners.
- Perform operational check of position lights (PARA 9-2-14).

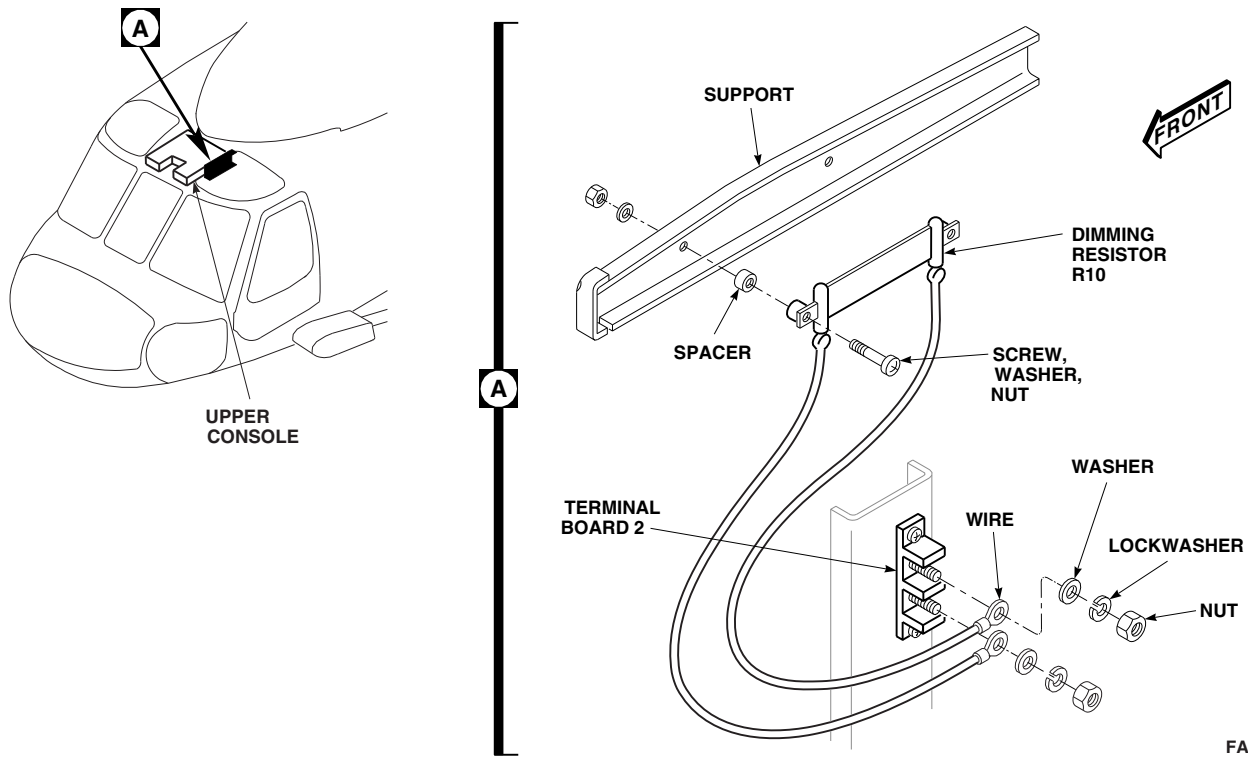


FIGURE 9-4-109-1. REPLACE DIMMING RESISTOR, R10

9-4-110 AUXILIARY SEARCHLIGHT BRACKET INCANDESCENT POSITION LIGHT LAMPS AND LENSES.

PARAGRAPH BREAKDOWN

	PAGE
9-4-110.1 Replace Auxiliary Searchlight Bracket Incandescent Position Light Lamps and Lenses	9-4-110-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

References

- PARA 1-6-16
- PARA 9-2-14

9-4-110.1 Replace Auxiliary Searchlight Bracket Incandescent Position Light Lamps and Lenses.

NOTE

Replacement of left and right auxiliary searchlight bracket incandescent position light lamps and lenses is the same.

9-4-110.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Remove screw and rubber washer securing lens to incandescent position light (Figure 9-4-110-1, Detail A).

- Remove lamp from incandescent position light.

9-4-110.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).
- Install lamp (Figure 9-4-110-1, Detail A).

NOTE

Right side lens is green. Left side lens is red.

- Install lens on incandescent position light and secure with rubber washer and screw.
- Perform operational check of position lights (PARA 9-2-14).

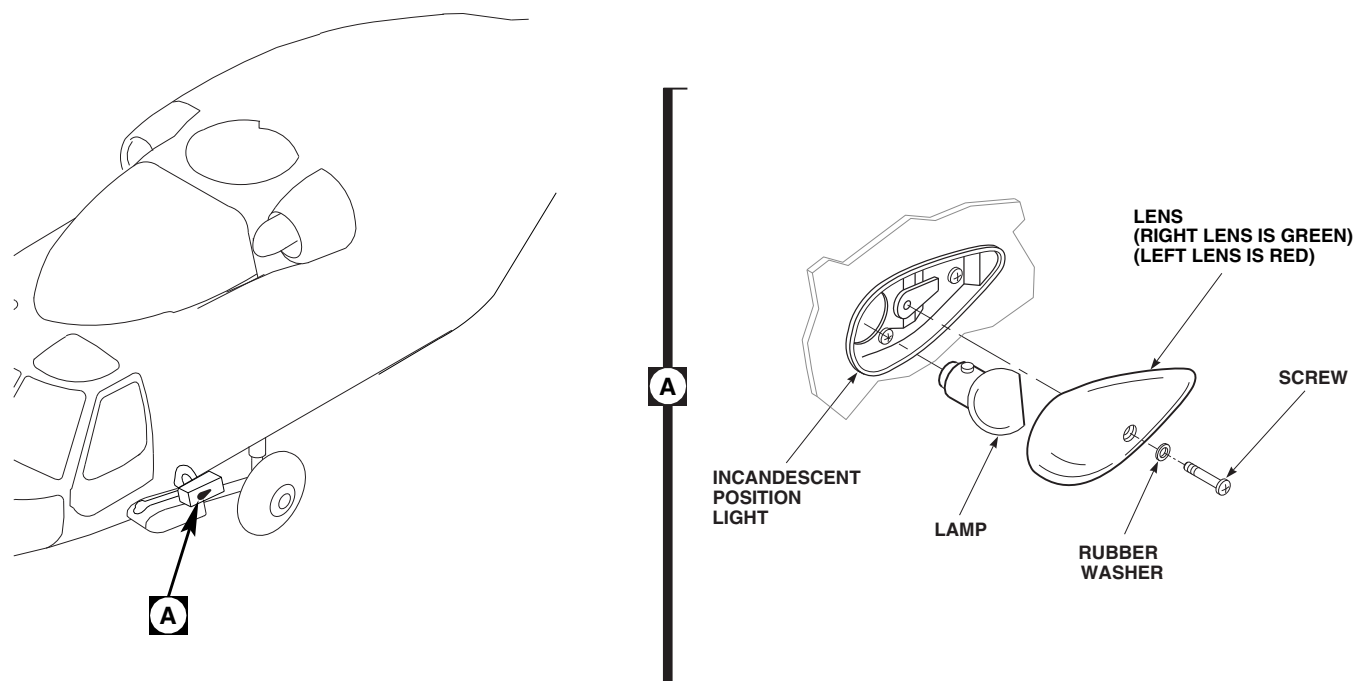
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FIGURE 9-4-110-1. REPLACE AUXILIARY SEARCHLIGHT BRACKET INCANDESCENT POSITION LIGHT LAMPS AND LENSES

9-4-111 **AUXILIARY SEARCHLIGHT BRACKET INCANDESCENT POSITION LIGHT.**

PARAGRAPH BREAKDOWN

	PAGE
9-4-111.1 Replace Auxiliary Searchlight Bracket Incandescent Position Light	9-4-111-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

Materials/Parts

- Protective Caps and Plugs
- Sealing Compound, Item 292, Appendix D
- Tape, Item 317C, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 9-2-14
- PARA 9-4-34
- PARA 9-4-110
- PARA 9-4-106

9-4-111.1 **Replace Auxiliary Searchlight Bracket Incandescent Position Light.**

9-4-111.1.1. **Remove.**

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Disconnect electrical connector, P340A from J340A, and electrical connector, P254A from J254A (Figure 9-4-111-1, Detail A).
- Install protective caps and plugs on electrical connectors.

- Remove screws and washers securing cover assembly to auxiliary searchlight bracket. Remove cover assembly.
- Remove screw, washers, nut, and clamp securing wiring to cover assembly.
- Remove screws, washers, and nuts securing electrical connector, J254A.
- Remove tape securing slack in incandescent and infrared position light wiring.
- Remove auxiliary searchlight bracket incandescent position light lamp and lens (PARA 9-4-110).

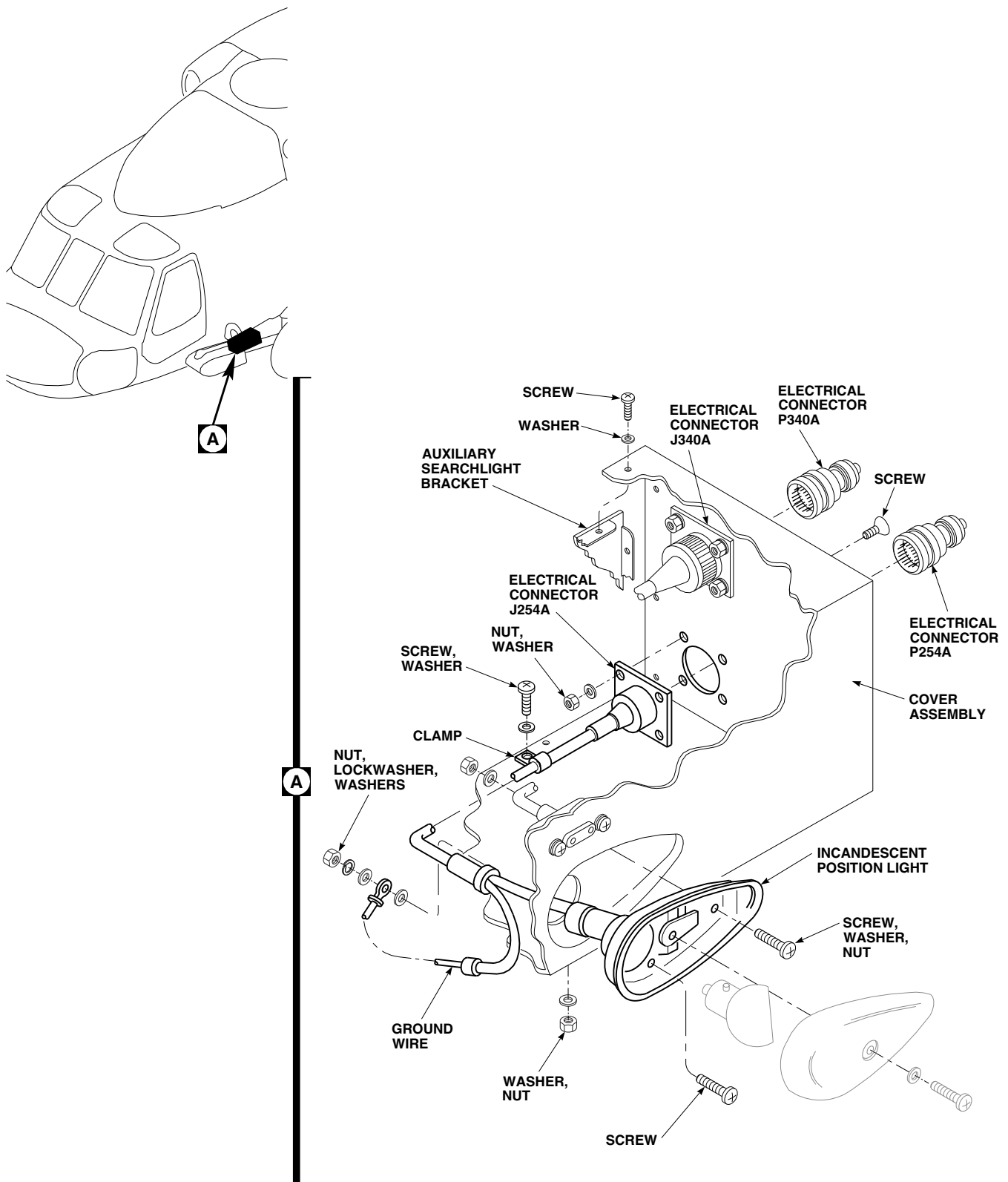
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FIGURE 9-4-111-1 REPLACE AUXILIARY SEARCHLIGHT BRACKET INCANDESCENT POSITION LIGHT

9-4-111-2

- j. Remove screw, washers, lockwasher, and nut securing ground wire and incandescent position light to cover assembly.
- k. Remove screw, washer, nut securing incandescent position light to cover assembly.
- l. Remove incandescent position light.

9-4-111.1.2. Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Install incandescent position light to cover assembly and secure with screw, washer, and nut (Figure 9-4-111-1, Detail A).
- c. Apply sealing compound, Item 292, Appendix D, around exterior edge of incandescent position light and cover assembly.
- d. Install ground wire and secure ground wire and incandescent position light to cover assembly with screw, washers, lockwasher, and nut.
- e. Install clamp securing wiring to cover assembly and secure with screw, washers, and nut.
- f. Install auxiliary searchlight bracket incandescent position light lamp and lens (PARA 9-4-110).
- g. Remove protective caps and plugs from electrical connector.
- h. Inspect and treat electrical connector (PARA 1-7-20).
- i. Install electrical connector, J254A, to cover assembly and secure with screws, washers, and nuts.
- j. Secure slack, as required, in incandescent and infrared position light wiring using tape, Item 317C, Appendix D.
- k. Make sure area is free of foreign objects before installing cover assembly.
- l. Install cover assembly on auxiliary searchlight bracket and secure with screws and washers.
- m. Remove protective caps and plugs from electrical connectors.
- n. Inspect and treat electrical connectors (PARA 1-7-20).
- o. Connect electrical connector, P340A to J340A, and electrical connector, P254A to J254A.
- p. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- q. Perform operational check of position lights (PARA 9-2-14).

9-4-112 **AUXILIARY SEARCHLIGHT BRACKET INFRARED POSITION LIGHT.**

PARAGRAPH BREAKDOWN

	PAGE
9-4-112.1 Replace Auxiliary Searchlight Bracket Infrared Position Light	9-4-112-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit

Materials/Parts

- Protective Caps and Plugs
- Sealing Compound, Item 292, Appendix D
- Tape, Item 317C, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 9-2-14
- PARA 9-4-34
- PARA 9-4-106

9-4-112.1 **Replace Auxiliary Searchlight Bracket Infrared Position Light.**

9-4-112.1.1. **Remove.**

- Turn off all electrical power (PARA 1-6-16).
- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- Disconnect electrical connector, P340A from J340A, and electrical connector, P254A from J254A (Figure 9-4-112-1, Detail A).
- Install protective caps and plugs on electrical connectors.

- Remove screws and washers securing cover assembly to auxiliary searchlight bracket. Remove cover assembly.
- Remove screws, washers, and nuts securing electrical connector, J340A, to cover assembly.
- Remove tape securing slack in incandescent and infrared position light wiring.
- Remove screws and washers securing infrared position light to cover assembly.
- Remove infrared position light.

9-4-112.1.2. **Install.**

- Turn off all electrical power (PARA 1-6-16).

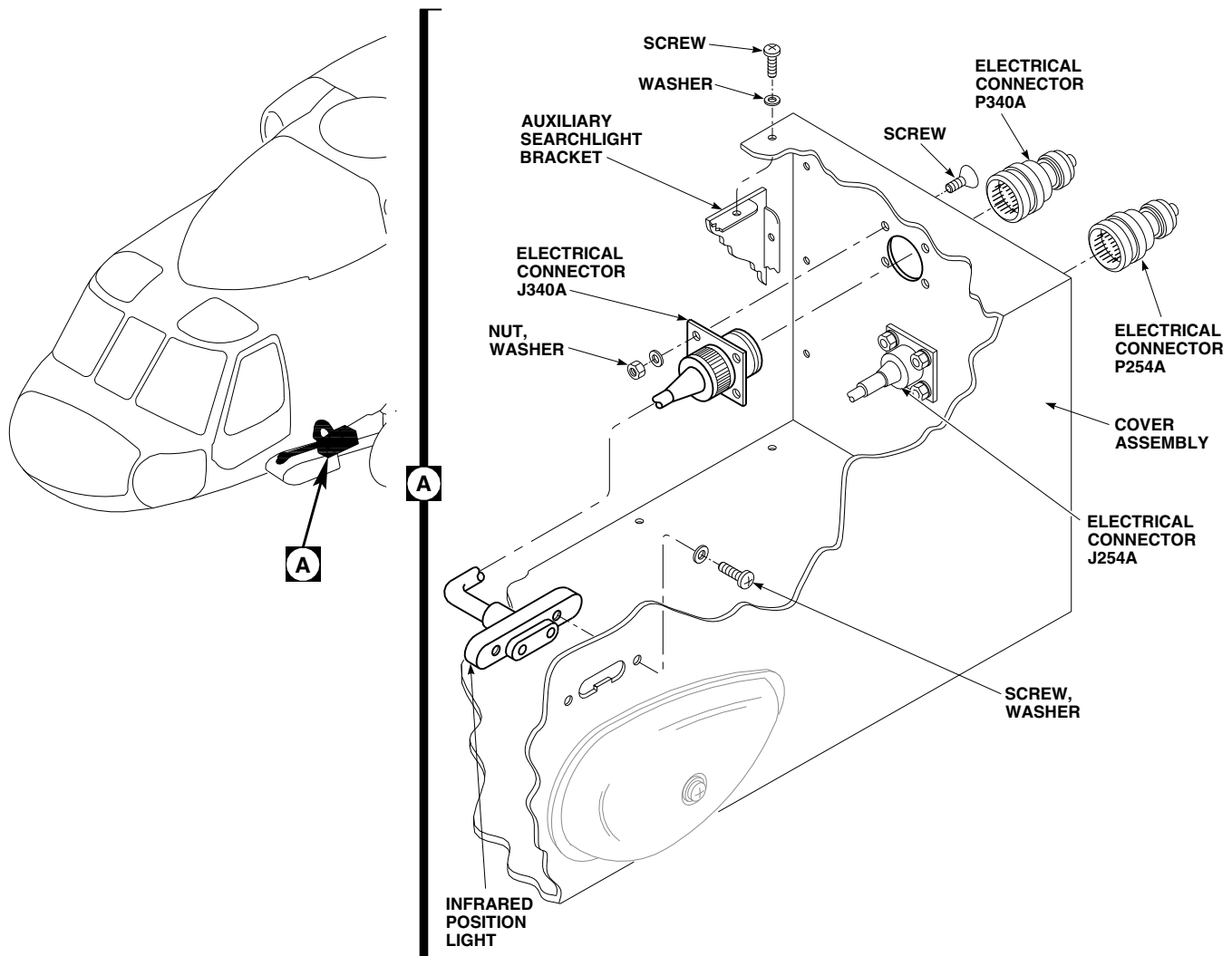
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FIGURE 9-4-112-1 REPLACE AUXILIARY SEARCHLIGHT BRACKET INFRARED POSITION LIGHT

- b. Install infrared position light on cover assembly and secure with screws and washers (Figure 9-4-112-1, Detail A).
- c. Apply sealing compound, Item 292, Appendix D, around exterior edge of infrared position light and cover assembly.
- d. Remove protective caps and plugs from electrical connector.
- e. Inspect and treat electrical connector (PARA 1-7-20).
- f. Install electrical connector, J340A, to cover assembly and secure with screws, washers, and nuts.
- g. Secure slack, as required, in incandescent and infrared position light wiring using tape, Item 317C, Appendix D.
- h. Make sure area is free of foreign objects before installing cover assembly.
- i. Install cover assembly onto auxiliary searchlight bracket and secure with screws and washers.

9-4-112-2

- j. Remove protective caps and plugs from electrical connectors.
- k. Inspect and treat electrical connectors (PARA 1-7-20).
- l. Connect electrical connector, P340A to J340A, and electrical connector, P254A to J254A.
- m. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- n. Perform operational check of position lights (PARA 9-2-14).

SECTION 9-5

MAINTENANCE PROCEDURES (BENCH)

SECTION OVERVIEW

PARAGRAPH	TITLE
9-5-1	Left Relay Panel Left Pitot Heat Transformer, T1 (BENCH)
9-5-2	Left Relay Panel Searchlight Relay, K41 (BENCH)
9-5-3	Left Relay Panel Searchlight Relay, K51 (BENCH)
9-5-4	Left Relay Panel Searchlight Relay, K52 (BENCH)
9-5-5	Left Relay Panel No. 1 Engine Start Relay, K45 (BENCH)
9-5-6	Left Relay Panel Indicator Lights Dimming Control Relay, K40 (BENCH)
9-5-7	Right Relay Panel Current Sensor Right Pitot Heat Transformer, T1 (BENCH)
9-5-8	Right Relay Panel DC Essential Bus Fail Relay, K20 (BENCH)
9-5-9	Right Relay Panel Windshield Anti-Ice Lockout Relay, K21 (BENCH)
9-5-10	Right Relay Panel APU Surge Relay, K22 (BENCH)
9-5-11	Right Relay Panel No. 2 Engine Start Relay, K26 (BENCH)
9-5-12	Right Relay Panel Hydraulic Accumulator Relay, K32 (BENCH)
9-5-13	Miscellaneous Switch Panel Pushbutton Switches (BENCH)
9-5-14	Miscellaneous Switch Panel Toggle Switch (BENCH)
9-5-15	Miscellaneous Switch Panel Electrical Connectors (BENCH)
9-5-16	Miscellaneous Switch Panel Printed Wiring Board (BENCH)
9-5-17	Fuel Boost Pump Control Panel Toggle Switches (BENCH)
9-5-18	Fuel Boost Pump Control Panel Indicator Lights (BENCH)

PARAGRAPH	TITLE
9-5-19	Fuel Boost Pump Control Panel Information Plate Lamps (BENCH)
9-5-20	Fuel Boost Pump Control Panel Electrical Connector (BENCH)
9-5-21	Battery Harness (BENCH)
9-5-22	Battery Electrical Receptacle, J2 (BENCH)
9-5-23	Battery Temperature Sensors (BENCH)
9-5-24	Battery Overtemperature Switches (BENCH)
9-5-25	Battery Clamping Diode (BENCH)
9-5-26	Battery Cells (BENCH)
9-5-27	Battery Test Vent Cap (BENCH)
9-5-28	Pressure Test Battery Cell Case (BENCH)
9-5-29	Battery Case and Cover (BENCH)
9-5-30	Pressure Test Battery Case (BENCH)
9-5-31	Recondition Battery (BENCH)
9-5-32	Deep Cycle Battery (BENCH)
9-5-33	New Battery (BENCH)
9-5-34	Battery Long-Term Storage (BENCH)
9-5-35	Battery Short-Term Storage (BENCH)
9-5-36	Controllable Searchlight (BENCH)
9-5-37	Indicator Lights Dimmer Box (BENCH)

9-5-1 LEFT RELAY PANEL LEFT PITOT HEAT TRANSFORMER, T1 (BENCH).

PARAGRAPH BREAKDOWN

	PAGE
9-5-1.1 Replace Left Relay Panel Left Pitot Heat Transformer, T1 (BENCH)	9-5-1-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit
- Protection, Eye and Skin
- Soldering Iron

Materials/Parts

- Solder, Item 313, Appendix D
- Tags

References

- Appendix D
- PARA 9-2-20

9-5-1.1 Replace Left Relay Panel Left Pitot Heat Transformer, T1 (BENCH).

9-5-1.1.1. Remove.

- Remove screws and washers that secure bottom cover to left relay panel (Figure 9-5-1-1).
- Remove nuts that secure left pitot heat transformer, T1, to left relay panel.
- Carefully lift left pitot heat transformer, T1, from left relay panel to gain access to wires.

- injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.
- Tag wires. Using soldering iron, unsolder wires from left pitot heat transformer, T1.
 - Remove left pitot heat transformer, T1.

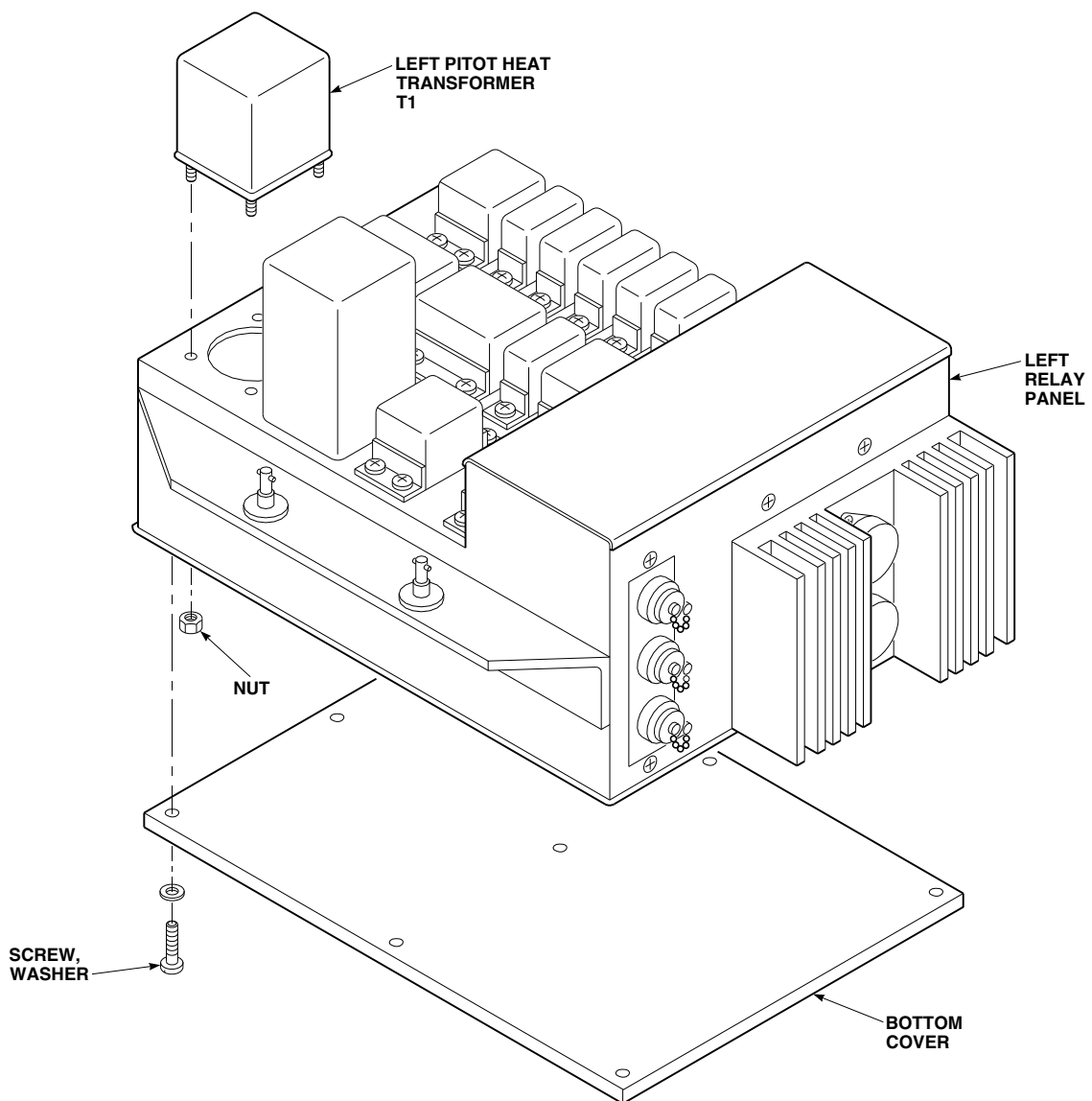
WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and

9-5-1.1.2. Install.**WARNING**

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- a. Using soldering iron and solder, Item 313, Appendix D, solder wires to left pitot heat transformer, T1. Remove tags.
- b. Install left pitot heat transformer, T1, in left relay panel and secure with nuts (Figure 9-5-1-1).
- c. Make sure area is clean and free of foreign material.
- d. Install bottom cover to left relay panel and secure with screws and washers.
- e. Perform operational check of left relay panel (PARA 9-2-20).



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FIGURE 9-5-1-1. REPLACE LEFT RELAY PANEL LEFT PITOT HEAT TRANSFORMER, T1 (BENCH)

9-5-2 LEFT RELAY PANEL SEARCHLIGHT RELAY, K41 (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
9-5-2.1	Replace Left Relay Panel Searchlight Relay, K41 (BENCH)	9-5-2-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit
- Protection, Eye and Skin
- Soldering Iron

Materials/Parts

- Solder, Item 313, Appendix D
- Tags

References

- Appendix D
- PARA 9-2-20

9-5-2.1 Replace Left Relay Panel Searchlight Relay, K41 (BENCH).

9-5-2.1.1. Remove.

- Remove screws and washers that secure bottom cover to left relay panel (Figure 9-5-2-1).
- Remove screws, washers, and nuts that secure searchlight relay, K41, to left relay panel.
- Carefully lift searchlight relay, K41, from left relay panel to gain access to wires.

injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- Tag wires. Using soldering iron, unsolder wires from searchlight relay, K41.
- Remove searchlight relay, K41.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and

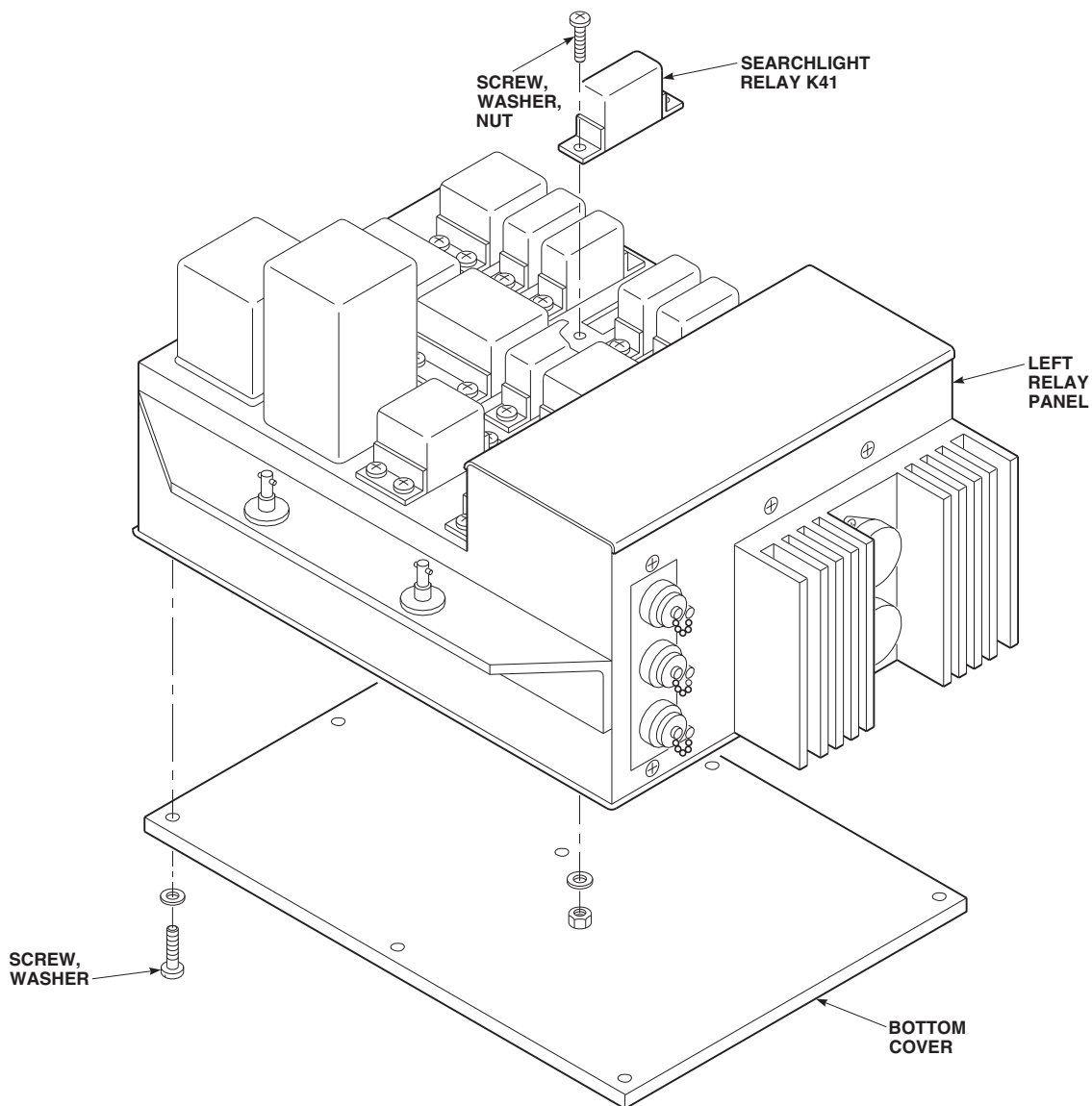
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FIGURE 9-5-2-1. REPLACE LEFT RELAY PANEL SEARCHLIGHT RELAY, K41 (BENCH)

9-5-2.1.2. Install.**WARNING**

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- a. Using soldering iron and solder, Item 313, Appendix D, solder wires to searchlight relay, K41. Remove tags.
- b. Install searchlight relay, K41, in left relay panel and secure with screws, washers, and nuts (Figure 9-5-2-1).
- c. Make sure area is clean and free of foreign material.
- d. Install bottom cover to left relay panel and secure with screws and washers.

9-5-2-2

- e. Perform operational check of left relay panel (PARA 9-2-20).

9-5-3 LEFT RELAY PANEL SEARCHLIGHT RELAY, K51 (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
9-5-3.1	Replace Left Relay Panel Searchlight Relay, K51 (BENCH)	9-5-3-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit
- Protection, Eye and Skin
- Soldering Iron

Materials/Parts

- Solder, Item 313, Appendix D
- Tags

References

- Appendix D
- PARA 9-2-20

9-5-3.1 Replace Left Relay Panel Searchlight Relay, K51 (BENCH).

9-5-3.1.1. Remove.

- Remove screws and washers that secure bottom cover to left relay panel (Figure 9-5-3-1).
- Remove screws, washers, and nuts that secure searchlight relay, K51, to left relay panel.
- Carefully lift searchlight relay, K51, from left relay panel to gain access to wires.

injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- Tag wires. Using soldering iron, unsolder wires from searchlight relay, K51.
- Remove searchlight relay, K51.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and

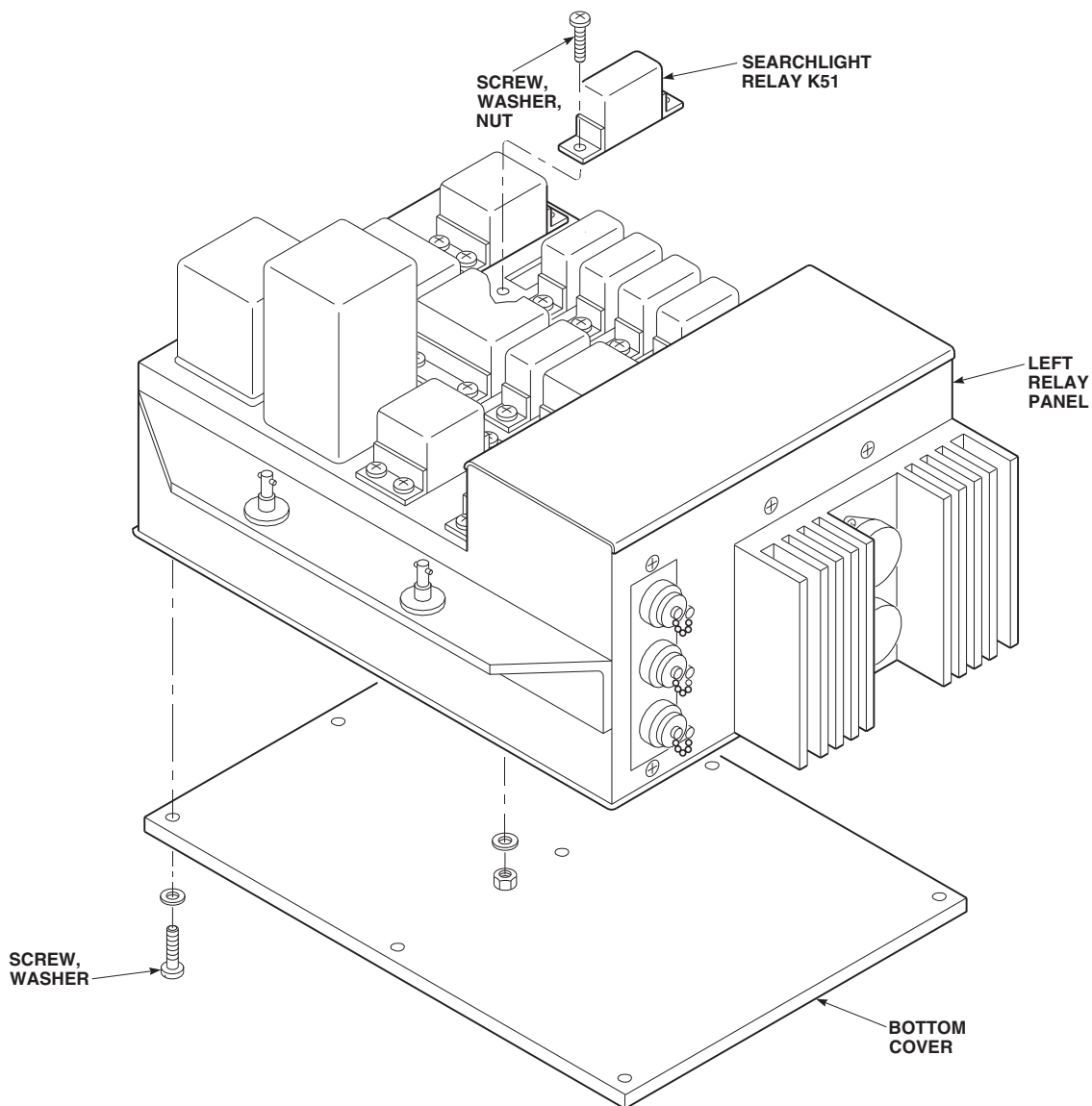
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FIGURE 9-5-3-1. REPLACE LEFT RELAY PANEL SEARCHLIGHT RELAY, K51 (BENCH)

9-5-3.1.2. Install.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- a. Using soldering iron and solder, Item 313, Appendix D, solder wires to searchlight relay, K51. Remove tags.
- b. Install searchlight relay, K51, in left relay panel and secure with screws, washers, and nuts (Figure 9-5-3-1).
- c. Make sure area is clean and free of foreign material.
- d. Install bottom cover to left relay panel and secure with screws and washers.

9-5-3-2

- e. Perform operational check of left relay panel (PARA 9-2-20).

9-5-4 LEFT RELAY PANEL SEARCHLIGHT RELAY, K52 (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
9-5-4.1	Replace Left Relay Panel Searchlight Relay, K52 (BENCH)	9-5-4-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit
- Protection, Eye and Skin
- Soldering Iron

Materials/Parts

- Solder, Item 313, Appendix D
- Tags

References

- Appendix D
- PARA 9-2-20

9-5-4.1 Replace Left Relay Panel Searchlight Relay, K52 (BENCH).

9-5-4.1.1. Remove.

- Remove screws and washers that secure bottom cover to left relay panel (Figure 9-5-4-1).
- Remove screws, washers, and nuts that secure searchlight relay, K52, to left relay panel.
- Carefully lift searchlight relay, K52, from left relay panel to gain access to wires.

injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- Tag wires. Using soldering iron, unsolder wires from searchlight relay, K52.
- Remove searchlight relay, K52.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and

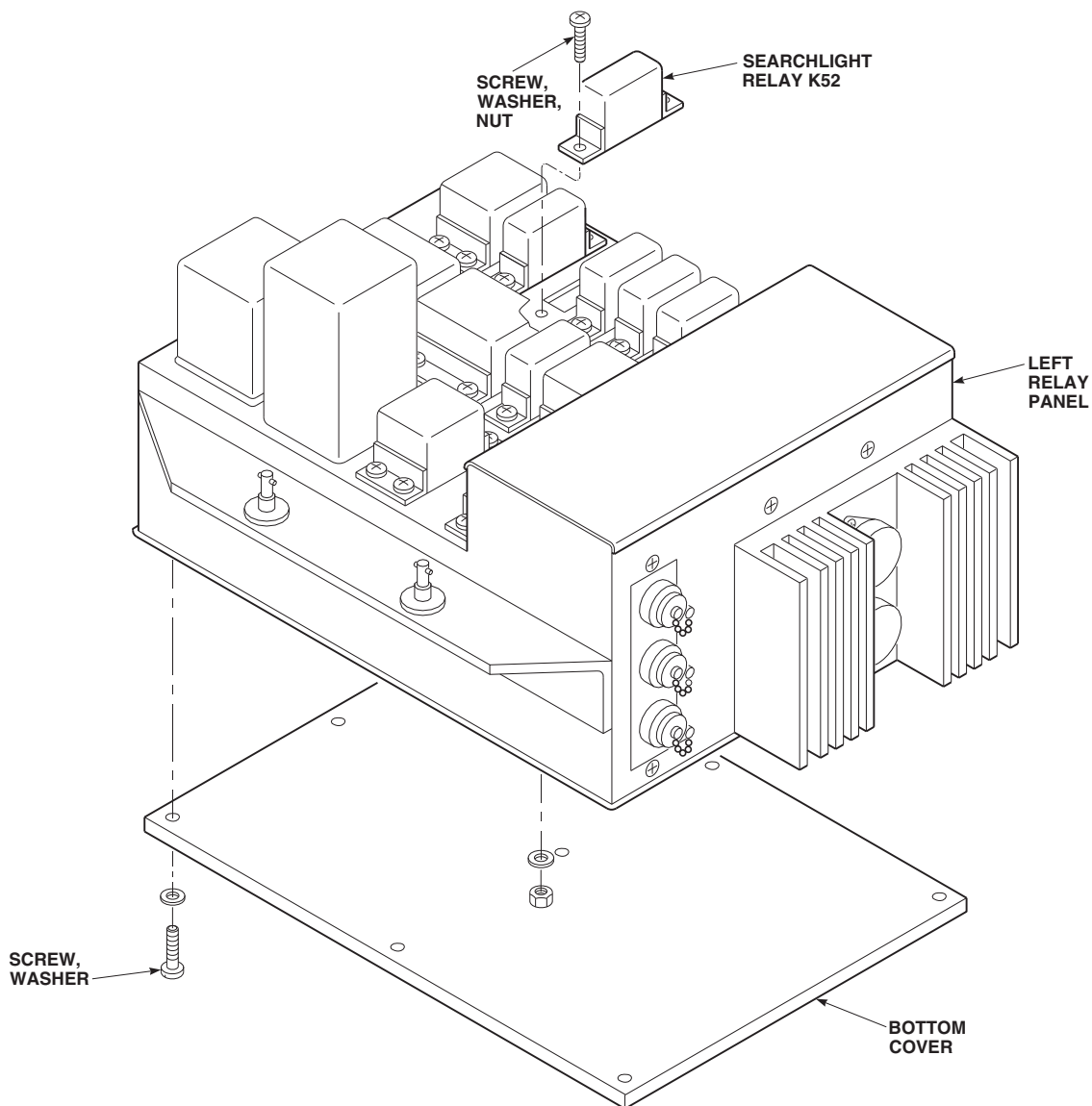
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FIGURE 9-5-4-1. REPLACE LEFT RELAY PANEL SEARCHLIGHT RELAY, K52 (BENCH)

9-5-4.1.2. Install.**WARNING**

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- a. Using soldering iron and solder, Item 313, Appendix D, solder wires to searchlight relay, K52. Remove tags.
- b. Install searchlight relay, K52, in left relay panel and secure with screws, washers, and nuts (Figure 9-5-4-1).
- c. Make sure area is clean and free of foreign material.
- d. Install bottom cover to left relay panel and secure with screws and washers.

9-5-4-2

- e. Perform operational check of left relay panel (PARA 9-2-20).

9-5-5 LEFT RELAY PANEL NO. 1 ENGINE START RELAY, K45 (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
9-5-5.1	Replace Left Relay Panel No. 1 Engine Start Relay, K45 (BENCH)	9-5-5-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit
- Protection, Eye and Skin
- Soldering Iron

Materials/Parts

- Solder, Item 313, Appendix D
- Tags

References

- Appendix D
- PARA 9-2-20

9-5-5.1 Replace Left Relay Panel No. 1 Engine Start Relay, K45 (BENCH).

9-5-5.1.1. Remove.

- Remove screws and washers that secure bottom cover to left relay panel (Figure 9-5-5-1).
- Remove screws, washers, and nuts that secure No. 1 engine start relay, K45, to left relay panel.
- Carefully lift No. 1 engine start relay, K45, from relay panel to gain access to wires.

injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- Tag wires. Using soldering iron, unsolder wires from No. 1 engine start relay, K45.
- Remove No. 1 engine start relay, K45.

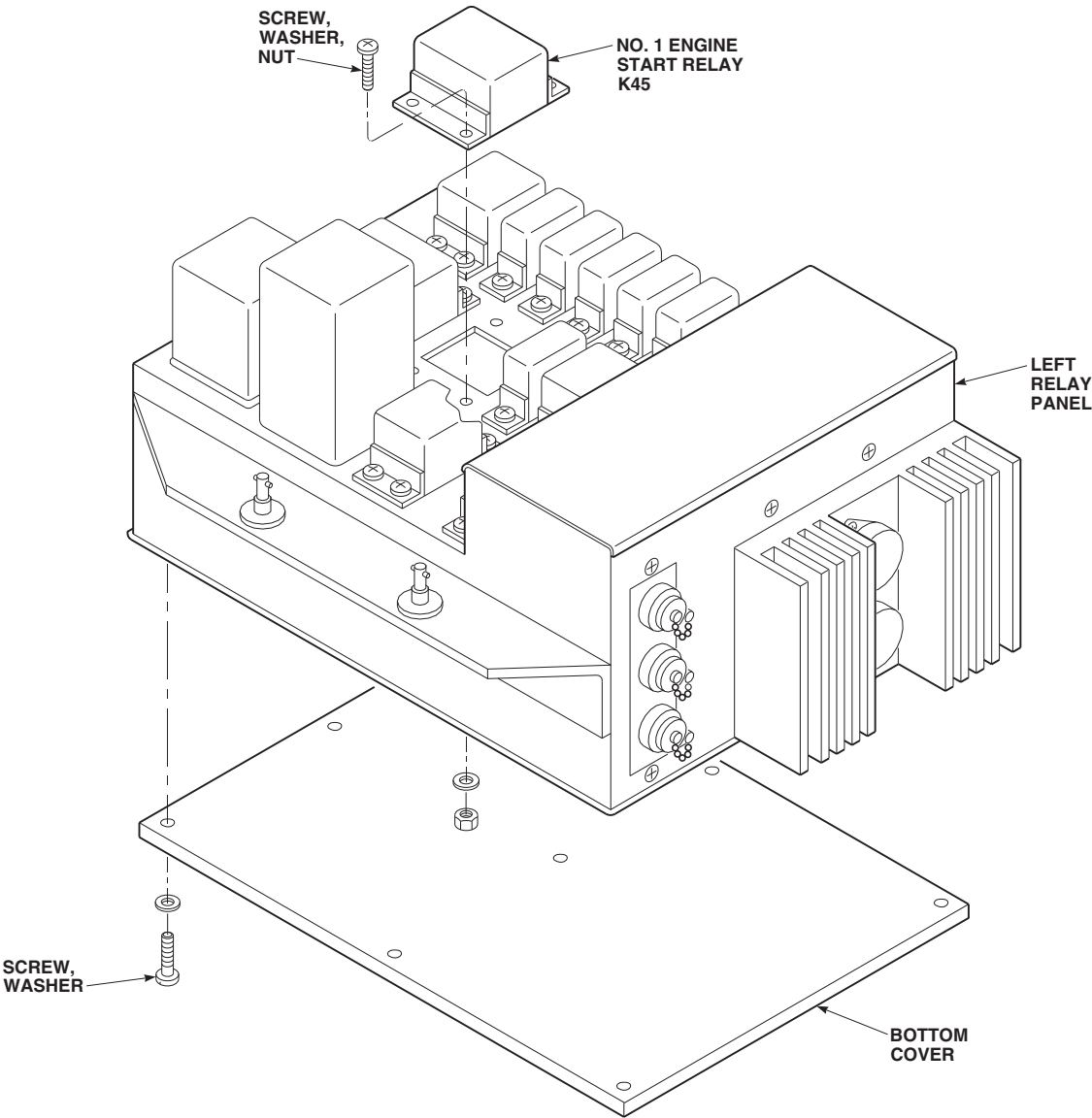
WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and

9-5-5.1.2. Install.**WARNING**

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- a. Using soldering iron and solder, Item 313, Appendix D, solder wires to No. 1 engine start relay, K45. Remove tags.
- b. Install No. 1 engine start relay, K45, in left relay panel and secure with screws, washers, and nuts (Figure 9-5-5-1).
- c. Make sure area is clean and free of foreign material.
- d. Install bottom cover to left relay panel and secure with screws and washers.
- e. Perform operational check of left relay panel (PARA 9-2-20).



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FIGURE 9-5-5-1. REPLACE LEFT RELAY PANEL NO. 1 ENGINE START RELAY, K45 (BENCH)

9-5-6 LEFT RELAY PANEL INDICATOR LIGHTS DIMMING CONTROL RELAY, K40 (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
9-5-6.1	Replace Left Relay Panel Indicator Lights Dimming Control Relay, K40 (BENCH)	9-5-6-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit
- Protection, Eye and Skin
- Soldering Iron

Materials/Parts

- Solder, Item 313, Appendix D
- Tags

References

- Appendix D
- PARA 9-2-20

9-5-6.1 Replace Left Relay Panel Indicator Lights Dimming Control Relay, K40 (BENCH).

9-5-6.1.1. Remove.

- Remove screws and washers that secure bottom cover to left relay panel (Figure 9-5-6-1).
- Remove screws, washers, and nuts that secure indicator lights dimming control relay, K40, to left relay panel.
- Carefully lift indicator lights dimming control relay, K40, from left relay panel to gain access to wires.



Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- Tag wires. Using soldering iron, unsolder wires from indicator lights dimming control relay, K40.

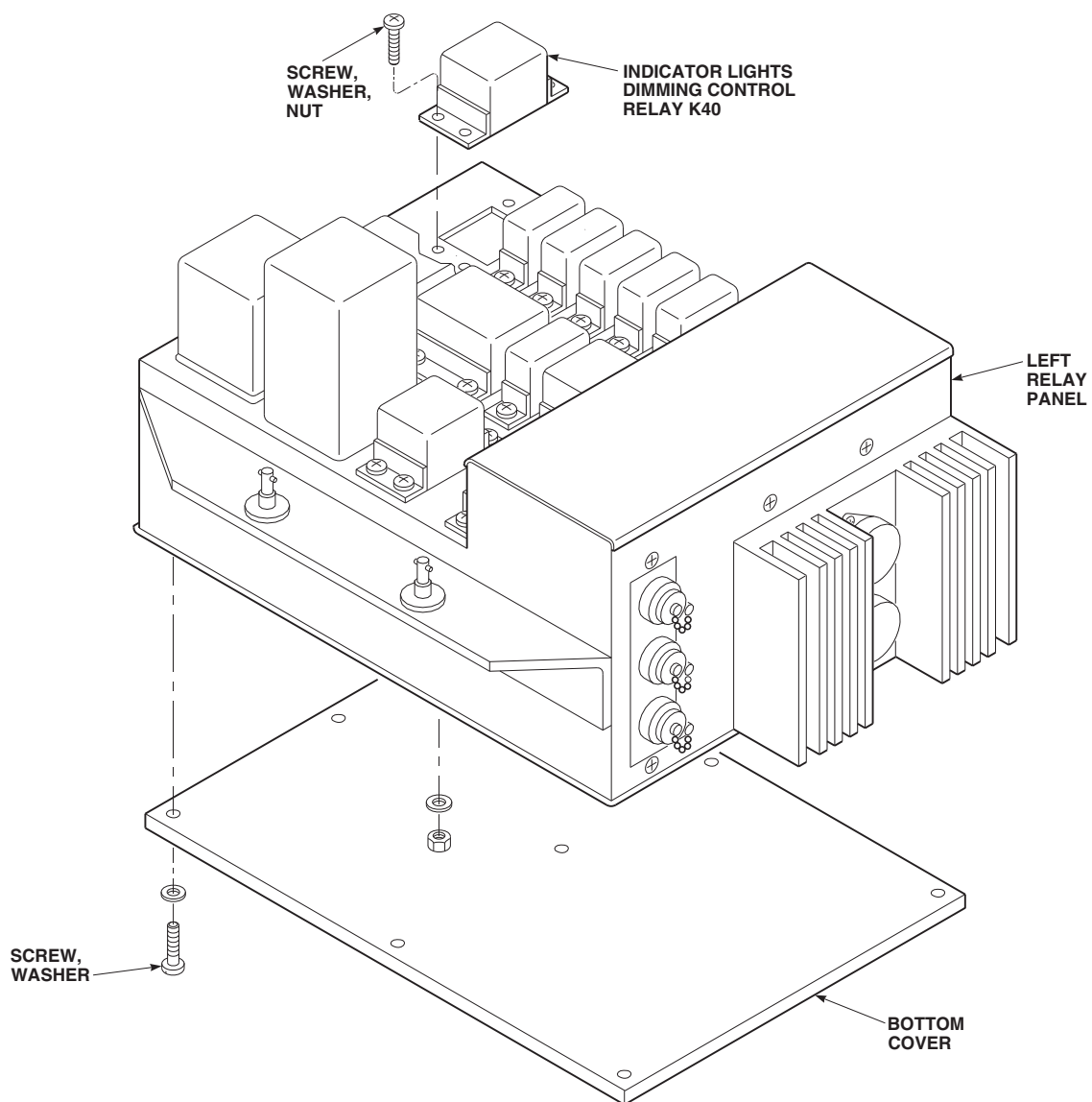
- e. Remove indicator lights dimming control relay, K40.

9-5-6.1.2. Install.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- a. Using soldering iron and solder, Item 313, Appendix D, solder wires to indicator lights dimming control relay, K40. Remove tags.
- b. Install indicator lights dimming control relay, K40, in left relay panel and secure with screws, washers, and nuts (Figure 9-5-6-1).
- c. Make sure area is clean and free of foreign material.
- d. Install bottom cover to left relay panel and secure with screws and washers.
- e. Perform operational check of left relay panel (PARA 9-2-20).



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FIGURE 9-5-6-1. REPLACE LEFT RELAY PANEL INDICATOR LIGHTS DIMMING CONTROL RELAY, K40 (BENCH)

9-5-7 RIGHT RELAY PANEL CURRENT SENSOR RIGHT PITOT HEAT TRANSFORMER, T1 (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
9-5-7.1	Replace Right Relay Panel Current Sensor Right Pitot Heat Transformer, T1 (BENCH)	9-5-7-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit
- Protection, Eye and Skin
- Soldering Iron

Materials/Parts

- Solder, Item 313, Appendix D
- Tags

References

- Appendix D
- PARA 9-2-19

9-5-7.1 Replace Right Relay Panel Current Sensor Right Pitot Heat Transformer, T1 (BENCH).

9-5-7.1.1. Remove.

- Remove screws and washers that secure bottom cover to right relay panel (Figure 9-5-7-1).
- Remove nuts that secure current sensor right pitot heat transformer, T1, to right relay panel.
- Carefully lift current sensor right pitot heat transformer, T1, from right relay panel to gain access to wires.



Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- Tag wires. Using soldering iron, unsolder wires from current sensor right pitot heater transformer, T1.

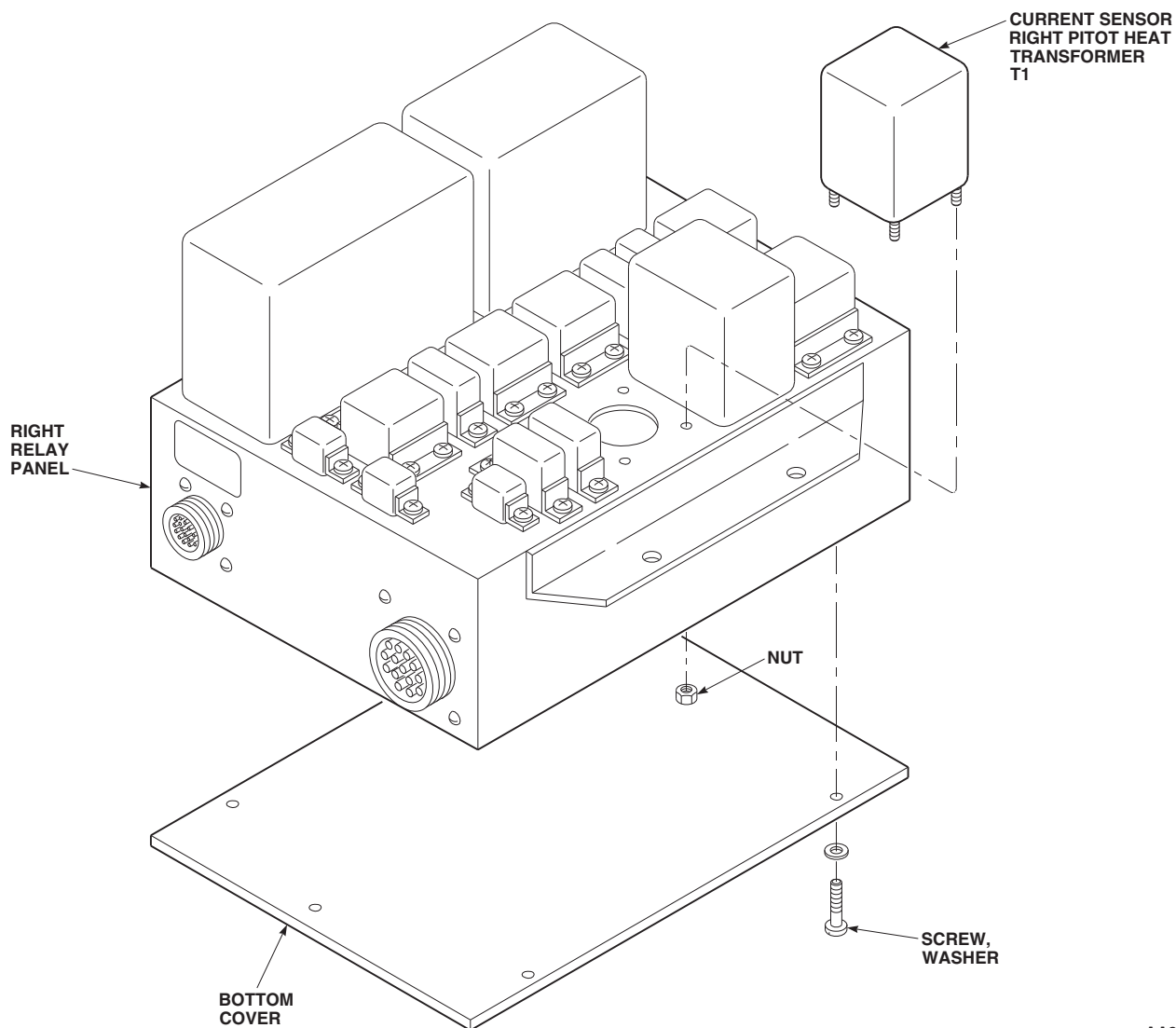
- e. Remove current sensor right pitot heater transformer, T1.

9-5-7.1.2. Install.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- a. Using soldering iron and solder, Item 313, Appendix D, solder wires to current sensor right pitot heat transformer, T1. Remove tags.
- b. Install current sensor right pitot heat transformer, T1, in right relay panel and secure with nuts (Figure 9-5-7-1).
- c. Make sure area is clean and free of foreign material.
- d. Install bottom cover to right relay panel and secure with screws and washers.
- e. Perform operational check of right relay panel (PARA 9-2-19).



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FIGURE 9-5-7-1. REPLACE RIGHT RELAY PANEL CURRENT SENSOR RIGHT PITOT HEAT TRANSFORMER, T1 (BENCH)

9-5-8 RIGHT RELAY PANEL DC ESSENTIAL BUS FAIL RELAY, K20 (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
9-5-8.1	Replace Right Relay Panel DC Essential Bus Fail Relay, K20 (BENCH)	9-5-8-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit
- Protection, Eye and Skin
- Soldering Iron

Materials/Parts

- Solder, Item 313, Appendix D
- Tags

References

- Appendix D
- PARA 9-2-19

9-5-8.1 Replace Right Relay Panel DC Es-
sential Bus Fail Relay, K20 (BENCH).

9-5-8.1.1. Remove.

- Remove screws and washers that secure bot-
tom cover to right relay panel (Figure 9-5-8-1).
- Remove screws, washers, and nuts that secure
dc essential bus fail relay, K20, to right relay
panel.
- Carefully lift dc essential bus fail relay, K20,
from right relay panel to gain access to wires.



Hot metal droppings occur when solder-
ing, causing both a fire hazard and
injury to personnel. Use protective
clothing, including eye protection while
soldering. Make sure hot metal drop-
pings do not come in contact with skin
or clothing.

- Tag wires. Using soldering iron, unsolder wires
from dc essential bus fail relay, K20.

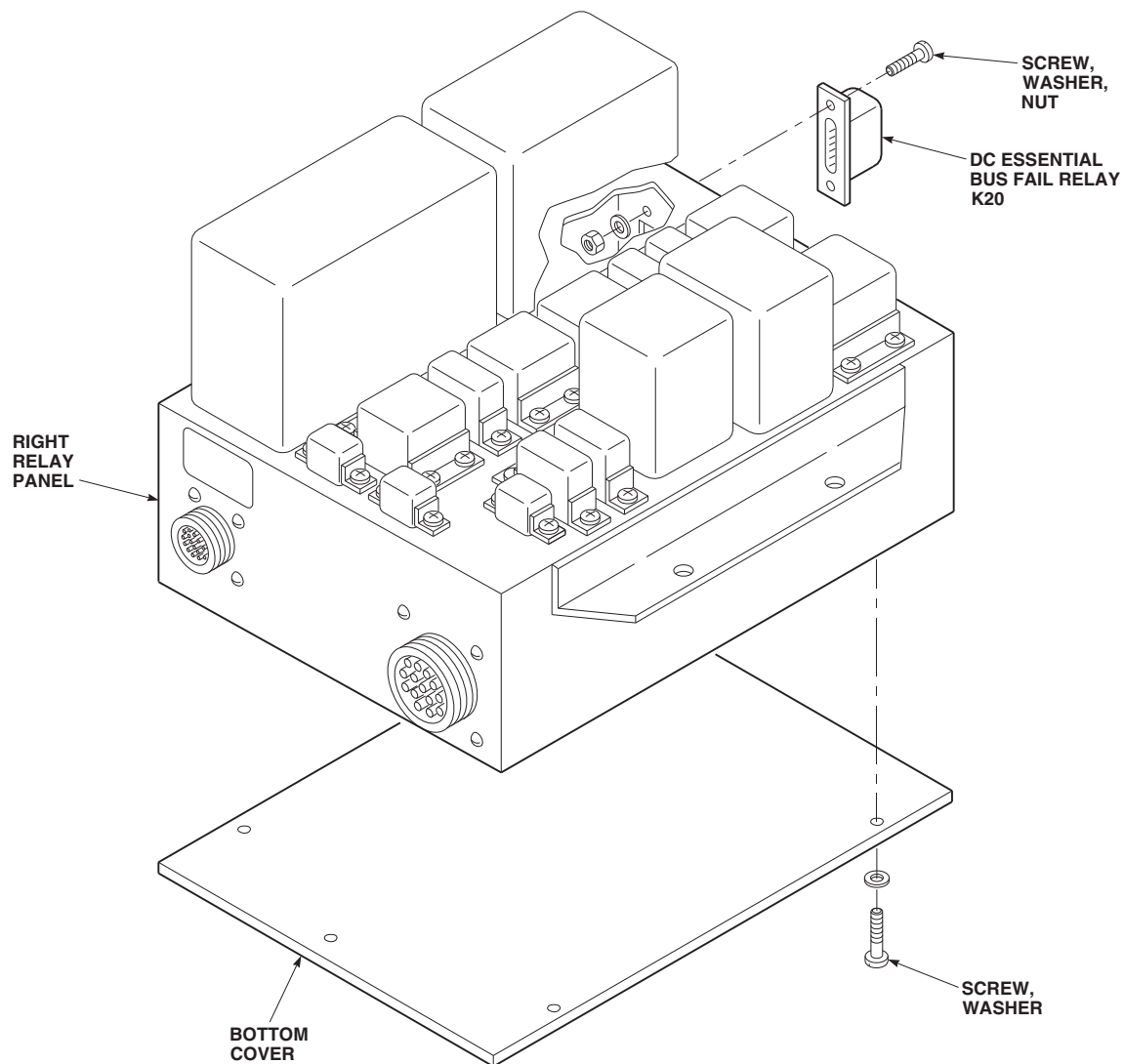
- e. Remove dc essential bus fail relay, K20.

9-5-8.1.2. Install.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- a. Using soldering iron and solder, Item 313, Appendix D, solder wires to dc essential bus fail relay, K20. Remove tags.
- b. Install dc essential bus fail relay, K20, in right relay panel and secure with screws, washers, and nuts (Figure 9-5-8-1).
- c. Make sure area is clean and free of foreign material.
- d. Install bottom cover to right relay panel and secure with screws and washers.
- e. Perform operational check of right relay panel (PARA 9-2-19).



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FIGURE 9-5-8-1. REPLACE RIGHT RELAY PANEL DC ESSENTIAL BUS FAIL RELAY, K20 (BENCH)

9-5-8-3/(9-5-8-4 Blank)

9-5-9 RIGHT RELAY PANEL WINDSHIELD ANTI-ICE LOCKOUT RELAY, K21 (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
9-5-9.1	Replace Right Relay Panel Windshield Anti-Ice Lockout Relay, K21 (BENCH)	9-5-9-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit
- Protection, Eye and Skin
- Soldering Iron

Materials/Parts

- Solder, Item 313, Appendix D
- Tags

References

- Appendix D
- PARA 9-2-19

9-5-9.1 Replace Right Relay Panel Windshield Anti-Ice Lockout Relay, K21 (BENCH).

9-5-9.1.1. Remove.

- Remove screws and washers that secure bottom cover to right relay panel (Figure 9-5-9-1).
- Remove screws, washers, and nuts that secure windshield anti-ice lockout relay, K21, to right relay panel.
- Carefully lift windshield anti-ice lockout relay, K21, from right relay panel to gain access to wires.



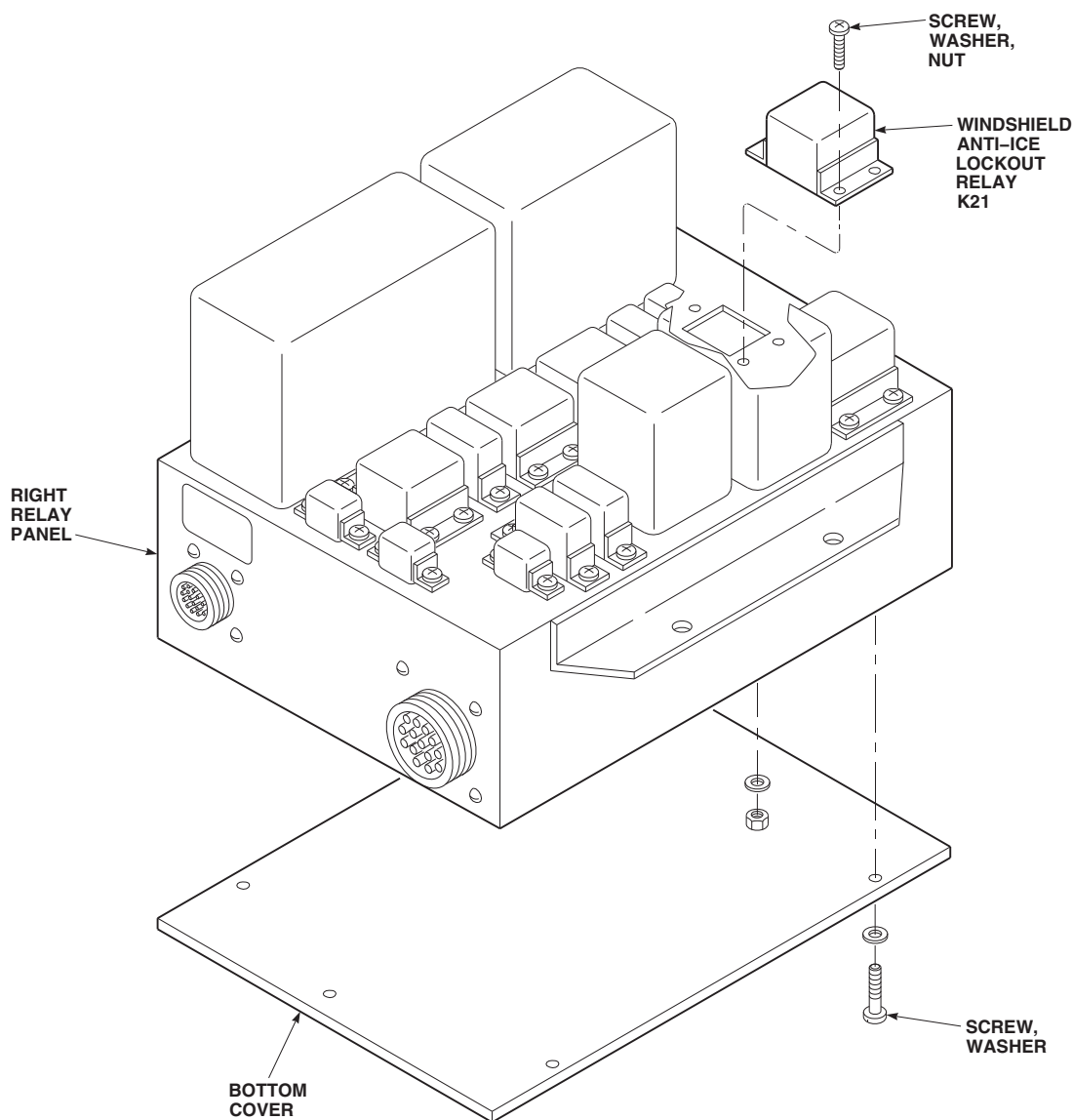
Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- Tag wires. Using soldering iron, unsolder wires from windshield anti-ice lockout relay, K21.
- Remove windshield anti-ice lockout relay, K21.

9-5-9.1.2. Install.**WARNING**

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- a. Using soldering iron and solder, Item 313, Appendix D, solder wires to windshield anti-ice lockout relay, K21. Remove tags.
- b. Install windshield anti-ice lockout relay, K21, in right relay panel and secure with screws, washers, and nuts (Figure 9-5-9-1).
- c. Make sure area is clean and free of foreign material.
- d. Install bottom cover to right relay panel and secure with screws and washers.
- e. Perform operational check of right relay panel (PARA 9-2-19).



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FIGURE 9-5-9-1. REPLACE RIGHT RELAY PANEL WINDSHIELD ANTI-ICE LOCKOUT RELAY, K21 (BENCH)

9-5-10 RIGHT RELAY PANEL APU SURGE RELAY, K22 (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
9-5-10.1	Replace Right Relay Panel APU Surge Relay, K22 (BENCH)	9-5-10-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit
- Protection, Eye and Skin
- Soldering Iron

Materials/Parts

- Solder, Item 313, Appendix D
- Tags

References

- Appendix D
- PARA 9-2-19

9-5-10.1 Replace Right Relay Panel APU Surge Relay, K22 (BENCH).

9-5-10.1.1. Remove.

- Remove screws and washers that secure bottom cover to right relay panel (Figure 9-5-10-1).
- Remove screws, washers, and nuts that secure APU surge relay, K22, to right relay panel.
- Carefully lift APU surge relay, K22, from right relay panel to gain access to wires.

injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- Tag wires. Using soldering iron, unsolder wires from APU surge relay, K22.
- Remove APU surge relay, K22.

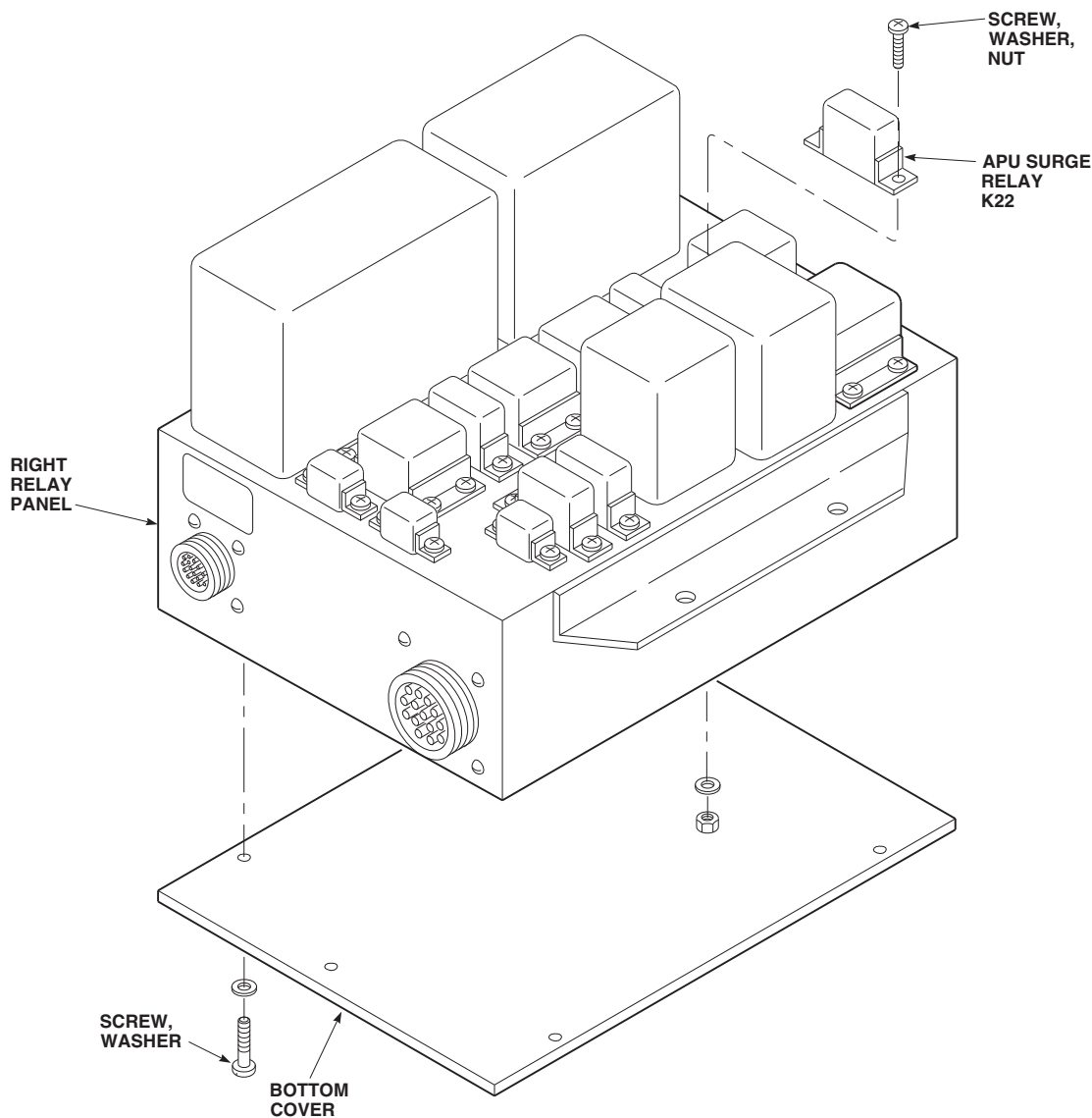
WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and

9-5-10.1.2. Install.**WARNING**

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- a. Using soldering iron and solder, Item 313, Appendix D, solder wires to APU surge relay, K22. Remove tags.
- b. Install APU surge relay, K22, in right relay panel and secure with screws, washers, and nuts (Figure 9-5-10-1).
- c. Make sure area is clean and free of foreign material.
- d. Install bottom cover to right relay panel and secure with screws and washers.
- e. Perform operational check of right relay panel (PARA 9-2-19).



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FIGURE 9-5-10-1. REPLACE RIGHT RELAY PANEL APU SURGE RELAY, K22 (BENCH)

9-5-10-3/(9-5-10-4 Blank)

9-5-11 RIGHT RELAY PANEL NO. 2 ENGINE START RELAY, K26 (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
9-5-11.1	Replace Right Relay Panel No. 2 Engine Start Relay, K26 (BENCH)	9-5-11-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit
- Protection, Eye and Skin
- Soldering Iron

Materials/Parts

- Solder, Item 313, Appendix D
- Tags

References

- Appendix D
- PARA 9-2-19

9-5-11.1 Replace Right Relay Panel No. 2 Engine Start Relay, K26 (BENCH).

9-5-11.1.1. Remove.

- Remove screws and washers that secure bottom cover to right relay panel (Figure 9-5-11-1).
- Remove screws, washers, and nuts that secure No. 2 engine start relay, K26, to right relay panel.
- Carefully lift No. 2 engine start relay, K26, from right relay panel to gain access to wires.



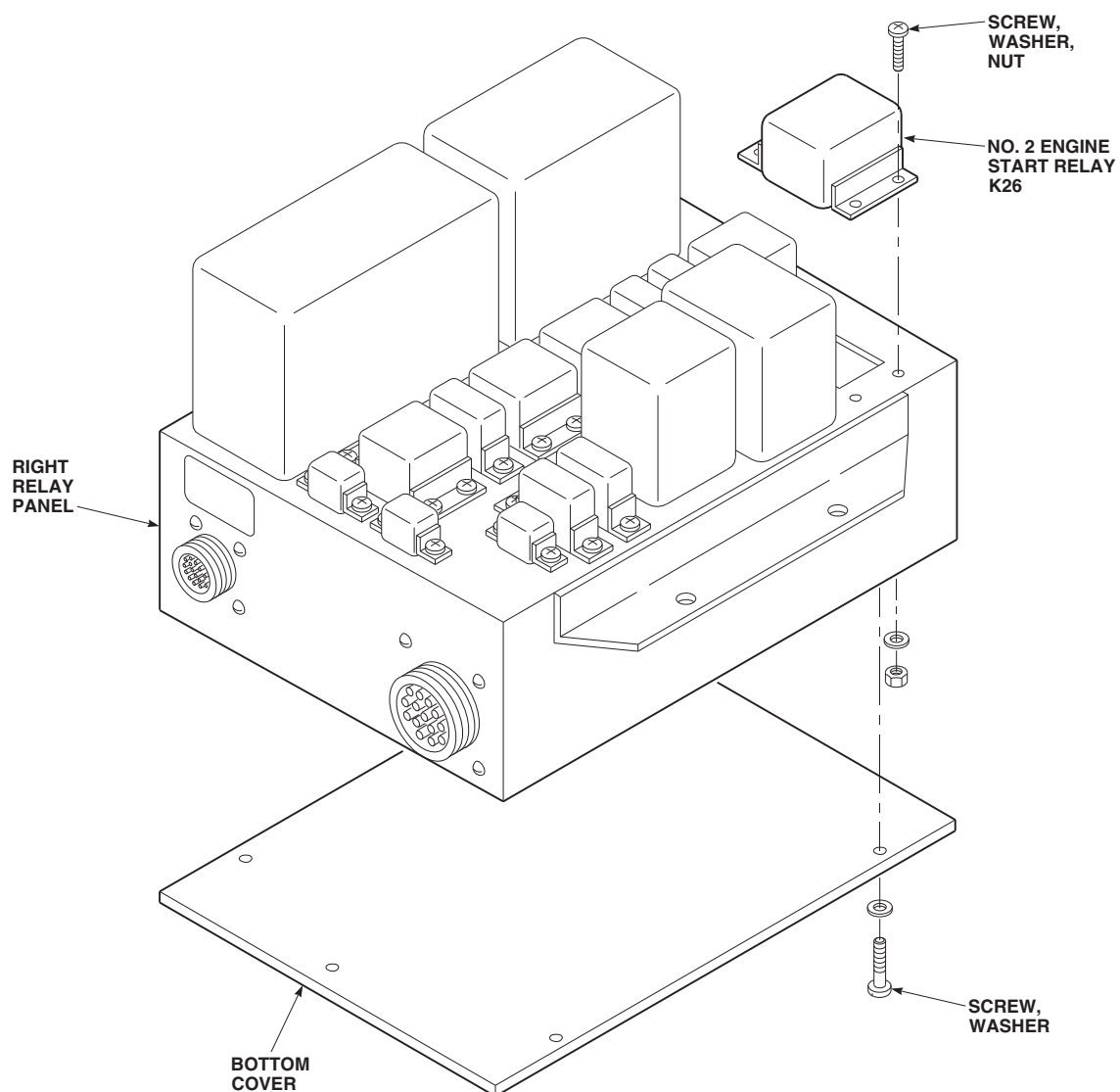
Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- Tag wires. Using soldering iron, unsolder wires from No. 2 engine start relay, K26.
- Remove No. 2 engine start relay, K26.

9-5-11.1.2. Install.**WARNING**

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- a. Using soldering iron and solder, Item 313, Appendix D, solder wires to No. 2 engine start relay, K26. Remove tags.
- b. Install No. 2 engine start relay, K26, in right relay panel and secure with screws, washers, and nuts (Figure 9-5-11-1).
- c. Make sure area is clean and free of foreign material.
- d. Install bottom cover to right relay panel and secure with screws and washers.
- e. Perform operational check of right relay panel (PARA 9-2-19).



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FIGURE 9-5-11-1. REPLACE RIGHT RELAY PANEL NO. 2 ENGINE START RELAY, K26 (BENCH)

9-5-11-3/(9-5-11-4 Blank)

9-5-12 RIGHT RELAY PANEL HYDRAULIC ACCUMULATOR RELAY, K32 (BENCH).

PARAGRAPH BREAKDOWN

	PAGE
9-5-12.1 Replace Right Relay Panel Hydraulic Accumulator Relay, K32 (BENCH)	9-5-12-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit
- Protection, Eye and Skin
- Soldering Iron

Materials/Parts

- Solder, Item 313, Appendix D
- Tags

References

- Appendix D
- PARA 9-2-19

9-5-12.1 Replace Right Relay Panel Hydraulic Accumulator Relay, K32 (BENCH).

9-5-12.1.1. Remove.

- Remove screws and washers that secure bottom cover to right relay panel (Figure 9-5-12-1).
- Remove screws, washers, lockwashers, and nuts that secure hydraulic accumulator relay, K32, to right relay panel.
- Carefully lift hydraulic accumulator relay, K32, from right relay panel to gain access to wires.

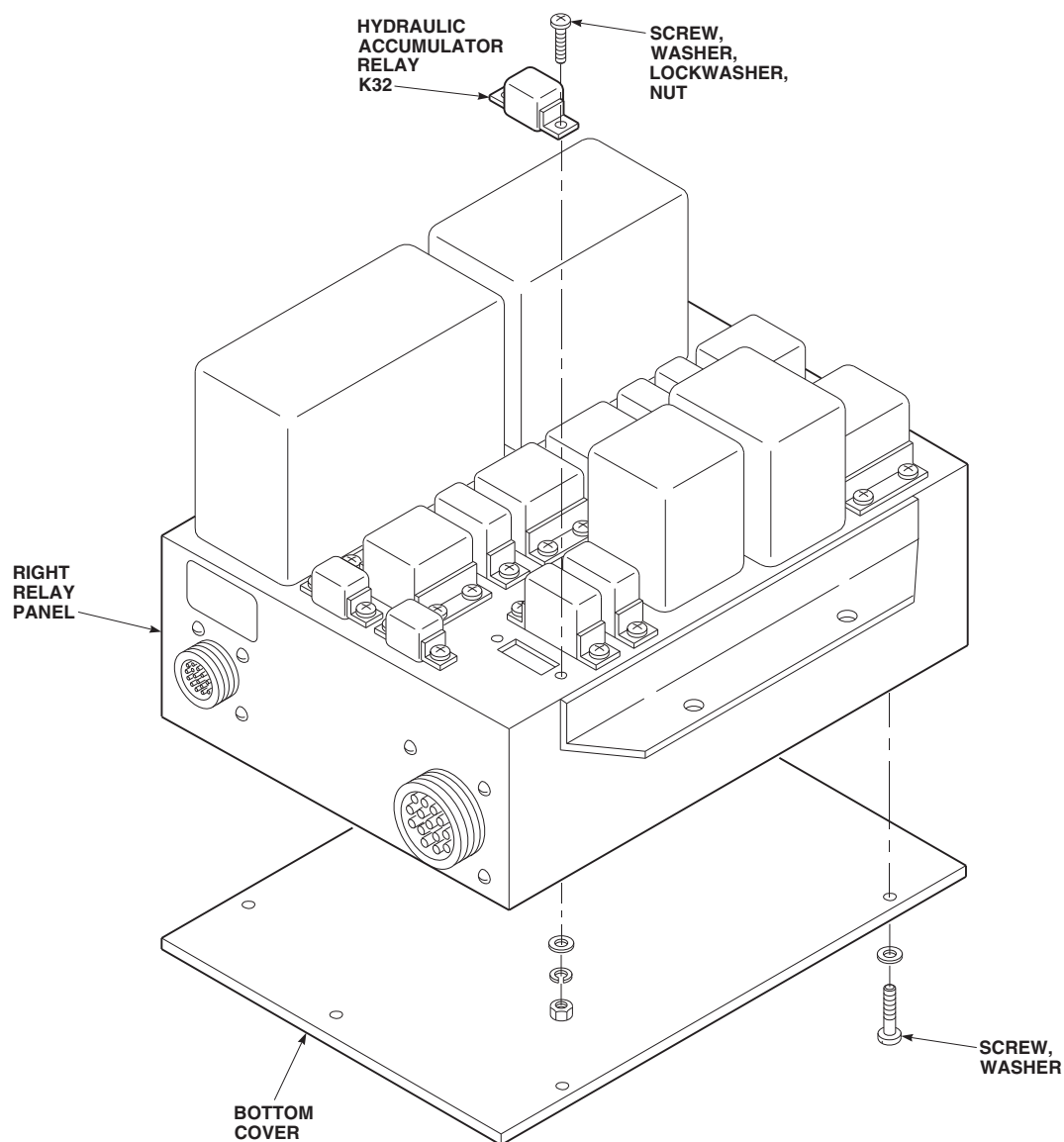


- Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.
- Tag wires. Using soldering iron, unsolder wires from hydraulic accumulator relay, K32.
 - Remove hydraulic accumulator relay, K32.

9-5-12.1.2. Install.**WARNING**

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- a. Using soldering iron and solder, Item 313, Appendix D, solder wires to hydraulic accumulator relay, K32. Remove tags.
- b. Install hydraulic accumulator relay, K32, in right relay panel and secure with screws, washers, lockwashers, and nuts (Figure 9-5-12-1).
- c. Make sure area is clean and free of foreign material.
- d. Install bottom cover to right relay panel and secure with screws and washers.
- e. Perform operational check of right relay panel (PARA 9-2-19).



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FIGURE 9-5-12-1. REPLACE RIGHT RELAY PANEL HYDRAULIC ACCUMULATOR RELAY, K32 (BENCH)

9-5-12-3/(9-5-12-4 Blank)

9-5-13 MISCELLANEOUS SWITCH PANEL PUSHBUTTON SWITCHES (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
9-5-13.1	Replace Miscellaneous Switch Panel Pushbutton Switches (BENCH)	9-5-13-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit
- Protection, Eye and Skin
- Soldering Iron

Materials/Parts

- Sealing Compound, Item 280, Appendix D
- Solder, Item 313, Appendix D

- Tags

References

- Appendix D
- PARA 3-2-2
- PARA 7-2-1
- PARA 9-2-5
- PARA 10-2-1
- TM 11-70-23

9-5-13.1 Replace Miscellaneous Switch Panel Pushbutton Switches (BENCH).

NOTE

Replacement of all miscellaneous switch panel pushbutton switches is the same.

9-5-13.1.1. Remove.

- Remove screws that secure cover to miscellaneous switch panel (Figure 9-5-13-1).

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective

clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- Tag wires. Using soldering iron, unsolder wiring to pushbutton switch.
- Remove indicator button from pushbutton switch (Figure 9-5-13-1).
- Loosen adjusting screws on pushbutton switch to unlock mounting cam tabs and separate mounting sleeve from pushbutton switch (Figure 9-5-13-1, Detail A).
- Pull pushbutton switch from miscellaneous switch panel.

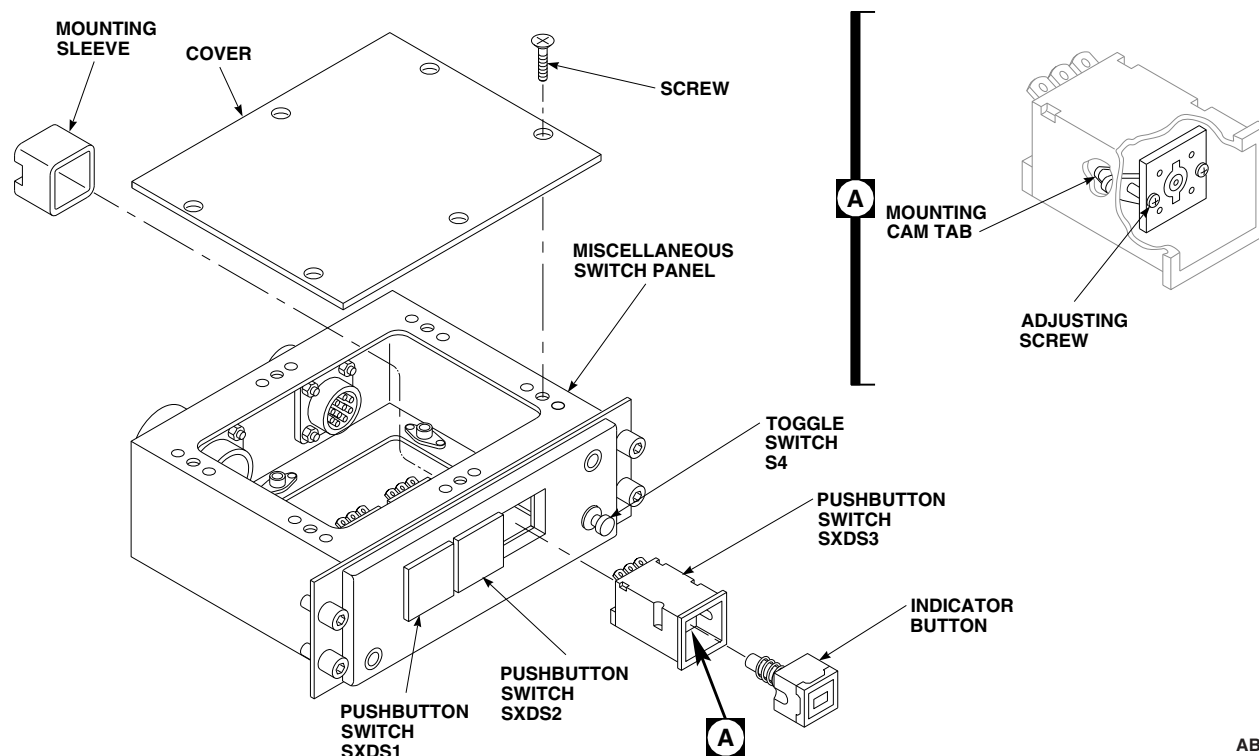
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SA

FIGURE 9-5-13-1. REPLACE MISCELLANEOUS SWITCH PANEL PUSHBUTTON SWITCHES (BENCH)

9-5-13.1.2. Install.

- Position pushbutton switch through opening in front of miscellaneous switch panel (Figure 9-5-13-1).
- Slide mounting sleeve fully onto rear of pushbutton switch and lock mounting cam tabs of pushbutton switch by turning adjusting screws to right (Figure 9-5-13-1, Detail A).
- Apply sealing compound, Item 280, Appendix D, to mounting cam tab and adjusting screws.

NOTE

Indicator button is marked TOP. Make sure TOP is upright when inserting indicator button into pushbutton switch.

- Install indicator button into pushbutton switch (Figure 9-5-13-1).

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- Using soldering iron and solder, Item 313, Appendix D, solder wires to pushbutton switch. Remove tags.
- Make sure area is clean and free of foreign material.
- Install cover on miscellaneous switch panel and secure with screws.
- Perform operational check of the following:
 - Fuel Quantity System (PARA 10-2-1).

- | | |
|---|--|
| <ul style="list-style-type: none"> (2) Tailwheel Lock System (PARA 3-2-2). (3) Attitude Indicating System (TM 11-70-23). (4) Tail Rotor Servo in No. 1 Hydraulic and Backup System (PARA 7-2-1). | <ul style="list-style-type: none"> (5) Upper and Lower Console Lights (PARA 9-2-5). |
|---|--|

9-5-14 MISCELLANEOUS SWITCH PANEL TOGGLE SWITCH (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
9-5-14.1	Replace Miscellaneous Switch Panel Toggle Switch (BENCH)	9-5-14-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit
- Protection, Eye and Skin
- Soldering Iron

Materials/Parts

- Solder, Item 313, Appendix D
- Tags

References

- Appendix D
- PARA 3-2-2
- PARA 7-2-1
- PARA 9-2-5
- PARA 9-4-9
- PARA 10-2-1
- TM 11-70-23

9-5-14.1 Replace Miscellaneous Switch Panel Toggle Switch (BENCH).

9-5-14.1.1. Remove.

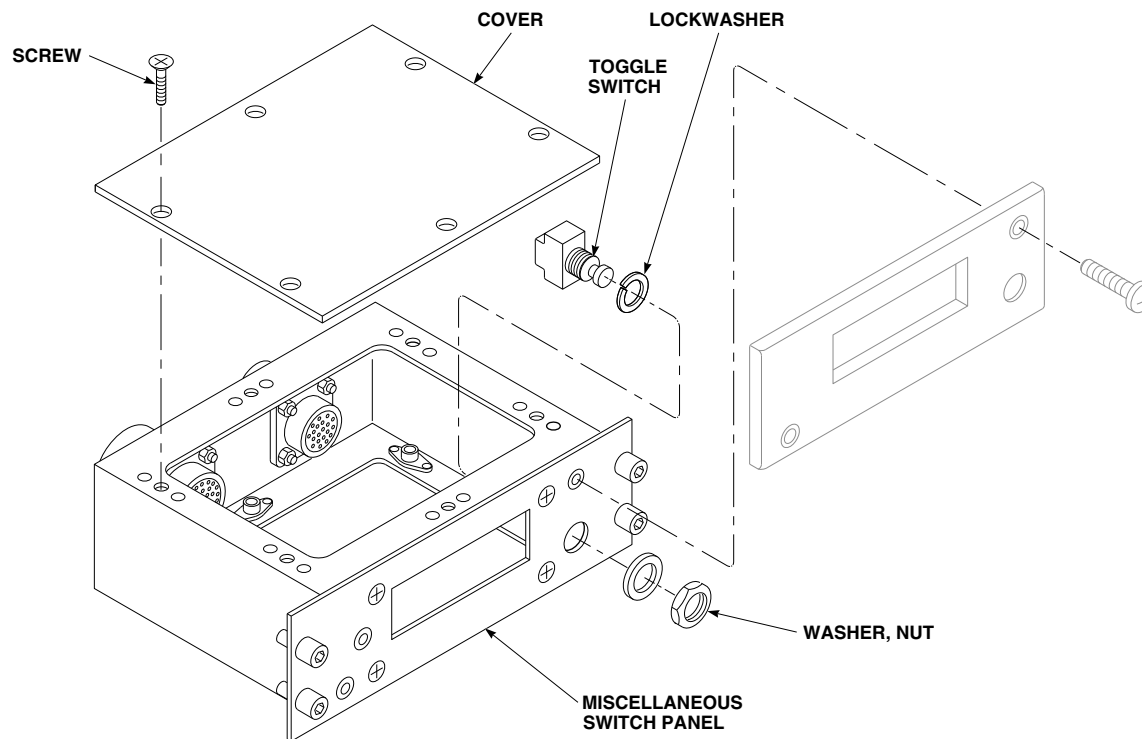
- Remove information plate (PARA 9-4-9).
- Remove screws that secure cover to miscellaneous switch panel (Figure 9-5-14-1).

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and

injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- Tag wires. Using soldering iron, unsolder wiring to toggle switch.
- Remove nut, washer, and lockwasher that secures toggle switch to miscellaneous switch panel.
- Remove toggle switch.

FA2025
SA**FIGURE 9-5-14-1. REPLACE MISCELLANEOUS SWITCH PANEL TOGGLE SWITCH (BENCH)****9-5-14.1.2. Install.**

- a. Install toggle switch through opening in front of miscellaneous switch panel and secure with nut, washer, and lockwasher (Figure 9-5-14-1).

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- b. Using soldering iron and solder, Item 313, Appendix D, solder wires to toggle switch. Remove tags.

- c. Make sure area is clean and free of foreign material.
- d. Install cover on miscellaneous switch panel and secure with screws.
- e. Install information plate (PARA 9-4-9).
- f. Perform operational check of the following:
 - (1) Fuel Quantity System (PARA 10-2-1).
 - (2) Tailwheel Lock System (PARA 3-2-2).
 - (3) Attitude Indicating System (TM 11-70-23).
 - (4) Tail Rotor Servo in No. 1 Hydraulic and Backup System (PARA 7-2-1).
 - (5) Upper and Lower Console Lights (PARA 9-2-5).

9-5-15 MISCELLANEOUS SWITCH PANEL ELECTRICAL CONNECTORS (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
9-5-15.1	Replace Miscellaneous Switch Panel Electrical Connectors (BENCH)	9-5-15-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit
- Insert/Extract Tool

Materials/Parts

- Tags

References

- PARA 1-7-20
- PARA 3-2-2

- PARA 7-2-1
- PARA 9-2-5
- PARA 10-2-1
- TM 11-70-23

9-5-15.1 Replace Miscellaneous Switch Panel Electrical Connectors (BENCH).

NOTE

Replacement of both miscellaneous switch panel electrical connectors is the same.

9-5-15.1.1. Remove.

- Remove screws that secure cover to miscellaneous switch panel (Figure 9-5-15-1).
- Tag wires. Using insert/extract tool, remove wires from electrical connector.
- Remove screws, washers, and nuts that secure electrical connector to miscellaneous switch panel.

- Remove electrical connector.

9-5-15.1.2. Install.

- Inspect and treat electrical connector (PARA 1-7-20).
- Install electrical connector to miscellaneous switch panel and secure with screws, washers, and nuts (Figure 9-5-15-1).
- Using insert/extract tool, install wires into electrical connector. Remove tags.
- Make sure area is clean and free of foreign material.
- Install cover on miscellaneous switch panel and secure with screws.

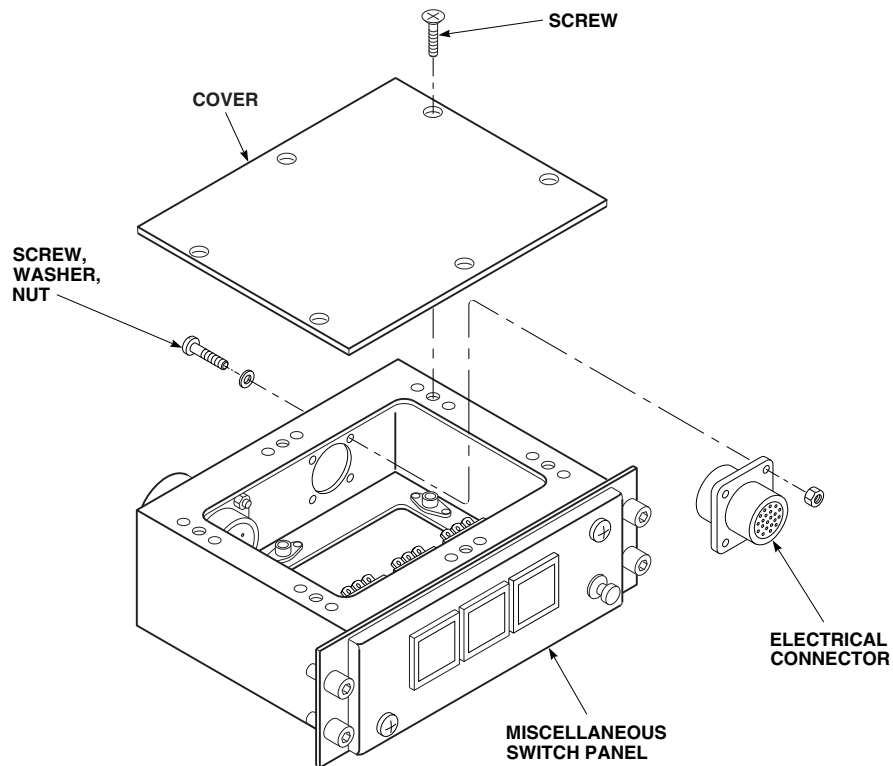
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SA

FIGURE 9-5-15-1. REPLACE MISCELLANEOUS SWITCH PANEL ELECTRICAL CONNECTORS (BENCH)

- f. Perform operational check of the following:
- (1) Fuel Quantity System (PARA 10-2-1).
 - (2) Tailwheel Lock System (PARA 3-2-2).
 - (3) Attitude Indicating System (TM 11-70-23).
 - (4) Tail Rotor Servo in No. 1 Hydraulic and Backup System (PARA 7-2-1).
 - (5) Upper and Lower Console Lights (PARA 9-2-5).

9-5-16 MISCELLANEOUS SWITCH PANEL PRINTED WIRING BOARD (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
9-5-16.1	Replace Miscellaneous Switch Panel Printed Wiring Board (BENCH)	9-5-16-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit
- Protection, Eye and Skin
- Soldering Iron

Materials/Parts

- Solder, Item 313, Appendix D
- Tags

References

- Appendix D
- PARA 3-2-2
- PARA 7-2-1
- PARA 9-2-5
- PARA 10-2-1
- TM 11-70-23

9-5-16.1 Replace Miscellaneous Switch Panel Printed Wiring Board (BENCH).

9-5-16.1.1. Remove.

- Remove screws that secure cover to miscellaneous switch panel (Figure 9-5-16-1).
- Remove nuts, washers, spacers, and screws that secure printed wiring board to housing of miscellaneous switch panel.

injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- Tag wires. Using soldering iron, unsolder wires from printed wiring board.
- Remove printed wiring board.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and

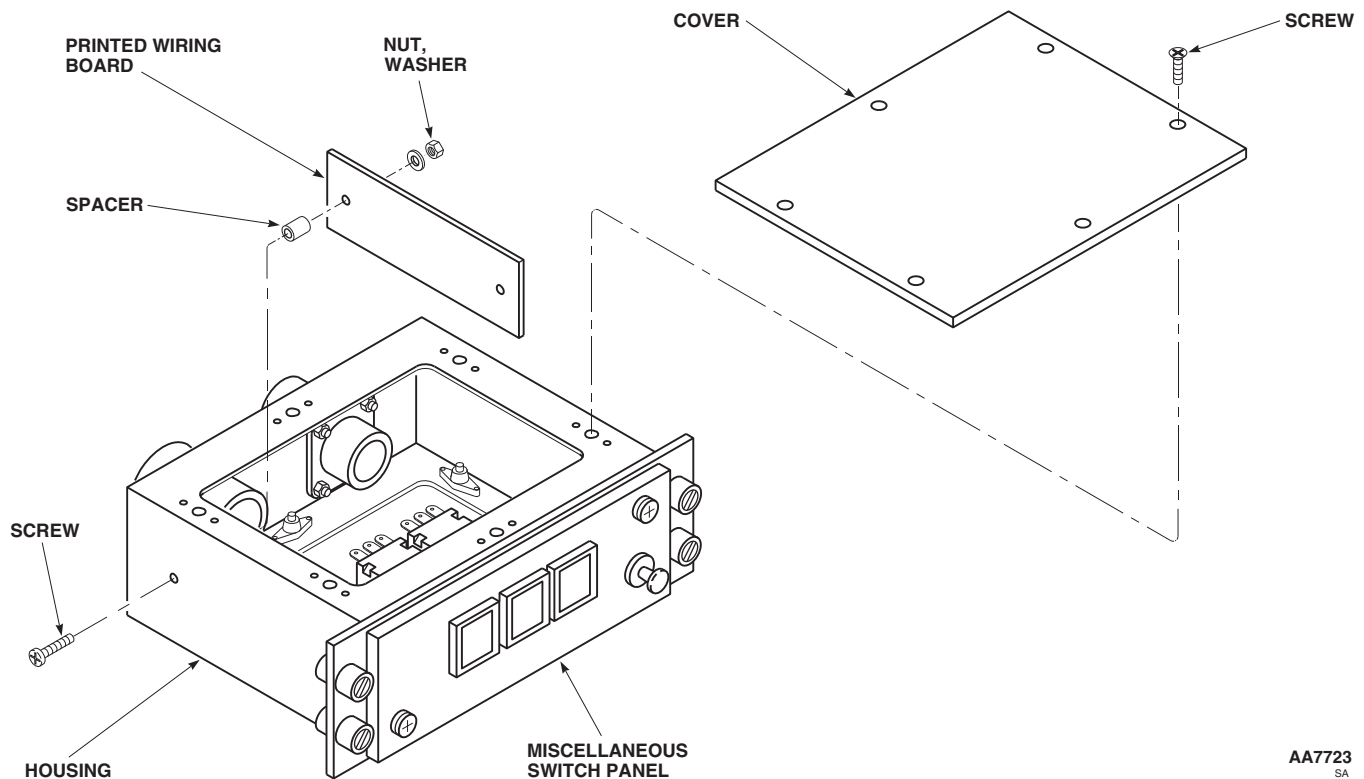
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SA

FIGURE 9-5-16-1. REPLACE MISCELLANEOUS SWITCH PANEL PRINTED WIRING BOARD (BENCH)

9-5-16.1.2. Install.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- a. Using soldering iron and solder, Item 313, Appendix D, solder wires to printed wiring board. Remove tags.
- b. Install printed wiring board to housing of miscellaneous switch panel and secure with screws, spacers, washers and nuts (Figure 9-5-16-1).
- c. Make sure area is clean and free of foreign material.
- d. Install cover on miscellaneous switch panel and secure with screws.
- e. Perform operational check of the following:
 - (1) Fuel Quantity System (PARA 10-2-1).
 - (2) Tailwheel Lock System (PARA 3-2-2).
 - (3) Attitude Indicating System (TM 11-70-23).
 - (4) Tail Rotor Servo in No. 1 Hydraulic and Backup System (PARA 7-2-1).
 - (5) Upper and Lower Console Lights (PARA 9-2-5).

9-5-17 FUEL BOOST PUMP CONTROL PANEL TOGGLE SWITCHES (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
9-5-17.1	Replace Fuel Boost Pump Control Panel Toggle Switches (BENCH)	9-5-17-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit

Materials/Parts

- Tags

References

- PARA 9-2-18
- PARA 9-4-41

9-5-17.1 Replace Fuel Boost Pump Control Panel Toggle Switches (BENCH).

NOTE

Replacement of both fuel boost pump control panel toggle switches is the same.

9-5-17.1.1. Remove.

- Remove fuel boost pump control panel information plate (PARA 9-4-41).
- Remove screws that secure cover to fuel boost pump control panel (Figure 9-5-17-1).
- Remove nut, lockwashers, and lockring that secure toggle switch to fuel boost pump control panel.
- Tag wires. Remove screws and lockwashers that secure wiring to toggle switch.

- Remove toggle switch.

9-5-17.1.2. Install.

- Install wiring to terminals of toggle switch and secure with screws and lockwashers (Figure 9-5-17-1). Remove tags.
- Install toggle switch through opening in front of fuel boost pump control panel and secure with nut, lockwashers, and lockring.
- Make sure area is clean and free of foreign material.
- Install cover on fuel boost pump control panel and secure with screws.
- Install fuel boost pump control panel information plate (PARA 9-4-41).
- Perform operational check of fuel boost pump control panel (PARA 9-2-18).

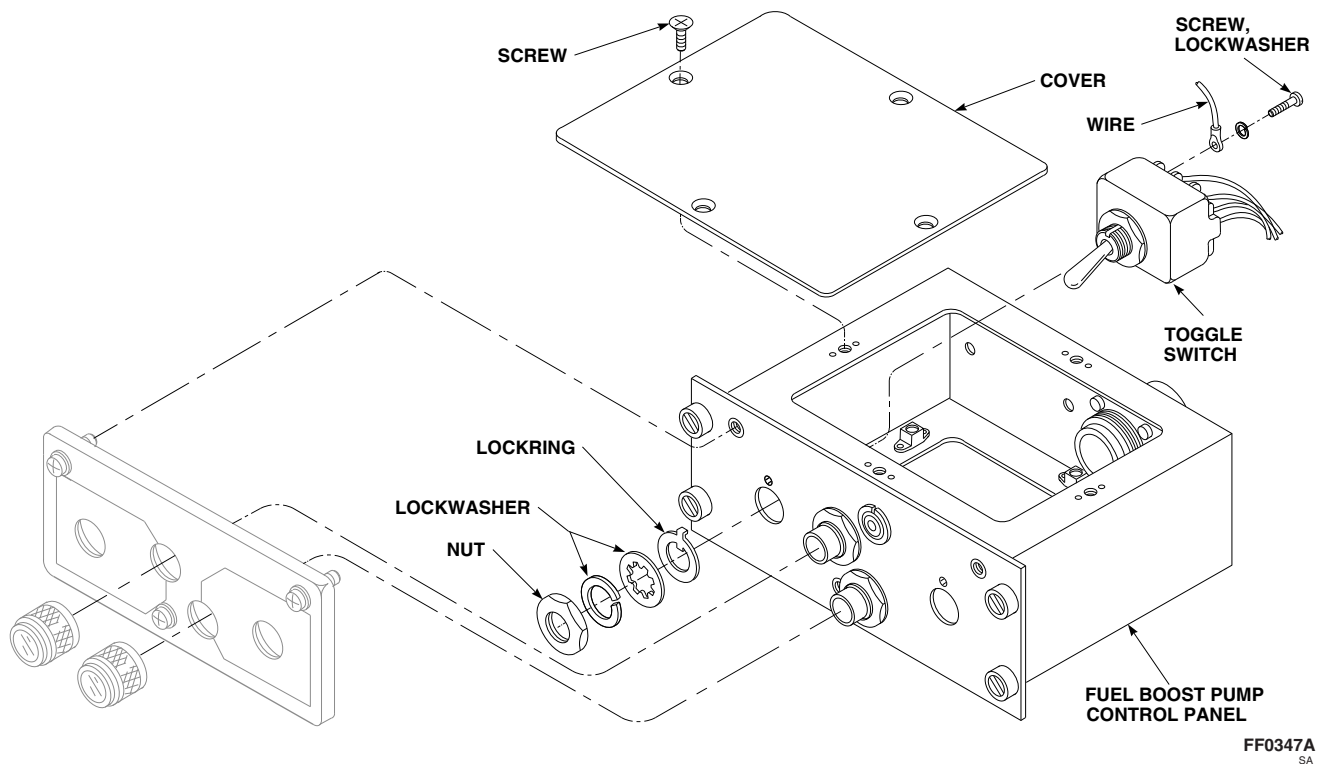


FIGURE 9-5-17-1. REPLACE FUEL BOOST PUMP CONTROL PANEL TOGGLE SWITCHES (BENCH)

9-5-18 FUEL BOOST PUMP CONTROL PANEL INDICATOR LIGHTS (BENCH).

PARAGRAPH BREAKDOWN

	PAGE
9-5-18.1 Replace Fuel Boost Pump Control Panel Indicator Lights (BENCH)	9-5-18-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit
- Protection, Eye and Skin
- Soldering Iron

Materials/Parts

- Solder, Item 313, Appendix D
- Tags

References

- Appendix D
- PARA 9-2-18
- PARA 9-4-41

9-5-18.1 Replace Fuel Boost Pump Control Panel Indicator Lights (BENCH).

NOTE

Replacement of both fuel boost pump control panel indicator lights is the same.

9-5-18.1.1. Remove.

- Remove fuel boost pump control panel information plate (PARA 9-4-41).
- Remove screws that secure cover to fuel boost pump control panel (Figure 9-5-18-1).



Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- Tag wires. Using soldering iron, unsolder wiring to indicator light.
- Remove nut, lockwashers, and indicator light from fuel boost pump control panel.

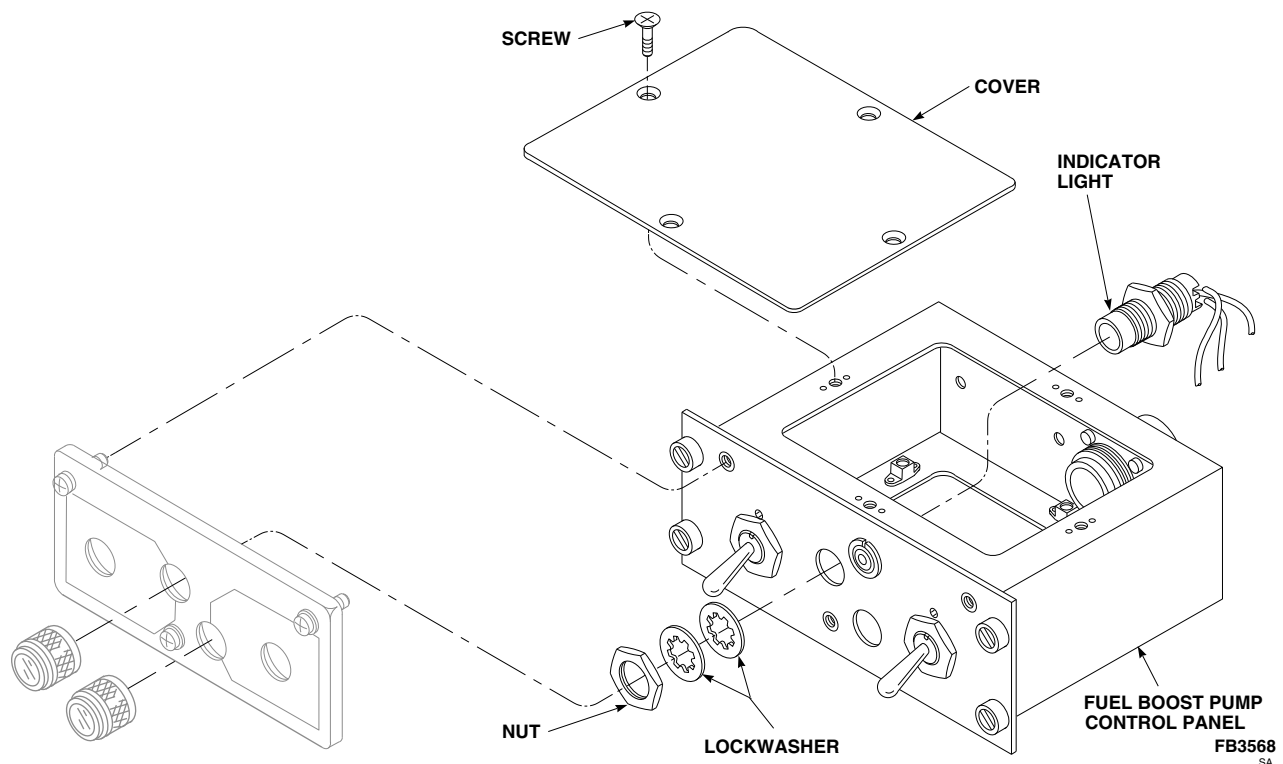


FIGURE 9-5-18-1. REPLACE FUEL BOOST PUMP CONTROL PANEL INDICATOR LIGHTS (BENCH)

- e. Remove indicator light.

9-5-18.1.2. Install.

- a. Install indicator light through opening in front of fuel boost pump control panel and secure with nut and lockwashers (Figure 9-5-18-1).

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal drop-

pings do not come in contact with skin or clothing.

- b. Using soldering iron and solder, Item 313, Appendix D, solder wires to indicator light. Remove tags.
- c. Make sure area is clean and free of foreign material.
- d. Install cover on fuel boost pump control panel and secure with screws.
- e. Install fuel boost pump control panel information plate (PARA 9-4-41).
- f. Perform operational check of fuel boost pump control panel (PARA 9-2-18).

9-5-19 FUEL BOOST PUMP CONTROL PANEL INFORMATION PLATE LAMPS (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
9-5-19.1	Replace Fuel Boost Pump Control Panel Information Plate Lamps (BENCH)	9-5-19-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit

References

- PARA 9-2-18
- PARA 9-4-41

9-5-19.1 Replace Fuel Boost Pump Control Panel Information Plate Lamps (BENCH).

NOTE

Replacement of all fuel boost pump control panel information plate lamps is the same.

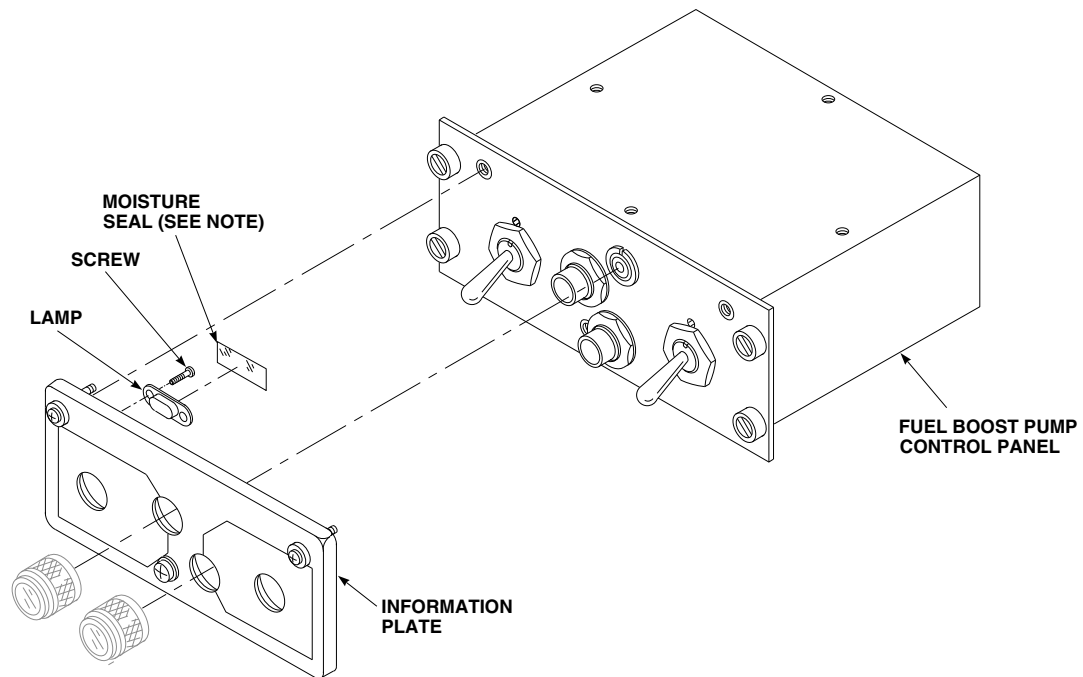
9-5-19.1.1 Remove.

- Remove fuel boost pump control panel information plate (PARA 9-4-41).
- On fuel boost pump control panel, 70902-01801-102, remove moisture seal from back of information plate. Discard moisture seal (Figure 9-5-19-1).
- Remove screws securing lamp to information plate.

- Remove lamp.

9-5-19.1.2 Install.

- Install lamp in information plate and secure with screws (Figure 9-5-19-1).
- On fuel boost pump control panel, 70902-01801-102, install new moisture seal over lamp on back of information plate.
- Make sure area is clean and free of foreign material.
- Install fuel boost pump control panel information plate (PARA 9-4-41).
- Perform operational check of fuel boost pump control panel (PARA 9-2-18).

**NOTE**

MOISTURE SEAL INSTALLED
ON FUEL BOOST PUMP CONTROL
PANEL, 70902-01801-102.

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FIGURE 9-5-19-1. REPLACE FUEL BOOST PUMP CONTROL PANEL INFORMATION PLATE LAMPS (BENCH)

9-5-20 FUEL BOOST PUMP CONTROL PANEL ELECTRICAL CONNECTOR (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
9-5-20.1	Replace Fuel Boost Pump Control Panel Electrical Connector (BENCH)	9-5-20-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit
- Insert/Extract Tool

Materials/Parts

- Tags

References

- PARA 1-7-20
- PARA 9-2-18

9-5-20.1 Replace Fuel Boost Pump Control Panel Electrical Connector (BENCH).

9-5-20.1.1. Remove.

- Remove screws that secure bottom cover to fuel boost pump control panel (Figure 9-5-20-1).
- Remove screws, washers, and nuts that secure electrical connector to fuel boost pump control panel.
- Tag wires. Using insert/extract tool, remove wires from electrical connector.
- Remove electrical connector.

9-5-20.1.2. Install.

- Inspect and treat electrical connector (PARA 1-7-20).
- Using insert/extract tool, install wires into electrical connector. Remove tags.
- Install electrical connector to fuel boost pump control panel and secure with screws, washers, and nuts (Figure 9-5-20-1).
- Make sure area is clean and free of foreign material.
- Install bottom cover to fuel boost pump control panel and secure with screws.
- Perform operational check of fuel boost pump control panel (PARA 9-2-18).

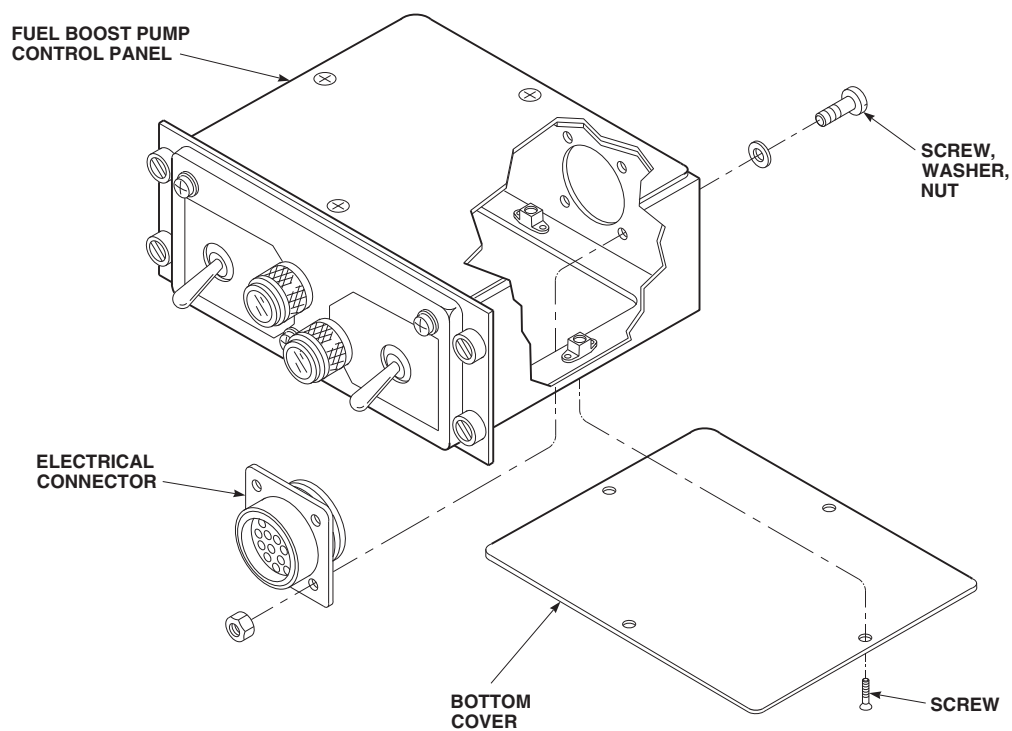
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FIGURE 9-5-20-1. REPLACE FUEL BOOST PUMP CONTROL PANEL ELECTRICAL CONNECTOR (BENCH)

9-5-21 BATTERY HARNESS (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
9-5-21.1	Replace Battery Harness (BENCH)	9-5-21-2

INITIAL SETUP

Tools

- Apron, Impermeable, Battery Worker’s
- Electronic Equipment Toolkit
- Face Shield, Industrial
- Torque Wrench, 0 - 30 in. lbs

Materials/Parts

- Corrosion Preventive Compound, Item 110, Appendix D
- Distilled Water, Item 123, Appendix D
- Protective Caps and Plugs

- Rubber Gloves, Item 265, Appendix D
- Tags
- Tiedown Strap, Item 328, Appendix D

References

- Appendix D
- PARA 9-2-22
- PARA 9-5-25
- PARA 9-5-26
- PARA 9-5-34
- PARA 9-5-35

9-5-21.1 Replace Battery Harness (BENCH).**WARNING**

- Injury to personnel or damage to battery may result if battery cells have not been shorted for discharging at least 3 hours.
- Serious burns will result if contact with the electrolyte (potassium hydroxide) is made. Wear protective clothing (battery worker's impermeable apron, industrial face shield, and rubber gloves, Item 265, Appendix D) when working around opened batteries. In case of accident, wash with distilled water, Item 123, Appendix D. Do not wear jewelry that might contact battery cell terminals.

CAUTION

- Damage to nickel-cadmium battery will result if maintenance is performed on a nickel-cadmium battery in a lead-acid battery shop. Never use any tools or materials which have been used on lead-acid batteries.
- Electrolyte is absorbed into battery cell plates and battery will appear dry, when battery is completely discharged. Do not add water.

9-5-21.1.1 Remove.

- Inspect, clean, current leakage check, and discharge battery (PARA 9-2-22).
- Make sure protective caps are installed over electrical receptacles, J1 and J2 (Figure 9-5-21-1).
- Unlatch and remove battery cover.
- Remove tiedown straps securing battery harness.

- Tag wires. Remove capscrews and Belleville washers that secure wires to electrical receptacle, J2, terminal risers.
- Remove battery cells near electrical receptacle, J1 (PARA 9-5-26).
- Remove nuts and Belleville washers that secure battery temperature sensors (No. 1 and No. 2) and battery overtemperature switches to intercell connectors to battery cell terminals.
- Remove jamnut and packing from electrical receptacle, J1 (Figure 9-5-21-1, Detail A). Discard packing.
- Carefully pull electrical receptacle, J1, from battery case.

NOTE

Do not unsolder cathode terminal wire of clamping diode unless it is suspected of malfunctioning.

- Remove clamping diode from battery cell terminals (PARA 9-5-25).
- Remove battery harness.

9-5-21.1.2 Install.

- Perform battery inspection (PARA 9-2-22).
- Position battery harness in battery (Figure 9-5-21-1, Detail A).
- Install clamping diode on battery cell terminal (PARA 9-5-25).
- Insert electrical receptacle, J1, through battery case and secure with packing and jamnut.
- Install all removed battery cells (PARA 9-5-26).

NOTE

Belleville washers are to be installed with concave side down.

- Connect wires of battery harness to terminals of electrical receptacle, J2, and secure with

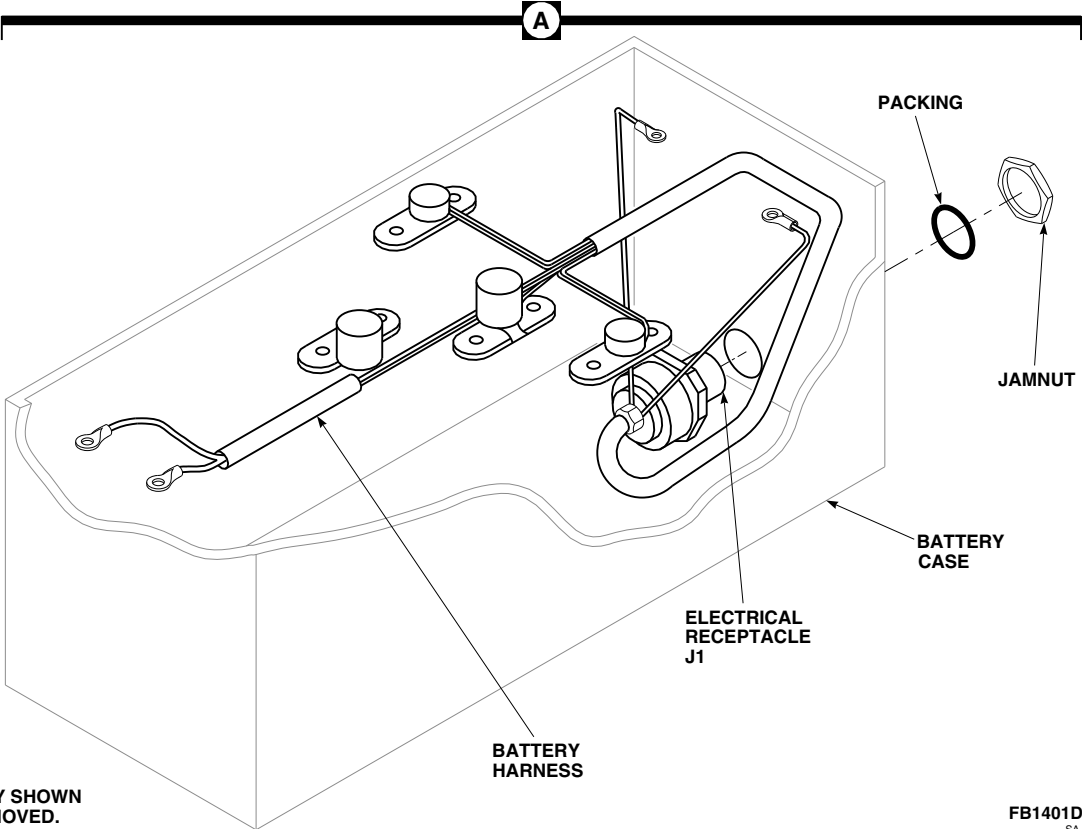
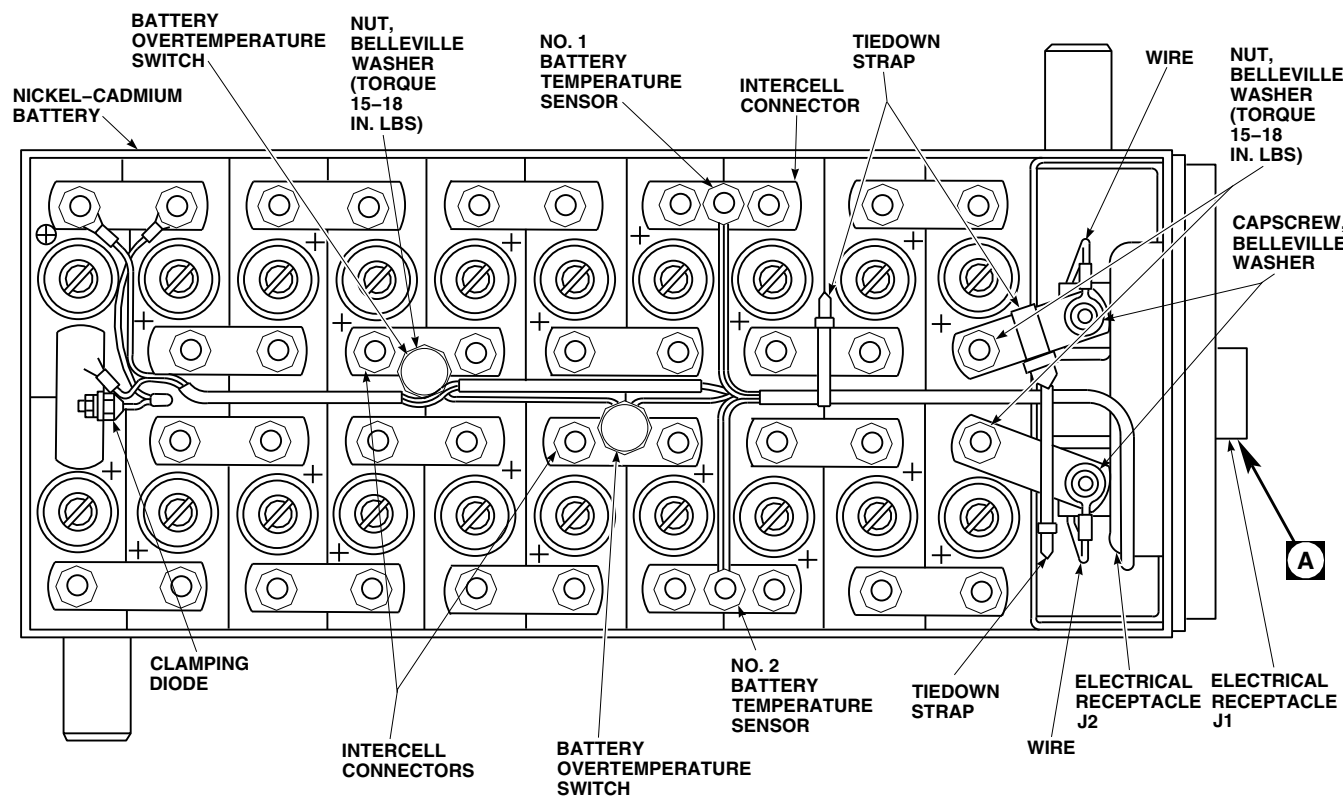


FIGURE 9-5-21-1. REPLACE BATTERY HARNESS (BENCH)

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- capscrews and Belleville washers (Figure 9-5-21-1). Remove tags.
- g. Install battery temperature sensors (No. 1 and No. 2) and battery overtemperature switches with nuts and Belleville washers to battery cell terminals.
 - h. **TORQUE NUTS TO 15 - 18 INCH-POUNDS.**
 - i. Install tiedown straps, Item 328, Appendix D, to secure harness in place.
 - j. Perform battery inspection (PARA 9-2-22).
 - k. Apply a light coat of corrosion preventive compound, Item 110, Appendix D, on battery cell terminals and intercell connectors, as required.
- l. Make sure shorting jumpers are installed on all battery cells.
 - m. Make sure area is clean and free of foreign material.
 - n. Install and latch battery cover.
 - o. Determine if battery is to be placed in service or storage. Proceed as follows:
 - (1) If battery is to be placed in service, perform 100-hour battery inspection (PARA 9-2-22).
 - (2) If battery is to be placed in short-term storage, store battery (PARA 9-5-35).
 - (3) If battery is to be placed in long-term storage, store battery (PARA 9-5-34).

9-5-22 BATTERY ELECTRICAL RECEPTACLE, J2 (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
9-5-22.1	Replace Battery Electrical Receptacle, J2 (BENCH)	9-5-22-2

INITIAL SETUP

Tools

- Apron, Impermeable, Battery Workers
- Digital Multimeter
- Electronic Equipment Toolkit
- Face Shield, Industrial
- Torque Wrench, 0 - 30 in. lbs

Materials/Parts

- Corrosion Preventive Compound, Item 110, Appendix D

- Distilled Water, Item 123, Appendix D
- Protective Caps and Plugs
- Rubber Gloves, Item 265, Appendix D
- Tags
- Tiedown Strap, Item 328, Appendix D

References

- Appendix D
- PARA 9-2-22

9-5-22.1 Replace Battery Electrical Receptacle, J2 (BENCH).

WARNING

- Injury to personnel or damage to battery may result if battery cells have not been shorted for discharging at least 3 hours.
- Serious burns will result if contact with the electrolyte (potassium hydroxide) is made. Wear protective clothing (battery workers impermeable apron, industrial face shield, and rubber gloves, Item 265, Appendix D) when working around opened batteries. In case of accident, wash with distilled water, Item 123, Appendix D. Do not wear jewelry that might contact battery cell terminals.

CAUTION

- Damage to nickel-cadmium battery will result if maintenance is performed on a nickel-cadmium battery in a lead-acid battery shop. Never use any tools or materials which have been used on lead-acid batteries.
- Electrolyte is absorbed into battery cell plates and battery will appear dry, when battery is completely discharged. Do not add water.

9-5-22.1.1. Remove.

- Inspect, clean, current leakage check, and discharge battery (PARA 9-2-22).
- Make sure protective caps are installed over electrical receptacles, J1 and J2 (Figure 9-5-22-1).
- Unlatch and remove battery cover.
- Tag wires. Remove capscrews and Belleville washers that secure wires to electrical receptacle, J2, terminal risers (Figure 9-5-22-1).

- Remove nuts and Belleville washers, that secure intercell connector, from battery cell terminals to terminals of electrical receptacle, J2.
- Remove screws that secure electrical receptacle, J2, from battery. Remove electrical receptacle, J2, and rectangular ring.
- Inspect electrical receptacle, J2, for corrosion, burns, discoloration, and excessive wear of silver plating on front main terminals, replace as required.

NOTE

Marbleizing is the graining, veining, or staining, to imitate marble, of plastic, acrylic, or rubber material through chemical or electrical action.

- Inspect for corrosion or powdering between rear terminal risers and green plastic molding, or marbleizing of green plastic molding.

WARNING

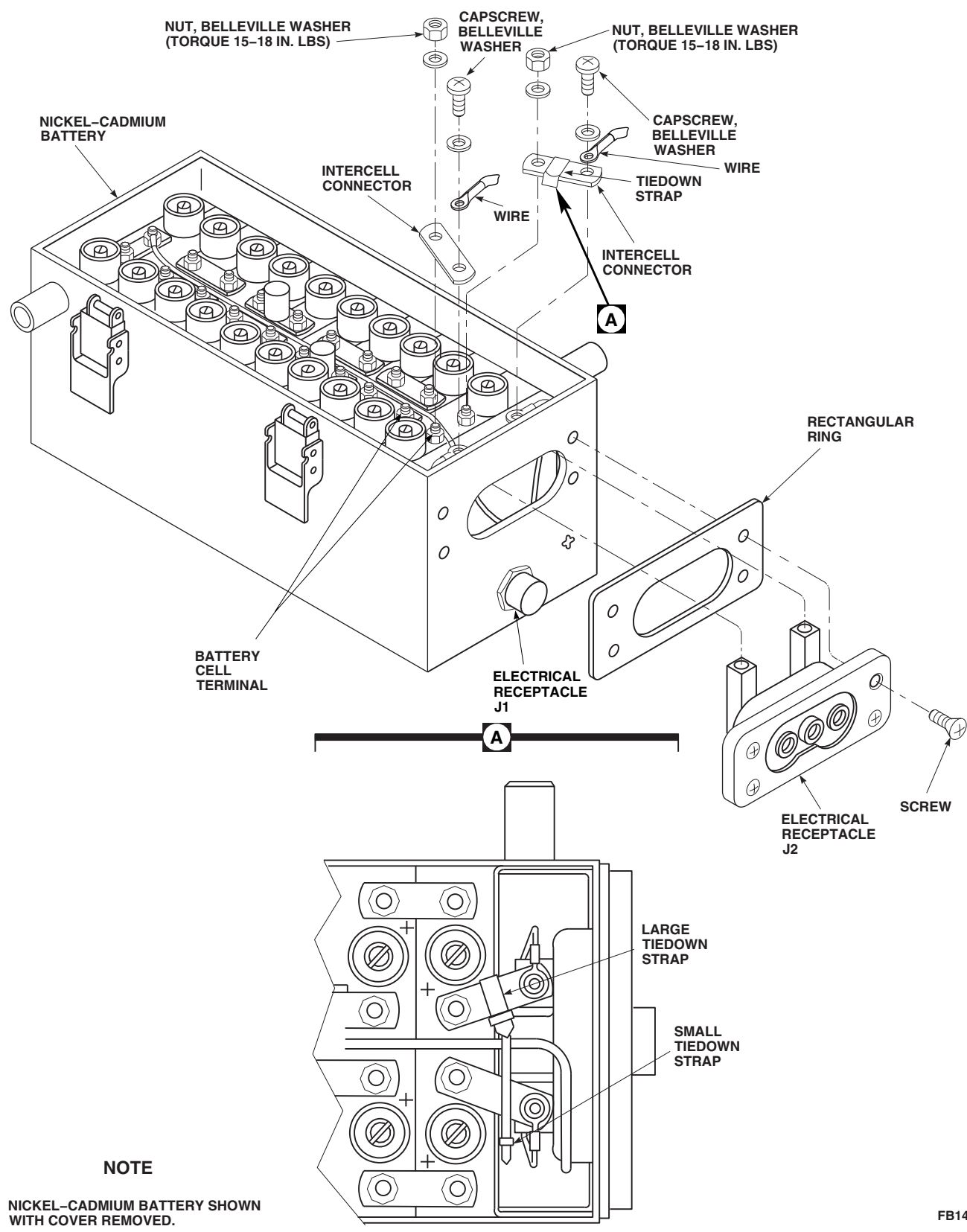
Avoid skin contact with water used to wet electrical receptacle, J2. Water may become caustic.

- If marbleizing or corrosion is found, place electrical receptacle, J2, in a container of water for a few seconds and remove.
- Using digital multimeter set to the lowest resistance scale, check continuity between front main terminals and rear terminal risers. **IF METER INDICATES MORE THAN 0.5 OHMS OF RESISTANCE, REPLACE ELECTRICAL RECEPTACLE, J2.**

9-5-22.1.2. Install.

- Install rectangular ring and electrical receptacle, J2, on battery and install with screws (Figure 9-5-22-1).

9-5-22-2



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FIGURE 9-5-22-1. REPLACE BATTERY ELECTRICAL RECEPTACLE, J2 (BENCH)

CAUTION

Damage to battery will result if battery cell/connector cavity divider contacts intercell connector. Battery will be shorted. Make sure large tiedown strap, Item 328, Appendix D, is attached to one intercell connector. This will prevent battery cell/connector cavity divider from touching intercell connector.

- b. Install large tiedown strap, Item 328, Appendix D, to one intercell connector (Figure 9-5-22-1, Detail A).

NOTE

Belleville washers are to be installed with concave side down.

- c. Install intercell connectors on battery cell terminals (Figure 9-5-22-1). Secure with nuts and Belleville washers.
- d. **TORQUE NUTS TO 15 - 18 INCH-POUNDS.**
- e. Install wires of battery harness to terminals of electrical receptacle, J2, and secure with cap-screws and Belleville washers. Remove tags.
- f. Perform battery inspection (PARA 9-2-22).
- g. Apply a light coat of corrosion preventive compound, Item 110, Appendix D, on terminals of electrical receptacle, J2, and intercell connectors, as required.
- h. Make sure area is clean and free of foreign material.
- i. Install and latch battery cover.

9-5-23 BATTERY TEMPERATURE SENSORS (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
9-5-23.1	Replace Battery Temperature Sensors (BENCH)	9-5-23-2

INITIAL SETUP

Tools

- Apron, Impermeable, Battery Worker’s
- Battery Repair Kit, No. 1 Battery Temperature Sensor
- Battery Repair Kit, No. 2 Battery Temperature Sensor
- Electronic Equipment Toolkit
- Face Shield, Industrial
- Heat Gun
- Torque Wrench, 0 - 30 in. lbs

Materials/Parts

- Corrosion Preventive Compound, Item 110, Appendix D

- Distilled Water, Item 123, Appendix D
- Heat-Shrink Insulation Sleeving
- Protective Caps and Plugs
- Rubber Gloves, Item 265, Appendix D

- Splice

References

- Appendix D
- PARA 9-2-22
- PARA 9-5-21
- PARA 9-5-34
- PARA 9-5-35

9-5-23.1 Replace Battery Temperature Sensors (BENCH).

WARNING

- Injury to personnel or damage to battery may result if battery cells have not been shorted for discharging at least 3 hours.
- Serious burns will result if contact with the electrolyte (potassium hydroxide) is made. Wear protective clothing (battery worker's impermeable apron, industrial face shield, and rubber gloves, Item 265, Appendix D) when working around opened batteries. In case of accident, wash with distilled water, Item 123, Appendix D. Do not wear jewelry that might contact battery cell terminals.

CAUTION

- Damage to nickel-cadmium battery will result if maintenance is performed on a nickel-cadmium battery in a lead-acid battery shop. Never use any tools or materials which have been used on lead-acid batteries.
- Electrolyte is absorbed into battery cell plates and battery will appear dry, when battery is completely discharged. Do not add water.

NOTE

- Replacement of both battery temperature sensors is the same.
- Replacement of No. 1 battery temperature sensor is shown.

9-5-23.1.1 Remove.

- Inspect, clean, current leakage check, and discharge battery (PARA 9-2-22).

- Make sure protective caps are installed over electrical receptacles, J1 and J2 (Figure 9-5-23-1).
- Unlatch and remove battery cover.
- Remove nuts and Belleville washers that secure battery temperature sensor and intercell connector to battery cell terminals (Figure 9-5-23-1, Detail A). If necessary, remove battery harness (PARA 9-5-21).
- Remove nut that secures defective battery temperature sensor to intercell connector.
- Cut wire leads of defective battery temperature sensor (Figure 9-5-23-1).

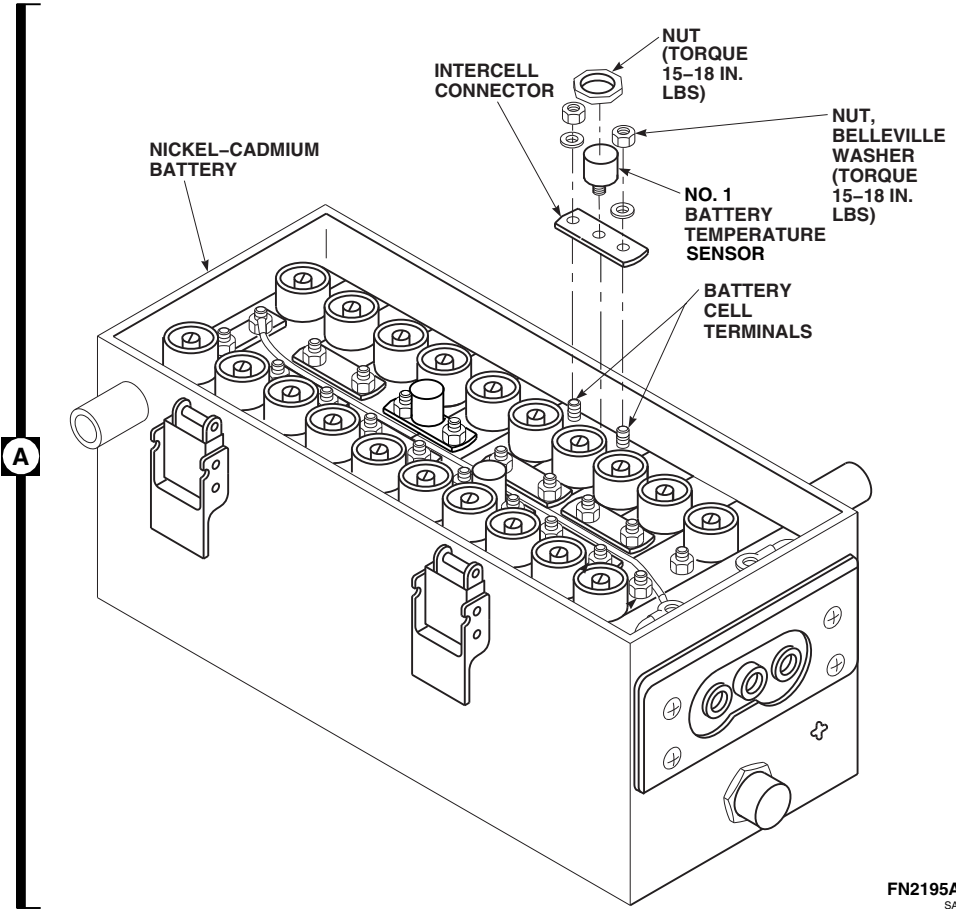
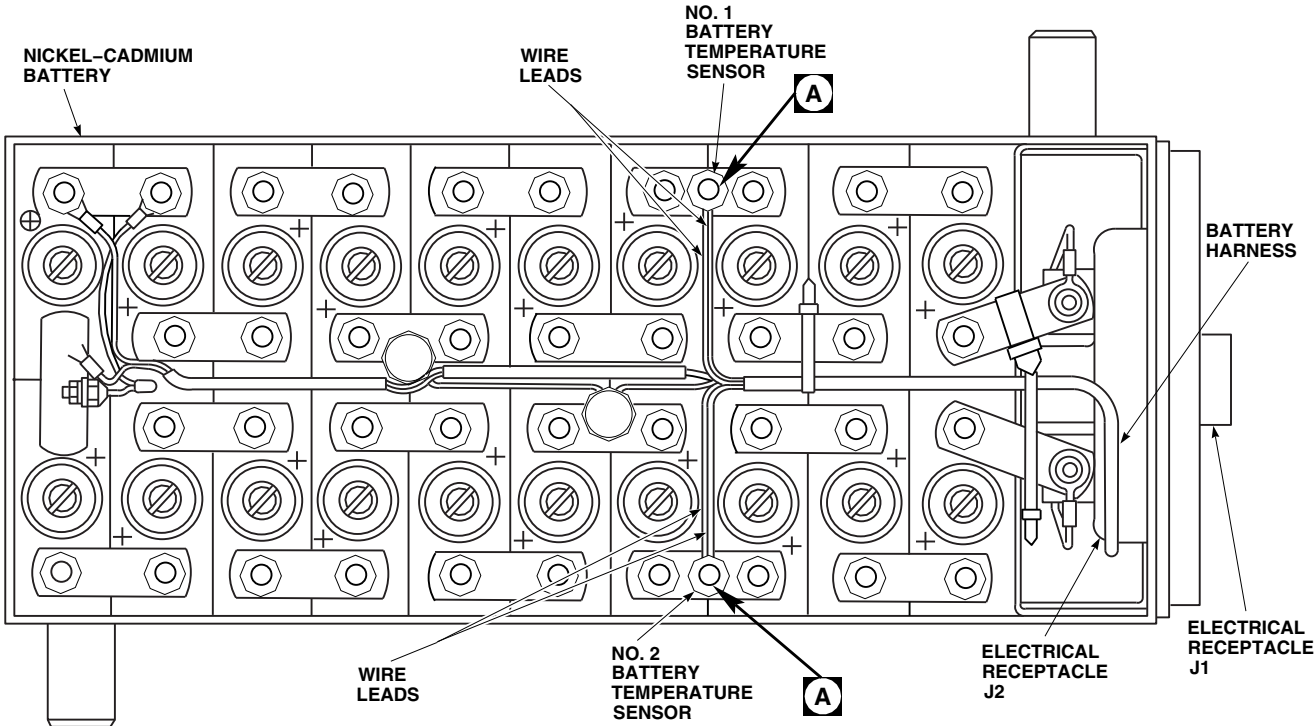
9-5-23.1.2 Install.

- Slide a 1-inch long heat-shrink insulation sleeving onto each wire lead (Figure 9-5-23-1).
- Strip 1/8-inch insulation off each wire lead.
- Using splice, secure wire leads of battery harness to wires of new battery temperature sensor.
- Slide heat-shrink insulation sleeving over splices.
- Using heat gun, shrink heat-shrink insulation sleeving.
- Place threaded terminal of new battery temperature sensor into intercell connector. Install with nut (Figure 9-5-23-1, Detail A).
- TORQUE NUT TO 15 - 18 INCH-POUNDS.**

NOTE

Belleville washers are to be installed with concave side down.

- If removed, install battery harness (PARA 9-5-21).
- Install battery temperature sensor and intercell connector to battery cell terminals and secure with nuts and Belleville washers.



NOTE
NICKEL-CADMIUM BATTERY SHOWN
WITH BATTERY COVER REMOVED.

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FIGURE 9-5-23-1. REPLACE BATTERY TEMPERATURE SENSORS (BENCH)

- j. **TORQUE NUTS TO 15 - 18 INCH-POUNDS.**
- k. Perform battery inspection (PARA 9-2-22).
- l. Apply a light coat of corrosion preventive compound, Item 110, Appendix D, on battery cell terminals and intercell connectors, as required.
- m. Make sure shorting jumpers are installed on all battery cells.
- n. Make sure area is clean and free of foreign material.
- o. Install and latch battery cover.
- p. Determine if battery is to be placed in service or storage. Proceed as follows:
 - (1) If battery is to be placed in service, perform 100-hour battery inspection (PARA 9-2-22).
 - (2) If battery is to be placed in short-term storage, store battery (PARA 9-5-35).
 - (3) If battery is to be placed in long-term storage, store battery (PARA 9-5-34).

9-5-24 BATTERY OVERTEMPERATURE SWITCHES (BENCH).

PARAGRAPH BREAKDOWN

	PAGE
9-5-24.1 Replace Battery Overtem- perature Switches (BENCH)	9-5-24-2

INITIAL SETUP

Tools

- Apron, Impermeable, Battery Worker’s
- Battery Repair Kit, Battery Overtemperature Switch
- Electronic Equipment Toolkit
- Face Shield, Industrial
- Heat Gun
- Torque Wrench, 0 - 30 in. lbs

Materials/Parts

- Corrosion Preventive Compound, Item 110, Appendix D

- Distilled Water, Item 123, Appendix D
- Heat-Shrink Insulation Sleeving
- Protective Caps and Plugs
- Rubber Gloves, Item 265, Appendix D
- Splice

References

- Appendix D
- PARA 9-2-22
- PARA 9-5-21
- PARA 9-5-34
- PARA 9-5-35

9-5-24.1 Replace Battery Overtemperature Switches (BENCH).

WARNING

- Injury to personnel or damage to battery may result if battery cells have not been shorted for discharging at least 3 hours.
- Serious burns will result if contact with the electrolyte (potassium hydroxide) is made. Wear protective clothing (battery worker's impermeable apron, industrial face shield, and rubber gloves, Item 265, Appendix D) when working around opened batteries. In case of accident, wash with distilled water, Item 123, Appendix D. Do not wear jewelry that might contact battery cell terminals.

CAUTION

- Damage to nickel-cadmium battery will result if maintenance is performed on a nickel-cadmium battery in a lead-acid battery shop. Never use any tools or materials which have been used on lead-acid batteries.
- Electrolyte is absorbed into battery cell plates and battery will appear dry, when battery is completely discharged. Do not add water.

NOTE

- Replacement of both battery overtemperature switches is the same.
- Replacement of the rear battery overtemperature switch is shown.

9-5-24.1.1 Remove.

- Inspect, clean, current leakage check, and discharge battery (PARA 9-2-22).

- Make sure protective caps are installed over electrical receptacles, J1 and J2 (Figure 9-5-24-1).
- Unlatch and remove battery cover.
- Remove nuts and Belleville washers that secure battery overtemperature switch and intercell connector to battery cell terminals (Figure 9-5-24-1, Detail A). If necessary, remove battery harness (PARA 9-5-21).
- Remove nut that secures defective battery overtemperature switch to intercell connector.
- Cut defective battery overtemperature switch short wire lead in half and long wire lead 3-inches from battery overtemperature switch (Figure 9-5-24-1).

9-5-24.1.2 Install.

- Slide 1-inch long heat-shrink insulation sleeving onto each wire lead (Figure 9-5-24-1).
- Strip 1/8-inch insulation off each wire lead.
- Using splice, secure wire leads of battery harness to wires of new battery overtemperature switch.
- Slide heat-shrink insulation sleeving over splices.
- Using heat gun, shrink heat-shrink insulation sleeving.
- Place threaded terminal of new battery overtemperature switch into intercell connector. Install with nut (Figure 9-5-24-1, Detail A).
- TORQUE NUT TO 15 - 18 INCH-POUNDS.**

NOTE

Belleville washers are to be installed with concave side down.

- If removed, install battery harness (PARA 9-5-21).
- Install battery overtemperature switch to battery cell terminals and secure with nuts and Belleville washers.

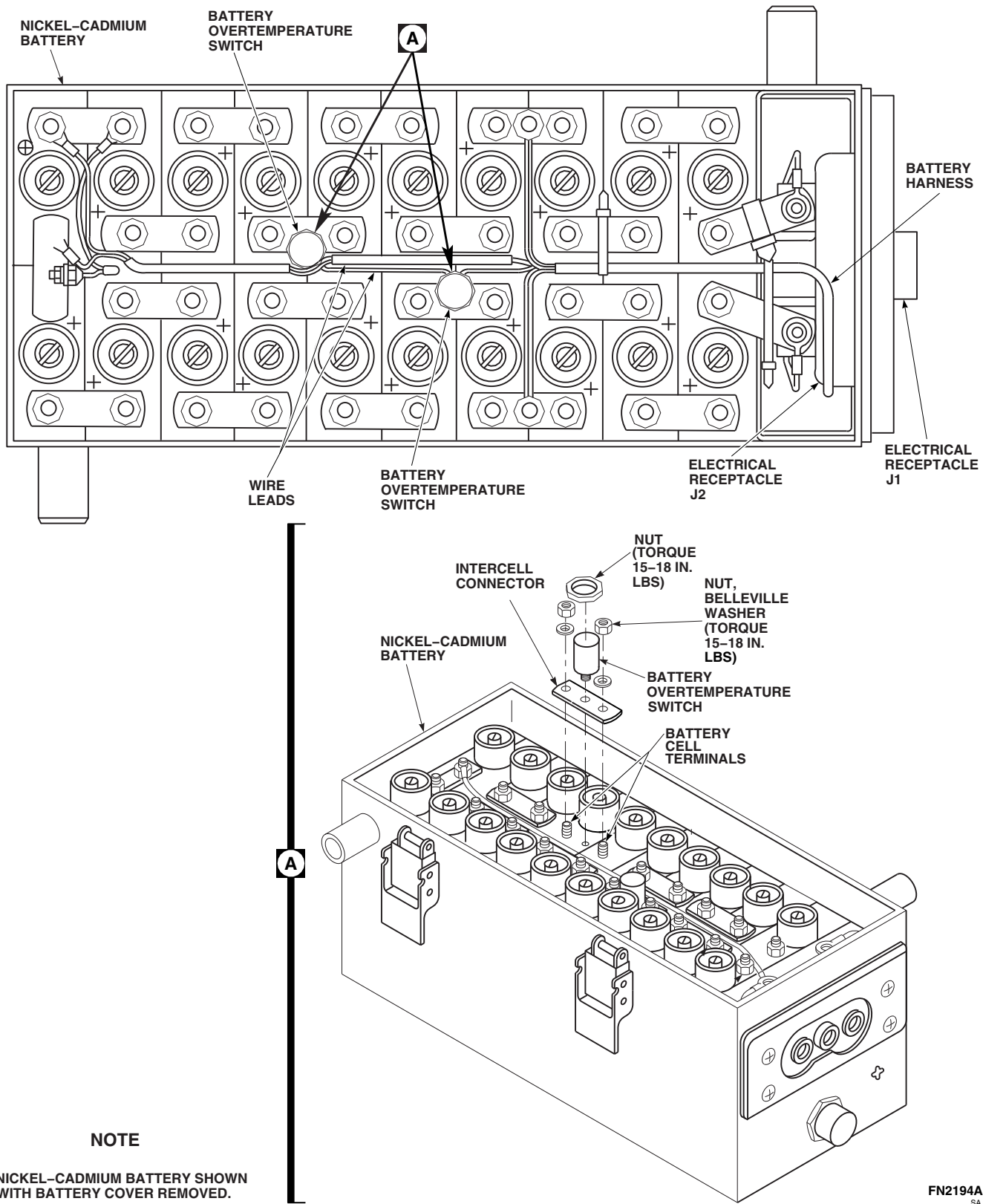


FIGURE 9-5-24-1. REPLACE BATTERY OVERTEMPERATURE SWITCHES (BENCH)

- j. **TORQUE NUTS TO 15 - 18 INCH-POUNDS.**
- k. Perform battery inspection (PARA 9-2-22).
- l. Apply a light coat of corrosion preventive compound, Item 110, Appendix D, on battery cell terminals and intercell connectors, as required.
- m. Make sure shorting jumpers are installed on all battery cells.
- n. Make sure area is clean and free of foreign material.
- o. Install and latch battery cover.
- p. Determine if battery is to be placed in service or storage. Proceed as follows:
 - (1) If battery is to be placed in service, perform 100-hour battery inspection (PARA 9-2-22).
 - (2) If battery is to be placed in short-term storage, store battery (PARA 9-5-35).
 - (3) If battery is to be placed in long-term storage, store battery (PARA 9-5-34).

9-5-25 BATTERY CLAMPING DIODE (BENCH).

PARAGRAPH BREAKDOWN

	PAGE
9-5-25.1 Replace Battery Clamping Diode (BENCH)	9-5-25-2

INITIAL SETUP

Tools

- Apron, Impermeable, Battery Worker’s
- Electronic Equipment Toolkit
- Face Shield, Industrial
- Heat Gun
- Soldering Iron
- Torque Wrench, 0 - 30 in. lbs

Materials/Parts

- Distilled Water, Item 123, Appendix D
- Heat-Shrink Insulation Sleeving

- Protective Caps and Plugs
- Rubber Gloves, Item 265, Appendix D
- Solder, Item 313, Appendix D
- Tags

References

- Appendix D
- PARA 9-2-22
- PARA 9-5-34
- PARA 9-5-35

9-5-25.1 Replace Battery Clamping Diode (BENCH).

WARNING

- Injury to personnel or damage to battery may result if battery cells have not been shorted for discharging at least 3 hours.
- Serious burns will result if contact with the electrolyte (potassium hydroxide) is made. Wear protective clothing (battery worker's impermeable apron, industrial face shield, and rubber gloves, Item 265, Appendix D) when working around opened batteries. In case of accident, wash with distilled water, Item 123, Appendix D. Do not wear jewelry that might contact battery cell terminals.

CAUTION

- Damage to nickel-cadmium battery will result if maintenance is performed on a nickel-cadmium battery in a lead-acid battery shop. Never use any tools or materials which have been used on lead-acid batteries.
- Electrolyte is absorbed into battery cell plates and battery will appear dry, when battery is completely discharged. Do not add water.

9-5-25.1.1 Remove.

- Inspect, clean, current leakage check, and discharge battery (PARA 9-2-22).
- Make sure protective caps are installed over electrical receptacles, J1 and J2 (Figure 9-5-25-1).
- Unlatch and remove battery cover.
- Tag wires of battery harness necessary to remove clamping diode.

- Remove nuts and Belleville washers that secure wires to intercell connectors.
- Remove nuts and Belleville washers from intercell connector (Figure 9-5-25-1, Detail A).
- Lift clamping diode from terminals.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- Tag wires. Using soldering iron, unsolder diode cathode terminal wire from terminal lug.
- Remove nut and washer that secures clamping diode to diode mounting link. Remove clamping diode.

9-5-25.1.2 Install.

- Install replacement clamping diode's anode terminal in intercell connector/diode mounting link and secure with washer and nut (Figure 9-5-25-1, Detail A).
- TORQUE NUT TO 20 - 25 INCH-POUNDS.**
- Slide 1-inch long heat-shrink insulation sleeving onto diode cathode terminal wire.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- Using soldering iron and solder, Item 313, Appendix D, solder diode cathode terminal wire to terminal lug. Remove tags.

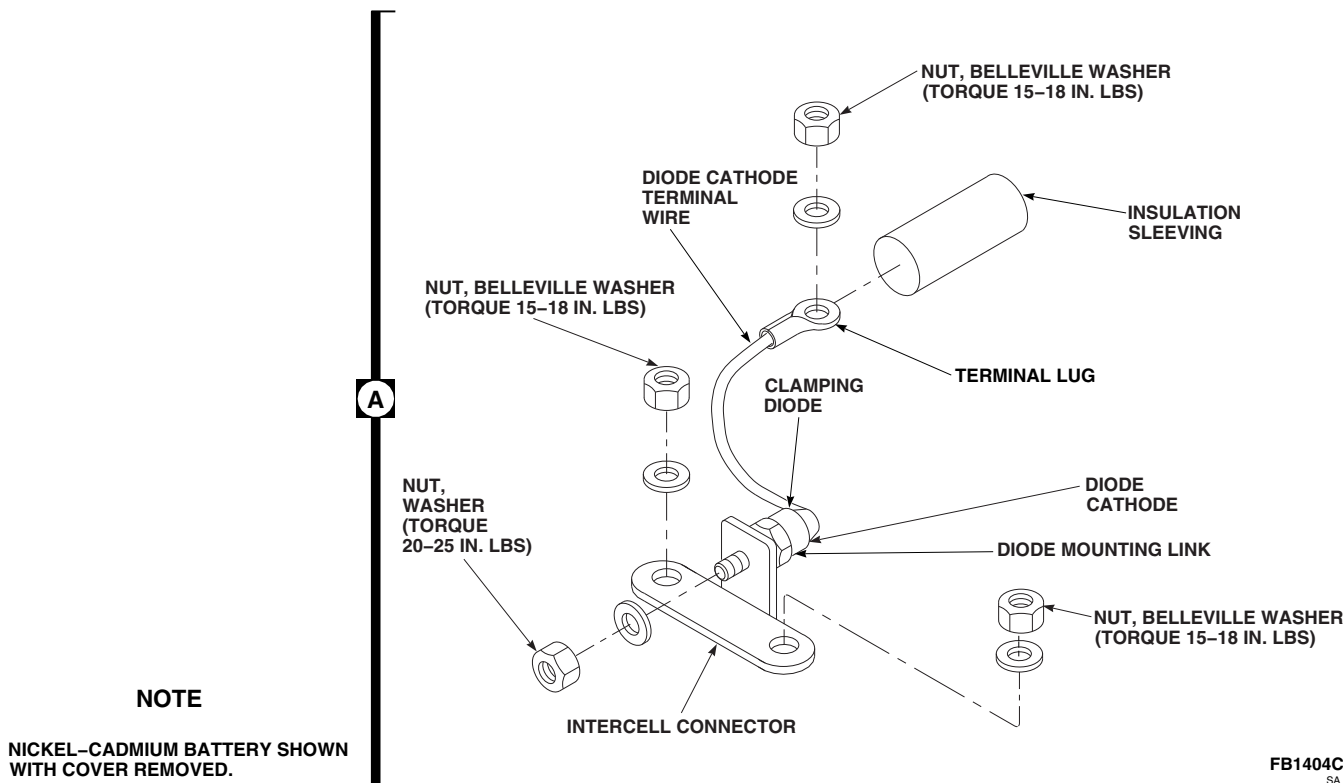
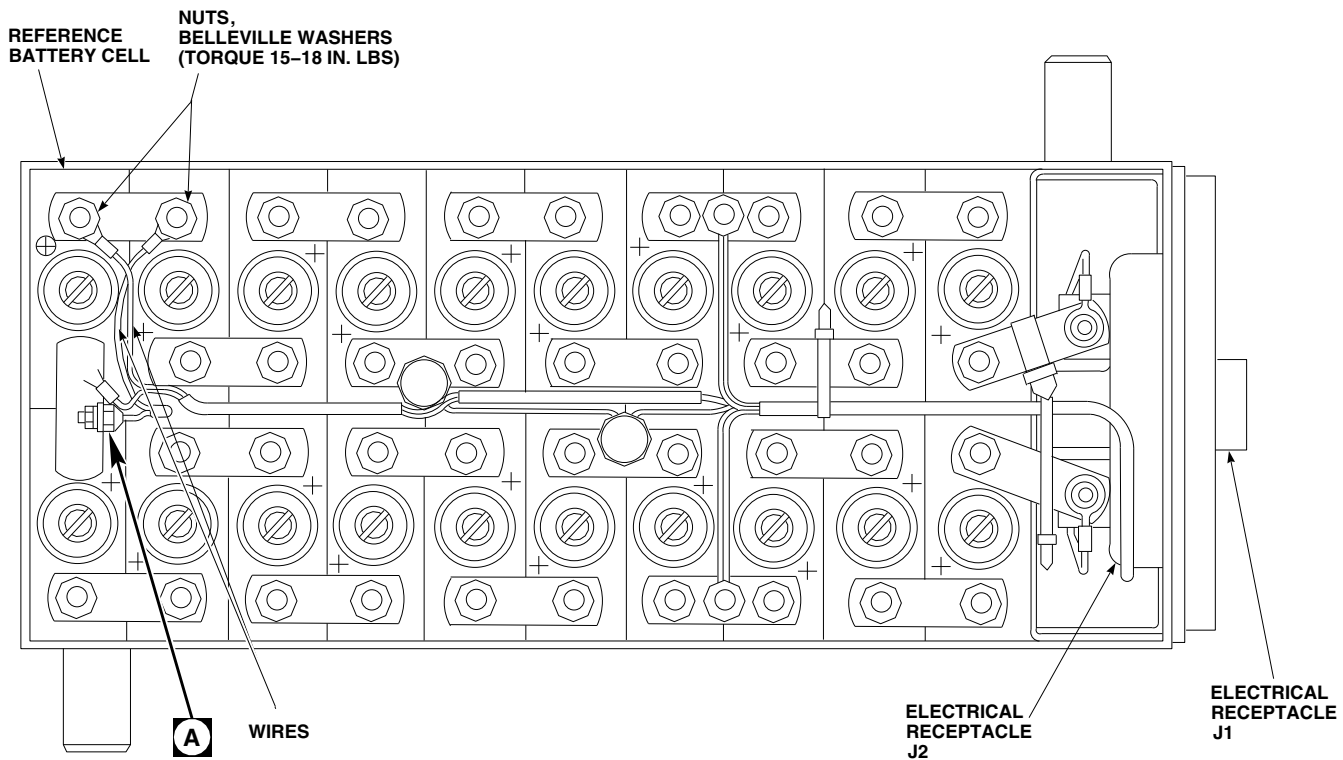


FIGURE 9-5-25-1. REPLACE BATTERY CLAMPING DIODE (BENCH)

- e. Slide heat-shrink insulation sleeving over splices.
- f. Using heat gun, shrink heat-shrink insulation sleeving.

NOTE

Belleville washers are to be installed with concave side down.

- g. Install intercell connector in battery and secure with nuts and Belleville washers (Figure 9-5-25-1 and Detail A).
- h. **TORQUE NUTS TO 15 - 18 INCH-POUNDS.**
- i. Install all wires of battery harness, previously removed, to intercell connectors and secure with nuts and Belleville washers (Figure 9-5-25-1). Remove tags.
- j. **TORQUE NUTS TO 15 - 18 INCH-POUNDS.**
- k. Make sure shorting jumpers are installed on all battery cells.
- l. Make sure area is clean and free of foreign material.
- m. Install and latch battery cover.
- n. Determine if battery is to be placed in service or storage. Proceed as follows:
 - (1) If battery is to be placed in service, perform 100-hour battery inspection (PARA 9-2-22).
 - (2) If battery is to be placed in short-term storage, store battery (PARA 9-5-35).
 - (3) If battery is to be placed in long-term storage, store battery (PARA 9-5-34).

9-5-26 BATTERY CELLS (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
9-5-26.1	Replace Battery Cells (BENCH)	9-5-26-2

INITIAL SETUP

Tools

- Apron, Impermeable, Battery Worker’s
- Cell Puller, Locally-Made, Figure H-47, Appendix H
- Face Shield, Industrial
- Shorting Jumpers
- Torque Wrench, 0 - 30 in. lbs
- Vent Cap Wrench

Materials/Parts

- Brush, Item 83, Appendix D
- Cheesecloth, Item 90, Appendix D
- Corrosion Preventive Compound, Item 110, Appendix D
- Distilled Water, Item 123, Appendix D
- Petrolatum, Item 234, Appendix D

- Plastic Strip, Item 237A, Appendix D
- Protective Caps and Plugs
- Rubber Gloves, Item 265, Appendix D
- Soap
- Tags

References

- Appendix D
- Appendix H
- PARA 9-2-22
- PARA 9-5-21
- PARA 9-5-23
- PARA 9-5-24
- PARA 9-5-25
- PARA 9-5-34
- PARA 9-5-35

9-5-26.1 Replace Battery Cells (BENCH).**WARNING**

- Injury to personnel or damage to battery may result if battery cells have not been shorted for discharging at least 3 hours.
- Serious burns will result if contact with the electrolyte (potassium hydroxide) is made. Wear protective clothing (battery worker's impermeable apron, industrial face shield, and rubber gloves, Item 265, Appendix D) when working around opened batteries. In case of accident, wash with distilled water, Item 123, Appendix D. Do not wear jewelry that might contact battery cell terminals.

CAUTION

- Damage to nickel-cadmium battery will result if maintenance is performed on a nickel-cadmium battery in a lead-acid battery shop. Never use any tools or materials which have been used on lead-acid batteries.
- Electrolyte is absorbed into battery cell plates and battery will appear dry, when battery is completely discharged. Do not add water.

NOTE

Replacement of all battery cells is the same.

9-5-26.1.1 Remove.

- Inspect, clean, current leakage check, and discharge battery (PARA 9-2-22).
- Make sure protective caps are installed over electrical receptacles, J1 and J2 (Figure 9-5-26-1).
- Unlatch and remove battery cover.

- Remove shorting jumpers from all battery cells.
- Remove nuts and Belleville washers securing intercell connectors to battery cell terminal. Remove intercell connector.
- If necessary, tag wiring. Disconnect battery harness enough to replace battery cell (PARA 9-5-21).
- If necessary, remove battery overtemperature switches, battery temperature sensors, and/or clamping diode (PARA 9-5-24, PARA 9-5-23, and PARA 9-5-25).
- Using vent cap wrench, loosen vent cap of battery cell to be removed and adjacent battery cells to relieve battery cell internal pressure.
- USING VENT CAP WRENCH, TIGHTEN VENT CAPS.**
- Locally make cell puller (Figure H-47, Appendix H).
- Using cell puller, connect to battery cell terminals, remove battery cell.

WARNING

Avoid skin contact with water used to wash battery cells. Water may become caustic.

CAUTION

Damage to battery cell will result if water is allowed to enter the battery cell. Electrolyte will become contaminated.

- Using water and soap, wash sides and bottom of battery cell.
- Using brush, Item 83, Appendix D, remove all potassium carbonate deposits.
- Rinse battery cell under running water to remove soap and deposits.

9-5-26-2 Change 6

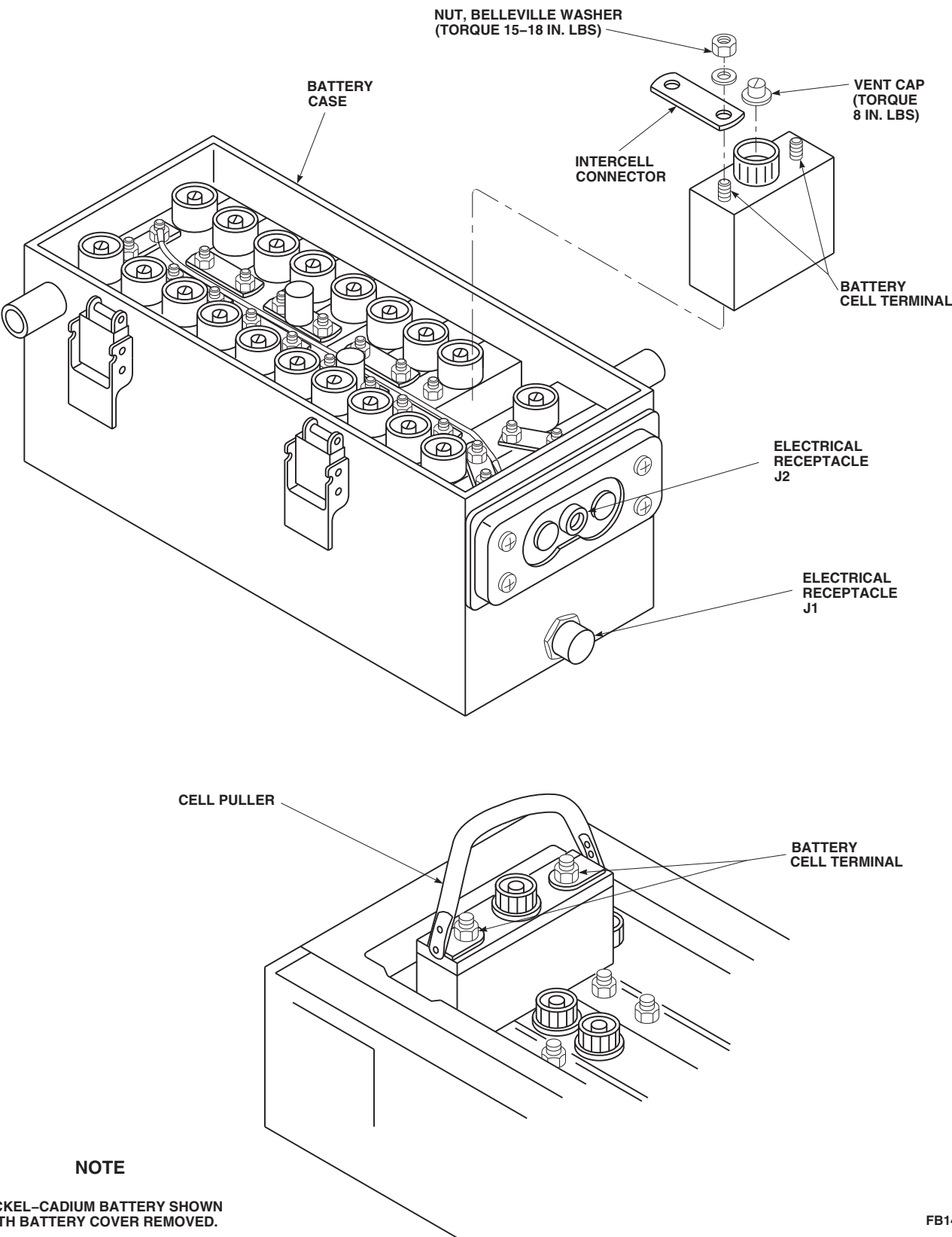


FIGURE 9-5-26-1. REPLACE BATTERY CELLS (BENCH)

- o. Using clean, lint-free cheesecloth, Item 90, Appendix D, dry battery cell.
- p. Install shorting jumper between terminals.

9-5-26.1.2 Install.

WARNING

Injury to personnel will result if electrolyte contacts eye or skin when battery cells are pressed into place. Make sure protective clothing is worn to protect from electrolyte spray.

CAUTION

- New battery cells and used battery cells cannot be mixed within a battery. Battery cells should be grouped together by known usage, dates of manufacture, contract number, etc.
 - Replace all battery cells, if battery cells cannot be grouped together by usage, manufacture date, manufacturer, contract number, etc.
 - All battery cells shall be from the same manufacturer, have the same part number, and ampere-hour rating.
 - All battery cells shall have the same color-code (if used) on the positive terminal.
- a. Loosen vent cap on replacement battery cell and on battery cells in the same row of the battery cell to be replaced (Figure 9-5-26-1).
 - b. Make sure replacement battery cell has shorting jumper installed.
 - c. If necessary, rub a light coat of petrolatum, Item 234, Appendix D, on the side of the replacement battery cell to ease installation.
 - d. Observing battery cell polarity, install replacement battery cell.

- e. If the replacement battery cell fits loose in the battery case additional shims made from plastic strip, Item 237A, Appendix D, may be inserted.
- f. Make sure all vent caps are tight.
- g. **TORQUE VENT CAPS TO 8 INCH-POUNDS.**
- h. If necessary, install battery harness (PARA 9-5-21). Remove tags.
- i. If necessary, install battery overtemperature switches, battery temperature sensors, and/or clamping diode (PARA 9-5-24, PARA 9-5-23, and PARA 9-5-25).

NOTE

Belleville washers are to be installed with concave side down.

- j. Install intercell connectors to battery cell terminals with Belleville washers and nuts.
- k. **TORQUE NUTS TO 15 - 18 INCH-POUNDS.**
- l. Perform battery inspection (PARA 9-2-22).
- m. Apply a light coat of corrosion preventive compound, Item 110, Appendix D, on battery cell terminals and intercell connectors, as required.
- n. Make sure shorting jumpers are installed on all battery cells.
- o. Make sure area is clean and free of foreign material.
- p. Install and latch battery cover.
- q. Determine if battery is to be placed in service or storage. Proceed as follows:
 - (1) If battery is to be placed in service, perform 100-hour battery inspection (PARA 9-2-22).
 - (2) If battery is to be placed in short-term storage, store battery (PARA 9-5-35).

- (3) If battery is to be placed in long-term storage, store battery (PARA 9-5-34).

9-5-27 BATTERY TEST VENT CAP (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
9-5-27.1	Pressure Test Battery Vent Cap (BENCH)	9-5-27-1

INITIAL SETUP

Tools

- Apron, Impermeable, Battery Workers
- Battery Test Cell, Locally-Made, Figure H-238, Appendix H
- Electronic Equipment Toolkit
- Face Shield, Industrial
- Pressure Gage, 0 - 21 psig
- Torque Wrench, 0 - 30 in. lbs
- Vent Cap Wrench

Materials/Parts

- Distilled Water, Item 123, Appendix D
- Protective Caps
- Rubber Gloves, Item 265, Appendix D

References

- Appendix D
- Appendix H
- PARA 9-2-22
- PARA 9-5-26

9-5-27.1 Pressure Test Battery Vent Cap (BENCH).

WARNING

- Injury to personnel or damage to battery may result if battery cells have not been shorted for discharging at least 3 hours.

- Serious burns will result if contact with the electrolyte (potassium hydroxide) is made. Wear protective clothing (battery workers impermeable apron, industrial face shield, and rubber gloves, Item 265, Appendix D) when working around opened batteries. In case of accident, wash with distilled water, Item 123, Appendix D. Do not wear jewelry that might contact battery cell terminals.

CAUTION

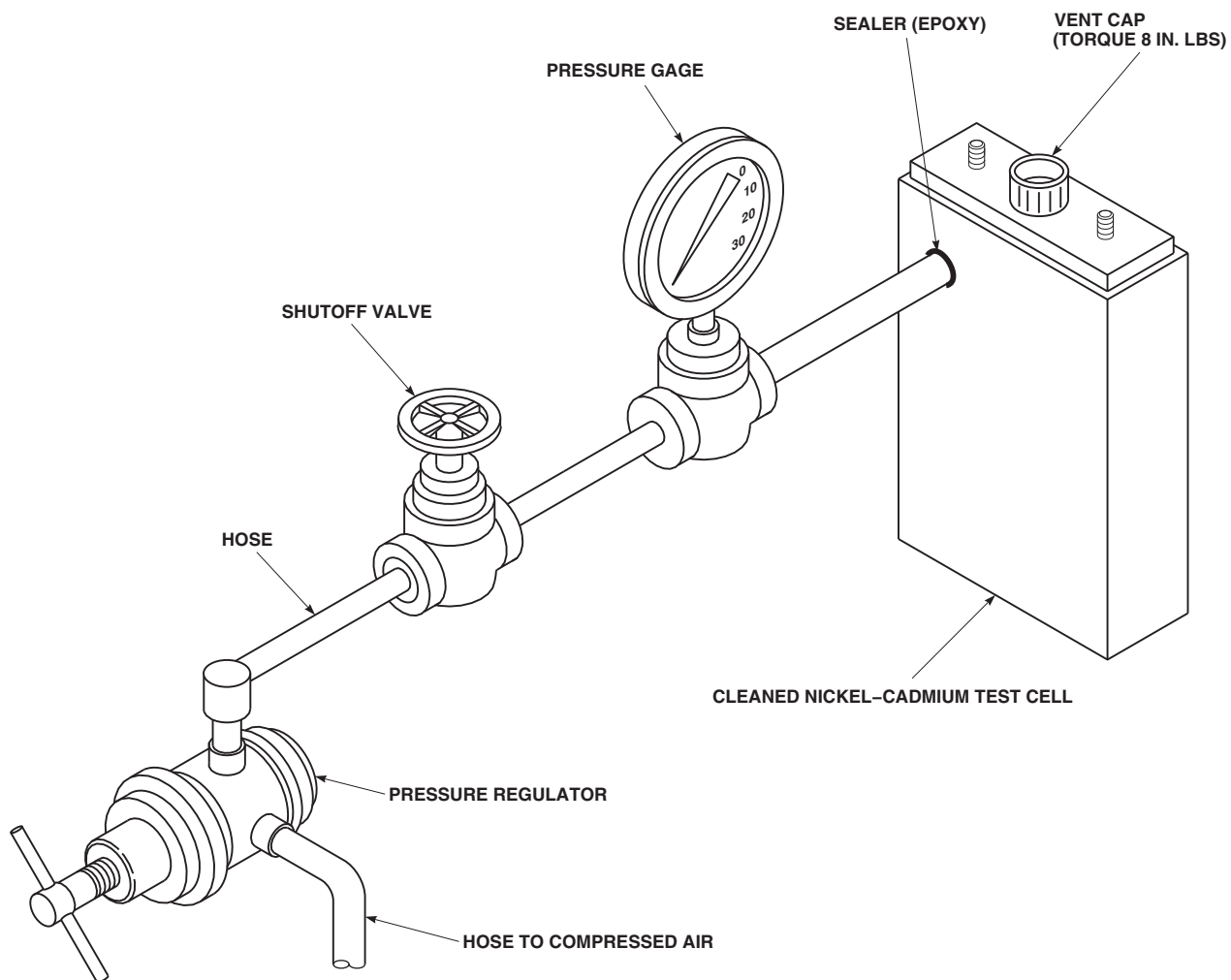
- Damage to nickel-cadmium battery will result if maintenance is performed on a nickel-cadmium battery in a lead-acid battery shop. Never use any tools or materials which have been used on lead-acid batteries.
- Electrolyte is absorbed into battery cell plates and battery will appear dry, when battery is completely discharged. Do not add water.

NOTE

Pressure tests of all battery cell vent caps is the same.

- a. Inspect, clean, current leakage check, and discharge battery (PARA 9-2-22).
- b. Make sure protective caps are installed over electrical receptacles, J1 and J2.
- c. Remove battery cell (PARA 9-5-26).

- d. Using vent cap wrench, remove vent cap (Figure 9-5-27-1).
- e. Inspect vent cap and packing for damage.
- f. Clean vent cap by sloshing in warm water, and allow to air dry.
- g. Locally make battery test cell (Figure H-238, Appendix H).
- h. Using battery test cell connect to an air source and gage.
- i. Install vent cap to be tested into battery test cell.
- j. **TORQUE VENT CAP TO 8 INCH-POUNDS.**
- k. Using pressure gage to monitor air pressure, slowly increase air source to no greater than 10 psi. Vent in vent cap shall open between 2 and 10 psi. If required, replace vent cap.
- l. Turn off air source. Pressure in locally-made battery test cell shall not drop below 2 psi. If required, replace vent cap.
- m. Install battery cell (PARA 9-5-26).



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FIGURE 9-5-27-1. PRESSURE TEST BATTERY VENT CAP (BENCH)

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9-5-28 PRESSURE TEST BATTERY CELL CASE (BENCH).

PARAGRAPH BREAKDOWN

	PAGE
9-5-28.1 Pressure Test Battery Cell Case (BENCH)	9-5-28-2

INITIAL SETUP

Tools

- Apron, Impermeable, Battery Workers
- Battery Test Vent Cap and Tee, Locally-Made, Figure H-239, Appendix H
- Container, 5-Gallon
- Electronic Equipment Toolkit
- Face Shield, Industrial
- Pressure Gage, 0 - 21 psig
- Vent Cap Wrench

Materials/Parts

- Distilled Water, Item 123, Appendix D
- Epoxy Coating Kit, Item 126A, Appendix D
- Nitrogen, Item 221, Appendix D
- Rubber Gloves, Item 265, Appendix D
- Sandpaper, Fine-Grit
- Soap

References

- Appendix D
- Appendix H
- PARA 9-5-26

9-5-28.1 Pressure Test Battery Cell Case (BENCH).

WARNING

- Injury to personnel or damage to battery may result if battery cells have not been shorted for discharging at least 3 hours.
- Serious burns will result if contact with the electrolyte (potassium hydroxide) is made. Wear protective clothing (battery workers impermeable apron, industrial face shield, and rubber gloves, Item 265, Appendix D) when working around opened batteries. In case of accident, wash with distilled water, Item 123, Appendix D. Do not wear jewelry that might contact battery cell terminals.

CAUTION

- Damage to nickel-cadmium battery will result if maintenance is performed on a nickel-cadmium battery in a lead-acid battery shop. Never use any tools or materials which have been used on lead-acid batteries.
- Electrolyte is absorbed into battery cell plates and battery will appear dry, when battery is completely discharged. Do not add water.

NOTE

Pressure tests of all battery cell cases is the same.

- Remove battery cell (PARA 9-5-26).
- Using vent cap wrench, remove vent cap.
- Inspect vent cap and packing for damage.
- Clean vent cap assembly by sloshing in warm water, and allow to air dry.

- Locally make battery test vent cap and tee (Figure H-239, Appendix H).
- Connect battery test vent cap and tee to battery cell (Figure 9-5-28-1).

CAUTION

Compressed air will contaminate battery cell. Use dry nitrogen only to pressurize battery cells. Do not exceed 2.0 psi.

- Connect nitrogen, Item 221, Appendix D, to battery test vent cap and tee.

WARNING

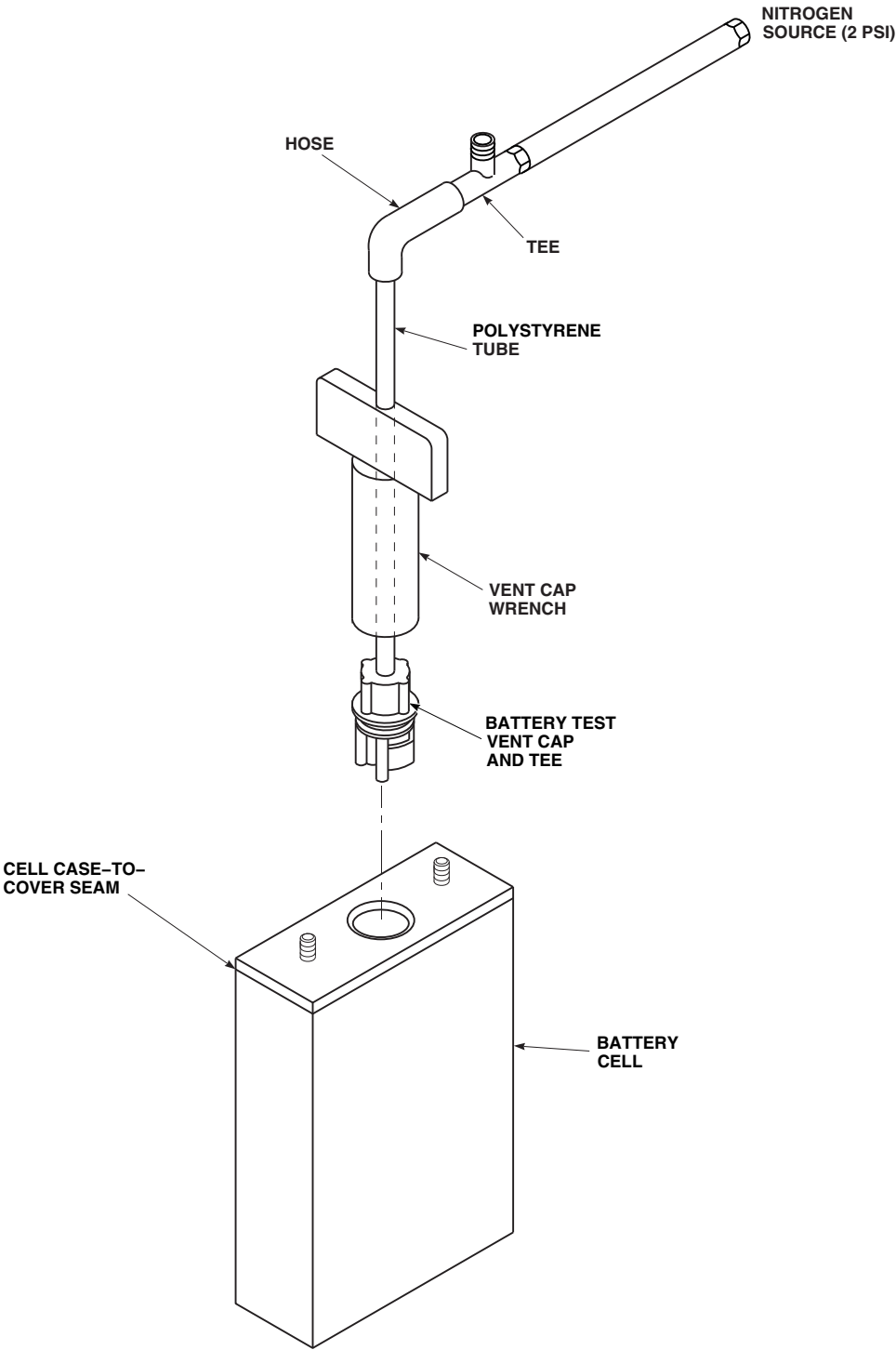
Injury to personnel will result if electrolyte contacts eyes or skin when battery cell is pressurized. Make sure protective clothing is worn to protect from electrolyte spray.

CAUTION

Damage to battery cell will result if water is allowed to enter the battery cell. Electrolyte will become contaminated.

- Using pressure gage to monitor nitrogen pressure, adjust nitrogen source regulator for 2.0 psi output, place a finger over the open side of the tee and install battery test vent cap and tee onto battery cell to be tested. Make sure it is an airtight seal.
- Submerge the battery cell into a 5-gallon container of water, making sure not to submerge the tee.
- Observe battery cell for 30 seconds. **NO BUBBLES, INDICATING LEAKAGE, ARE ALLOWED.**
- If battery cell has leakage around the seam, where the battery cell case attaches to the battery cell cover, perform the following:

9-5-28-2



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FIGURE 9-5-28-1. PRESSURE TEST BATTERY CELL CASE (BENCH)

- (1) Remove battery cell from water.
- (2) Remove finger from tee.
- (3) Using soap and water, clean the outside of battery cell.
- (4) Rinse and dry outside of battery cell thoroughly.
- (5) If possible, using fine-grit sandpaper, roughen the area around the leak.

NOTE

Do not allow epoxy to contact the battery cell terminals.

- (6) Using epoxy coating kit, Item 126A, Appendix D, apply a thin coat of epoxy to the area of the leak, and allow epoxy to harden.
 - (7) Retest battery cell case.
 - (8) If battery cell leakage cannot be stopped after several attempts, discard battery cell.
1. If battery cell has leakage any place other than the seam, discard battery cell.

9-5-29 BATTERY CASE AND COVER (BENCH).

PARAGRAPH BREAKDOWN

	PAGE
9-5-29.1 Repair/Clean Battery Case and Cover (BENCH)	9-5-29-2

INITIAL SETUP

Tools

- Apron, Impermeable, Battery Workers
- Electronic Equipment Toolkit
- Face Shield, Industrial
- Nonmetallic Scraper, Locally-Made
- Oven

Materials/Parts

- Adhesive, Rubber-to-Metal
- Brush, Item 83, Appendix D
- Cheesecloth, Item 90, Appendix D

- Distilled Water, Item 123, Appendix D
- Protective Caps and Plugs
- Rubber Gloves, Item 265, Appendix D
- Rubber Sheet
- Soap

References

- Appendix D
- PARA 9-2-22
- PARA 9-5-21
- PARA 9-5-22
- PARA 9-5-26

9-5-29.1 Repair/Clean Battery Case and Cover (BENCH).

WARNING

- Injury to personnel or damage to battery may result if battery cells have not been shorted for discharging at least 3 hours.
- Serious burns will result if contact with the electrolyte (potassium hydroxide) is made. Wear protective clothing (battery workers impermeable apron, industrial face shield, and rubber gloves, Item 265, Appendix D) when working around opened batteries. In case of accident, wash with distilled water, Item 123, Appendix D. Do not wear jewelry that might contact battery cell terminals.

CAUTION

- Damage to nickel-cadmium battery will result if maintenance is performed on a nickel-cadmium battery in a lead-acid battery shop. Never use any tools or materials which have been used on lead-acid batteries.
- Electrolyte is absorbed into battery cell plates and battery will appear dry, when battery is completely discharged. Do not add water.

9-5-29.1.1. Disassemble.

- Inspect, clean, current leakage check, and discharge battery (PARA 9-2-22).
- Make sure protective caps are installed over electrical receptacles, J1 and J2.
- Remove battery harness (PARA 9-5-21).
- Remove all battery cells (PARA 9-5-26).

WARNING

Avoid skin contact with water used to wash battery cells, battery, or internal parts. Water may become caustic.

- If necessary, remove shims, and electrical receptacle, J2 (PARA 9-5-22).
 - Using soap and water, wash components.
 - Using cheesecloth, Item 90, Appendix D, wipe clean.
 - Allow to air dry.
- If necessary, using nonmetallic scraper, remove cover liner and gasket, and case liner.

9-5-29.1.2. Clean.

WARNING

Avoid skin contact with water used to wash battery cells, or internal parts. Water may become caustic.

- Wash battery case and/or cover under running water.
- Using brush, Item 83, Appendix D, remove potassium carbonate deposits.
- Using cheesecloth, Item 90, Appendix D, wipe case clean.
- Allow battery case and/or cover to air dry thoroughly. If an oven is available, case and/or cover may be dried for 4 hours at 120° - 130°F (48.9° - 54.4°C).
- Check all hardware for corrosion or damage. Clean or replace as required.

9-5-29-2

NOTE

Repair and refinishing of battery case and/or cover is normally beyond the capabilities of the battery shop, and should be accomplished by the paint shop, or replaced.

- f. Check battery case and cover for finish coat cracking and/or peeling. Repair or replace as required.

9-5-29.1.3. Assemble.

- a. If cover liner was removed, replace as follows:
 - (1) Using old cover liner or one from the same type cover, as a pattern, trace the outline on rubber sheet of the same thickness.
 - (2) Cut out the new cover liner.
 - (3) Cut $\frac{3}{8}$ -inch wide strips to make raised pads and perimeter gasket. The thickness of the new cover liner and thickness of the original cover liner.

CAUTION

Raised pads are to exert a slight pressure on the battery cell vents to prevent battery cell movement when cover is installed. Make sure the raised pads do not completely cover the battery cell vents or excess pressure could build in the battery cells.

NOTE

- Observe the connector markings on the cover to make sure proper placement of liner.
- Proper placement of raised strips on the cover liner can be checked by looking

for vent cap impressions in the rubber after the cover has been installed and removed.

- (4) Using rubber-to-metal adhesive, cement the new cover liner and strips in place. Allow at least 8 hours for drying before reassembling battery.

WARNING

Pressure will build inside the battery and may result in it exploding if external vent holes are covered. Do not close off the two external vent holes on the sides of the battery case when installing case liners.

- b. If rubber case liner was removed, replace the same as cover liner per steps a. (1), a. (2), and a. (4).
- c. If cover gasket was removed, replace as follows:
 - (1) Make sure cover is clean of electrolyte, old gasket, and cement.
 - (2) Make sure new cover gasket is clean and dry.
 - (3) Using rubber-to-metal adhesive, apply to one side of cover gasket, and cover. Carefully place cover gasket inside of cover.
 - (4) Press cover gasket firmly into place and allow at least 8 hours to dry before installing cover.
- d. Install shims and/or electrical receptacle, J2, as required (PARA 9-5-22).
- e. Install all battery cells (PARA 9-5-26).
- f. Install battery harness (PARA 9-5-21).

9-5-30 PRESSURE TEST BATTERY CASE (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
9-5-30.1	Pressure Test Battery Case (BENCH)	9-5-30-2

INITIAL SETUP

Tools

- Apron, Impermeable, Battery Worker’s
- Electronic Equipment Toolkit
- Face Shield, Industrial
- Pressure Gage, 0 - 21 psig

Materials/Parts

- Distilled Water, Item 123, Appendix D
- Protective Caps and Plugs

- Rubber Gloves, Item 265, Appendix D

References

- Appendix D
- PARA 9-2-22
- PARA 9-5-22
- PARA 9-5-29
- PARA 9-5-34
- PARA 9-5-35

9-5-30.1 Pressure Test Battery Case (BENCH).**WARNING**

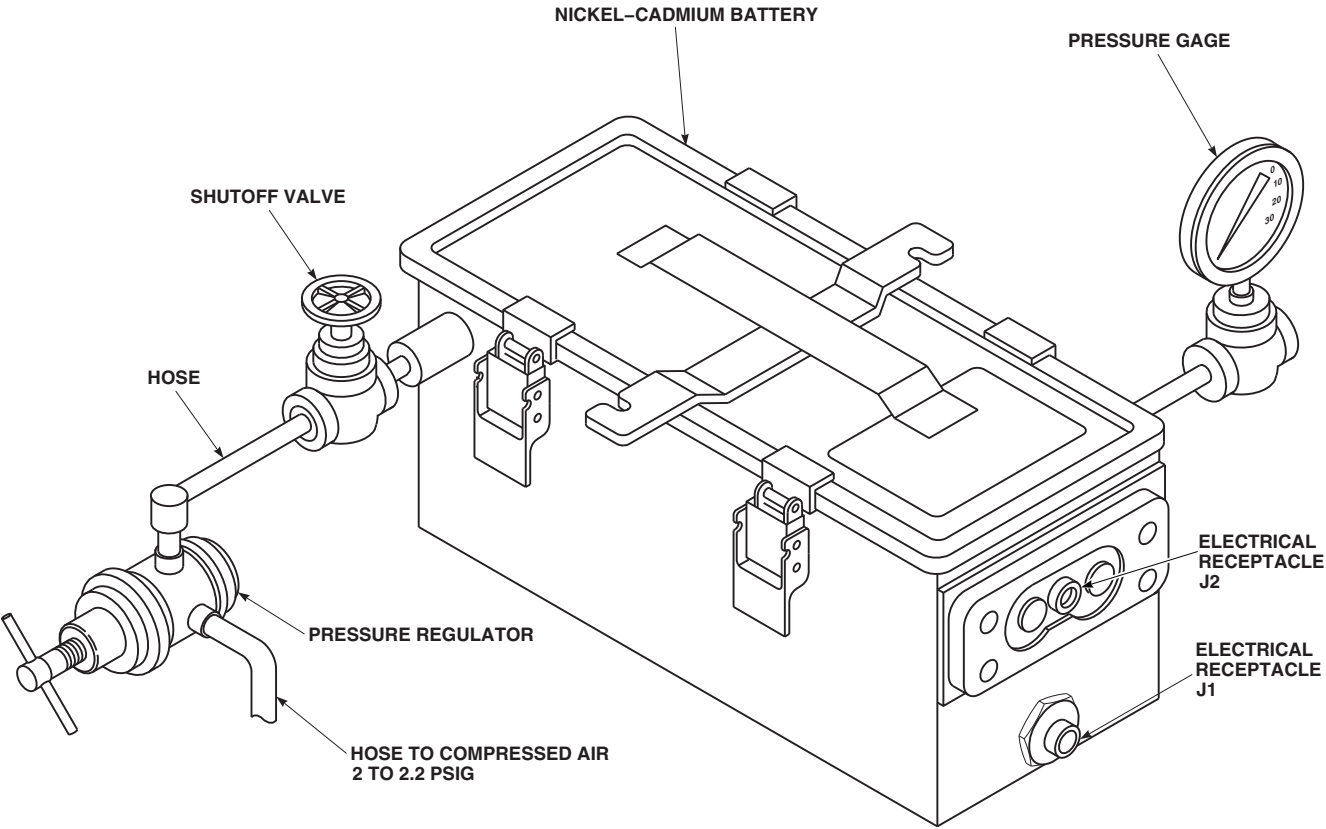
- Injury to personnel or damage to battery may result if battery cells have not been shorted for discharging at least 3 hours.
- Serious burns will result if contact with the electrolyte (potassium hydroxide) is made. Wear protective clothing (battery worker's impermeable apron, industrial face shield, and rubber gloves, Item 265, Appendix D) when working around opened batteries. In case of accident, wash with distilled water, Item 123, Appendix D. Do not wear jewelry that might contact battery cell terminals.

CAUTION

- Damage to nickel-cadmium battery will result if maintenance is performed on a nickel-cadmium battery in a lead-acid battery shop. Never use any tools or materials which have been used on lead-acid batteries.
- Electrolyte is absorbed into battery cell plates and battery will appear dry, when

battery is completely discharged. Do not add water.

- a. Inspect, clean, current leakage check, and discharge battery (PARA 9-2-22).
- b. Make sure protective caps are installed over electrical receptacles, J1 and J2 (Figure 9-5-30-1).
- c. Connect battery to test equipment for pressure test.
- d. Using pressure gage to monitor air pressure, slowly increase air pressure to 2.0 to 2.2 psig. Close shutoff valve.
- e. Pressure in battery shall not decrease to less than 1 psig after 2 minutes. Replace cover gasket (PARA 9-5-29) or electrical receptacle, J2, rectangular ring (PARA 9-5-22) as necessary.
- f. Determine if battery is to be placed in service or storage. Proceed as follows:
 - (1) If battery is to be placed in service, perform 100-hour battery inspection (PARA 9-2-22).
 - (2) If battery is to be placed in short-term storage, store battery (PARA 9-5-35).
 - (3) If battery is to be placed in long-term storage, store battery (PARA 9-5-34).



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FIGURE 9-5-30-1. PRESSURE TEST BATTERY CASE (BENCH)

9-5-31 RECONDITION BATTERY (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
9-5-31.1	Recondition Battery (BENCH)	9-5-31-2

INITIAL SETUP

Tools

- Apron, Impermeable, Battery Worker’s
- Battery Charger/Analyzer, PCA-131 50/60
- Electrical Power Source, 115 VAC, 50 - 60 Hz, Single Phase (or equivalent per Battery Charger/Analyzer)
- Electronic Equipment Toolkit
- Face Shield, Industrial

Materials/Parts

- Distilled Water, Item 123, Appendix D
- Rubber Gloves, Item 265, Appendix D

References

- Appendix D
- PARA 9-2-22
- PARA 9-5-32
- PARA 9-5-34
- PARA 9-5-35

9-5-31.1 Recondition Battery (BENCH).**WARNING**

- Serious burns will result if contact with the electrolyte (potassium hydroxide) is made. Wear protective clothing (battery worker's impermeable apron, industrial face shield, and rubber gloves, Item 265, Appendix D) when working around opened batteries. In case of accident, wash with distilled water, Item 123, Appendix D. Do not wear jewelry that might contact battery cell terminals.
- Explosive gases may be released during charging. Make sure that charging area is well ventilated. Do not use matches or open flame. Avoid short circuits, since arcs may cause explosion. Do not disconnect charging cable from battery until charger is off.

CAUTION

- Damage to nickel-cadmium battery will result if maintenance is performed on a nickel-cadmium battery in a lead-acid battery shop. Never use any tools or materials which have been used on lead-acid batteries.
- Electrolyte is absorbed into battery cell plates and battery will appear dry, when battery is completely discharged. Do not add water.
- Charge battery at an ambient temperature between 40° - 90°F (4.4° - 32.2°C).
- Battery case should never be more than slightly warm to the touch (100°F (37.8°C)) during charging. If battery exhibits overtemperature condition during charging, stop charging, discharge battery, and start charging again.

- Do not loosen or remove vent caps during charging. Open battery cells may cause formation of potassium carbonate.
- Never charge batteries with battery cells connected in parallel.

NOTE

Battery charger/analyzer is set up to operate with 115 VAC, 50 - 60 Hz, single phase. For other operating voltages refer to instructions on inside top lid of battery charger/analyzer. When 110 - 120 VAC is used, a 12 amp fuse must also be used.

- If installed, remove all shorting jumpers from battery cells.
- Inspect, clean, and current leakage check battery (PARA 9-2-22).
- Connect battery charger/analyzer to electrical power source.
- Turn SELECTOR switch to MA-500H.
- Connect battery cable, P2B, between battery and battery charger/analyzer.
- Set AC POWER switch to ON.
- Set AUTOMATIC CYCLE-MANUAL DISCHARGE switch to AUTOMATIC.
- Turn TIMER control to TOPPING charge. Record start time.
- Periodically monitor total battery cell voltage reading on battery charger/analyzer. Stop charging when voltage reaches 30 VDC or 30 minutes of charging elapses.
- If battery does not reach 30 VDC within 30 minutes, check individual battery cell voltage to make sure none are shorted. Reference battery cell voltage shall be 1.53 to 1.75 VDC, and regular battery cells voltage shall be 1.55 to 1.75 VDC. Replace as required.
- Perform electrolyte level check/adjustment (PARA 9-2-22).

- l. Discharge battery (PARA 9-2-22).
- m. If battery charger/analyzer BATTERY LOW light goes on during discharge, perform deep cycle procedure (PARA 9-5-32).
- n. Charge battery (PARA 9-2-22).
- o. Determine if battery is to be placed in service or storage. Proceed as follows:
 - (1) If battery is to be placed in service, perform 100-hour battery inspection (PARA 9-2-22).
 - (2) If battery is to be placed in short-term storage, store battery (PARA 9-5-35).
 - (3) If battery is to be placed in long-term storage, store battery (PARA 9-5-34).

9-5-32 DEEP CYCLE BATTERY (BENCH).

PARAGRAPH BREAKDOWN

	PAGE
9-5-32.1 Deep Cycle Battery (BENCH)	9-5-32-2

INITIAL SETUP

Tools

- Apron, Impermeable, Battery Worker’s
- Digital Multimeter
- Electronic Equipment Toolkit
- Face Shield, Industrial
- Shorting Jumper, Locally-Made, Figure H-246, Appendix H

Materials/Parts

- Distilled Water, Item 123, Appendix D
- Rubber Gloves, Item 265, Appendix D

References

- Appendix D
- Appendix H
- PARA 9-2-22
- PARA 9-5-34
- PARA 9-5-35

9-5-32.1 Deep Cycle Battery (BENCH).

WARNING

- Injury to personnel or damage to battery may result if battery cells have not been shorted for discharging at least 3 hours.
- Serious burns will result if contact with the electrolyte (potassium hydroxide) is made. Wear protective clothing (battery worker's impermeable apron, industrial face shield, and rubber gloves, Item 265, Appendix D) when working around opened batteries. In case of accident, wash with distilled water, Item 123, Appendix D. Do not wear jewelry that might contact battery cell terminals.

CAUTION

- Damage to nickel-cadmium battery will result if maintenance is performed on a nickel-cadmium battery in a lead-acid battery shop. Never use any tools or materials which have been used on lead-acid batteries.
 - Electrolyte is absorbed into battery cell plates and battery will appear dry, when battery is completely discharged. Do not add water.
- a. Charge battery (PARA 9-2-22).
 - b. Locally make shorting jumpers (Figure H-246, Appendix H).
 - c. Discharge battery (PARA 9-2-22).
 - d. When the discharge is complete, keep the AUTOMATIC CYCLE - MANUAL DISCHARGE switch on the battery conditioner/analyzer in MANUAL DISCHARGE position.

NOTE

Reference battery cell (black top), has a lower capacity than the regular battery cells, and will discharge at a faster rate. When monitoring battery cell voltages in the following steps, check reference battery cell often. As reference battery cell approaches +0.3 VDC, monitor it more closely.

- e. Using digital multimeter, monitor individual battery cell voltages.
- f. When reference battery cell reaches +0.3 VDC, connect shorting jumpers across battery cell terminals.
- g. When individual regular battery cells reach +0.5 VDC, connect shorting jumpers across battery cell terminals.
- h. Using shorting jumpers, continue discharging until 75% (15) of battery cells are shorted. As the remaining battery cells reach +0.5 VDC, connect 1 ohm shorting jumpers across battery cell terminals.
- i. When all battery cells are shorted (discharged) turn off and disconnect battery charger/analyzer.
- j. Allow battery to remain shorted (discharged), for at least 3 hours.
- k. Charge battery (PARA 9-2-22).
- l. Determine if battery is to be placed in service or storage. Proceed as follows:
 - (1) If battery is to be placed in service, perform 100-hour battery inspection (PARA 9-2-22).
 - (2) If battery is to be placed in short-term storage, store battery (PARA 9-5-35).
 - (3) If battery is to be placed in long-term storage, store battery (PARA 9-5-34).

9-5-33 NEW BATTERY (BENCH).

PARAGRAPH BREAKDOWN

	PAGE
9-5-33.1 Prepare New Battery for Service (BENCH)	9-5-33-1

INITIAL SETUP

Tools

- Apron, Impermeable, Battery Workers
- Electronic Equipment Toolkit
- Face Shield, Industrial
- Shorting Spring

Materials/Parts

- Distilled Water, Item 123, Appendix D
- Rubber Gloves, Item 265, Appendix D

References

- Appendix D
- PARA 9-2-22

9-5-33.1 Prepare New Battery for Service (BENCH).

WARNING

Serious burns will result if contact with the electrolyte (potassium hydroxide) is made. Wear protective clothing (battery workers impermeable apron, industrial face shield, and rubber gloves, Item 265, Appendix D) when working around opened batteries. In case of accident, wash with distilled water, Item 123, Appendix D. Do not wear jewelry that might contact battery cell terminals.

CAUTION

- Damage to nickel-cadmium battery will result if maintenance is performed on a

nickel-cadmium battery in a lead-acid battery shop. Never use any tools or materials which have been used on lead-acid batteries.

- Electrolyte is absorbed into battery cell plates and battery will appear dry, when battery is completely discharged. Do not add water.
 - Make sure shorting spring (strap) is removed from main power receptacle terminals.
 - Make sure shipping plugs are removed and replaced with proper vent plugs prior to performing any maintenance on the battery.
 - Perform 100-hour battery inspection (PARA 9-2-22).

9-5-34 BATTERY LONG-TERM STORAGE (BENCH).

PARAGRAPH BREAKDOWN

	PAGE
9-5-34.1 Battery Long-Term Storage (BENCH)	9-5-34-2

INITIAL SETUP

Tools

- Apron, Impermeable, Battery Workers
- Battery Charger/Analyzer, PCA-131 50/60
- Digital Multimeter
- Electrical Power Source, 115 vac, 50 - 60 Hz, Single Phase (or equivalent per Battery Charger/Analyzer)
- Electronic Equipment Toolkit
- Face Shield, Industrial
- Shorting Jumpers, Locally-Made, Figure H-246, Appendix H

- Shorting Spring

Materials/Parts

- Corrosion Preventive Compound, Item 110, Appendix D
- Distilled Water, Item 123, Appendix D
- Rubber Gloves, Item 265, Appendix D

References

- Appendix D
- Appendix H
- PARA 9-2-22

9-5-34.1 Battery Long-Term Storage (BENCH).

WARNING

- Injury to personnel or damage to battery may result if battery cells have not been shorted for discharging at least 3 hours.
- Serious burns will result if contact with the electrolyte (potassium hydroxide) is made. Wear protective clothing (battery workers impermeable apron, industrial face shield, and rubber gloves, Item 265, Appendix D) when working around opened batteries. In case of accident, wash with distilled water, Item 123, Appendix D. Do not wear jewelry that might contact battery cell terminals.

CAUTION

- Damage to nickel-cadmium battery will result if maintenance is performed on a nickel-cadmium battery in a lead-acid battery shop. Never use any tools or materials which have been used on lead-acid batteries.
- Electrolyte is absorbed into battery cell plates and battery will appear dry, when battery is completely discharged. Do not add water.

NOTE

Nickel-cadmium batteries may be stored in a noncorrosive environment for an unlimited period. Storage temperature should be between -65° - 120°F (-54° - 49°C). Upper limit may be extended to 160°F (71°C) for short-term storage (less than 90 days). The lower the storage temperature, the longer battery charge retention.

9-5-34.1.1. Charging During Long-Term Storage.

WARNING

Explosive gases may be released during charging. Make sure that charging area is well ventilated. Do not use matches or open flame. Avoid short circuits, since arcs may cause explosion. Do not disconnect charging cable from battery until charger is off.

NOTE

- A trickle charge of 15 milliamperes (3 milliamperes/ampere hour rating) is recommended. Make sure to check electrolyte level periodically.
- It is recommended that if battery is not to be charging during long-term storage, it be stored in a discharged and shorted-out state.
 - a. Make sure battery has been thoroughly inspected and cleaned (PARA 9-2-22).
 - b. Charge battery to full capacity (PARA 9-2-22).
 - c. Unlatch and remove battery cover.
 - d. Apply a light coat of corrosion preventive compound, Item 110, Appendix D, on receptacle terminals, battery cell terminals, and intercell connectors, as required.
 - e. Install and latch battery cover.
 - f. Place battery in storage facility and connect trickle charging equipment.

9-5-34.1.2. Shorted-Out Long-Term Storage.

- a. Make sure battery has been thoroughly inspected and cleaned (PARA 9-2-22).
- b. Locally make shorting jumpers (Figure H-246, Appendix H).

- c. Unlatch and remove battery cover.
- d. Discharge battery (PARA 9-2-22).
- e. When discharge is complete, keep AUTOMATIC CYCLE-MANUAL DISCHARGE switch in MANUAL DISCHARGE position.

NOTE

Reference battery cell (black top) has a lower capacity than the regular battery cells, and will discharge at a faster rate. When monitoring battery cell voltages in the following steps, check reference battery cell often. As reference battery cell approaches +0.3 vdc, monitor it more closely.

- f. Using digital multimeter, monitor individual battery cell voltages.
- g. When reference battery cell reaches +0.3 vdc, connect shorting jumpers across battery cell terminals.
- h. When individual regular battery cells reach +0.5 vdc, connect shorting jumpers across battery cell terminals.
- i. Using shorting jumpers, continue discharging until 75% (15) of battery cells are shorted. As the remaining battery cells reach +0.5 vdc, connect 1 ohm shorting jumpers across battery cell terminals.
- j. When all battery cells are shorted, turn off and disconnect battery charger/analyzer.
- k. Allow battery to be discharged with battery cells shorted for 24 hours.
- l. Install shorting spring (strap) across main power receptacle terminals.

- m. Remove battery cell shorting clips.
- n. Apply a light coat of corrosion preventive compound, Item 110, Appendix D, on receptacle terminals, battery cell terminals and intercell connectors, as required.
- o. Install shorting spring across electrical receptacle, J2, main terminals.
- p. Install and latch battery cover.
- q. Place battery in storage facility.

9-5-34.1.3. Return to Service.

- a. If battery was stored in a trickle charging facility, perform the following:
 - (1) Turn off and disconnect trickle charging equipment.
 - (2) Perform 50-hour battery inspection (PARA 9-2-22).

WARNING

- Serious burns will result if contact with the electrolyte (potassium hydroxide) is made. Wear protective clothing when working around opened batteries. In case of accident, wash with distilled water, Item 123, Appendix D. Do not wear jewelry that might contact battery cell terminals.
- Explosive gases may be released during charging. Make sure that charging area is well ventilated. Do not use matches or open flame. Avoid short circuits, since arcs may cause explosion. Do not disconnect charging cable from battery until charger is off.

CAUTION

- Damage to nickel-cadmium battery will result if maintenance is performed on a nickel-cadmium battery in a lead-acid battery shop. Never use any tools or materials which have been used on lead-acid batteries.
- Electrolyte is absorbed into battery cell plates and battery will appear dry, when battery is completely discharged. Do not add water.
- Charge battery at an ambient temperature between 40° - 90°F (4.4° - 32.2°C).
- Battery case should never be more than slightly warm to the touch 100°F (37.8°C) during charging. If battery exhibits overtemperature condition during charging, stop charging, discharge battery, and start charging again.
- Do not loosen or remove vent caps during charging. Open battery cells may cause formation of potassium carbonate.
- Never charge batteries with battery cells connected in parallel.

NOTE

Battery charger/analyzer is set up to operate with 115 vac, 50 - 60 Hz, single phase. For other operating voltages refer to instructions on inside top lid of battery charger/analyzer. When 110 - 120 vac is used, a 12 amp fuse must also be used.

- (3) Connect battery charger/analyzer to electrical power source.
- (4) Turn SELECTOR switch to MA-500H

- (5) Connect battery cable, P2B, between battery and battery charger/analyzer.
 - (6) Set AC POWER switch to ON.
 - (7) Set AUTOMATIC CYCLE-MANUAL DISCHARGE switch to AUTOMATIC.
 - (8) Turn TIMER control to TOPPING charge. Record start time.
 - (9) Periodically monitor total battery cell voltage reading on battery charger/analyzer. Stop charging when voltage reaches 30 vdc or 30 minutes of charging elapses.
 - (10) If battery does not reach 30 vdc within 30 minutes, check individual battery cell voltage to make sure none are shorted. Reference battery cell voltage shall be 1.53 to 1.75 vdc. Replace as required.
 - (11) Perform electrolyte level check/adjustment (PARA 9-2-22).
 - (12) Return battery to service.
- b. If battery was stored in a shorted-out condition, perform the following:
- (1) Remove shorting spring from electrical receptacle, J2, main terminals.
 - (2) Inspect and clean battery (PARA 9-2-22).
 - (3) Perform 100-hour battery inspection (PARA 9-2-22).

NOTE

It may be necessary to deep cycle battery to obtain a full charge.

- (4) Perform 50-hour battery inspection (PARA 9-2-22).
- (5) Return battery to service.

9-5-35 BATTERY SHORT-TERM STORAGE (BENCH).

PARAGRAPH BREAKDOWN

	PAGE
9-5-35.1 Battery Short-Term Storage (BENCH)	9-5-35-1

INITIAL SETUP

Tools

- Apron, Impermeable, Battery Workers
- Battery Charger/Analyzer, PCA-131 50/60
- Digital Multimeter
- Electrical Power Source, 115 vac, 50 - 60 Hz, Single Phase (or equivalent per Battery Charger/Analyzer)
- Electronic Equipment Toolkit
- Face Shield, Industrial
- Shorting Jumpers, Locally-Made, Figure H-246, Appendix H

- Shorting Spring

Materials/Parts

- Corrosion Preventive Compound, Item 110, Appendix D
- Distilled Water, Item 123, Appendix D
- Rubber Gloves, Item 265, Appendix D

References

- Appendix D
- Appendix H
- PARA 9-2-22
- PARA 9-5-31

9-5-35.1 Battery Short-Term Storage (BENCH).

9-5-35.1.1. Short-Term Charged Storage.

NOTE

Nickel-cadmium batteries may be stored in a noncorrosive environment for an unlimited period.

Storage temperature should be between -65° - 120°F (-54° - 49°C) for short-term storage (less than 90 days). The lower the storage temperature, the longer the battery charge retention.

9-5-35.1.2. Shorted-Out Short-Term Storage.

- Make sure battery has been thoroughly inspected and cleaned (PARA 9-2-22).
- Locally make shorting jumpers (Figure H-246, Appendix H).
- Unlatch and remove battery cover.
- Discharge battery (PARA 9-2-22).
- When the discharge is complete, keep the AUTOMATIC CYCLE - MANUAL DISCHARGE switch on the battery conditioner/analyzer in MANUAL DISCHARGE position.

NOTE

Reference battery cell (black top), has a lower capacity than the regular battery cells, and will discharge at a faster rate. When monitoring battery cell voltages in the following steps, check reference battery cell approaches +0.3 vdc, monitor it more closely.

- f. Using digital multimeter, monitor individual battery cell voltages.
- g. When reference battery cell reaches +0.3 vdc, connect shorting jumpers across battery cell terminals.
- h. When individual regular battery cells reach +0.5 vdc, connect shorting jumpers across battery cell terminals.
- i. Using shorting jumpers, continue discharging until 75% (15) of battery cells are shorted. As the remaining battery cells reach +0.5 vdc, connect 1 ohm shorting jumpers across battery cell terminals.
- j. When all battery cells are shorted, turn off and disconnect battery charger/analyzer.
- k. Allow battery to be discharged with battery cells shorted for 24 hours.
- l. Remove shorting jumpers.
- m. Apply a light coat of corrosion preventive compound, Item 110, Appendix D, on receptacle terminals, battery cell terminals, and intercell connectors, as required.
- n. Install shorting spring across electrical receptacle, J2, main terminals.
- o. Install and latch battery cover.
- p. Place battery in storage facility.

9-5-35.1.3. Return to Service.

- a. If battery was stored in a charged state (less than 90 days), perform the following:
 - (1) Allow battery temperature to stabilize between 60°- 80°F (15° - 27°C).
 - (2) Perform 50-hour battery inspection (PARA 9-2-22).

WARNING

- Serious burns will result if contact with the electrolyte (potassium hydroxide) is made. Wear protective clothing (battery workers impermeable apron, industrial face shield, and rubber gloves, Item 265, Appendix D) when working around opened batteries. In case of accident, wash with distilled water, Item 123, Appendix D. Do not wear jewelry that might contact battery cell terminals.
- Explosive gases may be released during charging. Make sure that charging area is well ventilated. Do not use matches or open flame. Avoid short circuits, since arcs may cause explosion. Do not disconnect charging cable from battery until charger is off.

CAUTION

- Damage to nickel-cadmium battery will result if maintenance is performed on a nickel-cadmium battery in a lead-acid battery shop. Never use any tools or materials which have been used on lead-acid batteries.
- Electrolyte is absorbed into battery cell plates and battery will appear dry, when battery is completely discharged. Do not add water.
- Charge battery at an ambient temperature between 40° - 90°F (4.4° - 32.2°C).
- Battery case should never be more than slightly warm to the touch 100°F (37.8°C) during charging. If battery exhibits overtemperature condition during charging, stop charging, discharge battery, and start charging again.
- Do not loosen or remove vent caps during charging. Open battery cells may cause formation of potassium carbonate.
- Never charge batteries with battery cells connected in parallel.

NOTE

Battery charger/analyzer is set up to operate with 115 vac, 50 - 60 Hz, single phase. For other operating voltages refer to instructions on inside top lid of battery charger/analyzer. When 110 - 120 vac is used, a 12 amp fuse must also be used.

- (3) Connect battery charger/analyzer to electrical power source.
- (4) Turn SELECTOR switch to MA-500H.

- (5) Connect battery cable, P2B, between battery and battery charger/analyzer.
- (6) Set AC POWER switch to ON.
- (7) Set AUTOMATIC CYCLE-MANUAL DISCHARGE switch to AUTOMATIC.
- (8) Turn TIMER control to TOPPING charge. Record start time.
- (9) Periodically monitor total battery cell voltage reading on battery charger/analyzer. Stop charging when voltage reaches 30 vdc or 30 minutes of charging elapses.
- (10) If battery does not reach 30 vdc within 30 minutes, recharge battery (PARA 9-5-31).

NOTE

It may be necessary to perform battery reconditioning up to 3 times, on a stored battery, before replacing battery cells.

- (11) Perform 50-hour battery inspection (PARA 9-2-22).
 - (12) Return battery to service.
- b. If battery was stored in a shorted-out condition, perform the following:

- (1) Inspect and clean battery (PARA 9-2-22).
- (2) Perform 100-hour battery inspection (PARA 9-2-22).

NOTE

It may be necessary to deep cycle battery to obtain a full charge.

- (3) Perform 50-hour battery inspection (PARA 9-2-22).
- (4) Return battery to service.

9-5-36 CONTROLLABLE SEARCHLIGHT (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
9-5-36.1	Repair Controllable Searchlight (BENCH)	9-5-36-2

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Arbor Press
- Bearing Puller
- Digital Multimeter
- Drill Bit, 1⁄16-inch diameter
- Drill Press
- Electrical Repairer’s Toolkit
- Magnifying Glass
- Nonmetallic Scraper, Locally-Made
- Power Supply, 28 VDC
- Protection, Eye and Skin
- Searchlight Test Stand, Figure H-249, Appendix H
- Soldering Iron

- Spanner Wrench
- Stopwatch

Materials/Parts

- Acetone, Item 25, Appendix D
- Adhesive, Item 43A, Appendix D
- Grease, Item 149, Appendix D
- Lubricating Oil, Item 205A, Appendix D
- Machinery Towel, Item 211, Appendix D
- Rubber Gloves, Item 265, Appendix D
- Sealing Compound, Item 276, Appendix D
- Solder, Item 313, Appendix D
- Tags

References

- Appendix D
- Appendix H
- MIL-I-6868

9-5-36.1 Repair Controllable Searchlight (BENCH).

9-5-36.1.1 Disassemble Controllable Searchlight.

- a. Remove screws that secure infrared lens and retaining ring to canopy (Figure 9-5-36-1, Sheet 1, Detail B). Remove infrared lens and retaining ring.
- b. Pull lamp away from canopy.
- c. Tag wires. Remove screws and washers securing wires to lamp. Remove wires.
- d. Remove screw and washer securing wire to canopy. Remove wire.
- e. Remove setscrews securing canopy to rotating mechanism. Remove canopy.
- f. Using nonmetallic scraper, remove sealing compound from around edges of extend gear box cover.
- g. Remove screws and washers securing extend gear box cover to extend gear box (Figure 9-5-36-1, Sheet 2, Detail C). Remove extend gear box cover.
- h. Remove screws and washers securing clamp to rotating mechanism. Remove clamp.
- i. Remove nut and washers securing short eyelet, long eyelet, spring, and bumper to baseplate. Remove short eyelet, long eyelet, spring, and bumper.
- j. Remove screw and bumper from baseplate. Remove bumper.
- k. Remove screws, washers, and nut securing extend gear box to baseplate.
- l. Remove screws that secure relay to baseplate. Remove relay.
- m. Remove screws, washers, and nuts securing shaft and block assembly to baseplate. Remove baseplate.

- n. Separate extend gear box, rotating mechanism, and shaft and block assembly.
- o. Remove extend pivot spacers from rotating mechanism.

9-5-36.1.2 Disassemble Controllable Searchlight Rotating Mechanism.

- a. Using nonmetallic scraper, remove sealing compound from around edges of rotating mechanism.
- b. Remove screws and washers securing rotating mechanism cover to rotating mechanism (Figure 9-5-36-1, Sheet 2, Detail C). Remove rotating mechanism cover.
- c. Remove screws securing plate to rotating mechanism (Figure 9-5-36-1, Sheet 3, Detail D). Remove plate.
- d. Remove screws and washers securing sector gear to rotating mechanism. Remove sector gear.

NOTE

Remove actuator only if replacement is required.

- e. If necessary, remove actuator as follows:
 - (1) Remove screws and washers securing actuator to sector gear (only if being replaced).
 - (2) Remove actuator.
 - (3) Remove screw and button from nut on actuator.
- f. Remove screw and washer securing electrical lead to bus bar. Remove electrical lead.
- g. Install screw and washer, after electrical lead is removed, to retain bus bar and insulator until pivot shaft is removed.

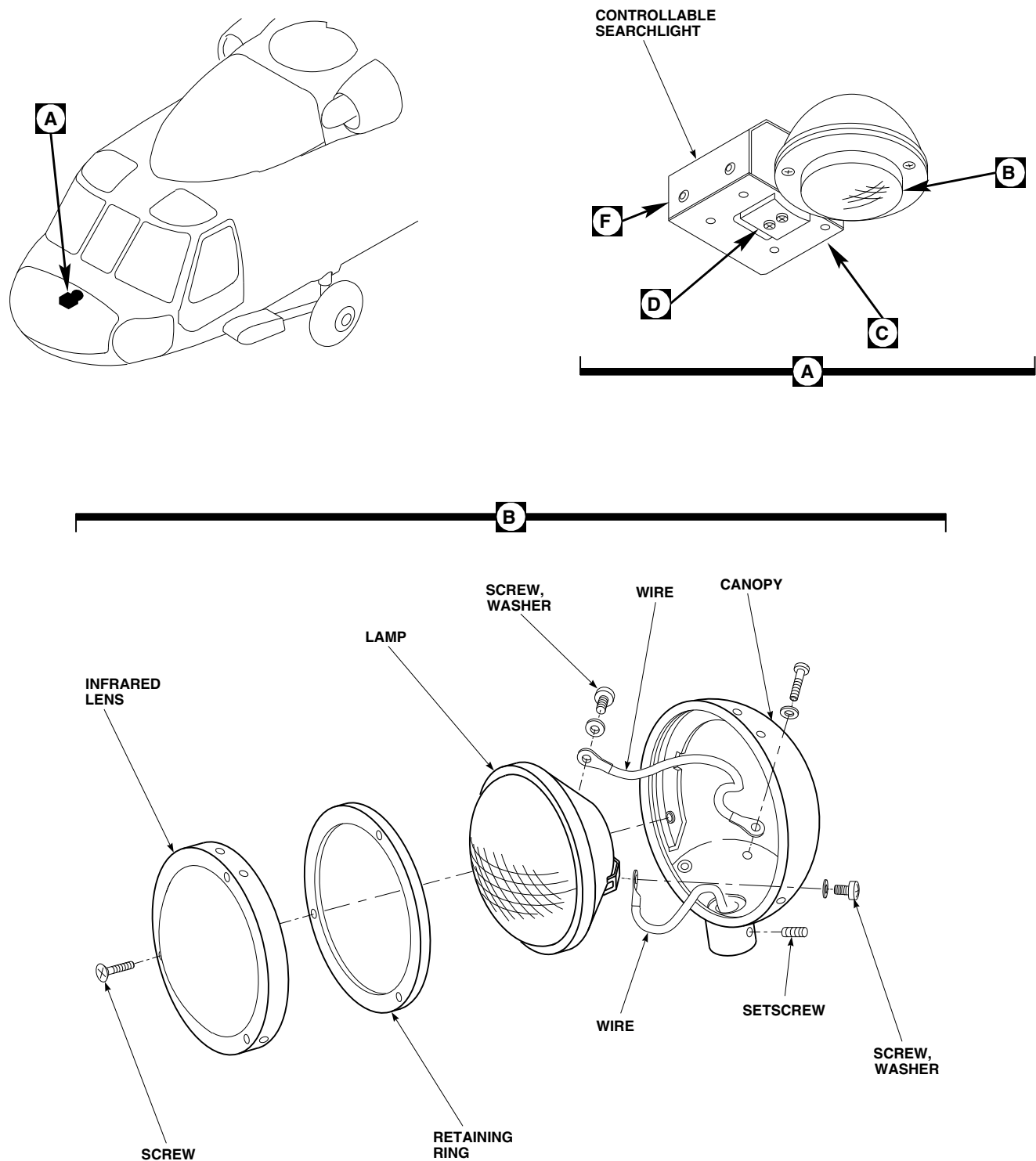


FIGURE 9-5-36-1. REPAIR CONTROLLABLE SEARCHLIGHT (BENCH) (SHEET 1 OF 5)

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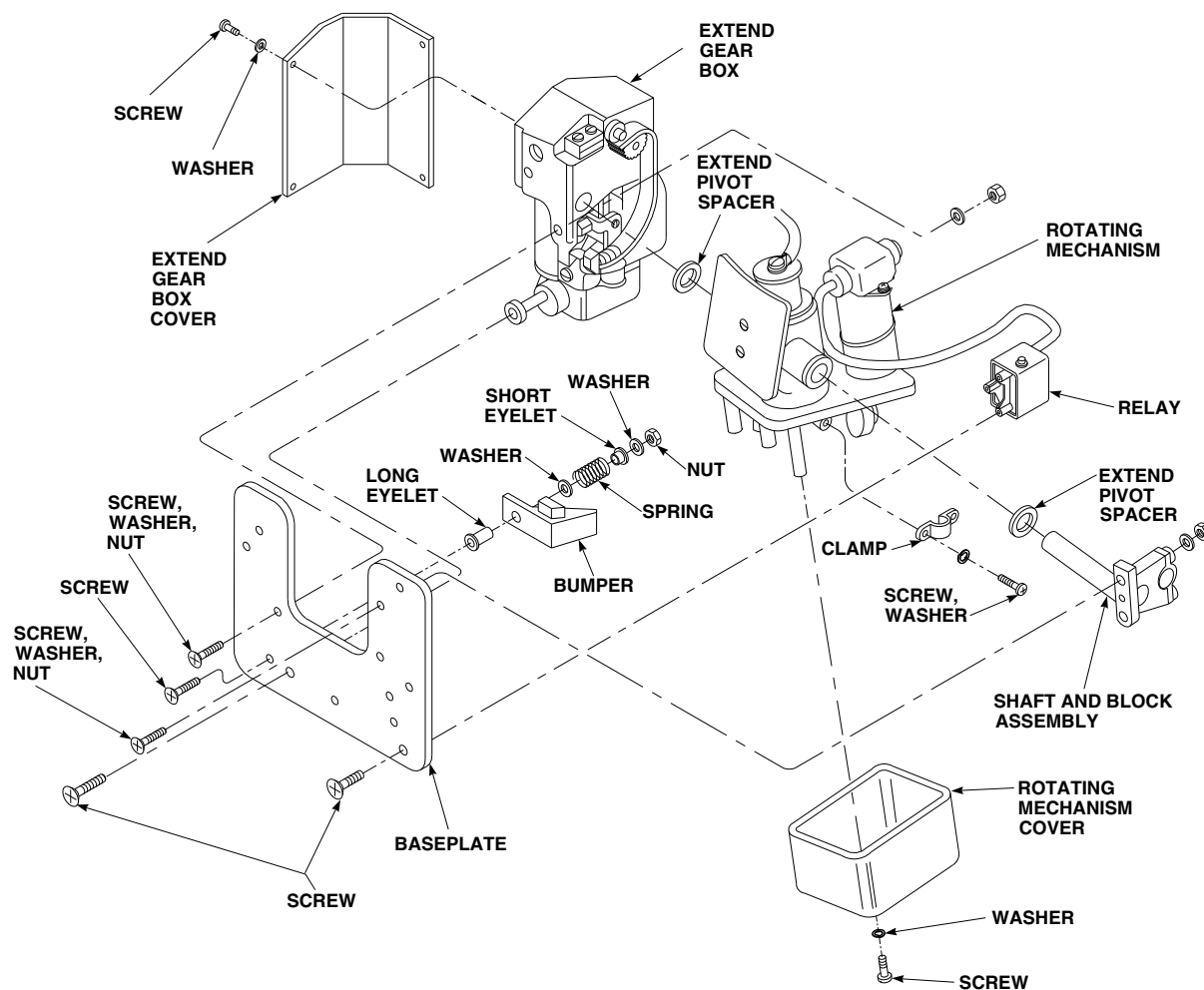
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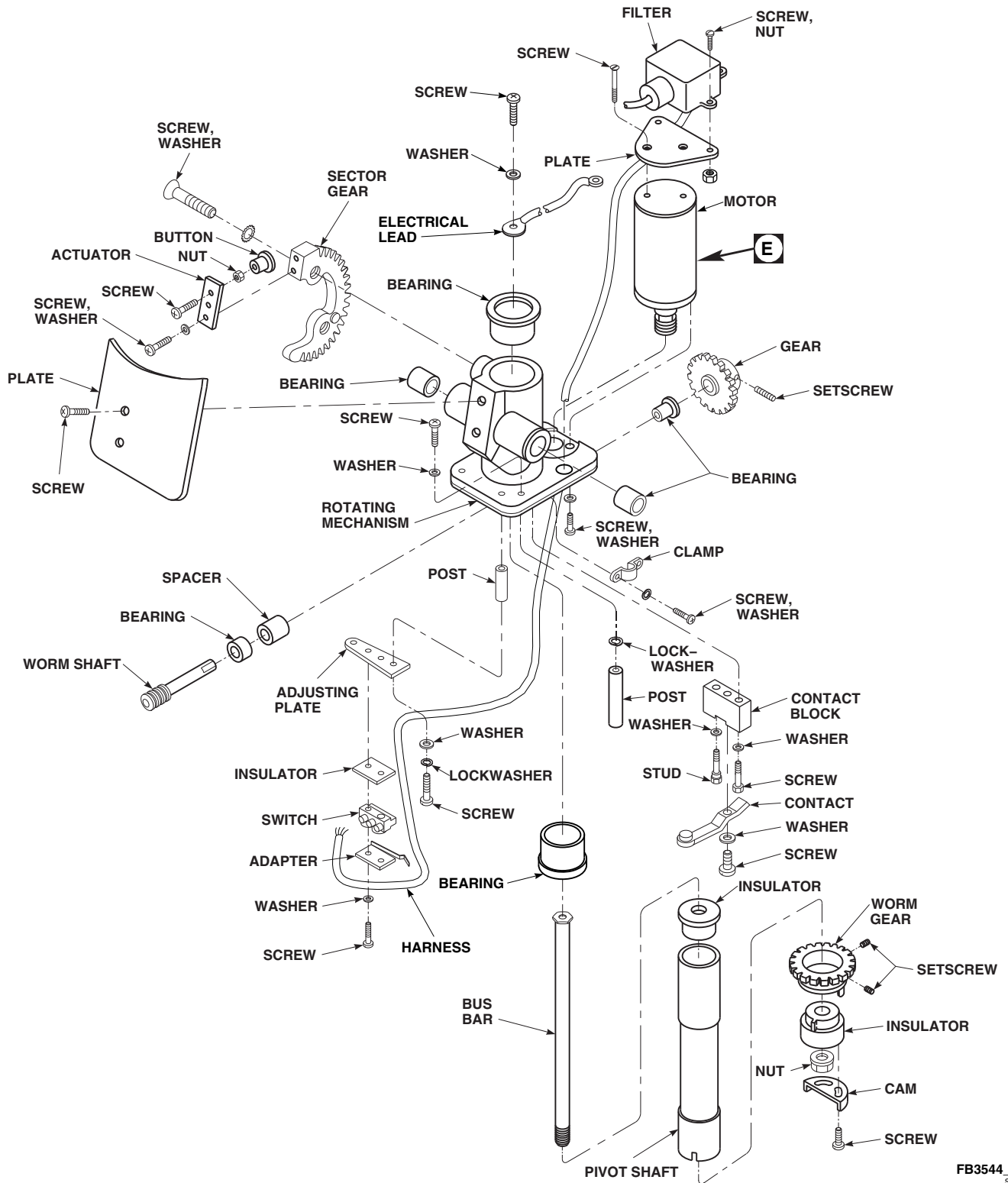
FIGURE 9-5-36-1. REPAIR CONTROLLABLE SEARCHLIGHT (BENCH) (SHEET 2 OF 5)

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- h. Tag wires. Using soldering iron, unsolder four wires from switch. Pull harness through hole in rotating mechanism.
- i. Remove screws and washers securing motor to rotating mechanism. Remove motor.
- j. Using nonmetallic scraper, remove sealing compound from around edges of plate. Remove screws securing filter and plate to motor. Remove filter and plate.
- k. Remove screws and washers securing adapter, switch, and insulator to adjusting plate. Remove adapter, switch, and insulator.
- l. Remove screws, lockwashers, and washers securing adjusting plate and posts to rotating mechanism. Remove adjusting plate and posts.

D



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FIGURE 9-5-36-1. REPAIR CONTROLLABLE SEARCHLIGHT (BENCH) (SHEET 3 OF 5)

E

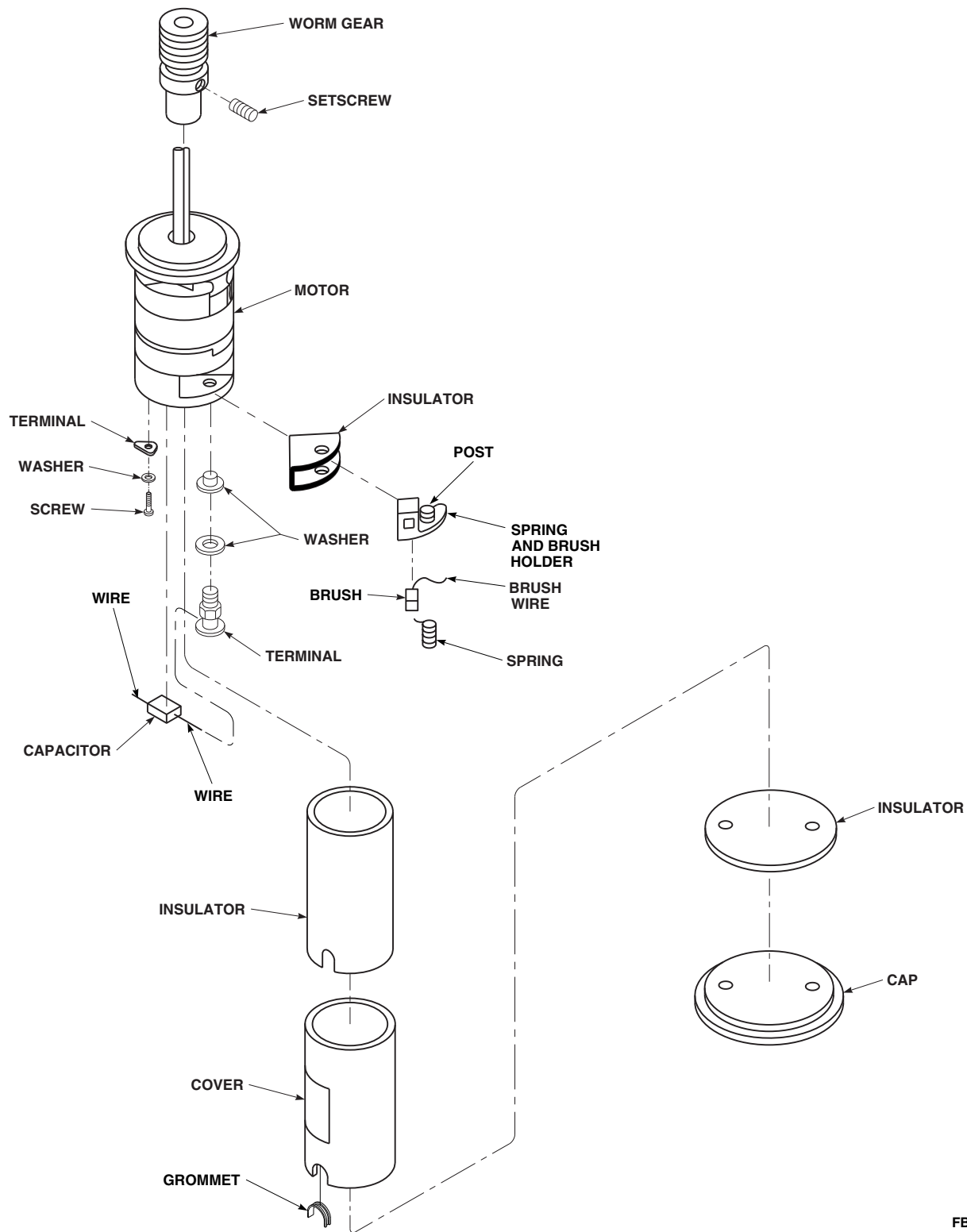
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FIGURE 9-5-36-1. REPAIR CONTROLLABLE SEARCHLIGHT (BENCH) (SHEET 4 OF 5)

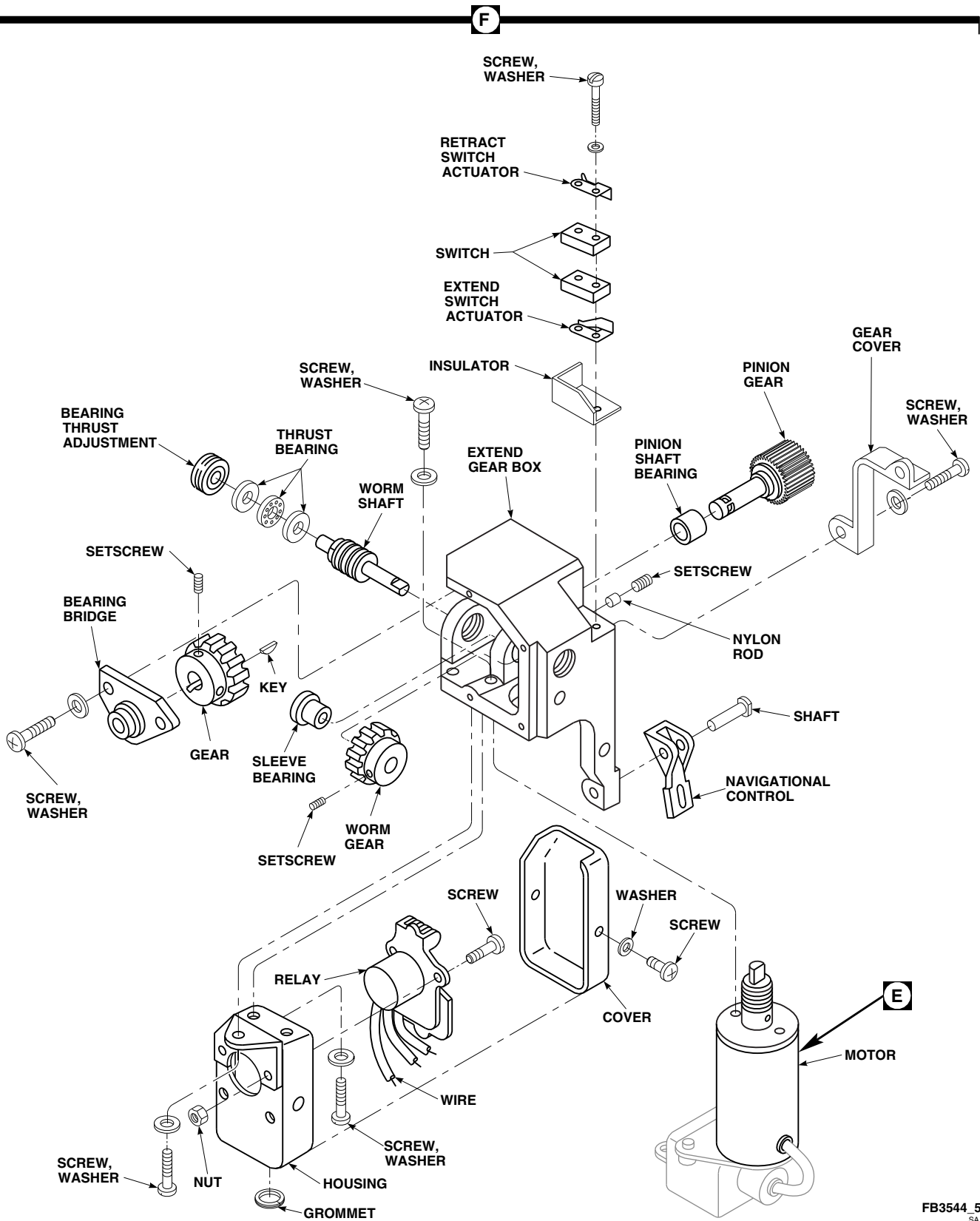


FIGURE 9-5-36-1. REPAIR CONTROLLABLE SEARCHLIGHT (BENCH) (SHEET 5 OF 5)

- m. Remove screw and washer securing contact to contact block. Remove contact.
- n. Remove screw, stud, and washers securing contact block to rotating mechanism. Remove contact block.
- o. Remove screw, lockwasher, and washer securing post to rotating mechanism. Remove post.
- p. Remove screws and washers securing clamp and harness to rotating mechanism. Remove clamp.
- q. Remove setscrews securing gear to worm shaft. Remove gear.
- r. Remove worm shaft from rotating mechanism.
- s. Using bearing puller, remove bearings from rotating mechanism.
- t. Using bearing puller, remove bearing and spacer from rotating mechanism.
- u. Remove screws securing cam to insulator. Remove cam.
- v. Remove nut securing insulators and bus bar to pivot shaft. Remove insulators and bus bar. Make sure not to damage insulators.
- w. Remove setscrews securing worm gear to pivot shaft. Remove worm gear.

9-5-36.1.3 Disassemble Controllable Searchlight Motor.

NOTE

Disassembly of both controllable searchlight motors is the same.

- a. Remove screws and nuts securing filter to plate (Figure 9-5-36-1, Sheet 3, Detail D). Remove filter.

- b. Remove setscrews securing secure worm gear to motor (Figure 9-5-36-1, Sheet 4, Detail E). Remove worm gear.
- c. Remove cap and insulator from cover (Figure 9-5-36-1, Sheet 3, Detail D and Sheet 4, Detail E).
- d. Remove grommet from cover.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

NOTE

Remove capacitor only if replacement is required.

- e. If necessary, remove capacitor as follows (Figure 9-5-36-1, Sheet 4, Detail E):
 - (1) Tag wires.
 - (2) Using soldering iron, unsolder wires from terminals.
 - (3) Remove capacitor.
- f. Remove cover from motor.
- g. Remove insulator from inside of cover.
- h. Remove terminals from motor.
- i. Remove washers, insulators, and spring and brush holders from motor.

- j. Remove screw and washer securing terminal to motor. Remove terminal.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- k. Tag wires. Using soldering iron, unsolder brush wires from posts on spring and brush holders.
- l. Remove brushes and springs.

9-5-36.1.4 Disassemble Controllable Searchlight Extend Gear Box.

- a. Remove screw, washer, retract switch actuator, switches, extend switch actuator, and insulator from extend gear box (Figure 9-5-36-1, Sheet 5, Detail F). Remove retract switch actuator, switches, extend switch actuator, and insulator.
- b. Remove screws and washers securing gear cover on extend gear box. Remove gear cover.
- c. Remove screws and washers securing motor to extend gear box. Remove motor.

NOTE

Remove navigational control only if replacement is required.

- d. If necessary, remove navigational control as follows:
 - (1) Using drill press with 1/16-inch diameter drill bit, drill out shaft.
 - (2) Remove shaft and navigational control.
- e. Remove screws and washers securing bearing bridge. Remove bearing bridge.

- f. Remove setscrews securing gear to extend gear box. Remove gear, and remove key from gear.

NOTE

Remove pinion shaft bearing only if replacement is required.

- g. If necessary, remove pinion shaft bearing as follows:
 - (1) Remove pinion gear.
 - (2) Using arbor press, remove pinion shaft bearing from extend gear box.

NOTE

Remove nylon rod only if replacement is required.

- h. If necessary, remove nylon rod as follows:
 - (1) Remove setscrew securing nylon rod to extend gear box.
 - (2) Remove nylon rod.
- i. Using spanner wrench, remove bearing thrust adjustment from extend gear box.
- j. Remove worm shaft from extend gear box.
- k. Remove thrust bearing (three part assembly) from end of worm shaft.
- l. Remove setscrew securing worm gear in extend gear box and bearing. Remove worm gear.

NOTE

Remove sleeve bearing only if replacement is required.

- m. If necessary, using bearing puller, remove sleeve bearing from extend gear box.
- n. Using nonmetallic scraper, remove sealing compound from around edges of housing.
- o. Remove screws and washers securing cover to housing. Remove cover.

- p. Remove screws and washers securing housing to extend gear box. Remove housing.
- q. Using nonmetallic scraper, remove sealing compound from around edges of relay.
- r. Remove screws and nuts securing relay to housing. Remove relay.
- s. Tag wires. Identify which side of housing wires route through.
- t. Remove grommets from housing.

9-5-36.1.5 Inspect.**NOTE**

- Parts that fail any part of inspection shall be repaired, if possible, or replaced.
- If necessary, a magnifying glass shall be used to perform visual inspections.
- a. Perform inspections of harness as follows:
 - (1) Serviceability (Visual)
 - (a) No loose or damaged electrical connector pins permitted.
 - (b) No cracks or other structural damage to electrical connector housing permitted.
 - (c) No corrosion permitted.
 - (d) No cracked, brittle, or otherwise damaged wire insulation permitted.
 - (e) No cracks or other structural damage in switch bodies permitted.
 - (f) No frayed or corroded shielding permitted.
 - (2) Continuity (Test with digital multimeter)
 - (a) No breaks in continuity permitted.
- b. Perform inspections of the sector gear as follows:

- (1) Serviceability (Visual, and if necessary MIL-I-6868)
 - (a) No cracks or distortion permitted.
 - (b) No other evidence of structural damage permitted.
- (2) Condition of surfaces (Visual)
 - (a) No gouges, cuts, corrosion, burrs, or nicks in sector gear teeth permitted.
 - (b) No other evidence of structural damage permitted.
- (3) Wear limits of gears (Visual)
 - (a) No visible wear indicated by surface pitting, scoring, or a nonuniform sector gear tooth profile that would cause too much backlash permitted.
 - (b) No visible wear indicated by surface pitting, scoring, or a nonuniform sector gear tooth profile that would cause too much sector gear wear permitted.
- (4) Proper location of rivets (Visual)
 - (a) Rivets are secure.
 - (b) Rivets are undamaged.
- c. Perform inspection of the electrical connector as follows:
 - (1) Serviceability (Visual)
 - (a) No damage or distortion permitted.
 - (b) No bent or broken pins permitted.
- d. Perform inspection of the noise filters as follows:
 - (1) Serviceability (Visual)
 - (a) No damage permitted.
 - (b) No distortion permitted.

- e. Perform inspection of the relay as follows:
 - (1) Serviceability (Visual)
 - (a) No damage permitted.
 - (b) No distortion permitted.
- f. Perform inspection of the motors as follows:
 - (1) Serviceability (Visual)
 - (a) No damage permitted.
 - (b) No distortion permitted.
- g. Perform inspection of the baseplate as follows:
 - (1) Serviceability (Visual)
 - (a) No damage or distortion to mounting holes permitted.
 - (b) No damage or distortion to baseplate permitted.
- h. Perform inspection of the canopy as follows:
 - (1) Serviceability (Visual)
 - (a) No damage which would impair form, fit, or function permitted.
 - (b) No distortion which would impair form, fit, or function permitted.
- i. Perform inspection of the extend gear box as follows:
 - (1) Serviceability (Visual)
 - (a) No damage which would impair form, fit, or function permitted.
 - (b) No distortion which would impair form, fit, or function permitted.
- j. Perform inspection of the rotating mechanism cover as follows:
 - (1) Serviceability (Visual)
 - (a) No damage which would impair form, fit, or function permitted.
 - (b) No distortion which would impair form, fit, or function permitted.
- k. Perform inspection of the lens assembly as follows:
 - (1) Serviceability (Visual)
 - (a) No damage which would impair form, fit, or function permitted.
 - (b) No distortion which would impair form, fit, or function permitted.

9-5-36.1.6 Assemble Controllable Searchlight Motor.

NOTE

Assembly of both controllable searchlight motors is the same.

- a. Install two springs in spring and brush holders (Figure 9-5-36-1, Sheet 4, Detail E).
- b. Install two brushes in spring and brush holders.
- c. Guide brush wires through spring slots in brush and spring and brush holders.
- d. Using soldering iron and solder, Item 313, Appendix D, solder two wires to ends of posts on spring and brush holders.
- e. Install terminal on motor and secure with screw and washer.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- f. Install two spring and brush holders and insulators in motor.
- g. Install two terminals and washers.
- h. If removed, connect capacitor across terminals. Do not solder.
- i. Install insulator inside cover and install insulator and cover on motor.
- j. Install grommet in opening at end of cover.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- k. Using soldering iron and solder, Item 313, Appendix D, solder motor wires and capacitor to terminals. Remove tags.
- l. Using soldering iron and solder, Item 313, Appendix D, solder two shields to terminal.
- m. Guide motor wires through grommet.
- n. Install insulator and cap on cover and secure with screws.
- o. Install worm gear and secure with setscrews.
- p. Using acetone, Item 25, Appendix D, machinery towel, Item 211, Appendix D, and rubber gloves, Item 265, Appendix D, clean edges of cap, plate, and grommet.
- q. Using rubber gloves, Item 265, Appendix D, apply sealing compound, Item 276, Appendix D, to seal edges of cap, plate, and grommet.
- r. Install filter on plate and secure with screws and nuts (Figure 9-5-36-1, Sheet 3, Detail D).

9-5-36.1.7 Assemble Controllable Searchlight Rotating Mechanism.

- a. Install worm gear onto indexed end of pivot shaft (Figure 9-5-36-1, Sheet 3, Detail D).
- b. Install worm gear in rotating mechanism and secure with two setscrews.
- c. Install top insulator onto pivot shaft.
- d. Install bus bar into pivot shaft.
- e. Install bottom insulator onto pivot shaft and secure with nut.
- f. Install cam onto insulator and secure with two screws.
- g. Using arbor press, install two bearings in rotating mechanism.
- h. Using arbor press, install bearings and spacer in rotating mechanism.
- i. Using lubricating oil, Item 205A, Appendix D, lightly coat pivot shaft and install in rotating mechanism.
- j. Using lubricating oil, Item 205A, Appendix D, lightly coat worm shaft and install in rotating mechanism.
- k. Using lubricating oil, Item 205A, Appendix D, lightly coat gear and install on end of worm shaft and secure with two setscrews.
- l. Install contact block in rotating mechanism and secure with screw, stud, and washers.
- m. Install contact on contact block and secure with screw and washer.
- n. Install post in rotating mechanism and secure with screw, lockwasher, and washer.
- o. Install clamp on rotating mechanism and secure with screws and washers.
- p. Install two posts and adjusting plate on rotating mechanism and secure with screws, lockwashers, and washers.

- q. Install switch, adapter, and insulator and secure with screws and washers.
- r. Guide harness through hole and clamp in rotating mechanism and clamp. Install motor on rotating mechanism and secure with screws and washers.
- s. Position plate on motor and secure with screws.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- t. Using soldering iron and solder, Item 313, Appendix D, solder four wires of harness to switch. Remove tags.
- u. Remove screw and washer securing electrical lead to bus bar on rotating mechanism. Position electrical lead to top of bus bar and secure with screw and washer.
- v. If removed, install actuator as follows:
 - (1) Install screw, nut, and button on actuator.
 - (2) Install actuator to sector gear and secure with screws and washers.
- w. Install sector gear to rotating mechanism and secure with screws and washers.
- x. Install plate to rotating mechanism and secure with screws.
- y. Install rotating mechanism cover to rotating mechanism and secure with screws and washers (Figure 9-5-36-1, Sheet 2, Detail C).

9-5-36.1.8 Assemble Controllable Searchlight Extend Gear Box.

- a. Install two rubber grommets in housing (Figure 9-5-36-1, Sheet 5, Detail F).

- b. Install relay in housing and secure with screws and nuts. Guide the three wires from relay through two rubber grommets in housing. Remove tags.
- c. Install housing on extend gear box and secure with screws and washers.
- d. Install cover on housing and secure with screws and washers.
- e. If removed, using arbor press, install sleeve bearing and pinion shaft bearing in extend gear box.
- f. Install thrust bearing (three part assembly) on end of worm shaft.
- g. Apply grease, Item 149, Appendix D, to worm gear and install worm gear in extend gear box and hold in position. Install worm shaft into extend gear box, through sleeve bearing and into worm gear.
- h. Install bearing thrust adjustment on worm shaft. **USING SPANNER WRENCH, TIGHTEN SO THAT WORM SHAFT TURNS FREELY WITH MINIMAL END PLAY.**
- i. If removed, install nylon rod in extend gear box and secure with setscrew.
- j. Install setscrews in worm gear and secure to end of worm shaft.
- k. Apply grease, Item 149, Appendix D, to pinion gear and install pinion gear in extend gear box.
- l. Apply grease, Item 149, Appendix D, to gear and install gear, with key installed, on pinion shaft and secure with setscrews.
- m. Install bearing bridge in extend gear box and secure with screws and washers.
- n. If removed, install navigational control as follows:
 - (1) Install navigational control and shaft in extend gear box.

- (2) Using arbor press, crimp end of shaft.
- o. Install motor in extend gear box and secure with screws and washers.
- p. Apply grease, Item 149, Appendix D, to gear and install gear cover on extend gear box and secure with screws and washers.
- q. Install insulator, extend switch actuator, switches, and retract switch actuator on extend gear box and secure with screws and washers.
- r. Install extend gear box cover and secure with screws and washers (Figure 9-5-36-1, Sheet 2, Detail C).
- s. Using acetone, Item 25, Appendix D, machinery towel, Item 211, Appendix D, and rubber gloves, Item 265, Appendix D, clean mating edges of extend gear box, extend gear box cover, housing, relay, and two grommets (Figure 9-5-36-1, Sheet 2, Detail C and Sheet 5, Detail F).
- t. Using rubber gloves, Item 265, Appendix D, apply sealing compound, Item 276, Appendix D, to seal edges of extend gear box cover and housing, and edges of relay which extend through housing.
- f. Install extend gear box to baseplate and secure with three screws, washers, and nuts.
- g. Guide wires from rotating mechanism through shaft and block assembly.
- h. Guide wires from extend gear box on baseplate to relay. Position wires along relay with wires from conduit bracket.
- i. Install clamp on shaft and block assembly and secure with screw and washer.
- j. Install extend gear box cover and secure with four screws and washers.
- k. Using rubber gloves, Item 265, Appendix D, apply sealing compound, Item 276, Appendix D, to seal edges of extend gear box cover.
- l. Install screw through front of baseplate and hold in place. On the back side of baseplate install the following on screw: long eyelet, bumper, washer, spring, short eyelet, and washer and secure with nut.
- m. Install canopy on rotating mechanism with wire inside canopy and secure with setscrews (Figure 9-5-36-1, Sheet 1, Detail B).
- n. Connect wire to canopy and secure with screw and washer.
- o. Position lamp in canopy, connect wires to lamp and secure with screws and washers. Remove tags.
- p. Install retaining ring and infrared lens on canopy and secure with screws.

9-5-36.1.9 Assemble Controllable Searchlight.

- a. Install shaft and block assembly and two extend pivot spacers on rotating mechanism (Figure 9-5-36-1, Sheet 2, Detail C).
- b. Install extend gear box on end of shaft and block assembly.
- c. Install extend gear box, rotating mechanism, shaft and block assembly, and relay on baseplate.
- d. Install relay to baseplate and secure with three screws. Using adhesive, Item 43A, Appendix D, seal screws.
- e. Install shaft and block assembly to baseplate and secure with two screws, washers, and nuts.

9-5-36.1.10 Test and Adjust Controllable Searchlight.

- a. Locally make searchlight test stand (Figure H-249, Appendix H).
- b. Connect searchlight test stand to controllable searchlight.
- c. Turn off electrical power. Adjust power supply output to 28 VDC.

WARNING

Injury to personnel may result if personnel look directly into infrared light. Do not look into lighted NVG controllable searchlight. Infrared light output can seriously injure eyes.

NOTE

To verify that NVG controllable searchlight is lit, check for heat output.

- d. Set power switch to ON. Check lamp. Lamp shall come on.
- e. Set power switch to OFF. Check lamp. Lamp shall go off.
- f. Set control switch to EXTEND. Check controllable searchlight. Controllable searchlight shall fully extend and stop.
- g. Measure length of spring. Record length.
- h. Set control switch to RETRACT. Check controllable searchlight. Controllable searchlight shall fully retract and stop.
- i. Check bumper spring. Bumper shall have compressed bumper spring $\frac{1}{16}$ -inch maximum. If bumper spring is not compressed, adjust controllable searchlight as follows:
 - (1) Remove two screws and cover plate.
 - (2) Turn adjustment screw counterclockwise to increase spring compression. Turn adjustment screw clockwise to decrease spring compression.
 - (3) Set control switch to EXTEND and RETRACT three times.
 - (4) Check bumper. Bumper shall have compressed bumper spring $\frac{1}{16}$ -inch maximum. If necessary, readjust.
- (5) Position cover plate and install screws.
- j. Set control switch to RIGHT. Check controllable searchlight. Controllable searchlight shall rotate to the right. Using stopwatch, record time of revolution. Complete revolution to right shall take no longer than 40 seconds.
- k. Set control switch to LEFT. Check controllable searchlight. Controllable searchlight shall rotate to the left. Using stopwatch, record time of revolution. Complete revolution to left shall take no longer than 40 seconds.
- l. Set control switch to EXTEND. Check controllable searchlight. Controllable searchlight shall fully extend and stop. Using stopwatch, record time of extension. Complete extension shall take no longer than 20 seconds.
- m. Set control switch to RETRACT. Check controllable searchlight. Controllable searchlight shall fully retract and stop. Using stopwatch, record time of retraction. Complete retraction shall take no longer than 20 seconds.
- n. Set control switch to EXTEND and fully extend controllable searchlight.
- o. Set control switch to RIGHT or LEFT and partially rotate controllable searchlight.
- p. Set control switch to RETRACT. Check controllable searchlight. Controllable searchlight shall fully retract and rotate to stowed position.
- q. Disconnect power supply.
- r. Disconnect searchlight test stand.
- s. Using rubber gloves, Item 265, Appendix D, apply sealing compound, Item 276, Appendix D, to seal exposed areas.

9-5-37 INDICATOR LIGHTS DIMMER BOX (BENCH).

PARAGRAPH BREAKDOWN

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9-5-37.1 Replace Indicator Lights Dimmer Box (BENCH)	9-5-37-1

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit
- Heat Gun
- Insert/Extract Tool
- Soldering Iron

Materials/Parts

- Heat-Shrink Insulation Sleeving
- Solder, Item 313, Appendix D

- Tags

References

- Appendix D
- PARA 9-2-10

9-5-37.1 Replace Indicator Lights Dimmer Box (BENCH).

9-5-37.1.1. Disassemble.

- a. To remove cover, perform the following:
 - (1) Remove screws, lockwashers, and washers securing cover to indicator lights dimmer box (Figure 9-5-37-1).
 - (2) Remove cover.
- b. To remove plate, perform the following:
 - (1) Remove cover per step a.
 - (2) Tag ground wire and remove screw, washers, and nut securing ground wire to inside of indicator lights dimmer box. Move ground wire out of the way.

- (3) Remove screws, washers, and nuts securing plate to indicator lights dimmer box.
- (4) Remove plate.
- c. To remove plug-in socket, perform the following:
 - (1) Remove cover per step a.
 - (2) Remove dust cap and remove screws securing transistor, Q1, insulator plate, and plug-in socket to indicator lights dimmer box.
 - (3) Remove screws, washers, and nuts securing plug-in socket to indicator lights dimmer box.
 - (4) Tag wires and using soldering iron, unsolder wires from plug-in socket (Figure 9-5-37-2).

- (5) Remove fuse cartridge, F1.
- d. To remove electrical connector, J393, perform the following:
 - (1) Remove cover per step a.
 - (2) Tag wires and using insert/extract tool, remove wires from electrical connector, J393 (Figure 9-5-37-2).
 - (3) Remove screws, washers, and nuts securing electrical connector, J393, to indicator lights dimmer box (Figure 9-5-37-1).
 - (4) Remove electrical connector, J393.
- e. To remove fuse cartridge, F1, perform the following:
 - (1) Tag wires. Remove heat-shrinkable insulation sleeving from terminals on fuse cartridge, F1.
 - (2) Using soldering iron, unsolder wires from fuse cartridge, F1.
 - (3) Push and turn fuse holder cap counterclockwise and remove fuse holder cap (Figure 9-5-37-1, Detail A).
 - (4) Remove fuse from fuse holder.
- f. To remove resistor, R1, perform the following:
 - (1) Remove cover per step a.
 - (2) Remove screws, washers, and nuts securing resistor, R1, to indicator lights dimmer box.
 - (3) Tag wires and using soldering iron, unsolder wires from resistor, R1 (Figure 9-5-37-2).
 - (4) Remove resistor, R1.
- g. To remove resistor, R2, or semiconductors, VR1 and VR2, perform the following:

NOTE

Remove procedure for resistor, R2, or semiconductors, VR1 and VR2 is the same.

- (1) Remove cover per step a.
- (2) Tag resistor or semiconductor leads and using a soldering iron, unsolder resistor or semiconductor leads from terminals (Figure 9-5-37-2).
- (3) Remove resistor or semiconductor.

9-5-37.1.2. Assemble.**NOTE**

Install procedure for resistor, R2, or semiconductors, VR1 and VR2 is the same.

- a. To install resistor, R2, or semiconductors, VR1 and VR2, perform the following:
 - (1) Install resistor or semiconductor between terminals on back of cover (Figure 9-5-37-1, Detail B).
 - (2) Using soldering iron and solder, Item 313, Appendix D, solder resistor or semiconductor leads to terminals (Figure 9-5-37-2). Remove tags.
 - (3) Make sure area is clean and free of foreign material.
 - (4) Install cover per step g.
- b. To install resistor, R1, perform the following:
 - (1) Using soldering iron and solder, Item 313, Appendix D, solder wires to resistor, R1 (Figure 9-5-37-2). Remove tags.
 - (2) Install resistor, R1, in indicator lights dimmer box and secure with screws, washers, and nuts (Figure 9-5-37-1, Detail A).
 - (3) Make sure area is clean and free of foreign material.
 - (4) Install cover per step g.
- c. To install fuse cartridge, F1, perform the following:

- (1) Install fuse in fuse holder (Figure 9-5-37-1, Detail A).
 - (2) Using soldering iron and solder, Item 313, Appendix D, solder wires to fuse cartridge, F1. Remove tags.
 - (3) Using appropriate heat-shrink insulation sleeving, cover solder terminals.
 - (4) Using heat gun, shrink heat-shrink insulation sleeving.
 - (5) Install fuse holder cap on fuse holder by pushing fuse holder cap and turning it clockwise.
- d. To install electrical connector, J393, perform the following:
- (1) Using insert/extract tool, install wires into electrical connector, J393 (Figure 9-5-37-2). Remove tags.
 - (2) Install electrical connector, J393, to indicator lights dimmer box and secure with screws, washers, and nuts (Figure 9-5-37-1).
 - (3) Make sure area is clean and free of foreign material.
 - (4) Install cover per step g.
- e. To install plug-in socket, perform the following:
- (1) Using soldering iron and solder, Item 313, Appendix D, solder wires to fuse cartridge, F1 (Figure 9-5-37-2). Remove tags.
 - (2) Install plug-in socket (to inside of indicator lights dimmer box) and secure with screws, washers, and nuts (Figure 9-5-37-1).
 - (3) Install insulator plate and transistor, Q1, to indicator lights dimmer box and secure with screws. Install dust cap.
 - (4) Make sure area is clean and free of foreign material.
 - (5) Install cover per step g.
- f. To install plate, perform the following:
- (1) Install plate on indicator lights dimmer box and secure with screws, washers, and nuts.
 - (2) Install ground wire to inside of indicator lights dimmer box and secure with screw, washers, and nut. Remove tag.
 - (3) Make sure area is clean and free of foreign material.
 - (4) Install cover per step g.
- g. To install cover, perform the following:
- (1) Make sure area is clean and free of foreign material.
 - (2) Install cover on indicator lights dimmer box and secure with screws, lockwashers, and washers.
- h. Do an operational check of dimming control (PARA 9-2-10).

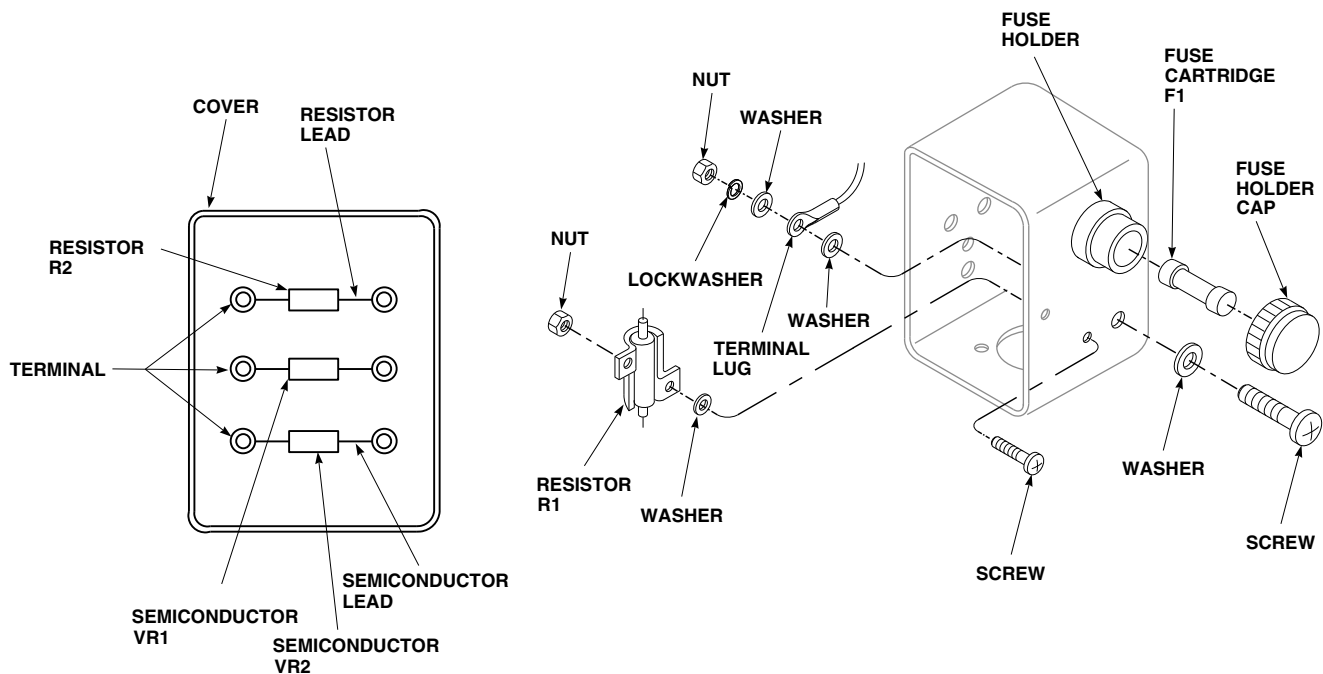
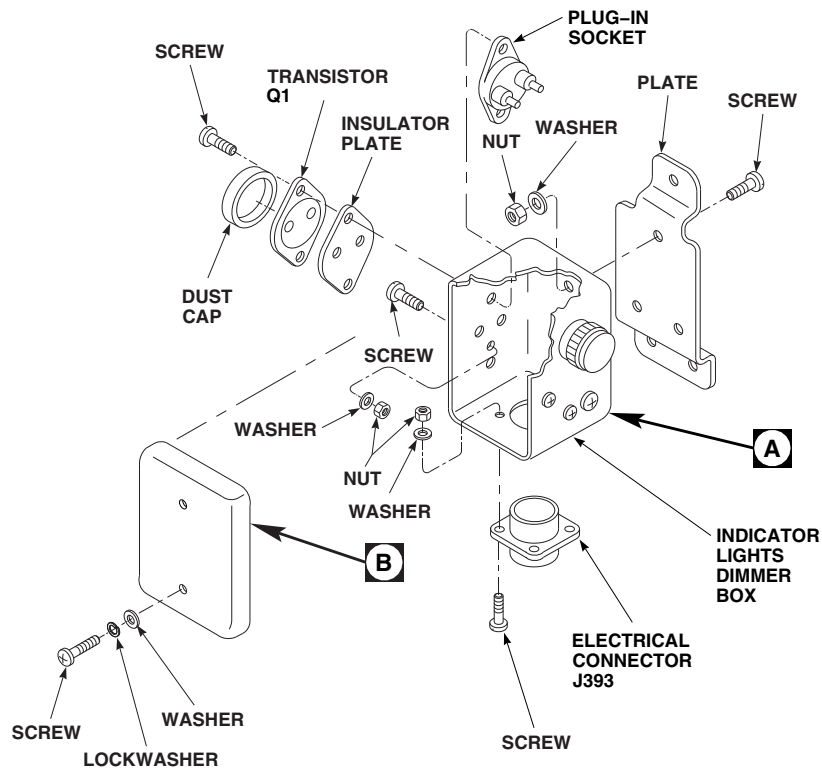
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FIGURE 9-5-37-1. REPLACE INDICATOR LIGHTS DIMMER BOX (BENCH)

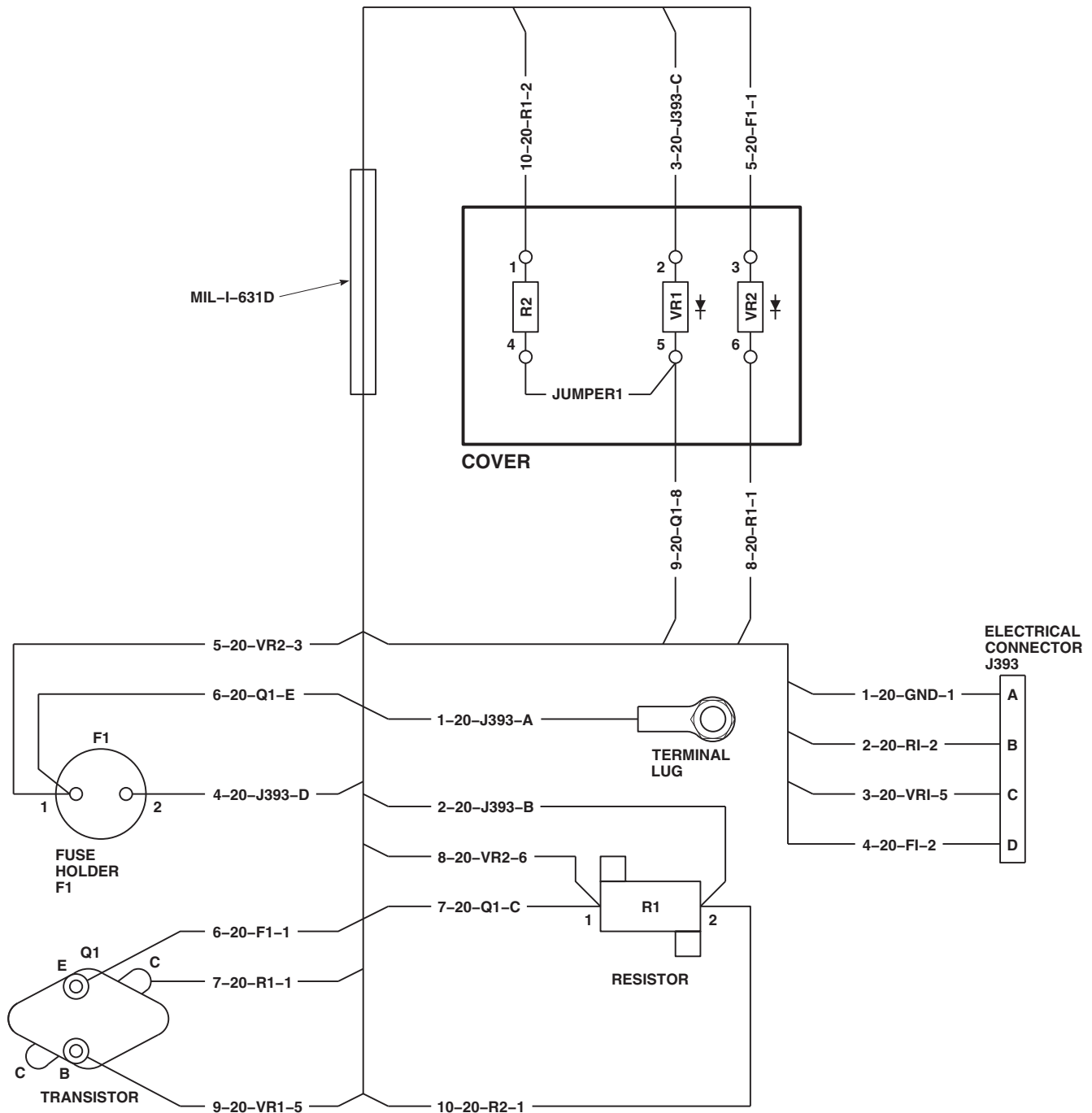


FIGURE 9-5-37-2. REPLACE INDICATOR LIGHTS DIMMER BOX (BENCH)

9-5-37-5/(9-5-37-6 Blank)

